DesignWare® Cores Enterprise Ethernet PCS

Databook

DWC Enterprise Ether PCS – Product Code: 5528-0
DWC AC Ethernet 25G PCS – Product Code: H644-0
DWC Data Integrity for 5528-0 Add-On – Product Code: H444-0

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Revision History

Revision history for DesignWare® Cores DWC_xlgpcs Databook.

Version Number	Date	Description
2.20a	October 2023	Added: ■ Chapter 3, “Registers Overview” ■ The following to Chapter 1, “Product Overview”: - Section “Area and Power Numbers” on page 24 - Section “Controller Requirements” on page 24 ■ The following to Chapter 2, “Architecture”: - Figure 2-14 - Sub-section “RS(544,514) Encoder” to “Reed Solomon FEC Encoder” on page 56 - Section “100G Interleaved RS-FEC Bit Ordering” on page 72 - Section “MDIO Interface” on page 109 - Section “Timestamping Accuracy” on page 151 - Section “Debug Bus” on page 154 ■ The following to Chapter 8, “Programming Guidelines”: - Section “Switching to 100GBASE-KR1 or 100GBASE-CR1 (PAM4) Mode” on page 1323 - Section “Switching to 5G Speed Mode” on page 1324 Updated: ■ Product information on cover page ■ “Preface” on page 11 ■ Chapter 1, “Product Overview” ■ Chapter 2, “Architecture” ■ Note on page 172 ■ Chapter 5, “Parameter Descriptions”, Chapter 6, “Signal Descriptions”, Chapter 7, “Register Descriptions”, and Appendix D, “Internal parameter descriptions” ■ Chapter 8, “Programming Guidelines” ■ Appendix A, “Area, Power, and Latency” ■ Appendix B, “DWC_xlgpcs Automotive Safety”

Version Number	Date	Description
2.10a	July 2022	Added: ■ Chapter 4, “Automotive Safety Features” ■ Section “50GBASE-KR” on page 37 ■ Section “50GBASE-CR” on page 37 ■ Section “100GBASE-KR2” on page 38 ■ Section “100GBASE-CR2” on page 38 Updated: ■ Section “Initializing the DWC_xlgpcs Controller” on page 1284 ■ Section “Switching to 10G Speed Mode” on page 1286 ■ Section “Switching to 40G Speed Mode” on page 1290 ■ Section “Switching to 20G Speed Mode” on page 1295 ■ Section “Switching to 25G Speed Mode” on page 1298 ■ Section “Switching to 50G Speed Mode” on page 1309 ■ Section “” on page 1315 ■ Section “Switching to 5G Speed Mode” on page 1324
2.00a	August 2017	First version of the book

Preface

This document describes Synopsys' DesignWare Enterprise Ethernet PCS Controller, which extends the operational distance of the 5G, 10G, 20G, 25G, 40G, 50G, and 100G Media-Independent Interface while reducing the number of interface signals required, in conformity with the IEEE 802.3 specification.

The DWC Enterprise Ethernet PCS corresponds to DWC Enterprise Ether PCS in the SolvNetPlus database.

This chapter has the following sections:

- "Databook Organization" on page 12
- "Web Resources" on page 13
- "Reference Documentation" on page 15
- "STAR on the Web (SotW)" on page 14
- "Reference Documentation" on page 15
- "Synopsys Statement on Inclusivity and Diversity" on page 16
- "Customer Support" on page 17

Databook Organization

The chapters of this databook are organized as follows:

- Chapter 1, "Product Overview," describes overall architecture, provides an overview of the XLGPCS protocol, and lists product features, deliverables, and general requirements.
- Chapter 2, "Architecture," describes in detail the blocks and data paths of the DWC_xlgpcs IP.
- Chapter 3, "Registers Overview," describes the register access types and the writing conventions used in field descriptions.
- Chapter 4, "Automotive Safety Features," provides details of the automotive safety features.
- Chapter 5, "Parameter Descriptions," provides detailed descriptions for all configuration options and parameters.
- Chapter 6, "Signal Descriptions," provides the pinout and describes the I/O for the DWC_xlgpcs IP.
- Chapter 7, "Register Descriptions," describes the DWC_xlgpcs register address space.
- Chapter 8, "Programming Guidelines," provides instructions for switching from one speed mode to another.
- Appendix A, "Area, Power, and Latency," provides area and power numbers for various configurations/features.
- Appendix B, "DWC_xlgpcs Automotive Safety," provides details of the automotive safety packages.
- Appendix C, "External Memory Requirements," provides details of external memories required for various configurations.
- Appendix D, "Internal Parameter Descriptions," lists the internal parameters of the DWC_xlgpcs IP.
- Glossary, "Glossary," provides the list of terms used and their meaning.

Web Resources

For more information, check the following Synopsys online resources:

- DesignWare IP product information: https://www.synopsys.com/designware-ip.html
- Your custom DesignWare IP page: https://www.synopsys.com/dw/mydesignware.php
- Documentation through SolvNetPlus: https://solvnetplus.synopsys.com (Synopsys password required)
- Synopsys Common Licensing (SCL): https://www.synopsys.com/support/licensing-installation-computeplatforms/licensing.html

STAR on the Web (SotW)

You must review all STARs on the Web (SotWs) associated with your product. SotWs are considered a part of the Synopsys documentation suite, and show critical information related to your product. To review product SotWs, refer to the DesignWare IP product information:

https://www.synopsys.com/designware-ip.html

You can review a list of STARs for DWC_xlgpcs at:

https://www.synopsys.com/dw/star.php?c=DWC_ether_xlgpcs

Reference Documentation

- *IEEE 802.3 Interpretation Document*, Nos. 1-0304, 1-0798, 2-0702, and 5-1103, October 2005.
- IEEE Std 802.3-2018, Clause 45, 46, 49, 69, 72, 73, 74, 81, 82, 83, 91, 92, and 93 (Depending on your configuration).
- IEEE 802.3az-2010
- IEEE 802.3ba-2010
- IEEE P802.3by June 2016
- IEEE 802.3cd-2018
- 25G consortium specification 1.6 October 2015
- *Reuse Methodology Manual for System-on-a-Chip Designs* by Michael Keating and Pierre Bricaud (Editors). Springer; 3rd edition, June 30, 2002.
- *AMBA 3 APB Protocol*, version 1.0, ARM Limited

Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

Customer Support

To obtain support for your product, prepare the required files and contact the support center using one of the methods described:

- Prepare the following debug information, if applicable:
  - For environment setup problems or failures with configuration, simulation, or synthesis that occur within coreConsultant or coreAssembler, select the following menu:
    - **File > Build Debug Tar-file**

    Check all the boxes in the dialog box that apply to your issue. This option gathers all the Synopsys product data needed to begin debugging an issue and writes it to the <*core tool startup directory*>/debug.tar.gz file.
  - For simulation issues outside of coreConsultant or coreAssembler:
    - Create a waveform file (such as VPD, VCD, or FSDB).
    - Identify the hierarchy path to the DesignWare instance.
    - Identify the timestamp of any signals or locations in the waveforms that are not understood.
- For the fastest response, enter a case through SolvNetPlus:

  a. https://solvnetplus.synopsys.com

  
  Note SolvNetPlus does not support Internet Explorer. Use a supported browser such as Microsoft Edge, Google Chrome, Mozilla Firefox, or Apple Safari.

  b. Click the **Cases** menu and then click **Create a New Case** (below the list of cases).

  c. Complete the mandatory fields that are marked with an asterisk and click **Save**.

  Make sure to include the following:
  - **Product L1:** DesignWare Cores
  - **Product L2:** Enterprise Ethernet

  d. After creating the case, attach any debug files you created.

  For more information about general usage information, refer to the following article in SolvNetPlus:

  https://solvnetplus.synopsys.com/s/article/SolvNetPlus-Usage-Help-Resources
- Or, send an e-mail message to mailto:support_center@synopsys.com (your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
  - Include the Product L1 and Product L2 names, and Version number in your e-mail so it can be routed correctly.
  - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood.
  - Attach any debug files you created.
- Or, telephone your local support center:

- North America:
  Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
- All other countries:
  https://www.synopsys.com/support/global-support-centers.html

1 Product Overview

This chapter introduces the Synopsys DesignWare Cores Enterprise Ethernet PCS (DWC_xlgpcs) features and deliverables. It contains the following sections:

- "General Product Description" on page 20
- "Standards Compliance" on page 20
- "List of Features" on page 21
- "Area and Power Numbers" on page 24
- "Controller Requirements" on page 24
- "Deliverables" on page 24
- "PCS Layer Overview" on page 24
- "Link Types Supported Through PMA Layer" on page 34

1.1 General Product Description

The Physical Coding Sublayer (PCS) layer provides an interface between the Media Access Control (MAC) and Physical Medium Attachment Sublayer (PMA) through a Media independent interface. The DWC_xlgpcs IP supports the following functions as per IEEE 802.3 and 25G consortium specifications (for 25G and 50G modes).

- PCS sub layer functions
- PMA sub layer functions
- FEC functions
- Link Training
- Auto Negotiation
- Host Interface via APB-3

PCS sub layer functions include 64/66b line coding and scrambler operations. The 64/66b encoding aids clock and data recovery. Scrambling is a PRBS function which ensures no continuous stream of 1's and 0's is present in the 64/66b encoded data.

PMA sub layer functions typically include various transmit and receive test pattern functions and local loopback operations.

FEC functions are supported for erroneous channels. Typically for links whose BER is < 10^{-12}, FEC functions need not be used. For links whose BER is less than 10^{-8} and greater than 10^{-12}, fire code based FEC function is supported as per the standard. For links whose BER is less than 10^{-5} and greater than 10^{-8}, Reed Solomon FEC function is supported as per the standard. Link Training functions are defined for back plane applications in which the link partners adjust their equalizer taps' coefficients based on the protocol defined.

Auto-negotiation functions help link partners resolve speed and FEC capabilities.

APB-3 protocol based Host interface is provided. The host can control the functioning of DWC_xlgpcs input through programming of the standard registers (as per the standard) and some vendor specific registers which are defined for providing additional control functions and status monitoring.

Note DWC_xlgpcs supports the automotive safety feature only for 25G configurations (namely, configurations in which the DWCXLP_MAIN_MODE parameter is 0).

1.2 Standards Compliance

The DWC_xlgpcs complies with the following IEEE Standards:

- IEEE Std 802.3-2018 Clause 45, 46, 49, 69, 72, 73, 74, 81, 82, 83, 91, 92, and 93 (Depending on your configuration)
- IEEE Std 802.3cd-2022
- 25G Consortium specifications for 25G & 50G r1.6 (Oct 15, 2015)

AMBA 3 APB Protocol, version 1.0, ARM Limited.

1.3 List of Features

The following features are implemented in the DWC_xlgpcs:

- System Level Features
  - **Ethernet PCS:**
    - 25G_Only configuration
    - 25G to 50G configuration
    - 40G_Only configuration
    - 100G configuration - all lower speed modes supported in 100G configuration

    10G and 5G support is enabled by default for all the above configurations.
  - Clause-74 BASE-R FEC (Optional)
  - Clause-108 (Clause 91) RS FEC (Optional)
  - PAM4 mode
  - Interleaved RS-FEC mode as per IEEE802.3ck specification
  - Clause-72/92/93 Link Training support (Optional)
  - Clause-73 auto-negotiation (Optional)
  - Clause-78 Energy Efficient Ethernet
  - Auto-negotiation MMD
  - PMA and PMD MMD
  - PCS MMD
- MAC Interface:
  - For 100G XLGMII interface:
    - Single clock edge 781.25 MHz 128-bit data, 16 control line interface for 100GBASE-KR4 and 100GBASE-CR4 modes
    - Single clock edge 804.92 MHz 128-bit data, 16 control line interface for 100GBASE-KR2,100GBASE-CR2, 100GAUI2, 100GBASE-KR1, and 100GBASE-CR1 modes
  - For 40G (40GBASE-R, KR4) XLGMII interface:
    - Single clock edge 312.5 MHz 128-bit data, 16 control line interface
  - For 25G (25GBASE-CR, 25GBASE-KR) XLGMII interface:
    - Single clock edge 390.625 MHz 128-bit data, 16 control line interface with back pressure on alternate clocks
    - Single clock edge 390.625 MHz 64-bit data, 8 control line interface on every clock
    - Single clock edge 78.125 MHz 32-bit data, 4 control line interface on every clock
  - For 50G XLGMII interface:
    - Single clock edge 390.625 MHz 128-bit data, 16 control line interface for 50GBASE-KR2/50GBASE-CR2 modes
    - Single clock edge 402.4621 MHz 128-bit data, 16 control line interface for 50GBASE-CR2, 50GBASE-KR2, or 50GAUI-2 modes

- For 10G (10GBASE-R, 10GBASE-KR) XGMII interface:
  - Single clock edge 312.5 MHz 32-bit data, 4 control line interface
- For 5G (5GBASE-R, 5GBASE-KR) XGMII interface:
  - Single clock edge 156.25 MHz 32-bit data, 4 control line interface

- PHY interface:
  - Supports 32-bit or 40-bit interface for configurations involving 50G or 100G speed modes.
  - Supports 32-bit or 40-bit or 66-bit interface for 25G_Only configuration.
  - Supports 16-bit, 20-bit, 32-bit, 33-bit, 40-bit, 64-bit, 66-bit, or 80-bit PHY interface for 40G configuration.
  - When operating in the 50GBASE-KR1 or 100GBASE-KR2 PAM4 modes, two 32-bit or two 40-bit XLGPCS lanes are clubbed together as one 64-bit or one 80-bit PHY lane, respectively.
  - When operating in 100GBASE-KR1 or CR1 PAM4 mode, four 32-bit or four 40-bit XLGPCS lanes are clubbed as one 128-bit or one 160-bit PHY lane, respectively.
- Programmable Lane Control support for multiple modes:

  DWC_xlgpcs supports the following combinations:
  - In 25G configuration, DWC_xlgpcs supports the following combinations:
    - Single lane support: 5G, 10G, or 25G Gbps data rate, based on clock setting
  - In 25G-50G configuration, DWC_xlgpcs supports the following combinations:
    - Single lane support: 5G, 10G, or 25G Gbps data rate, based on clock setting
    - Two lanes support: 20 Gbps data rate, if the DWCXLP_SINGLE_LANE parameter is 0
    - Four lanes support: 40 Gbps data rate, if the DWCXLP_SINGLE_LANE parameter is 0
    - Two lanes support: 50 Gbps data rate, if the DWCXLP_SINGLE_LANE parameter is 0
    - One lane support: 50G Gbps data rate, if the DWCXLP_SINGLE_LANE parameter is 1 or the DWCXLP_PAM4 parameter is 1
  - In 40G configuration, DWC_xlgpcs supports the following combinations:
    - Single lane support: 5G, 10G, or 25G Gbps data rate, based on clock setting
    - Two lanes support: 20 Gbps data rate
    - Four lanes support: 40 Gbps data rate (default mode)
  - In 100G configuration, DWC_xlgpcs supports the following combinations:
    - Single lane support: 5G, 10G, or 25G Gbps data rate, based on clock setting
    - Two lanes support: 20 Gbps data rate, if the DWCXLP_SINGLE_LANE parameter is 0
    - Four lanes support: 40 Gbps data rate, if the DWCXLP_SINGLE_LANE parameter is 0
    - Two lanes support: 50 Gbps data rate, if the DWCXLP_SINGLE_LANE parameter is 0
    - Four lanes support: 100 Gbps data rate, if the DWCXLP_SINGLE_LANE parameter is 0
    - Single lane support: 50Gbps data rate, if the DWCXLP_PAM4 parameter is 1
    - Two lanes support: 100Gbps data rate, if the DWCXLP_PAM4 parameter is 1
    - Single lane support: 100Gbps data rate, if the DWCXLP_PAM4 parameter is 1

- Configurable PHY support
  - Non-Synopsys PHY
  - Multi-protocol 12G PHY
  - Multi-protocol 16G PHY
  - Multi-protocol 25G PHY
  - Multi-protocol 32G PHY
  - Enterprise 56G PHY
  - Enterprise 112G PHY
- Configurable Device and Package Identifier (OUI numbers)
- Vendor-specific Register (VR) sets to control the PCS functions and Synopsys PHY functionality
- Lane-to-lane deskewing range (180ns) in the 20GBASE-R, 40GBASE-R, 50GBASE-R, or 100GBASE-R modes
- PRBS 31, PRBS 9, pseudo-random, and square wave test pattern generation in the 10G, 20G, 25G, 40G, 50G, or 100GBASE-R modes
- PRBS-31, PRBS-9, and pseudo-random test pattern checking support in the 10G, 20G, 25G, 40G, 50G, or 100GBASE-R modes
- Scrambled Idle Test pattern generator and checker for 25G, 40G, 50G, and 100GBASE-R
- Clock rate compensation of 200ppm (+/- 100 ppm) on the Receive MAC interface
- Energy efficient Ethernet support
- Automotive safety features

1.3.1 Monitoring, Test, and Debugging Support

DWC_xlgpcs provides the following features to support monitoring, testing, and debugging:

- Supports the following MDIO Manageable Devices (MMD):
  - PCS
  - PMA/PMD
  - Vendor specific control
  - Auto-negotiation
- Supports loopback control from the PHY Transmit to the PHY Receive lane.
- Supports internal loopback on XLGMII for debugging through Receive to Transmit digital path.
- Supports various error status signals and statistics for monitoring and debugging.
- Provides programmable control to reduce the simulation time by reducing the timer values defined in the IEEE standard.
- Provides programmable Device or Package identifiers (OUI numbers as defined in IEEE Std 802.3).

- Provides programmable features to dynamically enable or disable MMD devices present in some configurations.

1.4 Area and Power Numbers

For information on the area, power numbers, and latency values for various features and configurations, see Appendix A, "Area, Power, and Latency".

1.5 Controller Requirements

For information on clocks required for operation and clock resets, see chapter "*Clocks and Resets*" in the *DesignWare Cores Enterprise Ethernet PCS User Guide.*

1.6 Deliverables

The DWC_xlgpcs is packaged as a .run file. The license file, required to use this Synopsys DesignWare IP, is delivered separately. You use the Synopsys coreConsultant tool to configure, synthesize, and simulate the DWC_xlgpcs core. The DWC_xlgpcs image includes the following:

- Verilog RTL source code
- A Verilog test suite that integrates the DUT (which consists of the DWC_xlgpcs, integrated with a pin-compatible behavioral model of the PHY) and the Ethernet VIP
- Synopsys DesignWare Ethernet Verification IP and Synopsys Discovery Ethernet Verification IP that can be reused in your Verilog, VHDL, or Vera Verification environment
- Synthesis scripts for Synopsys Design Compiler and Synplify Pro
- Regression scripts for Synopsys VCS
- Spyglass lint checker rules used for linting and example report
- Support for creating gate-level Device Under Test (DUT) models using a GTECH library

1.6.1 Licensing Features

The licenses provided for DWC_xlgpcs are described in the *DesignWare Cores Enterprise Ethernet PCS Installation Guide*, available at:

https://www.synopsys.com/dw/ipdir.php?c=dwc_ether_xlgpcs

1.7 PCS Layer Overview

The Physical Coding Sublayer (PCS) layer provides an interface between the Media Access Control (MAC) and Physical Medium Attachment Sublayer (PMA) through a Media independent interface.

To maintain transition density and DC balancing, line decoding and scrambling/de-scrambling techniques are done in PCS layer.

Optionally, Forward Error Correction is added to mitigate the errors introduced by the channel medium.

DWC_xlgpcs supports 5GBASE-R, 10G BASE-R, 25GBASE-R, 40G BASE-R, 50G BASE-R, and 100GBASE-R modes of operation as specified in IEEE 802.3-2022 and IEEE 802.3cd-2018.

DWC_xlgpcs also supports 50GBASE-R as per 25G Consortium specification 1.6 - using 2X25G mode of operations.

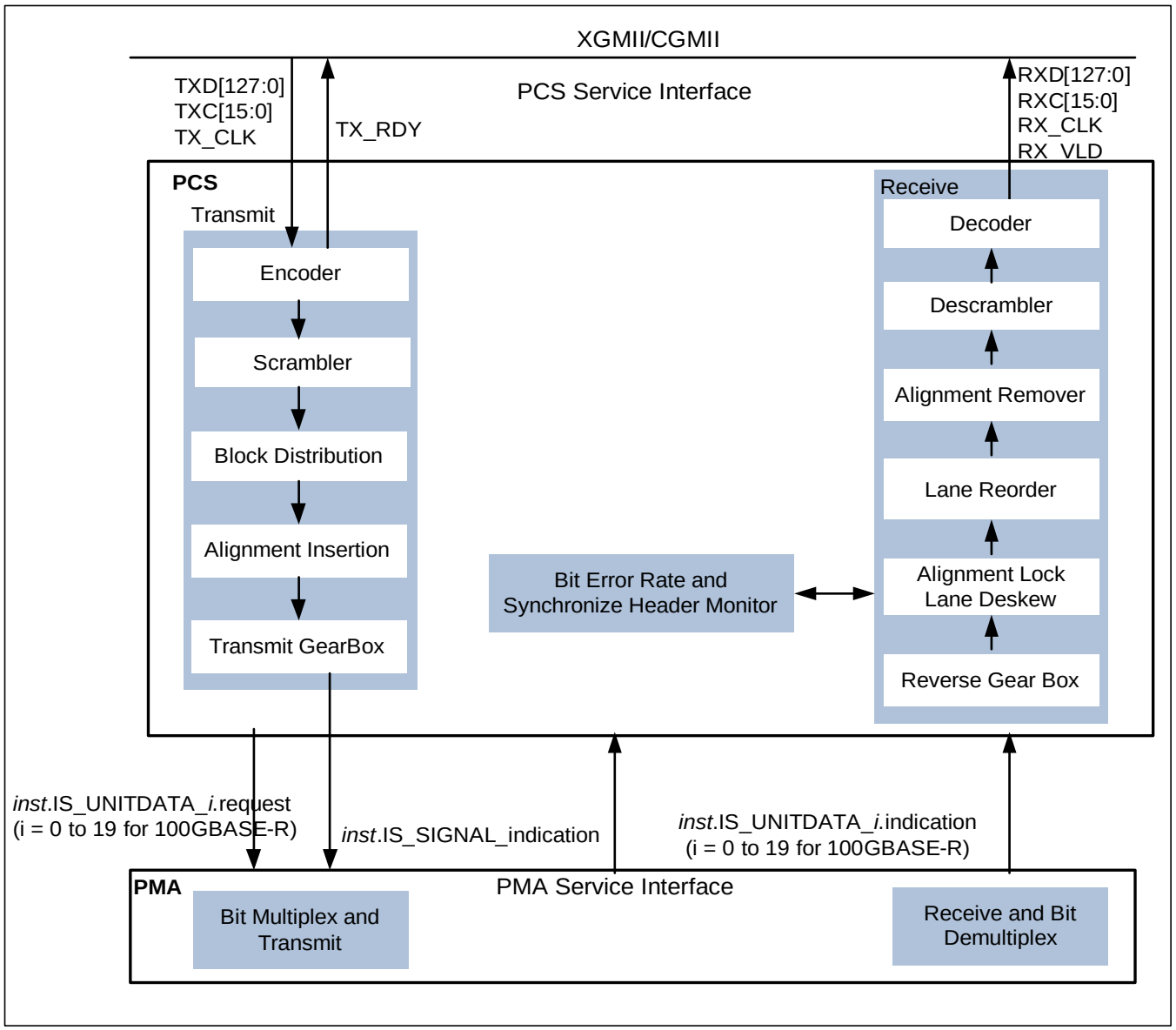
The following subsections briefly describe PCS layer operations under the supported modes.

1.7.1 100GBASE-R PCS Layer

The 100GBASE-R PCS layer has the following main blocks:

- **Transmit:** The Transmit block interfaces with the XLGMII 128-bit data and sixteen control lines at the clock rate of 781.25 MHz. The Transmit block converts two XLGMII data blocks into four 72-bit blocks (64-bit data + 8-bit control) at the rate of 390.625 MHz. The Transmit block performs 64B/66B encoding for each of the 64-bit data. The data part of encoded four 66-bit block is scrambled to achieve required transition density. The encoded and scrambled data is sent to twenty virtual (PCS) lanes. Once after block distribution to all twenty lanes, each lane inserts alignment marker after every 16383 66bit blocks. 66-bit data from 5 PCS lanes are bit multiplexed and passed to the PMA (SerDes). The PMA serializes the data on each lane at the data rate of 25.78125 Gbps.

Figure 1-1 DWC_xlgpcs Functional Block Diagram: 100GBASE-R PHY



- **Receive:** The Receive block receives the serialized self-clocked data from the per lane PMA (SerDes). The clock and data recovery are done in the PMA followed by bit demultiplexing to retrieve unaligned data of 5 virtual (PCS) lanes. The unaligned data is presented to the PCS Receive block. In the PCS Receive block. Boundary detection on each virtual lane is done by successive validation of sync-header pattern of the 66-bit encoded data. The block aligned data is re-ordered based on lane

detection through AM lock detection and that re-ordered data is descrambled and decoded through 64/66 decoders. The aligned and decoded data is forwarded over XLGMII Interface (128-bit data and sixteen control lines and one valid signal).

Note

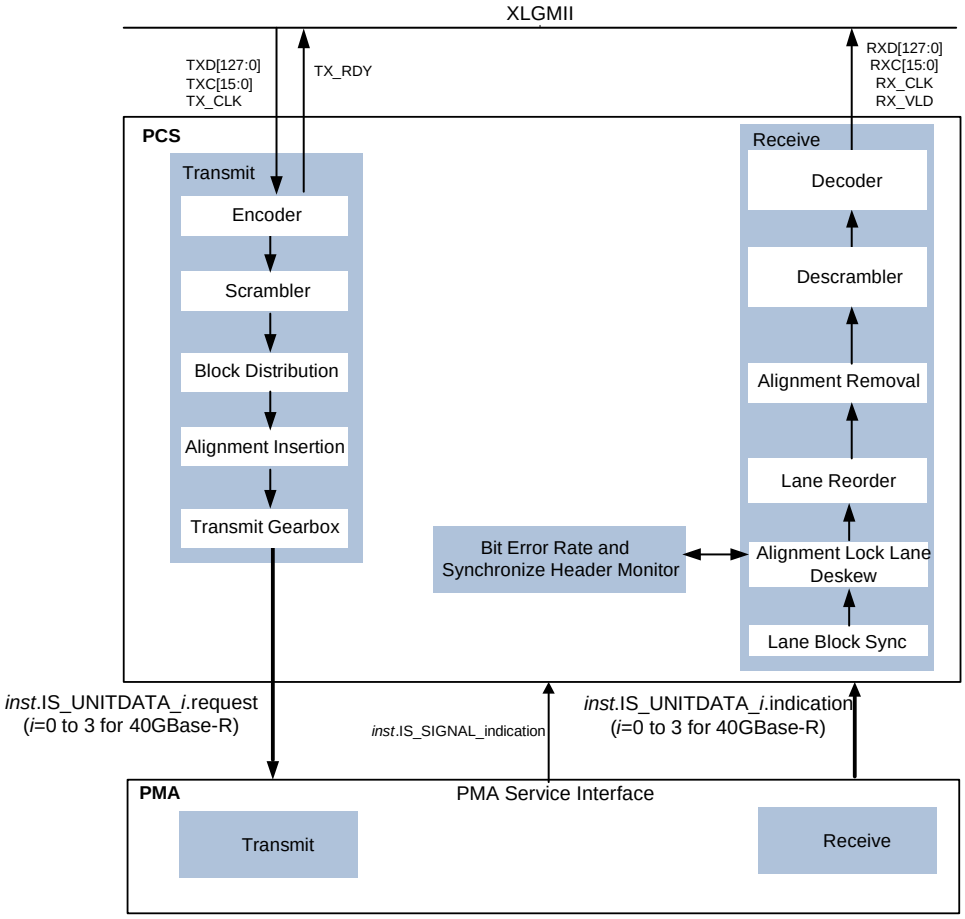
- When 100GBASE-R PCS is used to implement a
  - 100GAUI-4 link, the data rate on each of the four lanes is 26.5625Gbps.
  - 100GBASE-KR2/CR2 link, two PAM4 serial lanes each operating at 53.125Gbps rate exist.
  - 100GBASE-KR1/CR1 link, one PAM4 serial lane operating at 106.25 Gbps exists.
- When operating as 100GAUI-4 link, 100GBASE-KR2 link, 100GBASE-CR2 link, 100GBASE-KR1 link, or 100GBASE-CR1 link, the XLGMII interface operates at 804.9242 MHz clock instead of 781.25 MHz.

1.7.2 40GBASE-R PCS Layer

The 40GBASE-R PCS layer has the following main blocks:

- **Transmit:** The Transmit block interfaces with the XLGMII 128-bit data and sixteen control lines and one back pressure signal at the clock rate of 312.5 MHz. The Transmit block converts two XLGMII data blocks into four 72-bit blocks (64-bit data + 8-bit control) at the rate of 156.25 MHz. The Transmit block performs 64B/66B encoding for each of the 64-bit data. The data part of encoded four 66-bit block is scrambled to achieve required transition density. The encoded and scrambled data is sent to four lanes and passes to the PMA (SerDes). The PMA serializes the parallel data on each lane at the data rate of 10.3125 Gbps. Once after block distribution to all four lanes, each lane inserts alignment marker after every 16383 66bit blocks.
- **Receive:** The Receive block receives the serialized self-clocked data from the per lane PMA (SerDes). The clock and data recovery are done in the PMA and the unaligned data is presented to the PCS Receive block. In the PCS Receive block, per lane 66-bit data boundary detection is done by successive validation of sync-header pattern of the 66-bit encoded data. The block aligned data is reordered based on lane detection through AM lock detection and that re-ordered data is descrambled and decoded through four 64/66 decoder. The aligned and decoded data is forwarded over XLGMII Interface (128-bit data and sixteen control lines and one valid signal). This section contains the description of the PCS layer, including auto-negotiation.

Figure 1-2 DWC_xlgpcs Functional Block Diagram: 40GBASE-R PHY



1.7.3 50GBASE-R PCS Layer

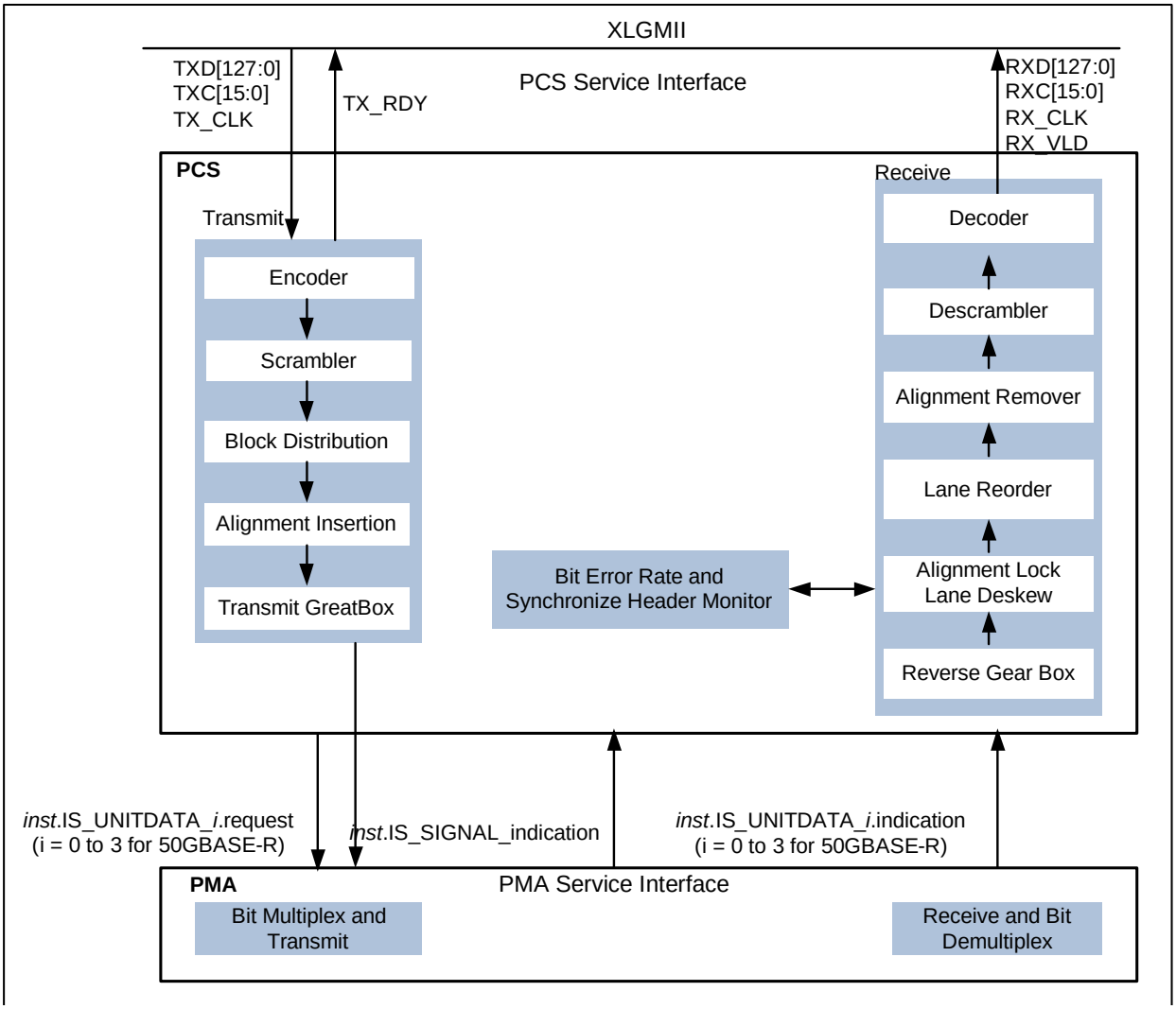
The 50GBASE-R PCS layer has the following main blocks:

- Transmit: The Transmit block interfaces with the XLGMII 128-bit data and sixteen control lines at the clock rate of 390.625 MHz. PCS inserts back-pressure on alternate clock cycles to match the 50Gbps data rate. The Transmit block converts XLGMII data blocks into four 72-bit blocks (64-bit data + 8-bit control). The Transmit block performs 64B/66B encoding for each of the 64-bit data. The data part of encoded four 66-bit blocks is scrambled to achieve required transition density. The PCS layer 64/66b encoding and scrambling process are compliant to the PCS functions defined in Clause 133. After block distribution to all four lanes, each lane inserts alignment marker after every 16383 66bit blocks. The encoded and scrambled data from two PCS Lanes is bit multiplexed on to one lane. The bit multiplexed data is sent over two lanes. The PMA serializes the 32-bit or 40-bit data on each lane at the data rate of 25.78125 Gbps.
- Receive: The Receive block receives the serialized self-clocked data from the per lane PMA (SerDes). The clock and data recovery are done in the PMA. The received data from each physical lane is bit de multiplexed on to two PCS lanes. This unaligned data is presented to the PCS Receive block. In the PCS Receive block, data bits received from two per lane 66-bit data boundary detection is done by successive validation of sync-header pattern of the 66-bit encoded data. The block aligned data is reordered based on lane detection through AM lock detection and that re-ordered data is descrambled and decoded through four 64/66 decoder. The PCS layer 64/66b decoding and descrambling process are compliant to the PCS functions defined in Clause 133. The aligned and decoded data is forwarded over XLGMII Interface (128-bit data and sixteen control lines and one valid signal).

Note

- When 50GBASE-R PCS is used to implement a
  - 50GAUI-2 link, the data rate on each of 2 lanes is 26.5625Gbps
  - 50GBASE-KR/CR link, there exists a single PAM4 serial lane at 53.125Gbps rate.
- When operating as 50GAUI-2/50GBASE-KR/50GBASE-CR link, the XLGMII interface operates at 402.4621 MHz clock, instead of a 390.625 MHz clock.

Figure 1-3 DWC_xlgpcs Functional Block Diagram: 50GBASE-R PHY



1.7.4 25GBASE-R PCS Layer

The 25GBASE-R PCS layer has the following main blocks:

- **Transmit:** When DWCXLP_XLGMII_INT_WIDTH is 0, the Transmit block interfaces with the XLGMII 128-bit data and sixteen control lines at the clock rate of 390.625MHz with back pressure asserted on alternate clocks to match the data rate.

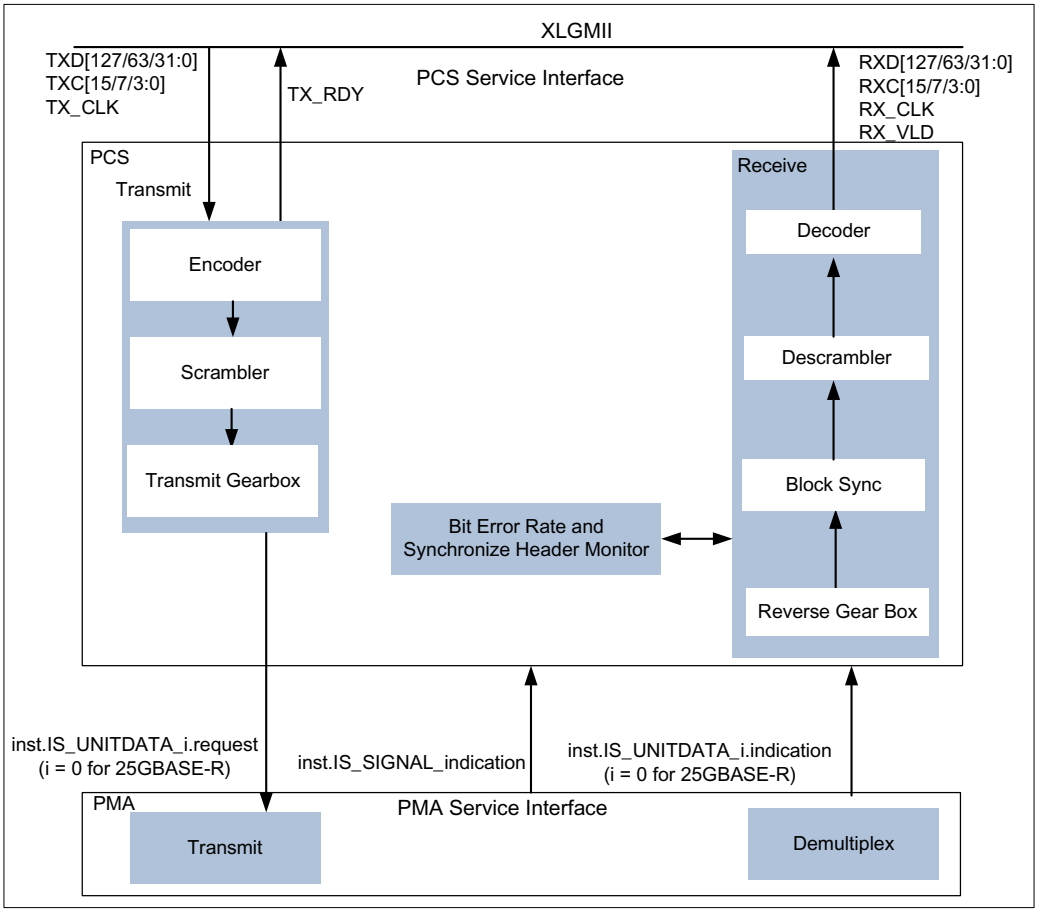
  When DWCXLP_XLGMII_INTF_WIDTH is 1, the transmit block interfaces with XLGMII 64-bit data and 8 control lines at a clock rate of 390.625 MHz.

  When DWCXLP_XLGMII_INTF_WIDTH is 2, the transmit block interfaces with XLGMII 32-bit data and 4 control lines at a clock rate of 781.25 MHz.

  The data part of encoded 66-bit block is scrambled to achieve required transition density. The PCS layer 64/66b encoding and scrambling process are compliant to the PCS functions defined in Clause 49. The encoded and scrambled data is passed to the PMA (SerDes). The PMA serializes this data at 25.78125 Gbps data rate.

- Receive: The Receive block receives the serialized self-clocked data from the PMA (SerDes). The clock and data recovery are done in the PMA and the unaligned data is presented to the PCS Receive block. In the PCS Receive block, 66-bit data boundary detection is done by successive validation of sync-header pattern of the 66-bit encoded data. The block aligned data is decoded through 64/66 decoder. The PCS layer 64/66b decoding and descrambling process are compliant to the PCS functions defined in Clause 49. The decoded data is forwarded over the XLGMII Interface on alternate clocks when DWCXLP_XLGMII_INTF_WIDTH is 0, or on every clock when DWCXLP_XLGMII_INTF_WIDTH is 1 or 2.

Figure 1-4 DWC_xlgpcs Functional Block Diagram: 25GBASE-R PHY



1.7.5 10GBASE-R and 5GBASE-R PCS Layer

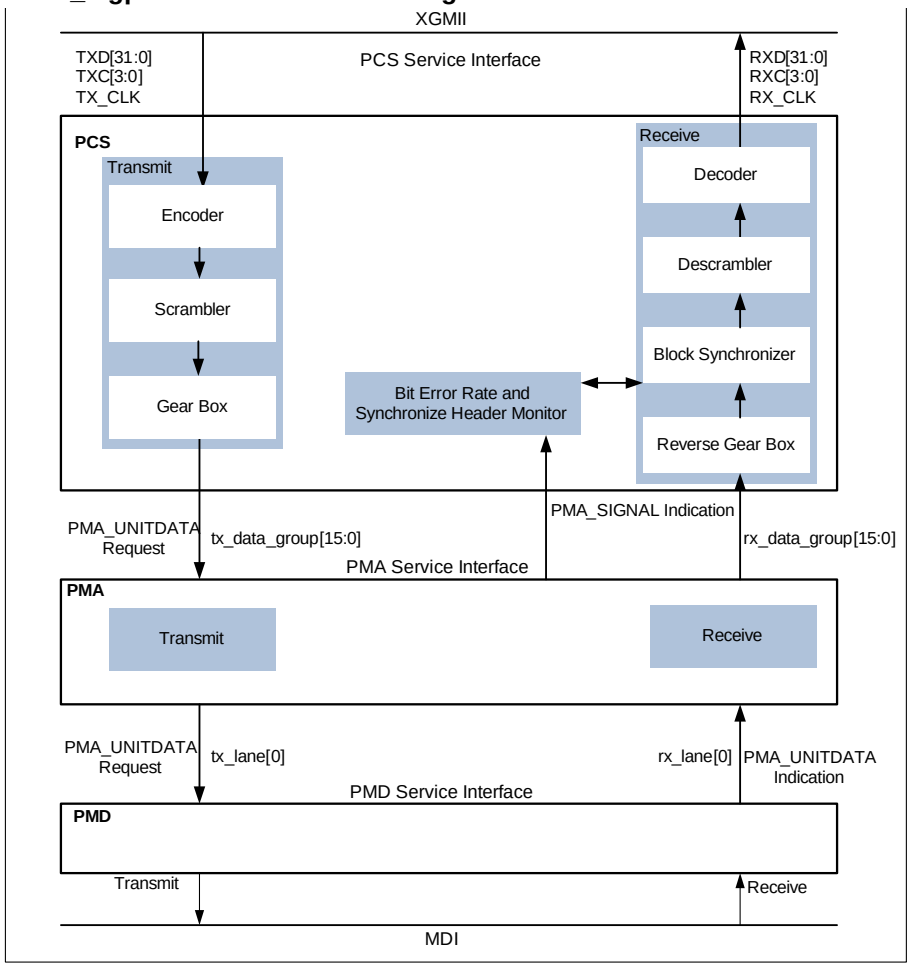
The 10GBASE-R PCS layer has the following main blocks:

- **Transmit**: The Transmit block interfaces with the XGMII 32-bit data and four control lines at the clock rate of 312.5 MHz for 10GBASE-R PCS, and 156.25 MHz for 5GBASE-R PCS. The Transmit block converts two XGMII data blocks into 66-bit blocks (64-bit data + 2-bit Sync Header) at the rate of 156.25 MHz for 10GBASE-R PCS, and 78.125 MHz for 5GBASE-R PCS. The data part of encoded 66-bit block is scrambled to achieve required transition density. The encoded and scrambled data is passed to the PMA (SerDes). The PMA serializes this data at 10.3125 Gbps data rate for 10GBASE-R PCS, and 5.15625 Gbps data rate for 5GBASE-R PCS.

- **Receive**: The Receive block receives the serialized self-clocked data from the PMA (SerDes). The clock and data recovery are done in the PMA and the unaligned data is presented to the PCS Receive block. In the PCS Receive block, 66-bit data boundary detection is done by successive validation of sync-header pattern of the 66-bit encoded data. The block aligned data is decoded through 64/66 decoder. The aligned and decoded data is forwarded over XGMII Interface (32-bit data and four control lines).

Figure 1-5 displays the functional block diagram for 10GBASE-R and 5GBASE-R physical layer.

Figure 1-5 DWC_xlgpcs Functional Block Diagram: 10GBASE-R or 5GBASE-R PHY



1.7.6 20G BASE-R PCS Layer (Non-Standard, Vendor Specific)

This is a non-standard vendor specific speed mode.

The 20GBASE-R PCS layer makes use of the following main blocks:

- Transmit

  The Transmit block interfaces with the XLGMII 128-bit data and sixteen control lines at the clock rate of 312.5 MHz. To match the 20G rate, PCS layer asserts back pressure signal on alternate cycles of the transmit clock.

The Transmit block converts XLGMII data blocks into four 72-bit blocks (64-bit data + 8-bit control) at every alternate 156.25 MHz clock. The Transmit block performs 64B/66B encoding for each of the 64-bit data. The data part of encoded four 66-bit block is scrambled to achieve required transition density. The encoded and scrambled data sent to two PMA lanes. The PMA serializes the data on each lane at the data rate of 10.3125 Gbps. Once after block distribution to two lanes, each lane inserts alignment marker after every 16383 66bit blocks.

- Receive

  The Receive block receives the serialized self-clocked data from the per lane PMA (SerDes). The clock and data recovery are done in the PMA and the unaligned data is presented to the PCS Receive block. In the PCS Receive block, per lane 66-bit data boundary detection is done by successive validation of sync-header pattern of the 66-bit encoded data. The block aligned data is reordered based on lane detection through AM lock detection and that re-ordered data is descrambled and decoded through four 64/66 decoder. The aligned and decoded data is forwarded over XLGMII Interface (128-bit data and sixteen control lines and one valid signal).

1.7.7 Auto-Negotiation Overview

Auto-negotiation enables a device to understand capabilities and status of the remote-end device. Auto-negotiation allows devices to advertise and share information, including speed, modes, fault signaling, and other control information.

- Auto-negotiation is defined in Clause 28 for any speed BASE-T (twisted-pair copper) PHY device and Clause 36 and 37 for 1000BASE-X (based on PCS /C/ order set). Auto-negotiation is not defined for 10GBASE-X and BASE-R.
- Backplane Ethernet is defined to include the extension of 10GBASE-R, 25GBASE-R, 40GBASE-R, 50GBASE-R and 100GBASE-R called 10GBASE-KR, 25GBASE-KR/CR, 25GBASE-KR/CR-S, 40GBASE-KR4/CR4, 50GBASE-KR2/CR2/KR/CR and 100GBASE-KR4/CR4/KR2/KR1/CR1 respectively.

  During auto-negotiation, the information is exchanged through serial Differential Manchester Encoding (DME) data bits. The serial data pages, called Link Code Words, are sent through newly defined DME pages (as defined in IEEE Std 802.3ap, Clause 73).

1.7.8 PCS Layer with Clause 73 Auto-Negotiation Support

Clause 73 of the IEEE Std 802.3 defines a new common auto-negotiation protocol which uses the signaling that is independent of the standard speed modes (10GBASE-KR, 25GBASE-KR/CR, 25GBASE-KR-S/CR-S, 40GBASE-KR4/CR4, 50GBASE-KR2/CR2/KR/CR and 100GBASE-KR4/CR4/KR2/CR2/KR1/CR1). The links can communicate with each other to determine the link speeds during auto-negotiation. There is a provision in the auto-negotiation to detect the legacy links which do not support auto-negotiation. The following links are defined for the Backplane applications using Ethernet technology:

- 10GBASE-KR is compatible with the 10GBASE-R link.
- 25GBASE-KR/CR is compatible with the 25GBASE-R link.
- 25GBASE-KR-S/CR-S is compatible with the 25GBASE-R link without RS-FEC capability.
- 40GBASE-KR4/CR4 is compatible with the 40GBASE-R links.
- 50GBASE-KR2/CR2/KR/CR is compatible with the 50GBASE-R links.

- 100GBASE-KR4/CR4/CR2/KR2/KR1/CR1 is compatible with the 100GBASE-R links.

Note Auto negotiation in 25G Consortium mode is also supported to resolve the capabilities in 25G and 50G modes of operation.

The IEEE Std 802.3 also defines special equalization protocol called Clause 72 KR Training and a Forward Error Correction (FEC) blocks for links at standard speeds (10G, 25G, 40G, 50G, and 100G) to maximize the link reliability. This is because the links work at a very high speed (>=10.3125 Gbps) on an electrical media.

To support the Backplane Ethernet auto-negotiation without any additional functions in the PHY, the PCS layer implements the following additional functions:

- Clause 73 auto-negotiation (Receive, Transmit, and Arbiter) state machine at the parallel PHY Interface
- DME page decoding and encoding at the parallel PHY Interface
- Auto-negotiation Manageable Device (MMD) for Management Interface register sets to handle auto-negotiation
- Switching between modes based on the auto-negotiation outcome when supporting multiple operating modes (with software control)
- Parallel detection for legacy devices which do not support Clause 73 auto-negotiation

Note In Clause 73 auto-negotiation, the parallel detection for legacy devices has the following limitations:

- The detection of Remote End legacy 10GBASE-KR compatible (10GBASE-R) devices is supported only when the PCS is programmed to work in 10GBASE-KR mode.
- The detection of Remote End legacy 40GBASE-KR4/CR4 compatible (40GBASE-R) devices is supported only when the PCS is programmed to work in 40GBASE-KR4/CR4 mode.
- The detection of Remote End 25GBASE-CR/25GBASE-CR-S/25GBASE-KR/25GBASE-KR-S compatible (25GBASE-R) devices is supported only when the PCS is programmed to work in 25G mode.
- The detection of Remote End 50GBASE-CR2/50GBASE-KR2/50GBASE-CR/50GBASE-KR compatible (50GBASE-R) devices is supported only when the PCS is programmed to work in 50G mode.
- The detection of Remote End 100GBASE-CR4/100GBASE-KR4/100GBASE-CR2/100GBASE-KR2/CR1/KR1 compatible (100GBASE-R) devices is supported only when the PCS is programmed to work in 100G mode.

This is because the DWC_xlgpcs assumes that the PMA layer is common for all speed modes, and it supports a single speed mode at a time.

1.8 Link Types Supported Through PMA Layer

The DWC_xlgpcs supports the following link types through PMA layer:

- 10GBASE-R or 5GBASE-R

- 25GBASE-CR
- 25GBASE-CR-S
- 25GBASE-KR
- 25GBASE-KR-S
- 40GBASE-R
- 40GBASE-KR4
- 50GBASE-CR2
- 50GBASE-KR2
- 50GBASE-KR
- 50GBASE-CR
- 100GBASE-CR4
- 100GBASE-KR4
- 100GBASE-KR2
- 100GBASE-CR2
- 100GBASE-KR1
- 100GBASE-CR1
- 50GAUI-2
- 100GAUI-4
- 50GAUI-1
- 100GAUI-2

1.8.1 10GBASE-R or 5GBASE-R

The 10GBASE-R or 5GBASE-R PCS layer is defined mainly for the optical fiber media. The 10GBASE-R or 5GBASE-R link is deployed for 10 Gbps or 5Gbps LAN or WAN applications. The DWC_xlgpcs provides the 10GBASE-R or 5GBASE-R PCS functionality to interface with Ethernet MAC (through XGMII interface) on one side and with the PHY (supporting 10GBASE-R or 5GBASE-R serial link) on the other side. The 10GBASE-LR, 10GBASE-SR, 10GBASE-ER, and 10GBASE-LRM are various derivatives of 10GBASE-R using different wave lengths on the serial link.

The DWC_xlgpcs supports the 10GBASE-SR, 10GBASE-LR, 10GBASE-ER, and 10GBASE-LRM serial interfaces through appropriate PHY to achieve the serial data rate of 10.3125 Gbps.

1.8.2 25GBASE-CR

The 25GBASE-CR and 25GBASE-CR-S is the 25G link defined for the Twin axial copper, which use the Ethernet technology. These are derivatives of the 25GBASE-R PCS based on IEEE 802.3 Clause 49.

The 25GBASE-CR and 25BASE-CR-S interface to a single PHY lane at 25.78125Gbps.

Additionally, 25GBASE-CR can operated in RS-FEC, BASE-R FEC, or no-FEC mode and 25GBASE-CR-S can operated in BASE-R FEC, or no-FEC mode only.

1.8.3 25GBASE-CR-S

The 25GBASE-CR-S is the 25G link defined for the Twin axial copper, which use the Ethernet technology. The 25GBASE-CR-S does not have the RS FEC capability. These are derivatives of the 25GBASE-R PCS based on IEEE 802.3 Clause 49. 25GBASE-CR interfaces to a single PHY lane at 25.78125Gbps.

1.8.4 25GBASE-KR

The 25GBASE-KR and 25GBASE-KR-S are the 25G links and are defined for the Backplanes applications , which use the Ethernet technology. The electrical characteristics of the Backplane Ethernet are different to support distance of more than 1 meter on a printed circuit board (including two connectors). 25GBASE-KR interfaces to a single PHY lane at 25.78125Gbps

1.8.5 25GBASE-KR-S

The 25GBASE-KR-S are the 25G links are defined for the Backplanes applications, which use the Ethernet technology. The 25GBASE-KR-S does not have the RS FEC capability. The electrical characteristics of the Backplane Ethernet are different to support distance of more than 1 meter on a printed circuit board (including two connectors). 25GBASE-KR-S interfaces to a single PHY lane at 25.78125Gbps

1.8.6 40GBASE-R

The 40GBASE-R PCS layer is defined mainly for the optical fiber media. The 40GBASE-R link is deployed for 40 Gbps LAN or WAN applications. DWC_xlgpcs provides the 40GBASE-R PCS functionality to interface with XLGMAC (through XLGMII interface) on one side and with the PHY (supporting 40GBASE-R serial link) on the other side. The 40GBASE-CR4, 40GBASE-LR4 and 40GBASE-SR4 are various derivatives of 40GBASE-R using different wave lengths on the serial link. DWC_xlgpcs supports the 40GBASE-CR4, 40GBASE-LR4 and 40GBASE-SR4 serial interfaces through appropriate PHY to achieve the serial data rate of 10.3125 Gbps. DWC_xlgpcs can support any one of the following interfaces to the PHY for a 40G configuration:

- 16-bit data at 644.53 MHz clock rate
- 20-bit data at 515.625 MHz clock rate
- 32-bit data at 322.265 MHz clock rate
- 33-bit data at 312.5 MHz clock rate
- 40-bit data at 257.8125 MHz clock rate
- 64-bit data at 161.1328125 MHz clock rate
- 66-bit data at 156.25 MHz clock rate

1.8.7 40GBASE-KR4

The 40GBASE-KR4 and 10GBASE-KR links are defined for the Backplane applications which use the Ethernet technology. The 40GBASE-KR4 link is compatible with the 40GBASE-R links, 10GBASE-KR link is

compatible with the 10GBASE-R links. The electrical characteristics of the Backplane Ethernet are different to support distance of more than 1 meter on a printed circuit board (including two connectors).

The IEEE Std 802.3, Clause 73 defines a new common auto-negotiation protocol which uses the DME signaling. The DME signaling is independent of any of the standard speed modes (40GBASE-KR4, 10GBASE-KR). Therefore, the links can communicate with each other to determine the link speeds during auto-negotiation. There is also a provision in the auto-negotiation to detect the legacy links which do not support the Clause 73 auto-negotiation.

The IEEE Std 802.3 also defines a special equalization protocol called Clause 72 KR Training and Forward Error Correction block for 10GBASE-KR and 40GBASE-KR4 links to maximize the link reliability. This is because KR and KR4 need to work at a very high speed (10.3125 Gbps) on an electrical media.

1.8.8 50GBASE-CR2

The 50GBASE-CR2 are the 50Gbps links defined by 25G Ethernet Consortium Group. It is defined for Twin axial copper medium which use the Ethernet technology. The 50GBASE-CR2 link is compatible with the 50GBASE-R links. The IEEE Std 802.3, Clause 73 can be used by the link partners to detect 50GBASE-CR2 link. There is also a provision in the auto-negotiation to detect the legacy links which do not support the Clause 73 auto-negotiation.

50GBASE-CR2 interfaces with two PHY lanes each working at 25.78125Gbps

1.8.9 50GBASE-KR2

The 50GBASE-KR2 are the 50Gbps links defined by 25G Ethernet Consortium Group. It is defined for Backplane medium which use the Ethernet technology. The 50GBASE-KR2 link is compatible with the 50GBASE-R links. The IEEE Std 802.3, Clause 73 can be used by the link partners to detect 50GBASE-KR2 link. There is also a provision in the auto-negotiation to detect the legacy links which do not support the Clause 73 auto-negotiation.

50GBASE-KR2 also interfaces with two PHY lanes each working at 25.78125Gbps

1.8.10 50GBASE-KR

The 50GBASE-KR is a PAM4 link operating at 53.125Gbps defined by IEEE802.3cd specification. It is defined for Backplane medium which uses the Ethernet technology. The IEEE Std 802.3, Clause 73 can be used by the link partners to detect 50GBASE-KR link.

1.8.11 50GBASE-CR

50GBASE-CR is a 50Gbps link defined by IEEE802.3cd specification. It is defined for Twin axial copper medium and employs a single lane at 53.125Gbps with PAM4 encoding.

1.8.12 100GBASE-CR4

The 100GBASE-CR4 are the 100Gbps links defined for Twin axial copper medium which use the Ethernet technology. The 100GBASE-CR4 link consists of 4 serial lanes carrying data at 25.78125Gbps and compatible with the 100GBASE-R links. The IEEE Std 802.3, Clause 73 can be used by the link partners to detect 100GBASE-CR4 link. There is also a provision in the auto-negotiation to detect the legacy links which do not support the Clause 73 auto-negotiation.

1.8.13 100GBASE-KR4

The 100GBASE-KR4 are the 100Gbps links defined for Backplane medium which use the Ethernet technology. The 100GBASE-KR4 link also consists of 4 serial lanes carrying data at 25.78125Gbps and compatible with the 100GBASE-R links. The IEEE Std 802.3, Clause 73 can be used by the link partners to detect 100GBASE-KR4 link. There is also a provision in the auto-negotiation to detect the legacy links which do not support the Clause 73 auto-negotiation.

1.8.14 100GBASE-KR2

The 100GBASE-KR2 is a PAM4 link defined for Backplane medium which use the Ethernet technology. The 100GBASE-KR2 link consists of 2 PAM4 serial lanes carrying data at 53.125Gbps rate. The IEEE Std 802.3, Clause 73 can be used by the link partners to detect 100GBASE-KR2 link.

1.8.15 100GBASE-CR2

This is a 100Gbps link defined for Twin axial copper medium. This link consists of 2 serial PAM4 lanes carrying data at 53.125 Gbps rate each. The IEEE standard 802.3, Clause 73 can be used by link partners to detect a 100GBASE-CR2 link.

1.8.16 100GBASE-KR1

The 100GBASE-KR1 is a PAM4 link defined for Backplane medium which uses the Ethernet technology. This link consists of a PAM4 serial lane which carries data at 106.25 Gbps rate. Link partners can use the IEEE Std 802.3, Clause 73 to detect a 100GBASE-KR1 link.

1.8.17 100GBASE-CR1

This is a 100Gbps link defined for Twin axial copper medium. This link consists of a PAM4 serial lane which carries data at 106.25 Gbps rate. Link partners can use the IEEE standard 802.3, Clause 73 to detect a 100GBASE-CR1 link.

1.8.18 50GAUI-2

50GAUI-2 C2C and 50GAUI-2 C2M are optional chip-to-chip or chip-to-module 50Gbps two-lane attachment unit interface. These are physical instantiations of the PMA service interface between the FEC and PMD. These links use the NRZ signaling. The nominal signaling rate for each lane is 26.5625 Gbps.

1.8.19 100GAUI-4

100GAUI-4 C2C and 100GAUI-4 C2M are optional chip-to-chip or chip-to-module 100Gbps four-lane attachment unit interface. These are physical instantiations of the PMA service interface between the FEC and PMD. These links use the NRZ signaling. The nominal signaling rate for each lane is 26.5625 Gbps.

1.8.20 50GAUI-1

50GAUI-1 C2C and 50GAUI-1 C2M are optional chip-to-chip or chip-to-module 50Gbps one-lane attachment unit interface. These are physical instantiations of the PMA service interface between the FEC and PMD. These links use the PAM4 signaling. The nominal signaling rate for each lane is 26.5625 GBd (Giga Baud).

1.8.21 100GAUI-2

100GAUI-2 C2C and 100GAUI-2 C2M are optional chip-to-chip or chip-to-module 100Gbps two-lane attachment unit interface. These are physical instantiations of the PMA service interface between the FEC and PMD. These links use the PAM4 signaling. The nominal signaling rate for each lane is 26.5625 GBd (Giga Baud)

2 Architecture

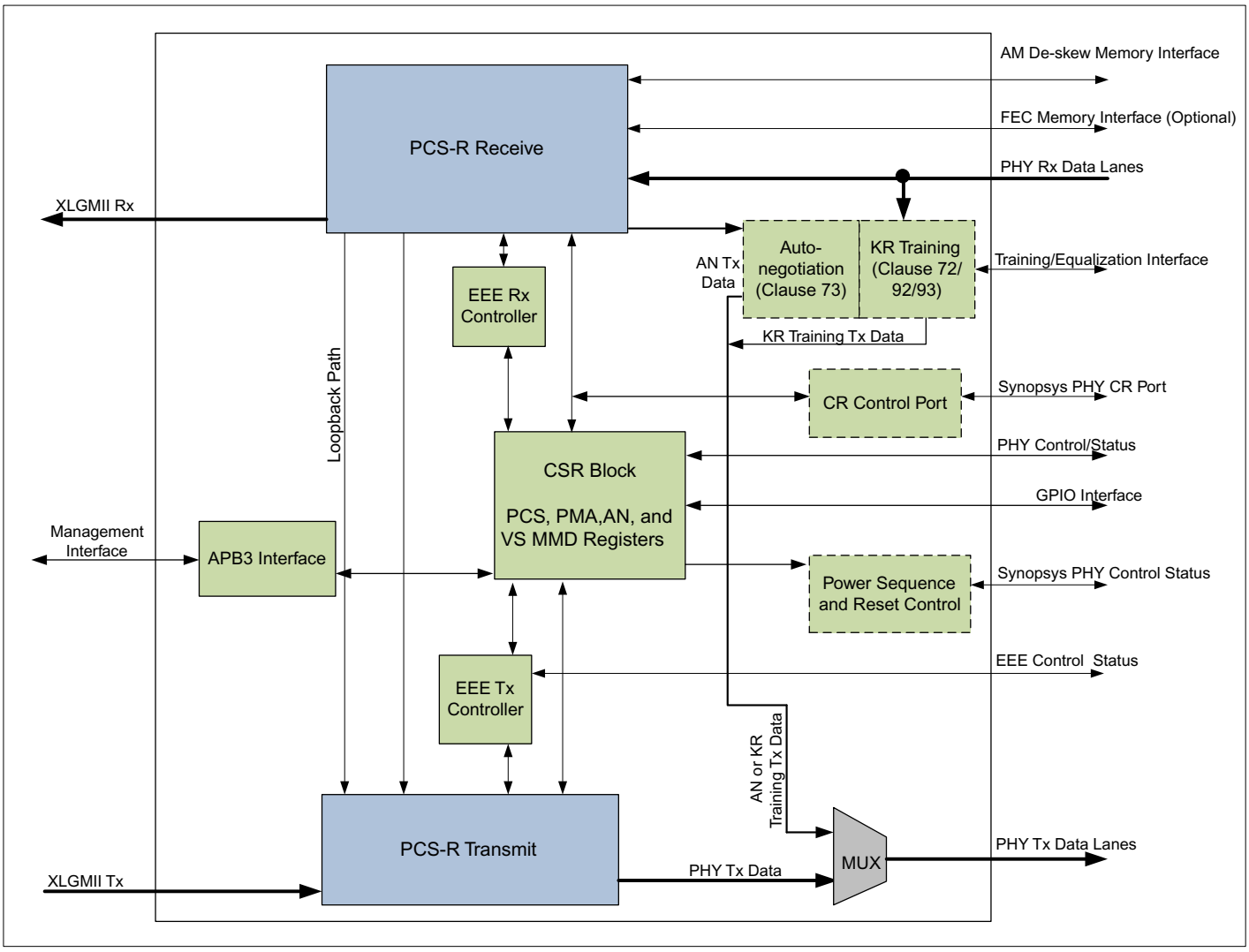
This chapter describes the DWC_xlgpcs architecture in functional blocks. It contains the following sections:

- "Functional Overview" on page 42
- "Data Path" on page 42
- "EEE Controller" on page 96
- "Command and Status Register Block" on page 104
- "Management Interface" on page 106
- "Power Sequence and Reset Generation Block" on page 113
- "Control Register Port Controller" on page 115
- "Clause 73 Auto-Negotiation Block" on page 115
- "Link KR Training" on page 131
- "Timestamping Accuracy" on page 151
- "Debug Bus" on page 154
- "Timing Diagrams" on page 154

2.1 Functional Overview

Figure 2-1 displays the DWC_xlgpcs as functional blocks.

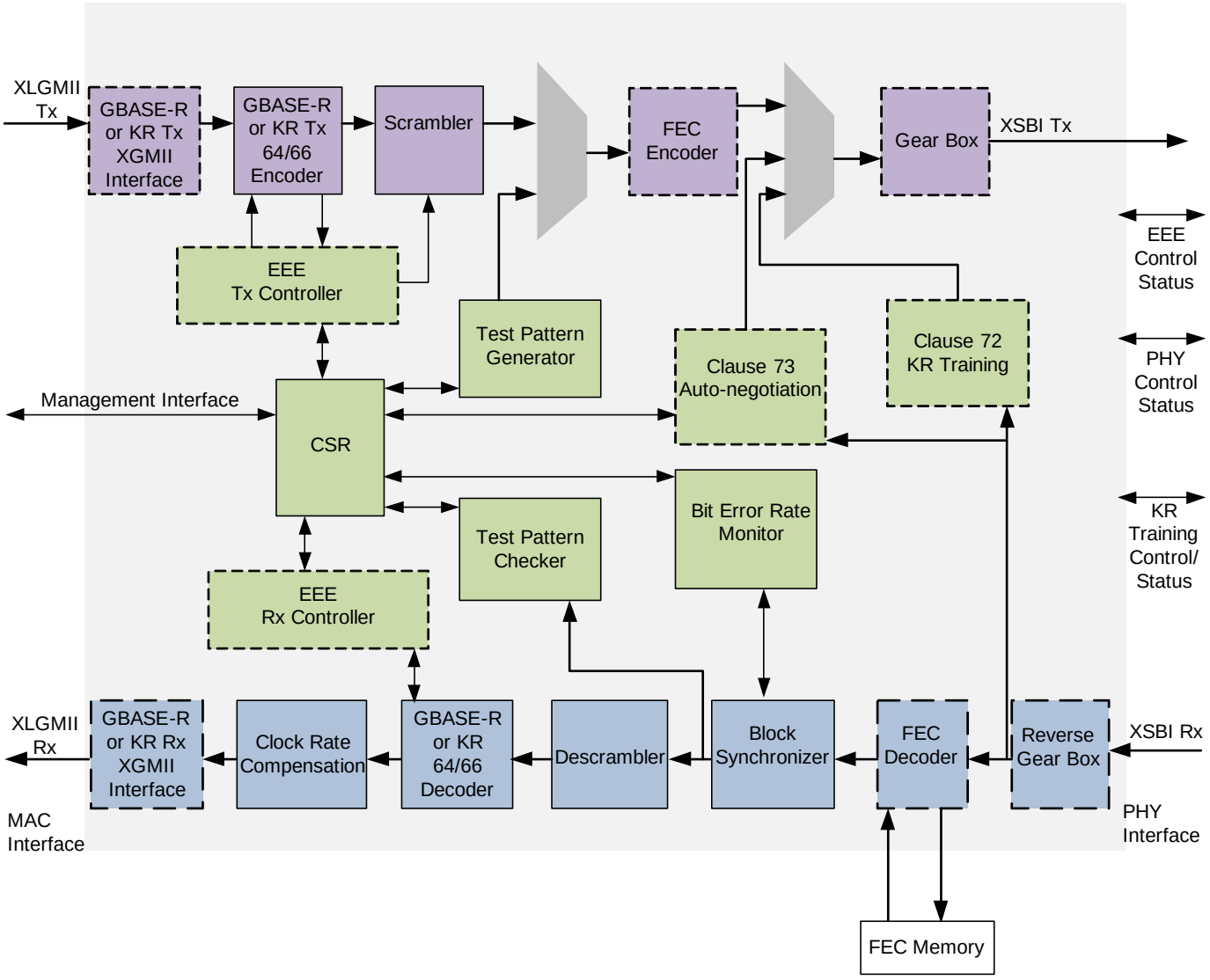
Figure 2-1 DWC_xlgpcs Block Diagram



2.2 Data Path

Figure 2-5 displays the components of the DWC_xlgpcs data path.

Figure 2-2 Data Path Components



Note See "When error forwarding is disabled, Reed-Solomon decoder counts the number of symbol errors detected in consecutive non-overlapping blocks of 8192 codewords. When the number of symbol errors in a block of 8192 codewords exceeds 417, the Reed-Solomon decoder makes synchronization header bits of each subsequent 66-bit block to 2'b11." for details of the Memory requirements for different FEC configurations.

2.2.1 Transmit Block

The following are the Transmit block features:

- Implements the following interfaces:
  - XLGMII interface towards the MAC

  - 5GBASE-R, 10G BASE-R, 25G BASE-R, 40GBASE-R, 50G BASE-R and 100GBASE-R parallel interface towards the PHY
- Performs 64B/66B encoding for block boundary determination
- Performs optional Forward Error Correction (FEC) encoding using CL74 FEC (BASE-R FEC) or Clause 91/Clause 108 RS FEC/Clause 161 interleaved RS-FEC
- Performs scrambling to ensure transition density
- Performs Insertion of Alignment marker with programmable Alignment marker timer
- In Multi lane configurations provides lane disable control for Lane 3, Lane 2 and Lane 1 either through programming the register or through external lane disable controls
- Generates the following test patterns:
  - Pseudo Random for 10G/25GBASE-R
  - PRBS 31 for 10G/25GBASE-R
  - PRBS 9 for 10G/25GBASE-R
  - Square wave for 10G/25GBASE-R
  - PRBS 31 for 25G/40G/50G/100GBASE-R PMA per lane
  - PRBS 9 for 25G/40G/50G/100GBASE-R PMA per lane
  - Square wave for 25G/40G/50G/100GBASE-R PMA per lane
  - Scrambled Idle TP for 25G/40G/50G/100GBASE-R

2.2.1.1 Transmit 5G or 10GBASE-R XGMII Interface

The Transmit 5GBASE-R, 10GBASE-R, or 10GBASE-KR XGMII block interfaces with the MAC. This block converts the XGMII Transmit data interface, operating in the SDR mode (32-bit data and 5-bit control at 312.5 MHz or 156.25 MHz), to the DDW mode (64-bit data and 9-bit control) required for 5GBASE-R or 10GBASE-R.

2.2.1.2 Transmit 25GBASE-R XLGMII Interface

The XLGMII block interface is used between the Transmit 25GBASE-R or its variants and the MAC. When the DWCXLP_XLGMII_INTF_WIDTH parameter takes a value of 0, this block converts the XLGMII Transmit data interface, operating with 128-bit data and 17-bit control at 390.625 MHz and with back pressure asserted on alternate clock cycles, to the 64-bit data and 9-bit control format required for single BASE-R PCS lane mode of operation.

For 25G_Only configurations with DWCXLP_XLGMII_INTF_WIDTH chosen as 1, this block doesn't exist, as the MAC interface data path width matches the internal requirements. For 25G_Only configurations with DWCXLP_XLGMII_INTF_WDITH chosen as 2, this block converts the 32-bit data and 5-bit control signal at 781.25 MHz to 64-bit and 9-bit control format required for internal transmit data path.

2.2.1.3 Transmit 50GBASE-R XLGMII Interface

The XLGMII block interface s is used between the Transmit 50GBASE-R or its variants and the MAC. This block converts the XLGMII Transmit data interface, operating in the SDR mode (128-bit data and 17-bit control at 390.625 MHz or 402.4621 MHz), to the DDW mode (256-bit data and 33-bit control) required for 50GBASE-R operation.

2.2.1.4 Transmit 100GBASE-R XLGMII Interface

The XLGMII block interface is used between the Transmit 100GBASE-R or its variants and the MAC. This block converts the XLGMII Transmit data interface, operating in the SDR mode (128-bit data and 17-bit control at 781.25MHz or 804.9242 MHz), to the DDW mode (256-bit data and 33-bit control at 390.625 MHz or 402.4621 MHz) required for 100GBASE-R operation.

2.2.1.5 Transmit 40GBASE-R XLGMII Interface

The Transmit 40GBASE-R XLGMII block interfaces with the MAC. This block converts the XLGMII Transmit data interface, operating in the SDR mode (128-bit data and 17-bit control at 312.5 MHz), to the DDW mode (256-bit data and 33-bit control) required for 40GBASE-R.

2.2.1.6 5G, 10G, or 25GBASE-R 64/66 Encoder

The 64/66 encoder block encodes the incoming data in DDW format (64-bit data and 8-bit control at 156.25 MHz) to encoded 66-bit data. This is done by adding the 2-bit Sync Header and following the encoding rules defined in the IEEE Std 802.3, Clause 49.

If the XGMII input has all the data (of format D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7), the encoding is done by just adding the Sync Header of 2'b01 to form 66-bits.

Input XGMII Data	Encoded Output Data	
	Sync Header	Data Encoding
D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7	2'b01	D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7

where, D_n = XGMII Data Character in n^{th} byte on xlgmii_txd_i[] with xgmii_txc_i[n] programmed to 0.

If the XGMII input has the non-data (that is, having format other than D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7), encoding is done in the following way:

1. By adding 2-bit Sync Header with the value 2'b10
2. By adding 8-bit Block Type field depending on the following factors:
   - Start of Frame control character (|S|) and location of |S| on 64-bit data
   - Terminate control character (|T|) and location of |T| on 64-bit data
   - Order set |O| (Link Message) and location of |O| on 64-bit data
   - Error control character |E|
   - Idle control character |I|

Figure 2-3 shows how the Encoder maps input XGMII control characters to output.

Figure 2-3 Encoded Output Data for Input XGMII Control Characters

XGMII Input Control Character	Encoded Output Data									
	SH	Block Type	Control Code							
$C_0 C_1 C_2 C_3 / C_4 C_5 C_6 C_7$	10	0x1e	C_0	C_1	C_2	C_3	C_4	C_5	C_6	C_7
$C_0 C_1 C_2 C_3 / O_4 D_5 D_6 D_7$	10	0x2d	C_0	C_1	C_2	C_3	O_4	D_5	D_6	D_7
$C_0 C_1 C_2 C_3 / S_4 D_5 D_6 D_7$	10	0x33	C_0	C_1	C_2	C_3		D_5	D_6	D_7
$O_0 D_1 D_2 D_3 / S_4 D_5 D_6 D_7$	10	0x66	D_1	D_2	D_3	O_0		D_5	D_6	D_7
$O_0 D_1 D_2 D_3 / O_4 D_5 D_6 D_7$	10	0x55	D_1	D_2	D_3	O_0	O_4	D_5	D_6	D_7
$S_0 D_1 D_2 D_3 / D_4 D_5 D_6 D_7$	10	0x78	D_1	D_2	D_3	D_4		D_5	D_6	D_7
$O_0 D_1 D_2 D_3 / C_4 C_5 C_6 C_7$	10	0x4b	D_1	D_2	D_3	O_0	C_4	C_5	C_6	C_7
$T_0 C_1 C_2 C_3 / C_4 C_5 C_6 C_7$	10	0x87		C_1	C_2	C_3	C_4	C_5	C_6	C_7
$D_0 T_1 C_2 C_3 / C_4 C_5 C_6 C_7$	10	0x99	D_0		C_2	C_3	C_4	C_5	C_6	C_7
$D_0 D_1 T_2 C_3 / C_4 C_5 C_6 C_7$	10	0xaa	D_0	D_1		C_3	C_4	C_5	C_6	C_7
$D_0 D_1 D_2 T_3 / C_4 C_5 C_6 C_7$	10	0xb4	D_0	D_1	D_2		C_4	C_5	C_6	C_7
$D_0 D_1 D_2 D_3 / T_4 C_5 C_6 C_7$	10	0xcc	D_0	D_1	D_2	D_3		C_5	C_6	C_7
$D_0 D_1 D_2 D_3 / D_4 T_5 C_6 C_7$	10	0xd2	D_0	D_1	D_2	D_3	D_4		C_6	C_7
$D_0 D_1 D_2 D_3 / D_4 D_5 T_6 C_7$	10	0xe1	D_0	D_1	D_2	D_3	D_4	D_5		C_7
$D_0 D_1 D_2 D_3 / D_4 D_5 D_6 T_7$	10	0xff	D_0	D_1	D_2	D_3	D_4	D_5	D_6	

where in XGMII Input Data Notations:

- C_n = XGMII Control Character in n^{th} byte (xlgmii_txc_i[n] programmed to 1)
- D_n = XGMII Data Character in n^{th} byte on xlgmii_txd_i[] with xgmii_txc_i[n] programmed to 0
- $S_{0/4}$ = XGMII Start Control Character in 0/4 byte (XGMII_DATA[] = 8'hFB and xgmii_txc_i [0/4] = 1)
- T_n = XGMII Terminate Character in n^{th} byte (XGMII_DATA[] = 8'hFD and xgmii_txc_i[n] = 1)
- $O_{0/4}$ = XGMII Link Message Control Character in 0/4 byte (xlgmii_txd_i[0/4] = 8'h9C and xgmii_txc_i [0/4] = 1)

and in Output Notations:

- C_n = Control Code in n^{th} byte (7-bit field with 7'h00 value for Idle and 7'h06 value for |LPI|)
- $O_{0/4}$ = Message Control Code in 0th or 4th byte (4-bit field with 4'h0 value)
- D_n = XGMII Data Character in n^{th} byte on xlgmii_txd_i[]
- || = 1-bit data with value 0 (1-bit field with 1'b0 value)

The output encoded control codes (C_n, $O_{0/4}$, |S|, |T|, |E|, and other reserved control characters) are shown in Table 2-1. If EEE_EN = Enabled, the DWC_xlgpcs also encodes the XGMII |LI| (LPI) characters as shown in Table 2-1.

Table 2-1 XGMII Output Control Characters

Control Character	Notation	XGMII Control Code	25GBASE-R Control Code	25GBASE-R O Code
Idle	/I/	0x07	0x00	-
Start	/S/	0xfb	Encoded by the Block Type field	-
Terminate	/T/	0xfd	Encoded by the Block Type field	-
LPI	/LI/	0x06	0x06	-
Error	/E/	0xfe	0x1e	-
Sequence ordered_set	/Q/	0x9c	Encoded by the Block Type field plus O code	0x0
reserved0	/R/	0x1c	0x2d	-
reserved1	-	0x3c	0x33	-
reserved2	/A/	0x7c	0x4b	-
reserved3	/K/	0xbc	0x55	-
reserved4	-	0xdc	0x66	-
reserved5	-	0xf7	0x78	-
Signal ordered_set	/Fsig/	0x5c	Encoded by the Block Type field	0xF

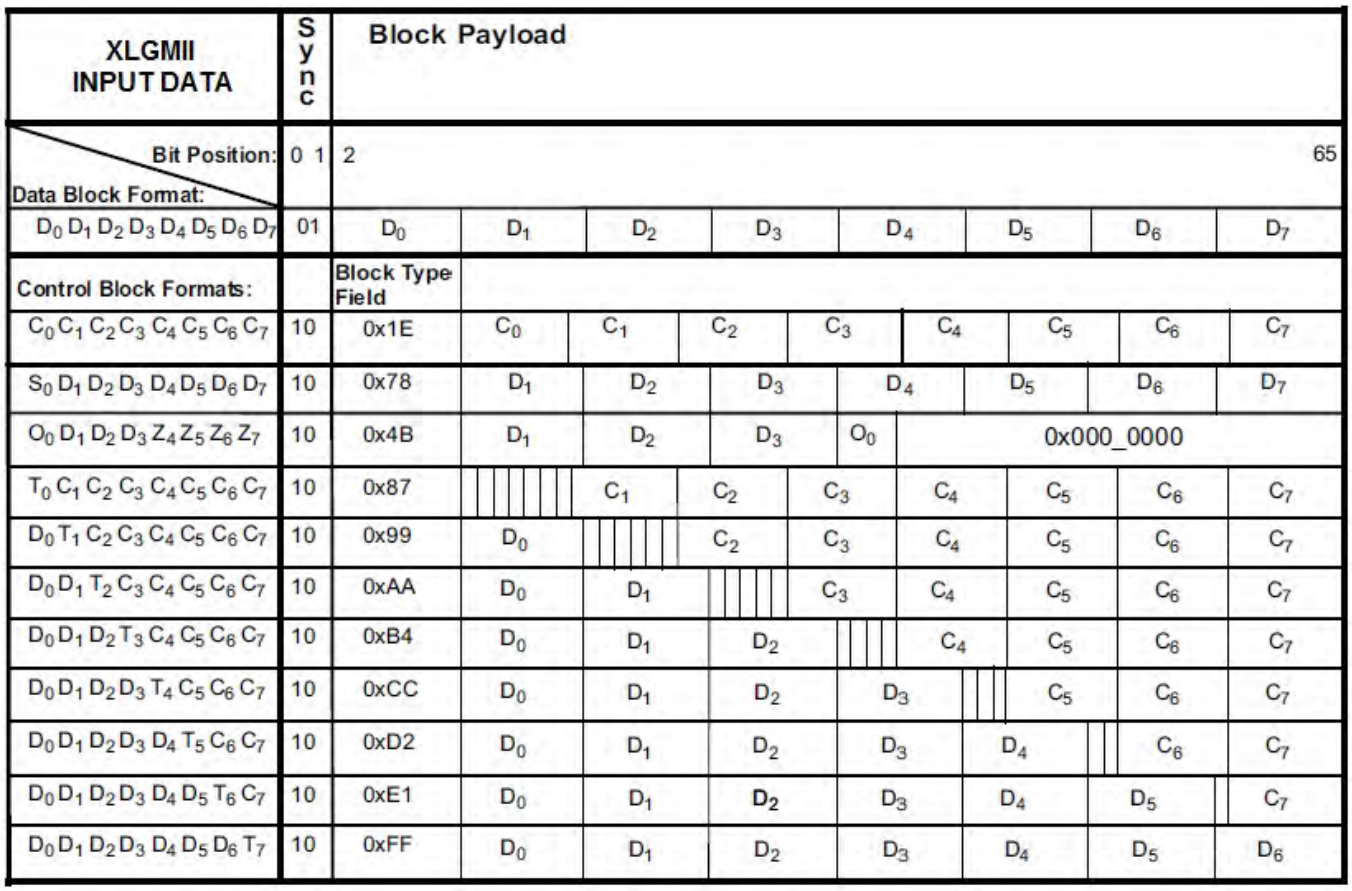
2.2.1.7 40/50G/100GBASE-R 64/66 Encoder

The 64/66 encoder block encodes the incoming data in DDW format (256-bit data and 32-bit control at 156 MHz for 40G, or 390.625 MHz for 50G/100G NRZ modes of 402.4621 MHz for 50G/100G PAM4 modes) to encoded 264-bit data. This is done by four 64/66 encoder blocks. Each of 64/66 encoder encodes the incoming 64 bit data and 8-bit control. This is done by adding the 2-bit Sync Header and following the encoding rules defined in the IEEE Std 802.3, Clause 82.

If DWC_xlgpcs input has all the data (of format D0 D1 D2 D3 D4 D5 D6 D7), the encoding is done by just adding the Sync Header of 2'b01 to form 66-bits.

Figure 2-4 shows how the Encoder maps input XLGMII data and control characters to output.

Figure 2-4 Encoded Output Data for Input XLGMII Control Characters



The output encoded control codes are shown in Table 2-2.

Table 2-2 XLGMII Output Control Characters

Control Character	Notation	XLGMII or CGMII Control Code	40/100GBASE-RO Code	40GBASE-R O Code
Idle	/I/	0x07	-	0x00
Start	/S/	0xFB	-	Encoded by the Block Type field
Terminate	/T/	0xFD	-	Encoded by the Block Type field
Error	/E/	0xFE	-	0x1E
Sequence ordered_set	/Q/	0x9c	0x0	Encoded by the Block Type 0x4B field plus O code, control codes are programmed to 0x00

Table 2-2 XLGMII Output Control Characters (Continued)

Control Character	Notation	XLGMII or CGMII Control Code	40/100GBASE-RO Code	40GBASE-R O Code
Signal ordered_set	/Fsig/	0x5c	0xF	Encoded by the Block Type 0x4B field plus O code, control codes are programmed to 0x00

2.2.1.8 Scrambler

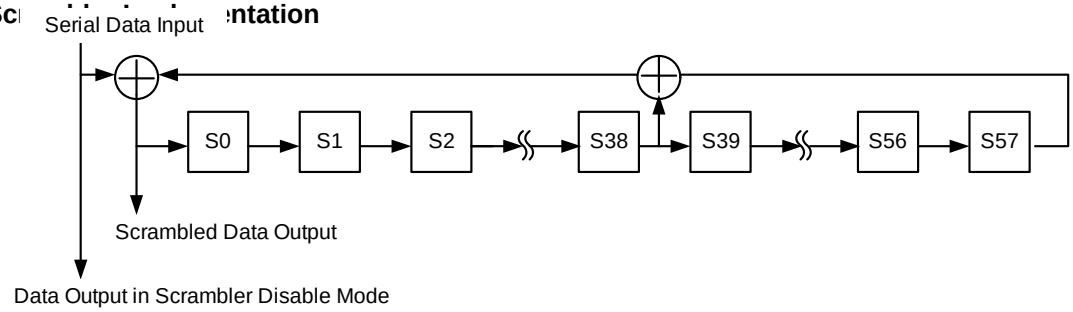
The encoded data is passed through the scrambler. The scrambler takes 256-bit data in 40G, 50G, or 100G mode and 64-bit data in 5G, 10G, or 25G mode and generates the scrambled data high in transition density. The scrambler does not alter the valid 2-bit Sync header for 5G, 10G, or 25G and 8-bit Sync header for 40G/50G/100G because it is used for Block alignment at the receiver and it must not be scrambled.

The scrambling is done using the following polynomial:

$G(x) = 1+ x^{39} + x^{58}$

The initial seed for this scrambler can be any value. Figure 2-5 shows the serial implementation of the scrambler. The scrambler is bypassed depending on bit 8 of the VR_PCS_KR_CTRL register. If DWCXLP_EEE_EN is enabled and DWCXLP_FEC_EN is also enabled, the scrambler is bypassed during low-power mode of the EEE protocol if bit 5 of the VR_PCS_KR_CTRL register is programmed.

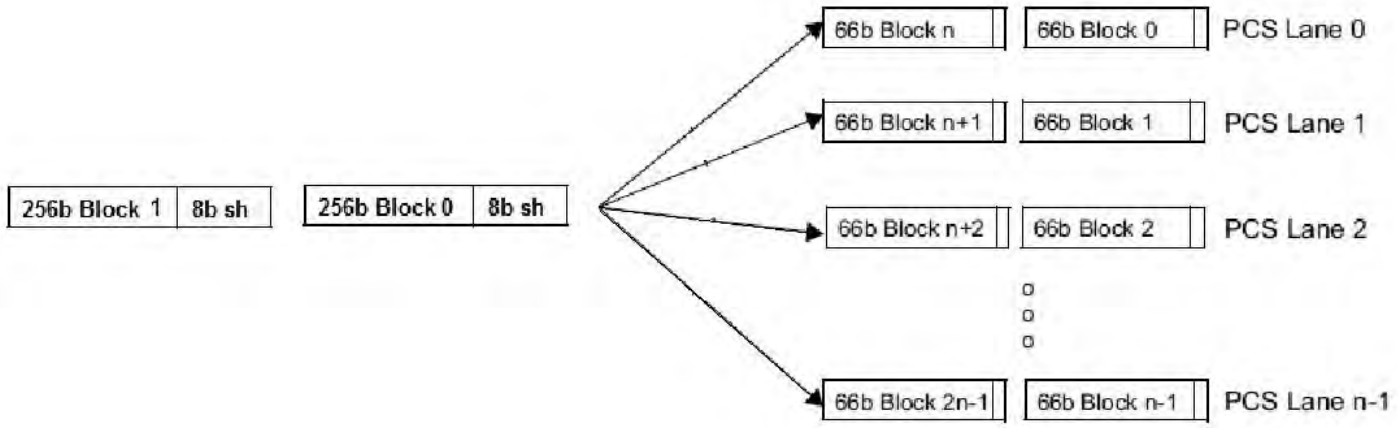
Figure 2-5 Sc ntation



2.2.1.9 Block Distribution

Block distribution is applicable to speed modes involving multiple PHY lanes (40G,50G and 100G speed modes). After the 256-bit data is encoded and scrambled, it is re-arranged to multiple PCS lanes of 66-bit blocks. This allows DWC_xlgpcs to support multiple physical lanes. 40G/50GBASE-R PCS distribute the 66-bit blocks to 4 PCS (virtual) lanes. 100GBASE-R PCS distributes the 66-bit blocks to 20 PCS (virtual) lanes. Figure 2-6 illustrates the block distribution process.

Figure 2-6 Block Distribution



2.2.1.10 Alignment Marker Insertion

To support deskew and reordering of individual PCS lanes at the receive PCS, alignment markers are added periodically to each PCS lane. The alignment marker has the form of a specially defined 66-bit block with a control block sync header. These markers interrupt any data transfer that is already in progress. This allows alignment markers to be inserted into all PCS lanes at the same time. The alignment markers shall be inserted after every 16383 66-bit blocks on each PCS lane. Figures 2-7 and 2-8 illustrate the alignment marker insertion process.

Figure 2-7 Alignment Marker Insertion (For 100G n = 19 corresponds to 19 Lanes)

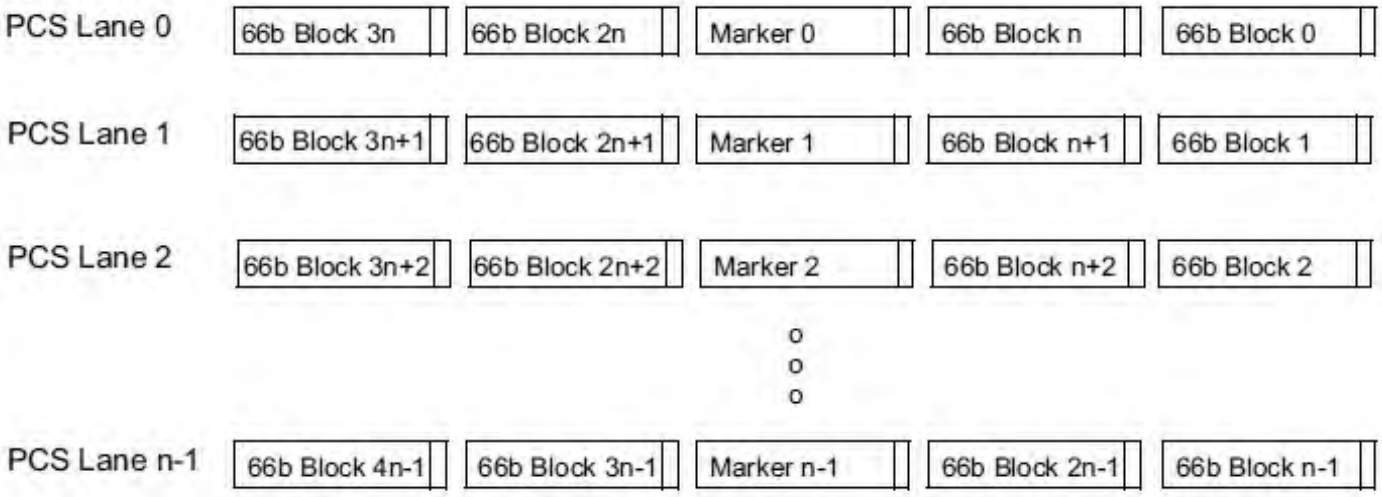


Figure 2-8 Alignment Marker Insertion Period

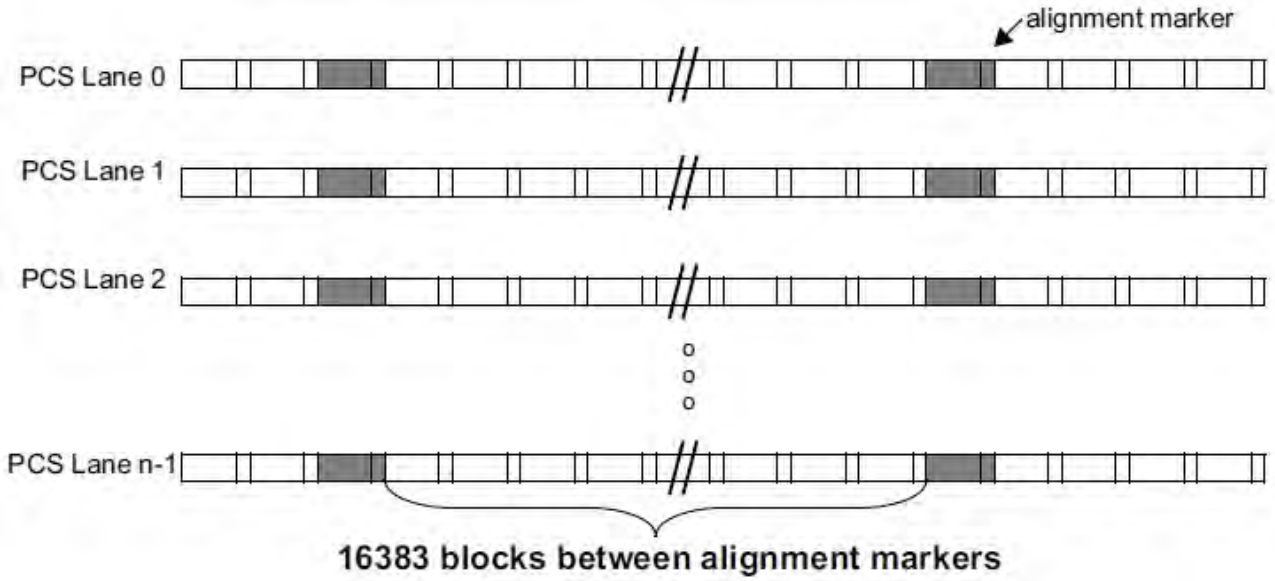
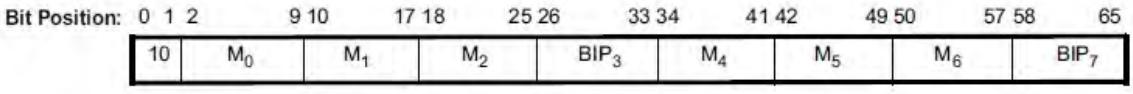


Figure 2-9 illustrates the format of the alignment marker.

Figure 2-9 Alignment Marker Format



The contents depend on the PCS lane number and the octet number.

Note M4 through M6 are the bit-wise inversion of M0 through M2, respectively. Similarly, BIP7 is the bit-wise inversion of BIP3. This property allows the alignment markers to be DC balanced.

Table 2-3 lists the 40GBASE-R and 20GBASE-R alignment marker encodings.

Table 2-3 40GBASE-R Alignment Marker Encodings

PCS Lane Number	Encoding for 40GBASE-R[1] {M_0, M_1, M_2, BIP_3, M_4, M_5, M_6, BIP_7}	Encoding for 20GBASE-R[2] {M_0, M_1, M_2, BIP_3, M_4, M_5, M_6, BIP_7}
0	0x90, 0x76, 0x47, BIP_3, 0x6F, 0x89, 0xB8, BIP_7	0x90, 0x76, 0x47, BIP_3, 0x6F, 0x89, 0xB8, BIP_7
1	0xF0, 0xC4, 0xE6, BIP_3, 0x0F, 0x3B, 0x19, BIP_7	0xF0, 0xC4, 0xE6, BIP_3, 0x0F, 0x3B, 0x19, BIP_7
2	0xC5, 0x65, 0x9B, BIP_3, 0x3A, 0x9A, 0x64, BIP_7	Not applicable
3	0xA2, 0x79, 0x3D, BIP_3, 0x5D, 0x86, 0xC2, BIP_7	Not applicable

1. Each octet is transmitted LSB to MSB.
2. Each octet is transmitted LSB to MSB.

Note In RS FEC mode, alignment marker format for 50G is same as that of 40G

Table 2-4 lists the 100GBASE-R Alignment marker encodings.

Table 2-4 100GBASE-R Alignment Marker Encodings

PCS Lane Number	Encoding[1] {M_0, M_1, M_2, BIP_3, M_4, M_5, M_6, BIP_7}
0	0xC1, 0x68, 0x21, BIP3, 0x3E, 0x97, 0xDE, BIP7
1	0x9D, 0x71, 0x8E, BIP3, 0x62, 0x8E, 0x71, BIP7
2	0x59, 0x4B, 0xE8, BIP3, 0xA6, 0xB4, 0x17, BIP7
3	0x4D, 0x95, 0x7B, BIP3, 0xB2, 0x6A, 0x84, BIP7
4	0xF5, 0x07, 0x09, BIP3, 0x0A, 0xF8, 0xF6, BIP7
5	0xDD, 0x14, 0xC2, BIP3, 0x22, 0xEB, 0x3D, BIP7
6	0x9A, 0x4A, 0x26, BIP3, 0x65, 0xB5, 0xD9, BIP7
7	0x7B, 0x45, 0x66, BIP3, 0x84, 0xBA, 0x99, BIP7
8	0xA0, 0x24, 0x76, BIP3, 0x5F, 0xDB, 0x89, BIP7
9	0x68, 0xC9, 0xFB, BIP3, 0x97, 0x36, 0x04, BIP7
10	0xFD, 0x6C, 0x99, BIP3, 0x02, 0x93, 0x66, BIP7
11	0xB9, 0x91, 0x55, BIP3, 0x46, 0x6E, 0xAA, BIP7
12	0x5C, 0x B9, 0xB2, BIP3, 0xA3, 0x46, 0x4D, BIP7
13	0x1A, 0xF8, 0xBD, BIP3, 0xE5, 0x07, 0x42, BIP7

Table 2-4 100GBASE-R Alignment Marker Encodings

PCS Lane Number	Encoding[1] {M_0, M_1, M_2, BIP_3, M_4, M_5, M_6, BIP_7}
14	0x83, 0xC7, 0xCA, BIP3, 0x7C, 0x38, 0x35, BIP7
15	0x35, 0x36, 0xCD, BIP3, 0xCA, 0xC9, 0x32, BIP7
16	0xC4, 0x31, 0x4C, BIP3, 0x3B, 0xCE, 0xB3, BIP7
17	0xAD, 0xD6, 0xB7, BIP3, 0x52, 0x29, 0x48, BIP7
18	0x5F, 0x66, 0x2A, BIP3, 0xA0, 0x99, 0xD5, BIP7
19	0xC0, 0xF0, 0xE5, BIP3, 0x3F, 0x0F, 0x1A, BIP7

1. Each octet is transmitted LSB to MSB

2.2.1.11 BIP Calculations

A PCS lane Bit Interleaved Parity (BIP) field is carried in each PCS Lane alignment marker. This allows an accurate and fast measure of the bit error ratio of a given PCS Lane. This information is used to update error counters; no state machines use this information.

Each alignment marker has two BIP fields: BIP_3 and BIP_7. BIP_7 is a bit-wise inversion of BIP_3 in order to keep the alignment marker DC balanced. The BIP_3 field contains the result of a bit interleaved parity calculation. Each bit in the BIP field is an even parity calculation over all of the previous specified bits of a given PCS Lane, from and including the previous alignment marker, but not including the current alignment marker.

Using the bit definitions shown in Table 2-3, Table 2-5 lists the bit assignments for each BIP_3 bit. As an example, BIP_3 bit 0 contains the result of XORing 131072 bits from 16384 66-bit words. BIP_3 bit 3 and bit 4 also include one sync header bit from each 66-bit word. Bit[3] and bit[4] each contain the result of XORing 147456 bits.

Table 2-3 lists the 40GBASE-R alignment marker encodings.

Table 2-5 BIP_3 Bit Assignments

BIP_3 Bit Number	Assigned 66-bit Word Bits
0	2, 10, 18, 26, 34, 42, 50, 58
1	3, 11, 19, 27, 35, 43, 51, 59
2	4, 12, 20, 28, 36, 44, 52, 60
3	0, 5, 13, 21, 29, 37, 45, 53, 61
4	1, 6, 14, 22, 30, 38, 46, 54, 62
5	7, 15, 23, 31, 39, 47, 55, 63
6	8, 16, 24, 32, 40, 48, 56, 64

Table 2-5 BIP_3 Bit Assignments

BIP_3 Bit Number	Assigned 66-bit Word Bits
7	9, 17, 25, 33, 41, 49, 57, 65

Note

- To speed up the simulation, you can reduce the Alignment Maker (AM) insertion period to lower values instead of the standard defined 16383. This enables the Remote End Receiver to Link UP faster. You can program the VR_PCS_AM_CNT register to use the reduced value of the AM insertion Interval.
- When you change the alignment maker insertion in DWC_xlgpcs, you must change the Alignment marker insertion interval in the remote end device also for the same value.

2.2.1.12 BASE-R FEC Encoder

The Forward Error Correction (FEC) encoder block is an optional block. This block is present only when the Add CL74 Forward Error Correction Support option is selected. This block enables the error correction mechanism on the Receive path. You can select this option to reduce the effective bit error rate (BER).

This block is needed for erroneous channels, especially when BASE-R type serial data is transmitted or received on copper (Backplane Ethernet applications called BASE-KR). This block uses one of the Cyclic Codes which has both Error Detection and Error Correcting capabilities. It can detect and correct up to 11-bit burst errors.

BASE-R FEC Encoder is per PCS lane:

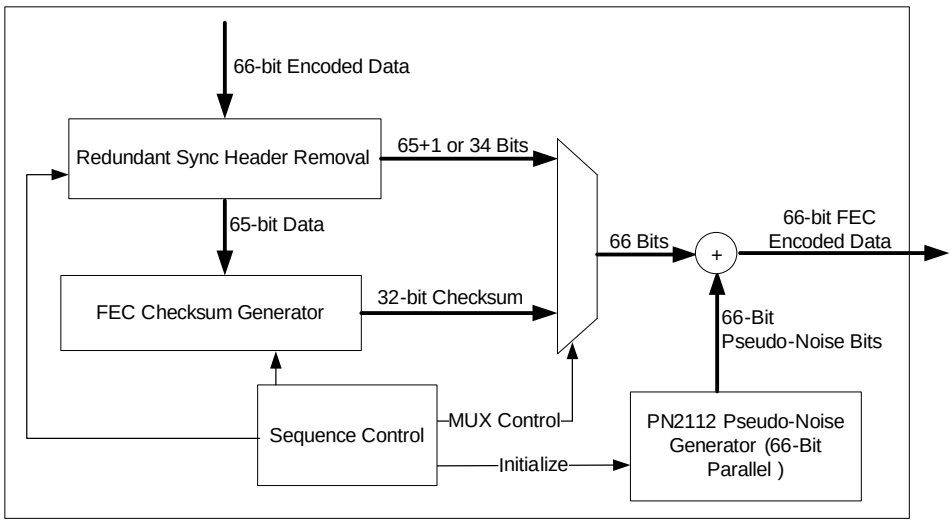
- 50GBASE-R: 4 lanes (4*12.890625Gbps)
- 40GBASE-R: 4 lanes (4*10.31232Gbps)
- 25GBASE-R: 1 lane (1*25.78125Gbps)
- 20GBASE-R: 2 lanes (2*10.31232Gbps)
- 10GBASE-R: 1 lane (1*10.31232Gbps)

Note

- The FEC capabilities are exchanged between the local device and the link partner through Clause 73 auto-negotiation. If both the local device and the link partner support FEC, the FEC feature can be enabled in the local device. You can obtain the FEC ability status through the SR_PMA_KR_FEC_ABL register and you can control the FEC features through the SR_PMA_KR_FEC_CTRL register.
- BASE-R FEC is not supported for 5GBASE-R and 100GBASE-R modes.

Figure 2-10 shows the FEC encoding process.

Figure 2-10 BASE-R FEC Generation Process



If FEC is enabled in your configuration and the FEC feature needs to be auto-negotiated, you need to enable the FEC by programming bit 0 of the SR_PMA_KR_FEC_CTRL register. The FEC is considered to be successfully auto-negotiated if the following settings are negotiated during base page exchange:

- Bit[14] of the SR_AN_ADV3 register is programmed to 1, indicating that the local device has the FEC capability.
- Bit[15] of the SR_AN_ADV3 register is programmed to 1, indicating that the local device is requesting the FEC mode.
- Bit[14] of the SR_AN_LP_ABL3 register is programmed to 1, indicating that the link partner has the FEC capability.
- Bit[15] of the SR_AN_LP_ABL3 register is programmed to 1, indicating that the link partner is requesting the FEC mode.

In addition, if CL72_EN is enabled, you need to program as follows:

1. The host must enable the Clause 72 start-up protocol by writing to bit 1 of the SR_PMA_KR_PMD_CTRL register.
2. The host must restart the Clause 72 start-up protocol by writing to bit 0 of the SR_PMA_KR_PMD_CTRL register.

2.2.1.12.1 FEC Checksum Generator

This block removes one of the redundant bits of Sync Header of 32-blocks (each block is of 66-bit) data and adds 32-bit checksum at the end. It uses the redundant bit (of Sync Header portion) of 64/66 encoded data to add the checksum so that the effective data rate does not change.

The polynomial used for FEC encoder is:

$g(x) = x^{32} + x^{23} + x^{21} + x^{11} + x^{2} + 1$

If $m(x)$ is the message polynomial and $g(x)$ is the generator polynomial, the checksum $p(x)$ is calculated by dividing the message polynomial with the generated polynomial. The remainder is the checksum.

$p(x) = x^{32}m(x) \bmod g(x)$

Note The CRC polynomial is of Cyclic Code (42,987, 42,955) which means the code length is 42,987 bits. This Cyclic Code has error correcting capability of up to 11-bit burst error. In DWC_xlgpcs, it is used in the Shortened form of Code (2112, 2080) to detect up to 11-bit burst for each 2112 (65x33) block size. This makes the error correcting capability of 11-bit burst error for every 65x33 data block size.

The FEC block `c(x)` is generated by appending or adding the 32-blocks (65 bits) `m(x)` with checksum *P(x)*:

$c(x) = x^{32}m(x) + p(x)$

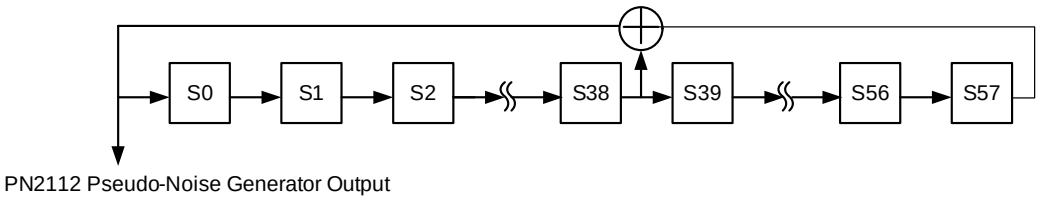
2.2.1.12.2 PN2112 Pseudo-Noise Generator

The output of the FEC Encoder block is mixed with the PN2112 pseudo-noise generator to ensure transition density for the FEC encapsulated block. The polynomial used for pseudo-noise generator is:

$r(x) = 1 + x^{39} + x^{58}$

Figure 2-11 shows the serial implementation of the PN2112 pseudo-noise generator.

Figure 2-11 Transmit Pseudo-Noise Generator



The following is the initial state of the PN2112 pseudo-noise generator:

```
S[57:0] = {29{2'b10}};
```

Note The pseudo-noise generator is initialized with this value before each FEC block (32-block of 66-bit encoded input).

2.2.1.13 Reed Solomon FEC Encoder

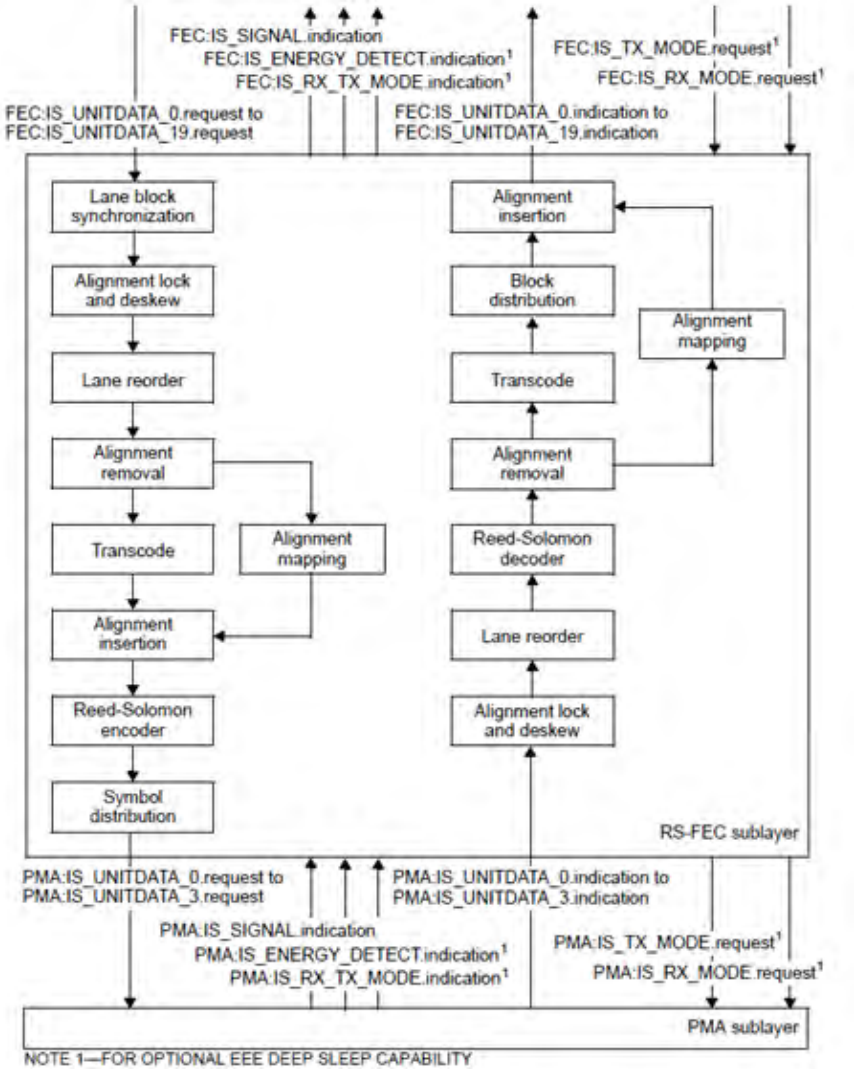
The Reed Solomon Forward Error Correction (FEC) encoder block is an optional block. Figure 2-12 shows the functional block diagram of Reed Solomon FEC encoder. This block is present only when the “Add RS Forward Error Correction Support” option is selected. This block enables the error correction mechanism on the Receive path. You can select this option to reduce the effective bit error rate (BER). Reed Solomon FEC is a non-binary Forward Error Correction code and the error detection capability is specified in terms of symbols.

The Reed Solomon Forward Error Correction (RS-FEC) scheme provides capability to detect and correct symbols (multi-bit errors) on an impaired channel. This is an optional function which is enabled (if present

in the core) based on the channel condition. The functions described below are applicable to both 100G (Clause 91, Clause 161) modes, 25G (Clause 108) and 50G (Clause91) modes, unless mentioned specifically.

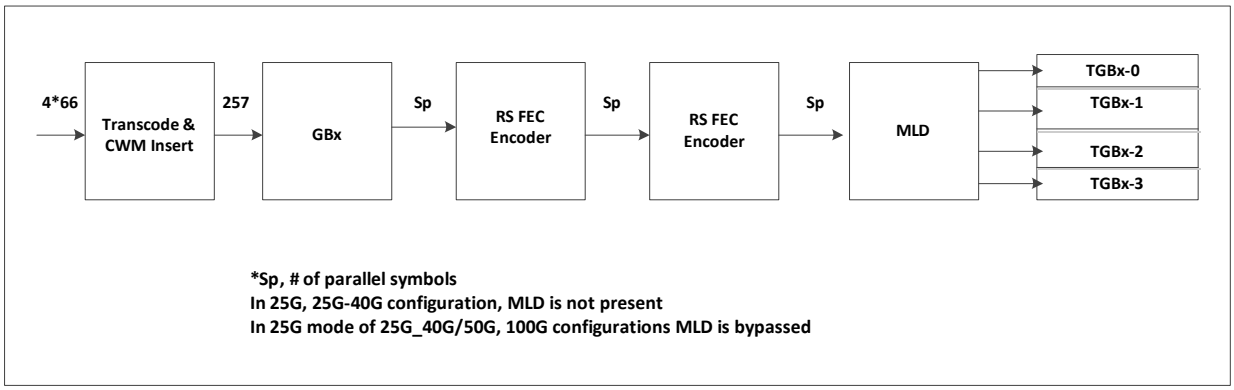
Note The RS-FEC capabilities are exchanged between the local device and the link partner through Clause 73 auto-negotiation. If both local device and link partner support RS-FEC, the RS-FEC feature can be enabled in the local device. You can obtain the RS RS-FEC ability status through SR_PMA_RS_FEC_STATUS and you can control the RS FEC features through SR_PMA_RS_FEC_CTRL.

Figure 2-12 Functional Block Diagram of the Reed Solomon FEC Sub Layer



The functionality of RS FEC sub layer in 25G/50G/100G speed modes is more or less same except that symbol distribution is applicable only in 50G and 100G configurations (Symbol distribution is not applicable in 25G modes as there is only one PHY lane involved). Figure 2-13 shows the sub blocks of the RS FEC data path.

Figure 2-13 Sub Blocks of Reed Solomon FEC Data Path



Note

- To speed up the simulation, you can reduce the Alignment Maker (AM)/Code Word Marker (CWM) insertion period to lower values instead of the default value. This enables the Remote End Receiver to Link UP faster. You can program the VR_PCS_AM_CNT register to use the reduced value of the AM insertion Interval.
- In RS FEC mode, the contents of the VR_PCS_AM_CNT register are interpreted as the interval between Alignment Markers/Code Word markers in terms of number of RS FEC code words.
- In Non-RS FEC mode, the contents of the VR_PCS_AM_CNT register are interpreted as the AM/CWM interval in terms of number of 66-bit blocks.
- In non RS FEC modes, value of AM interval is in terms of 66-bit blocks. An AM interval value of 16384 is equivalent to
  - 81920 clocks for 100G
  - 16384 clocks for 40G
  - 32768 clocks for 50G

In RS FEC enabled modes, value of RS FEC Alignment marker/Code word marker interval is in terms of RS FEC code words. An interval of 1024 results in AMs inserted every 655360 bytes once.

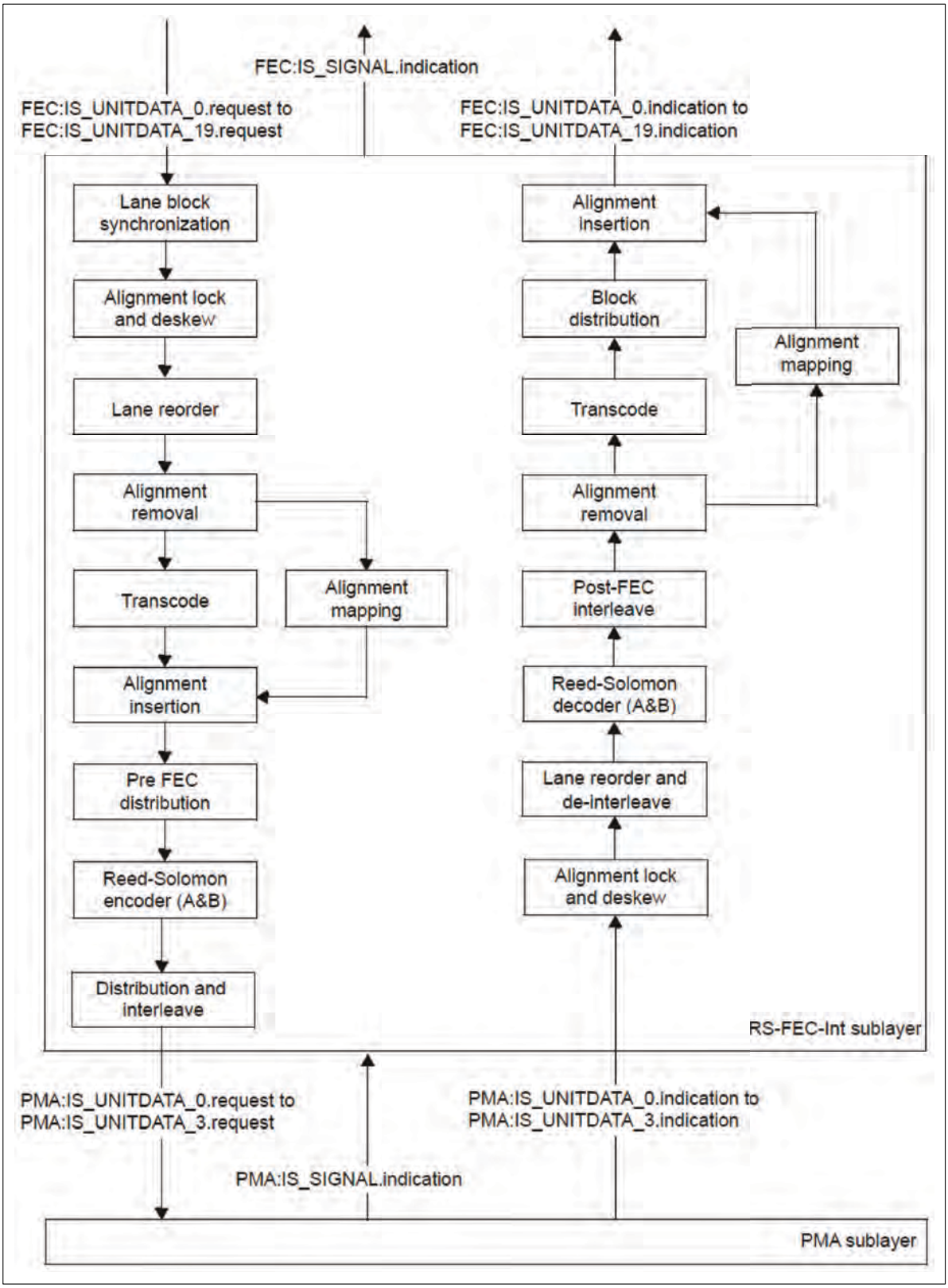
As the default mode is non-RS FEC mode, the default value of 16383 refers to AM interval in terms of number of 66-bit blocks.

When AM interval is to be changed, it has to be ensured that both the local device and the link partner have to be programmed with the same value of AM interval. Otherwise, behavior of the Enterprise PCS is undefined.

For more details, see the Note in “CWM lock: Code Word Marker Lock” on page 79.

When operating in 100GBASE-KR1 or 100GBASE-CR1 modes, DWC_xlgpcs can additionally support codeword interleaved RS-FEC mode as defined in IEEE802.3ck specification. Figure 2-14 shows the functional block diagram of codeword interleaved RS-FEC sub-layer.

Figure 2-14 Codeword Interleaved RS-FEC Sub-layer



2.2.1.13.1 Transcoder (64b/66b to 256b/257b Conversion)

The transcoder constructs a 257-bit block from a group of 4 x 66-bit blocks from the PCS sub layer based on sub-clause 91.5.2.5 for 50G/100G and on sub-clause 108.5.2.3 for 25G to allocate bandwidth for the parity check symbols added by the Reed-Solomon encoder. This relation satisfies the bandwidth requirement when RS(528,514) is used. Seven bits are removed from 264-bit input to form 257 bit output. Hence with twenty 66-bit blocks, 140 bits from input are removed to accommodate 140 bits of parity (14 parity symbols).

2.2.1.13.2 Code Word Marker/Alignment Marker Insert and Map

One group of aligned and reordered Code Word/alignment markers are mapped periodically based on the speed mode.

In 25G mode of operation, alignment markers appear once every 81920 number of 66-bit blocks (equivalently once every 1024 RS FEC code words).

As per the 25G/50G specification, in 50G mode of operation, alignment markers appear once every 20480 number of 66-bit on each of the four PCS lanes (which corresponds to every 1024 RS FEC code words).

2.2.1.13.3 Code Word/Alignment marker map in 100G

In 100G mode of operation, the fixed bytes of the alignment markers corresponding to PCS lanes 1, 2, and 3 are substituted with the fixed bytes for the alignment marker corresponding to PCS lane 0. Similarly, the fixed bytes of the alignment markers corresponding to PCS lanes 17, 18, and 19 are substituted with the fixed bytes for the alignment marker corresponding to PCS lane 16. The variable bytes BIP or CD are unchanged.

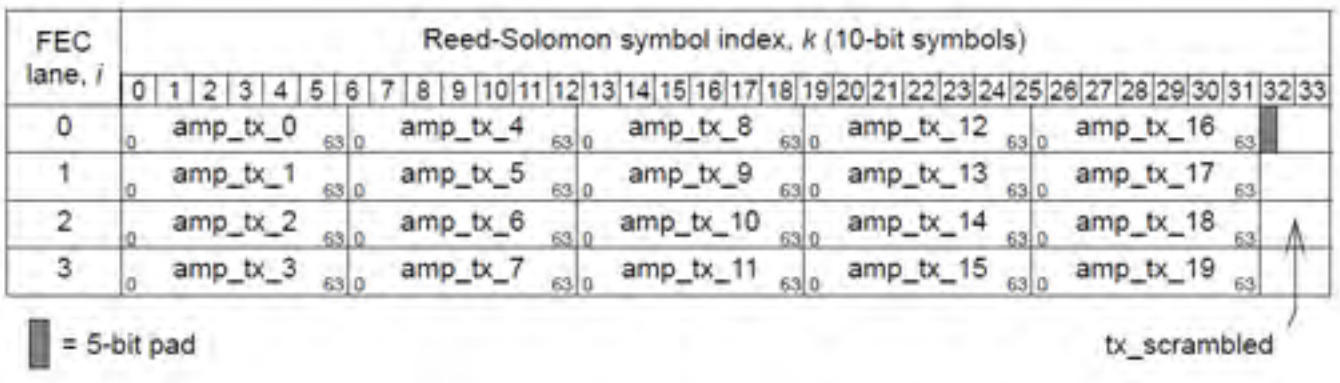
The alignment marker payloads corresponding to PCS lanes 0, 4, 8, 12, and 16 are transmitted on FEC lane 0, the alignment marker payloads corresponding to PCS lanes 0, 5, 9, 13, and 16 are transmitted on FEC lane 1, and so on.

The alignment marker mapping function is the reverse of the symbol distribution which is done after the RS FEC encoding.

The AM mapping function applicable to 100G mode of operation is:

1. Construct a matrix of 4 rows and 320 columns, am_txpayloads, as shown in Figure 2-15.

Figure 2-15 Alignment Marker MAP for RS-FEC



2. Given i=0 to 3, j=0 to 4, and x=i+4j, the matrix is derived per the following expression:

am_txpayloads<i, (64j+63):64j> = amp_tx_x<63:0>

3. Given i=0 to 3, k=0 to 31, and y=i+4k,

   am_txmapped<(10y+9):10y> = am_txpayloads<i, (10k+9):10k>

2.2.1.13.4 Code Word/Alignment Marker Map in 25G Mode

As 25G mode of operation has only 1 lane, the 4 numbers of the 64-bit alignment marker patterns are inserted as listed in .

Table 2-6 Alignment Marker MAP for RS-FEC in 25G Mode

AM Number	Encoding[a] {M_0, M_1, M_2, BIP_3, M_4, M_5, M_6, BIP_7}
0	0xC1, 0x68, 0x21, BIP3, 0x3E, 0x97, 0xDE, BIP7
1	0xF0, 0xC4, 0xE6, BIP3, 0x0F, 0x3B, 0x19, BIP7
2	0xC5, 0x65, 0x9B, BIP3, 0x3A, 0x9A, 0x64, BIP7
3	0xA2, 0x79, 0x3D, BIP3, 0x5D, 0x86, 0xC2, BIP7

[a] Each octet is transmitted LSB to MSB

The BIP3 field is programmed to the constant value 0x33 and the BIP7 field is programmed to the constant value 0xCC. Following is the alignment marker mapping function used in 25G mode of operation:

am_txmapped <256:0> is constructed as follows:

For x=0 to 3

am_txmapped<(64x+63):64x> = am_tx_x <65:2>

am_txmapped<256> = 1b'1 (pad bit).

2.2.1.13.5 Code Word/Alignment Marker Map in 50G Mode

provides the Alignment markers used in 50G mode of operation.

Table 2-7 Alignment Marker MAP for RS-FEC in 50G Mode

AM Number	Encoding[a] {M_0, M_1, M_2, BIP_3, M_4, M_5, M_6, BIP_7}
0	0x90, 0x76, 0x47, BIP3, 0x6F, 0x89, 0xB8, BIP7
1	0xF0, 0xC4, 0xE6, BIP3, 0x0F, 0x3B, 0x19, BIP7
2	0xC5, 0x65, 0x9B, BIP3, 0x3A, 0x9A, 0x64, BIP7
3	0xA2, 0x79, 0x3D, BIP3, 0x5D, 0x86, 0xC2, BIP7

[a] Each octet is transmitted LSB to MSB

The alignment marker mapping function for 50G mode of operation is as follows:

Given i=0 to 1, k=0 to 12, and y=i+2k, am_txmapped can then be derived from am_txpayloads as per the following expression:

If (y < 25) then

am_txmapped<(10y+9):10y> = am_txpayloads<i, (10k+9):10k>

am_txmapped<255:250> = am_txpayloads<1, 125:120>

am_txmapped<256> = 1b'1 (pad bit)

2.2.1.13.6 Reed Solomon Gearbox

The Transmit gearbox converts 257-bit wide input to required number of parallel RS FEC symbols based on the configuration.

2.2.1.13.7 Reed Solomon FEC Encoder

The RS-FEC sublayer employs a Reed-Solomon code operating over the Galois Field GF(2^10) where the symbol size is 10 bits. DWC_xlgpcs supports two types of Reed-Solomon codes: RS(528, 514) and RS(544, 514). Depending on the operating mode, approach option is selected. In 50GBASE-KR, 50GBASE-CR, 100GBASE-KR2, 100GBASE-CR2, 100GBASE-KR1, 100GBASE-CR1, 50GAUI-2, or 100GAUI-4 modes, RS(544, 514) is used. In all other modes, RS(528, 514) is used.

RS(528, 514) Encoder

The encoder processes 514 message symbols to generate 14parity symbols, which are then appended to the message to produce a codeword of 528=514+14 symbols. This RS code can detect 14 symbol errors and correct up to 7 symbol errors. Each symbol is 10-bit wide. Each 514-symbol message corresponds to twenty 257-bit blocks produced by the transcoder. The encoder introduces a latency of 1 clock cycle.

The generator polynomial RS(528,514) are the following:

n = 528, k = 514 and 2t = 14

g(x) = x14 + 904x13 + 6x12 + 701x11 + 32x10 + 656x9 + 925x8 + 900x7 + 614x6 + 391x5 + 592x4 + 265x3 + 945x2 + 290 x+ 432

RS(544,514) Encoder

This encoder processes 514 message symbols to generate 30parity symbols, which are then appended to the message to produce a codeword of 544=514+30 symbols. This RS code can correct up to 15 symbol errors. Each symbol is 10-bit wide. Each 514-symbol message corresponds to twenty 257-bit blocks produced by the transcoder.

The generator polynomial RS(544,514) are the following:

n = 544, k = 514 and 2t = 30

$g(x) = x^{30} + 575*x^{29} + 552*x^{28} + 187*x^{27} + 230*x^{26} + 552*x^{25} + x^{24} + 108*x^{23} + 565*x^{22} + 282*x^{21} + 249*x^{20} + 593*x^{19} + 132*x^{18} + 94*x^{17} + + 720*x^{16} + 495*x^{15} + 385*x^{14} + 942*x^{13} + 503*x^{12} + 883*x^{11} + 361*x^{10} + 788*x^{9} + 610*x^{8} + 193*x^{7} + 392*x^{6} + 127*x^{5} + 185*x^{4} + 158*x^{3} + 128*x^{2} + 834*x + 523$

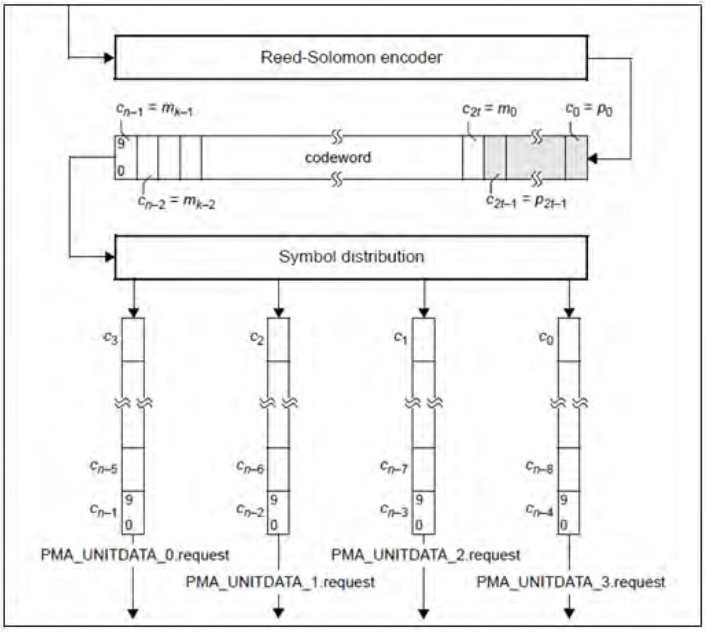
When operating in 100GBASE-KR1 and 100GBASE-CR1 modes, DWC_xlgpcs optionally supports a dual-encoder scheme as defined in the IEEE802.3ck specification. In this mode, the data from the transcoder (post alignment marker insertion) is distributed to two instances of the RS(544,514) encoder.

2.2.1.13.8 FEC Symbol distribution/ Multi-Lane Distribution (MLD)

In a multi-lane configuration (50G and 100G modes), once the data has been Reed Solomon-encoded, the symbols are distributed in a round robin fashion.

In 100G non-interleaved mode, symbols are distributed to 4 FEC lanes, one 10-bit symbol at a time in a round robin distribution, from the lowest to the highest numbered FEC lane as shown in Figure 2-16 below.

Figure 2-16 FEC Symbol Distribution in 100G Non-interleaved Mode



In 100G codeword interleaved RS-FEC mode, once the data has been FEC-encoded, two FEC codewords (cA<543:0> and cB<543:0>) are interleaved on a 10-bit basis before the data is distributed to each FEC lane. The interleaving of the two codewords is done as follows:

For all k=0 to 271

For all j=0 to 1

if even(k)

tx_out<4k+2j> = cA<543-2k-j>

tx_out<4k+2j+1> = cB<543-2k-j>

else

tx_out<4k+2j> = cB<543-2k-j>

tx_out<4k+2j+1> = cA<543-2k-j>

Once the data has been Reed Solomon-encoded and interleaved, it is distributed to four FEC lanes, one 10-bit symbol at a time, from the lowest to the highest FEC lane. The first bit transmitted from each 10-bit symbol is bit 0.

In 50G mode, symbols are distributed to 2 FEC lanes, one 10-bit symbol at a time in a round robin distribution from the lowest to the highest numbered FEC lane.

In 25G mode of operation, the RS-FEC encoded data is sent over only one physical lane. (Hence symbol distribution is not applicable for 25G RS-FEC mode of operation).

2.2.1.14 Gearbox

The gearbox block is an optional block. This block is not present when PCS-R PHY Interface Type is selected with 66-bit width.

2.2.1.14.1 40G Configuration

The scrambled or BASE-R FEC-encoded data goes to the gearbox which converts the data, from 66-bit data clocking at 156.25 MHz, to one of the following data formats:

- 16-bit data clocking at 644.53125 MHz (Standard XSBI interface format)
- 20-bit data clocking at 515.625MHz (20-bit XSBI interface format)
- 32-bit data clocking at 322.265625 MHz (32-bit XSBI interface format)
- 33-bit data clocking at 312.5 MHz
- 64-bit data clocking at 161.1328125 MHz
- 80-bit data clocking at 128.90625 MHz

The 33-bit data format conversion does not require the rate adaptation because the effective data rate of input and output is same. However, the gearbox handles the phase difference between two clocks. The operating frequency and width of the output data depends on the PCS-R PHY interface type.

There are four such gear boxes present in the Transmit data path, one for each lane.

- 5GBASE-R or 10GBASE-R: Only one of the four gear boxes are active.
- 40GBASE-R: All the four gear boxes are active.



When operating in 5GBASE-R mode, the relevant frequencies are half of the values mentioned in this section.

2.2.1.14.2 25G-Enabled Configurations (25G, 25G-20G,100G)

In 25G enabled configurations, the scrambled or BASE-R FEC encoded 66-bit wide data goes to the gearbox.

In 25G enabled configurations the gearbox always converts input data in to a 32-bit data (if DWCXCLP_R_SRSDS_INTF is programmed to 32) or 40-bit data (if DWCXCLP_R_SRSDS_INTF is programmed to 40) output data. Table 2-8 shows the relation between the data and clock frequencies at the input and output of the gearbox for different speed modes and FEC options.

Table 2-8 Gearbox Speed Mode - Bit Width - Frequency Table

Speed Mode	Input Data Width Bit X Frequency	Output Data Width Bit X Frequency	
		40-bit PHY Width	32-bit PHY Width
5G in no FEC mode	66bit x 78.125 MHz	40 bit X 128.90625 MHz	32 bit x 161.1328125 MHz
10G in no FEC/BASE-R FEC mode	66bit X 156.25 MHz	40bit X 257.8125MHz	32bit X 322.265625MHz

Table 2-8 Gearbox Speed Mode - Bit Width - Frequency Table

Speed Mode	Input Data Width Bit X Frequency	Output Data Width Bit X Frequency	
		40-bit PHY Width	32-bit PHY Width
25G in no FEC/BASE-R FEC mode	66bit X 390.625 MHz	40bit X 644.53125MHz	32bit X 805.6640625MHz
40G in no FEC/BASE-R FEC mode	66bit X 156.25 MHz	40bit X 257.8125MHz	32bit X 322.265625MHz
50G in no FEC/BASE-R FEC mode	66bit X 390.625 MHz	40bit X 644.53125MHz	32bit X 805.6640625MHz
100G in no FEC/BASE-R FEC mode	66bit X 390.625 MHz	40bit X 644.53125MHz	32bit X 805.6640625MHz
25G/50G/100G in RS FEC mode	80bit X 390.625 MHz	40bit X 644.53125MHz	32bit X 805.6640625MHz

Note In 25G-enabled configurations, the transmit and receive gear boxes take care of short term jitter up to 1 clock period (of 644.53125MHz in 40-bit interface or 805.6640625 MHz in 32-bit PHY interface). However, the average clock ratio is to be maintained at 1.65 (equal to 66/40) for 40bit PHY interface and at 2.0625 (equal to ratio of 66/32) for 32-bit PHY interface.

In 25G configuration, only one instance of gearbox is present. In 25G-50G single-lane configuration, two instances of gearbox are present. In other configurations (40G, 25G-50G multi-lane, 100G), four gearbox instances are present in the Transmit data path one for each lane:

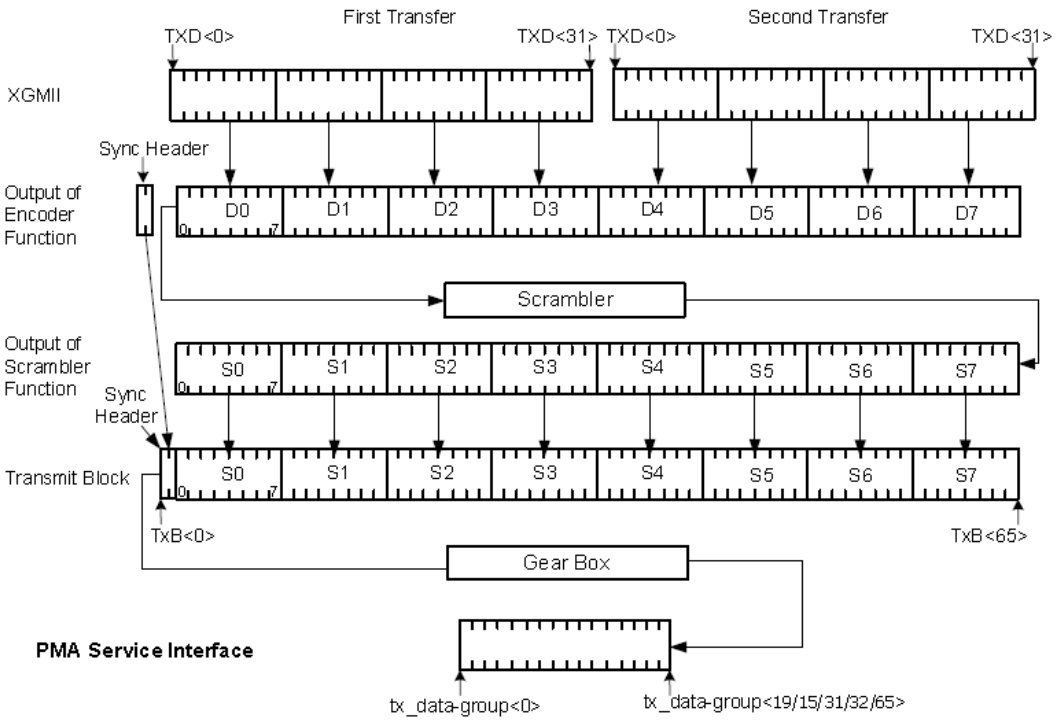
- In 10G/25G speed modes: Only one of the four gear boxes are active.
- In 20G, 50G speed modes: Two of the four gear boxes are active.
- In 40G, 100G speed modes: All the four gear boxes are active.

2.2.1.15 5G/10G/25G PCS - R Transmit Bit Ordering

In 64/66 PCS Layer, the transmit bit ordering is implemented as defined in IEEE Std 802.3, Clause 49. The Sync Header data is transmitted first, followed by the Block Type and Data or Control sub-blocks, as shown in Figure 2-17.

In Figure 2-17, the tx_data-group<0> is expected to be transmitted first. If TX_LSB_FIRST = Enabled, this bit corresponds to rpcs_tx_data_o[0]. If TX_LSB_FIRST = Disabled, this bit corresponds to rpcs_tx_data_o[15/31/32/65].

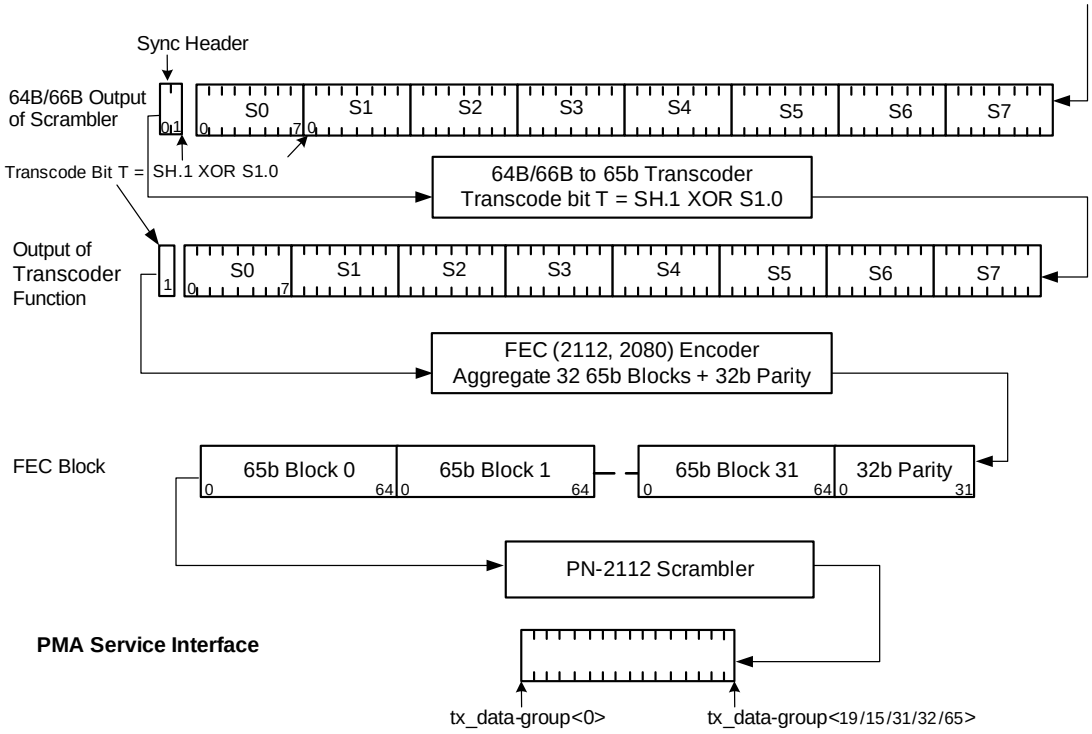
Figure 2-17 5G/10G/25G Speed Mode Transmit Bit Ordering in 64/66 PCS Layer



In 5G/10G/25G BASE-R FEC mode, the transmit bit ordering is implemented as defined in IEEE Std 802.3, Clause 74. 2.3 shows the transmit bit ordering in the FEC block.

In 2.3, the tx_data-group<0> is expected to be transmitted first. If DWCXLP_TX_LSB_FIRST = Enabled, this bit corresponds to rpcs_tx_data_o[0]. If DWCXLP_TX_LSB_FIRST = Disabled, this bit corresponds to rpcs_tx_data_o[31/39/65].

Figure 2-18 Transmit Bit Ordering in the FEC Block



2.2.1.16 40G/50G/100G PCS-R Transmit Bit Ordering

In 64/66 PCS Layer, the transmit bit ordering is implemented as defined in IEEE Std 802.3, Clause 82. The Sync Header data is transmitted first, followed by the Block Type and Data or Control sub-blocks. Figure 2-19 shows the Block bit transmission order. Figure 2-19 shows the mapping from XLGMII/CGMII to 64B/66B block for a block containing 8 data characters.

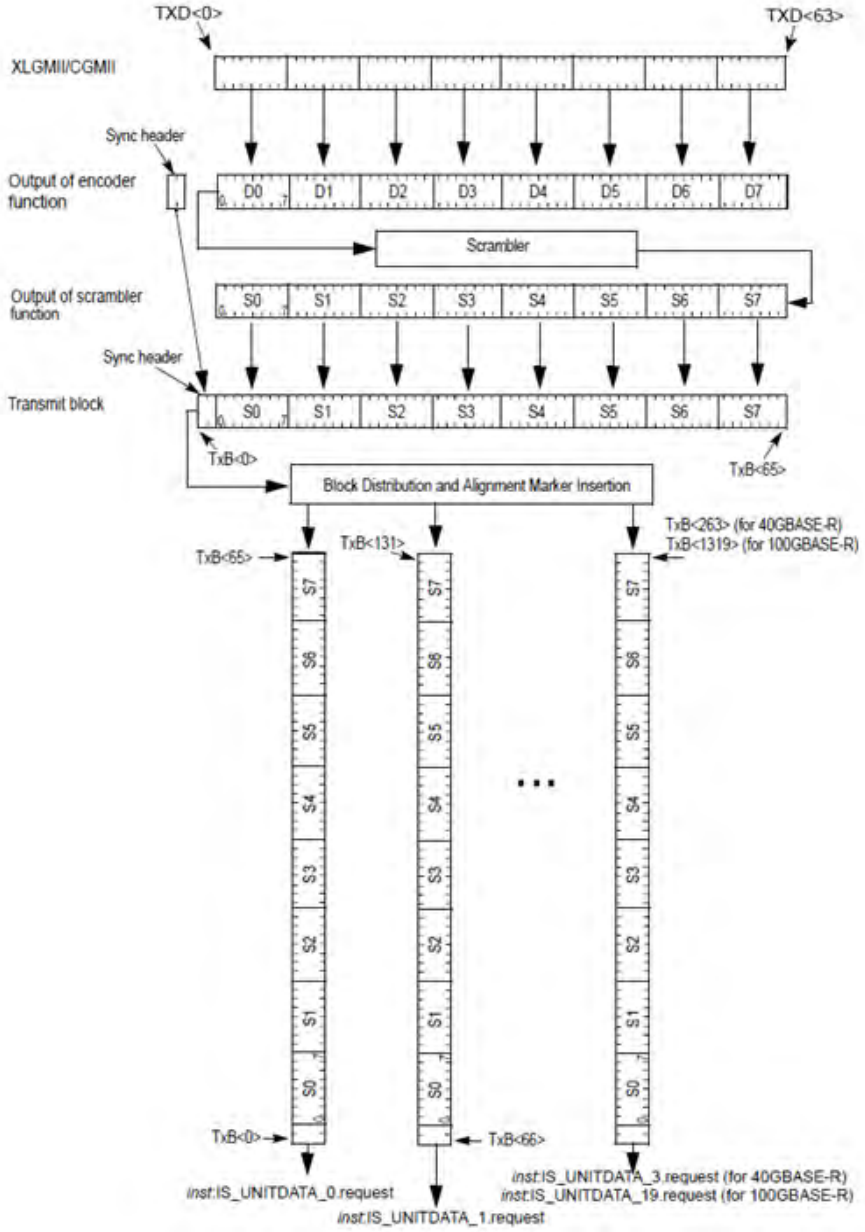
In Figure 2-19, the TxB<66*i+0> is expected to be transmitted first on lane 'i' where 'i' = PCS lane number. If TX_LSB_FIRST = Enabled, this bit corresponds to rpcs_tx_data_o[0]. If TX_LSB_FIRST = Disabled, this bit corresponds to rpcs_tx_data_o[15/31/32/39/65].

Note In 100G mode, the PCS coded data corresponding to 20 PCS/Virtual lanes are bit multiplexed such that data from 5 of 20 PCS lanes drive one PHY lane

The PMA provides bit-level multiplexing in 100G BASE-R (no FEC case). This involves multiplexing the PCS lanes data from 5 PCS lanes on to one physical lane. As the standard does not mandate bit ordering (the PCS lanes to be used in multiplexing), the following scheme is implemented.

- Bit multiplexed data from PCS lanes 0, 1, 2, 3 and 4 to be sent on PHY lane-0
- Bit multiplexed data from PCS lanes 5, 6, 7, 8 and 9 to be sent on PHY lane-1
- Bit multiplexed data from PCS lanes 10, 11, 12, 13 and 14 to be sent on PHY lane-2
- Bit multiplexed data from PCS lanes 15, 16, 17, 18 and 19 to be sent on PHY lane-3

Figure 2-19 Mapping From XLGMII/CGMII to 64B/66B Block for a Block Containing 8 Data Characters



Note Additionally, in 50G mode, data from each set of two PCS lanes are bit multiplexed on to a physical lane (MLD4:2 is done).

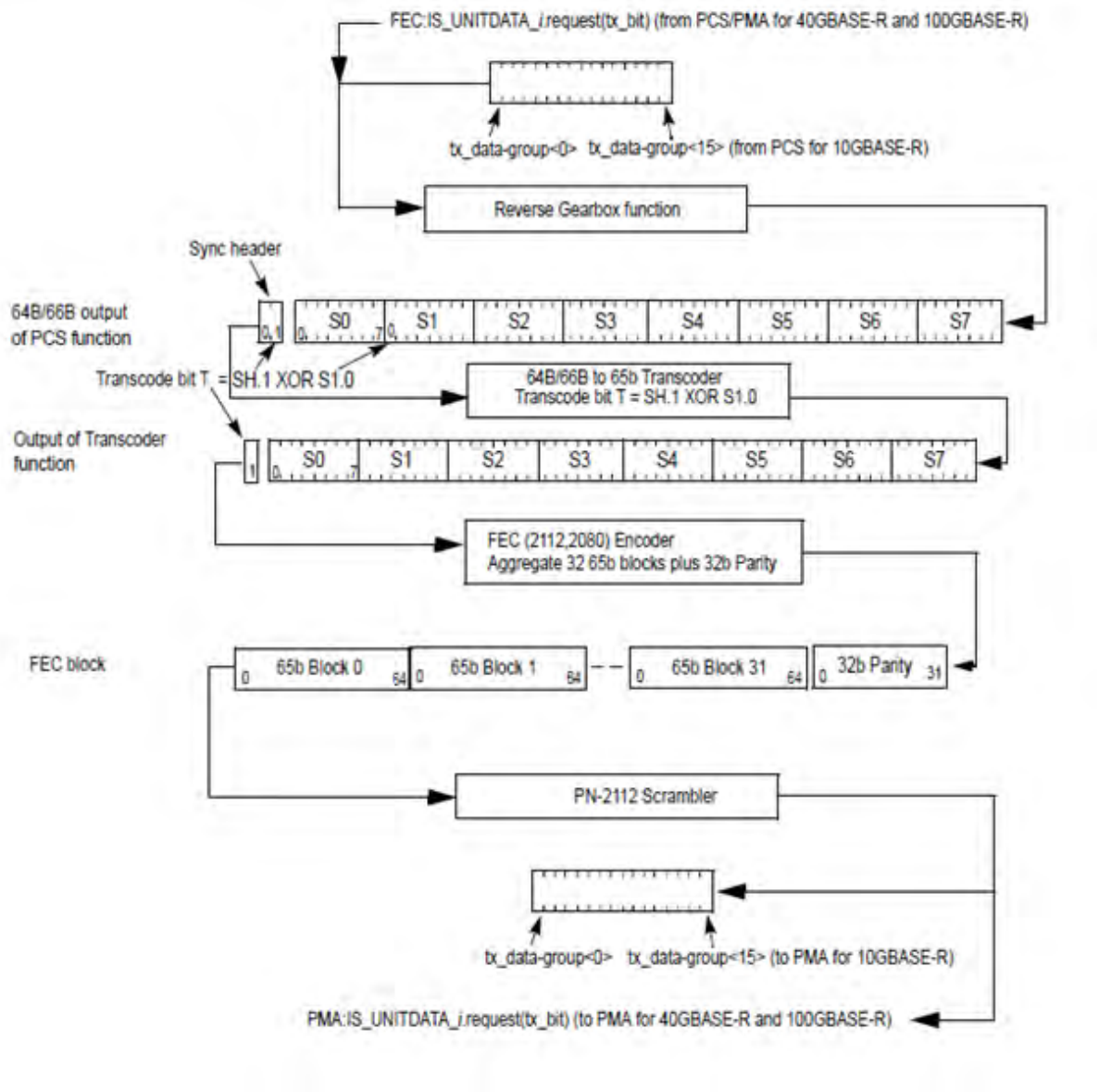
2.2.1.17 40G/50G BASE-R FEC Transmit Bit Ordering

Figure 2-20 shows the format of the FEC block and the transmit bit ordering in 40G/50G/100G mode.

Note

- Additionally, in 50G mode, data from each set of two PCS lanes after BASE-R FEC coding are bit multiplexed on to a physical lane (MLD4:2 is done).
- In 100G mode, BASE-R FEC is not supported.

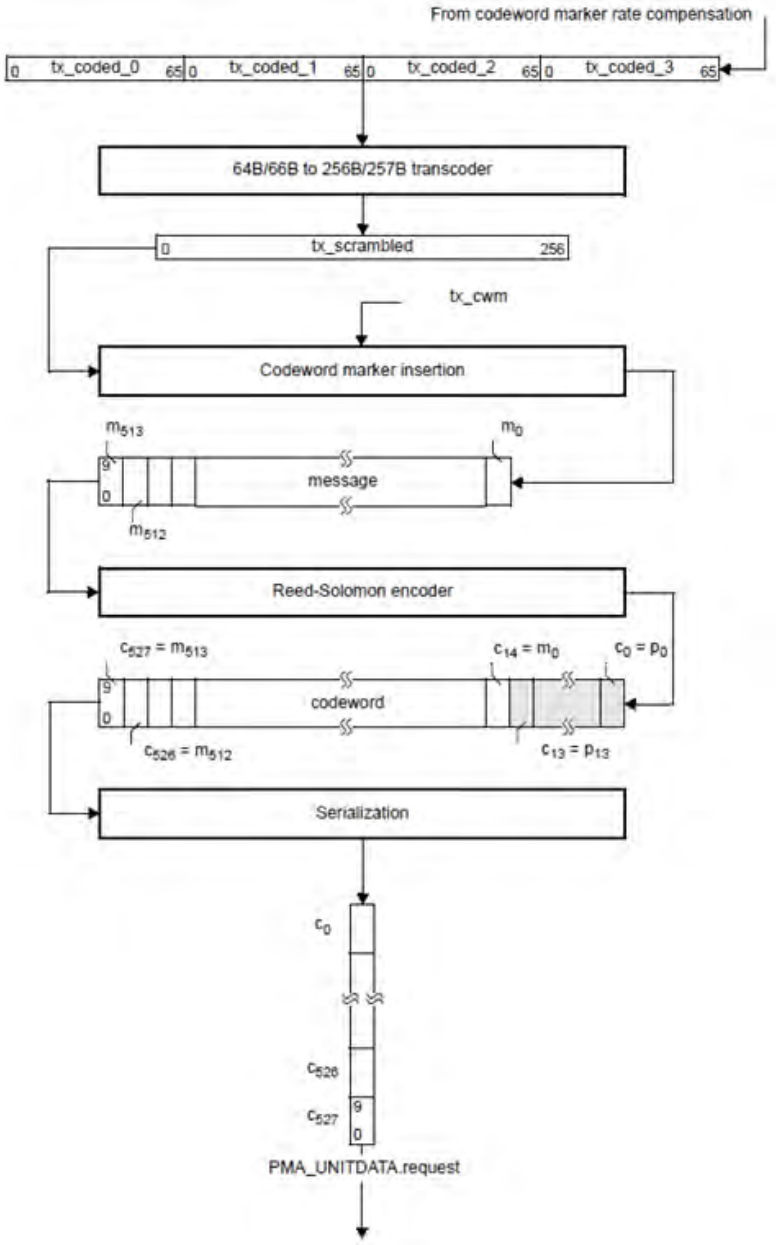
Figure 2-20 FEC Block Format and Transmit Bit Ordering



2.2.1.18 25G RS-FEC Transmit Bit Ordering

Figure 2-21 shows the bit ordering in RS-FEC enabled 25G mode of operation. From the RS FEC coded output, the message symbols starting with bit-0 are sent first followed by the generate parity symbols.

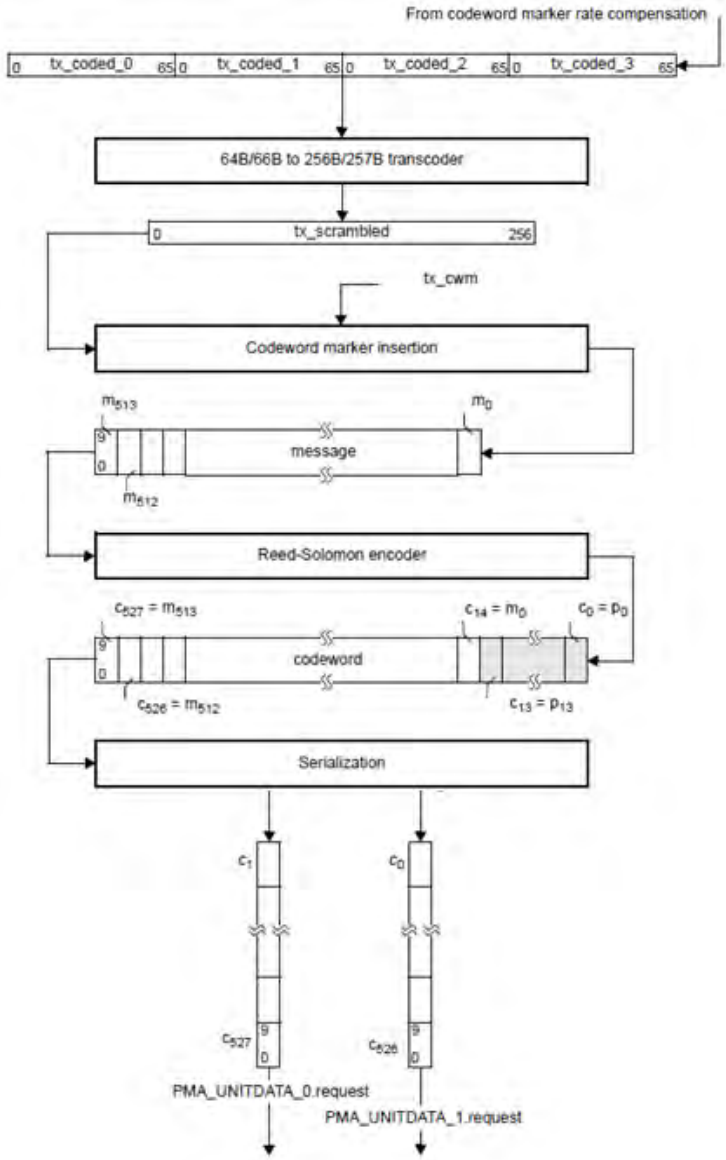
Figure 2-21 25G RS-FEC Transmit Bit Ordering



2.2.1.19 50G RS-FEC Transmit Bit Ordering

As shown in Figure 2-22, in the 50G RS-FEC enabled mode, the order of sending data is similar to that of 25G RS-FEC mode of operation, except that the symbols are distributed over two lanes. The even symbols are sent over lane-0 and odd symbols are sent on lane-1 with the transmission on each lane starting with bit-0 of RS FEC coded symbols.

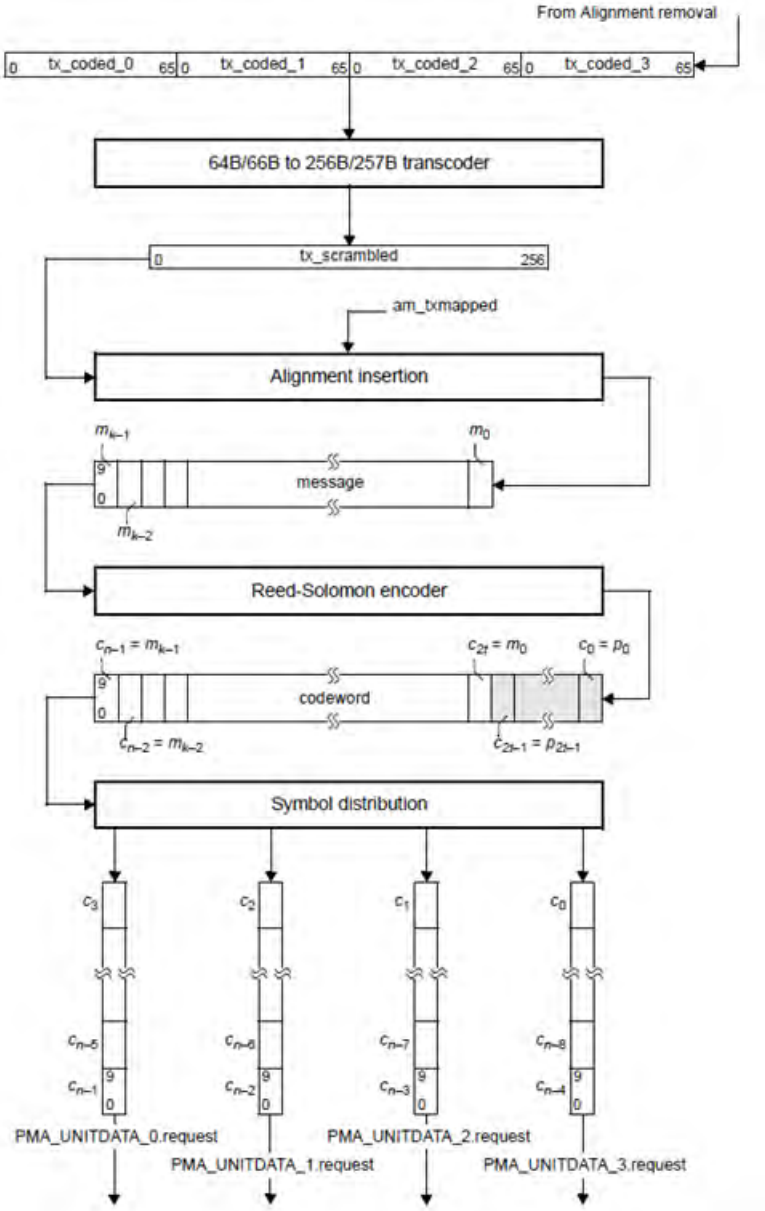
Figure 2-22 50G RS-FEC Transmit Bit Ordering



2.2.1.20 100G Non-interleaved RS-FEC Bit Ordering

The order of sending data is similar to that of 50G RS FEC mode of operation except that the symbols are distributed over four lanes. The symbols are sent over lane-0 to lane-3 in a round robin fashion starting from lane-0. The transmission on each lane starts with bit-0 of RS FEC coded symbols appearing on their respective lanes.

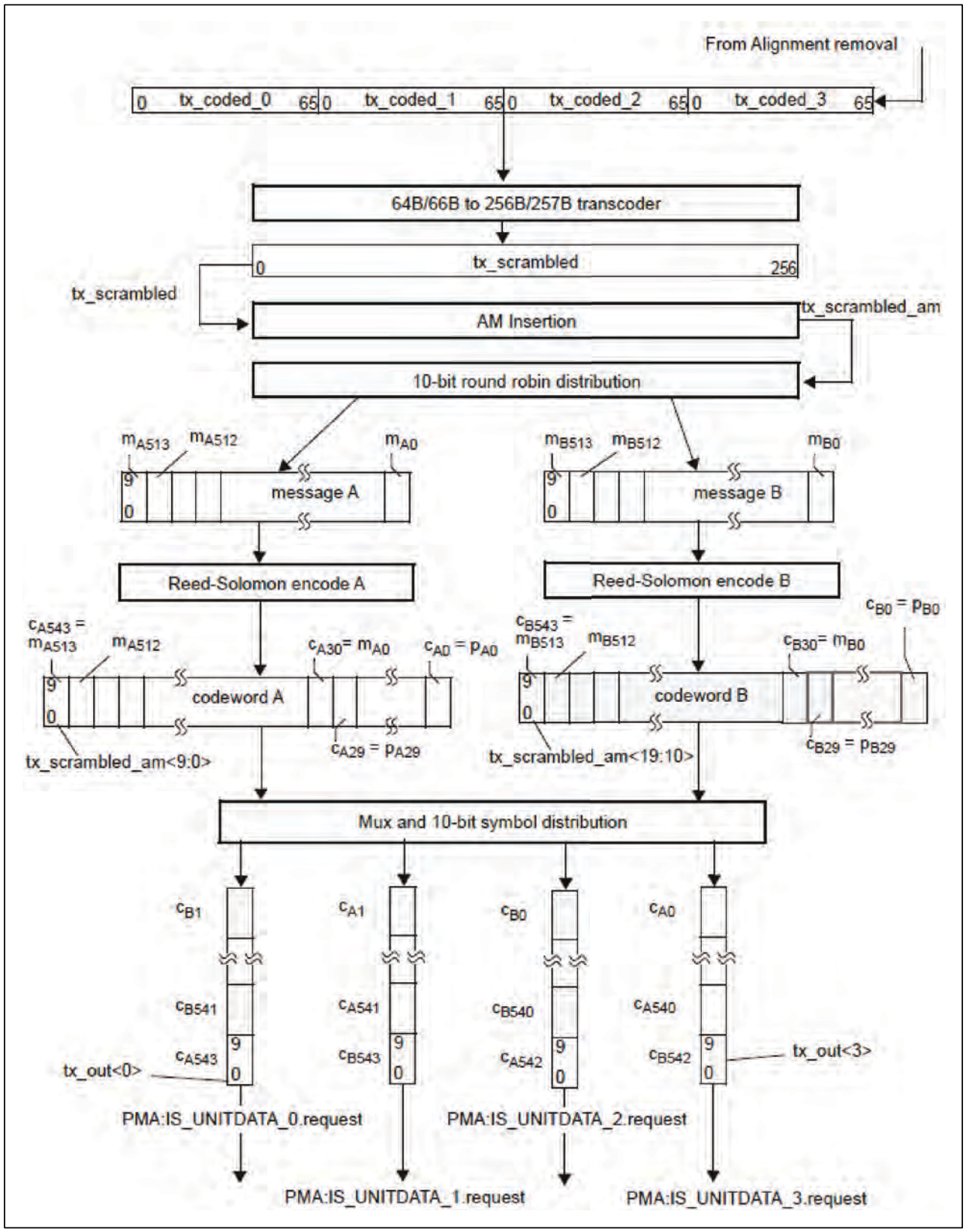
Figure 2-23 100G RS-FEC Transmit Bit Ordering



2.2.1.21 100G Interleaved RS-FEC Bit Ordering

In 100G interleaved mode, the symbols from the two RS-FEC encoders are distributed over four lanes. Figure 2-24 shows the bit ordering process.

Figure 2-24 100G Interleaved RS-FEC Transmit Bit Ordering



2.2.2 Receive Block

The following are the Receive block features:

- Implements the following interfaces:
  - XGMI, XLGMII interfaces towards the MAC
  - Parallel interfaces from the PHY
- Performs 66-bit block alignment
- Performs optional BASE-R FEC decoding
- Performs optional RS-FEC decoding
- Descrambles data to recover the original unscrambled data
- Performs 64B/66B decoding
- Checks the following test patterns:
  - Pseudo Random for 10G/25GBASE-R
  - PRBS 31 for 10G/25GBASE-R
  - PRBS 9 for 10GBASE-R
  - PRBS 31 and PRBS 9 for PMA per lane in 40G/100G BASE-R
  - Scrambled Idle TP Checker for 25G/40G/50G/100G BASE-R PCS
- Performs Clock Rate compensation (can handle a ppm difference of +/- 100) between the local and recovered clock
- Performs Removal of Alignment Markers
- Accepts lane-to-lane skew up to 180ns

2.2.2.1 Reverse Gearbox

In 40G configurations, the reverse gearbox converts the data from 16-bit, 20-bit, 32-bit, 33-bit, 64-bit, or 80-bit data for BASE-R (clocking at the rate of 644.53 MHz, 515.625 MHz, 312.5 MHz, 161.1328125 MHz, or 128.90625 MHz, respectively) to 66-bit data clocking at 156.25 MHz.

In 25G-enabled configurations (25G, 25G-50G, and 100G configurations), the reverse gearbox converts the data from 32-bit or 40-bit (based DWCXLP_R_SRDF_INTF is programmed to 40 or 32) to 66-bit data.

The input and output clock frequencies are programmed based on the speed mode as shown in Table 2-9.

Table 2-9 Input and Output Clock Frequency for Various Speed Modes

Speed Mode	Input Data Width Bit X Frequency		Output Data Width Bit X Frequency
	40-Bit PHY Width	32-Bit PHY Width	
10G in no FEC/BASE-R FEC mode	40bit X 257.8125 MHz	32bit X 322.265625 MHz	66bit X 156.25 MHz
25G in no FEC/BASE-R FEC mode	40bit X 644.53125 MHz	32bit X 805.6640625 MHz	66bit X 390.625 MHz
40G in no FEC/BASE-R FEC mode	40bit X 257.8125 MHz	32bit X 322.265625 MHz	66bit X 156.25 MHz

Table 2-9 Input and Output Clock Frequency for Various Speed Modes

Speed Mode	Input Data Width Bit X Frequency		Output Data Width Bit X Frequency
	40-Bit PHY Width	32-Bit PHY Width	
50G in no FEC/BASE-R FEC mode	40bit X 644.53125 MHz	32bit X 805.6640625 MHz	66bit X 390.625 MHz
100G in no FEC/BASE-R FEC mode	40bit X 644.53125 MHz	32bit X 805.6640625 MHz	66bit X 390.625 MHz
25G/50G/100G in RS FEC mode	40bit X 644.53125 MHz	32bit X 805.6640625 MHz	66bit X 390.625 MHz

This block performs the reverse process of the gearbox block. In 25G configuration only one gearbox is present. In multi-lane configurations (25G_40G, 25G-50G, and 100G configurations) there are four such reverse gear boxes present in the Receive data path one for each lane.

In a multi lane configuration, during 5G/10G/25G speed modes, only one gearbox is active. In 20G/50G speed modes, two gear boxes are active and in 40G/100G speed modes all four gear boxes are active.

Note In 50G non-RS-FEC mode, reverse gearbox-0 and reverse gearbox-2 are active.
In 50G RS-FEC mode, reverse gearbox-0 and reverse gearbox-1 are active.

2.2.2.2 BASE-R FEC Decoder

The Receive Forward Error Correction (FEC) Decoder block is an optional block. It is present only when the "Add Forward Error Correction Support" option is selected. This option enables the error correction mechanism on the Receive path. This block performs the following functions on the 66-bit data output from the reverse gearbox:

- FEC block synchronization
- FEC decoding (One FEC Decoder for one PCS lane)
- Error correction and error forwarding

If FEC_ERROR_FWD = Enabled, the FEC decoder block provides additional information about uncorrectable errors to the PCS-X block. You can enable the FEC Error Indication control bit by programming bit 1 of the `SR_PMA_KR_FEC_CTRL` register. You can find the FEC correctable error and uncorrectable error count information through following registers:

- SR PMA MMD 10/25GBASE-KR Lane0 FEC Correctable Error Block Counter 1 register
- SR PMA MMD 10/25GBASE-KR Lane0 FEC Correctable Error Block Counter 2 register
- SR PMA MMD 10/25GBASE-KR Lane0 FEC Uncorrectable Error Block Counter 1 register
- SR PMA MMD 10/25GBASE-KR Lane0 FEC Uncorrectable Error Block Counter 2 register

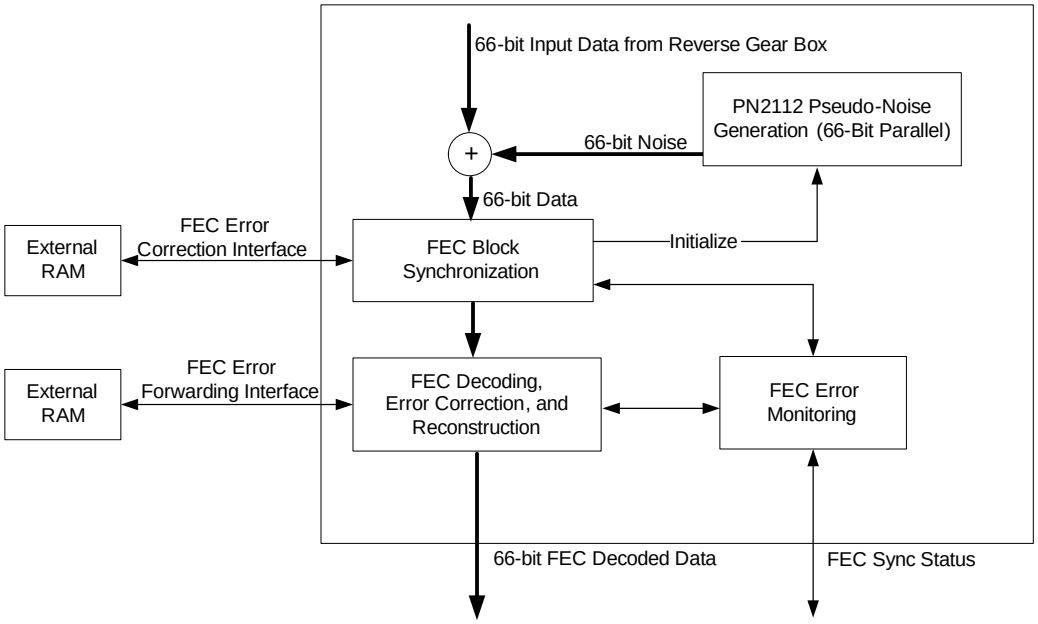
You can get the FEC ability status through SR_PMA_KR_FEC_ABL. You can control the FEC features through `SR_PMA_KR_FEC_ABL`.

Note An external dual-port RAM (or register array) of 32 depth and 66-bit width (32x66) is required for FEC decoding per lane. If FEC error forwarding feature is enabled, two such blocks per lane are required. Also, if both RS FEC and BASE-R FEC are enabled in the configuration, they share the memories and as a result error forwarding feature of the BASE-R FEC is always present in such configurations. However the BASE-R FEC Error forwarding can be disabled via programmable control.

2.2.2.2.1 FEC Decoding Process

Figure 2-25 shows the FEC decoding process.

Figure 2-25 FEC Decoding Process



The FEC block synchronization is done in the following way:

1. Test a potential candidate block-start position by following these steps:
   a. Descramble the block by using the PN2112 pseudo-noise generator to derive the raw data.
   b. Evaluate CRC check for the potential block and do one of the following:
      - If the CRC check matches, go to step 2.
      - If the CRC check does not match (that is, the received CRC does not match the computed CRC), shift start-candidate by 1 bit position and repeat step 1.
2. Validate that the potential block start position has a good CRC for *n* consecutive blocks. Do one of the following:
   - If *n* (where n = 4) consecutive blocks are received with good CRC, the FEC block synchronization is established.
   - If *m* (where m = 8) consecutive blocks are received with bad CRC, drop the FEC block synchronization and again repeat step 1.

This procedure can be repeated maximum 2,111 times for all bits positions in the 2,112 (32 x 66) block. An FEC decoding error is indicated to the CSR and the 64/66 decoder blocks of the Receive path. After successful decoding, the FEC block synchronizer performs the following:

1. Adds the additional redundant bit to form the original 66-bit Sync Header
2. Removes the CRC bits so that the effective data rate does not change

The input to the FEC block is 32*65 bit data + 32 bit checksum. The data is reconstructed by removing the 32-bit checksum and adding the redundant bit of the Sync Header to all 32-blocks (without affecting the total number of bits).

The DWC_xlgpcs indicates the decoding errors (error forwarding option) through Sync Header.

1. If decoding is successful (because the CRC matches or the FEC block is correctable) and the descrambled received transcode bit (T) is 1, the sync bits take a value of {SH.0,SH.1} = 01. If the descrambled received transcode bit (T) is 0, the sync bits take a value of {SH.0,SH.1} = 10.
2. If the FEC Error Forwarding is enabled (`SR_PMA_KR_FEC_CTRL` bit 1 = 1) to indicate errors to the descrambler and the received FEC block has uncorrectable errors, the 1st, 9th, 17th, 25th, and 32nd Sync Header bits for 32 decoded blocks take a value of {SH.0,SH.1} = 11. The Sync Header bits for all remaining blocks take values as described in step 1.

   -or-

   If the FEC Error Forwarding is disabled (`SR_PMA_KR_FEC_CTRL` bit 1 = 0) and the received FEC block has uncorrectable errors, the sync bits take a value as described in step 1.

Block Alignment

The FEC block synchronizer uses the following method to determine the block boundary of 66-bit encoded data based on the 2-bit Sync Header:

1. The block synchronizer looks for valid Sync Header (2'b10 or 2'b01) on 66-bit data.
2. The block synchronizer monitors the subsequent blocks to validate further consecutive 66-bit data based on the Sync Header (2'b10 or 2'b01).

   If the invalid block count is NORMAL, it locks to the boundary chosen in step 1 and the block alignment is declared as successful. If the invalid block count is 'Not NORMAL', it goes back to step 1 and tries to lock it to a new boundary on 64-bit data which has a valid Sync Header.

PN2122 Pseudo-Noise Generation

The PN2122 pseudo-noise generator block generates the pseudo noise which is mixed with the received data to recover the raw FEC encoded data. The pseudo-noise generator uses the following polynomial:

$r(x) = 1+x^{39}+x^{58}$

The Linear Feedback Shift Register (LFSR) implementation of this polynomial and the initial seed used in this block are similar to the one used in the transmitter. For more information, see "PN2112 Pseudo-Noise Generator" on page 56.

2.2.2.2.2 FEC Memory Interfaces

Figure 2-26 shows the timing diagram for FEC write interface -lane 0. When the rpcs_ec_wr_en0_o signal goes high and the rpcs_ec_wr_csn0_o signal goes low, the data on the rpcs_ec_wr_data0_o signal is written to the memory location indicated by the rpcs_ec_wr_addr0_o signal. For example, as shown in Figure 2-26, Ox2_10fc_7981_fc8c_ba95 data is written to 0x0 memory location and 0x0_239d_72f7_3ce7_7914 is written to 0x1 memory location.

Figure 2-26 FEC Write Interface

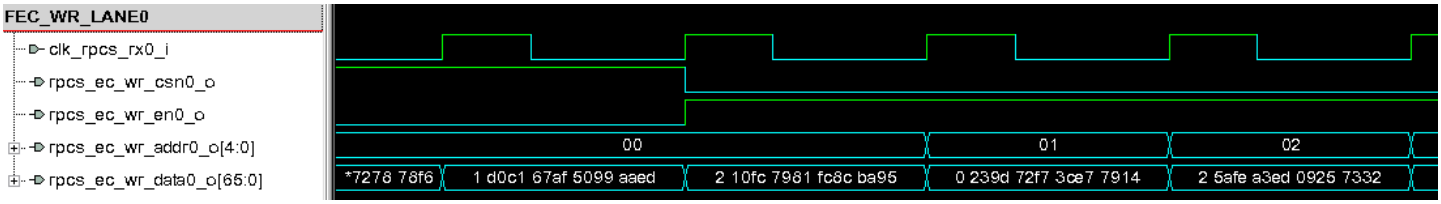
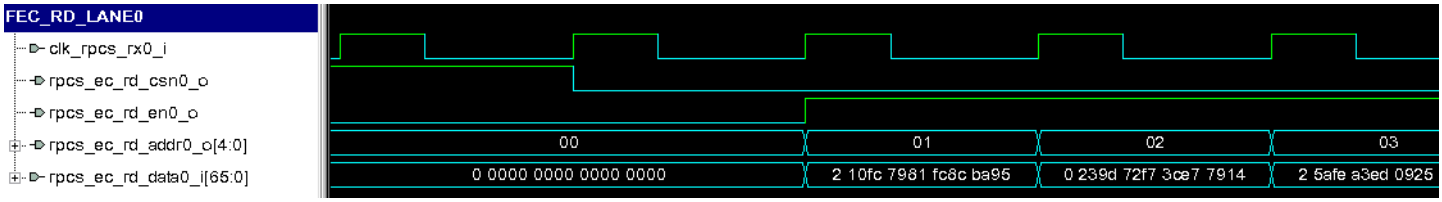


Figure 2-27 shows the timing diagram for FEC read interface - lane 0. The rpcs_ec_rd_csn0_o signal goes low to indicate the start of the read operation. In the next clock edge, the rpcs_ec_rd_en0_o signal goes high and the data (rpcs_ec_rd_data0_i) from the memory location, indicated by value of rpcs_ec_rd_addr_o at previous clock edge (when rpcs_ec_rd_csn0_o goes low), is available.

The address indicated by the rpcs_ec_rd_addr0_o signal is valid as long as the rpcs_ec_rd_csn0_o signal is low. For example, as shown in Figure 2-27, 0x2_10fc_7981_fc8c_ba95 data is read from 0x0 memory location and 0x0_239d_72f7_3ce7_7914 is read from 0x1 memory location.

Figure 2-27 FEC Read Interface



2.2.2.3 Reed Solomon FEC Decoder

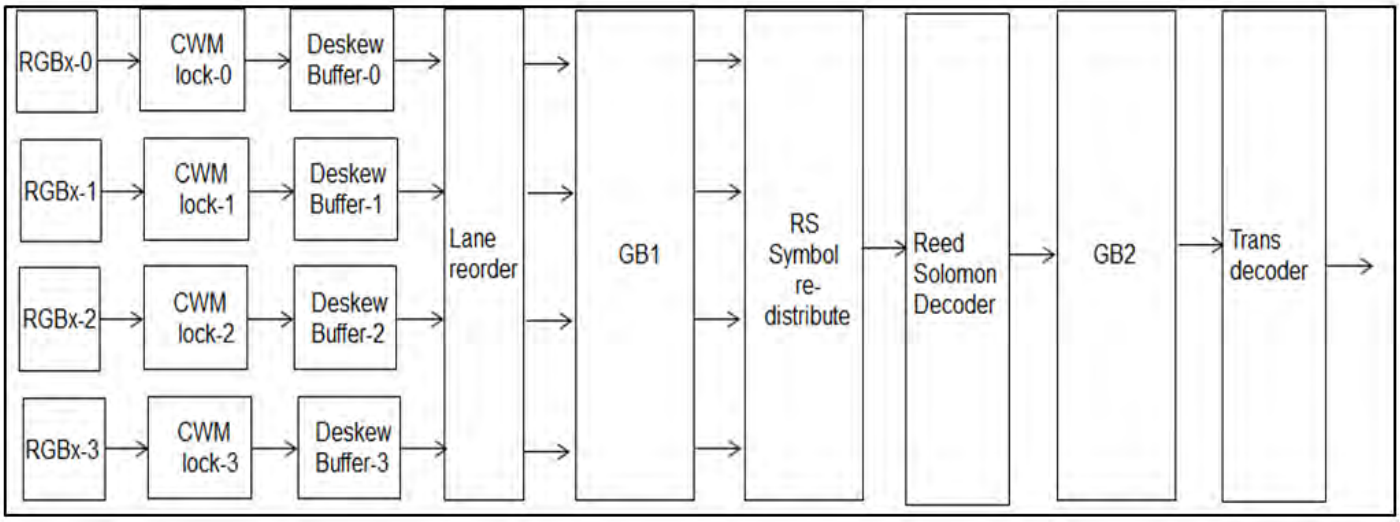
The Receive Reed Solomon Forward Error Correction (FEC) Decoder block is an optional block. It is present only when the “Add RS Forward Error Correction Support” option is selected. This option enables the error correction mechanism on the Receive path. This block performs the following functions on the 66-bit data output from the reverse gearbox:

- RS FEC Code Word Marker synchronization
- RS FEC decoding
  - Syndrome Calculation to detect Errors
  - Error correction and error forwarding
- Transdecoding

Figure 2-28 shows the block diagram of the Reed Solomon FEC Decoder for configurations with DWCXLP_IRSFEC=0, with other support functions.

For configurations with DWCXLP_IRSFEC=1, DWC_xlgpcs supports codeword interleaved RS-FEC decoder. In these configurations, there are two 'Reed Solomon decoder' instead of a single instance, as shown in Figure 2-28.

Figure 2-28 Reed Solomon FEC Decoder Block Diagram



2.2.2.3.1 CWM lock: Code Word Marker Lock

The DWC_xlgpcs implements the code word marker/Alignment Marker lock logic as per clause 108 for 25G, clause 91 for 50G/100G, or Clause 161 for 100G RS-FEC-interleaved modes. Bits from receive gearbox are concatenated in the order they are received. The concatenated data is compared with the 64-bit CWM/AM patterns. CWM/AM lock is declared when two identical CWM/AM patterns are detected between specified number of RS FEC code words.

Code word Marker/Alignment Marker lock is declared as per the FEC synchronization state diagram based on Clause 108, Clause 91, or Clause 161.

- For 25G RS FEC mode of operation, only one lane is active (lane-0) and hence only one instance of CWM lock logic is used.
- For 50G RS FEC mode of operation, only two lanes are active (lane0, 1) and hence two instances of CWM/AM lock logic is used (on lane-0, 1).
- For 100G RS FEC mode of operation, all four lanes are active and hence four instances of CWM/AM lock logic is used (for lanes 0, 1, 2 and 3).

For 50G/100G configurations the FEC lane mapping is also found out by comparing the CWM/AM markers received against the actual marker values.

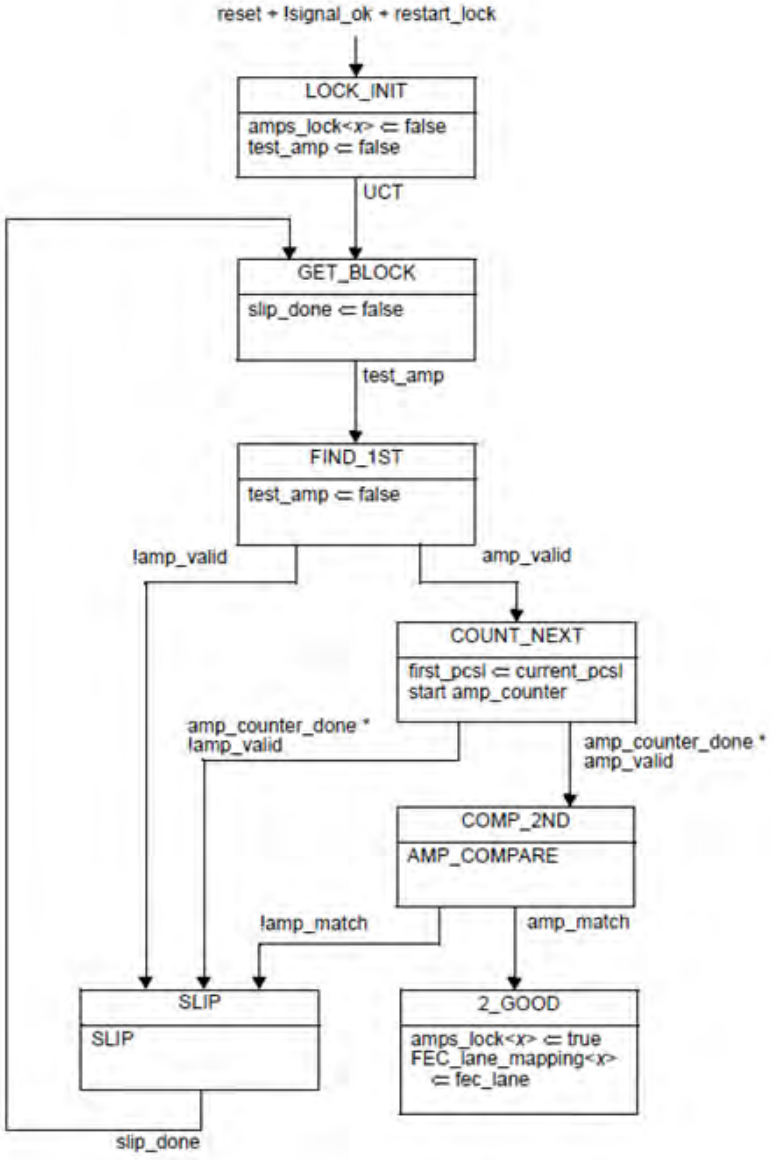
DWC_xlgpcs provides an option to program the 64-bits of Code word marker (tx_cwm<63:0>). The Vendor specific PMA registers `VR_PMA_CWM00-03` hold the CWM pattern to be used in 25G RS FEC mode.

Note

- To speed up the simulation, you can reduce the Code Word Marker (CWM)/Alignment Maker (AM) insertion period to lower values instead of the standard defined 1024 RS FEC code words. This enables the Remote End Receiver to Link UP faster. You can program the VR_PCS_AM_CNT register to use the reduced value of the CWM/AM insertion Interval. In RS FEC mode, the VR_PCS_AM_CNT register content is interpreted as number of RS FEC code words between two Code Word Markers/Alignment Markers.

  Each RS FEC code word corresponds to 80 X 66-bit blocks. For Example, if in RS FEC mode, the VR_PCS_AM_CNT register is programmed to 512, then
  - In 25G mode, CWM interval corresponds to 512 * 80 = 40960 number of 66-bit blocks
  - In 50G mode, AM interval corresponds to 512 * 80/4 = 10240 number of 66-bit blocks per PCS lane
  - In 100G mode, AM interval corresponds to 512 * 80/20 = 2048 number of 66-bit blocks per PCS lane
- When you change the alignment maker insertion in DWC_xlgpcs, you must change the Alignment marker insertion interval in the remote end device also for the same value.

Figure 2-29 Reed Solomon FEC Synchronization FSM

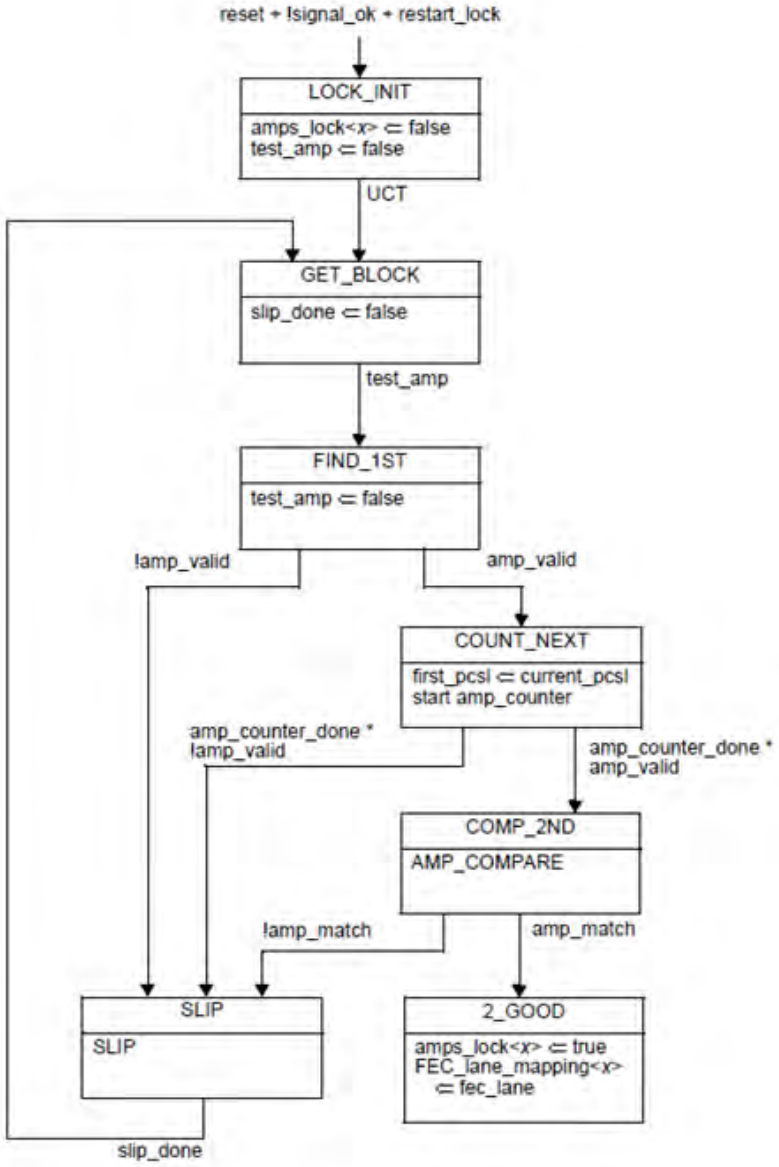


2.2.2.3.2 Lane-Deskew

After CWM/alignment marker lock is achieved on all four lanes, all inter-lane skew is removed as specified by the FEC alignment state diagram shown in Figure 2-30.

The FEC receive-logic supports a maximum Skew of 180 ns between FEC lanes and a maximum Skew Variation of 4 ns.

Figure 2-30 RS FEC Alignment State Diagram



2.2.2.3.3 Lane Reorder

RS FEC lanes can be received on different lanes of the service interface from which they were originally transmitted. The FEC receive function orders the FEC lanes according to the FEC lane number. The FEC lane number is defined by the sequence of alignment markers that are mapped to each FEC lane. After all FEC lanes are aligned, deskewed, and reordered, the FEC lanes are multiplexed together in the proper order to reconstruct the original stream of FEC codewords.

- In 25G RS FEC mode of operation Lane reorder is not applicable
- In 50G RS FEC mode of operation lane reorder is done between lane-0 and lane-1
- In 100G RS FEC mode of operation lane reorder is done among lane-0 to lane-3

Post lane-reordering the data is sent through a gearbox, shown as GB1 in Figure 2-28, to generate parallel RS FEC symbols. The parallel RS FEC symbols are reverse-distributed and fed to RS-FEC Decoder.

2.2.2.3.4 Reed-Solomon Decoder

Reed-Solomon Decoder works with parallel symbols to meet the required data rate and latency requirements. Depending on the operating speed, the Reed-Solomon decoder can operate in the RS(528,514) or RS(544,514) mode.

The decoder extracts the message symbols from the codeword, corrects them as necessary, and discards the parity symbols. The message symbols correspond to 20 transcode blocks.

The RS(528,514) sublayer is capable of correcting any combination of up to 7 symbol errors in a codeword. The RS(528,514) decoder is also capable of indicating when a codeword is non correctable (that is, when the number of symbols errors in a code word is more than 7).

The RS(544,514) decoder is capable of correcting any combination of up to 15 symbol errors in a codeword. The RS(544,514) decoder is also capable of indicating when a codeword is non correctable (that is, when the number of symbols errors in a code word is more than 15).

To reduce the delay contributed by the decoding, it also provides a programmable control to perform error detection without error correction by programming bit-0 in the RS-FEC control register.


Note Do not use this option when RS-FEC sublayer is used to form part of a 25GBASE-SR/100GBASE-SR PHY.

As per the standard, the Reed-Solomon decoder supports Error forwarding which indicates errors to the PCS sublayer by intentionally corrupting 66-bit block synchronization headers.

When the decoder determines that a codeword contains errors (when correction is bypassed) or the code word is uncorrectable (when the bypass correction feature is not supported or not enabled), it ensures that, for every other 257-bit block within the codeword starting with the first (1st, 3rd, 5th, etc.), the synchronization header for the first 66-bit block at the output of the 256B/257B to 64B/66B transcoder, rx_coded_0<1:0>, is programmed to 11. In addition, it ensures that rx_coded_0<1:0> corresponding to the second 257-bit block in the codeword are programmed to 11.

When the Error forwarding feature is selected during configuration, the RS-FEC decoder needs an additional external memory of size 257 bit x 20 locations when DWCXLP_IRSFEC=0, or of size 257 bit x 40 locations when DWCXLP_IRSFEC=1.

Note

When RS FEC is present in the configuration, Error indication ability is always present irrespective of whether RS-FEC Error forwarding feature and hence FEC bypass indication ability bit (bit-1) in the RS-FEC status register is programmed to 1.

However, when Error Forwarding feature is not enabled in RS FEC configuration, FEC bypass indication enable (bit-1) in RS FEC control register is Read-Only and its default value is programmed to 1. This implies Error Indication is not done in these configurations. RS FEC Error Forwarding feature is user selectable during configuration to trade off latency v/s accurate error indication XLGMII Receive interface.

When Error Forwarding feature is enabled in the configuration, the combination of RS FEC bypass Error correction (bit-0 in RS-FEC control register and RS FEC bypass Error indication (bit-1 in RS-FEC control register) controls Error forwarding.

When Error Forwarding feature is disabled in the configuration, error correction cannot be bypassed.

Table 2-10 Truth Table for Error Forwarding

RS FEC Bypass Error Correction Enable (bit-0 of RS FEC control register)	RS FEC bypass Error indication Enable (bit-1 of RS FEC control register)	Error Forwarding Function
0	0	Error forwarding is enabled
0	1	Error forwarding is disabled
1	0	Error forwarding is enabled
1	1	Error forwarding is enabled

When error forwarding is disabled, Reed-Solomon decoder counts the number of symbol errors detected in consecutive non-overlapping blocks of 8192 codewords. When the number of symbol errors in a block of 8192 codewords exceeds 417, the Reed-Solomon decoder makes synchronization header bits of each subsequent 66-bit block to 2'b11.

Note

The programming of sync bits to 2'b11 eventually sets hi_ber to true. When Auto Negotiation is supported and enabled, this event causes Auto Negotiation to restart.

2.2.2.3.5 Reed-Solomon Decoder Memory Interface

As a result of large decoding latency associated with Reed Solomon decoder, the input to the Decoder is stored in an external Dual Port Memory before applying error correction.

The RS-FEC Error correction memory organization is configuration dependent and is tabulated in Table 2-11.

Table 2-11 Reed Solomon FEC Error Correction Memory Organization

Configuration	BASE-R FEC	RS-FEC	Interleaved RS-FEC	Memory Organization (Instances X Depth X Width) Depth in Locations and Width in Bits
25G	Disabled	Enabled	Not Applicable	1 X 85 X 80
	Enabled	Enabled	Not Applicable	1 X 64 X 80 1 X 32 X 80#
25G-50G multi-lane	Disabled	Enabled	Not Applicable	1 X 64 X 160
	Enabled	Enabled	Not Applicable	4 X 32 X 66 4 X 32 X 66#
25G-50G single-lane	Disabled	Enabled	Not Applicable	1 X 64 X 160
	Enabled*	Enabled	Not Applicable	2 X 32 X 160
40G	Disabled	Not Applicable	Not Applicable	Nil
	Enabled	Not Applicable	Not Applicable	4 X 32 X 66 4 X 32 X 66$
100G multi-lane	Disabled	Enabled	Disabled	1 X 32 X 320
	Enabled@	Enabled	Disabled	4 X 32 X 66 4 X 32 X 66#
	Disabled	Enabled	Enabled	1 X 67 X 320
	Enabled@	Enabled	Enabled	4 X 67 X 66 4 X 67 X 66#
100G single-lane	Disabled	Enabled	Disabled	1 X 32 X 320
	Disabled	Enabled	Enabled	1 X 67 X 320
	Enabled*	Enabled	Disabled	2 X 32 X 160
	Enabled*	Enabled	Enabled	2 X 67 X 160

When both RS-FEC and BASE-R FEC are enabled, the BASE-R FEC Error Forwarding feature and associated memory interface are enabled by default.

$ In 40G configuration, you can optionally select or deselect the BASE-R FEC Error Forwarding feature during RTL configuration.

@ If the BASE-R FEC feature is enabled in a 100G multi-lane configuration, it can be used only for 10G, 20G, 25G, 40G, and 50G modes.

* If the BASE-R FEC feature is enabled in a 25G-50 single-lane configuration or in a 100G single-lane configuration, it can be used only for 10G and 25G modes.

2.2.2.3.6 RS-FEC Error Forwarding

The RS-FEC decoder supports the Error Forwarding mechanism as described in sub clause-91.5.3.3.

The RS-FEC Error Forwarding feature is always present in all RS FEC-enabled configurations. This feature requires an external Dual Port Memory of size 257-bit X 20 deep, corresponding to one RS-FEC code word size. For configurations with the DWCXLP_IRSFEC parameter enabled, the depth of the error forwarding memory is 40, as two RS-FEC code words need to be stored.

Error forwarding logic writes the decoded code word (in terms of 257-bits) in to error forwarding memory. The read initiation of the code word from memory is dependent on whether error forwarding is enabled or not.

When error forwarding is disabled (via programming), the reading of decoded data from error forwarding memory follows immediately after the beginning of the write operations. This adds a latency of

- 4 clock cycles (10.24ns) in 100G configurations.
- 5 clock cycles (12.8ns) in 25G_40G/50G configurations.
- 6 clock cycles (15.36ns) in 25G and 25G_40G configurations.

When error forwarding is enabled via programming, the reading of decoded data from error forwarding memory follows only after the complete code word is written. This adds a latency of

- 22 clock cycles (56.32ns) in 100G configurations.
- 34 clock cycles (87.04ns) in 25G_40G/50G configurations.
- 64 clock cycles (163.84ns) in 25G and 25G_40G configurations.

The large latency associated with error forwarding can be disabled through programming. For more details, see Table 2-11.

2.2.2.3.7 Code Word Marker Removal

The parallel symbols from RS FEC decoded output is converted to 257-bit data using gearbox GB2 as shown in Figure 2-28. The first 257 message bits in every 1024th code word (or after the number of RS FEC code words specified in the `VR_PCS_AM_CNT` register) correspond to Code Word Marker/Alignment Marker and is removed before prior to transdecoding.

2.2.2.3.8 256B/257B to 64B/66B Transdecoder

The transcoder extracts a group of four 66-bit blocks, from each 257-bit block received from gearbox2. The transcoder function is implemented as per sub clause 91.5.3.5 from IEEE802.3-2022 specification.

In 25G mode of operation the generation of h<3:0> is derived by cross-referencing to g<3:0> using the following block types defined in Figure 2-3.

In 50G/100G mode of operation, the generation of h<3:0> is derived by cross-referencing to g<3:0> using the following block types defined in Figure 2-4.

For every 257-bit input, the transdecoder generates 264-bit output corresponding to four 66-bit blocks of PCS data and is fed to the Descrambler.

Additionally, in 25G mode of operation, the four parallel 66-bit wide data is 66-bit serialized to feed the descrambler every clock cycle matching the data rate.

2.2.2.3.9 RS-FEC Debug

RS-FEC decoder provides debug status via registers as per Clause-91 (for 50G, 100G modes) and Clause-108 (for 25G modes).

Note The values of symbol error counters are not updated when the RS FEC code word is un-correctable (RS FEC code word is uncorrectable when the there are more than 7 symbol errors in the RS FEC code word)

2.2.2.4 Block Synchronization

A block lock process operates independently on each lane. Each block lock process looks for 64 valid sync headers in a row to declare lock. A valid sync header is either a 01 or a 10. Once in lock, the lock process looks for 65 invalid sync headers within a 1024 sync window to declare out of lock. An invalid sync header is a 11 or 00. Once block lock is achieved on a lane, then the alignment marker process starts.

2.2.2.5 Multi-Lane Alignment Lock Synchronization

An alignment marker lock process operates independently on each lane. Each alignment marker lock process looks for two valid alignment markers 16384 66-bit blocks apart to gain alignment marker lock. On a given lane of the service interface, lane markers must match each other and an entry from IEEE 802.3 for 40GBASE-R. Figures 2-31 and 2-32 show the alignment marker memory write and read interfaces.

Note The BIP_3 and BIP_7 fields are excluded from the markers when making a match to each other or the tables. Once in lock, a lane goes out of alignment marker lock if four markers are received in a row that do not match the alignment marker that the lane is currently locked to.

Figure 2-31 Alignment Marker Memory Write Interface

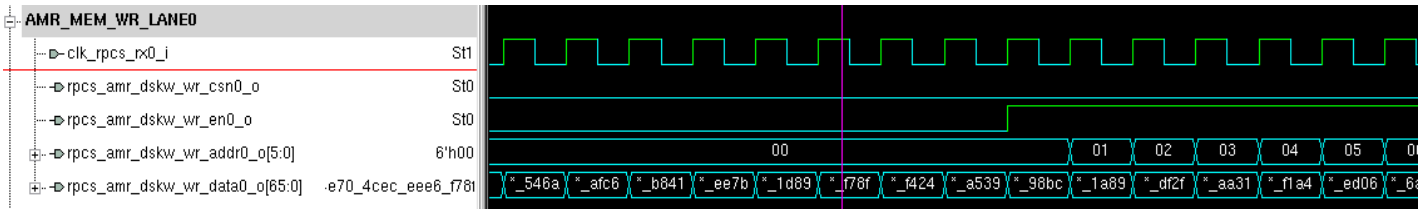
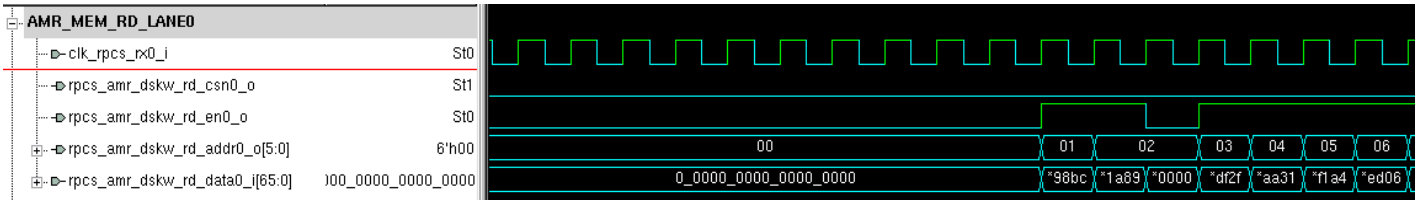


Figure 2-32 Alignment Marker Memory Read Interface



2.2.2.6 Multi-Lane Deskew

In multi-lane configuration, the deskew process ensures that any skew among the lanes is removed if the skew is within the maximum allowable skew. There are four deskew buffers corresponding to each physical lane, in multi-lane configurations.

Table 2-12 lists the maximum allowable skew and Table 2-13 shows the deskew buffer depths for all multi-lane configurations.

In addition to the lane-to-lane deskew, in 100G non RS FEC mode, inter PCS lanes deskew of up to 29 ns is also handled.

Table 2-12 Multi-Lane BASE-R PCS Lane Deskew

PCS	Maximum Skew
multi-lane PCS	180 ns

Table 2-13 Deskew Buffer Depth

Configuration	De-Skew Buffer Depth Needed
25G	None
25_50G (1-lane)	66-bit wide X 83 location deep X 2 instances.
40G, 25G-50G without RS-FEC	66-bit wide X 53 location deep X 4 instances.
25G-50G with RS-FEC	66-bit wide X 64 locations deep X 2 instances. 66-bit wide X 53 locations deep X 2 instances. When operating in RS-FEC mode, two memories (64 locations from one memory and 19 locations from another) are logically stitched together to operated as a single memory of 83 locations deep.
100G	66-bit wide X 95 locations deep X 4 instances.

2.2.2.7 Multi-Lane Reorder

The data received on a PCS lane can correspond to the data of some other lane (of the remote end transmitter) because of multiplexing or de-multiplexing that is performed in the PMA layer. The receive PCS re-orders the received PCS lanes according to the PCS lane number (which is embedded in the Alignment Marker inserted by the Transmitter). This block re-orders PCS lanes based on the Received Alignment Marker content.

- In 100G RS FEC mode, lane-reordering is done on data received from 4 PHY lanes (by looking at the Clause 91 AM or Clause 161 patterns).

  In the non-RS FEC mode, lane-reordering is done on 20PCS lanes (by looking at the AM patterns)
- In 40G/50G modes lane-reordering is done on 4PCS lanes
- In 20G mode lane reordering is done on 2 PCS lanes

2.2.2.8 Multi-Lane Alignment Marker Removal

After all the PCS lanes are aligned, de-skewed, and re-ordered based on the received Alignment Marker (AM), the AM is removed from the incoming data stream. The rate difference from the deleted AMs is compensated by inserting IDLE (de-asserting xlgmii_rx_vld_o). This removal has a dependency on Clock Rate Compensation while transferring received data from the recovered clock to local clock towards the XLGMII interface.

2.2.2.9 Bit Error Rate Monitor

To determine if the locked input data is showing high bit error rate (BER), the BER monitor continuously monitors the input data and validates whether it has a valid Sync Header.

In 10G mode of operation, If less than 16 erroneous blocks are detected in 125µs duration, it is considered as LOW BER. If 16 or more erroneous blocks are detected in 125µ s duration, it is considered as HIGH BER.

In 40G mode operation, DWC_xlgpcs enters HIGH BER state if 97 invalid 66-bit sync headers are detected within a 1.25 ms window. DWC_xlgpcs exits HIGH BER state if there are less than 97 invalid sync headers in the same window.

In 25G mode operation, DWC_xlgpcs enters HIGH BER state if 97 invalid 66-bit sync headers are detected within a 2ms window. DWC_xlgpcs exits HIGH BER state if there are less than 97 invalid sync headers in the same window.

In 50G mode operation, DWC_xlgpcs enters HIGH BER state if 97 invalid 66-bit sync headers are detected within a 1ms window. DWC_xlgpcs exits HIGH BER state if there are less than 97 invalid sync headers in the same window.

In 100G mode operation, DWC_xlgpcs enters HIGH BER state if 97 invalid 66-bit sync headers are detected within a 500us window. DWC_xlgpcs exits HIGH BER state if there are less than 97 invalid sync headers in the same window.



Note

In 5G or 10GBASE-R, for declaring the Receive link to be up (indicated by bit 2 of SR_PCS_STS1), the block alignment must be successful and HIGH BER must be low.

In 20/40/50/100GBASE-R, for declaring the Receive link to be up (indicated by bit 2 of SR_PCS_STS1), the align status must be successful and HIGH BER must be low.

2.2.2.10 Descrambler

This block performs the reverse process of the scrambler, described in "Scrambler" on page 49. If bit 9 of the `VR_PCS_KR_CTRL` register is programmed, the descrambler is bypassed. This feature can be used for debugging purpose.

25G configuration uses only one instance of Descrambler while other configurations use 4 instances of descrambler.

2.2.2.11 64/66 Decoder

This block performs the decoding of the 66-bit descrambled data to 64-bit data and 8-bit control as defined in IEEE Std 802.3, Clause 49. This block performs exactly inverse operation of the 5G, 10G, or 25GBASE-R 64/66 Encoder. When it receives any invalid 66-bit data, it maps the data to |E| (error character) in the XGMII format.

For 5G, 10G, or 25G speed modes, this block descrambles the boundary aligned data by using the polynomial defined in "Scrambler" on page 49. This block performs the reverse process of the scrambler and recovers the original data. If bit 9 of the `VR PCS 10GBASE-R` Control register is programmed, the descrambler is bypassed. This feature can be used for debugging purpose.

This block performs the decoding of the 66-bit descrambled data to 64-bit data and 8-bit control as defined in IEEE Std 802.3, Clause 49. This block performs exactly inverse operation of the 64/66 Encoder. When it receives any invalid 66-bit data, it maps the data to |E| (error character) in the XGMII format.

For multi-lane modes, this block performs the decoding of the 264-bit descrambled data to 256-bit data and 32-bit control as taking input of 66-bit data to 64-bit data and 8-bit control on each lane as defined in IEEE Std 802.3, Clause 82. This block performs exactly inverse operation of the 40GBASE-R or 40GBASEKR 64/66 Encoder. When it receives any invalid 66-bit data, it maps the data to |E| (error character).

2.2.2.12 Clock Rate Compensation

This block performs the data transfer from the Receive clock domain to the Transmit clock domain. It removes any clock frequency jitter of +/-100ppm between the Transmit and Receive clocks for the maximum frame size of 9,022 bytes. The rate is matched by inserting or deleting the IDLE, LPI, or Sequence Order Sets as applicable during IFG or IDLE period. This is done only if the IFG or IDLE period meets the minimum IFG requirement on XGMII or GMII.

This block only removes IDLE, LPI, or Sequence Order Sets. In order to insert IDLEs, the valid signal at XLGMII interface must get de-asserted.



Note The clock-compensation logic can support packets of 16k-bytes size for 25G_Only configurations.

2.2.2.13 5GBASE-R or 10GBASE-R Receive XGMII Interface

This block performs the data format conversion from the DDW format (64-bit data with 8-bit control clocking at 156.25 MHz for 10G mode, or 78.125 MHz for 5G mode) to SDR format (32-bit data with 4-bit control clocking at 312.5 MHz for 10G mode, or 156.25 MHz for 5G mode).

2.2.2.14 Start Control Character (SCC) Alignment Logic

The encoding in 25G mode is as per XGMII specification but DWC_xlgpcs uses XLGMII interface for 25G mode of operation. Hence SCC (equal to 0xFB) can appear on any of lane-0, 4, 8 and 12 on XLGMII interface. This requires MAC side's receive logic to do SCC hunt on all of the 4 lanes -0, 4, 8 and 12.

The SCC alignment automatically aligns the Start of Control character to either byte-0 or to byte-8 on XLGMII Receive interface when operating in 25G speed mode. Thus requiring MAC's receive side to hunt for SCC on lane-0 and lane-8 only.

This feature is not applicable in 40G configuration. SCC alignment logic can be enabled in all other configurations (25G, 25G-50G, and 100G configurations) by programming the configuration parameter DWCXLP_SCC_ALGN_EN to 1. When the SCC alignment feature is enabled during configuration, the alignment logic is applicable to 25G speed mode only and is always active (Alignment logic cannot be disabled when operating in 25G mode).

SCC alignment logic assumes link partner's Transmit (Remote end MAC Transmit) to insert Inter-frame gap (using IDLE control characters) as per Deficit IDLE count (DIC) algorithm. This implies,

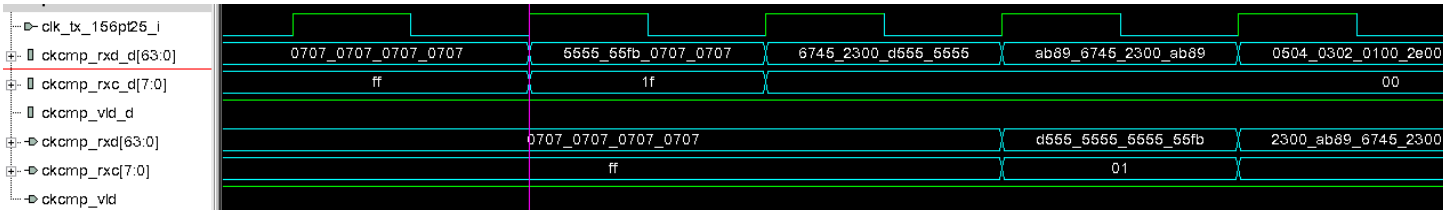
- The average value of the IPG between the frames is assumed to be 12
- Minimum value of IFG is assumed to be 9 and
- Maximum value of IFG is assumed to be 15
- IFG greater than 15 does not cause any harm to the functionality.
- IPG being less than 9 results in frame corruption.

- SCC alignment logic adds 1-clock latency.

Note When the DWCXLP_XLGMII_INTF_WIDTH parameter is chosen as 2 (32-bit), then the SCC alignment logic is not added irrespective of the value of the DWCXLP_SCC_ALGN_EN parameter.

Figure 2-33 shows the timing diagram of data before and after SCC alignment.

Figure 2-33 Timing Diagram of Data Before and After SCC Alignment



2.2.2.15 25/40/50/100GBASE-R Receive XLGMII Interface

In 25G_Only configuration, this block performs the data format conversion from the 64-bit data with 8-bit control format to the following format:

- 128-bit data with 16-bit control, when DWCXLP_XLGMII_INTF_WIDTH is 0.
- 32-bit data with 4-bit control, when DWCXLP_XLGMII_INTF_WIDTH is 2.

In 25G/40G/50G/100G speed modes of other configurations, this block performs the data format conversion from the DDW format (256-bit data with 32-bit control) to the following format:

- 128-bit data with 16-bit control clocking

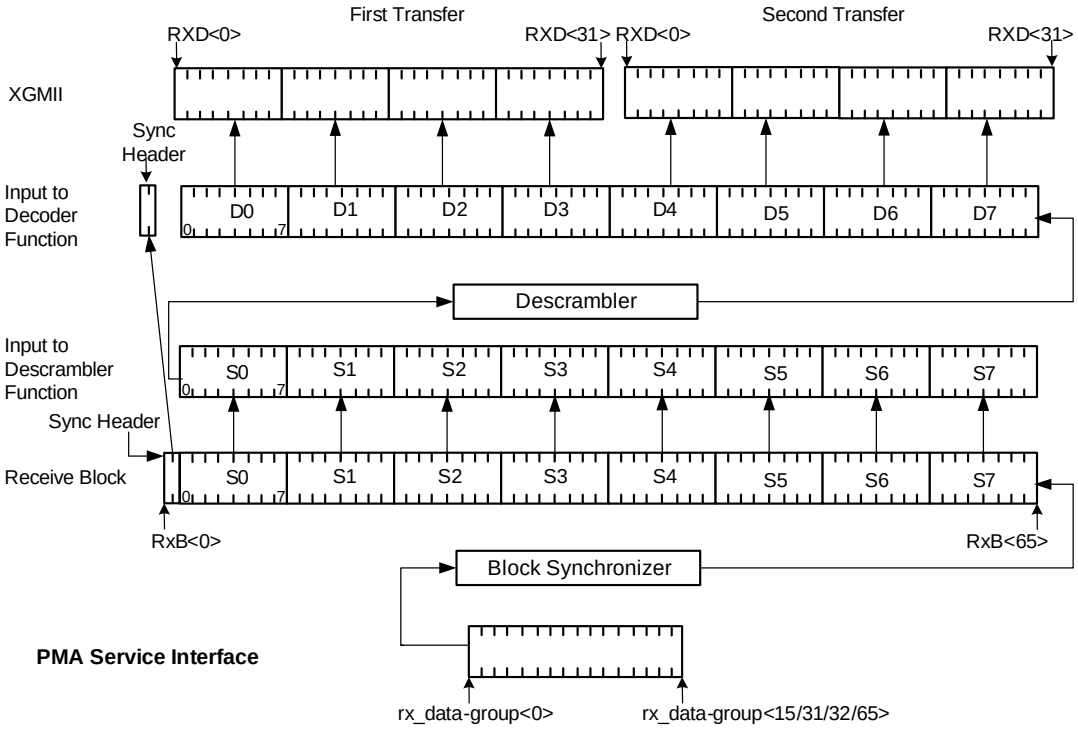
In 25G speed modes, a back pressure signal is asserted on the XLGMII interface to match the data rate

2.2.2.16 5G or 10G PCS-R Receive Bit Ordering

In 64/66 PCS Layer, the receive bit ordering is implemented as defined in IEEE Std 802.3, Clause 49. Figure 2-34 shows the receive bit ordering in the 64/66 PCS Layer.

In Figure 2-34, the rx_data-group<0> is expected to be received first. If RX_LSB_FIRST = Enabled, this bit corresponds to rpcs_rx_data_i[0]. If RX_LSB_FIRST = Disabled, this bit corresponds to rpcs_rx_data_o[15/31/32/65].

Figure 2-34 Receive Bit Ordering in 64/66 PCS Layer

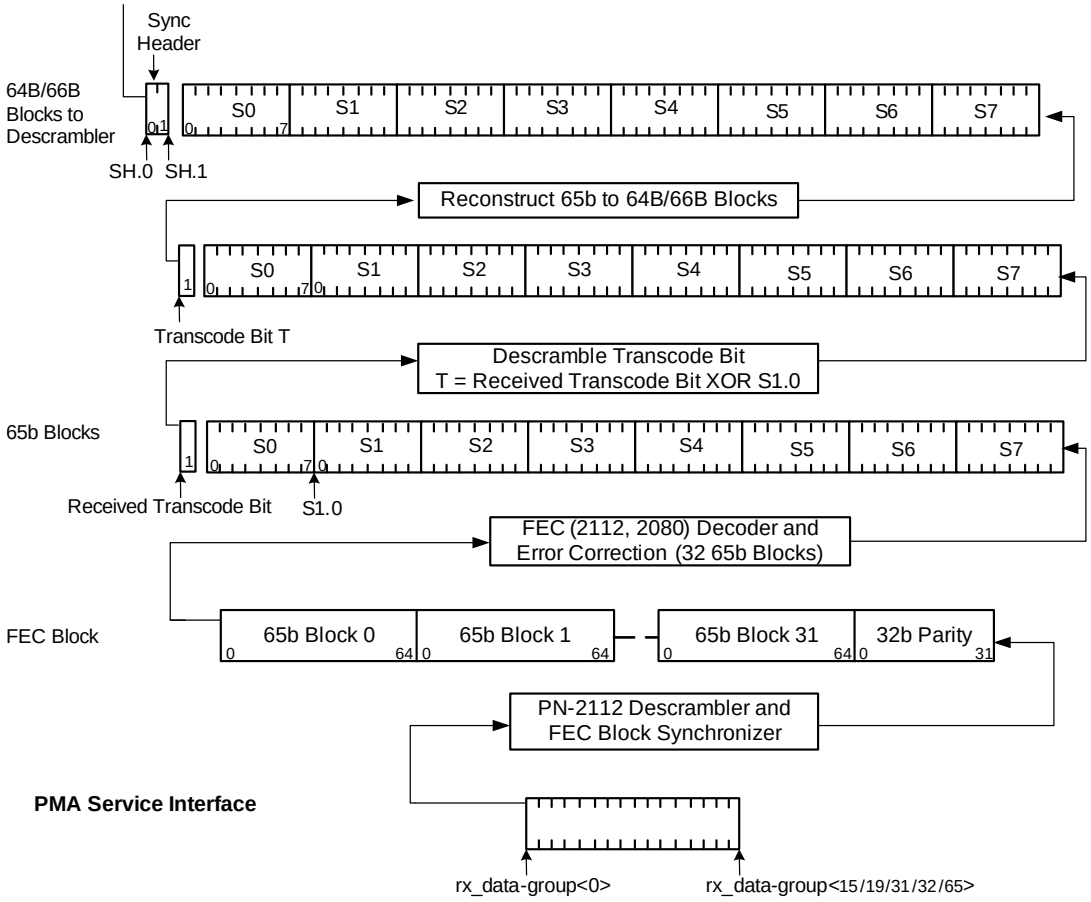


In the FEC block, the 10GBASE-KR Receive bit ordering is implemented as defined in the IEEE Std 802.3, *Clause 74*. Figure 2-35 shows the receive bit ordering in the FEC block.

In Figure 2-35, the rx_data-group<0> is expected to be received first. If RX_LSB_FIRST = Enabled, this bit corresponds to rpcs_rx_data_i[0]. If RX_LSB_FIRST = Disabled, this bit corresponds to rpcs_rx_data_o[15/31/32/65].

For 40/50GBASE-R, one FEC Decoder per one PCS lane, the FEC block, the receive bit ordering is implemented as defined in the IEEE Std 802.3ba, Clause 74. In Figure 2-35, the rx_data-group<0> is expected to be received first. If RX_LSB_FIRST = Enabled, this bit corresponds to rpcs_rx_data_i[0]. If RX_LSB_FIRST = Disabled, this bit corresponds to rpcs_rx_data_o[19].

Figure 2-35 Receive Bit Ordering in the FEC Block



2.2.3 10G/25G Test Pattern Generator and Checker Block

This block generates or checks the Test pattern defined in the IEEE Std 802.3, Clause 49. This block can generate or check the following Test patterns:

- Square Wave

  In the Transmit path, the DWC_xlgpcs sends the repeating pattern of *n* number of ones followed by *n* number of zeros (on serial line) where *n* can be any number between 4 and 11. You can program the value of *n* in bits [6:4] of the `VR_PCS_KR_CTRL` register.

  The DWC_xlgpcs does not support detection of this pattern in the Receive path.

- Pseudo Random

  In the Transmit path, the scrambler generates the test pattern for a given set of Seeds and Data inputs. The management can program the seeds and the data. The scrambler is loaded with a seed or its invert at the start of every 128th block. The seeds are loaded in the following pattern:

  a. Seed A
  b. Seed A Invert
  c. Seed B
  d. Seed B Invert

  You can select the data patterns for these seeds by using the SR PCS 10GBASE-R Test Pattern Seed A and B registers. After loading Seed A or Seed B, the scrambler input is driven with the data pattern. After loading the Seed A invert or Seed B Invert, the scrambler input is driven with the inverse of the data pattern. In pseudo-random pattern, the Sync Header is the control Sync Header (10). Therefore, the pseudo-random test pattern is a series of blocks with a control Sync Header and a pseudo-random payload.

  In the Receive path, the DWC_xlgpcs performs the test pattern checking by monitoring the received descrambled data every 128 clocks.

- PRBS31 or PRBS9

  In the Transmit path, the output of this block directly drives the gearbox. In PRBS31 test pattern, the DWC_xlgpcs generates the pseudo-random sequence corresponding to the following polynomial:

  $G(x) = 1+x^{28}+x^{31}$

  In PRBS9 test pattern, the DWC_xlgpcs generates the pseudo-random sequence corresponding to the following polynomial:

  $G(x) = 1+x^{5}+x^{9}$

  In the Receive path, the DWC_xlgpcs descrambles the PRBS31 or PRBS9 test pattern. If there are no errors, the descrambled output is expected to be 0.

If there is any error, the test pattern mismatch counter is incremented in the SR_PCS_TP_ERRCTR. You can control the test pattern generation or checking through SR_PCS_TP_CTRL and VR_PCS_KR_CTRL. When Test Pattern checking is enabled in the Receive path, you can use the error counters to monitor or test the link reliability and perform jitter testing.

Additionally, DWC_xlgpcs supports PRBS9, PRBS31, and scrambled idle test pattern for 25G mode of operation.

2.2.4 40G/50G/100G Test Pattern Generator and Checker Block

- Square Wave

  Square Wave is supported in PMA per lane

  PRBS31 or PRBS9 are supported in PMA per lane

  If there is any error, the test pattern mismatch counter is incremented in the following registers:

  - SR PMA MMD 40GBASE-KR PRBS Receive Error Counter, Lane0
  - SR PMA MMD 40GBASE-KR PRBS Receive Error Counter, Lane1
  - SR PMA MMD 40GBASE-KR PRBS Receive Error Counter, Lane2
  - SR PMA MMD 40GBASE-KR PRBS Receive Error Counter, Lane3

  You can control the test pattern generation or checking through SR PMA MMD 40GBASE-KR Test Pattern Control register and VR PCS 10GBASE-R Control register. When Test Pattern checking is enabled in the Receive path, you can use the error counters to monitor or test the link reliability and perform jitter testing.

- Scrambled Idle

  When a scrambled idle pattern is enabled in the Transmit path, the test pattern is generated by the scrambler. No seeding of the scrambler is required during test-pattern operation. The input to the scrambler is a control block (block type=0x1E) with all idles.

  In the Receive path, when align_status is true and the scrambled idle receive test-pattern mode is active, the scrambled idle test-pattern checker observes the sync header and the output from the descrambler. When the sync header and the output of the descrambler is the all idle pattern, a match is detected.

  If there is any error, the test pattern mismatch counter is incremented in the SR_PCS_TP_ERRCTR. You can control the Scrambled Idle test pattern generation or checking through SR_PCS_TP_CTRL. When test pattern checking is enabled in the Receive path, you can use the error counters to monitor or test the link reliability and perform jitter testing.

Note	Scrambled Idle Test pattern, the sync headers and alignment markers are added to the stream so that the receive PCS can align and deskew the PCS lanes. When the transmit channel is operating in test-pattern mode, the encoded bit stream is distributed to the PCS Lanes as in normal operation

2.3 EEE Controller

Energy Efficient Ethernet, specified in IEEE Std 802.3-2015, is disabled by default in DWC_xlgpcs. You can enable this feature by programming the DWCXLP_EEE_EN configuration parameter to 1.

In single lane mode (10G/25G speed modes), EEE controller is as per IEEE 802.3 Clause-49 and in multi lane modes (20G,40G,50G and 100G modes), EEE controller is as per IEEE 802.3 Clause-82

The EEE feature helps to save energy or reduce power consumption in the PHY when no data transmission or reception is required for a very long duration. The basic principle of this feature is that when the MAC transmitter indicates that it does not have any data to transmit, the local transmitter (which includes the PCS and the PHY) and the remote-end PHY receiver can be put in the power-save or sleep mode to save energy.



The hardware allows the selection of the DWCXLP_EEE_EN parameter only for configurations with DWCXLP_MAIN_MODE=0 (25G configurations), and configurations with DWCXLP_MAIN_MODE=2 (40G configurations).

2.3.1 EEE Transmit Controller

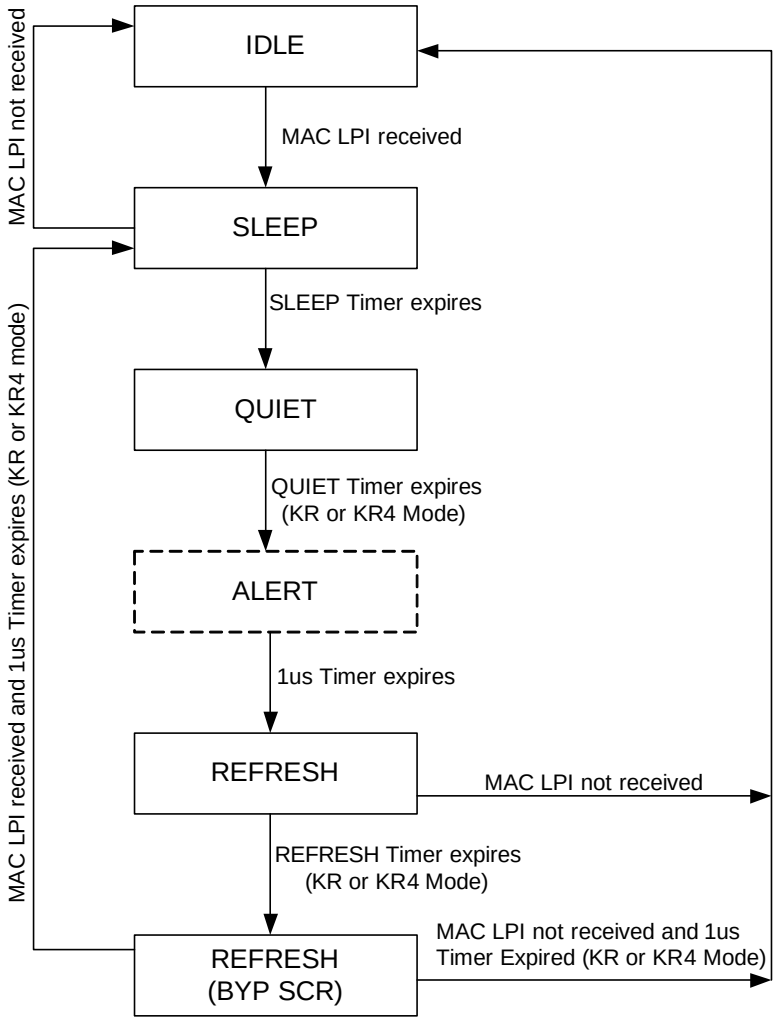
The DWC_xlgpcs Transmit path implements the following functions when the EEE support is selected:

- Monitor the XLGMII interface for Low Power Idle (LPI) pattern
- Maintain the following EEE states in the Transmit path:
  - Sleep (LPI request monitoring state)
  - Quiet (Power-save state)
  - Alert (Pre-wakeup state) in the PCS-R mode
  - Wakeup (Refresh state)
  - Bypass Scrambler state (to enable FEC Fast Synchronization)

Figure 2-46 shows the simplified version of the Transmit LPI state machine. The Transmit LPI flow is triggered when the MAC drives the LPI signals on the XLGMII interface. The DWC_xlgpcs maintains various states in which it periodically includes refresh signaling and freezing the transmitter in the quiet state. By maintaining this predefined sequence, the protocol ensures that the link is alive and at the same time power is saved. When the MAC stops sending the LPI pattern (and starts sending the IDLE pattern), the EEE Transmit controller also comes out of this sequence and reverts to the IDLE (or normal transmit) state.

The timers used in the EEE Transmit Controller have the default values given in the IEEE standard. You can change these default values (within a range) according to your requirement.

Figure 2-36 Transmit LPI flow

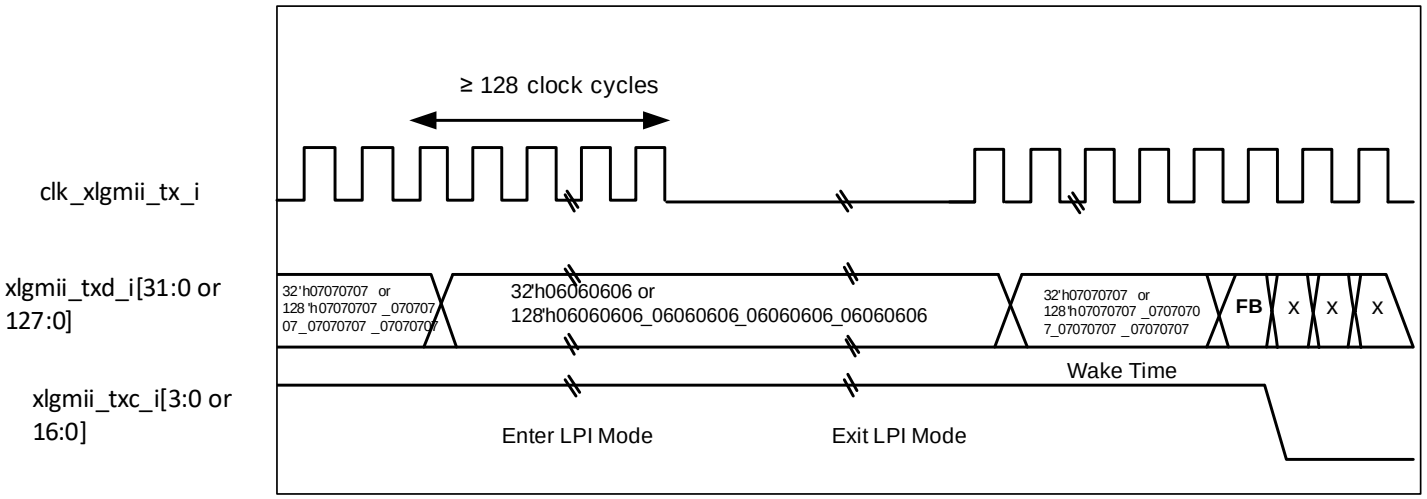


Note In 25G, 50G, and 100G modes with fast wake up, DWC_xlgpcs does not enter Quiet and refresh states.

2.3.1.1 MAC Transmit LPI

Figure 2-37 describes the Transmit LPI flow on the XGMII Interface. The MAC triggers the LPI mode in the DWC_xlgpcs by driving the xlgmii_txd_i[31:0 or 127:0] = 32'h06060606 or 128'h06060606_06060606_06060606_06060606 and xlgmii_txc_i[3:0 or 15:0] = 4'hF or 16'hFFFF.

Figure 2-37 40G MAC Transmit LPI



In configurations in which the clk_xlgmii_tx_i clock is used to sample the xlgmii_txd_i[] or xlgmii_txc_i[] signals in the DWC_xlgpcs, the MAC can also gate the clock (clk_xlgmii_tx_i) after 128 clocks of asserting the LPI Idle to save additional power in the PCS layer. In such configurations, the XLGMII interface is also directly sampled with the reference EEE controller clock (clk_eee_i) to monitor the LPI pattern in the wake-up cycle.

The MAC can come out of the LPI mode at any time by driving Normal Idle instead of LPI Idle with values xlgmii_txd_i[31:0 or 63:0] = 32'h07070707 or 128'h07070707_07070707_07070707_07070707 and xgmii_txc_i[3:0 or 15:0] = 4'hF or 16'hFFFF.

After coming out of the LPI mode, the MAC waits for the wake-time period before it starts sending the frames. The wake-time limit depends on various factors such as the following:

- Local transmitter PCS or PHY layer latency in coming to normal IDLE state
- Pre-wakeup timing (ALERT)
- Remote Link PHY Latency
- Remote PCS or MAC wake-up requirements

The IEEE standard provides an option to exchange the wake-up time requirements, by using DDLP link layer protocol with the link partner, before any of the nodes goes into the power-save mode.

2.3.1.2 PHY Transmit LPI Control

The DWC_xlgpcs generates the xlgpcs_lpitx_quiet_o signal when the Transmit LPI state machine is in the Quiet state. When this signal is asserted, the PHY can save power on the Transmit path. The power-saving strategy in the PHY is implementation dependent. When the Transmit LPI state machine is in the Quiet state, the PHY must freeze the Transmit serial lines or drive these to high-impedance. The PHY can switch off the power (or gate off the clocks) to some of its internal components only if these components can be restored quickly when the xlgpcs_lpitx_quiet_o signal is de-asserted. For example, you can gate off the PHY Transmit Interface clock inputs by using the xlgpcs_lpitx_quiet_o signal. For more information about these options and their dependencies, see chapter *"Integration"* in the *DesignWare Cores Enterprise Ethernet PCS User Guide.*

When the DWC_xlgpcs is in the Transmit Quiet state, it carries out the actions specified in Table 2-14.

Table 2-14 Actions Taken in Transmit Quiet State

PHY	Actions Taken in Transmit Quiet State
Non-Synopsys PHY	Drive xlgpcs_tx_en_o{lane} to 0

Note

- DWC_xlgpcs has recovery time after it comes out of the LPI mode and before it is ready to send the frames. The worst-case recovery time is equal to the following:

  `4 Clocks of clk_eee_i period + 3 clocks of clk_xlgmii_tx_i, or clk_rpcs_tx_i` (whichever is slower among these three clock)
- DWC_xlgpcs is ready to transmit the packets after it comes out of the LPI mode. However, the reliable packet transmission over the link occurs only after the WAKE TIMER (maintained in MAC layer as defined in IEEE802.3az standard) expires.

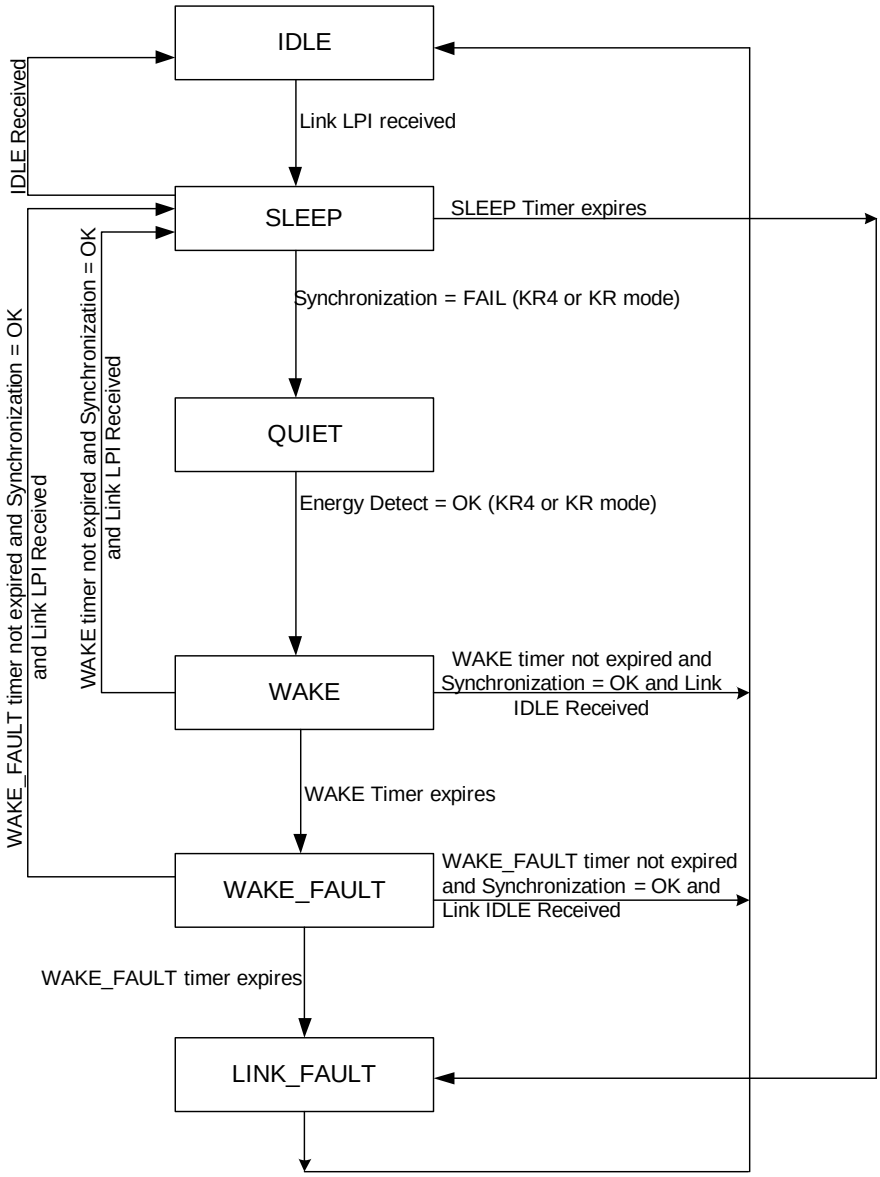
2.3.2 EEE Receive Controller

The PCS Receive path implements the following functions when the EEE support is selected:

- Decode the received LPI code group from the link partner.
- Maintain the following EEE states in the Receive path:
  - Sleep (LPI request monitoring state)
  - Quiet (Power-Save state)
  - Wake (Refresh state)
  - Wake-up Fault (Error Monitoring)

Figure 2-38 describes the simplified version of the LPI Receive state machine. The sequence in the state machine helps in saving the power in the PHY or PCS because some of the functional blocks can be switched off in the Quiet state. In addition, during Refresh state, the Receive path is re-synchronized so that it can monitor the link status. If FEC is enabled in KR mode, the Receive path also implements the Rapid Block Synchronization to minimize the FEC block synchronization time. The timers used in the EEE Receive Controller have the default values as given in the IEEE standard. You can change these default values (within a range) according to your requirement.

Figure 2-38 Receive LPI Flow



Note In 25G,50G 100G modes with fast wake up, DWC_xlgpcs does not enter the Quiet state.

2.3.2.1 PHY LPI Control

The Receive low power mode is triggered when the DWC_xlgpcs receives the LPI Idle code group on the link. It exits the LPI mode when the link partner stops transmitting the LPI Idle code groups.

The DWC_xlgpcs enters the Quiet state after successfully detecting LPI from the link partner followed by SIGNAL detect logic which indicates 'No Signal' on the Receive serial line. In normal flow, the DWC_xlgpcs is in the Quiet state as long as the SIGNAL is not detected on the serial line. The DWC_xlgpcs exits the Quiet

state when SIGNAL is detected (assertion of energy_detect_i) and waits for the PCS synchronization before decoding the incoming code groups. If the received code groups are LPIs, the cycle is repeated and the DWC_xlgpcs again enters the Quiet state. If the incoming code groups are normal Idle, the DWC_xlgpcs exits the LPI mode.

The DWC_xlgpcs asserts the xlgpcs_lpirx_quiet_o signal when the Receive LPI state machine is in the QUIET state. When this signal is asserted, the PHY can save power on its Receive path. The power-saving strategy in the PHY is implementation dependent. For example, the PHY Receive interface clocks can be gated by using the xlgpcs_lpirx_quiet_o signal. For more information about these options and their dependencies, see chapter "*Integration of the Controller*" in the *DesignWare Cores Enterprise Ethernet PCS User Guide*.

When the Receive LPI is in the Quiet state, the PHY can switch off some of the Receive functional blocks except the SIGNAL detect logic. However, the PHY must do it only if the switched-off blocks can be restored quickly when the xlgpcs_lpirx_quiet_o signal is de-asserted. Otherwise, the PCS receiver may not get sufficient time to synchronize successfully.

For configurations with Synopsys multi-protocol 12G/16G/25G/32G PHY, DWC_xlgpcs drives xlgpcs_rx{lane}_pstate_0 to P0s power state when it is in Receive Quiet State. When DWC_xlgpcs comes out of Receive Quiet state, xlgpcs_rx{lane}_pstate_o is driven to whatever value is programmed in the `VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL` register.

2.3.2.2 MAC Receive LPI

Note

DWC_xlgpcs needs some recovery time after it comes out of the LPI mode and before it is ready to receive the frames.

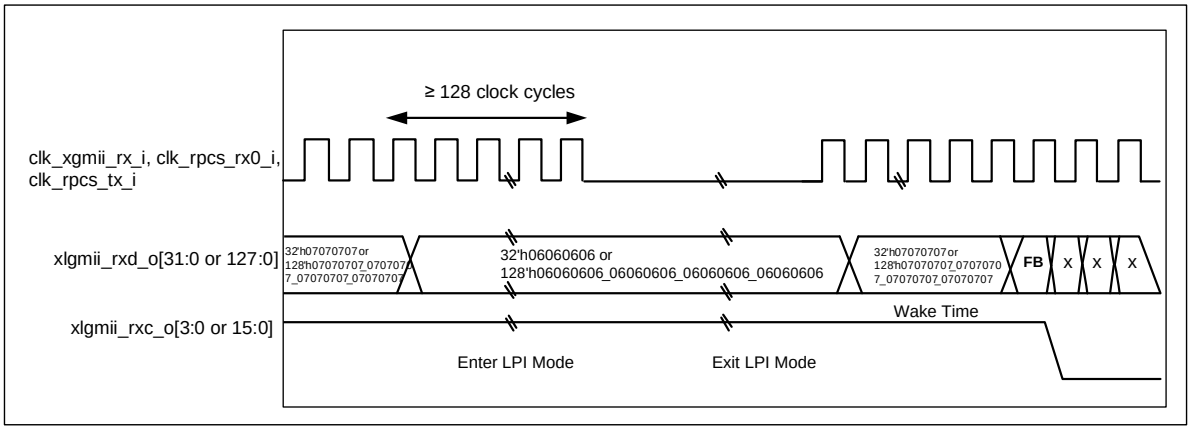
The worst-case recovery time is equal to the following:

11 Clocks of clk_eee_i period + 3 clocks of clk_xgmii_rx_i, clk_xgxs_rx_i, or clk_rpcs_rx_i (whichever is slower among these three clocks).

Figure 2-43 describes the Receive LPI flow on the XLGMII Interface. The Receive LPI mode is triggered when the DWC_xlgpcs receives the LPI from the link partner. After successful decoding, the controller enters the Sleep state and drives the LPI patterns on the Receive XLGMII signals as shown in the Figure 2-43 (xlgmii_rxd_i[31:0 or 127:0] = 32'h06060606 or 128'h06060606_06060606_06060606_06060606 and xgmii_rxc_i[3:0 or 15:0] = 4'hF or 16'hFFFF).

The xlgpcs_lrx_clk_gat_o signal is asserted after 128 clocks of driving the LPI pattern on the XLGMII interface. The xlgpcs_lrx_clk_gat_o signal can be used to gate the receiver clock towards the MAC. The xlgpcs_lrx_clk_gat_o signal is de-asserted when the DWC_xlgpcs fails to detect the LPI Idle during the Wake state or when the Sleep timer or the Wake timer expires and the controller enters the Idle or Fault states.

Figure 2-39 MAC Receive LPI



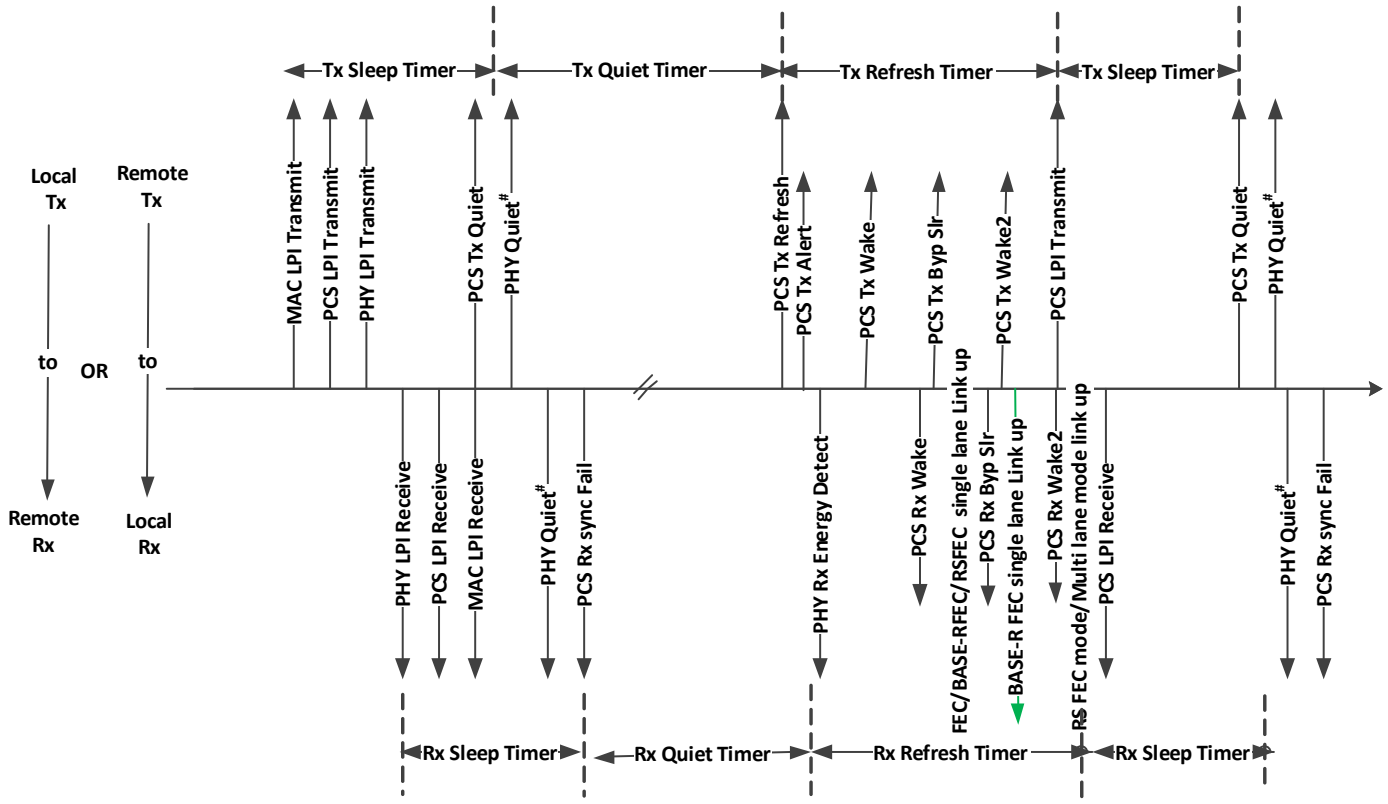
Note The 128 clock duration is the default setting. The host can increase this duration by programming bits[15:12] of the VR_PCS_EEE_MCTRL register.

2.3.3 Reference EEE Flow

Figure 2-46 provides a summary of the sequence of events (in time) that occur in the EEE mode in the local PCS Transmitter and remote Receiver (or vice versa). You can use this figure to determine the time margins available in the system for recovery during refresh periods. The details of these latencies (in DWC_xlgpcs) are given in Table 2-15.

The DWC_xlgpcs Transmit sends the LPI IDLE code groups during the Sleep and Refresh periods (except during Transmit Alert period in KR mode). During the Quiet period, the PHY transmitter drives the serial lines to HiZ.

Figure 2-40 EEE Flow



Note

- In IEEE 802.3 clause-83 GAUI mode, when shut down is not permitted in Quiet mode and alert mode, PRBS patterns corresponding to quiet state and alert state are sent over the PHY lanes.
- Bypass scrambler state is present only in BASE-R FEC enabled modes
- WAKE2 state is present in multi lane modes only
- In fast wake-up mode (applicable to 25G, 50G and 100G modes), DWC_xlgpcs does not enter into Quiet and Refresh cycles

Table 2-15 gives the DWC_xlgpcs Receive timing. The DWC_xlgpcs meets all transmitter timings requirements specified in IEEE Std 802.3.

Table 2-15 DWC_xlgpcs EEE Receive Timing

Parameter	IEEE 802.3 Specification for Transmitter	IEEE 802.3 Specification for Receiver	DWC_xlgpcs Receive Timing
Sleep Timer	■ Single lane modes: 4.9 µs to 5.1 µs ■ Multilane modes: 0.9 µs to 1.1 µs	■ Both single and multi lane modes: 2 ms to 3 ms **Note**: This is the maximum time for which the Receive tries to detect the Signal Detect or Synchronization FAIL.	Meets the Receive requirements

Table 2-15 DWC_xlgpcs EEE Receive Timing (Continued)

Parameter	IEEE 802.3 Specification for Transmitter	IEEE 802.3 Specification for Receiver	DWC_xlgpcs Receive Timing
Quiet Timer (Power-save mode)	■ Both single lane and multi lane modes: 1.7 to 1.8 ms	N/A Receive QUIET time = Transmit QUIET Time – (Signal Detect FAIL delay + PCS Synchronization FAIL delay) + Signal Detect OK delay These Receive delays must be as minimum as possible to Save maximum power in Receive path.	Maximum PCS Synchronization FAIL delay assuming incoming data is INVALID. ■ KR4/KR without FEC: 230.4 ns ■ TBD ■ KR4/KR with FEC and EF: 2329 ns
Refresh (Refresh period during which the Receive Link must be synchronized)	■ ALERT signalling: 1.15 µs to 1.3 µs (in both single and multilane modes) ■ Wake - Single lane: 10.9 to 11.1 µs - multilane: 1.5 to 1.6 µs ■ BYPASS Scrambler (When BASE-R FEC is enabled) - single lane: 1.1 to 1.3 µs - multilane : 0.9 to 1.1µs for 20G/40G and 1.9 to 2.1 µs for 50G ■ WAKE2 state - single lane: Not applicable - multilane: 2.4 to 2.5 µs	■ single lane modes: 11 µs in BASE-R FEC disabled mode and 13.7 µs in BASE-R FEC mode ■ multilane modes: 4.5 µs BASE-R FEC disabled mode and 5.5 µs in BASE-R FEC enabled mode. The Signal Detect or Energy Detect FAIL delay + PCS Synchronization OK delay must meet this value.	The minimum PCS Synchronization OK delay assuming that the DWC_xlgpcs received reliable data ■ KR4/KR with FEC: 1100 ns ■ KR4/KR with FEC and EF: 1510 ns

2.4 Command and Status Register Block

The Command and Status Registers (CSR) block performs the following functions:

- Maintains the following MMDs register sets (depending on the configuration and current operating mode):
  - PCS
  - PMA or PMD
  - Auto-negotiation
  - Vendor-specific Control MMD
- Interacts with all blocks to provide the control and status information

2.4.1 CSR Block Overview

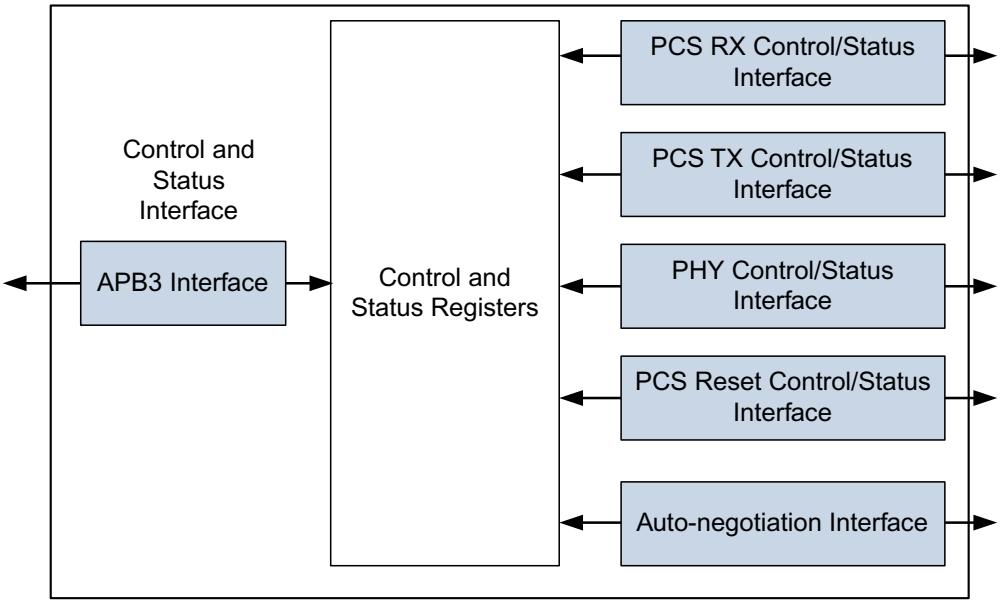
The CSR register sets are used as Command and Status registers. A management agent can issue commands to set attributes supported by the DWC_xlgpcs, or to read the status of the device or command. This block implements all register sets required for implementing the core and complying with IEEE std, Clause 45.

Along with the standard register sets, this block implements a vendor-specific register set to deploy additional features. Vendor-specific register sets provide flexibility to the management agent to use features and attributes supported by the DWC_xlgpcs and not defined in the Standard.

The host can access the CSR block through the AMBA APB 3 Interface.

Figure 2-41 displays the CSR block.

Figure 2-41 CSR Interface Block Diagram



2.4.2 CSR Addressing

The CSR register set addressing includes the following factors:

- MMD Device Address
- Register Address

2.4.2.1 MMD Device Address

The 5-bit MMD device address specifies which MMD device is being addressed by the host. The DWC_xlgpcs supports various MMDs depending on the configuration.

The DWC_xlgpcs supports the MMDs based on the configuration and current mode of operation. For more details about the Device Address map for various MMDs see the *"Register Descriptions"* chapter.

2.4.2.2 Register Address

The 16-bit register address indicates which register in selected MMD device is being addressed. If the 16th bit is zero, it indicates that the lower bits are addressing the Standard register set. If the 16th bit is one, it indicates that the lower bits are addressing a vendor-specific register set.

2.5 Management Interface

The APB3 or MDIO block functions as the management interface implementing the APB3 or MDIO protocols. All CSR register sets can be accessed through these interfaces.

2.5.1 APB3 Interface

This topic contains the following sections:

- "Overview of APB3 Interface" on page 106
- "APB3 Interface without Indirect Access" on page 106
- "APB3 Interface with Indirect Access" on page 108

2.5.1.1 Overview of APB3 Interface

The AMBA APB 3 interface is compliant with the *AMBA 3 APB Protocol Specification*, v1.0. It is a parallel interface to access the CSR registers in DWC_xlgpcs. The method in which the APB3 interface accesses the registers depends on the INDIRECT_ACCESS parameter.

DWC_xlgpcs also supports the following additional variants of the APB3 interface, in addition to the legacy APB3 interface:

- APB3_16

  In this configuration, though the registers are 16-bit wide, addresses of registers are byte-aligned, which means that the address value is in multiples of 2.
- APB3_32

  In this configuration, registers are 32-bit wide, and addresses of registers are multiples of 4.

In the legacy AB3 configuration, DWC_xlgpcs registers are 16-bit wide and addresses of registers are not byte-aligned. As a result, addresses are multiples of 1.

2.5.1.2 APB3 Interface without Indirect Access

If INDIRECT_ACCESS = Disabled, the APB3 interface accesses the registers in the DWC_xlgpcs in one phase. The width of the address bus without indirect option is as follows:

- 22 bits for APB3_16
- 23 bits for APB3_32
- 21 bits for legacy APB3

The following list describes the flow of register access:

- The assertion of psel_i signal indicates the start of the register access. When psel_i signal is asserted, the pwrite_i and paddr_i[20/21/22:0] signals are valid. The assertion of psel_i signal also indicates that the pwdata_i[15:0] signal is valid during write operation. When the pwrite_i signal is high, it indicates the write access; otherwise, it indicates the read access. Table 2-16 shows the signal used to indicate MMD Device address and Register address in APB3 variants.

Table 2-16 Signals Indicating the Device and Register Addresses in the APB3 Variants

APB3 Variant	MMD Device Address	Register Address
APB3_32	paddr_i[22:18]	paddr_i[17:0]
APB3_16	paddr_i[21:17]	paddr_i[16:0]
Legacy APB3	paddr_i[20:16]	paddr_i[15:0]

- Based on the type of APb3 interface, DWC_xlgpcs drives one of the following signals, which contains the data of the register being addressed, during read-address.

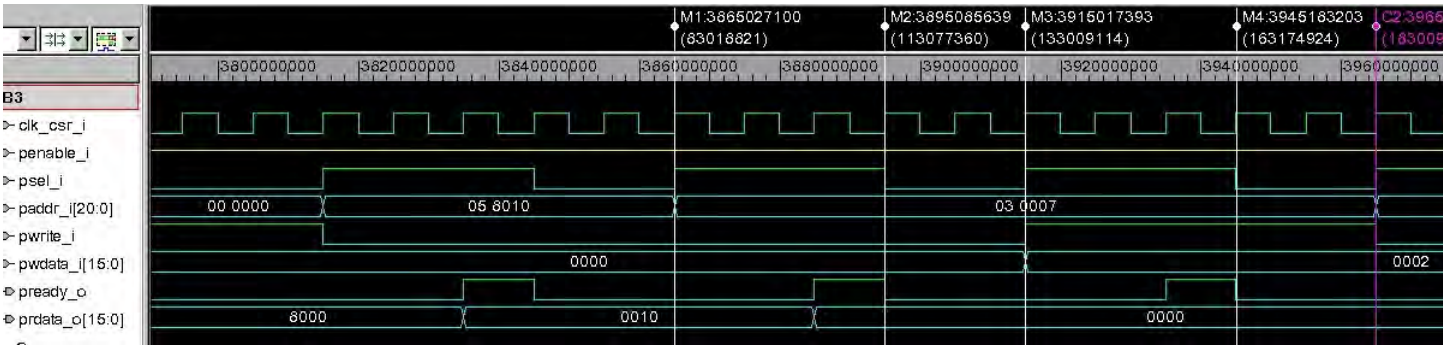
Table 2-17 Signal Driven by DWC_xlgpcs for APB3 Variants

APB3 Variant	Signal Driven by DWC_xlgpcs
APB3_32	prdata_o[31:0] prdata_o[31:16] always indicates 0. Though the registers are 32-bit wide, upper 16-bits are always reserved and read-only.
APB3_16	prdata_o[15:0]
Legacy APB3	

- DWC_xlgpcs drives the pready_o signal high which indicates the completion of the register access. In addition, it indicates that the prdata_o[15/31:0] has valid data during read access. The host must de-assert the psel_i signal when the DWC_xlgpcs asserts the pready_o signal. If the psel_i signal continues to be asserted, it is taken as next read or write access. If the host accesses a register with a reserved or unmapped address, the DWC_xlgpcs returns "all-zero" data during reads and ignores the data during write transfers.

Figure 2-42 shows an example of register read or write access.

Figure 2-42 APB3 Interface Register Access (without Indirect Access)



In Figure 2-42, the marker M1 indicates the start of the read access at address 21'b030007. The marker M2 indicates the end of the read access. The data read is 16'h0000. Similarly, the marker M3 indicates the start of the write access at address 21'h030007 with write data 16'h0002. The marker M4 indicates the end of the write access.

2.5.1.3 APB3 Interface with Indirect Access

If INDIRECT_ACCESS = Enabled, the APB3 interface accesses the registers in the DWC_xlgpcs in two phases. With indirect access, the memory range in the DWC_xlgpcs is reduced to:

- 256 register set for legacy APB3
- 512 register set for APB3_16
- 1024 register set for APB3_32

Therefore, the width of address bus is 8 bits for legacy APB3, 9 bits for APB3_16, or 10 bits for APB3_32. The APB3 interface accesses the CSR registers in the following phases:

- **Address Phase**

  In this phase, the host performs the write operation at register address 8'hFF for legacy APB3, 9'h1FE for APB3_16, or 10'h3FC for APB3_32 of any of the supported MMDs in DWC_xlgpcs. The write data must contain the base address of the 21-bit, 22-bit, or 23-bit address. The base address corresponds to address bits[20:8] for legacy APB3, bits[21:9] for APB3_16, or bits[22:10] for APB3_32 of 21-bit, 22-bit, or 23-bit full address. The base address is driven on the pwdata_i[12:0] bus.

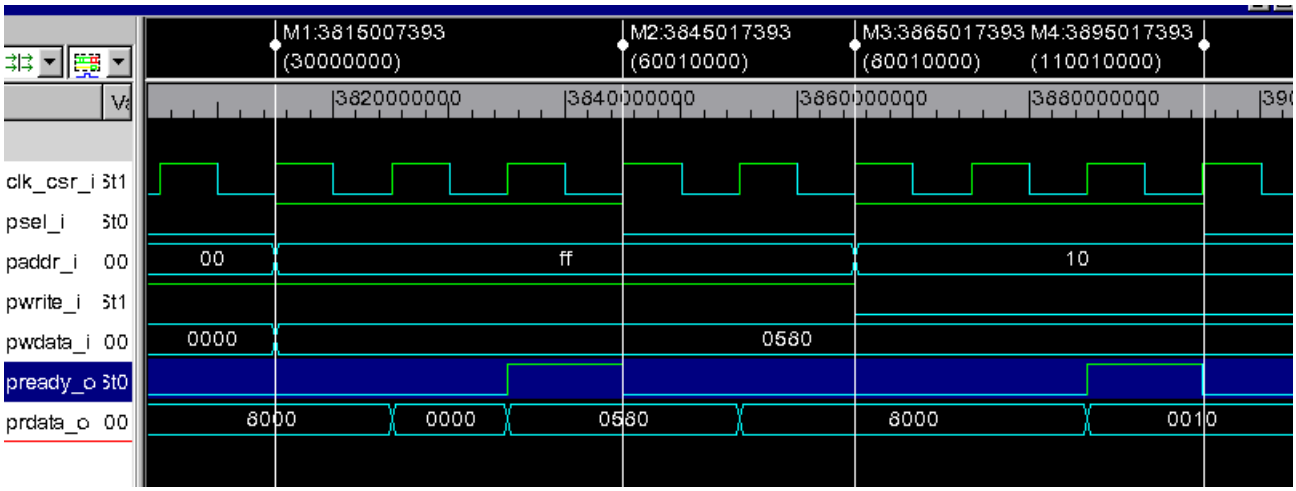
- **Data Phase**

  In this phase, the management provides the offset address of the 21-bit, 22-bit, or 23-bit address. The offset address corresponds to address bits[7/8/9:0] of the 21-bit, 22-bit, or 23-bit full address, and it is driven on the paddr_i[7/8/9:0] bus.

  DWC_xlgpcs derives the 21-bit, 22-bit, or 23-bit address from the base address of the Address phase and from the offset address of the Data phase, and performs the read or write access in the Data phase. You can perform multiple or consecutive Data phase transfers if the base address remains the same.

Figure 2-43 shows an example of indirect register Read access.

Figure 2-43 APB3 Interface Register Access (with Indirect Access)



In Figure 2-43, the marker M1 indicates the start of the Address phase. It is a write access at address 8'hFF with base address 13'h0580, which corresponds to base address[20:8] of the full address. The marker M2 indicates the end of the Address phase.

The marker M3 indicates the start of the Data phase. It is a read access to the offset address of 8'h10 which corresponds to offset address[7:0] of the full address. Therefore, the full register address is 21'h058010. The DWC_xlgpcs returns the read data (for this full address) with value 16'h0010. The marker M4 indicates the end of Data phase.

2.5.2 MDIO Interface

This topic contains the following sections:

- "Overview of MDIO Interface" on page 109
- "MDIO Register Access" on page 110
- "MDIO Timing Specification" on page 110
- "Clause 45 MDIO Frame Structure" on page 111
- "Signals Related to the MDIO Interface" on page 112

2.5.2.1 Overview of MDIO Interface

The IEEE Std 802.3 defines the MDIO Management interface, which serves to access the Management register of 802.3 compliant devices. This is a two-line interface including MDC (clock) and MDIO (bi-directional data). The standard defines two protocols that are associated with the same Management interface.

- Clause 22 defines access for the MII or GMII register sets.
- Clause 45 defines access for the MDIO Manageable Device (MMD) devices register sets.

Note

- DWC_xlgpcs supports only Clause 45 MDIO interfaces.
- You can select the MDIO interface only for Synopsys multi-protocol 12G, 16G, 25G, or 32G PHY configurations.

2.5.2.2 MDIO Register Access

The MDIO register access happens in the following phases:

- Address Write Phase: In this phase, the MAC provides the following:
  - Device address of 5-bits in the control field, indicating the MMD devices address.
  - Register address of 16-bits in the data field, indicating the Register implemented in an MMD.
- Data Access Phase: In this phase, the MAC accesses the data from the device.
  - Read Operation:
    - The MAC drives the device address in the Control field.
    - DWC_xlgpcs drives the data of the addressed resister in the Data field.
  - Write Operation:
    - The MAC drives the device address in the Control field.
    - The MAC drives the data of the addressed register in the Data field.

Note DWC_xlgpcs discards the first MDIO transaction after the power-on reset. However, this does not impact the functionality because, as per DWC_xlgpcs initialization sequence, after power-on-reset, the software must poll bit[15] of the `SR_XS_PCS_CTRL1/SR_MII_CTRL` register until it becomes 0, before proceeding to any other programming task. As a result, this behavior of MDIO is not visible to the software.

2.5.2.3 MDIO Timing Specification

DWC_xlgpcs assumes that the MDIO timing is compatible with the Clause 45 of the IEEE 802.3 standard. When DWC_xlgpcs is sampling the MDIO line, it assumes that the data is available 10 ns before the MDC positive edge (Setup time) and 10 ns after the MDC positive edge (Hold time) as described in Figure 2-44. Similarly, when the DWC_xlgpcs is driving the data on MDIO, it meets the hold time requirements as given in the Figure 2-45. Additionally, the frame structure and access methods are different. Figure 2-44 and Figure 2-45 show the timing relationship between MDIO and MDC.

Note If the MDIO timing is not compatible with the standard in your system, you need to change input or output delay constraints for the mdi_i and mdo_o signals during synthesis.

Figure 2-44 MDIO Driven by STA (Management)

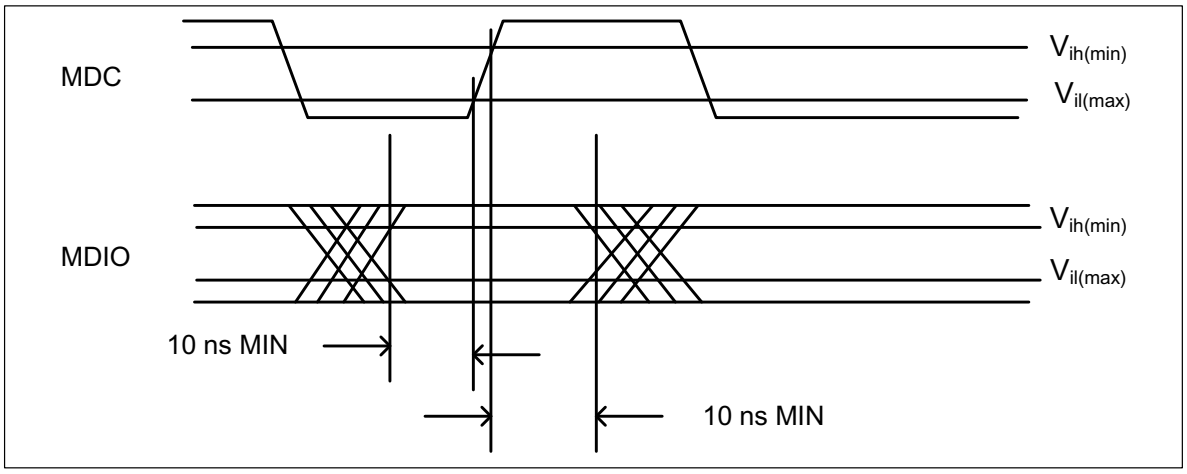
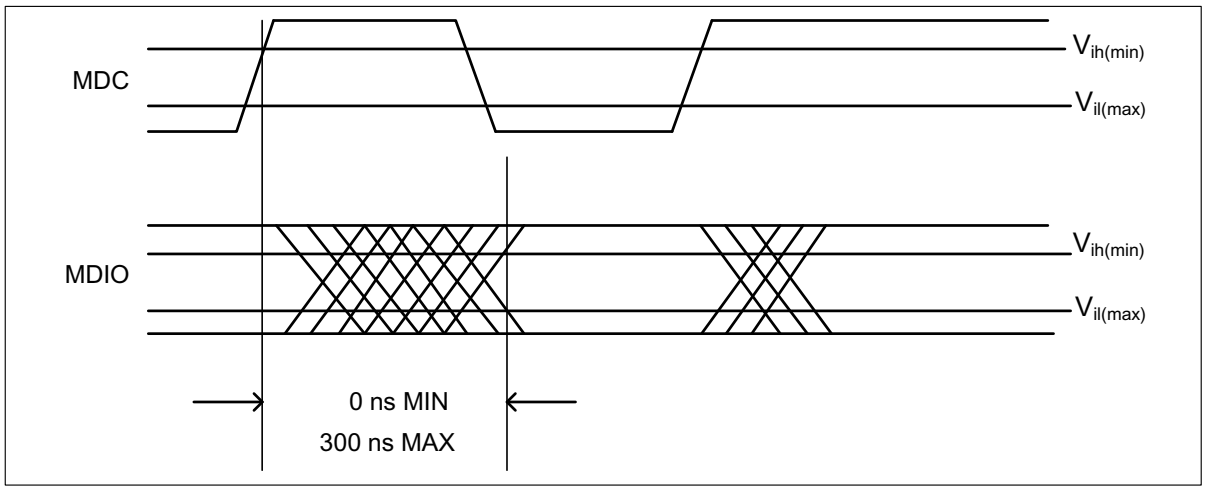


Figure 2-45 MDIO Driven by Device



2.5.2.4 Clause 45 MDIO Frame Structure

Table 2-18 describes the MDIO Frame structure.

Table 2-18 Clause 45 MDIO Frame Structure

	Management FrameFields							
	PREAMBLE	SOF	OPcode	Port Address	Device Address	Turnaround Time	Address/Data	IDLE
Address	1... 1	00	00	PPPPP	EEEEE	10	AAAAAAAAAAAAAAAA	Z
Write	1... 1	00	01	PPPPP	EEEEE	10	DDDDDDDDDDDDDDDD	Z
Read	1... 1	00	11	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z
Post-read increment address	1... 1	00	10	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z

Table 2-19 describes the MDIO Frame fields.

Note All fields are ordered with MSb first.

Table 2-19 MDIO Frame Field Descriptions

Field	Description
Idle	This is when MDIO Line is high impedance and no body is driving the MDIO line. The pull-up resistor keeps the MDIO line in high state.
Preamble	Consists of 32-bit contiguous ones.
SOF	Start of Frame delimiter. This has a pattern 00, indicating start of the frame.
OPcode	Operation Code. Indicates a read, write, or address phase. ■ 00: Address phase ■ 01: Data Write phase ■ 10: Data Read phase with automatic address increment ■ 11: Data Read phase
DEVAD	Devices address. Allows addressing a specific MMD such as PCS, XS, PMA, and auto-negotiation.
Port Address	PHY Address field. This 5-bit address corresponds to port_id_i[4:0]. If this field is not equal to port_id_i[4:0], the DWC_xlgpcs does not respond to any read operation and write operation is ignored.
Turnaround Time	This is 2-bit field. ■ Read operation: ❑ The first bit is not driven. ❑ The DWC_xlgpcs drives the next immediate bit with 0. ■ Write operation: ❑ The Management drives the first bit with 1. ❑ The Management drives the second bit with 0.
Address/Data	This field specifies the address or data. ■ OPcode field = 00 specifies the register address management wants to read or write. ■ Any other value specifies the register read or write contents. ❑ Read: The DWC_xlgpcs drives the data. ❑ Write: The Management drives the data.

2.5.2.5 Signals Related to the MDIO Interface

See the *"Signal Descriptions"* chapter.

2.6 Power Sequence and Reset Generation Block

The power sequence and reset generation block performs the following functions:

- Controls and tracks the power-up sequence of the Synopsys PHY.
- Generates the reset to various blocks of the DWC_xlgpcs.

2.6.1 Power Sequence

The power sequence and reset generation block controls the power-on and power-off sequence of the Synopsys multi-protocol 12G/16G/25G/32G PHY. The PCS logic starts functioning only when the power-up sequence state is in the normal state. This ensures stable Transmit clock propagation and proper data transmission. The DWC_xlgpcs supports the following power-off modes:

- **Power Save**: In this mode, PHY is put in a low-power state equivalent to P1 by asserting the xlgpcs_tx_lpd_o/xlgpcs_rx_lpd_o port.

  You can enable the power-save mode by programming bit 11 of any of the following registers:

  - `VR_PMA_DIG_CTRL1`
  - `VR_PCS_DIG_CTRL1`
  - `VR_AN_DIG_CTRL`

- **Power Down**: In this mode, PHY is put in a low power state equivalent to P2 by asserting xlgpcs_tx_reset_o/xlgpcs_rx_reset_o port. Powering up from this state requires asserting a reset.

  You can enable the power-down mode by programming bit 11 of any of the following registers:

  - `SR_PMA_CTRL1`
  - `SR_PCS_CTRL1`

You can program both modes by using the CSR registers.

2.6.2 Reset Generation

The reset generation block generates the reset for other blocks of the DWC_xlgpcs. The DWC_xlgpcs supports the following types of reset:

- Hardware Reset

  You can issue the hardware reset by asserting the pwr_on_rst_n signal. When this reset is asserted, all internal blocks of the DWC_xlgpcs are reset. In addition, the xlgpcs_reset_n_o signal is asserted to reset the PHY. The pwr_on_rst_n signal must be asserted long enough to propagate reset to all clock domains. Therefore, the minimum width must at least be equal to the clock period of the slowest clock in your configuration.

- Standard Soft Reset

  You can issue the standard soft reset by setting bit 15 of any of the following registers:

  - `SR_PMA_CTRL1`
  - `SR_PCS_CTRL1`
  - `SR_AN_CTRL`

For APB3 interface, the DWC_xlgpcs clears this bit after the following:

- **Non-Synopsys PHY configuration**: 32 clocks cycles of the clk_csr_i clock
- **Synopsys PHY configuration**: After PHY powers up

When this reset bit is programmed, all internal blocks of DWC_xlgpcs, except the Management Interface block, get reset. In addition, the xlgpcs_reset_n_o signal is asserted to reset the PHY.

- Vendor- Specific Soft Reset

  You can issue the vendor-specific soft reset by programming bit 15 of any of the following registers:

  - `VR_PMA_DIG_CTRL1`
  - `VR_PCS_DIG_CTRL1`
  - `VR_AN_DIG_CTRL`

  For MCI interface or APB3 interface, the DWC_xlgpcs clears this bit after the following:

  - **Non-Synopsys PHY configuration**: 32 clocks cycles of the clk_csr_i clock
  - **Synopsys PHY configuration**: After PHY powers up

  When this reset bit is programmed, all internal blocks of the DWC_xlgpcs are reset, except the Management Interface block and the CSR block. In addition, the xlgpcs_reset_n_o signal is asserted to reset the PHY.

- Power-Down mode

  You can issue the power-down mode by programming bit 11 of the any of the following registers:

  - `VR_PMA_DIG_CTRL1`
  - `VR_PCS_DIG_CTRL1`
  - `VR_AN_DIG_CTRL`

  When this bit is programmed, the host must clear this bit when it wants to power up DWC_xlgpcs. When this bit is programmed, all internal blocks of DWC_xlgpcs are reset, except the Management Interface block and the CSR block. The xlgpcs_reset_n_o output is not asserted during the power-down mode.

2.7 Control Register Port Controller

The Control Register (CR) Port Controller block implements the CR Port Interface protocol to access the internal register sets of the Synopsys PHY. This block interfaces with the CSR block of the DWC_xlgpcs through data, address, read or write, and miscellaneous control signals. Upon request from the CSR block, the CR block generates a read or write transaction on the asynchronous port of the Synopsys PHY.

This block is activated when the Station Management wants to access the internal registers set of the Synopsys PHY through the following registers:

- `VR_PMA_MP_25G_16G_CR_CTRL`
- `VR_PMA_MP_25G_16G_CR_ADDR`
- `VR_PMA_MP_25G_16G_CR_DATA`

The programming guidelines for CR port read and write operations are covered in "Switching to 5G Speed Mode" on page 1324.

The following list provides a brief overview of the read or write transfer protocol specific to Synopsys PHYs on the CR port:

Write protocol

1. Supply the register address on xgxs_cr_reg_address_o.
2. Supply the data on xgxs_cr_data_o.
3. Assert xgxs_cr_write_o (for one clock cycle) to initiate a write transfer to the PHY register.
4. Wait for xgxs_cr_ack_i to assert. This indicates the completion of write operation.

Read protocol

1. Supply the register address on xgxs_cr_reg_address_o.
2. Assert xgxs_cr_read_o (for one clock cycle) to initiate a read transfer to the PHY register.
3. Sample the data on 'xgxs_cr_data_i' when xgxs_cr_ack_i is asserted.



The format of the CR data for all registers is compatible with the Synopsys PHY specification.

2.8 Clause 73 Auto-Negotiation Block

The Clause 73 auto-negotiation block is an optional block. This block is available when you select the coreConsultant parameter 'Add CL73 Autoneg Support' during RTL configuration. This feature is used mainly for applications with Backplane and Copper cable assembly.

The auto-negotiation block implements the following functions:

- Recognizing the Differential Manchester Encoded (DME) Base page and Next pages on the Receive path and generating the DME pages on the Transmit path during the auto-negotiation process.

- Connecting and disconnecting the Transmit and Receive lanes (Lane 1, Lane 2, and Lane 3) based on outcome of auto-negotiation process.
- Implementing Parallel detection to detect legacy 10GBASE-R, 25GBASE-R, 40GBASE-R,50GBASE-R and 100GBASE-R Backplane devices (KR4, CR4, KR2, CR2, KR or CR) without Clause 73 auto-negotiation.

Auto-negotiation allows devices to advertise and share information for detecting capabilities and status of remote devices such as:

- Speed
- Modes
- Fault Signaling
- Legacy device detection
- PAUSE abilities
- Duplex information
- Next Page ability
- Technology ability
- Forward Error Correction (FEC)

2.8.1 Auto-Negotiation Format: Differential Manchester Encoding

A common auto-negotiation process is independent of any specific technology (KR4, CR4, KR2, CR2, KR or CR). The auto-negotiation information is exchanged through serial DME. In 10G, 40G speed modes the DME pages are exchanged at 312.5 MHz. In 25G, 50G and 100G, the DME pages are exchanged at 390.625 MHz.

The serial data, called Link Code Words, is sent through encoded pages called Differential Manchester Encoding (DME) pages

2.8.1.1 DME Pseudo-Random Number Generator

This block generates the pseudo-random numbers. For generating DME page random numbers, the following polynomial is used:

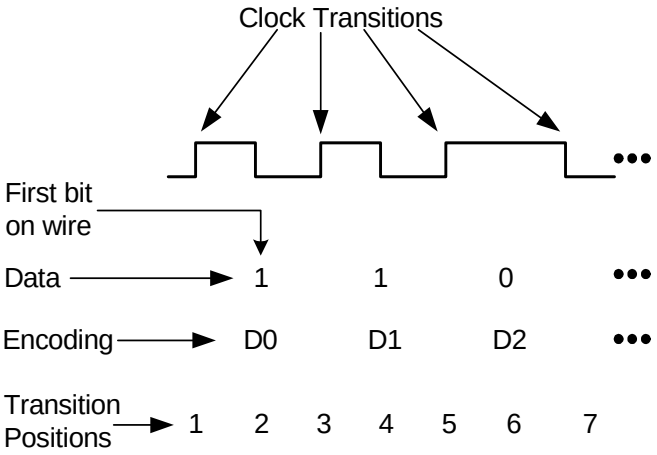
```
X7+X3+1
```

The pseudo-random number is used on the DME page as a last data bit. The same polynomial is used in the arbiter to generate Transmit Nonce field of the Base Page.

2.8.1.2 DME Page Timing Specification

Figure 2-46 shows the example encoding.

Figure 2-46 Data Bit Encoding with DME Pages



2.8.1.3 DME Page Transition Timing

Figure 2-47 and Figure 2-48 show the DME page transition timing.

Figure 2-47 DME Page Timing

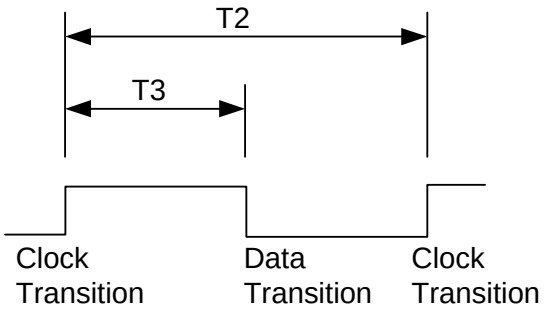


Figure 2-48 DME Page Transition

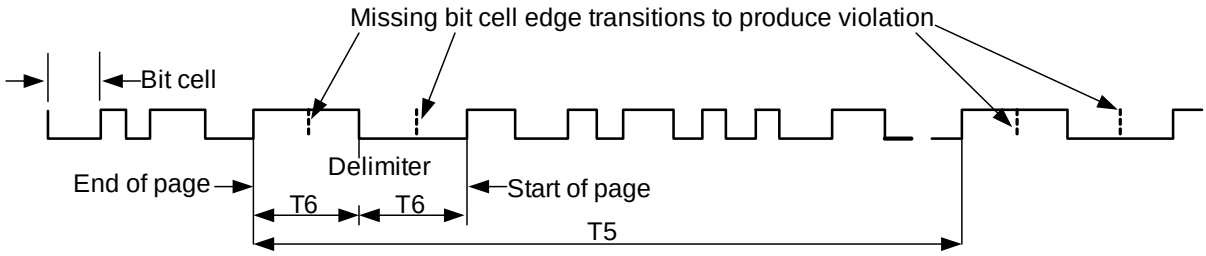


Table 2-20 describes the DME page transition timing.

Table 2-20 DME Page Transmit Timing as Specified by IEEE Std 802.3, Clause 73

#	Parameter	Min	Typical[1]	Max	Unit
T1	Transition position spacing	3.2 +/- 0.01%	3.2	3.2 +/- 0.01%	ns

Table 2-20 DME Page Transmit Timing as Specified by IEEE Std 802.3, Clause 73

#	Parameter	Min	Typical[1]	Max	Unit
T2	Clock Transition to Clock Transition	6.2	6.4	6.6	ns
T3	Clock Transition to Data Transition	3.0	3.2	3.4	ns
T4	Transitions on a DME page	51	-	100	-
T5	DME Page Width	338.8	339.2	339.6	ns
T6	DME MVD Width	12.6	12.8	13.0	ns

1. The DWC_xlgpcs Transmit supports timing as defined the Typical column.

2.8.1.4 DME Page Structure

A DME page contains 106 evenly-spaced transition positions, contained in 53 bits of 3.2 ns bit time. These 53 bits include the following:

- **Manchester's Violation Delimiter (MVD):** The first eight transition positions (4 bits) contain MVD, which marks the beginning of the page. The Manchester violation contains the transition at position 1 and 5, and no transition in the remaining transition position. Two consecutive MVDs are used for page synchronization.
- **The Auto-Negotiation page:** The auto-negotiation page is 96 transition positions (48 bits) corresponding to the Link Code word (described in section 2.8.2) encoded in the DME format. The following list describes the encoding rules:
  - Odd numbered transition positions contain clock information. Odd-numbered transition positions always contain a transition.
  - Even-numbered transition positions contain data information with the following transition rules:
    - Transitions present represent the data corresponding to logical 1.
    - Transitions absent represent the data corresponding to logical 0.
- **A pseudo-random bit**: The 53rd bit of the DME page contains a pseudo-random data bit.

2.8.2 Base Link Code Word Format

Figure 2-49 and Table 2-21 define the format of the Link Code Word used with AN Advertisement registers 1-3.

Figure 2-49 Base Page: Link Code Word Format

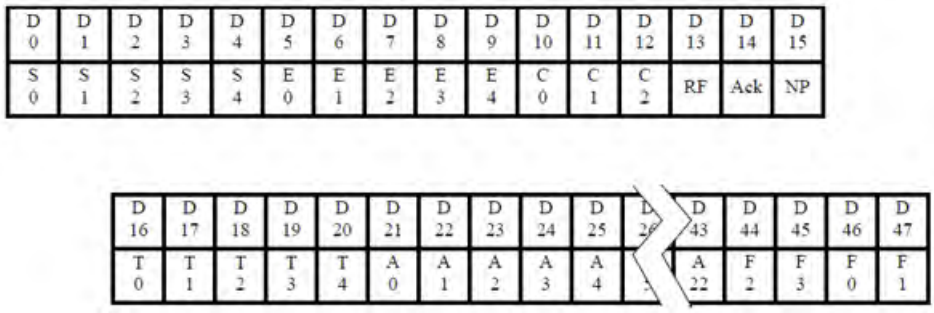


Table 2-21 Base Link Code Word Format Data

Item	Definition
NP	Next page to be sent or received
ACK	Code received successfully from the link partner
RF	Remote fault
C0	SYNC_PAUSE_ABILITY
C1	ASYNC_PAUSE_ABILITY
C2	Reserved
E[4:0] Echoed Nonce Field	Received Nonce field containing Transmitted Nonce field of remote device
S[4:0] Selector Field	IEEE standard with which this device is compliant
T[4:0]	Transmitted Nonce field.

Table 2-21 Base Link Code Word Format Data (Continued)

Item	Definition
A[14:0]	Technology Ability field indicates the supported technologies: ■ A0: When this bit is programmed to 1, the 1000BASE-KX technology is supported. Otherwise, the 1000BASE-KX technology is not supported. ■ A1: When this bit is programmed to 1, the 10GBASE-KX4 technology is supported. Otherwise, the 10GBASE-KX4 technology is not supported. ■ A2: When this bit is programmed to 1, the 10GBASE-KR technology is supported. Otherwise, the 10GBASE-KR technology is not supported. ■ A3: When this bit is programmed to 1, the 40GBASE-KR technology is supported. Otherwise, the 40GBASE-KR technology is not supported. ■ A4: When this bit is programmed to 1, the 40GBASE-CR technology is supported. Otherwise, the 40GBASE-CR technology is not supported. ■ A5: 100GBASE-CR 10 support programmed to 0, as this configuration is not supported. ■ A6: 100GBASE-KP4 support. Programmed to 0, as this configuration is not supported. ■ A7: 100GBASE-CR4 support. When this bit is programmed to 1, the 100GBASE-KR4 technology is supported. Otherwise, the 100GBASE-KR4 technology is not supported. ■ A8: 100GBASE-KR4 support. When this bit is programmed to 1, the 100GBASE-KR4 technology is supported. Otherwise, the 100GBASE-KR4 technology is not supported. ■ A9: 25GBASE-KR-S or 25GBASE-CR-S support refers to 25GBASE-CR without support for the RS-FEC sublayer specified in Clause 108. When this bit is programmed to 1, 25GBASE-CR/KR without the RS FEC as per Clause 108 is supported. Otherwise, 25GBASE-CR/KR with RS FEC as per Clause 108 is supported. ■ A10: 25GBASE-KR or 25GBASE-CR support. When this bit is programmed to 1, the 25GBASE-KR or 25GBASE-CR technology is supported. Otherwise, the 25GBASE-KR or 25GBASE-CR technology is not supported. ■ A11: When this bit is programmed to 1, 2.5GBASE-KX is supported. Otherwise it is not supported. ■ A12: When this bit is programmed to 1, 5GBASE-KR is supported. Otherwise it is not supported. ■ A13: When this bit is programmed to 1, 50GBASE-KR or 50GBASE-CR is supported. Otherwise it is not supported. ■ A14: When this bit is programmed to 1, 100GBASE-KR2 or 100GBASE-CR2 is supported. Otherwise it is not supported. ■ A15: Reserved ■ A16: When this bit is programmed to 1, 100GBASE-KR1 or 100GBASE-CR1 is supported. Otherwise it is not supported.
A[24:17]	Reserved.

Table 2-21 Base Link Code Word Format Data (Continued)

Item	Definition
F0	BASE-F FEC ability. When this bit is programmed, it indicates that BASE-R FEC is supported. When this bit is reset, it indicates no support for BASE-R FEC.
F1	BASE-R FEC Request. When this bit is programmed, it indicates that BASE-R FEC is requested. When this bit is reset, it indicates no request for BASE-R FEC.
F2	RS-FEC Request in 25G mode of operation. When this bit is programmed, it indicates that RS-FEC support is requested. When this bit is reset, it i indicates no request for RS-FEC.
F3	BASE-R FEC Request in 25G mode of operation. When this bit is programmed, it indicates that BASE-R FEC is requested. When this bit is reset, it indicates no request for BASE-R FEC operation.
F4	100G RS-FEC-Interleaved mode request. When programmed, this bit indicates that 100G RS-FEC codeword interleaved mode of operation is requested. When reset, it indicates no request for 100G RS-FEC codeword interleaved mode operation.

2.8.3 Next Page Link Code Word Format

Figures 2-50 and 2-51, along with Table 2-22, define the format of the Next Page Link Code Word used with SR_AN_XNP_TX1–3.

Figure 2-50 Message Next Page

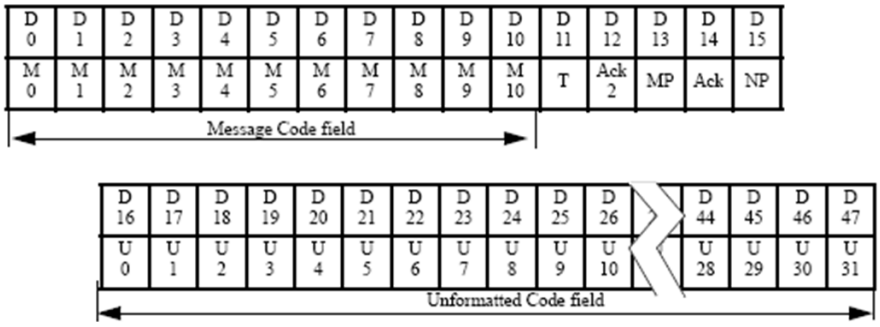


Figure 2-51 Unformatted Next Page

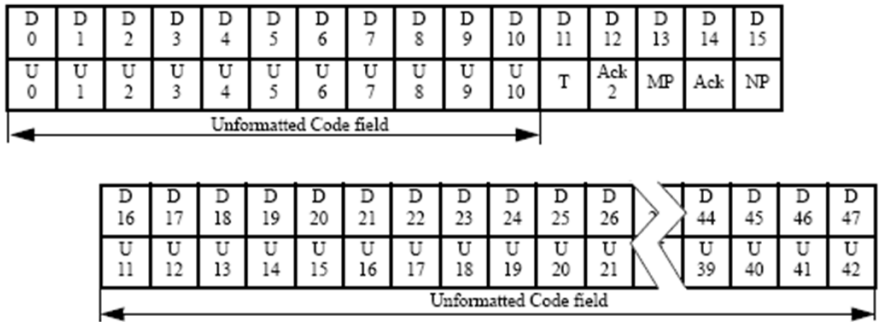


Table 2-22 Next Page Link Code Word Format Data

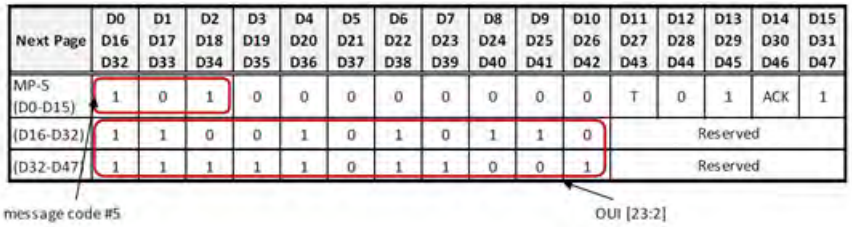
Item	Definition
NP	Next page to be sent or received
ACK	Code received successfully from link partner
MP	1: Formatted Message Page 0: Unformatted Message Page
ACK2	Code received successfully and act upon the information defined on the received message
T	Toggle bit
M[10:0]	Message field as defined in the Annex 73A of IEEE Std 802.3ap
U[31:0]	Unformatted code field on the formatted message page
U[42:0]	Unformatted code received on the Unformatted message page

2.8.3.1 Auto-Negotiation in 25G/50G Consortium Mode

Under Consortium 25 Gb/s and 50 Gb/s operation, the Auto-negotiation base pages are exchanged between the two ends of the backplane channel, with the exchange taking place on physical lane 0. After the exchange of the base page, the link partners exchange an OUI tagged formatted Next Page (using message code #5) and then the link partners exchange an OUI tagged unformatted Next Page with an extended technology abilities field.

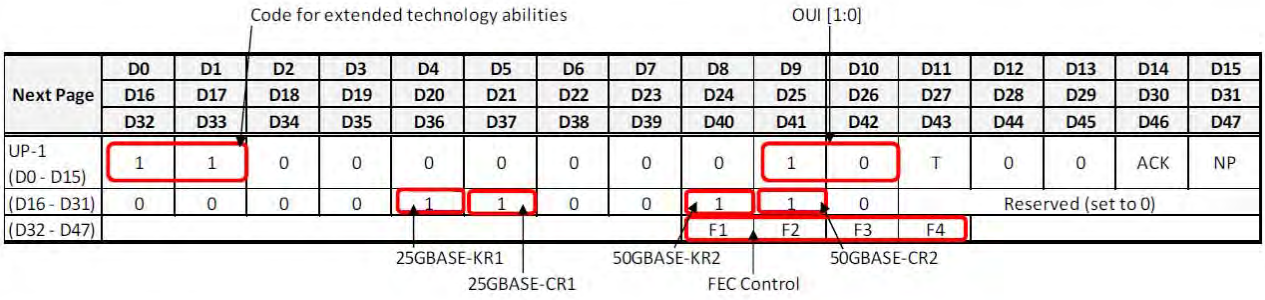
The Organization Unique Identifier (OUI) tagged formatted Next Page, with the OUI set to the 25G Ethernet Consortium CID of 0x6A737D(h) and a message code of 5 as shown in Figure 2-52.

Figure 2-52 Auto-Negotiation Next Page - OUI Extended



The OUI tagged unformatted Next Page, with the OUI set to the 25G Ethernet Consortium CID of 0x6A737D(h), a code for extended technology abilities, and the appropriate code bits for 25 Gb per second and 50 Gb per second operation, as well as two FEC control bits, is shown Figure 2-54.

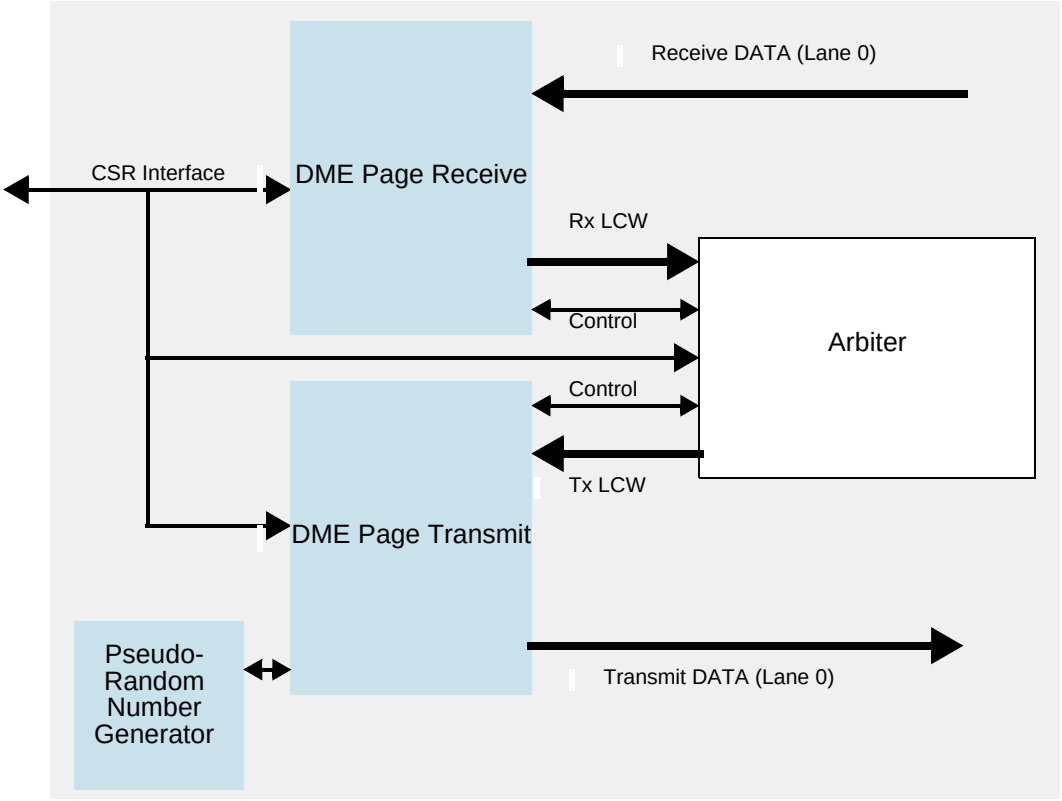
Figure 2-53 Auto-Negotiation OUI Extended Next Page Codes



2.8.4 Auto-Negotiation Block: Functional Components

Figure 2-54 displays the components of the auto-negotiation block.

Figure 2-54 Backplane-PCS Auto-Negotiation Block Detail



2.8.4.1 Arbiter

The arbiter component controls the DME page transmitter and DME page receiver. The arbiter implements the following features:

- Transmit lane disabling or enabling
- Receive lane disabling or enabling and scan for carrier operation
- Parallel detection
- Next Page transmission or reception
- Link code word decoding

The arbiter contains various timers as defined in the IEEE Std 802.3, Clause 73. This block is controlled by the Management requests to start, abort, and restart auto-negotiation. The arbiter also implements the state machine as defined in the IEEE Std 802.3, Clause 73, Figure 73-11.

Parallel Detection in the Arbiter

The parallel detection is done during the Clause 73 auto-negotiation to detect the legacy devices which do not support Clause 73 auto-negotiation or the devices which disable Clause 73 auto-negotiation. When the auto-negotiation is going on, the received data from the PHY is given to both Clause 73 auto-negotiation and KR4, CR4, KR2, CR2, KR, CR receive logic. While expecting the Clause 73 compliance DME page, if the arbiter state machine sees the Link Up status from the receive logic, it means one of the following:

- If the current speed mode is 10G-KR, the remote link is 10GBASE-R.

- If the current speed mode is 40G-KR4/CR4, the remote link is 40GBASE-R.
- If the current speed mode is 25G, the remote link is 25GBASE-R.
- If the current speed mode is 50G, the remote link is 50GBASE-R.
- If the current speed mode is 100G, the remote link is 100GBASE-R.



The DWC_xlgpcs assumes that the PHY is configured to operate in the same speed mode as the DWC_xlgpcs. Therefore, the parallel detection is done only in the active speed mode.

If DWC_xlgpcs is configured to have multiple speed modes and the auto-negotiation fails (AN Incompatible Link received) in the current speed mode, the management must switch to some other supported speed mode to detect the compatible device.

Mode	Compatible Legacy Speed Mode Device Detectable Through Parallel Detection
25GBASE-CR/KR	25GBASE-R
50GBASE-CR2/KR2/CR/KR	50GBASE-R
100G BASE-CR4/KR4/CR2/KR2	100GBASE-R
40GBASE-CR4/KR4	40GBASE-R
10GBASE-KR (KR)	10GBASE-R

2.8.4.2 DME Page Transmitter

When requested by the arbiter, the DME page transmitter implements the logic related to the DME page encoding of the Link Code Word (LCW) to be transmitted. The logic implemented for DME page encoding varies based on the current speed mode. This is because although the DME timings are defined for a serial interface which is common across all speed modes, the bit time for each speed mode (10G, 40G, 25G, 50G, and 100G modes) is different.

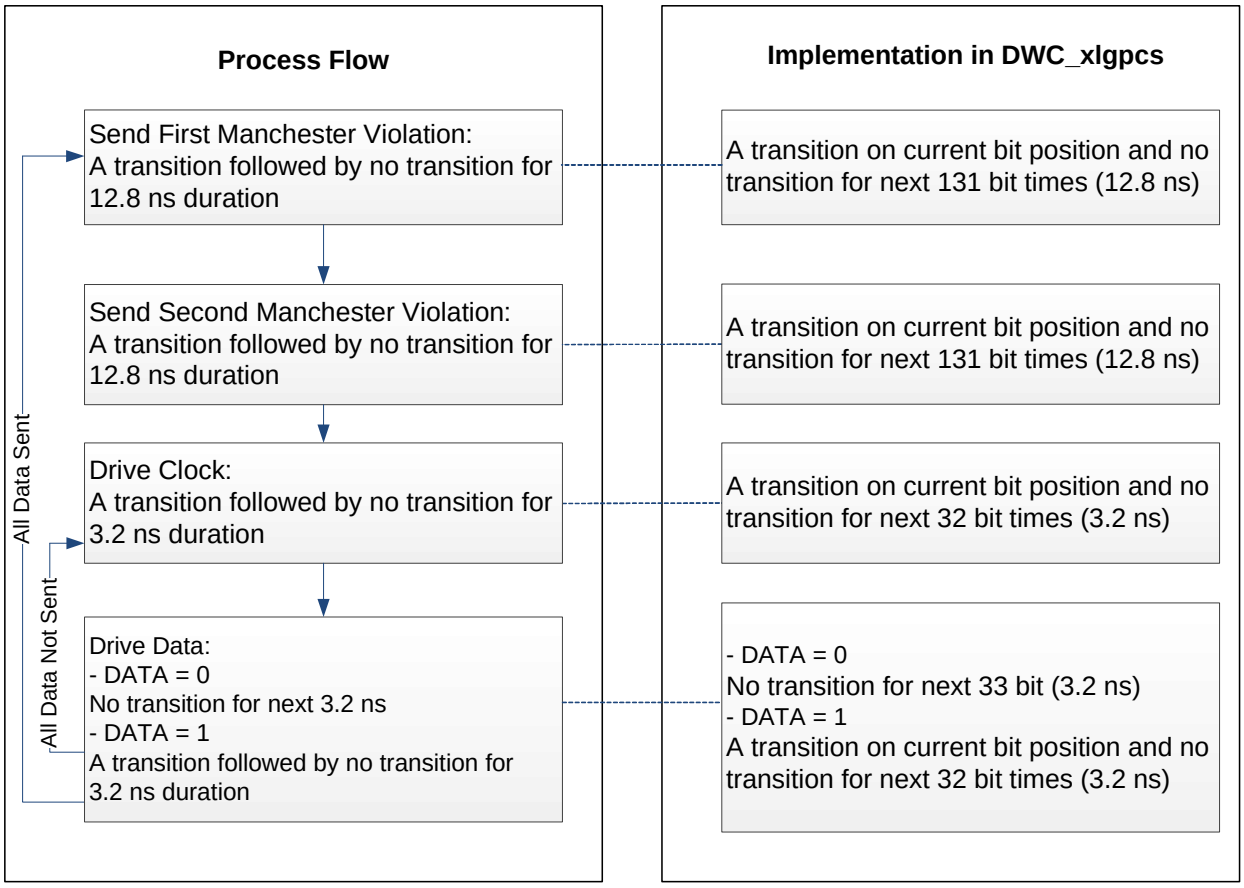
The DME page encoding has the following stages:

1. MV Delimiter: Two transitions separated by T6 (12.8 ns) time period
2. Clock: Occurs at every T2 (6.4 ns) time period
3. Data: Occurs at every T3 (3.2 ns) time period from the previous clock transition

2.8.4.2.1 DME Page Encoding for 10G BASE-KR/CR or 40G BASE-KR4/CR4 Mode

In the 10G BASE-KR/CR or 40G BASE-KR4/CR4 mode, 1 bit time corresponds to 0.09696 ns. Figure 2-55 shows the process for DME page encoding for10G BASE-KR/CR or 40G BASE-KR4/CR4 mode and also how this process is implemented in DWC_xlgpcs.

Figure 2-55 DME Page Encoding Process for 10G/40G Speed Modes

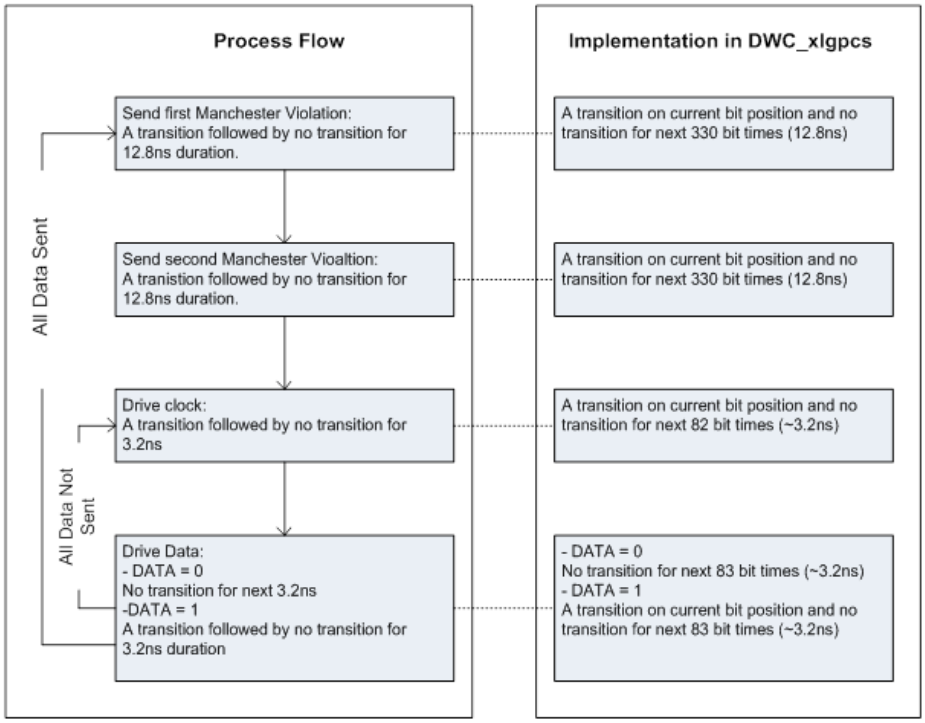


2.8.4.2.2 DME Page Encoding for 25G BASE-KR/CR, 50G BASE-KR2/CR2, and 100G BASE-KR4/CR4

In the 10G BASE-KR/CR or 40G BASE-KR4/CR4, 1 bit time corresponds to 38.78 ps.

Figure 2-56 shows the process for DME page encoding for 25G/50G/100G speed modes and also how this process is implemented in DWC_xlgpcs.

Figure 2-56 DME Page Encoding Process for 25G/50G/100G Speed Modes



Note DME page transmission/reception is not supported, when DWC_xlgpcs is programmed to 50GBASE-KR/CR or 100GBASE-KR2/CR2 speeds. For configurations supporting these speeds, auto-negotiation is initiated after switching DWC_xlgpcs to 10G speed mode.

2.8.4.3 DME Page Receiver

The DME page receiver implements the logic related to the DME page decoding of the Link Code Word. The logic implemented for DME page encoding varies based on the current speed mode. This is because although the DME timings are defined for a serial interface which is common across all speed modes, the bit time for each speed mode (10G, 25G, 40G, 50G or 100G) is different.

The DME page decoding timing is different from the encoding timing. The decoding is implemented based on the timing range as described in Table 2-23.

Note In KR speed mode, the DWC_xlgpcs implements the DME page decoding as per KR with Reduced Granularity timing (described in the last column in Table 2-23).

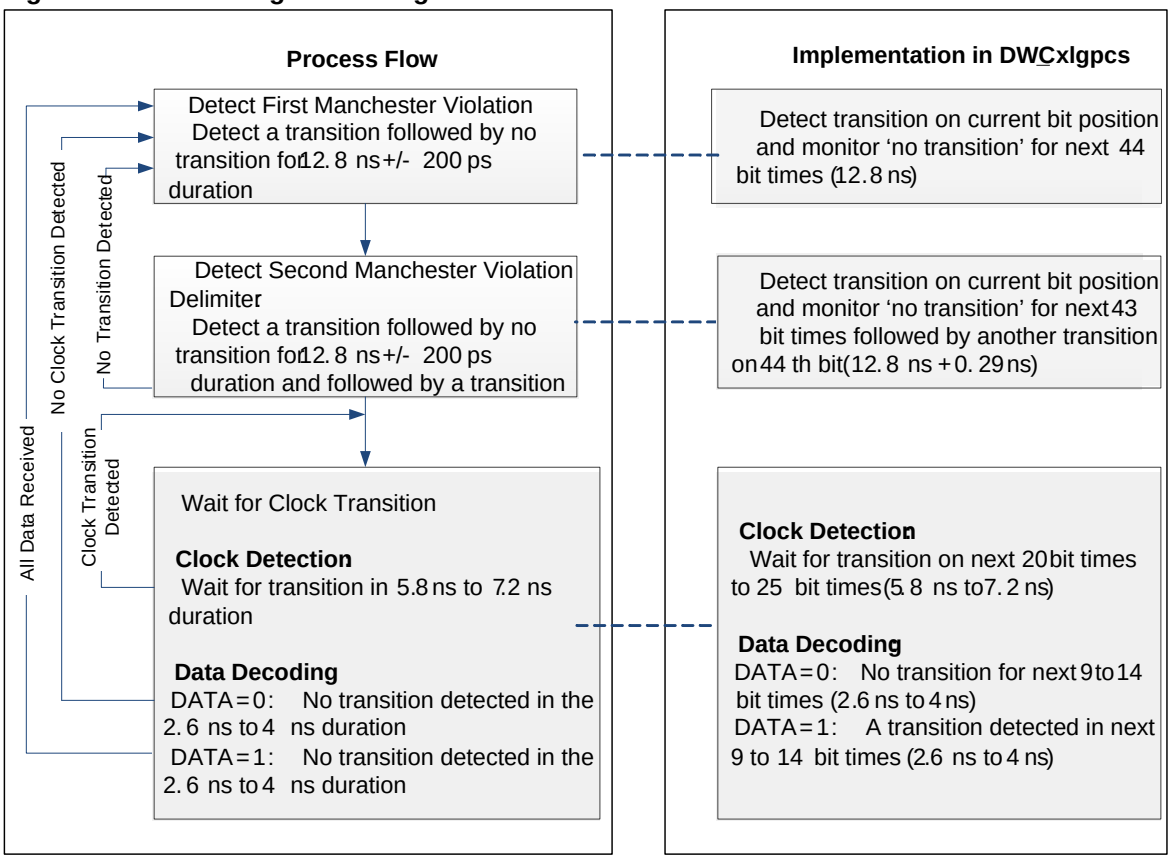
Table 2-23 DME Page Decoding Timing

Parameter	IEEE Value Range	10G BASE-KR/CR or 40G BASE-KR4/CR4 (0.096 ns Bit Period)	10G BASE-KR/CR or 40G BASE-KR4/CR4 (Reduced Granularity) (Bit Period of 0.2909 ns after merging 3 Bits of 10G/40G Speed)	25G/50G/100G Mode (0.038ns Bit Period on 66-Bit, 390.625 MHz
Clock detect min Timer	4.8 ns to 6.2 ns	Value = 5.8 ns Number of bits = 60	Value = 5.8 ns Number of bits = 20	Value = 5.6ns Number of bits = 144
Clock detect max Timer	6.6 ns to 8.0 ns	Value = 7.2 ns Number of bits = 75	Value = 7.2 ns Number of bits = 25	Value = 7.2ns Number of bits = 186
Data detect Min Timer	1.6ns to 3.0 ns	Value = 2.424 ns Number of bits = 25	Value = 2.6 ns Number of bits = 9	Value = 2.4ns Number of bits = 61
Data detect Max Timer	3.4ns to 4.8 ns	Value = 4.0 ns Number of bits = 42	Value = 4.0 ns Number of bits = 14	Value = 4.0ns Number of bits = 103
Page Test Min Timer	305 ns to 330 ns	Value = 305 ns Number of bits = 3196	Value = 305 ns Number of bits = 1065	Value = 305ns Number of bits = 7863
Page Test Max Timer	350 ns to 375 ns	Value = 370ns Number of bits = 3814	Value = 370 ns Number of bits = 1272	Value = 375ns Number of bits = 9667

2.8.4.3.1 DME Page Decoding for 10G/40G Mode

In the 10G or 40G mode, 1 bit time corresponds to 0.09696 ns. However, in DWC_xlgpcs, the implementation is done on reduced granularity by combining 3 bits of 10G or 40G mode speed into 1 bit. Therefore, the granularity is 0.29 ns. Figure 2-57 shows the process for DME page decoding for the 10G or 40G mode and also how this process is implemented in the DWC_xlgpcs. The +200 ps or – 200 ps resolution is implemented by considering 0.09696 ns bit granularity.

Figure 2-57 DME Page Decoding Process for 10G BASE-KR/CR or 40G BASE-KR4/CR4 Mode

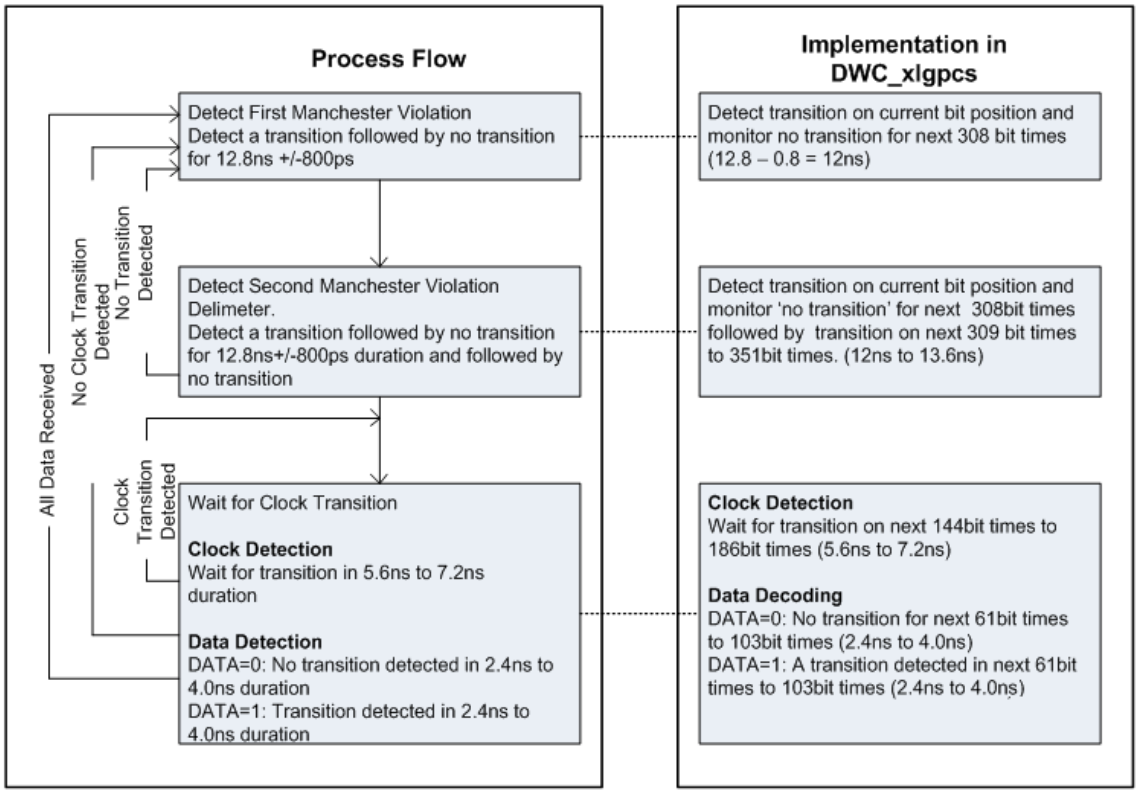


2.8.4.3.2 DME Page Decoding for 25G/50G/100G Modes

In the 25G BASE-KR/CR or 50G BASE-KR2/CR2, and 100G BASE-KR4/CR4 modes, 1 bit time corresponds to 0.003878 ns. Figure 2-58 shows the process for DME page decoding for the 25G/50G/100G modes and also how this process is implemented in the DWC_xlgpcs.

The +200 ps or – 200 ps resolution is implemented by considering 38.7878 ps bit granularity.

Figure 2-58 DMA Page Decoding for 25G/50G/100G Modes



2.8.5 Auto-Negotiation CSR Interface

The auto-negotiation CSR interface block interfaces with the CSR block to send or receive various types of information, such as the following:

- Enabling or disabling auto-negotiation
- Auto-negotiation status (PASS or FAIL)
- Lane enabling or disabling
- DME page Error indication
- Link code word to be transmitted or received
- Parallel detection Link Status
- Priority Resolution among 10GBASE-KR, 40GBASE-CR4/KR4, 25GBASE-CR/KR, 25GBASE-CR/KR-S, 50GBASE-CR2/KR2/CR/KR, and 100GBASE-CR4/KR4/CR2/KR2/KR1/CR1
- Test and Debug support
- Programmable auto-negotiation timers

The Clause 73 auto-negotiation specific registers are explained in the *"Register descriptions"* chapter.

2.9 Link KR Training

The Clause 72/92/93 KR training is an optional feature in the DWC_xlgpcs. You can enable the link training feature by selecting the Add Link Training Support option while configuring the DWC_xlgpcs. The IEEE Std 802.3, Clause 72/92/93 defines a start-up protocol between the local device and its link partner to facilitate the timing recovery and equalization tuning. It provides a mechanism through which a Local Receiver can tune the filter coefficients of the Remote Transmitter to optimize the signal quality over Backplane Interconnect. This is achieved by exchanging specialized frames called Training frames between the local device and its link partner.

The start-up protocol is initiated after exchanging all pages of Clause 73 auto-negotiation. The start-up protocol must be enabled only when the auto-negotiation priority resolution indicates that both Link partners are of one of the following types:

- KR or KR4 type
- CR or CR4 type

A training frame consists of a Preamble, a Control field, and a Training field as shown in Figure 2-59 on page 132. The Control field is encoded using Differential Manchester Encoding (DME) at lower speed, that is, one-fourth of 10.3125 Gbps (in 10G and 40G modes) and one-fourth of 25.78125 Gbps (in 25G, 50G, and 100G modes). For information about the DME encoding, see "Auto-Negotiation Format: Differential Manchester Encoding" on page 116. The Training field has the PRBS11 random sequence using standard PRBS11 sequence generator at 10.3125 Gbps (in 10G/40G modes) or at 25.78125 Gbps (in 25G/50G/100G modes). With this scheme, the Control field is always exchanged successfully without any need for equalization because it is transferred at a lower baud rate.

The Control field of a training frame has requests and responses. During the start-up protocol, the local device and the remote device simultaneously send or receive these requests or responses, respectively. The following list describes this:

- The local device sends requests to tune the Transmit filter coefficients of the remote device.
- The remote device sends requests to tune the Transmit filter coefficients of the local device.
- The local device receives status for requests given to the remote device.
- The remote device receives status for requests given to the local device.

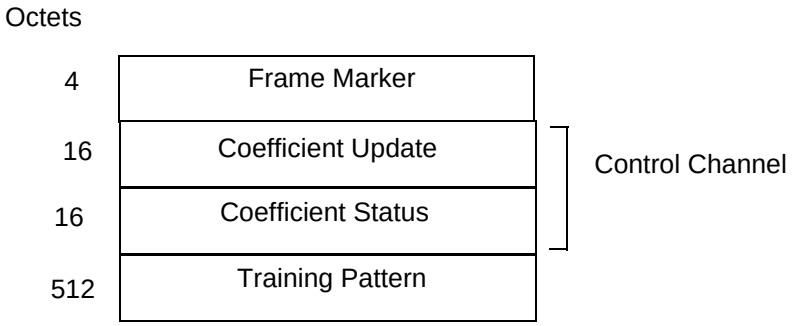
Note The information in the following subsections is provided from the perspective of a local device. This information is also applicable to a remote device.

Note For configurations with 50GBASE-KR/CR or 100GBASE-KR2/CR2/KR1/CR1 speed mode support, PHY must perform link-training. Link-training is not supported inside DWC_xlgpcs for such configurations.

2.9.1 Training Frame Structure

Figure 2-59 shows the frame structure specified in IEEE Std 802.3, Clause 72/92/93 based training.

Figure 2-59 Clause 72/92/93 KR Training Frame Structure



2.9.1.1 Frame Marker

The frame marker acts as the training frame delimiter. It has a fixed 32-bit pattern of 32'hFFFF0000. This 32-bit pattern does not appear in any other field (control channel or the training pattern).

2.9.1.2 Coefficient Update and Status (Control Channel)

The Coefficient Update and Status fields (control channel) are encoded with slower data rate by using the Differential Manchester Encoding (DME). Each DME Cell is of eight Unit Intervals (UI), that is, 8-bit duration.

- For 25G,50G and 100G modes, nominal value of UI corresponds to 38.787879 ps
- For 10G,40G modes of operation, nominal value of UI corresponds to 96.9696 ps

The middle bit (4th bit) indicates whether data is 1 (when there is an additional transition) or 0 (when there is no transition).

On a transmitted training frame, the Coefficient Update field has the requests from the local device to tune the remote transmitter. The Coefficient Status field has the status of the requests received from the remote device. On a received training frame, the Coefficient Update field has the requests from the remote device to tune the local transmitter. The Coefficient Status field has the status of the requests transmitted to remote device.

For detailed information about these fields, see "Coefficient Update and Coefficient Status Fields during Transmit and Receive Link Equalization" on page 137.

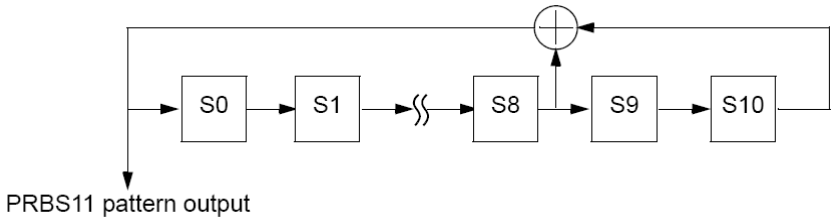
2.9.2 Training Pattern

In 10G/40G speed mode, the training pattern is 512 bytes of pseudo-random bit sequence generated through PRBS11 generator with initial random seed. It is not encoded by using the DME format. The following is the PRBS11 generator polynomial:

$G(x) = 1 + X^9 + X^{11}$

The PRBS11 pattern is sent or received at 10.3125 Gbps rate. During reception, this pattern can used to analyze the Receive link reliability or quality.

Figure 2-60 PRBS11 Pattern Output



In 25G speed mode, a programmable option to select either Clause 72 or Clause 92/93 based link training is provided. Bit[2] of the VR_PCS_DIG_CTRL3 register provides control to select between Clause-72- and Clause 92/93-based link training. When bit -2 of the CL72_EN control register field (BIT) is programmed, it enables Clause-72-based training; when it is reset, Clause 92/93 based link training is enabled.

By Default Clause-92/Clause-93-based training is enabled. When Clause 92/93 based link training is enabled, the training pattern for each lane consists of 4094 bits from the output of a pseudo-random bit sequence (PRBS) generator followed by two zeros. The PRBS generator for each lane implements each of the four generator polynomials (polynomial_n where n goes from 0 to 3) given in Table 3-24. The polynomial selection is done via the SR_PMA_LNK_TRN_PAT_{Lane i} registers (where 'i' is the lane number).

By default Physical lane-i uses Polynomial_i (where i is the physical lane number). At the start of the training pattern, the state of the generator is set to the value in seed_i (where i is the lane number), with the default values given in . The first 32 bits of the training pattern for each polynomial is also provided in .

Figure 2-61 PRBS11 Generator for Polynomial_0

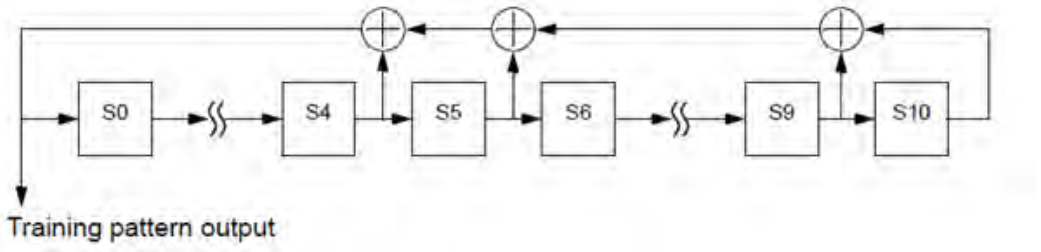


Table 2-24 PRBS Parameters for Each Physical Lane

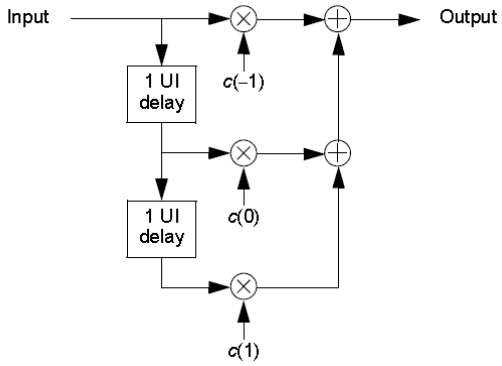
n	Polynomial_n, G(x)	Default Seed Bits, S0 is the Left Most Bit	Initial Output[a]
0	$1 + x^5 + x^6 + x^{10} + x^{11}$	10101111110	fbf1cb3e
1	$1 + x^5 + x^6 + x^9 + x^{11}$	11001000101	fbb1e665
2	$1 + x^4 + x^6 + x^8 + x^{11}$	11100101101	f3fdae46
3	$1 + x^4 + x^6 + x^7 + x^{11}$	11110110110	f2ffa46b

[a]The first 32 bits of the training pattern are presented in a hexadecimal representation where the hexadecimal symbols are transmitted from left to right and the most significant bit of each hexadecimal symbol is transmitted first.

2.9.3 Transmit Filter Structure

Figure 2-62 shows the filter structure that a transmitter can have at the serial outputs as defined in IEEE Std 802.3, Clause 72. There are three Transmit filter coefficients c(-1), c(0), and c(1). The equalization is achieved by tuning these Transmit filter coefficients. The tuning information and the corresponding status are exchanged through control channel of the training frame.

Figure 2-62 Transmit Filter Structure



2.9.4 Start-Up Protocol Flow

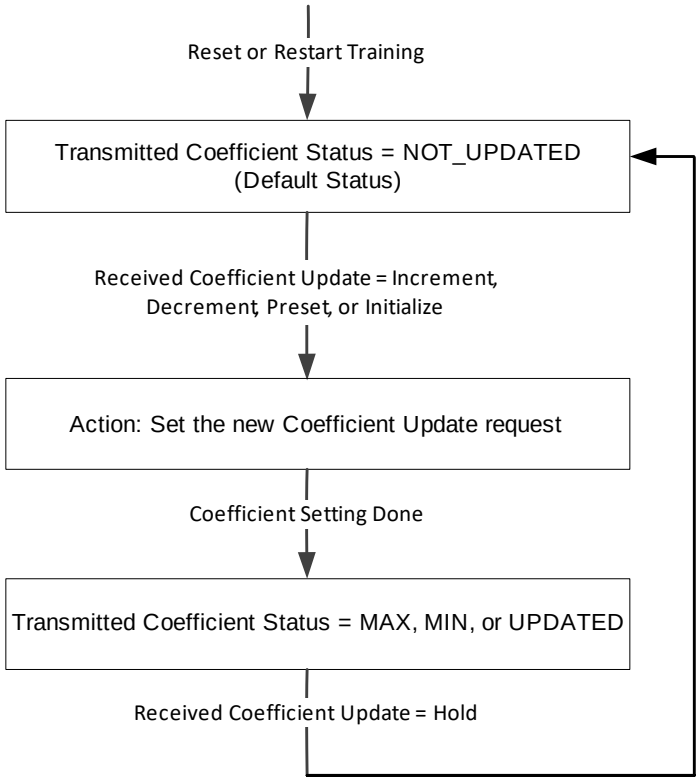
This section describes how the Transmit and Receive link equalization is achieved.

Transmit Link Equalization Flow

The Transmit link equalization is achieved by tuning the local transmitter as per the requests in the Control field of the received training frame. During start-up protocol, the remote device first initializes the local Transmit filter coefficients and then issues the increment or decrement request to tune these Transmit filter coefficients.

Figure 2-63 shows the Transmit link equalization flow. This flow is applicable for filter coefficients c(1), c(0), and c(-1). All three filter coefficients can be tuned simultaneously or independently.

Figure 2-63 Transmit Link Equalization Flow



The following list describes the flow shown in Figure 2-63:

1. The local device sends the status as Not_Updated (default response) on the transmitted training frame. The local device waits for the increment, decrement, preset, or initialize request in the incoming frame.
2. On receiving the request, the local device initiates the requested action by setting its transmitter filter coefficients.
3. After doing the requested settings, the local device responds with Max, Min, or Updated status in the transmitted training frame. Any of these responses acts as an acknowledgment to the remote device.
4. On receiving the acknowledgment, the remote device starts sending the hold request. At the same time, the remote device also analyzes its Receive link reliability and decides on further tuning of the local transmitter filters.
5. On receiving the hold request, the local device sends the Not_Updated response to indicate that it is ready for processing the next request.

This cycle continues until the remote receiver is satisfied with the quality of the received data (in such case, it continues to send the hold requests) or a timer as specified by the standard runs out.

The actual Transmit equalization is to tune the filters of the local PHY transmitter. The DWC_xlgpcs provides the following interface signals to achieve Transmit link equalization in the PHY:

- Local Transmit Filter Coefficient Update Interface

This output interface of the DWC_xlgpcs provides the filter tuning requests received from the remote device to the PHY. The requests are issued in the form of increment, decrement, preset, or initialize requests. The hold requests are not given out to the PHY because it is not required.

- Local Transmit Filter Coefficient Status Interface

  This input interface of the DWC_xlgpcs is used to get the status of the tuning requests that are issued through Local Transmit Filter Coefficient Update Interface to the PHY. The DWC_xlgpcs reports this status back to the remote device through training frames generated in the DWC_xlgpcs.

All the top-level interface signals with the prefix `rpcs_ktx_*` are related to Transmit Link equalization. For more details, see the "*Signal Descriptions*" chapter.

2.9.4.1 Receive Link Equalization Flow

The Receive link equalization is achieved by requesting the remote transmitter to tune its Transmit filters based on the Receive Link reliability analysis. The flow for Receive link equalization is similar to the one explained in "Transmit Link Equalization Flow" on page 134. The only difference is that the roles of the local device and the remote device are swapped.

To achieve the Receive link equalization in the PHY, the DWC_xlgpcs provides the following interfaces to the PHY:

- Receive Filter Coefficient Update Interface

  This is the input interface to the DWC_xlgpcs. The Receive equalization algorithm in the PHY uses this interface to issue requests to tune Transmit filters of the remote device. The requests can be issued in the form of preset, initialize, increment, or decrement.

- Receive Filter Coefficient Status Interface

  This output interface of the DWC_xlgpcs provides the received status of the requests sent to the remote device through Receive Filter Coefficient Update interface. The status is provided in the form of updated, maximum, or minimum as per the received training frame from the remote device. The PHY also uses this status as acknowledgment of its requests.

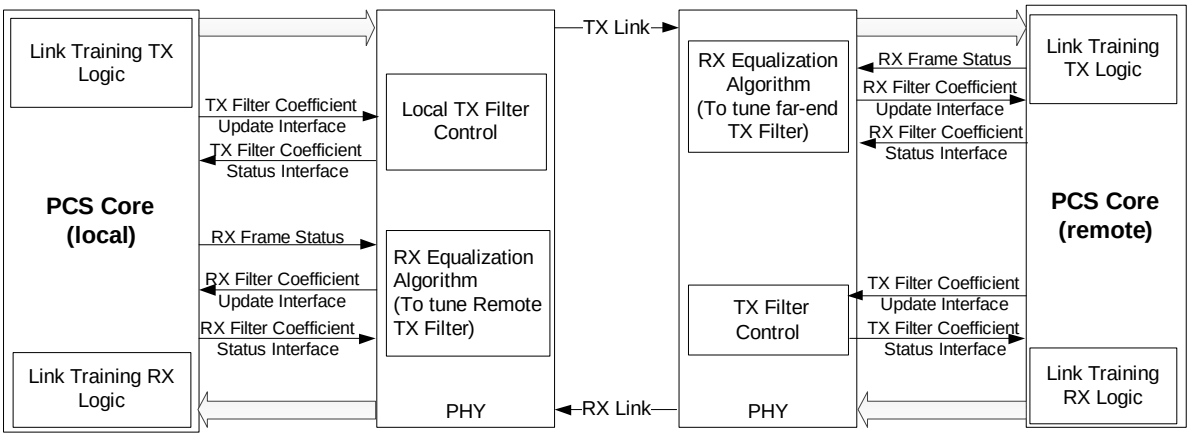
- Receive Frame Status Interface

  This output interface of the DWC_xlgpcs provides the status or quality of the received Training frame such as PRBS11 decoding status, frame lock status, or frame received event. The Receive equalization algorithm in the PHY can use this status to decide when to terminate or initiate further training.

All the top-level interface signals with prefix `rpcs_krx_*` are related to Receive link equalization. For more details, see the "*Signal Descriptions*" chapter.

Figure 2-64 shows all interfaces provided by the DWC_xlgpcs to the local PHY and equivalent implementation that exists in the remote device to achieve both Transmit and Receive link equalization.

Figure 2-64 Link Training Equalization Interfaces



2.9.4.2 Coefficient Update and Coefficient Status Fields during Transmit and Receive Link Equalization

Table 2-25 provides information about the coefficient update field during Transmit and Receive link equalization.

Table 2-25 Coefficient Update Field during Transmit and Receive Link Equalization

Field	Description	Generation	Reception
15:14	Reserved	N/A	N/A
13	Preset ■ 1: Preset ■ 0: No Preset	The Receive equalization process in the PHY generates this request. The PHY can generate this request as a first request, that is, the first request after the rpcs_ktr_restart_o output pulse is asserted. The PHY can also generate this request when it receives NOT_UPDATED as status from the remote device for all the filters. When the PHY issues the preset request, the DWC_xlgpcs continues to send the preset request until it receives updated or maximum as status from the remote device. When the DWC_xlgpcs receives the status, it updates this received status to the PHY and starts sending the hold request. **Note**: This request is usually sent in the beginning of the start-up protocol to analyze the link behavior when equalization is turned off.	When the DWC_xlgpcs receives this request, it requests the local PHY to perform preset. The PHY must preset its Transmit filter coefficients such that c(-1) and c(1) are programmed to zero and c(0) is programmed to the maximum value. After these settings are done, the PHY must update the status as updated or maximum to the DWC_xlgpcs. The DWC_xlgpcs then sends this status to the remote device.

Table 2-25 Coefficient Update Field during Transmit and Receive Link Equalization

Field	Description	Generation	Reception
12	Initialize ■ 1: Initialize ■ 0: No Initialize	The Receive equalization process in the PHY generates this request. The PHY can generate this request as a first request, that is, the first request after the rpcs_ktr_restart_o output pulse is asserted. The PHY can also generate this request when it receives NOT_UPDATED as status from the remote device for all the filters. When the initialize request is issued, the DWC_xlgpcs continues to send the initialize request until it receives updated, maximum, or minimum as status from the remote device. When the DWC_xlgpcs receives the status, it updates this received status to the PHY and starts sending the hold request. **Note**: This request is usually sent in the beginning of the start-up protocol to program the Transmit filters to predefined values before issuing the increment or decrement requests.	When the DWC_xlgpcs receives this request, it requests the local PHY to perform initialize.
11:6	Reserved	N/A	N/A
5:4	Coefficient update for c(1) filter ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold	The Receive equalization process in the PHY generates the increment or decrement request. This field is for c(1) filter. The DWC_xlgpcs automatically generates the hold request as part of the handshake and also when no request is issued by the PHY. The PHY can generate this request as a first request, that is, the first request after the rpcs_ktr_restart_o output pulse is asserted. The PHY can also generate this request when it receives NOT_UPDATED as status from the remote device for c(1) filter. When the PHY issue the increment or decrement request, the DWC_xlgpcs continues to send the increment or decrement request until it receives updated, maximum, or minimum as status from the remote device. When the DWC_xlgpcs receives the status, it updates this received status to the PHY and starts sending the hold request until the next request is received. **Note**: The generation of increment or decrement is based on the Receive equalization algorithm which is based on the Receive link reliability analysis.	When the DWC_xlgpcs receives the increment or decrement request, it sends the request to the local PHY for c(1) filter. The PHY must increment or decrement its c(1) Transmit filter coefficients as defined in the IEEE Std 802.3, Clause 72. The PHY must update the status as updated, maximum, or minimum to the DWC_xlgpcs. The DWC_xlgpcs then sends this status to the remote device.

Table 2-25 Coefficient Update Field during Transmit and Receive Link Equalization

Field	Description	Generation	Reception
3:2	Coefficient update for c(0) filter	This field is for c(0) filter. The information for this field is similar to the information given in bits[5:4] of this table for c(1) filter.	This field is for c(0) filter. The information for this field is similar to the information given in bits[5:4] of this table for c(1) filter.
1:0	Coefficient update for c(-1) filter	This field is for c(-1) filter. The information for this field is similar to the information given in bits[5:4] of this table for c(1) filter.	This field is for c(-1) filter. The information for this field is similar to the information given in bits[5:4] of this table for c(1) filter.

Table 2-26 provides information about the coefficient update field during Transmit and Receive link equalization.

Table 2-26 Coefficient Status Format

Field	Description	Generation	Reception
15	Receiver Ready	DWC_xlgpcs programs this bit to 1 when the local device Receive equalization process determines that the Receive link is ready for KR speed traffic. The Receive equalization process generates this status by asserting the rpcs_rx_trained_i input signal any time after the rpcs_ktr_restart_o output pulse is asserted. Otherwise, this bit is programmed to 0. **Note**: The Receive equalization process sets this input by monitoring the reliability of the Receive link based on successful decoding of training patterns (PRBS11) of the received training frame.	When the DWC_xlgpcs receives this status, it indicates that the remote device does not generate any more requests to tune the local transmitter filters. **Note**: This status is received in the DWC_xlgpcs. This status is not given to the PHY.
14:6	Reserved	N/A	N/A
5:4	Coefficient Status of c(1) ■ 11: Max ■ 10: Min ■ 01: Updated ■ 00: Not_Updated	The local PHY must generate this status for c(1) filter as a response to the preset, initialize, increment, or decrement request received for c(1) filter. **Note**: The DWC_xlgpcs automatically generates the Not_Updated status during handshaking. The DWC_xlgpcs transmits this status back to remote device. Therefore, the PHY does not need to give the Not_Updated status.	When the DWC_xlgpcs receives the max, min, or updated status, it sends the status to the PHY Receive equalization process. The DWC_xlgpcs receives this status as a response to the preset, initialize, increment, or decrement request sent to the remote device. When the DWC_xlgpcs receives this status, the Receive equalization can again evaluate the Receive link and issue further requests to tune the remote Transmit filters.

Table 2-26 Coefficient Status Format (Continued)

Field	Description	Generation	Reception
3:2	Coefficient Status of c(0)	This field is for c(0) filter. The information for this field is similar to the information given in bits[5:4] of this table for c(1) filter.	This field is for c(0) filter. The information for this field is similar to the information given in bits[5:4] of this table for c(1) filter.
1:0	Coefficient Status of c (-1)	This field is for c(-1) filter. The information for this field is similar to the information given in bits[5:4] of this table for c(1) filter.	This field is for c(-1) filter. The information for this field is similar to the information given in bits[5:4] of this table for c(1) filter.

2.9.4.3 Link Training Control Options

DWC_xlgpcs supports some custom features to control the mode of operation of Link Training.

2.9.4.3.1 PRBS31 Support

DWC_xlgpcs, by default, uses PRBS11 as the training pattern in the transmitted KRTR frames, as required by IEEE specification. However, DWC_xlgpcs has the capability to send and monitor PRBS31 pattern (instead of PRBS11) in the KRTR frames. This feature can be enabled by programming bit [1] (PRBS31_EN) of the `VR_PMA_KRTR_PRBS_CTRL0` register. If this mode is enabled, the 'Training Pattern' field of the transmitted CL72 Frame has PRBS31 instead of PRBS11. In addition, the PRBS checker in the receive path of DWC_xlgpcs monitors PRBS31 errors instead of PRBS11 errors.

Note

1. PRBS31 support is not available in CL-92/93 mode
2. PRBS31 support is available in CL-72 mode for 10G,20G,25G,40G and 50G speed modes only

2.9.4.3.2 PRBS Mode to Determine Successful Receive Equalization

DWC_xlgpcs has a control bit (bit [0] of the `VR_PMA_KRTR_PRBS_CTRL0` register) to enable PRBS mode to determine successful receive equalization. By default, this mode is not enabled, and DWC_xlgpcs determines successful Receive equalization based on the status of the primary input 'rpcs_rx_trained_i' (for non-Synopsys PHY configurations) or based on the expiry of an internal Receive-Train-Time timer (for Synopsys PHY configurations).

If PRBS mode is enabled, DWC_xlgpcs starts monitoring the received Link training Frames (after the assertion of 'rpcs_rx_trained_i' or after the expiry of Receive-Train-Time timer) for any PRBS errors, for an interval determined by the value programmed to the `VR_PMA_KRTR_PRBS_CTRL1` register. If the number of PRBS errors detected is less than or equal to the tolerance limit programmed to the `VR_PMA_KRTR_PRBS_CTRL2` register, then DWC_xlgpcs concludes that the receive equalization has been successful; otherwise this is treated as a 'TRAINING FAILURE'.

2.9.4.3.3 Sequential/Parallel Mode of Coefficient Update

During Receive equalization, DWC_xlgpcs makes Receive-link evaluation request to the PHY. In turn, PHY provides INC/DEC commands for the pre (C(-1)), post (C(1)) or main(C(0)) coefficients. In parallel mode (default mode), DWC_xlgpcs transmits the received commands for all the three coefficients (if all three have been provided by the PHY) to the remote-end device in the same frame.

In sequential mode, DWC_xlgpcs transmits the three coefficient update commands sequentially one after the other. DWC_xlgpcs starts with sending C(-1) update, then transmits C(0) update and ends with C(1) update. Prior to sending a new coefficient update, DWC_xlgpcs waits for an acknowledgment from the remote-end device for the previously transmitted coefficient update. This mode is useful if the remote-end device is not capable of handling multiple coefficient update request at the same time.

This feature is available only if the selected configuration has Synopsys Multi-protocol 12G/16G/25G/32G PHY. This mode is controlled through bit[8] (SEQ_EQ_EN) of VR_PMA_MP_25G_16G_RX_EQ_CTRL4.

2.9.4.3.4 Ping-Pong Mode of Receive Equalization

The ping-pong mode is valid only for Synopsys multi-protocol 12G/16G/25G/32G PHY configurations. To enable the ping-pong mode, program bit[9] (PING_PONG_EN) of the `VR_PMA_MP_25G_16G_RX_EQ_CTRL4` register.

In the Ping-Pong mode, DWC_xlgpcs sends exactly one of C(-1) or C(1) requests to the remote-end device, even if PHY gives the INC/DEC commands for both C(-1) and C(1) in a particular iteration of Receive equalization. DWC_xlgpcs alternates between C(-1) and C(1) between iterations; that is, if C(-1) is serviced in the nth iteration, DWC_xlgpcs serves C(1) (if any) in the (n+1)th iteration and vice versa.

In the ping-pong mode, DWC_xlgpcs

1. Checks for the validity of the coefficient update commands provided by the PHY.
2. Only if valid, DWC_xlgpcs processes and sends the command to the link partner.

An INC command for C(-1) is considered valid, only if the status returned by link partner for both C(-1) and C(0) coefficients are not MAX for the previous iteration. Similarly, an INC command for C(+1) is considered valid, only if the status returned by link partner for both C(+1) and C(0) coefficients are not MAX for the previous iteration.

Ping-Pong mode works well when automatic generation of main coefficient is enabled. To enable the auto-generation of main coefficient, program bit[10] (SELF_MAIN_EN) of the `VR_PMA_MP_25G_16G_RX_EQ_CTRL4` register. In the auto-generation of main coefficient mode, DWC_xlgpcs ignores the update command received from PHY for C(0) (if any) and transmits the INC/DEC based on the update command received for C(-1) or C(1).

If DWC_xlgpcs decides to serve C(-1) in a particular iteration and

- If C(-1) is an INC command, DWC_xlgpcs transmits the INC command for both C(-1) and C(0).
- If C(-1) is a DEC command, DWC_xlgpcs transmits the DEC command for both C(-1) and C(0).

Table 2-27 Command Sequence for C(-1) Command

Sequential Mode Enabled	Sequential Mode Not Enabled
INC Command	
INC Command Sequence 1. INC C(-1) 2. INC C(0)	Both INC C(0) and INC C(-1) are transmitted simultaneously.
DEC Command	

Table 2-27 Command Sequence for C(-1) Command

Sequential Mode Enabled	Sequential Mode Not Enabled
DEC Command Sequence 1. DEC C(0) 2. DEC C(-1)	Both DEC C(0) and DEC C(-1) are transmitted simultaneously.

If DWC_xlgpcs decides to serve C(1) in a particular iteration and

- If C(1) is an INC command, DWC_xlgpcs transmits the INC command for both C(1) and C(0).
- If C(1) is a DEC command, DWC_xlgpcs transmits the DEC command for both C(1) and C(0).

Table 2-28 Command Sequence for C(1) Command

Sequential Mode Enabled	Sequential Mode Not Enabled
INC Command	
INC Command Sequence 1. INC C(1) 2. INC C(0)	Both INC C(1) and INC C(0) are transmitted simultaneously.
DEC Command	
DEC Command Sequence 1. DEC C(1) 2. DEC C(0)	Both DEC C(1) and DEC C(0) are transmitted simultaneously.

2.9.4.3.5 Management Mode of Equalization

DWC_xlgpcs provides software mode of equalization. In this mode, the host software can program the equalization commands and status to both the local device PHY (LD) and the link partner PHY (LP). Vendor specific registers are provided to capture the command and status received from both the LP PHY and the LD PHY. Host can read these registers and can program the desired command and status to both LD PHY and/or LP PHY, thus making the entire process transparent.

By default, the DWC_xlgpcs controls the equalization (hardware mode). To bypass hardware equalization mode, host must enable the software mode of equalization.

In a multi lane mode, though equalization is done independently on each lanes, it is mandatory to run either hardware mode of equalization or software mode of equalization on all the lanes. Mixed mode (Some lanes working in software equalization mode) and remaining lanes working in hardware equalization mode) must not be done.

Management Mode of Transmit Equalization

Enable the software mode during Transmit equalization by programming bit [15] (TX_EQ_MM) of the `VR_PMA_KRTR_TX_EQ_STS_CTRL(i)` register (address – 0x80B4 +i) to 1 where i=0,1,2,3. Set this bit before enabling KR training, that is, before programming the TR_EN bit of the `SR_PMA_KR_PMD_CTRL` register to 1.

During the Transmit equalization process (of KR training), DUT receives training frames (from link partner) with Coefficient Update commands for the local PHY transmitter. These commands are reflected in the SR_PMA_KR_LP_CEU register in single lane mode and in the SR_PMA_MLN_KR_LP_CEU_LANE<i> i=0,1,2,3 register in the multi lane modes.

In hardware mode, these coefficient update commands get driven to the local PHY Transmit filter inputs. The PHY responds to these commands and provides the corresponding 'Status' information. This 'status' information is sent back to the link-partner. The 'Status' information sent to the link-partner is reflected in the SR_PMA_MLN_KR_LD_CESTS_LANE<i> i =0,1,2,3 register.

In the software mode, the Transmit coefficient update commands driven to the PHY Transmit filter inputs are determined by the Software. Software can program the desired coefficient update commands to the 'VR_PMA_KRTR_TX_EQ_CFF_CTRL<i>' register (address – 0x180B0+i) ,i=0,1,2,3. In addition, the 'Status' that is returned to the link-partner is also determined by the software. The 'status' values programmed to the 'VR_PMA_KRTR_TX_EQ_STS_CTRL<i> register (address – 0x180B4 +i) to 1 where i=0,1,2,3 are packed into the training frames sent by DWC_xlgpcs. The value programmed to this register gets reflected in the SR_PMA_MLN_KR_LD_CESTS_LANE<i> i =0,1,2,3 register, as it indicates the 'status' information in the most recently transmitted frame.

The 'Status' returned by the local PHY (for the Transmit coefficient update commands) is captured in the VR_PMA_KRTR_TX_EQ_STS(i) - address (0x180B8+i, i=0,1,2,3 register.

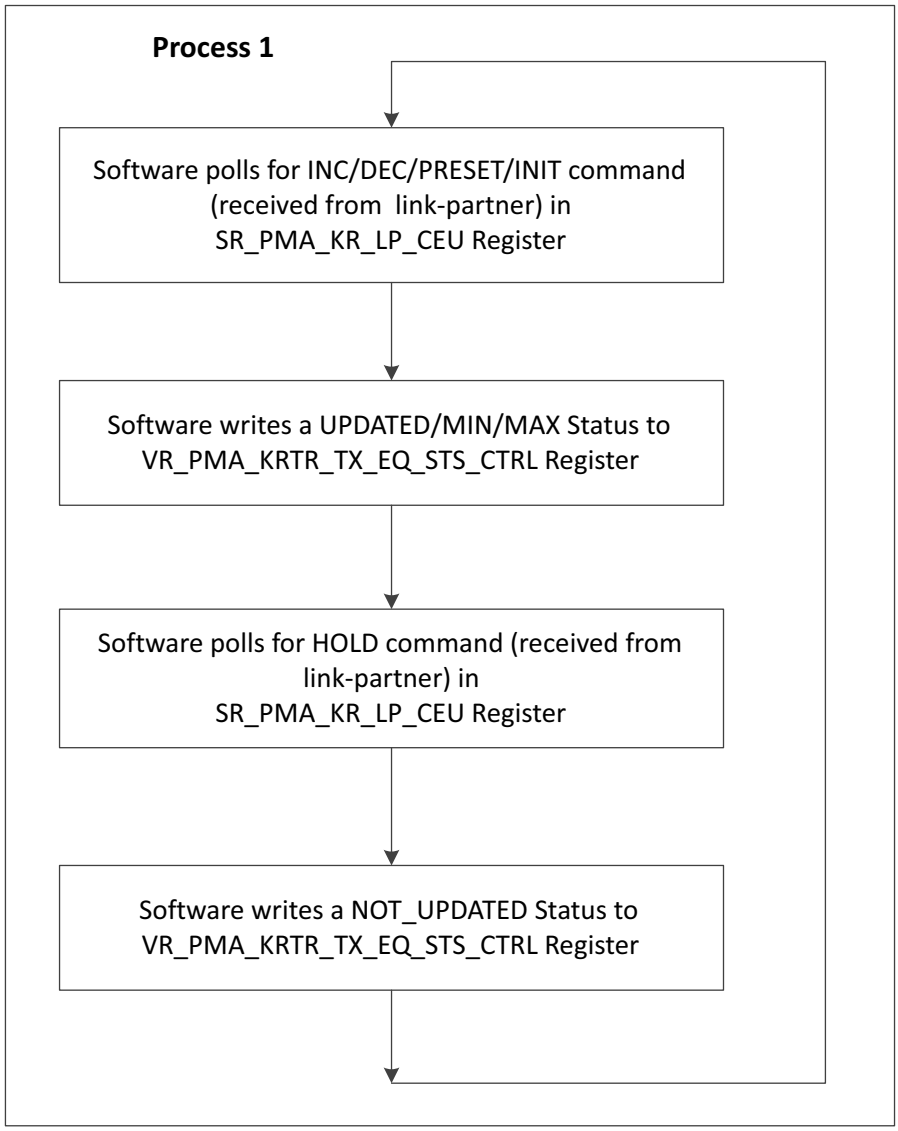
Main Features of Transmit Equalization

The main features of the Transmit equalization are:

- The Transmit coefficient command driven to the local PHY can be totally independent of the Transmit command received from the link-partner.
- The Transmit status returned to the link-partner can be totally independent of the 'status' provided by the local PHY.
- To ensure a logic sequence of command & status on the serial link, software must ensure the following:
  - If the command received from link-partner is PRESET/INIT/INC/DEC, then it must program 'UPDATED/MIN/MAX' status to the VR_PMA_KRTR_TX_EQ_STS_CTRL<i> register, i=0,1,2,3. (If command is PRESET/INIT, software must program the status values for all the three coefficients).
  - If the command received from link-partner is HOLD, then it must program 'NOT_UPDATED' status to the VR_PMA_KRTR_TX_EQ_STS_CTRL<i> register, i=0,1,2,3.

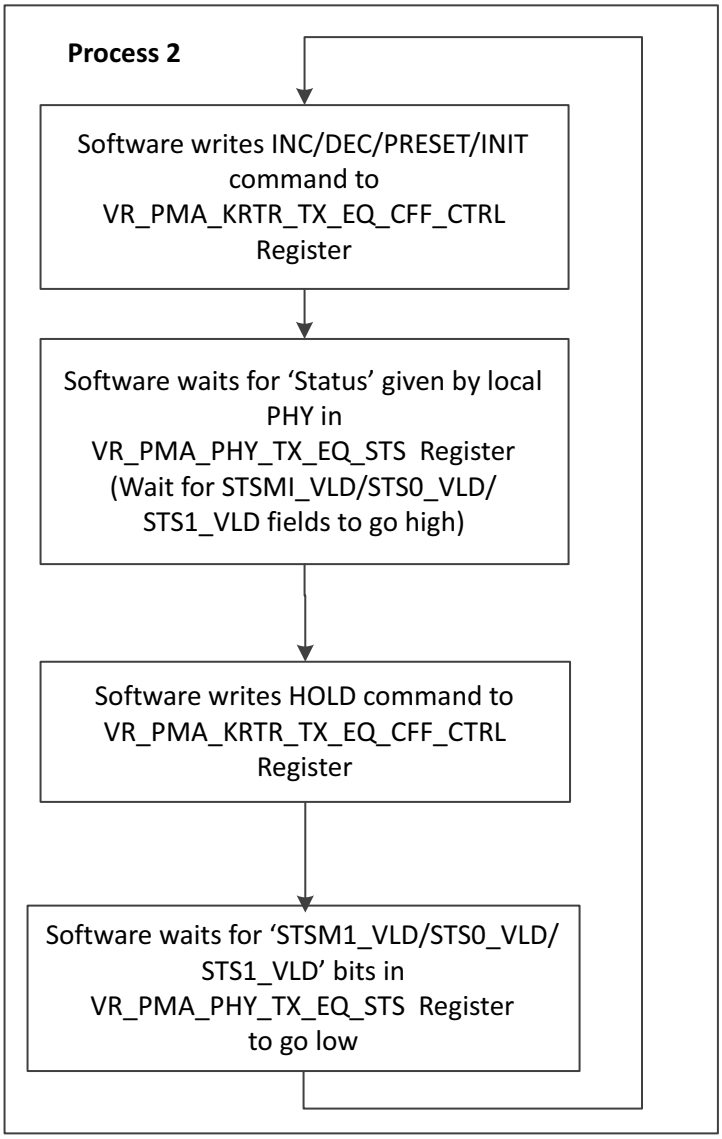
Software mode of Transmit equalization has two independent processes under the hood as explained by the following diagrams:

Figure 2-65 Software Mode of Transmit Equalization (Process 1 of 2)



Note When INIT/PRESET command is received in the SR_PMA_KR_LP_CEU register in single lane mode and the SR_PMA_MLN_KR_LP_CEU_LANE(i) i=0,1,2,3 register for multi lane modes, software must return the UPDATED/MIN/MAX status for all the three coefficients.

Figure 2-66 Software Mode of Transmit Equalization (Process 2 of 2)

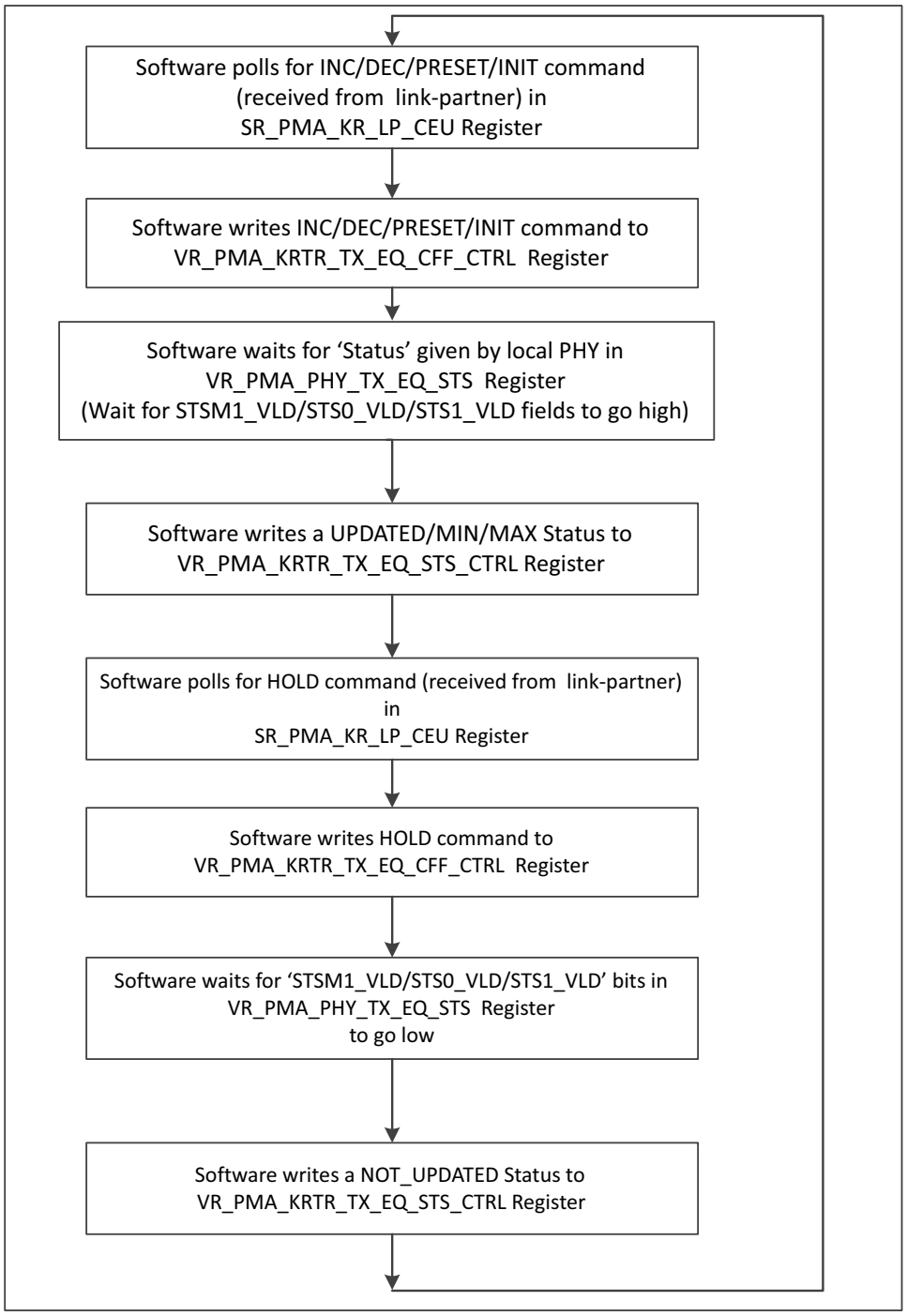


Note

- In case, the third party PHY does not provide an I/O interface to control the Transmit filter coefficients (and to reflect the status), then process 1 alone would suffice.
- Whenever INIT or PRESET commands are programed to VR_PMA_KRTR_TX_EQ_CFF_CTRL(i), 'Status' values are returned for all the three coefficients. Hence, software must write HOLD commands, thereafter, for all the 3 coefficients.

Though the two processes in Figure 2-65 and Figure 2-66 can be performed independent of each other (while training is enabled), software can optionally decide to perform Transmit equalization in the sequence (which is more logical) shown in Figure 2-67:

Figure 2-67 Transmit Equalization Sequence



Management Mode of Receive Equalization

Software mode during Receive equalization can be enabled by programming bit[15] (the RX_EQ_MM field) of the VR_PMA_KRTR_RX_EQ_CTRL<i> register i=0,1,2,3 (address -0x180BC+i) to 1. This bit must be programmed prior to enabling KR training, that is, before programming the TR_EN bit of the SR_PMA_KR_PMD_CTRL register to 1.

During Receive equalization process (of KR training), DUT sends training frames (to link partner) with Coefficient Update commands to tune the filter coefficients of the link-partner. These commands are reflected in the SR_PMA_KR_LD_CEU register in single lane mode and in the SR_PMA_MLN_KR_LD_CEU_LANE<i> register in multi lane modes where I =0,1,2,3.

These coefficient update commands are generated by the local PHY receiver and is provided to DWC_xlgpcs via its input ports (rpcs_krx_ceu{filter}_vld_i for Non-Synopsys configurations, rpcs_krx_{pre,main,post}_dir_i for Synopsys PHY configurations). In hardware mode of Receive equalization, DWC_xlgpcs transmits these coefficient update commands to the far-end.

However, in software mode, the management can program desired coefficient update commands to the VR_PMA_KRTR_RX_EQ_CTRL<i> register i=0,1,2,3. The values programmed to this register are transmitted to the link partner via training frames. The values programmed by the software to this register also reflects in the SR_PMA_KR_LD_CEU register in single lane mode and in the SR_PMA_MLN_KR_LD_CEU_LANE<i> i=0,1,2,3 registers in multi lane modes.

The link-partner, in turn, sends frames with the 'Status' information (for the coefficient update commands sent) to DWC_xlgpcs. This 'Status' information is available in the SR_PMA_KR_LP_CESTS register for single lane mode and in the SR_PMA_MLN_KR_LD_CESTS_LANE<i> i=0,1,2,3 register for multi lane modes.

During hardware mode, this status information is passed on the local PHY receiver via the 'rpcs_krx_ceu{filter}_sts_o' port of DWC_xlgpcs (Only for Non-Synopsys PHY configurations). However, in software mode, management can choose to provide any desired status to the local PHY by programming suitable values to the 'VR_PMA_KRTR_RX_EQ_STS_CTRL<i>' register (address - 0x180C0+i), where i = 0,1,2,3.

Though the coefficient update commands given by the local PHY are ignored by DWC_xlgpcs, in software mode, this information is captured in the 'VR_PMA_PHY_RX_EQ_CEU<i>' register (address - 0x180c4+i), where i=0,1,2, and 3.

In addition, software can program the 'RR_RDY' field of the VR_PMA_KRTR_RX_EQ_CTRL<i> (address - 0x180BC+i) register to 1, once it decides to send no more coefficient update commands to the link-partner. During this mode, DWC_xlgpcs ignores the value of the 'rpcs_rx_trained_i' port of DWC_xlgpcs.

Software mode of Receive equalization has two independent processes under the hood as explained by the following diagrams:

Figure 2-68 Software Mode of Receive Equalization (Process 1 of 2)

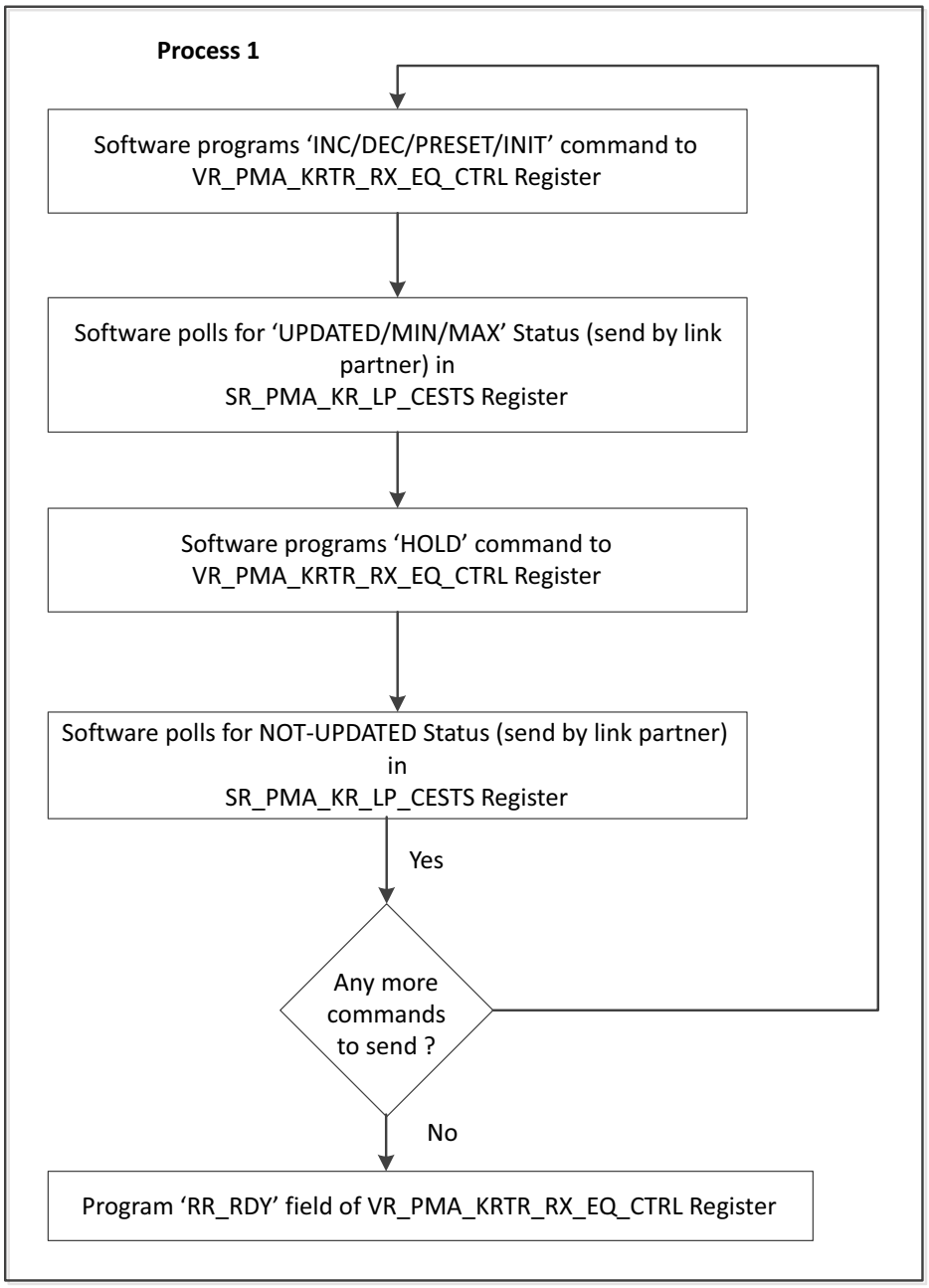
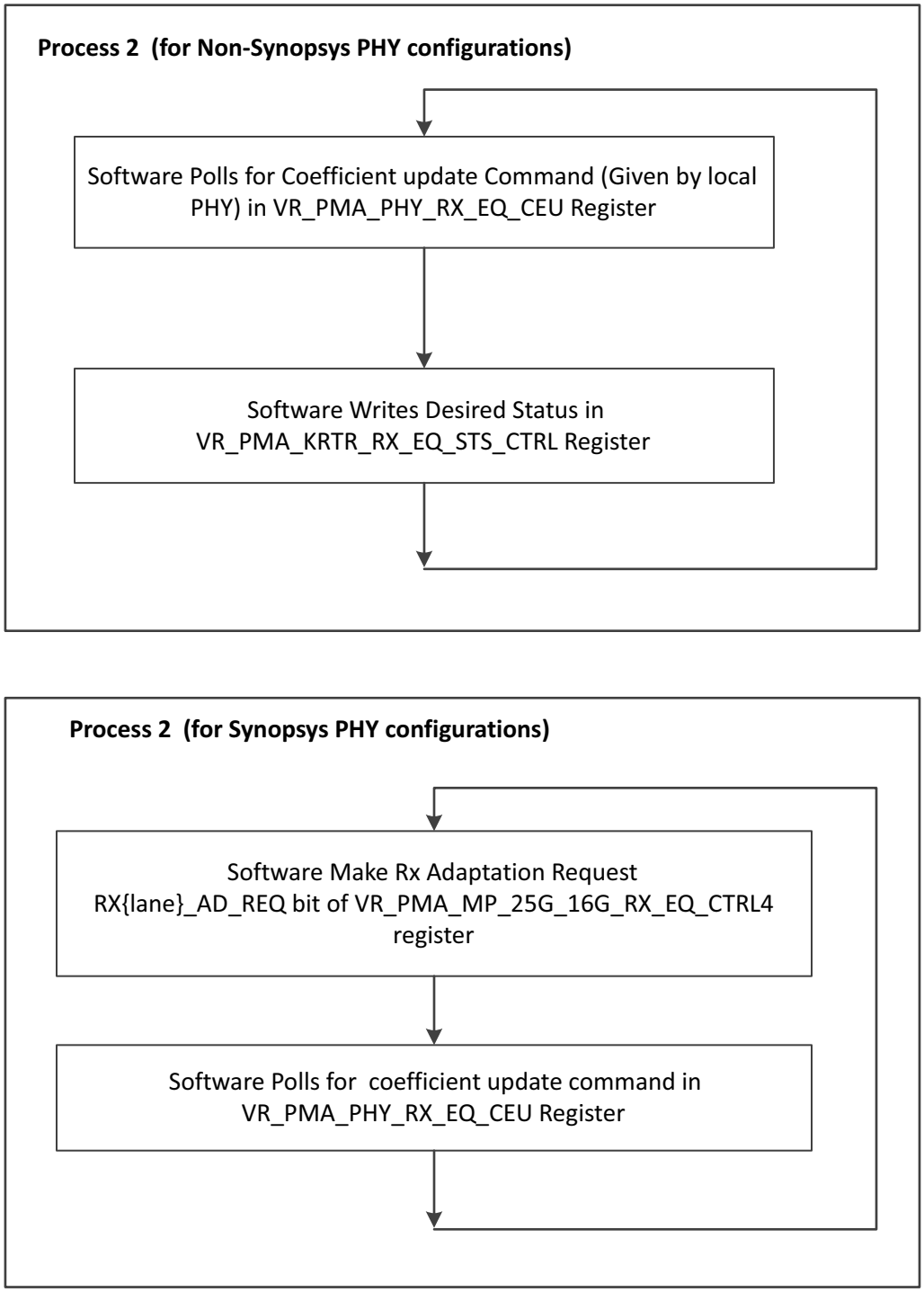
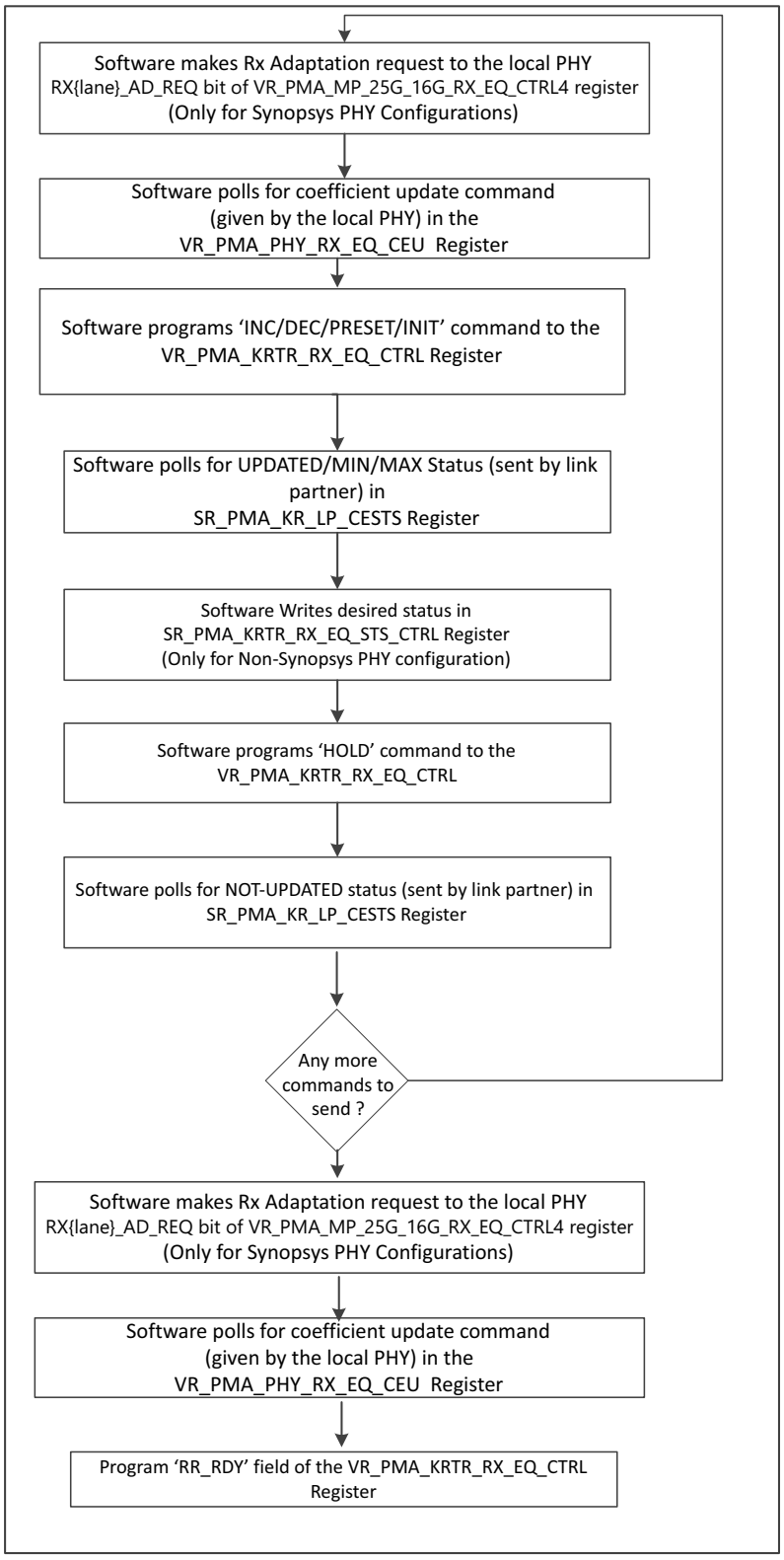


Figure 2-69 Software Mode of Receive Equalization (Process 2 of 2)



Though the two processes in Figure 2-68 and Figure 2-69 can be performed independent of each other (while training is enabled), software can optionally decide to perform Receive equalization in the sequence shown Figure 2-70:

Figure 2-70 Receive Equalization Sequence



2.9.4.3.6 Programmable Timers

Clause 72 defines two timers, max_wait_timer and wait_timer. As per the specification, the value of

- max_wait_timer must be
  - 3.1 seconds for 50GBASE-KR/CR2 or 100GBASE-KR2/CR2 modes
  - 500ms for other modes.
- wait_timer must be between 100 and 300 training frames.

DWC_xlgpcs offers the flexibility to program the values of these timers. Application can program the following registers to update the values of these timers:

- `VR_PMA_KRTR_TIMER_CTRL0`
- `VR_PMA_KRTR_TIMER_CTRL1`

The default values of these registers are such that the timers run for the duration specified by IEEE specification.

For configurations with Synopsys multi-protocol 12G/16G/25G/32G PHY, an additional timer determines the duration for receive equalization. This timer is programmable through the `VR_PMA_KRTR_TIMER_CTRL2` register.

Any change to these programmable timer registers must be done before enabling the KR training operation.



To reduce the gate count, correctness of the training pattern bits from bit 0 to bit 65 and bit 4029 to bit-4095 is not checked. Therefore, any error in these bit positions are not reported.

2.10 Timestamping Accuracy

DWC_xlgpcs supports IEEE802.3cx-based timestamping accuracy. The intent of the Timestamping Accuracy feature is to provide an accurate indication of the transmission and reception initiation times of all packets, as required to support various time synchronization protocols like IEEE 1588 or IEE802.1AS.

On a high-level, DWC_xlgpcs provides the following features to comply with the IEE802.3cx specification:

- An output port both in Transmit and in Receive, to indicate the detection of Data Delay Measurement point (DDMP).
- Registers indicating the maximum and minimum delay inside the DWC_xlgpcs IP, indicating nanosecond- and sub-nanosecond resolution.
- An output port both in Transmit and Receive, to indicate the dynamic change in latency for each packet.

The feature is enabled when the TS_MODE' bit [11] of the `VR_PCS_DIG_CTRL3` register is programmed to 1 by the host software.



This feature is currently supported only for configurations in which the DWCXLP_MAIN_MODE is 0 (25G configurations) and the PHY interface width (DWCXLP_R_SRDS_INTF) is 32 or 66 bits.

2.10.1 Data Delay Measurement Point (DDMP)

The Data Delay Measurement Point is the point in the packet at which timestamp is taken. This point can take the following options:

- SFD point
- First symbol after SFD

  This option is recommended as per the IEEE specification.

The DDMP_SEL field (bit[13]) of the `SR_PCS_TIME_SYNC_CFG` register controls the selection of the DDMP.

2.10.2 DDMP Indication Output Ports

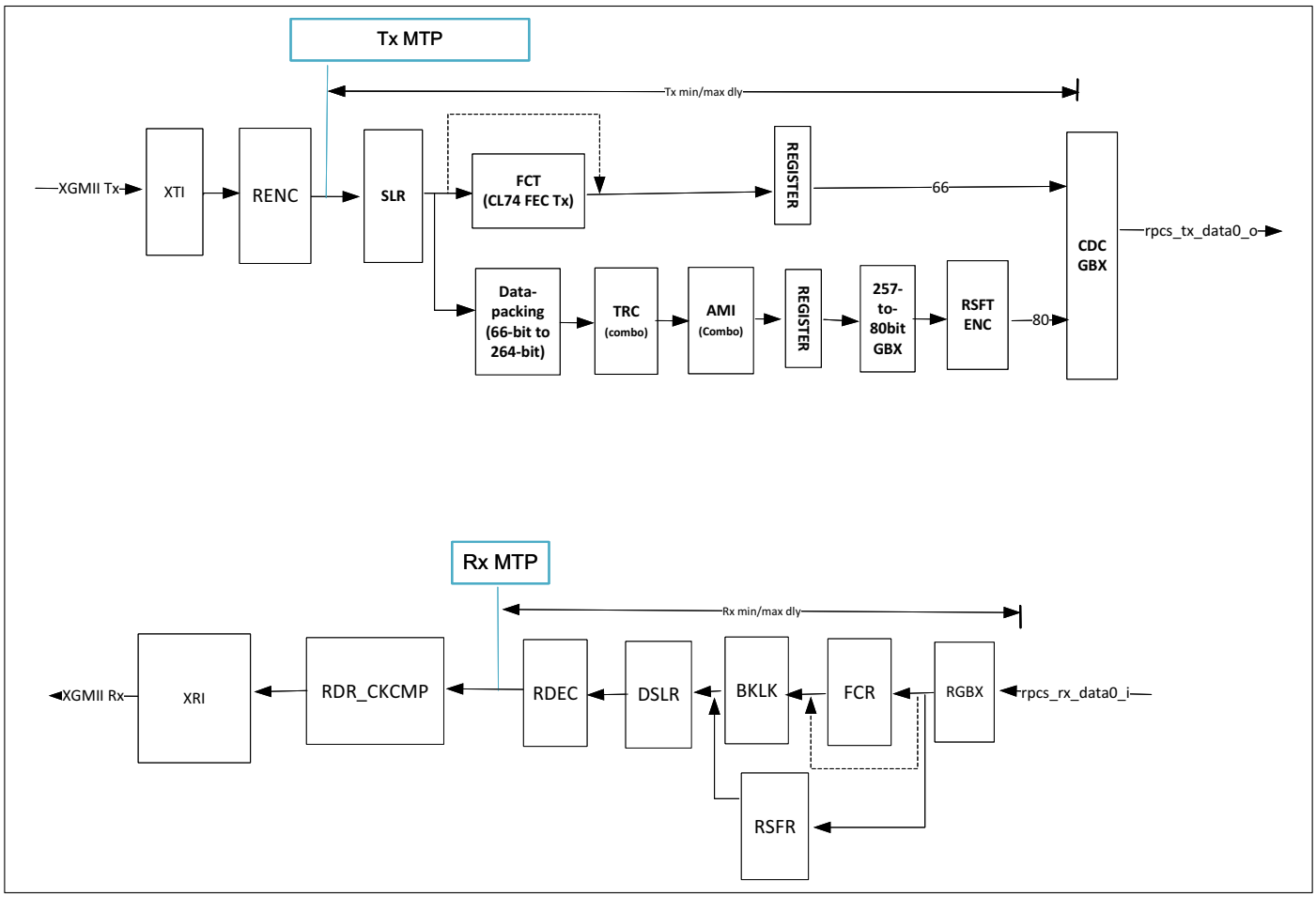
DWC_xlgpcs detects DDMP at a suitable internal point in its Transmit data-path. Figure 2-71 shows this point. Whenever DDMP is detected in its transmit data path, DWC_xlgpcs generates a pulse on the xlgpcs_tx_ddmp_o output port. This pulse can be sampled by the MAC Timesync client to calculate the timestamp. In addition to the pulse signal, DWC_xlgpcs also implements the xlgpcs_tx_ddmp_tgl_o toggle signal, which toggles every time a DDMP is detected in the transmit data path.

A similar approach is used in the Receiver. DDMP is detected in DWC_xlgpcs at a suitable internal point in its receive data-path, as shown in Figure 2-71. Whenever DDMP is detected in its Receive data path, DWC_xlgpcs generates a pulse on the xlgpcs_rx_ddmp_o output port. This pulse can be sampled by the MAC to capture the timestamp. In addition to the pulse signal, DWC_xlgpcs also implements the xlgpcs_rx_ddmp_tgl_o toggle signal, which toggles every time a DDMP is detected in the Receive data path.

For more information on the xlgpcs_tx_ddmp_tgl_o and xlgpcs_rx_ddmp_tgl_o signals, see the "*Signal descriptions*" chapter.

Figure 2-71 shows the internal data path point at which DWC_xlgpcs detects or indicates a DDMP for a 25G_Only configuration.

Figure 2-71 Internal Data Path Point for DDMP in DWC_xlgpcs



2.10.3 Delay Registers

DWC_xlgpcs implements several registers to indicate the minimum and maximum delay in its transmit and receive path, as follows:

- Transmit delay registers, which indicate the delay in DWC_xlgpcs from the internal point at which the DDMP detection pulse is generated to the rpcs_tx_data{lane}_o final Transmit output port.
- Receive delay registers, which indicate the delay in DWC_xlgpcs from the rpcs_rx_data{lane}_i Receive input port to the internal point at which the DDMP detection pulse is generated.

The registers indicate the Transmit or Receive minimum or maximum delay value in nano-seconds and sub-nano-seconds. The registers are the following:

- `SR_PCS_TIME_SYNC_TX_MAX_DLY_LWR`
- `SR_PCS_TIME_SYNC_TX_MIN_DLY_LWR`
- `SR_PCS_TIME_SYNC_TX_MAX_FINE_DLY`
- `SR_PCS_TIME_SYNC_TX_MIN_FINE_DLY`

- SR_PCS_TIME_SYNC_RX_MAX_DLY_LWR
- SR_PCS_TIME_SYNC_RX_MIN_DLY_LWR
- SR_PCS_TIME_SYNC_RX_MAX_FINE_DLY
- SR_PCS_TIME_SYNC_RX_MIN_FINE_DLY

The values in these registers change depending on the current operating mode. For more information on these registers, see the *"Register Descriptions"* chapter.

2.10.4 Dynamic Delay Variation Indication Ports

To indicate the dynamic variation in Transmit or Receive delay for an Ethernet packet, DWC_xlgpcs implements the following output ports:

- xlgpcs_tx_num_bit_chg_o[15:0]
- xlgpcs_rx_num_bit_chg_o[15:0]

Dynamic variation is indicated with respect to the average of the minimum and maximum delay register values.

Transmit

The xlgpcs_tx_num_bit_chg_o port indicates how many bits of delay change have occurred in the DWC_xlgpcs transmitter. The delay is indicated in two's compliment format and can thus range from -32768 to +32767. A value is indicated on this port for each packet that is received from the MAC Transmitter. If there is no dynamic variation, then these ports indicate zero. A value on the xlgpcs_tx_num_bit_chg_o port is sampled only when xlgpcs_tx_ddmp_o is asserted.

Receive

The xlgpcs_rx_num_bit_chg_o port indicates how many bits of delay change have occurred in the DWC_xlgpcs receiver for the incoming packet. The delay is indicated in two's compliment format and can thus range from -32768 to +32767. A value is indicated on this port for each packet that is received from the PHY Receive. If there is no dynamic variation, then these ports indicate zero. xlgpcs_rx_num_bit_chg_o[15:0] is sampled only when xlgpcs_rx_ddmp_o pulse is asserted.

For more information on the xlgpcs_tx_ddmp_o and xlgpcs_rx_ddmp_o signals, see the *"Signal Descriptions"* chapter.

2.11 Debug Bus

When the DWCXLP_DBG_BUS parameter is enabled, DWC_xlgpcs adds support for debug bus output ports. DWC_xlgpcs implements various debug status registers. In addition to these, DWC_xlgpcs provides additional debug information via debug output ports. A debug bus exists in each clock domain to indicate debug information from that domain. All the debug bus ports have xlgpcs_dbg* in their name. For more information on debug bus ports, see the *"Signal descriptions"* chapter.

2.12 Timing Diagrams

This section contains timing information for XLGMII Interface while operating in 100G, 50G, 40G, 25G, 10G speed modes.

2.12.1 XLGMII (40G speed mode)

The XLGMII Transmit waveform when operating in 40G speed mode is shown in Figure 2-72. It has 128-bit data and 16-control bits. The clock frequency is 312.5 MHz.

Figures 2-72 and 2-73 illustrates start of normal frame transmission.

Figure 2-72 Start of Normal Frame (Upper 64-Bit) Transmission

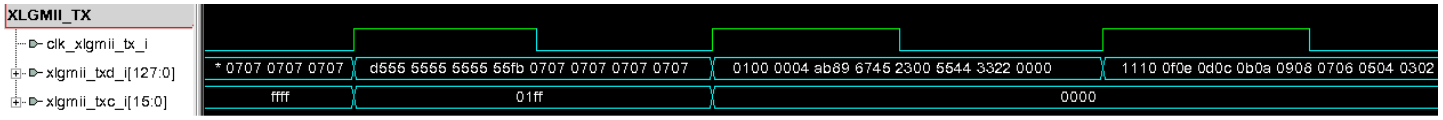
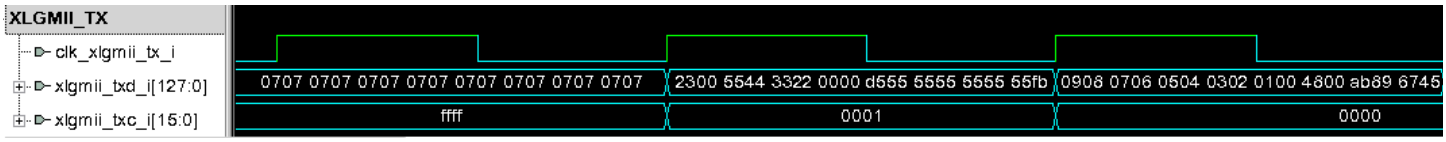


Figure 2-73 Start of Normal Frame (Lower 64-Bit) Transmission



The following two SOF are valid:

- xlgmii_txc_i[0]=0 and xlgmii_txd_i[7:0] = 8'hFB
- xlgmii_txc_i[8]=1 and xlgmii_txd_i[71:64] = 8'hFB

Figures 2-74 and 2-75 illustrates start of normal frame reception at XLGMII Receive Interface.

Figure 2-74 Start of Normal Frame (Upper 64-Bit) Reception

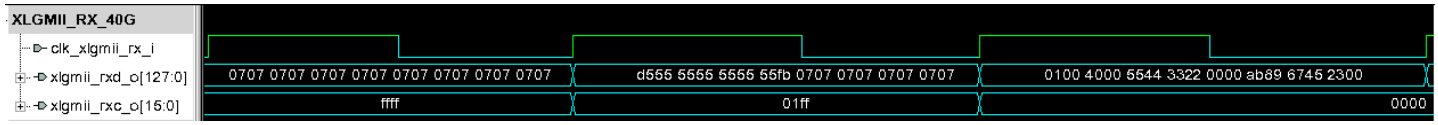
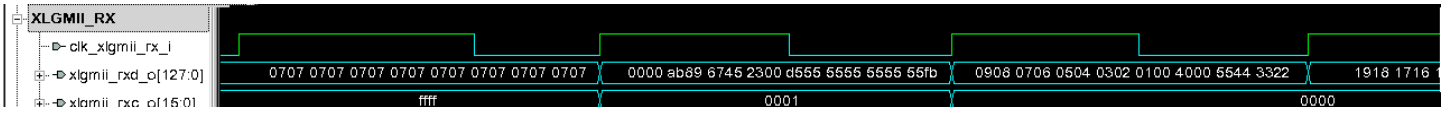


Figure 2-75 Start of Normal Frame (Lower 64-Bit) Reception



2.12.2 XGMII (10G Speed Mode)

The XGMII waveform when operating in 10G speed mode is shown in Figure 2-76. It has 32-bit data and 4-control bits and clock frequency is 312.5 MHz.

Figure 2-76 illustrates start of the normal frame transmission. The XGMII IDLE (with xgmii_txc_i[] = 1 and xlgmii_txd_i[] = 8'h07) followed by SOF (xlgmii_txc_i[0] = 1 with xlgmii_txd_i[7:0] = 8'hFB), and then the frame DATA (with xlgmii_txc_i[] = 0 and xlgmii_txd_i[] = Data).

Figure 2-76 Start of Normal Frame Transmission

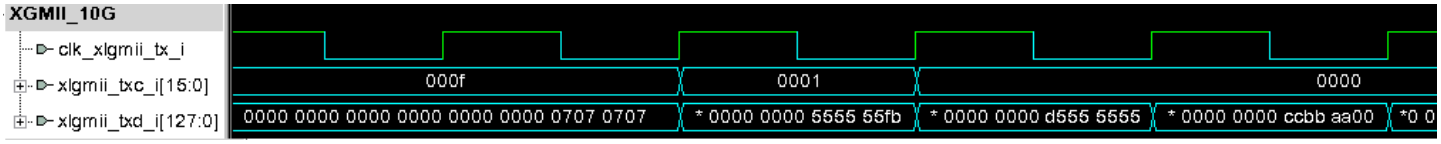


Figure 2-77 illustrates end of the normal frame transmission. The XGMII frame DATA (with xlgmii_txc_i[] = 0 and xlgmii_txd_i[] = Data), followed by XGMII EOF(xlgmii_txc_i[2] = 1 and xlgmii_txd_i[23:16] = 8'hFD), followed in turn by XGMII IDLE (with xlgmii_txc_i[] = 1 and xlgmii_txd_i[] = 8'h07). The EOF (End of Frame) can be inserted on any of the four lanes. In Figure 2-77, it is inserted on lane 2.

Figure 2-77 End of Normal Frame Transmission

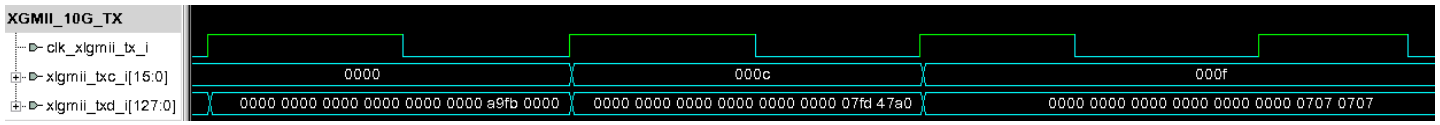


Figure 2-78 shows how an ERROR character can be inserted into a frame that is being sent. The ERROR character can be inserted by programming xlgmii_txc_i[] = 1 with xlgmii_txd_i[7:0] = 8'hFE. Similarly, any of the other XGMII control characters can be inserted in between Frame/Idle transmission.

The reception is similar to that of transmission on XGMII Receive interface (xlgmii_rxc_o[3:0] and xlgmii_rxd_o[31:0]).

Figure 2-78 ERROR Character Insertion During Frame Transmission

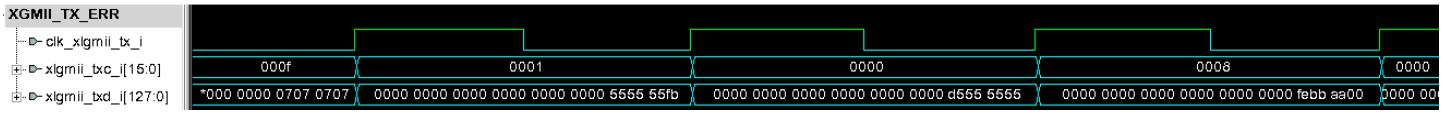


Figure 2-79 Start of Normal Frame Reception

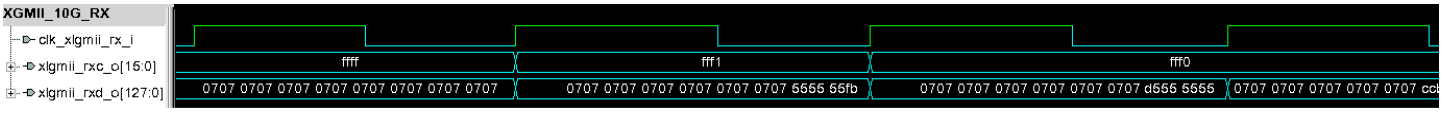
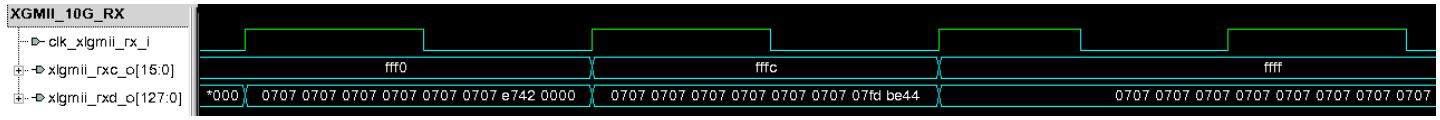


Figure 2-80 End of Normal Frame Reception



2.12.3 25G Speed Mode

The XLGMII waveform when operating in 25G speed mode is shown inFigure 2-81 and Figure 2-82. It has 128-bit data and 16- control bits and clock frequency is 390.625MHz with back pressure signal asserted on alternate clock cycles.

Figure 2-81 XLGMII 25G Transmit

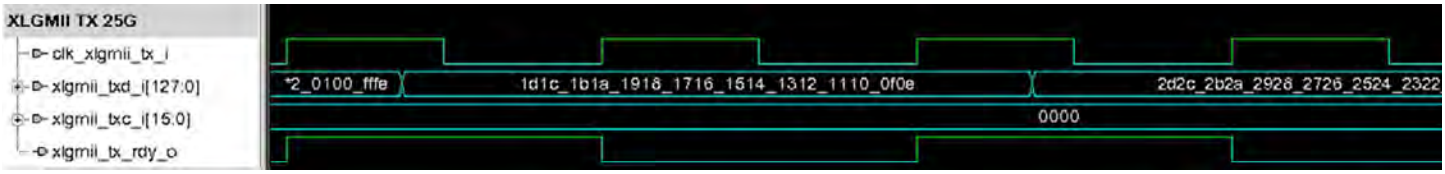


Figure 2-82 XLGMII 25G Receive



2.12.4 50G Speed Mode

The XLGMII waveform when operating in 50G speed mode is shown in Figure 2-83 and Figure 2-84. It has 128-bit data and 16- control bits and clock frequency is 390.625MHz.

Figure 2-83 XLGMII 50G Transmit

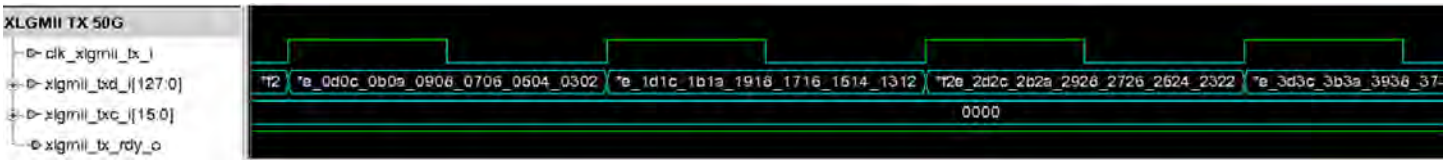
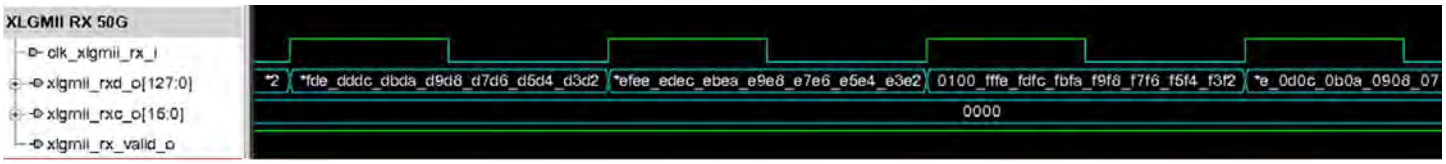


Figure 2-84 XLGMII 50G Receive



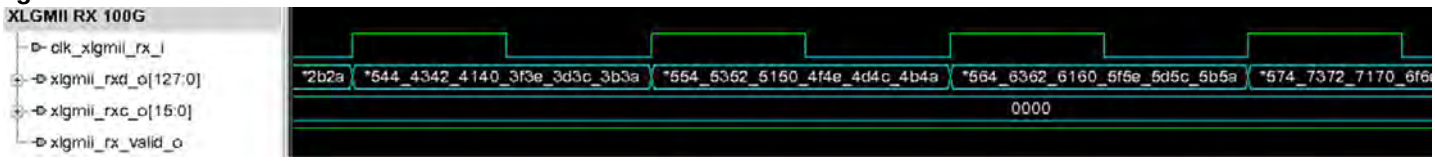
2.12.5 100G Speed Mode

The XLGMII waveform when operating in 100G speed mode is shown in Figure 2-85 and Figure 2-86. It has 128-bit data and 16- control bits and clock frequency is 781.25 MHz.

Figure 2-85 XLGMII 100G Transmit



Figure 2-86 XLGMII 100G Receive



3 Registers Overview

This section describes the register access types and the writing conventions used in field descriptions. The DWC_xlgpcs implements the standard register sets for the different MDIO Manageable Devices (MMD) as defined in Clause 45 of the IEEE 802.3 Standard. The DWC_xlgpcs additionally includes vendor-specific register sets:

- "Special Case Register Reset" on page 160
- "Read or Write of Unsupported Addresses" on page 161
- "Special Case Register Access for Multi-Protocol 12G, 16G, 25G, or 32G PHY" on page 161
- "VR_PMA_MP_12G_16G_25G_SRAM" on page 161
- "Conventions" on page 161

3.1 Special Case Register Reset

This section describes what happens when a register is accessed during a software-triggered reset, or when an address is not implemented or supported.

3.1.1 Read or Write During Software Reset

The software reset period is the duration between the following:

- The time at which the host resets any of the soft reset bits. The following bits can trigger a soft reset:
  - Bit[15] (RST field) of the SR_PMA_CTRL1, SR_PCS_CTRL1, or SR_AN_CTRL register.
  - Bit[15] (VR_RST field) of the VR_PMA_DIG_CTRL1, VR_PCS_DIG_CTRL1, or VR_AN_DIG_CTRL register.
- The time at which the soft reset bit gets self-cleared after completion of reset in all clock domains.

Note When the host programs any soft reset bit to 1 to initiate reset, the host must wait until DWC_xlgpcs clears the bit that was programmed (resets the bit to 0).

During a software reset period, the following is the behavior:

- Writes to any register are ignored.
- Reads to any register return the following:
  - If DWCXLP_SNPS_XAUI_PHY = Enabled, the following values are returned:
    - Bits[4:2] of any register indicate bits[4:2] of the VR_PCS_DIG_STS register.
    - Bit[15] of any register returns 1.
  - If DWCXLP_SNPS_XAUI_PHY = Disabled, reads to any register return 8000H.

Note Preferably, do not program the RST or VR_RST fields while LPI patterns are exchanged.

3.1.2 Read or Write During Power-Down or Power-Save Mode

To trigger

- Power-down mode, program the LPM field (bit 11) of the SR_PMA_CTRL1, SR_PCS_CTRL1 or SR_AN_CTRL register to 1.
- Power-save mode, program the PWRSV field (bit 11) of the VR_PMA_DIG_CTRL1, VR_PCS_DIG_CTRL1, or VR_AN_DIG_CTRL register to 1.

Note To initiate the power-down or power-save mode, the host must program one of these register bits to 1. When the host needs to power up DWC_xlgpcs, it must clear the bit that was programmed (reset the bit to 0).

In the power-down or power-save mode, the behavior is as follows:

- Write operations to any register are ignored.
- Read operations to any register return the following:
  - If DWCXLP_SNPS_XAUI_PHY = Enabled, the following values are returned:
    - Bits[4:2] of any register indicate bits[4:2] of the `VR_PCS_DIG_STS` register.
    - Bit[15] of any register returns 1.
  - If DWCXLP_SNPS_XAUI_PHY = Disabled, reads to any register return an undefined value.

Note Preferably, do not program the LPM or PWRSV bits while LPI patterns are exchanged.

3.2 Read or Write of Unsupported Addresses

If an address is not supported in the current configuration, the following happens:

- Writes are ignored.
- Reads return 16'h0000.

3.3 Special Case Register Access for Multi-Protocol 12G, 16G, 25G, or 32G PHY

DWC_xlgpcs generally does not allow Read or Write to its registers until the power-up or reset sequence (during hard-reset, soft-reset, power-down, or power-save) is complete. However, for configurations with Synopsys Multi-Protocol 12G, 16G, 25G, or 32G PHY, the following registers are allowed to be accessed even if the power-up or reset sequence is in progress:

- `VR_PMA_SNPS_CR_CTRL`
- `VR_PMA_SNPS_CR_ADDR`
- `VR_PMA_SNPS_CR_DATA`
- `VR_PMA_MP_12G_16G_25G_SRAM`

3.4 Conventions

This section describes the conventions used in register and bit-field descriptions.

3.4.1 Register Types

Registers are grouped into the categories shown in Table 3-1.

Table 3-1 Register Types

Prefix	Description
SR	Standard register set for PCS or PMA MMD
VR	Vendor-specific register set

Prefix	Description
AN	Standard AN MMD register set
VR AN	Vendor-specific AN MMD register set

3.4.2 Register Access

Table 3-2 displays the access types used in the description of register fields.

Table 3-2 Access Types

Access	Description
LH	Latched High bit. The default value is 0. DWC_xlgpcs programs the bit to 1 based on the described event. When this bit is programmed, it is not changed until the host reads the register.After the host reads the register, this bit indicates the current status of the event.
LL	Latched Low bit. The default value is 1. DWC_xlgpcs clears the bit to 0 based on the described event. When this bit is cleared, it does not change until the host reads the register. After the host reads the register, this bit indicates the current status of the event.
RO	Read-Only bit
RW	Read and Write bit
SC	Self-Clearing bit. The host programs this bit. DWC_xlgpcs clears this bit.
SS	Self-Setting bit. DWC_xlgpcs programs this bit. In some cases, DWC_xlgpcs clears it based on the functional requirements.
WS	Write Set. The host can program this bit only during write operations.Write of 0 is ignored.
COR	Clear On Read bit. This bit is cleared when the host reads the register containing this bit.

Table 3-3 Device Address Map for MMDs

Device Address	MMD	Implemented Configuration
00	Reserved	N/A
01	PMA or PMD	All configurations
02	WIS	N/A
03	PCS	All configurations
04	PHY XS	N/A
05	DTE XS	N/A
06	TC	N/A
07	Auto-negotiation (AN)	Configurations with the DWCXLP_CL73_AN parameter programmed to 1
08-1C	Reserved	N/A

Device Address	MMD	Implemented Configuration
1A	Vendor-specific MII_1 MMD	N/A
1B	Vendor-specific MII_2 MMD	N/A
1C	Vendor-specific MII_3 MMD	N/A
1D	Clause 22 extension	N/A
1E	Vendor-Specific 1 (Control MMD)	All configurations Note: This Control MMD is used to control other MMDs present in your configuration. This MMD is also used to enable fast simulation during gate-level simulation. For more information, see bit 4 of the SR_VSMMD_CTRL register.
1F	Vendor-Specific MII MMD (VR MII MMD)	N/A

4 Automotive Safety Features

This chapter describes the automotive safety features supported by DWC_xlgpcs and enabled by the DWCXLP_ASP parameter when you have the corresponding license.

This chapter has the following sections:

- "Error Correction Code (ECC) Protection for Memories" on page 166
- "On-Chip Data Path Parity Protection" on page 169
- "FSM Parity and Timeout Protection" on page 171
- "Interface Timeout Protection" on page 173
- "CSR Protection" on page 175

Note Automotive safety features are supported only for 25G configurations.

4.1 Error Correction Code (ECC) Protection for Memories

4.1.1 Overview

The Error Correction Code (ECC) block can correct single-bit error and detect double-bit error.

4.1.2 Description of ECC Protection for Memories

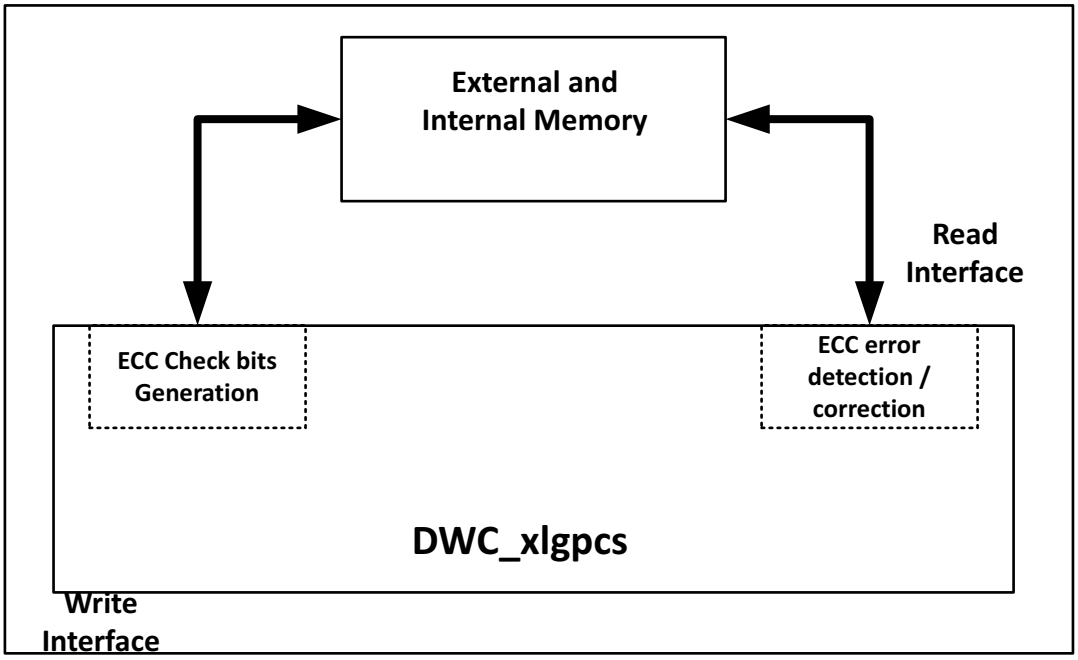
The Error Correction Code (ECC) block can correct single-bit error and detect double-bit error.

At the write interface, ECC check bits are generated by computing ECC on the contents of the data bus and respective address is appended with the data that is written to the memory.

At the read interface, ECC check bits are recomputed on the content of the read data and the respective address is compared with the received check bits in the memory.

Figure 4-1 shows the block diagram of ECC protection for memories

Figure 4-1 Block Diagram of ECC Protection for Memories



The number of ECC-check bits generated for each data block depends on the size of the data bus and address bus. Table 4-1 describes the memory bus widths, ECC code size and the generated check bits size.

Table 4-1 Memory Bus Width, ECC Bits Size

Memory Block	Data Width (Without ECC)	Address Width	ECC Check Bits	Data Width (with ECC)
BASE-R FEC/RS-FEC Error Correction Memory Instance 0 (2 options depending on configuration)	80 bits	6 bits or 7 bits (depends on configuration)	8 bits	88 bits
	66 bits	5 bits	8 bits	74 bits

Memory Block	Data Width (Without ECC)	Address Width	ECC Check Bits	Data Width (with ECC)
BASE-R FEC/RS-FEC Error Correction Memory Instance 1 (2 options depending on configuration)	80 bits	5 bits	8 bits	88 bits
	66 bits	5 bits	8 bits	74 bits
RS-FEC Error Forwarding Memory	257 bits	5 bits	10 bits	267 bits

4.1.2.1 Handling the Address Mismatch

ECC is calculated over the combination of memory data and memory address. However only data is written in the memory along with the ECC. (address is excluded).

On the read interface, the ECC is checked over the combination of memory DATA and address. When an ECC correctable error is detected in the address bit position, it is treated as an uncorrectable error. This is because data might have been read from a different location.

4.1.2.2 Diagnostic Support for the Error Management

Following statistics are provided for monitoring the error behavior on each of the memory blocks.

- Error status provided to management
  - Uncorrectable error status is reported in the `VR_PCS_SFTY_UE_INTR1` register. Separate status bits are available for each memory.
  - Correctable error status is reported in `VR_PCS_SFTY_CE_INTR` register. Separate status bits are available for each memory.
  - The most recent address location at which correctable error is detected is reported in the `VR_PCS_RSFECEF_ECC_STS`, `VR_PCS_FEC_ECC_STS{0-7}`, `VR_PCS_DSKW_ECC_STS{0-3}` registers. In addition, these registers indicate the number of ECC code-words that are corrected.
- Interrupts provided to management
  - Separate interrupts are generated for correctable and uncorrectable errors.
  - xlgpcs_sfty_ue_intr_o interrupt is generated when uncorrectable errors are detected and these errors cannot be masked. Application must reset DWC_xlgpcs IP on detecting uncorrectable errors.
  - xlgpcs_sfty_ce_intr_o interrupt is generated when correctable errors are detected.

For more details, see the "*Register Descriptions*" chapter.

4.1.2.3 ECC Error Injection Capabilities

ECC error protection in DWC_xlgpcs supports a hardware debug mode where the hardware injects error at the memory interface and reports it to the application through a safety interrupt. Failure to report Error/Safety Interrupt indicates an Error in the ECC Safety Mechanism.

To enable ECC Error Injection program the EC_EI field of the `VR_PCS_SFTY_ERR_INJ1` register. The memory in which error must be injected is selected by programming suitable values to ECC_SEL field of the `VR_ PCS_SFTY_ERR_INJ1` register.

DWC_xlgpcs supports the following types of error injection:

- 1-bit data error

- 2-bit data error

This can be controlled by programming the EI_MODE fields of the VR_PCS_SFTY_DBG_CTRL register.

Error can be introduced either in the data bits or the parity bits feeding the EC checker. Error is introduced only in the logic path feeding the checker. This error injection mode does not impact the normal data path. To control the error position (bit position) program the ERR_LOC field of the VR_PCS_SFTY_DBG_CTRL register. For the ECC parity bit calculation, both data bits and address bits are considered as 'data' field, address errors can be introduced by programming appropriate values in the ERR_LOC field. For more details, see ERR_LOC field in the "*Register Descriptions*" chapter.

4.1.3 Enabling ECC Protection for Memories

To enable the ECC support in the RTL, select the 'Add Automotive Safety Support' parameter, in the 'Specify Configuration' activity in CoreConsultant.

The ECC protection feature is enabled by default. However, you can disable it by programming the ECC_DIS bit of the VR_PCS_SFTY_DISABLE register to 1.

For details about these options, see Automotive Safety Features Parameters in the "*Parameter Descriptions*" chapter.



See the "Enabled" field of the parameter descriptions to understand the dependencies.

4.1.4 Signals Related to ECC Protection for Memories

- xlgpcs_sfty_ue_intr_o
- xlgpcs_sfty_ce_intr_o

For a description of signals related to ECC Protection for Memories, see the "*Signal Descriptions*" chapter.

4.1.5 Registers Related to ECC Protection for Memories

- VR_PCS_SFTY_UE_INTR1
- VR_PCS_SFTY_CE_INTR
- VR_PCS_FEC_ECC_STS0
- VR_PCS_FEC_ECC_STS1
- VR_PCS_FEC_ECC_STS2
- VR_PCS_FEC_ECC_STS3
- VR_PCS_FEC_ECC_STS4
- VR_PCS_FEC_ECC_STS5
- VR_PCS_FEC_ECC_STS6
- VR_PCS_FEC_ECC_STS7
- VR_PCS_RSFECEF_ECC_STS

- VR_PCS_DSKW_ECC_STS0
- VR_PCS_DSKW_ECC_STS1
- VR_PCS_DSKW_ECC_STS2
- VR_PCS_DSKW_ECC_STS3
- VR_PCS_SFTY_DISABLE
- VR_PCS_SFTY_DBG_CTRL
- VR_PCS_SFTY_ERR_INJ1

For more details, see the "*Register Descriptions*" chapter.

4.1.6 Programming Guidelines for ECC Protection for Memories

For more details about programming guidelines for ECC protection for memories, see "Programming Guidelines for ECC Protection for Memories" on page 1343.

4.2 On-Chip Data Path Parity Protection

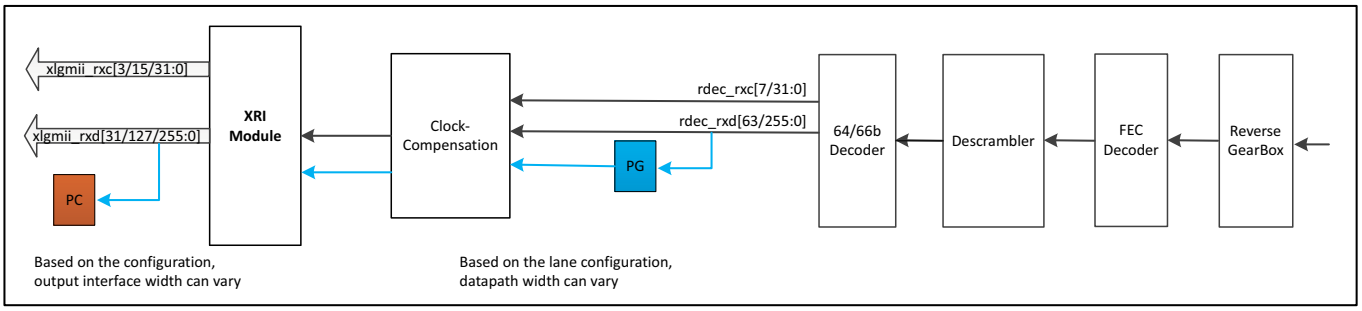
4.2.1 Overview

On-Chip Data Path Parity Protection supports protection for Transmit and Receive data paths, where feasible. For every byte of data, one-bit parity is generated. Only limited data path parity protection is implemented in DWC_xlgpcs as data gets manipulated inside various sub-modules of DWC_xlgpcs (due to encoding, decoding operations and so on). There is no parity protection at all in the transmit path. Error introduced in the DWC_xlgpcs Transmit data path is detected by the far-end MAC. Similarly, any error in the DWC_xlgpcs receiver is detected by the local end MAC.


Note DWC_xlgpcs implements an odd parity.

Figure 4-2 shows a high-level information on the Data-path parity protection scheme implemented in DWC_xlgpcs. Parity bits are computed on the 64/66b decoded data. The data and parity bits then pass through various blocks and final reach the XLGMII interface. A parity check is performed on the final data that is driven out on XLGMII Receive port.

Figure 4-2 Data Path Parity Generator and Checker



4.2.2 Parity Error Injection

The data path parity protection feature supports a hardware debug mode where the hardware injects an error and upon detection of an error, reports it to the application through a safety interrupt. Failure to report error/safety interrupt indicates an error in the data path parity safety mechanism.

4.2.3 Interrupt for Parity Error

DWC_xlgpcs generates an uncorrectable interrupt, xlgpcs_sfty_ue_intr_o to indicate to the application if any parity error is detected. In addition, it programs the DPP_INT field of the `VR_PCS_SFTY_UE_INTR0` register.

Note A parity error is uncorrectable, and in most cases the operation of the DWC_xlgpcs is not guaranteed after the parity error. A hardware reset is required to safely recover from the parity error reported by the assertion of xlgpcs_sfty_ue_intr_o signal.

4.2.4 Enabling On-Chip Data Path Parity Protection

To enable the On-chip data path parity protection support in the RTL, select the 'Add Automotive Safety Support' parameter, in the 'Specify Configuration' activity in CoreConsultant.

DWC_xlgpcs enables the data path parity protection by default. You can disable this feature by programming the DPP_DIS bit of the VR_PCS_SFTY_DISABLE to 1.

For details about these options, see Automotive Safety Features Parameters in the "*Parameter Descriptions*" chapter.

Note See the "Enabled" field of the parameter descriptions to understand the dependencies.

4.2.5 Signals Related to On-Chip Data Path Parity Protection

- xlgpcs_sfty_ue_intr_o

For a description of signals related to On-Chip Data Path Protection, see the "*Signal Descriptions*" chapter.

4.2.6 Registers Related to On-Chip Data Path Parity Protection

- VR_PCS_SFTY_UE_INTR0
- VR_PCS_SFTY_ERR_INJ0
- VR_PCS_SFTY_DBG_CTRL
- VR_PCS_SFTY_DISABLE

For more details, see the "*Register Descriptions*" chapter.

4.2.7 Programming Guidelines for On-Chip Data Path Parity Protection

For details about programming guidelines for on-chip data path parity protection, see "Programming Guidelines for On-chip Data Path Parity Protection" on page 1344.

4.3 FSM Parity and Timeout Protection

4.3.1 Overview

The FSM protection feature supports FSM parity and time out protection.

4.3.1.1 Description of FSM State Parity Protection

DWC_xlgpcs implements one-hot encoding scheme for its various internal FSMs. A one-hot encoding check is implemented on all the FSM state register bits. FSMs are monitored for every clock, after the reset is de-asserted. When a bit flips due to transient errors or permanent faults, the erroneous FSM is set to its default state and an uncorrectable error is indicated.

FSM state parity protection is enabled by default. When more than one bit in an FSM state variable becomes high, a parity error is detected and DWC_xlgpcs asserts the xlgpcs_sfty_ue_intr_o interrupt port.

FSM parity error information is provided as interrupt-status bits in the VR_PCS_SFTY_UE_INTR0 register. Individual FSM status information is not provided; instead only status information is provided 'per clock-domain'. FSM parity error status bits are provided for the following clock-domains:

- Transmit clock domain
- Receive clock domain
- CSR clock domain
- EEE clock domain

Though this protection feature checks for one-hot encoding, for simplicity, this is referred to as a parity check.

Error Injection mode is also supported for FSM parity error check. An additional register bit is added to each FSM state register bus. When error injection mode is enabled, this additional register bit (called as parity bit) gets corrupted. Program the Error Injection enable for the respective clock domain denoted by [3:0] bits of VR_PCS_SFTY_ERR_INJ0 register.

Note The FSM Parity error is classified as non-recoverable to recover. Therefore, to recover from this erroneous state, host must hard-reset the IP.

4.3.1.2 Description of FSM Timeout Protection

FSM timeout protection feature supports a mechanism to ensure that all the FSMs complete a transaction and returns to a known completion state within a programmed timeout value. If an FSM is hung in a state for a duration longer than expected, it indicates to the application layer by asserting xlgpcs_sfty_ue_intr_o interrupt.

To reduce area overhead, individual FSM timers are not implemented; instead generic 'tic' timers are implemented. Application must program SFTY_1US_MULT field of the VR_PCS_SFTY_TMR_CTRL register with a value that indicates the number of CSR clock cycles required to generate a 1 microsecond tic. This microsecond tic is internally used to generate various timeout values. DWC_xlgpcs allocates appropriate timeout values for various FSMs. The timeout values are judiciously determined based on the design intent and based on operating speed mode (contact Synopsys Support in case you require exact details of timeout values associated with various FSMs). However, DWC_xlgpcs provides an option to scale the timers by a factor of 0.1 or 10 if desired. This can be performed by programming suitable values to FSM_TO_SEL field of the VR_PCS_SFTY_TMR_CTRL register.

At every timer tick, all the FSMs are monitored for being in active state (non-IDLE state, which indicates FSM is active processing the data or handshakes) and an FSM active flag bit is programmed. The active flag bit is implemented for every FSM. Whenever the FSM reaches a safe states or safe conditions (for FSMs which does not have any safe state), the active flag for that FSM is reset. At the subsequent timer tick, the active flag bit is monitored and if it still programmed, it is treated as a timeout error. This process of setting the flag and checking at subsequent timer tick is repeated for every tic.

Contact Synopsys Support for details of the safe states or safe conditions associated with each FSM.

FSM parity error information is provided as interrupt-status bits in the VR_PCS_SFTY_UE_INTR0 register. Individual FSM status information is not provided; instead only status information is provided ‘per clock-domain’. FSM parity error status bits are provided for the following clock-domains:

- Transmit clock domain
- Receive clock domain
- CSR clock domain
- EEE clock domain

Timeout error injection feature is also supported in DWC_xlgpcs to verify the sanity of timeout feature. Per clock-domain timeout errors can be introduced by programming bits [8:5] of the VR_PCS_SFTY_ERR_INJ0.



- Based on the time at which the FSM enters the active state relative to the timer ticks the timeout period could be any value between the programmed timeout and 2x programmed timeout.
- The FSM timeout error is classified as non-recoverable error. Therefore, to recover from this erroneous state, the host must hard-reset the IP.
- If the FASTSIM bit is programmed to 1, FSM timeout interrupts can be generated incorrectly. This is because timeout values are deliberately shortened to small values for the internal testing of the FSM timeout feature of various FSMs within the Synopsys verification environment. In a real use case, a user never programs the FASTSIM bit to 1.

4.3.1.3 Enabling FSM Parity and Timeout Protection

To enable the FSM protection feature in the RTL, select the 'Add Automotive Safety Support' parameter, in the 'Specify Configuration' activity in CoreConsultant.

By default, FSM timeout feature is enabled. However, you can disable this feature by programming FT_DIS bit of VR_PCS_SFTY_DISABLE to 1.

By default, FSM parity protection feature is enabled. However, you can disable this feature by programming FP_DIS bit of VR_PCS_SFTY_DISABLE to 1.

For details about these options, see Automotive Safety Features Parameters in the "*Parameter Descriptions*" chapter.

Note See the "Enabled" field of the parameter descriptions to understand the dependencies.

4.3.1.4 Signals Related to FSM Parity and Timeout Protection

- xlgpcs_stfy_ue_intr_o

For a description of signals related to FSM Parity and Timeout Protection, see the "*Signal Descriptions*" chapter.

4.3.1.5 Registers Related to FSM Parity and Timeout Protection

- `VR_PCS_SFTY_UE_INTR0`
- `VR_PCS_SFTY_ERR_INJ0`
- `VR_PCS_SFTY_TMR_CTRL`
- `VR_PCS_SFTY_DISABLE`

For more details, see the "*Register Descriptions*" chapter.

4.3.1.6 Programming Guidelines for FSM Parity and Timeout Protection

For details about programming guidelines for FSM protection, see "Programming Guidelines for FSM Parity Protection" on page 1344.

4.4 Interface Timeout Protection

4.4.1 Overview

This feature provides timeout protection to the various Interfaces of DWC_xlgpcs.

4.4.2 Description of Interface Timeout Protection

All the interfaces which has the handshake mechanism in DWC_xlgpcs are monitored for potential hangs due to the lack of response for initiated requests. After the request is initiated the response arrival interval is monitored. If the response does not arrive within a programmed time (controlled by IFT_SEL bit of VR_PCS_SFTY_TMR_CTRL) the timeout is triggered, using similar trigger generation as explained in the Section FSM protection.

The hardware does not attempt to recover from the hang and relies on software to take appropriate corrective action, by resetting DWC_xlgpcs.

The timeout status is provided in the [11:10] field of the VR_PCS_SFTY_UE_INTR0 register. The safety interrupt, xlgpcs_sfty_ue_intr_o is asserted when timeout status is set.

Note

- Based on the time at which an interface transfer is initiated relative to the timer ticks the timeout period could be any value between the programmed timeout and 2x programmed timeout.
- Interface timeout is classified as non-recoverable error. Therefore, to recover from this erroneous state, host must hard-reset the IP.

4.4.3 PHY (CR) Master Interface

Synopsys PHYs provide a CR port interface to allow the application to access its internal registers. To access a PHY registers, DWC_xlgpcs drives the address/data on the corresponding buses and asserts the read/write enable signals. The corresponds responds with a ‘cr_ack’ to indicate a successful completion of the read/write operation. Normally the ‘cr_ack’ is asserted in a few clock cycles. If the acknowledgment from the PHY, is not generated within the programmed timeout, DWC_xlgpcs generates a timeout interrupt. This feature is available only for Synopsys PHY configurations. The Timeout status is reflected in the PHY_IF_INT field of the VR_PCS_SFTY_UE_INTR0 register.

Note This error is an uncorrectable error. Application must issue a hard-reset to facilitate a recovery.

4.4.4 APB3 Interface

DWC_xlgpcs supports a APB3 interface to allow access of its internal registers. When the application makes a write or read requests, it completes the transaction and responds with an acknowledgment within a few clock cycles.

If DWC_xlgpcs fails to generate the 'ack' within the programmed interval, an interface timeout error is flagged.

Note To recover for APB3 interface timeout, host must reset the APB3 master device in addition to resetting the DWC_xlgpcs IP

4.4.5 Enabling Interface Timeout Protection

To enable the Interface timeout protection feature in the RTL, select the 'Add Automotive Safety Support' parameter, in the 'Specify Configuration' activity in CoreConsultant.

By default, interface timeout protection feature is enabled. However, it can be disabled by programming the IFT_DIS bit of VR_PCS_SFTY_DISABLE to 1.

4.4.6 Signals Related to Interface Timeout Protection

Following is the main signal related to interface timeout protection feature:

- xlgpcs_sfty_ue_intr_o

For a description of signals related to Interface Timeout Protection, see the "*Signal Descriptions*" chapter.

4.4.7 Registers Related to Interface Timeout Protection

Following are the registers related to interface timeout protection feature:

- `VR_PCS_SFTY_UE_INTR0`
- `VR_ PCS_SFTY_ERR_INJ0`
- `VR_PCS_SFTY_TMR_CTRL`
- `VR_ PCS_SFTY_DISABLE`

For more details, see the "*Register Descriptions*" chapter.

4.4.8 Programming Guidelines for Interface Timeout Protection

See "Programming Guidelines for Interface Timeout Protection" on page 1346 for details.

4.5 CSR Protection

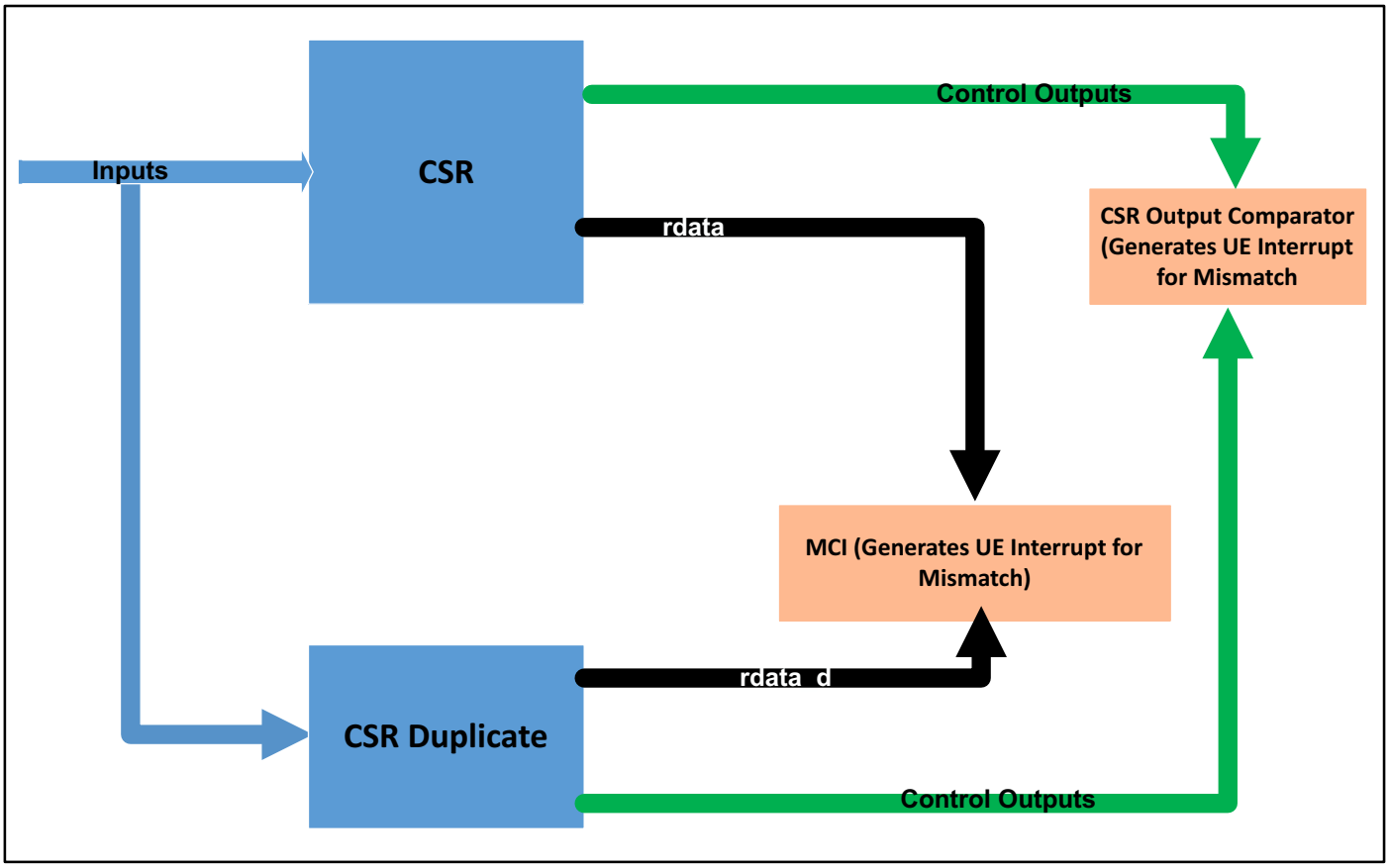
4.5.1 Overview

DWC_xlgpcs CSR defines several registers which control the operating mode of DWC_xlgpcs. The status information from various modules are provided in status registers.

4.5.2 Description of CSR Protection

Figure 4-3 shows the block diagram of CSR protection.

Figure 4-3 CSR Protection Block Diagram



4.5.2.1 Replication of Registers

CSR registers control the transmitter and receiver operations of DWC_xlgpcs. Any accidental bit-flip to the configuration register bit can alter the operating mode of the IP. To prevent such mishaps, DWC_xlgpcs replicates the CSR module. A comparator continuously checks the output ports of both, CSR module and its replica. DWC_xlgpcs asserts the xlgpcs_sfty_ue_intr_o interrupt if any discrepancy is detected.

For status register, when the application tries to read the register, it must get the correct value. To ensure correct value is read by the application, when DWC_xlgpcs receives a read request from the application, the final output of CSR block (rdata) and the data read from CSR Duplicate (rdata_d) are checked.

4.5.3 Enabling CSR Protection

To enable the CSR protection feature in the RTL, select the 'Add Automotive Safety Support' parameter, in the 'Specify Configuration' activity in CoreConsultant.

DWC_xlgpcs enables the CSR protection feature by default. You can disable this feature by programming the CSRP_DIS bit of the VR_PCS_SFTY_DISABLE to 1. For more details, see the "*Parameter Descriptions*" chapter.

4.5.4 Signals Related to CSR Protection

Following is one of the main signal related to CSR protection.

- xlgpcs_sfty_ue_intr_o

For a description of signals related to CSR Protection, see the "*Signal Descriptions*" chapter.

4.5.5 Registers Related to CSR Protection

- `VR_PCS_SFTY_UE_INTR0`
- `VR_ PCS_SFTY_ERR_INJ0`
- `VR_ PCS_SFTY_DISABLE`

For more details, see the "*Register Descriptions*" chapter.

4.5.6 Programming Guidelines for CSR Protection

See "Programming Guidelines for CSR Protection" on page 1346 for details.

5 Parameter Descriptions

This chapter details all the configuration parameters. **You can use the coreConsultant GUI configuration reports to determine the actual configured state of the controller.** Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>)** that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

The parameter descriptions in this chapter include the **Enabled:** attribute which indicates the values required to be set on other parameters before you can change the value of this parameter.

These tables define all of the configuration options for this component.

- Features / System Configurations Settings on page 180
- Features / PHY Interface on page 185
- Features / MAC Interface on page 188
- Registers / IEEE / PORT ID Defaults in CSR on page 189

5.1 Features / System Configurations Settings Parameters

Table 5-1 Features / System Configurations Settings Parameters

Label	Description
Mode of Operation	Configures the mode of operation of the DWC_xlgpcs. ■ 25G: RTL configuration supports 25G and 10G modes. ■ 25G-50G: RTL configuration supports 50G, 25G, and 10G modes. If 'Single-lane Only configuration' is not selected, the configuration additionally supports 40G mode. ■ 40G: RTL configuration supports 40G and 10G modes. ■ 100G: RTL configuration supports 100G, 50G, 40G, 25G, and 10G modes. **Values:** ■ 25G (0) ■ 25G-50G (1) ■ 40G (2) ■ 100G (3) **Default Value:** 40G **Enabled:** Always **Parameter Name:** DWCXLP_MAIN_MODE
Single-lane Only configuration	Configures the RTL to a single-lane Only Configuration. ■ If 'Mode of Operation' is chosen as '25G-50G' and if this option is selected, the selected RTL configuration will be a single-lane configuration supporting 50Gx1 (PAM4), 25Gx1 (NRZ) and 10Gx1(NRZ) modes. ■ If 'Mode of Operation' is chosen as '100G' and if this option is selected, the selected RTL configuration will be a single-lane configuration supporting 100Gx1 (PAM4), 50Gx1 (PAM4), 25Gx1 (NRZ) and 10Gx1(NRZ) modes. **Values:** 0, 1 **Default Value:** DWCXLP_MAIN_MODE==0 **Enabled:** (DWCXLP_MAIN_MODE == 1)||(DWCXLP_MAIN_MODE == 3) **Parameter Name:** DWCXLP_SINGLE_LANE
Add PAM4 Mode Support	Adds Support for PAM4 Mode of operation as per IEEE802.3cd (50Gx1 or 50Gx2) or IEEE802.3ck specification (100Gx1). **Values:** 0, 1 **Default Value:** DWCXLP_50G_25G||DWCXLP_100G_1LANE **Enabled:** (DWCXLP_MAIN_MODE == 1||DWCXLP_MAIN_MODE == 3)&&!DWCXLP_SINGLE_LANE **Parameter Name:** DWCXLP_PAM4
Add CL73 Autoneg Support	Enable CL73 Auto-negotiation. **Values:** 0, 1 **Default Value:** !(DWCXLP_50G_25G||DWCXLP_100G_1LANE) **Enabled:** Always **Parameter Name:** DWCXLP_CL73_AN

Table 5-1 Features / System Configurations Settings Parameters (Continued)

Label	Description
Support Alternate Autonegotiation flow	Adds Support for Alternate Clause 73 Auto-negotiation flow. As per this alternate flow, AN is performed only once with serdes configured to 10G NRZ mode. Switch to the negotiated speed is initiated in AN_GOOD_CHECK state without restarting the auto-negotiation. If this parameter is enabled, vendor-specific parallel-detect feature will work only for 10G NRZ mode. If this parameter is not selected, the legacy auto-negotiation flow, which involves two rounds of auto-negotiation will be supported and the vendor-specific parallel-detect feature can work in all the supported modes. User selection of this parameter is not allowed if the chosen configuration has PAM4 support or MDIO interface or Link-training support. **Values:** 0, 1 **Default Value:** DWCXLP_PAM4&&DWCXLP_CL73_AN **Enabled:** !DWCXLP_PAM4&&DWCXLP_CL73_AN&&!DWCXLP_SNPS_XAUI_PHY&&!DWCXLP_MDIO_INTF **Parameter Name:** DWCXLP_CL73_ALT
Add Timestamping Accuracy Support	Enables support for IEEE802.3cx timestamping feature. **Values:** 0, 1 **Default Value:** 0 **Enabled:** DWCXLP_25G_ONLY **Parameter Name:** DWCXLP_TS_EN
Management Interface	Specifies the serial Management Data Input/Output (MDIO) interface, or APB3 interface (3 types) with the Control and Status registers. For 'APB3 Interface (legacy)' interface, DWC_xlgpcs registers are 16-bit wide and addresses of registers are multiples of 1. This is the legacy APB3 interface supported till 2.10a. For 'APB3 Interface (16-bit register)' interface, DWC_xlgpcs registers are 16-bit wide but addresses of registers are multiples of 2. For 'APB3 Interface (32-bit register)' interface, DWC_xlgpcs registers are 32-bit wide but addresses of registers are multiples of 4. **Values:** ■ MDIO Interface (0) ■ MicroController Interface (1) ■ APB3 Interface (legacy) (2) ■ APB3 Interface (16-bit register) (3) ■ APB3 Interface (32-bit register) (4) **Default Value:** APB3 Interface (legacy) **Enabled:** Always **Parameter Name:** DWCXLP_CSR_INTERFACE

Table 5-1 Features / System Configurations Settings Parameters (Continued)

Label	Description
Enable Indirect Register Access	Enables indirect access to registers in the DWC_xlgpcs. By default, each DWC_xlgpcs register is accessible through 21-bit/22-bit(APB_16)/23-bit(APB3_32) wide address. By enabling this mode, the upper 13-bit of the address can be written to a pre-defined register so that only 8-bit or 9-bit (APB_16) or 10-bit (APB3_32) address bus is used for all register accesses in the DWC_xlgpcs. **Values:** 0, 1 **Default Value:** DWCXLP_CSR_INTERFACE!=0 **Enabled:** DWCXLP_CSR_INTERFACE!=0 **Parameter Name:** DWCXLP_INDIRECT_ACCESS
Add CL74 Forward Error Correction Support	Enables the support for the CL74 Forward Error Correction (FEC) sub layer embedded in the PCS. This FEC can detect and correct single burst error up to 11 bits. This feature is useful for erroneous channel. The CL74 FEC will not be used in 100G Speed mode. **Values:** 0, 1 **Default Value:** 0 **Enabled:** Always **Parameter Name:** DWCXLP_FEC_EN
Add CL74 FEC Error Forwarding Support	Enables the CL74 FEC block to forward error information to the Block Synchronization module. This option adds more design latency and hardware in the Receive path. When you enable this option, if there are uncorrectable errors, then the error reporting ability is accurate on the XLGMII receive interface. **Values:** 0, 1 **Default Value:** DWCXLP_FEC_EN **Enabled:** DWCXLP_FEC_EN&&!DWCXLP_RSFEC_EN **Parameter Name:** DWCXLP_FEC_ERROR_FWD
Add (528,514) RS FEC Error Correction Support	Enables the support for the Reed-Solomon Forward Error Correction (FEC) RS(528,514) sub layer as per IEEE 802.3 clause-91/clause 108 embedded in the PCS for serial lane speeds at 25Gbps. This feature is not applicable for 10G/20G/40G mode of operation. RS-FEC Error-forwarding feature also gets added when you select this option **Values:** 0, 1 **Default Value:** DWCXLP_50G_25G||DWCXLP_100G_1LANE **Enabled:** (DWCXLP_MAIN_MODE != 2)&&!(DWCXLP_50G_25G||DWCXLP_100G_1LANE) **Parameter Name:** DWCXLP_RSFEC_EN
Add (544,514) RS FEC Error Correction Support	Enables the support for the Reed-Solomon Forward Error Correction (FEC) RS(544,514) sub layer along with Error-Forwarding Support. This feature is applicable for 100Gx1, 50Gx1 and 50Gx2 modes. **Values:** 0, 1 **Default Value:** DWCXLP_PAM4 **Enabled:** 0 **Parameter Name:** DWCXLP_RS544

Table 5-1 Features / System Configurations Settings Parameters (Continued)

Label	Description
Add Interleaved RS(544,514) FEC Support	Enables the support for interleaved RS(544,514) FEC as per IEEE802.3ck specification for 100Gx1 mode. **Values:** 0, 1 **Default Value:** 0 **Enabled:** DWCXLP_100G_1LANE||DWCXLP_ADD_100G_1LANE **Parameter Name:** DWCXLP_IRSFEC
Add Link Training Support	Enables the support for Link training protocol based on Clause-72 and Clause-92/Clause-93 in the PCS layer. Clause-92/93 based link training are not applicable for 10G/40G mode of operation. For PAM4 configurations, Link training is performed by the PHY and not by DWC_xlgpcs. **Values:** 0, 1 **Default Value:** !DWCXLP_RS544 **Enabled:** !DWCXLP_RS544 **Parameter Name:** DWCXLP_LNK_TRAIN_EN
Add Clock Comp support	Enables the support for the Clock compensation operation to account for ppm differences between local and recovered clock. **Values:** 0, 1 **Default Value:** 1 **Enabled:** Always **Parameter Name:** DWCXLP_CLK_CMP_EN
Add SCC Align Support	Alignment of Start control character of frame on lane0/lane8 on xlgmii rx interface for 25g speed mode. **Values:** 0, 1 **Default Value:** 0 **Enabled:** DWCXLP_MAIN_MODE != 2 **Parameter Name:** DWCXLP_SCC_ALGN_EN
Add EEE Support	Enables support for Energy Efficient Ethernet (EEE). **Values:** 0, 1 **Default Value:** 0 **Enabled:** (DWCXLP_E56==0)&&(DWCXLP_E112==0)&&!DWCXLP_50G_25G&&!DWCXLP_100G_1LANE&&!DWCXLP_KR4_50G **Parameter Name:** DWCXLP_EEE_EN
Add Automotive Safety Support	Enables support for Automotive Safety features like FSM Parity, FSM Time out, Data Path Parity, ECC for Memory and Interface Time Out. **Values:** 0, 1 **Default Value:** 0 **Enabled:** DWCXLP_MAIN_MODE==0 **Parameter Name:** DWCXLP_ASP

Table 5-1 Features / System Configurations Settings Parameters (Continued)

Label	Description
Add ECC protection for memories	Enables support for ECC protection of memories. **Values:** 0, 1 **Default Value:** DWCXLP_ASP&&(DWCXLP_RSFEC_EN||DWCXLP_FEC_EN) **Enabled:** !DWCXLP_ASP&&(DWCXLP_RSFEC_EN||DWCXLP_FEC_EN)&&(DWCXLP_MAIN_MODE==0) **Parameter Name:** DWCXLP_ADD_ECC_PROT
Add Debug Bus	Enables support for Output Debug Bus. **Values:** 0, 1 **Default Value:** 0 **Enabled:** Always **Parameter Name:** DWCXLP_DBG_BUS

5.2 Features / PHY Interface Parameters

Table 5-2 Features / PHY Interface Parameters

Label	Description
PHY Receives LSb First	Indicates that the PHY first receives the least significant bit of the rpcs_rx_data{lane}_i[rpcs_phw:0]. **Values:** 0, 1 **Default Value:** DWCXLP_E56||DWCXLP_E112 **Enabled:** !DWCXLP_SNPS_XAUI_PHY&&!(DWCXLP_E56||DWCXLP_E112) **Parameter Name:** DWCXLP_RX_LSB_FIRST
PHY Transmits LSb First	Indicates that the PHY first transmits the least significant bit of rpcs_tx_data{lane}_o[rpcs_phw:0]. **Values:** 0, 1 **Default Value:** DWCXLP_E56||DWCXLP_E112 **Enabled:** !DWCXLP_SNPS_XAUI_PHY&&!(DWCXLP_E56||DWCXLP_E112) **Parameter Name:** DWCXLP_TX_LSB_FIRST
PHY Type	Select the Phy Type. Appropriate logic and I/O signals gets added to DWC_xlgpcs based on the selected PHY. **Values:** ■ Non-Synopsys PHY (0) ■ Multi-protocol 25G (1) ■ Multi-protocol 16G (2) ■ Multi-protocol 32G (3) ■ Multi-protocol 12G (4) ■ Enterprise 56G (5) ■ Enterprise 112G (6) **Default Value:** Non-Synopsys PHY **Enabled:** Always **Parameter Name:** DWCXLP_PHY_TYPE
Reference Clock Frequency	Indicates the reference clock frequency for Synopsys PHY to program the default MPLL parameters of PHY during power up. This parameter has no impact on DWC_xlgpcs RTL when 'Enterprise 56G' or 'Enterprise 112G" PHY is selected **Values:** ■ 156.25 MHz (0) ■ 78.125MHz (1) ■ 312.5MHz (2) **Default Value:** 156.25 MHz **Enabled:** DWCXLP_SNPS_XAUI_PHY **Parameter Name:** DWCXLP_REF_CLK_FREQ

Table 5-2 Features / PHY Interface Parameters (Continued)

Label	Description
Reference Clock Source	Selects the source of reference clock for Synopsys PHY. This parameter has no impact on DWC_xlgpcs RTL when 'Enterprise 56G' or 'Enterprise 112G" PHY is selected **Values:** ■ Pad Clk (0) ■ Alt Clk (1) ■ Alt 1 Clk (2) **Default Value:** Pad Clk **Enabled:** DWCXLP_SNPS_XAUI_PHY **Parameter Name:** DWCXLP_REF_CLK_SRC
Common Context Select Value	Default value of Common context select value that is driven to PHY. Set this to appropriate value for the default operating speed mode of DWC_xlgpcs as per the PHY ROM/SRAM image used **Values:** 0x0, ..., 0xff **Default Value:** (DWCXLP_KR4_KR||!DWCXLP_E32) ? 0x00 : 0x01 **Enabled:** DWCXLP_E32 **Parameter Name:** DWCXLP_CM_CNTX_SEL
Rx Context Select Value	Default value of Rx context select value that is driven to PHY. Set this to appropriate value for the default operating speed mode of DWC_xlgpcs as per the PHY ROM/SRAM image used **Values:** 0x0, ..., 0xff **Default Value:** (DWCXLP_KR4_KR||!DWCXLP_E32) ? 0x00 : 0x01 **Enabled:** DWCXLP_E32 **Parameter Name:** DWCXLP_RX_CNTX_SEL
Tx Context Select Value	Default value of Tx context select value that is driven to PHY. Set this to appropriate value for the default operating speed mode of DWC_xlgpcs as per the PHY ROM/SRAM image used **Values:** 0x0, ..., 0xff **Default Value:** (DWCXLP_KR4_KR||!DWCXLP_E32) ? 0x00 : 0x01 **Enabled:** DWCXLP_E32 **Parameter Name:** DWCXLP_TX_CNTX_SEL
Add signals to indicate PHY Ready	PHY Ready input signal. This will enable inclusion of input ports tx_phy_rdy_i and rx_phy_rdy_i, which when asserted indicates that all clocks from PHY are stable and normal operation can be started. **Values:** 0, 1 **Default Value:** DWCXLP_E56||DWCXLP_E112||DWCXLP_CL73_ALT **Enabled:** !DWCXLP_SNPS_XAUI_PHY&&!DWCXLP_E56&&!DWCXLP_E112&&!DWCXLP_CL73_ALT **Parameter Name:** DWCXLP_ADD_PHY_RDY

Table 5-2 Features / PHY Interface Parameters (Continued)

Label	Description
Add per-lane XSBI Tx clock	Enables Support for Separate XSBI Tx clock for each lane. This feature is enabled only if the SerDes does not take a transmit clock per-lane as input, but instead samples the input tx{lane} data with an internal per-lane clock. Note that, even though DWC_xlgpcs allows support of separate XSBI tx clock per-lane, it cannot tolerate any ppm difference in frequency between the various XSBI Tx clocks. It can only tolerate differences in jitter. Moreover, the internal datapath still operates with a single clk_rpcs_tx_i. **Values:** 0, 1 **Default Value:** 0 **Enabled:** (DWCXLP_R_SRDS_INTF!=3)&&!DWCXLP_SINGLE_LANE&&(DWCXLP_PHY_TYPE==0) **Parameter Name:** DWCXLP_XSBI_TX_CLK_PER_LN
RPCS PHY Interface Width in NRZ Mode	Specifies the data width of the data interface (rpcs_tx_data{lane}_o[rpcs_phw:0] and rpcs_rx_data{lane}_i[rpcs_phw:0]) to the BASE-R PHY when operating in NRZ mode. When operating in PAM4 modes, effective data-width will be double the chosen width, as 2 lanes of DWC_xlgpcs get connected to a single lane of the PHY. **Values:** ■ 16 (0) ■ 32 (1) ■ 33 (2) ■ 66 (3) ■ 20 (4) ■ 64 (5) ■ 80 (6) ■ 40 (7) **Default Value:** DWCXLP_PHY_TYPE==1) ? 0 : ((DWCXLP_MAIN_MODE!=2) ? 7:0 **Enabled:** Always **Parameter Name:** DWCXLP_R_SRDS_INTF
Enable GPIO	Enables the General Purpose Inputs/Outputs (GPIO). This option can be enabled if you want some side-band signals for control/status purpose. **Values:** 0, 1 **Default Value:** DWCXLP_E56 **Enabled:** !DWCXLP_E56 **Parameter Name:** DWCXLP_GPIO_EN

5.3 Features / MAC Interface Parameters

Table 5-3 Features / MAC Interface Parameters

Label	Description
XLGMII interface data-width	Specifies the data width of the MAC interface (xlgmii_txd_i and xlgmii_rxd_o) as 128-bit or 64-bit or 32-bit. This selection is available only for 25G_Only configurations. When the chosen option is '64', both 25G and 10G modes operate with 64-bit wide interface. When the chosen option is '32', both 25G and 10G modes operate with 32-bit wide interface. If the chosen option is '128', a 128-bit wide interface is used for all modes except 10G mode. In 10G mode, only lower 32-bits of the 128-bit interface will be active. **Values:** ■ 128 (0) ■ 64 (1) ■ 32 (2) **Default Value:** 128 **Enabled:** DWCXLP_MAIN_MODE == 0 **Parameter Name:** DWCXLP_XLGMII_INTF_WIDTH

5.4 Registers / IEEE / PORT ID Defaults in CSR Parameters

Table 5-4 Registers / IEEE / PORT ID Defaults in CSR Parameters

Label	Description
Support to Modify Default Values IEEE Registers	Makes the IEEE Device or Package Identifier registers programmable through Vendor-Specific Control MMD. **Values:** 0, 1 **Default Value:** 1 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_REG_WR_SUPPORT
XS/PCS Device OUI (24-bit)	Configures the default value of the 24-bit IEEE XGXS/PCS Device Organizationally Unique Identifier. **Values:** 0x0, ..., 0xffffff **Default Value:** 0xcda679 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_DEV_OUI
XS/PCS Device Model Number (6-bit)	Configures the default value of the 6-bit IEEE XGXS/PCS Device Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x2d **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_DEV_MODEL_NO
XS/PCS Device Revision Number (4-bit)	Configures the default value of the 4-bit IEEE XGXS/PCS Device Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_DEV_REV_NO
Package OUI (24-bit)	Configures the default value of the 24-bit IEEE Package Organizationally Unique Identifier. **Values:** 0x0, ..., 0xffffff **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_PKG_OUI
Package Model Number (6-bit)	Configures the default value of the 6-bit IEEE Package Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_PKG_MODEL_NO

Table 5-4 Registers / IEEE / PORT ID Defaults in CSR Parameters (Continued)

Label	Description
Package Revision Number (4-bit)	Configures the default value of the 4-bit IEEE Package Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_PKG_REV_NO
PMA Device OUI (24-bit)	Configures the default value of the 24-bit IEEE PMA Device Organizationally Unique Identifier. **Values:** 0x0, ..., 0xffffff **Default Value:** 0x0 **Enabled:** DWCXLP_PMA_MMD == 1 **Parameter Name:** DWCXLP_IEEE_PMA_DEV_OUI
PMA Device Model Number (6-bit)	Configures the default value of the 6-bit IEEE PMA Device Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x0 **Enabled:** DWCXLP_PMA_MMD == 1 **Parameter Name:** DWCXLP_IEEE_PMA_DEV_MODEL_NO
PMA Device Revision Number (4-bit)	Configures the default value of the 4-bit IEEE PMA Device Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** DWCXLP_PMA_MMD == 1 **Parameter Name:** DWCXLP_IEEE_PMA_DEV_REV_NO
AN Device OUI (24-bit)	Configures the default value of 24-bit IEEE AN Device Organizationally Unique Identifier (OUI). **Values:** 0x0, ..., 0xffffff **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_AN_DEV_OUI
AN Device Model Number (6-bit)	Configures the default value of 6-bit IEEE AN Device Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_AN_DEV_MODEL_NO
AN Device Revision Number (4-bit)	Configures the default value of 4-bit IEEE AN Device Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_AN_DEV_REV_NO

Table 5-4 Registers / IEEE / PORT ID Defaults in CSR Parameters (Continued)

Label	Description
Vendor-Specific MMD1 Device OUI (24-bit)	Configures the default value of 24-bit IEEE Vendor-Specific MMD1 Device Organizationally Unique Identifier (OUI). **Values:** 0x0, ..., 0xffffff **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_VS_DEV_OUI
Vendor-Specific MMD1 Device Model Number (6-bit)	Configures the default value of 6-bit IEEE Vendor-Specific MMD1 Device Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_VS_DEV_MODEL_NO
Vendor Specific MMD1 Device Revision Number (4-bit)	Configures the default value of 4-bit IEEE Vendor-Specific MMD1 Device Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** DWCXLP_IEEE_VS_DEV_REV_NO

6
Signal Descriptions

This chapter details all possible I/O signals in the controller. For configurable IP titles, your actual configuration might not contain all of these signals.

Inputs are on the left of the signal diagrams; outputs are on the right.

Attention: For configurable IP titles, do not use this document to determine the exact I/O footprint of the controller. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the I/O signals for your actual configuration at workspace/report/IO.html or workspace/report/IO.xml after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the I/O signals that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the widths might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>)** that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

In addition to describing the function of each signal, the signal descriptions in this chapter include the following information:

Active State: Indicates whether the signal is active high or active low. When a signal is not intended to be used in a particular application, then this signal needs to be tied or driven to the inactive state (opposite of the active state).

Registered: Indicates whether or not the signal is registered directly inside the IP boundary without intervening logic (excluding simple buffers). A value of No does not imply that the signal is not synchronous, only that there is some combinatorial logic between the signal's origin or destination register and the boundary of the controller. A value of N/A indicates that this information is not provided for this IP title.

Synchronous to: Indicates which clocks in the IP sample this input (drive for an output) when considering all possible configurations. A particular configuration might not have all of the clocks listed. This clock might not be the same as the clock that your application logic should use to clock (sample/drive) this pin. For more details, consult the clock section in the databook. The presence of the postfix SuperList indicates a list of all possible clocks over all possible configs. Consult coreConsultant report for which clock applies to your specific configuration.

Exists: Names of configuration parameters that populate this signal in your configuration.

Validated by: Assertion or de-assertion of signals that validates the signal being described.

The I/O signals are grouped as follows:

- System Interface on page 196
- XLGMII Transmit Interface on page 198
- XLGMII Receive Interface on page 202
- MDIO Management Interface on page 206
- APB3 Management Interface on page 208
- Miscellaneous Interface on page 211
- Non-Synopsys PHY Link Training on page 217
- Synopsys PHY CR Port Interface on page 265
- Miscellaneous Control Interface on page 267
- GPIO Interface on page 268
- PCS-R PHY Interface on page 269
- Custom1 PCS-PHY Interface on page 286
- FEC Error Correction Memory Interface on page 292
- FEC Error Forward Memory Interface on page 301
- RS-FEC Memory Interface on page 310
- Alignment Marker Removal Memory Interface on page 314
- Energy Efficient Ethernet PHY Interface on page 323
- Non-Synopsys PHY Generic on page 325
- Link Enable on page 327
- Link Status on page 329
- Synopsys PHY-Specific Tx Control and Status Interface on page 331
- Synopsys PHY-Specific Rx Control and Status Interface on page 350
- Synopsys PHY Miscellaneous Interface on page 399
- Synopsys PHY-Specific MPLL Control and Status Interface on page 403
- Synopsys PHY Link Training on page 434
- Synopsys PHY Specific- Status and Control Interface on page 447
- Synopsys PHY Specific MPLL Control and Status Interface on page 448
- Debug Interface on page 450
- Synopsys Enterprise PHY-Specific Tx Status and Control Interface on page 463

- Synopsys PHY-Specific Control and Status Interface on page 464
- Timestamping Interface on page 466
- Debug InterfaceRS-FEC AM Lock Indication. This signal indicates the Alignment Marker (Code-word marker) lock status when operating in RS-FEC Mode. on page 469

6.1 System Interface Signals

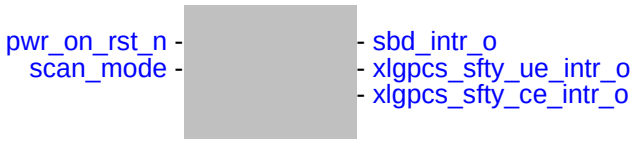


Table 6-1 System Interface Signals

Port Name	I/O	Description
pwr_on_rst_n	I	Power On Reset. Power reset input. **Exists:** Always **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** Low
scan_mode	I	Scan Mode. This signal is used to get the good Automatic Test Pattern Generator (ATPG) test vector coverage. When asserted, this signal bypasses the internal reset generation and directly connects the pwr_on_rst_n signal to all internally-generated resets. **Exists:** Always **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** System Test Clock
sbd_intr_o	O	Interrupt Output. This signal indicates the following events of Clause 73 . ■ Page Received (for Clause 73 auto-negotiation) ■ Auto-negotiation is completed (for Clause 73) ■ Incompatible link partner (for Clause 73 auto-negotiation) This interrupt generation is disabled by default. The Management must enable interrupt generation by writing to the Interrupt Mask register. **Exists:** DWCXLP_CL73_AN **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-1 System Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_sfty_ue_intr_o	O	Uncorrectable Error Safety Interrupt **Exists:** (DWCXLP_ASP||SNPS_RSVDPARAM_3||SNPS_RSVDPARAM_20||SNPS_RSVDPARAM_17||SNPS_RSVDPARAM_16||SNPS_RSVDPARAM_19) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_sfty_ce_intr_o	O	Correctable Error Safety Interrupt **Exists:** SNPS_RSVDPARAM_18||SNPS_RSVDPARAM_3 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.2 XLGMII Transmit Interface Signals



Table 6-2 XLGMII Transmit Interface Signals

Port Name	I/O	Description
clk_xlgmii_tx_i	I	XLGMII Transmit Clock. The XLGMII Tx Interface operates with this clock. The frequency of this clock based on operating speed mode is as follows: ■ 100G mode (NRZ) : 781.25MHz ■ 100G PAM4 Mode/100GAUI-4 Mode: 804.924242 MHz ■ 50G PAM4 Mode/50GAUI-2 Mode : 402.462121 MHz ■ 25G/50G (NRZ) mode : 390.625 MHz ■ 40G mode : 312.5MHz ■ 20G mode : 312.5MHz ■ 10G mode : 312.5MHz (if DWCXLP_XLGMII_INTF_WIDTH==0) or 156.25 MHz (if DWCXLP_XLGMII_INTF_WIDTH==1) This signal registers the following XLGMII input signals: ■ xlgmii_txd_i[127:0] ■ xlgmii_txc_i[15:0] ■ xlgmii_tx_rdy_o **Exists:** !DWCXLP_XLGMII_64BIT **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-2 XLGMII Transmit Interface Signals (Continued)

Port Name	I/O	Description
xlgmii_txd_i[127:0]	I	XLGMII Transmit Data Lanes. This port is the data lane of XLGMII, indicating the data or control character to be transmitted. **If DWCXLP_32BIT_INTF = Enabled:** ■ The data width of this port is 32-bit. **If DWCXLP_32BIT_INTF = Disabled:** ■ The data width of this port is 128-bit. **If SNPS_RSVDPARAM_5 = Disabled and If DWCXLP_32BIT_INTF = Disabled:** ■ In 10G (or 1-lane) mode, only lower 32-bits of this bus are valid. In other modes, all 128-bits are valid. **If SNPS_RSVDPARAM_5 = Enabled:** ■ Entire 128-bit bus is valid in all the 3 modes (4-lane, 2-lane, 1-lane) **Exists:** Always **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_xlgmii_tx_i - in 10G/40G/100G modes , clk_rpcs_tx_i in 25G/50G modes or when XLGMII interface data-width is 64 bit

Table 6-2 XLGMII Transmit Interface Signals (Continued)

Port Name	I/O	Description
xlgmii_txc_i[15:0]	I	XLGMII Transmit Control. These are the control lines of the XLGMII interface to indicate whether xlgmii_txd_i is a data or control character. **If DWCXLP_32BIT_INTF = Enabled:** ■ The control width of this port is 4-bit. **If DWCXLP_32BIT_INTF = Disabled:** ■ The control width of this port is 16-bit. **If SNPS_RSVDPARAM_5 = Disabled and If DWCXLP_32BIT_INTF = Disabled:** ■ In 10G (or 1-lane) mode, only lower 4-bits are valid. In other modes, all 16-bits are valid. **If SNPS_RSVDPARAM_5 = Enabled:** ■ All 16-bits are valid in all the 3 lanes (4-lane,2-lane,1-lane). **xgmii_txc_i[0]:** ■ 0: xlgmii_txd_i[7:0] is a data character. ■ 1: xlgmii_txd_i[7:0] is a control character. **xlgmii_txc_i[1]:** ■ 0: xlgmii_txd_i[15:8] is a data character. ■ 1: xlgmii_txd_i[15:8] is a control character. ... . **xlgmii_txc_i[15]:** ■ 0: xlgmii_txd_i[127:120] is a data character. ■ 1: xlgmii_txd_i[127:120] is a control character. **Exists:** Always **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_xlgmii_tx_i - in 10G/40G/100G modes , clk_rpcs_tx_i in 25G/50G modes or when XLGMII interface data-width is 64-bit

Table 6-2 XLGMII Transmit Interface Signals (Continued)

Port Name	I/O	Description
xlgmii_tx_rdy_o	O	Transmit Ready signal. This signal is asserted by DWC_xlgpcs when its transmitter is ready to receive a new data on xlgmii_txd_i[127:0]/xlgmii_txc_i[15:0]. This signal will be normally high in 40G/50G/100G mode (and in 20G Mode-if SNPS_RSVDPARAM_5=Enabled), except when DWC_xlgpcs inserts Alignment Marker in the transmit data stream. In 20G mode, (if SNPS_RSVDPARAM_5=Disabled), this signal will be low every alternate clk_xlgmii_tx_i clock, in addition to Alignment Maker insertion interval, to attain 20G data rate with 128-bit interface at 312.5MHz In 25G mode, this signal will be low every alternate clk_xlgmii_tx_i clock, in addition to Alignment Maker insertion interval, to attain 25G data rate with 128-bit interface at 390.625MHz. In 25G mode, when XLGMII interface width is 64, this signal is always high. In 10G mode, this signal will be always high. **Exists:** Always **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_xlgmii_tx_i in 10G/40G/100G modes , clk_rpcs_tx_i in 25G/50G modes or when 64-bit XLGMII interface is used

6.3 XLGMII Receive Interface Signals



Table 6-3 XLGMII Receive Interface Signals

Port Name	I/O	Description
clk_xlgmii_rx_i	I	XLGMII Receive Clock. The XLGMII Rx Interface operates with this clock. The frequency of this clock based on operating mode is as follows: ■ 100G mode (NRZ) : 781.25MHz ■ 100G PAM4 Mode/100GAUI-4 Mode: 804.924242 MHz ■ 50G PAM4 Mode/50GAUI-2 Mode : 402.462121 MHz ■ 25G/50G (NRZ) mode : 390.625 MHz ■ 40G mode : 312.5MHz ■ 20G mode : 312.5MHz ■ 10G mode : 312.5MHz (if DWCXLP_XLGMII_INTF_WIDTH==0) or 156.25 MHz (if DWCXLP_XLGMII_INTF_WIDTH==1) This clock input is used to drive the following XLGMII signals: ■ xlgmii_rxd_o[127:0] ■ xlgmii_rxc_o[15:0] ■ xlgmii_rx_valid_o **Exists:** !DWCXLP_XLGMII_64BIT **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-3 XLGMII Receive Interface Signals (Continued)

Port Name	I/O	Description
xlgmii_rxd_o[127:0]	O	XLGMII Receive Data Lanes. This port is the data lane of XLGMII, indicating received data. **If DWCXLP_32BIT_INTF = Enabled:** ■ The data width of this port is 32-bit. **If DWCXLP_XLGMII_64BIT = Enabled:** ■ The data width of this port is 64-bit. In all modes (including 10G), entire 64-bit is valid **If SNPS_RSVDPARAM_5 = Enabled:** ■ The data width of this port is 128-bit. Entire 128-bit bus is valid in all the 3 modes (4-lane,2-lane,1-lane). **If DWCXLP_32BIT_INTF = Disabled AND DWCXLP_XLGMII_64BIT = Disabled AND SNPS_RSVDPARAM_5 = Disabled:** ■ The data width of this port is 128-bit. ■ In 10G (or 1-lane) mode, only lower 32-bits of this bus are valid. In all other modes, all 128-bits are valid **Exists:** Always **Synchronous To:** (!DWCXLP_XLGMII_64BIT) ? "clk_xlgmii_rx_i" : ((DWCXLP_CLK_CMP_EN) ? "clk_rpcs_tx_i" : "clk_rpcs_rx0_i") **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-3 XLGMII Receive Interface Signals (Continued)

Port Name	I/O	Description
xlgmii_rxc_o[15:0]	O	XLGMII Receive Control Lines. These are the control lines of the XGMII interface to indicate whether xlgmii_rxd_o is a data or control character each lane. Any bit going low this bus indicates that corresponding data byte xlgmii_rxd_o is a data character. **If DWCXLP_32BIT_INTF = Enabled:** ■ The control width of this port is 4-bit. **If DWCXLP_XLGMII_64BIT = Enabled:** ■ The control width of this port is 8-bit. In all modes (including 10G), entire 8-bit is valid **If SNPS_RSVDPARAM_5 = Enabled:** ■ The control width of this port is 16-bit. All 16-bits are valid in all the 3 modes (4-lane,2-lane,1-lane). **If DWCXLP_32BIT_INTF = Disabled AND DWCXLP_XLGMII_64BIT = Disabled AND SNPS_RSVDPARAM_5 = Disabled:** ■ The control width of this port is 16-bit. ■ In 10G (or 1-lane) mode, only lower 4-bits are valid. In other modes, all 16-bits are valid. **xlgmii_rxc_o[0]:** ■ 0: xlgmii_rxd_o[7:0] is a data character. ■ 1: xlgmii_rxd_o[7:0] is a control character. **xlgmii_rxc_o[1]:** ■ 0: xlgmii_rxd_o[15:8] is a data character. ■ 1: xlgmii_rxd_o[15:8] is a control character. ..... **xlgmii_rxc_o[15]:** ■ 0: xlgmii_rxd_o[127:120] is a data character. ■ 1: xlgmii_rxd_o[127:120] is a control character. **Exists:** Always **Synchronous To:** (!DWCXLP_XLGMII_64BIT) ? "clk_xlgmii_rx_i" : ((DWCXLP_CLK_CMP_EN) ? "clk_rpcs_tx_i" : "clk_rpcs_rx0_i") **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-3 XLGMII Receive Interface Signals (Continued)

Port Name	I/O	Description
xlgmii_rx_valid_o	O	Receive Data Valid signal. This signal is asserted to indicate that the data on xlgmii_rxd_o and xlgmii_rxc_o are valid. This is a back-pressure signal that gets deasserted when DWC_xlgpcs deletes Alignment Marker from the receive data-stream and also due to clock-compensation. Additionally if DWCXLP_128BIT_INT=Disabled, in 20G mode, this signal toggles every clk_xlgmii_rx_i clock to maintain a data rate of 20G with 128-bit interface at 312.5MHz. Additionally in 25G mode, this signal will be low every alternate clk_xlgmii_rx_i clock to maintain a data rate of 25G with 128-bit interface at 390.625MHz. **Exists:** !SNPS_RSVDPARAM_13 **Synchronous To:** (!DWCXLP_XLGMII_64BIT) ? "clk_xlgmii_rx_i" : ((DWCXLP_CLK_CMP_EN) ? "clk_rpcs_tx_i" : "clk_rpcs_rx0_i") **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.4 MDIO Management Interface Signals



Table 6-4 MDIO Management Interface Signals

Port Name	I/O	Description
clk_mdc_i	I	Management Clock. This clock input is a management clock to do the following: ■ To register management data mdi_i ■ To drive data mdo_o The IEEE Std 802.3, Clause 45 defines the clk_mdc_i as 2.5 MHz. **Exists:** DWCXLP_MDIO_INTF **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
mdi_i	I	Management Data Input. This signal is a serial bit input to program internal device registers. **Exists:** DWCXLP_MDIO_INTF **Synchronous To:** clk_mdc_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
mdo_o	O	Management Data Output. This output is the serialized version of the register that is being read. This output is valid only when the mdo_en_o signal is asserted. **Exists:** DWCXLP_MDIO_INTF **Synchronous To:** clk_mdc_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-4 MDIO Management Interface Signals (Continued)

Port Name	I/O	Description
mdo_en_o	O	Management Data IO Enable. This signal enables selection of input or output mode because the management data pin is a bidirectional pin. **Values:** ■ 0: The DWC_xlgpcs is sampling the data mdi_i. ■ 1: The DWC_xlgpcs is driving valid data mdo_o. **Exists:** DWCXLP_MDIO_INTF **Synchronous To:** clk_mdc_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.5 APB3 Management Interface Signals

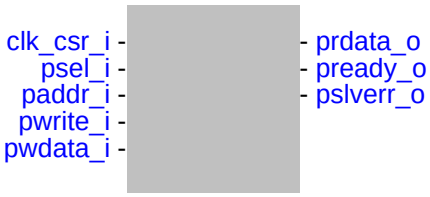


Table 6-5 APB3 Management Interface Signals

Port Name	I/O	Description
clk_csr_i	I	CSR Clock. This clock is used for the CSR block. The allowed range of clk_csr_i is 10MHz-300 MHz. For Synopsys PHY configurations, you need to additionally ensure that frequency of clk_csr_i is less than 5.3 times clock frequency of clk_cko_alive_i. **Exists:** DWCXLP_SNPS_XAUI_PHY||!DWCXLP_MDIO_INTF **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
psel_i	I	APB Slave Select. When this signal is asserted, it indicates the start of transaction and validity of the paddr_i[] and pwrite_i signals. In addition, during a write operation, this signal indicates validity of the pwdata_i[] write data. **Exists:** DWCXLP_APB3_INTF||DWCXLP_APB3_16||DWCXLP_APB3_32 **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-5 APB3 Management Interface Signals (Continued)

Port Name	I/O	Description
paddr_i[(DWCXLP_ADDRS_WIDTH-1):0]	I	APB Address. This signal carries the register address of the CSR module. The address is valid only when the psel_i signal is asserted. -If DWCXLP_INDIRECT_ACCESS = Disabled: -- The paddr_i[20:16] signal indicates the MMD Device Address -- The paddr_i[15:0] signal indicates the register address of the addressed MMD device -If DWCXLP_INDIRECT_ACCESS = Enabled, the value of the paddr_i signal is decoded as follows: ■ 0xFF (APB3 legacy) or 0x1FE (APB_16) or 0x3FC (APB_32): Access to the Base address register ■ Any other value: Access to the register (decoded by paddr_i[7/8/9:0]) within the group, indicated by the value in Base address register **Exists:** DWCXLP_APB3_INTF||DWCXLP_APB3_16||DWCXLP_APB3_32 **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
pwrite_i	I	APB Read or Write. When high, this signal indicates a write operation. When low, it indicates a read operation. **Exists:** DWCXLP_APB3_INTF||DWCXLP_APB3_16||DWCXLP_APB3_32 **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
pwdata_i[(DWCXLP_APB_DATA_W-1):0]	I	Write Data. This signal carries the write data from the host for the CSR module Control registers. **Exists:** DWCXLP_APB3_INTF||DWCXLP_APB3_16||DWCXLP_APB3_32 **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-5 APB3 Management Interface Signals (Continued)

Port Name	I/O	Description
prdata_o[(DWCXLP_APB_DATA_W-1):0]	O	Read Data. This port carries the read data from the DWC_xlgpcs CSR to the host. This data is valid only when the pready_o signal is asserted. **Exists:** DWCXLP_APB3_INTF||DWCXLP_APB3_16||DWCXLP_APB3_32 **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
pready_o	O	Read Data Ready. When high, this signal indicates the following: ■ Data the prdata_o signal is valid. The host can accept the data. ■ End of write transfer **Exists:** DWCXLP_APB3_INTF||DWCXLP_APB3_16||DWCXLP_APB3_32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
pslverr_o	O	APB3 Slave Error. DWC_xlgpcs asserts this signal to indicate that the APB3 transaction has completed with an error. If this signal is asserted for a read transaction, then the data on prdata_o is spurious and should be ignored. **Exists:** DWCXLP_APB3_INTF||DWCXLP_APB3_16||DWCXLP_APB3_32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.6 Miscellaneous Interface Signals

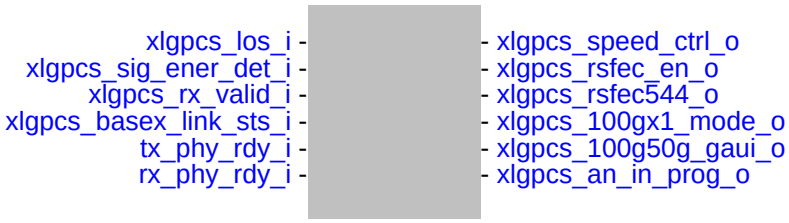


Table 6-6 Miscellaneous Interface Signals

Port Name	I/O	Description
xlgpcs_speed_ctrl_o[2:0]	O	PHY Speed Control. This signal indicates the current operating speed/mode. Values: ■ 111 : 20Gb/s ■ 110 : 50Gb/s ■ 101 : 25Gb/s ■ 100 : 100Gb/s ■ 011 : 40Gb/s ■ 010 : Reserved ■ 001 : 5Gb/s ■ 000 : 10Gb/s **Exists:** !SNPS_RSVDPARAM_10 **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rsfec_en_o	O	RSFEC enable indication. This will be asserted when Bit-2 in RSFEC Control Register is set. **Exists:** DWCXLP_RSFEC_EN **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-6 Miscellaneous Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rsfec544_o	O	544-RSFEC enable indication. When this signal is high, it indicates that RS-FEC (544,514) mode is active in DWC_xlgpcs. **Exists:** DWCXLP_RS544 **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_100gx1_mode_o	O	100G Single lane (PAM4) Mode Indicator. When this signal is high, it indicates that DWC_xlgpcs is configured to operate in 100GBASE-KR1/100GBASE-CR1 mode. **Exists:** (DWCXLP_ADD_100G_1LANE||DWCXLP_100G_1LANE) **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_100g50g_gaui_o	O	This signal will be high, whenever DWC_xlgpcs is operating in 100GAUI-4 or 50GAUI-2 Mode. **Exists:** DWCXLP_RS544&&DWCXLP_MULTI_LANE **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_los_i[(DWCXLP_NUM_LANES-1):0]	I	Loss of Signal Indication. This signal is the inverse of the signal-detect signal (defined in IEEE Std 802.3) from the PMA or PMD Layer. When asserted, this signal indicates that the PMA or PMD layer has detected an error condition the physical connectivity. Otherwise, it indicates the normal condition **Exists:** Always **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with clk_csr_i and clk_rpcs_rx{lane}_i

Table 6-6 Miscellaneous Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_sig_ener_det_i	I	Signal-Detect/Energy-Detect Indication from PMD(PHY) This signal is used by the PHY to indicate successful completion of Link Training. This signal should be low when auto-negotiation (AN page exchanges) or Link Training is ongoing and it should be asserted after 'Link Training' is completed. If link-training is not performed by the PHY (after AN page exchanges), this signal should be made high after PHY has completed the speed switch (while in AN_GOOD_CHECK state) to the negotiated link-speed. If Auto-negotiation is not enabled/performed by software, then this should be driven high. If PHY supports EEE capability, then this signal is made low by the PHY when there is no data on Rx serial inputs of the PHY(which corresponds to EEE QUIET state). Note that Rx Block-Lock/FEC-lock logic of DWC_xlgpcs will be inactive/disabled when this signal is low (unless PDET_EN bit of VR_AN_MODE_CTRL is programmed to 1). **Exists:** (!DWCXLP_SNPS_XAUI_PHY&&DWCXLP_EEE_EN)||(DWCXLP_CL73_ALT) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_valid_i[(DWCXLP_NUM_LANES-1):0]	I	Receiver Valid. This signal indicates if the PHY CDR is locked and if the Rx data is valid Values: ■ 0: Unlocked ■ 1: Locked This signal controls the RLU bit of SR PMA MMD Status 1 Register. **Exists:** Always **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with clk_csr_i

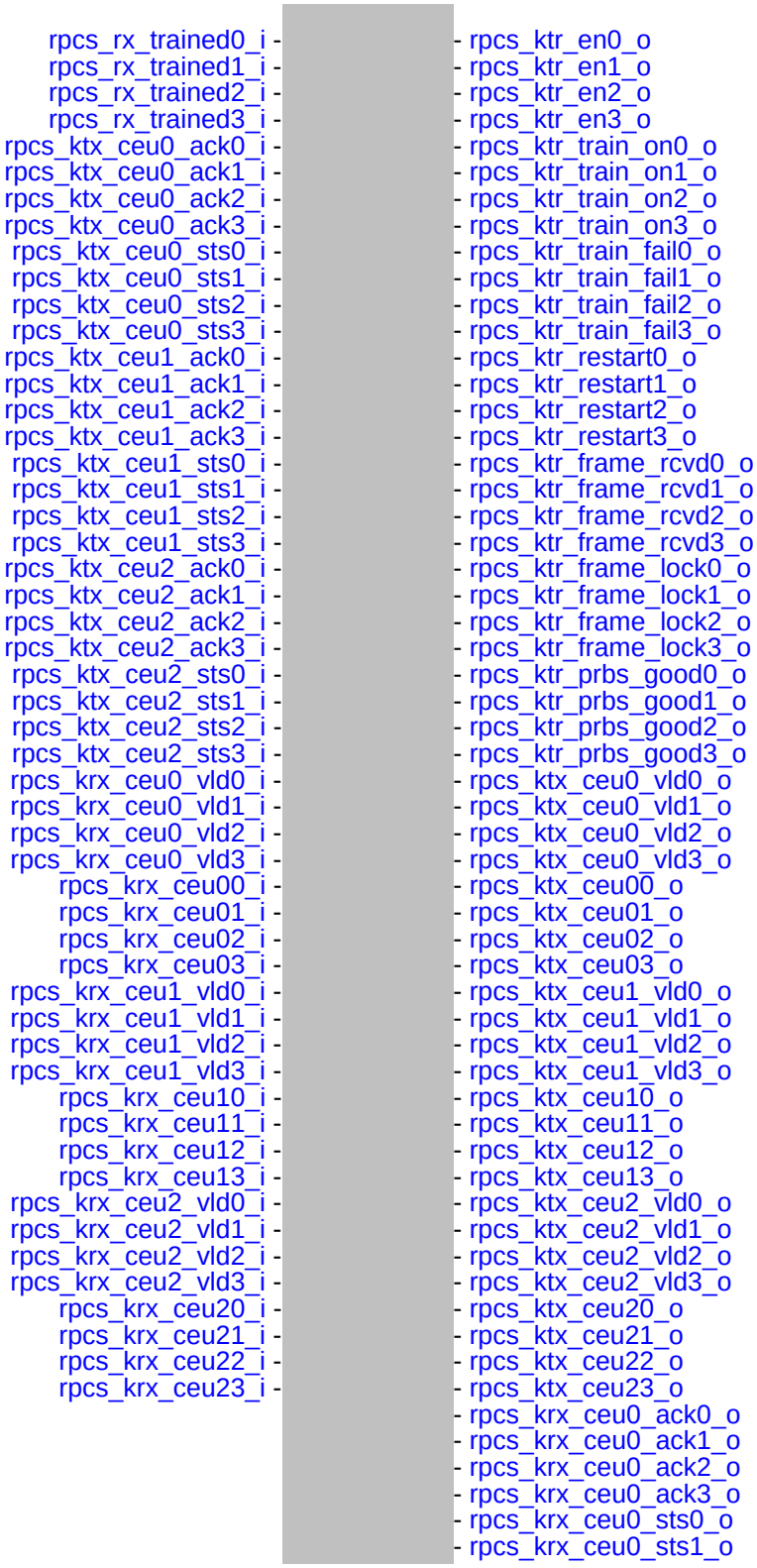
Table 6-6 Miscellaneous Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_an_in_prog_o	O	Auto-negotiation in Progress This signal is asserted when auto-negotiation is in progress (not in AN_GOOD_CHECK or AN_GOOD or AN_ENABLE state). This signal can be used to put the PHY in reference tracking mode so that low frequency auto-negotiation pages can be received correctly by PHY Receive CDR circuit. SoC can even make use of this signal to switch the PHY to 10G NRZ mode when auto-negotiation is started/restarted. **Exists:** DWCXLP_CL73_ALT **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_basex_link_sts_i	I	Link Status Input from BASE-X PCS (KX/KX4) This signal is useful in applications where both XLGPCS and an external BASE-X PCS (say Synopsys XPCS IP) connects to the same SERDES In such applications, we expect application to perform Clause 73 auto-negotiation in XLGPCS (which is connected to SERDES). After all AN page exchanges are complete (ie, AN FSM in XLGPCS reaches AN_GOOD_CHECK state) and if the negotiated speed (HCD) is KX/KX4, application connects external BASE-X PCS to the SERDES. Once external BASE-X PCS attains link-up in KX/KX4 mode, application is expected to make this xlgpcs_basex_link_sts_i input pin high. This ensures that AN FSM in XLGPCS can successfully transition to AN_GOOD state, to indicate successful completion of AN. Whenever BASE-X PCS link goes down, this input is expected to be made low. For applications where external BASE-X PCS is not used in conjunction with XLGPCS, you can tie-off this pin to 0. **Exists:** DWCXLP_CL73_AN&&DWCXLP_RS544 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-6 Miscellaneous Interface Signals (Continued)

Port Name	I/O	Description
tx_phy_rdy_i	I	Ready indication of Tx side PHY. This signal indicates the ready state of TX PHY (indication that Tx clocks are stable and normal operation can be done). When this input is not asserted, then DWC_xlgpcs Tx logic will be put under reset. Whenever Tx side of PHY is getting reset or during PHY speed switch, this signal should be de-asserted. This signal should be asserted back only after Tx path of the PHY is ready in target speed mode and Tx clocks are stable (at the least after 10 stable Tx clocks) **Exists:** DWCXLP_ADD_PHY_RDY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rx_phy_rdy_i	I	Ready Indication of Rx side PHY. This signal indicates the ready state of RX PHY (indication that Rx clocks are stable and normal operation can be done). When this input is not asserted, then DWC_xlgpcs Rx logic will be put under reset Whenever Rx side of PHY is getting reset or during PHY speed switch, this signal should be de-asserted. This signal should be asserted back only after Rx path of the PHY is ready in target speed mode and Rx clocks are stable (at the least after 10 stable Rx clocks) **Exists:** DWCXLP_ADD_PHY_RDY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.7 Non-Synopsys PHY Link Training Signals



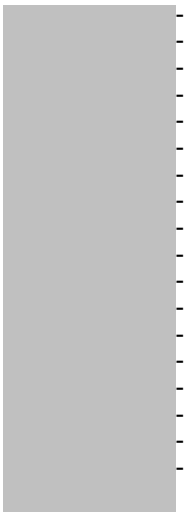


Table 6-7 Non-Synopsys PHY Link Training Signals

Port Name	I/O	Description
rpcs_rx_trained0_i	I	Local Receiver is Trained lane 0. When asserted, this signal indicates that the local KR adaptation process has determined that the local receiver and remote transmit equalization have been optimized to operate at BASE-KR speed. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_rx_trained1_i	I	Local Receiver is Trained lane 1. When asserted, this signal indicates that the local KR adaptation process has determined that the local receiver and remote transmit equalization have been optimized to operate at BASE-KR speed. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_rx_trained2_i	I	Local Receiver is Trained lane 2. When asserted, this signal indicates that the local KR adaptation process has determined that the local receiver and remote transmit equalization have been optimized to operate at BASE-KR speed. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_rx_trained3_i	I	Local Receiver is Trained lane 3. When asserted, this signal indicates that the local KR adaptation process has determined that the local receiver and remote transmit equalization have been optimized to operate at BASE-KR speed. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktr_en0_o	O	BASE-KR Training Enable lane 0. The DWC_xlgpcs asserts this signal when the host enables the Link Training by programming Bit 1 of SR PMA MMD BASE-KR PMD Control Register. This signal can be used in the KR PMA receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktr_en1_o	O	BASE-KR Training Enable lane 1. The DWC_xlgpcs asserts this signal when the host enables the Link Training by programming Bit 1 of SR PMA MMD BASE-KR PMD Control Register. This signal can be used in the KR PMA receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_en2_o	O	BASE-KR Training Enable lane 2. The DWC_xlgpcs asserts this signal when the host enables the Link Training by programming Bit 1 of SR PMA MMD BASE-KR PMD Control Register. This signal can be used in the KR PMA receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_en3_o	O	BASE-KR Training Enable lane 3. The DWC_xlgpcs asserts this signal when the host enables the Link Training by programming Bit 1 of SR PMA MMD BASE-KR PMD Control Register. This signal can be used in the KR PMA receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktr_train_on0_o	O	Training is ON lane 0. This signal is asserted when the host has enabled the Link Training process and the training process is not yet complete lane 0. This signal is de-asserted when the Link training is successfully completed. This signal can be used in the KR receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_train_on1_o	O	Training is ON lane 1. This signal is asserted when the host has enabled the Link Training process and the training process is not yet complete lane 1. This signal is de-asserted when the Link training is successfully completed. This signal can be used in the KR receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_train_on2_o	O	Training is ON lane 2. This signal is asserted when the host has enabled the Link Training process and the training process is not yet complete lane 2. This signal is de-asserted when the Link training is successfully completed. This signal can be used in the KR receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktr_train_on3_o	O	Training is ON lane 3. This signal is asserted when the host has enabled the Link Training process and the training process is not yet complete lane 3. This signal is de-asserted when the Link training is successfully completed. This signal can be used in the KR receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_train_fail0_o	O	Training Fail Status lane 0. This signal is asserted when the Link training process fails. The Link training process fails when both local and remote adaptation processes fail to establish reliable KR link through Transmit filter coefficient settings within 500 ms time. This signal is asserted until the management restarts the KR training process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_train_fail1_o	O	Training Fail Status lane 1. This signal is asserted when the Link training process fails. The Link training process fails when both local and remote adaptation processes fail to establish reliable KR link through Transmit filter coefficient settings within 500 ms time. This signal is asserted until the management restarts the KR training process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktr_train_fail2_o	O	Training Fail Status lane 2. This signal is asserted when the Link training process fails. TheLink training process fails when both local and remote adaptation processes fail to establish reliable KR link through Transmit filter coefficient settings within 500 ms time. This signal is asserted until the management restarts the KR training process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_train_fail3_o	O	Training Fail Status lane 3. This signal is asserted when the Link training process fails. TheLink training process fails when both local and remote adaptation processes fail to establish reliable KR link through Transmit filter coefficient settings within 500 ms time. This signal is asserted until the management restarts the KR training process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_restart0_o	O	Restart Training lane 0. The DWC_xlgpcs asserts this signal when the host restarts the Link Training by programming Bit 0 of SR PMA MMD BASE-KR PMD Control Register. **Note:** This signal is a pulse of 1 clock period of the clk_rpcs_tx_i clock. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktr_restart1_o	O	Restart Training lane 1. The DWC_xlgpcs asserts this signal when the host restarts the Link Training by programming Bit 0 of SR PMA MMD BASE-KR PMD Control Register. **Note:** This signal is a pulse of 1 clock period of the clk_rpcs_tx_i clock. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_restart2_o	O	Restart Training lane 2. The DWC_xlgpcs asserts this signal when the host restarts the Link Training by programming Bit 0 of SR PMA MMD BASE-KR PMD Control Register. **Note:** This signal is a pulse of 1 clock period of the clk_rpcs_tx_i clock. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_restart3_o	O	Restart Training lane 3. The DWC_xlgpcs asserts this signal when the host restarts the Link Training by programming Bit 0 of SR PMA MMD BASE-KR PMD Control Register. **Note:** This signal is a pulse of 1 clock period of the clk_rpcs_tx_i clock. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktr_frame_rcvd0_o	O	Training Frame Received lane 0. This signal indicates that the new training frame is received. This (pulse) signal is asserted for 1 clock period of the clk_rpcs_tx_i clock. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_frame_rcvd1_o	O	Training Frame Received lane 1. This signal indicates that the new training frame is received. This (pulse) signal is asserted for 1 clock period of the clk_rpcs_tx_i clock. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_frame_rcvd2_o	O	Training Frame Received lane 2. This signal indicates that the new training frame is received. This (pulse) signal is asserted for 1 clock period of the clk_rpcs_tx_i clock. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_frame_rcvd3_o	O	Training Frame Received lane 3. This signal indicates that the new training frame is received. This (pulse) signal is asserted for 1 clock period of the clk_rpcs_tx_i clock. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktr_frame_lock0_o	O	Training Frame Synchronization Status lane 0. This signal is asserted when the Received Training frame synchronization is successful lane 0. Otherwise, this signal is de-asserted. This signal can be used in the KR receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_frame_lock1_o	O	Training Frame Synchronization Status lane 1. This signal is asserted when the Received Training frame synchronization is successful lane 1. Otherwise, this signal is de-asserted. This signal can be used in the KR receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_frame_lock2_o	O	Training Frame Synchronization Status lane 2. This signal is asserted when the Received Training frame synchronization is successful lane 2. Otherwise, this signal is de-asserted. This signal can be used in the KR receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktr_frame_lock3_o	O	Training Frame Synchronization Status lane 3. This signal is asserted when the Received Training frame synchronization is successful lane 3. Otherwise, this signal is de-asserted. This signal can be used in the KR receiver adaptation process. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_prbs_good0_o	O	PRBS Good Indication lane 0. This signal indicates the PRBS data decoding status of the currently received training frame when the rpcs_ktr_frame_rcvd0_o signal is asserted. ■ 0: Decoding is not successful ■ 1: Decoding is successful **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_prbs_good1_o	O	PRBS Good Indication lane 1. This signal indicates the PRBS data decoding status of the currently received training frame when the rpcs_ktr_frame_rcvd1_o signal is asserted. ■ 0: Decoding is not successful ■ 1: Decoding is successful **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktr_prbs_good2_o	O	PRBS Good Indication lane 2. This signal indicates the PRBS data decoding status of the currently received training frame when the rpcs_ktr_frame_rcvd2_o signal is asserted. ■ 0: Decoding is not successful ■ 1: Decoding is successful **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_prbs_good3_o	O	PRBS Good Indication lane 3. This signal indicates the PRBS data decoding status of the currently received training frame when the rpcs_ktr_frame_rcvd3_o signal is asserted. ■ 0: Decoding is not successful ■ 1: Decoding is successful **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu0_vld0_o	O	Local Transmitter Coefficient Update Valid for C(-1) lane 0. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu00_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu0_ack0_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu0_vld1_o	O	Local Transmitter Coefficient Update Valid for C(-1) lane 1. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu01_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu0_ack1_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu0_vld2_o	O	Local Transmitter Coefficient Update Valid for C(-1) lane 2. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu02_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu0_ack2_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu0_vld3_o	O	Local Transmitter Coefficient Update Valid for C(-1) lane 3. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu03_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu0_ack3_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu00_o[1:0]	O	Local Device Filter Coefficient Update Request for C(-1) Lane 0. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu01_o[1:0]	O	Local Device Filter Coefficient Update Request for C(-1) Lane 1. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu02_o[1:0]	O	Local Device Filter Coefficient Update Request for C(-1) Lane 2. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu03_o[1:0]	O	Local Device Filter Coefficient Update Request for C(-1) Lane 3. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu0_ack0_i	I	Acknowledgement for rpcs_ktx_ceu0_vld0_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu0_vld0_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu0_vld0_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu0_sts0_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu0_ack1_i	I	Acknowledgement for rpcs_ktx_ceu0_vld1_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu0_vld1_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu0_vld1_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu0_sts1_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu0_ack2_i	I	Acknowledgement for rpcs_ktx_ceu0_vld2_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu0_vld2_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu0_vld2_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu0_sts2_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu0_ack3_i	I	Acknowledgement for rpcs_ktx_ceu0_vld3_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu0_vld3_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu0_vld3_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu0_sts3_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu0_sts0_i[1:0]	I	Status for rpcs_ktx_ceu0_vld0_o Request. This signal indicates the status of the rpcs_ktx_ceu0_vld0_o request. The status is valid only when the rpcs_ktx_ceu0_ack0_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu0_sts1_i[1:0]	I	Status for rpcs_ktx_ceu0_vld1_o Request. This signal indicates the status of the rpcs_ktx_ceu0_vld1_o request. The status is valid only when the rpcs_ktx_ceu0_ack1_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu0_sts2_i[1:0]	I	Status for rpcs_ktx_ceu0_vld2_o Request. This signal indicates the status of the rpcs_ktx_ceu0_vld2_o request. The status is valid only when the rpcs_ktx_ceu0_ack2_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu0_sts3_i[1:0]	I	Status for rpcs_ktx_ceu0_vld3_o Request. This signal indicates the status of the rpcs_ktx_ceu0_vld3_o request. The status is valid only when the rpcs_ktx_ceu0_ack3_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu1_vld0_o	O	Local Transmitter Coefficient Update Valid for C(0) lane 0. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu10_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu1_ack0_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu1_vld1_o	O	Local Transmitter Coefficient Update Valid for C(0) lane 1. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu11_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu1_ack1_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu1_vld2_o	O	Local Transmitter Coefficient Update Valid for C(0) lane 2. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu12_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu1_ack2_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu1_vld3_o	O	Local Transmitter Coefficient Update Valid for C(0) lane 3. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu13_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu1_ack3_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu10_o[1:0]	O	Local Device Filter Coefficient Update Request for C(0) Lane 0. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu11_o[1:0]	O	Local Device Filter Coefficient Update Request for C(0) Lane 1. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu12_o[1:0]	O	Local Device Filter Coefficient Update Request for C(0) Lane 2. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu13_o[1:0]	O	Local Device Filter Coefficient Update Request for C(0) Lane 3. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu1_ack0_i	I	Acknowledgement for rpcs_ktx_ceu1_vld0_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu1_vld0_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu1_vld0_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu1_sts0_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu1_ack1_i	I	Acknowledgement for rpcs_ktx_ceu1_vld1_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu1_vld1_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu1_vld1_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu1_sts1_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu1_ack2_i	I	Acknowledgement for rpcs_ktx_ceu1_vld2_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu1_vld0_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu1_vld2_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu1_sts2_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu1_ack3_i	I	Acknowledgement for rpcs_ktx_ceu1_vld3_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu1_vld3_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu1_vld3_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu1_sts3_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu1_sts0_i[1:0]	I	Status for rpcs_ktx_ceu1_vld0_o Request. This signal indicates the status of the rpcs_ktx_ceu1_vld0_o request. The status is valid only when the rpcs_ktx_ceu1_ack0_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu1_sts1_i[1:0]	I	Status for rpcs_ktx_ceu1_vld0_o Request. This signal indicates the status of the rpcs_ktx_ceu1_vld1_o request. The status is valid only when the rpcs_ktx_ceu1_ack1_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu1_sts2_i[1:0]	I	Status for rpcs_ktx_ceu1_vld2_o Request. This signal indicates the status of the rpcs_ktx_ceu1_vld2_o request. The status is valid only when the rpcs_ktx_ceu1_ack2_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu1_sts3_i[1:0]	I	Status for rpcs_ktx_ceu1_vld3_o Request. This signal indicates the status of the rpcs_ktx_ceu1_vld3_o request. The status is valid only when the rpcs_ktx_ceu1_ack3_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu2_vld0_o	O	Local Transmitter Coefficient Update Valid for C(+1) lane 0. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu20_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu2_ack0_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu2_vld1_o	O	Local Transmitter Coefficient Update Valid for C(+1) lane 1. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu21_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu2_ack1_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu2_vld2_o	O	Local Transmitter Coefficient Update Valid for C(+1) lane 2. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu22_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu2_ack2_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu2_vld3_o	O	Local Transmitter Coefficient Update Valid for C(+1) lane 3. This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed on the rpcs_ktx_ceu23_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu2_ack3_i signal is asserted from the Local KR PMA. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu20_o[1:0]	O	Local Device Filter Coefficient Update Request for C(+1) Lane 0. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu21_o[1:0]	O	Local Device Filter Coefficient Update Request for C(+1) Lane 1. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu22_o[1:0]	O	Local Device Filter Coefficient Update Request for C(+1) Lane 2. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu23_o[1:0]	O	Local Device Filter Coefficient Update Request for C(+1) Lane 3. This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu2_ack0_i	I	Acknowledgement for rpcs_ktx_ceu2_vld0_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu2_vld0_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu2_vld0_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu2_sts0_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu2_ack1_i	I	Acknowledgement for rpcs_ktx_ceu2_vld1_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu2_vld1_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu2_vld1_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu2_sts1_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu2_ack2_i	I	Acknowledgement for rpcs_ktx_ceu2_vld2_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu2_vld2_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu2_vld2_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu2_sts2_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu2_ack3_i	I	Acknowledgement for rpcs_ktx_ceu2_vld3_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu2_vld3_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu2_vld3_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu2_sts3_i signal is valid. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu2_sts0_i[1:0]	I	Status for rpcs_ktx_ceu2_vld0_o Request. This signal indicates the status of the rpcs_ktx_ceu2_vld0_o request. The status is valid only when the rpcs_ktx_ceu2_ack0_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu2_sts1_i[1:0]	I	Status for rpcs_ktx_ceu2_vld1_o Request. This signal indicates the status of the rpcs_ktx_ceu2_vld1_o request. The status is valid only when the rpcs_ktx_ceu2_ack1_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu2_sts2_i[1:0]	I	Status for rpcs_ktx_ceu2_vld2_o Request. This signal indicates the status of the rpcs_ktx_ceu2_vld2_o request. The status is valid only when the rpcs_ktx_ceu2_ack2_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_ceu2_sts3_i[1:0]	I	Status for rpcs_ktx_ceu2_vld3_o Request. This signal indicates the status of the rpcs_ktx_ceu2_vld3_o request. The status is valid only when the rpcs_ktx_ceu2_ack3_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_vld0_i	I	Request from the Local KR Adaptation Process for C(-1) to the Link Partner Transmitter lane 0. When this signal is asserted, the rpcs_krx_ceu00_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu0_ack0_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_vld1_i	I	Request from the Local KR Adaptation Process for C(-1) to the Link Partner Transmitter lane 1. When this signal is asserted, the rpcs_krx_ceu01_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu0_ack1_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu0_vld2_i	I	Request from the Local KR Adaptation Process for C(-1) to the Link Partner Transmitter lane 2. When this signal is asserted, the rpcs_krx_ceu02_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu0_ack2_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_vld3_i	I	Request from the Local KR Adaptation Process for C(-1) to the Link Partner Transmitter lane 3. When this signal is asserted, the rpcs_krx_ceu03_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu0_ack3_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu00_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(-1) lane 0. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu01_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(-1) lane 1. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu02_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(-1) lane 2. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu03_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(-1) lane 3. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_ack0_o	O	Acknowledgement for the rpcs_krx_ceu0_vld0_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts0_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu0_vld0_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_ack1_o	O	Acknowledgement for the rpcs_krx_ceu0_vld1_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts1_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu0_vld1_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu0_ack2_o	O	Acknowledgement for the rpcs_krx_ceu0_vld2_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts2_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu0_vld2_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_ack3_o	O	Acknowledgement for the rpcs_krx_ceu0_vld3_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts3_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu0_vld3_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_sts0_o[1:0]	O	Status for the rpcs_krx_ceu00_i[1:0] Request. This signal is valid when the rpcs_krx_ceu0_ack0_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu00_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu0_sts1_o[1:0]	O	Status for the rpcs_krx_ceu01_i[1:0] Request. This signal is valid when the rpcs_krx_ceu0_ack1_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu01_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_sts2_o[1:0]	O	Status for the rpcs_krx_ceu02_i[1:0] Request. This signal is valid when the rpcs_krx_ceu0_ack2_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu02_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu0_sts3_o[1:0]	O	Status for the rpcs_krx_ceu03_i[1:0] Request. This signal is valid when the rpcs_krx_ceu0_ack3_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu03_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_vld0_i	I	Request from the Local KR Adaptation Process for C(0) to the Link Partner Transmitter lane 0. When this signal is asserted, the rpcs_krx_ceu10_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu1_ack0_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_vld1_i	I	Request from the Local KR Adaptation Process for C(0) to the Link Partner Transmitter lane 0. When this signal is asserted, the rpcs_krx_ceu11_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu1_ack1_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu1_vld2_i	I	Request from the Local KR Adaptation Process for C(0) to the Link Partner Transmitter lane 2. When this signal is asserted, the rpcs_krx_ceu12_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu1_ack2_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_vld3_i	I	Request from the Local KR Adaptation Process for C(0) to the Link Partner Transmitter lane 3. When this signal is asserted, the rpcs_krx_ceu13_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu1_ack3_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu10_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(0) lane 0. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu11_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(0) lane 1. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu12_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(0) lane 2. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu13_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(0) lane 3. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_ack0_o	O	Acknowledgement for the rpcs_krx_ceu1_vld0_i Request. When this signal is asserted, the rpcs_krx_ceu1_sts0_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu1_vld0_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_ack1_o	O	Acknowledgement for the rpcs_krx_ceu1_vld1_i Request. When this signal is asserted, the rpcs_krx_ceu1_sts1_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu1_vld1_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : "Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu1_ack2_o	O	Acknowledgement for the rpcs_krx_ceu1_vld2_i Request. When this signal is asserted, the rpcs_krx_ceu1_sts2_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu1_vld2_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : "Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_ack3_o	O	Acknowledgement for the rpcs_krx_ceu1_vld3_i Request. When this signal is asserted, the rpcs_krx_ceu1_sts3_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu1_vld3_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_sts0_o[1:0]	O	Status for the rpcs_krx_ceu10_i[1:0] Request. This signal is valid when the rpcs_krx_ceu1_ack0_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu10_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu1_sts1_o[1:0]	O	Status for the rpcs_krx_ceu11_i[1:0] Request. This signal is valid when the rpcs_krx_ceu1_ack1_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu11_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : "Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_sts2_o[1:0]	O	Status for the rpcs_krx_ceu12_i[1:0] Request. This signal is valid when the rpcs_krx_ceu1_ack2_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu12_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu1_sts3_o[1:0]	O	Status for the rpcs_krx_ceu13_i[1:0] Request. This signal is valid when the rpcs_krx_ceu1_ack3_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu13_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu2_vld0_i	I	Request from the Local KR Adaptation Process for C(+1) to the Link Partner Transmitter lane 0. When this signal is asserted, the rpcs_krx_ceu20_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu2_ack0_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu2_vld1_i	I	Request from the Local KR Adaptation Process for C(+1) to the Link Partner Transmitter lane 1. When this signal is asserted, the rpcs_krx_ceu21_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu2_ack1_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu2_vld2_i	I	Request from the Local KR Adaptation Process for C(+1) to the Link Partner Transmitter lane 2. When this signal is asserted, the rpcs_krx_ceu22_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu2_ack2_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu2_vld3_i	I	Request from the Local KR Adaptation Process for C(+1) to the Link Partner Transmitter lane 3. When this signal is asserted, the rpcs_krx_ceu23_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu2_ack3_o signal is asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu20_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(1) lane 0. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu21_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(1) lane 1. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu22_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(1) lane 2. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu23_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(1) lane 3. The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. **Values:** ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu2_ack0_o	O	Acknowledgement for the rpcs_krx_ceu2_vld0_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts0_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu2_vld0_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu2_ack1_o	O	Acknowledgement for the rpcs_krx_ceu2_vld1_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts1_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu2_vld1_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu2_ack2_o	O	Acknowledgement for the rpcs_krx_ceu2_vld2_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts2_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu2_vld2_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu2_ack3_o	O	Acknowledgement for the rpcs_krx_ceu2_vld3_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts3_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu2_vld3_i signal is de-asserted. **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu2_sts0_o[1:0]	O	Status for the rpcs_krx_ceu20_i[1:0] Request. This signal is valid when the rpcs_krx_ceu2_ack0_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu20_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu2_sts1_o[1:0]	O	Status for the rpcs_krx_ceu21_i[1:0] Request. This signal is valid when the rpcs_krx_ceu2_ack1_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu21_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_KR4_50G ? "No" : " Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu2_sts2_o[1:0]	O	Status for the rpcs_krx_ceu22_i[1:0] Request. This signal is valid when the rpcs_krx_ceu2_ack22_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu22_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-7 Non-Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_ceu2_sts3_o[1:0]	O	Status for the rpcs_krx_ceu23_i[1:0] Request. This signal is valid when the rpcs_krx_ceu2_ack3_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu23_i[1:0] request. This information is generated based the Receive training from the link partner. **Values:** ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable (Not expected from PHY) **Exists:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.8 Synopsys PHY CR Port Interface Signals

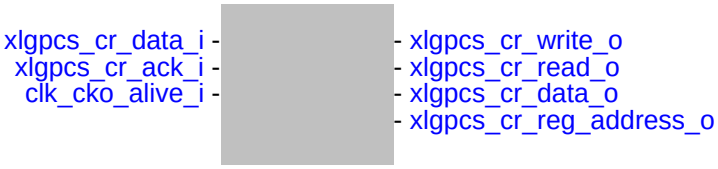


Table 6-8 Synopsys PHY CR Port Interface Signals

Port Name	I/O	Description
xlgpcs_cr_data_i[15:0]	I	CR Data. This signal transmits the data read from the Synopsys PHY. **Exists:** DWCXLP_SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Registered:** Yes, with clk_csr_i
xlgpcs_cr_ack_i	I	Acknowledge for CR port access. This signal acknowledges the assertion of xlgpcs_cr_read_o or xlgpcs_cr_write_o for Synopsys PHY **Exists:** DWCXLP_SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with clk_csr_i
xlgpcs_cr_write_o	O	CR Write. This signal is used to write the content of the xlgpcs_cr_data_o[15:0] signal into the dereferenced address register of the Synopsys PHY. **Exists:** DWCXLP_SNPS_XAUI_PHY **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-8 Synopsys PHY CR Port Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_cr_read_o	O	CR Read. This signal is asserted to read the data from dereferenced address register of the Synopsys PHY the xlgpcs_cr_data_i signal. **Exists:** DWCXLP_SNPS_XAUI_PHY **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_cr_data_o[15:0]	O	Write data for management interface. This is the data to be written to the PHY register through the management interface. **Exists:** DWCXLP_SNPS_XAUI_PHY **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_cr_reg_address_o[15:0]	O	Register Address. Specifies the address of the PHY register that is being accessed **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_cko_alive_i	I	Synopsys PHY Alive Clock. You must connect this input to ref_dig_fr_clk clock output of the Synopsys PHY. **Exists:** DWCXLP_SNPS_XAUI_PHY **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.9 Miscellaneous Control Interface Signals



Table 6-9 Miscellaneous Control Interface Signals

Port Name	I/O	Description
xlgpcs_reset_n_o	O	PHY Reset. Active Low Reset signal to PHY. **Exists:** Always **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** Low
xlgpcs_xlgmii_lb_en_o	O	Rx to Tx XLGMII Loopback enable. This signal indicates that the host programmed the device to enable Rx to Tx loopback at the XLGMII interface inside DWC_xlgpcs. The loopback mode is enabled through any of the following bits: ■ Bit 14 of VR PCS MMD Digital Control 1 Register **Exists:** DWCXLP_LOOPBACK_EN **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.10 GPIO Interface Signals



Table 6-10 GPIO Interface Signals

Port Name	I/O	Description
xlgpcs_gpo_o[7:0]	O	General Purpose Output. This output indicates the content of Bits[15:8] of VR PCS MMD GPIO Register. **Exists:** DWCXLP_GPIO_EN **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_gpo_i[7:0]	I	General Purpose Input. The contents of this input can be read by reading Bits[7:0] of VR PCS MMD GPIO Register. **Exists:** DWCXLP_GPIO_EN **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.11 PCS-R PHY Interface Signals

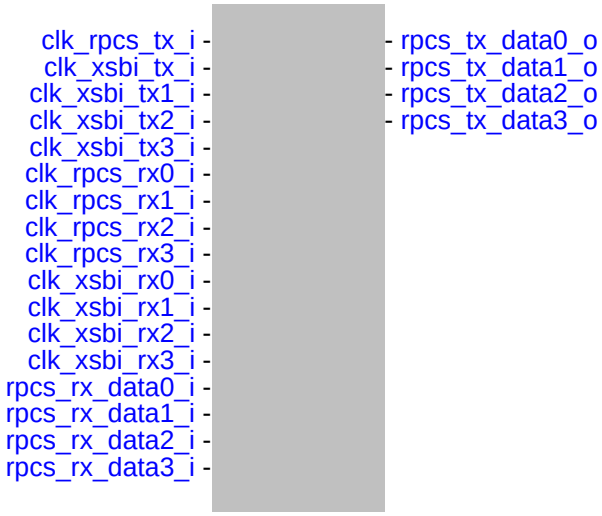


Table 6-11 PCS-R PHY Interface Signals

Port Name	I/O	Description
clk_rpcs_tx_i	I	PCS-R Transmit Data Path Clock. This clock is used in the Transmit data path of the PCS-R block. If DWCXLP_R_SRDS_INTF = 3, this clock is also used to drive the rpcs_tx_data{lane}_o[rpcs_phw:0] data. The frequency of this clock based on operating speed mode is as follows: ■ 10G/20G/40G mode : 156.25MHz ■ 25G/50G/100G mode (NRZ) : 390.625MHz ■ 50G PAM4/100G PAM4/100GAUI-4/50GAUI-2 mode: 402.462121 MHz **Exists:** Always **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_xsbi_tx_i	I	XSBI Transmit Clock. If DWCXLP_R_SRDS_INTF != 3, this clock drives the rpcs_tx_data{lane}_o[rpcs_phw:0] data. The frequency of this clock depends the PCS-R PHY Interface Type and operating mode. If DWCXLP_MAIN_MODE == 2, then ■ DWCXLP_R_SRDS_INTF = 0: 644.53125MHz ■ DWCXLP_R_SRDS_INTF = 4: 515.625MHz ■ DWCXLP_R_SRDS_INTF = 1: 322.265MHz ■ DWCXLP_R_SRDS_INTF = 2: 312.5MHz ■ DWCXLP_R_SRDS_INTF = 5: 161.132MHz ■ DWCXLP_R_SRDS_INTF = 3: 156.25MHz ■ DWCXLP_R_SRDS_INTF = 6: 128.906MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 7, then ■ 10G/20G/40G mode: 257.8125MHz ■ 25G/50G/100G mode: 644.53125MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 1, then ■ 10G/20G/40G mode: 322.265625MHz ■ 25G/50G/100G mode: 805.6640625MHz **Exists:** (DWCXLP_R_SRDS_INTF!=3) **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_xsbi_tx1_i	I	XSBI Transmit Clock for lane 1. If DWCXLP_R_SRDS_INTF != 3, this clock drives the rpcs_tx_data1_o[rpcs_phw:0] data. The frequency of this clock depends the PCS-R PHY Interface Type and operating mode. If DWCXLP_MAIN_MODE == 2, then ■ DWCXLP_R_SRDS_INTF = 0: 644.53125MHz ■ DWCXLP_R_SRDS_INTF = 4: 515.625MHz ■ DWCXLP_R_SRDS_INTF = 1: 322.265MHz ■ DWCXLP_R_SRDS_INTF = 2: 312.5MHz ■ DWCXLP_R_SRDS_INTF = 5: 161.132MHz ■ DWCXLP_R_SRDS_INTF = 3: 156.25MHz ■ DWCXLP_R_SRDS_INTF = 6: 128.906MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 7, then ■ 10G/20G/40G mode: 257.8125MHz ■ 25G/50G/100G mode: 644.53125MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 1, then ■ 10G/20G/40G mode: 322.265625MHz ■ 25G/50G/100G mode: 805.6640625MHz **Exists:** DWCXLP_XSBI_TX_CLK_PER_LN **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_xsbi_tx2_i	I	XSBI Transmit Clock for lane 2. If DWCXLP_R_SRDS_INTF != 3, this clock drives the rpcs_tx_data2_o[rpcs_phw:0] data. The frequency of this clock depends the PCS-R PHY Interface Type and operating mode. If DWCXLP_MAIN_MODE == 2, then ■ DWCXLP_R_SRDS_INTF = 0: 644.53125MHz ■ DWCXLP_R_SRDS_INTF = 4: 515.625MHz ■ DWCXLP_R_SRDS_INTF = 1: 322.265MHz ■ DWCXLP_R_SRDS_INTF = 2: 312.5MHz ■ DWCXLP_R_SRDS_INTF = 5: 161.132MHz ■ DWCXLP_R_SRDS_INTF = 3: 156.25MHz ■ DWCXLP_R_SRDS_INTF = 6: 128.906MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 7, then ■ 10G/20G/40G mode: 257.8125MHz ■ 25G/50G/100G mode: 644.53125MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 1, then ■ 10G/20G/40G mode: 322.265625MHz ■ 25G/50G/100G mode: 805.6640625MHz **Exists:** DWCXLP_XSBI_TX_CLK_PER_LN **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_xsbi_tx3_i	I	XSBI Transmit Clock for lane 3. If DWCXLP_R_SRDS_INTF != 3, this clock drives the rpcs_tx_data3_o[rpcs_phw:0] data. The frequency of this clock depends the PCS-R PHY Interface Type and operating mode. If DWCXLP_MAIN_MODE == 2, then ■ DWCXLP_R_SRDS_INTF = 0: 644.53125MHz ■ DWCXLP_R_SRDS_INTF = 4: 515.625MHz ■ DWCXLP_R_SRDS_INTF = 1: 322.265MHz ■ DWCXLP_R_SRDS_INTF = 2: 312.5MHz ■ DWCXLP_R_SRDS_INTF = 5: 161.132MHz ■ DWCXLP_R_SRDS_INTF = 3: 156.25MHz ■ DWCXLP_R_SRDS_INTF = 6: 128.906MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 7, then ■ 10G/20G/40G mode: 257.8125MHz ■ 25G/50G/100G mode: 644.53125MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 1, then ■ 10G/20G/40G mode: 322.265625MHz ■ 25G/50G/100G mode: 805.6640625MHz **Exists:** DWCXLP_XSBI_TX_CLK_PER_LN **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
rpcs_tx_data0_o[(DWCXLP_PHY_DW-1):0]	O	XSBI Transmit data for lane 0. This is the Transmit data the PCS-R PHY interface for lane 0. The width of this signal depends the PCS-R PHY Interface Type: If DWCXLP_MAIN_MODE == 2, then ■ If DWCXLP_R_SRDS_INTF = 0, Width = 16 ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 2, Width = 33 ■ If DWCXLP_R_SRDS_INTF = 3, Width = 66 ■ If DWCXLP_R_SRDS_INTF = 4, Width = 20 ■ If DWCXLP_R_SRDS_INTF = 5, Width = 64 ■ If DWCXLP_R_SRDS_INTF = 6, Width = 80 If DWCXLP_MAIN_MODE!=2 then, ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 7, Width = 40 This port will be active during all speed modes **Exists:** Always **Synchronous To:** (DWCXLP_R_SRDS_INTF!=3) ? "clk_xsbi_tx_i" : "clk_rpcs_tx_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
rpcs_tx_data1_o[(DWCXLP_PHY_DW-1):0]	O	XSBI Transmit data for lane 1. This is the Transmit data the PCS-R PHY interface for lane 1. The width of this signal depends the PCS-R PHY Interface Type: If DWCXLP_MAIN_MODE == 2, then ■ If DWCXLP_R_SRDS_INTF = 0, Width = 16 ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 2, Width = 33 ■ If DWCXLP_R_SRDS_INTF = 3, Width = 66 ■ If DWCXLP_R_SRDS_INTF = 4, Width = 20 ■ If DWCXLP_R_SRDS_INTF = 5, Width = 64 ■ If DWCXLP_R_SRDS_INTF = 6, Width = 80 If DWCXLP_MAIN_MODE!=2 then, ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 7, Width = 40 ■ This port will be driven during 20G, 40G,50G and 100G speed modes only ■ This port will not be driven during 10G and 25G speed modes **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** (DWCXLP_R_SRDS_INTF!=3) ? (DWCXLP_XSBI_TX_CLK_PER_LN ? "clk_xsbi_tx1_i" : "clk_xsbi_tx_i") : "clk_rpcs_tx_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
rpcs_tx_data2_o[(DWCXLP_PHY_DW-1):0]	O	XSBI Transmit data for lane 2. This is the Transmit data the PCS-R PHY interface for lane 2. The width of this signal depends the PCS-R PHY Interface Type: If DWCXLP_MAIN_MODE == 2, then ■ If DWCXLP_R_SRDS_INTF = 0, Width = 16 ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 2, Width = 33 ■ If DWCXLP_R_SRDS_INTF = 3, Width = 66 ■ If DWCXLP_R_SRDS_INTF = 4, Width = 20 ■ If DWCXLP_R_SRDS_INTF = 5, Width = 64 ■ If DWCXLP_R_SRDS_INTF = 6, Width = 80 If DWCXLP_MAIN_MODE!=2 then, ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 7, Width = 40 ■ This port will be driven during 40G and 100G speed modes only ■ This port will be not be driven during 10G, 20G, 25G and 50G speed modes **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** (DWCXLP_R_SRDS_INTF!=3) ? (DWCXLP_XSBI_TX_CLK_PER_LN ? "clk_xsbi_tx2_i" : "clk_xsbi_tx_i") : "clk_rpcs_tx_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
rpcs_tx_data3_o[(DWCXLP_PHY_DW-1):0]	O	XSBI Transmit data for lane 3. This is the Transmit data the PCS-R PHY interface for lane 3. The width of this signal depends the PCS-R PHY Interface Type: If DWCXLP_MAIN_MODE == 2, then ■ If DWCXLP_R_SRDS_INTF = 0, Width = 16 ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 2, Width = 33 ■ If DWCXLP_R_SRDS_INTF = 3, Width = 66 ■ If DWCXLP_R_SRDS_INTF = 4, Width = 20 ■ If DWCXLP_R_SRDS_INTF = 5, Width = 64 ■ If DWCXLP_R_SRDS_INTF = 6, Width = 80 If DWCXLP_MAIN_MODE!=2 then, ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 7, Width = 40 ■ This port will be driven during 40G and 100G speed modes only ■ This port will be not be driven during 10G, 20G, 25G and 50G speed modes **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** (DWCXLP_R_SRDS_INTF!=3) ? (DWCXLP_XSBI_TX_CLK_PER_LN ? "clk_xsbi_tx3_i" : "clk_xsbi_tx_i") : "clk_rpcs_tx_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_rpcs_rx0_i	I	PCS-R Receive Data Path Clock for lane 0. This clock is used in the Receive data path of the PCS-R block for lane 0. The rpcs_rx_data0_i[rpcs_phw:0] signal is synchronous to this clock when DWCXLP_R_SRDS_INTF = 3 . The frequency of this based on operating speed mode is as follows as: ■ 10G/20G/40G mode : 156.25MHz ■ 25G/50G/100G mode (NRZ) : 390.625MHz ■ 50G PAM4/100G PAM4/100GAUI-4/50GAUI-2 mode: 402.462121 MHz **Exists:** Always **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_rpcs_rx1_i	I	PCS-R Receive Data Path Clock for lane 1. This clock is used in the Receive data path of the PCS-R block for lane 1. The rpcs_rx_data1_i[rpcs_phw:0] signal is synchronous to this clock when DWCXLP_R_SRDS_INTF = 3 . This Clock will be present only in multi lane configuration. The frequency of this based on operating speed mode is as follows as: ■ 20G/40G mode : 156.25MHz ■ 50G/100G mode (NRZ) : 390.625MHz ■ 50G PAM4/100G PAM4/100GAUI-4/50GAUI-2 mode: 402.462121 MHz ■ This clock need not be driven in 10G, 25G speed modes **Exists:** DWCXLP_MULTI_LANE **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_rpcs_rx2_i	I	PCS-R Receive Data Path Clock for lane 2. This clock is used in the Receive data path of the PCS-R block for lane 2. The rpcs_rx_data2_i[rpcs_phw:0] signal is synchronous to this clock when DWCXLP_R_SRDS_INTF = 3 . This Clock will be present only in multi lane configuration. The frequency of this based on operating speed mode is as follows as: ■ 40G mode : 156.25MHz ■ 50G/100G mode (NRZ) : 390.625MHz ■ 50G PAM4/100G PAM4/100GAUI-4/50GAUI-2 mode: 402.462121 MHz ■ This clock need not be driven in 10G, 20G and 25G speed modes **Exists:** DWCXLP_MULTI_LANE **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_rpcs_rx3_i	I	PCS-R Receive Data Path Clock for lane 3. This clock is used in the Receive data path of the PCS-R block for lane 3. The rpcs_rx_data3_i[rpcs_phw:0] signal is synchronous to this clock when DWCXLP_R_SRDS_INTF = 3 . This Clock will be present only in multi lane configuration. The frequency of this based on operating speed mode is as follows as: ■ 40G mode : 156.25MHz ■ 50G/100G mode (NRZ) : 390.625MHz ■ 50G PAM4/100G PAM4/100GAUI-4/50GAUI-2 mode: 402.462121 MHz ■ This clock need not be driven in 10G, 20G and 25G speed modes **Exists:** DWCXLP_MULTI_LANE **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_xsbi_rx0_i	I	XSBI Receive Clock for lane 0. This clock samples the rpcs_rx_data0_i[rpcs_phw:0] data for all PCS-R PHY Interface types except DWCXLP_R_SRDS_INTF = 3 . The frequency of this clock depends the PCS-R PHY Interface Type and operating mode. If DWCXLP_MAIN_MODE == 2, then ■ DWCXLP_R_SRDS_INTF=0 : 644.53125MHz ■ DWCXLP_R_SRDS_INTF=4 : 515.625MHz ■ DWCXLP_R_SRDS_INTF=1 : 322.265MHz ■ DWCXLP_R_SRDS_INTF=2 : 312.5MHz ■ DWCXLP_R_SRDS_INTF=5 : 161.132MHz ■ DWCXLP_R_SRDS_INTF=3 : 156.25MHz ■ DWCXLP_R_SRDS_INTF=6 : 128.906MHz If DWCXLP_MAIN_MODE!=2 and DWCXLP_R_SRDS_INTF=7, then ■ 10G/20G/40G mode : 257.8125MHz ■ 25G/50G/100G mode : 644.53125MHz If DWCXLP_MAIN_MODE!=2 and DWCXLP_R_SRDS_INTF=1, then ■ 10G/20G/40G mode : 322.265625MHz ■ 25G/50G/100G mode : 805.6640625MHz **Exists:** (DWCXLP_R_SRDS_INTF!=3) **Synchronous To:** N/A **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_xsbi_rx1_i	I	XSBI Receive Clock for lane 1. This clock samples the rpcs_rx_data1_i[rpcs_phw:0] data for all PCS-R PHY Interface types except DWCXLP_R_SRDS_INTF = 3 . The frequency of this clock depends the PCS-R PHY Interface Type and operating mode. If DWCXLP_MAIN_MODE == 2, then ■ DWCXLP_R_SRDS_INTF=0: 644.53125MHz ■ DWCXLP_R_SRDS_INTF=4: 515.625MHz ■ DWCXLP_R_SRDS_INTF=1: 322.265MHz ■ DWCXLP_R_SRDS_INTF=2: 312.5MHz ■ DWCXLP_R_SRDS_INTF=5: 161.132MHz ■ DWCXLP_R_SRDS_INTF=3: 156.25MHz ■ DWCXLP_R_SRDS_INTF=6: 128.906MHz If DWCXLP_MAIN_MODE!=2 and DWCXLP_R_SRDS_INTF = 7, then ■ 20G/40G mode : 257.8125MHz ■ 50G/100G mode : 644.53125MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 1, then ■ 20G/40G mode : 322.265625MHz ■ 50G/100G mode : 805.6640625MHz ■ This clock need not be driven in 10G and 25G speed modes **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_R_SRDS_INTF!=3) **Synchronous To:** N/A **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_xsbi_rx2_i	I	XSBI Receive Clock for lane 2. This clock samples the rpcs_rx_data2_i[rpcs_phw:0] data for all PCS-R PHY Interface types except DWCXLP_R_SRDS_INTF = 3. The frequency of this clock depends the PCS-R PHY Interface Type and operating mode. If DWCXLP_MAIN_MODE == 2, then ■ DWCXLP_R_SRDS_INTF=0: 644.53125MHz ■ DWCXLP_R_SRDS_INTF=4: 515.625MHz ■ DWCXLP_R_SRDS_INTF=1: 322.265MHz ■ DWCXLP_R_SRDS_INTF=2: 312.5MHz ■ DWCXLP_R_SRDS_INTF=5: 161.132MHz ■ DWCXLP_R_SRDS_INTF=3: 156.25MHz ■ DWCXLP_R_SRDS_INTF=6: 128.906MHz If DWCXLP_MAIN_MODE!=2 and DWCXLP_R_SRDS_INTF = 7, then ■ 40G mode : 257.8125MHz ■ 50G/100G mode : 644.53125MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 1, then ■ 40G mode : 322.265625MHz ■ 50G/100G mode : 805.6640625MHz ■ This clock need not be driven in 10G, 20G and 25G speed modes **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_R_SRDS_INTF!=3) **Synchronous To:** N/A **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
clk_xsbi_rx3_i	I	XSBI Receive Clock for lane 3. This clock samples the rpcs_rx_data3_i[rpcs_phw:0] data for all PCS-R PHY Interface types except DWCXLP_R_SRDS_INTF = 3. The frequency of this clock depends the PCS-R PHY Interface Type and operating mode. If DWCXLP_MAIN_MODE == 2, then ■ DWCXLP_R_SRDS_INTF=0: 644.53125MHz ■ DWCXLP_R_SRDS_INTF=4: 515.625MHz ■ DWCXLP_R_SRDS_INTF=1: 322.265MHz ■ DWCXLP_R_SRDS_INTF=2: 312.5MHz ■ DWCXLP_R_SRDS_INTF=5: 161.132MHz ■ DWCXLP_R_SRDS_INTF=3: 156.25MHz ■ DWCXLP_R_SRDS_INTF=6: 128.906MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 7, then ■ 40G mode : 257.8125MHz ■ 100G mode : 644.53125MHz If DWCXLP_MAIN_MODE != 2 and DWCXLP_R_SRDS_INTF = 1, then ■ 40G mode : 322.265625MHz ■ 100G mode : 805.6640625MHz ■ This clock need not be driven in 10G, 20G, 25G and 50G speed modes **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_R_SRDS_INTF!=3) **Synchronous To:** N/A **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
rpcs_rx_data0_i[(DWCXLP_PHY_DW-1):0]	I	PCS-R Receive Data for lane 0. This is the Receive data the PCS-R PHY interface for lane 0. The width of this signal (rpcs_phw) depends the PCS-R PHY Interface Type: If DWCXLP_MAIN_MODE == 2, then ■ If DWCXLP_R_SRDS_INTF = 0, Width = 16 ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 2, Width = 33 ■ If DWCXLP_R_SRDS_INTF = 3, Width = 66 ■ If DWCXLP_R_SRDS_INTF = 4, Width = 20 ■ If DWCXLP_R_SRDS_INTF = 5, Width = 64 If DWCXLP_MAIN_MODE != 2 then, ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 7, Width = 40 **Exists:** Always **Synchronous To:** (DWCXLP_R_SRDS_INTF!=3) ? "clk_xsbi_rx0_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_rx_data1_i[(DWCXLP_PHY_DW-1):0]	I	PCS-R Receive Data for lane 1. This is the Receive data the PCS-R PHY interface for lane 1. The width of this signal (rpcs_phw) depends the PCS-R PHY Interface Type: If DWCXLP_MAIN_MODE == 2, then ■ If DWCXLP_R_SRDS_INTF = 0, Width = 16 ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 2, Width = 33 ■ If DWCXLP_R_SRDS_INTF = 3, Width = 66 ■ If DWCXLP_R_SRDS_INTF = 4, Width = 20 ■ If DWCXLP_R_SRDS_INTF = 5, Width = 64 If DWCXLP_MAIN_MODE != 2 then, ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 7, Width = 40 **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** (DWCXLP_R_SRDS_INTF!=3) ? "clk_xsbi_rx1_i" : "clk_rpcs_rx1_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-11 PCS-R PHY Interface Signals (Continued)

Port Name	I/O	Description
rpcs_rx_data2_i[(DWCXLP_PHY_DW-1):0]	I	PCS-R Receive Data for lane 2. This is the Receive data the PCS-R PHY interface for lane 2. The width of this signal (rpcs_phw) depends the PCS-R PHY Interface Type: If DWCXLP_MAIN_MODE == 2, then ■ If DWCXLP_R_SRDS_INTF = 0, Width = 16 ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 2, Width = 33 ■ If DWCXLP_R_SRDS_INTF = 3, Width = 66 ■ If DWCXLP_R_SRDS_INTF = 4, Width = 20 ■ If DWCXLP_R_SRDS_INTF = 5, Width = 64 If DWCXLP_MAIN_MODE!=2 then, ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 7, Width = 40 **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** (DWCXLP_R_SRDS_INTF!=3) ? "clk_xsbi_rx2_i" : "clk_rpcs_rx2_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_rx_data3_i[(DWCXLP_PHY_DW-1):0]	I	PCS-R Receive Data for lane 3. This is the Receive data the PCS-R PHY interface for lane 3. The width of this signal (rpcs_phw) depends the PCS-R PHY Interface Type: If DWCXLP_MAIN_MODE==2, then ■ If DWCXLP_R_SRDS_INTF = 0, Width = 16 ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 2, Width = 33 ■ If DWCXLP_R_SRDS_INTF = 3, Width = 66 ■ If DWCXLP_R_SRDS_INTF = 4, Width = 20 ■ If DWCXLP_R_SRDS_INTF = 5, Width = 64 If DWCXLP_MAIN_MODE!=2 then, ■ If DWCXLP_R_SRDS_INTF = 1, Width = 32 ■ If DWCXLP_R_SRDS_INTF = 7, Width = 40 **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** (DWCXLP_R_SRDS_INTF!=3) ? "clk_xsbi_rx3_i" : "clk_rpcs_rx3_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.12 Custom1 PCS-PHY Interface Signals

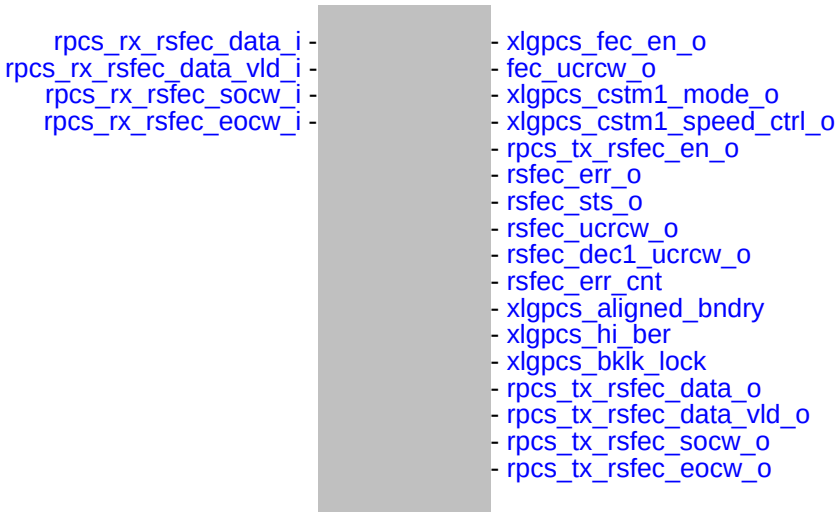


Table 6-12 Custom1 PCS-PHY Interface Signals

Port Name	I/O	Description
rpcs_rx_rsfec_data_i[(DWCXLP_RS_EC_MEM_DW-1):0]	I	PCS-R Receive Data when RS FEC is enabled in custom configuration(SNPS_RSVDPARAM_11). This is the Receive data on the PCS-R PHY interface for RSFEC path. **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_rx0_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_rx_rsfec_data_vld_i	I	PCS-R Receive Data Valid when RS FEC is enabled in custom configuration(SNPS_RSVDPARAM_11). This is the Receive data valid on the PCS-R PHY interface for RS FEC path and it qualifies the receive data on the RS-FEC path. **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_rx0_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-12 Custom1 PCS-PHY Interface Signals (Continued)

Port Name	I/O	Description
rpcs_rx_rsfec_socw_i	I	RS-FEC Start of code word when RS FEC is enabled in custom configuration(SNPS_RSVDPARAM_11). This is the control input on the PCS-R PHY interface for RS FEC path and it indicates the start of FEC codeword **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_rx0_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_rx_rsfec_eocw_i	I	RS-FEC End of code word when RS FEC is enabled in custom configuration(SNPS_RSVDPARAM_11). This is the control input on the PCS-R PHY interface for RS FEC path and it indicates the end of FEC codeword **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_rx0_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_fec_en_o	O	BASE-R FEC enable control. Signal to indicate BASE-R FEC is enabled . When asserted indicates BASE-R FEC is enabled. When de-asserted, indicates BASE-RFEC is disabled **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_FEC_EN) **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
fec_ucrcw_o	O	Uncorrected code word indication pulse in BASE-R FEC Mode . Signal to indicate detection of an uncorrected code word in BASE-R FEC mode . When asserted indicates detection of an un corrected code word in BASE-R FEC mode. **Exists:** (SNPS_RSVDPARAM_11||SNPS_RSVDPARAM_12)&&DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-12 Custom1 PCS-PHY Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_cstm1_mode_o	O	cstm1 Mode select control. Signal to indicate cstm1 mode or Ethernet mode of operation. When asserted indicates DWC_xlgpcs is working in cstm1 mode. When de-asserted, indicates Ethernet mode of operation **Exists:** (SNPS_RSVDPARAM_11) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_cstm1_speed_ctrl_o[1:0]	O	PHY Speed Control. When signal indicates the current operating speed/mode. Values: [1][0] --- 0 0 9.8304G 0 1 11.79648G 1 X 23.59296G **Exists:** (SNPS_RSVDPARAM_11) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_tx_rsfec_en_o	O	When asserted indicates that the DWC_xlgpcs transmit datapath is working in custom1 RS FEC Mode. This signal is asserted when both it-2 of RSFEC control register and cipri_tx_rsfec_en_i are set. **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rsfec_err_o	O	RS-FEC Error Indication. When high, it indicates that the RS-FEC code word block had errors. It is valid when the rsfec_sts_o is high. **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-12 Custom1 PCS-PHY Interface Signals (Continued)

Port Name	I/O	Description
rsfec_sts_o	O	RS-FEC code word status pulse which qualifies code word correction status. **Exists:** (SNPS_RSVDPARAM_11)&&DWCXLP_RSFEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rsfec_ucrcw_o	O	RS-FEC UnCorrected code word status. When high, it indicates that, code word is uncorrectable. When low, it indicates that the RSFEC code word was error-free or fully corrected by the Decoder. For configurations with SNPS_RSVDPARAM_11=1, this signal is valid only when the rsfec_sts_o is high. **Exists:** (SNPS_RSVDPARAM_11||SNPS_RSVDPARAM_12)&&DWCXLP_RSFEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rsfec_dec1_ucrcw_o	O	RS-FEC UnCorrected code word status from Decoder 1 when operating in interleaved RS-FEC Mode. When high, it indicates that, code word is uncorrectable. When low, it indicates that the RSFEC code word was error-free or fully corrected by the Decoder. This signal is valid only when interleaved RS-FEC mode is enabled by the host by programming relevant CSR bit. **Exists:** SNPS_RSVDPARAM_12&&DWCXLP_IRSFEC **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rsfec_err_cnt[2:0]	O	RS-FEC Corrected Error count. It gives the number of symbols corrected by the RS FEC in a code word block. It is valid only when rsfec_sts_o and rsfec_err_o are high and rsfec_ucrcw_o is low. **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** NA

Table 6-12 Custom1 PCS-PHY Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_aligned_bndry[6:0]	O	Block Lock Boundary. When xlgpcs_bklk_lock is high, it indicates the bit boundary at which the block synchronization is achieved. This signal is applicable to all custom speed modes and Ethernet 10G/25G speed modes as well. This provides bit boundary where the block sync is achieved by the PCS Receiver. **Exists:** (SNPS_RSVDPARAM_11) **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** NA
xlgpcs_hi_ber	O	High Bit Error ratio status. When high, it indicates Bit Error rate on the receive link has exceeded the threshold corresponding to the speed mode. ■ For 23G speed custom1 mode, 25G Ethernet mode, this is asserted if the number of sync bit errors exceeds 97 in a time window of ~2ms. ■ For 9G/11GG speed custom1 mode, 10G Ethernet mode, this is asserted if the number of sync bit errors exceeds 16 in a time window of ~125us **Exists:** (SNPS_RSVDPARAM_11||SNPS_RSVDPARAM_12) **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_bklk_lock	O	When high, it indicates the Rx block synchronization is achieved. This signal status is applicable only in non-RS FEC modes. This is asserted if PCS Receive logic receives 64 successive valid sync bits and is applicable to all custom1 speed modes and Ethernet 10G/25G speed modes as well **Exists:** (SNPS_RSVDPARAM_11||SNPS_RSVDPARAM_12) **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_tx_rsfec_data_o[(DWCXLP_RS_EC_MEM_DW-1):0]	O	This is the Transmit data from the PCS-R data path for RS FEC modes in custom configuration(SNPS_RSVDPARAM_11). **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-12 Custom1 PCS-PHY Interface Signals (Continued)

Port Name	I/O	Description
rpcs_tx_rsfec_data_vld_o	O	This signal qualifies the transmit data from the PCS-R data path for RS FEC modes in custom configuration(SNPS_RSVDPARAM_11). **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_tx_rsfec_socw_o	O	This signal indicates the Start of RS FEC code word in custom configuration(SNPS_RSVDPARAM_11). **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_tx_rsfec_eocw_o	O	This signal indicates the End of RS FEC code word in custom configuration(SNPS_RSVDPARAM_11) **Exists:** (SNPS_RSVDPARAM_11&&DWCXLP_RSFEC_EN) **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.13 FEC Error Correction Memory Interface Signals

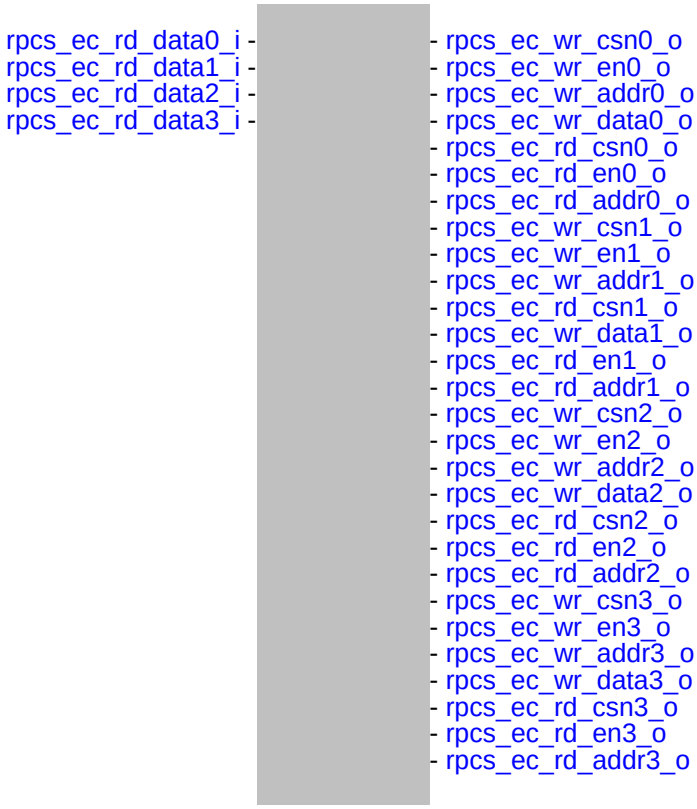


Table 6-13 FEC Error Correction Memory Interface Signals

Port Name	I/O	Description
rpcs_ec_wr_csn0_o	O	FEC Error Correction Memory Chip Select Signal of the Write Port - lane 0. This signal qualifies the rpcs_ec_wr_addr0_o signal. **Exists:** DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low

Table 6-13 FEC Error Correction Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ec_wr_en0_o	O	FEC Error Correction Memory Enable Signal of the Write Port -lane 0. When high, this signal indicates that the rpcs_ec_wr_data0_o data is valid and it is written to the memory. **Exists:** DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ec_wr_addr0_o[(DWCXLP_EC_MEM_AW-1):0]	O	FEC Error Correction Memory Address Bus of the Write Port -lane 0. This signal is the memory address bus of the write port. It is valid when the rpcs_ec_wr_csn0_o signal is asserted. **Exists:** DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ec_wr_data0_o[((ECC_RPCS_EC_CHKBW0+DWCXLP_EC_MEM_DW)-1):0]	O	FEC Error Correction Memory Data Bus of the Write Port- lane 0. This signal carries the data to be written into the memory. It is valid only when the rpcs_ec_wr_en0_o signal is asserted. **Exists:** DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ec_rd_csn0_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port-lane 0. This signal qualifies the rpcs_ec_rd_addr0_o signal. **Exists:** DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_ec_rd_en0_o	O	FEC Error Forward Memory Enable Signal of the Read Port -lane 0. When high, this signal indicates that the rpcs_ec_rd_data0_i signal contains valid data, and it sampled by the DWC_xlgpcs. **Exists:** DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-13 FEC Error Correction Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ec_rd_addr0_o[(DWCXLP_EC_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Read Port -lane 0. This signal is the memory address bus of the read port. It is valid when the rpcs_ec_rd_csn0_o signal is asserted. **Exists:** DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ec_rd_data0_i[((ECC_RPCS_EC_CHKBW0+DWCXLP_EC_MEM_DW)-1):0]	I	FEC Error Forward Memory Data Bus of the Read Port -lane 0. This signal carries the read data from the FEC error correction memory. It is valid when the rpcs_ec_rd_en0_o signal is asserted. **Exists:** DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ec_wr_csn1_o	O	FEC Error Correction Memory Chip Select Signal of the Write Port -lane 1. This signal qualifies the rpcs_ec_wr_addr1_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_wr_en1_o	O	FEC Error Correction Memory Enable Signal of the Write Port -lane 1. When high, this signal indicates that the rpcs_ec_wr_data1_o data is valid and it is written to the memory. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-13 FEC Error Correction Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ec_wr_addr1_o[(DWCXLP_EC_MEM_AW-1):0]	O	FEC Error Correction Memory Address Bus of the Write Port -lane 1. This signal is the memory address bus of the write port. It is valid when the rpcs_ec_wr_csn1_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_rd_csn1_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port-lane 1. This signal qualifies the rpcs_ec_rd_addr1_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_wr_data1_o[(ECC_RPCS_EC_CHKBW1+65):0]	O	FEC Error Correction Memory Data Bus of the Write Port- lane 1. This signal carries the data to be written into the memory. It is valid only when the rpcs_ec_wr_en1_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_rd_en1_o	O	FEC Error Forward Memory Enable Signal of the Read Port -lane 1. When high, this signal indicates that the rpcs_ec_rd_data1_i signal contains valid data, and it sampled by the DWC_xlgpcs. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-13 FEC Error Correction Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ec_rd_addr1_o[(DWCXLP_EC_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Read Port -lane 1. This signal is the memory address bus of the read port. It is valid when the rpcs_ec_rd_csn1_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_rd_data1_i[(ECC_RPCS_EC_CHKBW1+65):0]	I	FEC Error Forward Memory Data Bus of the Read Port -lane 1. This signal carries the read data from the FEC error correction memory. It is valid when the rpcs_ec_rd_en1_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_wr_csn2_o	O	FEC Error Correction Memory Chip Select Signal of the Write Port -lane 2. This signal qualifies the rpcs_ec_wr_addr2_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_wr_en2_o	O	FEC Error Correction Memory Enable Signal of the Write Port -lane 2. When high, this signal indicates that the rpcs_ec_wr_data2_o data is valid and it is written to the memory. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-13 FEC Error Correction Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ec_wr_addr2_o[(DWCXLP_EC_MEM_AW-1):0]	O	FEC Error Correction Memory Address Bus of the Write Port -lane 2. This signal is the memory address bus of the write port. It is valid when the rpcs_ec_wr_csn2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_wr_data2_o[(ECC_RPCS_EC_CHKBW2+65):0]	O	FEC Error Correction Memory Data Bus of the Write Port- lane 2. This signal carries the data to be written into the memory. It is valid only when the rpcs_ec_wr_en2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_rd_csn2_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port-lane 2. This signal qualifies the rpcs_ec_rd_addr2_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_rd_en2_o	O	FEC Error Forward Memory Enable Signal of the Read Port -lane 2. When high, this signal indicates that the rpcs_ec_rd_data2_i signal contains valid data, and it sampled by the DWC_xlgpcs. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-13 FEC Error Correction Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ec_rd_addr2_o[(DWCXLP_EC_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Read Port -lane 2. This signal is the memory address bus of the read port. It is valid when the rpcs_ec_rd_csn2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_rd_data2_i[(ECC_RPCS_EC_CHKBW2+65):0]	I	FEC Error Forward Memory Data Bus of the Read Port -lane 2. This signal carries the read data from the FEC error correction memory. It is valid when the rpcs_ec_rd_en2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_wr_csn3_o	O	FEC Error Correction Memory Chip Select Signal of the Write Port -lane 3. This signal qualifies the rpcs_ec_wr_addr3_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_wr_en3_o	O	FEC Error Correction Memory Enable Signal of the Write Port -lane 3. When high, this signal indicates that the rpcs_ec_wr_data3_o data is valid and it is written to the memory. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-13 FEC Error Correction Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ec_wr_addr3_o[(DWCXLP_EC_MEM_AW-1):0]	O	FEC Error Correction Memory Address Bus of the Write Port -lane 3. This signal is the memory address bus of the write port. It is valid when the rpcs_ec_wr_csn3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_wr_data3_o[(ECC_RPCS_EC_CHKBW3+65):0]	O	FEC Error Correction Memory Data Bus of the Write Port- lane 3. This signal carries the data to be written into the memory. It is valid only when the rpcs_ec_wr_en3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_rd_csn3_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port-lane 3. This signal qualifies the rpcs_ec_rd_addr3_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_rd_en3_o	O	FEC Error Forward Memory Enable Signal of the Read Port -lane 3. When high, this signal indicates that the rpcs_ec_rd_data3_i signal contains valid data, and it sampled by the DWC_xlgpcs. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-13 FEC Error Correction Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ec_rd_addr3_o[(DWCXLP_EC_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Read Port -lane 3. This signal is the memory address bus of the read port. It is valid when the rpcs_ec_rd_csn3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ec_rd_data3_i[(ECC_RPCS_EC_CHKBW3+65):0]	I	FEC Error Forward Memory Data Bus of the Read Port -lane 3. This signal carries the read data from the FEC error correction memory. It is valid when the rpcs_ec_rd_en3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

6.14 FEC Error Forward Memory Interface Signals

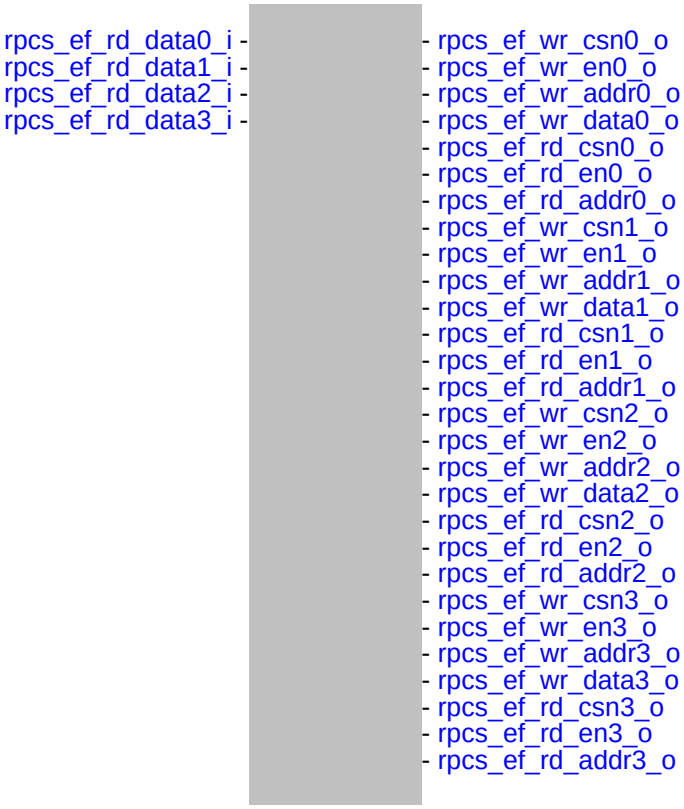


Table 6-14 FEC Error Forward Memory Interface Signals

Port Name	I/O	Description
rpcs_ef_wr_csn0_o	O	FEC Error Forward Memory Chip Select Signal of the Write Port-lane 0. This signal qualifies the rpcs_ef_wr_addr0_o signal. **Exists:** DWCXLP_FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low

Table 6-14 FEC Error Forward Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ef_wr_en0_o	O	FEC Error Forward Memory Enable Signal of the Write Port -lane 0. When high, this signal indicates that the rpcs_ef_wr_data0_o data is valid and it is written in the memory. **Exists:** DWCXLP_FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ef_wr_addr0_o[(DWCXLP_EF_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Write Port -lane 0. This signal is the memory address bus of the write port. It is valid when the rpcs_ef_wr_csn0_o signal is asserted. **Exists:** DWCXLP_FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ef_wr_data0_o[((ECC_RPCS_EF_CHKBW0+DWCXLP_EF_MEM_DW)-1):0]	O	FEC Error Forward Memory Data Bus of the Write Port -lane 0. This signal carries the data to be written into the memory. It is valid only when the rpcs_ef_wr_en0_o signal is asserted. **Exists:** DWCXLP_FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ef_rd_csn0_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port -lane 0. This signal qualifies the rpcs_ef_rd_addr0_o signal. **Exists:** DWCXLP_FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_ef_rd_en0_o	O	FEC Error Forward Memory Enable Signal of the Read Port -lane 0. When high, this signal indicates that the rpcs_ef_rd_data0_i signal contains valid data, and it is sampled by the DWC_xlgpcs. **Exists:** DWCXLP_FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-14 FEC Error Forward Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ef_rd_addr0_o[(DWCXLP_EF_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Read Port -lane 0. This signal is the memory address bus of the read port. It is valid when the rpcs_ef_rd_csn0_o signal is asserted. **Exists:** DWCXLP_FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ef_rd_data0_i[((ECC_RPCS_EF_CHKBW0+DWCXLP_EF_MEM_DW)-1):0]	I	FEC Error Forward Memory Data Bus of the Read Port -lane 0. This signal carries the read data from the FEC error forward memory. It is valid when the rpcs_ef_rd_en0_o signal is asserted. **Exists:** DWCXLP_FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ef_wr_csn1_o	O	FEC Error Forward Memory Chip Select Signal of the Write Port-lane 1. This signal qualifies the rpcs_ef_wr_addr1_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_wr_en1_o	O	FEC Error Forward Memory Enable Signal of the Write Port -lane 1. When high, this signal indicates that the rpcs_ef_wr_data1_o data is valid and it is written in the memory. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-14 FEC Error Forward Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ef_wr_addr1_o[(DWCXLP_EF_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Write Port -lane 1. This signal is the memory address bus of the write port. It is valid when the rpcs_ef_wr_csn1_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_wr_data1_o[(ECC_RPCS_EF_CHKBW1+65):0]	O	FEC Error Forward Memory Data Bus of the Write Port -lane 1. This signal carries the data to be written into the memory. It is valid only when the rpcs_ef_wr_en1_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_rd_csn1_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port -lane 1. This signal qualifies the rpcs_ef_rd_addr1_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_rd_en1_o	O	FEC Error Forward Memory Enable Signal of the Read Port -lane 1. When high, this signal indicates that the rpcs_ef_rd_data1_i signal contains valid data, and it is sampled by the DWC_xlgpcs. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-14 FEC Error Forward Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ef_rd_addr1_o[(DWCXLP_EF_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Read Port -lane 1. This signal is the memory address bus of the read port. It is valid when the rpcs_ef_rd_csn1_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_rd_data1_i[(ECC_RPCS_EF_CHKBW1+65):0]	I	FEC Error Forward Memory Data Bus of the Read Port -lane 1. This signal carries the read data from the FEC error forward memory. It is valid when the rpcs_ef_rd_en1_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx1_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_wr_csn2_o	O	FEC Error Forward Memory Chip Select Signal of the Write Port-lane 2. This signal qualifies the rpcs_ef_wr_addr2_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_wr_en2_o	O	FEC Error Forward Memory Enable Signal of the Write Port -lane 2. When high, this signal indicates that the rpcs_ef_wr_data2_o data is valid and it is written in the memory. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-14 FEC Error Forward Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ef_wr_addr2_o[(DWCXLP_EF_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Write Port -lane 2. This signal is the memory address bus of the write port. It is valid when the rpcs_ef_wr_csn2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_wr_data2_o[(ECC_RPCS_EF_CHKBW2+65):0]	O	FEC Error Forward Memory Data Bus of the Write Port -lane 2. This signal carries the data to be written into the memory. It is valid only when the rpcs_ef_wr_en2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_rd_csn2_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port -lane 2. This signal qualifies the rpcs_ef_rd_addr2_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_rd_en2_o	O	FEC Error Forward Memory Enable Signal of the Read Port -lane 2. When high, this signal indicates that the rpcs_ef_rd_data2_i signal contains valid data, and it is sampled by the DWC_xlgpcs. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-14 FEC Error Forward Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ef_rd_addr2_o[(DWCXLP_EF_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Read Port -lane 2. This signal is the memory address bus of the read port. It is valid when the rpcs_ef_rd_csn2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_rd_data2_i[(ECC_RPCS_EF_CHKBW2+65):0]	I	FEC Error Forward Memory Data Bus of the Read Port -lane 2. This signal carries the read data from the FEC error forward memory. It is valid when the rpcs_ef_rd_en2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx2_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_wr_csn3_o	O	FEC Error Forward Memory Chip Select Signal of the Write Port-lane 3. This signal qualifies the rpcs_ef_wr_addr3_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_wr_en3_o	O	FEC Error Forward Memory Enable Signal of the Write Port -lane 3. When high, this signal indicates that the rpcs_ef_wr_data3_o data is valid and it is written in the memory. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-14 FEC Error Forward Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ef_wr_addr3_o[(DWCXLP_EF_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Write Port -lane 3. This signal is the memory address bus of the write port. It is valid when the rpcs_ef_wr_csn3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_wr_data3_o[(ECC_RPCS_EF_CHKBW3+65):0]	O	FEC Error Forward Memory Data Bus of the Write Port -lane 3. This signal carries the data to be written into the memory. It is valid only when the rpcs_ef_wr_en3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_rd_csn3_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port -lane 3. This signal qualifies the rpcs_ef_rd_addr3_o signal. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_rd_en3_o	O	FEC Error Forward Memory Enable Signal of the Read Port -lane 3. When high, this signal indicates that the rpcs_ef_rd_data3_i signal contains valid data, and it is sampled by the DWC_xlgpcs. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

Table 6-14 FEC Error Forward Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_ef_rd_addr3_o[(DWCXLP_EF_MEM_AW-1):0]	O	FEC Error Forward Memory Address Bus of the Read Port -lane 3. This signal is the memory address bus of the read port. It is valid when the rpcs_ef_rd_csn3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)
rpcs_ef_rd_data3_i[(ECC_RPCS_EF_CHKBW3+65):0]	I	FEC Error Forward Memory Data Bus of the Read Port -lane 3. This signal carries the read data from the FEC error forward memory. It is valid when the rpcs_ef_rd_en3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE&&DWCXLP_FEC_ERROR_FWD **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** clk_rpcs_rx3_i in BASE-R FEC mode, clk_rpcs_rx0_i in RSFEC mode (if present)

6.15 RS-FEC Memory Interface Signals

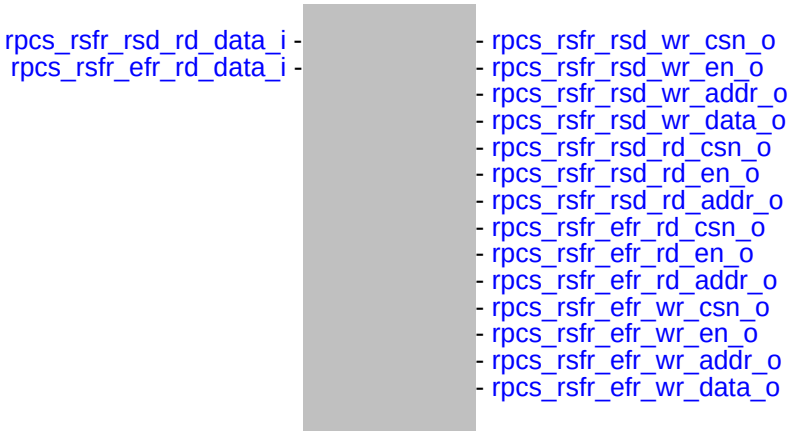


Table 6-15 RS-FEC Memory Interface Signals

Port Name	I/O	Description
rpcs_rsfr_rsd_wr_csn_o	O	RS-FEC Error Correction Memory Write Chip select signal. **Exists:** DWCXLP_RSFEC_EN&&!DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_rsfr_rsd_wr_en_o	O	RS-FEC Error Correction Memory Write enable. **Exists:** DWCXLP_RSFEC_EN&&!DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_rsfr_rsd_wr_addr_o[(DWCXLP_RS_EC_MEM_AW-1):0]	O	RS-FEC Error Correction Memory Write Address . **Exists:** DWCXLP_RSFEC_EN&&!DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** NA

Table 6-15 RS-FEC Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_rsfr_rsd_wr_data_o[((ECC_RSFR_EC_CHKBW+DWCXLP_RS_EC_MEM_DW)-1):0]	O	RS-FEC Error Correction Memory Write Data Bus. **Exists:** DWCXLP_RSFEC_EN&&!DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** NA
rpcs_rsfr_rsd_rd_csn_o	O	RS-FEC Error Correction Memory Read Chip Select signal. **Exists:** DWCXLP_RSFEC_EN&&!DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_rsfr_rsd_rd_en_o	O	RS-FEC Error Correction Memory Read enable. **Exists:** DWCXLP_RSFEC_EN&&!DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_rsfr_rsd_rd_addr_o[(DWCXLP_RS_EC_MEM_AW-1):0]	O	RS-FEC Error Correction Memory Read address. **Exists:** DWCXLP_RSFEC_EN&&!DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** NA
rpcs_rsfr_rsd_rd_data_i[((ECC_RSFR_EC_CHKBW+DWCXLP_RS_EC_MEM_DW)-1):0]	I	RS-FEC Error Correction Memory Read Data. **Exists:** DWCXLP_RSFEC_EN&&!DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** NA
rpcs_rsfr_efr_rd_csn_o	O	RS-FEC Error Forwarding Memory Chip select for the Read Port. **Exists:** DWCXLP_RSFEC_ERFW_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low

Table 6-15 RS-FEC Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_rsfr_efr_rd_en_o	O	RS-FEC Error Forwarding Memory enable for the Read Port. **Exists:** DWCXLP_RSFEC_ERFW_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_rsfr_efr_rd_addr_o[(DWCXLP_RSFR_EFR_MEM_AW-1):0]	O	RS-FEC Error Forwarding Memory address of the Read Port. **Exists:** DWCXLP_RSFEC_ERFW_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** NA
rpcs_rsfr_efr_rd_data_i[((ECC_RSFR_EF_CHKBW+DWCXLP_RSFR_EFR_MEM_DW)-1):0]	I	RS-FEC Error Forwarding Memory Data from Read Port. **Exists:** DWCXLP_RSFEC_ERFW_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** NA
rpcs_rsfr_efr_wr_csn_o	O	RS-FEC Error Forwarding Memory Chip select of the Write Port. **Exists:** DWCXLP_RSFEC_ERFW_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_rsfr_efr_wr_en_o	O	RS-FEC Error Forwarding Memory enable of the Write Port. **Exists:** DWCXLP_RSFEC_ERFW_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_rsfr_efr_wr_addr_o[(DWCXLP_RSFR_EFR_MEM_AW-1):0]	O	RS-FEC Error Forwarding Memory address of the Write Port. **Exists:** DWCXLP_RSFEC_ERFW_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low

Table 6-15 RS-FEC Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_rsfr_efr_wr_data_o[((ECC_RSFR_EF_CHKBW+DWCXLP_RSFR_EFR_MEM_DW)-1):0]	O	RS-FEC Error Correction Error Forwarding Write Data. **Exists:** DWCXLP_RSFEC_ERFW_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** NA

6.16 Alignment Marker Removal Memory Interface Signals

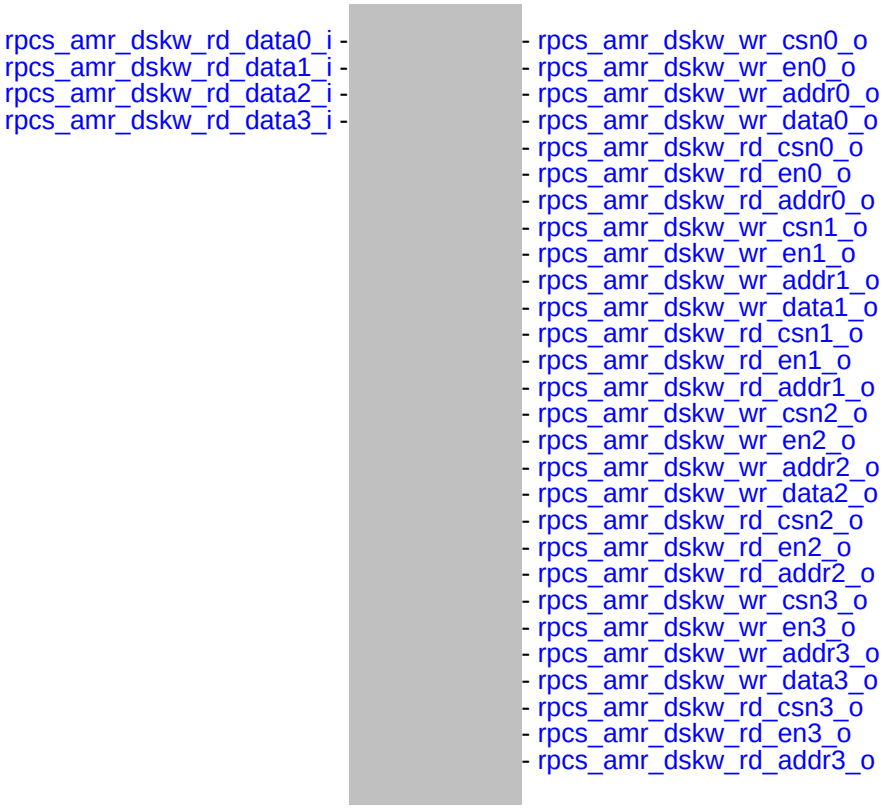


Table 6-16 Alignment Marker Removal Memory Interface Signals

Port Name	I/O	Description
rpcs_amr_dskw_wr_csn0_o	O	AM Removal and Deskew Memory Chip Select Signal of the Write Port -lane 0. This signal qualifies the rpcs_amr_dskw_wr_addr0_o signal. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-16 Alignment Marker Removal Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_amr_dskw_wr_en0_o	O	AM Removal and Deskew Memory Enable Signal of the Write Port - lane 0. When high, this signal indicates that the rpcs_amr_dskw_wr_data0_o data is valid and it is written to the memory. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_amr_dskw_wr_addr0_o[(DWCXLP_DSKW_BF_AW-1):0]	O	AM Removal and Deskew Memory Address Bus of the Write Port - lane 0. This signal is the memory address bus of the write port. It is valid when the rpcs_amr_dskw_wr_csn0_o signal is asserted. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_wr_data0_o[(ECC_DSKW_MEM_CHKBW0+65):0]	O	AM Removal and Deskew Memory Data Bus of the Write Port -lane 0. This signal carries the data to be written into the memory. It is valid only when the rpcs_amr_dskw_wr_en0_o signal is asserted. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_rd_csn0_o	O	AM Removal and Deskew Memory Chip Select Signal of the Read Port -lane 0. This signal qualifies the rpcs_ef_rd_addr0_o signal. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low

Table 6-16 Alignment Marker Removal Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_amr_dskw_rd_en0_o	O	AM Removal and Deskew Memory Enable Signal of the Read Port - lane 0. When high, this signal indicates that the rpcs_amr_dskw_rd_data0_i signal contains valid data, and it is sampled by the DWC_xlgpcs. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_amr_dskw_rd_addr0_o[(DWCXLP_DSKW_BF_AW-1):0]	O	AM Removal and Deskew Memory Address Bus of the Read Port - lane 0. This signal is the memory address bus of the read port. It is valid when the rpcs_amr_dskw_rd_csn0_o signal is asserted. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_rd_data0_i[(ECC_DSKW_MEM_CHKBW0+65):0]	I	AM Removal and Deskew Memory Data Bus of the Read Port -lane 0. This signal carries the read data from the AM Removal and Deskew Memory memory. It is valid when the rpcs_amr_dskw_rd_en0_o signal is asserted. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_wr_csn1_o	O	AM Removal and Deskew Memory Chip Select Signal of the Write Port -lane 1. This signal qualifies the rpcs_amr_dskw_wr_addr1_o signal. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** clk_rpcs_rx1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-16 Alignment Marker Removal Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_amr_dskw_wr_en1_o	O	AM Removal and Deskew Memory Enable Signal of the Write Port - lane 1. When high, this signal indicates that the rpcs_amr_dskw_wr_data1_o data is valid and it is written to the memory. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** clk_rpcs_rx1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_amr_dskw_wr_addr1_o[(DWCXLP_DSKW_BF_AW-1):0]	O	AM Removal and Deskew Memory Address Bus of the Write Port - lane 1. This signal is the memory address bus of the write port. It is valid when the rpcs_amr_dskw_wr_csn1_o signal is asserted. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** clk_rpcs_rx1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_wr_data1_o[(ECC_DSKW_MEM_CHKBW1+65):0]	O	AM Removal and Deskew Memory Data Bus of the Write Port -lane 1. This signal carries the data to be written into the memory. It is valid only when the rpcs_amr_dskw_wr_en1_o signal is asserted. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** clk_rpcs_rx1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_rd_csn1_o	O	AM Removal and Deskew Memory Chip Select Signal of the Read Port -lane 1. This signal qualifies the rpcs_ef_rd_addr1_o signal. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low

Table 6-16 Alignment Marker Removal Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_amr_dskw_rd_en1_o	O	AM Removal and Deskew Memory Enable Signal of the Read Port - lane 1. When high, this signal indicates that the rpcs_amr_dskw_rd_data1_i signal contains valid data, and it is sampled by the DWC_xlgpcs. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_amr_dskw_rd_addr1_o[(DWCXLP_DSKW_BF_AW-1):0]	O	AM Removal and Deskew Memory Address Bus of the Read Port - lane 1. This signal is the memory address bus of the read port. It is valid when the rpcs_amr_dskw_rd_csn1_o signal is asserted. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_rd_data1_i[(ECC_DSKW_MEM_CHKBW1+65):0]	I	AM Removal and Deskew Memory Data Bus of the Read Port -lane 1. This signal carries the read data from the AM Removal and Deskew Memory memory. It is valid when the rpcs_amr_dskw_rd_en1_o signal is asserted. **Exists:** !DWCXLP_25G_ONLY **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_wr_csn2_o	O	AM Removal and Deskew Memory Chip Select Signal of the Write Port -lane 2. This signal qualifies the rpcs_amr_dskw_wr_addr2_o signal. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** clk_rpcs_rx2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-16 Alignment Marker Removal Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_amr_dskw_wr_en2_o	O	AM Removal and Deskew Memory Enable Signal of the Write Port - lane 2. When high, this signal indicates that the rpcs_amr_dskw_wr_data2_o data is valid and it is written to the memory. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** clk_rpcs_rx2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_amr_dskw_wr_addr2_o[(DWCXLP_DSKW_BF_AW-1):0]	O	AM Removal and Deskew Memory Address Bus of the Write Port - lane 2. This signal is the memory address bus of the write port. It is valid when the rpcs_amr_dskw_wr_csn2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** clk_rpcs_rx2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_wr_data2_o[(ECC_DSKW_MEM_CHKBW2+65):0]	O	AM Removal and Deskew Memory Data Bus of the Write Port -lane 2. This signal carries the data to be written into the memory. It is valid only when the rpcs_amr_dskw_wr_en2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** clk_rpcs_rx2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_rd_csn2_o	O	AM Removal and Deskew Memory Chip Select Signal of the Read Port -lane 2. This signal qualifies the rpcs_ef_rd_addr2_o signal. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low

Table 6-16 Alignment Marker Removal Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_amr_dskw_rd_en2_o	O	AM Removal and Deskew Memory Enable Signal of the Read Port - lane 2. When high, this signal indicates that the rpcs_amr_dskw_rd_data2_i signal contains valid data, and it is sampled by the DWC_xlgpcs. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_amr_dskw_rd_addr2_o[(DWCXLP_DSKW_BF_AW-1):0]	O	AM Removal and Deskew Memory Address Bus of the Read Port - lane 2. This signal is the memory address bus of the read port. It is valid when the rpcs_amr_dskw_rd_csn2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_rd_data2_i[(ECC_DSKW_MEM_CHKBW2+65):0]	I	AM Removal and Deskew Memory Data Bus of the Read Port -lane 2. This signal carries the read data from the AM Removal and Deskew Memory memory. It is valid when the rpcs_amr_dskw_rd_en2_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_wr_csn3_o	O	AM Removal and Deskew Memory Chip Select Signal of the Write Port -lane 3. This signal qualifies the rpcs_amr_dskw_wr_addr3_o signal. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** clk_rpcs_rx3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-16 Alignment Marker Removal Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_amr_dskw_wr_en3_o	O	AM Removal and Deskew Memory Enable Signal of the Write Port - lane 3. When high, this signal indicates that the rpcs_amr_dskw_wr_data3_o data is valid and it is written to the memory. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** clk_rpcs_rx3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_amr_dskw_wr_addr3_o[(DWCXLP_DSKW_BF_AW-1):0]	O	AM Removal and Deskew Memory Address Bus of the Write Port - lane 3. This signal is the memory address bus of the write port. It is valid when the rpcs_amr_dskw_wr_csn3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** clk_rpcs_rx3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_wr_data3_o[(ECC_DSKW_MEM_CHKBW3+65):0]	O	AM Removal and Deskew Memory Data Bus of the Write Port -lane 3. This signal carries the data to be written into the memory. It is valid only when the rpcs_amr_dskw_wr_en3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** clk_rpcs_rx3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_rd_csn3_o	O	AM Removal and Deskew Memory Chip Select Signal of the Read Port -lane 3. This signal qualifies the rpcs_ef_rd_addr3_o signal. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low

Table 6-16 Alignment Marker Removal Memory Interface Signals (Continued)

Port Name	I/O	Description
rpcs_amr_dskw_rd_en3_o	O	AM Removal and Deskew Memory Enable Signal of the Read Port - lane 3. When high, this signal indicates that the rpcs_amr_dskw_rd_data3_i signal contains valid data, and it is sampled by the DWC_xlgpcs. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_amr_dskw_rd_addr3_o[(DWCXLP_DSKW_BF_AW-1):0]	O	AM Removal and Deskew Memory Address Bus of the Read Port - lane 3. This signal is the memory address bus of the read port. It is valid when the rpcs_amr_dskw_rd_csn3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_amr_dskw_rd_data3_i[(ECC_DSKW_MEM_CHKBW3+65):0]	I	AM Removal and Deskew Memory Data Bus of the Read Port -lane 3. This signal carries the read data from the AM Removal and Deskew Memory memory. It is valid when the rpcs_amr_dskw_rd_en3_o signal is asserted. **Exists:** DWCXLP_MULTI_LANE||DWCXLP_100G_1LANE **Synchronous To:** SNPS_RSVDPARAM_5 ? "clk_rpcs_2x_rx_i" : "clk_rpcs_rx0_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.17 Energy Efficient Ethernet PHY Interface Signals

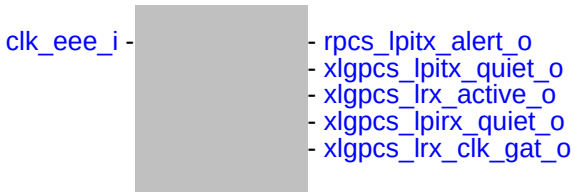


Table 6-17 Energy Efficient Ethernet PHY Interface Signals

Port Name	I/O	Description
clk_eee_i	I	Clock for the Energy Efficient Ethernet block. This is a free running clock, irrespective of current state of the Transmit and Receive. The frequency of this clock can be from 30 MHz to 200 MHz **Exists:** DWCXLP_EEE_EN **Synchronous To:** N/A **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_lpitx_alert_o	O	This signal when asserted indicates that the PCS core is sending EEE Alert Pattern. **Exists:** DWCXLP_EEE_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_lpitx_quiet_o	O	Transmit Path in Quiet State. This signal is asserted when the EEE controller in Tx path is in the Quiet (power-save) state. When this signal is asserted, you can disable the Tx data path of the PHY or the PMD. In addition, you can use this signal to stop the Transmit clocks to the DWC_xlgpcs and the PHY. **Exists:** DWCXLP_EEE_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-17 Energy Efficient Ethernet PHY Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_lrx_active_o	O	Receive Path in EEE Mode. This signal is asserted when the DWC_xlgpcs receives LPI Idle the Rx path. It is deactivated when it receives the normal Idle or Data. This signal provides an additional status to the PHY or the PMD. **Exists:** DWCXLP_EEE_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_lpirx_quiet_o	O	Receive Path in Quiet State. This signal is asserted when the EEE controller in Rx path is in the Quiet (power-save) state. When this signal is asserted, you can disable the Rx path of the PHY or the PMD. In addition, you can use this signal to stop the Receive clocks to the DWC_xlgpcs. **Exists:** DWCXLP_EEE_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_lrx_clk_gat_o	O	MAC Clock Gating Enable Control. This signal can be used for gating the XGMII/XLGMII Rx Clock towards the MAC. This signal is asserted when the DWC_xlgpcs receives the LPI Idle Rx path for a period equal to or greater than the number of clk_rpcs_rx0_i cycles as programmed in Bits[15:12] of VR PCS MMD EEE Mode Control Register and Bit 10 of SR PCS MMD Control1 Register is set. **Exists:** DWCXLP_EEE_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.18 Non-Synopsys PHY Generic Signals



Table 6-18 Non-Synopsys PHY Generic Signals

Port Name	I/O	Description
xlgpcs_loopback_en_o	O	Loopback enable in the PHY. This signal indicates that the host programmed the device to enable loopback mode in the PHY. The loopback mode is enabled through any of the following bits: ■ Bit 0 of SR PMA MMD Control 1 Register ■ Bit 14 of SR PCS MMD Control 1 Register ■ Bit 14 of SR MII MMD Control Register **Exists:** !DWCXLP_SNPS_XAUI_PHY **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_pdown_o	O	Power Down Enable. This signal indicates that the host has programmed the device for power-down mode **Exists:** !DWCXLP_SNPS_XAUI_PHY **Synchronous To:** !DWCXLP_SNPS_XAUI_PHY&&DWCXLP_MDIO_INTF ? "clk_mdc_i" : "clk_csr_i" **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-18 Non-Synopsys PHY Generic Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx_en_o[(DWCXLP_NUM_LANES-1):0]	O	Transmit Lane Enable or Disable Control. This signal controls the transmit data path of the PHY Values: ■ 0: Corresponding Transmit Lane is disabled. ■ 1: Corresponding Transmit Lane is enabled (default). This signal is controlled by either of: ■ Bits[4:1] of SR PMA MMD Transmit Disable Register ■ Bits[7:4] of VR PMA MMD Digital Control 1 Register ■ Bits[7:4] of VR PCS MMD Digital Control 1 Register **Exists:** !DWCXLP_SNPS_XAUI_PHY **Synchronous To:** (!DWCXLP_EEE_EN) ? "clk_csr_i" : "None" **Registered:** (!DWCXLP_EEE_EN) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_en_o[(DWCXLP_NUM_LANES-1):0]	O	Receive Lane Enable or Disable Control. This signal controls the receive data path of the PHY Values: ■ 0: Corresponding Receive Lane is disabled. ■ 1: Corresponding Receive Lane is enabled (default). This signal cannot be controlled through any register. It is internally controlled based the status of EEE or Auto-negotiation operation. For configurations without EEE or CL73 Auto-negotiation support, this signal will be always high. **Exists:** !DWCXLP_SNPS_XAUI_PHY **Synchronous To:** (!DWCXLP_EEE_EN) ? "clk_csr_i" : "None" **Registered:** (!DWCXLP_EEE_EN) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.19 Link Enable Signals

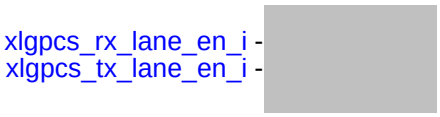


Table 6-19 Link Enable Signals

Port Name	I/O	Description
xlgpcs_rx_lane_en_i[(DWCXLP_NUM_LANES-1):0]	I	Receive Lane Enable Control Signal. This input port is to be set to indicate the active receive lanes corresponding to the speed mode, if LANE_CTRL_EXT bit of VR PCS MMD Digital Control 1 Register is set to 1. Valid input combinations are: ■ For multilane configuration: - 4'b1111: For 100G/40G - 4'b0011: For 50G/20G - 4'b0001: For 25G/10G - Other values: Reserved ■ For Single lane configuration: - 1'b1: For 25G/10G - 1'b0: Reserved The actual speed mode is differentiated based on the clock frequency clk_rpcs_rx_i[i]. where i = 0 to 3 **Exists:** DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes,with clk_csr_i

Table 6-19 Link Enable Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx_lane_en_i[(DWCXLP_NUM_LANES-1):0]	I	Transmit Lane Enable Control Signal . This input port is to be set to indicate the active transmit lanes corresponding to the speed mode, if LANE_CTRL_EXT bit of VR PCS MMD Digital Control 1 Register is set to 1. Valid input combinations are: ■ For multilane configuration: - 4'b1111: For 100G/40G - 4'b0011: For 50G/20G - 4'b0001: For 25G/10G - Other values- Reserved ■ For Single lane configuration: - 1'b1: For 25G/10G - 1'b0: Reserved The actual speed mode is differentiated based on the clock frequency clk_rpcs_tx_i[i] where i = 0 to 3 **Exists:** DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes,with clk_csr_i

6.20 Link Status Signals



Table 6-20 Link Status Signals

Port Name	I/O	Description
xlgpcs_rx_lane_sts_o	O	Receive Link Status Indicator. This signal is asserted if DWC_xlgpcs receiver has properly aligned/synchronized to the received data stream. **Exists:** Always **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_lane_sts_o	O	Transmit Link Status Signal. This signal is de-asserted when the transmitter detects a Remote Fault ordered set on XLGMII Tx interface. Otherwise, this signal will be high. **Exists:** Always **Synchronous To:** clk_rpcs_tx_i **Registered:** DWCXLP_25G_ONLY ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.21 Synopsys PHY-Specific Tx Control and Status Interface Signals

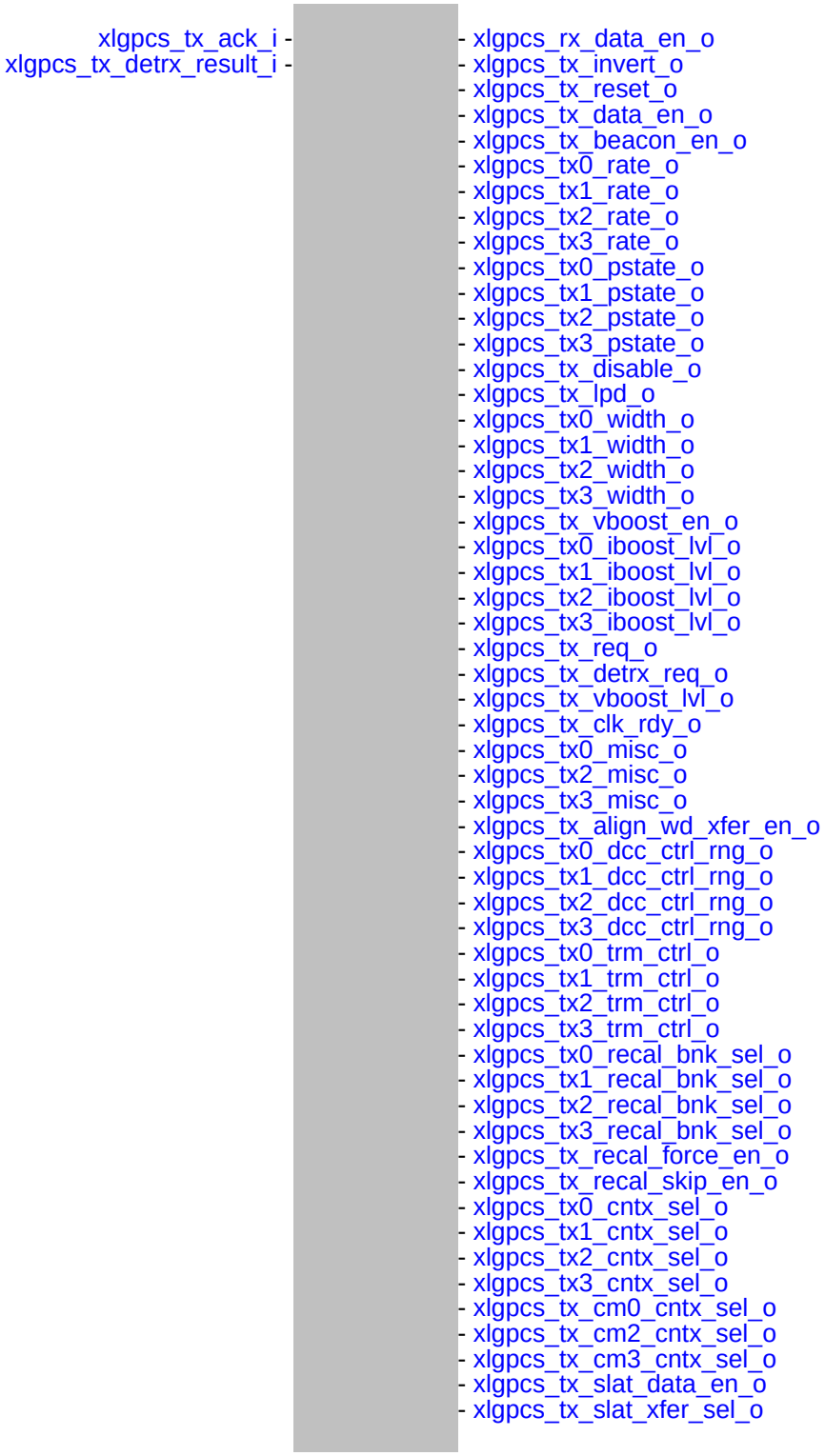


Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals

Port Name	I/O	Description
xlgpcs_rx_data_en_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver Data Enable. This signal enables the receiver data output from PHY. This signal is driven from RX_DT_EN_0/RX_DT_EN_3_1 fields of VR_PMA_MP_25G_16G_12G_RX_GENCTRL0 Register. This signal should be connected to rxX_data_en input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** (!DWCXLP_EEE_EN) ? "clk_csr_i" : "None" **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_invert_o[(DWCXLP_NUM_LANES-1):0]	O	Invert logical sense of Tx serial outputs of the PHY. This signal is driven from TX_INV_X fields of VR_PMA_MP_25G_16G_TX_GENCTRL0 Register. This signal should be connected to txX_invert input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_xsbi_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_reset_o[(DWCXLP_NUM_LANES-1):0]	O	PHY Transmitter reset. This signal is driven from the TX_RST_X fields of the VR_PMA_MP_25G_16G_TX_GENCTRL0 Register. This signal internally resets the Tx data-path, all PHY Tx settings and state machines. Also, this signal is controlled by LPM, bit[11] of SR PCS MMD Control1 Register. This signal must be connected to the txX_reset signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx_data_en_o[(DWCXLP_NUM_LANES-1):0]	O	Transmitter Data Enable. This signal is driven from TX_DT_EN_X fields of VR_PMA_MP_25G_16G_TX_GENCTRL0 Register. This signal should be connected to txX_data_en input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_xsbi_tx_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_beacon_en_o[(DWCXLP_NUM_LANES-1):0]	O	Transmitter Beaconing/LFPS Enable. This signal is driven from the TXBCN_EN_X fields of the VR_PMA_MP_25G_16G_TX_GENCTRL0 Register. This signal enables and disables transmitter beaconing. The period of transmit pulses is between 20-50ns. This signal must be connected to the txX_beacon_en input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx0_rate_o[2:0]	O	Transmit data rate - Lane 0. This signal is driven TX0_RATE fields of VR_PMA_MP_25G_16G_TX_RATE_CTRL Register. This signal controls the data rate PHY transmit serial lines. This signal should be connected to tx0_rate[2:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx1_rate_o[2:0]	O	Transmit data rate - Lane 1. This signal is driven TX1_RATE fields of VR_PMA_MP_25G_16G_TX_RATE_CTRL Register. This signal controls the data rate PHY transmit serial lines. This signal should be connected to tx1_rate[2:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx2_rate_o[2:0]	O	Transmit data rate - Lane 2. This signal is driven TX2_RATE fields of VR_PMA_MP_25G_16G_TX_RATE_CTRL Register. This signal controls the data rate PHY transmit serial lines. This signal should be connected to tx2_rate[2:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx3_rate_o[2:0]	O	Transmit data rate - Lane 3. This signal is driven TX3_RATE fields of VR_PMA_MP_25G_16G_TX_RATE_CTRL Register. This signal controls the data rate PHY transmit serial lines. This signal should be connected to tx3_rate[2:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx0_pstate_o[1:0]	O	Transmit Power state -Lane 0. This signal is driven from TX0_PSTATE fields of VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL Register. This signal controls the power-state of PHY Transmitter. This signal should be connected to tx0_pstate[1:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx1_pstate_o[1:0]	O	Transmit Power state -Lane 1. This signal is driven from TX1_PSTATE fields of VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL Register. This signal controls the power-state of PHY Transmitter. This signal should be connected to tx1_pstate[1:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx2_pstate_o[1:0]	O	Transmit Power state -Lane 2. This signal is driven from TX2_PSTATE fields of VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL Register. This signal controls the power-state of PHY Transmitter. This signal should be connected to tx2_pstate[1:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx3_pstate_o[1:0]	O	Transmit Power state -Lane 3. This signal is driven from TX3_PSTATE fields of VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL Register. This signal controls the power-state of PHY Transmitter. This signal should be connected to tx3_pstate[1:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_disable_o[(DWCXLP_NUM_LANES-1):0]	O	Transmitter Disable. This signal is driven from the TX_DISABLE fields of VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL Register. This signal must be connected to the txX_disable input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_lpd_o[(DWCXLP_NUM_LANES-1):0]	O	Transmitter Lane Power Down. This signal is driven from the TX_LPD_3_1 and TX_LPD_0 fields of VR_PMA_MP_25G_16G_TX_GENCTRL2 Register. Also, this signal is controlled by PWRSV bit[11] of VR PCS MMD Digital Control1 Register. This signal powers down the PHY transmitter to a power state equivalent to P1. This signal should be connected to txX_lpd input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx0_width_o[(TX_WIDTH_W-1):0]	O	Tx Data Width lane 0. This signal is driven from the TX0_WIDTH fields of VR_PMA_MP_25G_TX_WIDTH_CTRL Register. This signal selects the width of the input transmit data to the PHY. This signal must be connected to the tx0_width input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx1_width_o[(TX_WIDTH_W-1):0]	O	Tx Data Width lane 1. This signal is driven from the TX1_WIDTH fields of VR_PMA_MP_25G_TX_WIDTH_CTRL Register. This signal selects the width of the input transmit data to the PHY. This signal must be connected to the tx1_width input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx2_width_o[(TX_WIDTH_W-1):0]	O	Tx Data Width lane 2. This signal is driven from the TX2_WIDTH fields of VR_PMA_MP_25G_TX_WIDTH_CTRL Register. This signal selects the width of the input transmit data to the PHY. This signal must be connected to the tx2_width input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx3_width_o[(TX_WIDTH_W-1):0]	O	Tx Data Width lane 3. This signal is driven from the TX3_WIDTH fields of VR_PMA_MP_25G_TX_WIDTH_CTRL Register. This signal selects the width of the input transmit data to the PHY. This signal must be connected to the tx3_width input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_vboost_en_o[(DWCXLP_NUM_LANES-1):0]	O	Tx Voltage Boost Enable. This signal is driven from the VBOOST_EN_X field of VR_PMA_MP_25G_16G_TX_GENCTRL2 Register. This signal enables the current mode Tx Swing boost. This signal must be connected to the txX_vboost_en input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx0_iboost_lvl_o[3:0]	O	Tx Current Boost Level -Lane 0. This signal is driven from the VR_PMA_MP_25G_16G_TX_BOOST_CTRL Register. This signal controls the Tx swing level Lane 0 . This signal must be connected to the tx0_iboost_lvl input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx1_iboost_lvl_o[3:0]	O	Tx Current Boost Level -Lane 1. This signal is driven from the VR_PMA_MP_25G_16G_TX_BOOST_CTRL Register. This signal controls the Tx swing level Lane 1 . This signal must be connected to the tx1_iboost_lvl input signal of the PHY. Active State:N/A **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx2_iboost_lvl_o[3:0]	O	Tx Current Boost Level -Lane 2. This signal is driven from the VR_PMA_MP_25G_16G_TX_BOOST_CTRL Register. This signal controls the Tx swing level Lane 2 . This signal must be connected to the tx2_iboost_lvl input signal of the PHY. Active State:N/A **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx3_iboost_lvl_o[3:0]	O	Tx Current Boost Level -Lane 3. This signal is driven from the VR_PMA_MP_25G_16G_TX_BOOST_CTRL Register. This signal controls the Tx swing level Lane 3 . This signal must be connected to the tx3_iboost_lvl input signal of the PHY. Active State:N/A **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx_req_o[(DWCXLP_NUM_LANES-1):0]	O	Transmitter Request. This signal is driven from the TX_REQ fields of the VR_PMA_MP_25G_16G_TX_GENCTRL2 Register. This signal indicates a new transmitter setting request to the PHY. This signal must be connected to the txX_req input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_ack_i[(DWCXLP_NUM_LANES-1):0]	I	Transmitter Acknowledge. The value of this signal can be read from the TX_ACK fields of VR_PMA_MP_25G_16G_TX_STS Register. When this signal is asserted, it indicates that the request transmitter setting is complete or the requested Rx detection operation is complete. This signal must be connected to the txX_ack output signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_detrx_req_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver Detection Request. This signal is asserted to make a receiver detection request to the PHY. This signal is driven from DET_RX_REQ_X fields of VR_PMA_MP_25G_16G_TX_GENCTRL2 Register. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx_detrx_result_i[(DWCXLP_NUM_LANES-1):0]	I	Receiver Detection Result. This signal indicates the status of receiver detection operation. If this signal is high, it implies that a remote-end receiver has been successfully detected. This signal should be connected to tx{lane}_detrx_result output port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_vboost_lvl_o[2:0]	O	Tx Voltage Boost Maximum Level. This signal is driven from the VBOOST_LVL field of the VR_PMA_MP_25G_16G_TX_GENCTRL2 Register. This signal must be connected to the tx_vboost_lvl[2:0] input signal of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_clk_rdy_o[(DWCXLP_NUM_LANES-1):0]	O	Transmitter Input Clock Ready. This signal is driven from TX_CLK_RDY fields of VR_PMA_MP_25G_16G_TX_GENCTRL2 Register. This signal should be connected to txX_clk_rdy input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx0_misc_o[7:0]	O	TX Miscellaneous Controls along lane0. It controls miscellaneous settings in the TX analog block of EPHY for lane0 Tx path. This signal is driven from TX0_MISC fields of VR_PMA_MP_25G_16G_TX_MISC_CTRL0 Register. This signal should be connected to tx0_misc input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx2_misc_o[7:0]	O	TX Miscellaneous Controls along lane2. It controls miscellaneous settings in the TX analog block of EPHY for lane2 Tx path. This signal is driven from TX2_MISC fields of VR_PMA_MP_25G_16G_TX_MISC_CTRL1 Register. This signal should be connected to tx2_misc input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx3_misc_o[7:0]	O	TX Miscellaneous Controls along lane3. It controls miscellaneous settings in the TX analog block of EPHY for lane3 Tx path. This signal is driven from TX3_MISC fields of VR_PMA_MP_25G_16G_TX_MISC_CTRL1 Register. This signal should be connected to tx3_misc input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx_align_wd_xfer_en_o[(DWCXLP_NUM_LANES-1):0]	O	Transmitter Alignment Wide Transfer Enable. When asserted, the internal data transfer from the txX_clk domain to the internal clock domain is performed using a width of 32 bits (instead of 16 bits). Assert this signal only when the data rate is > 16 Gbps. This signal is driven from TX_ALIGN_WT fields of VR_PMA_MP_25G_16G_TX_MISC_CTRL1 Register. This signal should be connected to txX_align_wd_xfer_en input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx0_dcc_ctrl_rng_o[3:0]	O	TX Internal Clock Duty Cycle Control Range- Lane0. Sets the range of internal clock duty cycle correction circuitry. This signal is driven from TX0_DCC_CTRL_RNG fields of VR_PMA_MP_25G_TX_DCC_CTRL Register. This signal should be connected to tx0_dcc_ctrl_range input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx1_dcc_ctrl_rng_o[3:0]	O	TX Internal Clock Duty Cycle Control Range- Lane1. Sets the range of internal clock duty cycle correction circuitry. This signal is driven from TX1_DCC_CTRL_RNG fields of VR_PMA_MP_25G_TX_DCC_CTRL Register. This signal should be connected to tx1_dcc_ctrl_range input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx2_dcc_ctrl_rng_o[3:0]	O	TX Internal Clock Duty Cycle Control Range- Lane2. Sets the range of internal clock duty cycle correction circuitry. This signal is driven from TX2_DCC_CTRL_RNG fields of VR_PMA_MP_25G_TX_DCC_CTRL Register. This signal should be connected to tx2_dcc_ctrl_range input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx3_dcc_ctrl_rng_o[3:0]	O	TX Internal Clock Duty Cycle Control Range- Lane3. Sets the range of internal clock duty cycle correction circuitry. This signal is driven from TX1_DCC_CTRL_RNG fields of VR_PMA_MP_25G_TX_DCC_CTRL Register. This signal should be connected to tx3_dcc_ctrl_range input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx0_trm_ctrl_o[2:0]	O	Transmit Termination Control- Lane0. This signal sets the target transmitter termination impedance. This signal is driven from TX0_TRM_CTRL fields of VR_PMA_MP_25G_TX_TERM_CTRL Register. This signal should be connected to tx0_term_ctrl input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx1_trm_ctrl_o[2:0]	O	Transmit Termination Control- Lane1. This signal sets the target transmitter termination impedance. This signal is driven from TX1_TRM_CTRL fields of VR_PMA_MP_25G_TX_TERM_CTRL Register. This signal should be connected to tx1_term_ctrl input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx2_trm_ctrl_o[2:0]	O	Transmit Termination Control- Lane2. This signal sets the target transmitter termination impedance. This signal is driven from TX2_TRM_CTRL fields of VR_PMA_MP_25G_TX_TERM_CTRL Register. This signal should be connected to tx2_term_ctrl input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx3_trm_ctrl_o[2:0]	O	Transmit Termination Control- Lane3. This signal sets the target transmitter termination impedance. This signal is driven from TX3_TRM_CTRL fields of VR_PMA_MP_25G_TX_TERM_CTRL Register. This signal should be connected to tx3_term_ctrl input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx0_recal_bnk_sel_o[1:0]	O	Selects the bank to store the RX duty cycle correction (DCC) values for Lane0 **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx1_recal_bnk_sel_o[1:0]	O	Selects the bank to store the RX duty cycle correction (DCC) values for Lane1 **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx2_recal_bnk_sel_o[1:0]	O	Selects the bank to store the RX duty cycle correction (DCC) values for Lane2 **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx3_recal_bnk_sel_o[1:0]	O	Selects the bank to store the RX duty cycle correction (DCC) values for Lane3 **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_recal_force_en_o[(DWCXLP_NUM_LANES-1):0]	O	Forces a re-calibration of RX DCC in the selected bank regardless of the state of the corresponding done signal **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
xlgpcs_tx_recal_skip_en_o[(DWCXLP_NUM_LANES-1):0]	O	Skips re-calibration of TX DCC in the selected bank regardless of the state of the corresponding done signal **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low

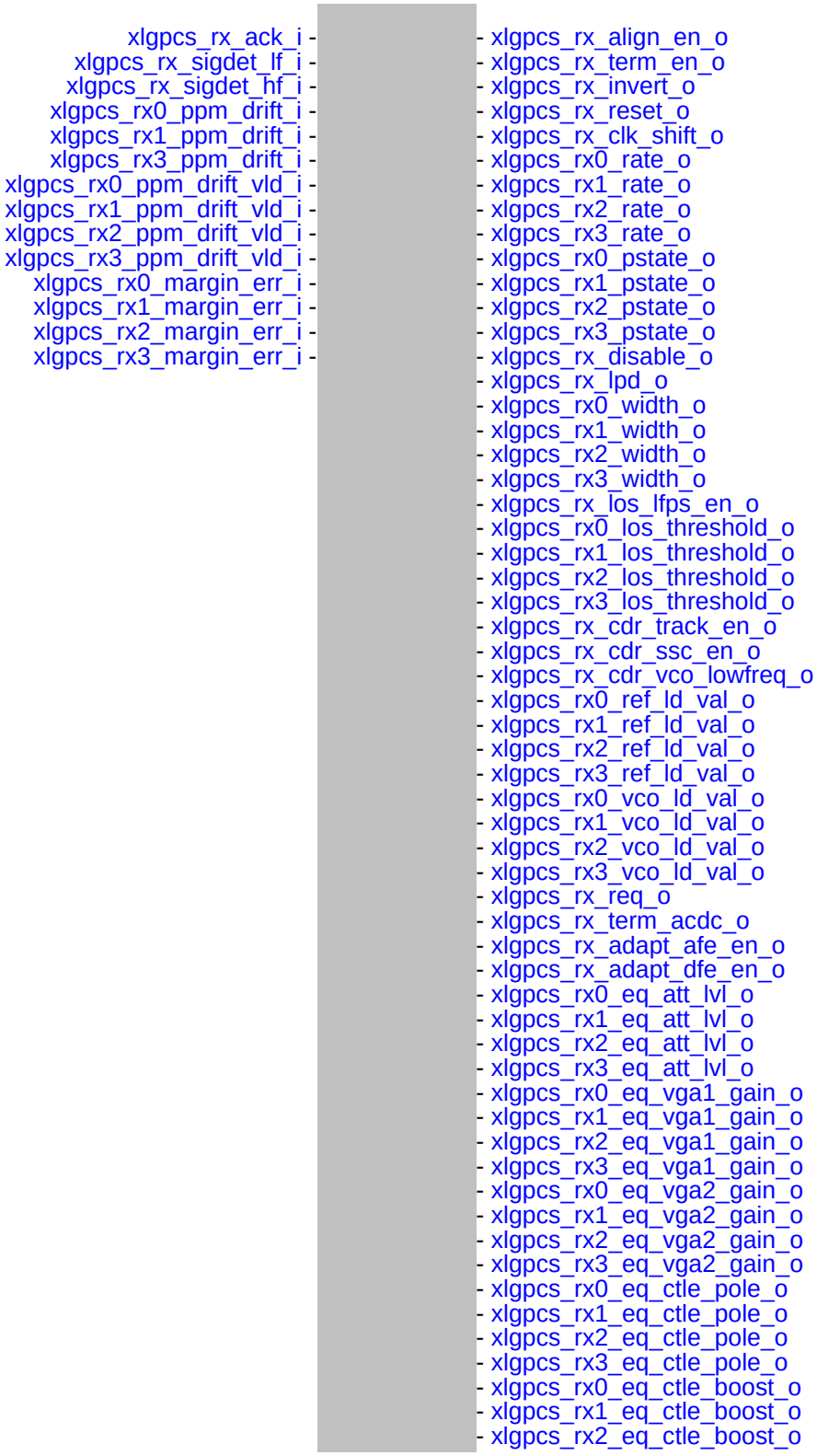
Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx0_cntx_sel_o[7:0]	O	Per-lane TX context select lane0. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx1_cntx_sel_o[7:0]	O	Per-lane TX context select lane1. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx2_cntx_sel_o[7:0]	O	Per-lane TX context select lane2. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx3_cntx_sel_o[7:0]	O	Per-lane TX context select lane3. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_cm0_cntx_sel_o[7:0]	O	Per-lane TX cmn context select lane0. A change in tx0_cmn_cntx_sel along with tx0_req triggers changes in MPLLA/B, RTUNE etc. So to avoid conflict, shared MPLL and ext RTUNE are disabled when this changes. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-21 Synopsys PHY-Specific Tx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx_cm2_cntx_sel_o[7:0]	O	Per-lane TX cmn context select lane2. A change in tx2_cmn_cntx_sel along with tx2_req triggers changes in MPLLA/B, RTUNE etc. So to avoid conflict, shared MPLL and ext RTUNE are disabled when this changes. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_cm3_cntx_sel_o[7:0]	O	Per-lane TX cmn context select lane3. A change in tx3_cmn_cntx_sel along with tx2_req triggers changes in MPLLA/B, RTUNE etc. So to avoid conflict, shared MPLL and ext RTUNE are disabled when this changes. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_slat_data_en_o[(DWCXLP_NUM_LANES-1):0]	O	Per-lane Transmitter data enable for short latency mode When asserted and the TX lane is in P0, then the TX output driver of PHY is enabled. If de-asserted or the TX lane is not in P0, then the TX output driver is disabled. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_slat_xfer_sel_o[(DWCXLP_NUM_LANES-1):0]	O	Per-lane TX short latency transfer enable. Static input signal that selects between short latency Tx mode and normal latency Tx mode. txX_reset or phy_reset is required after changing this port of PHY. Tie low if unused. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.22 Synopsys PHY-Specific Rx Control and Status Interface Signals



- xlgpcs_rx3_eq_ctle_boost_o
- xlgpcs_rx0_eq_dfe_tap1_o
- xlgpcs_rx1_eq_dfe_tap1_o
- xlgpcs_rx2_eq_dfe_tap1_o
- xlgpcs_rx3_eq_dfe_tap1_o
- xlgpcs_rx_adapt_cont_o
- xlgpcs_rx_offcan_cont_o
- xlgpcs_rx_vref_ctrl_o
- xlgpcs_rx_div16p5_clk_en_o
- xlgpcs_rx_sigdet_hf_en_o
- xlgpcs_rx_sigdet_lfps_flt_en_o
- xlgpcs_rx0_sigdet_lf_th_o
- xlgpcs_rx1_sigdet_lf_th_o
- xlgpcs_rx2_sigdet_lf_th_o
- xlgpcs_rx3_sigdet_lf_th_o
- xlgpcs_rx0_sigdet_hf_th_o
- xlgpcs_rx1_sigdet_hf_th_o
- xlgpcs_rx2_sigdet_hf_th_o
- xlgpcs_rx3_sigdet_hf_th_o
- xlgpcs_rx0_cdr_ppm_max_o
- xlgpcs_rx1_cdr_ppm_max_o
- xlgpcs_rx2_cdr_ppm_max_o
- xlgpcs_rx3_cdr_ppm_max_o
- xlgpcs_rx0_vco_cfg_o
- xlgpcs_rx1_vco_cfg_o
- xlgpcs_rx2_vco_cfg_o
- xlgpcs_rx3_vco_cfg_o
- xlgpcs_rx0_misc_o
- xlgpcs_rx1_misc_o
- xlgpcs_rx2_misc_o
- xlgpcs_rx3_misc_o
- xlgpcs_rx0_eq_afe_rate_o
- xlgpcs_rx1_eq_afe_rate_o
- xlgpcs_rx2_eq_afe_rate_o
- xlgpcs_rx3_eq_afe_rate_o
- xlgpcs_rx_dfe_byp_o
- xlgpcs_rx_adpt_sel_o
- xlgpcs_rx_adpt_in_prog_o
- xlgpcs_rx0_delta_iq_o
- xlgpcs_rx1_delta_iq_o
- xlgpcs_rx2_delta_iq_o
- xlgpcs_rx3_delta_iq_o
- xlgpcs_rx0_margin_iq_o
- xlgpcs_rx1_margin_iq_o
- xlgpcs_rx2_margin_iq_o
- xlgpcs_rx3_margin_iq_o
- xlgpcs_rx0_dcc_ctrl_rng_o
- xlgpcs_rx1_dcc_ctrl_rng_o
- xlgpcs_rx2_dcc_ctrl_rng_o
- xlgpcs_rx3_dcc_ctrl_rng_o
- xlgpcs_rx0_trm_ctrl_o
- xlgpcs_rx1_trm_ctrl_o
- xlgpcs_rx2_trm_ctrl_o
- xlgpcs_rx3_trm_ctrl_o
- xlgpcs_rx_recal_force_en_o
- xlgpcs_rx_recal_skip_en_o
- xlgpcs_rx0_recal_bank_sel_o
- xlgpcs_rx1_recal_bank_sel_o
- xlgpcs_rx2_recal_bank_sel_o
- xlgpcs_rx3_recal_bank_sel_o
- xlgpcs_vco_temp_comp_en_o
- xlgpcs_vco_step_ctrl_o
- xlgpcs_vco_fqband0_o

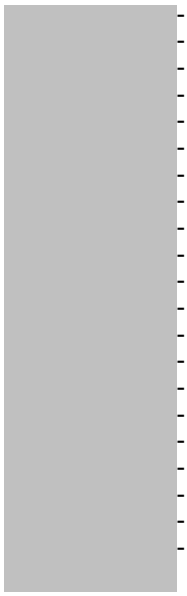


Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals

Port Name	I/O	Description
xlgpcs_rx_align_en_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver Data Alignment Enable. This signal is driven from RX_ALIGN_EN fields of VR_PMA_MP_25G_16G_12G_RX_GENCTRL0 Register. This signal should be connected to rxX_align_en input port of the PHY. **Exists:** (DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_term_en_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver Termination Enable. This signal is driven from RX_TERM_EN fields of VR_PMA_MP_25G_16G_12G_RX_GENCTRL0 Register. If this signal is high, PHY Rx is terminated with nominal 50 ohm resistance, else high-impedance. This signal should be connected to rxX_term_en input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_invert_o[(DWCXLP_NUM_LANES-1):0]	O	Invert logical sense of PHY Rx serial inputs. This signal is driven from RX_INV fields of VR_PMA_MP_25G_16G_RX_GENCTRL1 Register. This signal should be connected to rxX_invert input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_reset_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver Reset. This signal is driven from RX_RST fields of VR_PMA_MP_25G_16G_RX_GENCTRL1 Register. This signal resets the PHY receiver. This signal should be connected to rxX_reset input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_clk_shift_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver Clock shift. This signal is driven from RX_CLK_SFT fields of VR_PMA_MP_25G_16G_12G_RX_GENCTRL0 Register. This signal should be connected to rxX_clk_shift input port of the PHY. **Exists:** (DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_rate_o[(RX_RATE_W-1):0]	O	Receiver Data Rate -Lane 0. This signal selects the baud-rate for the receive path. This signal is driven from RX_RATE_0 field of VR_PMA_MP_25G_RX_RATE_CTRL Register. This signal should be connected to rx0_rate[2:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx1_rate_o[(RX_RATE_W-1):0]	O	Receiver Data Rate -Lane 1. This signal selects the baud-rate for the receive path. This signal is driven from RX1_RATE field of VR_PMA_MP_25G_RX_RATE_CTRL Register. This signal should be connected to rx1_rate[2:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_rate_o[(RX_RATE_W-1):0]	O	Receiver Data Rate -Lane 2. This signal selects the baud-rate for the receive path. This signal is driven from RX2_RATE field of VR_PMA_MP_25G_RX_RATE_CTRL Register. This signal should be connected to rx2_rate[2:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_rate_o[(RX_RATE_W-1):0]	O	Receiver Data Rate -Lane 3. This signal selects the baud-rate for the receive path. This signal is driven from RX3_RATE field of VR_PMA_MP_25G_RX_RATE_CTRL Register. This signal should be connected to rx3_rate[2:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_pstate_o[1:0]	O	Receiver Power state -Lane 0. This signal controls the power state of the receiver.This signal is driven from RX0_PSTATE field of VR_PMA_MP_25G_16G_RX_POWER_STATE_CTRL Register. This signal should be connected to rx0_pstate[1:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** (!DWCXLP_EEE_EN) ? "clk_csr_i" : "None" **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_pstate_o[1:0]	O	Receiver Power state -Lane 1. This signal controls the power state of the receiver.This signal is driven from RX1_PSTATE field of VR_PMA_MP_25G_16G_RX_POWER_STATE_CTRL Register. This signal should be connected to rx1_pstate[1:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_pstate_o[1:0]	O	Receiver Power state -Lane 2. This signal controls the power state of the receiver. This signal is driven from RX2_PSTATE field of VR_PMA_MP_25G_16G_RX_POWER_STATE_CTRL Register. This signal should be connected to rx2_pstate[1:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_pstate_o[1:0]	O	Receiver Power state -Lane 3. This signal controls the power state of the receiver. This signal is driven from RX3_PSTATE field of VR_PMA_MP_25G_16G_RX_POWER_STATE_CTRL Register. This signal should be connected to rx3_pstate[1:0] input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_disable_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver Disable. This signal is driven from RX_DISABLE fields of VR_PMA_MP_25G_16G_RX_POWER_STATE_CTRL Register. This signal should be connected to rxX_disable input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_lpd_o[(DWCXLP_NUM_LANES-1):0]	O	Receive Lane Power Down. This signal is driven from RX_LPD_3_1 and RX_LPD_0 fields of VR_PMA_MP_25G_16G_RX_GENCTRL1 Register. Also, this signal is controlled by PWRSV bit[11] of VR PCS MMD Digital Control1 Register This signal powers down the PHY receiver into a power state equivalent of P1. This signal should be connected to rxX_lpd input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_width_o[(RX_WIDTH_W-1):0]	O	Rx Data Width -Lane 0. This signal is driven from RX0_WIDTH field of VR_PMA_MP_25G_RX_WIDTH_CTRL Register. This signal selects the width of the receive data output from the PHY. This signal should be connected to rx0_width input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx1_width_o[(RX_WIDTH_W-1):0]	O	Rx Data Width -Lane 1. This signal is driven from RX1_WIDTH field of VR_PMA_MP_25G_RX_WIDTH_CTRL Register. This signal selects the width of the receive data output from the PHY. This signal should be connected to rx1_width input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx2_width_o[(RX_WIDTH_W-1):0]	O	Rx Data Width -Lane 2. This signal is driven from RX2_WIDTH field of VR_PMA_MP_25G_RX_WIDTH_CTRL Register. This signal selects the width of the receive data output from the PHY. This signal should be connected to rx2_width input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_width_o[(RX_WIDTH_W-1):0]	O	Rx Data Width -Lane 3. This signal is driven from RX3_WIDTH field of VR_PMA_MP_25G_RX_WIDTH_CTRL Register. This signal selects the width of the receive data output from the PHY. This signal should be connected to rx3_width input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx_los_lfps_en_o[(DWCXLP_NUM_LANES-1):0]	O	Rx LOS LFPS Enable. This signal is driven from LOS_LFPS_EN fields of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal enables the Low Frequency Periodic Signal (LFPS) filter in the PHY. This signal should be connected to rxX_los_lfps_en input pin of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_los_threshold_o[2:0]	O	Rx Loss-of-Signal Threshold Lane 0. This signal is driven from LOS_TRSHLD_0 field of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal sets the LOS threshold trip point of the PHY receiver. This signal should be connected to rx0_los_threshold input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx1_los_threshold_o[2:0]	O	Rx Loss-of-Signal Threshold Lane 1. This signal is driven from LOS_TRSHLD_1 field of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal sets the LOS threshold trip point of the PHY receiver. This signal should be connected to rx1_los_threshold input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_los_threshold_o[2:0]	O	Rx Loss-of-Signal Threshold Lane 2. This signal is driven from LOS_TRSHLD_2 field of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal sets the LOS threshold trip point of the PHY receiver. This signal should be connected to rx2_los_threshold input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_los_threshold_o[2:0]	O	Rx Loss-of-Signal Threshold Lane 3. This signal is driven from LOS_TRSHLD_3 field of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal sets the LOS threshold trip point of the PHY receiver. This signal should be connected to rx3_los_threshold input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_cdr_track_en_o[(DWCXLP_NUM_LANES-1):0]	O	Rx CDR Tracking Enable. This signal is driven from RX_CDR_TRACK_EN fields of VR_PMA_MP_12G_RX_GENCTRL5 Register. This signal is used to enable CDR tracking of receive data in the PHY. This signal should be connected to rxX_cdr_track_en input pin of the PHY. **Exists:** (DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_cdr_ssc_en_o[(DWCXLP_NUM_LANES-1):0]	O	Rx CDR SSC-Mode Enable. This signal is driven from CDR_SSC_EN fields of VR_PMA_MP_25G_RX_GEN_CTRL4 Register. This signal controls the CDR tracking gains and duration of the PHY. This signal should be asserted when Rx data has SS Clock. This signal should be connected to rxX_cdr_ssc_en input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_cdr_vco_lowfreq_o[(DWCXLP_NUM_LANES-1):0]	O	Rx VCO lower frequency band mode. This signal is driven from RX_CDR_VCO_LOWFREQ fields of VR_PMA_MP_12G_RX_GENCTRL5 Register. This signal controls the frequency of the Rx VCO to a lower frequency operating band. This signal should be connected to rxX_cdr_vco_lowfreq input pin of the PHY. **Exists:** (DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_ref_ld_val_o[(RF_LD_W-1):0]	O	Rx VCO calibration reference load value -Lane 0. This signal is driven from VCO_REF_LD_0 field of VR_PMA_MP_25G_16G_VCO_CAL_REF0 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal should be connected to rx0_ref_ld_val input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx1_ref_ld_val_o[(RF_LD_W-1):0]	O	Rx VCO calibration reference load value -Lane 1. This signal is driven from VCO_REF_LD_1 field of VR_PMA_MP_25G_16G_VCO_CAL_REF0 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal should be connected to rx1_ref_ld_val input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx2_ref_ld_val_o[(RF_LD_W-1):0]	O	Rx VCO calibration reference load value -Lane 2. This signal is driven from VCO_REF_LD_2 field of VR_PMA_MP_25G_16G_VCO_CAL_REF1 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal should be connected to rx2_ref_ld_val input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_ref_ld_val_o[(RF_LD_W-1):0]	O	Rx VCO calibration reference load value -Lane 3. This signal is driven from VCO_REF_LD_3 field of VR_PMA_MP_25G_16G_VCO_CAL_REF1 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal should be connected to rx3_ref_ld_val input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx0_vco_ld_val_o[12:0]	O	Rx VCO calibration load value -Lane 0. This signal is driven from VCO_LD_VAL_0 fields of VR_PMA_MP_25G_16G_VCO_CAL_LD0 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal should be connected to rx0_vco_ld_val input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx1_vco_ld_val_o[12:0]	O	Rx VCO calibration load value -Lane 1. This signal is driven from VCO_LD_VAL_1 fields of VR_PMA_MP_25G_16G_VCO_CAL_LD1 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal should be connected to rx1_vco_ld_val input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2_vco_ld_val_o[12:0]	O	Rx VCO calibration load value -Lane 2. This signal is driven from VCO_LD_VAL_2 fields of VR_PMA_MP_25G_16G_VCO_CAL_LD2 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal should be connected to rx2_vco_ld_val input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx3_vco_ld_val_o[12:0]	O	Rx VCO calibration load value -Lane 3. This signal is driven from VCO_LD_VAL_3 fields of VR_PMA_MP_25G_16G_VCO_CAL_LD3 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal should be connected to rx3_vco_ld_val input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx_req_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver Request. This signal is driven from RX_REQ fields of VR_PMA_MP_25G_16G_RX_GENCTRL2 Register. This signal indicates a new receive setting request. This signal should be connected to rxX_req input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** (!DWCXLP_EEE_EN) ? "clk_csr_i" : "None" **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_ack_i[(DWCXLP_NUM_LANES-1):0]	I	Receive Acknowledge. The value of this signal can be read from RX_ACK fields of VR_PMA_MP_25G_16G_RX_STS register. Assertion of this signal indicates that the request receiver setting is complete. This signal should be connected to rxX_ack output pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_term_acdc_o[(DWCXLP_NUM_LANES-1):0]	O	Rx termination control. This signal is driven from RX_TERM_ACDC fields of VR_PMA_MP_25G_16G_RX_GENCTRL1 Register. This controls the PHY receiver termination. This signal should be connected to rxX_term_acdc. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx_adapt_afe_en_o[(DWCXLP_NUM_LANES-1):0]	O	Rx AFE Enable. This signal is driven from AFE_EN fields of VR PMA Synopsys MultiProtocol 12G PHY Rx General Control Register 5. This signal enables the Rx adaption circuitry and applies the input receiver AFE equalization settings. This signal should be connected to rxX_adapt_afe_en input pin of the PHY. **Exists:** (DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_adapt_dfe_en_o[(DWCXLP_NUM_LANES-1):0]	O	Rx DFE Enable. This signal is driven from DFE_EN fields of VR PMA Synopsys MultiProtocol 12G PHY Rx General Control Register 5. This signal enables the Rx adaption and decision feedback equalization (DFE) circuitry and applies the input setting for DFE Tap1. This signal should be connected to irxX_adapt_dfe_en input pin of the PHY. **Exists:** (DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_eq_att_lvl_o[2:0]	O	Rx Equalization Attenuation Level -Lane 0. This signal is driven from RX0_EQ_ATT_LVL field of VR_PMA_MP_25G_16G_RX_ATTN_CTRL Register. This signal controls the AFE attenuation level. This signal should be connected to rx0_eq_att_lvl input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx1_eq_att_lvl_o[2:0]	O	Rx Equalization Attenuation Level -Lane 1. This signal is driven from RX1_EQ_ATT_LVL field of VR_PMA_MP_25G_16G_RX_ATTN_CTRL Register. This signal controls the AFE attenuation level. This signal should be connected to rx1_eq_att_lvl input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2_eq_att_lvl_o[2:0]	O	Rx Equalization Attenuation Level -Lane 2. This signal is driven from RX2_EQ_ATT_LVL field of VR_PMA_MP_25G_16G_RX_ATTN_CTRL Register. This signal controls the AFE attenuation level. This signal should be connected to rx2_eq_att_lvl input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx3_eq_att_lvl_o[2:0]	O	Rx Equalization Attenuation Level -Lane 3. This signal is driven from RX3_EQ_ATT_LVL field of VR_PMA_MP_25G_16G_RX_ATTN_CTRL Register. This signal controls the AFE attenuation level. This signal should be connected to rx3_eq_att_lvl input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx0_eq_vga1_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain1 -Lane 0. This signal is driven from VGA1_GAIN_0 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL0 Register. This signal controls the AFE first stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx0_eq_vga1_gain input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx1_eq_vga1_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain1 -Lane 1. This signal is driven from VGA1_GAIN_1 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL1 Register. This signal controls the AFE first stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx1_eq_vga1_gain input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx2_eq_vga1_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain1 -Lane 2. This signal is driven from VGA1_GAIN_2 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL2 Register. This signal controls the AFE first stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx2_eq_vga1_gain input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx3_eq_vga1_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain1 -Lane 3. This signal is driven from VGA1_GAIN_0 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL3 Register. This signal controls the AFE first stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx3_eq_vga1_gain input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_eq_vga2_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain2 -Lane 0. This signal is driven from VGA2_GAIN_0 fields of VR_PMA_MP_25G_16G_RX_EQ_CTRL0 Register. This signal controls the AFE second stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx0_eq_vga2_gain input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx1_eq_vga2_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain2 -Lane 1. This signal is driven from VGA2_GAIN_1 fields of VR_PMA_MP_25G_16G_RX_EQ_CTRL1 Register. This signal controls the AFE second stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx1_eq_vga2_gain input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx2_eq_vga2_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain2 -Lane 2. This signal is driven from VGA2_GAIN_2 fields of VR_PMA_MP_25G_16G_RX_EQ_CTRL2 Register. This signal controls the AFE second stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx2_eq_vga2_gain input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_eq_vga2_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain2 -Lane 3. This signal is driven from VGA2_GAIN_3 fields of VR_PMA_MP_25G_16G_RX_EQ_CTRL3 Register. This signal controls the AFE second stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx3_eq_vga2_gain input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx0_eq_ctle_pole_o[(POL_W-1):0]	O	Rx Equalization CTLE Pole -Lane 0. This signal is driven from CTLE_POLE_0 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL0 Register. This signal controls the continuous time linear equalization (CTLE) boost pole location of the PHY. This signal should be connected to rx0_eq_ctle_pole input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx1_eq_ctle_pole_o[(POL_W-1):0]	O	Rx Equalization CTLE Pole -Lane 1. This signal is driven from CTLE_POLE_1 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL1 Register. This signal controls the continuous time linear equalization (CTLE) boost pole location of the PHY. This signal should be connected to rx1_eq_ctle_pole input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2_eq_ctle_pole_o[(POL_W-1):0]	O	Rx Equalization CTLE Pole -Lane 2. This signal is driven from CTLE_POLE_2 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL2 Register. This signal controls the continuous time linear equalization (CTLE) boost pole location of the PHY. This signal should be connected to rx2_eq_ctle_pole input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx3_eq_ctle_pole_o[(POL_W-1):0]	O	Rx Equalization CTLE Pole -Lane 3. This signal is driven from CTLE_POLE_3 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL3 Register. This signal controls the continuous time linear equalization (CTLE) boost pole location of the PHY. This signal should be connected to rx3_eq_ctle_pole input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx0_eq_ctle_boost_o[4:0]	O	Rx Equalization CTLE Boost -Lane 0. This signal is driven from CTLE_BOOST_0 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL0 Register. This signal controls the CTLE Boost Level of the PHY. This signal should be connected to rx0_eq_ctle_boost input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx1_eq_ctle_boost_o[4:0]	O	Rx Equalization CTLE Boost -Lane 1. This signal is driven from CTLE_BOOST_1 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL1 Register. This signal controls the CTLE Boost Level of the PHY. This signal should be connected to rx1_eq_ctle_boost input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx2_eq_ctle_boost_o[4:0]	O	Rx Equalization CTLE Boost -Lane 2. This signal is driven from CTLE_BOOST_2 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL2 Register. This signal controls the CTLE Boost Level of the PHY. This signal should be connected to rx2_eq_ctle_boost input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx3_eq_ctle_boost_o[4:0]	O	Rx Equalization CTLE Boost -Lane 3. This signal is driven from CTLE_BOOST_3 field of VR_PMA_MP_25G_16G_RX_EQ_CTRL3 Register. This signal controls the CTLE Boost Level of the PHY. This signal should be connected to rx3_eq_ctle_boost input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_eq_dfe_tap1_o[(RX_TAP_W-1):0]	O	Rx Equalization DFE Tap1 -Lane 0. This signal is driven from DFE_TAP1_0 field of VR_PMA_MP_25G_16G_DFE_TAP_CTRL0 Register. This signal is used to control the value of DFE data Tap 1 of the PHY. This signal should be connected to rx0_eq_dfe_tap1 input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx1_eq_dfe_tap1_o[(RX_TAP_W-1):0]	O	Rx Equalization DFE Tap1 -Lane 1. This signal is driven from DFE_TAP1_1 field of VR_PMA_MP_25G_16G_DFE_TAP_CTRL0 Register. This signal is used to control the value of DFE data Tap 1 of the PHY. This signal should be connected to rx1_eq_dfe_tap1 input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx2_eq_dfe_tap1_o[(RX_TAP_W-1):0]	O	Rx Equalization DFE Tap1 -Lane 2. This signal is driven from DFE_TAP1_2 field of VR_PMA_MP_25G_16G_DFE_TAP_CTRL1 Register. This signal is used to control the value of DFE data Tap 1 of the PHY. This signal should be connected to rx2_eq_dfe_tap1 input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_eq_dfe_tap1_o[(RX_TAP_W-1):0]	O	Rx Equalization DFE Tap1 -Lane 3. This signal is driven from DFE_TAP1_3 field of VR_PMA_MP_25G_16G_DFE_TAP_CTRL1 Register. This signal is used to control the value of DFE data Tap 1 of the PHY. This signal should be connected to rx3_eq_dfe_tap1 input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx_adapt_cont_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver Adaption Continuous Operation. If this signal is high, PHY receiver adapts continuously. This signal should be connected to rxX_adapt_cont input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_offcan_cont_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver offset cancellation continuous operation. If this signal is high, continuous receiver offset cancellation in PHY is carried out. Otherwise, offset cancellation runs only when receiver exits the P2 power state. This signal should be connected to rxX_offcan_cont input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_vref_ctrl_o[4:0]	O	Rx Biasing Current Control. This signal is driven from RX_VREF_CTRL field of VR_PMA_MP_25G_16G_MISC_CTRL0 Register. This signal should be connected to rx_vref_ctrl input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_div16p5_clk_en_o[(DWCXLP_NUM_LANES-1):0]	O	Rx Divide 16.5 output clock Enable. This signal is driven from RX_DIV16P5_CLK_EN fields of VR_PMA_MP_25G_16G_RX_GENCTRL1 Register. This signal should be connected to rxX_div16p5_clk_en input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_sigdet_hf_en_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver High-Frequency Signal Detection Enable. Enables the high-frequency signal detection circuit. By default, this input (to EPHY) should be set to 1. This signal is driven from HF_EN fields of VR_PMA_MP_25G_RX_GENCTRL5 Register. This signal should be connected to rxX_sigdet_hf_en input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_sigdet_lfps_flt_en_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver LFPS Signal Detection Filter Enable. Enables the LFPS filter on the low-frequency signal detection circuit by reducing the cutoff frequency from 500 MHz to 50 MHz. Assert this signal only if rx_sigdet_lf_en is also asserted. This signal is driven from LFPS_FILT_EN fields of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal should be connected to rxX_sigdet_lfps_flt_en input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_sigdet_lf_th_o[2:0]	O	Receiver Low-Frequency Signal Detection Threshold (Lane0). Sets the low-frequency signal detection threshold level trip point. This signal is driven from LF_TH_0 fields of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal should be connected to rx0_sigdet_lf_th input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_sigdet_lf_th_o[2:0]	O	Receiver Low-Frequency Signal Detection Threshold (Lane1). Sets the low-frequency signal detection threshold level trip point. This signal is driven from LF_TH_1 fields of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal should be connected to rx1_sigdet_lf_th input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_sigdet_lf_th_o[2:0]	O	Receiver Low-Frequency Signal Detection Threshold (Lane2). Sets the low-frequency signal detection threshold level trip point. This signal is driven from LF_TH_2 fields of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal should be connected to rx2_sigdet_lf_th input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_sigdet_lf_th_o[2:0]	O	Receiver Low-Frequency Signal Detection Threshold (Lane3). Sets the low-frequency signal detection threshold level trip point. This signal is driven from LF_TH_3 fields of VR_PMA_MP_25G_RX_GENCTRL3 Register. This signal should be connected to rx3_sigdet_lf_th input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_sigdet_hf_th_o[2:0]	O	Receiver High-Frequency Signal Detection Threshold (Lane0). Sets the high-frequency signal detection threshold level trip point. This signal is driven from HF_TH_0 fields of VR_PMA_MP_25G_RX_GENCTRL5 Register. This signal should be connected to rx0_sigdet_hf_th input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_sigdet_hf_th_o[2:0]	O	Receiver High-Frequency Signal Detection Threshold (Lane1). Sets the high-frequency signal detection threshold level trip point. This signal is driven from HF_TH_1 fields of VR_PMA_MP_25G_RX_GENCTRL5 Register. This signal should be connected to rx1_sigdet_hf_th input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2_sigdet_hf_th_o[2:0]	O	Receiver High-Frequency Signal Detection Threshold (Lane2). Sets the high-frequency signal detection threshold level trip point. This signal is driven from HF_TH_2 fields of VR_PMA_MP_25G_RX_GENCTRL5 Register. This signal should be connected to rx2_sigdet_hf_th input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_sigdet_hf_th_o[2:0]	O	Receiver High-Frequency Signal Detection Threshold (Lane3). Sets the high-frequency signal detection threshold level trip point. This signal is driven from HF_TH_3 fields of VR_PMA_MP_25G_RX_GENCTRL5 Register. This signal should be connected to rx3_sigdet_hf_th input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_sigdet_lf_i[(DWCXLP_NUM_LANES-1):0]	I	Receive Low-Frequency Signal Detection Output. When asserted, indicates that the receiver is receiving low-frequency signals. This signal is valid only for low-frequency data on rxX_p/rxX_m. This signal should be connected to rxX_sigdet_lf output port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_sigdet_hf_i[(DWCXLP_NUM_LANES-1):0]	I	Receive High-Frequency Signal Detection Output. When asserted, indicates that the receiver is receiving high-frequency signals. This signal is valid only for high-frequency data on rxX_p/rxX_m. This signal should be connected to rxX_sigdet_hf output port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx0_cdr_ppm_max_o[4:0]	O	Maximum Allowed PPM on the RX CDR Clock(Along Lane0). Enables the high-frequency signal detection circuit. By default, this input should be set to 1. This signal is driven from CDR_PPM_MAX0 fields of VR_PMA_MP_25G_16G_RX_PPM_CTRL0 Register. This signal should be connected to rx0_cdr_ppm_max input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_cdr_ppm_max_o[4:0]	O	Maximum Allowed PPM on the RX CDR Clock(Along Lane1). This signal specifies the maximum PPM drift to be allowed on the lane1 RX clock (relative to ideal desired frequency) before the CDR recovery circuit is engaged. This signal is driven from CDR_PPM_MAX1 fields of VR_PMA_MP_25G_16G_RX_PPM_CTRL0 Register. This signal should be connected to rx1_cdr_ppm_max input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2_cdr_ppm_max_o[4:0]	O	Maximum Allowed PPM on the RX CDR Clock(Along Lane2). This signal specifies the maximum PPM drift to be allowed on the lane2 RX clock (relative to ideal desired frequency) before the CDR recovery circuit is engaged. This signal is driven from CDR_PPM_MAX2 fields of VR_PMA_MP_25G_16G_RX_PPM_CTRL1 Register. This signal should be connected to rx2_cdr_ppm_max input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_cdr_ppm_max_o[4:0]	O	Maximum Allowed PPM on the RX CDR Clock(Along Lane3). This signal specifies the maximum PPM drift to be allowed on the lane3 RX clock (relative to ideal desired frequency) before the CDR recovery circuit is engaged. This signal is driven from CDR_PPM_MAX3 fields of VR_PMA_MP_25G_16G_RX_PPM_CTRL1 Register. This signal should be connected to rx3_cdr_ppm_max input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_ppm_drift_i[5:0]	I	RX CDR PPM Drift (Along lane0). This value represents the amount of ppm on the rx0_clk(of PHY) with respect to the ideal desired frequency. This input is valid when xlgpcs_rx0_ppm_drift_vld_i is asserted. This signal is driven by rx0_ppm_drift output port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx1_ppm_drift_i[5:0]	I	RX CDR PPM Drift (Along lane1). This value represents the amount of ppm on the rx1_clk(of PHY) with respect to the ideal desired frequency. This input is valid when xlgpcs_rx1_ppm_drift_vld_i is asserted. This signal is driven by rx1_ppm_drift output port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_ppm_drift_i[5:0]	I	RX CDR PPM Drift (Along lane3). This value represents the amount of ppm on the rx3_clk(of PHY) with respect to the ideal desired frequency. This input is valid when xlgpcs_rx3_ppm_drift_vld_i is asserted. This signal is driven by rx3_ppm_drift output port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_ppm_drift_vld_i	I	RX CDR PPM Drift Valid(Along lane0). Indicates when the xlgpcs_rx0_ppm_drift_i input is valid and can be sampled. This signal is driven by rx0_ppm_drift_vld output port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx1_ppm_drift_vld_i	I	RX CDR PPM Drift Valid(Along lane1). Indicates when the xlgpcs_rx1_ppm_drift_i input is valid and can be sampled. This signal is driven by rx1_ppm_drift_vld output port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_ppm_drift_vld_i	I	RX CDR PPM Drift Valid(Along lane2). Indicates when the xlgpcs_rx2_ppm_drift_i input is valid and can be sampled. This signal is driven by rx2_ppm_drift_vld output port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_ppm_drift_vld_i	I	RX CDR PPM Drift Valid(Along lane3). Indicates when the xlgpcs_rx3_ppm_drift_i input is valid and can be sampled. This signal is driven by rx3_ppm_drift_vld output port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_vco_cfg_o[11:0]	O	RX CDR VCO Configuration Bus(Along lane0). Controls biasing, temperature compensation, DAC settings, and frequency range for the RX CDR VCO. This signal is driven from VCO_CONFIG fields of VR_PMA_MP_25G_RX_VCO_CFG_0 Register. This signal should be connected to rx0_vco_cfg input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_vco_cfg_o[11:0]	O	RX CDR VCO Configuration Bus(Along lane1). Controls biasing, temperature compensation, DAC settings, and frequency range for the RX CDR VCO. This signal is driven from VCO_CONFIG fields of VR_PMA_MP_25G_RX_VCO_CFG_1 Register. This signal should be connected to rx1_vco_cfg input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_vco_cfg_o[11:0]	O	RX CDR VCO Configuration Bus(Along lane2). Controls biasing, temperature compensation, DAC settings, and frequency range for the RX CDR VCO. This signal is driven from VCO_CONFIG fields of VR_PMA_MP_25G_RX_VCO_CFG_2 Register. This signal should be connected to rx2_vco_cfg input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_vco_cfg_o[11:0]	O	RX CDR VCO Configuration Bus(Along lane3). Controls biasing, temperature compensation, DAC settings, and frequency range for the RX CDR VCO. This signal is driven from VCO_CONFIG fields of VR_PMA_MP_25G_RX_VCO_CFG_3 Register. This signal should be connected to rx3_vco_cfg input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_misc_o[7:0]	O	RX Miscellaneous Controls along lane0. It controls miscellaneous settings in the TX analog block of EPHY for lane0 Rx path. This signal is driven from RX0_MISC fields of VR_PMA_MP_25G_16G_RX_MISC_CTRL0 Register. This signal should be connected to rx0_misc input port of the E16PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_misc_o[7:0]	O	TX Miscellaneous Controls along lane1. It controls miscellaneous settings in the TX analog block of EPHY for lane1 Rx path. This signal is driven from RX1_MISC fields of VR_PMA_MP_25G_16G_RX_MISC_CTRL0 Register. This signal should be connected to rx1_misc input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2_misc_o[7:0]	O	RX Miscellaneous Controls along lane2. It controls miscellaneous settings in the TX analog block of EPHY for lane2 Rx path. This signal is driven from RX2_MISC fields of VR_PMA_MP_25G_16G_RX_MISC_CTRL1 Register. This signal should be connected to rx2_misc input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_misc_o[7:0]	O	RX Miscellaneous Controls along lane3. It controls miscellaneous settings in the TX analog block of EPHY for lane3 Rx path. This signal is driven from RX3_MISC fields of VR_PMA_MP_25G_16G_RX_MISC_CTRL1 Register. This signal should be connected to rx3_misc input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_eq_afe_rate_o[2:0]	O	RX Equalization AFE Rate(Along lane0). Controls the amount of parasitic boost provided by the negative-C circuit in the Equalizer and VGA stages. This signal is driven from RX0_EQ_ARE_RATE fields of VR_PMA_MP_25G_RX_AFE_RATE_CTRL Register. This signal should be connected to rx0_eq_afe_rate input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx1_eq_afe_rate_o[2:0]	O	RX Equalization AFE Rate(Along lane1). Controls the amount of parasitic boost provided by the negative-C circuit in the Equalizer and VGA stages. This signal is driven from RX1_EQ_ARE_RATE fields of VR_PMA_MP_25G_RX_AFE_RATE_CTRL Register. This signal should be connected to rx1_eq_afe_rate input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_eq_afe_rate_o[2:0]	O	RX Equalization AFE Rate(Along lane2). Controls the amount of parasitic boost provided by the negative-C circuit in the Equalizer and VGA stages. This signal is driven from RX2_EQ_ARE_RATE fields of VR_PMA_MP_25G_RX_AFE_RATE_CTRL Register. This signal should be connected to rx2_eq_afe_rate input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_eq_afe_rate_o[2:0]	O	RX Equalization AFE Rate(Along lane3). Controls the amount of parasitic boost provided by the negative-C circuit in the Equalizer and VGA stages. This signal is driven from RX3_EQ_ARE_RATE fields of VR_PMA_MP_25G_RX_AFE_RATE_CTRL Register. This signal should be connected to rx3_eq_afe_rate input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_dfe_byp_o[(DWCXLP_NUM_LANES-1):0]	O	RX DFE By-pass Enable. Indicates by-passing of the decision feedback equalization (DFE) circuitry of E16 PHY. This signal is driven from RX_DFE_BYP fields of VR_PMA_MP_25G_RX_GEN_CTRL4 Register. This signal should be connected to rxX_dfe_byp input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_adpt_sel_o[(DWCXLP_NUM_LANES-1):0]	O	Select Storage Bank for RX Adaptation. The PHY supports two internal register storage banks for saving adaptation results-this signal specifies which one is selected. This signal is driven from RX_ADPT_SEL fields of VR_PMA_MP_25G_16G_ADPT_CTRL0 Register. This signal should be connected to rxX_adapt_sel input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_adpt_in_prog_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver adaptation in progress. For Multi-protocol 25G PHY configurationss, this signal can be controlled from RX_ADPT_IN_PROG_0/3_1 fields of VR_PMA_MP_25G_16G_ADPT_CTRL0 Register. For Multi-protocol 32G PHY configurationss, this signal can be controlled from AD_IP_0/3_1 fields of VR_PMA_MP_32G_RX_ADPT_CTRL0 Register. For Multi-protocol 32G PHY configurations, when link training is enabled, training hardware logic controls this signal, over-riding the CSR control. This signal should be connected to rxX_adapt_in_prog input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_delta_iq_o[3:0]	O	IQ offset value (Along Rx Lane0). Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This signal is driven from RX_DELTA_IQ_0 fields of VR_PMA_MP_25G_16G_RX_IQ_CTRL0 Register. This signal should be connected to rx0_delta_iq[3:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_delta_iq_o[3:0]	O	IQ offset value (Along Rx Lane1). Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This signal is driven from RX_DELTA_IQ_1 fields of VR_PMA_MP_25G_16G_RX_IQ_CTRL0 Register. This signal should be connected to rx1_delta_iq[3:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_delta_iq_o[3:0]	O	IQ offset value (Along Rx Lane2). Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This signal is driven from RX_DELTA_IQ_2 fields of VR_PMA_MP_25G_16G_RX_IQ_CTRL2 Register. This signal should be connected to rx2_delta_iq[3:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_delta_iq_o[3:0]	O	IQ offset value (Along Rx Lane3). Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This signal is driven from RX_DELTA_IQ_3 fields of VR_PMA_MP_25G_16G_RX_IQ_CTRL3 Register. This signal should be connected to rx3_delta_iq[3:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_margin_iq_o[6:0]	O	Value for RX IQ margining (Along Rx Lane0). This is a signed value for skewing the IQ separation for the purpose of RX margining. This signal is driven from RX_MARGIN_IQ_0 fields of VR_PMA_MP_25G_16G_RX_IQ_CTRL0 Register. This signal should be connected to rx0_margin_iq[6:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_margin_iq_o[6:0]	O	Value for RX IQ margining (Along Rx Lane1). This is a signed value for skewing the IQ separation for the purpose of RX margining. This signal is driven from RX_MARGIN_IQ_1 fields of VR_PMA_MP_25G_16G_RX_IQ_CTRL1 Register. This signal should be connected to rx1_margin_iq[6:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2_margin_iq_o[6:0]	O	Value for RX IQ margining (Along Rx Lane2). This is a signed value for skewing the IQ separation for the purpose of RX margining. This signal is driven from RX_MARGIN_IQ_2 fields of VR_PMA_MP_25G_16G_RX_IQ_CTRL2 Register. This signal should be connected to rx2_margin_iq[6:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_margin_iq_o[6:0]	O	Value for RX IQ margining (Along Rx Lane3). This is a signed value for skewing the IQ separation for the purpose of RX margining. This signal is driven from RX_MARGIN_IQ_3 fields of VR_PMA_MP_25G_16G_RX_IQ_CTRL3 Register. This signal should be connected to rx3_margin_iq[6:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_dcc_ctrl_rng_o[3:0]	O	RX Internal Clock Duty Cycle Control Range- Lane0. Sets the range of internal clock duty cycle correction circuitry. This signal is driven from RX0_DCC_CTRL_RNG fields of VR_PMA_MP_25G_RX_DCC_CTRL Register. This signal should be connected to rx0_dcc_ctrl_range input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx1_dcc_ctrl_rng_o[3:0]	O	RX Internal Clock Duty Cycle Control Range- Lane1. Sets the range of internal clock duty cycle correction circuitry. This signal is driven from RX1_DCC_CTRL_RNG fields of VR_PMA_MP_25G_RX_DCC_CTRL Register. This signal should be connected to rx1_dcc_ctrl_range input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_dcc_ctrl_rng_o[3:0]	O	RX Internal Clock Duty Cycle Control Range- Lane2. Sets the range of internal clock duty cycle correction circuitry. This signal is driven from RX2_DCC_CTRL_RNG fields of VR_PMA_MP_25G_RX_DCC_CTRL Register. This signal should be connected to rx2_dcc_ctrl_range input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_dcc_ctrl_rng_o[3:0]	O	RX Internal Clock Duty Cycle Control Range- Lane3. Sets the range of internal clock duty cycle correction circuitry. This signal is driven from RX1_DCC_CTRL_RNG fields of VR_PMA_MP_25G_RX_DCC_CTRL Register. This signal should be connected to rx3_dcc_ctrl_range input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_trm_ctrl_o[2:0]	O	Receive Termination Control- Lane0. This signal sets the target receiver termination impedance. This signal is driven from RX0_TRM_CTRL fields of VR_PMA_MP_25G_RX_TERM_CTRL Register. This signal should be connected to rx0_term_ctrl input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_trm_ctrl_o[2:0]	O	Receive Termination Control- Lane1. This signal sets the target receiver termination impedance. This signal is driven from RX1_TRM_CTRL fields of VR_PMA_MP_25G_RX_TERM_CTRL Register. This signal should be connected to rx1_term_ctrl input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx2_trm_ctrl_o[2:0]	O	Receive Termination Control- Lane2. This signal sets the target receiver termination impedance. This signal is driven from RX2_TRM_CTRL fields of VR_PMA_MP_25G_RX_TERM_CTRL Register. This signal should be connected to rx2_term_ctrl input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_trm_ctrl_o[2:0]	O	Receive Termination Control- Lane3. This signal sets the target receiver termination impedance. This signal is driven from RX3_TRM_CTRL fields of VR_PMA_MP_25G_RX_TERM_CTRL Register. This signal should be connected to rx3_term_ctrl input port of the EPHY. **Exists:** (DWCXLP_E25)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_recal_force_en_o[(DWCXLP_NUM_LANES-1):0]	O	Receive Force Re-calibration **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_recal_skip_en_o[(DWCXLP_NUM_LANES-1):0]	O	Receive Skip Re-Calibration **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_recal_bank_sel_o[1:0]	O	Receive Re-calibration Bank Select -lane 0 **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_recal_bank_sel_o[1:0]	O	Receive Re-calibration Bank Select -lane 1 **Exists:** (DWCXLP_E25||DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2_recal_bank_sel_o[1:0]	O	Receive Re-calibration Bank Select -lane 2 **Exists:** (DWCXLP_E25||DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_recal_bank_sel_o[1:0]	O	Receive Re-calibration Bank Select -lane 3 **Exists:** (DWCXLP_E25||DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_vco_temp_comp_en_o[(DWCXLP_NUM_LANES-1):0]	O	Value for RX CDR Temperature Compensation Enable. Control of CDR VCO temperature compensation circuit based on data rate. This signal is driven from CDR_VCO_TEMP_CTRL_EN_0/CDR_VCO_TEMP_CTRL_EN_3_1 fields of VR_PMA_MP_16G_RX_CDR_CTRL Register. This signal should be connected to rxX_cdr_vco_temp_comp_en input ports of the PHY. **Exists:** (DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_vco_step_ctrl_o[(DWCXLP_NUM_LANES-1):0]	O	RX CDR VCO Step Control . Controls CDR VCO biasing based on data rate. This signal is driven from CDR_VCO_STEP_CTRL0/CDR_VCO_STEP_CTRL_3_1 fields of VR_PMA_MP_16G_RX_CDR_CTRL Register. This signal should be connected to rxX_cdr_vco_step_ctrl input ports of the PHY. **Exists:** (DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_vco_fqband0_o[1:0]	O	RX CDR VCO Frequency Band (Along Lane0 Rx Path). Controls Lane0 coarse-DAC step-size based on data rate and VPH level. This signal is driven from CDR_VCO_FREQBAND_0 fields of VR_PMA_MP_16G_RX_CDR_CTRL Register. This signal should be connected to rx0_cdr_vco_freqband input ports of the PHY. **Exists:** (DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_vco_fqband1_o[1:0]	O	RX CDR VCO Frequency Band (Along Lane1 Rx Path). Controls Lane1 coarse-DAC step-size based on data rate and VPH level. This signal is driven from CDR_VCO_FREQBAND_1 fields of VR_PMA_MP_16G_RX_CDR_CTRL Register. This signal should be connected to rx1_cdr_vco_freqband input ports of the PHY. **Exists:** (DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_vco_fqband2_o[1:0]	O	RX CDR VCO Frequency Band (Along Lane2 Rx Path). Controls Lane2 coarse-DAC step-size based on data rate and VPH level. This signal is driven from CDR_VCO_FREQBAND_2 fields of VR_PMA_MP_16G_RX_CDR_CTRL Register. This signal should be connected to rx2_cdr_vco_freqband input ports of the PHY. **Exists:** (DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_vco_fqband3_o[1:0]	O	RX CDR VCO Frequency Band (Along Lane3 Rx Path). Controls Lane3 coarse-DAC step-size based on data rate and VPH level. This signal is driven from CDR_VCO_FREQBAND_3 fields of VR_PMA_MP_16G_RX_CDR_CTRL Register. This signal should be connected to rx3_cdr_vco_freqband input ports of the PHY. **Exists:** (DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_adpt_mode_o[(ADPT_MD_W-1):0]	O	RX Adaptation mode select (Along Rx Lane0). Selects what RX adaptation mode will be performed along Rx Lane0 of PHY. This signal is driven from RX_ADPT_MODE_0 fields of VR_PMA_MP_25G_16G_ADPT_CTRL0 Register. This signal should be connected to rx0_adapt_mode[1:0] input port of the E16PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_adpt_mode_o[(ADPT_MD_W-1):0]	O	RX Adaptation mode select (Along Rx Lane1). Selects what RX adaptation mode will be performed along Rx Lane1 of EPHY. This signal is driven from RX_ADPT_MODE_1 fields of VR_PMA_MP_25G_16G_ADPT_CTRL0 Register. This signal should be connected to rx1_adapt_mode[1:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2_adpt_mode_o[(ADPT_MD_W-1):0]	O	RX Adaptation mode select (Along Rx Lane2). Selects what RX adaptation mode will be performed along Rx Lane2 of EPHY. This signal is driven from RX_ADPT_MODE_2 fields of VR_PMA_MP_25G_16G_ADPT_CTRL0 Register. This signal should be connected to rx2_adapt_mode[1:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx3_adpt_mode_o[(ADPT_MD_W-1):0]	O	RX Adaptation mode select (Along Rx Lane3). Selects what RX adaptation mode will be performed along Rx Lane3 of EPHY. This signal is driven from RX_ADPT_MODE_3 fields of VR_PMA_MP_25G_16G_ADPT_CTRL0 Register. This signal should be connected to rx3_adapt_mode[1:0] input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_term_offset_o[4:0]	O	Offset for RX Termination. Specifies an additional fixed offset to calibrated RX termination value. This signal is driven from RX_TERM_OFFSET fields of VR_PMA_MP_25G_RX_GEN_CTRL4 Register. This signal should be connected to rx_term_offset input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx0_margin_err_i[5:0]	I	Error counter to indicate the number of bit-errors during RX margining - Lane 0. This output of the PHY is continuously updated when xlgpcs_rx_margin_in_prog_o is high except when xlgpcs_rx_ack_i is high **Exists:** (DWCXLP_E32) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx_margin_err_clr_o[(DWCXLP_NUM_LANES-1):0]	O	Clear the rxX_margin_error counter **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx1_margin_err_i[5:0]	I	Error counter to indicate the number of bit-errors during RX margining - Lane 1. This output of the PHY is continuously updated when xlgpcs_rx_margin_in_prog_o is high except when xlgpcs_rx_ack_i is high **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx2_margin_err_i[5:0]	I	Error counter to indicate the number of bit-errors during RX margining - Lane 2. This output of the PHY is continuously updated when xlgpcs_rx_margin_in_prog_o is high except when xlgpcs_rx_ack_i is high **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx3_margin_err_i[5:0]	I	Error counter to indicate the number of bit-errors during RX margining - Lane 3. This output of the PHY is continuously updated when xlgpcs_rx_margin_in_prog_o is high except when xlgpcs_rx_ack_i is high **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_margin_in_prog_o[(DWCXLP_NUM_LANES-1):0]	O	Receiver margining in progress: This indicates that RX margining is in progress. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_margin_vdac_o[8:0]	O	Value for RX voltage margining for lane0. This is a signed value. It covers a range from -50 mV to +50 mV **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx1_margin_vdac_o[8:0]	O	Value for RX voltage margining for lane1. This is a signed value. It covers a range from -50 mV to +50 mV **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx2_margin_vdac_o[8:0]	O	Value for RX voltage margining for lane2. This is a signed value. It covers a range from -50 mV to +50 mV **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx3_margin_vdac_o[8:0]	O	Value for RX voltage margining for lane3. This is a signed value. It covers a range from -50 mV to +50 mV **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_crse_adpt_skip_en_o[(DWCXLP_NUM_LANES-1):0]	O	Skips coarse adaptation in the selected bank (xlgpcs_rx0_adpt_sel_o) and mode (xlgpcs_rx0_adpt_mode_o) regardless of the state of the corresponding done signal **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_crse_adpt_frc_en_o[(DWCXLP_NUM_LANES-1):0]	O	Forces a coarse adaptation in the selected bank (xlgpcs_rx0_adpt_sel_o) and mode (xlgpcs_rx0_adpt_mode_o) regardless of the state of the corresponding done signal **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx0_cntx_sel_o[7:0]	O	Per-lane RX context select lane0. Reserved; tied to low using a programmable register. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx1_cntx_sel_o[7:0]	O	Per-lane RX context select lane1. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx2_cntx_sel_o[7:0]	O	Per-lane RX context select lane2. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-22 Synopsys PHY-Specific Rx Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx3_cntx_sel_o[7:0]	O	Per-lane RX context select lane3. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_cm1_cntx_sel_o[7:0]	O	Per-lane TX cmn context select lane1. A change in tx1_cmn_cntx_sel along with tx1_req triggers changes in MPLLA/B, RTUNE etc. So to avoid conflict, shared MPLL and ext RTUNE are disabled when this changes. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.23 Synopsys PHY Miscellaneous Interface Signals



Table 6-23 Synopsys PHY Miscellaneous Interface Signals

Port Name	I/O	Description
xlgpcs_rtune_req_o	O	Resistor Tune Request. This signal is driven from RTUNE_REQ field of VR_PMA_MP_25G_16G_MISC_CTRL0 Register. This signal should be connected to rtune_req input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rtune_ack_i	I	Resistor Tune acknowledge. The value of this signal is captured in RTUNE_ACK field of VR_PMA_MP_25G_16G_12G_MISC_STS0 Register. This signal should be connected to rtune_ack input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with clk_csr_i

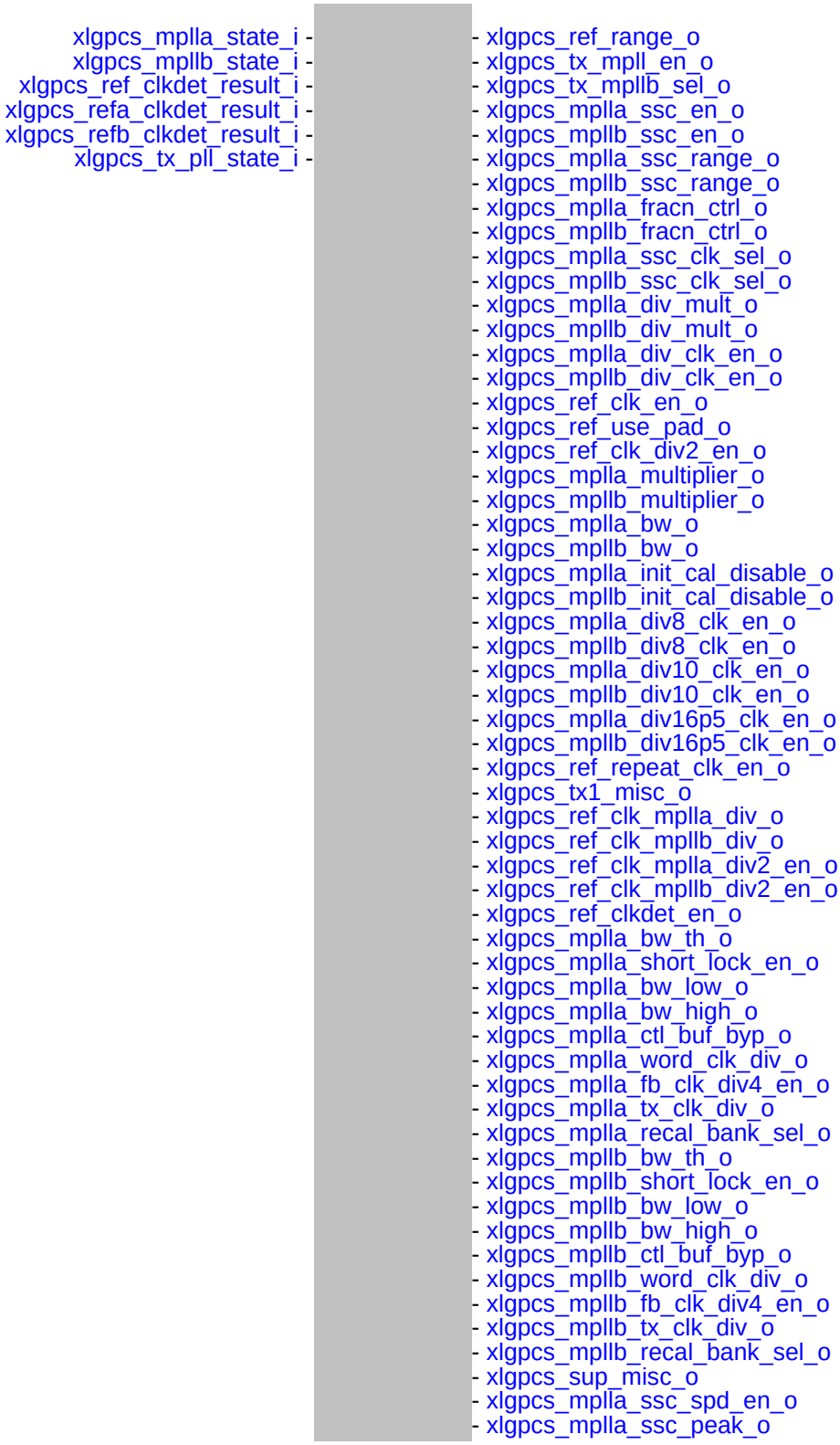
Table 6-23 Synopsys PHY Miscellaneous Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx2tx_par_lb_en_o[(DWCXLP_NUM_LANES-1):0]	O	Parallel (Rx to Tx) loopback enable. This signal is driven from RX2TX_LB_EN fields of VR_PMA_MP_25G_16G_MISC_CTRL0 Register. When this signal is asserted, recovered parallel data from the PHY receiver is looped back to the PHY transmit serializer. This signal should be connected to laneX_rx2tx_par_lb_en input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx2rx_ser_lb_en_o[(DWCXLP_NUM_LANES-1):0]	O	Analog serial (Tx to Rx) Loopback Enable. This signal is driven from TX2RX_LB_EN fields of VR_PMA_MP_25G_16G_MISC_CTRL0 Register. When this signal is asserted, data from Tx pre-driver in PHY is looped back to Rx analog front end. This signal should be connected to laneX_tx2rx_ser_lb_en input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_cr_para_sel_o	O	CR parallel port Select. This signal is driven from CR_PARA_SEL field of VR_PMA_MP_25G_16G_MISC_CTRL0 Register. This signal should be connected to cr_para_sel input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-23 Synopsys PHY Miscellaneous Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_sram_bootload_byp_o	O	SRAM Boot-Load Bypass. This signal is driven from BTLD_BYP field of VR_PMA_MP_25G_16G_SRAM Register. This signal can be connected to 'sram_bootload_bypass' input pin of the PHY (if available). **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_sram_ext_ld_done_o	O	Control to indicate that external loading of SRAM is Done. This signal is driven from EXT_LD_DN field of VR_PMA_MP_25G_16G_SRAM Register. Software should program this register field after it has completed the (optional) external loading of SRAM connected to Synopsys PHY. Only after this step, the PHY proceeds to complete its initialization sequence. This signal should be connected to 'sram_ext_ld_done' input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_sram_init_done_i	I	Status from PHY indicates that SRAM initialization is Complete. Value of this signal is reflected in INIT_DN field of VR_PMA_MP_25G_16G_SRAM Register. Whenever power-on-reset/soft-reset is asserted, software should first poll for this signal to go high. Thereafter, it should program 'EXT_LD_DN' bit (after completing the optional SRAM loading operation). Only after these steps are completed, the reset sequence of PHY (& in turn DWC_xlgpcs) gets over. This signal should be connected to 'sram_init_done' output pin of the PHY. In case, the PHY does not have SRAM interface, tie this input pin to zero. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with clk_csr_i

6.24 Synopsys PHY-Specific MPLL Control and Status Interface Signals



- xlgpcs_mplla_ssc_step_size_o
- xlgpcs_mplla_frac_en_o
- xlgpcs_mplla_frac_quot_o
- xlgpcs_mplla_frac_rem_o
- xlgpcs_mplla_frac_den_o
- xlgpcs_mplla_ssc_freq_cnt_init_o
- xlgpcs_mplla_ssc_freq_cnt_pk_o
- xlgpcs_mpllb_ssc_spd_en_o
- xlgpcs_mpllb_ssc_peak_o
- xlgpcs_mpllb_ssc_step_size_o
- xlgpcs_mpllb_frac_en_o
- xlgpcs_mpllb_frac_quot_o
- xlgpcs_mpllb_frac_rem_o
- xlgpcs_mpllb_frac_den_o
- xlgpcs_mpllb_ssc_freq_cnt_init_o
- xlgpcs_mpllb_ssc_freq_cnt_pk_o
- xlgpcs_txup_term_offset_o
- xlgpcs_txdn_term_offset_o
- xlgpcs_refa_clk_en_o
- xlgpcs_refb_clk_en_o
- xlgpcs_refa_clk_div2_en_o
- xlgpcs_refb_clk_div2_en_o
- xlgpcs_refa_range_o
- xlgpcs_refb_range_o
- xlgpcs_ref_rpt_clk_sel_o
- xlgpcs_refa_clk_sel_o
- xlgpcs_refb_clk_sel_o
- xlgpcs_refa_clkdet_en_o
- xlgpcs_refb_clkdet_en_o
- xlgpcs_lane_ref_sel_o
- xlgpcs_tx_pll_en_o
- xlgpcs_mplla_wrd_clk_en_o
- xlgpcs_mpllb_wrd_clk_en_o
- xlgpcs_refa_raw_clk_div2_en_o
- xlgpcs_refb_raw_clk_div2_en_o

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals

Port Name	I/O	Description
xlgpcs_ref_range_o[2:0]	O	Input Reference Clock Range. This signal is driven from REF_RANGE field of VR_PMA_MP_25G_16G_REF_CLK_CTRL Register. This signal is used to specify the range of input reference clock supplied to the PHY. This signal should be connected to ref_range input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx_mpll_en_o[(DWCXLP_NUM_LANES-1):0]	O	Tx MPLL Enable. This signal is driven from MPLL_EN fields of VR_PMA_MP_25G_16G_MPLL_CMN_CTRL Register. This signal should be connected to txX_mpll_en input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_mpllb_sel_o[(DWCXLP_NUM_LANES-1):0]	O	Tx MPLLB select. This signal is driven from MPLLB_SEL fields of VR_PMA_MP_25G_16G_MPLL_CMN_CTRL Register. This signal should be connected to txX_mpllb_sel input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_ssc_en_o	O	MPLLA Spread Spectrum Enable. This signal is driven from the MPLLA_SSC_EN fields of VR_PMA_MP_16G_MPLLA_CTRL1 Register. This signal is used to enable spread-spectrum clock generation the mplla_div_clk output of the PHY. This signal should be connected to mplla_ssc_en input of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_ssc_en_o	O	MPLLB Spread Spectrum Enable. This signal is driven from the MPLLB_SSC_EN fields of VR_PMA_MP_16G_MPLLB_CTRL1 Register. This signal is used to enable spread-spectrum clock generation the 'mpllb_div_clk' output of the PHY. This signal should be connected to mpllb_ssc_en input of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_ssc_range_o[2:0]	O	MPLLA Spread Spectrum Clock Range. This signal is driven from MPLLA_SSC_RANGE field of VR_PMA_MP_12G_MPLLA_CTRL1 Register. This signal is used to control the fixed offset applied to the phase accumulator in the PHY. This signal should be connected to mplla_ssc_range input of the PHY. **Exists:** (DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_mpllb_ssc_range_o[2:0]	O	MPLLB Spread Spectrum Clock Range. This signal is driven from MPLLB_SSC_RANGE field of VR_PMA_MP_12G_MPLLB_CTRL1 Register. This signal is used to control the fixed offset applied to the phase accumulator in the PHY. This signal should be connected to mpllb_ssc_range input of the PHY. **Exists:** (DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_fracn_ctrl_o[(FR_W-1):0]	O	MPLLA Fractional Control. This signal is driven from MPLLA_FRACN_CTRL fields of VR_PMA_MP_16G_MPLLA_CTRL1 Register. This signal should be connected to mplla_fracn_ctrl input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_mpllb_fracn_ctrl_o[(FR_W-1):0]	O	MPLLB Fractional Control. This signal is driven from MPLLB_FRACN_CTRL fields of VR_PMA_MP_16G_MPLLB_CTRL1 Register. This signal should be connected to mpllb_fracn_ctrl input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_mplla_ssc_clk_sel_o[(SSC_W-1):0]	O	MPLLA Spread Spectrum Clock Select. This signal is driven from MPLLA_SSC_CLK_SEL field of VR_PMA_MP_16G_MPLLA_CTRL1 Register. This signal selects the clock for generating spread spectrum generation in the PHY. This signal should be connected to mplla_ssc_clk_sel input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_ssc_clk_sel_o[(SSC_W-1):0]	O	MPLLB Spread Spectrum Clock Select. This signal is driven from MPLLB_SSC_CLK_SEL field of VR_PMA_MP_16G_MPLLB_CTRL1 Register. This signal selects the clock for generating spread spectrum generation in the PHY. This signal should be connected to mpllb_ssc_clk_sel input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_mplla_div_mult_o[(DIV_MUL_W-1):0]	O	MPLLA Output Frequency Multiplier Control. This signal is driven from MPLLA_DIV_MULT field of VR_PMA_MP_16G_MPLLA_CTRL2 Register. This frequency multiplication factor is used to control MPLL clock generation from the reference clock input. This signal should be connected to mplla_div_multiplier input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_mpllb_div_mult_o[(DIV_MUL_W-1):0]	O	MPLLB Output Frequency Multiplier Control. This signal is driven from MPLLB_DIV_MULT field of VR_PMA_MP_16G_MPLLB_CTRL2 Register. This frequency multiplication factor is used to control MPLL clock generation from the reference clock input. This signal should be connected to mpllb_div_multiplier input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_div_clk_en_o	O	MPLLA Divide Clock Enable. This signal is driven from MPLLA_DIV_CLK_EN field of VR_PMA_MP_16G_MPLLA_CTRL2 Register. If this signal is asserted, frequency of mplla_div_clk output from PHY will be MPLLA frequency divided by mplla_div_multiplier. This signal should be connected to mplla_div_clk_en input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_div_clk_en_o	O	MPLLB Divide Clock Enable. This signal is driven from MPLLB_DIV_CLK_EN field of VR_PMA_MP_16G_MPLLB_CTRL2 Register. If this signal is asserted, frequency of mpllb_div_clk output from PHY will be MPLLA frequency divided by mpllb_div_multiplier. This signal should be connected to mpllb_div_clk_en input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_ref_clk_en_o	O	Reference Clock Enable. This signal is driven from REF_CLK_EN field of VR_PMA_MP_25G_16G_REF_CLK_CTRL Register. This signal is used to enable reference clock to the PHY. This signal should be connected to ref_clk_en input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_ref_use_pad_o	O	Select Pad Reference Clock. This signal is driven from REF_USE_PAD field of VR_PMA_MP_25G_16G_REF_CLK_CTRL Register. This signal is used to select the reference clock connected to ref_pad_clk_p/m input pins of the PHY. This signal should be connected to ref_use_pad input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_ref_clk_div2_en_o	O	PHY Reference clock divider control. This signal is driven form REF_CLK_DIV2 field of VR_PMA_MP_25G_16G_REF_CLK_CTRL Register. When this signal is asserted, input reference clock to the PHY is divided by 2. This signal should be connected to ref_clk_div2_en input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_multiplier_o[(MPLL_MULT_W-1):0]	O	MPLLA Frequency Multiplier Control. This signal is driven from MPLLA_MULTIPLIER field of VR_PMA_MP_25G_MPLLA_CTRL0 Register. This signal is used to multiply the reference clock to a frequency suitable for intended operating speed. This signal should be connected to mplla_multiplier input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_multiplier_o[(MPLL_MULT_W-1):0]	O	MPLLB Frequency Multiplier Control. This signal is driven from MPLLB_MULTIPLIER field of VR_PMA_MP_25G_MPLLB_CTRL0 Register. This signal is used to multiply the reference clock to a frequency suitable for intended operating speed. This signal should be connected to mpllb_multiplier input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_mplla_bw_o[(MPLL_BW_W-1):0]	O	MPLLA Bandwidth Control. This signal is driven from MPLLA_BANDWIDTH field of VR_PMA_MP_25G_MPLLA_CTRL0 Register. This signal is set appropriately to control the bandwidth of MPLLA. This signal should be connected to mplla_bandwidth input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_mpllb_bw_o[(MPLL_BW_W-1):0]	O	MPLLB Bandwidth Control. This signal is driven from MPLLB_BANDWIDTH field of VR_PMA_MP_25G_MPLLB_CTRL0 Register. This signal is set appropriately to control the bandwidth of MPLLB. This signal should be connected to mpllb_bandwidth input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_init_cal_disable_o	O	MPLLA Calibration Disable. This signal is driven from MPLLA_CAL_DISABLE field of VR_PMA_MP_25G_MPLLA_CTRL0 Register. This signal is used to disable the calibration operation of MPLLA, that is performed by the PHY firmware. This signal should be connected to the mplla_init_cal_disable input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_init_cal_disable_o	O	MPLLA Calibration Disable. This signal is driven from MPLLB_CAL_DISABLE field of VR_PMA_MP_25G_MPLLB_CTRL0 Register. This signal is used to disable the calibration operation of MPLLB, that is performed by the PHY firmware. This signal should be connected to the mpllb_init_cal_disable input port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_div8_clk_en_o	O	MPLLA Divide by 8 Enable. This signal is driven from MPLLA_DIV8_CLK_EN field of VR_PMA_MP_16G_MPLLA_CTRL2 Register. When asserted, the frequency of mplla_word_clk output from PHY will be MPLLA frequency divided by 8. This signal should be connected to mplla_div8_clk_en input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_div8_clk_en_o	O	MPLLB Divide by 8 Enable. This signal is driven from MPLLB_DIV8_CLK_EN field of VR_PMA_MP_16G_MPLLB_CTRL2 Register. When asserted, the frequency of mpllb_word_clk output from PHY will be MPLLB frequency divided by 8. This signal should be connected to mpllb_div8_clk_en input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_div10_clk_en_o	O	MPLLA Divide by 10 Enable. This signal is driven from MPLLA_DIV10_CLK_EN field of VR_PMA_MP_16G_MPLLA_CTRL2 Register. When asserted, the frequency of mplla_word_clk output from PHY will be MPLLA frequency divided by 10. This signal should be connected to mplla_div10_clk_en input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_div10_clk_en_o	O	MPLLB Divide by 10 Enable. This signal is driven from MPLLB_DIV8_CLK_EN field of VR_PMA_MP_16G_MPLLB_CTRL2 Register. When asserted, the frequency of mpllb_word_clk output from PHY will be MPLLB frequency divided by 10. This signal should be connected to mpllb_div10_clk_en input port of the PHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_div16p5_clk_en_o	O	MPLLA Divided by 16.5 clock. This signal is driven from MPLLA_DIV16P5_CLK_EN field of VR_PMA_MP_16G_MPLLA_CTRL2 Register. This signal is used to enable the output clocks derived from MPLLA based 16.5,33 and 66 division ratios. This signal should be connected to mplla_div16p5_clk_en input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_div16p5_clk_en_o	O	MPLLB Divided by 16.5 clock. This signal is driven from MPLLB_DIV16P5_CLK_EN field of VR_PMA_MP_16G_MPLLB_CTRL2 Register. This signal is used to enable the output clocks derived from MPLLB based 16.5,33 and 66 division ratios. This signal should be connected to mpllb_div16p5_clk_en input pin of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_ref_repeat_clk_en_o	O	Enable Repeat Reference Clock from PHY. This signal is driven from REF_RPT_CLK_EN field of VR_PMA_MP_25G_16G_REF_CLK_CTRL Register. This signal should be connected to ref_repeat_clk_en input of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_state_i	I	MPLLA state. This signal indicates the status of MPLLA. The value of this signal is available in MPLLA_STS field of VR_PMA_MP_25G_16G_12G_MISC_STS0 Register. This input port of DWC_xlgpcs should be connected to mplla_state output of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with clk_csr_i
xlgpcs_mpllb_state_i	I	MPLLB state. This signal indicates the status of MPLLB. The value of this signal is available in MPLLB_STS field of VR_PMA_MP_25G_16G_12G_MISC_STS0 Register. This input port of DWC_xlgpcs should be connected to mpllb_state output of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with clk_csr_i
xlgpcs_tx1_misc_o[7:0]	O	TX Miscellaneous Controls along lane1. It controls miscellaneous settings in the TX analog block of EPHY for lane1 Tx path. This signal is driven from TX1_MISC fields of VR_PMA_MP_25G_16G_TX_MISC_CTRL0 Register. This signal should be connected to tx1_misc input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_ref_clk_mplla_div_o[2:0]	O	MPLLA Reference Clock Divider Control. The reference clock used for MPLLA calibration and locking can be divided by 2/4/8/16. This signal is driven from REF_CLK_MPLLA_DIV fields of VR_PMA_MP_25G_16G_REF_CLK_CTRL Register. This signal should be connected to ref_clk_mplla_div input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_ref_clk_mpllb_div_o[2:0]	O	MPLLB Reference Clock Divider Control. The reference clock used for MPLLA calibration and locking can be divided by 2/4/8/16. This signal is driven from REF_CLK_MPLLB_DIV fields of VR_PMA_MP_25G_16G_REF_CLK_CTRL Register. This signal should be connected to ref_clk_mpllb_div input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_ref_clk_mplla_div2_en_o	O	MPLLA Reference Clock Divider Control. The reference clock used for MPLLA calibration and locking can be divided by 2. This signal is driven from REF_MPLLA_DIV2 fields of VR_PMA_MP_25G_16G_REF_CLK_CTRL Register. This signal should be connected to ref_clk_mplla_div2_en input port of the EPHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_ref_clk_mpllb_div2_en_o	O	MPLLB Reference Clock Divider Control. The reference clock used for MPLLA calibration and locking can be divided by 2. This signal is driven from REF_MPLLB_DIV2 fields of VR_PMA_MP_25G_16G_REF_CLK_CTRL Register. This signal should be connected to ref_clk_mpllb_div2_en input port of the EPHY. **Exists:** (DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_ref_clkdet_en_o	O	Enable Reference Clock Detection. Enables detection of the reference clock on the pads input. This signal is driven from REF_CLKDET_EN fields of VR_PMA_MP_25G_16G_MISC_CTRL2 Register. This signal should be connected to ref_clkdet_en input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_ref_clkdet_result_i	I	Reference Clock Detection result. Indicates the presence of the reference clock on the pads input of EPHY. The value of this signal is available in REF_CLKDET_RESULT field of VR_PMA_MP_25G_16G_12G_MISC_STS0 Register. This signal is driven by ref_clkdet_result output port of EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_bw_th_o[(MPLL_TH_W-1):0]	O	MPLLA Bandwidth Threshold. Bandwidth threshold control for MPLLA . This value controls the threshold that the PHY uses to select between the mplla_bw_low and mpllb_bw_high values. This signal is driven from MPLLA_BW_TH_25G fields of VR_PMA_MP_32G_25G_MPLLA_CTRL1 Register. This signal should be connected to mplla_bw_threshold input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_short_lock_en_o	O	MPLLA short lock enable. This input indicates that a shared reference clock configuration is being used. When asserted, the MPLLA locking time will be reduced. Note: This signal can be changed only when the txX_mpll_en input for all lanes is de-asserted. This signal is driven from MPLLA_SRT_LK fields of VR_PMA_MP_32G_25G_MPLLA_CTRL1 Register. This signal should be connected to mplla_short_lock_en input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_bw_low_o[15:0]	O	MPLLA Bandwidth Control Lower limit. This signal is driven from MPLLA_LOW_BW fields of VR_PMA_MP_25G_MPLLA_CTRL3 Register. This signal should be connected to mplla_bw_low input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_bw_high_o[15:0]	O	MPLLA Bandwidth Control Upper limit. This signal is driven from MPLLA_HIGH_BW fields of VR_PMA_MP_25G_MPLLA_CTRL4 Register. This signal should be connected to mplla_bw_high input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_ctl_buf_byp_o	O	MPLLA Control Buffer Bypass. Control buffer bypass enable for MPLLA. This signal is driven from MPLLA_CTL_BUF_BYP fields of VR_PMA_MP_25G_MPLLA_CTRL0 Register. This signal should be connected to mplla_ctl_buf_byp input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_word_clk_div_o[1:0]	O	MPLLA Word Clock Divider Control. Controls the MPLLA word clock dividers for generating the mplla_word_clk output. These inputs can be changed only when the txX_mpll_en input is deasserted. This signal is driven from MPLLA_WCLK_DIV fields of VR_PMA_MP_25G_MPLLA_CTRL2 Register. This signal should be connected to mplla_word_clk_div input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_fb_clk_div4_en_o	O	MPLLA DIV Clock Output Frequency Divider Control. Additional frequency division factor used to generate mpll_div_clk clock output from the MPLLA VCO. This signal controls whether the MPLL VCO is divided by 2 (when mplla_fb_clk_div4_en is de-asserted) or 4 (when mplla_fb_clk_div4_en is asserted) before being divided down by mplla_div_multiplier[7:0]. This input can be changed only when the txX_mpll_en input for all lanes is de-asserted. This signal is driven from MPLLA_FB_CLK_DIV4 fields of VR_PMA_MP_25G_MPLLA_CTRL2 Register. This signal should be connected to mplla_fb_clk_div4 input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_tx_clk_div_o[(MPLL_CLK_DIV_W-1):0]	O	MPLLA TX Clock Divider. .Sets the divide ratio on the MPLL clock sent to the TX. Note: This signal can be changed only when the txX_mpll_en input for all lanes is de-asserted. This signal is driven from MPLLA_TX_CLK_DIV fields of VR_PMA_MP_25G_MPLLA_CTRL2 Register. This signal should be connected to mplla_tx_clk_div input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_recal_bank_sel_o[1:0]	O	MPLLA Re-calibration Bank Select. Selects the bank to store the MPLL calibration value. *Note:* This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2 in PHY(E25). This signal is driven from MPLLA_RECAL_BANK_SEL fields of VR_PMA_MP_25G_MPLLA_CTRL2 Register. This signal should be connected to mplla_recal_bank_sel input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_bw_th_o[(MPLL_TH_W-1):0]	O	MPLLB Bandwidth Threshold. Bandwidth threshold control for MPLLB . This value controls the threshold that the PHY uses to select between the mpllb_bw_low and mpllb_bw_high values. This signal is driven from MPLLB_BW_TH_25G fields of VR_PMA_MP_32G_25G_MPLLB_CTRL1 Register. This signal should be connected to mpllb_bw_threshold input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_short_lock_en_o	O	MPLLB short lock enable. This input indicates that a shared reference clock configuration is being used. When asserted, the MPLLB locking time will be reduced. Note: This signal can be changed only when the txX_mpll_en input for all lanes is de-asserted. This signal is driven from MPLLB_SRT_LK fields of VR_PMA_MP_32G_25G_MPLLB_CTRL1 Register. This signal should be connected to mpllb_short_lock_en input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_bw_low_o[15:0]	O	MPLLB Bandwidth Control Lower limit. This signal is driven from MPLLB_LOW_BW fields of VR_PMA_MP_25G_MPLLB_CTRL3 Register. This signal should be connected to mpllb_bw_low input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_bw_high_o[15:0]	O	MPLLB Bandwidth Control Upper limit. This signal is driven from MPLLB_HIGH_BW fields of VR_PMA_MP_25G_MPLLB_CTRL4 Register. This signal should be connected to mpllb_bw_high input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_ctl_buf_byp_o	O	MPLLB Control Buffer Bypass. Control buffer bypass enable for MPLLB. This signal is driven from MPLLB_CTL_BUF_BYP fields of VR_PMA_MP_25G_MPLLB_CTRL0 Register. This signal should be connected to mpllb_ctl_buf_byp input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_word_clk_div_o[1:0]	O	MPLLB Word Clock Divider Control. Controls the MPLLB word clock dividers for generating the mpllb_word_clk output. These inputs can be changed only when the txX_mpll_en input is deasserted. This signal is driven from MPLLB_WCLK_DIV fields of VR_PMA_MP_25G_MPLLB_CTRL2 Register. This signal should be connected to mpllb_word_clk_div input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_fb_clk_div4_en_o	O	MPLLB DIV Clock Output Frequency Divider Control. Additional frequency division factor used to generate mpll_div_clk clock output from the MPLLA VCO. This signal controls whether the MPLL VCO is divided by 2 (when mpllb_fb_clk_div4_en is de-asserted) or 4 (when mpllb_fb_clk_div4_en is asserted) before being divided down by mpllb_div_multiplier[7:0]. This input can be changed only when the txX_mpll_en input for all lanes is de-asserted. This signal is driven from MPLLB_FB_CLK_DIV4 fields of VR_PMA_MP_25G_MPLLB_CTRL2 Register. This signal should be connected to mpllb_fb_clk_div4 input port of the EPHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_tx_clk_div_o[(MPLL_CLK_DIV_W-1):0]	O	MPLLB TX Clock Divider. .Sets the divide ratio on the MPLL clock sent to the TX. Note: This signal can be changed only when the txX_mpll_en input for all lanes is de-asserted. This signal is driven from MPLLB_TX_CLK_DIV fields of VR_PMA_MP_25G_MPLLB_CTRL2 Register. This signal should be connected to mpllb_tx_clk_div input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_recal_bank_sel_o[1:0]	O	MPLLB Re-calibration Bank Select. Selects the bank to store the MPLL calibration value. *Note:* This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2 in PHY(E25). This signal is driven from MPLLB_RECAL_BANK_SEL fields of VR_PMA_MP_25G_MPLLA_CTRL2 Register. This signal should be connected to mpllb_recal_bank_sel input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_sup_misc_o[7:0]	O	Support Miscellaneous Controls. This input controls miscellaneous settings in the Support block of E16 PHY. This signal is driven from SUP_MISC fields of VR_PMA_MP_25G_16G_MISC_CTRL2 Register. This signal should be connected to sup_misc input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_ssc_spd_en_o	O	MPLLA SSC Up Spread Enable. When asserted, the SSC is applied in upwards direction (positive ppm) and when deasserted the SSC is applied downwards (negative ppm). Note: This input can be changed only when the MPLL is powered down. This signal is driven from MPLLA_SSC_UP_SPD fields of VR_PMA_MP_32G_25G_MPLLA_CTRL1 Register. This signal should be connected to mplla_ssc_up_spread input of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_ssc_peak_o[19:0]	O	MPLLA Spread Spectrum Peak Value. Controls the ppm shift amplitude of the spread-spectrum profile. This signal is driven from MPLLA_SSC_PEAK fields of VR_PMA_MP_25G_MPLLA_CTRL5 & VR_PMA_MP_25G_MPLLA_CTRL6 Register. This signal should be connected to mplla_ssc_peak input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_ssc_step_size_o[20:0]	O	MPLLA Spread Spectrum Step Size Value. Controls the spread-spectrum profile step. This signal is driven from MPLLA_SSC_STEP_SZ fields of VR_PMA_MP_25G_MPLLA_CTRL7 & VR_PMA_MP_25G_MPLLA_CTRL8 Register. This signal should be connected to mplla_ssc_step_size input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_frac_en_o	O	Reserved. This signal is driven from MPLLA_FRAC_EN fields of VR_PMA_MP_32G_25G_MPLLA_CTRL1 Register. This signal should be connected to mplla_frac_en input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_frac_quot_o[15:0]	O	Reserved. This signal is driven from MPLLA_FRAC_QUO fields of VR_PMA_MP_25G_MPLLA_CTRL9 Register. This signal should be connected to mplla_frac_quot input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_frac_rem_o[15:0]	O	Reserved. This signal is driven from MPLLA_FRAC_REM fields of VR_PMA_MP_32G_25G_MPLLA_CTRL10 Register. This signal should be connected to mplla_frac_rem input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_frac_den_o[15:0]	O	Reserved. This signal is driven from MPLLA_FRAC_DEN fields of VR_PMA_MP_32G_25G_MPLLA_CTRL11 Register. This signal should be connected to mplla_frac_den input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_ssc_freq_cnt_init_o[11:0]	O	MPLLA SSC Frequency Counter Initialization. Sets the initialization value for the SSC frequency counter. This signal is driven from MPLLA_SSC_FREQ_CNT_INT fields of VR_PMA_MP_25G_MPLLA_CTRL4 Register. This signal should be connected to mplla_ssc_freq_cnt_init input port of the PHY. **Exists:** (DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_ssc_freq_cnt_pk_o[7:0]	O	MPLLA SSC Frequency Counter Peak. Sets the peak value for the SSC frequency counter. This signal is driven from MPLLA_SSC_FRQ_CNT_PK fields of VR_PMA_MP_25G_MPLLA_CTRL5 Register. This signal should be connected to mplla_ssc_frq_cnt_peak[7:0] input port of the PHY. **Exists:** (DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_ssc_spd_en_o	O	MPLLB SSC Up Spread Enable. When asserted, the SSC is applied in upwards direction (positive ppm) and when deasserted the SSC is applied downwards (negative ppm). Note: This input can be changed only when the MPLL is powered down. This signal is driven from MPLLB_UP_SPD fields of VR_PMA_MP_32G_25G_MPLLB_CTRL1 Register. This signal should be connected to mpllb_ssc_up_spread input of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_ssc_peak_o[19:0]	O	MPLLB Spread Spectrum Peak Value. Controls the ppm shift amplitude of the spread-spectrum profile. This signal is driven from MPLLB_SSC_PEAK fields of VR_PMA_MP_25G_MPLLB_CTRL5 & VR_PMA_MP_25G_MPLLB_CTRL6 Register. This signal should be connected to mpllb_ssc_peak input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_ssc_step_size_o[20:0]	O	MPLLB Spread Spectrum Step Size Value. Controls the spread-spectrum profile step. This signal is driven from MPLLB_SSC_STEP_SZ fields of VR_PMA_MP_25G_MPLLB_CTRL7 & VR_PMA_MP_25G_MPLLB_CTRL8 Register. This signal should be connected to mpllb_ssc_step_size input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_frac_en_o	O	Reserved. This signal is driven from MPLLB_FRAC_EN fields of VR_PMA_MP_32G_25G_MPLLB_CTRL1 Register. This signal should be connected to mpllb_frac_en input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_frac_quot_o[15:0]	O	Reserved. This signal is driven from MPLLB_FRAC_QUO fields of VR_PMA_MP_25G_MPLLB_CTRL9 Register. This signal should be connected to mpllb_frac_quot input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_frac_rem_o[15:0]	O	Reserved. This signal is driven from MPLLB_FRAC_REM fields of VR_PMA_MP_25G_MPLLB_CTRL10 Register. This signal should be connected to mpllb_frac_rem input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_frac_den_o[15:0]	O	Reserved. This signal is driven from MPLLB_FRAC_DEN fields of VR_PMA_MP_25G_MPLLB_CTRL11 Register. This signal should be connected to mpllb_frac_den input of the PHY. **Exists:** (DWCXLP_E25) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_ssc_freq_cnt_init_o[11:0]	O	MPLLA SSC Frequency Counter Initialization. Sets the initialization value for the SSC frequency counter. This signal is driven from MPLLB_SSC_FREQ_CNT_INT fields of VR_PMA_MP_25G_MPLLB_CTRL4 Register. This signal should be connected to mpllb_ssc_freq_cnt_init input port of the PHY. **Exists:** (DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_ssc_freq_cnt_pk_o[7:0]	O	MPLLA SSC Frequency Counter Peak. Sets the peak value for the SSC frequency counter. This signal is driven from MPLLB_SSC_FRQ_CNT_PK fields of VR_PMA_MP_25G_MPLLB_CTRL5 Register. This signal should be connected to mpllb_ssc_frq_cnt_peak[7:0] input port of the PHY. **Exists:** (DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_txup_term_offset_o[8:0]	O	Offset for TX Up Termination. Specifies an additional fixed offset to calibrated TX up termination value. This signal is driven from TXUP_TERM_OFFSET fields of Synopsys PHY TX_GEN_CTRL3 Register. This signal should be connected to txup_term_offset input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_txdn_term_offset_o[8:0]	O	Offset for TX Down Termination. Specifies an additional fixed offset to calibrated TX down termination value. This signal is driven from TXDN_TERM_OFFSET fields of Synopsys PHY TX_GEN_CTRL4 Register. This signal should be connected to txdn_term_offset input port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E16) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_refa_clk_en_o	O	Reference clock enable for refa Enables the reference clock for the PHY. The xlgpcs_refa_clk_en_o signal must remain de-asserted until the reference clock is running at the appropriate frequency **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_refb_clk_en_o	O	Reference clock enable for refb Enables the reference clock for the PHY. The xlgpcs_refb_clk_en_o signal must remain de-asserted until the reference clock is running at the appropriate frequency **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_refa_clk_div2_en_o	O	Output reference clock divider control for refa When asserted, the reference clock is divided by 2 **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_refb_clk_div2_en_o	O	Output reference clock divider control for refb When asserted, the reference clock is divided by 2 **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_refa_range_o[2:0]	O	Output reference clock frequency range for refa Specifies the frequency range of the PHY's input reference clock. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_refb_range_o[2:0]	O	Output reference clock frequency range for refb Specifies the frequency range of the PHY's input reference clock. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_ref_rpt_clk_sel_o[1:0]	O	Selects the input reference clock for the CML output clocks xlgpcs_ref_repeat_clk_(p,m). **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_refa_clk_sel_o[1:0]	O	xlgpcs_refa_clk_sel - Select reference clock source from xlgpcs_ref_pad_clk_p/m or xlgpcs_ref_alt0/1_clk_p/m **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_refb_clk_sel_o[1:0]	O	xlgpcs_refb_clk_sel - Select reference clock source from xlgpcs_ref_pad_clk_p/m or xlgpcs_ref_alt0/1_clk_p/m **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_refa_clkdet_en_o	O	Enable reference clock detection for refa **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_refb_clkdet_en_o	O	Enable reference clock detection for refb **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_lane_ref_sel_o[(DWCXLP_NUM_LANES-1):0]	O	Selects the refa or refb reference clock - 0: Select refa clock - 1: Select refb clock **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
xlgpcs_refa_clkdet_result_i	I	Reference clock detection result for refa **Exists:** (DWCXLP_E32) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_refb_clkdet_result_i	I	Reference clock detection result for refb **Exists:** (DWCXLP_E32) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_pll_state_i[(DWCXLP_NUM_LANES-1):0]	I	Indicates the state of PLL for the lane. This signal is asserted when lane PLL is powered up and phase-locked. **Exists:** (DWCXLP_E32) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_tx_pll_en_o[(DWCXLP_NUM_LANES-1):0]	O	TX PLL Enable Powers up or powers down the ROPLL or MPLLA or MPLLB. For normal operation, this output is set to 1 when txX_pstate[1:0] is in P0 or P0s state. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-24 Synopsys PHY-Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mplla_wrd_clk_en_o	O	MPLLA word clock output enable Synopsys UPCS uses xlgpcs_txX_ropll_word clock instead of mpll(a,b)_word_clk. So, this is set to 0 in functional mode. In scan mode, it is set to 1. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_wrd_clk_en_o	O	MPLLB word clock output enable Synopsys UPCS uses xlgpcs_txX_ropll_word clock instead of mpll(a,b)_word_clk. So, this is set to 0 in functional mode. In scan mode, it is set to 1. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_refa_raw_clk_div2_en_o	O	Reference clock divider control for refa clock in RAW PCS. Asserted to divide refa clock by 2 in RAW PCS. phy_reset should be asserted only after this signal is changed **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_refb_raw_clk_div2_en_o	O	Reference clock divider control for refb clock in RAW PCS. Asserted to divide refb clock by 2 in RAW PCS. phy_reset should be asserted only after this signal is changed **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.25 Synopsys PHY Link Training Signals

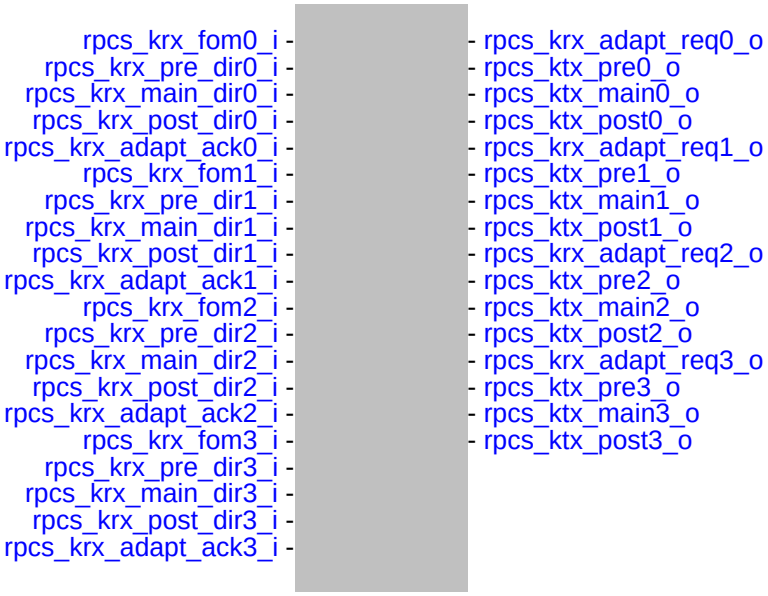


Table 6-25 Synopsys PHY Link Training Signals

Port Name	I/O	Description
rpcs_krx_fom0_i[7:0]	I	Receiver Figure of Merit for lane 0. This signal indicates the quality of the received data eye. The value of this signal is available at FOM0 field of VR_PMA_MP_25G_16G_MISC_STS1 Register. This signal should be connected to rx0_adapt_fom port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_pre_dir0_i[1:0]	I	Direction Change for far-end Tx Coefficient C(-1) for lane 0. This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx0_txpre_dir port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_main_dir0_i[1:0]	I	Direction Change for far-end Transmitter Coefficient C(0) for lane 0. This signal provides the recommended direction change for far-end Tx equalization settings. The encoding is as follows: ■ 2'b00: IDLE ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: HOLD This signal should be connected to rx0_txmain_dir port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_post_dir0_i[1:0]	I	Direction Change for far-end Tx Coefficient C(1) for lane 0. This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx0_txpost_dir port of the PHY **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_adapt_req0_o	O	Receiver Adaptation Request lane 0. This signal indicates a request for the receiver to adapt to incoming data. This signal should be connected to rx0_adapt_req port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_krx_adapt_ack0_i	I	Receiver adaptation acknowledge for lane 0. The assertion of this signal indicates the completion of receiver adaptation sequence and the values rx0_txmain_dir[1:0], rx0_txpre_dir[1:0] and rx0_txpost_dir[1:0] are valid. This signal should be connected to rx0_adapt_ack port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_pre0_o[5:0]	O	Transmitter Equalization filter coefficient C(-1) for lane 0. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_PRE field of VR PMA Synopsys PHY Transmit Equalization Control 0 -lane 0 Register. This signal should be connected to tx0_eq_pre port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_main0_o[5:0]	O	Transmitter Equalization filter coefficient C(0) for lane 0. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_MAIN field of VR PMA Synopsys PHY Transmit Equalization Control 0 -lane 0 Register. This signal should be connected to tx0_eq_main port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_post0_o[5:0]	O	Transmitter Equalization filter coefficient C(1) for lane 0. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_POST field of VR PMA Synopsys PHY Transmit Equalization Control 1 -lane 0 Register. This signal should be connected to tx0_eq_post port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_fom1_i[7:0]	I	Receiver Figure of Merit for lane 1. This signal indicates the quality of the received data eye. The value of this signal is available at FOM1 field of VR_PMA_MP_25G_16G_MISC_STS1 Register. This signal should be connected to rx0_adapt_fom port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_pre_dir1_i[1:0]	I	Direction Change for far-end Tx Coefficient C(-1) for lane 1. This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx1_txpre_dir port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_main_dir1_i[1:0]	I	Direction Change for far-end Transmitter Coefficient C(0) for lane 1. This signal provides the recommended direction change for far-end Tx equalization settings. The encoding is as follows: ■ 2'b00: IDLE ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: HOLD This signal should be connected to rx1_txmain_dir port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_post_dir1_i[1:0]	I	Direction Change for far-end Tx Coefficient C(1) for lane 1. This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx1_txpost_dir port of the PHY **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_adapt_req1_o	O	Receiver Adaptation Request lane 1. This signal indicates a request for the receiver to adapt to incoming data. This signal should be connected to rx1_adapt_req port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_krx_adapt_ack1_i	I	Receiver adaptation acknowledge for lane 1. The assertion of this signal indicates the completion of receiver adaptation sequence and the values rx1_txmain_dir[1:0], rx1_txpre_dir[1:0] and rx1_txpost_dir[1:0] are valid. This signal should be connected to rx1_adapt_ack port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_pre1_o[5:0]	O	Transmitter Equalization filter coefficient C(-1) for lane 1. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_PRE field of VR PMA Synopsys PHY Transmit Equalization Control 0 -lane 1 Register. This signal should be connected to tx1_eq_pre port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_main1_o[5:0]	O	Transmitter Equalization filter coefficient C(0) for lane 1. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_MAIN field of VR PMA Synopsys PHY Transmit Equalization Control 0 -lane 1 Register. This signal should be connected to tx1_eq_main port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_post1_o[5:0]	O	Transmitter Equalization filter coefficient C(1) for lane 1. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_POST field of VR PMA Synopsys PHY Transmit Equalization Control 1 -lane 1 Register. This signal should be connected to tx1_eq_post port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_fom2_i[7:0]	I	Receiver Figure of Merit for lane 2. This signal indicates the quality of the received data eye. The value of this signal is available at FOM2 field of VR_PMA_MP_25G_16G_MISC_STS2 Register. This signal should be connected to rx0_adapt_fom port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_pre_dir2_i[1:0]	I	Direction Change for far-end Tx Coefficient C(-1) for lane 2. This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx2_txpre_dir port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_main_dir2_i[1:0]	I	Direction Change for far-end Transmitter Coefficient C(0) for lane 2. This signal provides the recommended direction change for far-end Tx equalization settings. The encoding is as follows: ■ 2'b00: IDLE ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: HOLD This signal should be connected to rx2_txmain_dir port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_post_dir2_i[1:0]	I	Direction Change for far-end Tx Coefficient C(1) for lane 2. This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx2_txpost_dir port of the PHY **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_adapt_req2_o	O	Receiver Adaptation Request lane 2. This signal indicates a request for the receiver to adapt to incoming data. This signal should be connected to rx2_adapt_req port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_krx_adapt_ack2_i	I	Receiver adaptation acknowledge for lane 2. The assertion of this signal indicates the completion of receiver adaptation sequence and the values rx2_txmain_dir[1:0], rx2_txpre_dir[1:0] and rx2_txpost_dir[1:0] are valid. This signal should be connected to rx2_adapt_ack port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_pre2_o[5:0]	O	Transmitter Equalization filter coefficient C(-1) for lane 2. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_PRE field of VR PMA Synopsys PHY Transmit Equalization Control 0 -lane 2 Register. This signal should be connected to tx2_eq_pre port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_main2_o[5:0]	O	Transmitter Equalization filter coefficient C(0) for lane 2. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_MAIN field of VR PMA Synopsys PHY Transmit Equalization Control 0 -lane 2 Register. This signal should be connected to tx2_eq_main port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_post2_o[5:0]	O	Transmitter Equalization filter coefficient C(1) for lane 2. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_POST field of VR PMA Synopsys PHY Transmit Equalization Control 1 -lane 2 Register. This signal should be connected to tx2_eq_post port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_fom3_i[7:0]	I	Receiver Figure of Merit for lane 3. This signal indicates the quality of the received data eye. The value of this signal is available at FOM3 field of VR_PMA_MP_25G_16G_MISC_STS2 Register. This signal should be connected to rx3_adapt_fom port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_pre_dir3_i[1:0]	I	Direction Change for far-end Tx Coefficient C(-1) for lane 3. This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx3_txpre_dir port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_main_dir3_i[1:0]	I	Direction Change for far-end Transmitter Coefficient C(0) for lane 3. This signal provides the recommended direction change for far-end Tx equalization settings. The encoding is as follows: ■ 2'b00: IDLE ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: HOLD This signal should be connected to rx3_txmain_dir port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_post_dir3_i[1:0]	I	Direction Change for far-end Tx Coefficient C(1) for lane 3. This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx3_txpost_dir port of the PHY **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_CL72_EN&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_krx_adapt_req3_o	O	Receiver Adaptation Request lane 3. This signal indicates a request for the receiver to adapt to incoming data. This signal should be connected to rx3_adapt_req port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_krx_adapt_ack3_i	I	Receiver adaptation acknowledge for lane 3. The assertion of this signal indicates the completion of receiver adaptation sequence and the values rx3_txmain_dir[1:0], rx3_txpre_dir[1:0] and rx3_txpost_dir[1:0] are valid. This signal should be connected to rx3_adapt_ack port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_pre3_o[5:0]	O	Transmitter Equalization filter coefficient C(-1) for lane 3. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_PRE field of VR PMA Synopsys PHY Transmit Equalization Control 0 -lane 3 Register. This signal should be connected to tx3_eq_pre port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-25 Synopsys PHY Link Training Signals (Continued)

Port Name	I/O	Description
rpcs_ktx_main3_o[5:0]	O	Transmitter Equalization filter coefficient C(0) for lane 3. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_MAIN field of VR PMA Synopsys PHY Transmit Equalization Control 0 -lane 3 Register. This signal should be connected to tx3_eq_main port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_post3_o[5:0]	O	Transmitter Equalization filter coefficient C(1) for lane 3. The value of this signal is computed based KR Training algorithm when DWCXLP_CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_POST field of VR PMA Synopsys PHY Transmit Equalization Control 1 -lane 3 Register. This signal should be connected to tx3_eq_post port of the PHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&& DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_tx_i **Registered:** !DWCXLP_LNK_TRAIN_EN ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.26 Synopsys PHY Specific- Status and Control Interface Signals

xlgpcs_rx2_ppm_drift_i -

Table 6-26 Synopsys PHY Specific- Status and Control Interface Signals

Port Name	I/O	Description
xlgpcs_rx2_ppm_drift_i[5:0]	I	RX CDR PPM Drift (Along lane2). This value represents the amount of ppm on the rx2_clk(of PHY) with respect to the ideal desired frequency. This input is valid when xlgpcs_rx2_ppm_drift_vld_i is asserted. This signal is driven by rx2_ppm_drift output port of the EPHY. **Exists:** (DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE **Synchronous To:** Asynchronous **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.27 Synopsys PHY Specific MPLL Control and Status Interface Signals



Table 6-27 Synopsys PHY Specific MPLL Control and Status Interface Signals

Port Name	I/O	Description
xlgpcs_mplla_recal_force_en_o	O	MPLLA Force Re-calibration. This signal forces recalibration of MPLLA and TX DCC in the selected bank. **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mplla_recal_skip_en_o	O	MPLLA Skip Re-calibration This signal skips re-calibration of MPLLA and TX DCC in the selected bank **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_mpllb_recal_force_en_o	O	MPLLB Force Re-calibration. This signal forces recalibration of MPLLB and TX DCC in the selected bank. **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-27 Synopsys PHY Specific MPLL Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_mpllb_recal_skip_en_o	O	MPLLB Skip Re-calibration This signal skips re-calibration of MPLLB and TX DCC in the selected bank **Exists:** (DWCXLP_E25||DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.28 Debug Interface Signals

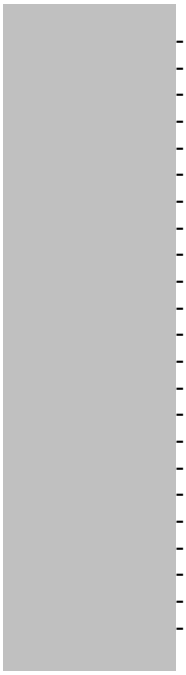


Table 6-28 Debug Interface Signals

Port Name	I/O	Description
xlgpcs_sh_err_o	O	This is the status signal indicating error in the synchronization bits (when sync bits is 00 or 11) of 66-bit PCS data. When xlgpcs_bklk_o is low, xlgpcs_sh_err_o indicates the sync header errors seen before achieving the block sync. When xlgpcs_bklk_lock is high, xlgpcs_sh_err_o indicates the sync header errors after achieving the block sync. This status is applicable in no-FEC and BASE-R FEC mode of operations only. This signal can be a level or a pulse depending on the input data. **Exists:** (SNPS_RSVDPARAM_11||SNPS_RSVDPARAM_12||SNPS_RSVDPARAM_15) **Synchronous To:** clk_rpcs_rx0_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_bklk_o	O	Rx block-lock indication. This signal indicates the block-lock status and is valid in non-FEC and BASE-R FEC modes **Exists:** SNPS_RSVDPARAM_15 **Synchronous To:** clk_rpcs_rx0_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rsfec_amlk_o	O	RS-FEC AM Lock Indication. This signal indicates the Alignment Marker (Code-word marker) lock status when operating in RS-FEC Mode. **Exists:** SNPS_RSVDPARAM_15&&DWCXLP_RSFEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_bitshift_o[6:0]	O	Rx data bit-shift. This field indicates the bit-shift done by DWC_xlgpcs in order to align the input Rx data to attain block-lock/FEC-lock/AM-lock. This field is valid only after xlgpcs_rx_lane_sts_o is high. This signal is valid in non-FEC, BASE-R FEC and RS-FEC Modes **Exists:** SNPS_RSVDPARAM_15 **Synchronous To:** clk_rpcs_rx0_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rdec_err_o	O	PCS Decoder (64/66b) error indication. This is a status signal indicating error in the 66b block seen by PCS decoder. When xlgpcs_rx_lane_sts_o is low, xlgpcs_rdec_err_o indicates the decoding errors seen before achieving Rx lock (hence this shall be ignored) . When xlgpcs_rx_lane_sts_o is high, xlgpcs_rdec_err_o indicates the decoding errors after achieving Rx lock. This signal can be a level or a pulse depending on the input 66bit data. **Exists:** SNPS_RSVDPARAM_15 **Synchronous To:** clk_rpcs_rx0_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_baserfec_ucrcw_o	O	BASE-R FEC UnCorrected code word Indicator pulse. When DWC_xlgpcs detects an uncorrectable BASER-R FEC code word, this signal is asserted for 1 clock. This signal is valid only when operating in BASE-R FEC Mode **Exists:** SNPS_RSVDPARAM_15&&DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rsfec_ucrcw_o	O	RS-FEC UnCorrected code word Indicator pulse. When DWC_xlgpcs detects an uncorrectable RS-FEC code word, this signal is asserted for 1 clock. This signal is valid only when operating in RS-FEC Mode **Exists:** SNPS_RSVDPARAM_15&&DWCXLP_RSFEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_dbg_bus_clk_rpcs_tx_o[67:0]	O	Debug Bus Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[4:0] : Write pointer of Tx Gearbox FIFO instance 0. For configurations without Gear-box, these bits are invalid and indicate 0 ■ Bits[9:5] : Write pointer of Tx Gearbox FIFO instance 1. For configurations without Gear-box or for 25G-Only configurations, these bits are invalid and indicate 0 ■ Bits[14:10] : Write pointer of Tx Gearbox FIFO instance 2. For configurations without Gear-box or for 25G-Only configurations, these bits are invalid and indicate 0 ■ Bits[19:15] : Write pointer of Tx Gearbox FIFO instance 3. For configurations without Gear-box or for 25G-Only configurations, these bits are invalid and indicate 0 ■ Bits[29:20] : Autonegotiation Arbiter FSM state. These bits are invalid and indicates 0 if configuration doesn't have auto-negotiation support or if the configurations has DWCXLP_CL73_ALT as 1. Interpretation of these bits are as follows: - 10'h1 : AN_ENABLE - 10'h2 : TRANSMIT_DISABLE - 10'h4 : ABILITY_DETECT - 10'h8 : ACKNOWLEDGE_DETECT - 10'h10 : COMPLETE_ACKNOWLEDGE - 10'h20 : NEXT_PAGE_WAIT - 10'h40 : LINK_STATUS_CHECK - 10'h80 : PARALLEL_DETECT_FAULT - 10'h100: AN_GOOD_CHECK - 10'h200: AN_GOOD ■ Bits[36:30]: Link Training FSM state for lane 0. These bits are synchronous to clk_rpcs_tx_i. These bits are invalid and indicates 0 if configuration doesn't have link-training support. Interpretation of these bits are as follows: - 7'h1 : INITIALIZE - 7'h2 : SEND_TRAINING - 7'h4 : TRAIN_LOCAL - 7'h8 : TRAIN_REMOTE - 7'h10 : LINK_READY - 7'h20 : SEND_DATA - 7'h40 : TRAINING_FAIL ■ Bits[43:37]: Link Training FSM state for lane 1. These bits are invalid and indicates 0 if configuration doesn't have multi-lane link-training support. Interpretation of these bits are identical to that of lane 0. ■ Bits[50:44]: Link Training FSM state for lane 2. These bits are invalid and indicates 0 if configuration doesn't have multi-lane link-training support. Interpretation of these bits are identical to that of lane 0.

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
		■ Bits[57:51]: Link Training FSM state for lane 3. These bits are invalid and indicates 0 if configuration doesn't have multi-lane link-training support. Interpretation of these bits are identical to that of lane 0. ■ Bits[67:58]: Read pointer of Clock-Compensation FIFO. These bits are valid only for configurations with DWCXLP_CLK_CMP_EN==1 AND DWCXLP_XLGMII_INTF_WIDTH=1 (64-bit). For other configurations, these bits always indicate 0 **Exists:** DWCXLP_DBG_BUS **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_dbg_bus_clk_xsbi_tx_o[19:0]	O	Debug Bus in clk_xsbi_tx_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[4:0] : Read pointer of Tx Gearbox FIFO instance 0. For configurations without Gear-box, these bits are invalid and indicate 0 ■ Bits[9:5] : Read pointer of Tx Gearbox FIFO instance 1. For configurations without Gear-box or for 25G-Only configurations or configurations with DWCXLP_XSBI_TX_CLK_PER_LN as 1, these bits are invalid and indicate 0 ■ Bits[14:10] : Read pointer of Tx Gearbox FIFO instance 2. For configurations without Gear-box or for 25G-Only configurations or configurations with DWCXLP_XSBI_TX_CLK_PER_LN as 1, these bits are invalid and indicate 0 ■ Bits[19:15] : Read pointer of Tx Gearbox FIFO instance 3. For configurations without Gear-box or for 25G-Only configurations or configurations with DWCXLP_XSBI_TX_CLK_PER_LN as 1, these bits are invalid and indicate 0 **Exists:** DWCXLP_DBG_BUS&&(DWCXLP_R_SRDS_INTF!=3) **Synchronous To:** clk_xsbi_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_dbg_bus_clk_xsbi_tx1_o[4:0]	O	Debug Bus in clk_xsbi_tx1_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[4:0] : Read pointer of Tx Gearbox FIFO instance 1. **Exists:** DWCXLP_DBG_BUS&&DWCXLP_XSBI_TX_CLK_PER_LN **Synchronous To:** clk_xsbi_tx1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_dbg_bus_clk_xsbi_tx2_o[4:0]	O	Debug Bus in clk_xsbi_tx2_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[4:0] : Read pointer of Tx Gearbox FIFO instance 2. **Exists:** DWCXLP_DBG_BUS&&DWCXLP_XSBI_TX_CLK_PER_LN **Synchronous To:** clk_xsbi_tx2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_dbg_bus_clk_xsbi_tx3_o[4:0]	O	Debug Bus in clk_xsbi_tx3_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[4:0] : Read pointer of Tx Gearbox FIFO instance 3. **Exists:** DWCXLP_DBG_BUS&&DWCXLP_XSBI_TX_CLK_PER_LN **Synchronous To:** clk_xsbi_tx3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_dbg_bus_clk_rpcs_rx0_o[56:0]	O	Debug Bus in clk_rpcs_rx0_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[5:0] : Read pointer of Rx Reverse Gearbox FIFO instance 0. For configurations without Gear-box, these bits are invalid and indicate 0 ■ Bits[9:5] : Read pointer of Rx Reverse Gearbox FIFO instance 1. These bits are valid only for single-lane configurations with 50G/100G support. For other configurations, these bits are invalid and indicate 0 ■ Bits[14:10]: Read pointer of Rx Reverse Gearbox FIFO instance 2. These bits are valid only for single-lane configurations with 100G support. For other configurations, these bits are invalid and indicate 0 ■ Bits[19:15]: Read pointer of Rx Reverse Gearbox FIFO instance 3. These bits are valid only for single-lane configurations with 100G support. For other configurations, these bits are invalid and indicate 0 ■ Bits[29:20]: Write pointer of Clock-Compensation FIFO ■ Bit[30] : Uncorrected Code-word seen by BASE-R FEC Decoder for lane 0. For configurations without BASE-R FEC support, this bit always indicates 0. ■ Bit[31] : Uncorrected Code-word seen by RS-FEC Decoder. This signal is a pulse. For configurations without RS-FEC support this bit always indicates 0. ■ Bit[32] : Rx Lock status from RS-FEC AM lock engine for RSFEC lane 0. For configurations without RSFEC support, this bit always indicates 0. ■ Bit[33] : Rx Lock status from RS-FEC AM lock engine for RSFEC lane 1. This bit is present only single-lane configurations with 50G/100G support. For other configurations, this bit always indicates 0. ■ Bit[34] : Rx Lock status from RS-FEC AM lock engine for RSFEC lane 2. This bit is present only single-lane configurations with 100G support. For other configurations, this bit always indicates 0. ■ Bit[35] : Rx Lock status from RS-FEC AM lock engine for RSFEC lane 3. This bit is present only single-lane configurations with 100G support. For other configurations, this bit always indicates 0. ■ Bit[36] : BASE-R FEC Lock Indication for lane 0. This bit is valid only in BASE-R FEC Mode. For configurations without BASE-R FEC support, this bit always indicates 0.

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
		■ Bit[37] : Start of Code-word (SOCW) seen at RSFEC Decoder output. For configurations without RSFEC support, this bit always indicates 0. ■ Bit[38] : End of Code-word (EOCW) seen at RSFEC Decoder output. For configurations without RSFEC support, this bit always indicates 0. ■ Bits[45:39]: EEE Receive FSM. These bits are invalid and indicate 0 if the configuration has DWCXLP_EEE_EN as 0. Interpretation of these bits are as follows: - 7'h1 : ACTIVE - 7'h2 : SLEEP - 7'h4 : QUIET - 7'h8 : WAKE - 7'h10: WAKE_TIME_FAULT - 7'h20: LINK_FAIL - 7'h40: FAST_WAKE ■ Bits[55:46] : Read pointer of Clock-Compensation FIFO. These bits are valid only for configurations with DWCXLP_CLK_CMP_EN==0 AND DWCXLP_XLGMII_INTF_WIDTH=1 (64). For other configurations, these bits always indicate 0 ■ Bit[56] : Uncorrected Code-word seen by RS-FEC Decoder B (for configurations with dual-decoder). This signal is a pulse. For configurations without interleaved RSFEC support this bit always indicates 0. **Exists:** DWCXLP_DBG_BUS **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_dbg_bus_clk_xsbi_rx0_o[19:0]	O	Debug Bus in clk_xsbi_rx0_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[4:0] : Write pointer of Rx Reverse Gearbox FIFO instance 0. For configurations without Gear-box, these bits are invalid and indicate 0 ■ Bits[9:5] : Write pointer of Rx Reverse Gearbox FIFO instance 1. These bits are valid only for single-lane configurations with 50G/100G support. For other configurations, these bits are invalid and indicate 0 ■ Bits[14:10] : Write pointer of Rx Reverse Gearbox FIFO instance 2. These bits are valid only for single-lane configurations with 100G support. For other configurations, these bits are invalid and indicate 0 ■ Bits[19:15] : Write pointer of Rx Reverse Gearbox FIFO instance 3. These bits are valid only for single-lane configurations with 100G support. For other configurations, these bits are invalid and indicate 0 **Exists:** DWCXLP_DBG_BUS&&(DWCXLP_R_SRDS_INTF!=3) **Synchronous To:** clk_xsbi_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_dbg_bus_clk_rpcs_rx1_o[7:0]	O	Debug Bus in clk_rpcs_rx1_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bit[0] : Uncorrected Code-word seen by BASE-R FEC Decoder for lane 1. This bit is a pulse. For configurations without BASE-R FEC support or for single-lane configurations, this bit always indicates 0. ■ Bit[1] : Rx Lock status from RS-FEC AM lock engine for RSFEC lane 1. For configurations without RSFEC support or for 25G_Only configurations, this bit always indicates 0. ■ Bit[2] : BASE-R FEC Lock Indication for lane 1. This bit is valid only in BASE-R FEC Mode. For configurations without BASE-R FEC support or for single-lane configurations, this bit always indicates 0. ■ Bits[7:3] : Read pointer of Rx Reverse Gearbox FIFO instance 1. For configurations without Gear-box or for single-lane configurations, these bits are invalid and indicate 0 **Exists:** DWCXLP_DBG_BUS&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_rx1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_dbg_bus_clk_xsbi_rx1_o[4:0]	O	Debug Bus in clk_xsbi_rx1_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[4:0] : Write pointer of Rx Reverse Gearbox FIFO instance 1. For configurations without Gear-box or for single-lane configurations, these bits are invalid and indicate 0 **Exists:** DWCXLP_DBG_BUS&&(DWCXLP_R_SRDS_INTF!=3)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_rx1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_dbg_bus_clk_rpcs_rx2_o[7:0]	O	Debug Bus in clk_rpcs_rx2_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bit[0] : Uncorrected Code-word seen by BASE-R FEC Decoder for lane 2. This bit is a pulse. For configurations without BASE-R FEC support or for single-lane configurations, this bit always indicates 0. ■ Bit[1] : Rx Lock status from RS-FEC AM lock engine for RSFEC lane 2. For configurations without RSFEC support or for configurations without 100G support, this bit always indicates 0. ■ Bit[2] : BASE-R FEC Lock Indication for lane 2. This bit is valid only in BASE-R FEC Mode. For configurations without BASE-R FEC support or for single-lane configurations, this bit always indicates 0. ■ Bits[7:3] : Read pointer of Rx Reverse Gearbox FIFO instance 2. For configurations without Gear-box or for single-lane configurations, these bits are invalid and indicate 0 **Exists:** DWCXLP_DBG_BUS&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_rx2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_dbg_bus_clk_xsbi_rx2_o[4:0]	O	Debug Bus in clk_xsbi_rx2_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[4:0] : Write pointer of Rx Reverse Gearbox FIFO instance 2. For configurations without Gear-box or for single-lane configurations, these bits are invalid and indicate 0 **Exists:** DWCXLP_DBG_BUS&&(DWCXLP_R_SRDS_INTF!=3)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_rx2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_dbg_bus_clk_rpcs_rx3_o[7:0]	O	Debug Bus in clk_rpcs_rx3_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bit[0] : Uncorrected Code-word seen by BASE-R FEC Decoder for lane 3. This bit is a pulse. For configurations without BASE-R FEC support or for single-lane configurations, this bit always indicates 0. ■ Bit[1] : Rx Lock status from RS-FEC AM lock engine for RSFEC lane 3. For configurations without RSFEC support or for configurations without 100G support, this bit always indicates 0. ■ Bit[2] : BASE-R FEC Lock Indication for lane 3. This bit is valid only in BASE-R FEC Mode. For configurations without BASE-R FEC support or for single-lane configurations, this bit always indicates 0. ■ Bits[7:3] : Read pointer of Rx Reverse Gearbox FIFO instance 3. For configurations without Gear-box or for single-lane configurations, these bits are invalid and indicate 0 **Exists:** DWCXLP_DBG_BUS&&DWCXLP_MULTI_LANE **Synchronous To:** clk_rpcs_rx3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_dbg_bus_clk_xsbi_rx3_o[4:0]	O	Debug Bus in clk_xsbi_rx3_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[4:0] : Write pointer of Rx Reverse Gearbox FIFO instance 3. For configurations without Gear-box or for single-lane configurations, these bits are invalid and indicate 0 **Exists:** DWCXLP_DBG_BUS&&(DWCXLP_R_SRDS_INTF!=3)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_xsbi_rx3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_dbg_bus_clk_xlgmii_rx_o[9:0]	O	Debug Bus in clk_xlgmii_rx_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[9:0] : Read pointer of Clock-Compensation FIFO. These bits are valid only for configurations with DWCXLP_XLGMII_INTF_WIDTH!=1. For configurations with DWCXLP_XLGMII_INTF_WIDTH as 1, these bits always indicate 0 **Exists:** DWCXLP_DBG_BUS **Synchronous To:** clk_xlgmii_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_dbg_bus_clk_eee_o[6:0]	O	Debug Bus in clk_eee_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[6:0]: EEE Transmit FSM. These bits are invalid and indicate 0 if the configuration has DWCXLP_EEE_EN as 0. Interpretation of these bits are as follows: - 7'h1 : ACTIVE - 7'h2 : SLEEP - 7'h4 : QUIET - 7'h8 : ALERT - 7'h10: REFRESH - 7'h20: SCRAMBLER_BYPASS - 7'h40: WAKE2 **Exists:** DWCXLP_DBG_BUS&&DWCXLP_EEE_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-28 Debug Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_dbg_bus_clk_csr_o[22:0]	O	Debug Bus in clk_csr_i clock domain Various bits of this output bus provide various debug information. Details are as follows: ■ Bits[12:0]: Power-up Sequence FSM. These bits are invalid only for Synopsys Multi-protocol 12G/16G/25G/32G PHY configurations and indicate 0 for other configurations. Interpretation of these bits are as follows: - 13'h1 : WF_ACK_H0 - 13'h2 : WF_ACK_L0 - 13'h4 : WF_ACK_H1 - 13'h8 : WF_ACK_L1 - 13'h10: POWER_GOOD - 13'h20: POWER_SAVE - 13'h40: POWER_DOWN - 13'h80: WF_RX_VLD0 - 13'h100: WF_ACK_H2 - 13'h200: WF_ACK_L2 - 13'h400: WF_RX_VLD1 - 13'h800: WF_ADPT_ACK_H - 13'h1000:WF_ADPT_ACK_L ■ Bits[22:13]: Autonegotiation Arbiter FSM state when DWCXLP_CL73_ALT is 1. These bits are invalid and indicate 0 if DWCXLP_CL73_ALT is 0. Interpretation of these bits are as follows: - 10'h1 : AN_ENABLE - 10'h2 : TRANSMIT_DISABLE - 10'h4 : ABILITY_DETECT - 10'h8 : ACKNOWLEDGE_DETECT - 10'h10 : COMPLETE_ACKNOWLEDGE - 10'h20 : NEXT_PAGE_WAIT - 10'h40 : LINK_STATUS_CHECK - 10'h80 : PARALLEL_DETECT_FAULT - 10'h100: AN_GOOD_CHECK - 10'h200: AN_GOOD **Exists:** DWCXLP_DBG_BUS **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.29 Synopsys Enterprise PHY-Specific Tx Status and Control Interface Signals


- xlgpcs_tx_ropll_wrd_clk_en_o

Table 6-29 Synopsys Enterprise PHY-Specific Tx Status and Control Interface Signals

Port Name	I/O	Description
xlgpcs_tx_ropll_wrd_clk_en_o[(DWCXLP_NUM_LANES-1):0]	O	TX ROPLL word clock output enable **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.30 Synopsys PHY-Specific Control and Status Interface Signals

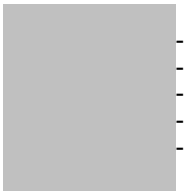


Table 6-30 Synopsys PHY-Specific Control and Status Interface Signals

Port Name	I/O	Description
xlgpcs_lane0_link_num_o[3:0]	O	Per-lane context link number. Ensures that lanes with the same lane0_link_num, change rate, simultaneously. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_lane1_link_num_o[3:0]	O	Per-lane context link number. Ensures that lanes with the same lane1_link_num, change rate, simultaneously. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_lane2_link_num_o[3:0]	O	Per-lane context link number. Ensures that lanes with the same lane2_link_num, change rate, simultaneously. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_lane3_link_num_o[3:0]	O	Per-lane context link number. Ensures that lanes with the same lane3_link_num, change rate, simultaneously. **Exists:** (DWCXLP_E32)&&DWCXLP_MULTI_LANE **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-30 Synopsys PHY-Specific Control and Status Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_lane_cntx_en_o[(DWCXLP_NUM_LANES-1):0]	O	Per-lane context enable. This signal is changed only during phy_reset. Here, tied off to low using programmable register. **Exists:** (DWCXLP_E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.31 Timestamping Interface Signals

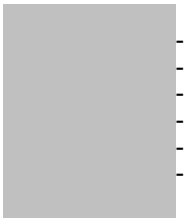


Table 6-31 Timestamping Interface Signals

Port Name	I/O	Description
xlgpcs_tx_ddmp_o	O	Transmit DDMP pulse This is a single clock-cycle pulse in clk_rpcs_tx_i that gets generated whenever PCS transmitter detects a Data Delay Measurement point (DDMP) in the data-path, whenever TS_MODE bit of VR_PCS_DIG_CTRL3 register is set to 1. When TS_MODE bit is 0, this output port is invalid and it always indicates 0. For packets with error, this pulse may or may not be generated. In case this pulse is generated, MAC shall ignore it **Exists:** DWCXLP_TS_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xlgpcs_rx_ddmp_o	O	Receive DDMP pulse This is a single clock-cycle pulse in clk_rpcs_rx0_i that gets generated whenever PCS receiver detects a Data Delay Measurement point (DDMP) in the data-path, whenever TS_MODE bit of VR_PCS_DIG_CTRL3 register is set to 1. When TS_MODE bit is 0, this output port is invalid and it always indicates 0. For packets received with error, this pulse may or may not be generated. In case this pulse is generated, MAC shall ignore it **Exists:** DWCXLP_TS_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Table 6-31 Timestamping Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_tx_ddmp_tgl_o	O	Transmit DDMP Toggle This signal toggles whenever PCS transmitter detects a Data Delay Measurement point (DDMP) in the data-path, whenever TS_MODE bit of VR_PCS_DIG_CTRL3 register is set to 1. When TS_MODE bit is 0, this output port is invalid and it always indicates 0. User can make use of this signal (instead of the xlgpcs_tx_ddmp_o pulse) to generate a pulse in the clock-domain required by your MAC. **Exists:** DWCXLP_TS_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_rx_ddmp_tgl_o	O	Receive DDMP Toggle This signal toggles whenever PCS receiver detects a Data Delay Measurement point (DDMP) in the data-path, whenever TS_MODE bit of VR_PCS_DIG_CTRL3 register is set to 1. When TS_MODE bit is 0, this output port is invalid and it always indicates 0. User can make use of this signal (instead of the xlgpcs_rx_ddmp_o pulse) to generate a pulse in the clock-domain required by your MAC. **Exists:** DWCXLP_TS_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xlgpcs_tx_num_bit_chg_o[15:0]	O	Number of Bit Change in Tx Delay for each DDMP This signal is an indication of dynamic delay change and it indicates how many bits of delay change has occurred inside the DWC_xlgpcs transmitter for each DDMP that is transmitted out. The delay is indicated in two's compliment format. The value on this port is valid only when xlgpcs_tx_ddmp_o pulse is high. **Exists:** DWCXLP_TS_EN **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Table 6-31 Timestamping Interface Signals (Continued)

Port Name	I/O	Description
xlgpcs_rx_num_bit_chg_o[15:0]	O	Number of Bit Change in Rx Delay for each DDMP This signal is an indication of dynamic delay change and it indicates how many bits of delay change has occurred inside the DWC_xlgpcs receiver for each DDMP that is received from PHY. The delay is indicated in two's compliment format. The value on this port is valid only when xlgpcs_rx_ddmp_o pulse is high. **Exists:** DWCXLP_TS_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.32 Debug InterfaceRS-FEC AM Lock Indication. This signal indicates the Alignment Marker (Code-word marker) lock status when operating in RS-FEC Mode. Signals



Table 6-32 Debug InterfaceRS-FEC AM Lock Indication. This signal indicates the Alignment Marker (Code-word marker) lock status when operating in RS-FEC Mode. Signals

Port Name	I/O	Description
xlgpcs_baserfec_lock_o	O	BASE-R FEC Lock Indication. This signal indicates if the BASE-R FEC receiver has identified FEC block boundary and has locked on the incoming Rx data as defined by Clause 74 of IEEE802.3. This signal is valid only in BASE-R FEC mode. **Exists:** SNPS_RSVDPARAM_15&&DWCXLP_FEC_EN **Synchronous To:** clk_rpcs_rx0_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High

7
Register Descriptions

This chapter details all possible registers in the IP. They are arranged hierarchically into maps and blocks (banks).Your actual configuration might not contain all of these registers.

Attention: For configurable IP titles, do not use this document to determine the exact attributes of your register map. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the register attributes for your actual configuration at workspace/report/ComponentRegisters.html or workspace/report/ComponentRegisters.xml after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the registers that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the Offset and Memory Access values might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>)** that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

Exists Expressions

These expressions indicate the combination of configuration parameters required for a register, field, or block to exist in the memory map. The expression is only valid in the local context and does not indicate the conditions for existence of the parent. For example, the expression for a bit field in a register assumes that the register exists and does not include the conditions for existence of the register.

Offset

The term *Offset* is synonymous with *Address.*

Memory Access Attributes

The Memory Access attribute is defined as <ReadBehavior>/<WriteBehavior> which are defined in the following table.

Table 7-1 Possible Read and Write Behaviors

Read (or Write) Behavior	Description
RC	A read clears this register field.
RS	A read sets this register field.
RM	A read modifies this register field in a manner not covered above.
Wo	You can only write once to this register field.
W1C	A write of 1 clears this register field.
W1S	A write of 1 sets this register field.
W1T	A write of 1 toggles this register field.
W0C	A write of 0 clears this register field.
W0S	A write of 0 sets this register field.
W0T	A write of 0 toggles this register field.
WC	Any write clears this register field.
WS	Any write sets this register field.
WM	A write modifies this register field in a manner not covered above.
no Read Behavior attribute	You cannot read this register. It is Write-Only.
no Write Behavior attribute	You cannot write to this register. It is Read-Only.

Table 7-2 Memory Access Examples

Memory Access	Description
R	Read-only register field.
W	Write-only register field.
R/W	Read/write register field.
R/W1C	You can read this register field. Writing 1 clears it.
RC/W1C	Reading this register field clears it. Writing 1 clears it.
R/Wo	You can read this register field. You can only write to it once.

Special Optional Attributes

Some register fields might use the following optional attributes.

Table 7-3 Optional Attributes

Attribute	Description
Volatile	As defined by the IP-XACT specification. If true, indicates in the case of a write followed by read, or in the case of two consecutive reads, there is no guarantee as to what is returned by the read on the second transaction or that this return value is consistent with the write or read of the first transaction. The element implies there is some additional mechanism by which this field can acquire new values other than by reads/writes/resets and other access methods known to IP-XACT. For example, when the core updates the register field contents.
Testable	As defined by the IP-XACT specification. Possible values are unconstrained, untestable, readOnly, writeAsRead, restore. Untestable means that this field is untestable by a simple automated register test. For example, the read-write access of the register is controlled by a pin or another register. readOnly means that you should not write to this register; only read from it. This might apply for a register that modifies the contents of another register.
Reset Mask	As defined by the IP-XACT specification. Indicates that this register field has an unknown reset value. For example, the reset value is set by another register or an input pin; or the register is implemented using RAM.
* Varies	Indicates that the memory access (or reset) attribute (read, write behavior) is not fixed. For example, the read-write access of the register is controlled by a pin or another register. Or when the access depends on some configuration parameter; in this case the post-configuration report in coreConsultant gives the actual access value.

Component Banks/Blocks

The following table shows the address blocks for each memory map. Follow the link for an address block to see a table of its registers.

Table 7-4 Address Banks/Blocks for Memory Map: DWC_xlgpcs_map

Address Block	Description
PMA_MMD on page 475	The PMA MMD Standard (SR) and Vendor Specific Register (VR) address block **Exists:** (DWCXLP_PMA_MMD)
PCS_MMD on page 1018	The PCS MMD Standard (SR) & Vendor Specific (VR) Registers address block **Exists:** Always

Table 7-4 Address Banks/Blocks for Memory Map: DWC_xlgpcs_map (Continued)

Address Block	Description
AN_MMD on page 1191	The Auto-Negotiation (AN) MMD Standard (SR) & Vendor Specific (VR) address block **Exists:** (DWCXLP_CL73_AN)
VS_MMD1 on page 1260	The Vendor Specific MMD1 Registers address block **Exists:** Always

7.1 DWC_xlgpcs_map/PMA_MMD Registers

The PMA MMD Standard (SR) and Vendor Specific Register (VR) address block. Follow the link for the register to see a detailed description of the register.

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD

Register	Offset	Description
SR_PMA_CTRL1 on page 492	* Varies	SR PMA MMD Control1 Register
SR_PMA_STATUS1 on page 496	* Varies	SR PMA MMD Status1 Register
SR_PMA_DEV_ID_1 on page 498	* Varies	SR PMA MMD Device Identifier Register 1
SR_PMA_DEV_ID_2 on page 499	* Varies	SR PMA MMD Device Identifier Register 2
SR_PMA_SPD_ABL on page 501	* Varies	SR PMA MMD Speed Ability Register
SR_PMA_DEV_PKG1 on page 503	* Varies	SR PMA MMD Devices in Package Register 1
SR_PMA_DEV_PKG2 on page 506	* Varies	SR PMA MMD Devices in Package Register 2
SR_PMA_CTRL2 on page 507	* Varies	SR PMA MMD Control2 Register
SR_PMA_STATUS2 on page 509	* Varies	SR PMA MMD Status2 Register
SR_PMA_TX_DIS on page 512	* Varies	SR PMA MMD Transmit Disable Register
SR_PMA_RX_SIG_DET on page 514	* Varies	SR PMA MMD Receive Signal Detect Register
SR_PMA_EXT_ABL on page 516	* Varies	SR PMA or PMD Extended Ability Register
SR_PMA_EXT_ABL1 on page 519	* Varies	SR PMA or PMD Extended Ability Register 1.
SR_PMA_PKG1 on page 522	* Varies	SR PMA MMD Package Identifier Register 1
SR_PMA_PKG2 on page 523	* Varies	SR PMA MMD Package Identifier Register 2
SR_EEE_ABL on page 525	* Varies	SR PMA MMD EEE capability Register
SR_PMA_EXT_ABL2 on page 527	* Varies	SR PMA or PMD Extended Ability Register 2
SR_PMA_2PT5G_5G_EXT_ABL on page 529	* Varies	SR PMA MMD 2.5G/5G Extended Ability Register
SR_PMA_40G_100G_EXT_ABL2 on page 531	* Varies	SR PMA or PMD 40G/100G Extended Ability Register 2.
SR_PMA_KR_PMD_CTRL on page 532	* Varies	SR PMA MMD BASE-KR PMD Control Register. This register is present only if Link Training Support...
SR_PMA_KR_PMD_STS on page 534	* Varies	SR PMA MMD BASE-KR Status Register This register is present only if CL72 Training Support is enabled...
SR_PMA_KR_LP_CEU on page 539	* Varies	SR PMA MMD BASE-KR LP Coefficient Update Register This register is present when the Link Training...

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
SR_PMA_KR_LP_CESTS on page 542	* Varies	SR PMA MMD BASE-KR LP Coefficient Status Register This register is present when the Link Training...
SR_PMA_KR_LD_CEU on page 544	* Varies	SR PMA MMD BASE-KR LD Coefficient Update Register This register is present when the Link Training...
SR_PMA_KR_LD_CESTS on page 547	* Varies	SR PMA MMD BASE-KR LD Coefficient Status Register This register is present only for the PMA...
SR_PMA_KR_FEC_ABL on page 549	* Varies	SR PMA MMD BASE-KR FEC Ability Register This register is present only if DWCXLP_FEC_EN =...
SR_PMA_KR_FEC_CTRL on page 550	* Varies	SR PMA MMD BASE-KR FEC Control Register This register is present only if DWCXLP_FEC_EN = Enabled....
SR_PMA_KR_FEC_CORR_BLK1 on page 552	* Varies	SR PMA MMD BASE-KR Lane0 FEC Correctable Error Block Counter 1 Register. This register is present...
SR_PMA_KR_FEC_CORR_BLK2 on page 553	* Varies	SR PMA MMD KR Lane0 FEC Correctable Error Block Counter 2 Register This register is present only...
SR_PMA_KR_FEC_UCORR_BLK1 on page 554	* Varies	SR PMA MMD BASE-KR Lane0 FEC Uncorrectable Error Block Counter 1 Register This register is present...
SR_PMA_KR_FEC_UCORR_BLK2 on page 555	* Varies	SR PMA MMD Lane0 FEC Uncorrectable Error Block Counter 2 Register This register is present only...
SR_PMA_RS_FEC_CTRL on page 556	* Varies	SR PMA MMD RS-FEC control register. This register provides control to bypass RS-FEC Error Indication...
SR_PMA_RS_FEC_STATUS on page 559	* Varies	SR PMA MMD RS-FEC status register. This register provides status of Alignment Marker Lock, bypass...
SR_PMA_RS_FEC_CORR_CW_CNTL on page 563	* Varies	SR PMA MMD RS-FEC corrected codewords lower (RO, COR Type). This register holds the lower 16 bits...
SR_PMA_RS_FEC_CORR_CW_CNTH on page 564	* Varies	SR PMA MMD RS-FEC corrected codewords upper (RO, COR Type). This register holds the upper 16 bits...
SR_PMA_RS_FEC_UCOR_CW_CNTL on page 565	* Varies	SR PMA MMD RS-FEC uncorrected codewords lower (RO, COR Type). This register holds the lower 16...
SR_PMA_RS_FEC_UCOR_CW_CNTH on page 566	* Varies	SR PMA MMD RS-FEC uncorrected codewords upper (RO, COR Type). This register holds the upper 16...
SR_PMA_RS_FEC_LANE_MAP on page 567	* Varies	SR PMA MMD RS-FEC lane mapping register. When the RS-FEC detects and locks the RS-FEC for PMA service...
SR_PMA_RS_FEC_CW_CNTL on page 569	* Varies	SR PMA MMD RS-FEC CodeWord Count Low Register
SR_PMA_RS_FEC_CW_CNTM on page 570	* Varies	SR PMA MMD RS-FEC CodeWord Count Middle Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
SR_PMA_RS_FEC_CW_CNTH on page 571	* Varies	SR PMA MMD RS-FEC CodeWord Count High Register
SR_PMA_RS_FEC_ERR_CNTL_L0 on page 572	* Varies	SR PMA MMD RS-FEC symbol error counter register (RO, COR Type). This register holds the lower 16...
SR_PMA_RS_FEC_ERR_CNTH_L0 on page 573	* Varies	SR PMA MMD RS-FEC symbol error counter register (RO, COR Type). This register holds the upper 16...
SR_PMA_RS_FEC_ERR_CNTL_L1 on page 574	* Varies	SR PMA MMD RS-FEC symbol error counter register (RO, COR Type). This Register holds the lower 16...
SR_PMA_RS_FEC_ERR_CNTH_L1 on page 575	* Varies	SR PMA MMD RS-FEC symbol error counter register (RO, COR Type). This register holds the upper 16...
SR_PMA_RS_FEC_ERR_CNTL_L2 on page 576	* Varies	SR PMA MMD RS-FEC symbol error counter register (RO, COR Type). This register holds the lower 16...
SR_PMA_RS_FEC_ERR_CNTH_L2 on page 577	* Varies	SR PMA MMD RS-FEC symbol error counter register (RO, COR Type). This register holds the upper 16...
SR_PMA_RS_FEC_ERR_CNTL_L3 on page 578	* Varies	SR PMA MMD RS-FEC symbol error counter register (RO, COR Type). This register holds the lower 16...
SR_PMA_RS_FEC_ERR_CNTH_L3 on page 579	* Varies	SR PMA MMD RS-FEC symbol error counter register (RO, COR Type). This register holds the upper 16...
SR_PMA_KR_FEC_CORR_BLK1_LANE (i) (for i = 0; i <= 3) on page 580	(2*i)+0x1012c	SR PMA MMD Multi lane FEC Correctable Error Block Counter 1 Register -Lane i This register is...
SR_PMA_KR_FEC_CORR_BLK2_LANE (i) (for i = 0; i <= 3) on page 581	(2*i)+0x1012d	SR PMA MMD Multi lane FEC Correctable Error Block Counter 2 Register -Lane i This register is...
SR_PMA_RS_FEC_CW_ERR_BIN1L on page 582	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin1 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN1H on page 583	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin1 High Register
SR_PMA_RS_FEC_CW_ERR_BIN2L on page 584	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin2 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN2H on page 585	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin2 High Register
SR_PMA_RS_FEC_CW_ERR_BIN3L on page 586	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin3 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN3H on page 587	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin3 High Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
SR_PMA_RS_FEC_CW_ERR_BIN4L on page 588	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin4 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN4H on page 589	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin4 High Register
SR_PMA_RS_FEC_CW_ERR_BIN5L on page 590	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin5 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN5H on page 591	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin5 High Register
SR_PMA_RS_FEC_CW_ERR_BIN6L on page 592	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin6 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN6H on page 593	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin6 High Register
SR_PMA_RS_FEC_CW_ERR_BIN7L on page 594	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin7 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN7H on page 595	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin7 High Register
SR_PMA_RS_FEC_CW_ERR_BIN8L on page 596	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin8 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN8H on page 597	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin8 High Register
SR_PMA_RS_FEC_CW_ERR_BIN9L on page 598	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin9 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN9H on page 599	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin9 High Register
SR_PMA_RS_FEC_CW_ERR_BIN10L on page 600	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin10 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN10H on page 601	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin10 High Register
SR_PMA_RS_FEC_CW_ERR_BIN11L on page 602	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin11 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN11H on page 603	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin11 High Register
SR_PMA_RS_FEC_CW_ERR_BIN12L on page 604	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin12 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN12H on page 605	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin12 High Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
SR_PMA_RS_FEC_CW_ERR_BIN13L on page 606	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin13 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN13H on page 607	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin13 High Register
SR_PMA_RS_FEC_CW_ERR_BIN14L on page 608	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin14 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN14H on page 609	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin14 High Register
SR_PMA_RS_FEC_CW_ERR_BIN15L on page 610	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin15 Low Register
SR_PMA_RS_FEC_CW_ERR_BIN15H on page 611	* Varies	SR PMA MMD RS-FEC CodeWord Error Bin15 High Register
SR_PMA_RS_FEC_DEGRADED_SER_ACT_THLD_0 on page 612	* Varies	SR PMA MMD RS-FEC Degraded SER Activate Threshold 0 Register
SR_PMA_RS_FEC_DEGRADED_SER_ACT_THLD_1 on page 613	* Varies	SR PMA MMD RS-FEC Degraded SER Activate Threshold 1 Register
SR_PMA_RS_FEC_DEGRADED_SER_DEACT_THLD_0 on page 614	* Varies	SR PMA MMD RS-FEC Degraded SER Deactivate Threshold 0 Register
SR_PMA_RS_FEC_DEGRADED_SER_DEACT_THLD_1 on page 615	* Varies	SR PMA MMD RS-FEC Degraded SER Deactivate Threshold 1 Register
SR_PMA_RS_FEC_DEGRADED_SER_INTRVL_0 on page 616	* Varies	SR PMA MMD RS-FEC Degraded SER Interval 0 Register
SR_PMA_RS_FEC_DEGRADED_SER_INTRVL_1 on page 617	* Varies	SR PMA MMD RS-FEC Degraded SER Interval 1 Register
SR_PMA_KR_FEC_UCORR_BLK1_LANE (i) (for i = 0; i <= 3) on page 618	(2*i)+0x102bc	SR PMA MMD Multi lane FEC Uncorrectable Error Block Counter 1 Register -Lane i This register is...
SR_PMA_KR_FEC_UCORR_BLK2_LANE (i) (for i = 0; i <= 3) on page 619	(2*i)+0x102bd	SR PMA MMD Multi lane FEC Uncorrectable Error Block Counter 2 Register -Lane i This register is...
SR_PMA_MLN_KR_LP_CEU_LANE (i) (for i = 0; i <= 3) on page 620	i+0x1044c	SR PMA MMD BASE-KR LP Coefficient Update Register-Lane i
SR_PMA_MLN_KR_LP_CESTS_LANE (i) (for i = 0; i <= 3) on page 623	i+0x104b0	SR PMA MMD BASE-KR LP Coefficient Status Register-Lane i This register is present when the Link...
SR_PMA_MLN_KR_LD_CEU_LANE (i) (for i = 0; i <= 3) on page 625	i+0x10514	SR PMA MMD BASE-KR LD Coefficient Update Register-Lane i This register is present when the Link...

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
SR_PMA_MLN_KR_LD_CESTS_LANE (i) (for i = 0; i <= 3) on page 628	i+0x10578	SR PMA MMD BASE-KR LD Coefficient Status Register- Lane i This register is present only for the...
SR_PMA_LNK_TRN_PAT on page 630	* Varies	SR PMA MMD Link Training Pattern Register This register is present only if Link Training is Enabled....
SR_PMA_LNK_TRN_PAT_LN1 on page 631	* Varies	SR PMA MMD Lane-1 Link Training Pattern Register This register is present only if DWCXLP_LNK_TRAIN_EN...
SR_PMA_LNK_TRN_PAT_LN2 on page 632	* Varies	SR PMA MMD Lane-2 Link Training Pattern Register. This register is present only if DWCXLP_LNK_TRAIN_EN...
SR_PMA_LNK_TRN_PAT_LN3 on page 633	* Varies	SR PMA MMD Lane-3 Link Training Pattern Register This register is present only if DWCXLP_LNK_TRAIN_EN...
SR_PMA_KR_TST_PAT_ABL on page 634	* Varies	SR PMA MMD 25GBASE-CR/KR, 40G/100GBASE-CR4/KR4 Test Pattern Ability Register
SR_PMA_KR_TST_PAT_CTRL on page 636	* Varies	SR PMA MMD 25GBASE-CR/KR, 40G/100GBASE-CG4/KR4 Test Pattern Control Register
SR_PMA_KR_SQUARE_WAVE_CTRL on page 638	* Varies	SR PMA MMD 25G/40G/100G Test Pattern Control Register
SR_PMA_KR_PRBS_TX_ERR_CNTR_LANE (i) (for i = 0; i <= 3) on page 640	i+0x10640	SR PMA MMD PRBS Transmit Error Counter Register- Lane i (RO, COR Type). Note: In single-lane...
SR_PMA_KR_PRBS_RX_ERR_CNTR_LANE (i) (for i = 0; i <= 3) on page 641	i+0x106a4	SR PMA MMD 40G/100GBASE-CR4/KR4 PRBS Receive Error Counter Register - Lane i (RO, COR Type)(RO,...
VR_PMA_DIG_CTRL1 on page 642	* Varies	VR PMA MMD Digital Control1 Register
VR_PMA_RSFEC_ERR_WIN on page 646	* Varies	VR PMA MMD RS-FEC Hi-SER Error Window Register
VR_PMA_RSFEC_ERR_THLD on page 647	* Varies	VR PMA MMD RS-FEC High-SER Error Threshold Register
VR_PMA_KRTR_PRBS_CTRL0 on page 648	* Varies	VR PMA MMD KRTR PRBS Control0 Register
VR_PMA_KRTR_PRBS_CTRL1 on page 650	* Varies	VR PMA MMD KRTR PRBS Control1 Register
VR_PMA_KRTR_PRBS_CTRL2 on page 651	* Varies	VR PMA MMD KRTR PRBS Control2 Register
VR_PMA_KRTR_TIMER_CTRL0 on page 652	* Varies	VR PMA MMD KRTR Timer Control 0 Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_KRTR_TIMER_CTRL1 on page 653	* Varies	VR PMA MMD KRTR Timer Control 1 Register
VR_PMA_KRTR_TIMER_CTRL2 on page 654	* Varies	VR PMA MMD KRTR Timer Control 2 Register
VR_PMA_DIG_STS on page 656	* Varies	VR PMA MMD Digital Status Register
VR_PMA_GPIO on page 658	* Varies	VR PMA MMD GPIO Register
VR_PMA_RX_LSTS on page 659	* Varies	VR PMA PHY Rx Lane Status Register
VR_PMA_MP_25G_16G_TX_GENCTRL0 on page 661	* Varies	VR PMA Synopsys Multi-protocol PHY Tx General Register 0
VR_PMA_MP_25G_16G_TX_GENCTRL1 on page 664	* Varies	VR PMA Synopsys Multi-protocol PHY Tx General Register 1
VR_PMA_MP_25G_16G_TX_GENCTRL2 on page 667	* Varies	VR PMA Synopsys Multi-protocol PHY Tx General Register 2
VR_PMA_MP_25G_16G_TX_BOOST_CTRL on page 669	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Tx Boost Control Register
VR_PMA_MP_25G_16G_TX_RATE_CTRL on page 671	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Tx Rate Control Register
VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL on page 673	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Power State Register
VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN0 on page 675	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 0 -Lane 0 Register
VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN0 on page 677	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 1 -Lane 0 Register
VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN1 on page 680	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 0 -Lane 1 Register
VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN1 on page 682	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 1 -Lane 1 Register
VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN2 on page 685	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 0 -Lane 2 Register
VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN2 on page 687	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 1 -Lane 2 Register
VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN3 on page 690	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 0 -Lane 3 Register
VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN3 on page 692	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 1 -Lane 3 Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_MP_25G_16G_TX_EQ_CTRL2 on page 695	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 2 Register
VR_PMA_MP_25G_16G_TX_EQ_CTRL3 on page 697	* Varies	VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 3 Register
VR_PMA_MP_25G_16G_TX_STS on page 699	* Varies	VR PMA Synopsys Multi-protocol 32G/25G/16G PHY Tx Status Register
VR_PMA_MP_25G_16G_TX_GEN_CTRL3 on page 701	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY TX General Control 3 Register
VR_PMA_MP_25G_16G_TX_GEN_CTRL4 on page 702	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY TX General Control 4 Register
VR_PMA_MP_25G_16G_TX_MISC_CTRL0 on page 703	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY TX Miscellaneous Control 0 Register
VR_PMA_MP_25G_16G_TX_MISC_CTRL1 on page 704	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY TX Miscellaneous Control 1 Register
VR_PMA_MP_25G_TX_GEN_CTRL5 on page 705	* Varies	VR PMA Synopsys Multi-protocol 25G PHY TX General Control 5 Register
VR_PMA_MP_32G_TX_GEN_CTRL5 on page 707	* Varies	VR PMA Synopsys Multi-protocol 32G PHY TX General Control 5 Register
VR_PMA_MP_25G_TX_DCC_CTRL on page 709	* Varies	VR PMA Synopsys Multi-protocol 25G PHY TX DCC Control Register
VR_PMA_MP_25G_TX_TERM_CTRL on page 711	* Varies	VR PMA Synopsys Multi-protocol 25G PHY TX Terminate Control Register
VR_PMA_MP_25G_TX_WIDTH_CTRL on page 713	* Varies	VR PMA Synopsys Multi-protocol 32G/25G PHY TX Width Control Register
VR_PMA_MP_16G_TX_WIDTH_CTRL on page 716	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY TX Width Control Register
VR_PMA_MP_25G_RX_VCO_CFG_0 on page 718	* Varies	VR PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Control0 Register
VR_PMA_MP_25G_RX_VCO_CFG_1 on page 719	* Varies	VR PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Control1 Register
VR_PMA_MP_25G_RX_VCO_CFG_2 on page 720	* Varies	VR PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Control2 Register
VR_PMA_MP_25G_RX_VCO_CFG_3 on page 721	* Varies	VR PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Control3 Register
VR_PMA_MP_25G_RX_WIDTH_CTRL on page 722	* Varies	VR PMA Synopsys Multi-protocol 32G/25G PHY RX Width Control Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_MP_16G_RX_WIDTH_CTRL on page 725	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY RX Width Control Register
VR_PMA_MP_25G_16G_12G_RX_GEN CTRL0 on page 727	* Varies	VR PMA Synopsys Multi-protocol PHY Rx General Control 0 Register
VR_PMA_MP_25G_16G_RX_GENCTRL 1 on page 730	* Varies	VR PMA Synopsys Multi-protocol PHY Rx General Control 1 Register
VR_PMA_MP_25G_16G_RX_GENCTRL 2 on page 733	* Varies	VR PMA Synopsys Multi-protocol PHY Rx General Control 2 Register
VR_PMA_MP_25G_RX_GENCTRL3 on page 735	* Varies	VR PMA Synopsys Multi-protocol 25G PHY Rx General Control 3 Register
VR_PMA_MP_16G_RX_GENCTRL3 on page 738	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY Rx General Control 3 Register
VR_PMA_MP_25G_RX_RATE_CTRL on page 741	* Varies	VR PMA Synopsys Multi-protocol 25G PHY Rx Rate Control Register
VR_PMA_MP_16G_RX_RATE_CTRL on page 744	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY Rx Rate Control Register
VR_PMA_MP_25G_16G_RX_POWER_S TATE_CTRL on page 746	* Varies	VR PMA Synopsys Multi-protocol PHY Rx Power State Register
VR_PMA_MP_25G_RX_GENCTRL5 on page 749	* Varies	VR PMA Synopsys Multi-protocol 25G PHY Rx General Control 5 Register
VR_PMA_MP_16G_RX_CDR_CTRL on page 751	* Varies	VR PMA Synopsys Multi-protocol 16G PHY Rx CDR Control Register
VR_PMA_MP_12G_RX_GENCTRL5 on page 754	* Varies	VR PMA Synopsys Multi-protocol 12G PHY Rx General Control Register 5
VR_PMA_MP_25G_16G_RX_ATTN_CTR L on page 759	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Rx Attenuation Control Register
VR_PMA_MP_25G_16G_RX_EQ_CTRL0 on page 761	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY Rx Equalization Control 0 Register
VR_PMA_MP_25G_16G_RX_EQ_CTRL1 on page 763	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Rx Equalization Control 1 Register
VR_PMA_MP_25G_16G_RX_EQ_CTRL2 on page 765	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Rx Equalization Control 2 Register
VR_PMA_MP_25G_16G_RX_EQ_CTRL3 on page 767	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Rx Equalization Control 3 Register
VR_PMA_MP_25G_16G_RX_EQ_CTRL4 on page 769	* Varies	VR PMA Synopsys Multi-protocol PHY Rx Equalization Control 4 Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_MP_25G_RX_AFE_RATE_CTRL on page 776	* Varies	VR PMA Synopsys Multi-protocol 25G PHY Rx AFE Rate Control Register
VR_PMA_MP_25G_16G_DFE_TAP_CTRL0 on page 778	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY DFE Tap Control 0 Register
VR_PMA_MP_25G_16G_DFE_TAP_CTRL1 on page 779	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY DFE Tap Control 1 Register
VR_PMA_MP_25G_16G_RX_STS on page 780	* Varies	VR PMA Synopsys Multi-protocol PHY Rx Status Register
VR_PMA_MP_25G_16G_RX_PPM_STS0 on page 783	* Varies	VR PMA Synopsys Multi-protocol 32G/25G/16G PHY Rx PPM Status 0 Register
VR_PMA_MP_25G_16G_RX_PPM_STS1 on page 785	* Varies	VR PMA Synopsys Multi-protocol 32G/25G/16G PHY Rx PPM Status 1 Register
VR_PMA_MP_25G_RX_RECAL_CTRL on page 787	* Varies	VR PMA Synopsys Multi-protocol 32G/25G PHY Rx Re-calibration Control Register
VR_PMA_MP_25G_16G_ADPT_CTRL0 on page 789	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY Rx Adaptation Control 0 Register
VR_PMA_MP_25G_16G_RX_PPM_CTRL0 on page 792	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY Rx PPM Control 0 Register
VR_PMA_MP_25G_16G_RX_PPM_CTRL1 on page 794	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY Rx PPM Control 1 Register
VR_PMA_MP_25G_RX_GEN_CTRL4 on page 796	* Varies	VR PMA Synopsys Multi-protocol PHY Rx General Control 4 Register
VR_PMA_MP_25G_16G_RX_MISC_CTRL0 on page 798	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY RX Miscellaneous Control 0 Register
VR_PMA_MP_25G_16G_RX_MISC_CTRL1 on page 799	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY RX Miscellaneous Control 1 Register
VR_PMA_MP_25G_16G_RX_IQ_CTRL0 on page 800	* Varies	VR PMA Synopsys Multi-protocol 32G/25G/16G PHY RX IQ Control 0 Register
VR_PMA_MP_25G_16G_RX_IQ_CTRL1 on page 802	* Varies	VR PMA Synopsys Multi-protocol 32G/25G/16G PHY RX IQ Control 1 Register
VR_PMA_MP_25G_16G_RX_IQ_CTRL2 on page 804	* Varies	VR PMA Synopsys Multi-protocol 32G/25G/16G PHY RX IQ Control 2 Register
VR_PMA_MP_25G_16G_RX_IQ_CTRL3 on page 806	* Varies	VR PMA Synopsys Multi-protocol 32G/25G/16G PHY RX IQ Control 3 Register
VR_PMA_MP_25G_RX_DCC_CTRL on page 808	* Varies	VR PMA Synopsys Multi-protocol 25G PHY RX DCC Control Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_MP_25G_RX_TERM_CTRL on page 810	* Varies	VR PMA Synopsys Multi-protocol 25G PHY RX Terminate Control Register
VR_PMA_MP_25G_16G_MPLL_CMN_CTRL on page 812	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY MPLL Common Control Register
VR_PMA_MP_32G_MPLL_CMN_CTRL on page 814	* Varies	VR PMA Synopsys Multi-protocol 32G PHY MPLL Common Control Register
VR_PMA_MP_25G_MPLLA_CTRL0 on page 816	* Varies	VR PMA Synopsys Multi-protocol 32G/25G PHY MPLLA Control 0 Register
VR_PMA_MP_16G_MPLLA_CTRL0 on page 818	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLA Control 0 Register
VR_PMA_MP_32G_25G_MPLLA_CTRL1 on page 820	* Varies	VR PMA Synopsys Multi-protocol 32G/25G PHY MPLLA Control 1 Register
VR_PMA_MP_16G_MPLLA_CTRL1 on page 822	* Varies	VR PMA Synopsys Multi-protocol 16G PHY MPLLA Control 1 Register
VR_PMA_MP_12G_MPLLA_CTRL1 on page 824	* Varies	VR PMA Synopsys Multi-Protocol 12G PHY MPLLA Control 1 Register
VR_PMA_MP_16G_MPLLA_CTRL2 on page 826	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLA Control 2 Register
VR_PMA_MP_25G_MPLLA_CTRL2 on page 829	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 2 Register
VR_PMA_MP_25G_MPLLB_CTRL0 on page 832	* Varies	VR PMA Synopsys Multi-protocol 25G/32G PHY MPLLB Control 0 Register
VR_PMA_MP_16G_MPLLB_CTRL0 on page 834	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLB Control 0 Register
VR_PMA_MP_32G_25G_MPLLB_CTRL1 on page 836	* Varies	VR PMA Synopsys Multi-protocol 32G/25G PHY MPLLB Control 1 Register
VR_PMA_MP_16G_MPLLB_CTRL1 on page 838	* Varies	VR PMA Synopsys Multi-protocol 16G PHY MPLLB Control 1 Register
VR_PMA_MP_12G_MPLLB_CTRL1 on page 840	* Varies	VR PMA Synopsys Multi-Protocol 12G PHY MPLLB Control 1 Register
VR_PMA_MP_16G_MPLLB_CTRL2 on page 842	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLB Control 2 Register
VR_PMA_MP_25G_MPLLB_CTRL2 on page 845	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 2 Register
VR_PMA_MP_16G_MPLLA_CTRL3 on page 848	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLA Control 3 Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_MP_25G_MPLLA_CTRL3 on page 849	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 3 Register
VR_PMA_MP_16G_MPLLB_CTRL3 on page 850	* Varies	VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLB Control 3 Register
VR_PMA_MP_25G_MPLLB_CTRL3 on page 851	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 3 Register
VR_PMA_MP_25G_MPLLA_CTRL4 on page 852	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 4 Register
VR_PMA_MP_16G_MPLLA_CTRL4 on page 853	* Varies	VR PMA Synopsys Multi-protocol 16G PHY MPLLA Control 4 Register
VR_PMA_MP_25G_MPLLB_CTRL4 on page 854	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 4 Register
VR_PMA_MP_16G_MPLLB_CTRL4 on page 855	* Varies	VR PMA Synopsys Multi-protocol 16G PHY MPLLB Control 4 Register
VR_PMA_MP_25G_MPLLA_CTRL5 on page 856	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 5 Register
VR_PMA_MP_16G_MPLLA_CTRL5 on page 857	* Varies	VR PMA Synopsys Multi-protocol 16G PHY MPLLA Control 5 Register
VR_PMA_MP_25G_MPLLA_CTRL6 on page 858	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 6 Register
VR_PMA_MP_25G_MPLLB_CTRL5 on page 859	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 5 Register
VR_PMA_MP_16G_MPLLB_CTRL5 on page 860	* Varies	VR PMA Synopsys Multi-protocol 16G PHY MPLLB Control 5 Register
VR_PMA_MP_25G_MPLLB_CTRL6 on page 861	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 6 Register
VR_PMA_MP_25G_16G_EQ_INIT_CTRL 0 on page 862	* Varies	VR PMA Synopsys Multi-protocol PHY Equalization Init Control 0 Register
VR_PMA_MP_25G_16G_EQ_INIT_CTRL 1 on page 864	* Varies	VR PMA Synopsys Multi-protocol PHY Equalization Init Control 1 Register
VR_PMA_MP_25G_MPLLA_CTRL7 on page 865	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 7 Register
VR_PMA_MP_25G_MPLLA_CTRL8 on page 866	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 8 Register
VR_PMA_MP_25G_MPLLB_CTRL7 on page 867	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 7 Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_MP_25G_MPLLB_CTRL8 on page 868	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 8 Register
VR_PMA_MP_25G_MPLLA_CTRL9 on page 869	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 9 Register
VR_PMA_MP_25G_MPLLB_CTRL9 on page 870	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 9 Register
VR_PMA_MP_25G_MPLLA_CTRL10 on page 871	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 10 Register
VR_PMA_MP_25G_MPLLB_CTRL10 on page 872	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 10 Register
VR_PMA_MP_25G_MPLLA_CTRL11 on page 873	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 11 Register
VR_PMA_MP_25G_MPLLB_CTRL11 on page 874	* Varies	VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 11 Register
VR_PMA_MP_EPHY_EQ_PRESET_CTRL0 on page 875	* Varies	VR PMA Synopsys Multi Protocol Enterprise PHY Equalization Preset Control 0 register
VR_PMA_MP_EPHY_EQ_PRESET_CTRL1 on page 877	* Varies	VR PMA Synopsys Multi Protocol Enterprise PHY Equalization Preset Control 1 register
VR_PMA_MP_25G_16G_MISC_CTRL0 on page 878	* Varies	VR PMA Synopsys Multi-protocol PHY Miscellaneous Control 0 Register
VR_PMA_MP_25G_16G_REF_CLK_CTRL on page 881	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Reference Control Register
VR_PMA_MP_32G_REF_CLK_CTRL0 on page 885	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Reference Clock Control0 Register
VR_PMA_MP_25G_16G_VCO_CAL_LD0 on page 887	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Load Register 0
VR_PMA_MP_25G_16G_VCO_CAL_LD1 on page 888	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Load Register 1
VR_PMA_MP_25G_16G_VCO_CAL_LD2 on page 889	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Load Register 2
VR_PMA_MP_25G_16G_VCO_CAL_LD3 on page 890	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Load Register 3
VR_PMA_MP_25G_16G_VCO_CAL_REF0 on page 891	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Reference Register 0
VR_PMA_MP_25G_16G_VCO_CAL_REF1 on page 893	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Reference Register 1

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_MP_25G_16G_MISC_CTRL1 on page 895	* Varies	VR PMA Synopsys Multi-protocol PHY Miscellaneous Control 1 Register
VR_PMA_MP_25G_16G_EEE_CTRL on page 896	* Varies	VR PMA Synopsys Multi-protocol PHY EEE Control Register This register is present only in EEE...
VR_PMA_MP_25G_16G_SRAM on page 897	* Varies	VR PMA Synopsys Multi-protocol PHY SRAM Register. Note: This is a special register which can be...
VR_PMA_MP_25G_16G_MISC_CTRL2 on page 899	* Varies	VR PMA Synopsys Multi-protocol 25G/16G PHY Miscellaneous Control 2 Register
VR_PMA_MP_32G_MISC_CTRL on page 900	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Miscellaneous Control Register
VR_PMA_MP_32G_REF_CLK_CTRL1 on page 902	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Reference Clock Control1 Register
VR_PMA_MP_25G_16G_12G_MISC_STS0 on page 904	* Varies	VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Miscellaneous Status 0 Register
VR_PMA_MP_32G_MISC_STS0 on page 906	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Miscellaneous Status0 Register
VR_PMA_MP_25G_16G_MISC_STS1 on page 908	* Varies	VR PMA Synopsys Multi-protocol PHY Miscellaneous Status 1 Register
VR_PMA_MP_25G_16G_MISC_STS2 on page 909	* Varies	VR PMA Synopsys Multi-protocol PHY Miscellaneous Status 2 Register
VR_PMA_MP_25G_16G_CR_CTRL on page 910	* Varies	VR PMA Synopsys Multi-protocol PHY CR Control Register
VR_PMA_MP_25G_16G_CR_ADDR on page 912	* Varies	VR PMA Synopsys Multi-protocol PHY CR Address Register
VR_PMA_MP_25G_16G_CR_DATA on page 913	* Varies	VR PMA Synopsys Multi-protocol PHY CR Data Register
VR_PMA_CWM00 on page 914	* Varies	VR PMA MMD RSFEC alignment marker bits[15:0]
VR_PMA_CWM01 on page 915	* Varies	VR PMA MMD RSFEC alignment marker bits[31:16]. See the note mention in the description of VR_PMA_CWM00...
VR_PMA_CWM02 on page 916	* Varies	VR PMA MMD RSFEC alignment marker bits[47:32] See the note mentioned in the description of the VR_PMA_CWM00...
VR_PMA_CWM03 on page 917	* Varies	VR PMA MMD RSFEC alignment marker bits[63:48] See the note mentioned in the description of VR_PMA_CWM00 Program...
VR_PMA_CTRL3 on page 918	* Varies	VR PMA MMD RS(544,514) auxiliary Control 3 Register This register provides additional parameter...

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_CTRL4 on page 920	* Varies	VR PMA MMD RS(544,514) auxiliary Control 4 Register This register provides additional parameter...
VR_PMA_KRTR_TX_EQ_CFF_CTRL0 on page 922	* Varies	VR PMA MMD KRTR Transmit Equalization Coefficient Control Register. This register is present...
VR_PMA_KRTR_TX_EQ_CFF_CTRL1 on page 925	* Varies	VR PMA MMD KRTR Transmit Equalization Coefficient Control Register. This register is present...
VR_PMA_KRTR_TX_EQ_CFF_CTRL2 on page 928	* Varies	VR PMA MMD KRTR Transmit Equalization Coefficient Control Register. This register is present...
VR_PMA_KRTR_TX_EQ_CFF_CTRL3 on page 931	* Varies	VR PMA MMD KRTR Transmit Equalization Coefficient Control Register. This register is present...
VR_PMA_KRTR_TX_EQ_STS_CTRL0 on page 934	* Varies	VR PMA MMD KRTR Transmit Equalization Status Control Register This register is present only if...
VR_PMA_KRTR_TX_EQ_STS_CTRL1 on page 937	* Varies	VR PMA MMD KRTR Transmit Equalization Status Control Register This register is present only if...
VR_PMA_KRTR_TX_EQ_STS_CTRL2 on page 940	* Varies	VR PMA MMD KRTR Transmit Equalization Status Control Register This register is present only if...
VR_PMA_KRTR_TX_EQ_STS_CTRL3 on page 943	* Varies	VR PMA MMD KRTR Transmit Equalization Status Control Register This register is present only if...
VR_PMA_PHY_TX_EQ_STS0 on page 946	* Varies	Lane-0 VR PMA MMD PHY Transmit Equalization Status Register This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled....
VR_PMA_PHY_TX_EQ_STS1 on page 948	* Varies	Lane-1 VR PMA MMD PHY Transmit Equalization Status Register This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled....
VR_PMA_PHY_TX_EQ_STS2 on page 950	* Varies	Lane-2 VR PMA MMD PHY Transmit Equalization Status Register This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled....
VR_PMA_PHY_TX_EQ_STS3 on page 952	* Varies	Lane-3 VR PMA MMD PHY Transmit Equalization Status Register This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled....
VR_PMA_KRTR_RX_EQ_CTRL0 on page 954	* Varies	VR PMA MMD KRTR Receive Equalization Control Register This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. ...
VR_PMA_KRTR_RX_EQ_CTRL1 on page 957	* Varies	VR PMA MMD KRTR Receive Equalization Control Register This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. ...

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_KRTR_RX_EQ_CTRL2 on page 960	* Varies	VR PMA MMD KRTR Receive Equalization Control Register This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. ...
VR_PMA_KRTR_RX_EQ_CTRL3 on page 963	* Varies	VR PMA MMD KRTR Receive Equalization Control Register This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. ...
VR_PMA_KRTR_RX_EQ_STS_CTRL0 on page 966	* Varies	VR PMA MMD KRTR Receive Equalization Status Control Register This register is present only if...
VR_PMA_KRTR_RX_EQ_STS_CTRL1 on page 969	* Varies	VR PMA MMD KRTR Receive Equalization Status Control Register This register is present only if...
VR_PMA_KRTR_RX_EQ_STS_CTRL2 on page 973	* Varies	VR PMA MMD KRTR Receive Equalization Status Control Register This register is present only if...
VR_PMA_KRTR_RX_EQ_STS_CTRL3 on page 977	* Varies	VR PMA MMD KRTR Receive Equalization Status Control Register This register is present only if...
VR_PMA_PHY_RX_EQ_CEU0 on page 981	* Varies	VR PMA MMD PHY Receive Equalization Coefficient Update This register is present when the Clause...
VR_PMA_PHY_RX_EQ_CEU1 on page 985	* Varies	VR PMA MMD PHY Receive Equalization Coefficient Update This register is present when the Clause...
VR_PMA_PHY_RX_EQ_CEU2 on page 989	* Varies	VR PMA MMD PHY Receive Equalization Coefficient Update This register is present when the Clause...
VR_PMA_PHY_RX_EQ_CEU3 on page 993	* Varies	VR PMA MMD PHY Receive Equalization Coefficient Update This register is present when the Clause...
VR_PMA_MP_32G_RECAL on page 997	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Tx Re-Calibration Control Register
VR_PMA_MP_32G_TX_CTRL0 on page 999	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Tx Control0 Register
VR_PMA_MP_32G_MARGIN_LANE0 on page 1001	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Margin Lane0 Register
VR_PMA_MP_32G_MARGIN_LANE1 on page 1002	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Margin Lane1 Register
VR_PMA_MP_32G_MARGIN_LANE2 on page 1003	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Margin Lane2 Register
VR_PMA_MP_32G_MARGIN_LANE3 on page 1004	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Margin Lane3 Register
VR_PMA_MP_32G_MARGIN_CTRL on page 1005	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Margin Control Register

Table 7-5 Registers for Address Block: DWC_xlgpcs_map/PMA_MMD (Continued)

Register	Offset	Description
VR_PMA_MP_32G_RX_ADPT_CTRL0 on page 1007	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Rx Adapt Control0 Register
VR_PMA_MP_32G_CNTX_LINK_NUM_CTRL on page 1009	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Context Link Number Control Register
VR_PMA_MP_32G_CNTX_CTRL on page 1011	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Context Control Register
VR_PMA_MP_32G_TX_CNTX_SEL0 on page 1012	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Tx Context Select0 Register
VR_PMA_MP_32G_TX_CNTX_SEL1 on page 1013	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Tx Context Select1 Register
VR_PMA_MP_32G_CM_CNTX_SEL0 on page 1014	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Common Context Select0 Register
VR_PMA_MP_32G_CM_CNTX_SEL1 on page 1015	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Common Context Select1 Register
VR_PMA_MP_32G_RX_CNTX_SEL0 on page 1016	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Rx Context Select0 Register
VR_PMA_MP_32G_RX_CNTX_SEL1 on page 1017	* Varies	VR PMA Synopsys Multi-protocol 32G PHY Rx Context Select1 Register

7.1.1 SR_PMA_CTRL1

- **Description:** SR PMA MMD Control1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

15	14	13	12	11	10:7	6	5:2	1	0
RST	Reserved_14	SS13	Reserved_12	LPM	Reserved_10_7	SS6	SS_5_2	Reserved_1	LB

Table 7-6 Fields for Register: SR_PMA_CTRL1

Bits	Name	Memory Access	Description
15	RST	R/W	Soft Reset (RW,SC Type) For Synopsys PHY configurations: When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared when PSEQ_STATE in VR PCS MMD Digital Status Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. For Non-Synopsys PHY configurations: When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. This bit is self-cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface Note: For information about read or write-access for registers during power-down mode, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	SS13	R	Speed Selection Bit 13 This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-6 Fields for Register: SR_PMA_CTRL1 (Continued)

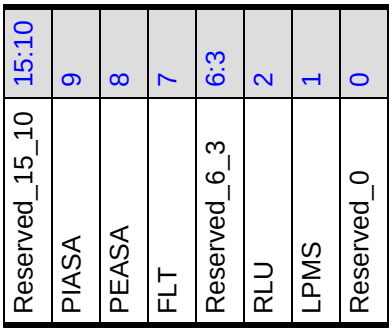
Bits	Name	Memory Access	Description
11	LPM	R/W	Low-Power Enable This bit controls the power-down mode of the DWC_xlgpcs. ■ 0: Normal Operation ■ 1: The DWC_xlgpcs enters power-down mode along with the PHY. For non-Synopsys PHY, DWC_xlgpcs asserts xlgpcs_pdown_o port is asserted. For Synopsys PHY, DWC_xlgpcs does the following to trigger the power-down mode: ■ Turns off the PHY Receiver and Transmitter ■ Switches off all the clocks When the host clears this bit, DWC_xlgpcs resumes normal operation. In a configuration with Synopsys PHY, after clearing this bit, the host must wait until Bits[4:2] of VR PCS MMD Digital Status Register indicate that DWC_xlgpcs is in normal state. **Note:** For information about read or write-access for registers during power-down mode, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always
10:7	Reserved_10_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	SS6	R	Speed Selection Bit 6 This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always

Table 7-6 Fields for Register: SR_PMA_CTRL1 (Continued)

Bits	Name	Memory Access	Description
5:2	SS_5_2	R/W	Speed Selection Bits [5:2] This field selects the speed of operation when both SS13 and SS6 are set. Speed encoding is as follows: ■ 0100 = 25Gbps ■ 0101 = 50Gbps ■ 0011 = 100Gbps ■ 0010 = 40Gbps ■ 0001 = 10PASS-TS/2BASE-TL ■ 0000 = 10Gbps ■ Other Values = Reserved **Value After Reset:** "DWCXLP_KR4_100G ? 0x3 : (DWCXLP_25G_ONLY||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 0x4 : 0x2" **Exists:** Always
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	LB	R/W	Loopback Enable This bit enables the loopback control of the PHY. ■ 0: The DWC_xlgpcs de-asserts the xlgpcs_loopback_en_o signal. ■ 1: The DWC_xlgpcs asserts the xlgpcs_loopback_en_o signal. **Value After Reset:** 0x0 **Exists:** Always

7.1.2 SR_PMA_STATUS1

- **Description:** SR PMA MMD Status1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1 : (DWCXLP_APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)



15:10	9	8	7	6:3	2	1	0
Reserved_15_10	PIASA	PEASA	FLT	Reserved_6_3	RLU	LPMS	Reserved_0

Table 7-7 Fields for Register: SR_PMA_STATUS1

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	PIASA	R	PMA Ingress AUI Stop Ability This bit indicates that The ability to support transition to a low power state in the Ingress direction. This bit is always set when EEE is enabled in the configuration. **Value After Reset:** "(DWCXLP_EEE_EN) ? 0x1 : 0x0" **Exists:** Always
8	PEASA	R	PMA Egress AUI Stop Ability This bit indicates that The ability to support transition to a low power state in the egress direction. This bit is always set when EEE is enabled in the configuration. **Value After Reset:** "(DWCXLP_EEE_EN) ? 0x1 : 0x0" **Exists:** Always

Table 7-7 Fields for Register: SR_PMA_STATUS1 (Continued)

Bits	Name	Memory Access	Description
7	FLT	R	Fault Condition Detected This bit indicates that a fault condition is detected in the PHY. This bit is set when the Rx path of the PHY encounters any of the following fault conditions: ■ Loss of signal detection. It is indicated when xlgpcs_los_i{lane} signal goes high (for all valid lanes) ■ Rx CDR failure. It is indicated when xlgpcs_rx_valid_i{lane} signal goes low (for all valid lanes) For Synopsys PHYs, xlgpcs_los_i and xlgpcs_rx_valid_i are unreliable. Hence, this bit must not be used to infer Fault condition in the PHY. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
6:3	Reserved_6_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	RLU	R	Rx Link Up (RO, LL Type) ■ 0: This bit goes low when the CDR status of the PHY goes low (indicated by xlgpcs_rx_valid_i) ■ 1: This bit is set when the CDR status of the PHY goes high (indicated by xlgpcs_rx_valid_i) **Value After Reset:** 0x0 **Exists:** Always
1	LPMS	R	Low-Power Mode Support ■ 1: Indicates that DWC_xlgpcs supports low-power mode. ■ 0: Indicates that DWC_xlgpcs does not support the low-power mode. **Value After Reset:** 0x1 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.1.3 SR_PMA_DEV_ID_1

- **Description:** SR PMA MMD Device Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x2 : (DWCXLP_APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)



Table 7-8 Fields for Register: SR_PMA_DEV_ID_1

Bits	Name	Memory Access	Description
15:0	PMA_DEV_OUI_3_18	R	Organizationally Unique Identifier[3:18] The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The value is configured in coreConsultant for each of the MMDs. This read-only field contains Bits[18:3] of 24-bit OUI of device manufacturer. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR Control MMD PMA Device Identifier Register 1 modifies the content of this register. **Value After Reset:** "(DWCXLP_PMA_MMD) ? DWCXLP_IEEE_PMA_DEV_OUI_3_18 : 0x0" **Exists:** Always

7.1.4 SR_PMA_DEV_ID_2

- **Description:** SR PMA MMD Device Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x3 : (DWCXLP_APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

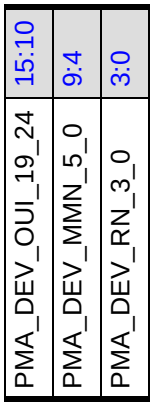


Table 7-9 Fields for Register: SR_PMA_DEV_ID_2

Bits	Name	Memory Access	Description
15:10	PMA_DEV_OUI_19_24	R	Organizationally Unique Identifier[19:24] The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. You can configure the value of each MMD using coreConsultant. This read-only field contains Bits[24:19] of 24-bit OUI of device manufacturer. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR Control MMD PMA Device Identifier Register 2 modifies the content of this register. **Value After Reset:** "(DWCXLP_PMA_MMD) ? DWCXLP_IEEE_PMA_DEV_OUI_19_24 : 0x0" **Exists:** Always

Table 7-9 Fields for Register: SR_PMA_DEV_ID_2 (Continued)

Bits	Name	Memory Access	Description
9:4	PMA_DEV_MMN_5_0	R	Model Number This read-only field contains the 6-bit Model number of the device. You can configure the value of each MMD using coreConsultant. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR Control MMD PMA Device Identifier Register 2 modifies the content of this register. **Value After Reset:** "(DWCXLP_PMA_MMD) ? DWCXLP_IEEE_PMA_DEV_MODEL_NO_0_5 : 0x0" **Exists:** Always
3:0	PMA_DEV_RN_3_0	R	Revision Number This read-only field contains the 4-bit Revision number of the device. You can configure the value of each MMD using coreConsultant. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR Control MMD PMA Device Identifier Register 2 modifies the content of this register. **Value After Reset:** "(DWCXLP_PMA_MMD) ? DWCXLP_IEEE_PMA_DEV_REV_NO_0_3 : 0x0" **Exists:** Always

7.1.5 SR_PMA_SPD_ABL

- **Description:** SR PMA MMD Speed Ability Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x4 : (DWCXLP_APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

15:12	11	10	9	8	7:5	4	3:1	0
Reserved_15_12	XXVGC	Reserved_10	CGC	XLGC	Reserved_7_5	GC	Reserved_3_1	XGC

Table 7-10 Fields for Register: SR_PMA_SPD_ABL

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	XXVGC	R	25G Capable ■ Indicates that the DWC_xlgpcs can support 25G mode. ■ Indicates that the DWC_xlgpcs does not support 25G mode. **Value After Reset:** "(!DWCXLP_KR4_KR) ? 0x1 : 0x0" **Exists:** Always
10	Reserved_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-10 Fields for Register: SR_PMA_SPD_ABL (Continued)

Bits	Name	Memory Access	Description
9	CGC	R	100G Capable ■ 1: Indicates that the DWC_xlgpcs can support 100G mode. ■ 0: Indicates that the DWC_xlgpcs does not support 100G mode. **Value After Reset:** "(DWCXLP_KR4_100G||DWCXLP_100G_1LANE) ? 0x1 : 0x0" **Exists:** Always
8	XLGC	R	40G Capable ■ 1: Indicates that the DWC_xlgpcs can support 40G mode. ■ 0: Indicates that the DWC_xlgpcs does not support 40G mode. **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x1 : 0x0" **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	GC	R	1G Capable This bit indicates that the DWC_xlgpcs can support 1G mode. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	XGC	R	10G Capable This bit indicates that the DWC_xlgpcs can support 10G mode. **Value After Reset:** 0x1 **Exists:** Always

7.1.6 SR_PMA_DEV_PKG1

- **Description:** SR PMA MMD Devices in Package Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x5 : (DWCXLP_APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

15:12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved_15_12	SPMA4	SPMA3	SPMA2	SPMA1	AN	TC	DTEXS	PHYXS	PCS	WIS	PMA_PMD	CLS22

Table 7-11 Fields for Register: SR_PMA_DEV_PKG1

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	SPMA4	R	separated pma 1 ■ 1: separated PMA (4) present in package ■ 0: separated PMA (4) not present in package **Value After Reset:** 0x0 **Exists:** Always
10	SPMA3	R	separated pma 1 ■ 1: separated PMA (3) present in package ■ 0: separated PMA (3) not present in package **Value After Reset:** 0x0 **Exists:** Always
9	SPMA2	R	separated pma 2 ■ 1: separated PMA (2) present in package ■ 0: separated PMA (2) not present in package **Value After Reset:** 0x0 **Exists:** Always

Table 7-11 Fields for Register: SR_PMA_DEV_PKG1 (Continued)

Bits	Name	Memory Access	Description
8	SPMA1	R	separated pma 1 ■ 1: separated PMA (1) present in package ■ 0: separated PMA (1) not present in package **Value After Reset:** 0x0 **Exists:** Always
7	AN	R	Auto-Negotiation MMD This bit indicates that an AN MMD is present in the package. **Default Value:** If 'Add CL73 Autoneg Support' option is selected, this bit returns 1. For all other configurations, this bit returns 0. **Value After Reset:** "(DWCXLP_CL73_AN) ? 0x1 : 0x0" **Exists:** Always
6	TC	R	TC MMD This bit indicates that the device has the TC MMDs in the package. This bit is always 0. **Value After Reset:** 0x0 **Exists:** Always
5	DTEXS	R	DTE XGXS MMD This bit indicates that the device has the DTE XS MMD in the package. For all configurations, this bit returns 0. **Value After Reset:** 0x0 **Exists:** Always
4	PHYXS	R	PHY XGXS MMD This bit indicates that the device has the PHY XS MMD in the package. **Value After Reset:** 0x0 **Exists:** Always
3	PCS	R	PCS MMD This bit indicates that the device has PCS MMD in the package. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always

Table 7-11 Fields for Register: SR_PMA_DEV_PKG1 (Continued)

Bits	Name	Memory Access	Description
2	WIS	R	WIS MMD This bit indicates that the device has the WIS MMD in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	PMA_PMD	R	PMA or PMD MMD This bit indicates that the device has the PMA or PMD MMD in the package. This bit always returns 1. **Value After Reset:** "(DWCXLP_PMA_MMD) ? 0x1 : 0x0" **Exists:** Always
0	CLS22	R	Clause 22 Register Support This bit indicates that the Clause 22 (1G) functional register sets are implemented in this package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

7.1.7 SR_PMA_DEV_PKG2

- **Description:** SR PMA MMD Devices in Package Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x6 : (DWCXLP_APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

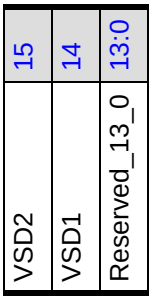


Table 7-12 Fields for Register: SR_PMA_DEV_PKG2

Bits	Name	Memory Access	Description
15	VSD2	R	Vendor-Specific Device 2 This bit indicates that a vendor-specific device 2 is present in the package. **Value After Reset:** 0x0 **Exists:** Always
14	VSD1	R	Vendor-Specific Device 1 This bit indicates that the vendor-specific device 1 is present in the package. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
13:0	Reserved_13_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.1.8 SR_PMA_CTRL2

- **Description:** SR PMA MMD Control2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x7 : (DWCXLP_APB3_16) ? (0x7*0x2) : (0x7*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

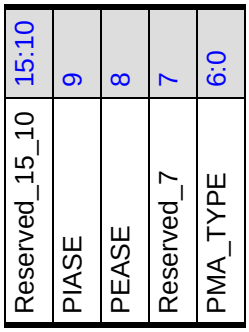


Table 7-13 Fields for Register: SR_PMA_CTRL2

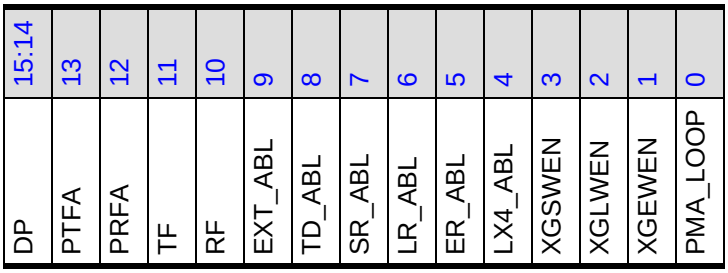
Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	PIASE	* Varies	PMA Ingress AUI Stop Enable ■ 1: Allows transition to a low-power state in the Ingress direction. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_EEE_EN) ? \"read-write\" : \"read-only\""
8	PEASE	* Varies	PMA Egress AUI Stop Enable ■ 1: Allows transition to a low-power state in the Egress direction. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_EEE_EN) ? \"read-write\" : \"read-only\""

Table 7-13 Fields for Register: SR_PMA_CTRL2 (Continued)

Bits	Name	Memory Access	Description
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	PMA_TYPE	R/W	PMA Type Select During initialization, the host must program an appropriate value depending upon the active PHY with which DWC_xlgpcs is interfacing. Encoding is as follows: ■ 1 0 0 0 1 1 0 : 100GBASE-CR1 PMA/PMD type ■ 1 0 0 0 1 1 1 : 100GBASE-KR1 PMA/PMD type ■ 1 0 0 0 0 0 1 : 50GBASE-CR PMA/PMD type ■ 1 0 0 0 0 0 0 : 50GBASE-KR PMA/PMD type ■ 0 1 1 1 0 1 0 : 25GBASE-SR PMA/PMD type ■ 0 1 1 1 0 0 1 : 25GBASE-KR or 25GBASE-KR-S PMA/PMD type ■ 0 1 1 1 0 0 0 : 25GBASE-CR or 25GBASE-CR-S PMA/PMD type ■ 0 1 0 1 1 1 1 : 100GBASE-SR4 PMA/PMD type ■ 0 1 0 1 1 1 0 : 100GBASE-CR4 PMA/PMD type ■ 0 1 0 1 1 0 1 : 100GBASE-KR4 PMA/PMD type ■ 0 1 0 1 0 1 1 : 100GBASE-ER4 PMA/PMD type ■ 0 1 0 1 0 1 0 : 100GBASE-LR4 PMA/PMD type ■ 1 0 0 1 0 0 0 : 100GBASE-KR2 PMA/PMD type ■ 1 0 0 1 0 0 1 : 100GBASE-CR2 PMA/PMD type ■ 0 1 0 0 1 0 1 : 40GBASE-ER4 PMA/PMD type ■ 0 1 0 0 1 0 0 : 40GBASE-FR PMA/PMD type ■ 0 1 0 0 0 1 1 : 40GBASE-LR4 PMA/PMD type ■ 0 1 0 0 0 1 0 : 40GBASE-SR4 PMA/PMD type ■ 0 1 0 0 0 0 1 : 40GBASE-CR4 PMA/PMD type ■ 0 1 0 0 0 0 0 : 40GBASE-KR4 PMA/PMD type ■ 0 0 0 1 0 1 1 : 10GBASE-KR PMA/PMD type ■ Other values : Reserved Depending on chosen RTL configuration, DWC_xlgpcs may or may not support all the above PMA/PMDs. **Value After Reset:** 0x0 **Exists:** Always

7.1.9 SR_PMA_STATUS2

- **Description:** SR PMA MMD Status2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x8 : (DWCXLP_APB3_16) ? (0x8*0x2) : (0x8*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)



15:14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DP	PTFA	PRFA	TF	RF	EXT_ABL	TD_ABL	SR_ABL	LR_ABL	ER_ABL	LX4_ABL	XGSWEN	XGLWEN	XGEWEN	PMA_LOOP

Table 7-14 Fields for Register: SR_PMA_STATUS2

Bits	Name	Memory Access	Description
15:14	DP	R	Device Present Status The value 2'b10 indicates that the PMA MMD is present and responding to this register address. Any other value indicates that the PMA MMD is not present or not functioning properly. This field is always set to 2'b10. **Value After Reset:** 0x2 **Exists:** Always
13	PTFA	R	PMA Tx Fault Ability This bit indicates that the PMA MMD is able to detect a Tx fault. DWC_xlgpcs does not detect the PMA faults on the Tx path. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
12	PRFA	R	PMA Rx Fault Ability This bit indicates that the PMA MMD is able to detect an Rx fault. DWC_xlgpcs supports PMA fault detection on the Rx path based on xlgpcs_los_i{lane} **Value After Reset:** 0x1 **Exists:** Always

Table 7-14 Fields for Register: SR_PMA_STATUS2 (Continued)

Bits	Name	Memory Access	Description
11	TF	R	Transmitter Fault (RO,LH Type) This bit indicates that DWC_xlgpcs encountered a remote fault condition on the PMA Tx path. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
10	RF	R	Receiver Fault (RO,LH Type) This bit indicates that the device encountered a fault condition on the Rx path. This bit is 1 when any of the xlgpcs_los_i{lane} inputs goes high (only valid lanes based on operating mode are checked). For Synopsys PHYs, the LOS indication is not reliable for the rates at which DWC_xlgpcs is used. Therefore, even if this bit is set to 1, it need not necessarily indicate a fault condition. **Value After Reset:** "((!DWCXLP_SNPS_XAUI_PHY)&&(DWCXLP_MDIO_INTF)) ? 0x1 : 0x0" **Exists:** Always **Testable:** untestable
9	EXT_ABL	R	PMA/PMD Extended Ability This bit indicates that the PMA or PMD has extended abilities listed in the SR PMA or PMD Extended Ability Register. **Value After Reset:** 0x1 **Exists:** Always
8	TD_ABL	R	PMA Tx Disable Ability This bit indicates that the PMD device is able to disable the Tx path. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
7	SR_ABL	R	10GBASE-SR Capable This bit indicates if the PMA or PMD can support the 10GBASE-SR operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
6	LR_ABL	R	10GBASE-LR Capable This bit indicates if the PMA or PMD can support the 10GBASE-LR operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-14 Fields for Register: SR_PMA_STATUS2 (Continued)

Bits	Name	Memory Access	Description
5	ER_ABL	R	10GBASE-ER Capable This bit indicates if the PMA or PMD can support the 10GBASE-ER operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
4	LX4_ABL	R	10GBASE-LX4 Capable This bit indicates if the PMA or PMD can support the 10GBASE-LX4 operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
3	XGSWEN	R	10GBASE-SW Capable This bit indicates if the PMA can support the 10GBASE-SW operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
2	XGLWEN	R	10GBASE-LW Capable This bit indicates if the PMA can support the 10GBASE-LW operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	XGEWEN	R	10GBASE-EW Capable This bit indicates if the PMA can support the 10GBASE-EW operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
0	PMA_LOOP	R	PMA Loopback Ability This bit indicates if the PMA has loopback ability. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always

7.1.10 SR_PMA_TX_DIS

- **Description:** SR PMA MMD Transmit Disable Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x9 : (DWCXLP_APB3_16) ? (0x9*0x2) : (0x9*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

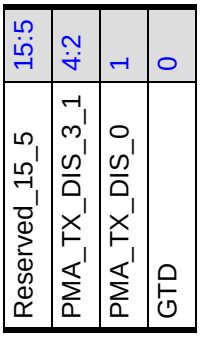


Table 7-15 Fields for Register: SR_PMA_TX_DIS

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:2	PMA_TX_DIS_3_1	* Varies	Transmit Disable on Lanes[3:1] In multi-lane configurations, this field disables Tx on Lane 3, 2, and 1. In single-lane configurations, these fields are reserved. These bits are shared with bits[7:5] of VR PMA MMD Digital Control 1 Register and bits[7:5] of VR PCS MMD Digital Control 1 Register. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-15 Fields for Register: SR_PMA_TX_DIS (Continued)

Bits	Name	Memory Access	Description
1	PMA_TX_DIS_0	R/W	Transmit Disable on Lane 0 This bit disables Tx on Lane 0. This bit is shared with bit[4] of VR PMA MMD Digital Control 1 Register and bit[4] of VR PCS MMD Digital Control 1 Register **Value After Reset:** 0x0 **Exists:** Always
0	GTD	R/W	Global Transmit Disable This bit disables all Tx lanes. ■ 1: Bits[4:1] of this register are set to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.11 SR_PMA_RX_SIG_DET

- **Description:** SR PMA MMD Receive Signal Detect Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | | DWCXLP_MDIO_INTF) ? 0xA : (DWCXLP_APB3_16) ? (0xA*0x2) : (0xA*0x4)" + "(DWCXLP_APB3_INTF | | DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

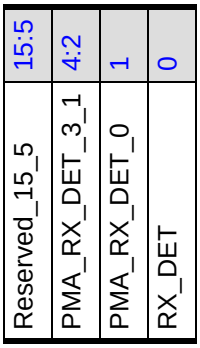


Table 7-16 Fields for Register: SR_PMA_RX_SIG_DET

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:2	PMA_RX_DET_3_1	R	Receive Signal Detect on Lanes[3:1] In multi-lane configurations, this field indicates signal detection on Lane 3, 2, and 1. In single-lane configurations, this field is reserved. **Value After Reset:** 0x0 **Exists:** Always
1	PMA_RX_DET_0	R	Receive Signal Detect on Lane 0 This bit indicates signal detection on Lane 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-16 Fields for Register: SR_PMA_RX_SIG_DET (Continued)

Bits	Name	Memory Access	Description
0	RX_DET	R	Global Receive Signal Detect This is a combined signal detect for all lanes. This bit is set only when signal detect on all the active lanes corresponding to a speed mode are high. Example: In 40G/100G modes, when signal detect on all 4 lanes is high, DWC_xlgpcs sets this bit to 1. In 50G mode, when signal detect on lane-0 and lane-1 is high, DWC_xlgpcs sets this bit to 1. In 10G/25G mode, when signal detect on lane-0 is high, DWC_xlgpcs sets this bit to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.12 SR_PMA_EXT_ABL

- **Description:** SR PMA or PMD Extended Ability Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0xB : (DWCXLP_APB3_16) ? (0xB*0x2) : (0xB*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

15:13	12	11:9	8	7	6	5	4	3	2	1	0
Reserved_15_13	R_25BR_ABL	Reserved_11_9	R_10BT_ABL	R_100BT_ABL	R_100BKX_ABL	GBT_ABL	XGBKR_ABL	XGKX4_ABL	XGBT_ABL	XGBLRM_ABL	XGBCX4_ABL

Table 7-17 Fields for Register: SR_PMA_EXT_ABL

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	R_25BR_ABL	R	25G Capable This bit indicates if the PMA or PMD can support the 25G operation. This bit always returns 1. **Value After Reset:** "(!DWCXLP_KR4_KR) ? 0x1 : 0x0" **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	R_10BT_ABL	R	10BASE-T Capable This bit indicates if the PMA or PMD can support the 10BASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-17 Fields for Register: SR_PMA_EXT_ABL (Continued)

Bits	Name	Memory Access	Description
7	R_100BT_ABL	R	100BASE-T Capable This bit indicates if the PMA or PMD can support the 100BASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
6	R_100BKX_ABL	R	1000BASE-KX Capable This bit indicates if the PMA or PMD can support the 1000BASE-KX operation. **Value After Reset:** 0x0 **Exists:** Always
5	GBT_ABL	R	1000BASE-T Capable This bit indicates if the PMA or PMD can support the 1000BASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
4	XGBKR_ABL	R	10GBASE-KR Capable This bit indicates if the PMA or PMD can support the 10GBASE-KR operation. **Value After Reset:** 0x1 **Exists:** Always
3	XGKX4_ABL	R	10GBASE-KX4 Capable This bit indicates if the PMA or PMD can support the 10GBASE-KX4 operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
2	XGBT_ABL	R	10GBASE-T Capable This bit indicates if the PMA or PMD can support the 10GBASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	XGBLRM_ABL	R	10GBASE-LRM Capable This bit indicates if the PMA or PMD can support the 10GBASE-LRM operation. **Value After Reset:** 0x0 **Exists:** Always

Table 7-17 Fields for Register: SR_PMA_EXT_ABL (Continued)

Bits	Name	Memory Access	Description
0	XGBCX4_ABL	R	10GBASE-CX4 Capable This bit indicates if the PMA or PMD can support the 10GBASE-CX4 operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

7.1.13 SR_PMA_EXT_ABL1

- **Description:** SR PMA or PMD Extended Ability Register 1.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | | DWCXLP_MDIO_INTF) ? 0xD : (DWCXLP_APB3_16) ? (0xD*0x2) : (0xD*0x4)" + "(DWCXLP_APB3_INTF | | DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PMA_RLB_ABL	CGBCR4_ABL	CGBKR4_ABL	CGBKP4_ABL	CGBER4_ABL	CGBLR4_ABL	CGBSR10_ABL	CGBCR10_ABL	CGBSR4_ABL	Reserved_6	XLGBER4_ABL	XLGBFR_ABL	XLGBLR4_ABL	XLGBSR4_ABL	XLGBCR4_ABL	XLGBKR4_ABL

Table 7-18 Fields for Register: SR_PMA_EXT_ABL1

Bits	Name	Memory Access	Description
15	PMA_RLB_ABL	R	PMA Remote Loopback Ability This bit indicates if the PMA can perform a remote loopback function. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
14	CGBCR4_ABL	R	100GBASE-CR4 Capable This bit indicates if the PMA or PMD can support the 100GBASE-CR4 operation. **Value After Reset:** "(DWCXLP_KR4_100G) ? 0x1 : 0x0" **Exists:** Always
13	CGBKR4_ABL	R	100GBASE-KR4 Capable This bit indicates if the PMA or PMD can support the 100GBASE-KR4 operation. **Value After Reset:** "(DWCXLP_KR4_100G) ? 0x1 : 0x0" **Exists:** Always
12	CGBKP4_ABL	R	100GBASE-KP4 Capable This bit indicates if the PMA or PMD can support the 100GBASE-KP4 operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-18 Fields for Register: SR_PMA_EXT_ABL1 (Continued)

Bits	Name	Memory Access	Description
11	CGBER4_ABL	R	100GBASE-ER4 Capable This bit indicates if the PMA or PMD can support the 100GBASE-ER4 operation. **Value After Reset:** "(DWCXLP_KR4_100G) ? 0x1 : 0x0" **Exists:** Always
10	CGBLR4_ABL	R	100GBASE-LR4 Capable This bit indicates if the PMA or PMD can support the 100GBASE-LR4 operation. **Value After Reset:** "(DWCXLP_KR4_100G) ? 0x1 : 0x0" **Exists:** Always
9	CGBSR10_ABL	R	100GBASE-SR10 Capable This bit indicates if the PMA or PMD can support the 100GBASE-SR10 operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
8	CGBCR10_ABL	R	100GBASE-CR10 Capable This bit indicates if the PMA or PMD can support the 100GBASE-CR10 operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
7	CGBSR4_ABL	R	100GBASE-SR4 Capable This bit indicates if the PMA or PMD can support the 100GBASE-SR4 operation. **Value After Reset:** "(DWCXLP_KR4_100G&&DWCXLP_RSFEC_EN) ? 0x1 : 0x0" **Exists:** Always
6	Reserved_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	XLGBER4_ABL	R	40GBASE-ER4 Capable This bit indicates if the PMA or PMD can support the 40GBASE-ER4 operation. This bit always returns 1 in a multi-lane configuration. **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x1 : 0x0" **Exists:** Always

Table 7-18 Fields for Register: SR_PMA_EXT_ABL1 (Continued)

Bits	Name	Memory Access	Description
4	XLGBFR_ABL	R	40GBASE-FR Capable This bit indicates if the PMA or PMD can support the 40GBASE-FR operation. This bit always returns 1 in a multi-lane configuration. **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x1 : 0x0" **Exists:** Always
3	XLGBLR4_ABL	R	40GBASE-LR4 Capable This bit indicates if the PMA or PMD can support the 40GBASE-LR4 operation. This bit always returns 1 in a multi-lane configuration. **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x1 : 0x0" **Exists:** Always
2	XLGBSR4_ABL	R	40GBASE-SR4 Capable This bit indicates if the PMA or PMD can support the 40GBASE-SR4 operation. This bit always returns 1 in a multi-lane configuration. **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x1 : 0x0" **Exists:** Always
1	XLGBCR4_ABL	R	40GBASE-CR4 Capable This bit indicates if the PMA or PMD can support the 40GBASE-CR4 operation. This bit always returns 1 in a multi-lane configuration. **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x1 : 0x0" **Exists:** Always
0	XLGBKR4_ABL	R	40GBASE-KR4 Capable This bit indicates if the PMA or PMD can support the 40GBASE-KR4 operation. **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x1 : 0x0" **Exists:** Always

7.1.14 SR_PMA_PKG1

- **Description:** SR PMA MMD Package Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0xE : (DWCXLP_APB3_16) ? (0xE*0x2) : (0xE*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)



Table 7-19 Fields for Register: SR_PMA_PKG1

Bits	Name	Memory Access	Description
15:0	PMA_PKG_OUI_3_18	R	Organizationally Unique Identifier The DWC_xlgpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. The value is configured within coreConsultant. This read-only field contains Bits[18:3] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS and AN) present in the package. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR Control MMD Package Identifier Register 1 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_OUI_3_18 **Exists:** Always

7.1.15 SR_PMA_PKG2

- **Description:** SR PMA MMD Package Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0xF : (DWCXLP_APB3_16) ? (0xF*0x2) : (0xF*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

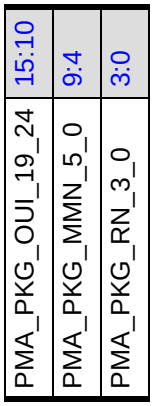


Table 7-20 Fields for Register: SR_PMA_PKG2

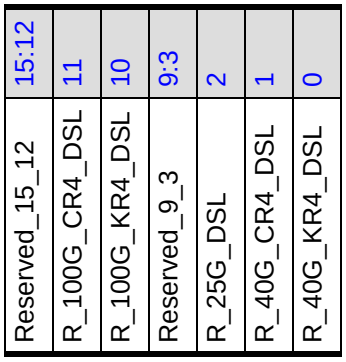
Bits	Name	Memory Access	Description
15:10	PMA_PKG_OUI_19_24	R	Organizationally Unique Identifier DWC_xlgpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. This value is configured by the host. This read-only field contains Bits[24:19] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS and AN) present in the package. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_OUI_19_24 **Exists:** Always

Table 7-20 Fields for Register: SR_PMA_PKG2 (Continued)

Bits	Name	Memory Access	Description
9:4	PMA_PKG_MMN_5_0	R	Model Number This read-only field contains the 5-bit Model Number of the Package in which the MMDs are installed. This value is configured by the host. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_MODEL_NO_0_5 **Exists:** Always
3:0	PMA_PKG_RN_3_0	R	Revision Number This read-only field contains the 4-bit Revision Number of the Package. This value is configured by the host. It is common for all the MMDs in the device. It is common for all MMDs (PMA, PCS and AN) present in the package. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_REV_NO_0_3 **Exists:** Always

7.1.16 SR_EEE_ABL

- **Description:** SR PMA MMD EEE capability Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x10 : (DWCXLP_APB3_16) ? (0x10*0x2) : (0x10*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_EEE_EN



15:12	11	10	9:3	2	1	0
Reserved_15_12	R_100G_CR4_DSL	R_100G_KR4_DSL	Reserved_9_3	R_25G_DSL	R_40G_CR4_DSL	R_40G_KR4_DSL

Table 7-21 Fields for Register: SR_EEE_ABL

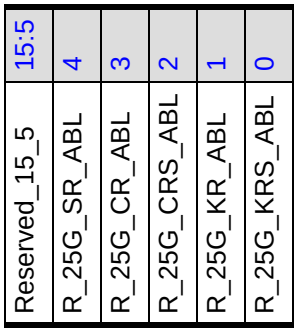
Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	R_100G_CR4_DSL	R	100GBASE-R CR4 Deep Sleep Support This bit indicates the EEE support in the 100GBASE-R CR4 mode: ■ 0: Deep Sleep mode is not supported in 100GBASE-R CR4 mode. ■ 1: Deep Sleep mode is supported in 100GBASE-R CR4 mode. **Value After Reset:** "(DWCXLP_EEE_EN&&DWCXLP_KR4_100G) ? 0x1 : 0x0" **Exists:** Always

Table 7-21 Fields for Register: SR_EEE_ABL (Continued)

Bits	Name	Memory Access	Description
10	R_100G_KR4_DSL	R	100GBASE-R KR4 Deep Sleep Support This bit indicates the EEE support in the 100GBASE-R KR4 mode: ■ 0: Deep Sleep mode is not supported in 100GBASE-R KR4 mode. ■ 1: Deep Sleep mode is supported in 100GBASE-R KR4 mode. **Value After Reset:** "(DWCXLP_EEE_EN&&DWCXLP_KR4_100G) ? 0x1 : 0x0" **Exists:** Always
9:3	Reserved_9_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	R_25G_DSL	R	25GBASE-R Deep Sleep Support This bit indicates the EEE support in the 25GBASE-R mode: ■ 0: Deep Sleep mode is not supported in 25GBASE-R mode. ■ 1: Deep Sleep mode is supported in 25GBASE-R mode. **Value After Reset:** "(DWCXLP_EEE_EN&&!DWCXLP_KR4_KR) ? 0x1 : 0x0" **Exists:** Always
1	R_40G_CR4_DSL	R	40G BASE-CR4 Deep Sleep Support This bit indicates the EEE support in the 40G BASE-CR4 mode: ■ 0: Deep Sleep mode is not supported in 40GBASE-CR4 mode. ■ 1: Deep Sleep mode is supported in 40GBASE-CR4 mode. **Value After Reset:** "(DWCXLP_EEE_EN&&!DWCXLP_25G_ONLY) ? 0x1 : 0x0" **Exists:** Always
0	R_40G_KR4_DSL	R	40G BASE-KR4 Deep Sleep Support This bit indicates the EEE support in the 40G BASE-KR4 mode: ■ 0: Deep Sleep mode is not supported in 40GBASE-KR4 mode. ■ 1: Deep Sleep mode is supported in 40GBASE-KR4 mode. **Value After Reset:** "(DWCXLP_EEE_EN&&!DWCXLP_25G_ONLY) ? 0x1 : 0x0" **Exists:** Always

7.1.17 SR_PMA_EXT_ABL2

- **Description:** SR PMA or PMD Extended Ability Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x13 : (DWCXLP_APB3_16) ? (0x13*0x2) : (0x13*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)



15:5	4	3	2	1	0
Reserved_15_5	R_25G_SR_ABL	R_25G_CR_ABL	R_25G_CRS_ABL	R_25G_KR_ABL	R_25G_KRS_ABL

Table 7-22 Fields for Register: SR_PMA_EXT_ABL2

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	R_25G_SR_ABL	R	25GBASE-SR Capable This bit indicates if the PMA or PMD can support the 25GBASE-SR operation. **Value After Reset:** DWCXLP_RSFEC_EN **Exists:** Always
3	R_25G_CR_ABL	R	25GBASE-CR Capable This bit indicates if the PMA or PMD can support the 25GBASE-CR operation. **Value After Reset:** !DWCXLP_KR4_KR&&DWCXLP_RSFEC_EN&&DWCXLP_FEC_EN **Exists:** Always

Table 7-22 Fields for Register: SR_PMA_EXT_ABL2 (Continued)

Bits	Name	Memory Access	Description
2	R_25G_CRS_ABL	R	25GBASE-CR-S Capable This bit indicates if the PMA or PMD can support the 25GBASE-CR-S operation. **Value After Reset:** !DWCXLP_KR4_KR&&!DWCXLP_RSFEC_EN&&DWCXLP_FEC_EN **Exists:** Always
1	R_25G_KR_ABL	R	25GBASE-KR Capable This bit indicates if the PMA or PMD can support the 25GBASE-KR operation. **Value After Reset:** !DWCXLP_KR4_KR&&DWCXLP_RSFEC_EN&&DWCXLP_FEC_EN **Exists:** Always
0	R_25G_KRS_ABL	R	25GBASE-KR-S Capable This bit indicates if the PMA or PMD can support the 25GBASE-KR-S operation **Note:** ■ A 25GBASE-CR PHY implements both BASE-R FEC sublayer (Clause 74) and the 25GBASE-R RS-FEC sublayer (Clause 108) ■ A 25GBASE-CR-S PHY implements the BASE-R FEC sublayer (Clause 74) only ■ A 25GBASE-KR PHY implements both the BASE-R FEC sublayer (Clause 74) and the 25GBASE-R RS-FEC sublayer (Clause 108) ■ A 25GBASE-KR-S PHY implements the BASE-R FEC sublayer (Clause 74) only. **Value After Reset:** !DWCXLP_KR4_KR&&!DWCXLP_RSFEC_EN&&DWCXLP_FEC_EN **Exists:** Always

7.1.18 SR_PMA_2PT5G_5G_EXT_ABL

- **Description:** SR PMA MMD 2.5G/5G Extended Ability Register
- **Size:** "(DWCXLP_APB3_32) ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x15 : (DWCXLP_APB3_16) ? (0x15*0x2) : (0x15*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD

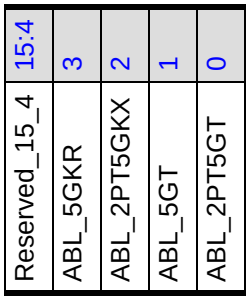


Table 7-23 Fields for Register: SR_PMA_2PT5G_5G_EXT_ABL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	ABL_5GKR	R	5GBASE-KR Capable This bit indicates if the PMA or PMD can support the 5GBASE-KR operation. **Value After Reset:** "DWCXLP_ADD_5G ? 0x1 : 0x0" **Exists:** Always
2	ABL_2PT5GKX	R	2.5GBASE-KX Capable This bit indicates if the PMA or PMD can support the 2.5GBASE-KX operation. **Value After Reset:** 0x0 **Exists:** Always
1	ABL_5GT	R	5GBASE-T Capable This bit indicates if the PMA or PMD can support the 5GBASE-T operation. For all configurations, this bit returns 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-23 Fields for Register: SR_PMA_2PT5G_5G_EXT_ABL (Continued)

Bits	Name	Memory Access	Description
0	ABL_2PT5GT	R	2.5GBASE-T Capable This bit indicates if the PMA or PMD can support the 2.5GBASE-T operation. For all configurations, this bit returns 0. **Value After Reset:** 0x0 **Exists:** Always

7.1.19 SR_PMA_40G_100G_EXT_ABL2

- **Description:** SR PMA or PMD 40G/100G Extended Ability Register 2.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x1A : (DWCXLP_APB3_16) ? (0x1A*0x2) : (0x1A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

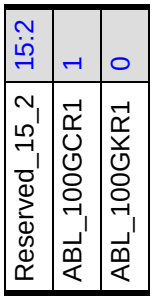


Table 7-24 Fields for Register: SR_PMA_40G_100G_EXT_ABL2

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	ABL_100GCR1	R	100GBASE-CR1 Capable This bit indicates if the PMA or PMD can support the 100GBASE-CR1 operation. **Value After Reset:** "(DWCXLP_100G_1LANE||DWCXLP_ADD_100G_1LANE) ? 0x1 : 0x0" **Exists:** Always
0	ABL_100GKR1	R	100GBASE-KR1 Capable This bit indicates if the PMA or PMD can support the 100GBASE-KR1 operation. **Value After Reset:** "(DWCXLP_100G_1LANE||DWCXLP_ADD_100G_1LANE) ? 0x1 : 0x0" **Exists:** Always

7.1.20 SR_PMA_KR_PMD_CTRL

- **Description:** SR PMA MMD BASE-KR PMD Control Register.

  This register is present only if Link Training Support is enabled in the selected configuration.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x96 : (DWCXLP_APB3_16) ? (0x96*0x2) : (0x96*0x4)" + "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

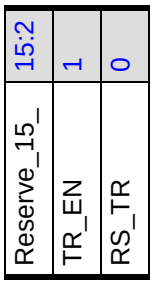


Table 7-25 Fields for Register: SR_PMA_KR_PMD_CTRL

Bits	Name	Memory Access	Description
15:2	Reserve_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	TR_EN	R/W	Training Enable When this bit is set, DWC_xlgpcs enables the Link training protocol (for optimizing the link reliability through Transmit equalization settings). ■ 1: Enable the Link training start-up protocol ■ 0: Disable the Link training start-up protocol If this bit is set before or during Clause 73 auto-negotiation, the training starts after all pages have been exchanged during Clause 73 auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always

Table 7-25 Fields for Register: SR_PMA_KR_PMD_CTRL (Continued)

Bits	Name	Memory Access	Description
0	RS_TR	R/W	Restart Training (RW,SC Type) When this bit is 1, DWC_xlgpcs restarts the Link Training protocol if Bit[1] of this register is 1. ■ 1: Restart the start-up protocol. ■ 0: Normal operation. This bit is self-cleared to 0 after restarting the startup protocol. **Value After Reset:** 0x0 **Exists:** Always

7.1.21 SR_PMA_KR_PMD_STS

- **Description:** SR PMA MMD BASE-KR Status Register

  This register is present only if CL72 Training Support is enabled in the selected configuration.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x97 : (DWCXLP_APB3_16) ? (0x97*0x2) : (0x97*0x4)" + "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR_FAIL3	SU_PR_DTD3	FRM_LCK3	RCV_STS3	TR_FAIL2	SU_PR_DTD2	FRM_LCK2	RCV_STS2	TR_FAIL1	SU_PR_DTD1	FRM_LCK1	RCV_STS1	TR_FAIL0	SU_PR_DTD0	FRM_LCK0	RCV_STS0

Table 7-26 Fields for Register: SR_PMA_KR_PMD_STS

Bits	Name	Memory Access	Description
15	TR_FAIL3	R	Training Failure for Lane 3. ■ 1: Training is completed with failure. ■ 0: Training is not yet complete. When this bit is set to 1, it indicates that the host enabled the Link training protocol but the training is not completed successfully. This can happen because of the following reasons: 1. The Link training protocol is not supported or enabled in the link partner. 2. A reliable link could not be established through Link training protocol within 500 ms. When the host gets the Training Failure status, it can do one of the following: ■ Disable the Link training protocol and rely on the default Transmit equalization settings for the link. ■ Restart the Link training protocol to try again. **Value After Reset:** 0x0 **Exists:** Always

Table 7-26 Fields for Register: SR_PMA_KR_PMD_STS (Continued)

Bits	Name	Memory Access	Description
14	SU_PR_DTD3	R	Start-Up Protocol Status for Lane 3 When this bit is set to 1, it indicates that the Link training protocol is either in progress or completed with failure. ■ 1: Start-up protocol in progress. ■ 0: Start-up protocol complete. **Value After Reset:** 0x0 **Exists:** Always
13	FRM_LCK3	R	Frame Lock for Lane 3 When this bit is set to 1, it indicates that the DWC_xlgpcs receiver successfully and consistently decoded the incoming Link training protocol frame from the link partner. ■ 1: Training frame delineation successfully detected. ■ 0: Training frame delineation not detected. **Value After Reset:** 0x0 **Exists:** Always
12	RCV_STS3	R	Receiver Status for Lane 3. When this bit is set to 1, it indicates that the Rx link is tuned for KR speed mode. The external equalization process indicates this based on the Rx link reliability status. ■ 1: Receiver trained and ready to receive data. ■ 0: Receiver training. **Value After Reset:** 0x0 **Exists:** Always
11	TR_FAIL2	R	Training Failure for lane 2. ■ 1: Training is completed with failure. ■ 0: Training is not yet complete. When this bit is set to 1, it indicates that the host enabled the Link training protocol but the training is not completed successfully. This can happen because of the following reasons: 1. The Link training protocol is not supported or enabled in the link partner. 2. A reliable link could not be established through Link training protocol within 500 ms. When the host gets the Training Failure status, it can do one of the following: ■ Disable the Link training protocol and rely on the default Transmit equalization settings for the link. ■ Restart the Link training protocol to try again. **Value After Reset:** 0x0 **Exists:** Always

Table 7-26 Fields for Register: SR_PMA_KR_PMD_STS (Continued)

Bits	Name	Memory Access	Description
10	SU_PR_DTD2	R	Start-Up Protocol Status for lane 2. When this bit is set to 1, it indicates that the Link training protocol is either in progress or completed with failure. ■ 1: Start-up protocol in progress. ■ 0: Start-up protocol complete. **Value After Reset:** 0x0 **Exists:** Always
9	FRM_LCK2	R	Frame Lock for lane 2 When this bit is set to 1, it indicates that the DWC_xlgpcs receiver successfully and consistently decoded the incoming Link training protocol frame from the link partner. ■ 1: Training frame delineation successfully detected. ■ 0: Training frame delineation not detected. **Value After Reset:** 0x0 **Exists:** Always
8	RCV_STS2	R	Receiver Status for lane 2 When this bit is set to 1, it indicates that the Rx link is tuned for KR speed mode. The external equalization process indicates this based on the Rx link reliability status. ■ 1: Receiver trained and ready to receive data. ■ 0: Receiver training. **Value After Reset:** 0x0 **Exists:** Always
7	TR_FAIL1	R	Training Failure for lane 1. ■ 1: Training is completed with failure. ■ 0: Training is not yet complete. When this bit is set to 1, it indicates that the host enabled the Link training protocol but the training is not completed successfully. This can happen because of the following reasons: 1. The Link training protocol is not supported or enabled in the link partner. 2. A reliable link could not be established through Link training protocol within 500 ms. When the host gets the Training Failure status, it can do one of the following: ■ Disable the Link training protocol and rely on the default Transmit equalization settings for the link. ■ Restart the Link training protocol to try again. **Value After Reset:** 0x0 **Exists:** Always

Table 7-26 Fields for Register: SR_PMA_KR_PMD_STS (Continued)

Bits	Name	Memory Access	Description
6	SU_PR_DTD1	R	Start-Up Protocol Status for lane 1 When this bit is set to 1, it indicates that the Link training protocol is either in progress or completed with failure. ■ 1: Start-up protocol in progress. ■ 0: Start-up protocol complete. **Value After Reset:** 0x0 **Exists:** Always
5	FRM_LCK1	R	Frame Lock for lane 1 When this bit is set to 1, it indicates that the DWC_xlgpcs receiver successfully and consistently decoded the incoming Link training protocol frame from the link partner. ■ 1: Training frame delineation successfully detected. ■ 0: Training frame delineation not detected. **Value After Reset:** 0x0 **Exists:** Always
4	RCV_STS1	R	Receiver Status for lane 1. When this bit is set to 1, it indicates that the Rx link is tuned for KR speed mode. The external equalization process indicates this based on the Rx link reliability status. ■ 1: Receiver trained and ready to receive data. ■ 0: Receiver training. **Value After Reset:** 0x0 **Exists:** Always
3	TR_FAIL0	R	Training Failure for lane 0. ■ 1: Training is completed with failure. ■ 0: Training is not yet complete. When this bit is set to 1, it indicates that the host enabled the Link training protocol but the training is not completed successfully. This can happen because of the following reasons: 1. The Link training protocol is not supported or enabled in the link partner. 2. A reliable link could not be established through Link training protocol within 500 ms. When the host gets the Training Failure status, it can do one of the following: ■ Disable the Link training protocol and rely on the default Transmit equalization settings for the link. ■ Restart the Link training protocol to try again. **Value After Reset:** 0x0 **Exists:** Always

Table 7-26 Fields for Register: SR_PMA_KR_PMD_STS (Continued)

Bits	Name	Memory Access	Description
2	SU_PR_DTD0	R	Start-Up Protocol Status for lane 0. When this bit is set to 1, it indicates that the Link training protocol is either in progress or completed with failure. ■ 1: Start-up protocol in progress. ■ 0: Start-up protocol complete. **Value After Reset:** 0x0 **Exists:** Always
1	FRM_LCK0	R	Frame Lock for lane 0. When this bit is 1, it indicates that the DWC_xlgpcs receiver successfully and consistently decoded the incoming Link training protocol frame from the link partner. ■ 1: Training frame delineation successfully detected. ■ 0: Training frame delineation not detected. **Value After Reset:** 0x0 **Exists:** Always
0	RCV_STS0	R	Receiver Status for lane 0. When this bit is set to 1, it indicates that the Rx link is tuned for KR speed mode. The external equalization process indicates this based on the Rx link reliability status. ■ 1: Receiver trained and ready to receive data. ■ 0: Receiver training. **Value After Reset:** 0x0 **Exists:** Always

7.1.22 SR_PMA_KR_LP_CEU

- **Description:** SR PMA MMD BASE-KR LP Coefficient Update Register

  This register is present when the Link Training support is enabled. This register indicates the most recently received training control field from the link partner (LP) when KR training is enabled. If KR training is disabled, the management can write to this register to tune the Transmit filter coefficients of local device.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x98 : (DWCXLP_APB3_16) ? (0x98*0x2) : (0x98*0x4)" + "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

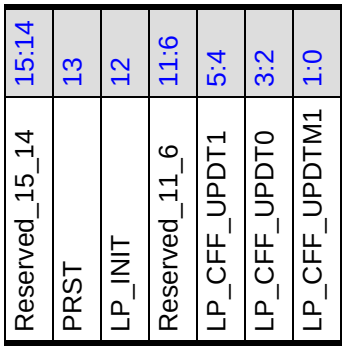


Table 7-27 Fields for Register: SR_PMA_KR_LP_CEU

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	PRST	R/W	Preset Coefficient When this bit is set to 1, it indicates that the link partner requested the transmitter filter coefficients of the local device to be set to a state where transmit equalization is turned off. This means that the filter coefficient value, corresponding to filter (-1) and filter (+1), should be zero and the filter coefficient filter, corresponding to filter (0), should be maximum. ■ 1: Pre-initialize coefficients ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always

Table 7-27 Fields for Register: SR_PMA_KR_LP_CEU (Continued)

Bits	Name	Memory Access	Description
12	LP_INIT	R/W	Initialize Coefficient When this bit is set to 1, it indicates that the link partner requested the transmitter filter coefficients of the local device to be initialized. The initialize requirements of the filter coefficient is defined in IEEE Std 802.3, Clause 72, section 72.6.10.4.2. This request is usually received at the beginning of the start-up protocol. ■ 1: Initialize the local device transmit filter coefficient ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	LP_CFF_UPDT1	R/W	Co-efficient Update (+1) When this bit is set to 1, it indicates that the link partner requested the local device to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	LP_CFF_UPDT0	R/W	Co-efficient Update (0) When this bit is set to 1, it indicates that the link partner requested the local device to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Table 7-27 Fields for Register: SR_PMA_KR_LP_CEU (Continued)

Bits	Name	Memory Access	Description
1:0	LP_CFF_UPDTM1	R/W	Coefficient Update (-1) When this bit is set to 1, it indicates that the link partner requested the local device to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.23 SR_PMA_KR_LP_CESTS

- **Description:** SR PMA MMD BASE-KR LP Coefficient Status Register

  This register is present when the Link Training support is enabled. This register indicates the most recently received training status field from the link partner (LP).

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x99 : (DWCXLP_APB3_16) ? (0x99*0x2) : (0x99*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

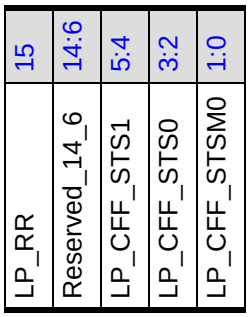


Table 7-28 Fields for Register: SR_PMA_KR_LP_CESTS

Bits	Name	Memory Access	Description
15	LP_RR	R	Receiver Ready When this bit is set to 1, it indicates that the transmit link is reliable to transmit the KR speed data. The link partner sends this information based on successful and consistent reception of the PRBS data field in the KR Training frame (transmitted by the local device). ■ 1: The LP receiver has determined that the training is complete, and it is ready to receive data. ■ 0: The LP receiver is requesting that the training should be continued. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-28 Fields for Register: SR_PMA_KR_LP_CESTS (Continued)

Bits	Name	Memory Access	Description
5:4	LP_CFF_STS1	R	Coefficient Status (+1) This field indicates the status from the link partner for the (increment, decrement, or hold) request sent by the local device for transmit filter (+1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	LP_CFF_STS0	R	Coefficient Status (0) This field indicates the status from the link partner for the (increment, decrement, or hold) request sent by the local device for transmit filter (0). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	LP_CFF_STSM0	R	Coefficient Status (-1) This field indicates the status from the link partner for the (increment, decrement, or hold) request sent by the local device for transmit filter (-1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

7.1.24 SR_PMA_KR_LD_CEU

- **Description:** SR PMA MMD BASE-KR LD Coefficient Update Register

  This register is present when the Link Training support is enabled. This register indicates the most-recently transmitted training control field from the local device (LD) to the link partner.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x9A : (DWCXLP_APB3_16) ? (0x9A*0x2) : (0x9A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

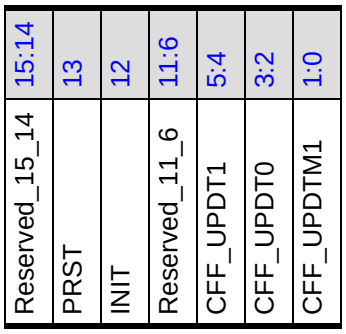


Table 7-29 Fields for Register: SR_PMA_KR_LD_CEU

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	PRST	R	Pre-initialize the Transmit Filter Coefficient of the Link Partner When this bit is set to 1, it indicates that the local device is requesting the transmitter filter coefficients of link partner to be set to a state where transmit equalization is turned off. This means the filter coefficient value, corresponding to filter (-1) and filter (+1), should be zero and the filter coefficient value, corresponding to filter (0), should be maximum. ■ 1: Preset coefficients ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always

Table 7-29 Fields for Register: SR_PMA_KR_LD_CEU (Continued)

Bits	Name	Memory Access	Description
12	INIT	R	Initialize the Transmit Filter Coefficient of the Link Partner When this bit is set to 1, it indicates that the local device requested the transmitter filter coefficients of link partner to be initialized. The initialize requirements of the filter coefficient is defined in IEEE Std 802.3, Clause 72, section 72.6.10.4.2. This request is usually received at the beginning of the start-up protocol. ■ 1: Initialize the local device transmit filter coefficient ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1	R	Filter Coefficient Update (+1) Request to the Link Partner Transmitter When this bit is set to 1, it indicates that the local device requested the link partner to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R	Filter Coefficient Update (0) Request to the Link Partner Transmitter When this bit is set to 1, it indicates that the local device requested the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Table 7-29 Fields for Register: SR_PMA_KR_LD_CEU (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R	Filter Coefficient Update (-1) Request to the Link Partner Transmitter When this bit is set to 1, it indicates that the local device requested the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.25 SR_PMA_KR_LD_CESTS

- **Description:** SR PMA MMD BASE-KR LD Coefficient Status Register

  This register is present only for the PMA MMD when the Link Training support is enabled. This register indicates the most-recently transmitted training status field from the local device (LD) to the link partner.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x9B : (DWCXLP_APB3_16) ? (0x9B*0x2) : (0x9B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

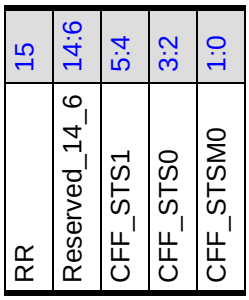


Table 7-30 Fields for Register: SR_PMA_KR_LD_CESTS

Bits	Name	Memory Access	Description
15	RR	R	Receiver Ready When this bit is set to 1, it indicates that the receive link is reliable to receive the KR speed data. The local device Rx adaptation process sets this bit based on successful and consistent reception of the PRBS data field in received training frame at KR speed. ■ 1: The receiver of local device has determined that the training is complete, and it is ready to receive data. ■ 0: The receiver of local device is requesting that the training should be continued. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-30 Fields for Register: SR_PMA_KR_LD_CESTS (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_STS1	R	Filter Coefficient Status (+1) from Link Partner Transmitter to Local Device This field indicates the status from the local device for the (increment, decrement, or hold) request received from the link partner for transmit filter (+1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R	Filter Coefficient Status (0) from Link Partner Transmitter to Local Device This field indicates the status from the local device for the (increment, decrement, or hold) request received from the link partner for transmit filter (0). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_STSM0	R	Filter Coefficient Status (-1) from Link Partner Transmitter to Local Device This field indicates the status from the local device for the (increment, decrement, or hold) request received from the link partner for transmit filter (-1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

7.1.26 SR_PMA_KR_FEC_ABL

- **Description:** SR PMA MMD BASE-KR FEC Ability Register

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xAA : (DWCXLP_APB3_16) ? (0xAA*0x2) : (0xAA*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN)

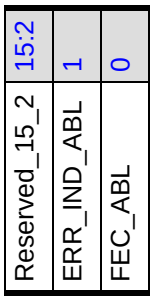


Table 7-31 Fields for Register: SR_PMA_KR_FEC_ABL

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	ERR_IND_ABL	R	BASE-KR FEC Error Indication Ability This bit indicates that the BASE-KR PHY is able to report FEC decoding errors to the PCS layer. Default Value: If DWCXLP_FEC_ERROR_FWD = Enabled, this bit returns 1. Otherwise, this bit returns 0. **Value After Reset:** "(DWCXLP_FEC_ERROR_FWD) ? 0x1 : 0x0" **Exists:** Always
0	FEC_ABL	R	BASE-KR FEC Ability This bit indicates that the BASE-KR PHY supports FEC. This bit always returns 1 indicating the FEC ability. **Value After Reset:** "(DWCXLP_FEC_EN) ? 0x1 : 0x0" **Exists:** Always

7.1.27 SR_PMA_KR_FEC_CTRL

- **Description:** SR PMA MMD BASE-KR FEC Control Register

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xAB : (DWCXLP_APB3_16) ? (0xAB*0x2) : (0xAB*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN)

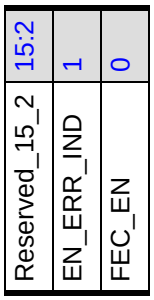


Table 7-32 Fields for Register: SR_PMA_KR_FEC_CTRL

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	EN_ERR_IND	* Varies	FEC Enable Error Indication When set, this bit configures the FEC decoder to indicate uncorrectable errors to the PCS layer. When reset, the FEC decoder does not pass the uncorrectable error information to the PCS layer. Access Type: If DWCXLP_FEC_ERROR_FWD = Enabled, this bit is RW. Otherwise, this bit is RO. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_FEC_ERROR_FWD) ? \"read-write\" : \"read-only\""

Table 7-32 Fields for Register: SR_PMA_KR_FEC_CTRL (Continued)

Bits	Name	Memory Access	Description
0	FEC_EN	R/W	FEC Enable When set, this bit enables BASE-KR FEC. When reset, this bit disables BASE-KR FEC. The FEC ability of the local device and the link partner is negotiated through Clause 73 auto-negotiation. The host should enable this bit only if during auto-negotiation, both link partner and local device indicate that they have the FEC ability. **Value After Reset:** 0x0 **Exists:** Always

7.1.28 SR_PMA_KR_FEC_CORR_BLK1

- **Description:** SR PMA MMD BASE-KR Lane0 FEC Correctable Error Block Counter 1 Register.

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with correctable errors after the FEC block lock is achieved. The total width of the error counter is 32-bit. This register holds the lower 16-bits of the 32-bits counter.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xAC : (DWCXLP_APB3_16) ? (0xAC*0x2) : (0xAC*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN)

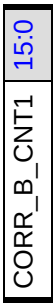


Table 7-33 Fields for Register: SR_PMA_KR_FEC_CORR_BLK1

Bits	Name	Memory Access	Description
15:0	CORR_B_CNT1	R	FEC Corrected Blocks Lower (RO, COR Type) This field indicates the FEC correctable error block counter. These are the lower 16-bits of the counter. This is a non-roll over counter. This field is cleared when this register is read. **Value After Reset:** 0x0 **Exists:** Always

7.1.29 SR_PMA_KR_FEC_CORR_BLK2

- **Description:** SR PMA MMD KR Lane0 FEC Correctable Error Block Counter 2 Register

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with correctable errors after the FEC block lock is achieved. The total width of the error counter is 32-bit. This register holds the higher 16-bits of the 32-bits counter.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xAD : (DWCXLP_APB3_16) ? (0xAD*0x2) : (0xAD*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN)

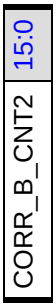


Table 7-34 Fields for Register: SR_PMA_KR_FEC_CORR_BLK2

Bits	Name	Memory Access	Description
15:0	CORR_B_CNT2	R	FEC Corrected Blocks Lower (RO,COR Type) This field indicates the FEC correctable error block counter. These are the higher 16-bits of the counter. This is a non-roll over counter. This field is cleared when SR PMA MMD KR FEC Correctable Error Block Counter 1 register is read. **Value After Reset:** 0x0 **Exists:** Always

7.1.30 SR_PMA_KR_FEC_UCORR_BLK1

- **Description:** SR PMA MMD BASE-KR Lane0 FEC Uncorrectable Error Block Counter 1 Register

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with uncorrectable errors after the FEC block lock is achieved. The total width of the Error counter is 32-bit. This register holds the lower 16-bits of the 32-bits counter.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xAE : (DWCXLP_APB3_16) ? (0xAE*0x2) : (0xAE*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN)

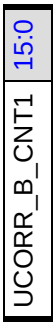


Table 7-35 Fields for Register: SR_PMA_KR_FEC_UCORR_BLK1

Bits	Name	Memory Access	Description
15:0	UCORR_B_CNT1	R	FEC Uncorrected Blocks Lower (RO, COR Type) This field indicates the FEC uncorrectable error block counter. These are the lower 16-bits of the counter. This is a non-roll over counter. This register is cleared when it is read. **Value After Reset:** 0x0 **Exists:** Always

7.1.31 SR_PMA_KR_FEC_UCORR_BLK2

- **Description:** SR PMA MMD Lane0 FEC Uncorrectable Error Block Counter 2 Register

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with uncorrectable errors after the FEC block lock is achieved. The total width of the Error counter is 32-bit. This register holds the higher 16-bits of the 32-bits counter.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xAF : (DWCXLP_APB3_16) ? (0xAF*0x2) : (0xAF*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN)

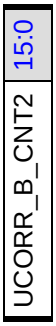


Table 7-36 Fields for Register: SR_PMA_KR_FEC_UCORR_BLK2

Bits	Name	Memory Access	Description
15:0	UCORR_B_CNT2	R	FEC Uncorrected Blocks Lower (RO, COR Type) This field indicates the FEC uncorrectable error block counter. These are the higher 16-bits of the counter. This is a non-roll over counter. This field is cleared when SR PMA MMD FEC Uncorrectable Error Block Counter 1 register is read. **Value After Reset:** 0x0 **Exists:** Always

7.1.32 SR_PMA_RS_FEC_CTRL

- **Description:** SR PMA MMD RS-FEC control register.

  This register provides control to bypass RS-FEC Error Indication and Error Correction
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xC8 : (DWCXLP_APB3_16) ? (0xC8*0x2) : (0xC8*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_RSFEC_EN)

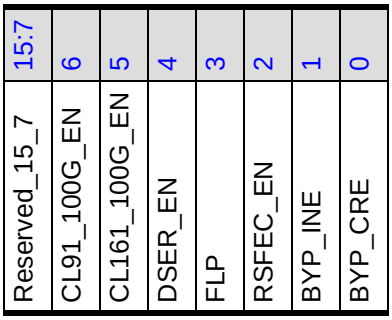


Table 7-37 Fields for Register: SR_PMA_RS_FEC_CTRL

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	CL91_100G_EN	* Varies	100G Non-interleaved RS-FEC Mode Enable For configurations with both Clause 91 and Clause 161 RSFEC support, whenever this bit is set, Clause 91 based RS-FEC is enabled in 100G Mode. When this bit is set, you must not set CL161_100G_EN to 1. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_IRSFEC ? \"read-write\": \"read-only\""

Table 7-37 Fields for Register: SR_PMA_RS_FEC_CTRL (Continued)

Bits	Name	Memory Access	Description
5	CL161_100G_EN	* Varies	100G Interleaved RS-FEC Mode Enable When this bit is set, Clause 161 code-word interleaved RS-FEC mode is enabled.This bit should not be programmed to 1 when operating in any speed mode other than 100G Mode. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_IRSFEC ? \"read-write\": \"read-only\""
4	DSER_EN	* Varies	FEC Degraded SER Enable When operating in RS-FEC (544,514) mode, set this bit to enable the 'Degraded Symbol Error Rate' feature. This field is reserved for configurations without RS-FEC (544,514) support. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_RS544 ? \"read-write\": \"read-only\""
3	FLP	* Varies	Four Lane PMD Indication Encoding of this field is as follows: ■ 0: four-lane-PMD is used. ■ 1: four-lane PMD is not used. **Note:** Encoding of this field is logical inverse of the four_lane_pmd variable defined in IEEE802.3 Clause 91.6.2a ■ 100GBASE-KR2 or 100GAUI-4 modes (which make use of RS544): Program this field to 1. ■ 100GBASE-KR4 mode (which makes use of RS528): Program this field to 0. In other speed modes, the value of this field is ignored. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_RS544 ? \"read-write\": \"read-only\""
2	RSFEC_EN	R/W	25G Base Reed-Solomon FEC Enable This bit enables the 25GBase-R Reed Solomon Forward Error Correction. ■ 1: The 25GBASE-R Reed-Solomon FEC is enabled. ■ 0: The 25GBASE-R Reed-Solomon FEC is disabled. **Value After Reset:** 0x0 **Exists:** Always

Table 7-37 Fields for Register: SR_PMA_RS_FEC_CTRL (Continued)

Bits	Name	Memory Access	Description
1	BYP_INE	* Varies	RS-FEC bypass indication enable This bit enables the RS-FEC decoder to bypass error indication to the upper layers (PCS) through the sync bits. ■ 1: RS-FEC decoder does not indicate errors to the PCS ■ 0: RS-FEC decoder indicates errors to the PCS layer This bit must be set/reset before the RSFEC_EN is enabled. In RS-FEC Error Forwarding disabled configurations (when DWCXLP_RSFEC_ERFW_EN==0), error indication is not done **Value After Reset:** "(DWCXLP_RSFEC_EN&&!DWCXLP_RSFEC_ERFW_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_RSFEC_EN&&DWCXLP_RSFEC_ERFW_EN) ? \"read-write\" : \"read-only\"""
0	BYP_CRE	R/W	RS-FEC bypass correction enable This bit enables the RS-FEC decoder to bypass Error Correction. ■ 1: RS-FEC decoder performs error detection without error correction ■ 0: RS-FEC decoder performs error detection and error correction This bit should be set/reset before the RSFEC_EN is enabled. **Value After Reset:** 0x0 **Exists:** Always

7.1.33 SR_PMA_RS_FEC_STATUS

- **Description:** SR PMA MMD RS-FEC status register.

  This register provides status of Alignment Marker Lock, bypass capabilities for RS-FEC Error Indication and Error Correction of RS-FEC Decoder

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xC9 : (DWCXLP_APB3_16) ? (0xC9*0x2) : (0xC9*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_RSFEC_EN)

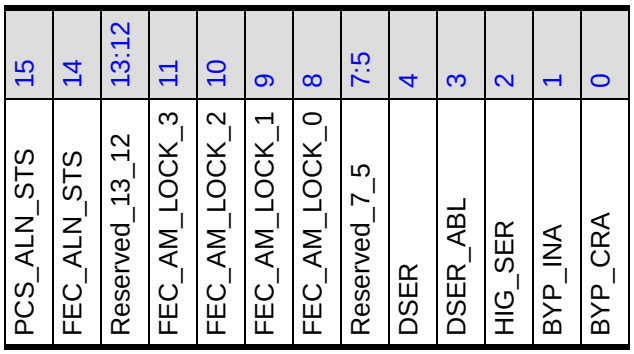


Table 7-38 Fields for Register: SR_PMA_RS_FEC_STATUS

Bits	Name	Memory Access	Description
15	PCS_ALN_STS	R	PCS lane alignment status This bit indicates the Status of PCS RSFEC encoders lane-to-lane alignment. A device that implements the RS-FEC status register but does not implement a separated RS-FEC must return value of 1. ■ 1: FEC encoder has locked and aligned all PCS lanes ■ 0: FEC encoder has locked and aligned all PCS lanes **Value After Reset:** 0x1 **Exists:** Always
14	FEC_ALN_STS	R	FEC lane alignment status This bit indicates the Status of FEC lane-to-lane alignment ■ 1: RS-FEC receive lanes locked and aligned ■ 0: RS-FEC receive lanes not locked and aligned **Value After Reset:** 0x0 **Exists:** Always

Table 7-38 Fields for Register: SR_PMA_RS_FEC_STATUS (Continued)

Bits	Name	Memory Access	Description
13:12	Reserved_13_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	FEC_AM_LOCK_3	R	FEC AM lock 3 This bit when set indicates whether the receiver has detected the location of the alignment marker payload sequence for FEC lane-3. ■ 1: RS-FEC receive lane 3 locked and aligned. ■ 0: RS-FEC receive lane 3 not locked and aligned. This field is available in 100G RS-FEC enabled configurations. In other configurations, this is a reserved field **Value After Reset:** 0x0 **Exists:** Always
10	FEC_AM_LOCK_2	R	FEC AM lock 2 This bit when set indicates whether the receiver has detected the location of the alignment marker payload sequence for FEC lane-2. ■ 1: RS-FEC receive lane 2 locked and aligned. ■ 0: RS-FEC receive lane 2 not locked and aligned. This field is available in 100G RS-FEC enabled configurations. In other configurations, this is a reserved field **Value After Reset:** 0x0 **Exists:** Always
9	FEC_AM_LOCK_1	R	FEC AM lock 1 This bit when set indicates whether the receiver has detected the location of the alignment marker payload sequence for FEC lane-1. ■ 1: RS-FEC receive lane 1 locked and aligned. ■ 0: RS-FEC receive lane 1 not locked and aligned. This field is available in 50G and 100G RS-FEC enabled configurations. In other configurations, this is a reserved field **Value After Reset:** 0x0 **Exists:** Always
8	FEC_AM_LOCK_0	R	FEC AM lock 0 This bit when set indicates whether the receiver has detected the location of the alignment marker payload sequence for FEC lane-0. ■ 1: RS-FEC receive lane 0 locked and aligned. ■ 0: RS-FEC receive lane 0 not locked and aligned. **Value After Reset:** 0x0 **Exists:** Always

Table 7-38 Fields for Register: SR_PMA_RS_FEC_STATUS (Continued)

Bits	Name	Memory Access	Description
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	DSER	R	RS-FEC Degraded SER ■ 1: Degraded Symbol Error Rate is present ■ 0: Degraded Symbol Error Rate is absent Degraded SER is flagged when the number of symbol errors detected in a window (determined by) is greater than the activation threshold specified by SR_PMA_RS_FEC_DEGRADED_SER_ACT_THLD_0 and SR_PMA_RS_FEC_DEGRADED_SER_ACT_THLD_1 registers. When asserted, degraded SER goes low, when the number of symbol errors detected in a block is lower than the deactivation threshold specified by SR_PMA_RS_FEC_DEGRADED_SER_DEACT_THLD_0 and SR_PMA_RS_FEC_DEGRADED_SER_DEACT_THLD_1 registers. This bit is valid only when operating in RS-FEC(544,514) mode. **Value After Reset:** 0x0 **Exists:** Always
3	DSER_ABL	R	RS-FEC Degraded SER ability This bit indicate if RS-FEC decoder can indicate presence of a degraded SER when operating in RS(544,514) mode. **Value After Reset:** "DWCXLP_RS544 ? 0x1 : 0x0" **Exists:** Always
2	HIG_SER	R	RS-FEC high SER (RO,LH Type) This bit indicates High Symbol Error (high_ser) status of RS-FEC decoder. High SER is asserted when the number of uncorrected RS-FEC code words in a window of 8192 code words exceeds 417. This condition is common across 25G,50G, and 100G RS-FEC enabled modes. ■ 1: RS-FEC errors have exceeded threshold. ■ 0: RS-FEC errors have not exceeded threshold. **Value After Reset:** 0x0 **Exists:** Always

Table 7-38 Fields for Register: SR_PMA_RS_FEC_STATUS (Continued)

Bits	Name	Memory Access	Description
1	BYP_INA	R	RS-FEC bypass indication ability This bit indicates the ability of RS-FEC decoder to bypass Error Indication. ■ 1: RS-FEC decoder has the ability to bypass error indication ■ 0: RS-FEC decoder does not have the ability to bypass error indication **Value After Reset:** 0x1 **Exists:** Always
0	BYP_CRA	R	RS-FEC Bypass Correction Ability This bit indicates ability of RS-FEC decoder to bypass error Correction ■ 1: RS-FEC decoder has the ability to bypass error correction ■ 0: RS-FEC decoder does not have the ability to bypass error correction **Value After Reset:** 0x1 **Exists:** Always

7.1.34 SR_PMA_RS_FEC_CORR_CW_CNTL

- **Description:** SR PMA MMD RS-FEC corrected codewords lower (RO, COR Type).

  This register holds the lower 16 bits of the Number of RS-FEC corrected code words.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xCA : (DWCXLP_APB3_16) ? (0xCA*0x2) : (0xCA*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_RSFEC_EN)



Table 7-39 Fields for Register: SR_PMA_RS_FEC_CORR_CW_CNTL

Bits	Name	Memory Access	Description
15:0	FEC_CCW_CNT_LOW	R	SR PMA MMD RS-FEC corrected codewords lower (RO, COR Type). This register holds the lower 16 bits of the Number of RS-FEC corrected code words. These bits must be held at all ones in case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.35 SR_PMA_RS_FEC_CORR_CW_CNTH

- **Description:** SR PMA MMD RS-FEC corrected codewords upper (RO, COR Type).

  This register holds the upper 16 bits of the Number of RS-FEC corrected code words.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xCB : (DWCXLP_APB3_16) ? (0xCB*0x2) : (0xCB*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_RSFEC_EN)



Table 7-40 Fields for Register: SR_PMA_RS_FEC_CORR_CW_CNTH

Bits	Name	Memory Access	Description
15:0	FEC_CCW_CNT_HIG	R	SR PMA MMD RS-FEC uncorrected codewords higher (RO, COR Type) This register holds the lower 16 bits of the Number of RS-FEC corrected code words. These bits must be held at all ones in case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.36 SR_PMA_RS_FEC_UCOR_CW_CNTL

- **Description:** SR PMA MMD RS-FEC uncorrected codewords lower (RO, COR Type).

  This register holds the lower 16 bits of the Number of RS-FEC uncorrected code words.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0xCC : (DWCXLP_APB3_16) ? (0xCC*0x2) : (0xCC*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_RSFEC_EN)



Table 7-41 Fields for Register: SR_PMA_RS_FEC_UCOR_CW_CNTL

Bits	Name	Memory Access	Description
15:0	FEC_UCW_CNT_LOW	R	SR PMA MMD RS-FEC uncorrected codewords lower (RO, COR Type) This register holds the lower 16 bits of the Number of RS-FEC uncorrected code words. These bits must be held at all ones in case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.37 SR_PMA_RS_FEC_UCOR_CW_CNTH

- **Description:** SR PMA MMD RS-FEC uncorrected codewords upper (RO, COR Type).

  This register holds the upper 16 bits of the Number of RS-FEC uncorrected code words.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xCD : (DWCXLP_APB3_16) ? (0xCD*0x2) : (0xCD*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_RSFEC_EN)



Table 7-42 Fields for Register: SR_PMA_RS_FEC_UCOR_CW_CNTH

Bits	Name	Memory Access	Description
15:0	FEC_UCW_CNT_HIG	R	SR PMA MMD RS-FEC uncorrected codewords upper (RO, COR Type) This register holds the lower 16 bits of the Number of RS-FEC uncorrected code words. These bits must be held at all ones in case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.38 SR_PMA_RS_FEC_LANE_MAP

- **Description:** SR PMA MMD RS-FEC lane mapping register.

  When the RS-FEC detects and locks the RS-FEC for PMA service interface lane 0, the detected RS-FEC lane number is recorded bits 1:0 in this register. Similarly, the detected RS-FEC lane numbers for PMA service lanes 1, 2, and 3 are recorded in register bits 3:2, 5:4, and 7:6, respectively.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xCE : (DWCXLP_APB3_16) ? (0xCE*0x2) : (0xCE*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_RSFEC_EN&&(DWCXLP_KR4_50G||DWCXLP_50G_25G||DWCXLP_100G_1LANE)

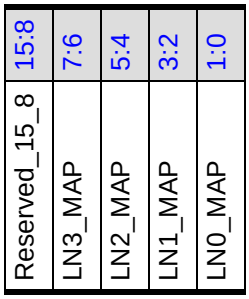


Table 7-43 Fields for Register: SR_PMA_RS_FEC_LANE_MAP

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:6	LN3_MAP	R	RS-FEC lane mapped to PMA lane 3 **Value After Reset:** 0x0 **Exists:** Always
5:4	LN2_MAP	R	RS-FEC lane mapped to PMA lane 2 **Value After Reset:** 0x0 **Exists:** Always
3:2	LN1_MAP	R	RS-FEC lane mapped to PMA lane 1 **Value After Reset:** 0x0 **Exists:** Always

Table 7-43 Fields for Register: SR_PMA_RS_FEC_LANE_MAP (Continued)

Bits	Name	Memory Access	Description
1:0	LN0_MAP	R	RS-FEC lane mapped to PMA lane 0 **Value After Reset:** 0x0 **Exists:** Always

7.1.39 SR_PMA_RS_FEC_CW_CNTL

- **Description:** SR PMA MMD RS-FEC CodeWord Count Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xCF : (DWCXLP_APB3_16) ? (0xCF*0x2) : (0xCF*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-44 Fields for Register: SR_PMA_RS_FEC_CW_CNTL

Bits	Name	Memory Access	Description
15:0	CW_CNTL	R	RS-FEC Code-word Counter lower 16-bits. DWC_xlgpcs implements a 48-bit counter, which increments whenever the RS-FEC receiver detects a code-word. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 48-bit count value. This register (and the 48-bit counter) is cleared on read. In order to get the entire 48-bit count value, user should first read this register, followed by reading the SR_PMA_RS_FEC_CW_CNTM and SR_PMA_RS_FEC_CW_CNTH registers **Value After Reset:** 0x0 **Exists:** Always

7.1.40 SR_PMA_RS_FEC_CW_CNTM

- **Description:** SR PMA MMD RS-FEC CodeWord Count Middle Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD0 : (DWCXLP_APB3_16) ? (0xD0*0x2) : (0xD0*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-45 Fields for Register: SR_PMA_RS_FEC_CW_CNTM

Bits	Name	Memory Access	Description
15:0	CW_CNTM	R	RS-FEC Code-word Counter Middle 16-bits. DWC_xlgpcs implements a 48-bit counter, which increments whenever the RS-FEC receiver detects a code-word. This counter saturates when it reaches it terminal value of all ones. This register holds the middle 16-bits of the 48-bit count value. This register (and the 48-bit counter) is cleared on read. **Value After Reset:** 0x0 **Exists:** Always

7.1.41 SR_PMA_RS_FEC_CW_CNTH

- **Description:** SR PMA MMD RS-FEC CodeWord Count High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD1 : (DWCXLP_APB3_16) ? (0xD1*0x2) : (0xD1*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-46 Fields for Register: SR_PMA_RS_FEC_CW_CNTH

Bits	Name	Memory Access	Description
15:0	CW_CNTH	R	RS-FEC Code-word Counter higher 16-bits. DWC_xlgpcs implements a 48-bit counter, which increments whenever the RS-FEC receiver detects a code-word. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 48-bit count value. This register (and the 48-bit counter) is cleared on read. **Value After Reset:** 0x0 **Exists:** Always

7.1.42 SR_PMA_RS_FEC_ERR_CNTL_L0

- **Description:** SR PMA MMD RS-FEC symbol error counter register (RO, COR Type).

  This register holds the lower 16 bits of the Number of RS-FEC symbol Errors of the 32 bit counter for RS-FEC lane-0.

  For uncorrectable CWs, the number of symbol errors detected will be a random value. Hence, the symbol error counter will increment by an arbitrary number.

  For correctable CWs, the number of symbol errors can be determined accurately and hence symbol error counter will increment deterministically.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD2 : (DWCXLP_APB3_16) ? (0xD2*0x2) : (0xD2*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_RSFEC_EN)

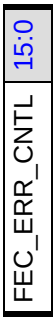


Table 7-47 Fields for Register: SR_PMA_RS_FEC_ERR_CNTL_L0

Bits	Name	Memory Access	Description
15:0	FEC_ERR_CNTL	R	This Register holds the lower 16 bits of the Number erroneous RS-FEC code words (RO, COR Type). These bits are held at all ones in case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.43 SR_PMA_RS_FEC_ERR_CNTH_L0

- **Description:** SR PMA MMD RS-FEC symbol error counter register (RO, COR Type).

  This register holds the upper 16 bits of the Number of RS-FEC symbol Errors of the 32 bit counter for RS-FEC lane-0

  For uncorrectable CWs, the number of symbol errors detected will be a random value. Hence, the symbol error counter will increment by an arbitrary number.

  For correctable CWs, the number of symbol errors can be determined accurately and hence symbol error counter will increment deterministically.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD3 : (DWCXLP_APB3_16) ? (0xD3*0x2) : (0xD3*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_RSFEC_EN)



Table 7-48 Fields for Register: SR_PMA_RS_FEC_ERR_CNTH_L0

Bits	Name	Memory Access	Description
15:0	FEC_ERR_CNTH	R	This Register holds the upper 16 bits of the Number erroneous RS-FEC code words (RO, COR Type). These bits are held at all ones in case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.44 SR_PMA_RS_FEC_ERR_CNTL_L1

- **Description:** SR PMA MMD RS-FEC symbol error counter register (RO, COR Type). This Register holds the lower 16 bits of the Number of RS-FEC symbol Errors of the 32 bit counter for RS-FEC lane-1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD4 : (DWCXLP_APB3_16) ? (0xD4*0x2) : (0xD4*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_KR4_50G||DWCXLP_50G_25G||DWCXLP_100G_1LANE)

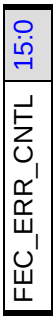


Table 7-49 Fields for Register: SR_PMA_RS_FEC_ERR_CNTL_L1

Bits	Name	Memory Access	Description
15:0	FEC_ERR_CNTL	R	This Register holds the lower 16 bits of the Number of erroneous RS-FEC symbols (RO, COR Type). These bits are held at all ones in case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.45 SR_PMA_RS_FEC_ERR_CNTH_L1

- **Description:** SR PMA MMD RS-FEC symbol error counter register (RO, COR Type).

  This register holds the upper 16 bits of the Number of RS-FEC symbol Errors of the 32 bit counter for RS-FEC lane-1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD5 : (DWCXLP_APB3_16) ? (0xD5*0x2) : (0xD5*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_KR4_50G||DWCXLP_50G_25G||DWCXLP_100G_1LANE)



Table 7-50 Fields for Register: SR_PMA_RS_FEC_ERR_CNTH_L1

Bits	Name	Memory Access	Description
15:0	FEC_ERR_CNTH	R	This Register holds the upper 16 bits of the Number of erroneous RS-FEC symbols (RO, COR Type). These bits are held at all ones in the case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.46 SR_PMA_RS_FEC_ERR_CNTL_L2

- **Description:** SR PMA MMD RS-FEC symbol error counter register (RO, COR Type).

  This register holds the lower 16 bits of the Number of RS-FEC symbol Errors of the 32 bit counter for RS-FEC lane-2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD6 : (DWCXLP_APB3_16) ? (0xD6*0x2) : (0xD6*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_KR4_100G||DWCXLP_100G_1LANE)

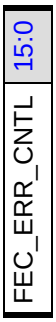


Table 7-51 Fields for Register: SR_PMA_RS_FEC_ERR_CNTL_L2

Bits	Name	Memory Access	Description
15:0	FEC_ERR_CNTL	R	This Register holds the lower 16 bits of the Number of erroneous RS-FEC symbols (RO, COR Type). These bits are held at all ones in case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.47 SR_PMA_RS_FEC_ERR_CNTH_L2

- **Description:** SR PMA MMD RS-FEC symbol error counter register (RO, COR Type).

  This register holds the upper 16 bits of the Number of RS-FEC symbol Errors of the 32 bit counter for RS-FEC lane-2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD7 : (DWCXLP_APB3_16) ? (0xD7*0x2) : (0xD7*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_KR4_100G||DWCXLP_100G_1LANE)



Table 7-52 Fields for Register: SR_PMA_RS_FEC_ERR_CNTH_L2

Bits	Name	Memory Access	Description
15:0	FEC_ERR_CNTH	R	This Register holds the upper 16 bits of the Number of erroneous RS-FEC symbols (RO, COR Type). These bits are held at all ones in the case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.48 SR_PMA_RS_FEC_ERR_CNTL_L3

- **Description:** SR PMA MMD RS-FEC symbol error counter register (RO, COR Type).

  This register holds the lower 16 bits of the Number of RS-FEC symbol Errors of the 32 bit counter for RS-FEC lane-3

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD8 : (DWCXLP_APB3_16) ? (0xD8*0x2) : (0xD8*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_KR4_100G||DWCXLP_100G_1LANE)

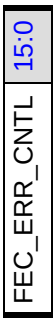


Table 7-53 Fields for Register: SR_PMA_RS_FEC_ERR_CNTL_L3

Bits	Name	Memory Access	Description
15:0	FEC_ERR_CNTL	R	This Register holds the lower 16 bits of the Number of erroneous RS-FEC symbols (RO, COR Type). These bits are held at all ones in the case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.49 SR_PMA_RS_FEC_ERR_CNTH_L3

- **Description:** SR PMA MMD RS-FEC symbol error counter register (RO, COR Type).

  This register holds the upper 16 bits of the Number of RS-FEC symbol Errors of the 32 bit counter for RS-FEC lane-3
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xD9 : (DWCXLP_APB3_16) ? (0xD9*0x2) : (0xD9*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_KR4_100G||DWCXLP_100G_1LANE)



Table 7-54 Fields for Register: SR_PMA_RS_FEC_ERR_CNTH_L3

Bits	Name	Memory Access	Description
15:0	FEC_ERR_CNTH	R	This Register holds the upper 16 bits of the Number of erroneous RS-FEC symbols (RO, COR Type). These bits are held at all ones in case of overflow. **Value After Reset:** 0x0 **Exists:** Always

7.1.50 SR_PMA_KR_FEC_CORR_BLK1_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD Multi lane FEC Correctable Error Block Counter 1 Register -Lane *i*

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with correctable errors after the FEC block lock is achieved. The total width of the error counter is 32-bit. This register holds the lower 16-bits of the 32-bits counter.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** (2*i)+0x1012c
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN&&DWCXLP_MULTI_LANE)

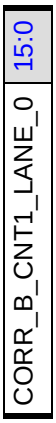


Table 7-55 Fields for Register: SR_PMA_KR_FEC_CORR_BLK1_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15:0	CORR_B_CNT1_LANE_0	R	FEC Corrected Blocks Lower (RO, COR Type) This field indicates the FEC correctable error block counter. These are the lower 16-bits of the counter. This is a non-roll over counter. This field is cleared when this register is read. **Value After Reset:** 0x0 **Exists:** Always

7.1.51 SR_PMA_KR_FEC_CORR_BLK2_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD Multi lane FEC Correctable Error Block Counter 2 Register -Lane *i*

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with correctable errors after the FEC block lock is achieved. The total width of the error counter is 32-bit. This register holds the higher 16-bits of the 32-bits counter.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** (2*i)+0x1012d
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN&&DWCXLP_MULTI_LANE)

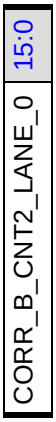


Table 7-56 Fields for Register: SR_PMA_KR_FEC_CORR_BLK2_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15:0	CORR_B_CNT2_LANE_0	R	FEC Corrected Blocks Lower (RO, COR Type) This field indicates the FEC correctable error block counter. These are the higher 16-bits of the counter. This is a non-roll over counter. This field is cleared when SR PMA MMD Multi lane FEC Correctable Error Block Counter 1 -Lane *i* register is read. **Value After Reset:** 0x0 **Exists:** Always

7.1.52 SR_PMA_RS_FEC_CW_ERR_BIN1L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin1 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x154 : (DWCXLP_APB3_16) ? (0x154*0x2) : (0x154*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-57 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN1L

Bits	Name	Memory Access	Description
15:0	BIN1L	R	RS-FEC Code-word Error Bin1 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with one symbol error. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.53 SR_PMA_RS_FEC_CW_ERR_BIN1H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin1 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x155 : (DWCXLP_APB3_16) ? (0x155*0x2) : (0x155*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-58 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN1H

Bits	Name	Memory Access	Description
15:0	BIN1H	R	RS-FEC Code-word Error Bin1 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with one symbol error. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.54 SR_PMA_RS_FEC_CW_ERR_BIN2L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin2 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x156 : (DWCXLP_APB3_16) ? (0x156*0x2) : (0x156*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

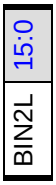


Table 7-59 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN2L

Bits	Name	Memory Access	Description
15:0	BIN2L	R	RS-FEC Code-word Error Bin2 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with two symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.55 SR_PMA_RS_FEC_CW_ERR_BIN2H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin2 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x157 : (DWCXLP_APB3_16) ? (0x157*0x2) : (0x157*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

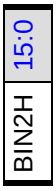


Table 7-60 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN2H

Bits	Name	Memory Access	Description
15:0	BIN2H	R	RS-FEC Code-word Error Bin2 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with two symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.56 SR_PMA_RS_FEC_CW_ERR_BIN3L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin3 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x158 : (DWCXLP_APB3_16) ? (0x158*0x2) : (0x158*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

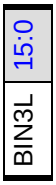


Table 7-61 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN3L

Bits	Name	Memory Access	Description
15:0	BIN3L	R	RS-FEC Code-word Error Bin3 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with three symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.57 SR_PMA_RS_FEC_CW_ERR_BIN3H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin3 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x159 : (DWCXLP_APB3_16) ? (0x159*0x2) : (0x159*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

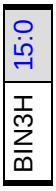


Table 7-62 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN3H

Bits	Name	Memory Access	Description
15:0	BIN3H	R	RS-FEC Code-word Error Bin3 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with three symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.58 SR_PMA_RS_FEC_CW_ERR_BIN4L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin4 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x15A : (DWCXLP_APB3_16) ? (0x15A*0x2) : (0x15A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

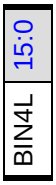


Table 7-63 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN4L

Bits	Name	Memory Access	Description
15:0	BIN4L	R	RS-FEC Code-word Error Bin4 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with four symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.59 SR_PMA_RS_FEC_CW_ERR_BIN4H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin4 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x15B : (DWCXLP_APB3_16) ? (0x15B*0x2) : (0x15B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-64 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN4H

Bits	Name	Memory Access	Description
15:0	BIN4H	R	RS-FEC Code-word Error Bin4 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with four symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.60 SR_PMA_RS_FEC_CW_ERR_BIN5L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin5 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x15C : (DWCXLP_APB3_16) ? (0x15C*0x2) : (0x15C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

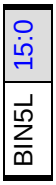


Table 7-65 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN5L

Bits	Name	Memory Access	Description
15:0	BIN5L	R	RS-FEC Code-word Error Bin5 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with five symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.61 SR_PMA_RS_FEC_CW_ERR_BIN5H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin5 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x15D : (DWCXLP_APB3_16) ? (0x15D*0x2) : (0x15D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

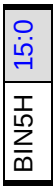


Table 7-66 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN5H

Bits	Name	Memory Access	Description
15:0	BIN5H	R	RS-FEC Code-word Error Bin5 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with five symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.62 SR_PMA_RS_FEC_CW_ERR_BIN6L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin6 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x15E : (DWCXLP_APB3_16) ? (0x15E*0x2) : (0x15E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

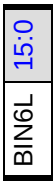


Table 7-67 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN6L

Bits	Name	Memory Access	Description
15:0	BIN6L	R	RS-FEC Code-word Error Bin6 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with six symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.63 SR_PMA_RS_FEC_CW_ERR_BIN6H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin6 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x15F : (DWCXLP_APB3_16) ? (0x15F*0x2) : (0x15F*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

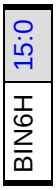


Table 7-68 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN6H

Bits	Name	Memory Access	Description
15:0	BIN6H	R	RS-FEC Code-word Error Bin6 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with six symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.64 SR_PMA_RS_FEC_CW_ERR_BIN7L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin7 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x160 : (DWCXLP_APB3_16) ? (0x160*0x2) : (0x160*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-69 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN7L

Bits	Name	Memory Access	Description
15:0	BIN7L	R	RS-FEC Code-word Error Bin7 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with seven symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.65 SR_PMA_RS_FEC_CW_ERR_BIN7H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin7 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x161 : (DWCXLP_APB3_16) ? (0x161*0x2) : (0x161*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-70 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN7H

Bits	Name	Memory Access	Description
15:0	BIN7H	R	RS-FEC Code-word Error Bin7 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with seven symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.66 SR_PMA_RS_FEC_CW_ERR_BIN8L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin8 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x162 : (DWCXLP_APB3_16) ? (0x162*0x2) : (0x162*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

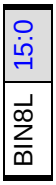


Table 7-71 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN8L

Bits	Name	Memory Access	Description
15:0	BIN8L	R	RS-FEC Code-word Error Bin8 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with eight symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.67 SR_PMA_RS_FEC_CW_ERR_BIN8H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin8 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x163 : (DWCXLP_APB3_16) ? (0x163*0x2) : (0x163*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

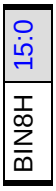


Table 7-72 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN8H

Bits	Name	Memory Access	Description
15:0	BIN8H	R	RS-FEC Code-word Error Bin8 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with eight symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.68 SR_PMA_RS_FEC_CW_ERR_BIN9L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin9 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x164 : (DWCXLP_APB3_16) ? (0x164*0x2) : (0x164*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-73 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN9L

Bits	Name	Memory Access	Description
15:0	BIN9L	R	RS-FEC Code-word Error Bin9 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with nine symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.69 SR_PMA_RS_FEC_CW_ERR_BIN9H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin9 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x165 : (DWCXLP_APB3_16) ? (0x165*0x2) : (0x165*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

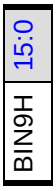


Table 7-74 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN9H

Bits	Name	Memory Access	Description
15:0	BIN9H	R	RS-FEC Code-word Error Bin9 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with nine symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.70 SR_PMA_RS_FEC_CW_ERR_BIN10L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin10 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x166 : (DWCXLP_APB3_16) ? (0x166*0x2) : (0x166*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-75 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN10L

Bits	Name	Memory Access	Description
15:0	BIN10L	R	RS-FEC Code-word Error Bin10 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with ten symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.71 SR_PMA_RS_FEC_CW_ERR_BIN10H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin10 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x167 : (DWCXLP_APB3_16) ? (0x167*0x2) : (0x167*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-76 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN10H

Bits	Name	Memory Access	Description
15:0	BIN10H	R	RS-FEC Code-word Error Bin10 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with ten symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.72 SR_PMA_RS_FEC_CW_ERR_BIN11L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin11 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x168 : (DWCXLP_APB3_16) ? (0x168*0x2) : (0x168*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-77 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN11L

Bits	Name	Memory Access	Description
15:0	BIN11L	R	RS-FEC Code-word Error Bin11 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with eleven symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.73 SR_PMA_RS_FEC_CW_ERR_BIN11H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin11 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x169 : (DWCXLP_APB3_16) ? (0x169*0x2) : (0x169*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-78 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN11H

Bits	Name	Memory Access	Description
15:0	BIN11H	R	RS-FEC Code-word Error Bin11 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with eleven symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.74 SR_PMA_RS_FEC_CW_ERR_BIN12L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin12 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x16A : (DWCXLP_APB3_16) ? (0x16A*0x2) : (0x16A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

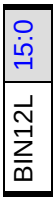


Table 7-79 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN12L

Bits	Name	Memory Access	Description
15:0	BIN12L	R	RS-FEC Code-word Error Bin12 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with twelve symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.75 SR_PMA_RS_FEC_CW_ERR_BIN12H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin12 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x16B : (DWCXLP_APB3_16) ? (0x16B*0x2) : (0x16B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

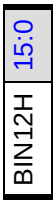


Table 7-80 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN12H

Bits	Name	Memory Access	Description
15:0	BIN12H	R	RS-FEC Code-word Error Bin12 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with twelve symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.76 SR_PMA_RS_FEC_CW_ERR_BIN13L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin13 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x16C : (DWCXLP_APB3_16) ? (0x16C*0x2) : (0x16C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

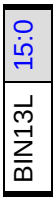


Table 7-81 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN13L

Bits	Name	Memory Access	Description
15:0	BIN13L	R	RS-FEC Code-word Error Bin13 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with thirteen symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.77 SR_PMA_RS_FEC_CW_ERR_BIN13H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin13 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x16D : (DWCXLP_APB3_16) ? (0x16D*0x2) : (0x16D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-82 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN13H

Bits	Name	Memory Access	Description
15:0	BIN13H	R	RS-FEC Code-word Error Bin13 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with thirteen symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.78 SR_PMA_RS_FEC_CW_ERR_BIN14L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin14 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x16E : (DWCXLP_APB3_16) ? (0x16E*0x2) : (0x16E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC

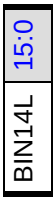


Table 7-83 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN14L

Bits	Name	Memory Access	Description
15:0	BIN14L	R	RS-FEC Code-word Error Bin14 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with fourteen symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.79 SR_PMA_RS_FEC_CW_ERR_BIN14H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin14 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x16F : (DWCXLP_APB3_16) ? (0x16F*0x2) : (0x16F*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-84 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN14H

Bits	Name	Memory Access	Description
15:0	BIN14H	R	RS-FEC Code-word Error Bin14 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with fourteen symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.80 SR_PMA_RS_FEC_CW_ERR_BIN15L

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin15 Low Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x170 : (DWCXLP_APB3_16) ? (0x170*0x2) : (0x170*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-85 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN15L

Bits	Name	Memory Access	Description
15:0	BIN15L	R	RS-FEC Code-word Error Bin15 lower 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with fifteen symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the lower 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.81 SR_PMA_RS_FEC_CW_ERR_BIN15H

- **Description:** SR PMA MMD RS-FEC CodeWord Error Bin15 High Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x171 : (DWCXLP_APB3_16) ? (0x171*0x2) : (0x171*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_IRSFEC



Table 7-86 Fields for Register: SR_PMA_RS_FEC_CW_ERR_BIN15H

Bits	Name	Memory Access	Description
15:0	BIN15H	R	RS-FEC Code-word Error Bin15 higher 16-bits. DWC_xlgpcs implements a 32-bit counter, which increments whenever the RS-FEC receiver detects a code-word with fifteen symbol errors. This counter saturates when it reaches it terminal value of all ones. This register holds the higher 16-bits of the 32-bit count value. This register (and the 32-bit counter) is cleared on read. This register is not valid when BYP_CRE is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.82 SR_PMA_RS_FEC_DEGRADED_SER_ACT_THLD_0

- **Description:** SR PMA MMD RS-FEC Degraded SER Activate Threshold 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x28A : (DWCXLP_APB3_16) ? (0x28A*0x2) : (0x28A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_RS544



Table 7-87 Fields for Register: SR_PMA_RS_FEC_DEGRADED_SER_ACT_THLD_0

Bits	Name	Memory Access	Description
15:0	ACT_TH_LWR	R/W	FEC degraded SER activate threshold lower This register controls the lower 16-bit of the 32-bit activate threshold for degraded SER feature when operating in RS(544,514) mode. **Value After Reset:** 0xffff **Exists:** Always

7.1.83 SR_PMA_RS_FEC_DEGRADED_SER_ACT_THLD_1

- **Description:** SR PMA MMD RS-FEC Degraded SER Activate Threshold 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x28B : (DWCXLP_APB3_16) ? (0x28B*0x2) : (0x28B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_RS544

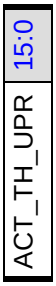


Table 7-88 Fields for Register: SR_PMA_RS_FEC_DEGRADED_SER_ACT_THLD_1

Bits	Name	Memory Access	Description
15:0	ACT_TH_UPR	R/W	FEC degraded SER activate threshold upper This register controls the upper 16-bit of the 32-bit activate threshold for degraded SER feature when operating in RS(544,514) mode. **Value After Reset:** 0x0 **Exists:** Always

7.1.84 SR_PMA_RS_FEC_DEGRADED_SER_DEACT_THLD_0

- **Description:** SR PMA MMD RS-FEC Degraded SER Deactivate Threshold 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x28C : (DWCXLP_APB3_16) ? (0x28C*0x2) : (0x28C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_RS544



Table 7-89 Fields for Register: SR_PMA_RS_FEC_DEGRADED_SER_DEACT_THLD_0

Bits	Name	Memory Access	Description
15:0	DEACT_TH_LWR	R/W	FEC degraded SER deactivate threshold lower This register controls the lower 16-bit of the 32-bit deactivate threshold for degraded SER feature when operating in RS(544,514) mode. **Value After Reset:** 0x1000 **Exists:** Always

7.1.85 SR_PMA_RS_FEC_DEGRADED_SER_DEACT_THLD_1

- **Description:** SR PMA MMD RS-FEC Degraded SER Deactivate Threshold 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x28D : (DWCXLP_APB3_16) ? (0x28D*0x2) : (0x28D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_RS544



Table 7-90 Fields for Register: SR_PMA_RS_FEC_DEGRADED_SER_DEACT_THLD_1

Bits	Name	Memory Access	Description
15:0	DEACT_TH_UPR	R/W	FEC degraded SER deactivate threshold upper This register controls the upper 16-bit of the 32-bit deactivate threshold for degraded SER feature when operating in RS(544,514) mode. **Value After Reset:** 0x0 **Exists:** Always

7.1.86 SR_PMA_RS_FEC_DEGRADED_SER_INTRVL_0

- **Description:** SR PMA MMD RS-FEC Degraded SER Interval 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x28E : (DWCXLP_APB3_16) ? (0x28E*0x2) : (0x28E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_RS544



Table 7-91 Fields for Register: SR_PMA_RS_FEC_DEGRADED_SER_INTRVL_0

Bits	Name	Memory Access	Description
15:0	INTRVL_LWR	R/W	FEC degraded SER interval lower This register controls the lower 16-bit of the 32-bit interval (in terms of FEC code-words) for degraded SER feature when operating in RS(544,514) mode. Note:When operating in Clause 161 based interleaved RS-FEC mode, only a even number must be programmed to this register. **Value After Reset:** 0xffff **Exists:** Always

7.1.87 SR_PMA_RS_FEC_DEGRADED_SER_INTRVL_1

- **Description:** SR PMA MMD RS-FEC Degraded SER Interval 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x28F : (DWCXLP_APB3_16) ? (0x28F*0x2) : (0x28F*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_RS544

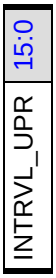


Table 7-92 Fields for Register: SR_PMA_RS_FEC_DEGRADED_SER_INTRVL_1

Bits	Name	Memory Access	Description
15:0	INTRVL_UPR	R/W	FEC degraded SER interval upper This register controls the upper 16-bit of the 32-bit interval (in terms of FEC code-words) for degraded SER feature when operating in RS(544,514) mode. **Value After Reset:** 0x0 **Exists:** Always

7.1.88 SR_PMA_KR_FEC_UCORR_BLK1_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD Multi lane FEC Uncorrectable Error Block Counter 1 Register -Lane *i*

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with uncorrectable errors after the FEC block lock is achieved. The total width of the Error counter is 32-bit. This register holds the lower 16-bits of the 32-bits counter.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** (2*i)+0x102bc
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN&&DWCXLP_MULTI_LANE)



Table 7-93 Fields for Register: SR_PMA_KR_FEC_UCORR_BLK1_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15:0	UCORR_B_CNT1_LANE_0	R	FEC Uncorrected Blocks Lower (RO, COR Type) This field indicates the FEC uncorrectable error block counter. These are the lower 16-bits of the counter. This is a non-roll over counter. This register is cleared when it is read. **Value After Reset:** 0x0 **Exists:** Always

7.1.89 SR_PMA_KR_FEC_UCORR_BLK2_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD Multi lane FEC Uncorrectable Error Block Counter 2 Register -Lane *i*

  This register is present only if DWCXLP_FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with uncorrectable errors after the FEC block lock is achieved. The total width of the Error counter is 32-bit. This register holds the higher 16-bits of the 32-bits counter.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** (2*i)+0x102bd
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_FEC_EN&&DWCXLP_MULTI_LANE)

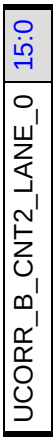


Table 7-94 Fields for Register: SR_PMA_KR_FEC_UCORR_BLK2_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15:0	UCORR_B_CNT2_LANE_0	R	FEC Uncorrected Blocks Lower (RO, COR Type) This field indicates the FEC uncorrectable error block counter. These are the lower 16-bits of the counter. This is a non-roll over counter. This register is cleared when SR PMA MMD 40GBASE-KR Lane *i* FEC Uncorrectable Error Block Counter 1 Register is read. **Value After Reset:** 0x0 **Exists:** Always

7.1.90 SR_PMA_MLN_KR_LP_CEU_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD BASE-KR LP Coefficient Update Register-Lane *i*
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** i+0x1044c
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE)

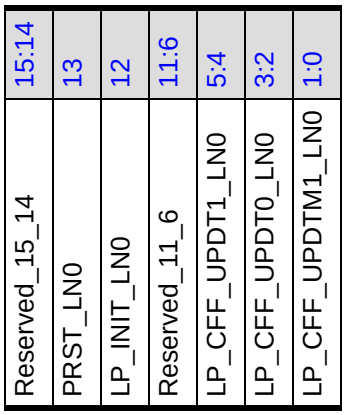


Table 7-95 Fields for Register: SR_PMA_MLN_KR_LP_CEU_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	PRST_LN0	R/W	Preset Coefficient When this bit is set to 1, it indicates that the link partner requested the transmitter filter coefficients of the local device to be set to a state where transmit equalization is turned off. This means that the filter coefficient value, corresponding to filter (-1) and filter (+1), should be zero and the filter coefficient filter, corresponding to filter (0), should be maximum. ■ 1: Pre-initialize coefficients ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always

Table 7-95 Fields for Register: SR_PMA_MLN_KR_LP_CEU_LANE (i) (for i = 0; i <= 3) (Continued)

Bits	Name	Memory Access	Description
12	LP_INIT_LN0	R/W	Initialize Coefficient When this bit is set to 1, it indicates that the link partner requested the transmitter filter coefficients of the local device to be initialized. The initialize requirements of the filter coefficient is defined in IEEE Std 802.3, Clause 72, section 72.6.10.4.2. This request is usually received at the beginning of the start-up protocol. ■ 1: Initialize the local device transmit filter coefficient ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	LP_CFF_UPDT1_LN0	R/W	Co-efficient Update (+1) When this bit is set to 1, it indicates that the link partner requested the local device to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	LP_CFF_UPDT0_LN0	R/W	Co-efficient Update (0) When this bit is set to 1, it indicates that the link partner requested the local device to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Table 7-95 Fields for Register: SR_PMA_MLN_KR_LP_CEU_LANE (i) (for i = 0; i <= 3) (Continued)

Bits	Name	Memory Access	Description
1:0	LP_CFF_UPDTM1_LN0	R/W	Coefficient Update (-1) When this bit is set to 1, it indicates that the link partner requested the local device to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.91 SR_PMA_MLN_KR_LP_CESTS_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD BASE-KR LP Coefficient Status Register-Lane *i*

  This register is present when the Link Training support is enabled. This register indicates the most recently received training status field from the link partner (LP).
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** i+0x104b0
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE)

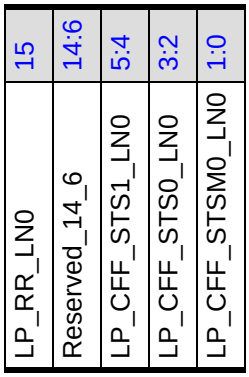


Table 7-96 Fields for Register: SR_PMA_MLN_KR_LP_CESTS_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15	LP_RR_LN0	R	Receiver Ready When this bit is set to 1, it indicates that the transmit link is reliable to transmit the KR speed data. The link partner sends this information based on successful and consistent reception of the PRBS data field in the KR Training frame (transmitted by the local device). ■ 1: The LP receiver has determined that the training is complete, and it is ready to receive data. ■ 0: The LP receiver is requesting that the training should be continued. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-96 Fields for Register: SR_PMA_MLN_KR_LP_CESTS_LANE (i) (for i = 0; i <= 3) (Continued)

Bits	Name	Memory Access	Description
5:4	LP_CFF_STS1_LN0	R	Coefficient Status (+1) This field indicates the status from the link partner for the (increment, decrement, or hold) request sent by the local device for transmit filter (+1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	LP_CFF_STS0_LN0	R	Coefficient Status (0) This field indicates the status from the link partner for the (increment, decrement, or hold) request sent by the local device for transmit filter (0). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	LP_CFF_STSM0_LN0	R	Coefficient Status (-1) This field indicates the status from the link partner for the (increment, decrement, or hold) request sent by the local device for transmit filter (-1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

7.1.92 SR_PMA_MLN_KR_LD_CEU_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD BASE-KR LD Coefficient Update Register-Lane *i*

  This register is present when the Link Training support is enabled. This register indicates the most-recently transmitted training control field from the local device (LD) to the link partner.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** i+0x10514
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE)

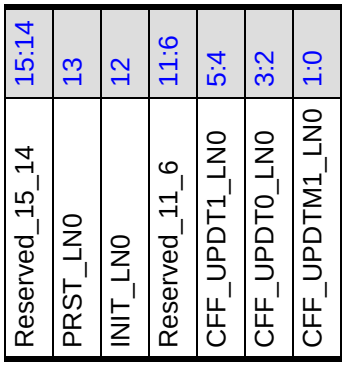


Table 7-97 Fields for Register: SR_PMA_MLN_KR_LD_CEU_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	PRST_LN0	R	Pre-initialize the Transmit Filter Coefficient of the Link Partner When this bit is set to 1, it indicates that the local device is requesting the transmitter filter coefficients of link partner to be set to a state where transmit equalization is turned off. This means the filter coefficient value, corresponding to filter (-1) and filter (+1), should be zero and the filter coefficient value, corresponding to filter (0), should be maximum. ■ 1: Preset coefficients ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always

Table 7-97 Fields for Register: SR_PMA_MLN_KR_LD_CEU_LANE (i) (for i = 0; i <= 3) (Continued)

Bits	Name	Memory Access	Description
12	INIT_LN0	R	Initialize the Transmit Filter Coefficient of the Link Partner When this bit is set to 1, it indicates that the local device requested the transmitter filter coefficients of link partner to be initialized. The initialize requirements of the filter coefficient is defined in IEEE Std 802.3, Clause 72, section 72.6.10.4.2. This request is usually received at the beginning of the start-up protocol. ■ 1: Initialize the local device transmit filter coefficient ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1_LN0	R	Filter Coefficient Update (+1) Request to the Link Partner Transmitter. When this bit is set to 1, it indicates that the local device requested the link partner to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0_LN0	R	Filter Coefficient Update (0) Request to the Link Partner Transmitter When this bit is set to 1, it indicates that the local device requested the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Table 7-97 Fields for Register: SR_PMA_MLN_KR_LD_CEU_LANE (i) (for i = 0; i <= 3) (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1_LN0	R	Filter Coefficient Update (-1) Request to the Link Partner Transmitter When this bit is set to 1, it indicates that the local device requested the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.93 SR_PMA_MLN_KR_LD_CESTS_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD BASE-KR LD Coefficient Status Register-Lane *i*

  This register is present only for the PMA MMD when the Link Training support is enabled. This register indicates the most-recently transmitted training status field from the local device (LD) to the link partner.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** i+0x10578
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_MULTI_LANE)

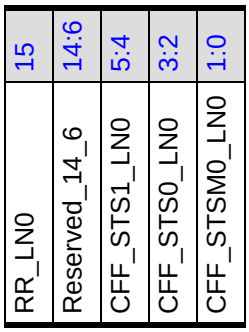


Table 7-98 Fields for Register: SR_PMA_MLN_KR_LD_CESTS_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15	RR_LN0	R	Receiver Ready When this bit is set to 1, it indicates that the receive link is reliable to receive the KR speed data. The local device Rx adaptation process sets this bit based on successful and consistent reception of the PRBS data field in received training frame at KR speed. ■ 1: The receiver of local device has determined that the training is complete, and it is ready to receive data. ■ 0: The receiver of local device is requesting that the training should be continued. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-98 Fields for Register: SR_PMA_MLN_KR_LD_CESTS_LANE (i) (for i = 0; i <= 3) (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_STS1_LN0	R	Filter Coefficient Status (+1) from Link Partner Transmitter to Local Device. This field indicates the status from the local device for the (increment, decrement, or hold) request received from the link partner for transmit filter (+1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0_LN0	R	Filter Coefficient Status (0) from Link Partner Transmitter to Local Device. This field indicates the status from the local device for the (increment, decrement, or hold) request received from the link partner for transmit filter (0). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_STSM0_LN0	R	Filter Coefficient Status (-1) from Link Partner Transmitter to Local Device This field indicates the status from the local device for the (increment, decrement, or hold) request received from the link partner for transmit filter (-1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

7.1.94 SR_PMA_LNK_TRN_PAT

- **Description:** SR PMA MMD Link Training Pattern Register

  This register is present only if Link Training is Enabled. Otherwise, this register is reserved. This register indicates the Polynomial identifier and Seed values for PMD training pattern for Lane-0.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x5AA : (DWCXLP_APB3_16) ? (0x5AA*0x2) : (0x5AA*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&!DWCXLP_KR4_KR

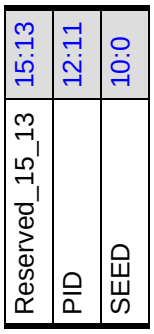


Table 7-99 Fields for Register: SR_PMA_LNK_TRN_PAT

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:11	PID	R/W	Polynomial Identifier PID[1:0]. ■ PID[1:0] = 00 : G(x) = 1 + x5 + x6 + x10 + x11 ■ PID[1:0] = 01 : G(x) = 1 + x5 + x6 + x9 + x11 ■ PID[1:0] = 10 : G(x) = 1 + x4 + x6 + x8 + x11 ■ PID[1:0] = 11 : G(x) = 1 + x4 + x6 + x7 + x11 **Value After Reset:** 0x0 **Exists:** Always
10:0	SEED	R/W	11 bit binary seed for sequence. **Value After Reset:** 0x3f5 **Exists:** Always

7.1.95 SR_PMA_LNK_TRN_PAT_LN1

- **Description:** SR PMA MMD Lane-1 Link Training Pattern Register

  This register is present only if DWCXLP_LNK_TRAIN_EN = Enabled. Otherwise, this register is reserved. This register indicates the Polynomial identifier and Seed values for PMD training pattern for Lane-1.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x5AB : (DWCXLP_APB3_16) ? (0x5AB*0x2) : (0x5AB*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_KR4_50G

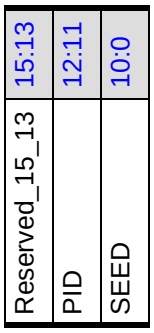


Table 7-100 Fields for Register: SR_PMA_LNK_TRN_PAT_LN1

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:11	PID	R/W	Polynomial Identifier PID[1:0]. ■ PID[1:0] = 00 : G(x) = 1 + x5 + x6 + x10 + x11 ■ PID[1:0] = 01 : G(x) = 1 + x5 + x6 + x9 + x11 ■ PID[1:0] = 10 : G(x) = 1 + x4 + x6 + x8 + x11 ■ PID[1:0] = 11 : G(x) = 1 + x4 + x6 + x7 + x11 **Value After Reset:** 0x0 **Exists:** Always
10:0	SEED	R/W	11 bit binary seed for sequence. **Value After Reset:** 0x513 **Exists:** Always

7.1.96 SR_PMA_LNK_TRN_PAT_LN2

- **Description:** SR PMA MMD Lane-2 Link Training Pattern Register.

  This register is present only if DWCXLP_LNK_TRAIN_EN = Enabled. Otherwise, this register is reserved. This register indicates the Polynomial identifier and Seed values for PMD training pattern for Lane-2.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x5AC : (DWCXLP_APB3_16) ? (0x5AC*0x2) : (0x5AC*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_KR4_100G

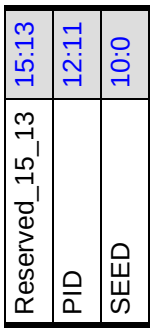


Table 7-101 Fields for Register: SR_PMA_LNK_TRN_PAT_LN2

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:11	PID	R/W	Polynomial Identifier PID[1:0]. ■ PID[1:0] = 00 : G(x) = 1 + x5 + x6 + x10 + x11 ■ PID[1:0] = 01 : G(x) = 1 + x5 + x6 + x9 + x11 ■ PID[1:0] = 10 : G(x) = 1 + x4 + x6 + x8 + x11 ■ PID[1:0] = 11 : G(x) = 1 + x4 + x6 + x7 + x11 **Value After Reset:** 0x0 **Exists:** Always
10:0	SEED	R/W	11 bit binary seed for sequence. **Value After Reset:** 0x5a7 **Exists:** Always

7.1.97 SR_PMA_LNK_TRN_PAT_LN3

- **Description:** SR PMA MMD Lane-3 Link Training Pattern Register

  This register is present only if DWCXLP_LNK_TRAIN_EN = Enabled. Otherwise, this register is reserved. This register indicates the Polynomial identifier and Seed values for PMD training pattern for Lane-3.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x5AD : (DWCXLP_APB3_16) ? (0x5AD*0x2) : (0x5AD*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&DWCXLP_KR4_100G

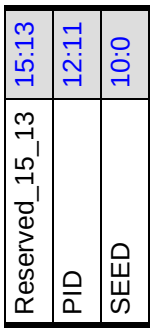


Table 7-102 Fields for Register: SR_PMA_LNK_TRN_PAT_LN3

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:11	PID	R/W	Polynomial Identifier PID[1:0]. ■ PID[1:0] = 00 : G(x) = 1 + x5 + x6 + x10 + x11 ■ PID[1:0] = 01 : G(x) = 1 + x5 + x6 + x9 + x11 ■ PID[1:0] = 10 : G(x) = 1 + x4 + x6 + x8 + x11 ■ PID[1:0] = 11 : G(x) = 1 + x4 + x6 + x7 + x11 **Value After Reset:** 0x0 **Exists:** Always
10:0	SEED	R/W	11 bit binary seed for sequence. **Value After Reset:** 0x36f **Exists:** Always

7.1.98 SR_PMA_KR_TST_PAT_ABL

- **Description:** SR PMA MMD 25GBASE-CR/KR, 40G/100GBASE-CR4/KR4 Test Pattern Ability Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x5dc : (DWCXLP_APB3_16) ? (0x5dc*0x2) : (0x5dc*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&(DWCXLP_MULTI_LANE||!DWCXLP_KR4_KR))

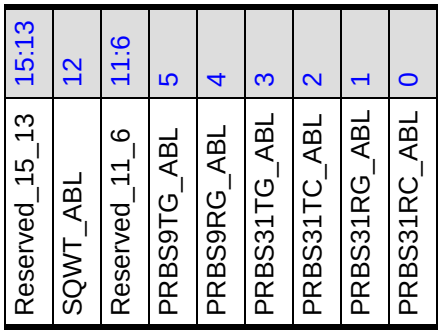


Table 7-103 Fields for Register: SR_PMA_KR_TST_PAT_ABL

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	SQWT_ABL	R	Square Wave Test Pattern Ability This bit indicates if PMA supports square wave test pattern. **Value After Reset:** 0x1 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	PRBS9TG_ABL	R	PRBS9 Tx Generator Ability This bit indicates if PMA supports transmit direction pattern generator. **Value After Reset:** 0x1 **Exists:** Always

Table 7-103 Fields for Register: SR_PMA_KR_TST_PAT_ABL (Continued)

Bits	Name	Memory Access	Description
4	PRBS9RG_ABL	R	PRBS9 Rx Generator Ability This bit indicates if PMA supports receive direction pattern generator. **Value After Reset:** 0x0 **Exists:** Always
3	PRBS31TG_ABL	R	PRBS31 Tx Generator Ability This bit indicates if PMA supports transmit direction pattern generator. **Value After Reset:** 0x1 **Exists:** Always
2	PRBS31TC_ABL	R	PRBS31 Tx Checker Ability This bit indicates if PMA supports transmit direction pattern checker. **Value After Reset:** 0x0 **Exists:** Always
1	PRBS31RG_ABL	R	PRBS31 Rx Generator Ability This bit indicates if PMA supports receive direction pattern generator. **Value After Reset:** 0x0 **Exists:** Always
0	PRBS31RC_ABL	R	PRBS31 Rx Checker Ability This bit indicates if PMA supports PRBS31 receive direction pattern checker. **Value After Reset:** 0x1 **Exists:** Always

7.1.99 SR_PMA_KR_TST_PAT_CTRL

- **Description:** SR PMA MMD 25GBASE-CR/KR, 40G/100GBASE-CG4/KR4 Test Pattern Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x5dd : (DWCXLP_APB3_16) ? (0x5dd*0x2) : (0x5dd*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&(DWCXLP_MULTI_LANE||!DWCXLP_KR4_KR))

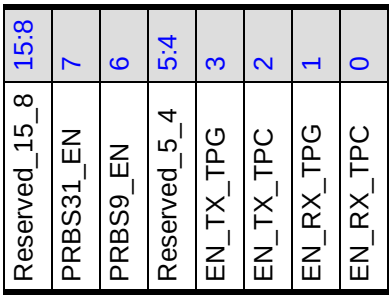


Table 7-104 Fields for Register: SR_PMA_KR_TST_PAT_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	PRBS31_EN	R/W	PRBS31 Test Pattern Enable ■ 1: Enable PRBS31 test pattern when bit 3 (EN_TX_TPG) is set to 1 ■ 0: Disable PRBS31 test pattern mode **Value After Reset:** 0x0 **Exists:** Always
6	PRBS9_EN	R/W	PRBS9 Test Pattern Enable ■ 1: Enable PRBS9 test pattern when bit 3 (EN_TX_TPG) is set to 1 ■ 0: Disable PRBS9 test pattern mode **Value After Reset:** 0x0 **Exists:** Always
5:4	Reserved_5_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-104 Fields for Register: SR_PMA_KR_TST_PAT_CTRL (Continued)

Bits	Name	Memory Access	Description
3	EN_TX_TPG	R/W	Transmit Generator Enable ■ 1: Enable transmit direction test-pattern generator. ■ 0: Disable transmit direction test-pattern generator. **Value After Reset:** 0x0 **Exists:** Always
2	EN_TX_TPC	R	Transmit Checker Enable ■ 1: Enable transmit direction test-pattern test-pattern checker ■ 0: Disable transmit direction test-pattern test-pattern checker **Value After Reset:** 0x0 **Exists:** Always
1	EN_RX_TPG	R	Rx Generator Enable ■ 1: Enable receive direction test-pattern generator ■ 0: Disable receive direction test-pattern generator **Value After Reset:** 0x0 **Exists:** Always
0	EN_RX_TPC	R/W	Rx Checker Enable ■ 1: Enable receive direction test-pattern checker ■ 0: Disable receive direction test-pattern checker **Value After Reset:** 0x0 **Exists:** Always

7.1.100 SR_PMA_KR_SQUARE_WAVE_CTRL

- **Description:** SR PMA MMD 25G/40G/100G Test Pattern Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x5e6 : (DWCXLP_APB3_16) ? (0x5e6*0x2) : (0x5e6*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&(DWCXLP_MULTI_LANE||!DWCXLP_KR4_KR))

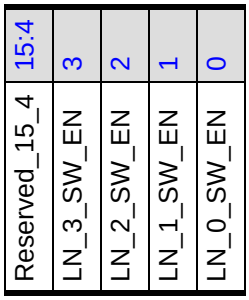


Table 7-105 Fields for Register: SR_PMA_KR_SQUARE_WAVE_CTRL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	LN_3_SW_EN	* Varies	Square Wave Enable on Lane 3. Bits [6:4] of VR PCS BASE-R Control Register specify the period of the square wave **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!DWCXLP_MULTI_LANE) ? \"read-only\" : \"read-write\""
2	LN_2_SW_EN	* Varies	Square Wave Enable on Lane 2. Bits [6:4] of VR PCS BASE-R Control Register specify the period of the square wave **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!DWCXLP_MULTI_LANE) ? \"read-only\" : \"read-write\""

Table 7-105 Fields for Register: SR_PMA_KR_SQUARE_WAVE_CTRL (Continued)

Bits	Name	Memory Access	Description
1	LN_1_SW_EN	* Varies	Square Wave Enable on Lane 1. Bits [6:4] of VR PCS BASE-R Control Register specify the period of the square wave **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!DWCXLP_MULTI_LANE) ? \"read-only\" : \"read-write\""
0	LN_0_SW_EN	R/W	Square Wave Enable on Lane 0. Bits [6:4] of VR PCS BASE-R Control Register specify the period of the square wave **Value After Reset:** 0x0 **Exists:** Always

7.1.101 SR_PMA_KR_PRBS_TX_ERR_CNTR_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD PRBS Transmit Error Counter Register- Lane *i* (RO, COR Type).

  Note: In single-lane configurations, only SR_PMA_KR_PRBS_TX_ERR_CNTR_LANE0 is present
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** i+0x10640
- **Exists:** (DWCXLP_PMA_MMD)

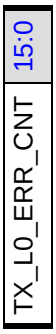


Table 7-106 Fields for Register: SR_PMA_KR_PRBS_TX_ERR_CNTR_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15:0	TX_L0_ERR_CNT	R	Transmit PRBS Error Counter for Lane *i* (RO, COR Type) **Value After Reset:** 0x0 **Exists:** Always

7.1.102 SR_PMA_KR_PRBS_RX_ERR_CNTR_LANE (i) (for i = 0; i <= 3)

- **Description:** SR PMA MMD 40G/100GBASE-CR4/KR4 PRBS Receive Error Counter Register - Lane *i* (RO, COR Type)(RO, COR Type).

  Note: In single lane configurations, only SR_PMA_KR_PRBS_RX_ERR_CNTR_LANE0 is present
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** i+0x106a4
- **Exists:** (DWCXLP_PMA_MMD)

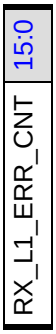


Table 7-107 Fields for Register: SR_PMA_KR_PRBS_RX_ERR_CNTR_LANE (i) (for i = 0; i <= 3)

Bits	Name	Memory Access	Description
15:0	RX_L1_ERR_CNT	R	Receive PRBS Error Counter for Lane *i* (RO, COR Type) **Value After Reset:** 0x0 **Exists:** Always

7.1.103 VR_PMA_DIG_CTRL1

- **Description:** VR PMA MMD Digital Control1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8000 : (DWCXLP_APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)

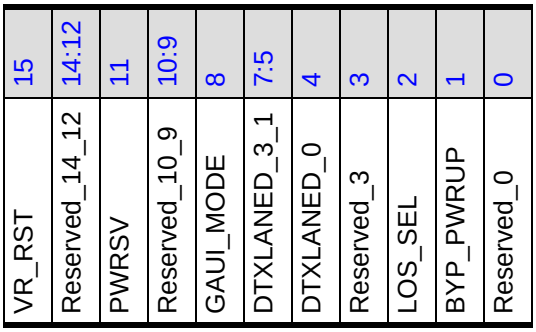


Table 7-108 Fields for Register: VR_PMA_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-specific Soft Reset (RW,SC Type) When the host sets this bit to 1, the DWC_xlgpcs initiates the reset for all the internal blocks except the Management Interface block and the CSR block. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: ■ For Synopsys PHY: This bit is self cleared when Bits[4:2] in VR PCS MMD Digital Status Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. ■ For Non-Synopsys PHY: This bit is self cleared after the following: 32 clk_csr_i clocks for the MCI interface; 1 MDC clock period for the MDIO interface **Value After Reset:** 0x0 **Exists:** Always
14:12	Reserved_14_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-108 Fields for Register: VR_PMA_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
11	PWRSV	* Varies	For Synopsys PHY configurations: Power Save This bit is present only when the DWC_xlgpcs is configured for Synopsys PHY. ■ 0: Normal operation. ■ 1: DWC_xlgpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xlgpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys PHY, DWC_xlgpcs asserts xlgpcs_tx_lpd_o{lane} and xlgpcs_rx_lpd_o{lane} when this bit is set. After clearing this bit, the host must wait until Bits[4:2] of the VR PCS MMD Digital Status Register indicate that the DWC_xlgpcs is in a normal state. For a description of read or write access for any register while this bit is set, see "Special Case Register Access" section. For non-Synopsys PHY configurations: Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\"""
10:9	Reserved_10_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	GAUI_MODE	* Varies	EEE GAUI mode enable In EEE enabled configurations, ■ 1: GAUI_mode is enabled and this is the default mode. The alert and quiet state outputs are based on the Figure 83-7 of IEEE-802.3 specification. ■ 0: non-GAUI_mode. Alert output is sent as 0xFF00. **Value After Reset:** "(DWCXLP_EEE_EN) ? 0x1 : 0x0 " **Exists:** Always **Memory Access:** "(DWCXLP_EEE_EN) ? \"read-write\" : \"read-only\""

Table 7-108 Fields for Register: VR_PMA_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
7:5	DTXLANED_3_1	* Varies	Tx Disable on Lanes[3:1] When any bits of this field are set, the DWC_xlgpcs disables the corresponding Tx lanes (3, 2, or 1) of the PHY. When reset, the DWC_xlgpcs enables the corresponding Tx lanes of the PHY.**Notes:** This field is shared with the following fields: ■ Bits[7:5] of VR PCS MMD Digital Control 1 Register ■ Bits[4:2] of SR PMA MMD Transmit Disable Register **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
4	DTXLANED_0	R/W	Tx Disable on Lane 0 When this bit is set, the DWC_xlgpcs disables the Tx Lane 0 of the PHY. When reset, the DWC_xlgpcs enables the Tx Lane 0 of the PHY. **Notes:** This bit is shared with the following bits: ■ Bit 4 of VR PCS MMD Digital Control1 Register ■ Bit 1 of SR PMA MMD Transmit Disable Register **Value After Reset:** 0x0 **Exists:** Always
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	LOS_SEL	* Varies	Loss-of-Signal Selection If this bit is programmed to 0, DWC_xlgpcs uses xlgpcs_los_i input to infer Loss-of-Signal condition from the PHY. If this bit is programmed to 1, DWC_xlgpcs uses xlgpcs_rx_sigdet_hf_i input to infer Loss-of-Signal condition from the PHY. This field is valid only for Multi-protocol 25G and 32G PHY configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E25||DWCXLP_E32) ? \"read-write\" : \"read-only\""

Table 7-108 Fields for Register: VR_PMA_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
1	BYP_PWRUP	* Varies	For Synopsys PHY configurations: Bypass Power-Up Sequence This bit is present only when the DWC_xlgpcs is configured for the Synopsys PHY. In all other configurations, this bit is reserved. When this bit is set to 1, the DWC_xlgpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xlgpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xlgpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xlgpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming the normal transmission and reception. **Note:** This bit is shared with Bit 1 of VR PCS MMD Digital Control1 Register and VR AN MMD Digital Control1 Register. Access Type: If DWCXLP_SNPS_XAUI_PHY= Enabled, this bit is RW. Otherwise, this bit is RO. For non-Synopsys PHY configurations: Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.1.104 VR_PMA_RSFEC_ERR_WIN

- **Description:** VR PMA MMD RS-FEC Hi-SER Error Window Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8001 : (DWCXLP_APB3_16) ? (0x8001*0x2) : (0x8001*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_RSFEC_EN

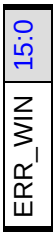


Table 7-109 Fields for Register: VR_PMA_RSFEC_ERR_WIN

Bits	Name	Memory Access	Description
15:0	ERR_WIN	R/W	RS-FEC High-SER Error Window When operating in RS-FEC mode (with bypass_indication enabled), a High Symbol Error Rate status information is provided by the RS-FEC decoder. The error-window (in terms of number of RS-FEC code-words) defined for High-SER generation is 8192. This register field provides a flexibility to change the error-window to any user specific value. Note that, for interleaved RSFEC mode, only even number value is allowed as there are two decoders operating in parallel. **Value After Reset:** 0x2000 **Exists:** Always

7.1.105 VR_PMA_RSFEC_ERR_THLD

- **Description:** VR PMA MMD RS-FEC High-SER Error Threshold Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8002 : (DWCXLP_APB3_16) ? (0x8002*0x2) : (0x8002*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_RSFEC_EN



Table 7-110 Fields for Register: VR_PMA_RSFEC_ERR_THLD

Bits	Name	Memory Access	Description
15:0	ERR_THLD	R/W	RS-FEC High-SER Symbol Error Threshold When operating in RS-FEC mode (with bypass_indication enabled), a High Symbol Error Rate status information is provided by the RS-FEC decoder. The error-threshold (in terms of number of symbol errors) defined by IEEE802.3 specification for High-SER generation is as follows: ■ 16'd417 for 100GBASE-CR4/KR4, 25GBASE-R, 50GBASE-CR2/KR2 ■ 16'd6380 for 50GBASE-CR1/KR1, 100GBASE-CR2/KR2, 100GBASE-CR1/KR1 with Clause 91 RS-FEC ■ 16'd5560 for 100GBASE-CR1/KR1 with Clause 161 interleaved RSFEC User should program appropriate values to this register field based on the operating mode. This register field also provides the flexibility to change the value to some other user-specific value, if required. **Value After Reset:** 0x1a1 **Exists:** Always

7.1.106 VR_PMA_KRTR_PRBS_CTRL0

- **Description:** VR PMA MMD KRTR PRBS Control0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8003 : (DWCXLP_APB3_16) ? (0x8003*0x2) : (0x8003*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

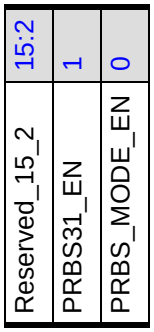


Table 7-111 Fields for Register: VR_PMA_KRTR_PRBS_CTRL0

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	PRBS31_EN	R/W	Enable PRBS31 as the KRTR Training Pattern If this bit is set, PCS sends PRBS31 as the training pattern during KR Training. It also checks for PRBS31 data (instead of PRBS11) in the received frames. **Value After Reset:** 0x0 **Exists:** Always

Table 7-111 Fields for Register: VR_PMA_KRTR_PRBS_CTRL0 (Continued)

Bits	Name	Memory Access	Description
0	PRBS_MODE_EN	R/W	Enable PRBS Mode to determine Rx Link Training Status. If this bit is set, PCS monitors the received KRTR frames for any PRBS errors for a specified interval. (This interval starts after 300ms of KR training in case of Synopsys PHYs OR after 'rpcs_rx_trained{lane}_i' is asserted for non-Synopsys PHY configurations). If the number of PRBS errors are less than or equal to the programmed threshold, then Rx link is declared as Good, otherwise it is considered as a training failure. If this bit is not set, Rx link is assumed to be successfully trained after 300ms of KR Training or after 'rpcs_trained_i' input is asserted. **Value After Reset:** 0x0 **Exists:** Always

7.1.107 VR_PMA_KRTR_PRBS_CTRL1

- **Description:** VR PMA MMD KRTR PRBS Control1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8004 : (DWCXLP_APB3_16) ? (0x8004*0x2) : (0x8004*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)



Table 7-112 Fields for Register: VR_PMA_KRTR_PRBS_CTRL1

Bits	Name	Memory Access	Description
15:0	PRBS_TIM_LMT	R/W	PRBS Timer Duration Control If PRBS Mode (bit[0] of VR PMA MMD KRTR PRBS Control 0 Register) is enabled to determine successful Rx Equalization, DWC_xlgpcs monitors PRBS errors for a certain duration at the end of Rx Equalization. In a 40G configuration, there is a 20-bit timer to control this monitoring interval and this timer runs at a clock frequency of 156.25 MHz. This field (PRBS_TIM_LMT) forms the upper 16-bit of the value to be loaded to the timer in 40G mode and the lower 4-bits are hard-coded as 4'h0. In 25G/50G/100G configuration, there is a 21-bit timer to control this monitoring interval and this timer forms the upper 16 bits and runs at a clock frequency of 390.625 MHz. This field has a default value such that this timer runs for duration of 3ms. Application should program a non-zero value to this field if PRBS Mode is enabled. **Value After Reset:** "(DWCXLP_KR4_KR) ? 0x7270 : 0x8F0D " **Exists:** Always

7.1.108 VR_PMA_KRTR_PRBS_CTRL2

- **Description:** VR PMA MMD KRTR PRBS Control2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8005 : (DWCXLP_APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

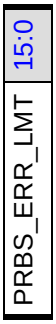


Table 7-113 Fields for Register: VR_PMA_KRTR_PRBS_CTRL2

Bits	Name	Memory Access	Description
15:0	PRBS_ERR_LMT	R/W	PRBS Tolerable Error Limit This field determines the maximum allowed PRBS errors in the received KRTR frames during the interval specified by 'PRBS_TIM_LMT' field of VR PMA MMD KRTR PRBS Control 1 Register. If the number of PRBS errors is greater than the value programmed in this field, training is considered as a failure. **Value After Reset:** 0x0 **Exists:** Always

7.1.109 VR_PMA_KRTR_TIMER_CTRL0

- **Description:** VR PMA MMD KRTR Timer Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8006 : (DWCXLP_APB3_16) ? (0x8006*0x2) : (0x8006*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)



Table 7-114 Fields for Register: VR_PMA_KRTR_TIMER_CTRL0

Bits	Name	Memory Access	Description
15:0	MAX_WAIT_TIME	R/W	Maximum Wait Time for KR Training. This field determines the maximum wait time (max_wait_timer as per IEEE clause 72.6.10.3.2) allowed for successful KR Training. In 10G/40G modes, there is a 27-bit timer running at a clock frequency of 156.25 MHz to serve this purpose. This field forms the upper 16-bits of the value to be loaded to this timer. The lower 11-bits are hard-coded as 11'h7C8. In 25G/50G/100G modes, there is a 28-bit timer running at a clock frequency of 390.625 MHz to serve the purpose. This field forms the upper 16-bits of the value to be loaded to this timer. The lower 12-bits are hard-coded as 12'hB74. The default value corresponds to 500ms as mandated by IEEE spec. **Value After Reset:** "DWCXLP_KR4_KR ? 0x9502 : 0xBA43 " **Exists:** Always

7.1.110 VR_PMA_KRTR_TIMER_CTRL1

- **Description:** VR PMA MMD KRTR Timer Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8007 : (DWCXLP_APB3_16) ? (0x8007*0x2) : (0x8007*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)



Table 7-115 Fields for Register: VR_PMA_KRTR_TIMER_CTRL1

Bits	Name	Memory Access	Description
15:0	WAIT_TIME	R/W	Wait Time during KR Training. This field determines the wait time (wait_timer as per IEEE clause 72.6.10.3.2) in the LINK_READY state of KR Training. There is a 16-bit time running at a clock frequency of 156.25 MHz to serve this purpose(runs at 390.625 MHz for 25G/50G and 100G modes). This field forms the value to be loaded to this timer. The default value corresponds to ~64us. **Value After Reset:** "DWCXLP_KR4_KR ? 0x26EB : 0x4dd7 " **Exists:** Always

7.1.111 VR_PMA_KRTR_TIMER_CTRL2

- **Description:** VR PMA MMD KRTR Timer Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8008 : (DWCXLP_APB3_16) ? (0x8008*0x2) : (0x8008*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)

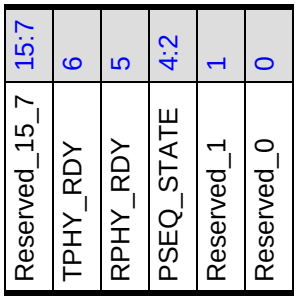


Table 7-116 Fields for Register: VR_PMA_KRTR_TIMER_CTRL2

Bits	Name	Memory Access	Description
15:0	RX_TRAIN_TIME	R/W	Timer interval for Rx Training. This field determines the duration for which Rx Equalization should be carried out for Synopsys PHY configurations. After this time interval, rx_trained (as per IEEE clause 72.6.10.31.1) is made high by PCS. . DWC_xlgpcs implements a 28-bit timer for this purpose and this timer runs at clk_rpcs_rx{lane}_i clock. The timer runs for a duration equal to {RX_TRAIN_TIME,11'h178}*6.4ns when operating in 10G/40G modes. The timer runs for a duration equal to {RX_TRAIN_TIME,12'h3AC}*2.56ns when operating in 25G/50G/100G modes. PCS would send coefficient update requests to remote-end partner during this time interval. There after it does not send any more coefficient update requests. When operating in 10G/40G modes, it is recommended to program this field to 16'h5968 and while operating in 25G/50G/100G modes, it is recommended to program this field to 16'h6FC2. The aforementioned values ensure that rx-train-timer runs for a duration of 300ms. **Value After Reset:** "DWCXLP_KR4_KR ? 0x5968 : 0x6FC2" **Exists:** Always

7.1.112 VR_PMA_DIG_STS

- **Description:** VR PMA MMD Digital Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8010 : (DWCXLP_APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD)



15:7	6	5	4:2	1	0
Reserved_15_7	TPHY_RDY	RPHY_RDY	PSEQ_STATE	Reserved_1	Reserved_0

Table 7-117 Fields for Register: VR_PMA_DIG_STS

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	TPHY_RDY	R	For Synopsys PHY configurations: This field is reserved For non-Synopsys PHY configurations with PHY RDY inputs: If DWCXLP_ADD_PHY_RDY!=0, This field reflects the status of tx_phy_rdy_i input pin. If DWCXLP_ADD_PHY_RDY==0, This field is reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Table 7-117 Fields for Register: VR_PMA_DIG_STS (Continued)

Bits	Name	Memory Access	Description
5	RPHY_RDY	R	For Synopsys PHY configurations: This field is reserved For non-Synopsys PHY configurations with PHY RDY inputs: If DWCXLP_ADD_PHY_RDY!=0, This field reflects the status of rx_phy_rdy_i input pin. If DWCXLP_ADD_PHY_RDY==0, This field is reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
4:2	PSEQ_STATE	R	For Synopsys PHY configurations: Power Up Sequence State This field indicates the state variable value of the power-up sequence module. For Synopsys PHY ■ 3'b000: Wait for ACK High0 ■ 3'b001: Wait for ACK Low0 ■ 3'b010: Wait for Tx ACK High1 ■ 3'b011: Wait for Rx ACK Low1 ■ 3'b100: Tx/Rx stable (Power_Good state) ■ 3'b101: Power Save state ■ 3'b110: Power Down state ■ 3'b111: Other intermediate states (valid only for Synopsys 32G PHY configurations) Dependency: This field is reserved if DWCXLP_SNPS_XAUI_PHY = Disabled. Power Up Sequence State For non-Synopsys PHY configurations: Reserved **Value After Reset:** 0x4 **Exists:** Always
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.1.113 VR_PMA_GPIO

- **Description:** VR PMA MMD GPIO Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8015 : (DWCXLP_APB3_16) ? (0x8015*0x2) : (0x8015*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_GPIO_EN)

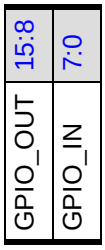
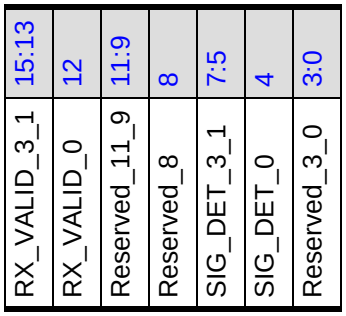


Table 7-118 Fields for Register: VR_PMA_GPIO

Bits	Name	Memory Access	Description
15:8	GPIO_OUT	R/W	GPIO Output The content written in this field is driven to the xlgpcs_gpo_o[7:0] output port.**Note:** This field is shared with Bits[15:8] of VR PCS MMD GPIO Register. **Value After Reset:** 0x0 **Exists:** Always
7:0	GPIO_IN	R	GPIO Input This field indicates the content of the xlgpcs_gpo_i[7:0] port. **Value After Reset:** 0x0 **Exists:** Always

7.1.114 VR_PMA_RX_LSTS

- **Description:** VR PMA PHY Rx Lane Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8020 : (DWCXLP_APB3_16) ? (0x8020*0x2) : (0x8020*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD



15:13	12	11:9	8	7:5	4	3:0
RX_VALID_3_1	RX_VALID_0	Reserved_11_9	Reserved_8	SIG_DET_3_1	SIG_DET_0	Reserved_3_0

Table 7-119 Fields for Register: VR_PMA_RX_LSTS

Bits	Name	Memory Access	Description
15:13	RX_VALID_3_1	R	DPLL Lock Status for Lanes[3:1] This field indicates that the DPLL in the PHY is locked on the serial data in the corresponding lane. ■ 3'b**1: Lane 1 DPLL bit locked ■ 3'b*1*: Lane 2 DPLL bit locked ■ 3'b1**: Lane 3 DPLL bit locked **Value After Reset:** 0x0 **Exists:** Always
12	RX_VALID_0	R	DPLL Lock Status for Lane 0 This field indicates that the DPLL in the PHY is locked on the serial data in Lane 0. The value 1'b1 indicates that Lane 0 DPLL bit is locked. **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-119 Fields for Register: VR_PMA_RX_LSTS (Continued)

Bits	Name	Memory Access	Description
8	Reserved_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	SIG_DET_3_1	R	Rx Signal Detect for Lanes[3:1] This field indicates that the Rx has detected the signal on the corresponding lanes. The bits in this field are the complement value of the signals input from PHY (xlgpcs_los_i{lane}) on the corresponding lane. ■ 3'b**1: Lane 1 signal detected ■ 3'b*1*: Lane 2 signal detected ■ 3'b1**: Lane 3 signal detected **Value After Reset:** 0x0 **Exists:** Always
4	SIG_DET_0	R	Rx Signal Detect for Lane 0 This bit indicates that the Rx detected the signal on Lane 0. This bit is the complement value of the signals input from PHY (xlgpcs_los_i[0]) on Lane 0. The value 1'b1 indicates that Lane 0 signal is detected. **Value After Reset:** 0x0 **Exists:** Always
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.1.115 VR_PMA_MP_25G_16G_TX_GENCTRL0

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx General Register 0
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8030 : (DWCXLP_APB3_16) ? (0x8030*0x2) : (0x8030*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

15:13	12	11:9	8	7:5	4	3:1	0
TX_DT_EN_3_1	TX_DT_EN_0	TX_RST_3_1	TX_RST_0	TX_INV_3_1	TX_INV_0	TXBCN_EN_3_1	TXBCN_EN_0

Table 7-120 Fields for Register: VR_PMA_MP_25G_16G_TX_GENCTRL0

Bits	Name	Memory Access	Description
15:13	TX_DT_EN_3_1	* Varies	**For Multi-lane configurations :** Transmit Data Enable on Lanes[3:1] This field controls the output port 'xlgpcs_tx_data_en_o[3:1]'. This field must be cleared when PHY Tx is not in P0 power state **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x7 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
12	TX_DT_EN_0	R/W	Tx Data Enable on lane 0 of Multi-protocol PHY When this bit is set, Transmit Output Driver in the PHY is enabled. This bit controls the output port 'xlgpcs_tx_data_en_o[0]'. This field should be cleared when PHY Tx is not in P0 power state **Value After Reset:** 0x1 **Exists:** Always

Table 7-120 Fields for Register: VR_PMA_MP_25G_16G_TX_GENCTRL0 (Continued)

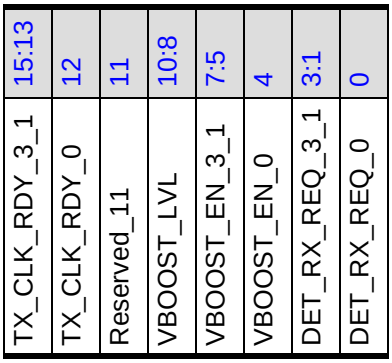
Bits	Name	Memory Access	Description
11:9	TX_RST_3_1	* Varies	**For Multi-lane configurations :** Transmit Reset on Lanes[3:1] This signal drives the output port 'xlgpcs_tx_reset_o[3:1]' when DWC_xlgpcs is in POWER_GOOD state. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
8	TX_RST_0	R/W	Tx Reset on lane 0 of Multi-protocol PHY When this bit is set, PHY transmitter is reset, including common-mode adjustment and receiver detection state machines. This signal drives the output port 'xlgpcs_tx_reset_o[0]' when DWC_xlgpcs is in POWER_GOOD state. **Value After Reset:** 0x0 **Exists:** Always
7:5	TX_INV_3_1	* Varies	**For Multi-lane configurations :** Transmit Data Invert on Lanes[3:1] This field drives the output port 'xlgpcs_tx_invert_o[3:1]'. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
4	TX_INV_0	R/W	Tx Invert on lane 0 of Multi-protocol PHY When this bit is set, the data on PHY Tx serial lines are logically inverted. This signal drives the output port 'xlgpcs_tx_invert_o[0]'. **Value After Reset:** 0x0 **Exists:** Always

Table 7-120 Fields for Register: VR_PMA_MP_25G_16G_TX_GENCTRL0 (Continued)

Bits	Name	Memory Access	Description
3:1	TXBCN_EN_3_1	* Varies	**For Multi-lane configurations :** Transmit Beacon Enable Control on Lanes[3:1] This field drives the output port 'xlgpcs_tx_beacon_en_o[3:1]'. When set to 1, this field enables the Tx beaconing. ■ TXBCN_EN_3_1[1]: Beacon enable control on Lane 1 ■ TXBCN_EN_3_1[2]: Beacon enable control on Lane 2 ■ TXBCN_EN_3_1[3]: Beacon enable control on Lane 3 **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	TXBCN_EN_0	R/W	Tx Beaconing Enable on lane 0 of Multi-protocol PHY ■ 1: PHY enables transmitter beaconing (LFPS). The period for transmit pulses is between 20-50ns. This field drives the output port 'xlgpcs_tx_beacon_en_o[0]'. **Value After Reset:** 0x0 **Exists:** Always

7.1.116 VR_PMA_MP_25G_16G_TX_GENCTRL1

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx General Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8031 : (DWCXLP_APB3_16) ? (0x8031*0x2) : (0x8031*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)



15:13	12	11	10:8	7:5	4	3:1	0
TX_CLK_RDY_3_1	TX_CLK_RDY_0	Reserved_11	VBOOST_LVL	VBOOST_EN_3_1	VBOOST_EN_0	DET_RX_REQ_3_1	DET_RX_REQ_0

Table 7-121 Fields for Register: VR_PMA_MP_25G_16G_TX_GENCTRL1

Bits	Name	Memory Access	Description
15:13	TX_CLK_RDY_3_1	* Varies	**For Multi-lane configurations :** Transmitter Input clock ready on Lanes[3:1] This field drives the output xlgpcs_tx_clk_rdy_o[3:1]. This field should remain high, as long as PHY Tx Clock is active. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x7 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
12	TX_CLK_RDY_0	R/W	Transmitter Input clock ready on lane 0 This field drives the output xlgpcs_tx_clk_rdy_o[0]. This field should remain high, as long as PHY Tx Clock is active **Value After Reset:** 0x1 **Exists:** Always

Table 7-121 Fields for Register: VR_PMA_MP_25G_16G_TX_GENCTRL1 (Continued)

Bits	Name	Memory Access	Description
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10:8	VBOOST_LVL	* Varies	Tx Voltage Boost Maximum Level This field controls the maximum achievable Tx swing in the PHY This field drives the output port 'xlgpcs_tx_vboost_lvl_o[2:0]'. **Value After Reset:** "DWCXLP_E32 ? 0x0 : DWCXLP_E16 ? 0x4 : 0x5 " **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""
7:5	VBOOST_EN_3_1	* Varies	**For Multi-lane configurations :** Transmitter Voltage Boost Enable on Lanes[3:1] This field drives the output port xlgpcs_tx_vboost_en_o[3:1] **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_E32 ? 0x0 : DWCXLP_25G_ONLY ? 0x0 : 0x7 " **Exists:** Always **Memory Access:** "(DWCXLP_E32 || DWCXLP_25G_ONLY) ? \"read-only\" : \"read-write\""
4	VBOOST_EN_0	* Varies	Tx voltage Boost Enable on lane 0 of Multi-protocol PHY When this bit is set, the current mode Tx Swing boost in the PHY is enabled. This bit drives the output port 'xlgpcs_tx_vboost_en_o[0]'. **Value After Reset:** "DWCXLP_E32 ? 0x0 : 0x1 " **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""

Table 7-121 Fields for Register: VR_PMA_MP_25G_16G_TX_GENCTRL1 (Continued)

Bits	Name	Memory Access	Description
3:1	DET_RX_REQ_3_1	* Varies	**For Multi-lane configurations :** Transmitter Rx-Detection Request on Lanes[3:1] This field drives the output port 'xlgpcs_tx_detrx_req_o[3:1]'. When this field is set, a receiver detection request is made towards the PHY on lanes 3 to 1. The result of the Rx Detection operation is available at field 'DETRX_RSLT_3_1' when 'TX_ACK_3_1' is read as high. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	DET_RX_REQ_0	R/W	Transmitter Rx-Detection request on lane 0 of Multi-protocol PHY. This field drives the output port 'xlgpcs_tx_detrx_req_o[0]'. When this bit is set, a receiver detection request is made towards the PHY on lane 0. After this bit is set, it should remain high till 'TX_ACK_0' bit is read as high. When 'TX_ACK_0' is read as high, the result of the Rx Detection operation is available at field 'DETRX_RSLT_0'. After obtaining the result of 'Rx-Detect operation', this bit should be cleared. Rx Detection operation is valid only when transmitter is in P1 power state **Value After Reset:** 0x0 **Exists:** Always

7.1.117 VR_PMA_MP_25G_16G_TX_GENCTRL2

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx General Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8032 : (DWCXLP_APB3_16) ? (0x8032*0x2) : (0x8032*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

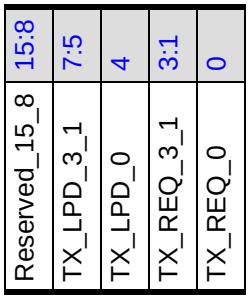


Table 7-122 Fields for Register: VR_PMA_MP_25G_16G_TX_GENCTRL2

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	TX_LPD_3_1	* Varies	**For Multi-lane configurations :** Transmitter Lane Power Down on Lanes[3:1] **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
4	TX_LPD_0	R/W	Transmitter Lane Power Down on lane 0 of Multi-protocol PHY **Value After Reset:** 0x0 **Exists:** Always

Table 7-122 Fields for Register: VR_PMA_MP_25G_16G_TX_GENCTRL2 (Continued)

Bits	Name	Memory Access	Description
3:1	TX_REQ_3_1	* Varies	**For Multi-lane configurations :** Transmitter Request on Lanes[3:1] These bits can be set by application and they are self-cleared once acknowledgement comes from the PHY. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	TX_REQ_0	R/W	Transmitter operation request on lane 0 of Multi-protocol PHY. This bit can be set by application and it is self-cleared once acknowledgement comes from the PHY. **Value After Reset:** 0x0 **Exists:** Always

7.1.118 VR_PMA_MP_25G_16G_TX_BOOST_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Tx Boost Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8033 : (DWCXLP_APB3_16) ? (0x8033*0x2) : (0x8033*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)

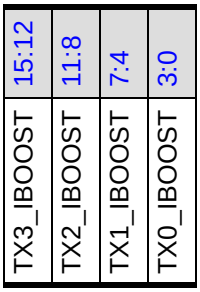


Table 7-123 Fields for Register: VR_PMA_MP_25G_16G_TX_BOOST_CTRL

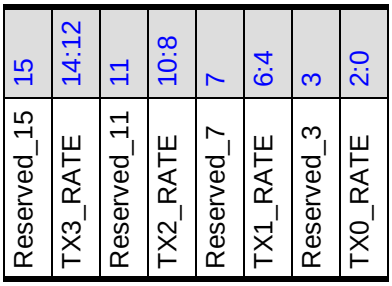
Bits	Name	Memory Access	Description
15:12	TX3_IBOOST	* Varies	**For Multi-lane configurations :** Tx current boost on lane 3 of the PHY This bit drives the output port 'xlgpcs_tx3_iboost_lvl_o[3:0]'. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0xf" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11:8	TX2_IBOOST	* Varies	**For Multi-lane configurations :** Tx current boost on lane 2 of the PHY This bit drives the output port 'xlgpcs_tx2_iboost_lvl_o[3:0]'. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0xf" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-123 Fields for Register: VR_PMA_MP_25G_16G_TX_BOOST_CTRL (Continued)

Bits	Name	Memory Access	Description
7:4	TX1_IBOOST	* Varies	**For Multi-lane configurations :** Tx current boost on lane 1 of the PHY This bit drives the output port 'xlgpcs_tx1_iboost_lvl_o[3:0]'. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0xf" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3:0	TX0_IBOOST	R/W	Tx current boost level on lane 0 of the PHY This bit drives the output port xlgpcs_tx0_iboost_lvl_o[3:0]. For details on how to set this field, refer PHY databook. **Value After Reset:** 0xf **Exists:** Always

7.1.119 VR_PMA_MP_25G_16G_TX_RATE_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Tx Rate Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8034 : (DWCXLP_APB3_16) ? (0x8034*0x2) : (0x8034*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)



15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	TX3_RATE	Reserved_11	TX2_RATE	Reserved_7	TX1_RATE	Reserved_3	TX0_RATE

Table 7-124 Fields for Register: VR_PMA_MP_25G_16G_TX_RATE_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	TX3_RATE	* Varies	**For Multi-lane configurations :** Tx date rate on lane 3 of the PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-124 Fields for Register: VR_PMA_MP_25G_16G_TX_RATE_CTRL (Continued)

Bits	Name	Memory Access	Description
10:8	TX2_RATE	* Varies	**For Multi-lane configurations :** Tx date rate on lane 2 of the PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	TX1_RATE	* Varies	**For Multi-lane configurations :** Tx date rate on lane 1 of the PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	TX0_RATE	R/W	Tx date rate on lane 0 of the PHY Data rate encoding is as follows: ■ 3'b000 : baud ■ 3'b001 : baud/2 ■ 3'b010 : baud/4 ■ 3'b011 : baud/8 ■ 3'b100 : baud/16.5 ■ 3'b101 : Not supported ■ 3'b110 : baud/12 ■ 3'b111 : baud/10 **Value After Reset:** 0x0 **Exists:** Always

7.1.120 VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Power State Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8035 : (DWCXLP_APB3_16) ? (0x8035*0x2) : (0x8035*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

15:12	11:9	8	7:6	5:4	3:2	1:0
Reserved_15_12	TX_DISABLE_3_1	TX_DISABLE_0	TX3_PSTATE	TX2_PSTATE	TX1_PSTATE	TX0_PSTATE

Table 7-125 Fields for Register: VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	TX_DISABLE_3_1	* Varies	**For Multi-lane configurations :** Transmitter Disable on lanes [3:1] **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
8	TX_DISABLE_0	R/W	Transmitter Disable on lane 0 **Value After Reset:** 0x0 **Exists:** Always

Table 7-125 Fields for Register: VR_PMA_MP_25G_16G_TX_POWER_STATE_CTRL (Continued)

Bits	Name	Memory Access	Description
7:6	TX3_PSTATE	* Varies	**For Multi-lane configurations :** Tx power state control for lane 3 of Multi-protocol 25G/16G PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
5:4	TX2_PSTATE	* Varies	**For Multi-lane configurations :** Tx power state control for lane 2 of Multi-protocol PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3:2	TX1_PSTATE	* Varies	**For Multi-lane configurations :** Tx power state control for lane 1 of Multi-protocol PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
1:0	TX0_PSTATE	R/W	Tx power state control for lane 0 of Multi-protocol PHY **Value After Reset:** 0x0 **Exists:** Always

7.1.121 VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN0

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 0 -Lane 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8036 : (DWCXLP_APB3_16) ? (0x8036*0x2) : (0x8036*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

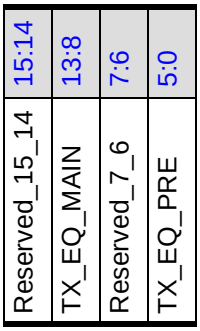


Table 7-126 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN0

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	TX_EQ_MAIN	R/W	Control for setting Tx driver output amplitude This field drives the output port 'rpcs_ktx_main0_o' if 'TX_EQ_OVR_RIDE' bit of 'VR PMA Synopsys Multi-protocol PHY Transmit Equalization Control 1 -Lane 0 Register' is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x28 : (DWCXLP_E32 ? 0x17 : 0x18)" **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-126 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN0 (Continued)

Bits	Name	Memory Access	Description
5:0	TX_EQ_PRE	R/W	Tx Pre-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient).This field drives the output port 'rpcs_ktx_pre0_o' if 'TX_EQ_OVR_RIDE' bit of VR PMA Synopsys Multi-protocol PHY Transmit Equalization Control 1 -Lane 0 Register is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** "(DWCXLP_E32 ? 0x4 : 0x0)" **Exists:** Always

7.1.122 VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN0

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 1 -Lane 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8037 : (DWCXLP_APB3_16) ? (0x8037*0x2) : (0x8037*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

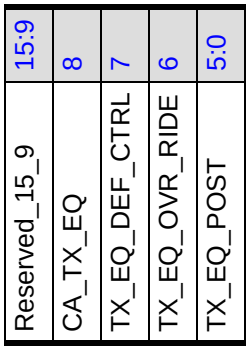


Table 7-127 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN0

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-127 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN0 (Continued)

Bits	Name	Memory Access	Description
8	CA_TX_EQ	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Enable 'Constant Amplitude' Mode of Transmit Equalization in lane0. When this mode is enabled, on receiving INC/DEC commands for C(+1) or C(-1), transmit equalization algorithm updates the Tx filter coefficients such that the amplitude of the signal transmitted by the PHY is held constant. This method is optimal for Synopsys PHY. However, it does not conform to the KR/BaseR voltage specifications mandated by IEEE. When this bit is low, the default transmit equalization conforming to IEEE specification gets selected. Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
7	TX_EQ_DEF_CTRL	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Transmit Equalization Default Settings Control in Lane0. This field controls the default values that are assigned to the Transmit Equalization coefficients (rpcs_ktx_pre0_o, rpcs_ktx_main0_o, rpcs_ktx_post0_o), when Link Training is restarted. The Link adaptation algorithm implemented in DWC_xlgpcs makes use of these default values to compute the new values of the coefficients, when it receives a coefficient update request from the link partner. **Value :** ■ 0: The default values corresponds to CL72/CL92/CL93 Link Training INIT ■ 1: The default values corresponds to CL72/CL92/CL93 Link Training PRESET For configurations with DWCXLP_LNK_TRAIN_EN=Disabled Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""

Table 7-127 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN0 (Continued)

Bits	Name	Memory Access	Description
6	TX_EQ_OVR_RIDE	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Over-ride Tx Eq setting of KRTR module with CSR values on lane 0 If this bit is set, values from CSR over-ride the values from KRT module. For configurations with DWCXLP_LNK_TRAIN_EN=Disabled Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
5:0	TX_EQ_POST	R/W	Tx Post-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient). This field drives the output port 'rpcs_ktx_post0_o' if 'TX_EQ_OVR_RIDE' bit is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** 0x0 **Exists:** Always

7.1.123 VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN1

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 0 -Lane 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8038 : (DWCXLP_APB3_16) ? (0x8038*0x2) : (0x8038*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

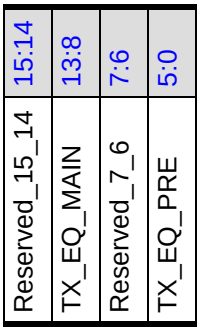


Table 7-128 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN1

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	TX_EQ_MAIN	R/W	Control for setting Tx driver output amplitude This field drives the output port 'rpcs_ktx_main1_o' if 'TX_EQ_OVR_RIDE' bit of 'VR PMA Synopsys Multi-protocol PHY Transmit Equalization Control 1 -Lane 1 Register' is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x28 : (DWCXLP_E32 ? 0x17 : 0x18)" **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-128 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN1 (Continued)

Bits	Name	Memory Access	Description
5:0	TX_EQ_PRE	R/W	Tx Pre-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient).This field drives the output port 'rpcs_ktx_pre1_o' if 'TX_EQ_OVR_RIDE' bit of VR PMA Synopsys Multi-protocol PHY Transmit Equalization Control 1 -Lane 1 Register is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** "DWCXLP_E32 ? 0x4 : 0x00" **Exists:** Always

7.1.124 VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN1

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 1 -Lane 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8039 : (DWCXLP_APB3_16) ? (0x8039*0x2) : (0x8039*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

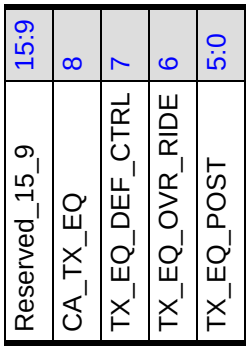


Table 7-129 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN1

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-129 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN1 (Continued)

Bits	Name	Memory Access	Description
8	CA_TX_EQ	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Enable 'Constant Amplitude' Mode of Transmit Equalization in lane1. When this mode is enabled, on receiving INC/DEC commands for C(+1) or C(-1), transmit equalization algorithm updates the Tx filter coefficients such that the amplitude of the signal transmitted by the PHY is held constant. This method is optimal for Synopsys PHY. However, it does not conform to the KR/BaseR voltage specifications mandated by IEEE. When this bit is low, the default transmit equalization conforming to IEEE specification gets selected. Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
7	TX_EQ_DEF_CTRL	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Transmit Equalization Default Settings Control in Lane1. This field controls the default values that are assigned to the Transmit Equalization coefficients (rpcs_ktx_pre1_o, rpcs_ktx_main1_o, rpcs_ktx_post1_o), when Link Training is restarted. The Link adaptation algorithm implemented in DWC_xlgpcs makes use of these default values to compute the new values of the coefficients, when it receives a coefficient update request from the link partner. **Value :** ■ 0: The default values corresponds to CL72/CL92/CL93 Link Training INIT ■ 1: The default values corresponds to CL72/CL92/CL93 Link Training PRESET For configurations with DWCXLP_LNK_TRAIN_EN=Disabled Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""

Table 7-129 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN1 (Continued)

Bits	Name	Memory Access	Description
6	TX_EQ_OVR_RIDE	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Over-ride Tx Eq setting of KRTR module with CSR values on lane 1 If this bit is set, values from CSR over-ride the values from KRT module. For configurations with DWCXLP_LNK_TRAIN_EN=Disabled Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
5:0	TX_EQ_POST	R/W	Tx Post-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient). This field drives the output port 'rpcs_ktx_post1_o' if 'TX_EQ_OVR_RIDE' bit is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** 0x0 **Exists:** Always

7.1.125 VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN2

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 0 -Lane 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x803A : (DWCXLP_APB3_16) ? (0x803A*0x2) : (0x803A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

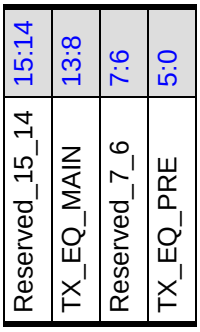


Table 7-130 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	TX_EQ_MAIN	R/W	Control for setting Tx driver output amplitude This field drives the output port 'rpcs_ktx_main2_o' if 'TX_EQ_OVR_RIDE' bit of 'VR PMA Synopsys Multi-protocol PHY Transmit Equalization Control 1 -Lane 2 Register' is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x28 : (DWCXLP_E32 ? 0x17 : 0x18)" **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-130 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN2 (Continued)

Bits	Name	Memory Access	Description
5:0	TX_EQ_PRE	R/W	Tx Pre-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient).This field drives the output port 'rpcs_ktx_pre2_o' if 'TX_EQ_OVR_RIDE' bit of VR PMA Synopsys Multi-protocol PHY Transmit Equalization Control 1 -Lane 2 Register is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** "DWCXLP_E32 ? 0x4 : 0x00" **Exists:** Always

7.1.126 VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN2

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 1 -Lane 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x803B : (DWCXLP_APB3_16) ? (0x803B*0x2) : (0x803B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

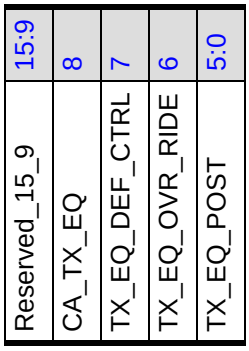


Table 7-131 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN2

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-131 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN2 (Continued)

Bits	Name	Memory Access	Description
8	CA_TX_EQ	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Enable 'Constant Amplitude' Mode of Transmit Equalization in lane2. When this mode is enabled, on receiving INC/DEC commands for C(+1) or C(-1), transmit equalization algorithm updates the Tx filter coefficients such that the amplitude of the signal transmitted by the PHY is held constant. This method is optimal for Synopsys PHY. However, it does not conform to the KR/BaseR voltage specifications mandated by IEEE. When this bit is low, the default transmit equalization conforming to IEEE specification gets selected. Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
7	TX_EQ_DEF_CTRL	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Transmit Equalization Default Settings Control in Lane2. This field controls the default values that are assigned to the Transmit Equalization coefficients (rpcs_ktx_pre2_o, rpcs_ktx_main2_o, rpcs_ktx_post2_o), when Link Training is restarted. The Link adaptation algorithm implemented in DWC_xlgpcs makes use of these default values to compute the new values of the coefficients, when it receives a coefficient update request from the link partner. **Value :** ■ 0: The default values corresponds to CL72/CL92/CL93 Link Training INIT ■ 1: The default values corresponds to CL72/CL92/CL93 Link Training PRESET For configurations with DWCXLP_LNK_TRAIN_EN=Disabled Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""

Table 7-131 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN2 (Continued)

Bits	Name	Memory Access	Description
6	TX_EQ_OVR_RIDE	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Over-ride Tx Eq setting of KRTR module with CSR values on lane 2 If this bit is set, values from CSR over-ride the values from KRT module. For configurations with DWCXLP_LNK_TRAIN_EN=Disabled Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
5:0	TX_EQ_POST	R/W	Tx Post-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient). This field drives the output port 'rpcs_ktx_post2_o' if 'TX_EQ_OVR_RIDE' bit is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** 0x0 **Exists:** Always

7.1.127 VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN3

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 0 -Lane 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x803C : (DWCXLP_APB3_16) ? (0x803C*0x2) : (0x803C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

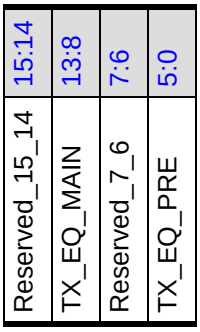


Table 7-132 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN3

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	TX_EQ_MAIN	R/W	Control for setting Tx driver output amplitude This field drives the output port 'rpcs_ktx_main3_o' if 'TX_EQ_OVR_RIDE' bit of 'VR PMA Synopsys Multi-protocol PHY Transmit Equalization Control 1 -Lane 3 Register' is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x28 : (DWCXLP_E32 ? 0x17 : 0x18)" **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-132 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL0_LN3 (Continued)

Bits	Name	Memory Access	Description
5:0	TX_EQ_PRE	R/W	Tx Pre-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient).This field drives the output port 'rpcs_ktx_pre3_o' if 'TX_EQ_OVR_RIDE' bit of VR PMA Synopsys Multi-protocol PHY Transmit Equalization Control 1 -Lane 3 Register is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** "DWCXLP_E32 ? 0x4 : 0x00" **Exists:** Always

7.1.128 VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN3

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 1 -Lane 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x803D : (DWCXLP_APB3_16) ? (0x803D*0x2) : (0x803D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

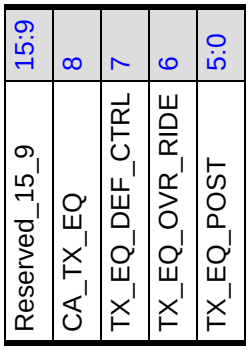


Table 7-133 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN3

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-133 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN3 (Continued)

Bits	Name	Memory Access	Description
8	CA_TX_EQ	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Enable 'Constant Amplitude' Mode of Transmit Equalization in lane3. When this mode is enabled, on receiving INC/DEC commands for C(+1) or C(-1), transmit equalization algorithm updates the Tx filter coefficients such that the amplitude of the signal transmitted by the PHY is held constant. This method is optimal for Synopsys PHY. However, it does not conform to the KR/BaseR voltage specifications mandated by IEEE. When this bit is low, the default transmit equalization conforming to IEEE specification gets selected. Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
7	TX_EQ_DEF_CTRL	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Transmit Equalization Default Settings Control in Lane3. This field controls the default values that are assigned to the Transmit Equalization coefficients (rpcs_ktx_pre3_o, rpcs_ktx_main3_o, rpcs_ktx_post3_o), when Link Training is restarted. The Link adaptation algorithm implemented in DWC_xlgpcs makes use of these default values to compute the new values of the coefficients, when it receives a coefficient update request from the link partner. **Value :** ■ 0: The default values corresponds to CL72/CL92/CL93 Link Training INIT ■ 1: The default values corresponds to CL72/CL92/CL93 Link Training PRESET For configurations with DWCXLP_LNK_TRAIN_EN=Disabled Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""

Table 7-133 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL1_LN3 (Continued)

Bits	Name	Memory Access	Description
6	TX_EQ_OVR_RIDE	* Varies	For configurations with DWCXLP_LNK_TRAIN_EN=Enabled: Over-ride Tx Eq setting of KRTR module with CSR values on lane 3 If this bit is set, values from CSR over-ride the values from KRT module. For configurations with DWCXLP_LNK_TRAIN_EN=Disabled Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
5:0	TX_EQ_POST	R/W	Tx Post-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient). This field drives the output port 'rpcs_ktx_post3_o' if 'TX_EQ_OVR_RIDE' bit is set or in configurations with DWCXLP_LNK_TRAIN_EN=Disabled. **Value After Reset:** 0x0 **Exists:** Always

7.1.129 VR_PMA_MP_25G_16G_TX_EQ_CTRL2

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x803E : (DWCXLP_APB3_16) ? (0x803E*0x2) : (0x803E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_LNK_TRAIN_EN

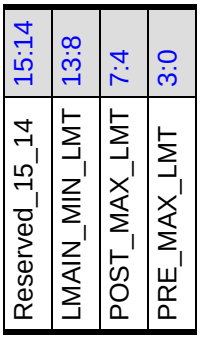


Table 7-134 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	LMAIN_MIN_LMT	R/W	Minimum Value of Lmain Transmit Equalization process makes use of a variable known as Lmain (8-bit value). This field can be used to configure the integer part of the minimum allowed value of Lmain (upper 6-bits). The fractional part (lower 2-bits) are always zero. The desired minimum limit of this variable (integer part) is 6'h16/D. However, this field provides flexibility to the application to change the limit. (If desired). **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x16 : DWCXLP_E32 ? 0x2 : 0xE" **Exists:** Always

Table 7-134 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL2 (Continued)

Bits	Name	Memory Access	Description
7:4	POST_MAX_LMT	R/W	Maximum Value of Post Cursor Value C(+1) This field can be suitably programmed to control the maximum possible value of Tx equalization coefficient C(+1). The adaptation process within DWC_xlgpcs is constrained so that C(+1) does not exceed the value programmed to this field. This field controls only the integer part (upper 4-bits). Fractional part (lower 2 bits) is always zero. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0xF : DWCXLP_E32 ? 0x9 : 0x8" **Exists:** Always
3:0	PRE_MAX_LMT	R/W	Maximum Value of Pre Cursor Value C(-1) This field can be suitably programmed to control the maximum possible value of Tx equalization coefficient C (-1). The adaptation process within DWC_xlgpcs is constrained so that C(-1) does not exceed the value programmed to this field. This field controls only the integer part (upper 4-bits). Fractional part (lower 2 bits) is always zero **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12||DWCXLP_E32) ? 0x7 : 0x3" **Exists:** Always

7.1.130 VR_PMA_MP_25G_16G_TX_EQ_CTRL3

- **Description:** VR PMA Synopsys Multi-protocol PHY Tx Equalization Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x803F : (DWCXLP_APB3_16) ? (0x803F*0x2) : (0x803F*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_LNK_TRAIN_EN

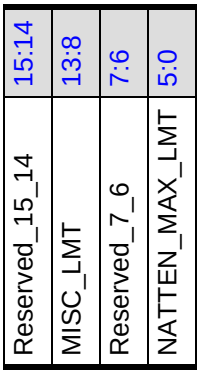


Table 7-135 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL3

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	MISC_LMT	R/W	Transmit Equalization Miscellaneous Limit Transmit Equalization process is carried out by DWC_xlgpcs subject to various rules, so that the computed values of Tx coefficients do not violate the IEEE standard requirements. The adaptation algorithm uses the following variables: - Lmain[7:0] - Lpost[5:0] - Lpre[5:0] This field is used in the following rule check: Lmain[7:0] Lpost[5:0] Lpre[5:0] >= (MISC_LMT) * 4 The desired value of MISC_LMT is 2/4. However, this field provides flexibility to the application to change the limit. (if desired). **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x4 : 0x2" **Exists:** Always

Table 7-135 Fields for Register: VR_PMA_MP_25G_16G_TX_EQ_CTRL3 (Continued)

Bits	Name	Memory Access	Description
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	NATTEN_MAX_LMT	R/W	Maximum Value of Natten[5:0] The Transmit Equalization process makes use of a variable called 'Natten[5:0]'. This field can be used to configure the maximum allowed value of Natten. The desired maximum limit of this variable is 6'h12/B. However, this field provides flexibility to the application to change the limit. (If desired). **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x12 : 0xA" **Exists:** Always

7.1.131 VR_PMA_MP_25G_16G_TX_STS

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G/16G PHY Tx Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8040 : (DWCXLP_APB3_16) ? (0x8040*0x2) : (0x8040*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

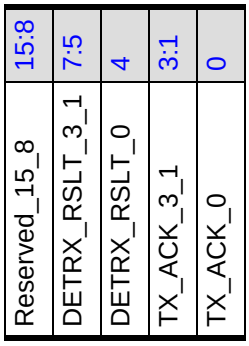


Table 7-136 Fields for Register: VR_PMA_MP_25G_16G_TX_STS

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	DETRX_RSLT_3_1	R	**For Multi-lane configurations :** Receiver Detection Result on Lanes[3:1] This field captures the value of the input port 'xlgpcs_tx_detrx_result_i[3:1]'. The value of this field is valid when 'TX_ACK_3_1' is high. ■ 1'b0 : Receiver not detected ■ 1'b1 : Receiver detected **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-136 Fields for Register: VR_PMA_MP_25G_16G_TX_STS (Continued)

Bits	Name	Memory Access	Description
4	DETRX_RSLT_0	R	Receiver Detection Result on lane 0 of Multi-protocol PHY This field captures the value of the input port 'xlgpcs_tx_detrx_result_i[0]'. The value of this field is valid when 'TX_ACK_0' is high. ■ 1'b0 : Receiver not detected ■ 1'b1 : Receiver detected **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_ACK_3_1	R	**For Multi-lane configurations :** Transmit Acknowledge on Lanes[3:1] This bit captures the value of the input port 'xlgpcs_tx_ack_i[3:1]'. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX_ACK_0	R	Tx Acknowledge on lane 0 of Multi-protocol PHY This bit captures the value of the input port 'xlgpcs_tx_ack_i[0]'. When this bit is read as high, it indicates that the requested transmitter setting is complete or the requested RX-detection operation is complete. After this bit is read high, it remains high till bit TX_REQ_0 or DET_RX_REQ_0 is cleared. **Value After Reset:** 0x0 **Exists:** Always

7.1.132 VR_PMA_MP_25G_16G_TX_GEN_CTRL3

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY TX General Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8042 : (DWCXLP_APB3_16) ? (0x8042*0x2) : (0x8042*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25||DWCXLP_E16))

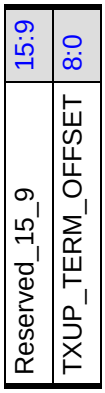


Table 7-137 Fields for Register: VR_PMA_MP_25G_16G_TX_GEN_CTRL3

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	TXUP_TERM_OFFSET	R/W	Offset for TX Up Termination. This field drives the output port xlgpcs_txup_term_offset_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.133 VR_PMA_MP_25G_16G_TX_GEN_CTRL4

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY TX General Control 4 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8043 : (DWCXLP_APB3_16) ? (0x8043*0x2) : (0x8043*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25||DWCXLP_E16))

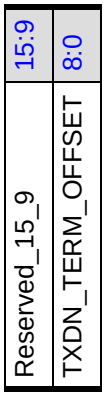


Table 7-138 Fields for Register: VR_PMA_MP_25G_16G_TX_GEN_CTRL4

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	TXDN_TERM_OFFSET	R/W	Offset for TX Down Termination. This field drives the output port xlgpcs_txdn_term_offset_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.134 VR_PMA_MP_25G_16G_TX_MISC_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY TX Miscellaneous Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8044 : (DWCXLP_APB3_16) ? (0x8044*0x2) : (0x8044*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25||DWCXLP_E16))

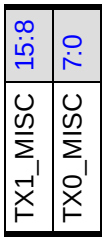


Table 7-139 Fields for Register: VR_PMA_MP_25G_16G_TX_MISC_CTRL0

Bits	Name	Memory Access	Description
15:8	TX1_MISC	* Varies	**For Multi-lane configurations:** TX Miscellaneous control for lane1. This field drives the output port xlgpcs_tx1_misc_o[7:0]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\" "
7:0	TX0_MISC	R/W	TX Miscellaneous control for lane0. This field drives the output port xlgpcs_tx0_misc_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.135 VR_PMA_MP_25G_16G_TX_MISC_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY TX Miscellaneous Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8045 : (DWCXLP_APB3_16) ? (0x8045*0x2) : (0x8045*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16)&&(DWCXLP_MULTI_LANE))

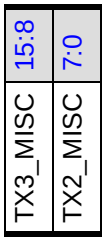


Table 7-140 Fields for Register: VR_PMA_MP_25G_16G_TX_MISC_CTRL1

Bits	Name	Memory Access	Description
15:8	TX3_MISC	R/W	TX Miscellaneous control for lane3. This field drives the output port xlgpcs_tx3_misc_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always
7:0	TX2_MISC	R/W	TX Miscellaneous control for lane2. This field drives the output port xlgpcs_tx2_misc_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.136 VR_PMA_MP_25G_TX_GEN_CTRL5

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY TX General Control 5 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8046 : (DWCXLP_APB3_16) ? (0x8046*0x2) : (0x8046*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25))

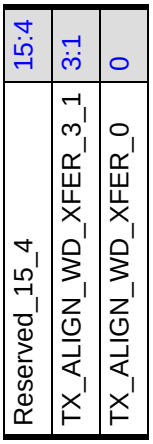


Table 7-141 Fields for Register: VR_PMA_MP_25G_TX_GEN_CTRL5

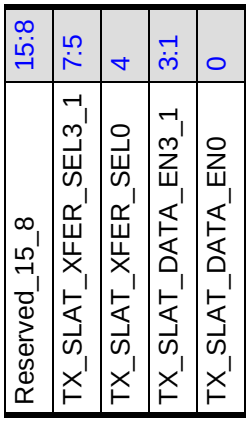
Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_ALIGN_WD_XFER_3_1	* Varies	**For Multi-lane configurations:** Transmitter Alignment Wide Transfer Enable - for Lane3 to 1. This field drives the output port xlgpcs_tx_align_wd_xfer_en_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : ((DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x7: 0x0) " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-141 Fields for Register: VR_PMA_MP_25G_TX_GEN_CTRL5 (Continued)

Bits	Name	Memory Access	Description
0	TX_ALIGN_WD_XFER_0	R/W	Transmitter Alignment Wide Transfer Enable - for Lane0. This field drives the output port xlgpcs_tx_align_wd_xfer_en_o[0]. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x1: 0x0 " **Exists:** Always

7.1.137 VR_PMA_MP_32G_TX_GEN_CTRL5

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY TX General Control 5 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8046 : (DWCXLP_APB3_16) ? (0x8046*0x2) : (0x8046*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E32))



15:8	7:5	4	3:1	0
Reserved_15_8	TX_SLAT_XFER_SEL3_1	TX_SLAT_XFER_SEL0	TX_SLAT_DATA_EN3_1	TX_SLAT_DATA_EN0

Table 7-142 Fields for Register: VR_PMA_MP_32G_TX_GEN_CTRL5

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	TX_SLAT_XFER_SEL3_1	* Varies	**For Multi-lane configurations:** Transmitter Short Latency Transfer Select - for Lane3 to 1. This field drives the output port xlgpcs_tx_slat_xfer_sel_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-142 Fields for Register: VR_PMA_MP_32G_TX_GEN_CTRL5 (Continued)

Bits	Name	Memory Access	Description
4	TX_SLAT_XFER_SEL0	R/W	Transmitter Short Latency Transfer Select - for Lane0. This field drives the output port xlgpcs_tx_slat_xfer_sel_o[0]. **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_SLAT_DATA_EN3_1	* Varies	**For Multi-lane configurations:** Transmitter Short Latency Data Enable - for Lane3 to 1. This field drives the output port xlgpcs_tx_slat_data_en_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	TX_SLAT_DATA_EN0	R/W	Transmitter Short Latency Data Enable - for Lane0. This field drives the output port xlgpcs_tx_slat_data_en_o[0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.138 VR_PMA_MP_25G_TX_DCC_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY TX DCC Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8047 : (DWCXLP_APB3_16) ? (0x8047*0x2) : (0x8047*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25))

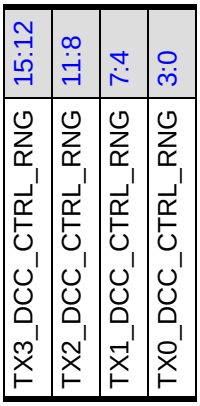


Table 7-143 Fields for Register: VR_PMA_MP_25G_TX_DCC_CTRL

Bits	Name	Memory Access	Description
15:12	TX3_DCC_CTRL_RNG	* Varies	**For Multi-lane configurations:** TX Internal Clock Duty Cycle Control Range - for Lane3. This field drives the output port xlgpcs_tx3_dcc_ctrl_rng_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0xC: 0xB " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-143 Fields for Register: VR_PMA_MP_25G_TX_DCC_CTRL (Continued)

Bits	Name	Memory Access	Description
11:8	TX2_DCC_CTRL_RNG	* Varies	**For Multi-lane configurations:** TX Internal Clock Duty Cycle Control Range - for Lane2. This field drives the output port xlgpcs_tx2_dcc_ctrl_rng_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0xC: 0xB " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7:4	TX1_DCC_CTRL_RNG	* Varies	**For Multi-lane configurations:** TX Internal Clock Duty Cycle Control Range - for Lane1. This field drives the output port xlgpcs_tx1_dcc_ctrl_rng_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0xC: 0xB" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3:0	TX0_DCC_CTRL_RNG	R/W	TX Internal Clock Duty Cycle Control Range - for Lane0. This field drives the output port xlgpcs_tx0_dcc_ctrl_rng_o. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0xC: 0xB " **Exists:** Always

7.1.139 VR_PMA_MP_25G_TX_TERM_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY TX Terminate Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8048 : (DWCXLP_APB3_16) ? (0x8048*0x2) : (0x8048*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25))

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	TX3_TRM_CTRL	Reserved_11	TX2_TRM_CTRL	Reserved_7	TX1_TRM_CTRL	Reserved_3	TX0_TRM_CTRL

Table 7-144 Fields for Register: VR_PMA_MP_25G_TX_TERM_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	TX3_TRM_CTRL	* Varies	**For Multi-lane configurations:** Transmit Termination Control - for Lane3. This field drives the output port xlgpcs_tx3_trm_ctrl_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-144 Fields for Register: VR_PMA_MP_25G_TX_TERM_CTRL (Continued)

Bits	Name	Memory Access	Description
10:8	TX2_TRM_CTRL	* Varies	**For Multi-lane configurations:** Transmit Termination Control - for Lane2. This field drives the output port xlgpcs_tx2_trm_ctrl_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	TX1_TRM_CTRL	* Varies	**For Multi-lane configurations:** Transmit Termination Control - for Lane1. This field drives the output port xlgpcs_tx1_trm_ctrl_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	TX0_TRM_CTRL	R/W	Transmit Termination Control - for Lane0. This field drives the output port xlgpcs_tx0_trm_ctrl_o. **Value After Reset:** 0x2 **Exists:** Always

7.1.140 VR_PMA_MP_25G_TX_WIDTH_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G PHY TX Width Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8049 : (DWCXLP_APB3_16) ? (0x8049*0x2) : (0x8049*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25||DWCXLP_E32))

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	TX3_WIDTH	Reserved_11	TX2_WIDTH	Reserved_7	TX1_WIDTH	Reserved_3	TX0_WIDTH

Table 7-145 Fields for Register: VR_PMA_MP_25G_TX_WIDTH_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	TX3_WIDTH	* Varies	**For Multi-lane configurations :** Tx Datapath Width on lane 3 of the PHY This field drives the output port xlgpcs_tx3_width_o[2:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==7)? 0x5 : (DWCXLP_R_SRDS_INTF==1)? 0x4 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-145 Fields for Register: VR_PMA_MP_25G_TX_WIDTH_CTRL (Continued)

Bits	Name	Memory Access	Description
10:8	TX2_WIDTH	* Varies	**For Multi-lane configurations :** Tx Datapath Width on lane 2 of the PHY This field drives the output port xlgpcs_tx2_width_o[2:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==7)? 0x5 : (DWCXLP_R_SRDS_INTF==1)? 0x4 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	TX1_WIDTH	* Varies	**For Multi-lane configurations :** Tx Datapath Width on lane 1 of the PHY This field drives the output port xlgpcs_tx1_width_o[2:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==7)? 0x5 : (DWCXLP_R_SRDS_INTF==1)? 0x4 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-145 Fields for Register: VR_PMA_MP_25G_TX_WIDTH_CTRL (Continued)

Bits	Name	Memory Access	Description
2:0	TX0_WIDTH	R/W	Tx Datapath Width on lane 0 of the PHY This field controls the width of output transmit data on lane 0. The encoding of the width is as follows: ■ 3'b000: 8-bit ■ 3'b001: 10-bit ■ 3'b010: 16-bit ■ 3'b011: 20-bit ■ 3'b100: 32-bit ■ 3'b101: 40-bit This field drives the output port xlgpcs_tx0_width_o[2:0]. **Value After Reset:** "(DWCXLP_R_SRDS_INTF==7)? 0x5 : (DWCXLP_R_SRDS_INTF==1)? 0x4 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always

7.1.141 VR_PMA_MP_16G_TX_WIDTH_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY TX Width Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8049 : (DWCXLP_APB3_16) ? (0x8049*0x2) : (0x8049*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E16||DWCXLP_E12))

15:14	13:12	11:10	9:8	7:6	5:4	3:2	1:0
Reserved_15_14	TX3_WIDTH	Reserved_11_10	TX2_WIDTH	Reserved_7_6	TX1_WIDTH	Reserved_3_2	TX0_WIDTH

Table 7-146 Fields for Register: VR_PMA_MP_16G_TX_WIDTH_CTRL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	TX3_WIDTH	* Varies	**For Multi-lane configurations :** Tx Datapath Width on lane 3 of the PHY This field drives the output port xlgpcs_tx3_width_o[1:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11:10	Reserved_11_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-146 Fields for Register: VR_PMA_MP_16G_TX_WIDTH_CTRL (Continued)

Bits	Name	Memory Access	Description
9:8	TX2_WIDTH	* Varies	**For Multi-lane configurations :** Tx Datapath Width on lane 2 of the PHY This field drives the output port xlgpcs_tx2_width_o[1:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	TX1_WIDTH	* Varies	**For Multi-lane configurations :** Tx Datapath Width on lane 1 of the PHY This field drives the output port xlgpcs_tx1_width_o[1:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1:0	TX0_WIDTH	R/W	Tx Datapath Width on lane 0 of the PHY This field controls the width of output transmit data on lane 0. The encoding of the width is as follows: ■ 2'b00: 8-bit ■ 2'b01: 10-bit ■ 2'b10: 16-bit ■ 2'b11: 20-bit This field drives the output port xlgpcs_tx0_width_o[1:0]. **Value After Reset:** "(DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always

7.1.142 VR_PMA_MP_25G_RX_VCO_CFG_0

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Control0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x804A : (DWCXLP_APB3_16) ? (0x804A*0x2) : (0x804A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

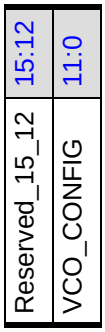


Table 7-147 Fields for Register: VR_PMA_MP_25G_RX_VCO_CFG_0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	VCO_CONFIG	R/W	RX CDR VCO Configuration Bus for Lane 0 This field drives the output port xlgpcs_rx0_vco_cfg_o[11:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.143 VR_PMA_MP_25G_RX_VCO_CFG_1

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Control1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x804B : (DWCXLP_APB3_16) ? (0x804B*0x2) : (0x804B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)&&DWCXLP_MULTI_LANE

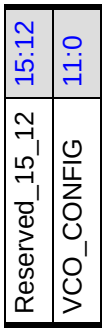


Table 7-148 Fields for Register: VR_PMA_MP_25G_RX_VCO_CFG_1

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	VCO_CONFIG	R/W	RX CDR VCO Configuration Bus for Lane 1 This field drives the output port xlgpcs_rx1_vco_cfg_o[11:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.144 VR_PMA_MP_25G_RX_VCO_CFG_2

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Control2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x804C : (DWCXLP_APB3_16) ? (0x804C*0x2) : (0x804C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)&&DWCXLP_MULTI_LANE

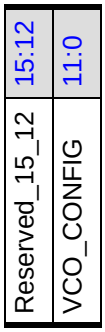


Table 7-149 Fields for Register: VR_PMA_MP_25G_RX_VCO_CFG_2

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	VCO_CONFIG	R/W	RX CDR VCO Configuration Bus for Lane 2 This field drives the output port xlgpcs_rx2_vco_cfg_o[11:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.145 VR_PMA_MP_25G_RX_VCO_CFG_3

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Control3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x804D : (DWCXLP_APB3_16) ? (0x804D*0x2) : (0x804D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)&&DWCXLP_MULTI_LANE

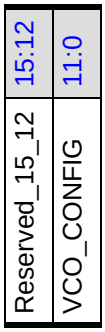
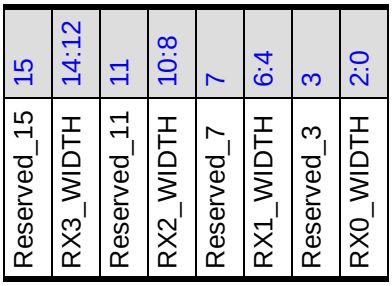


Table 7-150 Fields for Register: VR_PMA_MP_25G_RX_VCO_CFG_3

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	VCO_CONFIG	R/W	RX CDR VCO Configuration Bus for Lane 3 This field drives the output port xlgpcs_rx3_vco_cfg_o[11:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.146 VR_PMA_MP_25G_RX_WIDTH_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G PHY RX Width Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x804E : (DWCXLP_APB3_16) ? (0x804E*0x2) : (0x804E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25||DWCXLP_E32))



15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	RX3_WIDTH	Reserved_11	RX2_WIDTH	Reserved_7	RX1_WIDTH	Reserved_3	RX0_WIDTH

Table 7-151 Fields for Register: VR_PMA_MP_25G_RX_WIDTH_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	RX3_WIDTH	* Varies	**For Multi-lane configurations :** Rx Datapath Width on lane 3 of the PHY This field drives the output port xlgpcs_rx3_width_o[2:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==7)? 0x5 : (DWCXLP_R_SRDS_INTF==1)? 0x4 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-151 Fields for Register: VR_PMA_MP_25G_RX_WIDTH_CTRL (Continued)

Bits	Name	Memory Access	Description
10:8	RX2_WIDTH	* Varies	**For Multi-lane configurations :** Rx Datapath Width on lane 2 of the PHY This field drives the output port xlgpcs_rx2_width_o[2:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==7)? 0x5 : (DWCXLP_R_SRDS_INTF==1)? 0x4 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	RX1_WIDTH	* Varies	**For Multi-lane configurations :** Rx Datapath Width on lane 1 of the PHY This field drives the output port xlgpcs_rx1_width_o[2:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==7)? 0x5 : (DWCXLP_R_SRDS_INTF==1)? 0x4 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-151 Fields for Register: VR_PMA_MP_25G_RX_WIDTH_CTRL (Continued)

Bits	Name	Memory Access	Description
2:0	RX0_WIDTH	R/W	Rx Datapath Width on lane 0 of the PHY This field controls the width of input receive data on lane 0. The encoding of the width is as follows: ■ 3'b000: 8-bit ■ 3'b001: 10-bit ■ 3'b010: 16-bit ■ 3'b011: 20-bit ■ 3'b100: 32-bit ■ 3'b101: 40-bit This field drives the output port xlgpcs_rx0_width_o[2:0]. **Value After Reset:** "(DWCXLP_R_SRDS_INTF==7)? 0x5 : (DWCXLP_R_SRDS_INTF==1)? 0x4 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always

7.1.147 VR_PMA_MP_16G_RX_WIDTH_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY RX Width Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x804E : (DWCXLP_APB3_16) ? (0x804E*0x2) : (0x804E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E16||DWCXLP_E12))

15:14	13:12	11:10	9:8	7:6	5:4	3:2	1:0
Reserved_15_14	RX3_WIDTH	Reserved_11_10	RX2_WIDTH	Reserved_7_6	RX1_WIDTH	Reserved_3_2	RX0_WIDTH

Table 7-152 Fields for Register: VR_PMA_MP_16G_RX_WIDTH_CTRL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	RX3_WIDTH	* Varies	**For Multi-lane configurations :** Rx Datapath Width on lane 3 of the PHY This field drives the output port xlgpcs_rx3_width_o[1:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11:10	Reserved_11_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-152 Fields for Register: VR_PMA_MP_16G_RX_WIDTH_CTRL (Continued)

Bits	Name	Memory Access	Description
9:8	RX2_WIDTH	* Varies	**For Multi-lane configurations :** Rx Datapath Width on lane 2 of the PHY This field drives the output port xlgpcs_rx2_width_o[1:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	RX1_WIDTH	* Varies	**For Multi-lane configurations :** Rx Datapath Width on lane 1 of the PHY This field drives the output port xlgpcs_rx1_width_o[1:0] . **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1:0	RX0_WIDTH	R/W	Rx Datapath Width on lane 0 of the PHY. This field controls the width of input receive data on lane 0. The encoding of the width is as follows: ■ 2'b00: 8-bit ■ 2'b01: 10-bit ■ 2'b10: 16-bit ■ 2'b11: 20-bit This field drives the output port xlgpcs_rx0_width_o[1:0]. **Value After Reset:** "(DWCXLP_R_SRDS_INTF==4)? 0x3 : 0x2 " **Exists:** Always

7.1.148 VR_PMA_MP_25G_16G_12G_RX_GENCTRL0

- **Description:** VR PMA Synopsys Multi-protocol PHY Rx General Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8050 : (DWCXLP_APB3_16) ? (0x8050*0x2) : (0x8050*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

15:13	12	11:9	8	7:5	4	3:1	0
RX_CLK_SFT_3_1	RX_CLK_SFT_0	RX_DT_EN_3_1	RX_DT_EN_0	RX_ALIGN_EN_3_1	RX_ALIGN_EN_0	RX_TERM_EN_3_1	RX_TERM_EN_0

Table 7-153 Fields for Register: VR_PMA_MP_25G_16G_12G_RX_GENCTRL0

Bits	Name	Memory Access	Description
15:13	RX_CLK_SFT_3_1	* Varies	Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_E12)||!(DWCXLP_MULTI_LANE) ? \"read-only\" : \"read-write\""
12	RX_CLK_SFT_0	* Varies	Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_E12) ? \"read-only\" : \"read-write\""

Table 7-153 Fields for Register: VR_PMA_MP_25G_16G_12G_RX_GENCTRL0 (Continued)

Bits	Name	Memory Access	Description
11:9	RX_DT_EN_3_1	* Varies	**For Multi-lane configurations :** Rx Data Enable on Lanes[3:1] This bit should be set to enable the PHY receiver data output on lanes 3 to 1. This field drives the output port 'xlgpcs_rx_data_en_o[3:1]'. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
8	RX_DT_EN_0	R/W	Rx Data Enable on lane 0 of Multi-protocol PHY This bit should be set to enable the PHY receiver data output on lane 0. This bit drives the output port 'xlgpcs_rx_data_en_o[0]'. **Value After Reset:** 0x1 **Exists:** Always
7:5	RX_ALIGN_EN_3_1	* Varies	Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_E12)||!(DWCXLP_MULTI_LANE) ? \"read-only\" : \"read-write\""
4	RX_ALIGN_EN_0	* Varies	Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_E12) ? \"read-only\" : \"read-write\""

Table 7-153 Fields for Register: VR_PMA_MP_25G_16G_12G_RX_GENCTRL0 (Continued)

Bits	Name	Memory Access	Description
3:1	RX_TERM_EN_3_1	* Varies	**For Multi-lane configurations :** Rx Termination Enable on Lanes[3:1] When these bits are set, PHY Rx is terminated with a nominal 50 ohm resistance. Otherwise, the termination is in high impedance. This field drives the output port 'xlgpcs_rx_term_en_o[3:1]'. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	RX_TERM_EN_0	R/W	Rx Termination Enable on lane 0 of Multi-protocol PHY When this bit is set, PHY Rx is terminated with a nominal 50 ohm resistance. Otherwise, the termination is in high impedance. This field drives the output port 'xlgpcs_rx_term_en_o[0]'. **Value After Reset:** 0x1 **Exists:** Always

7.1.149 VR_PMA_MP_25G_16G_RX_GENCTRL1

- **Description:** VR PMA Synopsys Multi-protocol PHY Rx General Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8051 : (DWCXLP_APB3_16) ? (0x8051*0x2) : (0x8051*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

15:13	12	11:9	8	7:5	4	3:1	0
RX_DIV16P5_CLK_EN_3_1	RX_DIV16P5_CLK_EN_0	RX_TERM_ACDC_3_1	RX_TERM_ACDC_0	RX_RST_3_1	RX_RST_0	RX_INV_3_1	RX_INV_0

Table 7-154 Fields for Register: VR_PMA_MP_25G_16G_RX_GENCTRL1

Bits	Name	Memory Access	Description
15:13	RX_DIV16P5_CLK_EN_3_1	* Varies	**For Multi-lane configurations :** Receiver Divide by 16.5 output clock on lanes [3:1] This field enables rxX_div16p5_clk on lanes 3 to 1. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY||DWCXLP_E32 ? 0x0 : 0x7 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY||DWCXLP_E32 ? \"read-only\" : \"read-write\""

Table 7-154 Fields for Register: VR_PMA_MP_25G_16G_RX_GENCTRL1 (Continued)

Bits	Name	Memory Access	Description
12	RX_DIV16P5_CLK_EN_0	* Varies	Receiver Divide by 16.5 output clock on lane 0 When this bit is set, it enables the divide by 16.5 Rx VCO recovered clock. This clock is available at rx0_div16p5_clk output port of PHY. This bit must be set when operating in RPCS mode **Value After Reset:** "DWCXLP_E32 ? 0x0 : 0x1 " **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""
11:9	RX_TERM_ACDC_3_1	* Varies	**For Multi-lane configurations :** Rx Termination control on Lanes[3:1] This field controls the termination of PHY receiver ■ 0: Floating Rx Termination (DC coupled PHY) ■ 1: Grounded Rx Termination (AC coupled PHY) This field drives the output port 'xlgpcs_rx_term_acdc_o[3:1]'. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY||DWCXLP_E32 ? 0x0 : 0x7 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY||DWCXLP_E32 ? \"read-only\" : \"read-write\""
8	RX_TERM_ACDC_0	* Varies	Rx Termination control on lane 0 of Multi-protocol PHY. ■ 0: DC Termination (Floating Rx) ■ 1: AC Termination (Grounded Rx) This field drives the output port xlgpcs_rx_term_acdc_o[0]. **Value After Reset:** "DWCXLP_E32 ? 0x0 : 0x1 " **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""

Table 7-154 Fields for Register: VR_PMA_MP_25G_16G_RX_GENCTRL1 (Continued)

Bits	Name	Memory Access	Description
7:5	RX_RST_3_1	* Varies	**For Multi-lane configurations :** Rx reset on Lanes[3:1] When this field is set, RX data path, all the receiver settings and state machines of the PHY (on lanes 3 to 1) are reset This field drives the output port xgxs_rx_reset_o[3:1] when DWC_xlgpcs is in POWER_GOOD state. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
4	RX_RST_0	R/W	Rx reset on lane 0 of Multi-protocol PHY When this bit is set, RX data path, all the receiver settings and state machines of the PHY are reset This field drives the output port xgxs_rx_reset_o[0] when DWC_xlgpcs is in POWER_GOOD state. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_INV_3_1	* Varies	**For Multi-lane configurations :** Rx Data Invert on Lanes[3:1] This field drives the output port xlgpcs_rx_invert_o[3:1] **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	RX_INV_0	R/W	Rx Data Invert on lane 0 of Multi-protocol PHY When this bit is set, the data on PHY Rx serial lines are logically inverted. This signal drives the output port xlgpcs_rx_invert_o[0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.150 VR_PMA_MP_25G_16G_RX_GENCTRL2

- **Description:** VR PMA Synopsys Multi-protocol PHY Rx General Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8052 : (DWCXLP_APB3_16) ? (0x8052*0x2) : (0x8052*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

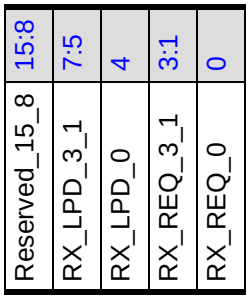


Table 7-155 Fields for Register: VR_PMA_MP_25G_16G_RX_GENCTRL2

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	RX_LPD_3_1	* Varies	**For Multi-lane configurations :** Receiver Lane Power Down on Lanes[3:1] This bit can be set to power down the receiver to a power state equivalent to that of P1. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-155 Fields for Register: VR_PMA_MP_25G_16G_RX_GENCTRL2 (Continued)

Bits	Name	Memory Access	Description
4	RX_LPD_0	R/W	Receiver Lane Power Down on lane 0 of Multi-protocol PHY #RAL : ONLY_RST_TEST Reason: This is self-clear bit. So, it might get cleared before read as 1 set_register_field_attribute ${design}_map/PMA_MMD/VR_PMA_MP_25G_16G_RX_GENCTRL2/RX_REQ_3_1 RALTest\[bit_bash\] {"(!@DWCXLP_25G_ONLY)? 1 : 0"} This bit can be set to power down the receiver to a power state equivalent to that of P1. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_REQ_3_1	* Varies	**For Multi-lane configurations :** Receiver Request on Lanes[3:1] These bits can be set by application to make a Rx operation request.This bit drives the output port xlgpcs_rx_req_o[3:1]. Completion of the request is indicated by fields'RX_ACK_3_1'. These bit are self-cleared when the acknowledgement is asserted. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	RX_REQ_0	R/W	Receiver operation request on lane 0 of Multi-protocol PHY. This bit can be set to 1 by application.This bit is self-cleared when 'xlgpcs_tx_ack_i[0]' is asserted. When this bit is set, a new receiver setting request is made towards the PHY. This bit drives the output port xlgpcs_rx_req_o[0]. A successful completion of receiver request operation on lane 0 is indicated by bit RX_ACK_0 **Value After Reset:** 0x0 **Exists:** Always

7.1.151 VR_PMA_MP_25G_RX_GENCTRL3

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY Rx General Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8053 : (DWCXLP_APB3_16) ? (0x8053*0x2) : (0x8053*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

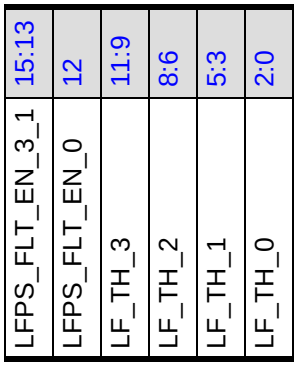


Table 7-156 Fields for Register: VR_PMA_MP_25G_RX_GENCTRL3

Bits	Name	Memory Access	Description
15:13	LFPS_FLT_EN_3_1	* Varies	**For Multi-lane configurations :** Receiver LFPS Signal Detection Filter Enable on lane [3:1] of the PHY This field drives the output port xlgpcs_rx_sigdet_lfps_flt_en_o[3:1] to enable the LFPS filter on lanes 3 to 1 of the PHY. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
12	LFPS_FLT_EN_0	R/W	Receiver LFPS Signal Detection Filter Enable on lane 0 of the PHY. This field drives the output port xlgpcs_rx_sigdet_lfps_flt_en_o[0] to enable the LFPS filter on lane 0 of the PHY. **Value After Reset:** 0x0 **Exists:** Always

Table 7-156 Fields for Register: VR_PMA_MP_25G_RX_GENCTRL3 (Continued)

Bits	Name	Memory Access	Description
11:9	LF_TH_3	* Varies	**For Multi-lane configurations :** Receiver Low-Frequency Signal Detection Threshold on Lane 3 This field drives the output port xlgpcs_rx3_sigdet_lf_th_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE && !DWCXLP_E32) ? 0x4 : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
8:6	LF_TH_2	* Varies	**For Multi-lane configurations :** Receiver Low-Frequency Signal Detection Threshold on Lane 2 This field drives the output port xlgpcs_rx2_sigdet_lf_th_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE && !DWCXLP_E32) ?0x4 : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
5:3	LF_TH_1	* Varies	**For Multi-lane configurations :** Receiver Low-Frequency Signal Detection Threshold on Lane 1 This field drives the output port xlgpcs_rx1_sigdet_lf_th_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE && !DWCXLP_E32) ? 0x4 : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-156 Fields for Register: VR_PMA_MP_25G_RX_GENCTRL3 (Continued)

Bits	Name	Memory Access	Description
2:0	LF_TH_0	R/W	Receiver Low-Frequency Signal Detection Threshold on lane 0 of Multi-protocol 25G PHY. This field drives the output port xlgpcs_rx0_sigdet_lf_th_o[2:0]. Threshold voltages for various values are as follows: ■ 3'b000 : 100mVpp ■ 3'b001 : 120 mVpp ■ 3'b010 : 140 mVpp ■ 3'b011 : 160 mVpp ■ 3'b100 : 180 mVpp ■ 3'b101 : 210 mVpp ■ 3'b110 : 240 mVpp ■ 3'b111 : 270 mVpp **Value After Reset:** "DWCXLP_E32 ? 0x0 : 0x4" **Exists:** Always

7.1.152 VR_PMA_MP_16G_RX_GENCTRL3

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY Rx General Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8053 : (DWCXLP_APB3_16) ? (0x8053*0x2) : (0x8053*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16||DWCXLP_E12)

15:13	12	11:9	8:6	5:3	2:0
LOS_LFPS_EN_3_1	LOS_LFPS_EN_0	LOS_TRSHLD_3	LOS_TRSHLD_2	LOS_TRSHLD_1	LOS_TRSHLD_0

Table 7-157 Fields for Register: VR_PMA_MP_16G_RX_GENCTRL3

Bits	Name	Memory Access	Description
15:13	LOS_LFPS_EN_3_1	* Varies	**For Multi-lane configurations :** Receiver LOS LFPS Enable on lane [3:1] of the PHY This field drives the output port xlgpcs_rx_los_lfps_en_o[3:1] to enable the LFPS filter on lanes 3 to 1 of the PHY. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
12	LOS_LFPS_EN_0	R/W	Receiver LOS LFPS Filter Enable on lane 0 of the PHY This field drives the output port xlgpcs_rx_los_lfps_en_o[0] to enable the LFPS filter on lane 0 of the PHY. **Value After Reset:** 0x0 **Exists:** Always

Table 7-157 Fields for Register: VR_PMA_MP_16G_RX_GENCTRL3 (Continued)

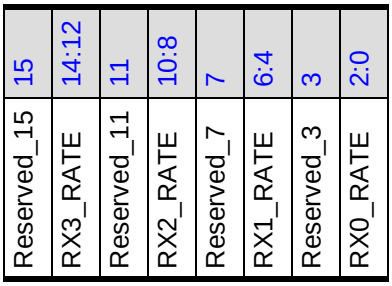
Bits	Name	Memory Access	Description
11:9	LOS_TRSHLD_3	* Varies	**For Multi-lane configurations :** Receiver Loss of signal threshold on Lane 3 This field drives the output port xlgpcs_rx3_los_threshold_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_MULTI_LANE ? (DWCXLP_E12 ? 0x6 : 0x7) : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
8:6	LOS_TRSHLD_2	* Varies	**For Multi-lane configurations :** Receiver Loss of signal threshold on Lane 2 This field drives the output port xlgpcs_rx2_los_threshold_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_MULTI_LANE ? (DWCXLP_E12 ? 0x6 : 0x7) : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
5:3	LOS_TRSHLD_1	* Varies	**For Multi-lane configurations :** Receiver Loss of signal threshold on Lane 1 This field drives the output port xlgpcs_rx1_los_threshold_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_MULTI_LANE ? (DWCXLP_E12 ? 0x6 : 0x7) : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-157 Fields for Register: VR_PMA_MP_16G_RX_GENCTRL3 (Continued)

Bits	Name	Memory Access	Description
2:0	LOS_TRSHLD_0	R/W	Receiver Loss of signal threshold on lane 0 of Multi-protocol 16G/12G PHY. This field drives the output port xlgpcs_rx0_los_threshold_o[2:0]. Threshold voltages for various values are as follows: ■ 3'b000 : 100mVpp ■ 3'b001 : 120 mVpp ■ 3'b010 : 140 mVpp ■ 3'b011 : 160 mVpp ■ 3'b100 : 180 mVpp ■ 3'b101 : 210 mVpp ■ 3'b110 : 240 mVpp ■ 3'b111 : 270 mVpp **Value After Reset:** "DWCXLP_E12 ? 0x6 : 0x7" **Exists:** Always

7.1.153 VR_PMA_MP_25G_RX_RATE_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY Rx Rate Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8054 : (DWCXLP_APB3_16) ? (0x8054*0x2) : (0x8054*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	RX3_RATE	Reserved_11	RX2_RATE	Reserved_7	RX1_RATE	Reserved_3	RX0_RATE

Table 7-158 Fields for Register: VR_PMA_MP_25G_RX_RATE_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	RX3_RATE	* Varies	**For Multi-lane configurations :** Rx date rate on lane 3 of the PHY **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x0: 0x1" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-158 Fields for Register: VR_PMA_MP_25G_RX_RATE_CTRL (Continued)

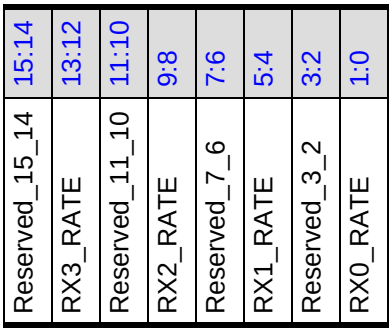
Bits	Name	Memory Access	Description
10:8	RX2_RATE	* Varies	**For Multi-lane configurations :** Rx date rate on lane 2 of the PHY **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x0: 0x1" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	RX1_RATE	* Varies	**For Multi-lane configurations :** Rx date rate on lane 1 of the PHY **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x0: 0x1" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-158 Fields for Register: VR_PMA_MP_25G_RX_RATE_CTRL (Continued)

Bits	Name	Memory Access	Description
2:0	RX0_RATE	R/W	Rx date rate on lane 0 of the PHY Data Rate Encoding is as follows: ■ 3'b000: baud ■ 3'b001: baud/2 ■ 3'b010: baud/4 ■ 3'b011: baud/8 ■ 3'b100: baud/16 ■ 3'b101: Reserved ■ 3'b110: Reserved ■ 3'b111: Reserved **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x0: 0x1" **Exists:** Always

7.1.154 VR_PMA_MP_16G_RX_RATE_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY Rx Rate Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8054 : (DWCXLP_APB3_16) ? (0x8054*0x2) : (0x8054*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16||DWCXLP_E12)



15:14	13:12	11:10	9:8	7:6	5:4	3:2	1:0
Reserved_15_14	RX3_RATE	Reserved_11_10	RX2_RATE	Reserved_7_6	RX1_RATE	Reserved_3_2	RX0_RATE

Table 7-159 Fields for Register: VR_PMA_MP_16G_RX_RATE_CTRL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	RX3_RATE	* Varies	**For Multi-lane configurations :** Rx date rate on lane 3 of the PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11:10	Reserved_11_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-159 Fields for Register: VR_PMA_MP_16G_RX_RATE_CTRL (Continued)

Bits	Name	Memory Access	Description
9:8	RX2_RATE	* Varies	**For Multi-lane configurations :** Rx date rate on lane 2 of the PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	RX1_RATE	* Varies	**For Multi-lane configurations :** Rx date rate on lane 1 of the PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1:0	RX0_RATE	R/W	Rx date rate on lane 0 of the PHY. Data Rate Encoding is as follows: ■ 2'b00: baud ■ 2'b01: baud/2 ■ 2'b10: baud/4 ■ 2'b11: baud/8 **Value After Reset:** 0x0 **Exists:** Always

7.1.155 VR_PMA_MP_25G_16G_RX_POWER_STATE_CTRL

- **Description:** VR PMA Synopsys Multi-protocol PHY Rx Power State Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8055 : (DWCXLP_APB3_16) ? (0x8055*0x2) : (0x8055*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

15:13	12	11:9	8	7:6	5:4	3:2	1:0
Reserved_15_13	EEE_OVR_RIDE	RX_DISABLE_3_1	RX_DISABLE_0	RX3_PSTATE	RX2_PSTATE	RX1_PSTATE	RX0_PSTATE

Table 7-160 Fields for Register: VR_PMA_MP_25G_16G_RX_POWER_STATE_CTRL

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-160 Fields for Register: VR_PMA_MP_25G_16G_RX_POWER_STATE_CTRL (Continued)

Bits	Name	Memory Access	Description
12	EEE_OVR_RIDE	* Varies	For configurations with EEE: Rx Power state over-ride control during EEE DWC_xlgpcs switches Rx power state to P0s by itself during Rx QUIET state of EEE to save power in the PHY. This bit can be set to 1, to over-ride the above behavior of DWC_xlgpcs. If this bit is set, Rx power state would remain unchanged during Rx EEE and would retain the values as programmed in 'RX{lane}_PSTATE' fields of this register. **Note:**When operating in GAUI modes, this bit should be mandatorily set to 1. For configurations without EEE: Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_EEE_EN) ? \"read-write\" : \"read-only\" "
11:9	RX_DISABLE_3_1	* Varies	**For Multi-lane configurations :** Receiver Disable on lanes [3:1] These bits can be set in P1 power state to put the receiver in a low power mode. This field drives the output port 'xlgpcs_rx_disable_o[3:1]'. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
8	RX_DISABLE_0	R/W	Receiver Disable on lane 0 This bit can be set in P1 power state to put the receiver in a low power mode. This field drives the output port 'xlgpcs_rx_disable_o[0]'. **Value After Reset:** 0x0 **Exists:** Always

Table 7-160 Fields for Register: VR_PMA_MP_25G_16G_RX_POWER_STATE_CTRL (Continued)

Bits	Name	Memory Access	Description
7:6	RX3_PSTATE	* Varies	**For Multi-lane configurations :** Rx power state control for lane 3 of Multi-protocol PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
5:4	RX2_PSTATE	* Varies	**For Multi-lane configurations :** Rx power state control for lane 2 of Multi-protocol PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3:2	RX1_PSTATE	* Varies	**For Multi-lane configurations :** Rx power state control for lane 1 of Multi-protocol PHY **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
1:0	RX0_PSTATE	R/W	Rx power state control for lane 0 of Multi-protocol PHY. Power state encoding is as follows: ■ 2'b00: P0 ■ 2'b01: P0s ■ 2'b10: P1 ■ 2'b11: P2 **Value After Reset:** 0x0 **Exists:** Always

7.1.156 VR_PMA_MP_25G_RX_GENCTRL5

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY Rx General Control 5 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8056 : (DWCXLP_APB3_16) ? (0x8056*0x2) : (0x8056*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

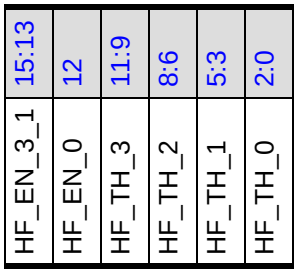


Table 7-161 Fields for Register: VR_PMA_MP_25G_RX_GENCTRL5

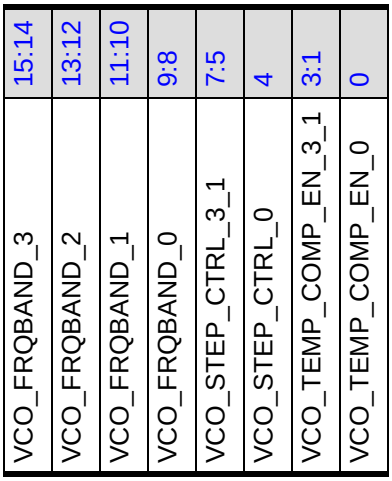
Bits	Name	Memory Access	Description
15:13	HF_EN_3_1	* Varies	**For Multi-lane configurations :** Receiver High-Pass Signal Detection Filter Enable on lane [3:1] of the PHY This field drives the output port xlgpcs_rx_sigdet_hf_en_o[3:1] to enable the High-Pass filter on lanes 3 to 1 of the PHY. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
12	HF_EN_0	R/W	Receiver High-Pass Signal Detection Filter Enable on lane 0 of the PHY This field drives the output port xlgpcs_rx_sigdet_hf_en_o[0] to enable the High-Pass filter on lane 0 of the PHY. **Value After Reset:** 0x0 **Exists:** Always

Table 7-161 Fields for Register: VR_PMA_MP_25G_RX_GENCTRL5 (Continued)

Bits	Name	Memory Access	Description
11:9	HF_TH_3	* Varies	**For Multi-lane configurations :** Receiver High-Frequency Signal Detection Threshold on Lane 3 This field drives the output port xlgpcs_rx3_sigdet_hf_th_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_MULTI_LANE ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
8:6	HF_TH_2	* Varies	**For Multi-lane configurations :** Receiver High-Frequency Signal Detection Threshold on Lane 2 This field drives the output port xlgpcs_rx2_sigdet_hf_th_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_MULTI_LANE ?0x2 : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
5:3	HF_TH_1	* Varies	**For Multi-lane configurations :** Receiver High-Frequency Signal Detection Threshold on Lane 1 This field drives the output port xlgpcs_rx1_sigdet_hf_th_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_MULTI_LANE ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
2:0	HF_TH_0	R/W	Receiver High-Frequency Signal Detection Threshold on lane 0 of Multi-protocol 25G PHY. This field drives the output port xlgpcs_rx0_sigdet_hf_th_o[2:0]. **Value After Reset:** 0x2 **Exists:** Always

7.1.157 VR_PMA_MP_16G_RX_CDR_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 16G PHY Rx CDR Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8056 : (DWCXLP_APB3_16) ? (0x8056*0x2) : (0x8056*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16)



15:14	13:12	11:10	9:8	7:5	4	3:1	0
VCO_FRQBAND_3	VCO_FRQBAND_2	VCO_FRQBAND_1	VCO_FRQBAND_0	VCO_STEP_CTRL_3_1	VCO_STEP_CTRL_0	VCO_TEMP_COMP_EN_3_1	VCO_TEMP_COMP_EN_0

Table 7-162 Fields for Register: VR_PMA_MP_16G_RX_CDR_CTRL

Bits	Name	Memory Access	Description
15:14	VCO_FRQBAND_3	* Varies	**For Multi-lane configurations:** RX CDR VCO Frequency Band lane3 This field drives the output port xlgpcs_vco_freqband3_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x1" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-162 Fields for Register: VR_PMA_MP_16G_RX_CDR_CTRL (Continued)

Bits	Name	Memory Access	Description
13:12	VCO_FRQBAND_2	* Varies	**For Multi-lane configurations:** RX CDR VCO Frequency Band lane2 This field drives the output port xlgpcs_vco_freqband2_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 :0x1" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\"""
11:10	VCO_FRQBAND_1	* Varies	**For Multi-lane configurations:** RX CDR VCO Frequency Band lane1 This field drives the output port xlgpcs_vco_freqband1_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x1" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\"""
9:8	VCO_FRQBAND_0	R/W	RX CDR VCO Frequency Band lane0 This field drives the output port xlgpcs_vco_freqband0_o[1:0]. **Value After Reset:** 0x1 **Exists:** Always
7:5	VCO_STEP_CTRL_3_1	* Varies	**For Multi-lane configurations:** RX_CDR VCO Step Control Lane 3 to 1 This field drives the output port xlgpcs_vco_step_ctrl_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY)? 0x0 : 0x7" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\"""

Table 7-162 Fields for Register: VR_PMA_MP_16G_RX_CDR_CTRL (Continued)

Bits	Name	Memory Access	Description
4	VCO_STEP_CTRL_0	R/W	RX_CDR VCO Step Control Lane 0 This field drives the output port xlgpcs_vco_step_ctrl_o[0]. **Value After Reset:** 0x1 **Exists:** Always
3:1	VCO_TEMP_COMP_EN_3_1	* Varies	**For Multi-lane configurations:** RX_CDR VCO Temperature Compensation Enable on Lane 3 to 1 This field drives the output port xlgpcs_vco_temp_comp_en_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY)? 0x0 :0x7" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
0	VCO_TEMP_COMP_EN_0	R/W	RX_CDR VCO Temperature Compensation Enable Lane 0 This field drives the output port xlgpcs_vco_temp_comp_en_o[0]. **Value After Reset:** 0x1 **Exists:** Always

7.1.158 VR_PMA_MP_12G_RX_GENCTRL5

- **Description:** VR PMA Synopsys Multi-protocol 12G PHY Rx General Control Register 5
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8056 : (DWCXLP_APB3_16) ? (0x8056*0x2) : (0x8056*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E12)

15:13	12	11:9	8	7:5	4	3:1	0
DFE_EN_3_1	DFE_EN_0	AFE_EN_3_1	AFE_EN_0	RX_CDR_VCO_LOW_FREQ_3_1	RX_CDR_VCO_LOW_FREQ_0	RX_CDR_TRACK_EN_3_1	RX_CDR_TRACK_EN_0

Table 7-163 Fields for Register: VR_PMA_MP_12G_RX_GENCTRL5

Bits	Name	Memory Access	Description
15:13	DFE_EN_3_1	* Varies	**For Multi-lane configurations:** Rx DFE Enable on lanes [3:1] of Synopsys Multi-Protocol 12G PHY This bit drives the output port xlgpcs_rx_adapt_dfe_en_o[3:1]. This bit can be set to enable Rx adaption and decision feedback equalization (DFE) circuitry and applies the input setting of DFE Tap1: -rx{lane}_eq_dfe_tap1[7:0]. If this bit is not set, the change on the input DFE settings can be applied using a 'rxX_req/rxX_ack' handshake. If this bit is set, DFE settings are initialized to the inputs and RX DFE adaption routines are executed to achieve optimal DFE settings **For single lane configurations:** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE) ? \"0x7\" : \"0x0\"" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
12	DFE_EN_0	R/W	Rx DFE Enable on lane 0 of Synopsys Multi-Protocol 12G PHY This bit drives the output port xlgpcs_rx_adapt_dfe_en_o[0]. This bit can be set to enable Rx adaption and decision feedback equalization (DFE) circuitry and applies the input setting of DFE Tap1: -rx0_eq_dfe_tap1[7:0]. If this bit is not set, the change on the input DFE settings can be applied using a 'rx0_req/rx0_ack' handshake. If this bit is set, DFE settings are initialized to the inputs and RX DFE adaption routines are executed to achieve optimal DFE settings. **Value After Reset:** 0x1 **Exists:** Always

Table 7-163 Fields for Register: VR_PMA_MP_12G_RX_GENCTRL5 (Continued)

Bits	Name	Memory Access	Description
11:9	AFE_EN_3_1	* Varies	**For Multi-lane configurations:** Rx Adaptation AFE Enable on lanes [3:1]of Synopsys Multi Protocol 12G PHY This bit drives the output port xlgpcs_rx_adapt_afe_en_o[3:1]. This bit can be set to enable Rx adaption circuitry and applies the following input receiver equalization settings to the PHY: ■ rx{lane}_eq_att_lvl[2:0] ■ rx{lane}_eq_vga1_gain[3:0] ■ rx{lane}_eq_vga2_gain[3:0] ■ rx{lane}_eq_ctle_pole[2:0] ■ rx{lane}_eq_ctle_boost[4:0] If this bit is not set, change on input AFE settings can be applied using rxX_req/rxX_ack handshake. If this bit is set, AFE equalization settings are initialized to the PHY inputs, but any further change on these inputs are ignored by PHY and RX AFE adaption routines are executed to achieve optimal AFE settings **For single lane configurations:** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE) ? \"0x7\" : \"0x0\"" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-163 Fields for Register: VR_PMA_MP_12G_RX_GENCTRL5 (Continued)

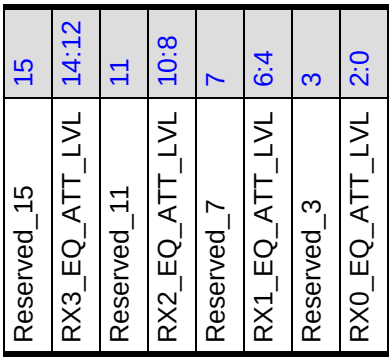
Bits	Name	Memory Access	Description
8	AFE_EN_0	R/W	Rx Adaptation AFE Enable on lane 0 of Synopsys Multi-Protocol 12G PHY This bit drives the output port xlgpcs_rx_adapt_afe_en_o[0]. This bit can be set to enable Rx adaption circuitry and applies the following input receiver equalization settings to the PHY: ■ rx0_eq_att_lvl[2:0] ■ rx0_eq_vga1_gain[3:0] ■ rx0_eq_vga2_gain[3:0] ■ rx0_eq_ctle_pole[2:0] ■ rx0_eq_ctle_boost[4:0] If this bit is not set, change on input AFE settings can be applied using rx0_req/rx0_ack handshake. If this bit is set, AFE equalization settings are initialized to the PHY inputs, but any further change on these inputs are ignored by PHY and RX AFE adaption routines are executed to achieve optimal AFE settings **Value After Reset:** 0x1 **Exists:** Always
7:5	RX_CDR_VCO_LOW_FREQ_3_1	* Varies	**For Multi-lane configurations:** Rx VCO lower frequency band mode on lanes [3:1] of the Synopsys MultiProtocol 12G PHY This field controls the frequency of the Rx VCO to a lower-frequency operating band. This field drives the output port xlgpcs_rx_cdr_vco_lowfreq_o[3:1]. **For single lane configurations:** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE) ? \"0x7\" : \"0x0\"" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
4	RX_CDR_VCO_LOW_FREQ_0	R/W	Rx VCO lower frequency band mode on lane 0 of the Synopsys MultiProtocol 12G PHY This field controls the frequency of the Rx VCO to a lower-frequency operating band. This field drives the output port xlgpcs_rx_cdr_vco_lowfreq_o[0]. **Value After Reset:** 0x1 **Exists:** Always

Table 7-163 Fields for Register: VR_PMA_MP_12G_RX_GENCTRL5 (Continued)

Bits	Name	Memory Access	Description
3:1	RX_CDR_TRACK_EN_3_1	* Varies	**For Multi-lane configurations:** Rx CDR Tracking Enable on Lane 0 of Synopsys 12G PHY This bit should be set to enable CDR tracking of receive data on lane 0 of the PHY. This bit drives the output port 'xlgpcs_rx_cdr_track_en_o[0]'. **For single lane configurations:** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE) ? \"0x7\" : \"0x0\"" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
0	RX_CDR_TRACK_EN_0	R/W	Rx CDR Tracking Enable on Lane 0 of Synopsys 12G PHY This bit should be set to enable CDR tracking of receive data on lane 0 of the PHY. This bit drives the output port 'xlgpcs_rx_cdr_track_en_o[0]'. **Value After Reset:** 0x1 **Exists:** Always

7.1.159 VR_PMA_MP_25G_16G_RX_ATTN_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Rx Attenuation Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8057 : (DWCXLP_APB3_16) ? (0x8057*0x2) : (0x8057*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)



15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	RX3_EQ_ATT_LVL	Reserved_11	RX2_EQ_ATT_LVL	Reserved_7	RX1_EQ_ATT_LVL	Reserved_3	RX0_EQ_ATT_LVL

Table 7-164 Fields for Register: VR_PMA_MP_25G_16G_RX_ATTN_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	RX3_EQ_ATT_LVL	* Varies	**For Multi-lane configurations :** Rx Equalization Attenuation level for lane 3 of the PHY This field drives the output port xlgpcs_rx3_eq_att_lvl_o[2:0]. This field controls the AFE attenuation level of the PHY. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_25G_ONLY) ? \"read-write\" : \"read-only\" "
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-164 Fields for Register: VR_PMA_MP_25G_16G_RX_ATTN_CTRL (Continued)

Bits	Name	Memory Access	Description
10:8	RX2_EQ_ATT_LVL	* Varies	**For Multi-lane configurations :** Rx Equalization Attenuation level for lane 2 of the PHY This field drives the output port xlgpcs_rx2_eq_att_lvl_o[2:0]. This field controls the AFE attenuation level of the PHY. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_25G_ONLY) ? \"read-write\" : \"read-only\" "
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	RX1_EQ_ATT_LVL	* Varies	**For Multi-lane configurations :** Rx Equalization Attenuation level for lane 1 of the PHY This field drives the output port xlgpcs_rx1_eq_att_lvl_o[2:0]. This field controls the AFE attenuation level of the PHY. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_25G_ONLY) ? \"read-write\" : \"read-only\" "
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_EQ_ATT_LVL	R/W	Rx Equalization Attenuation level for lane 0 of the PHY This field drives the output port xlgpcs_rx0_eq_att_lvl_o[2:0]. This field controls the AFE attenuation level of the PHY. **Value After Reset:** 0x0 **Exists:** Always

7.1.160 VR_PMA_MP_25G_16G_RX_EQ_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY Rx Equalization Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8058 : (DWCXLP_APB3_16) ? (0x8058*0x2) : (0x8058*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)

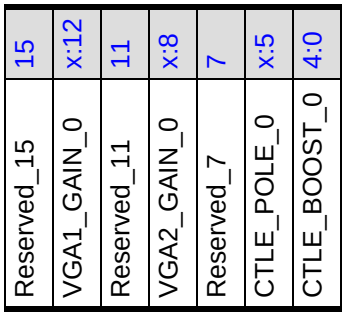


Table 7-165 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL0

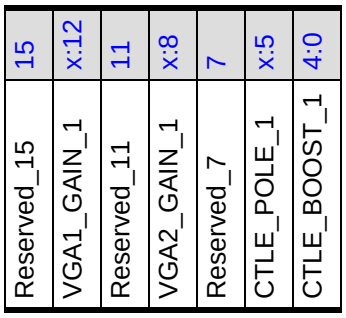
Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)
x:12	VGA1_GAIN_0	R/W	Rx Equalization VGA1 Gain on lane 0 of the PHY This field drives the output port xlgpcs_rx0_eq_vga1_gain_o. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "DWCXLP_E12 ? 0x7 : DWCXLP_E16 ? 0x3 : 0x5" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E16||DWCXLP_E25) ? \"3\" : \"4\" " + 11
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)

Table 7-165 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL0 (Continued)

Bits	Name	Memory Access	Description
x:8	VGA2_GAIN_0	R/W	Rx Equalization VGA2 Gain on lane 0 of the PHY This field drives the output port xlgpcs_rx0_eq_vga2_gain_o.This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "DWCXLP_E12 ? 0x7 : DWCXLP_E16 ? 0x3 : 0x5" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E16||DWCXLP_E25) ? \"3\" : \"4\" " + 7
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)
x:5	CTLE_POLE_0	R/W	Rx Equalization CTLE Pole Value on lane 0 of the PHY This field drives the output port xlgpcs_rx0_eq_ctle_pole_o.This field controls the continuous time linear equalizer boost pole location of the PHY. **Value After Reset:** "DWCXLP_E12 ? 0x2 : DWCXLP_E16 ? 0x0 : (DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x3 : 0x2 " **Exists:** Always **Range Variable[x]:** "!(DWCXLP_E12) ? \"2\" : \"3\" " + 4
4:0	CTLE_BOOST_0	R/W	Rx Equalization CTLE Boost value on lane 0 of the PHY This field drives the output port xlgpcs_rx0_eq_ctle_boost_o[4:0].This field controls the CTLE boost level. **Value After Reset:** "DWCXLP_E16||DWCXLP_E12 ? 0xA : (DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0xC : 0x11 " **Exists:** Always

7.1.161 VR_PMA_MP_25G_16G_RX_EQ_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Rx Equalization Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8059 : (DWCXLP_APB3_16) ? (0x8059*0x2) : (0x8059*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE



15	x:12	11	x:8	7	x:5	4:0
Reserved_15	VGA1_GAIN_1	Reserved_11	VGA2_GAIN_1	Reserved_7	CTLE_POLE_1	CTLE_BOOST_1

Table 7-166 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)&&DWCXLP_MULTI_LANE
x:12	VGA1_GAIN_1	R/W	Rx Equalization VGA1 Gain on lane 1 of the PHY This field drives the output port xlgpcs_rx1_eq_vga1_gain_o. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "DWCXLP_E12 ? 0x7 : DWCXLP_E16 ? 0x3 : 0x5" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E16||DWCXLP_E25) ? \"3\" : \"4\" " + 11

Table 7-166 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL1 (Continued)

Bits	Name	Memory Access	Description
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)&&DWCXLP_MULTI_LANE
x:8	VGA2_GAIN_1	R/W	Rx Equalization VGA2 Gain on lane 1 of the PHY This field drives the output port xlgpcs_rx1_eq_vga2_gain_o.This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "DWCXLP_E12 ? 0x7 : DWCXLP_E16 ? 0x3 : 0x5" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E16||DWCXLP_E25) ? \"3\" : \"4\" " + 7
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)&&DWCXLP_MULTI_LANE
x:5	CTLE_POLE_1	R/W	Rx Equalization CTLE Pole Value on lane 1 of the PHY This field drives the output port xlgpcs_rx1_eq_ctle_pole_o.This field controls the continuous time linear equalizer boost pole location of the PHY. **Value After Reset:** "DWCXLP_E12 ? 0x2 : DWCXLP_E16 ? 0x0 : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x3 : 0x2 " **Exists:** Always **Range Variable[x]:** "!(DWCXLP_E12) ? \"2\" : \"3\" " + 4
4:0	CTLE_BOOST_1	R/W	Rx Equalization CTLE Boost value on lane 1 of the PHY This field drives the output port xlgpcs_rx1_eq_ctle_boost_o[4:0].This field controls the CTLE boost level. **Value After Reset:** "DWCXLP_E16||DWCXLP_E12 ? 0xA : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0xc : 0x11 " **Exists:** Always

7.1.162 VR_PMA_MP_25G_16G_RX_EQ_CTRL2

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Rx Equalization Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x805A : (DWCXLP_APB3_16) ? (0x805A*0x2) : (0x805A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

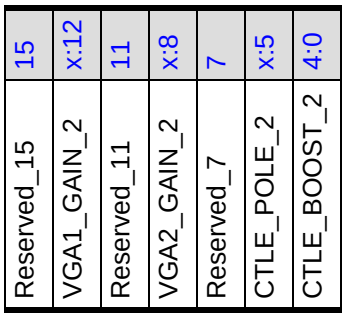


Table 7-167 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL2

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)&&DWCXLP_MULTI_LANE
x:12	VGA1_GAIN_2	R/W	Rx Equalization VGA1 Gain on lane 2 of the PHY This field drives the output port xlgpcs_rx2_eq_vga1_gain_o[]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "DWCXLP_E12 ? 0x7 : DWCXLP_E16 ? 0x3 : 0x5" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E16||DWCXLP_E25) ? \"3\" : \"4\" " + 11

Table 7-167 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL2 (Continued)

Bits	Name	Memory Access	Description
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)&&DWCXLP_MULTI_LANE
x:8	VGA2_GAIN_2	R/W	Rx Equalization VGA2 Gain on lane 2 of the PHY This field drives the output port xlgpcs_rx2_eq_vga2_gain_o[].This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "DWCXLP_E12 ? 0x7 : DWCXLP_E16 ? 0x3 : 0x5" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E16||DWCXLP_E25) ? \"3\" : \"4\" " + 7
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)&&DWCXLP_MULTI_LANE
x:5	CTLE_POLE_2	R/W	Rx Equalization CTLE Pole Value on lane 2 of the PHY This field drives the output port xlgpcs_rx2_eq_ctle_pole_o.This field controls the continuous time linear equalizer boost pole location of the PHY. **Value After Reset:** "DWCXLP_E12 ? 0x2 : DWCXLP_E16 ? 0x0 : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x3 : 0x2 " **Exists:** Always **Range Variable[x]:** "!(DWCXLP_E12) ? \"2\" : \"3\" " + 4
4:0	CTLE_BOOST_2	R/W	Rx Equalization CTLE Boost value on lane 2 of the PHY This field drives the output port xlgpcs_rx2_eq_ctle_boost_o[4:0].This field controls the CTLE boost level. **Value After Reset:** "DWCXLP_E16||DWCXLP_E12 ? 0xA : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0xC : 0x11 " **Exists:** Always

7.1.163 VR_PMA_MP_25G_16G_RX_EQ_CTRL3

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Rx Equalization Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x805B : (DWCXLP_APB3_16) ? (0x805B*0x2) : (0x805B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

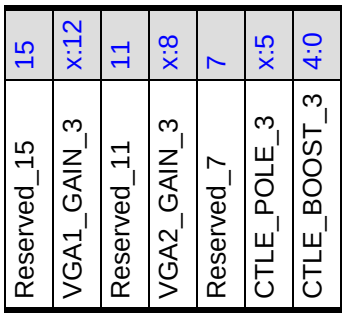


Table 7-168 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL3

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)&&DWCXLP_MULTI_LANE
x:12	VGA1_GAIN_3	R/W	Rx Equalization VGA1 Gain on lane 3 of the PHY. This field drives the output port xlgpcs_rx3_eq_vga1_gain_o[]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "DWCXLP_E12 ? 0x7 : DWCXLP_E16 ? 0x3 : 0x5" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E16||DWCXLP_E25) ? \"3\" : \"4\" " + 11

Table 7-168 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL3 (Continued)

Bits	Name	Memory Access	Description
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)&&DWCXLP_MULTI_LANE
x:8	VGA2_GAIN_3	R/W	Rx Equalization VGA2 Gain on lane 3 of the PHY. This field drives the output port xlgpcs_rx3_eq_vga2_gain_o[].This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "DWCXLP_E12 ? 0x7 : DWCXLP_E16 ? 0x3 : 0x5" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E16||DWCXLP_E25) ? \"3\" : \"4\" " + 7
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&!(DWCXLP_E12)&&DWCXLP_MULTI_LANE
x:5	CTLE_POLE_3	R/W	Rx Equalization CTLE Pole Value on lane 3 of the PHY. This field drives the output port xlgpcs_rx3_eq_ctle_pole_o[].This field controls the continuous time linear equalizer boost pole location of the PHY. **Value After Reset:** "DWCXLP_E12 ? 0x2 :DWCXLP_E16 ? 0x0 : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x3 : 0x2 " **Exists:** Always **Range Variable[x]:** "!(DWCXLP_E12) ? \"2\" : \"3\" " + 4
4:0	CTLE_BOOST_3	R/W	Rx Equalization CTLE Boost value on lane 3 of the PHY. This field drives the output port xlgpcs_rx3_eq_ctle_boost_o[4:0].This field controls the CTLE boost level. **Value After Reset:** "DWCXLP_E16||DWCXLP_E12 ? 0xA : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0xC : 0x11 " **Exists:** Always

7.1.164 VR_PMA_MP_25G_16G_RX_EQ_CTRL4

- **Description:** VR PMA Synopsys Multi-protocol PHY Rx Equalization Control 4 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x805C : (DWCXLP_APB3_16) ? (0x805C*0x2) : (0x805C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

15	14	13	12	11	10	9	8	7:5	4	3:1	0
RX3_AD_REQ	RX2_AD_REQ	RX1_AD_REQ	RX0_AD_REQ	RX_EQ_STRT_CTRL	SELF_MAIN_EN	PING_PONG_EN	SEQ_EQ_EN	CONT_OFF_CAN_3_1	CONT_OFF_CAN_0	CONT_ADAPT_3_1	CONT_ADAPT_0

Table 7-169 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL4

Bits	Name	Memory Access	Description
15	RX3_AD_REQ	* Varies	**For Multi-lane configurations:** Receive Adaptation Request Lane3 This bit drives the 'rpcs_krx_adapt_req3_o' port when Link training is disabled (TR_EN bit of SR PMA MMD BASE-KR PMD Control Register is 0). Software can set this bit to make an Receiver Adaptation Request to the PHY. The acknowledgement from the PHY (for the adaptation request made) is reflected in RX_AD_ACK3 of VR_PMA_MP_25G_16G_12G_MISC_STS0/VR_PMA_MP_32G_MISC_STS0 register. After receiving the acknowledgement, software should clear this register field. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE ? \"read-write\" : \"read-only\""
14	RX2_AD_REQ	* Varies	**For Multi-lane configurations:** Receive Adaptation Request Lane2 This bit drives the 'rpcs_krx_adapt_req2_o' port when Link training is disabled (TR_EN bit of SR PMA MMD BASE-KR PMD Control Register is 0). Software can set this bit to make an Receiver Adaptation Request to the PHY. The acknowledgement from the PHY (for the adaptation request made) is reflected in RX_AD_ACK2 of VR_PMA_MP_25G_16G_12G_MISC_STS0/VR_PMA_MP_32G_MISC_STS0 register. After receiving the acknowledgement, software should clear this register field. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE ? \"read-write\" : \"read-only\""

Table 7-169 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL4 (Continued)

Bits	Name	Memory Access	Description
13	RX1_AD_REQ	* Varies	**For Multi-lane configurations:** Receive Adaptation Request Lane1 This bit drives the 'rpcs_krx_adapt_req1_o' port when Link training is disabled (TR_EN bit of SR PMA MMD BASE-KR PMD Control Register is 0). Software can set this bit to make an Receiver Adaptation Request to the PHY. Software can set this bit to make an Receiver Adaptation Request to the PHY. The acknowledgement from the PHY (for the adaptation request made) is reflected in RX_AD_ACK1 of VR_PMA_MP_25G_16G_12G_MISC_STS0/VR_PMA_MP_32G_MISC_STS0 register. After receiving the acknowledgement, software should clear this register field. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE ? \"read-write\" : \"read-only\""
12	RX0_AD_REQ	* Varies	Receive Adaptation Request Lane0. This bit drives the 'rpcs_krx_adapt_req0_o' port when Link training is disabled (TR_EN bit of SR PMA MMD BASE-KR PMD Control Register is 0). Software can set this bit to make an Receiver Adaptation Request to the PHY. The acknowledgement from the PHY (for the adaptation request made) is reflected in RX_AD_ACK0 of VR_PMA_MP_25G_16G_12G_MISC_STS0/VR_PMA_MP_32G_MISC_STS0 register. After receiving the acknowledgement, software should clear this register field. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) ? \"read-write\" : \"read-only\""

Table 7-169 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL4 (Continued)

Bits	Name	Memory Access	Description
11	RX_EQ_STRT_CTRL	* Varies	**For configurations with DWCXLP_LNK_TRAIN_EN=Enabled:** Receive Equalization Start Control This bits controls the behavior of DWC_xlgpcs at the start of receive equalization process. This bit decides whether DWC_xlgpcs would send PRESET or INIT as the first coefficient update request to the link-partner when KR Training is enabled. There-after, DWC_xlgpcs would send INC/DEC as provided by the Multi-protocol PHY. Value: ■ 0: DWC_xlgpcs would send INIT first ■ 1: DWC_xlgpcs would send PRESET first This bit is present only in configurations with DWCXLP_LNK_TRAIN_EN Enabled. > **For configurations with DWCXLP_LNK_TRAIN_EN=Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
10	SELF_MAIN_EN	* Varies	**For configurations with DWCXLP_LNK_TRAIN_EN=Enabled:** Enable Automatic Generation of Main -C(0) If this bit is set, DWC_xlgpcs generates C(0) (Main) coefficient update requests by itself during Rx equalization process and ignores the update request given by PHY for C(0).This bit can be set to 1,when Ping Pong Method is selected for Rx equalization. If PRE/POST is a INC request, DWC_xlgpcs would generate a INC request for MAIN. If PRE/POST is a DEC request, DWC_xlgpcs would generate a INC request for MAIN. This bit is present only in configurations with DWCXLP_LNK_TRAIN_EN Enabled. **For configurations with DWCXLP_LNK_TRAIN_EN=Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""

Table 7-169 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL4 (Continued)

Bits	Name	Memory Access	Description
9	PING_PONG_EN	* Varies	**For configurations with DWCXLP_LNK_TRAIN_EN=Enabled:** Ping-Pong Mode Enable for Rx equalization In ping-pong method, only one of PRE/POST update request is sent to remote-end device during one cycle of Rx equalization and the other coefficient update is ignored. DWC_xlgpcs alternates (ping-pongs) between PRE and POST requests between iterations. This method works well 'SEQ_EQ_EN' and 'SELF_MAIN_EN' bits are set to 1. This bit is present only in configurations with DWCXLP_LNK_TRAIN_EN Enabled. **For configurations with DWCXLP_LNK_TRAIN_EN=Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""

Table 7-169 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL4 (Continued)

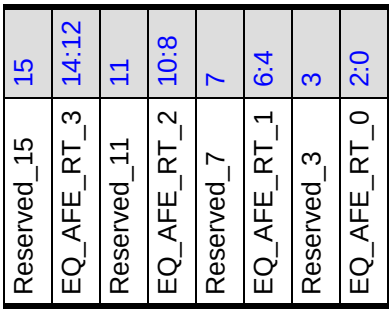
Bits	Name	Memory Access	Description
8	SEQ_EQ_EN	* Varies	**For configurations with DWCXLP_LNK_TRAIN_EN=Enabled:** Sequential Mode Enable for Rx equalization This bit controls the mode for Rx Equalization. ■ 1: Sequential Mode ■ 0: Parallel Mode In parallel mode, DWC_xlgpcs sends all the coefficient update requests (C(-1), C(0), and C(1)) received from PHY (in a particular evaluation iteration) to the remote-end device simultaneously (in the same frame). In sequential mode DWC_xlgpcs sends the coefficient update commands (received in a particular iteration) one by one. DWC_xlgpcs starts with sending C(-1). Then it waits for UPDATED/MAX/MINstatus for C(-1). Then it sends c(0) and waits for UPDATED/MAX/MIN status for C(0). Then it sends C(1) and wait for UPDATED/MAX/MIN status for C(1). This method can be used when the remote-end partner is not capable of receiving simultaneous coefficient update commands for all the three coefficients. This bit is present only in configurations with DWCXLP_LNK_TRAIN_EN Enabled. **For configurations with DWCXLP_LNK_TRAIN_EN=Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_LNK_TRAIN_EN ? \"read-write\" : \"read-only\""
7:5	CONT_OFF_CAN_3_1	* Varies	**For Multi-lane configurations :** Receiver offset cancellation continuous operation on lanes[3:1] **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_E32) ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&!DWCXLP_E32) ? \"read-write\" : \"read-only\" "

Table 7-169 Fields for Register: VR_PMA_MP_25G_16G_RX_EQ_CTRL4 (Continued)

Bits	Name	Memory Access	Description
4	CONT_OFF_CAN_0	* Varies	Receiver offset cancellation continuous operation on lane 0 This bit can be set if continuous receiver offset cancellation is required. If this bit is 0, offset cancellation runs when receiver exits P2 power state. **Value After Reset:** "(DWCXLP_E32) ? 0x0 : 0x1 " **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "
3:1	CONT_ADAPT_3_1	* Varies	Receiver Adaptation Continuous Operation on lanes[3:1] This bit can be set to enable continuous receiver adaptation in the PHY. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_E32) ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&!DWCXLP_E32) ? \"read-write\" : \"read-only\" "
0	CONT_ADAPT_0	* Varies	Receiver Adaptation Continuous Operation on lane 0 This bit can be set to enable continuous receiver adaptation in the PHY. **Value After Reset:** "(DWCXLP_E32) ? 0x0 : 0x1 " **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "

7.1.165 VR_PMA_MP_25G_RX_AFE_RATE_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY Rx AFE Rate Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x805D : (DWCXLP_APB3_16) ? (0x805D*0x2) : (0x805D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	EQ_AFE_RT_3	Reserved_11	EQ_AFE_RT_2	Reserved_7	EQ_AFE_RT_1	Reserved_3	EQ_AFE_RT_0

Table 7-170 Fields for Register: VR_PMA_MP_25G_RX_AFE_RATE_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	EQ_AFE_RT_3	* Varies	**For Multi-lane configurations :** RX Equalization AFE Rate on Lane 3 This field drives the output port xlgpcs_rx3_eq_afe_rate_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x0 : 0x4 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-170 Fields for Register: VR_PMA_MP_25G_RX_AFE_RATE_CTRL (Continued)

Bits	Name	Memory Access	Description
10:8	EQ_AFE_RT_2	* Varies	**For Multi-lane configurations:** RX Equalization AFE Rate on Lane 2 This field drives the output port xlgpcs_rx2_eq_afe_rate_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x0 : 0x4 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	EQ_AFE_RT_1	* Varies	**For Multi-lane configurations :** RX Equalization AFE Rate on Lane 1 This field drives the output port xlgpcs_rx1_eq_afe_rate_o[2:0]. **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x0 : 0x4 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	EQ_AFE_RT_0	R/W	RX Equalization AFE Rate on lane 0 of Multi-protocol 25G PHY. This field drives the output port xlgpcs_rx0_eq_afe_rate_o[2:0]. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x0 : 0x4 " **Exists:** Always

7.1.166 VR_PMA_MP_25G_16G_DFE_TAP_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY DFE Tap Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x805E : (DWCXLP_APB3_16) ? (0x805E*0x2) : (0x805E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)

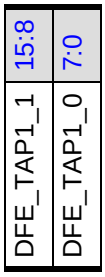


Table 7-171 Fields for Register: VR_PMA_MP_25G_16G_DFE_TAP_CTRL0

Bits	Name	Memory Access	Description
15:8	DFE_TAP1_1	* Varies	**For Multi-lane configurations :** Rx Equalization DFE Tap1 value on lane 1 of the PHY This field drives the output port xlgpcs_rx1_eq_dfe_tap1_o[7:0] **For Single-lane configurations :** Reserved **Value After Reset:** "(DWCXLP_E25)&&DWCXLP_MULTI_LANE ? 0xF : 0x0 " **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\" "
7:0	DFE_TAP1_0	R/W	Rx Equalization DFE Tap1 value on lane 0 of the PHY. This field drives the output port xlgpcs_rx0_eq_dfe_tap1_o[7:0] **Value After Reset:** "(DWCXLP_E25) ? 0xF : 0x0 " **Exists:** Always

7.1.167 VR_PMA_MP_25G_16G_DFE_TAP_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY DFE Tap Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x805F : (DWCXLP_APB3_16) ? (0x805F*0x2) : (0x805F*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

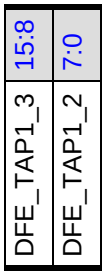


Table 7-172 Fields for Register: VR_PMA_MP_25G_16G_DFE_TAP_CTRL1

Bits	Name	Memory Access	Description
15:8	DFE_TAP1_3	R/W	Rx Equalization DFE Tap1 value on lane 3 of the PHY. This field drives the output port xlgpcs_rx3_eq_dfe_tap1_o[7:0]. **Value After Reset:** "(DWCXLP_E25) ? 0xF : 0x0 " **Exists:** Always
7:0	DFE_TAP1_2	R/W	Rx Equalization DFE Tap1 value on lane 2 of the PHY. This field drives the output port xlgpcs_rx2_eq_dfe_tap1_o[7:0] **Value After Reset:** "(DWCXLP_E25) ? 0xF : 0x0 " **Exists:** Always

7.1.168 VR_PMA_MP_25G_16G_RX_STS

- **Description:** VR PMA Synopsys Multi-protocol PHY Rx Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8060 : (DWCXLP_APB3_16) ? (0x8060*0x2) : (0x8060*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

15:12	11:9	8	7:5	4	3:1	0
Reserved_15_12	HF_SD_3_1	HF_SD_0	LF_SD_3_1	LF_SD_0	RX_ACK_3_1	RX_ACK_0

Table 7-173 Fields for Register: VR_PMA_MP_25G_16G_RX_STS

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	HF_SD_3_1	R	**For Multi-protocol 25G/32G PHY configurations:Receive High-Frequency Signal Detection on Lanes[3:1]** This bit captures the value of the input port xlgpcs_rx_sigdet_hf_i[3:1]. If this bit is set, it indicates that the receiver is receiving high-frequency signals. This signal is valid only for high-frequency data on rxX_p/rxX_m of Multi-protocol 25G/32G PHY. This field is valid only for multi-lane configurations. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-173 Fields for Register: VR_PMA_MP_25G_16G_RX_STS (Continued)

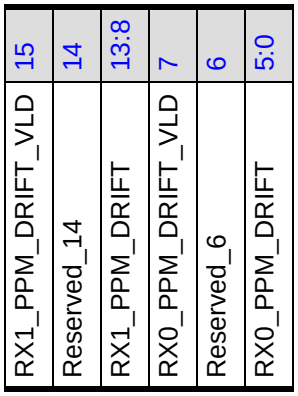
Bits	Name	Memory Access	Description
8	HF_SD_0	R	**For Multi-protocol 25G/32G PHY configurations:Receive High-Frequency Signal Detection on lane 0 of Multi-protocol 25G/32G PHY** This bit captures the value of the input port xlgpcs_rx_sigdet_hf_i[0]. If this bit is set, it indicates that the receiver is receiving high-frequency signals. This signal is valid only for high-frequency data on rxX_p/rxX_m of Multi-protocol 25G/32G PHY. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	LF_SD_3_1	R	**For Multi-protocol 25G/32G PHY configurations:** Receive Low-frequency Signal Detect on lanes 1-3. This field captures the value of the input port 'xlgpcs_rx_sigdet_lf_i[3:1]'. If this bit is set, it indicates that the receiver is receiving low-frequency signals. This signal is valid only for low-frequency data on rxX_p/rxX_m of Multi-protocol 25G/32G PHY. This field is valid only for multi-lane configurations. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
4	LF_SD_0	R	**For Multi-protocol 25G/32G PHY configurations:** Receive Low-Frequency Signal Detection on lane 0 of Multi-protocol 25G/32G PHY This bit captures the value of the input port xlgpcs_rx_sigdet_lf_i[0]. If this bit is set, it indicates that the receiver is receiving low-frequency signals. This signal is valid only for low-frequency data on rxX_p/rxX_m of Multi-protocol 25G/32G PHY. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-173 Fields for Register: VR_PMA_MP_25G_16G_RX_STS (Continued)

Bits	Name	Memory Access	Description
3:1	RX_ACK_3_1	R	**For Multi-lane configurations :** Receive Acknowledge on Lanes[3:1] This bit captures the value of the input port xlgpcs_rx_ack_i[3:1]. If this field is set, it indicates that the requested receiver setting is complete. This field forms a hand-shake with 'RX_REQ_3_1' field. Once this field is set, it stays high till 'RX_REQ_3_1' is de-asserted. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_ACK_0	R	Rx Acknowledge on lane 0 of Multi-protocol PHY. This bit captures the value of the input port xlgpcs_rx_ack_i[0]. If this bit is set, it indicates that the requested receiver setting is complete. This bit forms a hand-shake with 'RX_REQ_0' bit. Once this bit is set, it stays high till 'RX_REQ_0' is de-asserted. **Value After Reset:** 0x0 **Exists:** Always

7.1.169 VR_PMA_MP_25G_16G_RX_PPM_STS0

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G/16G PHY Rx PPM Status 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8061 : (DWCXLP_APB3_16) ? (0x8061*0x2) : (0x8061*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)



15	14	13:8	7	6	5:0
RX1_PPM_DRIFT_VLD	Reserved_14	RX1_PPM_DRIFT	RX0_PPM_DRIFT_VLD	Reserved_6	RX0_PPM_DRIFT

Table 7-174 Fields for Register: VR_PMA_MP_25G_16G_RX_PPM_STS0

Bits	Name	Memory Access	Description
15	RX1_PPM_DRIFT_VLD	R	**For Multi-lane configurations:** RX CDR PPM Drift Valid on lane1 This field is driven by the input port xlgpcs_rx1_ppm_drift_vld_i. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-174 Fields for Register: VR_PMA_MP_25G_16G_RX_PPM_STS0 (Continued)

Bits	Name	Memory Access	Description
13:8	RX1_PPM_DRIFT	R	**For Multi-lane configurations:** RX CDR PPM Drift on lane1 This field is driven by the input port xlgpcs_rx1_ppm_drift_i[5:0]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
7	RX0_PPM_DRIFT_VLD	R	RX CDR PPM Drift Valid on lane0. This field is driven by the input port xlgpcs_rx0_ppm_drift_vld_i. **Value After Reset:** 0x0 **Exists:** Always
6	Reserved_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	RX0_PPM_DRIFT	R	RX CDR PPM Drift on lane0 This field is driven by the input port xlgpcs_rx0_ppm_drift_i[5:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.170 VR_PMA_MP_25G_16G_RX_PPM_STS1

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G/16G PHY Rx PPM Status 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8062 : (DWCXLP_APB3_16) ? (0x8062*0x2) : (0x8062*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE

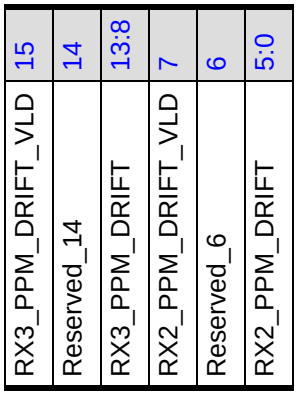


Table 7-175 Fields for Register: VR_PMA_MP_25G_16G_RX_PPM_STS1

Bits	Name	Memory Access	Description
15	RX3_PPM_DRIFT_VLD	R	RX CDR PPM Drift Valid on lane3. This field is driven by the input port xlgpcs_rx1_ppm_drift_vld_i. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	RX3_PPM_DRIFT	R	RX CDR PPM Drift on lane3. This field is driven by the input port xlgpcs_rx3_ppm_drift_i[5:0]. **Value After Reset:** 0x0 **Exists:** Always

Table 7-175 Fields for Register: VR_PMA_MP_25G_16G_RX_PPM_STS1 (Continued)

Bits	Name	Memory Access	Description
7	RX2_PPM_DRIFT_VLD	R	RX CDR PPM Drift Valid on lane2. This field is driven by the input port xlgpcs_rx2_ppm_drift_vld_i. **Value After Reset:** 0x0 **Exists:** Always
6	Reserved_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	RX2_PPM_DRIFT	R	RX CDR PPM Drift on lane2. This field is driven by the input port xlgpcs_rx2_ppm_drift_i[5:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.171 VR_PMA_MP_25G_RX_RECAL_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G PHY Rx Re-calibration Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8063 : (DWCXLP_APB3_16) ? (0x8063*0x2) : (0x8063*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32)

15:14	13:12	11:10	9:8	7:5	4	3:1	0
RX3_BNK_SEL	RX2_BNK_SEL	RX1_BNK_SEL	RX0_BNK_SEL	RCAL_SK_EN_3_1	RCAL_SK_EN_0	RCAL_FR_EN_3_1	RCAL_FR_EN_0

Table 7-176 Fields for Register: VR_PMA_MP_25G_RX_RECAL_CTRL

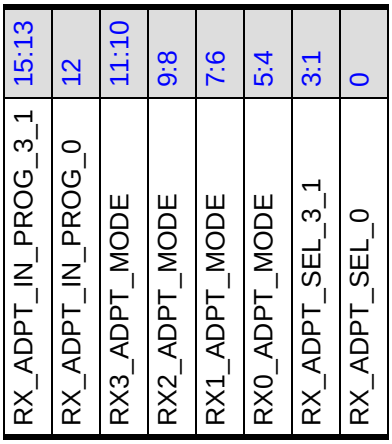
Bits	Name	Memory Access	Description
15:14	RX3_BNK_SEL	* Varies	Rx Recalibration Bank Select-lane 3. This field drives the xlgpcs_rx1_recal_bank_sel_o output port and selects the bank to store the Rx duty cycle correction and IQ correction calibration values This field is present only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_MULTI_LANE ? \"read-write\" : \"read-only\""
13:12	RX2_BNK_SEL	* Varies	Rx Recalibration Bank Select-lane 2. This field drives the xlgpcs_rx1_recal_bank_sel_o output port and selects the bank to store the Rx duty cycle correction and IQ correction calibration values This field is present only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_MULTI_LANE ? \"read-write\" : \"read-only\""

Table 7-176 Fields for Register: VR_PMA_MP_25G_RX_RECAL_CTRL (Continued)

Bits	Name	Memory Access	Description
11:10	RX1_BNK_SEL	* Varies	Rx Recalibration Bank Select-lane 1. This field drives the xlgpcs_rx1_recal_bank_sel_o output port and selects the bank to store the Rx duty cycle correction and IQ correction calibration values This field is present only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_MULTI_LANE ? \"read-write\" : \"read-only\""
9:8	RX0_BNK_SEL	R/W	Rx Recalibration Bank Select-lane 0. This field drives the xlgpcs_rx0_recal_bank_sel_o output port and selects the bank to store the Rx duty cycle correction and IQ correction calibration values **Value After Reset:** 0x0 **Exists:** Always
7:5	RCAL_SK_EN_3_1	* Varies	Rx Skip Re-calibration -lanes 3 to 1 This field drives the xlgpcs_rx_recal_skip_en_o[3:1] port. This field is valid only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_MULTI_LANE ? \"read-write\" : \"read-only\""
4	RCAL_SK_EN_0	R/W	Rx Skip Re-calibration -lane 0. This field drives the xlgpcs_rx_recal_skip_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always
3:1	RCAL_FR_EN_3_1	* Varies	Rx Force Re-calibration -lanes 3 to 1 This field drives the xlgpcs_rx_recal_force_en_o[3:1] port. This field is valid only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_MULTI_LANE ? \"read-write\" : \"read-only\""
0	RCAL_FR_EN_0	R/W	Rx Force Re-calibration -lane 0 This field drives the xlgpcs_rx_recal_force_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always

7.1.172 VR_PMA_MP_25G_16G_ADPT_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY Rx Adaptation Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8064 : (DWCXLP_APB3_16) ? (0x8064*0x2) : (0x8064*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16)



15:13	12	11:10	9:8	7:6	5:4	3:1	0
RX_ADPT_IN_PROG_3_1	RX_ADPT_IN_PROG_0	RX3_ADPT_MODE	RX2_ADPT_MODE	RX1_ADPT_MODE	RX0_ADPT_MODE	RX_ADPT_SEL_3_1	RX_ADPT_SEL_0

Table 7-177 Fields for Register: VR_PMA_MP_25G_16G_ADPT_CTRL0

Bits	Name	Memory Access	Description
15:13	RX_ADPT_IN_PROG_3_1	* Varies	**For Multi-protocol 25G configurations:** Receiver adaptation in progress on Lane 3 to 1 This field drives the output port xlgpcs_rx_adpt_in_prog_o[3:1]. This field is valid only for multi-lane configurations. **For Multi-protocol 16G PHY/Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE)&&(DWCXLP_E25) ? \"read-write\" : \"read-only\""

Table 7-177 Fields for Register: VR_PMA_MP_25G_16G_ADPT_CTRL0 (Continued)

Bits	Name	Memory Access	Description
12	RX_ADPT_IN_PROG_0	* Varies	**For Multi-protocol 25G PHY configurations:** Receiver adaptation in progress on Lane0 This field drives the output port xlgpcs_rx_adpt_in_prog_o[0]. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E25) ? \"read-write\" : \"read-only\""
11:10	RX3_ADPT_MODE	* Varies	**For Multi-lane configurations:** RX Adaptation for lane3 This field drives the output port xlgpcs_rx3_adpt_mode_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
9:8	RX2_ADPT_MODE	* Varies	**For Multi-lane configurations:** RX Adaptation for lane2 This field drives the output port xlgpcs_rx2_adpt_mode_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-177 Fields for Register: VR_PMA_MP_25G_16G_ADPT_CTRL0 (Continued)

Bits	Name	Memory Access	Description
7:6	RX1_ADPT_MODE	* Varies	**For Multi-lane configurations:** RX Adaptation for lane1 This field drives the output port xlgpcs_rx1_adpt_mode_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
5:4	RX0_ADPT_MODE	R/W	RX Adaptation for lane0. This field drives the output port xlgpcs_rx0_adpt_mode_o[1:0]. **Value After Reset:** 0x3 **Exists:** Always
3:1	RX_ADPT_SEL_3_1	* Varies	**For Multi-lane configurations:** Select Storage Bank for RX adaptation on Lane 3 to 1 This field drives the output port xlgpcs_rx_adpt_sel_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
0	RX_ADPT_SEL_0	R/W	Select Storage Bank for RX adaptation on Lane0. This field drives the output port xlgpcs_rx_adpt_sel_o[0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.173 VR_PMA_MP_25G_16G_RX_PPM_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY Rx PPM Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8065 : (DWCXLP_APB3_16) ? (0x8065*0x2) : (0x8065*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16)

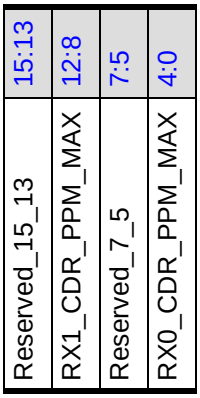


Table 7-178 Fields for Register: VR_PMA_MP_25G_16G_RX_PPM_CTRL0

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	RX1_CDR_PPM_MAX	* Varies	**For Multi-lane configurations:** Maximum Allowed PPM on the RX1 CDR Clock This field drives the output port xlgpcs_rx1_cdr_ppm_max_o[4:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "(DWCXLP_25G_ONLY)? 0x0 : DWCXLP_E16 ? 0x12 : ((DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x14 : 0x13) : (DWCXLP_REF_CLK_78 ? 0x13 : 0x12))" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-178 Fields for Register: VR_PMA_MP_25G_16G_RX_PPM_CTRL0 (Continued)

Bits	Name	Memory Access	Description
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:0	RX0_CDR_PPM_MAX	R/W	Maximum Allowed PPM on the RX0 CDR Clock. This field drives the output port xlgpcs_rx0_cdr_ppm_max_o[4:0]. **Value After Reset:** "DWCXLP_E16 ? 0x12 :(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x14 : 0x13) : (DWCXLP_REF_CLK_78 ? 0x13 : 0x12)" **Exists:** Always

7.1.174 VR_PMA_MP_25G_16G_RX_PPM_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY Rx PPM Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8066 : (DWCXLP_APB3_16) ? (0x8066*0x2) : (0x8066*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE

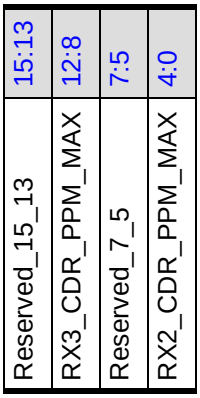


Table 7-179 Fields for Register: VR_PMA_MP_25G_16G_RX_PPM_CTRL1

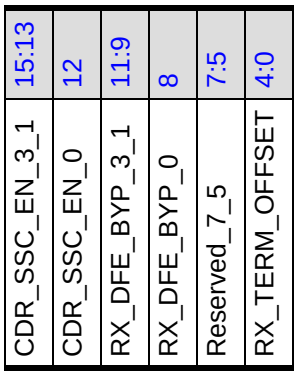
Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	RX3_CDR_PPM_MAX	R/W	Maximum Allowed PPM on the RX3 CDR Clock This field drives the output port xlgpcs_rx3_cdr_ppm_max_o[4:0]. **Value After Reset:** "DWCXLP_E16 ? 0x12 : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x14 : 0x13) : (DWCXLP_REF_CLK_78 ? 0x13 : 0x12)" **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-179 Fields for Register: VR_PMA_MP_25G_16G_RX_PPM_CTRL1 (Continued)

Bits	Name	Memory Access	Description
4:0	RX2_CDR_PPM_MAX	R/W	Maximum Allowed PPM on the RX2 CDR Clock This field drives the output port xlgpcs_rx2_cdr_ppm_max_o[4:0]. **Value After Reset:** "DWCXLP_E16 ? 0x12 : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x14 : 0x13) : (DWCXLP_REF_CLK_78 ? 0x13 : 0x12)" **Exists:** Always

7.1.175 VR_PMA_MP_25G_RX_GEN_CTRL4

- **Description:** VR PMA Synopsys Multi-protocol PHY Rx General Control 4 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8067 : (DWCXLP_APB3_16) ? (0x8067*0x2) : (0x8067*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)



15:13	12	11:9	8	7:5	4:0
CDR_SSC_EN_3_1	CDR_SSC_EN_0	RX_DFE_BYP_3_1	RX_DFE_BYP_0	Reserved_7_5	RX_TERM_OFFSET

Table 7-180 Fields for Register: VR_PMA_MP_25G_RX_GEN_CTRL4

Bits	Name	Memory Access	Description
15:13	CDR_SSC_EN_3_1	* Varies	**For Multi-lane configurations :** Rx CDR SSC-Mode Enable on lanes [3:1] of the PHY This field controls the CDR tracking gains and duration. This field should be set to 3'b111 when receive data (on lanes 3 to 1) has a spread spectrum clock and should be cleared if receive data does not have SSC. This bit drives the output port 'xlgpcs_rx_cdr_ssc_en_o[3:1]'. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\" "

Table 7-180 Fields for Register: VR_PMA_MP_25G_RX_GEN_CTRL4 (Continued)

Bits	Name	Memory Access	Description
12	CDR_SSC_EN_0	R/W	Rx CDR SSC Mode Enable on lane 0 of the PHY. This field controls the CDR tracking gains and duration. This bit should be set to 1 when receive data has a spread spectrum clock and should be cleared if receive data does not have SSC. This bit drives the output port 'xlgpcs_rx_cdr_ssc_en_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
11:9	RX_DFE_BYP_3_1	* Varies	**For Multi-lane configurations:** RX DFE Bypass Enable on Lane 3 to 1 This field drives the output port xlgpcs_rx_dfe_byp_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!DWCXLP_MULTI_LANE||DWCXLP_E32||DWCXLP_E12) ? \"read-only\" : \"read-write\""
8	RX_DFE_BYP_0	* Varies	RX DFE Bypass Enable on Lane o This field drives the output port xlgpcs_rx_dfe_byp_o[0]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E32||DWCXLP_E12) ? \"read-only\" : \"read-write\" "
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:0	RX_TERM_OFFSET	* Varies	Offset for RX Termination This field drives the output port xlgpcs_rx_term_offset_o[4:0]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E32||DWCXLP_E12) ? \"read-only\" : \"read-write\" "

7.1.176 VR_PMA_MP_25G_16G_RX_MISC_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY RX Miscellaneous Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8068 : (DWCXLP_APB3_16) ? (0x8068*0x2) : (0x8068*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16)

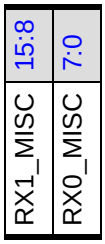


Table 7-181 Fields for Register: VR_PMA_MP_25G_16G_RX_MISC_CTRL0

Bits	Name	Memory Access	Description
15:8	RX1_MISC	* Varies	**For Multi-lane configurations:** RX Miscellaneous control for lane1. This field drives the output port xlgpcs_rx1_misc_o[7:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_E16&&DWCXLP_MULTI_LANE ? 0x2 : 0x0 " **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\" "
7:0	RX0_MISC	R/W	RX Miscellaneous control for lane0. This field drives the output port xlgpcs_rx0_misc_o[7:0]. **Value After Reset:** "DWCXLP_E16 ? 0x2 : 0x0 " **Exists:** Always

7.1.177 VR_PMA_MP_25G_16G_RX_MISC_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY RX Miscellaneous Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8069 : (DWCXLP_APB3_16) ? (0x8069*0x2) : (0x8069*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16)&&DWCXLP_MULTI_LANE

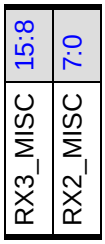


Table 7-182 Fields for Register: VR_PMA_MP_25G_16G_RX_MISC_CTRL1

Bits	Name	Memory Access	Description
15:8	RX3_MISC	R/W	RX Miscellaneous control for lane3. This field drives the output port xlgpcs_rx3_misc_o[7:0]. **Value After Reset:** "DWCXLP_E16 ? 0x2 : 0x0 " **Exists:** Always
7:0	RX2_MISC	R/W	RX Miscellaneous control for lane2. This field drives the output port xlgpcs_rx2_misc_o[7:0]. **Value After Reset:** "DWCXLP_E16 ? 0x2 : 0x0 " **Exists:** Always

7.1.178 VR_PMA_MP_25G_16G_RX_IQ_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G/16G PHY RX IQ Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x806A : (DWCXLP_APB3_16) ? (0x806A*0x2) : (0x806A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)

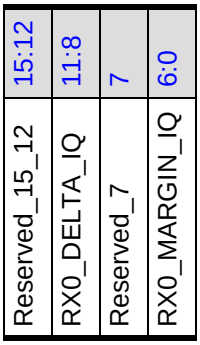


Table 7-183 Fields for Register: VR_PMA_MP_25G_16G_RX_IQ_CTRL0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	RX0_DELTA_IQ	* Varies	RX IQ Offset Value for lane0. This field drives the output port xlgpcs_rx0_delta_iq_o[3:0]. **Value After Reset:** "DWCXLP_E32 ? 0x0 : (!(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G)&&(DWCXLP_E25) ? 0x6 : 0x3) " **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-183 Fields for Register: VR_PMA_MP_25G_16G_RX_IQ_CTRL0 (Continued)

Bits	Name	Memory Access	Description
6:0	RX0_MARGIN_IQ	R/W	Value of RX IQ Margining on lane0. This field drives the output port xlgpcs_rx0_margin_iq_o[6:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.179 VR_PMA_MP_25G_16G_RX_IQ_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G/16G PHY RX IQ Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x806B : (DWCXLP_APB3_16) ? (0x806B*0x2) : (0x806B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE

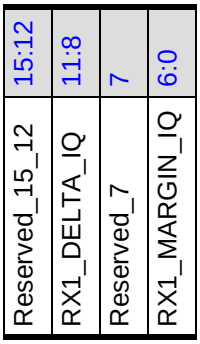


Table 7-184 Fields for Register: VR_PMA_MP_25G_16G_RX_IQ_CTRL1

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	RX1_DELTA_IQ	* Varies	RX IQ Offset Value for lane1. This field drives the output port xlgpcs_rx1_delta_iq_o[3:0]. **Value After Reset:** "DWCXLP_E32 ? 0x0 : (!(DWCXLP_KR4_50G||DWCXLP_KR4_100G)&&(DWCXLP_E25) ? 0x6 : 0x3) " **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-184 Fields for Register: VR_PMA_MP_25G_16G_RX_IQ_CTRL1 (Continued)

Bits	Name	Memory Access	Description
6:0	RX1_MARGIN_IQ	R/W	Value of RX IQ Margining on lane1. This field drives the output port xlgpcs_rx1_margin_iq_o[6:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.180 VR_PMA_MP_25G_16G_RX_IQ_CTRL2

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G/16G PHY RX IQ Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x806C : (DWCXLP_APB3_16) ? (0x806C*0x2) : (0x806C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE

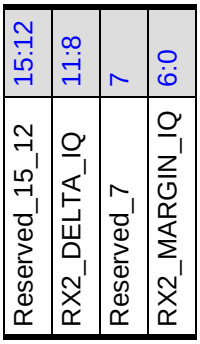


Table 7-185 Fields for Register: VR_PMA_MP_25G_16G_RX_IQ_CTRL2

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	RX2_DELTA_IQ	* Varies	RX IQ Offset Value for lane2. This field drives the output port xlgpcs_rx2_delta_iq_o[3:0]. **Value After Reset:** "DWCXLP_E32 ? 0x0 : (!(DWCXLP_KR4_50G||DWCXLP_KR4_100G)&&(DWCXLP_E25) ? 0x6 : 0x3) " **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-185 Fields for Register: VR_PMA_MP_25G_16G_RX_IQ_CTRL2 (Continued)

Bits	Name	Memory Access	Description
6:0	RX2_MARGIN_IQ	R/W	Value of RX IQ Margining on lane2. This field drives the output port xlgpcs_rx2_margin_iq_o[6:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.181 VR_PMA_MP_25G_16G_RX_IQ_CTRL3

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G/16G PHY RX IQ Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x806D : (DWCXLP_APB3_16) ? (0x806D*0x2) : (0x806D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16)&&DWCXLP_MULTI_LANE

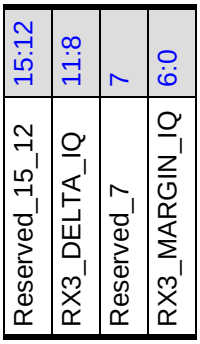


Table 7-186 Fields for Register: VR_PMA_MP_25G_16G_RX_IQ_CTRL3

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	RX3_DELTA_IQ	* Varies	RX IQ Offset Value for lane3. This field drives the output port xlgpcs_rx3_delta_iq_o[3:0]. **Value After Reset:** "DWCXLP_E32 ? 0x0 : (!(DWCXLP_KR4_50G||DWCXLP_KR4_100G)&&(DWCXLP_E25) ? 0x6 : 0x3) " **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-186 Fields for Register: VR_PMA_MP_25G_16G_RX_IQ_CTRL3 (Continued)

Bits	Name	Memory Access	Description
6:0	RX3_MARGIN_IQ	R/W	Value of RX IQ Margining on lane3. This field drives the output port xlgpcs_rx3_margin_iq_o[6:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.182 VR_PMA_MP_25G_RX_DCC_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY RX DCC Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x806E : (DWCXLP_APB3_16) ? (0x806E*0x2) : (0x806E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25))

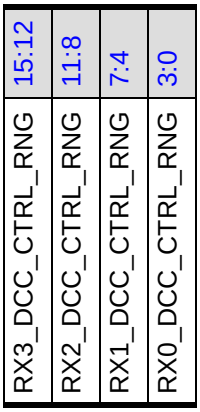


Table 7-187 Fields for Register: VR_PMA_MP_25G_RX_DCC_CTRL

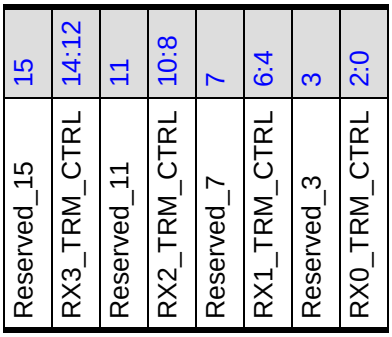
Bits	Name	Memory Access	Description
15:12	RX3_DCC_CTRL_RNG	* Varies	**For Multi-lane configurations:** RX Internal Clock Duty Cycle Control Range - for Lane3. This field drives the output port xlgpcs_rx3_dcc_ctrl_rng_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0xB " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-187 Fields for Register: VR_PMA_MP_25G_RX_DCC_CTRL (Continued)

Bits	Name	Memory Access	Description
11:8	RX2_DCC_CTRL_RNG	* Varies	**For Multi-lane configurations:** RX Internal Clock Duty Cycle Control Range - for Lane2. This field drives the output port xlgpcs_rx2_dcc_ctrl_rng_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0xB " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7:4	RX1_DCC_CTRL_RNG	* Varies	**For Multi-lane configurations:** RX Internal Clock Duty Cycle Control Range - for Lane1. This field drives the output port xlgpcs_rx1_dcc_ctrl_rng_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0xB " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3:0	RX0_DCC_CTRL_RNG	R/W	RX Internal Clock Duty Cycle Control Range - for Lane0. This field drives the output port xlgpcs_rx0_dcc_ctrl_rng_o. **Value After Reset:** 0xb **Exists:** Always

7.1.183 VR_PMA_MP_25G_RX_TERM_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY RX Terminate Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x806F : (DWCXLP_APB3_16) ? (0x806F*0x2) : (0x806F*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** ((DWCXLP_PMA_MMD)&&(DWCXLP_E25))



15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	RX3_TRM_CTRL	Reserved_11	RX2_TRM_CTRL	Reserved_7	RX1_TRM_CTRL	Reserved_3	RX0_TRM_CTRL

Table 7-188 Fields for Register: VR_PMA_MP_25G_RX_TERM_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	RX3_TRM_CTRL	* Varies	**For Multi-lane configurations:** Receive Termination Control - for Lane3. This field drives the output port xlgpcs_rx3_trm_ctrl_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-188 Fields for Register: VR_PMA_MP_25G_RX_TERM_CTRL (Continued)

Bits	Name	Memory Access	Description
10:8	RX2_TRM_CTRL	* Varies	**For Multi-lane configurations:** Receive Termination Control - for Lane2. This field drives the output port xlgpcs_rx2_trm_ctrl_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	RX1_TRM_CTRL	* Varies	**For Multi-lane configurations:** Receive Termination Control - for Lane1. This field drives the output port xlgpcs_rx1_trm_ctrl_o. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : 0x2 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_TRM_CTRL	R/W	Receive Termination Control - for Lane0. This field drives the output port xlgpcs_rx0_trm_ctrl_o. **Value After Reset:** 0x2 **Exists:** Always

7.1.184 VR_PMA_MP_25G_16G_MPLL_CMN_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY MPLL Common Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8070 : (DWCXLP_APB3_16) ? (0x8070*0x2) : (0x8070*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)

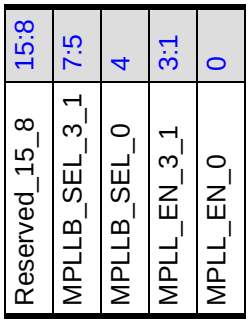


Table 7-189 Fields for Register: VR_PMA_MP_25G_16G_MPLL_CMN_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	MPLLB_SEL_3_1	* Varies	**For Multi-lane configurations:** Tx MPLLB select-lanes[3:1] When this bit is set, PHY selects MPLLB to generate Tx analog clocks. **For Single-lane configurations:** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x7 : 0x0 " **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-189 Fields for Register: VR_PMA_MP_25G_16G_MPLL_CMN_CTRL (Continued)

Bits	Name	Memory Access	Description
4	MPLLB_SEL_0	R/W	Tx MPLLB Select-lane 0. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x1 : 0x0" **Exists:** Always
3:1	MPLL_EN_3_1	* Varies	**For Multi-lane configurations:** Tx MPLL Enable-lanes[3:1] These bit should be set to power-up the MPLL. This bit must be 1, for normal operation. **For Single-lane configurations:** Reserved **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
0	MPLL_EN_0	R/W	Tx MPLL Enable-lane 0. This bit should be set to power-up the MPLL. This bit should be 1, for normal operation. **Value After Reset:** 0x1 **Exists:** Always

7.1.185 VR_PMA_MP_32G_MPLL_CMN_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY MPLL Common Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8070 : (DWCXLP_APB3_16) ? (0x8070*0x2) : (0x8070*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

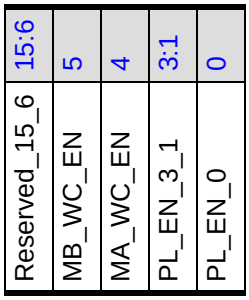


Table 7-190 Fields for Register: VR_PMA_MP_32G_MPLL_CMN_CTRL

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	MB_WC_EN	R/W	MPLLB word clock output enable. This field drives the output port 'xlgpcs_mpllb_wrd_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
4	MA_WC_EN	R/W	MPLLA word clock output enable. This field drives the output port 'xlgpcs_mplla_wrd_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
3:1	PL_EN_3_1	* Varies	TX PLL Enable Lane3_1. Powers up or powers down the ROPLL or MPLLA or MPLLB. This field drives the output port 'xlgpcs_tx_pll_en_o'. **Value After Reset:** "DWCXLP_MULTI_LANE ? 0x7 : 0x0 " **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-190 Fields for Register: VR_PMA_MP_32G_MPLL_CMN_CTRL (Continued)

Bits	Name	Memory Access	Description
0	PL_EN_0	R/W	TX PLL Enable Lane0. Powers up or powers down the ROPLL or MPLLA or MPLLB. This field drives the output port 'xlgpcs_tx_pll_en_o'. **Value After Reset:** 0x1 **Exists:** Always

7.1.186 VR_PMA_MP_25G_MPLLA_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G PHY MPLLA Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8071 : (DWCXLP_APB3_16) ? (0x8071*0x2) : (0x8071*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32)



Table 7-191 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL0

Bits	Name	Memory Access	Description
15	MPLLA_CAL_DISABLE	* Varies	MPLLA Calibration Disable This field can be programmed to 1, to disable calibration of MPLLA by PHY firmware. **Value After Reset:** "((DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G)&&DWCXLP_E25) ? 0x1 : 0x0 " **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "
14	MPLLA_CTL_BUF_BYP	* Varies	MPLLA Control Buffer Bypass. This field drives the output port 'xlgpcs_mplla_ctl_buf_byp_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "

Table 7-191 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL0 (Continued)

Bits	Name	Memory Access	Description
13:12	MPLLA_RECAL_BANK_SEL	R/W	MPLLA Re-calibration Bank Select. This field drives the output port 'xlgpcs_mplla_recal_bank_sel_o[1:0]'. **Value After Reset:** 0x0 **Exists:** Always
11:0	MPLLA_MULTIPLIER	* Varies	MPLLA frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed Any change in this field should be followed by a Vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** "DWCXLP_E32 ? 0x0 : 0x084" **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "

7.1.187 VR_PMA_MP_16G_MPLLA_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLA Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8071 : (DWCXLP_APB3_16) ? (0x8071*0x2) : (0x8071*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16||DWCXLP_E12)

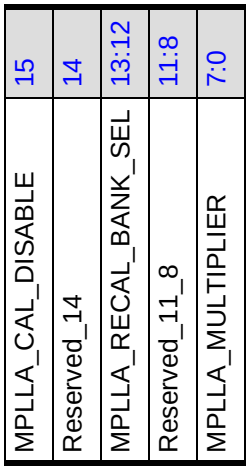


Table 7-192 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL0

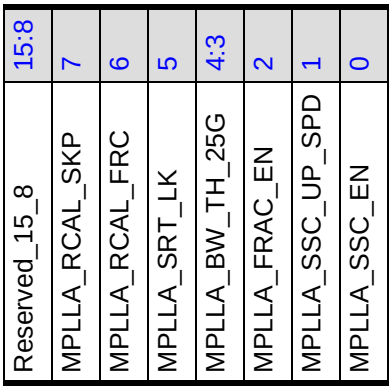
Bits	Name	Memory Access	Description
15	MPLLA_CAL_DISABLE	R/W	MPLLA Calibration Disable. Program this field to 1 to disable the calibration of MPLLA by PHY firmware. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	MPLLA_RECAL_BANK_SEL	* Varies	MPLLA Re-calibration Bank Select This field drives the output port 'xlgpcs_mplla_recal_bank_sel_o[1:0]'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E12) ? \"read-only\" : \"read-write\""

Table 7-192 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL0 (Continued)

Bits	Name	Memory Access	Description
11:8	Reserved_11_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	MPLLA_MULTIPLIER	R/W	MPLLA frequency Multiplier Control. This field controls the multiplication of reference clock to a frequency suitable for operating speed Any change in this field should be followed by a Vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** 0x21 **Exists:** Always

7.1.188 VR_PMA_MP_32G_25G_MPLLA_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G PHY MPLLA Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8072 : (DWCXLP_APB3_16) ? (0x8072*0x2) : (0x8072*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32)



15:8	7	6	5	4:3	2	1	0
Reserved_15_8	MPLLA_RCAL_SKP	MPLLA_RCAL_FRC	MPLLA_SRT_LK	MPLLA_BW_TH_25G	MPLLA_FRAC_EN	MPLLA_SSC_UP_SPD	MPLLA_SSC_EN

Table 7-193 Fields for Register: VR_PMA_MP_32G_25G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	MPLLA_RCAL_SKP	R/W	MPLLA Recalibration Skip Enable This field drives the xlgpcs_mplla_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always
6	MPLLA_RCAL_FRC	R/W	MPLLA Recalibration Force Enable. This drives the xlgpcs_mplla_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always

Table 7-193 Fields for Register: VR_PMA_MP_32G_25G_MPLLA_CTRL1 (Continued)

Bits	Name	Memory Access	Description
5	MPLLA_SRT_LK	* Varies	MPLLA short lock enable This field drives the output port xlgpcs_mplla_short_lock_en_o. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""
4:3	MPLLA_BW_TH_25G	* Varies	MPLLA Bandwidth Threshold for 32G PHY This field drives the output port xlgpcs_mplla_bw_th_o. **Value After Reset:** "(DWCXLP_E32) ? 0x0 : 0x3" **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""
2	MPLLA_FRAC_EN	* Varies	MPLLA SSC Clock Select This field drives the output port xlgpcs_mplla_frac_en_o. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""
1	MPLLA_SSC_UP_SPD	* Varies	MPLLA SSC Up Spread Enable This field drives the output port xlgpcs_mplla_ssc_spd_en_o. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""
0	MPLLA_SSC_EN	R/W	MPLLA SSC Enable. This field can be set to enable spread-spectrum generation on mplla_div_clk output. This bit drives the output port xlgpcs_mplla_ssc_en_o. **Value After Reset:** 0x0 **Exists:** Always

7.1.189 VR_PMA_MP_16G_MPLLA_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 16G PHY MPLLA Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8072 : (DWCXLP_APB3_16) ? (0x8072*0x2) : (0x8072*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16)

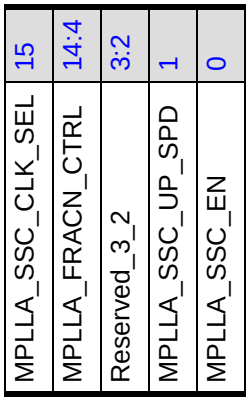


Table 7-194 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
15	MPLLA_SSC_CLK_SEL	R/W	MPLLA Spread Spectrum Clock Select. This field drives the output port 'xlgpcs_mplla_ssc_clk_sel_o' **Value After Reset:** 0x0 **Exists:** Always
14:4	MPLLA_FRACN_CTRL	R/W	MPLLA Fractional Control. This field drives the output port 'xlgpcs_mplla_fracn_ctrl_o'. **Value After Reset:** 0x0 **Exists:** Always
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	MPLLA_SSC_UP_SPD	R/W	MPLLA SSC Up Spread Enable. This field drives the output port 'xlgpcs_mplla_ssc_spd_en_o'. **Value After Reset:** 0x0 **Exists:** Always

Table 7-194 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL1 (Continued)

Bits	Name	Memory Access	Description
0	MPLLA_SSC_EN	R/W	MPLLA Spread Spectrum Enable. This field can be set to enable spread-spectrum generation on mplla_div_clk output. This bit drives the output port xlgpcs_mplla_ssc_en_o. **Value After Reset:** 0x0 **Exists:** Always

7.1.190 VR_PMA_MP_12G_MPLLA_CTRL1

- **Description:** VR PMA Synopsys Multi-Protocol 12G PHY MPLLA Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8072 : (DWCXLP_APB3_16) ? (0x8072*0x2) : (0x8072*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E12)

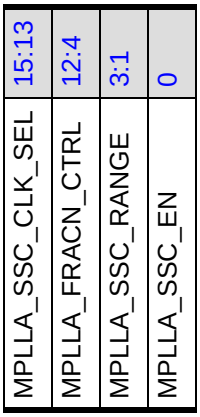


Table 7-195 Fields for Register: VR_PMA_MP_12G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
15:13	MPLLA_SSC_CLK_SEL	R/W	MPLLA Spread Spectrum Clock Select. This field drives the output port 'xlgpcs_mplla_ssc_clk_sel_o' **Value After Reset:** 0x0 **Exists:** Always
12:4	MPLLA_FRACN_CTRL	R/W	MPLLA Fractional Control. This field drives the output port 'xlgpcs_mplla_fracn_ctrl_o'. **Value After Reset:** 0x0 **Exists:** Always

Table 7-195 Fields for Register: VR_PMA_MP_12G_MPLLA_CTRL1 (Continued)

Bits	Name	Memory Access	Description
3:1	MPLLA_SSC_RANGE	R/W	MPLLA Spread Spectrum Range This field selects the modulation to insert. This field applies a fixed PPM offset to phase accumulator. mplla_ssc_range PPM Offset 3'b000 -4980 3'b001 -4492 3'b010 -4003 3'b011 -3027 3'b100 4980 3'b101 4492 3'b110 4003 3'b111 3027 This field drives the output port 'xlgpcs_mplla_ssc_range_o'. **Value After Reset:** 0x0 **Exists:** Always
0	MPLLA_SSC_EN	R/W	MPLLA Spread Spectrum Enable. This field can be set to enable spread-spectrum generation on mplla_div_clk output. This bit drives the output port xlgpcs_mplla_ssc_en_o. **Value After Reset:** 0x0 **Exists:** Always

7.1.191 VR_PMA_MP_16G_MPLLA_CTRL2

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLA Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8073 : (DWCXLP_APB3_16) ? (0x8073*0x2) : (0x8073*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16||DWCXLP_E12)

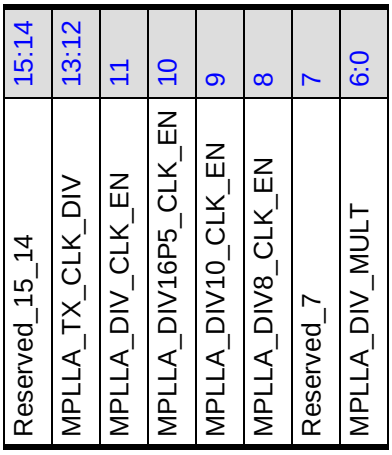


Table 7-196 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-196 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL2 (Continued)

Bits	Name	Memory Access	Description
13:12	MPLLA_TX_CLK_DIV	* Varies	MPLLA TX Clock Divider. It sets the divide ratio on the MPLL clock sent to the TX as follows: ■ 2'b00: div1 ■ 2'b01: div2 ■ 2'b10: div3 ■ 2'b11: div1 (duplicate state) This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2. This field drives the output port 'xlgpcs_mplla_tx_clk_div_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E12) ? \"read-only\" : \"read-write\""
11	MPLLA_DIV_CLK_EN	R/W	Enable mplla_div_clk from PHY When asserted, the frequency of mplla_div_clk output from PHY is MPLLA frequency divided by 'mplla_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always
10	MPLLA_DIV16P5_CLK_EN	R/W	MPLLA Divide by 16.5 Enable. Program this bit to 1 to enable output clocks derived from MPLLb based on 16.5,33 and 66 division ratios. **Value After Reset:** 0x1 **Exists:** Always
9	MPLLA_DIV10_CLK_EN	R/W	MPLLA Divide by 10 Enable. ■ 1: The frequency of the mplla_word_clk output clock from PHY is MPLLA frequency divided by 10. **Value After Reset:** "(DWCXLP_R_SRDS_INTF==4) ? 0x1 : 0x0" **Exists:** Always
8	MPLLA_DIV8_CLK_EN	R/W	MPLLA Divide by 8 Enable. ■ 1: The frequency of the mplla_word_clk output clock from PHY is MPLLA frequency divided by 8. **Value After Reset:** "(DWCXLP_R_SRDS_INTF==0) ? 0x1 : 0x0" **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-196 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL2 (Continued)

Bits	Name	Memory Access	Description
6:0	MPLLA_DIV_MULT	R/W	MPLLA Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLA clock output from the reference clock input as seen by the MPLL. **Value After Reset:** 0x0 **Exists:** Always

7.1.192 VR_PMA_MP_25G_MPLLA_CTRL2

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8073 : (DWCXLP_APB3_16) ? (0x8073*0x2) : (0x8073*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

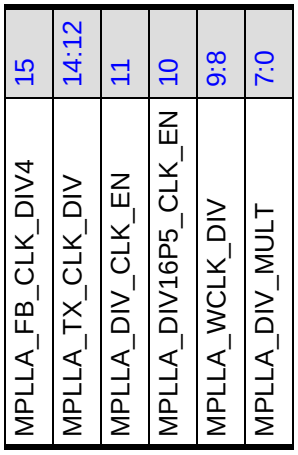


Table 7-197 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL2

Bits	Name	Memory Access	Description
15	MPLLA_FB_CLK_DIV4	R/W	MPLLA DIV Clock Output Frequency Divider Control. Additional frequency division factor used to generate mplla_div_clk clock output from the MPLLA VCO. This signal controls whether the MPLL VCO is divided by 2 (when mplla_div_clk_div4_en is de-asserted) or 4 (when mplla_div_clk_div4_en is asserted) before being divided down by mplla_div_multiplier[7:0]. This value can be changed only when the txX_mpll_en input for all lanes is de-asserted. This field drives the output port 'xlgpcs_mplla_fb_clk_div4_en_o'. **Value After Reset:** 0x1 **Exists:** Always

Table 7-197 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL2 (Continued)

Bits	Name	Memory Access	Description
14:12	MPLLA_TX_CLK_DIV	R/W	MPLLA TX Clock Divider. It sets the divide ratio on the MPLL clock sent to the TX as follows: ■ 3'b000: Reserved ■ 3'b001: div1 ■ 3'b010: div2 ■ 3'b011: div3 ■ 3'b100: div4 ■ 3'b101: div5 ■ 3'b110: Reserved ■ 3'b111: Reserved This field drives the output port 'xlgpcs_mplla_tx_clk_div_o'. **Value After Reset:** 0x2 **Exists:** Always
11	MPLLA_DIV_CLK_EN	R/W	Enable mplla_div_clk from PHY When asserted, the frequency of mpllb_div_clk output from PHY is MPLLA frequency divided by 'mplla_div_multiplier' This field drives the output port 'xlgpcs_mplla_div_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
10	MPLLA_DIV16P5_CLK_EN	R/W	MPLLA Divide by 16.5 Enable This bit can be set to enable output clocks derived from MPLLA based on 16.5,33 and 66 division ratios. This field drives the output port 'xlgpcs_mplla_div16p5_clk_en_o'. **Value After Reset:** 0x1 **Exists:** Always
9:8	MPLLA_WCLK_DIV	R/W	MPLLA Word Clock Divider Control. It Controls the MPLLA word clock dividers for generating the mplla_word_clk. These inputs can be changed only when the txX_mpll_en input to PHY is deasserted. ■ 2'b00: Divide by 4 ■ 2'b01: Divide by 5 ■ 2'b10: Divide by 8 ■ 2'b11: Divide by 10 This field drives the output port 'xlgpcs_mplla_word_clk_div_o'. **Value After Reset:** "(DWCXLP_R_SRDS_INTF==0||DWCXLP_R_SRDS_INTF==1) ? 0x2 : 0x3" **Exists:** Always

Table 7-197 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL2 (Continued)

Bits	Name	Memory Access	Description
7:0	MPLLA_DIV_MULT	R/W	MPLLA Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLA clock output from the reference clock input as seen by the MPLL. This field drives the output port 'xlgpcs_mplla_div_mult_o'. **Value After Reset:** 0x8 **Exists:** Always

7.1.193 VR_PMA_MP_25G_MPLLB_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 25G/32G PHY MPLLB Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8074 : (DWCXLP_APB3_16) ? (0x8074*0x2) : (0x8074*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32)



Table 7-198 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL0

Bits	Name	Memory Access	Description
15	MPLLB_CAL_DISABLE	* Varies	MPLLB Calibration Disable This field can be programmed to 1, to disable calibration of MPLLB by PHY firmware. **Value After Reset:** "(((DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G)&&DWCXLP_E25)||DWCXLP_E32) ? 0x0 : 0x1 " **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "
14	MPLLB_CTL_BUF_BYP	* Varies	MPLLB Control Buffer Bypass. This field drives the output port 'xlgpcs_mpllb_ctl_buf_byp_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "

Table 7-198 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL0 (Continued)

Bits	Name	Memory Access	Description
13:12	MPLLB_RECAL_BANK_SEL	R/W	MPLLB Re-calibration Bank Select This field drives the output port 'xlgpcs_mpllb_recal_bank_sel_o[1:0]'. **Value After Reset:** 0x0 **Exists:** Always
11:0	MPLLB_MULTIPLIER	* Varies	MPLLB frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed Any change in this field should be followed by a Vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** "DWCXLP_E32 ? 0x0 : 0x0A5" **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "

7.1.194 VR_PMA_MP_16G_MPLLB_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLB Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8074 : (DWCXLP_APB3_16) ? (0x8074*0x2) : (0x8074*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16||DWCXLP_E12)

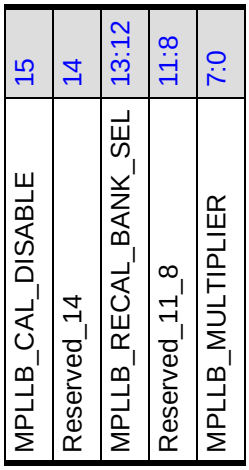


Table 7-199 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL0

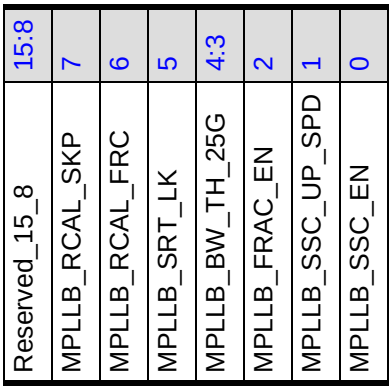
Bits	Name	Memory Access	Description
15	MPLLB_CAL_DISABLE	R/W	MPLLB Calibration Disable. Program this field to 1 to disable calibration of MPLLB by PHY firmware. **Value After Reset:** 0x1 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	MPLLB_RECAL_BANK_SEL	* Varies	MPLLB Re-calibration Bank Select This field drives the output port 'xlgpcs_mpllb_recal_bank_sel_o[1:0]'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E12) ? \"read-only\" : \"read-write\""

Table 7-199 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL0 (Continued)

Bits	Name	Memory Access	Description
11:8	Reserved_11_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	MPLLB_MULTIPLIER	R/W	MPLLB frequency Multiplier Control. This field controls the multiplication of reference clock to a frequency suitable for operating speed Any change in this field should be followed by a Vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** 0x0 **Exists:** Always

7.1.195 VR_PMA_MP_32G_25G_MPLLB_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 32G/25G PHY MPLLB Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8075 : (DWCXLP_APB3_16) ? (0x8075*0x2) : (0x8075*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32)



15:8	7	6	5	4:3	2	1	0
Reserved_15_8	MPLLB_RCAL_SKP	MPLLB_RCAL_FRC	MPLLB_SRT_LK	MPLLB_BW_TH_25G	MPLLB_FRAC_EN	MPLLB_SSC_UP_SPD	MPLLB_SSC_EN

Table 7-200 Fields for Register: VR_PMA_MP_32G_25G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	MPLLB_RCAL_SKP	R/W	MPLLB Recalibration Skip Enable. This field drives the xlgpcs_mpllb_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always
6	MPLLB_RCAL_FRC	R/W	MPLLB Recalibration Force Enable. This drives the xlgpcs_mpllb_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always
5	MPLLB_SRT_LK	* Varies	MPLLB short lock enable This field drives the output port xlgpcs_mpllb_short_lock_en_o. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""

Table 7-200 Fields for Register: VR_PMA_MP_32G_25G_MPLLB_CTRL1 (Continued)

Bits	Name	Memory Access	Description
4:3	MPLLB_BW_TH_25G	* Varies	MPLLB Bandwidth Threshold for 32G PHY This field drives the output port xlgpcs_mpllb_bw_th_o. **Value After Reset:** "(DWCXLP_E32) ? 0x0 : 0x3" **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""
2	MPLLB_FRAC_EN	* Varies	MPLLB Fraction Enable This field drives the output port 'xlgpcs_mpllb_frac_en_o' **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""
1	MPLLB_SSC_UP_SPD	* Varies	MPLLB SSC Up Spread Enable This field drives the output port 'xlgpcs_mpllb_ssc_spd_en_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_E32 ? \"read-only\" : \"read-write\""
0	MPLLB_SSC_EN	R/W	MPLLB Spread Spectrum Enable. This field can be set to enable spread-spectrum generation on mplla_div_clk output. This bit drives the output port xlgpcs_mpllb_ssc_en_o. **Value After Reset:** 0x0 **Exists:** Always

7.1.196 VR_PMA_MP_16G_MPLLB_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 16G PHY MPLLB Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8075 : (DWCXLP_APB3_16) ? (0x8075*0x2) : (0x8075*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16)

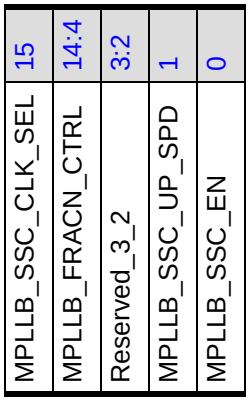


Table 7-201 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
15	MPLLB_SSC_CLK_SEL	R/W	MPLLB Spread Spectrum Clock Select. This field drives the output port 'xlgpcs_mpllb_ssc_clk_sel_o' **Value After Reset:** 0x0 **Exists:** Always
14:4	MPLLB_FRACN_CTRL	R/W	MPLLB Fractional Control This field drives the output port 'xlgpcs_mpllb_fracn_ctrl_o'. **Value After Reset:** 0x0 **Exists:** Always
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	MPLLB_SSC_UP_SPD	R/W	MPLLB SSC Up Spread Enable. This field drives the output port 'xlgpcs_mpllb_ssc_spd_en_o'. **Value After Reset:** 0x0 **Exists:** Always

Table 7-201 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL1 (Continued)

Bits	Name	Memory Access	Description
0	MPLLB_SSC_EN	R/W	MPLLB Spread Spectrum Enable. This field can be set to enable spread-spectrum generation on mplla_div_clk output. This bit drives the output port xlgpcs_mpllb_ssc_en_o. **Value After Reset:** 0x0 **Exists:** Always

7.1.197 VR_PMA_MP_12G_MPLLB_CTRL1

- **Description:** VR PMA Synopsys Multi-Protocol 12G PHY MPLLB Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8075 : (DWCXLP_APB3_16) ? (0x8075*0x2) : (0x8075*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E12)

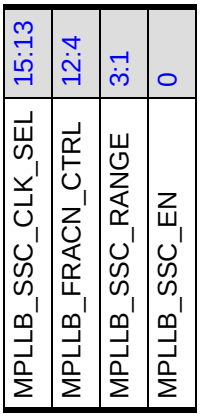


Table 7-202 Fields for Register: VR_PMA_MP_12G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
15:13	MPLLB_SSC_CLK_SEL	R/W	MPLLB Spread Spectrum Clock Select. This field drives the output port 'xlgpcs_mpllb_ssc_clk_sel_o' **Value After Reset:** 0x0 **Exists:** Always
12:4	MPLLB_FRACN_CTRL	R/W	MPLLB Fractional Control. This field drives the output port 'xlgpcs_mpllb_fracn_ctrl_o'. **Value After Reset:** 0x0 **Exists:** Always

Table 7-202 Fields for Register: VR_PMA_MP_12G_MPLLB_CTRL1 (Continued)

Bits	Name	Memory Access	Description
3:1	MPLLB_SSC_RANGE	R/W	MPLLB Spread Spectrum Range This field selects the modulation to insert. This field applies a fixed PPM offset to phase accumulator. mpllb_ssc_range PPM Offset 3'b000 -4980 3'b001 -4492 3'b010 -4003 3'b011 -3027 3'b100 4980 3'b101 4492 3'b110 4003 3'b111 3027 This field drives the output port 'xlgpcs_mpllb_ssc_range_o'. **Value After Reset:** 0x0 **Exists:** Always
0	MPLLB_SSC_EN	R/W	MPLLB Spread Spectrum Enable. This field can be set to enable spread-spectrum generation on mpllb_div_clk output. This bit drives the output port 'xlgpcs_mpllb_ssc_en_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.198 VR_PMA_MP_16G_MPLLB_CTRL2

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLB Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8076 : (DWCXLP_APB3_16) ? (0x8076*0x2) : (0x8076*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16||DWCXLP_E12)

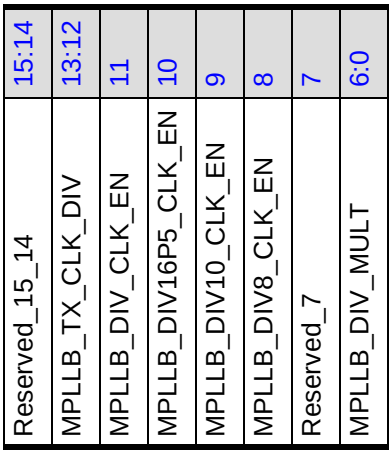


Table 7-203 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-203 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL2 (Continued)

Bits	Name	Memory Access	Description
13:12	MPLLB_TX_CLK_DIV	* Varies	MPLLB TX Clock Divider. It sets the divide ratio on the MPLL clock sent to the TX as follows: ■ 2'b00: div1 ■ 2'b01: div2 ■ 2'b10: div3 ■ 2'b11: div1 (duplicate state) This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2. This field drives the output port 'xlgpcs_mpllb_tx_clk_div_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E12) ? \"read-only\" : \"read-write\" "
11	MPLLB_DIV_CLK_EN	R/W	Enable mpllb_div_clk from PHY When asserted, the frequency of mpllb_div_clk output from PHY is MPLLB frequency divided by 'mpllb_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always
10	MPLLB_DIV16P5_CLK_EN	R/W	MPLLB Divide by 16.5 Enable This bit can be set to enable output clocks derived from MPLLb based on 16.5,33 and 66 division ratios. **Value After Reset:** 0x0 **Exists:** Always
9	MPLLB_DIV10_CLK_EN	R/W	MPLLB Divide by 10 Enable When this bit is set, the frequency of the mpllb_word_clk output clock from PHY is MPLLB frequency divided by 10. **Value After Reset:** 0x0 **Exists:** Always
8	MPLLB_DIV8_CLK_EN	R/W	MPLLB Divide by 8 Enable When this bit is set, the frequency of the mpllb_word_clk output clock from PHY is MPLLB frequency divided by 8. **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-203 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL2 (Continued)

Bits	Name	Memory Access	Description
6:0	MPLLB_DIV_MULT	R/W	MPLLB Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLB clock output from the reference clock input as seen by the MPLL. **Value After Reset:** 0x0 **Exists:** Always

7.1.199 VR_PMA_MP_25G_MPLLB_CTRL2

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8076 : (DWCXLP_APB3_16) ? (0x8076*0x2) : (0x8076*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

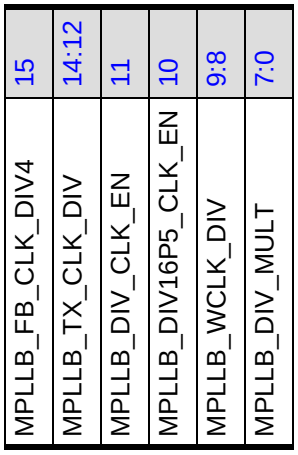


Table 7-204 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL2

Bits	Name	Memory Access	Description
15	MPLLB_FB_CLK_DIV4	R/W	MPLLB DIV Clock Output Frequency Divider Control. Additional frequency division factor used to generate mpllb_div_clk clock output from the MPLLB VCO. This signal controls whether the MPLL VCO is divided by 2 (when mpllb_div_clk_div4_en is de-asserted) or 4 (when mpllb_div_clk_div4_en is asserted) before being divided down by mpllb_div_multiplier[7:0]. This value can be changed only when the txX_mpll_en input for all lanes is de-asserted. This field drives the output port 'xlgpcs_mpllb_fb_clk_div4_en_o'. **Value After Reset:** 0x1 **Exists:** Always

Table 7-204 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL2 (Continued)

Bits	Name	Memory Access	Description
14:12	MPLLB_TX_CLK_DIV	R/W	MPLLB TX Clock Divider. It sets the divide ratio on the MPLL clock sent to the TX as follows: ■ 3'b000: Reserved ■ 3'b001: div1 ■ 3'b010: div2 ■ 3'b011: div3 ■ 3'b100: div4 ■ 3'b101: div5 ■ 3'b110: Reserved ■ 3'b111: Reserved This field drives the output port 'xlgpcs_mpllb_tx_clk_div_o'. **Value After Reset:** 0x1 **Exists:** Always
11	MPLLB_DIV_CLK_EN	R/W	Enable mpllb_div_clk from PHY When asserted, the frequency of mpllb_div_clk output from PHY is MPLLB frequency divided by 'mpllb_div_multiplier' This field drives the output port 'xlgpcs_mpllb_div_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
10	MPLLB_DIV16P5_CLK_EN	R/W	MPLLB Divide by 16.5 Enable Program this bit to 1 to enable output clocks derived from MPLLB based on 16.5,33 and 66 division ratios. This field drives the output port 'xlgpcs_mpllb_div16p5_clk_en_o'. **Value After Reset:** 0x1 **Exists:** Always
9:8	MPLLB_WCLK_DIV	R/W	MPLLB Word Clock Divider Control. It Controls the MPLLB word clock dividers for generating the mpllb_word_clk. These inputs can be changed only when the txX_mpll_en input to PHY is deasserted. ■ 2'b00: Divide by 4 ■ 2'b01: Divide by 5 ■ 2'b10: Divide by 8 ■ 2'b11: Divide by 10 This field drives the output port 'xlgpcs_mpllb_word_clk_div_o'. **Value After Reset:** "(DWCXLP_R_SRDS_INTF==0||DWCXLP_R_SRDS_INTF==1) ? 0x2 : 0x3" **Exists:** Always

Table 7-204 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL2 (Continued)

Bits	Name	Memory Access	Description
7:0	MPLLB_DIV_MULT	R/W	MPLLB Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLB clock output from the reference clock input as seen by the MPLL. This field drives the output port 'xlgpcs_mpllb_div_mult_o'. **Value After Reset:** 0x4 **Exists:** Always

7.1.200 VR_PMA_MP_16G_MPLLA_CTRL3

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLA Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8077 : (DWCXLP_APB3_16) ? (0x8077*0x2) : (0x8077*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16||DWCXLP_E12)

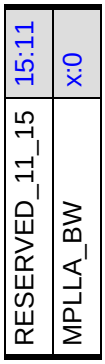


Table 7-205 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL3

Bits	Name	Memory Access	Description
15:11	RESERVED_11_15	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&DWCXLP_E12
x:0	MPLLA_BW	R/W	MPLLA Bandwidth Control This field drives the output port 'xlgpcs_mplla_bw_o'. **Value After Reset:** "(DWCXLP_E12) ? \"0x6F\" : \"0xA016\" " **Exists:** Always **Range Variable[x]:** "(DWCXLP_E12) ? \"11\" : \"16\" " - 1

7.1.201 VR_PMA_MP_25G_MPLLA_CTRL3

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8077 : (DWCXLP_APB3_16) ? (0x8077*0x2) : (0x8077*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-206 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLA_LOW_BW	R/W	MPLLA Low Frequency Bandwidth Control This field drives the output port 'xlgpcs_mplla_bw_low_o'. **Value After Reset:** 0x18 **Exists:** Always

7.1.202 VR_PMA_MP_16G_MPLLB_CTRL3

- **Description:** VR PMA Synopsys Multi-protocol 16G/12G PHY MPLLB Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8078 : (DWCXLP_APB3_16) ? (0x8078*0x2) : (0x8078*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16||DWCXLP_E12)

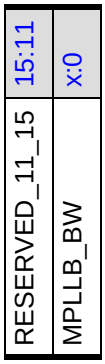


Table 7-207 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL3

Bits	Name	Memory Access	Description
15:11	RESERVED_11_15	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_PMA_MMD&&DWCXLP_E12
x:0	MPLLB_BW	R/W	MPLLB Bandwidth Control This field drives the output port 'xlgpcs_mpllb_bw_o'. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(DWCXLP_E12) ? \"11\" : \"16\" " - 1

7.1.203 VR_PMA_MP_25G_MPLLB_CTRL3

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8078 : (DWCXLP_APB3_16) ? (0x8078*0x2) : (0x8078*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-208 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLB_LOW_BW	R/W	MPLLB Low Frequency Bandwidth Control This field drives the output port 'xlgpcs_mpllb_bw_low_o'. **Value After Reset:** 0x18 **Exists:** Always

7.1.204 VR_PMA_MP_25G_MPLLA_CTRL4

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 4 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8079 : (DWCXLP_APB3_16) ? (0x8079*0x2) : (0x8079*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

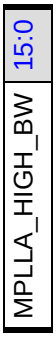


Table 7-209 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL4

Bits	Name	Memory Access	Description
15:0	MPLLA_HIGH_BW	R/W	MPLLA High Frequency Bandwidth Control This field drives the output port 'xlgpcs_mplla_bw_high_o'. **Value After Reset:** 0x18 **Exists:** Always

7.1.205 VR_PMA_MP_16G_MPLLA_CTRL4

- **Description:** VR PMA Synopsys Multi-protocol 16G PHY MPLLA Control 4 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8079 : (DWCXLP_APB3_16) ? (0x8079*0x2) : (0x8079*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16)



Table 7-210 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL4

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	MPLLA_SSC_FRQ_CNT_INT	R/W	MPLLA SSC Frequency Counter Initialization. This field drives the output port xlgpcs_mplla_ssc_freq_cnt_init_o[11:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.206 VR_PMA_MP_25G_MPLLB_CTRL4

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 4 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x807A : (DWCXLP_APB3_16) ? (0x807A*0x2) : (0x807A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

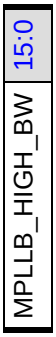


Table 7-211 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL4

Bits	Name	Memory Access	Description
15:0	MPLLB_HIGH_BW	R/W	MPLLB High Frequency Bandwidth Control This field drives the output port 'xlgpcs_mpllb_bw_high_o'. **Value After Reset:** 0x18 **Exists:** Always

7.1.207 VR_PMA_MP_16G_MPLLB_CTRL4

- **Description:** VR PMA Synopsys Multi-protocol 16G PHY MPLLB Control 4 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x807A : (DWCXLP_APB3_16) ? (0x807A*0x2) : (0x807A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16)



Table 7-212 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL4

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	MPLLB_SSC_FRQ_CNT_INT	R/W	MPLLB SSC Frequency Counter Initialization. This field drives the output port xlgpcs_mpllb_ssc_freq_cnt_init_o[11:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.208 VR_PMA_MP_25G_MPLLA_CTRL5

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 5 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x807B : (DWCXLP_APB3_16) ? (0x807B*0x2) : (0x807B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-213 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL5

Bits	Name	Memory Access	Description
15:0	MPLLA_SSC_PEAK_15_0	R/W	MPLLA Spread Spectrum Peak Value. This field drives the output port 'xlgpcs_mplla_ssc_peak_o[15:0]'. **Value After Reset:** 0x0 **Exists:** Always

7.1.209 VR_PMA_MP_16G_MPLLA_CTRL5

- **Description:** VR PMA Synopsys Multi-protocol 16G PHY MPLLA Control 5 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x807B : (DWCXLP_APB3_16) ? (0x807B*0x2) : (0x807B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16)

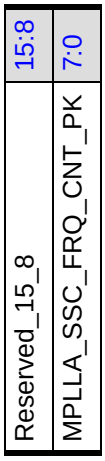


Table 7-214 Fields for Register: VR_PMA_MP_16G_MPLLA_CTRL5

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	MPLLA_SSC_FRQ_CNT_PK	R/W	MPLLA SSC Frequency Counter Peak. This field drives the output port xlgpcs_mplla_ssc_freq_cnt_pk_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.210 VR_PMA_MP_25G_MPLLA_CTRL6

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 6 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x807C : (DWCXLP_APB3_16) ? (0x807C*0x2) : (0x807C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

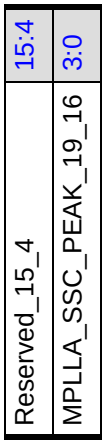


Table 7-215 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL6

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	MPLLA_SSC_PEAK_19_16	R/W	MPLLA Spread Spectrum Peak Value. This field drives the output port 'xlgpcs_mplla_ssc_peak_o[19:16]'. **Value After Reset:** 0x0 **Exists:** Always

7.1.211 VR_PMA_MP_25G_MPLLB_CTRL5

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 5 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x807D : (DWCXLP_APB3_16) ? (0x807D*0x2) : (0x807D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-216 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL5

Bits	Name	Memory Access	Description
15:0	MPLLB_SSC_PEAK_15_0	R/W	MPLLB Spread Spectrum Peak Value. This field drives the output port 'xlgpcs_mpllb_ssc_peak_o[15:0]'. **Value After Reset:** 0x0 **Exists:** Always

7.1.212 VR_PMA_MP_16G_MPLLB_CTRL5

- **Description:** VR PMA Synopsys Multi-protocol 16G PHY MPLLB Control 5 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x807D : (DWCXLP_APB3_16) ? (0x807D*0x2) : (0x807D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E16)

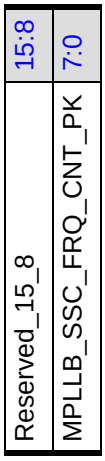


Table 7-217 Fields for Register: VR_PMA_MP_16G_MPLLB_CTRL5

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	MPLLB_SSC_FRQ_CNT_PK	R/W	MPLLB SSC Frequency Counter Peak. This field drives the output port xlgpcs_mpllb_ssc_freq_cnt_pk_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always

7.1.213 VR_PMA_MP_25G_MPLLB_CTRL6

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 6 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x807E : (DWCXLP_APB3_16) ? (0x807E*0x2) : (0x807E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

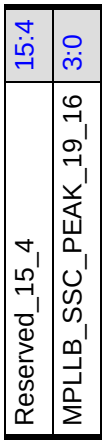


Table 7-218 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL6

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	MPLLB_SSC_PEAK_19_16	R/W	MPLLB Spread Spectrum Peak Value. This field drives the output port 'xlgpcs_mpllb_ssc_peak_o[19:16]'. **Value After Reset:** 0x0 **Exists:** Always

7.1.214 VR_PMA_MP_25G_16G_EQ_INIT_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol PHY Equalization Init Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8080 : (DWCXLP_APB3_16) ? (0x8080*0x2) : (0x8080*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_LNK_TRAIN_EN

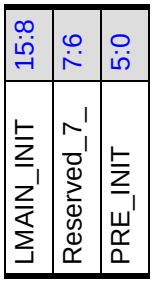


Table 7-219 Fields for Register: VR_PMA_MP_25G_16G_EQ_INIT_CTRL0

Bits	Name	Memory Access	Description
15:8	LMAIN_INIT	R/W	Value of LMAIN corresponding to KR INIT. DWC_xlgpcs considers the value programmed to this field as the 'LMAIN' value corresponding to 'INIT' command. The default value of this field is set to 8'd108/64. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x6C : 0x40" **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-219 Fields for Register: VR_PMA_MP_25G_16G_EQ_INIT_CTRL0 (Continued)

Bits	Name	Memory Access	Description
5:0	PRE_INIT	R/W	Value of PRE corresponding to KR INIT. PRE, LMAIN and POST are three parameters/coefficients which control the Synopsys Multi-protocol PHY Transmit Filter Values. As per the information provided in PHY databook, the values of these parameters corresponding to INIT are : PRE=8, LMAIN=108 and POST=44. DWC_xlpcs considers the value programmed to this field as the 'PRE' value corresponding to 'INIT' command. The default value of this field is set to 8/5. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x8 : DWCXLP_E32 ? 0x4 : 0x5" **Exists:** Always

7.1.215 VR_PMA_MP_25G_16G_EQ_INIT_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol PHY Equalization Init Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8081 : (DWCXLP_APB3_16) ? (0x8081*0x2) : (0x8081*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_LNK_TRAIN_EN

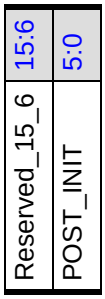


Table 7-220 Fields for Register: VR_PMA_MP_25G_16G_EQ_INIT_CTRL1

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	POST_INIT	R/W	Value of PRE corresponding to KR INIT. PRE, LMAIN and POST are three parameters/coefficients which control the Synopsys Multi-protocol PHY Transmit Filter Values. As per the information provided in PHY databook, the values of these parameters corresponding to INIT are : PRE=8, LMAIN=108 and POST=44. DWC_xlgpcs considers the value programmed to this field as the 'PRE' value corresponding to 'INIT' command. The default value of this field is set to 6'd44/26. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? 0x2C : DWCXLP_E32 ? 0x1C : 0x1A" **Exists:** Always

7.1.216 VR_PMA_MP_25G_MPLLA_CTRL7

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 7 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8082 : (DWCXLP_APB3_16) ? (0x8082*0x2) : (0x8082*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-221 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL7

Bits	Name	Memory Access	Description
15:0	MPLLA_SSC_STEP_SIZE_15_0	R/W	MPLLA Spread Spectrum Step Size Value. This field drives the output port 'xlgpcs_mplla_ssc_step_size_o[15:0]'. **Value After Reset:** 0x0 **Exists:** Always

7.1.217 VR_PMA_MP_25G_MPLLA_CTRL8

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 8 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8083 : (DWCXLP_APB3_16) ? (0x8083*0x2) : (0x8083*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

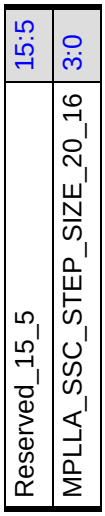


Table 7-222 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL8

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	MPLLA_SSC_STEP_SIZE_20_16	R/W	MPLLA Spread Spectrum Step Size Value. This field drives the output port 'xlgpcs_mplla_ssc_step_size_o[20:16]'. **Value After Reset:** 0x0 **Exists:** Always

7.1.218 VR_PMA_MP_25G_MPLLB_CTRL7

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 7 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8084 : (DWCXLP_APB3_16) ? (0x8084*0x2) : (0x8084*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-223 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL7

Bits	Name	Memory Access	Description
15:0	MPLLB_SSC_STEP_SIZE_15_0	R/W	MPLLB Spread Spectrum Step Size Value. This field drives the output port 'xlgpcs_mpllb_ssc_step_size_o[15:0]'. **Value After Reset:** 0x0 **Exists:** Always

7.1.219 VR_PMA_MP_25G_MPLLB_CTRL8

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 8 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8085 : (DWCXLP_APB3_16) ? (0x8085*0x2) : (0x8085*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

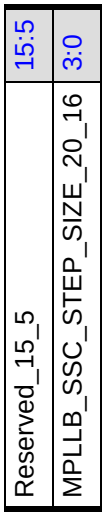


Table 7-224 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL8

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	MPLLB_SSC_STEP_SIZE_20_16	R/W	MPLLB Spread Spectrum Step Size Value. This field drives the output port 'xlgpcs_mpllb_ssc_step_size_o[20:16]'. **Value After Reset:** 0x0 **Exists:** Always

7.1.220 VR_PMA_MP_25G_MPLLA_CTRL9

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 9 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8086 : (DWCXLP_APB3_16) ? (0x8086*0x2) : (0x8086*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-225 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL9

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_QUOT	R/W	MPLLA Fraction Quotient. This field drives the output port 'xlgpcs_mplla_frac_quot_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.221 VR_PMA_MP_25G_MPLLB_CTRL9

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 9 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8087 : (DWCXLP_APB3_16) ? (0x8087*0x2) : (0x8087*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-226 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL9

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_QUOT	R/W	MPLLB Fraction Quotient. This field drives the output port 'xlgpcs_mpllb_frac_quot_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.222 VR_PMA_MP_25G_MPLLA_CTRL10

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 10 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8088 : (DWCXLP_APB3_16) ? (0x8088*0x2) : (0x8088*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

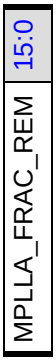


Table 7-227 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL10

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_REM	R/W	MPLLA Fraction Remainder. This field drives the output port 'xlgpcs_mplla_frac_rem_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.223 VR_PMA_MP_25G_MPLLB_CTRL10

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 10 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8089 : (DWCXLP_APB3_16) ? (0x8089*0x2) : (0x8089*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)

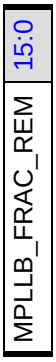


Table 7-228 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL10

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_REM	R/W	MPLLB Fraction Remainder. This field drives the output port 'xlgpcs_mpllb_frac_rem_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.224 VR_PMA_MP_25G_MPLLA_CTRL11

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLA Control 11 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x808A : (DWCXLP_APB3_16) ? (0x808A*0x2) : (0x808A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-229 Fields for Register: VR_PMA_MP_25G_MPLLA_CTRL11

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_DEN	R/W	MPLLA Fraction Denominator. This field drives the output port 'xlgpcs_mplla_frac_den_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.225 VR_PMA_MP_25G_MPLLB_CTRL11

- **Description:** VR PMA Synopsys Multi-protocol 25G PHY MPLLB Control 11 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x808B : (DWCXLP_APB3_16) ? (0x808B*0x2) : (0x808B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25)



Table 7-230 Fields for Register: VR_PMA_MP_25G_MPLLB_CTRL11

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_DEN	R/W	MPLLB Fraction Denominator. This field drives the output port 'xlgpcs_mpllb_frac_den_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.226 VR_PMA_MP_EPHY_EQ_PRESET_CTRL0

- **Description:** VR PMA Synopsys Multi Protocol Enterprise PHY Equalization Preset Control 0 register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x808C : (DWCXLP_APB3_16) ? (0x808C*0x2) : (0x808C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN

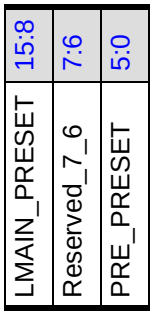


Table 7-231 Fields for Register: VR_PMA_MP_EPHY_EQ_PRESET_CTRL0

Bits	Name	Memory Access	Description
15:8	LMAIN_PRESET	R/W	Value of LMAIN corresponding to KR PRESET. PRE, LMAIN and POST are three parameters/coefficients which control the Synopsys Multi Protocol EPHY Transmit Filter values. DWC_xlgpcs considers the value programmed to this field as the 'LMAIN' value corresponding to 'PRESET' command. **Value After Reset:** "DWCXLP_E16||DWCXLP_E12 ? 0xA0 : DWCXLP_E32 ? 0x5C : 0x60" **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-231 Fields for Register: VR_PMA_MP_EPHY_EQ_PRESET_CTRL0 (Continued)

Bits	Name	Memory Access	Description
5:0	PRE_PRESET	R/W	Value of PRE corresponding to KR PRESET. PRE, LMAIN and POST are three parameters/coefficients which control the Synopsys Multi Protocol EPHY Transmit Filter values. DWC_xlgpcs considers the value programmed to this field as the 'PRE' value corresponding to 'PRESET' command. **Value After Reset:** "DWCXLP_E32 ? 0x4 : 0x0" **Exists:** Always

7.1.227 VR_PMA_MP_EPHY_EQ_PRESET_CTRL1

- **Description:** VR PMA Synopsys Multi Protocol Enterprise PHY Equalization Preset Control 1 register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x808D : (DWCXLP_APB3_16) ? (0x808D*0x2) : (0x808D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_SNPS_XAUI_PHY&&DWCXLP_LNK_TRAIN_EN

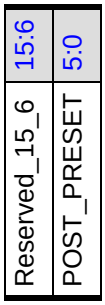
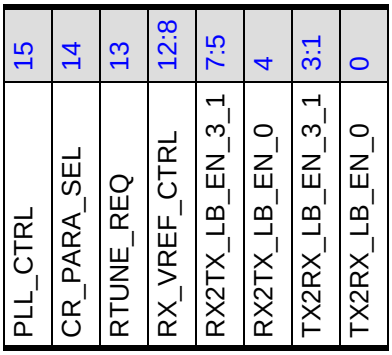


Table 7-232 Fields for Register: VR_PMA_MP_EPHY_EQ_PRESET_CTRL1

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	POST_PRESET	R/W	Value of POST corresponding to KR PRESET. PRE, LMAIN and POST are three parameters/coefficients which control the Synopsys Multi Protocol EPHY Transmit Filter values. DWC_xlgpcs considers the value programmed to this field as the 'POST' value corresponding to 'PRESET' command. **Value After Reset:** 0x0 **Exists:** Always

7.1.228 VR_PMA_MP_25G_16G_MISC_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol PHY Miscellaneous Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8090 : (DWCXLP_APB3_16) ? (0x8090*0x2) : (0x8090*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)



15	14	13	12:8	7:5	4	3:1	0
PLL_CTRL	CR_PARA_SEL	RTUNE_REQ	RX_VREF_CTRL	RX2TX_LB_EN_3_1	RX2TX_LB_EN_0	TX2RX_LB_EN_3_1	TX2RX_LB_EN_0

Table 7-233 Fields for Register: VR_PMA_MP_25G_16G_MISC_CTRL0

Bits	Name	Memory Access	Description
15	PLL_CTRL	R/W	PLL Re-initialization Control If the PHY PLL does not lock for a long time, DWC_xlgpcs would initiate a Rx power-state change from P0 to P1 and then back to P0, if this bit is set. This process is done to re-initialize the PLL. The time duration to wait before reinitializing the PLL is determined by 'VR PMA Synopsys Multi-protocol PHY Miscellaneous Control 1 Register **Value After Reset:** 0x0 **Exists:** Always
14	CR_PARA_SEL	R/W	Select CR Para Port This bit select the interface for accessing PHY registers as follows: ■ 0: JTAG ■ 1: CR parallel port This bit should be changed only after disabling 'jtag_tck'to PHY. **Value After Reset:** 0x1 **Exists:** Always

Table 7-233 Fields for Register: VR_PMA_MP_25G_16G_MISC_CTRL0 (Continued)

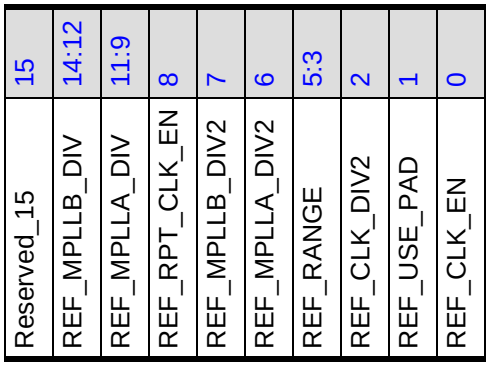
Bits	Name	Memory Access	Description
13	RTUNE_REQ	R/W	Resistor Tuning Request This bit can be set to trigger a resistor tune request to the PHY. This bit controls the 'xlgpcs_rtune_req_o' output port. **Value After Reset:** 0x0 **Exists:** Always
12:8	RX_VREF_CTRL	* Varies	Rx Biasing Current Control This field drives the output port xlgpcs_rx_vref_ctrl_o[4:0]. This field sets the Rx biasing current for Rx analog front end. **Value After Reset:** "(DWCXLP_E32) ? 0x0 : ((DWCXLP_E16) ? 0x11 : (DWCXLP_E12) ? 0xF : 0x5)" **Exists:** Always **Memory Access:** "!(DWCXLP_E32) ? \"read-write\" : \"read-only\" "
7:5	RX2TX_LB_EN_3_1	* Varies	**For Multi-lane configurations :** Enable Parallel Rx-to-Tx Loopback on lanes[3:1] **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
4	RX2TX_LB_EN_0	R/W	Enable Parallel Rx-to-Tx Loopback on lane 0 When this bit is set, recovered parallel data from PHY receiver is looped back to the transmit serializer. This loopback takes place internal to the PHY (not within DWC_xlgpcs). **Value After Reset:** 0x0 **Exists:** Always
3:1	TX2RX_LB_EN_3_1	* Varies	**For Multi-lane configurations :** Enable Analog Tx-to-Rx Serial Loopback on lanes[3:1] This field enable serial loopback in the PHY from Tx pre-driver to Rx analog front-end on lanes 3 to 1. **For Single-lane configurations :** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-233 Fields for Register: VR_PMA_MP_25G_16G_MISC_CTRL0 (Continued)

Bits	Name	Memory Access	Description
0	TX2RX_LB_EN_0	R/W	Enable Analog Tx-to-Rx Serial Loopback on lane 0 This bit can be set to enable serial loopback in the PHY from Tx pre-driver to Rx analog front-end. **Value After Reset:** 0x0 **Exists:** Always

7.1.229 VR_PMA_MP_25G_16G_REF_CLK_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Reference Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8091 : (DWCXLP_APB3_16) ? (0x8091*0x2) : (0x8091*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)



15	14:12	11:9	8	7	6	5:3	2	1	0
Reserved_15	REF_MPLLB_DIV	REF_MPLLA_DIV	REF_RPT_CLK_EN	REF_MPLLB_DIV2	REF_MPLLA_DIV2	REF_RANGE	REF_CLK_DIV2	REF_USE_PAD	REF_CLK_EN

Table 7-234 Fields for Register: VR_PMA_MP_25G_16G_REF_CLK_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-234 Fields for Register: VR_PMA_MP_25G_16G_REF_CLK_CTRL (Continued)

Bits	Name	Memory Access	Description
14:12	REF_MPLLB_DIV	* Varies	**Multi-protocol 25G configurations:** MPLLB Reference Clock Divider Control. The reference clock used for MPLLB calibration and locking can be divided by 2/4/8/16. Note that this division is applied independently of ref_clk_div2_en. ■ 3'b000: Divide by 1 ■ 3'b001: Divide by 2 ■ 3'b010: Divide by 4 ■ 3'b011: Divide by 8 ■ 3'b100: Divide by 16 ■ 3'b101: Reserved ■ 3'b110: Reserved ■ 3'b111: Reserved This field drives the output port xlgpcs_ref_clk_mpllb_div_o **For other configurations:** Reserved **Value After Reset:** "!(DWCXLP_E25) ? 0x0 : (DWCXLP_REF_CLK_FREQ==0) ? 0x1 : (DWCXLP_REF_CLK_FREQ==1) ? 0x0 : 0x2" **Exists:** Always **Memory Access:** "(DWCXLP_E25) ? \"read-write\" :\"read-only\""

Table 7-234 Fields for Register: VR_PMA_MP_25G_16G_REF_CLK_CTRL (Continued)

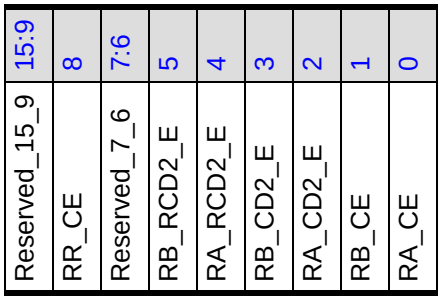
Bits	Name	Memory Access	Description
11:9	REF_MPLLA_DIV	* Varies	**Multi-protocol 25G configurations:** MPLLA Reference Clock Divider Control. The reference clock used for MPLLA calibration and locking can be divided by 2/4/8/16. Note that this division is applied independently of ref_clk_div2_en. ■ 3'b000: Divide by 1 ■ 3'b001: Divide by 2 ■ 3'b010: Divide by 4 ■ 3'b011: Divide by 8 ■ 3'b100: Divide by 16 ■ 3'b101: Reserved ■ 3'b110: Reserved ■ 3'b111: Reserved This field drives the output port xlgpcs_ref_clk_mplla_div_o **For other configurations:** Reserved **Value After Reset:** "!(DWCXLP_E25) ? 0x0 : (DWCXLP_REF_CLK_FREQ==0) ? 0x1 : (DWCXLP_REF_CLK_FREQ==1) ? 0x0 : 0x2" **Exists:** Always **Memory Access:** "(DWCXLP_E25) ? \"read-write\" :\"read-only\"""
8	REF_RPT_CLK_EN	R/W	Repeat Reference Clock Enable. This field drives the output port xlgpcs_ref_repeat_clk_en_o **Value After Reset:** 0x0 **Exists:** Always
7	REF_MPLLB_DIV2	* Varies	Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E16||DWCXLP_E12) ? \"read-write\" :\"read-only\"""
6	REF_MPLLA_DIV2	* Varies	Reserved **Value After Reset:** "((DWCXLP_E16||DWCXLP_E12)&&!DWCXLP_REF_CLK_78) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_E16||DWCXLP_E12) ? \"read-write\" :\"read-only\"""

Table 7-234 Fields for Register: VR_PMA_MP_25G_16G_REF_CLK_CTRL (Continued)

Bits	Name	Memory Access	Description
5:3	REF_RANGE	R/W	Input Reference Clock Range This field specifies the frequency range of the input reference clock (post ref_clk_div2_en division if any). The code mapping of REF_RANGE is as follows: ■ 3'b000: 20 - 26 MHz ■ 3'b001: 26.1 - 52 MHz ■ 3'b010: 52.1 - 78 MHz ■ 3'b011: 78.1 - 104 MHz ■ 3'b100: 104.1 - 130 MHz ■ 3'b101: 130.1 - 156 MHz ■ 3'b110: 156.1 - 182 MHz ■ 3'b111: 182.1 - 200 MHz This field should be set to appropriate values based on your reference clock frequency. **Value After Reset:** "(DWCXLP_REF_CLK_FREQ==1) ? 0x3 : 0x6" **Exists:** Always
2	REF_CLK_DIV2	R/W	Reference Clock divide by 2 If this bit is set, reference clock provided to PHY gets divided by 2 internally in the PHY. **Value After Reset:** "(DWCXLP_REF_CLK_FREQ==2) ? 0x1 : 0x0" **Exists:** Always
1	REF_USE_PAD	R/W	Use Pad Clk as Reference Clock If this bit is set, PHY selects the clock connected to ref_pad_clk_p/m as its reference clock; otherwise it selects the clock connected to ref_alt_clk_p/m. **Value After Reset:** "(DWCXLP_REF_CLK_SRC==0) ? 0x1 : 0x0" **Exists:** Always
0	REF_CLK_EN	R/W	Reference Clock Enable This bit should be set to enable reference clock to the PHY. This bit controls the 'xlgpcs_ref_clk_en_o' output port. **Value After Reset:** 0x1 **Exists:** Always

7.1.230 VR_PMA_MP_32G_REF_CLK_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Reference Clock Control0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8091 : (DWCXLP_APB3_16) ? (0x8091*0x2) : (0x8091*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)



15:9	8	7:6	5	4	3	2	1	0
Reserved_15_9	RR_CE	Reserved_7_6	RB_RCD2_E	RA_RCD2_E	RB_CD2_E	RA_CD2_E	RB_CE	RA_CE

Table 7-235 Fields for Register: VR_PMA_MP_32G_REF_CLK_CTRL0

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RR_CE	R/W	Ref Repeat Clock Enable. This field drives the output port 'xlgpcs_ref_repeat_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	RB_RCD2_E	R/W	Reference clock divider control for refb in RAW PCS. Asserted to divide refb clock by 2 in RAW PCS. This field drives the output port 'xlgpcs_refb_raw_clk_div2_en_o'. **Value After Reset:** "DWCXLP_REF_CLK_78 ? 0x0 : 0x1" **Exists:** Always

Table 7-235 Fields for Register: VR_PMA_MP_32G_REF_CLK_CTRL0 (Continued)

Bits	Name	Memory Access	Description
4	RA_RCD2_E	R/W	Reference clock divider control for refa in RAW PCS. Asserted to divide refa clock by 2 in RAW PCS. This field drives the output port 'xlgpcs_refa_raw_clk_div2_en_o'. **Value After Reset:** "DWCXLP_REF_CLK_78 ? 0x0 : 0x1" **Exists:** Always
3	RB_CD2_E	R/W	Reference clock divider control for refb. When asserted, the reference clock is divided by 2. This field drives the output port 'xlgpcs_refb_clk_div2_en_o'. **Value After Reset:** "DWCXLP_REF_CLK_312 ? 0x1 : 0x0" **Exists:** Always
2	RA_CD2_E	R/W	Reference clock divider control for refa. When asserted, the reference clock is divided by 2. This field drives the output port 'xlgpcs_refa_clk_div2_en_o'. **Value After Reset:** "DWCXLP_REF_CLK_312 ? 0x1 : 0x0" **Exists:** Always
1	RB_CE	R/W	Reference clock enable for refb. Enables the reference 'b' clock for the PHY. This field drives the output port 'xlgpcs_refb_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
0	RA_CE	R/W	Reference clock enable for refa. Enables the reference 'a' clock for the PHY. This field drives the output port 'xlgpcs_refa_clk_en_o'. **Value After Reset:** 0x1 **Exists:** Always

7.1.231 VR_PMA_MP_25G_16G_VCO_CAL_LD0

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Load Register 0
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8092 : (DWCXLP_APB3_16) ? (0x8092*0x2) : (0x8092*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)

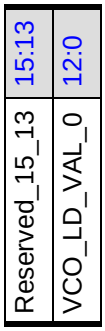


Table 7-236 Fields for Register: VR_PMA_MP_25G_16G_VCO_CAL_LD0

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:0	VCO_LD_VAL_0	R/W	Rx VCO calibration load value on lane 0 of the PHY This field is used to load internal calibration registers in the PHY to perform Rx VCO calibration on lane 0. This field drives the output port 'xlgpcs_rx0_vco_ld_val_o'. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? (DWCXLP_REF_CLK_78 ? 0x56A : 0x549) : (DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x5CD : 0x57B) : (DWCXLP_REF_CLK_78 ? 0x5AC : 0x56A)" **Exists:** Always

7.1.232 VR_PMA_MP_25G_16G_VCO_CAL_LD1

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Load Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8093 : (DWCXLP_APB3_16) ? (0x8093*0x2) : (0x8093*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

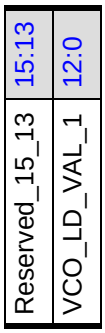


Table 7-237 Fields for Register: VR_PMA_MP_25G_16G_VCO_CAL_LD1

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:0	VCO_LD_VAL_1	R/W	Rx VCO calibration load value on lane 1 of the PHY This field is used to load internal calibration registers in the PHY to perform Rx VCO calibration on lane 1. This field drives the output port 'xlgpcs_rx1_vco_ld_val_o'. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? (DWCXLP_REF_CLK_78 ? 0x56A : 0x549) : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x5CD : 0x57B) : (DWCXLP_REF_CLK_78 ? 0x5AC : 0x56A)" **Exists:** Always

7.1.233 VR_PMA_MP_25G_16G_VCO_CAL_LD2

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Load Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8094 : (DWCXLP_APB3_16) ? (0x8094*0x2) : (0x8094*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

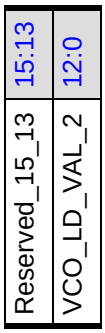


Table 7-238 Fields for Register: VR_PMA_MP_25G_16G_VCO_CAL_LD2

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:0	VCO_LD_VAL_2	R/W	Rx VCO calibration load value on lane 2 of the PHY This field is used to load internal calibration registers in the PHY to perform Rx VCO calibration on lane 2. This field drives the output port 'xlgpcs_rx2_vco_ld_val_o'. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? (DWCXLP_REF_CLK_78 ? 0x56A : 0x549) : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x5CD : 0x57B) : (DWCXLP_REF_CLK_78 ? 0x5AC : 0x56A)" **Exists:** Always

7.1.234 VR_PMA_MP_25G_16G_VCO_CAL_LD3

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Load Register 3
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8095 : (DWCXLP_APB3_16) ? (0x8095*0x2) : (0x8095*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

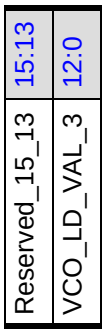


Table 7-239 Fields for Register: VR_PMA_MP_25G_16G_VCO_CAL_LD3

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:0	VCO_LD_VAL_3	R/W	Rx VCO calibration load value on lane 3 of the PHY This field is used to load internal calibration registers in the PHY to perform Rx VCO calibration on lane 3. This field drives the output port 'xlgpcs_rx3_vco_ld_val_o' **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? (DWCXLP_REF_CLK_78 ? 0x56A : 0x549) : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x5CD : 0x57B) : (DWCXLP_REF_CLK_78 ? 0x5AC : 0x56A)" **Exists:** Always

7.1.235 VR_PMA_MP_25G_16G_VCO_CAL_REF0

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Reference Register 0
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8096 : (DWCXLP_APB3_16) ? (0x8096*0x2) : (0x8096*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)

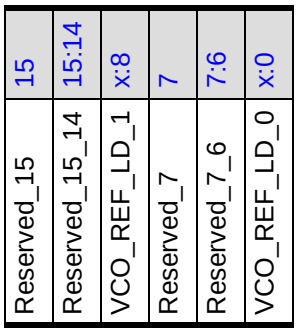


Table 7-240 Fields for Register: VR_PMA_MP_25G_16G_VCO_CAL_REF0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** !DWCXLP_E12
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_E12

Table 7-240 Fields for Register: VR_PMA_MP_25G_16G_VCO_CAL_REF0 (Continued)

Bits	Name	Memory Access	Description
x:8	VCO_REF_LD_1	* Varies	**For Multi-lane configurations :** Rx VCO calibration reference load value on lane 1 of the PHY This field controls the PHY's internal calibration registers used to perform Rx VCO calibration on lane 1. This field drives the output port 'xlgpcs_rx1_ref_ld_val_o[]'. **For Single-lane configurations :** Reserved **Value After Reset:** "DWCXLP_25G_ONLY ? 0x0 : (DWCXLP_E16||DWCXLP_E12) ? (DWCXLP_REF_CLK_78 ? 0x15 : 0x29) : ((DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? ((DWCXLP_REF_CLK_78 ? 0x9 : 0x11)) : (DWCXLP_REF_CLK_78 ? 0xB : 0x15))" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E12) ? \"6\" : \"7\" " + 7 **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** !DWCXLP_E12
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** DWCXLP_E12
x:0	VCO_REF_LD_0	R/W	Rx VCO calibration reference load value -lane 0 This field controls the PHY's internal calibration registers used to perform Rx VCO calibration on lane 0. This field drives the output port 'xlgpcs_rx0_ref_ld_val_o[]'. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? (DWCXLP_REF_CLK_78 ? 0x15 : 0x29) : (DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x9 : 0x11) : (DWCXLP_REF_CLK_78 ? 0xB : 0x15)" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E12) ? \"6\" : \"7\" " - 1

7.1.236 VR_PMA_MP_25G_16G_VCO_CAL_REF1

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY VCO Calibration Reference Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8097 : (DWCXLP_APB3_16) ? (0x8097*0x2) : (0x8097*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

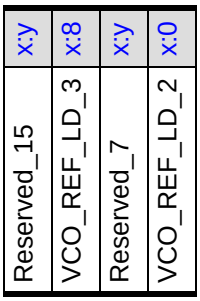


Table 7-241 Fields for Register: VR_PMA_MP_25G_16G_VCO_CAL_REF1

Bits	Name	Memory Access	Description
x:y	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(DWCXLP_E12) ? \"2\" :\"1\"" + "(DWCXLP_E12) ? \"14\" :\"15\"" - 1 **Range Variable[y]:** "(DWCXLP_E12) ? \"14\" :\"15\""
x:8	VCO_REF_LD_3	R/W	Rx VCO calibration reference load value on lane 3 of the PHY This field controls the PHY's internal calibration registers used to perform Rx VCO calibration on lane 3. This field drives the output port 'xlgpcs_rx3_ref_ld_val_o[]'. **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? (DWCXLP_REF_CLK_78 ? 0x15 : 0x29) : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x9 : 0x11) : (DWCXLP_REF_CLK_78 ? 0xB : 0x15)" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E12) ? \"6\" : \"7\" " + 7

Table 7-241 Fields for Register: VR_PMA_MP_25G_16G_VCO_CAL_REF1 (Continued)

Bits	Name	Memory Access	Description
x:y	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(DWCXLP_E12) ? \"2\" :\"1\"" + "(DWCXLP_E12) ? \"6\" :\"7\"" - 1 **Range Variable[y]:** "(DWCXLP_E12) ? \"6\" :\"7\""
x:0	VCO_REF_LD_2	R/W	Rx VCO calibration reference load value -lane 2 This field controls the PHY's internal calibration registers used to perform Rx VCO calibration on lane 2. This field drives the output port 'xlgpcs_rx2_ref_ld_val_o[]' **Value After Reset:** "(DWCXLP_E16||DWCXLP_E12) ? (DWCXLP_REF_CLK_78 ? 0x15 : 0x29) : (DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? (DWCXLP_REF_CLK_78 ? 0x9 : 0x11) : (DWCXLP_REF_CLK_78 ? 0xB : 0x15)" **Exists:** Always **Range Variable[x]:** "(DWCXLP_E12) ? \"6\" : \"7\" " - 1

7.1.237 VR_PMA_MP_25G_16G_MISC_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol PHY Miscellaneous Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8098 : (DWCXLP_APB3_16) ? (0x8098*0x2) : (0x8098*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

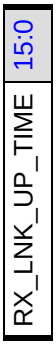


Table 7-242 Fields for Register: VR_PMA_MP_25G_16G_MISC_CTRL1

Bits	Name	Memory Access	Description
15:0	RX_LNK_UP_TIME	R/W	Wait Time before PLL Re-initialization This field determines the number of 'clk_csr_i' clock cycles that can elapse before the Rx link is established (after DWC_xlgpcs has reached POWER_GOOD state). This field is relevant only if 'PLL_CTRL' bit of VR PMA Synopsys Multi-protocol PHY Miscellaneous Control 0 Register is programmed to 1. If the receive link is not up (or if frame lock does not happen during CL72 Training or if AN page is not received), within this specified interval, DWC_xlgpcs would program the PHY Rx power state to P1 and would then bring it back to P0 (thus re-initializing the PHY PLL). **Value After Reset:** 0xffff **Exists:** Always

7.1.238 VR_PMA_MP_25G_16G_EEE_CTRL

- **Description:** VR PMA Synopsys Multi-protocol PHY EEE Control Register

  This register is present only in EEE configurations
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8099 : (DWCXLP_APB3_16) ? (0x8099*0x2) : (0x8099*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_EEE_EN



Table 7-243 Fields for Register: VR_PMA_MP_25G_16G_EEE_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	MIN_QUIET_COUNT	R/W	Minimum Rx Quiet Duration This field determines the minimum duration for which Rx LPI FSM remains in the RX_QUIET state.DWC_xlgpcs implements a 'MIN QUIET timer' in the RX_QUIET state of LPI state machine. This timer (running at 156.25 MHz) starts as soon as RX_QUIET state is reached and it times-out when it reaches the value mentioned as 'MIN QUIET COUNT'. DWC_xlgpcs start monitoring for 'energy_detect' only after the expiry of this timer. **Value After Reset:** 0x4f **Exists:** Always

7.1.239 VR_PMA_MP_25G_16G_SRAM

- **Description:** VR PMA Synopsys Multi-protocol PHY SRAM Register.

  Note: This is a special register which can be read/written even before DWC_xlgpcs completes its reset/initialization sequence.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x809A : (DWCXLP_APB3_16) ? (0x809A*0x2) : (0x809A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:**
  DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

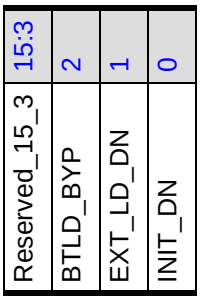


Table 7-244 Fields for Register: VR_PMA_MP_25G_16G_SRAM

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	BTLD_BYP	* Varies	SRAM Bootload Bypass. This field drives the xlgpcs_sram_bootload_byp_o port. **Value After Reset:** "(DWCXLP_E32&&!SNPS_RSVDPARAM_1) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_E25||DWCXLP_E32) ? \"read-write\" : \"read-only\""

Table 7-244 Fields for Register: VR_PMA_MP_25G_16G_SRAM (Continued)

Bits	Name	Memory Access	Description
1	EXT_LD_DN	R/W	SRAM External Loading Done. This field drives the output port 'xlgpcs_sram_ext_ld_done_i' of DWC_xlgpcs. Software should set this bit to 1, after PHY has completed its SRAM initialization and after software has completed the (optional) external loading of the SRAM attached to the Multi-protocol PHY. The PHY completes its initialization sequence only after this register bit is set to 1. **Notes:** ■ Once software has set this bit as 1, it should remain 1 throughout normal operation. ■ This bit gets cleared during standard soft reset (RST) ■ This bit does not get cleared during vendor soft reset (VR_RST). ■ When software asserts VR_RST, this bit should be cleared prior to that (in case, software wishes to alter the SRAM contents initialized by the PHY). **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
0	INIT_DN	R	SRAM Initialization Done. This register field reflects the value of the 'xlgpcs_sram_init_done_i' port of DWC_xlgpcs. When PHY gets reset, the boot-loader in the Raw PCS loads the code from the internal look-up table to the SRAM. After this SRAM initialization is complete, PHY asserts this signal. After the SRAM initialization is done, software can optionally change the SRAM contents using the CR port interface by addressing the SRAM address space in the register map of the PHY. During power-on-reset or when RST or VR_RST register bits are programmed to 1, the PHY also gets reset. Hence, software should first poll this register bit to become 1. Thereafter, it should set the 'EXT_LD_DN' bit to 1. Only then the PHY and the DWC_xlgpcs proceeds to complete its reset sequence. **Note:** Though default value of this register bit is given as 1, please be aware that this is the value provided by the PHY after the completion of SRAM initialization. **Value After Reset:** 0x1 **Exists:** Always

7.1.240 VR_PMA_MP_25G_16G_MISC_CTRL2

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G PHY Miscellaneous Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x809B : (DWCXLP_APB3_16) ? (0x809B*0x2) : (0x809B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16)

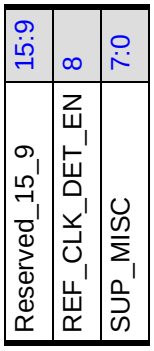


Table 7-245 Fields for Register: VR_PMA_MP_25G_16G_MISC_CTRL2

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	REF_CLK_DET_EN	R/W	Reference Clock Detect enable. This field drives the output port xlgpcs_ref_clkdet_en_o. **Value After Reset:** 0x0 **Exists:** Always
7:0	SUP_MISC	R/W	Support Miscellaneous Controls. This field drives the output port xlgpcs_sup_misc_o[7:0]. **Value After Reset:** "(DWCXLP_E25) ? 0x0 : 0x1" **Exists:** Always

7.1.241 VR_PMA_MP_32G_MISC_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Miscellaneous Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x809B : (DWCXLP_APB3_16) ? (0x809B*0x2) : (0x809B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

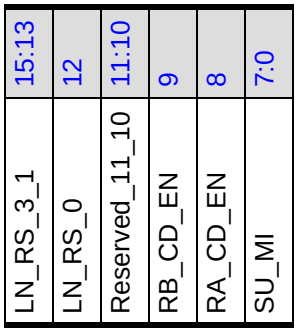


Table 7-246 Fields for Register: VR_PMA_MP_32G_MISC_CTRL

Bits	Name	Memory Access	Description
15:13	LN_RS_3_1	* Varies	Lane3_1 reference clock selection. Selects the refa or refb reference clock. This field drives the output port 'xlgpcs_lane_ref_sel_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
12	LN_RS_0	R/W	Lane0 reference clock selection. Selects the refa or refb reference clock. This field drives the output port 'xlgpcs_lane_ref_sel_o'. **Value After Reset:** 0x0 **Exists:** Always
11:10	Reserved_11_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-246 Fields for Register: VR_PMA_MP_32G_MISC_CTRL (Continued)

Bits	Name	Memory Access	Description
9	RB_CD_EN	R/W	Refb Clock Detection Enable. Enables reference clock detection for refb. This field drives the output port 'xlgpcs_refb_clkdet_en_o'. **Value After Reset:** 0x0 **Exists:** Always
8	RA_CD_EN	R/W	Refa Clock Detection Enable. Enables reference clock detection for refa. This field drives the output port 'xlgpcs_refa_clkdet_en_o'. **Value After Reset:** 0x0 **Exists:** Always
7:0	SU_MI	R/W	Support Miscellaneous Controls. Controls miscellaneous settings in the support block of the PHY. This field drives the output port 'xlgpcs_sup_misc_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.242 VR_PMA_MP_32G_REF_CLK_CTRL1

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Reference Clock Control1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x809C : (DWCXLP_APB3_16) ? (0x809C*0x2) : (0x809C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

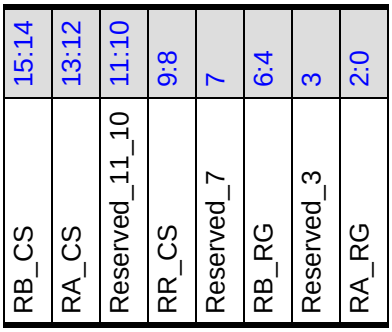


Table 7-247 Fields for Register: VR_PMA_MP_32G_REF_CLK_CTRL1

Bits	Name	Memory Access	Description
15:14	RB_CS	R/W	Refb Clock Select. Selects reference 'b' clock source. This field drives the output port 'xlgpcs_refb_clk_sel_o'. **Value After Reset:** "(DWCXLP_REF_CLK_SRC==1) ? 0x0 : (DWCXLP_REF_CLK_SRC==0) ? 0x1 : 0x2" **Exists:** Always
13:12	RA_CS	R/W	Refa Clock Select. Selects reference 'a' clock source. This field drives the output port 'xlgpcs_refa_clk_sel_o'. **Value After Reset:** "(DWCXLP_REF_CLK_SRC==1) ? 0x0 : (DWCXLP_REF_CLK_SRC==0) ? 0x1 : 0x2" **Exists:** Always
11:10	Reserved_11_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:8	RR_CS	R/W	Repeat reference clock select. Selects the reference clock for the CML clocks. This field drives the output port 'xlgpcs_ref_rpt_clk_sel_o'. **Value After Reset:** 0x0 **Exists:** Always

Table 7-247 Fields for Register: VR_PMA_MP_32G_REF_CLK_CTRL1 (Continued)

Bits	Name	Memory Access	Description
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	RB_RG	R/W	Reference clock frequency range for refb. Specifies the frequency range of the reference 'b' clock. This field drives the output port 'DWC_xlgpcs_refb_range_o'. **Value After Reset:** "DWCXLP_REF_CLK_78 ? 0x3 : 0x6" **Exists:** Always
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RA_RG	R/W	Reference clock frequency range for refa. Specifies the frequency range of the reference 'a' clock. This field drives the output port 'DWC_xlgpcs_refa_range_o'. **Value After Reset:** "DWCXLP_REF_CLK_78 ? 0x3 : 0x6" **Exists:** Always

7.1.243 VR_PMA_MP_25G_16G_12G_MISC_STS0

- **Description:** VR PMA Synopsys Multi-protocol 25G/16G/12G PHY Miscellaneous Status 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x809D : (DWCXLP_APB3_16) ? (0x809D*0x2) : (0x809D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E16||DWCXLP_E12)

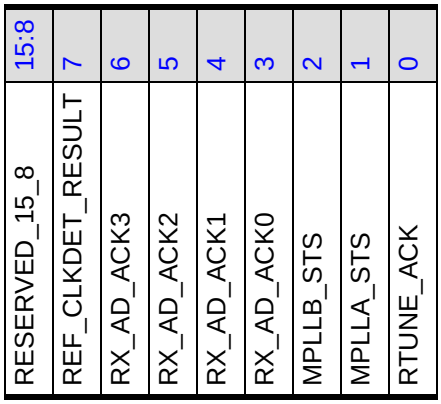


Table 7-248 Fields for Register: VR_PMA_MP_25G_16G_12G_MISC_STS0

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	REF_CLKDET_RESULT	R	Reference Clock Detect enable This field is driven by the input xlgpcs_ref_clkdet_result_i. **Value After Reset:** 0x0 **Exists:** Always
6	RX_AD_ACK3	R	**For multi-lane Configurations:** Receiver Adaptation Acknowledgement for lane-3. This field reflects the value of the input port 'rpcs_krx_adapt_ack3_i'. **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-248 Fields for Register: VR_PMA_MP_25G_16G_12G_MISC_STS0 (Continued)

Bits	Name	Memory Access	Description
5	RX_AD_ACK2	R	**For multi-lane Configurations:** Receiver Adaptation Acknowledgement for lane-2. This field reflects the value of the input port 'rpcs_krx_adapt_ack2_i'. **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
4	RX_AD_ACK1	R	**For multi-lane Configurations:** Receiver Adaptation Acknowledgement for lane-1. This field reflects the value of the input port 'rpcs_krx_adapt_ack1_i'. **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
3	RX_AD_ACK0	R	Receiver Adaptation Acknowledgement for lane-0. This field reflects the value of the input port 'rpcs_krx_adapt_ack0_i'. **Value After Reset:** 0x0 **Exists:** Always
2	MPLLB_STS	R	Status of MPLLB from PHY This bit denotes the value of xlgpcs_mpllb_state_i input. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x1 : 0x0" **Exists:** Always
1	MPLLA_STS	R	Status of MPLLA from PHY. This bit denotes the value of xlgpcs_mplla_state_i input **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_KR4_50G||DWCXLP_KR4_100G) ? 0x0 : 0x1" **Exists:** Always
0	RTUNE_ACK	R	Acknowledgement for Resistor Tune Request This bit denotes the value of 'xlgpcs_rtune_ack_i' input. **Value After Reset:** 0x0 **Exists:** Always

7.1.244 VR_PMA_MP_32G_MISC_STS0

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Miscellaneous Status0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x809D : (DWCXLP_APB3_16) ? (0x809D*0x2) : (0x809D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

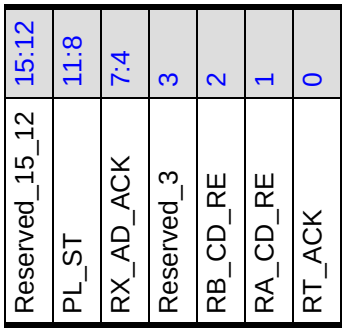


Table 7-249 Fields for Register: VR_PMA_MP_32G_MISC_STS0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	PL_ST	R	TX PLL State Indicator. Indicates the state of PLL for the lane. This field reflects the value of the input port 'xlgpcs_tx_pll_state_i'. **Value After Reset:** 0x0 **Exists:** Always
7:4	RX_AD_ACK	R	Receiver Adaptation Acknowledgement for lane-X. This field reflects the value of the input port 'rpcs_krx_adapt_ack0_i'. **Value After Reset:** 0x0 **Exists:** Always
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-249 Fields for Register: VR_PMA_MP_32G_MISC_STS0 (Continued)

Bits	Name	Memory Access	Description
2	RB_CD_RE	R	Reference clock detection result for refb. Indicates the presence of the reference clock. This field reflects the value of the input port 'xlgpcs_refb_clkdet_result_i'. **Value After Reset:** 0x0 **Exists:** Always
1	RA_CD_RE	R	Reference clock detection result for refa. Indicates the presence of the reference clock. This field reflects the value of the input port 'xlgpcs_refa_clkdet_result_i'. **Value After Reset:** 0x0 **Exists:** Always
0	RT_ACK	R	Resistor tune acknowledgement. Indicates that a resistor tune has completed. This field reflects the value of the input port 'xlgpcs_rtune_ack_i'. **Value After Reset:** 0x0 **Exists:** Always

7.1.245 VR_PMA_MP_25G_16G_MISC_STS1

- **Description:** VR PMA Synopsys Multi-protocol PHY Miscellaneous Status 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x809E : (DWCXLP_APB3_16) ? (0x809E*0x2) : (0x809E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)

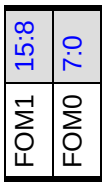


Table 7-250 Fields for Register: VR_PMA_MP_25G_16G_MISC_STS1

Bits	Name	Memory Access	Description
15:8	FOM1	R	For multi-lane Configurations: Figure of merit output from the PHY on lane 1 is captured in this field. For single-lane configurations: Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	FOM0	R	Figure of merit output from the PHY on lane 0 is captured in this field. **Value After Reset:** 0x0 **Exists:** Always

7.1.246 VR_PMA_MP_25G_16G_MISC_STS2

- **Description:** VR PMA Synopsys Multi-protocol PHY Miscellaneous Status 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x809F : (DWCXLP_APB3_16) ? (0x809F*0x2) : (0x809F*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12)&&DWCXLP_MULTI_LANE

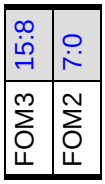


Table 7-251 Fields for Register: VR_PMA_MP_25G_16G_MISC_STS2

Bits	Name	Memory Access	Description
15:8	FOM3	R	Figure of merit output from the PHY on lane 3 is captured in this field. **Value After Reset:** 0x0 **Exists:** Always
7:0	FOM2	R	Figure of merit output from the PHY on lane 2 is captured in this field. **Value After Reset:** 0x0 **Exists:** Always

7.1.247 VR_PMA_MP_25G_16G_CR_CTRL

- **Description:** VR PMA Synopsys Multi-protocol PHY CR Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80a0 : (DWCXLP_APB3_16) ? (0x80a0*0x2) : (0x80a0*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12))

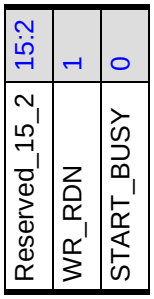


Table 7-252 Fields for Register: VR_PMA_MP_25G_16G_CR_CTRL

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	WR_RDN	R/W	Write or Read Indicator This bit indicates whether a read or write operation is to be performed to the Synopsys PHY registers: ■ 0: Read ■ 1: Write **Value After Reset:** 0x0 **Exists:** Always

Table 7-252 Fields for Register: VR_PMA_MP_25G_16G_CR_CTRL (Continued)

Bits	Name	Memory Access	Description
0	START_BUSY	R/W	Start CR Port Access or Busy Indicator (WS,SC Type) This bit indicates if CR port access is in progress: ■ 0: CR port not busy ■ 1: CR port busy The sequence is: 1. The host sets this bit to start a read or write transfer through the CR port to the Synopsys PHY registers. 2. This bit remains set during the CR port access. 3. The DWC_xlgpcs clears this bit when the CR port access is complete. Dependencies: The host must read this bit as 0 before writing to any of the following registers: ■ VR PMA Synopsys Multi-protocol PHY CR Control Register (this register) ■ VR PMA Synopsys Multi-protocol PHY CR Address Register ■ VR PMA Synopsys Multi-protocol PHY CR Data Register During read, the data from the CR port interface is placed into the VR PMA Synopsys Multi-protocol PHY CR Data Register. **Value After Reset:** 0x0 **Exists:** Always

7.1.248 VR_PMA_MP_25G_16G_CR_ADDR

- **Description:** VR PMA Synopsys Multi-protocol PHY CR Address Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80a1 : (DWCXLP_APB3_16) ? (0x80a1*0x2) : (0x80a1*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12))

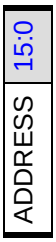


Table 7-253 Fields for Register: VR_PMA_MP_25G_16G_CR_ADDR

Bits	Name	Memory Access	Description
15:0	ADDRESS	R/W	CR Port Address This field indicates the address of the register to be accessed through the CR port. Dependency: ■ The host must not change this field when Bit 0 of the VR PMA Synopsys Multi-protocol PHY CR Control Register is set. ■ This field must be written before writing to any of the following registers: - VR PMA Synopsys Multi-protocol PHY CR Control Register - VR PMA Synopsys Multi-protocol PHY CR Address Register (this register) - VR PMA Synopsys Multi-protocol PHY CR Data Register **Value After Reset:** 0x0 **Exists:** Always

7.1.249 VR_PMA_MP_25G_16G_CR_DATA

- **Description:** VR PMA Synopsys Multi-protocol PHY CR Data Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80a2 : (DWCXLP_APB3_16) ? (0x80a2*0x2) : (0x80a2*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12))

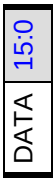


Table 7-254 Fields for Register: VR_PMA_MP_25G_16G_CR_DATA

Bits	Name	Memory Access	Description
15:0	DATA	R/W	CR Port Data This field contains the data for CR Port access. ■ During a write operation, the data in this register is written to the Synopsys PHY register pointed by the CR Port Address. ■ During a read operation, the data read from the Synopsys PHY is written to this register. The content becomes valid only when the DWC_xlgpcs clears Bit 0 of the VR PMA Synopsys Multi-protocol PHY CR Control Register. The host must not change this field when Bit 0 of VR PMA Synopsys Multi-protocol PHY CR Control Register is set. **Value After Reset:** 0x0 **Exists:** Always

7.1.250 VR_PMA_CWM00

- **Description:** VR PMA MMD RSFEC alignment marker bits[15:0]
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80a3 : (DWCXLP_APB3_16) ? (0x80a3*0x2) : (0x80a3*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_RSFEC_EN)

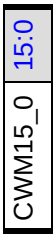


Table 7-255 Fields for Register: VR_PMA_CWM00

Bits	Name	Memory Access	Description
15:0	CWM15_0	R/W	This register provides the RSFEC Code Word Marker bits[15:0] to be in 25G mode of operation.**Note:** The purpose of registers VR_PMA_CWM00-03 is to allow flexibility in the definition of CWM[63:0]. So, program the CWM00-03 register to match either the AM pattern for PCS lane-0 defined in Table 82-2 or that in Table 82-3. Program the VR_PMA_CWM00 register with 0x68C1. Behavior of DWC_xlgpcs is undefined for any other value programmed **Value After Reset:** 0x7690 **Exists:** Always

7.1.251 VR_PMA_CWM01

- **Description:** VR PMA MMD RSFEC alignment marker bits[31:16]. See the note mention in the description of VR_PMA_CWM00 register.

  Program the VR_PMA_CWM01 register with 0x3321.

  Behavior of DWC_xlgpcs is undefined for any other value programmed.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80a4 : (DWCXLP_APB3_16) ? (0x80a4*0x2) : (0x80a4*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_RSFEC_EN)

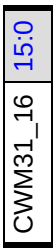


Table 7-256 Fields for Register: VR_PMA_CWM01

Bits	Name	Memory Access	Description
15:0	CWM31_16	R/W	This register provides the RSFEC Code Word Marker bits[31:16] to be in 25G mode of operation **Value After Reset:** 0x3347 **Exists:** Always

7.1.252 VR_PMA_CWM02

- **Description:** VR PMA MMD RSFEC alignment marker bits[47:32] See the note mentioned in the description of the VR_PMA_CWM00 register. Program the VR_PMA_CWM02 register with 0x973E. Behavior of DWC_xlgpcs is undefined for any other value programmed.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80a5 : (DWCXLP_APB3_16) ? (0x80a5*0x2) : (0x80a5*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_RSFEC_EN)

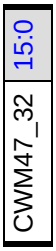


Table 7-257 Fields for Register: VR_PMA_CWM02

Bits	Name	Memory Access	Description
15:0	CWM47_32	R/W	This register provides the RSFEC Code Word Marker bits[47:32] to be in 25G mode of operation **Value After Reset:** 0x896f **Exists:** Always

7.1.253 VR_PMA_CWM03

- **Description:** VR PMA MMD RSFEC alignment marker bits[63:48] See the note mentioned in the description of VR_PMA_CWM00

  Program the VR_PMA_CWM03 register with 0xCCDE.

  Behavior of DWC_xlgpcs is undefined for any other value programmed.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80a6 : (DWCXLP_APB3_16) ? (0x80a6*0x2) : (0x80a6*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_RSFEC_EN)

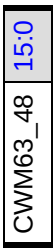


Table 7-258 Fields for Register: VR_PMA_CWM03

Bits	Name	Memory Access	Description
15:0	CWM63_48	R/W	This register provides the RSFEC Code Word Marker bits[63:48] to be in 25G mode of operation **Value After Reset:** 0xccb8 **Exists:** Always

7.1.254 VR_PMA_CTRL3

- **Description:** VR PMA MMD RS(544,514) auxiliary Control 3 Register

  This register provides additional parameter values for RS(544,514) PMA operation based on the AM interval.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80a8 : (DWCXLP_APB3_16) ? (0x80a8*0x2) : (0x80a8*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_RS544)

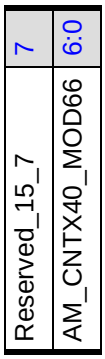


Table 7-259 Fields for Register: VR_PMA_CTRL3

Bits	Name	Memory Access	Description
7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-259 Fields for Register: VR_PMA_CTRL3 (Continued)

Bits	Name	Memory Access	Description
6:0	AM_CNTX40_MOD66	R/W	Depending on Operating mode, this field provides the value equal to: ■ 100G Mode (50Gx2 or 100Gx1): (PCS_AM_CNT x 40) MOD 66 ■ 50G Mode (50Gx1): (PCS_AM_CNT x 80) MOD 66 This value is needed by DWC_xlgpcs during AM boundary hunt when operating in RS (544,514) mode. In other modes, this field is not used. Example 1: In 50Gx2 mode, if VR_PCS_AM_CNT programmed is 10, then (10*40) mod 66 = 6'd4. Hence, value of 7'd4 has to be programmed to this field. Example 2: In 50Gx1 mode, if VR_PCS_AM_CNT programmed is 1024, then (1024*80) mod 66 = 7'd14. Hence, value of 7'd14 has to be programmed to this field. **Value After Reset:** "(DWCXLP_50G_25G||(DWCXLP_RS544&&!DWCXLP_KR4_100G&&!DWCXLP_100G_1LANE)) ? 0xE : 0x1C" **Exists:** Always

7.1.255 VR_PMA_CTRL4

- **Description:** VR PMA MMD RS(544,514) auxiliary Control 4 Register

  This register provides additional parameter values for RS(544,514) PMA operation based on the AM interval

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80a9 : (DWCXLP_APB3_16) ? (0x80a9*0x2) : (0x80a9*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_RS544)

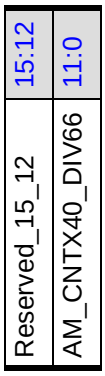


Table 7-260 Fields for Register: VR_PMA_CTRL4

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-260 Fields for Register: VR_PMA_CTRL4 (Continued)

Bits	Name	Memory Access	Description
11:0	AM_CNTX40_DIV66	R/W	Depending on Operating mode, this field provides the value equal to: ■ 100G Mode (50Gx2 or 100Gx1): - CEIL ((PCS_AM_CNT x 40) DIV 66) if ((PCS_AM_CNT x 40) MOD 66)!=0 - ((PCS_AM_CNT x 40) DIV 66)+1 if ((PCS_AM_CNT x 40) MOD 66)==0 ■ 50G Mode (50Gx1): - CEIL ((PCS_AM_CNT x 80) DIV 66) if ((PCS_AM_CNT x 80) MOD 66)!=0 - ((PCS_AM_CNT x 80) DIV 66)+1 if ((PCS_AM_CNT x 80) MOD 66)==0 This value is needed by DWC_xlgpcs during AM boundary hunt when operating in RS (544,514) mode. In other modes, this field is not used. Example 1: In 50Gx2 mode, if VR_PCS_AM_CNT programmed is 10, as (10*40) mod 66) !=0, then ceil ((10*40) div 66) = 12'd7. Hence, value of 12'd7 has to be programmed to this field. Example 2: In 50Gx1 mode, if VR_PCS_AM_CNT programmed is 1024, as (1024*80) mod 66) !=0, then ceil ((1024*80) div 66) = 12'd1242. Hence, value of 12'd1242 has to be programmed to this field. **Value After Reset:** "(DWCXLP_50G_25G||(DWCXLP_RS544&&!DWCXLP_KR4_100G&&!DWCXLP_100G_1LANE)) ? 0x4DA : 0x9B3" **Exists:** Always

7.1.256 VR_PMA_KRTR_TX_EQ_CFF_CTRL0

- **Description:** VR PMA MMD KRTR Transmit Equalization Coefficient Control Register.

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. This register can be used by management (software) to control the coefficient updates commands driven to the local PHY. This mode can be enabled by programming bit[15] of VR_PMA_KRTR_TX_EQ_STS_CTRL0 register to 1. During this mode, DWC_xlgpcs ignores the Tx coefficient update commands received from the link partner (reflected in SR_PMA_KR_LP_CEU Register). Instead, the values programmed by the management to the various fields of this register are considered as the Tx coefficient update commands that need to be driven to the local PHY.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B0 : (DWCXLP_APB3_16) ? (0x80B0*0x2) : (0x80B0*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

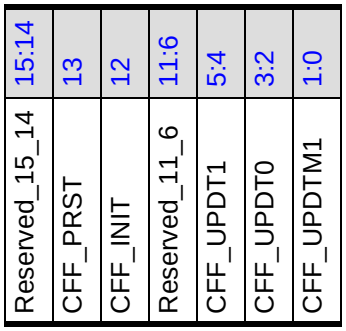


Table 7-261 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL0

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-261 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL0 (Continued)

Bits	Name	Memory Access	Description
13	CFF_PRST	R/W	Preset Coefficient Management can set this bit to 1 to request the local PHY to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Note:** If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always
12	CFF_INIT	R/W	Initialize Coefficient Management can set this bit to 1, to request the local PHY to initialize its transmitter filter coefficients **Note:** If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1	R/W	Co-efficient Update (+1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R/W	Co-efficient Update (0) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Table 7-261 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL0 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R/W	Coefficient Update (-1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.257 VR_PMA_KRTR_TX_EQ_CFF_CTRL1

- **Description:** VR PMA MMD KRTR Transmit Equalization Coefficient Control Register.

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. This register can be used by management (software) to control the coefficient updates commands driven to the local PHY. This mode can be enabled by programming bit[15] of VR_PMA_KRTR_TX_EQ_STS_CTRL1 register to 1. During this mode, DWC_xlgpcs ignores the Tx coefficient update commands received from the link partner (reflected in SR_PMA_KR_LP_CEU Register). Instead, the values programmed by the management to the various fields of this register are considered as the Tx coefficient update commands that need to be driven to the local PHY.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B1 : (DWCXLP_APB3_16) ? (0x80B1*0x2) : (0x80B1*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

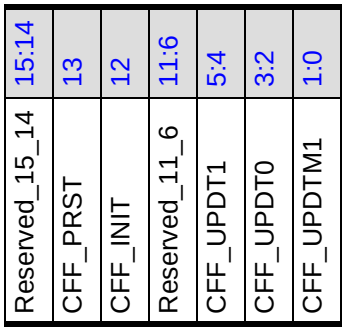


Table 7-262 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL1

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-262 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL1 (Continued)

Bits	Name	Memory Access	Description
13	CFF_PRST	R/W	Preset Coefficient Management can set this bit to 1 to request the local PHY to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Note:** If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always
12	CFF_INIT	R/W	Initialize Coefficient Management can set this bit to 1, to request the local PHY to initialize its transmitter filter coefficients **Note:** If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1	R/W	Co-efficient Update (+1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R/W	Co-efficient Update (0) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Table 7-262 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL1 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R/W	Coefficient Update (-1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.258 VR_PMA_KRTR_TX_EQ_CFF_CTRL2

- **Description:** VR PMA MMD KRTR Transmit Equalization Coefficient Control Register.

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. This register can be used by management (software) to control the coefficient updates commands driven to the local PHY. This mode can be enabled by programming bit[15] of VR_PMA_KRTR_TX_EQ_STS_CTRL2 register to 1. During this mode, DWC_xlgpcs ignores the Tx coefficient update commands received from the link partner (reflected in SR_PMA_KR_LP_CEU Register). Instead, the values programmed by the management to the various fields of this register are considered as the Tx coefficient update commands that need to be driven to the local PHY.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B2 : (DWCXLP_APB3_16) ? (0x80B2*0x2) : (0x80B2*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

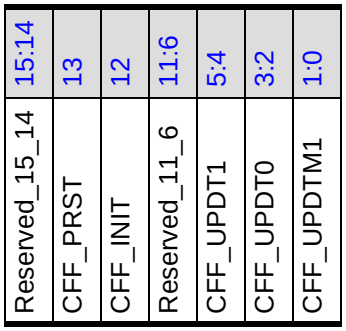


Table 7-263 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-263 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL2 (Continued)

Bits	Name	Memory Access	Description
13	CFF_PRST	R/W	Preset Coefficient Management can set this bit to 1 to request the local PHY to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Note:** If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always
12	CFF_INIT	R/W	Initialize Coefficient Management can set this bit to 1, to request the local PHY to initialize its transmitter filter coefficients **Note:** If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1	R/W	Co-efficient Update (+1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R/W	Co-efficient Update (0) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Table 7-263 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL2 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R/W	Coefficient Update (-1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.259 VR_PMA_KRTR_TX_EQ_CFF_CTRL3

- **Description:** VR PMA MMD KRTR Transmit Equalization Coefficient Control Register.

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. This register can be used by management (software) to control the coefficient updates commands driven to the local PHY. This mode can be enabled by programming bit[15] of VR_PMA_KRTR_TX_EQ_STS_CTRL3 register to 1. During this mode, DWC_xlgpcs ignores the Tx coefficient update commands received from the link partner (reflected in SR_PMA_KR_LP_CEU Register). Instead, the values programmed by the management to the various fields of this register are considered as the Tx coefficient update commands that need to be driven to the local PHY.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B3 : (DWCXLP_APB3_16) ? (0x80B3*0x2) : (0x80B3*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

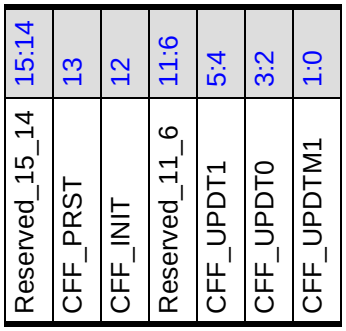


Table 7-264 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL3

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-264 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL3 (Continued)

Bits	Name	Memory Access	Description
13	CFF_PRST	R/W	Preset Coefficient Management can set this bit to 1 to request the local PHY to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Note:** If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always
12	CFF_INIT	R/W	Initialize Coefficient Management can set this bit to 1, to request the local PHY to initialize its transmitter filter coefficients **Note:** If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1	R/W	Co-efficient Update (+1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R/W	Co-efficient Update (0) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Table 7-264 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL3 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R/W	Coefficient Update (-1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

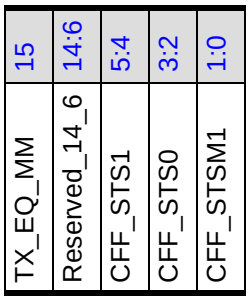
7.1.260 VR_PMA_KRTR_TX_EQ_STS_CTRL0

- **Description:** VR PMA MMD KRTR Transmit Equalization Status Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled.

  This register provides a mechanism to allow the software mode of Transmit Equalization. This mode can be enabled by programming bit [15] of this register to 1 (This bit should be programmed to the desired value prior to enabling KR training). During this mode the Tx coefficient update commands (driven towards local PHY) and status (sent to link-partner) are totally controlled by the software.

  The coefficient update commands received from the link partner are available in SR PMA MMD BASE-KR LP Coefficient Update Register. Irrespective of the command received from the link-partner, software can drive any desired Tx coefficient Update command towards the local PHY by writing suitable values to VR_PMA_KRTR_TX_EQ_CFF_CTRL0 register. In addition, the 'status' to be returned to the link-partner is controlled by the values programmed to VR_PMA_KRTR_TX_EQ_STS_CTRL0 register.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B4 : (DWCXLP_APB3_16) ? (0x80B4*0x2) : (0x80B4*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)



15	14:6	5:4	3:2	1:0
TX_EQ_MM	Reserved_14_6	CFF_STS1	CFF_STS0	CFF_STSM1

Table 7-265 Fields for Register: VR_PMA_KRTR_TX_EQ_STS_CTRL0

Bits	Name	Memory Access	Description
15	TX_EQ_MM	R/W	Management Mode of Transmit Equalization. This bit can be set to 1 to allow the software to decide the transmit equalization coefficient/status. During this mode, the coefficient update commands in the received training frames are not passed on to the local PHY. Instead the values programmed to VR_PMA_KRTR_TX_EQ_CFF_CTRL0 Register are driven towards the PHY. In addition, the status values (to be sent to the link partner) are also controlled by the software.DWC_xlgpcs ignores the coefficient status given by the local PHY (or computed internally by DWC_xlgpcs' training algorithm) during KR training operation. The values programmed by management to the CFF_STS1, CFF_STS0 and CFF_STSM1 fields of this register are transmitted to the link partner. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (+1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

Table 7-265 Fields for Register: VR_PMA_KRTR_TX_EQ_STS_CTRL0 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

7.1.261 VR_PMA_KRTR_TX_EQ_STS_CTRL1

- **Description:** VR PMA MMD KRTR Transmit Equalization Status Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled.

  This register provides a mechanism to allow the software mode of Transmit Equalization. This mode can be enabled by programming bit [15] of this register to 1 (This bit should be programmed to the desired value prior to enabling KR training). During this mode the Tx coefficient update commands (driven towards local PHY) and status (sent to link-partner) are totally controlled by the software.

  The coefficient update commands received from the link partner are available in SR PMA MMD BASE-KR LP Coefficient Update Register. Irrespective of the command received from the link-partner, software can drive any desired Tx coefficient Update command towards the local PHY by writing suitable values to VR_PMA_KRTR_TX_EQ_CFF_CTRL1 register. In addition, the 'status' to be returned to the link-partner is controlled by the values programmed to VR_PMA_KRTR_TX_EQ_STS_CTRL1 register.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B5 : (DWCXLP_APB3_16) ? (0x80B5*0x2) : (0x80B5*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

15	14:6	5:4	3:2	1:0
TX_EQ_MM	Reserved_14_6	CFF_STS1	CFF_STS0	CFF_STSM1

Table 7-266 Fields for Register: VR_PMA_KRTR_TX_EQ_STS_CTRL1

Bits	Name	Memory Access	Description
15	TX_EQ_MM	R/W	Management Mode of Transmit Equalization. This bit can be set to 1 to allow the software to decide the transmit equalization coefficient/status. During this mode, the coefficient update commands in the received training frames are not passed on to the local PHY. Instead the values programmed to VR_PMA_KRTR_TX_EQ_CFF_CTRL1 Register are driven towards the PHY. In addition, the status values (to be sent to the link partner) are also controlled by the software.DWC_xlgpcs ignores the coefficient status given by the local PHY (or computed internally by DWC_xlgpcs' training algorithm) during KR training operation. The values programmed by management to the CFF_STS1, CFF_STS0 and CFF_STSM1 fields of this register are transmitted to the link partner. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (+1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

Table 7-266 Fields for Register: VR_PMA_KRTR_TX_EQ_STS_CTRL1 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

7.1.262 VR_PMA_KRTR_TX_EQ_STS_CTRL2

- **Description:** VR PMA MMD KRTR Transmit Equalization Status Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled.

  This register provides a mechanism to allow the software mode of Transmit Equalization. This mode can be enabled by programming bit [15] of this register to 1 (This bit should be programmed to the desired value prior to enabling KR training). During this mode the Tx coefficient update commands (driven towards local PHY) and status (sent to link-partner) are totally controlled by the software.

  The coefficient update commands received from the link partner are available in SR PMA MMD BASE-KR LP Coefficient Update Register. Irrespective of the command received from the link-partner,software can drive any desired Tx coefficient Update command towards the local PHY by writing suitable values to VR_PMA_KRTR_TX_EQ_CFF_CTRL2 register. In addition, the 'status' to be returned to the link-partner is controlled by the values programmed to VR_PMA_KRTR_TX_EQ_STS_CTRL2 register.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B6 : (DWCXLP_APB3_16) ? (0x80B6*0x2) : (0x80B6*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

15	14:6	5:4	3:2	1:0
TX_EQ_MM	Reserved_14_6	CFF_STS1	CFF_STS0	CFF_STSM1

Table 7-267 Fields for Register: VR_PMA_KRTR_TX_EQ_STS_CTRL2

Bits	Name	Memory Access	Description
15	TX_EQ_MM	R/W	Management Mode of Transmit Equalization. This bit can be set to 1 to allow the software to decide the transmit equalization coefficient/status. During this mode, the coefficient update commands in the received training frames are not passed on to the local PHY. Instead the values programmed to VR_PMA_KRTR_TX_EQ_CFF_CTRL2 Register are driven towards the PHY. In addition, the status values (to be sent to the link partner) are also controlled by the software.DWC_xlgpcs ignores the coefficient status given by the local PHY (or computed internally by DWC_xlgpcs' training algorithm) during KR training operation. The values programmed by management to the CFF_STS1, CFF_STS0 and CFF_STSM1 fields of this register are transmitted to the link partner. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (+1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

Table 7-267 Fields for Register: VR_PMA_KRTR_TX_EQ_STS_CTRL2 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status to the Link Partner Transmitter. This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

7.1.263 VR_PMA_KRTR_TX_EQ_STS_CTRL3

- **Description:** VR PMA MMD KRTR Transmit Equalization Status Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled.

  This register provides a mechanism to allow the software mode of Transmit Equalization. This mode can be enabled by programming bit [15] of this register to 1 (This bit should be programmed to the desired value prior to enabling KR training). During this mode the Tx coefficient update commands (driven towards local PHY) and status (sent to link-partner) are totally controlled by the software.

  The coefficient update commands received from the link partner are available in SR PMA MMD BASE-KR LP Coefficient Update Register. Irrespective of the command received from the link-partner, software can drive any desired Tx coefficient Update command towards the local PHY by writing suitable values to VR_PMA_KRTR_TX_EQ_CFF_CTRL3 register. In addition, the 'status' to be returned to the link-partner is controlled by the values programmed to VR_PMA_KRTR_TX_EQ_STS_CTRL3 register.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B7 : (DWCXLP_APB3_16) ? (0x80B7*0x2) : (0x80B7*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

15	14:6	5:4	3:2	1:0
TX_EQ_MM	Reserved_14_6	CFF_STS1	CFF_STS0	CFF_STSM1

Table 7-268 Fields for Register: VR_PMA_KRTR_TX_EQ_STS_CTRL3

Bits	Name	Memory Access	Description
15	TX_EQ_MM	R/W	Management Mode of Transmit Equalization. This bit can be set to 1 to allow the software to decide the transmit equalization coefficient/status. During this mode, the coefficient update commands in the received training frames are not passed on to the local PHY. Instead the values programmed to VR_PMA_KRTR_TX_EQ_CFF_CTRL3 Register are driven towards the PHY. In addition, the status values (to be sent to the link partner) are also controlled by the software.DWC_xlgpcs ignores the coefficient status given by the local PHY (or computed internally by DWC_xlgpcs' training algorithm) during KR training operation. The values programmed by management to the CFF_STS1, CFF_STS0 and CFF_STSM1 fields of this register are transmitted to the link partner. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (+1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

Table 7-268 Fields for Register: VR_PMA_KRTR_TX_EQ_STS_CTRL3 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

7.1.264 VR_PMA_PHY_TX_EQ_STS0

- **Description:** Lane-0 VR PMA MMD PHY Transmit Equalization Status Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. This register indicates the Tx equalization status provided by the local PHY (on rpcs_ktx_ceu{filter}_sts{lane}_i ports). For Synopsys PHY configurations, this register indicates the Tx equalization status computed internally by the DWC_xlgpcs (based on the selected PHY algorithm).

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B8 : (DWCXLP_APB3_16) ? (0x80B8*0x2) : (0x80B8*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

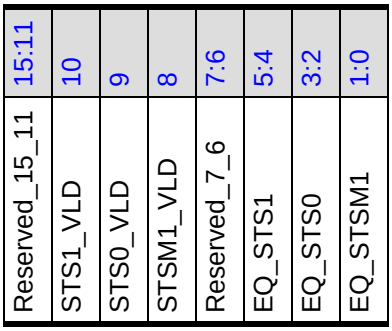


Table 7-269 Fields for Register: VR_PMA_PHY_TX_EQ_STS0

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	STS1_VLD	R	Filter Coefficient (1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS1 field is valid. **Value After Reset:** 0x0 **Exists:** Always
9	STS0_VLD	R	Filter Coefficient (0) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS0 field is valid. **Value After Reset:** 0x0 **Exists:** Always

Table 7-269 Fields for Register: VR_PMA_PHY_TX_EQ_STS0 (Continued)

Bits	Name	Memory Access	Description
8	STSM1_VLD	R	Filter Coefficient (-1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STSM1 field is valid. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	EQ_STS1	R	Filter Coefficient (1) Status given by local PHY. The status indicated in this field is valid when STS1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	EQ_STS0	R	Filter Coefficient (0) Status given by local PHY. The status indicated in this field is valid when STS0_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	EQ_STSM1	R	Filter Coefficient (-1) Status given by local PHY. The status indicated in this field is valid when STSM1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

7.1.265 VR_PMA_PHY_TX_EQ_STS1

- **Description:** Lane-1 VR PMA MMD PHY Transmit Equalization Status Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. This register indicates the Tx equalization status provided by the local PHY (on rpcs_ktx_ceu{filter}_sts{lane}_i ports). For Synopsys PHY configurations, this register indicates the Tx equalization status computed internally by the DWC_xlgpcs (based on the selected PHY algorithm).

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80B9 : (DWCXLP_APB3_16) ? (0x80B9*0x2) : (0x80B9*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

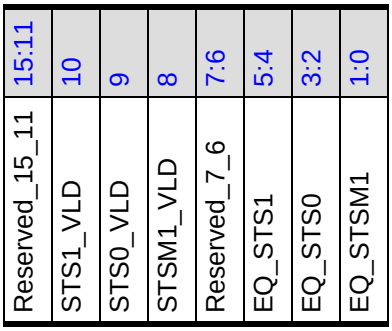


Table 7-270 Fields for Register: VR_PMA_PHY_TX_EQ_STS1

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	STS1_VLD	R	Filter Coefficient (1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS1 field is valid. **Value After Reset:** 0x0 **Exists:** Always
9	STS0_VLD	R	Filter Coefficient (0) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS0 field is valid. **Value After Reset:** 0x0 **Exists:** Always

Table 7-270 Fields for Register: VR_PMA_PHY_TX_EQ_STS1 (Continued)

Bits	Name	Memory Access	Description
8	STSM1_VLD	R	Filter Coefficient (-1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STSM1 field is valid. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	EQ_STS1	R	Filter Coefficient (1) Status given by local PHY. The status indicated in this field is valid when STS1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	EQ_STS0	R	Filter Coefficient (0) Status given by local PHY. The status indicated in this field is valid when STS0_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	EQ_STSM1	R	Filter Coefficient (-1) Status given by local PHY. The status indicated in this field is valid when STSM1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

7.1.266 VR_PMA_PHY_TX_EQ_STS2

- **Description:** Lane-2 VR PMA MMD PHY Transmit Equalization Status Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. This register indicates the Tx equalization status provided by the local PHY (on rpcs_ktx_ceu{filter}_sts{lane}_i ports). For Synopsys PHY configurations, this register indicates the Tx equalization status computed internally by the DWC_xlgpcs (based on the selected PHY algorithm).

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80BA : (DWCXLP_APB3_16) ? (0x80BA*0x2) : (0x80BA*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

15:11	10	9	8	7:6	5:4	3:2	1:0
Reserved_15_11	STS1_VLD	STS0_VLD	STSM1_VLD	Reserved_7_6	EQ_STS1	EQ_STS0	EQ_STSM1

Table 7-271 Fields for Register: VR_PMA_PHY_TX_EQ_STS2

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	STS1_VLD	R	Filter Coefficient (1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS1 field is valid. **Value After Reset:** 0x0 **Exists:** Always
9	STS0_VLD	R	Filter Coefficient (0) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS0 field is valid. **Value After Reset:** 0x0 **Exists:** Always

Table 7-271 Fields for Register: VR_PMA_PHY_TX_EQ_STS2 (Continued)

Bits	Name	Memory Access	Description
8	STSM1_VLD	R	Filter Coefficient (-1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STSM1 field is valid. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	EQ_STS1	R	Filter Coefficient (1) Status given by local PHY. The status indicated in this field is valid when STS1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	EQ_STS0	R	Filter Coefficient (0) Status given by local PHY. The status indicated in this field is valid when STS0_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	EQ_STSM1	R	Filter Coefficient (-1) Status given by local PHY. The status indicated in this field is valid when STSM1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

7.1.267 VR_PMA_PHY_TX_EQ_STS3

- **Description:** Lane-3 VR PMA MMD PHY Transmit Equalization Status Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled. This register indicates the Tx equalization status provided by the local PHY (on rpcs_ktx_ceu{filter}_sts{lane}_i ports). For Synopsys PHY configurations, this register indicates the Tx equalization status computed internally by the DWC_xlgpcs (based on the selected PHY algorithm).

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80BB : (DWCXLP_APB3_16) ? (0x80BB*0x2) : (0x80BB*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

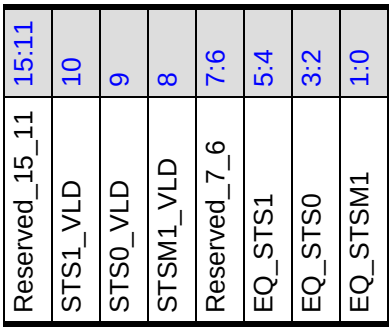


Table 7-272 Fields for Register: VR_PMA_PHY_TX_EQ_STS3

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	STS1_VLD	R	Filter Coefficient (1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS1 field is valid. **Value After Reset:** 0x0 **Exists:** Always
9	STS0_VLD	R	Filter Coefficient (0) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS0 field is valid. **Value After Reset:** 0x0 **Exists:** Always

Table 7-272 Fields for Register: VR_PMA_PHY_TX_EQ_STS3 (Continued)

Bits	Name	Memory Access	Description
8	STSM1_VLD	R	Filter Coefficient (-1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STSM1 field is valid. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	EQ_STS1	R	Filter Coefficient (1) Status given by local PHY. The status indicated in this field is valid when STS1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	EQ_STS0	R	Filter Coefficient (0) Status given by local PHY. The status indicated in this field is valid when STS0_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	EQ_STSM1	R	Filter Coefficient (-1) Status given by local PHY. The status indicated in this field is valid when STSM1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

7.1.268 VR_PMA_KRTR_RX_EQ_CTRL0

- **Description:** VR PMA MMD KRTR Receive Equalization Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled.

  This register can be used by management to send coefficient update requests to the link partner when KR Training is enabled. This mode can be enabled by programming bit [15] of this register to 1 (This bit should be programmed to the desired value prior to enabling KR training). During this mode, the coefficient updates (and rx_trained information) given by the PHY are ignored. For Synopsys PHY configurations, DWC_xlgpcs does not make the receiver requests to the PHY, that is, rpcs_krx_adapt_req_o/rpcs_krx_eq_eval_o' ports are not asserted. The values programmed to the RR_RDY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are sent by DWC_xlgpcs to the link partner.

  After programming a new coefficient update value (INC/DEC/PRST/INIT) to this register, application must wait till the response for the corresponding command is received (from link partner) in SR PMA MMD BASE-KR LP Coefficient Status Register. After receiving the 'UPDATED/MIN/MAX' status, application must send 'HOLD' command to the link partner by programming appropriate values to this register. Then, application should wait for 'NOT_UPDATED' status to be reflected in SR PMA MMD BASE-KR LP Coefficient Status Register.

  After a suitable number of training iterations, application can set the 'RR_RDY' field of this register to 1. At this point, DWC_xlgpcs infers that the local receiver is trained. Also, it passes this information to the link-partner by setting 'Receiver Ready' cell of the 'Status report' field of the KR training frame.

  Please note that DWC_xlgpcs ignores the 'rpcs_rx_trained{lane}_i' input port during this mode.

  Additionally, during this software mode of receive equalization, the 'status' information received from the link partner are not driven on 'rpcs_krx_ceu{filter}_sts{lane}_o' port of DWC_xlgpcs. Instead, the 'status' programmed to various fields of VR_PMA_KRTR_RX_EQ_STS_CTRL0 Registers are driven towards the PHY.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80BC : (DWCXLP_APB3_16) ? (0x80BC*0x2) : (0x80BC*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

15	14:9	8	7	6	5:4	3:2	1:0
RX_EQ_MM	Reserved_14_9	RR_RDY	PRST	INIT	CFF_UPDT1	CFF_UPDT0	CFF_UPDTM1

Table 7-273 Fields for Register: VR_PMA_KRTR_RX_EQ_CTRL0

Bits	Name	Memory Access	Description
15	RX_EQ_MM	R/W	Enable Rx Equalization by Management This bit can be set to 1, to bypass/ignore Rx coefficient updates given by PHY. During this mode, the values programmed by management to the RR_DY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are transmitted to the link partner. In addition, the 'status' values programmed to VR_PMA_KRTR_RX_EQ_STS_CTRL0 Register are driven on the 'rpcs_krx_ceu{filter}_sts{lane}_o' port of DWC_xlgpcs (towards the local PHY). **Value After Reset:** 0x0 **Exists:** Always
14:9	Reserved_14_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RR_RDY	R/W	Local Receiver Ready This bit can be set to 1, to indicate to the link partner that the local receiver is trained and is ready to accept data. **Value After Reset:** 0x0 **Exists:** Always
7	PRST	R/W	Pre-initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Value After Reset:** 0x0 **Exists:** Always
6	INIT	R/W	Initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to initialize its transmitter filter coefficients. **Value After Reset:** 0x0 **Exists:** Always

Table 7-273 Fields for Register: VR_PMA_KRTR_RX_EQ_CTRL0 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_UPDT1	R/W	Filter Coefficient Update (+1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R/W	Filter Coefficient Update (0) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_UPDTM1	R/W	Filter Coefficient Update (-1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.269 VR_PMA_KRTR_RX_EQ_CTRL1

- **Description:** VR PMA MMD KRTR Receive Equalization Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled.

  This register can be used by management to send coefficient update requests to the link partner when KR Training is enabled. This mode can be enabled by programming bit [15] of this register to 1 (This bit should be programmed to the desired value prior to enabling KR training). During this mode, the coefficient updates (and rx_trained information) given by the PHY are ignored. For Synopsys PHY configurations, DWC_xlgpcs does not make the receiver requests to the PHY, i.e, rpcs_krx_adapt_req_o/rpcs_krx_eq_eval_o' ports are not asserted. The values programmed to the RR_RDY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are sent by DWC_xlgpcs to the link partner.

  After programming a new coefficient update value (INC/DEC/PRST/INIT) to this register, application must wait till the response for the corresponding command is received (from link partner) in SR PMA MMD BASE-KR LP Coefficient Status Register. After receiving the 'UPDATED/MIN/MAX' status, application must send 'HOLD' command to the link partner by programming appropriate values to this register. Then, application should wait for 'NOT_UPDATED' status to be reflected in SR PMA MMD BASE-KR LP Coefficient Status Register.

  After a suitable number of training iterations, application can set the 'RR_RDY' field of this register to 1. At this point, DWC_xlgpcs infers that the local receiver is trained. Also, it passes this information to the link-partner by setting 'Receiver Ready' cell of the 'Status report' field of the KR training frame.

  Please note that DWC_xlgpcs ignores the 'rpcs_rx_trained{lane}_i' input port during this mode.

  Additionally, during this software mode of receive equalization, the 'status' information received from the link partner are not driven on 'rpcs_krx_ceu{filter}_sts{lane}_o' port of DWC_xlgpcs. Instead, the 'status' programmed to various fields of VR_PMA_KRTR_RX_EQ_STS_CTRL1 Registers are driven towards the PHY.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80BD : (DWCXLP_APB3_16) ? (0x80BD*0x2) : (0x80BD*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

15	14:9	8	7	6	5:4	3:2	1:0
RX_EQ_MM	Reserved_14_9	RR_RDY	PRST	INIT	CFF_UPDT1	CFF_UPDT0	CFF_UPDTM1

Table 7-274 Fields for Register: VR_PMA_KRTR_RX_EQ_CTRL1

Bits	Name	Memory Access	Description
15	RX_EQ_MM	R/W	Enable Rx Equalization by Management This bit can be set to 1, to bypass/ignore Rx coefficient updates given by PHY. During this mode, the values programmed by management to the RR_DY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are transmitted to the link partner. In addition, the 'status' values programmed to VR_PMA_KRTR_RX_EQ_STS_CTRL1 Register are driven on the 'rpcs_krx_ceu{filter}_sts{lane}_o' port of DWC_xlgpcs (towards the local PHY). **Value After Reset:** 0x0 **Exists:** Always
14:9	Reserved_14_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RR_RDY	R/W	Local Receiver Ready This bit can be set to 1, to indicate to the link partner that the local receiver is trained and is ready to accept data. **Value After Reset:** 0x0 **Exists:** Always
7	PRST	R/W	Pre-initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Value After Reset:** 0x0 **Exists:** Always
6	INIT	R/W	Initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to initialize its transmitter filter coefficients. **Value After Reset:** 0x0 **Exists:** Always

Table 7-274 Fields for Register: VR_PMA_KRTR_RX_EQ_CTRL1 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_UPDT1	R/W	Filter Coefficient Update (+1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R/W	Filter Coefficient Update (0) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_UPDTM1	R/W	Filter Coefficient Update (-1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.270 VR_PMA_KRTR_RX_EQ_CTRL2

- **Description:** VR PMA MMD KRTR Receive Equalization Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled.

  This register can be used by management to send coefficient update requests to the link partner when KR Training is enabled. This mode can be enabled by programming bit [15] of this register to 1 (This bit should be programmed to the desired value prior to enabling KR training). During this mode, the coefficient updates (and rx_trained information) given by the PHY are ignored. For Synopsys PHY configurations, DWC_xlgpcs does not make the receiver requests to the PHY, i.e, rpcs_krx_adapt_req_o/rpcs_krx_eq_eval_o' ports are not asserted. The values programmed to the RR_RDY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are sent by DWC_xlgpcs to the link partner.

  After programming a new coefficient update value (INC/DEC/PRST/INIT) to this register, application must wait till the response for the corresponding command is received (from link partner) in SR PMA MMD BASE-KR LP Coefficient Status Register. After receiving the 'UPDATED/MIN/MAX' status, application must send 'HOLD' command to the link partner by programming appropriate values to this register. Then, application should wait for 'NOT_UPDATED' status to be reflected in SR PMA MMD BASE-KR LP Coefficient Status Register.

  After a suitable number of training iterations, application can set the 'RR_RDY' field of this register to 1. At this point, DWC_xlgpcs infers that the local receiver is trained. Also, it passes this information to the link-partner by setting 'Receiver Ready' cell of the 'Status report' field of the KR training frame.

  Please note that DWC_xlgpcs ignores the 'rpcs_rx_trained{lane}_i' input port during this mode.

  Additionally, during this software mode of receive equalization, the 'status' information received from the link partner are not driven on 'rpcs_krx_ceu{filter}_sts{lane}_o' port of DWC_xlgpcs. Instead, the 'status' programmed to various fields of VR_PMA_KRTR_RX_EQ_STS_CTRL2 Registers are driven towards the PHY.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80BE : (DWCXLP_APB3_16) ? (0x80BE*0x2) : (0x80BE*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

15	14:9	8	7	6	5:4	3:2	1:0
RX_EQ_MM	Reserved_14_9	RR_RDY	PRST	INIT	CFF_UPDT1	CFF_UPDT0	CFF_UPDTM1

Table 7-275 Fields for Register: VR_PMA_KRTR_RX_EQ_CTRL2

Bits	Name	Memory Access	Description
15	RX_EQ_MM	R/W	Enable Rx Equalization by Management This bit can be set to 1, to bypass/ignore Rx coefficient updates given by PHY. During this mode, the values programmed by management to the RR_DY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are transmitted to the link partner. In addition, the 'status' values programmed to VR_PMA_KRTR_RX_EQ_STS_CTRL2 Register are driven on the 'rpcs_krx_ceu{filter}_sts{lane}_o' port of DWC_xlgpcs (towards the local PHY). **Value After Reset:** 0x0 **Exists:** Always
14:9	Reserved_14_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RR_RDY	R/W	Local Receiver Ready This bit can be set to 1, to indicate to the link partner that the local receiver is trained and is ready to accept data. **Value After Reset:** 0x0 **Exists:** Always
7	PRST	R/W	Pre-initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Value After Reset:** 0x0 **Exists:** Always
6	INIT	R/W	Initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to initialize its transmitter filter coefficients. **Value After Reset:** 0x0 **Exists:** Always

Table 7-275 Fields for Register: VR_PMA_KRTR_RX_EQ_CTRL2 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_UPDT1	R/W	Filter Coefficient Update (+1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R/W	Filter Coefficient Update (0) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_UPDTM1	R/W	Filter Coefficient Update (-1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.271 VR_PMA_KRTR_RX_EQ_CTRL3

- **Description:** VR PMA MMD KRTR Receive Equalization Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled.

  This register can be used by management to send coefficient update requests to the link partner when KR Training is enabled. This mode can be enabled by programming bit [15] of this register to 1 (This bit should be programmed to the desired value prior to enabling KR training). During this mode, the coefficient updates (and rx_trained information) given by the PHY are ignored. For Synopsys PHY configurations, DWC_xlgpcs does not make the receiver requests to the PHY, i.e, rpcs_krx_adapt_req_o/rpcs_krx_eq_eval_o' ports are not asserted. The values programmed to the RR_RDY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are sent by DWC_xlgpcs to the link partner.

  After programming a new coefficient update value (INC/DEC/PRST/INIT) to this register, application must wait till the response for the corresponding command is received (from link partner) in SR PMA MMD BASE-KR LP Coefficient Status Register. After receiving the 'UPDATED/MIN/MAX' status, application must send 'HOLD' command to the link partner by programming appropriate values to this register. Then, application should wait for 'NOT_UPDATED' status to be reflected in SR PMA MMD BASE-KR LP Coefficient Status Register.

  After a suitable number of training iterations, application can set the 'RR_RDY' field of this register to 1. At this point, DWC_xlgpcs infers that the local receiver is trained. Also, it passes this information to the link-partner by setting 'Receiver Ready' cell of the 'Status report' field of the KR training frame.

  Please note that DWC_xlgpcs ignores the 'rpcs_rx_trained{lane}_i' input port during this mode.

  Additionally, during this software mode of receive equalization, the 'status' information received from the link partner are not driven on 'rpcs_krx_ceu{filter}_sts{lane}_o' port of DWC_xlgpcs. Instead, the 'status' programmed to various fields of VR_PMA_KRTR_RX_EQ_STS_CTRL3 Registers are driven towards the PHY.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80BF : (DWCXLP_APB3_16) ? (0x80BF*0x2) : (0x80BF*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

15	14:9	8	7	6	5:4	3:2	1:0
RX_EQ_MM	Reserved_14_9	RR_RDY	PRST	INIT	CFF_UPDT1	CFF_UPDT0	CFF_UPDTM1

Table 7-276 Fields for Register: VR_PMA_KRTR_RX_EQ_CTRL3

Bits	Name	Memory Access	Description
15	RX_EQ_MM	R/W	Enable Rx Equalization by Management This bit can be set to 1, to bypass/ignore Rx coefficient updates given by PHY. During this mode, the values programmed by management to the RR_DY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are transmitted to the link partner. In addition, the 'status' values programmed to VR_PMA_KRTR_RX_EQ_STS_CTRL3 Register are driven on the 'rpcs_krx_ceu{filter}_sts{lane}_o' port of DWC_xlgpcs (towards the local PHY). **Value After Reset:** 0x0 **Exists:** Always
14:9	Reserved_14_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RR_RDY	R/W	Local Receiver Ready This bit can be set to 1, to indicate to the link partner that the local receiver is trained and is ready to accept data. **Value After Reset:** 0x0 **Exists:** Always
7	PRST	R/W	Pre-initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Value After Reset:** 0x0 **Exists:** Always
6	INIT	R/W	Initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to initialize its transmitter filter coefficients. **Value After Reset:** 0x0 **Exists:** Always

Table 7-276 Fields for Register: VR_PMA_KRTR_RX_EQ_CTRL3 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_UPDT1	R/W	Filter Coefficient Update (+1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R/W	Filter Coefficient Update (0) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_UPDTM1	R/W	Filter Coefficient Update (-1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

7.1.272 VR_PMA_KRTR_RX_EQ_STS_CTRL0

- **Description:** VR PMA MMD KRTR Receive Equalization Status Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled and if SNPS_XAUI_PHY=Disabled.

  This register allows the software to control the Rx equalization status driven towards the local PHY when RX_EQ_MM bit (of VR_PMA_KRTR_RX_EQ_CTRL0 Register) is set to 1.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80C0 : (DWCXLP_APB3_16) ? (0x80C0*0x2) : (0x80C0*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&!DWCXLP_SNPS_XAUI_PHY)

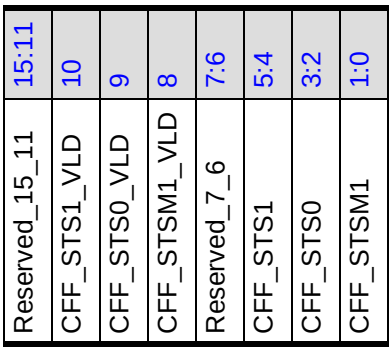


Table 7-277 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL0

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	CFF_STS1_VLD	R/W	Filter Coefficient (1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu2_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS1 field. It is recommended that software does not assert this bit, if CFF_STS1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always

Table 7-277 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL0 (Continued)

Bits	Name	Memory Access	Description
9	CFF_STS0_VLD	R/W	Filter Coefficient (0) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS0 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu1_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS0 field. It is recommended that software does not assert this bit, if CFF_STS0 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
8	CFF_STSM1_VLD	R/W	Filter Coefficient (-1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STSM1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu0_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STSM1 field. It is recommended that software does not assert this bit, if CFF_STSM1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu2_sts_o' port. The value programmed to this field is considered valid when bit[10] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

Table 7-277 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL0 (Continued)

Bits	Name	Memory Access	Description
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu1_sts_o' port. The value programmed to this field is considered valid when bit[9] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated The value programmed to this field is reflected on 'rpcs_krx_ceu0_sts_o' port. The value programmed to this field is considered valid when bit[8] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.273 VR_PMA_KRTR_RX_EQ_STS_CTRL1

- **Description:** VR PMA MMD KRTR Receive Equalization Status Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled and if SNPS_XAUI_PHY=Disabled.

  This register allows the software to control the Rx equalization status driven towards the local PHY when RX_EQ_MM bit (of VR_PMA_KRTR_RX_EQ_CTRL1 Register) is set to 1.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80C1 : (DWCXLP_APB3_16) ? (0x80C1*0x2) : (0x80C1*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&!DWCXLP_SNPS_XAUI_PHY)

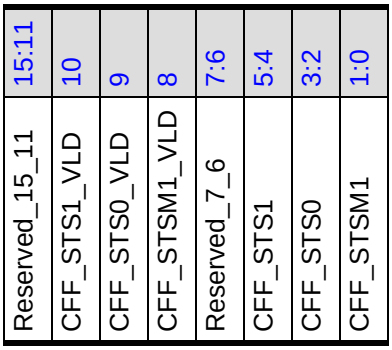


Table 7-278 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL1

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-278 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL1 (Continued)

Bits	Name	Memory Access	Description
10	CFF_STS1_VLD	R/W	Filter Coefficient (1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu2_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS1 field. It is recommended that software does not assert this bit, if CFF_STS1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
9	CFF_STS0_VLD	R/W	Filter Coefficient (0) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS0 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu1_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS0 field. It is recommended that software does not assert this bit, if CFF_STS0 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
8	CFF_STSM1_VLD	R/W	Filter Coefficient (-1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STSM1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu0_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STSM1 field. It is recommended that software does not assert this bit, if CFF_STSM1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-278 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL1 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu2_sts_o' port. The value programmed to this field is considered valid when bit[10] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu1_sts_o' port. The value programmed to this field is considered valid when bit[9] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

Table 7-278 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL1 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated The value programmed to this field is reflected on 'rpcs_krx_ceu0_sts_o' port. The value programmed to this field is considered valid when bit[8] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.274 VR_PMA_KRTR_RX_EQ_STS_CTRL2

- **Description:** VR PMA MMD KRTR Receive Equalization Status Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled and if SNPS_XAUI_PHY=Disabled.

  This register allows the software to control the Rx equalization status driven towards the local PHY when RX_EQ_MM bit (of VR_PMA_KRTR_RX_EQ_CTRL2 Register) is set to 1.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80C2 : (DWCXLP_APB3_16) ? (0x80C2*0x2) : (0x80C2*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&!DWCXLP_SNPS_XAUI_PHY)

15:11	10	9	8	7:6	5:4	3:2	1:0
Reserved_15_11	CFF_STS1_VLD	CFF_STS0_VLD	CFF_STSM1_VLD	Reserved_7_6	CFF_STS1	CFF_STS0	CFF_STSM1

Table 7-279 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL2

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-279 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL2 (Continued)

Bits	Name	Memory Access	Description
10	CFF_STS1_VLD	R/W	Filter Coefficient (1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu2_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS1 field. It is recommended that software does not assert this bit, if CFF_STS1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
9	CFF_STS0_VLD	R/W	Filter Coefficient (0) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS0 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu1_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS0 field. It is recommended that software does not assert this bit, if CFF_STS0 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
8	CFF_STSM1_VLD	R/W	Filter Coefficient (-1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STSM1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu0_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STSM1 field. It is recommended that software does not assert this bit, if CFF_STSM1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-279 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL2 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu2_sts_o' port. The value programmed to this field is considered valid when bit[10] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu1_sts_o' port. The value programmed to this field is considered valid when bit[9] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

Table 7-279 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL2 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated The value programmed to this field is reflected on 'rpcs_krx_ceu0_sts_o' port. The value programmed to this field is considered valid when bit[8] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.275 VR_PMA_KRTR_RX_EQ_STS_CTRL3

- **Description:** VR PMA MMD KRTR Receive Equalization Status Control Register

  This register is present only if DWCXLP_LNK_TRAIN_EN=Enabled and if SNPS_XAUI_PHY=Disabled.

  This register allows the software to control the Rx equalization status driven towards the local PHY when RX_EQ_MM bit (of VR_PMA_KRTR_RX_EQ_CTRL3 Register) is set to 1.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80C3 : (DWCXLP_APB3_16) ? (0x80C3*0x2) : (0x80C3*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN&&!DWCXLP_SNPS_XAUI_PHY)

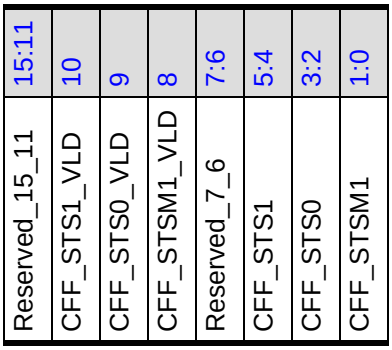


Table 7-280 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL3

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-280 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL3 (Continued)

Bits	Name	Memory Access	Description
10	CFF_STS1_VLD	R/W	Filter Coefficient (1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu2_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS1 field. It is recommended that software does not assert this bit, if CFF_STS1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
9	CFF_STS0_VLD	R/W	Filter Coefficient (0) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS0 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu1_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS0 field. It is recommended that software does not assert this bit, if CFF_STS0 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
8	CFF_STSM1_VLD	R/W	Filter Coefficient (-1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STSM1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu0_ack{lane}_o' port. Software should set this bit as 1, after programming either 01 or 10 or 11 to CFF_STSM1 field. It is recommended that software does not assert this bit, if CFF_STSM1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-280 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL3 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu2_sts_o' port. The value programmed to this field is considered valid when bit[10] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu1_sts_o' port. The value programmed to this field is considered valid when bit[9] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

Table 7-280 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL3 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated The value programmed to this field is reflected on 'rpcs_krx_ceu0_sts_o' port. The value programmed to this field is considered valid when bit[8] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

7.1.276 VR_PMA_PHY_RX_EQ_CEU0

- **Description:** VR PMA MMD PHY Receive Equalization Coefficient Update

  This register is present when the Clause 72 KR Training support is enabled. This register indicates the command update commands given by the local PHY.

  During normal (hardware) mode of Rx equalization, these coefficient update commands are transmitted to the link partner.Hence,the values appearing this register would reflect in SR_PMA_KR_LD_CEU Register too.

  If management mode of Rx equalization is enabled, the coefficient update commands given by local PHY are not transmitted to the link-partner; instead the values programmed to VR_PMA_KRTR_RX_EQ_CTRL0 Register is transmitted o the link partner. This register provides a means for the software to know the coefficient update commands given by the local PHY.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80C4 : (DWCXLP_APB3_16) ? (0x80C4*0x2) : (0x80C4*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

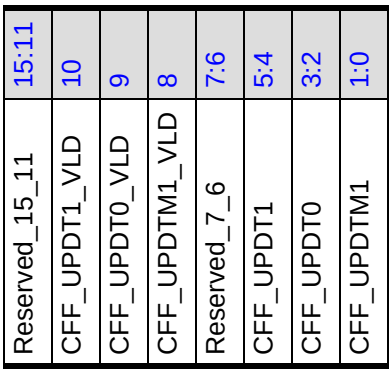


Table 7-281 Fields for Register: VR_PMA_PHY_RX_EQ_CEU0

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-281 Fields for Register: VR_PMA_PHY_RX_EQ_CEU0 (Continued)

Bits	Name	Memory Access	Description
10	CFF_UPDT1_VLD	R	Filter Coefficient Update (1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT1' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_vld_i' port. ■ For Synopsys 25G/16G PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
9	CFF_UPDT0_VLD	R	Filter Coefficient Update (0) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT0' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
8	CFF_UPDTM1_VLD	R	Filter Coefficient Update (-1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDTM1' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-281 Fields for Register: VR_PMA_PHY_RX_EQ_CEU0 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_UPDT1	R	Filter Coefficient Update (+1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_di2_i' port. The validity of this field is indicated by bit[10] of this register. **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R	Filter Coefficient Update (0) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_dir_i' port. The validity of this field is indicated by bit[9] of this register. **Value After Reset:** 0x0 **Exists:** Always

Table 7-281 Fields for Register: VR_PMA_PHY_RX_EQ_CEU0 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R	Filter Coefficient Update (-1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_pre_dir_i' port. The validity of this field is indicated by bit[8] of this register. **Value After Reset:** 0x0 **Exists:** Always

7.1.277 VR_PMA_PHY_RX_EQ_CEU1

- **Description:** VR PMA MMD PHY Receive Equalization Coefficient Update

  This register is present when the Clause 72 KR Training support is enabled. This register indicates the command update commands given by the local PHY.

  During normal (hardware) mode of Rx equalization, these coefficient update commands are transmitted to the link partner.Hence,the values appearing this register would reflect in SR_PMA_KR_LD_CEU Register too.

  If management mode of Rx equalization is enabled, the coefficient update commands given by local PHY are not transmitted to the link-partner; instead the values programmed to VR_PMA_KRTR_RX_EQ_CTRL1 Register is transmitted o the link partner. This register provides a means for the software to know the coefficient update commands given by the local PHY.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80C5 : (DWCXLP_APB3_16) ? (0x80C5*0x2) : (0x80C5*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

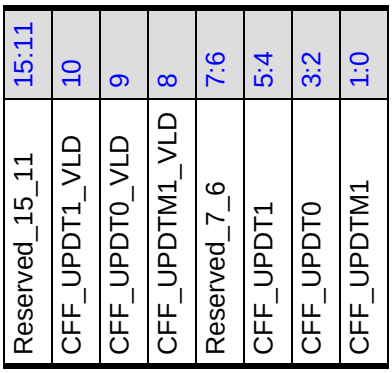


Table 7-282 Fields for Register: VR_PMA_PHY_RX_EQ_CEU1

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-282 Fields for Register: VR_PMA_PHY_RX_EQ_CEU1 (Continued)

Bits	Name	Memory Access	Description
10	CFF_UPDT1_VLD	R	Filter Coefficient Update (1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT1' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
9	CFF_UPDT0_VLD	R	Filter Coefficient Update (0) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT0' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
8	CFF_UPDTM1_VLD	R	Filter Coefficient Update (-1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDTM1' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-282 Fields for Register: VR_PMA_PHY_RX_EQ_CEU1 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_UPDT1	R	Filter Coefficient Update (+1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_di2_i' port. The validity of this field is indicated by bit[10] of this register. **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R	Filter Coefficient Update (0) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_dir_i' port. The validity of this field is indicated by bit[9] of this register. **Value After Reset:** 0x0 **Exists:** Always

Table 7-282 Fields for Register: VR_PMA_PHY_RX_EQ_CEU1 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R	Filter Coefficient Update (-1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_pre_dir_i' port. The validity of this field is indicated by bit[8] of this register. **Value After Reset:** 0x0 **Exists:** Always

7.1.278 VR_PMA_PHY_RX_EQ_CEU2

- **Description:** VR PMA MMD PHY Receive Equalization Coefficient Update

  This register is present when the Clause 72 KR Training support is enabled. This register indicates the command update commands given by the local PHY.

  During normal (hardware) mode of Rx equalization, these coefficient update commands are transmitted to the link partner.Hence,the values appearing this register would reflect in SR_PMA_KR_LD_CEU Register too.

  If management mode of Rx equalization is enabled, the coefficient update commands given by local PHY are not transmitted to the link-partner; instead the values programmed to VR_PMA_KRTR_RX_EQ_CTRL2 Register is transmitted o the link partner. This register provides a means for the software to know the coefficient update commands given by the local PHY.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80C6 : (DWCXLP_APB3_16) ? (0x80C6*0x2) : (0x80C6*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

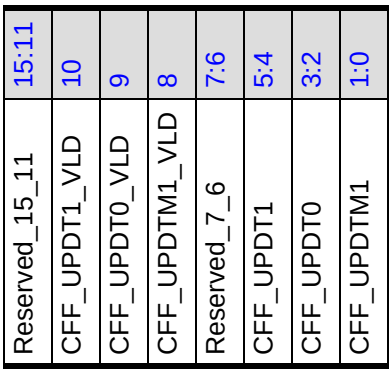


Table 7-283 Fields for Register: VR_PMA_PHY_RX_EQ_CEU2

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-283 Fields for Register: VR_PMA_PHY_RX_EQ_CEU2 (Continued)

Bits	Name	Memory Access	Description
10	CFF_UPDT1_VLD	R	Filter Coefficient Update (1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT1' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
9	CFF_UPDT0_VLD	R	Filter Coefficient Update (0) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT0' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
8	CFF_UPDTM1_VLD	R	Filter Coefficient Update (-1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDTM1' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-283 Fields for Register: VR_PMA_PHY_RX_EQ_CEU2 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_UPDT1	R	Filter Coefficient Update (+1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_di2_i' port. The validity of this field is indicated by bit[10] of this register. **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R	Filter Coefficient Update (0) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_dir_i' port. The validity of this field is indicated by bit[9] of this register. **Value After Reset:** 0x0 **Exists:** Always

Table 7-283 Fields for Register: VR_PMA_PHY_RX_EQ_CEU2 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R	Filter Coefficient Update (-1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_pre_dir_i' port. The validity of this field is indicated by bit[8] of this register. **Value After Reset:** 0x0 **Exists:** Always

7.1.279 VR_PMA_PHY_RX_EQ_CEU3

- **Description:** VR PMA MMD PHY Receive Equalization Coefficient Update

  This register is present when the Clause 72 KR Training support is enabled. This register indicates the command update commands given by the local PHY.

  During normal (hardware) mode of Rx equalization, these coefficient update commands are transmitted to the link partner. Therefore, the values appearing this register would reflect in SR_PMA_KR_LD_CEU Register too.

  If management mode of Rx equalization is enabled, the coefficient update commands given by local PHY are not transmitted to the link-partner; instead the values programmed to VR_PMA_KRTR_RX_EQ_CTRL3 Register is transmitted o the link partner. This register provides a means for the software to know the coefficient update commands given by the local PHY.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80C7 : (DWCXLP_APB3_16) ? (0x80C7*0x2) : (0x80C7*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_MULTI_LANE&&DWCXLP_PMA_MMD&&DWCXLP_LNK_TRAIN_EN)

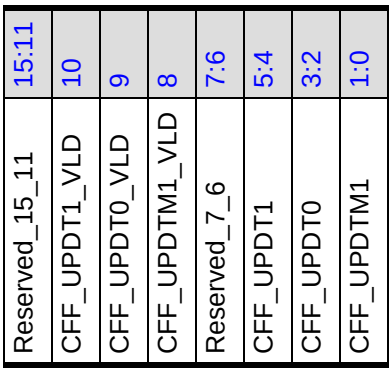


Table 7-284 Fields for Register: VR_PMA_PHY_RX_EQ_CEU3

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-284 Fields for Register: VR_PMA_PHY_RX_EQ_CEU3 (Continued)

Bits	Name	Memory Access	Description
10	CFF_UPDT1_VLD	R	Filter Coefficient Update (1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT1' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
9	CFF_UPDT0_VLD	R	Filter Coefficient Update (0) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT0' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
8	CFF_UPDTM1_VLD	R	Filter Coefficient Update (-1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDTM1' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_vld_i' port. ■ For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-284 Fields for Register: VR_PMA_PHY_RX_EQ_CEU3 (Continued)

Bits	Name	Memory Access	Description
5:4	CFF_UPDT1	R	Filter Coefficient Update (+1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_di2_i' port. The validity of this field is indicated by bit[10] of this register. **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R	Filter Coefficient Update (0) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_dir_i' port. The validity of this field is indicated by bit[9] of this register. **Value After Reset:** 0x0 **Exists:** Always

Table 7-284 Fields for Register: VR_PMA_PHY_RX_EQ_CEU3 (Continued)

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R	Filter Coefficient Update (-1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_pre_dir_i' port. The validity of this field is indicated by bit[8] of this register. **Value After Reset:** 0x0 **Exists:** Always

7.1.280 VR_PMA_MP_32G_RECAL

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Tx Re-Calibration Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80D0 : (DWCXLP_APB3_16) ? (0x80D0*0x2) : (0x80D0*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

15:14	13:12	11:10	9:8	7:5	4	3:1	0
RC_BS_3	RC_BS_2	RC_BS_1	RC_BS_0	RC_SK_EN_3_1	RC_SK_EN_0	RC_FR_EN_3_1	RC_FR_EN_0

Table 7-285 Fields for Register: VR_PMA_MP_32G_RECAL

Bits	Name	Memory Access	Description
15:14	RC_BS_3	* Varies	TX Re-Calibration Bank Select Lane 3. Selects the bank to store the RX duty cycle correction (DCC) values. This field drives the output port 'xlgpcs_tx3_recal_bnk_sel_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
13:12	RC_BS_2	* Varies	TX Re-Calibration Bank Select Lane 2. Selects the bank to store the RX duty cycle correction (DCC) values. This field drives the output port 'xlgpcs_tx2_recal_bnk_sel_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-285 Fields for Register: VR_PMA_MP_32G_RECAL (Continued)

Bits	Name	Memory Access	Description
11:10	RC_BS_1	* Varies	TX Re-Calibration Bank Select Lane 1. Selects the bank to store the RX duty cycle correction (DCC) values. This field drives the output port 'xlgpcs_tx1_recal_bnk_sel_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
9:8	RC_BS_0	R/W	TX Re-Calibration Bank Select Lane 0. Selects the bank to store the RX duty cycle correction (DCC) values. This field drives the output port 'xlgpcs_tx0_recal_bnk_sel_o'. **Value After Reset:** 0x0 **Exists:** Always
7:5	RC_SK_EN_3_1	* Varies	TX Skip Re-Calibration Lane3_1. Skips re-calibration of TX DCC in the selected bank regardless of the state of the corresponding done signal. This field drives the output port 'xlgpcs_tx_recal_skip_en_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
4	RC_SK_EN_0	R/W	TX Skip Re-Calibration Lane0. Skips re-calibration of TX DCC in the selected bank regardless of the state of the corresponding done signal. This field drives the output port 'xlgpcs_tx_recal_skip_en_o'. **Value After Reset:** 0x0 **Exists:** Always
3:1	RC_FR_EN_3_1	* Varies	TX Force Re-Calibration Lane3_1. Forces a re-calibration of TX DCC in the selected bank regardless of the state of the corresponding done signal. This field drives the output port 'xlgpcs_tx_recal_force_en_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	RC_FR_EN_0	R/W	TX Force Re-Calibration Lane0. Forces a re-calibration of TX DCC in the selected bank regardless of the state of the corresponding done signal. This field drives the output port 'xlgpcs_tx_recal_force_en_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.281 VR_PMA_MP_32G_TX_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Tx Control0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x80DF : (DWCXLP_APB3_16) ? (0x80DF*0x2) : (0x80DF*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

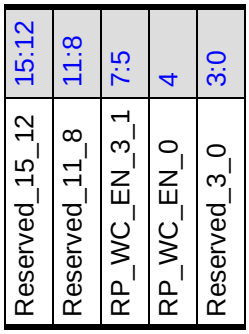


Table 7-286 Fields for Register: VR_PMA_MP_32G_TX_CTRL0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	Reserved_11_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	RP_WC_EN_3_1	* Varies	TX ROPLL word clock output enable Lane3_1. This field drives the output port 'xlgpcs_tx_ropll_wrd_clk_en_o'. **Value After Reset:** "DWCXLP_MULTI_LANE ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
4	RP_WC_EN_0	R/W	TX ROPLL word clock output enable Lane0. This field drives the output port 'xlgpcs_tx_ropll_wrd_clk_en_o'. **Value After Reset:** 0x1 **Exists:** Always

Table 7-286 Fields for Register: VR_PMA_MP_32G_TX_CTRL0 (Continued)

Bits	Name	Memory Access	Description
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.1.282 VR_PMA_MP_32G_MARGIN_LANE0

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Margin Lane0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80E4 : (DWCXLP_APB3_16) ? (0x80E4*0x2) : (0x80E4*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

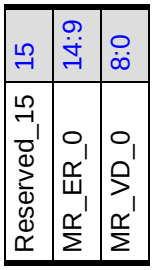


Table 7-287 Fields for Register: VR_PMA_MP_32G_MARGIN_LANE0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:9	MR_ER_0	R	Rx Margin Error Lane 0. Error counter to indicate the number of bit-errors during RX margining. This field reflects the value of the input port 'xlgpcs_rx0_margin_err_i'. **Value After Reset:** 0x0 **Exists:** Always
8:0	MR_VD_0	R/W	Rx Voltage Margining Lane 0. Provides the value for voltage margining. This field drives the output port 'xlgpcs_rx0_margin_vdac_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.283 VR_PMA_MP_32G_MARGIN_LANE1

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Margin Lane1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80E5 : (DWCXLP_APB3_16) ? (0x80E5*0x2) : (0x80E5*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

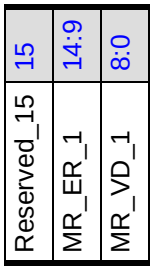


Table 7-288 Fields for Register: VR_PMA_MP_32G_MARGIN_LANE1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:9	MR_ER_1	R	Rx Margin Error Lane 1. Error counter to indicate the number of bit-errors during RX margining. This field reflects the value of the input port 'xlgpcs_rx1_margin_err_i'. **Value After Reset:** 0x0 **Exists:** Always
8:0	MR_VD_1	* Varies	Rx Voltage Margining Lane 1. Provides the value for voltage margining. This field drives the output port 'xlgpcs_rx1_margin_vdac_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

7.1.284 VR_PMA_MP_32G_MARGIN_LANE2

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Margin Lane2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80E6 : (DWCXLP_APB3_16) ? (0x80E6*0x2) : (0x80E6*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

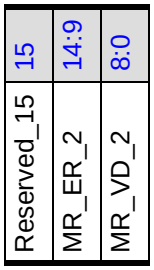


Table 7-289 Fields for Register: VR_PMA_MP_32G_MARGIN_LANE2

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:9	MR_ER_2	R	Rx Margin Error Lane 2. Error counter to indicate the number of bit-errors during RX margining. This field reflects the value of the input port 'xlgpcs_rx2_margin_err_i'. **Value After Reset:** 0x0 **Exists:** Always
8:0	MR_VD_2	* Varies	Rx Voltage Margining Lane 2. Provides the value for voltage margining. This field drives the output port 'xlgpcs_rx2_margin_vdac_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

7.1.285 VR_PMA_MP_32G_MARGIN_LANE3

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Margin Lane3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80E7 : (DWCXLP_APB3_16) ? (0x80E7*0x2) : (0x80E7*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

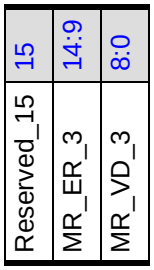


Table 7-290 Fields for Register: VR_PMA_MP_32G_MARGIN_LANE3

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:9	MR_ER_3	R	Rx Margin Error Lane 3. Error counter to indicate the number of bit-errors during RX margining. This field reflects the value of the input port 'xlgpcs_rx3_margin_err_i'. **Value After Reset:** 0x0 **Exists:** Always
8:0	MR_VD_3	* Varies	Rx Voltage Margining Lane 3. Provides the value for voltage margining. This field drives the output port 'xlgpcs_rx3_margin_vdac_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

7.1.286 VR_PMA_MP_32G_MARGIN_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Margin Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80E8 : (DWCXLP_APB3_16) ? (0x80E8*0x2) : (0x80E8*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

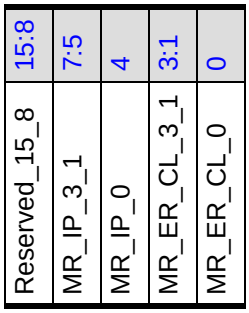


Table 7-291 Fields for Register: VR_PMA_MP_32G_MARGIN_CTRL

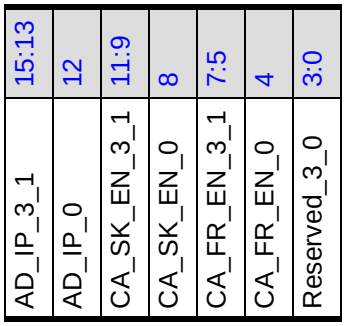
Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	MR_IP_3_1	* Varies	Rx Margining In Progress Lane3_1. Indicates if RX margining is in progress. This field drives the output port 'xlgpcs_rx_margin_in_prog_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
4	MR_IP_0	R/W	Rx Margining In Progress Lane0. Indicates if RX margining is in progress. This field drives the output port 'xlgpcs_rx_margin_in_prog_o'. **Value After Reset:** 0x0 **Exists:** Always

Table 7-291 Fields for Register: VR_PMA_MP_32G_MARGIN_CTRL (Continued)

Bits	Name	Memory Access	Description
3:1	MR_ER_CL_3_1	* Varies	Rx Voltage Margining Lane3_1. Provides the value for voltage margining. This field drives the output port 'xlgpcs_rx_margin_err_clr_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	MR_ER_CL_0	R/W	Rx Voltage Margining Lane0. Provides the value for voltage margining. This field drives the output port 'xlgpcs_rx_margin_err_clr_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.287 VR_PMA_MP_32G_RX_ADPT_CTRL0

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Rx Adapt Control0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80ED : (DWCXLP_APB3_16) ? (0x80ED*0x2) : (0x80ED*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)



15:13	12	11:9	8	7:5	4	3:0
AD_IP_3_1	AD_IP_0	CA_SK_EN_3_1	CA_SK_EN_0	CA_FR_EN_3_1	CA_FR_EN_0	Reserved_3_0

Table 7-292 Fields for Register: VR_PMA_MP_32G_RX_ADPT_CTRL0

Bits	Name	Memory Access	Description
15:13	AD_IP_3_1	* Varies	RX Adaptation in Progress Lane_3_1. This field drives the output port 'xlgpcs_rx_adpt_in_prog_o[3:1]'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
12	AD_IP_0	R/W	RX Adaptation in Progress Lane0. This field drives the output port 'xlgpcs_rx_adpt_in_prog_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
11:9	CA_SK_EN_3_1	* Varies	RX Skip Coarse Adaption Enable Lane_3_1. This field drives the output port 'xlgpcs_rx_crse_adpt_skip_en_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-292 Fields for Register: VR_PMA_MP_32G_RX_ADPT_CTRL0 (Continued)

Bits	Name	Memory Access	Description
8	CA_SK_EN_0	R/W	RX Skip Coarse Adaption Enable Lane0. This field drives the output port 'xlgpcs_rx_crse_adpt_skip_en_o'. **Value After Reset:** 0x0 **Exists:** Always
7:5	CA_FR_EN_3_1	* Varies	RX Force Coarse Adaption Enable Lane_3_1. This field drives the output port 'xlgpcs_rx_crse_adpt_frc_en_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
4	CA_FR_EN_0	R/W	RX Force Coarse Adaption Enable Lane0. This field drives the output port 'xlgpcs_rx_crse_adpt_frc_en_o'. **Value After Reset:** 0x0 **Exists:** Always
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.1.288 VR_PMA_MP_32G_CNTX_LINK_NUM_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Context Link Number Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F0 : (DWCXLP_APB3_16) ? (0x80F0*0x2) : (0x80F0*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

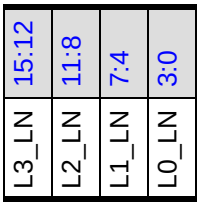


Table 7-293 Fields for Register: VR_PMA_MP_32G_CNTX_LINK_NUM_CTRL

Bits	Name	Memory Access	Description
15:12	L3_LN	* Varies	Lane3 Link Number. Ensures that lanes with the same lane3_link_num, change rate simultaneously. This field drives the output port 'xlgpcs_lane3_link_num_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
11:8	L2_LN	* Varies	Lane2 Link Number. Ensures that lanes with the same lane2_link_num, change rate simultaneously. This field drives the output port 'xlgpcs_lane2_link_num_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7:4	L1_LN	* Varies	Lane1 Link Number. Ensures that lanes with the same lane1_link_num, change rate simultaneously. This field drives the output port 'xlgpcs_lane1_link_num_o'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""

Table 7-293 Fields for Register: VR_PMA_MP_32G_CNTX_LINK_NUM_CTRL (Continued)

Bits	Name	Memory Access	Description
3:0	L0_LN	R/W	Lane0 Link Number. Ensures that lanes with the same lane0_link_num, change rate simultaneously. This field drives the output port 'xlgpcs_lane0_link_num_o'. **Value After Reset:** 0x0 **Exists:** Always

7.1.289 VR_PMA_MP_32G_CNTX_CTRL

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Context Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F1 : (DWCXLP_APB3_16) ? (0x80F1*0x2) : (0x80F1*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

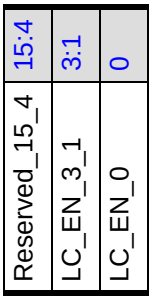


Table 7-294 Fields for Register: VR_PMA_MP_32G_CNTX_CTRL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:1	LC_EN_3_1	* Varies	Lane Context Enable Lane3_1. This field drives the output port 'xlgpcs_lane_cntx_en_o'. **Value After Reset:** "!DWCXLP_MULTI_LANE ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
0	LC_EN_0	R/W	Lane Context Enable Lane0. This field drives the output port 'xlgpcs_lane_cntx_en_o'. **Value After Reset:** 0x1 **Exists:** Always

7.1.290 VR_PMA_MP_32G_TX_CNTX_SEL0

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Tx Context Select0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8101 : (DWCXLP_APB3_16) ? (0x8101*0x2) : (0x8101*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

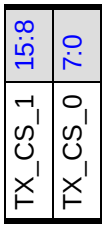


Table 7-295 Fields for Register: VR_PMA_MP_32G_TX_CNTX_SEL0

Bits	Name	Memory Access	Description
15:8	TX_CS_1	* Varies	TX Context Select Lane1. This field drives the output port 'xlgpcs_tx1_cntx_sel_o'. **Value After Reset:** "DWCXLP_MULTI_LANE ? DWCXLP_TX_CNTX_SEL : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7:0	TX_CS_0	R/W	TX Context Select Lane0. This field drives the output port 'xlgpcs_tx0_cntx_sel_o'. **Value After Reset:** DWCXLP_TX_CNTX_SEL **Exists:** Always

7.1.291 VR_PMA_MP_32G_TX_CNTX_SEL1

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Tx Context Select1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8102 : (DWCXLP_APB3_16) ? (0x8102*0x2) : (0x8102*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)&&DWCXLP_MULTI_LANE

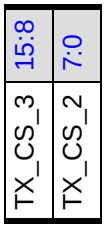


Table 7-296 Fields for Register: VR_PMA_MP_32G_TX_CNTX_SEL1

Bits	Name	Memory Access	Description
15:8	TX_CS_3	R/W	TX Context Select Lane3. This field drives the output port 'xlgpcs_tx3_cntx_sel_o'. **Value After Reset:** DWCXLP_TX_CNTX_SEL **Exists:** Always
7:0	TX_CS_2	R/W	TX Context Select Lane2. This field drives the output port 'xlgpcs_tx2_cntx_sel_o'. **Value After Reset:** DWCXLP_TX_CNTX_SEL **Exists:** Always

7.1.292 VR_PMA_MP_32G_CM_CNTX_SEL0

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Common Context Select0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8104 : (DWCXLP_APB3_16) ? (0x8104*0x2) : (0x8104*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

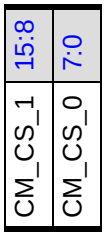


Table 7-297 Fields for Register: VR_PMA_MP_32G_CM_CNTX_SEL0

Bits	Name	Memory Access	Description
15:8	CM_CS_1	* Varies	TX Common Context Select Lane1. This field drives the output port 'xlgpcs_tx_cm1_cntx_sel_o'. **Value After Reset:** "DWCXLP_MULTI_LANE ? DWCXLP_CM_CNTX_SEL : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7:0	CM_CS_0	R/W	TX Common Context Select Lane0. This field drives the output port 'xlgpcs_tx_cm0_cntx_sel_o'. **Value After Reset:** DWCXLP_CM_CNTX_SEL **Exists:** Always

7.1.293 VR_PMA_MP_32G_CM_CNTX_SEL1

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Common Context Select1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8105 : (DWCXLP_APB3_16) ? (0x8105*0x2) : (0x8105*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)&&DWCXLP_MULTI_LANE

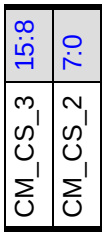


Table 7-298 Fields for Register: VR_PMA_MP_32G_CM_CNTX_SEL1

Bits	Name	Memory Access	Description
15:8	CM_CS_3	R/W	TX Common Context Select Lane3. This field drives the output port 'xlgpcs_tx_cm3_cntx_sel_o'. **Value After Reset:** DWCXLP_CM_CNTX_SEL **Exists:** Always
7:0	CM_CS_2	R/W	TX Common Context Select Lane2. This field drives the output port 'xlgpcs_tx_cm2_cntx_sel_o'. **Value After Reset:** DWCXLP_CM_CNTX_SEL **Exists:** Always

7.1.294 VR_PMA_MP_32G_RX_CNTX_SEL0

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Rx Context Select0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8110 : (DWCXLP_APB3_16) ? (0x8110*0x2) : (0x8110*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)

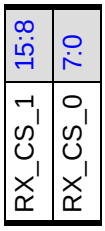


Table 7-299 Fields for Register: VR_PMA_MP_32G_RX_CNTX_SEL0

Bits	Name	Memory Access	Description
15:8	RX_CS_1	* Varies	RX Context Select Lane1. This field drives the output port 'xlgpcs_rx1_cntx_sel_o'. **Value After Reset:** "DWCXLP_MULTI_LANE ? DWCXLP_RX_CNTX_SEL : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_25G_ONLY ? \"read-only\" : \"read-write\""
7:0	RX_CS_0	R/W	RX Context Select Lane0. This field drives the output port 'xlgpcs_rx0_cntx_sel_o'. **Value After Reset:** DWCXLP_RX_CNTX_SEL **Exists:** Always

7.1.295 VR_PMA_MP_32G_RX_CNTX_SEL1

- **Description:** VR PMA Synopsys Multi-protocol 32G PHY Rx Context Select1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8111 : (DWCXLP_APB3_16) ? (0x8111*0x2) : (0x8111*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x010000 : (DWCXLP_APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** DWCXLP_PMA_MMD&&(DWCXLP_E32)&&DWCXLP_MULTI_LANE

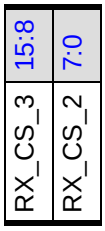


Table 7-300 Fields for Register: VR_PMA_MP_32G_RX_CNTX_SEL1

Bits	Name	Memory Access	Description
15:8	RX_CS_3	R/W	RX Context Select Lane3. This field drives the output port 'xlgpcs_rx3_cntx_sel_o'. **Value After Reset:** DWCXLP_RX_CNTX_SEL **Exists:** Always
7:0	RX_CS_2	R/W	RX Context Select Lane2. This field drives the output port 'xlgpcs_rx2_cntx_sel_o'. **Value After Reset:** DWCXLP_RX_CNTX_SEL **Exists:** Always

7.2 DWC_xlgpcs_map/PCS_MMD Registers

The PCS MMD Standard (SR) & Vendor Specific (VR) Registers address block. Follow the link for the register to see a detailed description of the register.

Table 7-301 Registers for Address Block: DWC_xlgpcs_map/PCS_MMD

Register	Offset	Description
SR_PCS_CTRL1 on page 1023	* Varies	SR PCS MMD Control1 Register
SR_PCS_STS1 on page 1027	* Varies	SR PCS MMD Status1 Register
SR_PCS_DEV_ID1 on page 1030	* Varies	SR PCS MMD Device Identifier Register 1
SR_PCS_DEV_ID2 on page 1031	* Varies	SR PCS MMD Device Identifier Register 2
SR_PCS_SPD_ABL on page 1033	* Varies	SR PCS MMD Speed Ability Register
SR_PCS_DEV_PKG1 on page 1035	* Varies	SR PCS MMD Devices in Package Register 1
SR_PCS_DEV_PKG2 on page 1038	* Varies	SR PCS MMD Devices in Package Register 2
SR_PCS_CTRL2 on page 1039	* Varies	SR PCS Control2 Register
SR_PCS_STS2 on page 1041	* Varies	SR PCS MMD Status2 Register
SR_PCS_PKG1 on page 1044	* Varies	SR PCS MMD Package Identifier Register 1
SR_PCS_PKG2 on page 1045	* Varies	SR PCS MMD Package Identifier Register 2
SR_PCS_EEE_ABL on page 1047	* Varies	SR PCS MMD EEE Capability Register
SR_PCS_EEE_WKERR on page 1051	* Varies	SR PCS MMD EEE Wake Error Counter Register
SR_PCS_BASER_STS1 on page 1052	* Varies	SR PCS MMD BASE-R Status1 Register
SR_PCS_BASER_STS2 on page 1054	* Varies	SR PCS MMD BASE-R Status2 Register
SR_PCS_TP_A0 on page 1056	* Varies	SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed A0 Register. This register contains Bits[15:0]...
SR_PCS_TP_A1 on page 1057	* Varies	SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed A1 Register
SR_PCS_TP_A2 on page 1058	* Varies	SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed A2 Register
SR_PCS_TP_A3 on page 1059	* Varies	SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed A3 Register
SR_PCS_TP_B0 on page 1060	* Varies	SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed B0 Register This register contains Bits[15:0]...
SR_PCS_TP_B1 on page 1061	* Varies	SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed B1 Register

Table 7-301 Registers for Address Block: DWC_xlgpcs_map/PCS_MMD (Continued)

Register	Offset	Description
SR_PCS_TP_B2 on page 1062	* Varies	SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed B2 Register
SR_PCS_TP_B3 on page 1063	* Varies	SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed B3 Register
SR_PCS_TP_CTRL on page 1064	* Varies	SR PCS BASE-R Test Pattern Control Register
SR_PCS_TP_ERRCTR on page 1067	* Varies	SR PCS BASE-R Test Pattern Error Counter Register
SR_PCS_BER_HI_ORDER_CNT on page 1068	* Varies	SR PCS BASE-R BER High Order Counter Register
SR_PCS_ERR_BLK_HI_ORDER_CNT on page 1069	* Varies	SR PCS BASE-R Errored Blocks High Order Counter Register
SR_PCS_MLN_ALIGN_STS1 on page 1070	* Varies	SR PCS BASE-R Multi-lane PCS Alignment Status 1 Register
SR_PCS_MLN_ALIGN_STS2 on page 1073	* Varies	SR PCS BASE-R Multi-lane PCS Alignment Status 2 Register. All the block lock status bits present...
SR_PCS_MLN_ALIGN_STS3 on page 1076	* Varies	SR PCS BASE-R Multi-lane PCS Alignment Status 3 Register
SR_PCS_MLN_ALIGN_STS4 on page 1079	* Varies	SR PCS BASE-R Multi-lane PCS Alignment Status 4 Register . The alignment status fields in this register...
SR_PCS_BIP_ERR_CNTR_LANE (i) (for i = 0; i <= 19) on page 1082	i+0x300c8	SR PCS BASE-R PCS BIP Error Counter Register -Lane i (RO, COR Type) Note: In 40G and 50G configurations...
SR_PCS_LANE (i) MAPPING (for i = 0; i <= 19) on page 1083	i+0x30190	SR PCS BASE-R Lane i Mapping Register
SR_PCS_TIME_SYNC_PCS_ABL on page 1084	* Varies	SR PCS MMD Time Sync Capability Register
SR_PCS_TIME_SYNC_TX_MAX_DLY_LWR on page 1087	* Varies	SR PCS MMD Time Sync Tx Max Delay Lower Register
SR_PCS_TIME_SYNC_TX_MAX_DLY_UPR on page 1088	* Varies	SR PCS MMD Time Sync Tx Max Delay Upper Register
SR_PCS_TIME_SYNC_TX_MIN_DLY_LWR on page 1089	* Varies	SR PCS MMD Time Sync Tx Min Delay Lower Register
SR_PCS_TIME_SYNC_TX_MIN_DLY_UPR on page 1090	* Varies	SR PCS MMD Time Sync Tx Min Delay Upper Register
SR_PCS_TIME_SYNC_RX_MAX_DLY_LWR on page 1091	* Varies	SR PCS MMD Time Sync Rx Max Delay Lower Register

Table 7-301 Registers for Address Block: DWC_xlgpcs_map/PCS_MMD (Continued)

Register	Offset	Description
SR_PCS_TIME_SYNC_RX_MAX_DLY_UPR on page 1092	* Varies	SR PCS MMD Time Sync Rx Max Delay Upper Register
SR_PCS_TIME_SYNC_RX_MIN_DLY_LWR on page 1093	* Varies	SR PCS MMD Time Sync Rx Min Delay Lower Register
SR_PCS_TIME_SYNC_RX_MIN_DLY_UPR on page 1094	* Varies	SR PCS MMD Time Sync Rx Min Delay Upper Register
SR_PCS_TIME_SYNC_TX_MAX_FINE_DLY on page 1095	* Varies	SR PCS MMD Time Sync Tx Max Fine Delay Register
SR_PCS_TIME_SYNC_TX_MIN_FINE_DLY on page 1096	* Varies	SR PCS MMD Time Sync Tx Min Fine Delay Register
SR_PCS_TIME_SYNC_RX_MAX_FINE_DLY on page 1097	* Varies	SR PCS MMD Time Sync Rx Max Fine Delay Register
SR_PCS_TIME_SYNC_RX_MIN_FINE_DLY on page 1098	* Varies	SR PCS MMD Time Sync Rx Min Fine Delay Register
SR_PCS_TIME_SYNC_CFG on page 1099	* Varies	SR PCS MMD Time Sync Configuration Register
VR_PCS_DIG_CTRL1 on page 1100	* Varies	VR PCS MMD Digital Control1 Register
VR_PCS_DIG_CTRL2 on page 1105	* Varies	VR PCS MMD Digital Control2 Register
VR_PCS_DIG_CTRL3 on page 1108	* Varies	VR PCS MMD Digital Control3 Register This register is applicable only when the DWC_xlgpcs is configured...
VR_PCS_DEBUG_CTRL on page 1112	* Varies	VR PCS MMD Debug Control Register
VR_PCS_EEE_MCTRL on page 1114	* Varies	VR PCS MMD EEE Mode Control Register This register enables or disables the Energy Efficient Ethernet...
VR_PCS_KR_CTRL on page 1118	* Varies	VR PCS BASE-R Control Register
VR_PCS_EEE_TXTIMER on page 1121	* Varies	VR PCS MMD EEE Tx Timer Register
VR_PCS_EEE_RXTIMER on page 1126	* Varies	VR PCS MMD EEE Rx Timer Register
VR_PCS_EEE_MISC_TIMER on page 1129	* Varies	VR PCS MMD EEE GAUI Timer Register
VR_PCS_DIG_STS on page 1130	* Varies	VR PCS MMD Digital Status Register
VR_PCS_GPIO on page 1134	* Varies	VR PCS MMD GPIO Register
VR_PCS_AM_CNT on page 1135	* Varies	VR PCS MMD Alignment Marker Counter Register
VR_PCS_SKW_OUT_OF_RANGE on page 1136	* Varies	VR PCS MMD Skew Out of Range Register

Table 7-301 Registers for Address Block: DWC_xlgpcs_map/PCS_MMD (Continued)

Register	Offset	Description
VR_PCS_ALT_AM_CNT on page 1138	* Varies	VR PCS MMD Alternate Alignment Marker Counter Register
VR_PCS_MISC_STS on page 1140	* Varies	VR PCS Miscellaneous Status Register
VR_PCS_BIT_BNDRY on page 1142	* Varies	VR PCS Block Synchronization bit boundary Status Register
VR_PCS_SFTY_DISABLE on page 1143	* Varies	VR PCS MMD Safety Disable Register
VR_PCS_SFTY_UE_INTR0 on page 1145	* Varies	VR PCS MMD Uncorrectable Safety Error Interrupt Register 0
VR_PCS_SFTY_UE_INTR1 on page 1149	* Varies	VR PCS MMD Uncorrectable Safety Error Interrupt Register 1
VR_PCS_SFTY_CE_INTR on page 1154	* Varies	VR PCS MMD Correctable Safety Error Interrupt Register
VR_PCS_RSFECEF_ECC_STS on page 1159	* Varies	VR PCS MMD RSFEC Error forwarding memory ECC status. This register is applicable only for configurations...
VR_PCS_FEC_ECC_STS0 on page 1160	* Varies	VR PCS MMD FEC ECC memory status for Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error...
VR_PCS_FEC_ECC_STS1 on page 1162	* Varies	VR PCS MMD FEC ECC memory status for Base-R FEC Error-Forwarding Memory for lane 0/RS-FEC Error...
VR_PCS_SFTY_DBG_CTRL on page 1164	* Varies	VR PCS MMD safety debug control register Note: Various fields in this register should be programmed...
VR_PCS_SFTY_TMR_CTRL on page 1169	* Varies	VR PCS MMD safety Timer Control register.
VR_PCS_SFTY_ERR_INJ0 on page 1171	* Varies	VR PCS MMD safety Error Injection register 0
VR_PCS_SFTY_ERR_INJ1 on page 1174	* Varies	VR PCS MMD safety Error Injection register 1
VR_PCS_FEC_ECC_STS2 on page 1177	* Varies	VR PCS MMD FEC ECC memory status for Base-R FEC Error-Correction Memory for lane 1. Note: This register...
VR_PCS_FEC_ECC_STS3 on page 1178	* Varies	VR PCS MMD FEC ECC memory status for Base-R FEC Error-Forward Memory for lane 1. Note: This register...
VR_PCS_FEC_ECC_STS4 on page 1179	* Varies	VR PCS MMD FEC ECC memory status for Base-R FEC Error-Correction Memory for lane 2. Note: This register...
VR_PCS_FEC_ECC_STS5 on page 1180	* Varies	VR PCS MMD FEC ECC memory status for Base-R FEC Error-Forward Memory for lane 2. Note: This register...
VR_PCS_FEC_ECC_STS6 on page 1181	* Varies	VR PCS MMD FEC ECC memory status for Base-R FEC Error-Correction Memory for lane 3. Note: This register...
VR_PCS_FEC_ECC_STS7 on page 1182	* Varies	VR PCS MMD FEC ECC memory status for Base-R FEC Error-Forward Memory for lane 3. Note: This register...

Table 7-301 Registers for Address Block: DWC_xlgpcs_map/PCS_MMD (Continued)

Register	Offset	Description
VR_PCS_DSKW_ECC_STS0 on page 1183	* Varies	De-skew memory status for lane 0. Note: This register is present only if (!DWCXLP_25G_ONLY) =...
VR_PCS_DSKW_ECC_STS1 on page 1185	* Varies	De-skew memory status for lane 1. Note: This register is present only if (!DWCXLP_25G_ONLY) =...
VR_PCS_DSKW_ECC_STS2 on page 1187	* Varies	De-skew memory status for lane 2. Note: This register is present only if (DWCXLP_MULTI_LANE) =...
VR_PCS_DSKW_ECC_STS3 on page 1189	* Varies	De-skew memory status for lane 3. Note: This register is present only if (DWCXLP_MULTI_LANE) =...

7.2.1 SR_PCS_CTRL1

- **Description:** SR PCS MMD Control1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

15	14	13	12	11	10	9:7	6	5:2	1:0
RST	Reserved_14	SS13	Reserved_12	LPM	CS_EN	Reserved_9_	SS6	SS_5_2	Reserved_1_

Table 7-302 Fields for Register: SR_PCS_CTRL1

Bits	Name	Memory Access	Description
15	RST	R/W	Soft Reset (RW,SC Type) For Synopsys PHY configurations: When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared when PSEQ_STATE in VR PCS MMD Digital Status Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. For Non-Synopsys PHY configurations: When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. This bit is self-cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For a description of read or write access for registers during the reset process, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	SS13	R	Speed Selection Bit 13 This bit always return 1. **Value After Reset:** 0x1 **Exists:** Always
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-302 Fields for Register: SR_PCS_CTRL1 (Continued)

Bits	Name	Memory Access	Description
11	LPM	R/W	This bit controls the power-down mode of DWC_xlgpcs. ■ 0: Normal Operation ■ 1: DWC_xlgpcs goes to power-down mode along with the PHY. The DWC_xlgpcs does the following to trigger powerdown mode: ■ Turns off the PHY Receiver and Transmitter ■ Switches off all the clocks When the host clears this bit, DWC_xlgpcs resumes normal operation. In a Synopsys PHY configuration, after clearing this bit, the host must wait until Bits[4:2] of VR PCS MMD Digital Status Register indicate that DWC_xlgpcs is in normal state.**Note:** For a description of read or write access for registers during the reset process, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always
10	CS_EN	* Varies	Clock Stop Enable This bit controls stopping the clock during LPI mode on the XGMII Rx interface: ■ 1: The PHY may stop the clock during LPI mode ■ 0: The clock cannot be stopped during LPI mode You must program this bit based on the capability of the XGMAC during Rx LPI mode. **Value After Reset:** "DWCXLP_EEE_EN ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "DWCXLP_EEE_EN ? \"read-write\" : \"read-only\""
9:7	Reserved_9_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	SS6	R	Speed Selection Bit 6 This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always

Table 7-302 Fields for Register: SR_PCS_CTRL1 (Continued)

Bits	Name	Memory Access	Description
5:2	SS_5_2	R/W	Speed Selection Bits [5:2] This field selects the speed of operation when both SS13 and SS6 are set. Speed encoding is as follows: ■ 1 0 0 0 = 5 Gb/s ■ 0 1 1 0 = 50 Gb/s ■ 0 1 0 1 = 25 Gb/s ■ 0 1 0 0 = 100 Gb/s ■ 0 0 1 1 = 40 Gb/s ■ 0 0 0 0 = 10 Gb/s ■ Other Values = Reserved **Value After Reset:** "DWCXLP_KR4_100G ? 0x4 : (DWCXLP_25G_ONLY||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 0x5 : 0x3" **Exists:** Always
1:0	Reserved_1_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.2.2 SR_PCS_STS1

- **Description:** SR PCS MMD Status1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1 : (DWCXLP_APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

15:12	11	10	9	8	7	6	5:3	2	1	0
Reserved_15_12	TXLPIR	RXLPIR	TXLPII	RXLPII	FLT	CSC	Reserved_5_3	RLU	LPMS	Reserved_0

Table 7-303 Fields for Register: SR_PCS_STS1

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	TXLPIR	R	Tx LPI Received (RO,LH Type) This bit indicates if the DWC_xlgpcs Tx received LPI in the Transmit path from the XGMII interface of the MAC or XGMAC: ■ 1: LPI received ■ 0: LPI not received **Value After Reset:** 0x0 **Exists:** Always
10	RXLPIR	R	Rx LPI Received (RO,LH Type) This bit indicates if the DWC_xlgpcs Rx received LPI in the Receive path from the link partner: ■ 1: LPI received ■ 0: LPI not received **Value After Reset:** 0x0 **Exists:** Always

Table 7-303 Fields for Register: SR_PCS_STS1 (Continued)

Bits	Name	Memory Access	Description
9	TXLPII	R	Tx LPI Indication This bit indicates if DWC_xlgpcs Tx is currently receiving LPI in the Transmit path from the XGMII interface of the MAC or XGMAC: ■ 1: DWC_xlgpcs Tx is currently receiving LPI ■ 0: DWC_xlgpcs Tx is currently not receiving LPI This bit is read-only. **Value After Reset:** 0x0 **Exists:** Always
8	RXLPII	R	Rx LPI Indication This bit indicates if DWC_xlgpcs Rx is currently receiving LPI in the Receive path from the link partner: ■ 1: DWC_xlgpcs Rx is currently receiving LPI ■ 0: DWC_xlgpcs Rx is currently not receiving LPI This bit is read-only. **Value After Reset:** 0x0 **Exists:** Always
7	FLT	R	Fault Condition Detected This bit indicates that a fault condition is detected in the DWC_xlgpcs. This bit is set when the Tx or Rx path of the DWC_xlgpcs encounters any of the following fault conditions: ■ Rx path did not align or synchronize ■ Tx fault encountered the remote fault signaling **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
6	CSC	R	Clock Stop Capable This bit indicates if the MAC can stop the clock during the LPI mode in the transmit path: ■ 1: The MAC can stop the clock during the LPI mode. ■ 0: The MAC cannot stop the clock during the LPI mode. **Value After Reset:** "DWCXLP_EEE_EN ? 0x1 : 0x0 " **Exists:** Always
5:3	Reserved_5_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-303 Fields for Register: SR_PCS_STS1 (Continued)

Bits	Name	Memory Access	Description
2	RLU	R	Rx Link Up (RO,LL Type) For BASE-R mode the Rx Link up is defined when hi_ber is low along with the following conditions, PCS MMD in 40G, 100G modes and 50G (no FEC and BASE-R FEC) mode ■ 0: When code group alignment, alignment lock and block lock are not successful. ■ 1: When code group alignment, alignment lock and block lock are successful. PCS MMD in 50G and 100G (with RS-FEC) mode ■ 0: When code group alignment, Code Word Marker lock, Lane alignments are not successful. ■ 1: When code group alignment, Code Word Marker lock, Lane alignments are successful. PCS MMD in 10G and 25G (no FEC or with BASER FEC) modes ■ 0: When code group alignment and block lock are not successful. ■ 1: When code group alignment and block lock are successful. PCS MMD in 25G (with RS-FEC) mode ■ 0: When code group alignment and RS-FEC Code Word Marker lock are not successful. ■ 1: When code group alignment and RS-FEC Code Word Marker lock are successful. **Value After Reset:** 0x0 **Exists:** Always
1	LPMS	R	Low-Power Mode Support This bit indicates that the DWC_xlgpcs supports the low-power mode. **Value After Reset:** 0x1 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.2.3 SR_PCS_DEV_ID1

- **Description:** SR PCS MMD Device Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x2 : (DWCXLP_APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-304 Fields for Register: SR_PCS_DEV_ID1

Bits	Name	Memory Access	Description
15:0	PCS_DEV_OUI_3_18	R	Organizationally Unique Identifier[3:18] The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. This value is configured in coreConsultant. This read-only field contains Bits[18:3] of the 24-bit OUI of device manufacturer. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR Control MMD PCS Device Identifier Register 1 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_DEV_OUI_3_18 **Exists:** Always

7.2.4 SR_PCS_DEV_ID2

- **Description:** SR PCS MMD Device Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x3 : (DWCXLP_APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

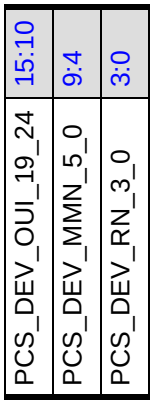


Table 7-305 Fields for Register: SR_PCS_DEV_ID2

Bits	Name	Memory Access	Description
15:10	PCS_DEV_OUI_19_24	R	Organizationally Unique Identifier[19:24] The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. You can configure the value of each MMD using coreConsultant. This read-only field contains Bits[24:19] of 24-bit OUI of the device manufacturer. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR Control MMD PCS Device Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_DEV_OUI_19_24 **Exists:** Always

Table 7-305 Fields for Register: SR_PCS_DEV_ID2 (Continued)

Bits	Name	Memory Access	Description
9:4	PCS_DEV_MMN_5_0	R	Model Number This read-only field contains the 6-bit Model number of the device. You can configure the value of each MMD using coreConsultant. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR Control MMD PCS Device Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_DEV_MODEL_NO_0_5 **Exists:** Always
3:0	PCS_DEV_RN_3_0	R	Revision Number This read-only field contains the 4-bit Revision number of the device. You can configure the value of each MMD using coreConsultant. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR Control MMD PCS Device Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_DEV_REV_NO_0_3 **Exists:** Always

7.2.5 SR_PCS_SPD_ABL

- **Description:** SR PCS MMD Speed Ability Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x4 : (DWCXLP_APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

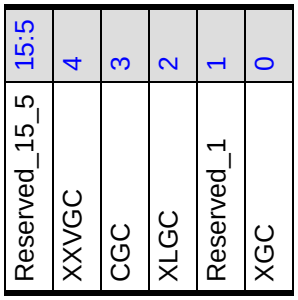


Table 7-306 Fields for Register: SR_PCS_SPD_ABL

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	XXVGC	R	25G Capable This bit indicates that DWC_xlgpcs can support 25G mode. **Value After Reset:** !DWCXLP_KR4_KR **Exists:** Always
3	CGC	R	100G Capable This bit indicates that DWC_xlgpcs can support 1000G mode. **Value After Reset:** "(DWCXLP_KR4_100G||DWCXLP_100G_1LANE) ? 0x1 : 0x0" **Exists:** Always
2	XLGC	R	40G Capable This bit indicates that DWC_xlgpcs can support 40G mode. **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x1 : 0x0" **Exists:** Always

Table 7-306 Fields for Register: SR_PCS_SPD_ABL (Continued)

Bits	Name	Memory Access	Description
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	XGC	R	10G Capable This bit indicates that DWC_xlgpcs can support 10G mode. **Value After Reset:** 0x1 **Exists:** Always

7.2.6 SR_PCS_DEV_PKG1

- **Description:** SR PCS MMD Devices in Package Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x5 : (DWCXLP_APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

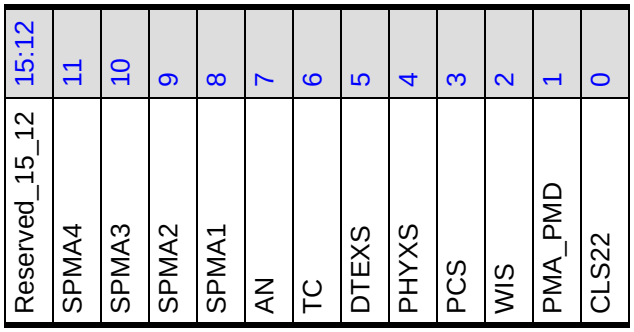


Table 7-307 Fields for Register: SR_PCS_DEV_PKG1

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	SPMA4	R	Separated PMA 1 ■ 1: Separated PMA (4) present in package ■ 0: Separated PMA (4) not present in package **Value After Reset:** 0x0 **Exists:** Always
10	SPMA3	R	Separated PMA 1 ■ 1: Separated PMA (3) present in package ■ 0: Separated PMA (3) not present in package **Value After Reset:** 0x0 **Exists:** Always

Table 7-307 Fields for Register: SR_PCS_DEV_PKG1 (Continued)

Bits	Name	Memory Access	Description
9	SPMA2	R	Separated PMA 2 ■ 1: Separated PMA (2) present in package ■ 0: Separated PMA (2) not present in package **Value After Reset:** 0x0 **Exists:** Always
8	SPMA1	R	Separated PMA 1 ■ 1: separated PMA (1) present in package ■ 0: separated PMA (1) not present in package **Value After Reset:** 0x0 **Exists:** Always
7	AN	R	Auto-Negotiation MMD This bit indicates that an AN MMD is present in the package. Default Value: If 'Add CL73 Autoneg Support' option is selected, this bit returns 1. For all other configurations, this bit returns 0. **Value After Reset:** "(DWCXLP_CL73_AN) ? \"0x1\" : \"0x0\"" **Exists:** Always
6	TC	R	TC MMD This bit indicates that the device has the TC MMDs in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
5	DTEXS	R	DTE XGXS MMD This bit indicates that the device has the DTE XS MMD in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
4	PHYXS	R	PHY XGXS MMD This bit indicates that the device has the PHY XS MMD in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
3	PCS	R	PCS MMD This bit indicates that the device has the PCS MMD in the package. **Value After Reset:** 0x1 **Exists:** Always

Table 7-307 Fields for Register: SR_PCS_DEV_PKG1 (Continued)

Bits	Name	Memory Access	Description
2	WIS	R	WIS MMD This bit indicates that the device has the WIS MMDs in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	PMA_PMD	R	PMA or PMD MMD This bit indicates that the device has PMA or PMD MMD in the package. This bit always returns 1. **Value After Reset:** "(DWCXLP_PMA_MMD) ? 0x1 : 0x0" **Exists:** Always
0	CLS22	R	Clause 22 Register Support This bit indicates that the Clause 22 (1G) functional register sets are implemented in this package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

7.2.7 SR_PCS_DEV_PKG2

- **Description:** SR PCS MMD Devices in Package Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x6 : (DWCXLP_APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

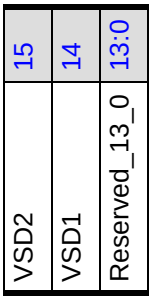


Table 7-308 Fields for Register: SR_PCS_DEV_PKG2

Bits	Name	Memory Access	Description
15	VSD2	R	Vendor-Specific Device 2 This bit indicates that a vendor-specific device 2 is present in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
14	VSD1	R	Vendor-specific Device 1 This bit indicates that the vendor-specific device 1 is present in the package. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
13:0	Reserved_13_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.2.8 SR_PCS_CTRL2

- **Description:** SR PCS Control2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x07 : (DWCXLP_APB3_16) ? (0x07*0x2) : (0x07*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

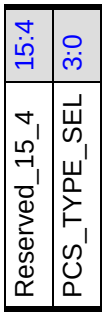


Table 7-309 Fields for Register: SR_PCS_CTRL2

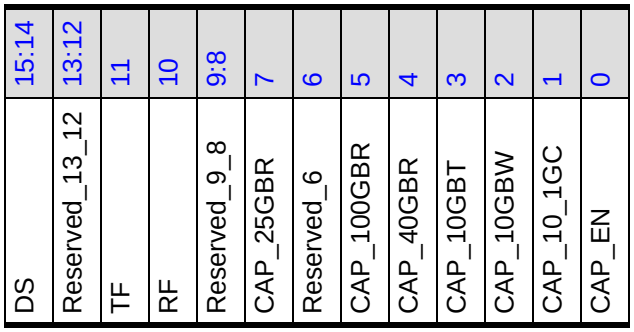
Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-309 Fields for Register: SR_PCS_CTRL2 (Continued)

Bits	Name	Memory Access	Description
3:0	PCS_TYPE_SEL	R/W	PCS Type Select This field is used to select the speed mode in DWC_xlgpcs when multiple speed modes are present . Mode encoding is as follows: ■ 1111: Select 5GBASE-R PCS type. ■ 0111: Select 25GBASE-R PCS type. ■ 0110: Select 40GBASE-T PCS type. ■ 1000: Select 50GBASE-R PCS type ■ 0101: Select 100GBASE-R PCS type. ■ 0100: Select 40GBASE-R PCS type. ■ 0011: Select 10GBASE-T PCS type. ■ 0010: Select 10GBASE-W PCS Type. ■ 0001: Select 10GBASE-X PCS Type ■ 0000: Select 10GBASE-R PCS Type **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 0x7 : DWCXLP_KR4_100G ? 0x5 : 0x4" **Exists:** Always

7.2.9 SR_PCS_STS2

- **Description:** SR PCS MMD Status2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x8 : (DWCXLP_APB3_16) ? (0x8*0x2) : (0x8*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



15:14	13:12	11	10	9:8	7	6	5	4	3	2	1	0
DS	Reserved_13_12	TF	RF	Reserved_9_8	CAP_25GBR	Reserved_6	CAP_100GBR	CAP_40GBR	CAP_10GBT	CAP_10GBW	CAP_10_1GC	CAP_EN

Table 7-310 Fields for Register: SR_PCS_STS2

Bits	Name	Memory Access	Description
15:14	DS	R	Device Present Status ■ 2'b10: Indicates that the MMD is present and responding to this register address. ■ Other values: Indicates the MMD is not present or not functioning properly. These bits are always set to 2'b10. **Value After Reset:** 0x2 **Exists:** Always
13:12	Reserved_13_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	TF	R	Transmitter Fault (RO,LH Type) This bit indicates that DWC_xlgpcs encountered a remote fault condition being transmitted on the Tx path. **Value After Reset:** 0x0 **Exists:** Always

Table 7-310 Fields for Register: SR_PCS_STS2 (Continued)

Bits	Name	Memory Access	Description
10	RF	R	Receiver Fault (RO,LH Type) This bit indicates that the device encountered a fault condition on the Rx path. This bit is set when the code group alignment or lane-to-lane deskew status is not successful in PCS-X mode and Block Lock is not successful in the PCS-R mode. This bit is also set when the Rx path receives the remote-fault message. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
9:8	Reserved_9_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	CAP_25GBR	R	25GBASE-R Capable This bit indicates if DWC_xlgpcs is capable of 25GBASE-R operation. **Value After Reset:** "!DWCXLP_KR4_KR ? 0x1: 0x0" **Exists:** Always
6	Reserved_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	CAP_100GBR	R	100GBASE-R Capable This bit indicates if DWC_xlgpcs is capable of 100GBASE-R operation. This bit always returns 1 when DWCXLP_KR4_100G is enabled. **Value After Reset:** "(DWCXLP_KR4_100G||DWCXLP_100G_1LANE) ? 0x1: 0x0" **Exists:** Always
4	CAP_40GBR	R	40GBASE-R Capable This bit indicates if DWC_xlgpcs is capable of 40GBASE-R operation. This bit always returns 1 in multi-lane configurations. In 25G only configuration, this bit returns 0. **Value After Reset:** "(DWCXLP_MULTI_LANE) ? 0x1 : 0x0" **Exists:** Always
3	CAP_10GBT	R	10GBASE-T Capable This bit indicates if DWC_xlgpcs is capable of 10GBASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-310 Fields for Register: SR_PCS_STS2 (Continued)

Bits	Name	Memory Access	Description
2	CAP_10GBW	R	10GBASE-W Capable This bit indicates if DWC_xlgpcs is capable of the 10GBASE-W operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	CAP_10_1GC	R	10GBASE-X Capable This bit indicates if DWC_xlgpcs is capable of the 10GBASE-X operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
0	CAP_EN	R	10GBASE-R Capable This bit indicates if DWC_xlgpcs is capable of 10GBASE-R operation. ■ 1: DWC_xlgpcs can support 10GBASE-R PCS type ■ 0: DWC_xlgpcs cannot support 10GBASE-R PCS type **Value After Reset:** 0x1 **Exists:** Always

7.2.10 SR_PCS_PKG1

- **Description:** SR PCS MMD Package Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xe : (DWCXLP_APB3_16) ? (0xe*0x2) : (0xe*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-311 Fields for Register: SR_PCS_PKG1

Bits	Name	Memory Access	Description
15:0	PCS_PKG_OUI_3_18	R	Organizationally Unique Identifier DWC_xlgpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. This value is configured in coreConsultant. This read-only field contains Bits[18:3] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS and AN) present in the package. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR Control MMD Package Identifier Register 1 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_OUI_3_18 **Exists:** Always

7.2.11 SR_PCS_PKG2

- **Description:** SR PCS MMD Package Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0xf : (DWCXLP_APB3_16) ? (0xf*0x2) : (0xf*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

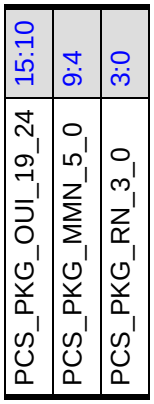


Table 7-312 Fields for Register: SR_PCS_PKG2

Bits	Name	Memory Access	Description
15:10	PCS_PKG_OUI_19_24	R	Organizationally Unique Identifier DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. This value is configured by the host. This read-only field contains Bits[24:19] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS and AN) present in the package. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR Control MMD Package Identifier Register 2 modifies the contents of this register. **Value After Reset:** DWCXLP_IEEE_PKG_OUI_19_24 **Exists:** Always

Table 7-312 Fields for Register: SR_PCS_PKG2 (Continued)

Bits	Name	Memory Access	Description
9:4	PCS_PKG_MMN_5_0	R	Model Number This read-only field contains the 5-bit model number of the package that is common for all the MMDs in the Device. This value is configured by the host. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_MODEL_NO_0_5 **Exists:** Always
3:0	PCS_PKG_RN_3_0	R	Revision Number This read-only field contains the 4-bit revision number of the package in which the MMDs are installed. The value is configured by the host. It is common for all the MMDs in the device. It is common for all MMDs (PMA, PCS and AN) present in the package. If DWCXLP_IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR Control MMD Package Identifier Register 2 modifies the content of this register. Dependencies: The value is configured by the host. This value is common for all the MMDs in the device. **Value After Reset:** DWCXLP_IEEE_PKG_REV_NO_0_3 **Exists:** Always

7.2.12 SR_PCS_EEE_ABL

- **Description:** SR PCS MMD EEE Capability Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x14 : (DWCXLP_APB3_16) ? (0x14*0x2) : (0x14*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_EEE_EN

15:14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved_15_14	CGDSL	CGFWK	XXVGDSL	XXVGFWK	XLGDSL	XLGFWK	XLG_EEE	XG_EEE	KX4EEE	KXEEE	TEEE	EN_1GTEEE	EN_100TEE	FWK_EN

Table 7-313 Fields for Register: SR_PCS_EEE_ABL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	CGDSL	R	100G Deep Sleep Support This bit indicates the Deep Sleep support in the 100G mode: ■ 0: Deep Sleep is not supported in the 100G mode ■ 1: Deep Sleep is supported in the 100G mode This bit always returns 1 when 100G mode is supported in the configuration. **Value After Reset:** "(DWCXLP_KR4_100G&&DWCXLP_EEE_EN) ? 0x1 : 0x0" **Exists:** Always

Table 7-313 Fields for Register: SR_PCS_EEE_ABL (Continued)

Bits	Name	Memory Access	Description
12	CGFWK	R	100G FastWake Up Support This bit indicates the Fast Wake Up support in the 100G mode: ■ 0: Fast Wake Up is not supported in the 100G mode ■ 1: Fast Wake Up is supported in the 100G mode This bit always returns 1 when 100G mode is supported in the configuration **Value After Reset:** "(DWCXLP_KR4_100G&&DWCXLP_EEE_EN) ? 0x1 : 0x0" **Exists:** Always
11	XXVGDSL	R	25G Deep Sleep Support This bit indicates the Deep Sleep support in the 25G mode: ■ 0: Deep Sleep is not supported in the 25G mode ■ 1: Deep Sleep is supported in the 25G mode This bit always returns 1 when 25G mode is supported in the configuration. **Value After Reset:** "(!DWCXLP_KR4_KR&&DWCXLP_EEE_EN) ? 0x1 : 0x0" **Exists:** Always
10	XXVGFWK	R	25G FastWake Up Support This bit indicates the Fast Wake Up support in the 25G mode: ■ 0: Fast Wake Up is not supported in the 25G mode ■ 1: Fast Wake Up is supported in the 25G mode This bit always returns 1 when 25G mode is supported in the configuration **Value After Reset:** "(!DWCXLP_KR4_KR&&DWCXLP_EEE_EN) ? 0x1 : 0x0" **Exists:** Always
9	XLGDSL	R	40G Deep Sleep Support This bit indicates the Deep Sleep support in the 40G mode: ■ 0: Deep Sleep is not supported in the 40G mode ■ 1: Deep Sleep is supported in the 40G mode This bit always returns 1 when 40G mode is supported in the configuration. **Value After Reset:** "(!DWCXLP_25G_ONLY&&DWCXLP_EEE_EN) ? 0x1 : 0x0" **Exists:** Always

Table 7-313 Fields for Register: SR_PCS_EEE_ABL (Continued)

Bits	Name	Memory Access	Description
8	XLGFWK	R	40G FastWake Up Support This bit indicates the Fast Wake Up support in the 40G mode: ■ 0: Fast Wake Up is not supported in the 40G mode ■ 1: Fast Wake Up is supported in the 40G mode This bit always returns 1. **Value After Reset:** "(!DWCXLP_25G_ONLY&&DWCXLP_EEE_EN) ? 0x1 : 0x0" **Exists:** Always
7	XLG_EEE	R	40GBASE-KR Mode EEE Support This bit indicates if DWC_xlgpcs supports Energy Efficient Ethernet (EEE) in the 40GBASE-KR mode: ■ 0: EEE is not supported in 40GBASE-KR mode ■ 1: EEE is supported in 40GBASE-KR mode **Value After Reset:** "(!DWCXLP_25G_ONLY) ? 0x1 : 0x0" **Exists:** Always
6	XG_EEE	R	10GBASE-KR Mode EEE Support This bit indicates if DWC_xlgpcs supports Energy Efficient Ethernet (EEE) in the 10GBASE-R PCS or 10GBASE-KR mode: ■ 0: EEE is not supported in 10GBASE-R PCS or 10GBASE-KR mode ■ 1: EEE is supported in 10GBASE-R PCS or 10GBASE-KR mode Default Value: In KR_Present configurations, this bit always returns 1. Otherwise, this bit always returns 0. **Value After Reset:** 0x1 **Exists:** Always
5	KX4EEE	R	10GBASE-KX4 Mode EEE Support This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
4	KXEEE	R	KX EEE Support This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-313 Fields for Register: SR_PCS_EEE_ABL (Continued)

Bits	Name	Memory Access	Description
3	TEEE	R	10GBASE-T Mode EEE Support This bit indicates the EEE support in the 10GBASE-T mode: ■ 0: EEE is not supported in the 10GBASE-T mode ■ 1: EEE is supported in the 10GBASE-T mode **Value After Reset:** 0x0 **Exists:** Always
2	EN_1GTEEE	R	1000BASE-T Mode EEE Support This bit indicates the EEE support in the 1000BASE-T mode: ■ 0: EEE is not supported in the 1000BASE-T mode ■ 1: EEE is supported in the 1000BASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	EN_100TEE	R	100BASE-T Mode EEE Support This bit indicates the EEE support in the 100BASE-T mode: ■ 0: EEE is not supported in the 100BASE-T mode ■ 1: EEE is supported in the 100BASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
0	FWK_EN	R/W	EEE Fast Wake Up Enable. This bit controls Deep Sleep or Fast Wake Up operation. Setting this bit to 1 enables Fast Wake Up, setting to zero enables Deep Sleep mode. ■ 0: Fast WakeUp Mode is enabled ■ 1: Deep Sleep Mode is enabled (and Fast wake up is disabled) **Value After Reset:** 0x1 **Exists:** Always

7.2.13 SR_PCS_EEE_WKERR

- **Description:** SR PCS MMD EEE Wake Error Counter Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x16 : (DWCXLP_APB3_16) ? (0x16*0x2) : (0x16*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_EEE_EN



Table 7-314 Fields for Register: SR_PCS_EEE_WKERR

Bits	Name	Memory Access	Description
15:0	EEE_WEC	R	EEE Wake Error Counter This field indicates the Rx wake up error counter in the EEE mode. This counter is incremented if the Refresh signaling is not received in time when DWC_xlgpcs is in low-power mode. When this counter reaches 16'hFFFF, it stops incrementing. This counter is cleared by reading this register. **Value After Reset:** 0x0 **Exists:** Always

7.2.14 SR_PCS_BASER_STS1

- **Description:** SR PCS MMD BASE-R Status1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x20 : (DWCXLP_APB3_16) ? (0x20*0x2) : (0x20*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

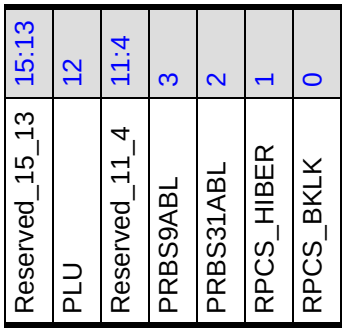


Table 7-315 Fields for Register: SR_PCS_BASER_STS1

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	PLU	R	RPCS Link Up When read as a one, this bit indicates that the PCS is in a fully operational state. When read as a zero, bit the PCS is not fully operational. When this bit is high, it indicates that the PCS-R 66-bit code group alignment is successful and the high bit rate condition is not encountered. If FEC is enabled, this bit also indicates that the FEC block lock is successful. ■ 1: BASE-R PCS receive link is up ■ 0: BASE-R PCS receive link is down **Value After Reset:** 0x0 **Exists:** Always
11:4	Reserved_11_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-315 Fields for Register: SR_PCS_BASER_STS1 (Continued)

Bits	Name	Memory Access	Description
3	PRBS9ABL	R	PRBS9 Testing Ability ■ 1: The DWC_xlgpcs supports the PRBS9 pattern testing ■ 0: The DWC_xlgpcs does not support the PRBS9 pattern testing This bit always returns 1 because the DWC_xlgpcs supports the PRBS9 testing ability. **Value After Reset:** 0x1 **Exists:** Always
2	PRBS31ABL	R	PRBS31 Testing Ability ■ 1: The DWC_xlgpcs supports the PRBS31 pattern testing ■ 0: The DWC_xlgpcs does not support the PRBS31 pattern testing This bit always returns 1 because the DWC_xlgpcs supports the PRBS31 testing ability. **Value After Reset:** 0x1 **Exists:** Always
1	RPCS_HIBER	R	BASE-R PCS High Bit Error Rate ■ 1: BASE-R PCS reporting a high BER ■ 0: BASE-R PCS not reporting a high BER When read as a one, this bit indicates that the 64B/66B receiver is detecting a BER of > 10^-4. When read as a zero, this bit indicates that the 64B/66B receiver is detecting a BER of < 10^-4. This bit is a direct reflection of the state of the hi_ber **Value After Reset:** 0x0 **Exists:** Always
0	RPCS_BKLK	R	BASE-R Block Lock ■ 1: BASE-R is locked (Block Alignment Success) to received blocks ■ 0: BASE-R is not locked to received blocks For a multi-lane PCS, this bit indicates that the receiver has both block lock and alignment for all lanes **Value After Reset:** 0x0 **Exists:** Always

7.2.15 SR_PCS_BASER_STS2

- **Description:** SR PCS MMD BASE-R Status2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x21 : (DWCXLP_APB3_16) ? (0x21*0x2) : (0x21*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

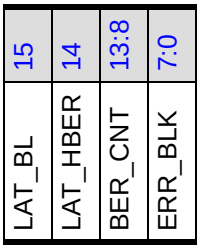


Table 7-316 Fields for Register: SR_PCS_BASER_STS2

Bits	Name	Memory Access	Description
15	LAT_BL	R	Latched Block Lock (RO,LL Type) ■ 1: BASE-R PCS achieved the 66-bit code group alignment ■ 0: BASE-R PCS did not achieve the 66-bit code group alignment **Value After Reset:** 0x0 **Exists:** Always
14	LAT_HBER	R	Latched High Bit Error Rate (RO,LH type) In 10G mode, High bit error is when more than 15 invalid 66-bit blocks are received in 125us duration. In 25G mode, High bit error is when more than 96 invalid 66-bit blocks are received in 2ms duration. In 40G mode, High bit error is when more than 96 invalid 66-bit blocks are received in 1.25ms duration. In 50G mode, High bit error is when more than 96 invalid 66-bit blocks are received in 1ms duration. In 100G mode, High bit error is when more than 96 invalid 66-bit blocks are received in 500us duration. ■ 1: BASE-R PCS is reporting a high BER ■ 0: BASE-R PCS is not reporting a high BER **Value After Reset:** 0x0 **Exists:** Always

Table 7-316 Fields for Register: SR_PCS_BASER_STS2 (Continued)

Bits	Name	Memory Access	Description
13:8	BER_CNT	R	Bit Error Rate Counter Value (RO, COR Type) This field indicates the high bit error count on the receive path. It increments for every invalid Sync Header. It stops incrementing when it reaches the value 6'h3F. This counter is cleared when read. This counter also gets cleared when 'ERR_BLK' field of this register is read. Hence byte-enable access to this register is not advisable. **Value After Reset:** 0x0 **Exists:** Always
7:0	ERR_BLK	R	Error Block Counter Value (RO, COR Type) This field indicates invalid 66-bit blocks in the receive path. It stops incrementing when it reaches the value 8'hFF. This counter is cleared when read. This counter also gets cleared when 'BER_CNT' field of this register is read. Hence byte-enable access to this register is not advisable. **Value After Reset:** 0x0 **Exists:** Always

7.2.16 SR_PCS_TP_A0

- **Description:** SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed A0 Register.

  This register contains Bits[15:0] of Test pattern seed A. The content of this register is used as an initial seed to load scrambler for generating the pseudo-random test patterns.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x22 : (DWCXLP_APB3_16) ? (0x22*0x2) : (0x22*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-317 Fields for Register: SR_PCS_TP_A0

Bits	Name	Memory Access	Description
15:0	TP_SA0	R/W	Test Pattern Seed A0 This field contains Bits[15:0] of Test pattern seed A. **Value After Reset:** 0x0 **Exists:** Always

7.2.17 SR_PCS_TP_A1

- **Description:** SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed A1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x23 : (DWCXLP_APB3_16) ? (0x23*0x2) : (0x23*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

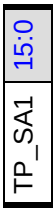


Table 7-318 Fields for Register: SR_PCS_TP_A1

Bits	Name	Memory Access	Description
15:0	TP_SA1	R/W	Test Pattern Seed A1 This field contains Bits[31:16] of Test pattern seed A. **Value After Reset:** 0x0 **Exists:** Always

7.2.18 SR_PCS_TP_A2

- **Description:** SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed A2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x24 : (DWCXLP_APB3_16) ? (0x24*0x2) : (0x24*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

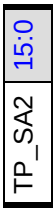


Table 7-319 Fields for Register: SR_PCS_TP_A2

Bits	Name	Memory Access	Description
15:0	TP_SA2	R/W	Test Pattern Seed A2 This field contains Bits[47:32] of Test pattern seed A. **Value After Reset:** 0x0 **Exists:** Always

7.2.19 SR_PCS_TP_A3

- **Description:** SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed A3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x25 : (DWCXLP_APB3_16) ? (0x25*0x2) : (0x25*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

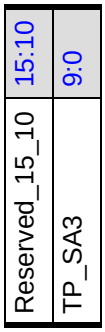


Table 7-320 Fields for Register: SR_PCS_TP_A3

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:0	TP_SA3	R/W	Test Pattern Seed A3 This field contains Bits[57:48] of Test pattern seed A. **Value After Reset:** 0x0 **Exists:** Always

7.2.20 SR_PCS_TP_B0

- **Description:** SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed B0 Register

  This register contains Bits[15:0] of Test pattern seed B. The content of this register is used as an initial seed to load scrambler for generating the pseudo-random test patterns.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x26 : (DWCXLP_APB3_16) ? (0x26*0x2) : (0x26*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-321 Fields for Register: SR_PCS_TP_B0

Bits	Name	Memory Access	Description
15:0	TP_SB0	R/W	Test Pattern Seed B0 This field contains Bits[15:0] of Test pattern seed B. **Value After Reset:** 0x0 **Exists:** Always

7.2.21 SR_PCS_TP_B1

- **Description:** SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed B1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x27 : (DWCXLP_APB3_16) ? (0x27*0x2) : (0x27*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

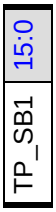


Table 7-322 Fields for Register: SR_PCS_TP_B1

Bits	Name	Memory Access	Description
15:0	TP_SB1	R/W	Test Pattern Seed B1 This field contains Bits[31:16] of Test pattern seed B. **Value After Reset:** 0x0 **Exists:** Always

7.2.22 SR_PCS_TP_B2

- **Description:** SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed B2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x28 : (DWCXLP_APB3_16) ? (0x28*0x2) : (0x28*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

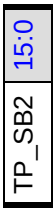


Table 7-323 Fields for Register: SR_PCS_TP_B2

Bits	Name	Memory Access	Description
15:0	TP_SB2	R/W	Test Pattern Seed B2 This field contains Bits[47:32] of Test pattern seed B. **Value After Reset:** 0x0 **Exists:** Always

7.2.23 SR_PCS_TP_B3

- **Description:** SR PCS MMD 10G/25GBASE-R Pseudo-Random Test Pattern Seed B3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x29 : (DWCXLP_APB3_16) ? (0x29*0x2) : (0x29*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

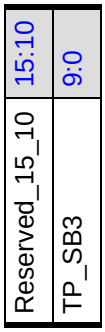


Table 7-324 Fields for Register: SR_PCS_TP_B3

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:0	TP_SB3	R/W	Test Pattern Seed B3 This field contains Bits[57:48] of Test pattern seed B. **Value After Reset:** 0x0 **Exists:** Always

7.2.24 SR_PCS_TP_CTRL

- **Description:** SR PCS BASE-R Test Pattern Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x2A : (DWCXLP_APB3_16) ? (0x2A*0x2) : (0x2A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

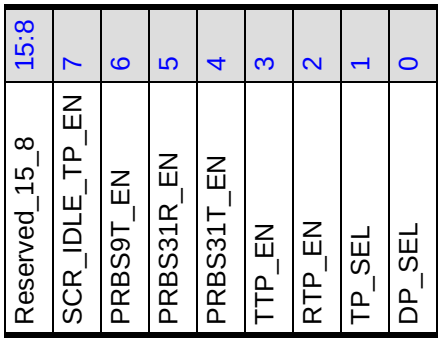


Table 7-325 Fields for Register: SR_PCS_TP_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	SCR_IDLE_TP_EN	R/W	Scrambled Idle Transmit Test Pattern Enable ■ 1: Enable scrambled idle test pattern on the Transmit path when Bit 3 (TTP_EN) is set to 1 and Bit 1 (TP_SEL) is set to 0 ■ 0: Disable scrambled idle test-pattern mode on the Transmit path **Value After Reset:** 0x0 **Exists:** Always
6	PRBS9T_EN	R/W	PRBS9 Transmit Test Pattern Enable ■ 1: Enable PRBS9 test pattern on the Transmit path when Bit 3 (TTP_EN), Bit 1 (TP_SEL), and Bit 4 and Bit 7 are set to 0 ■ 0: Disable PRBS9 test-pattern mode on the Transmit path **Value After Reset:** 0x0 **Exists:** Always

Table 7-325 Fields for Register: SR_PCS_TP_CTRL (Continued)

Bits	Name	Memory Access	Description
5	PRBS31R_EN	R/W	PRBS31 Receive Test Pattern Enable ■ 1: Enable PRBS31 test pattern checking on the Receive path when Bit 2 (RTP_EN) is set to 0 ■ 0: Disable PRBS31 test pattern checking on the Receive path **Value After Reset:** 0x0 **Exists:** Always
4	PRBS31T_EN	R/W	PRBS31 Transmit Pattern Enable ■ 1: Enable PRBS31 test pattern on the Transmit path when Bit 3 (TTP_EN) and Bit 1 (TP_SEL) of this register is set to 0 ■ 0: Disable PRBS31 test-pattern mode on the Transmit path **Value After Reset:** 0x0 **Exists:** Always
3	TTP_EN	R/W	Transmit Test-pattern Enable This bit must be set to send Pseudo-Random or Square Wave or Scrambled Idle Pattern in the transmit path ■ 1: Enables pseudo-random or square wave or scrambled idle test pattern depending on Bit 1 (TP_SEL) ■ 0: Disables pseudo-random or square wave or scrambled test patterns This bit must be set to 0 to enable PRBS31 or PRBS9 test pattern. You can enable the PRBS31 or PRBS9 test pattern by setting Bit 4 (PRBS31T_EN) or Bit 6 (PRBS9T_EN) to 1, respectively. **Value After Reset:** 0x0 **Exists:** Always
2	RTP_EN	R/W	Receive Test Pattern Enable ■ 1: Enables receive pseudo-random test pattern when Bit 1 (TP_SEL) is set to 0 ■ 0: Disables receive pseudo-random test pattern This field must be set to 0 to enable PRBS31 or PRBS9 test pattern in the Receive path. You can enable the PRBS31 or PRBS9 test pattern by setting Bit 5 (PRBS31R_EN) of this register or Bit 7 (PRBS9RXEN) of VR PCS BASE-R Control Register to 1, respectively. **Value After Reset:** 0x0 **Exists:** Always

Table 7-325 Fields for Register: SR_PCS_TP_CTRL (Continued)

Bits	Name	Memory Access	Description
1	TP_SEL	R/W	Transmit Test Pattern Select ■ 1: DWC_xlgpcs transmits the square wave test pattern. Bits [6:4] of VR PCS BASE-R Control Register specify the period of the square wave. ■ 0: DWC_xlgpcs transmits the pseudo random test pattern. **Value After Reset:** 0x0 **Exists:** Always
0	DP_SEL	R/W	Data Pattern Select for Pseudo-Random Testing This bit selects the data given to the scrambler when pseudo-random test pattern is enabled (Bit 3 (TTP_EN) is set). ■ 1: All-zero ■ 0: Local-fault pattern (LF) You can change the data pattern by programming Bits [3:1] of VR PCS KR Control Register. **Value After Reset:** 0x0 **Exists:** Always

7.2.25 SR_PCS_TP_ERRCTR

- **Description:** SR PCS BASE-R Test Pattern Error Counter Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x2B : (DWCXLP_APB3_16) ? (0x2B*0x2) : (0x2B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-326 Fields for Register: SR_PCS_TP_ERRCTR

Bits	Name	Memory Access	Description
15:0	TP_ERR_CNT	R	Test Pattern Error Counter (RO, COR Type) If checking for any test pattern is enabled in the Rx path, this field indicates the test pattern mismatch error count in terms of blocks. This counter stops when it becomes all-ones. It gets cleared when this register is read. **Value After Reset:** 0x0 **Exists:** Always

7.2.26 SR_PCS_BER_HI_ORDER_CNT

- **Description:** SR PCS BASE-R BER High Order Counter Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x2c : (DWCXLP_APB3_16) ? (0x2c*0x2) : (0x2c*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-327 Fields for Register: SR_PCS_BER_HI_ORDER_CNT

Bits	Name	Memory Access	Description
15:0	BER_HO	R	BER High Order Counter (RO, COR Type) If checking for bit error rate high order count in the receive path, this field indicates the bits [21:6] of BER counter. **Value After Reset:** 0x0 **Exists:** Always

7.2.27 SR_PCS_ERR_BLK_HI_ORDER_CNT

- **Description:** SR PCS BASE-R Errored Blocks High Order Counter Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x2d : (DWCXLP_APB3_16) ? (0x2d*0x2) : (0x2d*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

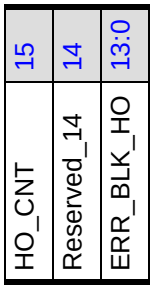
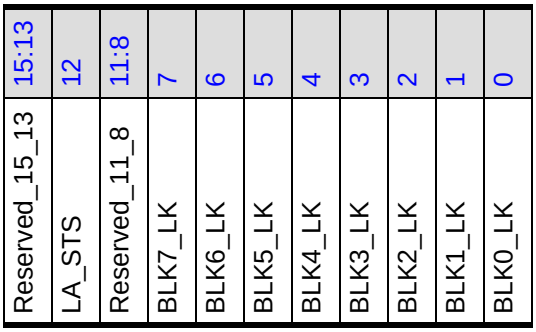


Table 7-328 Fields for Register: SR_PCS_ERR_BLK_HI_ORDER_CNT

Bits	Name	Memory Access	Description
15	HO_CNT	R	High Order Counter This bit always returns one. **Value After Reset:** 0x1 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:0	ERR_BLK_HO	R	Errored Blocks High Order Counter (RO, COR Type). If checking for Errored Blocks High Order Count in the receive path, this field indicates the bits [21:8] of Errored Blocks Counter. This counter gets cleared on read. **Value After Reset:** 0x0 **Exists:** Always

7.2.28 SR_PCS_MLN_ALIGN_STS1

- **Description:** SR PCS BASE-R Multi-lane PCS Alignment Status 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x32 : (DWCXLP_APB3_16) ? (0x32*0x2) : (0x32*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_MULTI_LANE||DWCXLP_50G_25G



15:13	12	11:8	7	6	5	4	3	2	1	0
Reserved_15_13	LA_STS	Reserved_11_8	BLK7_LK	BLK6_LK	BLK5_LK	BLK4_LK	BLK3_LK	BLK2_LK	BLK1_LK	BLK0_LK

Table 7-329 Fields for Register: SR_PCS_MLN_ALIGN_STS1

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	LA_STS	R	PCS Lane Alignment Status. ■ 1: PCS receive lanes locked and aligned ■ 0: PCS receive lanes not locked or not aligned **Value After Reset:** 0x0 **Exists:** Always
11:8	Reserved_11_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	BLK7_LK	R	PCS lane 7 block lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 7 has achieved block lock. ■ 0: PCS receive lane 7 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always

Table 7-329 Fields for Register: SR_PCS_MLN_ALIGN_STS1 (Continued)

Bits	Name	Memory Access	Description
6	BLK6_LK	R	PCS lane 6 block lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 6 has achieved block lock. ■ 0: PCS receive lane 6 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always
5	BLK5_LK	R	PCS lane 5 block lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 5 has achieved block lock. ■ 0: PCS receive lane 5 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always
4	BLK4_LK	R	PCS lane 4 block lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 4 has achieved block lock. ■ 0: PCS receive lane 4 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always
3	BLK3_LK	R	PCS lane 3 block lock status ■ 1: PCS receive lane 3 has achieved block lock. ■ 0: PCS receive lane 3 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always
2	BLK2_LK	R	PCS lane 2 block lock status ■ 1: PCS receive lane 2 has achieved block lock. ■ 0: PCS receive lane 2 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always
1	BLK1_LK	R	PCS lane 1 block lock status ■ 1: PCS receive lane 1 has achieved block lock. ■ 0: PCS receive lane 1 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always

Table 7-329 Fields for Register: SR_PCS_MLN_ALIGN_STS1 (Continued)

Bits	Name	Memory Access	Description
0	BLK0_LK	R	PCS lane 0 block lock status ■ 1: PCS receive lane 0 has achieved block lock. ■ 0: PCS receive lane 0 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always

7.2.29 SR_PCS_MLN_ALIGN_STS2

- **Description:** SR PCS BASE-R Multi-lane PCS Alignment Status 2 Register. All the block lock status bits present in this register are applicable to 100G no FEC mode of operation only.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x33 : (DWCXLP_APB3_16) ? (0x33*0x2) : (0x33*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_KR4_100G

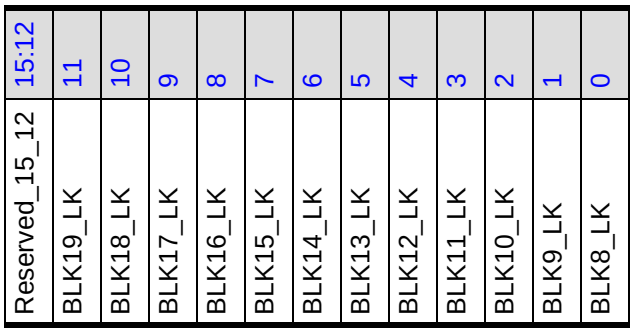


Table 7-330 Fields for Register: SR_PCS_MLN_ALIGN_STS2

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	BLK19_LK	R	PCS lane 19 block lock status ■ 1: PCS receive lane 19 has achieved block lock ■ 0: PCS receive lane 19 has not achieved block lock **Value After Reset:** 0x0 **Exists:** Always
10	BLK18_LK	R	PCS lane 18 block lock status ■ 1: PCS receive lane 18 has achieved block lock ■ 0: PCS receive lane 18 has not achieved block lock **Value After Reset:** 0x0 **Exists:** Always

Table 7-330 Fields for Register: SR_PCS_MLN_ALIGN_STS2 (Continued)

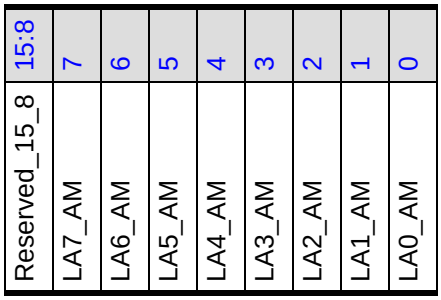
Bits	Name	Memory Access	Description
9	BLK17_LK	R	PCS lane 17 block lock status - 1: PCS receive lane 17 has achieved block lock - 0: PCS receive lane 17 has not achieved block lock **Value After Reset:** 0x0 **Exists:** Always
8	BLK16_LK	R	PCS lane 16 block lock status ■ 1: PCS receive lane 16 has achieved block lock ■ 0: PCS receive lane 16 has not achieved block lock **Value After Reset:** 0x0 **Exists:** Always
7	BLK15_LK	R	PCS lane 15 block lock status ■ 1: PCS receive lane 15 has achieved block lock ■ 0: PCS receive lane 15 has not achieved block lock **Value After Reset:** 0x0 **Exists:** Always
6	BLK14_LK	R	PCS lane 14 block lock status ■ 1: PCS receive lane 14 has achieved block lock ■ 0: PCS receive lane 14 has not achieved block lock **Value After Reset:** 0x0 **Exists:** Always
5	BLK13_LK	R	PCS lane 13 block lock status ■ 1: PCS receive lane 13 has achieved block lock ■ 0: PCS receive lane 13 has not achieved block lock **Value After Reset:** 0x0 **Exists:** Always
4	BLK12_LK	R	PCS lane 12 block lock status ■ 1: PCS receive lane 12 has achieved block lock ■ 0: PCS receive lane 12 has not achieved block lock **Value After Reset:** 0x0 **Exists:** Always
3	BLK11_LK	R	PCS lane 11 block lock status ■ 1: PCS receive lane 11 has achieved block lock. ■ 0: PCS receive lane 11 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always

Table 7-330 Fields for Register: SR_PCS_MLN_ALIGN_STS2 (Continued)

Bits	Name	Memory Access	Description
2	BLK10_LK	R	PCS lane 10 block lock status ■ 1: PCS receive lane 10 has achieved block lock. ■ 0: PCS receive lane 10 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always
1	BLK9_LK	R	PCS lane 9 block lock status ■ 1: PCS receive lane 9 has achieved block lock. ■ 0: PCS receive lane 9 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always
0	BLK8_LK	R	PCS lane 8 block lock status ■ 1: PCS receive lane 8 has achieved block lock. ■ 0: PCS receive lane 8 has not achieved block lock. **Value After Reset:** 0x0 **Exists:** Always

7.2.30 SR_PCS_MLN_ALIGN_STS3

- **Description:** SR PCS BASE-R Multi-lane PCS Alignment Status 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x34 : (DWCXLP_APB3_16) ? (0x34*0x2) : (0x34*0x4)" + "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_MULTI_LANE| |DWCXLP_50G_25G



15:8	7	6	5	4	3	2	1	0
Reserved_15_8	LA7_AM	LA6_AM	LA5_AM	LA4_AM	LA3_AM	LA2_AM	LA1_AM	LA0_AM

Table 7-331 Fields for Register: SR_PCS_MLN_ALIGN_STS3

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	LA7_AM	R	PCS lane 7 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 7 has achieved Alignment Marker lock ■ 0: PCS receive lane 7 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
6	LA6_AM	R	PCS lane 6 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 6 has achieved Alignment Marker lock ■ 0: PCS receive lane 6 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always

Table 7-331 Fields for Register: SR_PCS_MLN_ALIGN_STS3 (Continued)

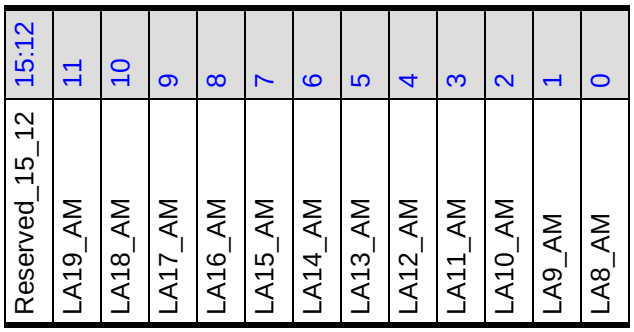
Bits	Name	Memory Access	Description
5	LA5_AM	R	PCS lane 5 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 5 has achieved Alignment Marker lock ■ 0: PCS receive lane 5 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
4	LA4_AM	R	PCS lane 4 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 4 has achieved Alignment Marker lock ■ 0: PCS receive lane 4 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
3	LA3_AM	R	PCS lane 3 Alignment Marker lock status ■ 1: PCS receive lane 3 has achieved Alignment Marker lock ■ 0: PCS receive lane 3 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
2	LA2_AM	R	PCS lane 2 Alignment Marker lock status ■ 1: PCS receive lane 2 has achieved Alignment Marker lock ■ 0: PCS receive lane 2 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
1	LA1_AM	R	PCS lane 1 block Alignment Marker status ■ 1: PCS receive lane 1 has achieved Alignment Marker lock ■ 0: PCS receive lane 1 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always

Table 7-331 Fields for Register: SR_PCS_MLN_ALIGN_STS3 (Continued)

Bits	Name	Memory Access	Description
0	LA0_AM	R	PCS lane 0 Alignment Marker lock status ■ 1: PCS receive lane 0 has achieved Alignment Marker lock ■ 0: PCS receive lane 0 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always

7.2.31 SR_PCS_MLN_ALIGN_STS4

- **Description:** SR PCS BASE-R Multi-lane PCS Alignment Status 4 Register . The alignment status fields in this register are applicable for 100G no FEC mode only
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x35 : (DWCXLP_APB3_16) ? (0x35*0x2) : (0x35*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_MULTI_LANE



15:12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved_15_12	LA19_AM	LA18_AM	LA17_AM	LA16_AM	LA15_AM	LA14_AM	LA13_AM	LA12_AM	LA11_AM	LA10_AM	LA9_AM	LA8_AM

Table 7-332 Fields for Register: SR_PCS_MLN_ALIGN_STS4

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	LA19_AM	R	PCS lane 19 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 19 has achieved Alignment Marker lock ■ 0: PCS receive lane 19 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
10	LA18_AM	R	PCS lane 18 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 18 has achieved Alignment Marker lock ■ 0: PCS receive lane 18 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always

Table 7-332 Fields for Register: SR_PCS_MLN_ALIGN_STS4 (Continued)

Bits	Name	Memory Access	Description
9	LA17_AM	R	PCS lane 17 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 17 has achieved Alignment Marker lock ■ 0: PCS receive lane 17 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
8	LA16_AM	R	PCS lane 16 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 16 has achieved Alignment Marker lock ■ 0: PCS receive lane 16 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
7	LA15_AM	R	PCS lane 15 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 15 has achieved Alignment Marker lock ■ 0: PCS receive lane 15 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
6	LA14_AM	R	PCS lane 14 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 14 has achieved Alignment Marker lock ■ 0: PCS receive lane 14 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
5	LA13_AM	R	PCS lane 13 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 13 has achieved Alignment Marker lock ■ 0: PCS receive lane 13 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always

Table 7-332 Fields for Register: SR_PCS_MLN_ALIGN_STS4 (Continued)

Bits	Name	Memory Access	Description
4	LA12_AM	R	PCS lane 12 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 12 has achieved Alignment Marker lock ■ 0: PCS receive lane 12 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
3	LA11_AM	R	PCS lane 11 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 11 has achieved Alignment Marker lock ■ 0: PCS receive lane 11 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
2	LA10_AM	R	PCS lane 10 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 10 has achieved Alignment Marker lock ■ 0: PCS receive lane 10 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
1	LA9_AM	R	PCS lane 9 block Alignment Marker status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 9 has achieved Alignment Marker lock ■ 0: PCS receive lane 9 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always
0	LA8_AM	R	PCS lane 8 Alignment Marker lock status This is applicable only in 100G no FEC mode. ■ 1: PCS receive lane 8 has achieved Alignment Marker lock ■ 0: PCS receive lane 8 has not achieved Alignment Marker lock **Value After Reset:** 0x0 **Exists:** Always

7.2.32 SR_PCS_BIP_ERR_CNTR_LANE (i) (for i = 0; i <= 19)

- **Description:** SR PCS BASE-R PCS BIP Error Counter Register -Lane *i* (RO, COR Type)

  Note: In 40G and 50G configurations only SR_PCS_BIP_ERR_CNTR_LANE-0,1,2,3 is present. In 100G configuration, SR_PCS_BIP_ERR_CNTR_LANE-0,1,2,...18.19 is present.

  Note: In 25G mode, there is no BIP. However, this field can indicate a random value. This value must be ignored.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** i+0x300c8
- **Exists:** DWCXLP_MULTI_LANE||DWCXLP_50G_25G||DWCXLP_100G_1LANE



Table 7-333 Fields for Register: SR_PCS_BIP_ERR_CNTR_LANE (i) (for i = 0; i <= 19)

Bits	Name	Memory Access	Description
15:0	BIP_ERR_CNTR0	R	BIP Error Counter for Lane *i* (RO, COR Type). This field indicates the number of BIP Error detected by PCS on lane *i*. **Value After Reset:** 0x0 **Exists:** Always

7.2.33 SR_PCS_LANE (i) MAPPING (for i = 0; i <= 19)

- **Description:** SR PCS BASE-R Lane *i* Mapping Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** i+0x30190
- **Exists:** DWCXLP_MULTI_LANE||DWCXLP_50G_25G

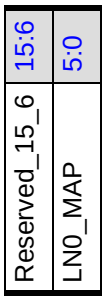
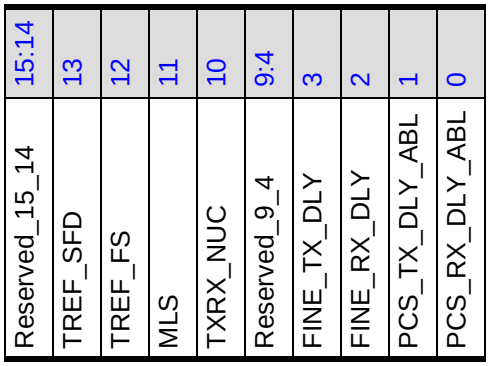


Table 7-334 Fields for Register: SR_PCS_LANE (i) MAPPING (for i = 0; i <= 19)

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	LN0_MAP	R	PCS lane received in service interface lane *i* . In 40G and 50G configurations only SR_PCS_LANE-0,1,2,3 is present. In 100G configurations SR_PCS_LANE-0,1,....,18,19 registers is present **Value After Reset:** 0x0 **Exists:** Always

7.2.34 SR_PCS_TIME_SYNC_PCS_ABL

- **Description:** SR PCS MMD Time Sync Capability Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x0708 : (DWCXLP_APB3_16) ? (0x0708*0x2) : (0x0708*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



15:14	13	12	11	10	9:4	3	2	1	0
Reserved_15_14	TREF_SFD	TREF_FS	MLS	TXRX_NUC	Reserved_9_4	FINE_TX_DLY	FINE_RX_DLY	PCS_TX_DLY_ABL	PCS_RX_DLY_ABL

Table 7-335 Fields for Register: SR_PCS_TIME_SYNC_PCS_ABL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	TREF_SFD	R	Timestamp reference at SFD - 0: DWC_xlgpcs does not support data delay measurement point at the beginning of SFD - 1: DWC_xlgpcs supports data delay measurement point at the beginning of SFD **Value After Reset:** "DWCXLP_TS_EN ? 0x1 : 0x0" **Exists:** Always
12	TREF_FS	R	Timestamp reference at first symbol after SFD - 0: DWC_xlgpcs does not support data delay measurement point at the beginning of the first symbol after SFD - 1: DWC_xlgpcs supports data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** "DWCXLP_TS_EN ? 0x1 : 0x0" **Exists:** Always

Table 7-335 Fields for Register: SR_PCS_TIME_SYNC_PCS_ABL (Continued)

Bits	Name	Memory Access	Description
11	MLS	R	Multi-lane Support. ■ 0: DWC_xlgpcs does not support reporting of multi-PCS lane path data delay ■ 1: DWC_xlgpcs supports reporting of multi-PCS lane path data delay **Value After Reset:** "DWCXLP_TS_EN&&!(DWCXLP_25G_ONLY||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 0x1 : 0x0" **Exists:** Always
10	TXRX_NUC	R	Transmit or Receive Num_Unit_Change Support Available. If this bit is set, it indicates that DWC_xlgpcs supports Transmit/Receive Num_Unit_Change indication capability **Value After Reset:** "DWCXLP_TS_EN ? 0x1 : 0x0" **Exists:** Always
9:4	Reserved_9_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	FINE_TX_DLY	R	Fine Resolution Transmit Path Data Delay Available. If this bit is set, it indicates that DWC_xlgpcs provides information on transmit data path delay with sub-ns resolution in SR_PCS_TIME_SYNC_TX_MAX_FINE_DLY and SR_PCS_TIME_SYNC_TX_MIN_FINE_DLY registers. **Value After Reset:** "DWCXLP_TS_EN ? 0x1 : 0x0" **Exists:** Always
2	FINE_RX_DLY	R	Fine Resolution Receive Path Data Delay Available. If this bit is set, it indicates that DWC_xlgpcs provides information on receive data path delay with sub-ns resolution in SR_PCS_TIME_SYNC_RX_MAX_FINE_DLY and SR_PCS_TIME_SYNC_RX_MIN_FINE_DLY registers. **Value After Reset:** "DWCXLP_TS_EN ? 0x1 : 0x0" **Exists:** Always
1	PCS_TX_DLY_ABL	R	PCS Transmit Path Data Delay Information Available. If this bit is set, it indicates that the information regarding the maximum and minimum transmit data delay of PCS is available in registers 3.1801 through 3.1804 (Register number, see the IEEE numbers) **Value After Reset:** 0x1 **Exists:** Always

Table 7-335 Fields for Register: SR_PCS_TIME_SYNC_PCS_ABL (Continued)

Bits	Name	Memory Access	Description
0	PCS_RX_DLY_ABL	R	PCS Receive Path Data Delay Information Available. If this bit is set, it indicates that the information regarding the maximum and minimum receive data delay of PCS is available in registers 3.1805 through 3.1808 (Register number, see the IEEE numbers) **Value After Reset:** 0x1 **Exists:** Always

7.2.35 SR_PCS_TIME_SYNC_TX_MAX_DLY_LWR

- **Description:** SR PCS MMD Time Sync Tx Max Delay Lower Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x0709 : (DWCXLP_APB3_16) ? (0x0709*0x2) : (0x0709*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-336 Fields for Register: SR_PCS_TIME_SYNC_TX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_LWR	R	PCS Transmit Max Delay Lower This field indicates the lower 16-bit of the 32-bit value which indicates integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xlgpcs. The value returned by this field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xlgpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 0x27 : (DWCXLP_KR4_KR) ? 0x5A : 0x2C" **Exists:** Always

7.2.36 SR_PCS_TIME_SYNC_TX_MAX_DLY_UPR

- **Description:** SR PCS MMD Time Sync Tx Max Delay Upper Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070A : (DWCXLP_APB3_16) ? (0x070A*0x2) : (0x070A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-337 Fields for Register: SR_PCS_TIME_SYNC_TX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_UPR	R	PCS Transmit Max Delay Upper This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xlgpcs. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xlgpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** 0x0 **Exists:** Always

7.2.37 SR_PCS_TIME_SYNC_TX_MIN_DLY_LWR

- **Description:** SR PCS MMD Time Sync Tx Min Delay Lower Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070B : (DWCXLP_APB3_16) ? (0x070B*0x2) : (0x070B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

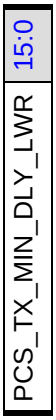


Table 7-338 Fields for Register: SR_PCS_TIME_SYNC_TX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_LWR	R	PCS Transmit Min Delay Lower This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xlgpcs. The value returned by this register field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xlgpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 0x1C : (DWCXLP_KR4_KR) ? 0x49 : 0x1C" **Exists:** Always

7.2.38 SR_PCS_TIME_SYNC_TX_MIN_DLY_UPR

- **Description:** SR PCS MMD Time Sync Tx Min Delay Upper Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x070C : (DWCXLP_APB3_16) ? (0x070C*0x2) : (0x070C*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-339 Fields for Register: SR_PCS_TIME_SYNC_TX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_UPR	R	PCS Transmit Min Delay Upper This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xlgpcs. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xlgpcs in the transmit path from the DDMP point until the final Tx data output port **Value After Reset:** 0x0 **Exists:** Always

7.2.39 SR_PCS_TIME_SYNC_RX_MAX_DLY_LWR

- **Description:** SR PCS MMD Time Sync Rx Max Delay Lower Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x070D : (DWCXLP_APB3_16) ? (0x070D*0x2) : (0x070D*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-340 Fields for Register: SR_PCS_TIME_SYNC_RX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_LWR	R	PCS Receive Max Delay Lower This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xlgpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xlgpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? (DWCXLP_CLK_CMP_EN ? 0x52: 0x3E) : (DWCXLP_KR4_KR) ? (DWCXLP_CLK_CMP_EN ? 0xFA: 0xED) : (DWCXLP_KR4_50G&&!DWCXLP_KR4_100G) ? (DWCXLP_CLK_CMP_EN ? 0x83: 0x79): (DWCXLP_CLK_CMP_EN ? 0x8A: 0x80)" **Exists:** Always

7.2.40 SR_PCS_TIME_SYNC_RX_MAX_DLY_UPR

- **Description:** SR PCS MMD Time Sync Rx Max Delay Upper Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070E : (DWCXLP_APB3_16) ? (0x070E*0x2) : (0x070E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-341 Fields for Register: SR_PCS_TIME_SYNC_RX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_UPR	R	PCS Receive Max Delay Upper This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xlgpcs. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xlgpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** 0x0 **Exists:** Always

7.2.41 SR_PCS_TIME_SYNC_RX_MIN_DLY_LWR

- **Description:** SR PCS MMD Time Sync Rx Min Delay Lower Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070F : (DWCXLP_APB3_16) ? (0x070F*0x2) : (0x070F*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-342 Fields for Register: SR_PCS_TIME_SYNC_RX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_LWR	R	PCS Receive Min Delay Lower This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xlgpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xlgpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** "(DWCXLP_25G_ONLY||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? (DWCXLP_CLK_CMP_EN ? 0x42: 0x2C) : (DWCXLP_KR4_KR) ? (DWCXLP_CLK_CMP_EN ? 0xD3 : 0xC6) : (DWCXLP_KR4_50G&&!DWCXLP_KR4_100G) ? (DWCXLP_CLK_CMP_EN ? 0x6E : 0x64) : (DWCXLP_CLK_CMP_EN ? 0x7D : 0x73)" **Exists:** Always

7.2.42 SR_PCS_TIME_SYNC_RX_MIN_DLY_UPR

- **Description:** SR PCS MMD Time Sync Rx Min Delay Upper Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x0710 : (DWCXLP_APB3_16) ? (0x0710*0x2) : (0x0710*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always



Table 7-343 Fields for Register: SR_PCS_TIME_SYNC_RX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_UPR	R	PCS Receive Min Delay Upper This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xlgpcs. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xlgpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** 0x0 **Exists:** Always

7.2.43 SR_PCS_TIME_SYNC_TX_MAX_FINE_DLY

- **Description:** SR PCS MMD Time Sync Tx Max Fine Delay Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x0711 : (DWCXLP_APB3_16) ? (0x0711*0x2) : (0x0711*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_TS_EN



Table 7-344 Fields for Register: SR_PCS_TIME_SYNC_TX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MAXDLY	R	Transmit Maximum Fine Delay This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xlgpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

7.2.44 SR_PCS_TIME_SYNC_TX_MIN_FINE_DLY

- **Description:** SR PCS MMD Time Sync Tx Min Fine Delay Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x0712 : (DWCXLP_APB3_16) ? (0x0712*0x2) : (0x0712*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_TS_EN



Table 7-345 Fields for Register: SR_PCS_TIME_SYNC_TX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MINDLY	R	Transmit Minimum Fine Delay This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xlgpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

7.2.45 SR_PCS_TIME_SYNC_RX_MAX_FINE_DLY

- **Description:** SR PCS MMD Time Sync Rx Max Fine Delay Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x0713 : (DWCXLP_APB3_16) ? (0x0713*0x2) : (0x0713*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_TS_EN



Table 7-346 Fields for Register: SR_PCS_TIME_SYNC_RX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MAXDLY	R	Receive Maximum Fine Delay This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xlgpcs receieve path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

7.2.46 SR_PCS_TIME_SYNC_RX_MIN_FINE_DLY

- **Description:** SR PCS MMD Time Sync Rx Min Fine Delay Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x0714 : (DWCXLP_APB3_16) ? (0x0714*0x2) : (0x0714*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_TS_EN

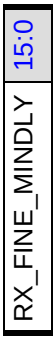


Table 7-347 Fields for Register: SR_PCS_TIME_SYNC_RX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MINDLY	R	Receive Minimum Fine Delay This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xlgpcs receieve path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

7.2.47 SR_PCS_TIME_SYNC_CFG

- **Description:** SR PCS MMD Time Sync Configuration Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x0715 : (DWCXLP_APB3_16) ? (0x0715*0x2) : (0x0715*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_TS_EN

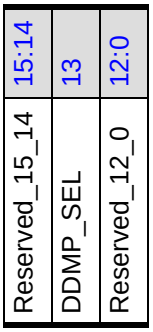


Table 7-348 Fields for Register: SR_PCS_TIME_SYNC_CFG

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	DDMP_SEL	R/W	Data Delay Measurement Point Selection ■ 0: DWC_xlgpcs is configured to use the data delay measurement point at the beginning of SFD ■ 1: DWC_xlgpcs is configured to use the data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** 0x0 **Exists:** Always
12:0	Reserved_12_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.2.48 VR_PCS_DIG_CTRL1

- **Description:** VR PCS MMD Digital Control1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8000 : (DWCXLP_APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

15	14	13	12	11	10	9	8	7:5	4	3	2	1	0
VR_RST	R2TLBE	EN_VSMMD1	CL37_BP	PWRSV	Reserved_10	RX_INIT	TX_INIT	DTXLANED_3_1	DTXLANED_0	CR_CJN	LANE_CTRL_EXT	BYP_PWRUP	DSKBYP

Table 7-349 Fields for Register: VR_PCS_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-Specific Soft Reset (RW,SC Type) When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block and CSR block. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: ■ For Synopsys PHY: This bit is self cleared when Bits[4:2] in VR PCS MMD Digital Status Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. ■ For Non-Synopsys PHY: This bit is self cleared after the following: - 32 clk_csr_i clocks for the MCI interface - 1 MDC clock period for the MDIO interface **Value After Reset:** 0x0 **Exists:** Always

Table 7-349 Fields for Register: VR_PCS_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
14	R2TLBE	R/W	Rx to Tx Loopback Enable This bit controls the loopback path from the XGMII Rx to the XGMII Tx at the XGMII interface. ■ 0: Loopback path is disabled. ■ 1: Loopback path is enabled. You can use this bit for debugging purpose. **Value After Reset:** 0x0 **Exists:** Always
13	EN_VSMMD1	R/W	Enable Vendor-Specific MMD1 When this bit is set to 1, the Vendor-Specific MMD1 (VSMMD1) is enabled. When this bit is set to zero, VSMMD1 is disabled. **Value After Reset:** 0x1 **Exists:** Always
12	CL37_BP	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	PWRSV	* Varies	For Synopsys PHY configurations: Power Save This bit is present only when the DWC_xlgpcs is configured for Synopsys PHY for all MMDs. ■ 0: Normal operation. ■ 1: DWC_xlgpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xlgpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys Multi-protocol PHY, DWC_xlgpcs asserts xlgpcs_tx_lpd_o{lane} and xlgpcs_rx_lpd_o{lane} when this bit is set. When the host clears this bit, DWC_xlgpcs resumes normal operation. After clearing this bit, the host must wait until Bits[4:2] of the VR PCS MMD Digital Status Register indicate that the DWC_xlgpcs is in a normal state. For non-Synopsys PHY configurations: ■ Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Table 7-349 Fields for Register: VR_PCS_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
10	Reserved_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	RX_INIT	R/W	Rx Datapath Initialization Control This bit can be set to flush/initialize the various FIFOs implemented in Receive data-path of DWC_xlgpcs. This is Self-Clear bit. After writing 1 to this bit, application should poll this bit continuously. Applicaition should proceed to any other operation, only after reading this bit as 0. **Value After Reset:** 0x0 **Exists:** Always
8	TX_INIT	* Varies	Tx Datapath Initialization Control This bit can be set to flush/initialize the various FIFOs implemented in Transmit data-path of DWC_xlgpcs. This is Self-Clear bit. After writing 1 to this bit, application should poll this bit continuously. Applicaition should proceed to any other operation, only after reading this bit as 0. Note:When XLGMII interface width is 64 bits, PHY interface width is 66 and RSFEC is not enabled, this field is reserved. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(DWCXLP_XLGMII_64BIT&&(DWCXLP_R_SRDS_INTF==3)&&!DWCXLP_RSFEC_EN) ? \"read-write\" : \"read-only\""
7:5	DTXLANED_3_1	* Varies	Tx Disable on Lanes[3:1] When any bits of this field are set, the DWC_xlgpcs disables the corresponding Tx lanes (3, 2, or 1) of the PHY. When reset, the DWC_xlgpcs enables the corresponding Tx lanes of the PHY. This field is shared with the following fields: ■ Bits[7:5] of VR PMA MMD Digital Control1 Register ■ Bits[4:2] of SR PMA MMD Transmit Disable Register **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-349 Fields for Register: VR_PCS_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
4	DTXLANED_0	R/W	Tx Disable on Lane 0 When this bit is set, the DWC_xlgpcs disables the Tx Lane 0 of the PHY. When reset, the DWC_xlgpcs enables the Tx Lane 0 of the PHY. **Note:** This bit is shared with the following bits: ■ Bit 4 of VR PMA MMD Digital Control1 Register ■ Bit 1 of SR PMA MMD Transmit Disable Register **Value After Reset:** 0x0 **Exists:** Always
3	CR_CJN	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	LANE_CTRL_EXT	* Varies	Lane Control External The transmit and receive lane disabling selection is based on the value programmed to this bit: ■ 1: The inputs of xlgpcs_rx_lane{lane}_en_i and xlgpcs_tx_lane{lane}_en_i are selected to enable/disable Tx and Rx lanes ■ 0: The values programmed in VR PCS MMD Digital Control2 Register bits [15:13] and bits [11:9] are selected to enable/disable Tx and Rx lanes. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-349 Fields for Register: VR_PCS_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
1	BYP_PWRUP	* Varies	For Synopsys PHY configurations: Bypass Power-Up Sequence This bit is present only when DWC_xlgpcs is configured for the Synopsys PHY. In all other configurations, this bit is reserved. When this bit is set to 1, the DWC_xlgpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xlgpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xlgpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xlgpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming normal transmission and reception. **Note:** This bit is shared with Bit 1 of the following registers: ■ VR PMA MMD Digital Control1 Register ■ VR AN MMD Digital Control1 Register Access Type: For configurations with Synopsys PHY, this bit is RW. Otherwise, this bit is RO. For non-Synopsys PHY configurations: ■ Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
0	DSKBYP	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.2.49 VR_PCS_DIG_CTRL2

- **Description:** VR PCS MMD Digital Control2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8001 : (DWCXLP_APB3_16) ? (0x8001*0x2) : (0x8001*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

15:13	12	11:9	8	7:5	4	3:1	0
PTX_LN_DIS_3_1	Reserved_12	PRX_LN_DIS_3_1	Reserved_8	TX_POL_INV_3_1	TX_POL_INV_0	RX_POL_INV_3_1	RX_POL_INV_0

Table 7-350 Fields for Register: VR_PCS_DIG_CTRL2

Bits	Name	Memory Access	Description
15:13	PTX_LN_DIS_3_1	* Varies	Tx Lane Disable This field indicates PCS Tx Lane disable on the corresponding Lane 1, Lane 2, or Lane 3. The following are the valid combinations: ■ 000: Lanes 0 to 3 are enabled (This setting is required for KR4/40G Mode of operation). ■ 110: Lanes 2 and 3 are disabled. Lanes 0 and 1 are operational (This setting is required for KR2/20G Mode of operation). ■ Other values: Reserved. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-350 Fields for Register: VR_PCS_DIG_CTRL2 (Continued)

Bits	Name	Memory Access	Description
11:9	PRX_LN_DIS_3_1	* Varies	Rx Lane Disable This field indicates PCS Rx Lane disable on the corresponding lane (Lane 1, 2, or 3). When this bit is set, the PHY data of corresponding lanes is ignored and synchronization and deskew/alignment is performed only among enabled lanes. The following are the valid combinations: ■ 000: Lanes 0, 1, 2, and 3 are enabled (This setting is required for 40G/100G Mode of operation). ■ 110: Lanes 2 and 3 are disabled. Lanes 0 and 1 are operational . (This setting is required for 20G Mode of operation). ■ Other values: Reserved. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
8	Reserved_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	TX_POL_INV_3_1	* Varies	Tx Polarity Invert on Lanes [3:1] When set to 1, this field reverses the data polarity on the Tx differential lines of the corresponding lanes. ■ 3'b**1: Lane 1 Tx data inverted ■ 3'b*1*: Lane 2 Tx data inverted ■ 3'b1**: Lane 3 Tx data inverted **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
4	TX_POL_INV_0	R/W	Tx Polarity Invert on Lane 0 When set to 1, this bit reverses the data polarity on the Tx differential lines of Lane 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-350 Fields for Register: VR_PCS_DIG_CTRL2 (Continued)

Bits	Name	Memory Access	Description
3:1	RX_POL_INV_3_1	* Varies	Rx Polarity Invert on Lanes [3:1] When set to 1, this field reverses the polarity on the data received from the PHY. ■ 3'b**1: Lane 1 Rx data inverted ■ 3'b*1*: Lane 2 Rx data inverted ■ 3'b1**: Lane 3 Rx data inverted **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
0	RX_POL_INV_0	R/W	Rx Polarity Invert on Lane 0 When set to 1, this bit reverses the polarity on the data received from the PHY. **Value After Reset:** 0x0 **Exists:** Always

7.2.50 VR_PCS_DIG_CTRL3

- **Description:** VR PCS MMD Digital Control3 Register This register is applicable only when the DWC_xlgpcs is configured to have 25G or 50G support. These register bits are applicable only when the DWC_xlgpcs is working in 25G or 50G speed mode.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8003 : (DWCXLP_APB3_16) ? (0x8003*0x2) : (0x8003*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** !DWCXLP_KR4_KR

15:12	11	10	9	8	2	1	0
reserved_15_12	TS_MODE	GAUI_544	BMX_544	MSK_PHY_RST	CL72_EN	EN_50G	CNS_EN

Table 7-351 Fields for Register: VR_PCS_DIG_CTRL3

Bits	Name	Memory Access	Description
15:12	reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-351 Fields for Register: VR_PCS_DIG_CTRL3 (Continued)

Bits	Name	Memory Access	Description
11	TS_MODE	* Varies	TimeStamp Mode. When this bit is 0, DWC_xlgpcs indicates End-to-End Tx/Rx min/max delay values in the time-delay registers. Also, DWC_xlgpcs drives a value of 0 on xlgpcs_tx/rx_num_unit_chg_o and xlgpcs_tx/rx_ddmp_o ports. When this bit is 1, DWC_xlgpcs generates DDMP indication pulses on xlgpcs_tx/rx_ddmp_o ports and indicates dynamic delay variation on xlgpcs_tx/rx_num_unit_chg_o ports. Also, in this mode, the tx/rx delay in time-delay registers (ns and sub-ns registers) indicate the delay inside the DWC_xlgpcs with respect to the DDMP generation points. **Value After Reset:** 0x0 **Exists:** DWCXLP_TS_EN **Memory Access:** "DWCXLP_TS_EN ? \"read-write\" : \"read-only\""
10	GAUI_544	* Varies	100GAUI-4 or 50GAUI-2 Mode that uses RS544 FEC This bit should be programmed to one, when desired operating mode is 100GAUI-4 or 50GAUI-2 This bit controls the output port xlgpcs_100g50g_gaui_o. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_RS544&&DWCXLP_MULTI_LANE ? \"read-write\" : \"read-only\""
9	BMX_544	* Varies	Enable PMA Bit-Mux in RS(544,514) Mode. When DWC_xlgpcs is used to implement a 50GBASE-KR or 100GBASE-KR2 link, this bit can be set to 1 to perform the PMA layer bit-multiplexing operation. If this bit is not set, bit-multiplexing operation should be done inside the PHY. **Note:**When operating in 100GAUI-4 or 50GAUI-2 modes, this bit should never be set. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_RS544 ? \"read-write\" : \"read-only\""

Table 7-351 Fields for Register: VR_PCS_DIG_CTRL3 (Continued)

Bits	Name	Memory Access	Description
8	MSK_PHY_RST	R/W	Mask PHY reset. Generally, when vendor-soft-reset (VR_RST) is programmed, DWC_xlgpcs asserts the xlgpcs_reset_n_o port (to reset the PHY). If this bit is set to 1, xlgpcs_reset_n_o does not get asserted during VR_RST. This mode is useful in bifurcation mode usage with Synopsys PHYs. **Value After Reset:** 0x0 **Exists:** Always
2	CL72_EN	* Varies	This control bit provides enabling Clause 72 based training. In 25G/50G mode When training is enabled ■ 0: Disables Clause-72 based link training and enables Clause-92/Clause-93 based link training. ■ 1: Enables Clause-72 based link training and disables Clause-92/Clause-93 based link training. By Default Clause-92/Clause-93 based training is enabled Dependencies: This bit is present in configurations having Link Training enabled in the configurations **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_LNK_TRAIN_EN) ? \"read-write\" : \"read-only\""
1	EN_50G	* Varies	This control bit provides enabling 25G or 50G mode of operation In 50G enabled configuration ■ CNS_EN ==1 EN_50G==1: enables Consortium 50G mode of operation, ■ CNS_EN ==1 EN_50G==0: enables Consortium 25G mode of operation. ■ CNS_EN ==0, status of EN_50G bit has no impact, both 25G/50G modes are disabled In 25G Only enabled configuration ■ Always 0 and is Read Only. ■ Hence when CNS_EN (bit-0) is set, the 25G mode is enabled. Dependencies: This bit is present in configurations having 50G modes. **Value After Reset:** "DWCXLP_KR4_100G ? 0x0 : DWCXLP_KR4_50G ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_KR4_50G) ? \"read-write\" : \"read-only\""

Table 7-351 Fields for Register: VR_PCS_DIG_CTRL3 (Continued)

Bits	Name	Memory Access	Description
0	CNS_EN	* Varies	This control bit provides enabling 25G or 50G mode of operation ■ In 25G Only enabled Configuration, - 1: enables Consortium 25G mode of operation - 0: disables Consortium 25G mode of operation ■ In 50G enabled configuration, - 1: enables Consortium 25G/50G mode of operation - 0: disables Consortium 25G/50G mode of operation The selection between 25G/50G modes is done via EN_50G (bit-1) Dependencies: This bit is present in configurations having 25G/50G modes **Value After Reset:** "DWCXLP_KR4_100G ? 0x0 : DWCXLP_KR4_50G ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(!DWCXLP_KR4_KR) ? \"read-write\" : \"read-only\""

7.2.51 VR_PCS_DEBUG_CTRL

- **Description:** VR PCS MMD Debug Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8005 : (DWCXLP_APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

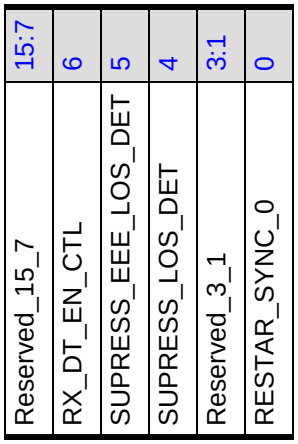


Table 7-352 Fields for Register: VR_PCS_DEBUG_CTRL

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-352 Fields for Register: VR_PCS_DEBUG_CTRL (Continued)

Bits	Name	Memory Access	Description
6	RX_DT_EN_CTL	* Varies	For Multi-protocol Multi-protocol PHY configurations: Rx Data Enable Control. This bit controls the behaviour of xlgpcs_rx_data_en_o{lane} output from DWC_xlgpcs. During normal operation DWC_xlgpcs de-asserts xlgpcs_rx_data_en_o on detecting of Loss-of-Signal. However, if this bit is set, DWC_xlgpcs does not de-assert xlgpcs_rx_data_en_o on detection of Loss-of-Signal. Instead, the value of RX_DT_EN_* bits from VR_PMA_MP_25G_16G_12G_RX_GENCTRL0 Register is driven on xlgpcs_rx_data_en_o output port. For other configurations: Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_E25||DWCXLP_E32||DWCXLP_E16||DWCXLP_E12) ? \"read-write\" : \"read-only\""
5	SUPRESS_EEE_LOS_DET	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	SUPRESS_LOS_DET	R/W	Suppress Loss of Signal Detection When this field is set to 1, Loss of Signal indicated by the PHY (using input port xlgpcs_los_i) is ignored by DWC_xlgpcs while evaluating the Receive link. Receive link is purely evaluated based the Rx data received on the 'xgxs_rx_data{lane}_i' or 'rpcs_rx_data{lane}_i' port of DWC_xlgpcs. When this field is set to 0, Loss of signal indicated by the PHY is considered by the DWC_xlgpcs while evaluating the Receive link status. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RESTAR_SYNC_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.2.52 VR_PCS_EEE_MCTRL

- **Description:** VR PCS MMD EEE Mode Control Register

  This register enables or disables the Energy Efficient Ethernet (EEE) support.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8006 : (DWCXLP_APB3_16) ? (0x8006*0x2) : (0x8006*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_EEE_EN

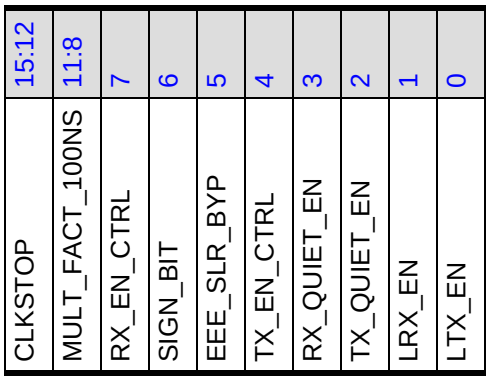


Table 7-353 Fields for Register: VR_PCS_EEE_MCTRL

Bits	Name	Memory Access	Description
15:12	CLKSTOP	R/W	Clock Stop This field holds the count value after which the Rx clock to the MAC can be stopped. The default value is 8. The host should program this value depending on the capability of the MAC. This field is used to control the assertion of the xlgpcs_lrx_clk_gat_o signal (to gate the Rx clock of the MAC). By default, the xlgpcs_lrx_clk_gat_o signal is asserted after the PCS Rx module detects the LPI pattern followed by the following: 120 clocks of clk_rpcs_rx0_i + Number of clk_rpcs_rx0_i clocks (specified in this field) + clk_eee_i synchronization delay + FIFO initialization delay **Value After Reset:** 0x8 **Exists:** Always

Table 7-353 Fields for Register: VR_PCS_EEE_MCTRL (Continued)

Bits	Name	Memory Access	Description
11:8	MULT_FACT_100NS	R/W	100 ns Clock Tic Multiplying Factor This bit is the multiplying factor to the clk_eee_i clock time period to make it closer to 100 ns. For example, if the clk_eee_i clock time period (clk_eee_i_time_period) is 10 ns, the value of this field is 9, that is, 1 less than 10. This value should be programmed such that the clk_eee_i_time_period * (MULT_FACT_100NS + 1) should be within 80 ns to 120 ns. The default value of this register is 9, assuming that the default clk_eee_i time period to be 10 ns. **Value After Reset:** 0x9 **Exists:** Always
7	RX_EN_CTRL	R/W	Rx Control Enable This bit controls the generation of the xlgpcs_rx_en_o{lane} signal. - When this bit is set to 1, the xlgpcs_rx_en_o signal is de-asserted when the EEE receive controller reaches the Quiet state. - When this bit is set to 0, the xlgpcs_rx_en_o signal is not de-asserted when the EEE receive controller reaches the Quiet state. In GAUI mode, ■ when shut-down is enabled: - RX_EN_CTRL = 1: xlgpcs_rx_en_o deasserts when the EEE receive controller reaches the Quiet state , - RX_EN_CTRL = 0: xlgpcs_rx_en_o does not deassert when the EEE receive controller reaches the Quiet state. ■ when shut-down is not enabled: - xlgpcs_rx_en_o always remains asserted independent of the RX_EN_CTRL signal. **Value After Reset:** 0x1 **Exists:** Always
6	SIGN_BIT	R/W	Effective 100 ns Tic Value The host should use this bit to fine tune the EEE timing requirement. The host should set this bit to 0 when the clk_eee_i_time_period * (MULT_FACT_100NS + 1) value is more than or equal to 100 ns. The host should set this bit to 1 when the clk_eee_i clock period * (MULT_FACT_100NS + 1) value is less than 100 ns. **Value After Reset:** 0x0 **Exists:** Always

Table 7-353 Fields for Register: VR_PCS_EEE_MCTRL (Continued)

Bits	Name	Memory Access	Description
5	EEE_SLR_BYP	R/W	Scrambler Bypass This bit is used to bypass the transmit scrambler to enable the Rapid Block synchronization. When this bit is set to 1, the scrambler is bypassed in PCS-R during EEE Tx mode. When this bit is set to 0, the scrambler is not bypassed. This bit is enabled only when DWCXLP_FEC_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always
4	TX_EN_CTRL	R/W	Tx Control Enable This bit controls the generation of xlgpcs_tx_en_o{lane} output for non-Synopsys PHY Configuration In non-GAUI mode, ■ When this bit is 1, the xlgpcs_tx_en_o{lane} signal is de-asserted when the EEE transmit controller reaches the Quiet state. ■ When this bit is 0, the xlgpcs_tx_en_o{lane} signal is not de-asserted when the EEE transmit controller reaches the Quiet state. In GAUI mode, ■ when shut-down is enabled - TX_EN_CTRL = 1: xlgpcs_tx_en_o deasserts when the EEE transmit controller reaches the Quiet state, - TX_EN_CTRL = 0: xlgpcs_tx_en_o does not deassert when the EEE transmit controller reaches the Quiet state. ■ when shut-down is not enabled - xlgpcs_tx_en_o always remains asserted independent of the TX_EN_CTRL signal. **Value After Reset:** 0x1 **Exists:** Always
3	RX_QUIET_EN	R/W	Rx Quiet Enable This bit controls the generation of the xlgpcs_lpirx_quiet_o output. When this bit is set to 1, the xlgpcs_lpirx_quiet_o output is set to 1 when the EEE receive controller reaches the Quiet state. When this bit set to 0, the xlgpcs_lpirx_quiet_o output is not set to 1. In GAUI mode with shut down disable, xlgpcs_lpirx_quiet_o is always de-asserted and state of RX_QUIET_EN does not have any effect **Value After Reset:** 0x1 **Exists:** Always

Table 7-353 Fields for Register: VR_PCS_EEE_MCTRL (Continued)

Bits	Name	Memory Access	Description
2	TX_QUIET_EN	R/W	Tx Quiet Enable This bit controls the generation of the xlgpcs_lpitx_quiet_o output. When this bit is set to 1, the xlgpcs_lpitx_quiet_o output is set to 1 when the EEE transmit controller reaches the Quiet state. When this bit set to 0, the xlgpcs_lpitx_quiet_o output is not set to 1. In GAUI mode with shut down disable, xlgpcs_lpitx_quiet_o is always deasserted and therefore state of this bit does not have any effect. **Note:** For configurations with DWCXLP_XGMII_64BIT == 1 and DWCXLP_CLK_CMP_EN == 1, this bit must be programmed to 0. This is required to prevent disruption in data reception (in DWC_xlgpcs receiver) when transmitter is in TX_QUIET state. **Value After Reset:** 0x1 **Exists:** Always
1	LRX_EN	R/W	LPI Rx Enable When set to 1, this bit enables the Energy Efficient Ethernet support in the DWC_xlgpcs Receive path. When set to 0, it disables the support for Energy Efficient Ethernet in the DWC_xlgpcs Receive path. **Value After Reset:** 0x0 **Exists:** Always
0	LTX_EN	R/W	LPI Tx Enable When set to 1, this bit enables the Energy Efficient Ethernet support in the DWC_xlgpcs Transmit path. When set to 0, it disables the support for Energy Efficient Ethernet in the DWC_xlgpcs Transmit path. **Value After Reset:** 0x0 **Exists:** Always

7.2.53 VR_PCS_KR_CTRL

- **Description:** VR PCS BASE-R Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8007 : (DWCXLP_APB3_16) ? (0x8007*0x2) : (0x8007*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

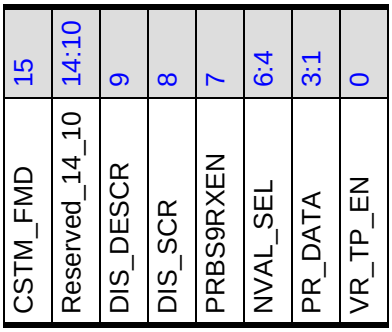


Table 7-354 Fields for Register: VR_PCS_KR_CTRL

Bits	Name	Memory Access	Description
15	CSTM_FMD	* Varies	Custom KR Training frame marker detect mode When this bit is set, DWC_xlgpcs additionally supports KR Training frame marker pattern with one bit and/or two bits tolerance at the transition boundaries along with the standard IEEE frame marker pattern (The standard pattern as per IEEE frame marker is 32'hFFFF0000) Encoding is as follows: ■ 0: Only 16 ones followed by 16 zeroes are treated as frame marker in the receiver(32'hFFFF0000) as per the standard IEEE pattern. ■ 1: Along with the standard frame marker defined by IEEE (32'hFFFF0000), the following four additional patterns are also supported * 15 ones followed by 17 zeroes(32'hFFF70000) * 17 ones followed by 15 zeroes(32'hFFFF1000) * 14 ones followed by 18 zeroes(32'hFFF30000) * 18 ones followed by 14 zeroes(32'hFFFF3000) **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_CL72_EN ? \"read-write\": \"read-only\""

Table 7-354 Fields for Register: VR_PCS_KR_CTRL (Continued)

Bits	Name	Memory Access	Description
14:10	Reserved_14_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	DIS_DESCR	R/W	Disable Descrambler When set to 1, this bit disables the BASE-R PCS descrambling function. You can use this bit for debugging purpose. **Value After Reset:** 0x0 **Exists:** Always
8	DIS_SCR	R/W	Disable Scrambler When set to 1, this bit disables the BASE-R PCS scrambling function. You can use this bit for debugging purpose. However, when PCS scrambler is bypassed in transmit side, the block sync logic may get locked on to wrong bit boundary and possibly result in hangs/errors in the PCS decoded data. To avoid this, ensure that XLGMII data and consequently 64/66b data blocks have sufficient/random transitions **Value After Reset:** 0x0 **Exists:** Always
7	PRBS9RXEN	R/W	Enable PRBS9 Testing in Receive Path When this bit is set to 1 and Bit[5] (PRBS31R_EN) and Bit[2] (RTP_EN) of SR PCS BASE-R Test Pattern Control Register are set to 0, this bit enables the PRBS9 test pattern checking in BASE-R PCS Receive path. **Value After Reset:** 0x0 **Exists:** Always

Table 7-354 Fields for Register: VR_PCS_KR_CTRL (Continued)

Bits	Name	Memory Access	Description
6:4	NVAL_SEL	R/W	Square Wave Control When square wave test pattern generation is enabled either in SR PCS BASE-R or SR PMA MMD KR SQUARE WAVE CTRL Test Pattern Control Register, this field indicates the number of bit times for which Ones and Zeros should be driven on the serial lines (square wave period). Encoding: ■ 3'b000: NVAL_SEL = 4; ■ 3'b001: NVAL_SEL = 5; ■ 3'b010: NVAL_SEL = 6; ■ 3'b011: NVAL_SEL = 7; ■ 3'b100: NVAL_SEL = 8; ■ 3'b101: NVAL_SEL = 9; ■ 3'b110: NVAL_SEL = 10; ■ 3'b111: NVAL_SEL = 11; **Value After Reset:** 0x0 **Exists:** Always
3:1	PR_DATA	R/W	Pseudo-Random Test Pattern Data When pseudo-random test pattern is enabled in Bit 3 (TTP_EN) of SR PCS BASE-R Test Pattern Control Register and Bit 0 (VR_TP_EN) of this register is set to 1, this field selects the data input for the scrambler. Encoding: ■ 3'b000: Data = All zero (0x00); IEEE specified ■ 3'b001: Data = XGMII local fault; IEEE specified ■ 3'b010: Data = All five (0x55); HFTP ■ 3'b011: Data = All A (0xAA); HFTP ■ 3'b100: Data = All ones (0xFF); LFTP ■ 3'b101: Data = All three (0x33); MFTP ■ 3'b110: Data = All C (0xCC); MFTP ■ 3'b111 **Value After Reset:** 0x0 **Exists:** Always
0	VR_TP_EN	R/W	PCS-R Vendor-Specific Test Pattern Enable When this bit set to 1, the data for the pseudo-random test pattern is selected as per the PR_DATA selection. **Value After Reset:** 0x0 **Exists:** Always

7.2.54 VR_PCS_EEE_TXTIMER

- **Description:** VR PCS MMD EEE Tx Timer Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8008 : (DWCXLP_APB3_16) ? (0x8008*0x2) : (0x8008*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_EEE_EN

15:13	12:8	7:6	5:0
TWL2_RES	TWL_RES	T1U_RES	TSL_RES

Table 7-355 Fields for Register: VR_PCS_EEE_TXTIMER

Bits	Name	Memory Access	Description
15:13	TWL2_RES	* Varies	TWL2 Resolution This field stores the resolution value for the TWL2 timer under multi-lane modes and for other modes, this . When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more counts (SIGN_BIT is low) or less counts (SIGN_BIT is high) than the ideal count to meet the range of the TWL timer. The DWC_xlgpcs maintains the default value of the TWL2 timer as 24, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1)is equal to 100 ns to produce 2400 ns (2.4 micro seconds) as per the TWL2 requirement in the IEEE standard. Example: In multi configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1) is 90, the default TWL2 timer produces 2160 (2.16micro seconds) which is lesser than the standard 2.400 micro seconds. To make it 2.4micro seconds, you should program this register such that (24 +/- TWL2_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 2.4micro seconds and less than 2.5micro seconds (max limit), that is, (24 + TWL2_RES)*90 = 24000 ns. TWL2_RES = 3 meets the requirement which produces 2.43 micro seconds. The SIGN_BIT should be programmed as 1. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_EEE_EN&&!DWCXLP_25G_ONLY) ? \"read-write\" : \"read-only\""

Table 7-355 Fields for Register: VR_PCS_EEE_TXTIMER (Continued)

Bits	Name	Memory Access	Description
12:8	TWL_RES	* Varies	TWL Resolution This field stores the resolution value for the TWL timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more counts (SIGN_BIT is low) or less counts (SIGN_BIT is high) than the ideal count to meet the range of the TWL timer. The DWC_xlgpcs maintains the default value of the TWL timer as 109, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1)is equal to 100 ns to produce 10900 ns (10.9micro seconds) as per the TWL requirement in the IEEE standard. Example: For KR_Present configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1) is 90, the default TWL timer produces 9810 ns (9.81micro seconds) which is lesser than the standard 10.9micro seconds. To make it 10.9micro seconds, you should program this register such that (109 +/- TWL_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 10.9micro seconds and less than 11micro seconds (max limit), that is, (109 + TWL_RES)*90 = 10900 ns. TWL_RES = 13 meets the requirement which produces 10.98micro seconds. The SIGN_BIT should be programmed as 1. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_EEE_EN) ? \"read-write\" : \"read-only\"""

Table 7-355 Fields for Register: VR_PCS_EEE_TXTIMER (Continued)

Bits	Name	Memory Access	Description
7:6	T1U_RES	R/W	T1u Resolution This field stores the resolution value for the T1u timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the T1u timer. The DWC_xlgpcs maintains the default value of the T1u timer as 11, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce 1100 ns (1.1 micro seconds) as per the T1u requirement in the IEEE standard. Example: For KR_Present configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1)is 90, the default T1u timer produces 990 ns (0.99micro seconds) which is lesser than the standard 1.1 micro seconds. To make it 1.1 micro seconds, you should program this register such that (11 +/- T1U_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 1.1 micro seconds and less than 1.3 micro seconds (max limit), that is, (11+ T1U_RES)*90 = 1.1 micro seconds. T1U_RES = 2 meets the requirement which produces 1.17 micro seconds. The SIGN_BIT should be programmed as 1. **Value After Reset:** 0x0 **Exists:** Always

Table 7-355 Fields for Register: VR_PCS_EEE_TXTIMER (Continued)

Bits	Name	Memory Access	Description
5:0	TSL_RES	R/W	TSL Resolution This field stores the resolution value for the TSL timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the TSL timer. The DWC_xlgpcs maintains the following default value of the TSL timer: ■ 49 for KR_present configurations (KR4/KR) ■ 199 for all other configurations This value is maintained assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce the following as per the TSL requirement in the IEEE standard: ■ 4900 ns (4.9 us) for KR_Present configurations (KR4/KR) ■ 19900 ns (19.9 us) for all other configurations Example: For KR_Present configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1) is 90, the default TSL timer produces 4410 ns (4.41micro seconds) which is lesser than the standard value 4.9micro seconds. To make it 4.9micro seconds, you should program this register such that (49 +/- TSL_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 4.9micro seconds and less than 5.1micro seconds (max limit), that is, (49 + TSL_RES)*90 = 4.9micro seconds. TSL_RES = 6 meets the requirement which produces 4.95micro seconds. The SIGN_BIT should be programmed as 1. **Value After Reset:** 0x0 **Exists:** Always

7.2.55 VR_PCS_EEE_RXTIMER

- **Description:** VR PCS MMD EEE Rx Timer Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8009 : (DWCXLP_APB3_16) ? (0x8009*0x2) : (0x8009*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_EEE_EN

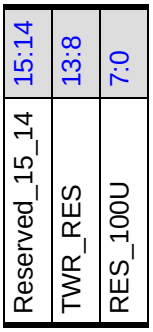


Table 7-356 Fields for Register: VR_PCS_EEE_RXTIMER

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-356 Fields for Register: VR_PCS_EEE_RXTIMER (Continued)

Bits	Name	Memory Access	Description
13:8	TWR_RES	R/W	TWR Resolution This field stores the resolution value for the TWR timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the TWR timer. Default Value: The following is the default value: ■ 11.5 us for KR mode when scrambler is not bypassed ■ 13.7 us for KR mode when scrambler is bypassed ■ 4.5us for KR4/CR4 modes when Scrambler is bypassed ■ 5.5us for KR4/CR4 modes when scrambler is not bypassed (Not applicable to 100G mode) These values are as per the requirements specified in the standard assuming that clk_eee_i_time_period * (MULT_FACT_100NS + 1) produces 100 ns tick. Example: For KR mode (when scrambler is not bypassed), the time requirement is 11.5micro seconds. The DWC_xlgpcs maintains the default value of the TWR timer as 115, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce 1150 ns (11.5micro seconds). If clk_eee_i_time_period * (MULT_FACT_100NS + 1) is 110, the default TWR timer produces 12650 ns (12.65micro seconds) which is more than 11.5microseconds. To make it 11.5micro seconds, you should program this register such that (115 +/- TWR_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) = 115micro seconds. Therefore, use the following equation: (115 - TWR_RES) * 110 <= 11500 ns TWR_RES = 11 meets the requirement which produces 11.44micro seconds. The SIGN_BIT should be programmed as 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-356 Fields for Register: VR_PCS_EEE_RXTIMER (Continued)

Bits	Name	Memory Access	Description
7:0	RES_100U	R/W	100 us Resolution This field stores the resolution value for the 100 us timer. If the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the 100 us timer. This field is used to control the generation of 100micro seconds time tick using the clk_eee_i clock and MULT_FACT_100NS field. By default, the 100micro seconds timer is generated assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns. Therefore, the DWC_xlgpcs maintains the default value of the 100micro seconds timer as 1000 to achieve 100000 ns (100micro seconds). You should use this field if clk_eee_i_time_period * (MULT_FACT_100NS + 1) is not equal to 100 ns. To program this field, use the following equation: (1000 +/- RES_100US)* ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) = 100micro seconds Note: When in Fast Simulation mode, the value obtained from above equation must be divided by 40 before programming. The SIGN_BIT should be programmed as 1 for (+) and 0 for (-). **Value After Reset:** 0x0 **Exists:** Always

7.2.56 VR_PCS_EEE_MISC_TIMER

- **Description:** VR PCS MMD EEE GAUI Timer Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x800A : (DWCXLP_APB3_16) ? (0x800A*0x2) : (0x800A*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_EEE_EN

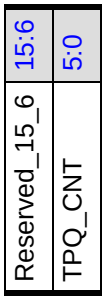


Table 7-357 Fields for Register: VR_PCS_EEE_MISC_TIMER

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	TPQ_CNT	R/W	Maximum count value to cater for the Tpq timeout. This field stores the maximum count value to count the Tpq timeout when operating in GAUI shut down mode. The default value of this register field corresponds to clk_eee_i running at 100MHz. The Time out value to be programmed is to be calculated as (Tpq in ns) / (EEE clock period in ns) and the Value of Tpq ranges from 200ns to 225ns as per IEEE clause 83.5.11. In EEE modes other than GAUI shut down, the value in this field is ignored. **Note:** This field must be programmed once to update the value even though the default value matches the requirement. **Value After Reset:** 0x15 **Exists:** Always

7.2.57 VR_PCS_DIG_STS

- **Description:** VR PCS MMD Digital Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8010 : (DWCXLP_APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** Always

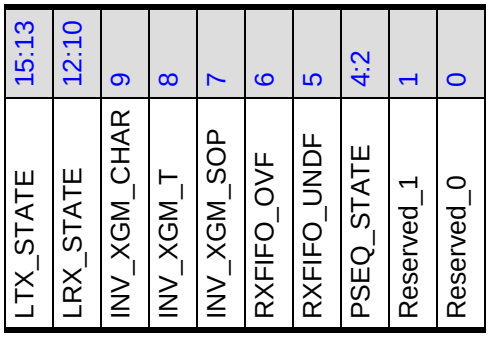


Table 7-358 Fields for Register: VR_PCS_DIG_STS

Bits	Name	Memory Access	Description
15:13	LTX_STATE	R	For EEE configurations: LPI Transmit State. This field indicates the current state of the LPI Transmit state Machine. This is a 2-bit field if the selected configuration does not have RPCS. Otherwise, it is a 3-bit field. ■ 000: LTX_ACTIVE ■ 001: LTX_SLEEP ■ 010: LTX_QUIET ■ 011: LTX_REF_WAKE ■ 100: LTX_ALERT ■ 101: LTX_SCR_BYP For other configurations: Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-358 Fields for Register: VR_PCS_DIG_STS (Continued)

Bits	Name	Memory Access	Description
12:10	LRX_STATE	R	For EEE configurations: LPI Receive State This field indicates the current state of the LPI Receive State Machine ■ 000: LRX_ACTIVE ■ 001: LRX_SLEEP ■ 010: LRX_QUIET ■ 011: LRX_WAKE ■ 100: LRX_WTF ■ 101: LRX_LINK_FAIL For other configurations: Reserved **Value After Reset:** 0x0 **Exists:** Always
9	INV_XGM_CHAR	R	Invalid XGMII Character (RO,LH Type) This bit indicates an invalid XGMII character on the Transmit path. ■ 0: No Invalid XGMII character ■ 1: Invalid character This bit is set when one of the following conditions is true on the XGMII interface: ■ |Q| is detected on a lane other than Lane 0. ■ |Q| is detected on Lane 0 and also any of the following conditions is true: - Lane 1 does not contain the data, 8'h00. - Lane 2 does not contain the data, 8'h00. - Lane 3 does not contain the data 8'h01 or 8'h02. ■ |E| or |T| is encountered in non-data phase and |Q| is encountered in data phase. ■ |D| is detected during the Idle phase (after Terminate and before valid SOF). ■ |I| is detected during the data phase (after SOF and before Terminate). **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Table 7-358 Fields for Register: VR_PCS_DIG_STS (Continued)

Bits	Name	Memory Access	Description
8	INV_XGM_T	R	Invalid XGMII T Character (RO,LH Type) This bit indicates that the transmit data received on XGMII has an invalid Terminate character. ■ 0: Normal operation ■ 1: Invalid Terminate character The following can cause an invalid Terminate character: ■ In Terminate column, before lane corresponds to the |T| character, there is a |I| or |Q| character. ■ In Terminate column, after lane corresponds to the |T| character, there is a non-|I| character **Value After Reset:** 0x0 **Exists:** Always
7	INV_XGM_SOP	R	Invalid XGMII Start Character (RO,LH Type) This bit indicates that the XGMII transmit frame contains invalid SOP character. ■ 0: Normal operation ■ 1: Invalid SOP character An invalid SOP event occurs when a |S| character is detected in a lane other than Lane 0. **Value After Reset:** 0x0 **Exists:** Always
6	RXFIFO_OVF	R	Rx FIFO Overflow (RO,LH Type) This bit indicates the clock rate compensation FIFO overflow. ■ 0: Normal operation ■ 1: FIFO overflow This is a latched-high bit. **Value After Reset:** 0x0 **Exists:** Always
5	RXFIFO_UNDF	R	Rx FIFO Underflow (RO,LH Type) This bit indicates the clock rate compensation FIFO underflow. ■ 0: Normal operation ■ 1: FIFO underflow This is a latched-high bit. **Value After Reset:** 0x0 **Exists:** Always

Table 7-358 Fields for Register: VR_PCS_DIG_STS (Continued)

Bits	Name	Memory Access	Description
4:2	PSEQ_STATE	R	For Synopsys PHY configurations: Power-Up Sequence State This field indicates the state variable value of the power-up sequence module. For Synopsys PHY: ■ 3'b000: Wait for ACK High0 ■ 3'b001: Wait for ACK Low0 ■ 3'b010: Wait for Tx ACK High1 ■ 3'b011: Wait for Rx ACK Low1 ■ 3'b100: Tx/Rx stable (Power_Good state) ■ 3'b101: Power Save state ■ 3'b110: Power Down state ■ 3'b111: Other intermediate states (valid only for Synopsys 32G PHY configurations) Dependency: This field is reserved for non-Synopsys PHY configurations. For non-Synopsys PHY configurations: ■ Reserved **Value After Reset:** 0x4 **Exists:** Always
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.2.58 VR_PCS_GPIO

- **Description:** VR PCS MMD GPIO Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8015 : (DWCXLP_APB3_16) ? (0x8015*0x2) : (0x8015*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_GPIO_EN

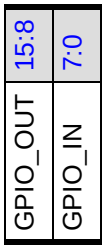


Table 7-359 Fields for Register: VR_PCS_GPIO

Bits	Name	Memory Access	Description
15:8	GPIO_OUT	R/W	GPIO Output The content written on this field is driven to the xlgpcs_gpo_o[7:0] output port. **Value After Reset:** 0x0 **Exists:** Always
7:0	GPIO_IN	R	GPIO Input This field indicates the content of the xlgpcs_gpo_i[7:0] port. **Value After Reset:** 0x0 **Exists:** Always

7.2.59 VR_PCS_AM_CNT

- **Description:** VR PCS MMD Alignment Marker Counter Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8018 : (DWCXLP_APB3_16) ? (0x8018*0x2) : (0x8018*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_MULTI_LANE||DWCXLP_RSFEC_EN

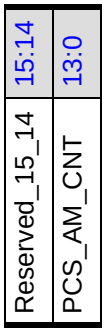


Table 7-360 Fields for Register: VR_PCS_AM_CNT

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:0	PCS_AM_CNT	R/W	Alignment Marker Count Value. This field determines the interval at which Alignment Markers are inserted into the data-stream. In non-RSFEC mode, the interval is defined as the number of 66-bit blocks between two alignment markers (without counting the alignment markers). In RSFEC mode this field determines the interval in terms of RSFEC code words at which Alignment Markers are inserted into the data-stream and it has to be programmed before enabling the RSFEC mode. The standard value in RSFEC mode is 1024 for RS-FEC enabled 25G and 50G modes, 4096 for RS-FEC enabled 100G modes . **Note:** Any change in the value of this field should be followed by a vendor-specific soft-reset **Value After Reset:** "DWCXLP_100G_1LANE ? 0x1000 : DWCXLP_50G_25G ? 0x400 : 0x3fff" **Exists:** Always

7.2.60 VR_PCS_SKW_OUT_OF_RANGE

- **Description:** VR PCS MMD Skew Out of Range Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8019 : (DWCXLP_APB3_16) ? (0x8019*0x2) : (0x8019*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_MULTI_LANE||DWCXLP_50G_25G||DWCXLP_100G_1LANE

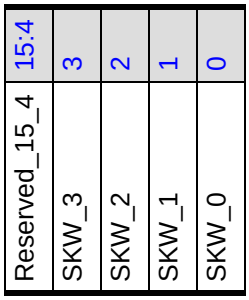


Table 7-361 Fields for Register: VR_PCS_SKW_OUT_OF_RANGE

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	SKW_3	R	Skew Out of Range Indication for Lane 3 (RO, COR Type) This field indicates if skew on lane 3 is more than the allowed value of 180ns. **Note:**This bit may get set initially during Rx link-up even if there is no actual skew (in cases where first AM is somehow missed by DWC_xlgpcs on any of the lanes). Hence,read and clear this bit initially. If this bit is persistently becoming 1, then it indicates an actual Skew out or Range Condition. **Value After Reset:** 0x0 **Exists:** Always

Table 7-361 Fields for Register: VR_PCS_SKW_OUT_OF_RANGE (Continued)

Bits	Name	Memory Access	Description
2	SKW_2	R	Skew Out of Range Indication for Lane 2 (RO, COR Type) This field indicates if skew on lane 2 is more than the allowed value of 180ns. **Note:**This bit may get set initially during Rx link-up even if there is no actual skew (in cases where first AM is somehow missed by DWC_xlgpcs on any of the lanes). Hence,read and clear this bit initially. If this bit is persistently becoming 1, then it indicates an actual Skew out or Range Condition. **Value After Reset:** 0x0 **Exists:** Always
1	SKW_1	R	Skew Out of Range Indication for Lane 1 (RO, COR Type) This field indicates if skew on lane 1 is more than the allowed value of 180ns. **Note:**This bit may get set initially during Rx link-up even if there is no actual skew (in cases where first AM is somehow missed by DWC_xlgpcs on any of the lanes). Hence,read and clear this bit initially. If this bit is persistently becoming 1, then it indicates an actual Skew out or Range Condition. **Value After Reset:** 0x0 **Exists:** Always
0	SKW_0	R	Skew Out of Range Indication for Lane 0 (RO, COR Type) This field indicates if skew on lane 0 is more than the allowed value of 180ns. **Note:**This bit may get set initially during Rx link-up even if there is no actual skew (in cases where first AM is somehow missed by DWC_xlgpcs on any of the lanes). Hence,read and clear this bit initially. If this bit is persistently becoming 1, then it indicates an actual Skew out or Range Condition. **Value After Reset:** 0x0 **Exists:** Always

7.2.61 VR_PCS_ALT_AM_CNT

- **Description:** VR PCS MMD Alternate Alignment Marker Counter Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x801A : (DWCXLP_APB3_16) ? (0x801A*0x2) : (0x801A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_EEE_EN&&(DWCXLP_MULTI_LANE||DWCXLP_RSFEC_EN)

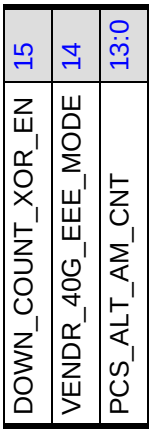


Table 7-362 Fields for Register: VR_PCS_ALT_AM_CNT

Bits	Name	Memory Access	Description
15	DOWN_COUNT_XOR_EN	* Varies	Controls the XOR of Down count and the M0 byte of Rapid Alignment Marker in 25G. When programmed to 1, enables the XOR of Down count and M0 byte of RAM in 25G speed mode. When 0, Down count value is inserted in CD3 field of RAM without performing XOR operation with the M0 byte of RAM. This control bit is applicable only in 25G speed mode. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_RSFEC_EN) ? \"read-write\" : \"read-only\""

Table 7-362 Fields for Register: VR_PCS_ALT_AM_CNT (Continued)

Bits	Name	Memory Access	Description
14	VENDR_40G_EEE_MODE	R/W	Vendor-Specific KR4 EEE mode Control. This field can be programmed to 1, to activate vendor-specific 40G EEE mode. When this bit is programmed to 1, Alignment Marker insertion happens at interval specified by bits[13:0] of this register, instead of the normal value of 16383, when DWC_xlgpcs in the Tx LPI mode. In addition, alignment marker lock module, checks for alignment marker in the received data at an interval specified by either bits [13:0] of this register or the one specified in VR PCS Alignment Maker Counter Register **Value After Reset:** 0x0 **Exists:** Always
13:0	PCS_ALT_AM_CNT	R/W	KR4 Alternate Alignment Marker Count Value. This field is set at a default value of 64, which is much lower than the normal value of 16383. If bit[14] of this register is programmed to 1,then the value specified by this field is used to compute alignment marker insertion interval while operating in KR4 EEE mode . In addition, alignment Marker Lock module in the Rx path checks for alignment marker at intervals specified by either this field or the one specified in VR PCS Alignment Maker Counter Register, when bit[14] of this register is programmed to 1. **Value After Reset:** 0x40 **Exists:** Always

7.2.62 VR_PCS_MISC_STS

- **Description:** VR PCS Miscellaneous Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8020 : (DWCXLP_APB3_16) ? (0x8020*0x2) : (0x8020*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_MULTI_LANE||DWCXLP_50G_25G||DWCXLP_100G_1LANE

15:8	7	6	5	4	3	2	1	0
Reserved_15_8	DSKW_FIFO_UNF3	DSKW_FIFO_UNF2	DSKW_FIFO_UNF1	DSKW_FIFO_UNF0	DSKW_FIFO_OVF3	DSKW_FIFO_OVF2	DSKW_FIFO_OVF1	DSKW_FIFO_OVF0

Table 7-363 Fields for Register: VR_PCS_MISC_STS

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	DSKW_FIFO_UNF3	R	Alignment Marker Deskew FIFO Underflow indication for lane 3 (RO,LH Type) **Value After Reset:** 0x0 **Exists:** Always
6	DSKW_FIFO_UNF2	R	Alignment Marker Deskew FIFO Underflow indication for lane 2 (RO,LH Type) **Value After Reset:** 0x0 **Exists:** Always
5	DSKW_FIFO_UNF1	R	Alignment Marker Deskew FIFO Underflow indication for lane 1 (RO,LH Type) **Value After Reset:** 0x0 **Exists:** Always

Table 7-363 Fields for Register: VR_PCS_MISC_STS (Continued)

Bits	Name	Memory Access	Description
4	DSKW_FIFO_UNF0	R	Alignment Marker Deskew FIFO Underflow indication for lane 0 (RO,LH Type) **Value After Reset:** 0x0 **Exists:** Always
3	DSKW_FIFO_OVF3	R	Alignment Marker Deskew FIFO Overflow indication for lane 3 (RO,LH Type) **Value After Reset:** 0x0 **Exists:** Always
2	DSKW_FIFO_OVF2	R	Alignment Marker Deskew FIFO Overflow indication for lane 2 (RO,LH Type) **Value After Reset:** 0x0 **Exists:** Always
1	DSKW_FIFO_OVF1	R	Alignment Marker Deskew FIFO Overflow indication for lane 1 (RO,LH Type) **Value After Reset:** 0x0 **Exists:** Always
0	DSKW_FIFO_OVF0	R	Alignment Marker Deskew FIFO Overflow indication for lane 0 (RO, LH Type) **Value After Reset:** 0x0 **Exists:** Always

7.2.63 VR_PCS_BIT_BNDRY

- **Description:** VR PCS Block Synchronization bit boundary Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8021 : (DWCXLP_APB3_16) ? (0x8021*0x2) : (0x8021*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** SNPS_RSVDPARAM_11||SNPS_RSVDPARAM_15

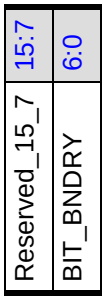


Table 7-364 Fields for Register: VR_PCS_BIT_BNDRY

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	BIT_BNDRY	R	Bit boundary from Block synchronization logic **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

7.2.64 VR_PCS_SFTY_DISABLE

- **Description:** VR PCS MMD Safety Disable Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F1 : (DWCXLP_APB3_16) ? (0x80F1*0x2) : (0x80F1*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_ASP||DWCXLP_ADD_ECC_PROT)

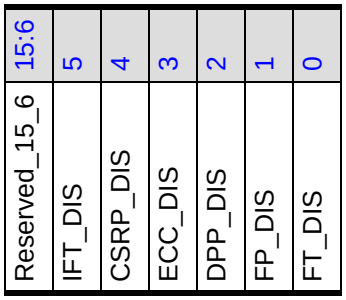


Table 7-365 Fields for Register: VR_PCS_SFTY_DISABLE

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	IFT_DIS	R/W	Interface Time-Out Disable. When this bit is set to 1, it disables the Interface time-out safety feature. **Value After Reset:** 0x0 **Exists:** Always
4	CSRP_DIS	R/W	CSR Protection Disable. When this bit is set to 1, it disables the CSR Protection safety feature. **Value After Reset:** 0x0 **Exists:** Always
3	ECC_DIS	R/W	Error Correction code protection (ECC) Disable. When this bit is set to 1, it disables the ECC safety feature. **Value After Reset:** 0x0 **Exists:** Always
2	DPP_DIS	R/W	Datapath Parity Protection (DPP) Disable. When this bit is set to 1, it disables the DPP safety feature. **Value After Reset:** 0x0 **Exists:** Always

Table 7-365 Fields for Register: VR_PCS_SFTY_DISABLE (Continued)

Bits	Name	Memory Access	Description
1	FP_DIS	R/W	FSM Parity Disable. When this bit is set to 1, it disables the FSM Parity safety feature. **Value After Reset:** 0x0 **Exists:** Always
0	FT_DIS	R/W	FSM Time-out Disable. When this bit is set to 1, it disables the FSM time-out safety feature. **Value After Reset:** 0x0 **Exists:** Always

7.2.65 VR_PCS_SFTY_UE_INTR0

- **Description:** VR PCS MMD Uncorrectable Safety Error Interrupt Register 0
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F2 : (DWCXLP_APB3_16) ? (0x80F2*0x2) : (0x80F2*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** DWCXLP_ASP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved_15	PSEQ_ERR	CSR_INT	DPP_INT	PHY_IFT_INT	APB3_IFT_INT	Reserved_9	EEEFT_INT	CSRFT_INT	RXFT_INT	TXFT_INT	Reserved_4	EEEFP_INT	CSRFP_INT	RXFP_INT	TXFP_INT

Table 7-366 Fields for Register: VR_PCS_SFTY_UE_INTR0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14	PSEQ_ERR	R	Error in Power-up Sequence FSM. This bit is read as 1 when the power-sequence FSM is stuck in a state other than 'POWER_GOOD'. This field is present only for Synopsys PHY configurations.**Note:** When power-up sequence FSM is any other state other than 'POWER_GOOD', read to any register returns bit[15] as 1 and bits [4:2] indicate the power-up sequence FSM state (PSEQ_STATE field of VR_PCS_DIG_STS register.) **Value After Reset:** 0x0 **Exists:** Always
13	CSR_INT	R/W	Control & Status register Error Interrupt . SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects error in Control and Status register. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-366 Fields for Register: VR_PCS_SFTY_UE_INTR0 (Continued)

Bits	Name	Memory Access	Description
12	DPP_INT	R/W	Datapath Parity Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a parity error in receive data-path logic. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always
11	PHY_IFT_INT	* Varies	PHY Interface Time-out Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a time-out Error at the PHY interface. Once this bit is set, it remains high till software clears it by writing 0.**Note:** This bit is valid only for SNPS PHY configs. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_SNPS_XAUI_PHY ? \"read-write\" : \"read-only\""
10	APB3_IFT_INT	* Varies	APB3/MCI Interface Time-out Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a time-out Error at the APB3/MCI interface. Once this bit is set, it remains high till software clears it by writing 0.**Note:** This bit is valid only for APB3/MCI supported configs. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_MDIO_INTF ? \"read-only\" :\"read-write\""
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	EEEFT_INT	* Varies	EEE FSM Time-out Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a Time-out Error in FSMs operating in clk_eee_i domain. Once this bit is set, it remains high till software clears it by writing 0.**Note:** This bit is valid only for configurations with EEE_EN=Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_EEE_EN ? \"read-write\" : \"read-only\""

Table 7-366 Fields for Register: VR_PCS_SFTY_UE_INTR0 (Continued)

Bits	Name	Memory Access	Description
7	CSRFT_INT	R/W	CSR FSM Time-out Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a Time-out Error in FSMs operating in CSR clock domain. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always
6	RXFT_INT	R/W	Rx FSM Time-out Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a Time-out Error in FSMs operating in Rx clock domain. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always
5	TXFT_INT	R/W	Tx FSM Time-out Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a Time-out Error in FSMs operating in Tx clock domain. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always
4	Reserved_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	EEEFP_INT	* Varies	EEE FSM Parity Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a parity Error in FSMs operating in clk_eee_i domain. Once this bit is set, it remains high till software clears it by writing 0. Once this bit is cleared by software, it remains zero even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_EEE_EN ? \"read-write\" : \"read-only\""

Table 7-366 Fields for Register: VR_PCS_SFTY_UE_INTR0 (Continued)

Bits	Name	Memory Access	Description
2	CSRFP_INT	R/W	CSR FSM Parity Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a parity Error in FSMs operating in CSR clock domain. Once this bit is set, it remains high till software clears it by writing 0. Once this bit is cleared by software, it remains zero even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always
1	RXFP_INT	R/W	Rx FSM Parity Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a parity Error in FSMs operating in Rx clock domain. Once this bit is set, it remains high till software clears it by writing 0. Once this bit is cleared by software, it remains zero even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always
0	TXFP_INT	R/W	Tx FSM Parity Error Interrupt. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects a parity Error in FSMs operating in Tx clock domain. Once this bit is set, it remains high till software clears it by writing 0. Once this bit is cleared by software, it remains zero even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always

7.2.66 VR_PCS_SFTY_UE_INTR1

- **Description:** VR PCS MMD Uncorrectable Safety Error Interrupt Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F3 : (DWCXLP_APB3_16) ? (0x80F3*0x2) : (0x80F3*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_ASP||DWCXLP_ADD_ECC_PROT)

15	14	13	12	11:9	8	7	6	5	4	3	2	1	0
DS_MEM3_UE_INT	DS_MEM2_UE_INT	DS_MEM1_UE_INT	DS_MEM0_UE_INT	Reserved_11_9	RSFECEF_UE_INT	FEC_MEM7_UE_INT	FEC_MEM6_UE_INT	FEC_MEM5_UE_INT	FEC_MEM4_UE_INT	FEC_MEM3_UE_INT	FEC_MEM2_UE_INT	FEC_MEM1_UE_INT	FEC_MEM0_UE_INT

Table 7-367 Fields for Register: VR_PCS_SFTY_UE_INTR1

Bits	Name	Memory Access	Description
15	DS_MEM3_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from De-Skew Memory for lane 3. This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in De-Skew Memory for lane 3. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
14	DS_MEM2_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from De-Skew Memory for lane 2. This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in De-Skew Memory for lane 2. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-367 Fields for Register: VR_PCS_SFTY_UE_INTR1 (Continued)

Bits	Name	Memory Access	Description
13	DS_MEM1_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from De-Skew Memory for lane 1. This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in De-Skew Memory for lane 1. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!DWCXLP_25G_ONLY) ? \"read-write\" : \"read-only\""
12	DS_MEM0_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from De-Skew Memory for lane 0. This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in De-Skew Memory for lane 0. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!DWCXLP_25G_ONLY) ? \"read-write\" : \"read-only\""
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RSFECEF_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from RS-FEC Error-Forwarding Memory. This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in RS-FEC Error-Forwarding Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_RSFEC_EN||DWCXLP_RS544) ? \"read-write\" : \"read-only\""

Table 7-367 Fields for Register: VR_PCS_SFTY_UE_INTR1 (Continued)

Bits	Name	Memory Access	Description
7	FEC_MEM7_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from Base-R FEC Error Forwarding Memory for lane 3/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in Base-R FEC Error Forwarding Memory for lane 3/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""
6	FEC_MEM6_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from Base-R FEC Error-Correction Memory for lane 2/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in Base-R FEC Error-Correction Memory for lane 2/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""
5	FEC_MEM5_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from Base-R FEC Error Forwarding Memory for lane 2/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in Base-R FEC Error Forwarding Memory for lane 2/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""

Table 7-367 Fields for Register: VR_PCS_SFTY_UE_INTR1 (Continued)

Bits	Name	Memory Access	Description
4	FEC_MEM4_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from Base-R FEC Error-Correction Memory for lane 1/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in Base-R FEC Error-Correction Memory for lane 1/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""
3	FEC_MEM3_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from Base-R FEC Error Forwarding Memory for lane 1/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in Base-R FEC Error Forwarding Memory for lane 1/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""
2	FEC_MEM2_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""

Table 7-367 Fields for Register: VR_PCS_SFTY_UE_INTR1 (Continued)

Bits	Name	Memory Access	Description
1	FEC_MEM1_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from Base-R FEC Error Forwarding Memory for lane 0/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in Base-R FEC Error Forwarding Memory lane 0/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_FEC_EN||DWCXLP_RSFEC_EN) ? \"read-write\" : \"read-only\""
0	FEC_MEM0_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects an uncorrectable ECC Error in Base-R FEC Error-Correction Memory lane 0/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_FEC_EN||DWCXLP_RSFEC_EN) ? \"read-write\" : \"read-only\""

7.2.67 VR_PCS_SFTY_CE_INTR

- **Description:** VR PCS MMD Correctable Safety Error Interrupt Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F4 : (DWCXLP_APB3_16) ? (0x80F4*0x2) : (0x80F4*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_ASP||DWCXLP_ADD_ECC_PROT)

15	14	13	12	11:9	8	7	6	5	4	3	2	1	0
DS_MEM3_CE_INT	DS_MEM2_CE_INT	DS_MEM1_CE_INT	DS_MEM0_CE_INT	Reserved_11_9	RSFECEF_CE_INT	FEC_MEM7_CE_INT	FEC_MEM6_CE_INT	FEC_MEM5_CE_INT	FEC_MEM4_CE_INT	FEC_MEM3_CE_INT	FEC_MEM2_CE_INT	FEC_MEM1_CE_INT	FEC_MEM0_CE_INT

Table 7-368 Fields for Register: VR_PCS_SFTY_CE_INTR

Bits	Name	Memory Access	Description
15	DS_MEM3_CE_INT	* Varies	Correctable ECC Error Interrupt from De-Skew Memory for lane 3. This bit gets set automatically when the safety feature in DWC_xlgpcs detects an correctable ECC Error in De-Skew Memory for lane 3. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""
14	DS_MEM2_CE_INT	* Varies	Correctable ECC Error Interrupt from De-Skew Memory for lane 2. This bit gets set automatically when the safety feature in DWC_xlgpcs detects an correctable ECC Error in De-Skew Memory for lane 2. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE) ? \"read-write\" : \"read-only\""

Table 7-368 Fields for Register: VR_PCS_SFTY_CE_INTR (Continued)

Bits	Name	Memory Access	Description
13	DS_MEM1_CE_INT	* Varies	Correctable ECC Error Interrupt from De-Skew Memory for lane 1. This bit gets set automatically when the safety feature in DWC_xlgpcs detects an correctable ECC Error in De-Skew Memory for lane 1. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!DWCXLP_25G_ONLY) ? \"read-write\" : \"read-only\""
12	DS_MEM0_CE_INT	* Varies	Correctable ECC Error Interrupt from De-Skew Memory for lane 0. This bit gets set automatically when the safety feature in DWC_xlgpcs detects an correctable ECC Error in De-Skew Memory for lane 0. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!DWCXLP_25G_ONLY) ? \"read-write\" : \"read-only\""
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RSFECEF_CE_INT	* Varies	Correctable ECC Error Interrupt from RS-FEC Error-Forwarding Memory. This bit gets set automatically when the safety feature in DWC_xlgpcs detects correctable ECC Error in RS-FEC Error-Forwarding Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_RSFEC_EN||DWCXLP_RS544) ? \"read-write\" : \"read-only\""

Table 7-368 Fields for Register: VR_PCS_SFTY_CE_INTR (Continued)

Bits	Name	Memory Access	Description
7	FEC_MEM7_CE_INT	* Varies	Correctable ECC Error Interrupt from Base-R FEC Error-Correction Memory for lane 3/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects correctable ECC Error in Base-R FEC Error-Correction Memory for lane 3/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""
6	FEC_MEM6_CE_INT	* Varies	Correctable ECC Error Interrupt from Base-R FEC Error-Correction Memory for lane 3/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects correctable ECC Error in Base-R FEC Error-Correction Memory for lane 2/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""
5	FEC_MEM5_CE_INT	* Varies	Correctable ECC Error Interrupt from Base-R FEC Error Forwarding Memory for lane 2/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects correctable ECC Error in Base-R FEC Error Forwarding Memory for lane 2/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""

Table 7-368 Fields for Register: VR_PCS_SFTY_CE_INTR (Continued)

Bits	Name	Memory Access	Description
4	FEC_MEM4_CE_INT	* Varies	Correctable ECC Error Interrupt from Base-R FEC Error-Correction Memory for lane 2/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects correctable ECC Error in Base-R FEC Error-Correction Memory for lane 1/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""
3	FEC_MEM3_CE_INT	* Varies	Correctable ECC Error Interrupt from Base-R FEC Error Forwarding Memory for lane 1/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects correctable ECC Error in Base-R FEC Error Forwarding Memory for lane 1/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""
2	FEC_MEM2_CE_INT	* Varies	Correctable ECC Error Interrupt from Base-R FEC Error-Correction Memory for lane 1/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects correctable ECC Error in Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_MULTI_LANE&&DWCXLP_FEC_EN) ? \"read-write\" : \"read-only\""

Table 7-368 Fields for Register: VR_PCS_SFTY_CE_INTR (Continued)

Bits	Name	Memory Access	Description
1	FEC_MEM1_CE_INT	* Varies	Correctable ECC Error Interrupt from Base-R FEC Error Forwarding Memory for lane 0/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects correctable ECC Error in Base-R FEC Error Forwarding Memory for lane 0/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_FEC_EN||DWCXLP_RSFEC_EN) ? \"read-write\" : \"read-only\""
0	FEC_MEM0_CE_INT	* Varies	Correctable ECC Error Interrupt from Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error Correction Memory. SS, WC (self-set, write-clear). This bit gets set automatically when the safety feature in DWC_xlgpcs detects correctable ECC Error in Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_FEC_EN||DWCXLP_RSFEC_EN) ? \"read-write\" : \"read-only\""

7.2.68 VR_PCS_RSFECEF_ECC_STS

- **Description:** VR PCS MMD RSFEC Error forwarding memory ECC status. This register is applicable only for configurations with DWCXLP_RSFEC_EN = Enabled
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F5 : (DWCXLP_APB3_16) ? (0x80F5*0x2) : (0x80F5*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_RSFEC_EN&&(DWCXLP_ASP||DWCXLP_ADD_ECC_PROT))

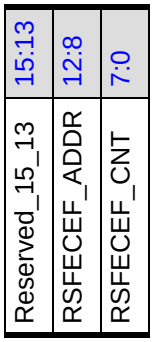


Table 7-369 Fields for Register: VR_PCS_RSFECEF_ECC_STS

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	RSFECEF_ADDR	R	Address of the RSFEC Error-Forwarding Memory (RO, COR Type) data that has been corrected by ECC logic of DWC_xlgpcs. In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always
7:0	RSFECEF_CNT	R	ECC Correctable Error Count from RSFEC Error-Forwarding Memory (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety. The value of this register saturates on reaching 8'hFF. **Value After Reset:** 0x0 **Exists:** Always

7.2.69 VR_PCS_FEC_ECC_STS0

- **Description:** VR PCS MMD FEC ECC memory status for Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error Correction Memory.**Note:** This register is present only if (DWCXLP_FEC_EN || DWCXLP_RSFEC_EN) =True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F6 : (DWCXLP_APB3_16) ? (0x80F6*0x2) : (0x80F6*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** ((DWCXLP_FEC_EN||DWCXLP_RSFEC_EN)&&(DWCXLP_ASP||DWCXLP_ADD_ECC_PROT))

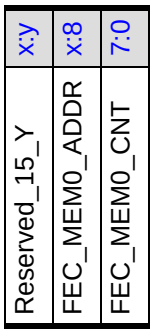


Table 7-370 Fields for Register: VR_PCS_FEC_ECC_STS0

Bits	Name	Memory Access	Description
x:y	Reserved_15_Y	R	Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(DWCXLP_FEC_EN ? (15-8-DWCXLP_EC_MEM_AW+1) : (!DWCXLP_FEC_EN&&DWCXLP_RSFEC_EN) ? (15-8-DWCXLP_RS_EC_MEM_AW+1) : (15-8-DWCXLP_EC_MEM_AW+1))" + "(DWCXLP_FEC_EN ? (8+DWCXLP_EC_MEM_AW) : (!DWCXLP_FEC_EN&&DWCXLP_RSFEC_EN) ? (8+DWCXLP_RS_EC_MEM_AW) : (8+DWCXLP_EC_MEM_AW))" - 1 **Range Variable[y]:** "(DWCXLP_FEC_EN ? (8+DWCXLP_EC_MEM_AW) : (!DWCXLP_FEC_EN&&DWCXLP_RSFEC_EN) ? (8+DWCXLP_RS_EC_MEM_AW) : (8+DWCXLP_EC_MEM_AW))"

Table 7-370 Fields for Register: VR_PCS_FEC_ECC_STS0 (Continued)

Bits	Name	Memory Access	Description
x:8	FEC_MEM0_ADDR	R	Address of the Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error Correction Memory data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(DWCXLP_FEC_EN ? DWCXLP_EC_MEM_AW : (!DWCXLP_FEC_EN&&DWCXLP_RSFEC_EN) ? DWCXLP_RS_EC_MEM_AW : DWCXLP_EC_MEM_AW)" + 7
7:0	FEC_MEM0_CNT	R	ECC Correctable Error Count from Base-R FEC Error-Correction Memory for lane 0/RS-FEC Error Correction Memory (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 0. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.70 VR_PCS_FEC_ECC_STS1

- **Description:** VR PCS MMD FEC ECC memory status for Base-R FEC Error-Forwarding Memory for lane 0/RS-FEC Error Forwarding Memory.**Note:** This register is present only if (DWCXLP_FEC_EN || DWCXLP_RSFEC_EN) =True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F7 : (DWCXLP_APB3_16) ? (0x80F7*0x2) : (0x80F7*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_FEC_ERROR_FWD&&(DWCXLP_ASP||DWCXLP_ADD_ECC_PROT))

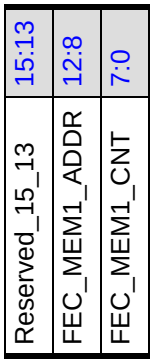


Table 7-371 Fields for Register: VR_PCS_FEC_ECC_STS1

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	FEC_MEM1_ADDR	R	Address of the Base-R FEC Error-Forwarding Memory for lane 0/RS-FEC Error Forwarding Memory data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always

Table 7-371 Fields for Register: VR_PCS_FEC_ECC_STS1 (Continued)

Bits	Name	Memory Access	Description
7:0	FEC_MEM1_CNT	R	ECC Correctable Error Count from Base-R FEC Error-Forwarding Memory for lane 0/RS-FEC Error Forwarding Memory (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 0. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.71 VR_PCS_SFTY_DBG_CTRL

- **Description:** VR PCS MMD safety debug control register**Note:** Various fields in this register should be programmed to desired values prior to enabling the error injection enable bits in VR_PCS_SFTY_ERR_INJ register. Changing the fields of this register, after enabling the error-injection bits, can yield ambiguous results.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80F8 : (DWCXLP_APB3_16) ? (0x80F8*0x2) : (0x80F8*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_ASP||DWCXLP_ADD_ECC_PROT)

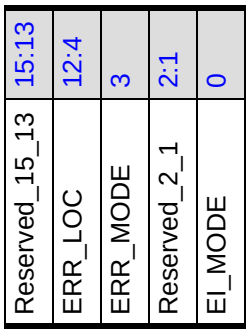


Table 7-372 Fields for Register: VR_PCS_SFTY_DBG_CTRL

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-372 Fields for Register: VR_PCS_SFTY_DBG_CTRL (Continued)

Bits	Name	Memory Access	Description
12:4	ERR_LOC	R/W	Error Location. This field controls the bit-location at which DPP error or ECC error should be introduced. This field is applicable for both DPP Error injection and ECC error injection features. Software should program judicious values to this field depending on the width of the data or parity bus applicable to the feature selected and depending on the current operating mode. If an invalid value is programmed to this field, then error injection behavior is undefined. Encoding of this field is as follows:**For ECC error injection:**If ECC_SEL = 0000 ■ 9'd0 to 9'dD : To introduce error in MEM0 data bits (when ERR_MODE = 0) ■ 9'd(D+1) to 9'd(D+1+A) : To introduce error in MEM0 address bits (when ERR_MODE = 0) ■ 9'd0 to 9'dP : To introduce error in MEM0 ECC bits (when ERR_MODE = 1) **Note:** ■ For 25G configs with RSFEC and no BASE-R FEC, D=79,A=6,P=7 ■ For 25G configs with BASE-R FEC and no RSFEC, D=65,A=4,P=7 ■ For 25G configs with BASE-R FEC and RSFEC: - D=65,A=4,P=7, when operating in BASE-R FEC Mode - D=79,A=5,P=7, when operating in RSFEC Mode If ECC_SEL = 0001 ■ 9'd0 to 9'D : To introduce error in MEM1 data bits (when ERR_MODE = 0) ■ 9'd(D+1) to 9'd(D+1+A) : To introduce error in MEM1 address bits (when ERR_MODE = 0) ■ 9'd0 to 9'P : To introduce error in MEM1 ECC bits (when ERR_MODE = 1) **Note:** ■ For 25G configs with BASE-R FEC (and BASE-R FEC error forwarding) and no RSFEC, D=65,A=4,P=7 ■ For 25G configs with BASE-R FEC and RSFEC: - D=65,A=4,P=7, when operating in BASE-R FEC Mode

Table 7-372 Fields for Register: VR_PCS_SFTY_DBG_CTRL (Continued)

Bits	Name	Memory Access	Description
			- D=79,A=4,P=7, when operating in RSFEC Mode If ECC_SEL = 0010 ■ 9'd0 to 9'd65 : To introduce error in FEC data bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd66 to 9'd70 : To introduce error in FEC address bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd0 to 9'd7 : Introduce error in FEC ECC bits (when ERR_MODE = 1) If ECC_SEL = 0011 ■ 9'd0 to 9'd65 : To introduce error in FEC data bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd66 to 9'd70 : To introduce error in FEC address bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd0 to 9'd7 : Introduce error in FEC ECC bits (when ERR_MODE = 1) If ECC_SEL = 0100 ■ 9'd0 to 9'd65 : To introduce error in FEC data bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd66 to 9'd70 : To introduce error in FEC address bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd0 to 9'd7 : Introduce error in FEC ECC bits (when ERR_MODE = 1) If ECC_SEL = 0101 ■ 9'd0 to 9'd65 : To introduce error in FEC data bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd66 to 9'd70 : To introduce error in FEC address bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd0 to 9'd7 : Introduce error in FEC ECC bits (when ERR_MODE = 1) If ECC_SEL = 0110 ■ 9'd0 to 9'd65 : To introduce error in FEC data bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd66 to 9'd70 : To introduce error in FEC address bits (when ERR_MODE = 0 and Base R FEC) ■ 9'd0 to 9'd7 : Introduce error in FEC ECC bits (when ERR_MODE = 1) If ECC_SEL = 0111 ■ 9'd0 to 9'd65 : To introduce error in FEC data bits (when

Table 7-372 Fields for Register: VR_PCS_SFTY_DBG_CTRL (Continued)

Bits	Name	Memory Access	Description
			■ 9'd0 to 9'd256 : To introduce error in FEC data bits (when ERR_MODE = 0) ■ 9'd257 to 9'd261 : To introduce error in FEC address bits (when ERR_MODE = 0) ■ 9'd0 to 9'd9 : Introduce error in FEC ECC bits (when ERR_MODE = 1) If ECC_SEL = 1001 ■ 9'd0 to 9'd319 : To introduce error in FEC data bits (when ERR_MODE = 0) ■ 9'dX to 9'dX : To introduce error in FEC address bits (when ERR_MODE = 0) ■ 9'd0 to 9'd9 : Introduce error in FEC ECC bits (when ERR_MODE = 1) If ECC_SEL = 1100 ■ 9'd0 to 9'd65 : To introduce error in De-skew data bits (when ERR_MODE = 0) ■ 9'd66 to 9'd71 : To introduce error in De-skew address bits (when ERR_MODE = 0) ■ 9'd0 to 9'd7 : Introduce error in De-skew ECC bits (when ERR_MODE = 1) If ECC_SEL = 1101 ■ 9'd0 to 9'd65 : To introduce error in De-skew data bits (when ERR_MODE = 0) ■ 9'd66 to 9'd71 : To introduce error in De-skew address bits (when ERR_MODE = 0) ■ 9'd0 to 9'd7 : Introduce error in De-skew ECC bits (when ERR_MODE = 1) If ECC_SEL = 1110 ■ 9'd0 to 9'd65 : To introduce error in De-skew data bits (when ERR_MODE = 0) ■ 9'd66 to 9'd71 : To introduce error in De-skew address bits (when ERR_MODE = 0) ■ 9'd0 to 9'd7 : Introduce error in De-skew ECC bits (when ERR_MODE = 1) If ECC_SEL = 1111 ■ 9'd0 to 9'd65 : To introduce error in De-skew data bits (when ERR_MODE = 0) ■ 9'd66 to 9'd71 : To introduce error in De-skew address bits (when ERR_MODE = 0) ■ 9'd0 to 9'd7 : Introduce error in De-skew ECC bits (when ERR_MODE = 1)

Table 7-372 Fields for Register: VR_PCS_SFTY_DBG_CTRL (Continued)

Bits	Name	Memory Access	Description
3	ERR_MODE	R/W	Error Mode This bit determines whether DPP error or ECC error should be introduced in the data-bits or parity-bits. This field is applicable for both DPP Error injection and ECC error injection features. The encoding of this field is as follows: Encoding of this field is as follows: ■ 0: Introduce error in the data bits ■ 1: Introduce error in the parity/ECC bits **Value After Reset:** 0x0 **Exists:** Always
2:1	Reserved_2_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	EI_MODE	R/W	Error Injection Mode for Selected memory This bit introduces whether single-bit or two-bit error should be introduced to the ECC logic catering to Selected Memory. Encoding of this field is as follows: ■ 0: 1-bit ECC data/parity bit corruption ■ 1: 2-bit ECC data/parity bit corruption For 1-bit error, the bit-location at which error should be introduced is controlled by ERR_LOC field of this register. For 2-bit error mode, bit [0] is always corrupted. The 2nd location at which error should be introduced is controlled by ERR_LOC field. Hence, software is expected to program a non-zero value to ERR_LOC field. In addition, ERR_MODE field of this register controls whether the data or parity bits should be corrupted. **Value After Reset:** 0x0 **Exists:** Always

7.2.72 VR_PCS_SFTY_TMR_CTRL

- **Description:** VR PCS MMD safety Timer Control register.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x80F9 : (DWCXLP_APB3_16) ? (0x80F9*0x2) : (0x80F9*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_ASP)

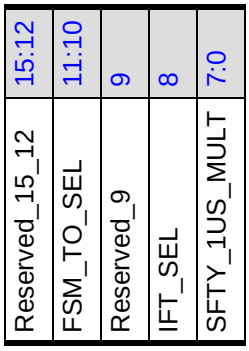


Table 7-373 Fields for Register: VR_PCS_SFTY_TMR_CTRL

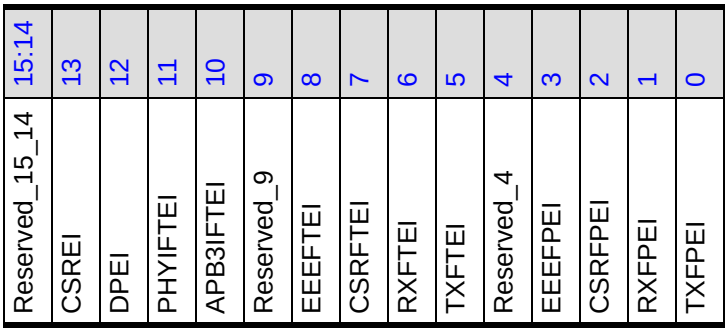
Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:10	FSM_TO_SEL	R/W	FSM Time-out Select. DWC_xlgpcs allocates appropriate time-out values to various FSMs inside it. These pre-allocated time-out values range from 10 micro-seconds to 1 second. This field provides an option to scale the time-out values if required Encoding of this field is as follows: ■ 2'b00: Timers get scaled by 0.1 ■ 2'b01: Timers run for standard duration ■ 2'b10: Timers get scaled by 10 ■ 2'b11: Reserved This field has no effect if FASTSIM bit is set to 1. **Value After Reset:** 0x1 **Exists:** Always

Table 7-373 Fields for Register: VR_PCS_SFTY_TMR_CTRL (Continued)

Bits	Name	Memory Access	Description
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	IFT_SEL	R/W	Interface Time-out Selection. Encoding of this field is as follows: ■ 0: Interface Time-out duration is 10us ■ 1: Interface Time-out duration is 50us **Value After Reset:** 0x0 **Exists:** Always
7:0	SFTY_1US_MULT	R/W	Safety 1us Clock Tic Multiplying Factor. This bit is the multiplying factor to the clk_csr_i clock time period to make it equal to 1us. This value should be programmed such that the (clk_csr_i time_period * 2* SFTY_1US_MULT) should be 1us or as close as possible to 1us. For example, if the clk_csr_i clock time period is 10 ns, the value of this field needs to be set to 50. The default value of this register is set as 50 corresponding to a 100 MHz clk_csr_i. **Value After Reset:** 0x32 **Exists:** Always

7.2.73 VR_PCS_SFTY_ERR_INJ0

- **Description:** VR PCS MMD safety Error Injection register 0
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x80FA : (DWCXLP_APB3_16) ? (0x80FA*0x2) : (0x80FA*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_ASP)



15:14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved_15_14	CSREI	DPEI	PHYIFTEI	APB3IFTEI	Reserved_9	EEEFTEI	CSRFTEI	RXFTEI	TXFTEI	Reserved_4	EEEFPEI	CSRFPEI	RXFPEI	TXFPEI

Table 7-374 Fields for Register: VR_PCS_SFTY_ERR_INJ0

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	CSREI	R/W	CSR Error Injection (R/W, SC Type). When this bit is set to 1, DWC_xlgpcs injects error into the CSR protection feature. Consequently, CSR_INT interrupt gets set. **Value After Reset:** 0x0 **Exists:** Always
12	DPEI	R/W	Datapath Parity Error Inject (R/W, SC Type). When this bit is set to 1, DWC_xlgpcs injects a parity error in its data-path for one clock cycle. This bit self-clears after data-path parity error interrupt is generated **Value After Reset:** 0x0 **Exists:** Always

Table 7-374 Fields for Register: VR_PCS_SFTY_ERR_INJ0 (Continued)

Bits	Name	Memory Access	Description
11	PHYIFTEI	* Varies	PHY Interface Time-out Inject (R/W, SC Type). This bit can be set to 1 to inject interface time-out error at the PHY Interface. This bit self-clears after error interrupt is generated. This bit is valid only for SNPS PHY supported configs. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_SNPS_XAUI_PHY ? \"read-write\" : \"read-only\""
10	APB3IFTEI	* Varies	APB3/MCI Interface Time-out Inject (R/W, SC Type). This bit can be set to 1 to inject interface time-out error at the MCI/APB3 Interface. This bit self-clears after error interrupt is generated. This bit is valid only for MCI/APB3 supported configs. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_MDIO_INTF ? \"read-only\" :\"read-write\""
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	EEEFTEI	* Varies	EEE FSM Parity Time-out Inject (R/W, SC Type). This bit can be set to 1 to inject time-out error in all the FSMs operating in clk_eee_i. This bit self-clears after error interrupt is generated. This bit is valid only for configurations with EEE_EN=Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_EEE_EN ? \"read-write\" : \"read-only\""
7	CSRFTEI	R/W	CSR FSM Parity Time-out Inject (R/W, SC Type). This bit can be set to 1 to inject time-out error in all the FSMs operating in clk_csr_i. This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
6	RXFTEI	R/W	RX FSM Parity Time-Out Inject (R/W, SC Type). This bit can be set to 1 to inject time-out error in all the FSMs operating in Receive Clock domains. This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

Table 7-374 Fields for Register: VR_PCS_SFTY_ERR_INJ0 (Continued)

Bits	Name	Memory Access	Description
5	TXFTEI	R/W	TX FSM Parity Time-out Inject (R/W, SC Type). This bit can be set to 1 to inject time-out error in all the FSMs operating in Transmit Clock domain(s). This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
4	Reserved_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	EEEFPEI	* Varies	EEE FSM Parity Error Inject (R/W, SC Type). This bit can be set to 1 to inject parity error in all the FSMs operating in clk_eee_i. This bit self-clears after error interrupt is generated. This bit is valid only for configurations with EEE_EN=Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_EEE_EN ? \"read-write\" : \"read-only\""
2	CSRFPEI	R/W	CSR FSM Parity Error Inject (R/W, SC Type). This bit can be set to 1 to inject parity error in all the FSMs operating in clk_csr_i. This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
1	RXFPEI	R/W	RX FSM Parity Error Inject (R/W, SC Type). This bit can be set to 1 to inject parity error in all the FSMs operating in Receive Clock domain(s). This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
0	TXFPEI	R/W	TX FSM Parity Error Inject (R/W, SC Type). This bit can be set to 1 to inject parity error in all the FSMs operating in Transmit Clock domain(s). This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

7.2.74 VR_PCS_SFTY_ERR_INJ1

- **Description:** VR PCS MMD safety Error Injection register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x80FB : (DWCXLP_APB3_16) ? (0x80FB*0x2) : (0x80FB*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_ASP||DWCXLP_ADD_ECC_PROT)

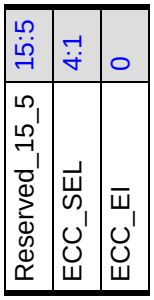


Table 7-375 Fields for Register: VR_PCS_SFTY_ERR_INJ1

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-375 Fields for Register: VR_PCS_SFTY_ERR_INJ1 (Continued)

Bits	Name	Memory Access	Description
4:1	ECC_SEL	R/W	ECC Memory Select for Error Injection. Encoding of this field is as follows: If DWCXLP_FEC_EN is Enabled. ■ 4'b0000: FEC Mem0 (Configure if RSFEC or Base R FEC is Enabled.) Else ■ 4'b0000: Reserved If DWCXLP_FEC_ERROR_FWD is Enabled. ■ 4'b0001: FEC Mem1 (Configure if Base R FEC Error Forwarding/RSFEC is Enabled.) Else ■ 4'b0001: Reserved If (DWCXLP_FEC_EN& DWCXLP_MULTI_LANE) is Enabled. ■ 4'b0010: FEC Mem2 (Configure if Base R FEC is Enabled.) Else ■ 4'b0010: Reserved If (DWCXLP_FEC_ERROR_FWD&&DWCXLP_MULTI_LANE) is Enabled. ■ 4'b0011: FEC Mem3 (Configure if Base R FEC Error Forwarding is Enabled.) Else ■ 4'b0011: Reserved If (DWCXLP_FEC_EN&&DWCXLP_MULTI_LANE) is Enabled. ■ 4'b0100: FEC Mem4 (Configure if Base R FEC is Enabled.) Else ■ 4'b0100: Reserved If (DWCXLP_FEC_ERROR_FWD&&DWCXLP_MULTI_LANE) is Enabled. ■ 4'b0101: FEC Mem5 (Configure if Base R FEC Error Forwarding is Enabled.) Else ■ 4'b0101: Reserved If (DWCXLP_FEC_EN&&DWCXLP_MULTI_LANE) is Enabled. ■ 4'b0110: FEC Mem6 (Configure if Base R FEC is Enabled.) Else

Table 7-375 Fields for Register: VR_PCS_SFTY_ERR_INJ1 (Continued)

Bits	Name	Memory Access	Description
			■ 4'b1001: Reserved. ■ 4'b1010-4'b1011: Reserved. If (!DWCXLP_25G_ONLY) is Enabled. ■ 4'b1100: De-skew Mem0 Else ■ 4'b1100: Reserved. If (!DWCXLP_25G_ONLY) is Enabled. ■ 4'b1101: De-skew Mem1 Else ■ 4'b1101: Reserved. If (DWCXLP_MULTI_LANE) is Enabled. ■ 4'b1110: De-skew Mem2 Else ■ 4'b1110: Reserved. If (DWCXLP_MULTI_LANE) is Enabled. ■ 4'b1111: De-skew Mem3 Else ■ 4'b1111: Reserved. **Value After Reset:** 0x0 **Exists:** Always
0	ECC_EI	R/W	Memory Error Injection enable. ■ 1: DWC_xlgpcs corrupts the data that is written to the memory selected by ECC_SEL. Type of data corruption is controlled by EI_MODE field of VR_PCS_SFTY_DBG_CTRL register. This bit self-clears after the corresponding error interrupt is generated. **Note:** Error is injected only for one clock cycle. **Value After Reset:** 0x0 **Exists:** Always

7.2.75 VR_PCS_FEC_ECC_STS2

- **Description:** VR PCS MMD FEC ECC memory status for Base-R FEC Error-Correction Memory for lane 1.**Note:** This register is present only if (DWCXLP_FEC_EN && DWCXLP_MULTI_LANE) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8100 : (DWCXLP_APB3_16) ? (0x8100*0x2) : (0x8100*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** ((DWCXLP_FEC_EN&&DWCXLP_MULTI_LANE)&&DWCXLP_ASP)

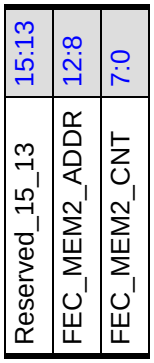


Table 7-376 Fields for Register: VR_PCS_FEC_ECC_STS2

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	FEC_MEM2_ADDR	R	Address of the Base-R FEC Error-Correction Memory for lane 1 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always
7:0	FEC_MEM2_CNT	R	ECC Correctable Error Count from Base-R FEC Error-Correction Memory for lane 1 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 1. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.76 VR_PCS_FEC_ECC_STS3

- **Description:** VR PCS MMD FEC ECC memory status for Base-R FEC Error-Forward Memory for lane 1.**Note:** This register is present only if (DWCXLP_FEC_ERROR_FWD && DWCXLP_MULTI_LANE) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8101 : (DWCXLP_APB3_16) ? (0x8101*0x2) : (0x8101*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** ((DWCXLP_FEC_ERROR_FWD&&DWCXLP_MULTI_LANE)&&DWCXLP_ASP)

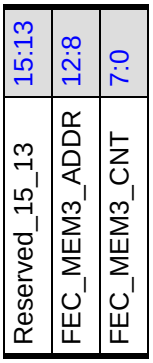


Table 7-377 Fields for Register: VR_PCS_FEC_ECC_STS3

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	FEC_MEM3_ADDR	R	Address of the Base-R FEC Error-Forward Memory for lane 1 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always
7:0	FEC_MEM3_CNT	R	ECC Correctable Error Count from Base-R FEC Error-Forward Memory for lane 1 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 1. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.77 VR_PCS_FEC_ECC_STS4

- **Description:** VR PCS MMD FEC ECC memory status for Base-R FEC Error-Correction Memory for lane 2.**Note:** This register is present only if (DWCXLP_FEC_EN && DWCXLP_MULTI_LANE) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8102 : (DWCXLP_APB3_16) ? (0x8102*0x2) : (0x8102*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** ((DWCXLP_FEC_EN&&DWCXLP_MULTI_LANE)&&DWCXLP_ASP)

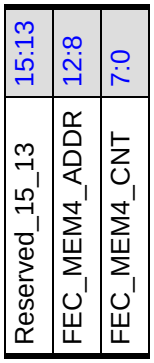


Table 7-378 Fields for Register: VR_PCS_FEC_ECC_STS4

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	FEC_MEM4_ADDR	R	Address of the Base-R FEC Error-Correction Memory for lane 2 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always
7:0	FEC_MEM4_CNT	R	ECC Correctable Error Count from Base-R FEC Error-Correction Memory for lane 2 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 2. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.78 VR_PCS_FEC_ECC_STS5

- **Description:** VR PCS MMD FEC ECC memory status for Base-R FEC Error-Forward Memory for lane 2.**Note:** This register is present only if (DWCXLP_FEC_ERROR_FWD && DWCXLP_MULTI_LANE) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8103 : (DWCXLP_APB3_16) ? (0x8103*0x2) : (0x8103*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** ((DWCXLP_FEC_ERROR_FWD&&DWCXLP_MULTI_LANE)&&DWCXLP_ASP)

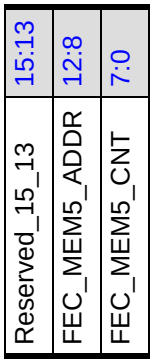


Table 7-379 Fields for Register: VR_PCS_FEC_ECC_STS5

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	FEC_MEM5_ADDR	R	Address of the Base-R FEC Error-Forward Memory for lane 2 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always
7:0	FEC_MEM5_CNT	R	ECC Correctable Error Count from Base-R FEC Error-Forward Memory for lane 2 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 2. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.79 VR_PCS_FEC_ECC_STS6

- **Description:** VR PCS MMD FEC ECC memory status for Base-R FEC Error-Correction Memory for lane 3.**Note:** This register is present only if (DWCXLP_FEC_EN && DWCXLP_MULTI_LANE) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8104 : (DWCXLP_APB3_16) ? (0x8104*0x2) : (0x8104*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** ((DWCXLP_FEC_EN&&DWCXLP_MULTI_LANE)&&DWCXLP_ASP)

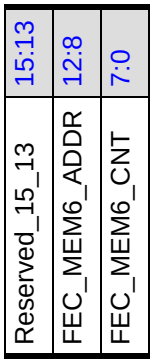


Table 7-380 Fields for Register: VR_PCS_FEC_ECC_STS6

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	FEC_MEM6_ADDR	R	Address of the Base-R FEC Error-Correction Memory for lane 3 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always
7:0	FEC_MEM6_CNT	R	ECC Correctable Error Count from Base-R FEC Error-Correction Memory for lane 3 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 3. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.80 VR_PCS_FEC_ECC_STS7

- **Description:** VR PCS MMD FEC ECC memory status for Base-R FEC Error-Forward Memory for lane 3.**Note:** This register is present only if (DWCXLP_FEC_ERROR_FWD && DWCXLP_MULTI_LANE) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8105 : (DWCXLP_APB3_16) ? (0x8105*0x2) : (0x8105*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** ((DWCXLP_FEC_ERROR_FWD&&DWCXLP_MULTI_LANE)&&DWCXLP_ASP)

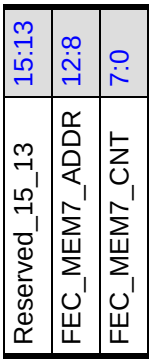


Table 7-381 Fields for Register: VR_PCS_FEC_ECC_STS7

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	FEC_MEM7_ADDR	R	Address of the Base-R FEC Error-Forward Memory for lane 3 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always
7:0	FEC_MEM7_CNT	R	ECC Correctable Error Count from Base-R FEC Error-Forward Memory for lane 3 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 3. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.81 VR_PCS_DSKW_ECC_STS0

- **Description:** De-skew memory status for lane 0.**Note:** This register is present only if (!DWCXLP_25G_ONLY) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x810A : (DWCXLP_APB3_16) ? (0x810A*0x2) : (0x810A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (!DWCXLP_25G_ONLY&&DWCXLP_ASP)

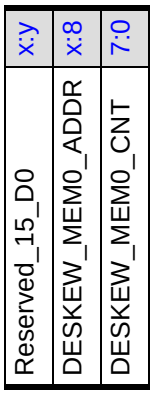


Table 7-382 Fields for Register: VR_PCS_DSKW_ECC_STS0

Bits	Name	Memory Access	Description
x:y	Reserved_15_D0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(15-8-DWCXLP_DSKW_BF_AW+1)" + "(8+DWCXLP_DSKW_BF_AW)" - 1 **Range Variable[y]:** "(8+DWCXLP_DSKW_BF_AW)"
x:8	DESKEW_MEM0_ADDR	R	Address of the De-skew Memory for lane 0 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWCXLP_DSKW_BF_AW + 7

Table 7-382 Fields for Register: VR_PCS_DSKW_ECC_STS0 (Continued)

Bits	Name	Memory Access	Description
7:0	DESKEW_MEM0_CNT	R	ECC Correctable Error Count from De-skew Memory for lane 0 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 0. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.82 VR_PCS_DSKW_ECC_STS1

- **Description:** De-skew memory status for lane 1.**Note:** This register is present only if (!DWCXLP_25G_ONLY) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x810B : (DWCXLP_APB3_16) ? (0x810B*0x2) : (0x810B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (!DWCXLP_25G_ONLY&&DWCXLP_ASP)

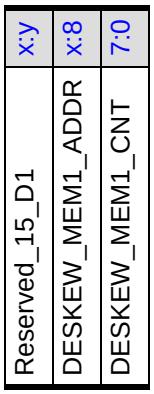


Table 7-383 Fields for Register: VR_PCS_DSKW_ECC_STS1

Bits	Name	Memory Access	Description
x:y	Reserved_15_D1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(15-8-DWCXLP_DSKW_BF_AW+1)" + "(8+DWCXLP_DSKW_BF_AW)" - 1 **Range Variable[y]:** "(8+DWCXLP_DSKW_BF_AW)"
x:8	DESKEW_MEM1_ADDR	R	Address of the De-skew Memory for lane 1 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWCXLP_DSKW_BF_AW + 7

Table 7-383 Fields for Register: VR_PCS_DSKW_ECC_STS1 (Continued)

Bits	Name	Memory Access	Description
7:0	DESKEW_MEM1_CNT	R	ECC Correctable Error Count from De-skew Memory for lane 1 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 1. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.83 VR_PCS_DSKW_ECC_STS2

- **Description:** De-skew memory status for lane 2.**Note:** This register is present only if (DWCXLP_MULTI_LANE) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x810C : (DWCXLP_APB3_16) ? (0x810C*0x2) : (0x810C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_ASP)

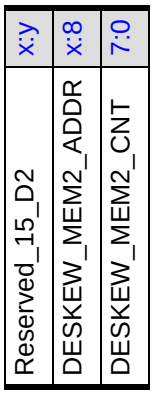


Table 7-384 Fields for Register: VR_PCS_DSKW_ECC_STS2

Bits	Name	Memory Access	Description
x:y	Reserved_15_D2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(15-8-DWCXLP_DSKW_BF_AW+1)" + "(8+DWCXLP_DSKW_BF_AW)" - 1 **Range Variable[y]:** "(8+DWCXLP_DSKW_BF_AW)"
x:8	DESKEW_MEM2_ADDR	R	Address of the De-skew Memory for lane 2 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWCXLP_DSKW_BF_AW + 7

Table 7-384 Fields for Register: VR_PCS_DSKW_ECC_STS2 (Continued)

Bits	Name	Memory Access	Description
7:0	DESKEW_MEM2_CNT	R	ECC Correctable Error Count from De-skew Memory for lane 2 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 2. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.2.84 VR_PCS_DSKW_ECC_STS3

- **Description:** De-skew memory status for lane 3.**Note:** This register is present only if (DWCXLP_MULTI_LANE) = True
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x810D : (DWCXLP_APB3_16) ? (0x810D*0x2) : (0x810D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x030000 : (DWCXLP_APB3_16) ? (0x030000*0x2) : (0x030000*0x4)"
- **Exists:** (DWCXLP_MULTI_LANE&&DWCXLP_ASP)

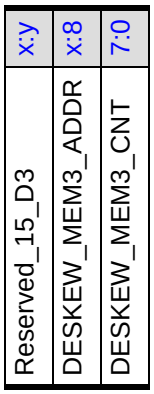


Table 7-385 Fields for Register: VR_PCS_DSKW_ECC_STS3

Bits	Name	Memory Access	Description
x:y	Reserved_15_D3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(15-8-DWCXLP_DSKW_BF_AW+1)" + "(8+DWCXLP_DSKW_BF_AW)" - 1 **Range Variable[y]:** "(8+DWCXLP_DSKW_BF_AW)"
x:8	DESKEW_MEM3_ADDR	R	Address of the De-skew Memory for lane 3 data that has been corrected by ECC logic of DWC_xlgpcs (RO, COR Type). In case ECC has corrected data from multiple memory locations, this field stores the address of the most recent location that has been corrected. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWCXLP_DSKW_BF_AW + 7

Table 7-385 Fields for Register: VR_PCS_DSKW_ECC_STS3 (Continued)

Bits	Name	Memory Access	Description
7:0	DESKEW_MEM3_CNT	R	ECC Correctable Error Count from De-skew Memory for lane 3 (RO, COR Type). This register indicates the number of ECC code-words that have been corrected by DWC_xlgpcs safety feature for lane 3. The value of this register saturates on reaching 8'hFF **Value After Reset:** 0x0 **Exists:** Always

7.3 DWC_xlgpcs_map/AN_MMD Registers

The Auto-Negotiation (AN) MMD Standard (SR) & Vendor Specific (VR) address block. Follow the link for the register to see a detailed description of the register.

Table 7-386 Registers for Address Block: DWC_xlgpcs_map/AN_MMD

Register	Offset	Description
SR_AN_CTRL on page 1193	* Varies	SR AN MMD Control Register
SR_AN_STS on page 1195	* Varies	SR AN MMD Status Register
SR_AN_DEV_ID1 on page 1198	* Varies	SR AN MMD Device Identifier Register 1
SR_AN_DEV_ID2 on page 1199	* Varies	SR AN MMD Device Identifier Register 2
SR_AN_DEV_PKG1 on page 1201	* Varies	SR AN MMD Devices in Package Register 1
SR_AN_DEV_PKG2 on page 1203	* Varies	SR AN MMD Devices in Package Register 2
SR_AN_PKG1 on page 1204	* Varies	SR AN MMD Package Identifier Register 1
SR_AN_PKG2 on page 1205	* Varies	SR AN MMD Package Identifier Register 2
SR_AN_ADV1 on page 1207	* Varies	SR AN MMD Advertisement Register 1 The AN Advertisement Register 1, 2, and 3 contain the Base...
SR_AN_ADV2 on page 1209	* Varies	SR AN MMD Advertisement Register 2
SR_AN_ADV3 on page 1210	* Varies	SR AN MMD Advertisement Register 3
SR_AN_LP_ABL1 on page 1212	* Varies	SR AN MMD LP Base Page Ability Register 1 The management uses the SR AN LP Base Page Ability...
SR_AN_LP_ABL2 on page 1214	* Varies	SR AN MMD LP Base Page Ability Register 2
SR_AN_LP_ABL3 on page 1215	* Varies	SR AN MMD LP Base Page Ability Register 3
SR_AN_XNP_TX1 on page 1216	* Varies	SR AN MMD XNP Transmit Register 1 This register holds the next page Link Code Word to be transmitted...
SR_AN_XNP_TX2 on page 1218	* Varies	SR AN MMD XNP Transmit Register 2
SR_AN_XNP_TX3 on page 1219	* Varies	SR AN MMD XNP Transmit Register 3
SR_AN_LP_XNP_ABL1 on page 1220	* Varies	SR AN MMD LP XNP Ability Register 1 The management uses the AN LP XNP Ability Register 1, 2, or...
SR_AN_LP_XNP_ABL2 on page 1222	* Varies	SR AN MMD LP XNP Ability Register 2
SR_AN_LP_XNP_ABL3 on page 1223	* Varies	SR AN MMD LP XNP Ability Register 3
SR_AN_COMP_STS on page 1224	* Varies	SR AN MMD Completion Status Register
SR_AN_COMP_STS2 on page 1228	* Varies	SR AN MMD Completion Status 2 Register

Table 7-386 Registers for Address Block: DWC_xlgpcs_map/AN_MMD (Continued)

Register	Offset	Description
SR_AN_EEE_ABL on page 1230	* Varies	SR AN MMD EEE Capability Register
SR_AN_EEE_LP_ABL on page 1234	* Varies	SR AN MMD EEE Link Partner Capability Register
SR_AN_EEE_ABL2 on page 1238	* Varies	SR AN MMD EEE Capability 2 Register
SR_AN_EEE_LP_ABL2 on page 1240	* Varies	SR AN MMD EEE Link Partner Capability 2 Register
VR_AN_DIG_CTRL1 on page 1241	* Varies	VR AN MMD Digital Control1 Register
VR_AN_INTR_MSK on page 1245	* Varies	VR AN MMD Interrupt Mask Register
VR_AN_INTR on page 1247	* Varies	VR AN MMD Interrupt Register Note: The interrupt port sbd_intr_o goes high when any bit of this...
VR_AN_MODE_CTRL on page 1250	* Varies	VR AN MMD KR Mode Control Register
VR_AN_TIMER_CTRL0 on page 1251	* Varies	VR AN MMD Timer Control 0 Register
VR_AN_TIMER_CTRL1 on page 1253	* Varies	VR AN MMD Timer Control 1 Register
VR_AN_TIMER_CTRL2 on page 1255	* Varies	VR AN MMD Timer Control 2 Register
VR_AN_TIMER_CTRL3 on page 1256	* Varies	VR AN MMD Timer Control 3 Register
VR_AN_DIG_STS on page 1257	* Varies	VR AN MMD Digital Status Register

7.3.1 SR_AN_CTRL

- **Description:** SR AN MMD Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

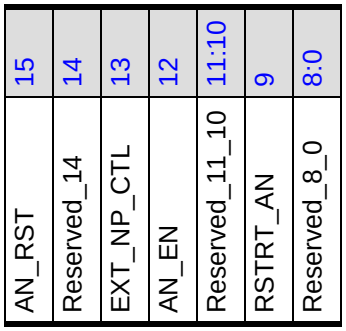


Table 7-387 Fields for Register: SR_AN_CTRL

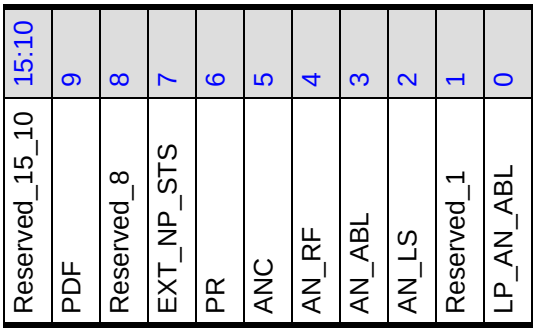
Bits	Name	Memory Access	Description
15	AN_RST	R/W	Auto-Negotiation Soft Reset (RW,SC Type) When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks, except the Management Interface block, are reset. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: ■ Synopsys PHY: This bit is self cleared when Bits[4:2] in VR PCS MMD Digital Status Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. ■ Non-Synopsys PHY: This bit is self cleared after the following: -- 32 clk_csr_i clocks for the MCI interface -- 1 MDC clock period for the MDIO interface **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-387 Fields for Register: SR_AN_CTRL (Continued)

Bits	Name	Memory Access	Description
13	EXT_NP_CTL	R/W	Extended Next Page Control ■ 0: Extended next page is disabled ■ 1: Extended next page is enabled **Value After Reset:** 0x1 **Exists:** Always
12	AN_EN	R/W	Auto-Negotiation Enable ■ 0: Auto-negotiation is disabled. The SR PMA MMD Control1 Register or SR PCS Control2 Register determines the link speed. ■ 1: The host determines the link speed based on the outcome of the Clause 73 auto-negotiation. **Value After Reset:** 0x1 **Exists:** Always
11:10	Reserved_11_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	RSTRT_AN	R/W	Restart Auto-Negotiation (RW,SC Type) ■ 0: Normal operation ■ 1: The auto-negotiation process is started or restarted, irrespective of the current state of the auto-negotiation process. **Value After Reset:** 0x0 **Exists:** Always
8:0	Reserved_8_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.3.2 SR_AN_STS

- **Description:** SR AN MMD Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1 : (DWCXLP_APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



15:10	9	8	7	6	5	4	3	2	1	0
Reserved_15_10	PDF	Reserved_8	EXT_NP_STS	PR	ANC	AN_RF	AN_ABL	AN_LS	Reserved_1	LP_AN_ABL

Table 7-388 Fields for Register: SR_AN_STS

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	PDF	R	Parallel Detection Fault (RO,LH Type) The DWC_xlgpcs sets this bit when a parallel detection fault is detected in the Clause 73 auto-negotiation ■ 0: No fault detected ■ 1: Parallel detection fault detected A parallel detection fault is declared when the following happens: 1. The device detects the legacy device by monitoring the Rx Link Up. 2. After Rx Link Up, the Rx Link Down is detected before 50 ms. **Value After Reset:** 0x0 **Exists:** Always
8	Reserved_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-388 Fields for Register: SR_AN_STS (Continued)

Bits	Name	Memory Access	Description
7	EXT_NP_STS	R	Extended Next Page Status ■ 0: Extended next page is not allowed if the local device or the link partner does not support extended next page. ■ 1: Extended next page format is used. It indicates that the local device and the link partner have indicated to use the extended next page support. **Value After Reset:** 0x0 **Exists:** Always
6	PR	R	Page Received (RO, LH Type) ■ 0: No page received ■ 1: Page is received and the corresponding Link Code Word is stored in the following: - SR AN MMD LP Base Page Ability Register 1, 2, or 3 for base page. - SR AN MMD XNP Transmit Register 1, 2, or 3 for next page. This bit is Latched-High type. **Value After Reset:** 0x0 **Exists:** Always
5	ANC	R	Auto-Negotiation Complete ■ 0: Indicates that the auto-negotiation process is not complete. ■ 1: Indicates that the auto-negotiation process is complete. **Value After Reset:** 0x0 **Exists:** Always
4	AN_RF	R	Auto-Negotiation Remote Fault (RO, LH Type) A remote fault occurs when the auto-negotiation state transitions to the TX_DISABLED state from the ACKNOWLEDGE_DETECT or NEXT_PAGE_WAIT state. This happens when a valid DME page reception is not successful. ■ 0: No fault detected ■ 1: Auto-negotiation process detected a remote fault This is a latched-high bit. **Value After Reset:** 0x0 **Exists:** Always

Table 7-388 Fields for Register: SR_AN_STS (Continued)

Bits	Name	Memory Access	Description
3	AN_ABL	R	Auto-Negotiation Ability ■ 0: The local device does not support auto-negotiation. ■ 1: The local device can perform auto-negotiation. This bit is always set to 1. **Value After Reset:** 0x1 **Exists:** Always
2	AN_LS	R	Auto-negotiation Link Status (R0,LL Type) This bit indicates the link status as determined by the auto-negotiation process. ■ 0: Clause 73 auto-negotiation process is not complete or the process could not determine a valid link. ■ 1: Clause 73 auto-negotiation process is complete and the process successfully determined a valid link. **Value After Reset:** 0x0 **Exists:** Always
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	LP_AN_ABL	R	Link Partner Auto-Negotiation Ability ■ 0: The link partner is not able to participate in auto-negotiation. ■ 1: The link partner is able to participate in auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always

7.3.3 SR_AN_DEV_ID1

- **Description:** SR AN MMD Device Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x2 : (DWCXLP_APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



Table 7-389 Fields for Register: SR_AN_DEV_ID1

Bits	Name	Memory Access	Description
15:0	AN_DEV_OUI_3_18	R	Organizationally Unique Identifier This field indicates the Organizationally Unique Identifier (OUI) of the manufacturer of AN device from bits 3 to 18. You can configure this value. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR Control MMD AN Device Identifier Register 1 modifies the content of this register **Value After Reset:** DWCXLP_IEEE_AN_DEV_OUI_3_18 **Exists:** Always

7.3.4 SR_AN_DEV_ID2

- **Description:** SR AN MMD Device Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x3 : (DWCXLP_APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

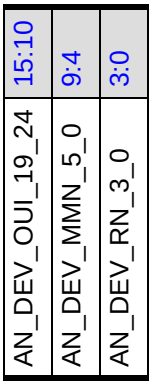


Table 7-390 Fields for Register: SR_AN_DEV_ID2

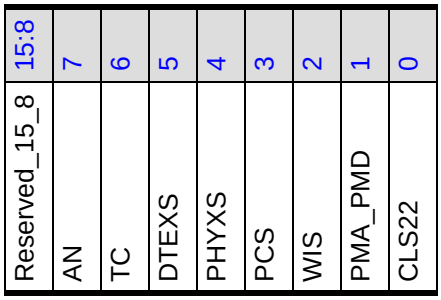
Bits	Name	Memory Access	Description
15:10	AN_DEV_OUI_19_24	R	Organizationally Unique Identifier This indicates the Organizationally Unique Identifier (OUI) of the AN device manufacturer from bits 19 to 24. You can configure this value. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR Control MMD AN Device Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_AN_DEV_OUI_19_24 **Exists:** Always
9:4	AN_DEV_MMN_5_0	R	Model Number The 5-bit Model Number of the AN Device. You can configure this value. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR Control MMD AN Device Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_AN_DEV_MODEL_NO_0_5 **Exists:** Always

Table 7-390 Fields for Register: SR_AN_DEV_ID2 (Continued)

Bits	Name	Memory Access	Description
3:0	AN_DEV_RN_3_0	R	Revision Number The 4-bit Revision Number of the AN Device. You can configure this value. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR Control MMD AN Device Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_AN_DEV_REV_NO_0_3 **Exists:** Always

7.3.5 SR_AN_DEV_PKG1

- **Description:** SR AN MMD Devices in Package Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x5 : (DWCXLP_APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



15:8	7	6	5	4	3	2	1	0
Reserved_15_8	AN	TC	DTEXS	PHYXS	PCS	WIS	PMA_PMD	CLS22

Table 7-391 Fields for Register: SR_AN_DEV_PKG1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	AN	R	Auto-Negotiation MMD This bit indicates that an AN MMD is present in the package. Default Value: If 'Add CL73 Autoneg Support' option is selected, this bit returns 1. For all other configurations, this bit returns 0. **Value After Reset:** "(DWCXLP_CL73_AN) ? 0x1 : 0x0" **Exists:** Always
6	TC	R	TC MMD This bit indicates that the device has the TC MMDs in the package. This bit is always 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-391 Fields for Register: SR_AN_DEV_PKG1 (Continued)

Bits	Name	Memory Access	Description
5	DTEXS	R	DTE XGXS MMD This bit indicates that the device has the DTE XS MMD in the package. **Value After Reset:** 0x0 **Exists:** Always
4	PHYXS	R	PHY XGXS MMD This bit indicates that the device has the PHY XS MMD in the package. **Value After Reset:** 0x0 **Exists:** Always
3	PCS	R	PCS MMD This bit indicates that the device has PCS MMD in the package. **Value After Reset:** 0x1 **Exists:** Always
2	WIS	R	WIS MMD This bit indicates that the device has the WIS MMD in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	PMA_PMD	R	PMA or PMD MMD This bit indicates that the device has the PMA or PMD MMD in the package. Default Value: ■ This bit is set to 1 for the following configurations: Backplane Ethernet PCS for Synopsys PHY ■ This bit returns 0 for all other configurations. **Value After Reset:** 0x1 **Exists:** Always
0	CLS22	R	Clause 22 Register Support This bit indicates that the Clause 22 (1G) functional register sets are implemented in this package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

7.3.6 SR_AN_DEV_PKG2

- **Description:** SR AN MMD Devices in Package Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x6 : (DWCXLP_APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

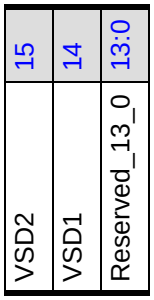


Table 7-392 Fields for Register: SR_AN_DEV_PKG2

Bits	Name	Memory Access	Description
15	VSD2	R	Vendor-Specific Device 2 This bit indicates that a vendor-specific device 2 is present in the package. Default Value: For configurations with CL37_AN = Enabled, this bit returns 1 indicating that a vendor-specific 1G PCS MMD is present. For all other configurations, this bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
14	VSD1	R	Vendor-Specific Device 1 This bit indicates that the vendor-specific device 1 is present in the package. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
13:0	Reserved_13_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.3.7 SR_AN_PKG1

- **Description:** SR AN MMD Package Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xe : (DWCXLP_APB3_16) ? (0xe*0x2) : (0xe*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



Table 7-393 Fields for Register: SR_AN_PKG1

Bits	Name	Memory Access	Description
15:0	AN_PKG_OUI_3_18	R	Organizationally Unique Identifier DWC_xlgpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. The value is configured within coreConsultant. This read-only field contains Bits[18:3] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS and AN) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR Control MMD Package Identifier Register 1 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_OUI_3_18 **Exists:** Always

7.3.8 SR_AN_PKG2

- **Description:** SR AN MMD Package Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0xf : (DWCXLP_APB3_16) ? (0xf*0x2) : (0xf*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

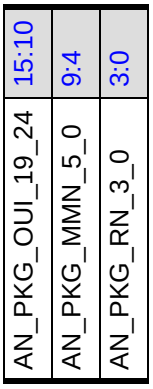


Table 7-394 Fields for Register: SR_AN_PKG2

Bits	Name	Memory Access	Description
15:10	AN_PKG_OUI_19_24	R	Organizationally Unique Identifier DWC_xlgpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. This value is configured by the host. This read-only field contains Bits[24:19] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS and AN) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_OUI_19_24 **Exists:** Always

Table 7-394 Fields for Register: SR_AN_PKG2 (Continued)

Bits	Name	Memory Access	Description
9:4	AN_PKG_MMN_5_0	R	Model Number This read-only field contains the 5-bit Model Number of the Package in which the MMDs are installed. This value is configured by the host. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_MODEL_NO_0_5 **Exists:** Always
3:0	AN_PKG_RN_3_0	R	Revision Number This read-only field contains the 4-bit Revision Number of the Package. This value is configured by the host. It is common for all the MMDs in the device. It is common for all MMDs (PMA, PCS and AN) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** DWCXLP_IEEE_PKG_REV_NO_0_3 **Exists:** Always

7.3.9 SR_AN_ADV1

- **Description:** SR AN MMD Advertisement Register 1

  The AN Advertisement Register 1, 2, and 3 contain the Base Page Link Code Word for the local device. The host can program these fields. However, the host must program the AN Advertisement Register 1, 2, and 3 within 60 ms after Power On Reset or Soft Reset.**Note:** Before writing to this register, the host must write to the AN Advertisement Register 2 and 3.
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x10 : (DWCXLP_APB3_16) ? (0x10*0x2) : (0x10*0x4)" + "(DWCXLP_APB3_INTF| |DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

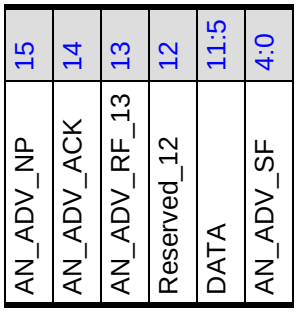


Table 7-395 Fields for Register: SR_AN_ADV1

Bits	Name	Memory Access	Description
15	AN_ADV_NP	R/W	Next Page ■ 0: The management does not intend to exchange the next page after the base page. If the link partner supports Next Page (indicated by Bit 15 of SR AN MMD LP Base Page Ability Register 1 being set to 1), the management performs the next page exchange by programming the SR AN MMD XNP Transmit Register 1, 2, or 3 with a NULL message. ■ 1: The management intends to exchange the next page after the base page. The management performs the next page exchange by programming the AN XNP Transmit Register 1, 2, or 3 with the message it intends to convey to its Link Partner. **Value After Reset:** 0x0 **Exists:** Always

Table 7-395 Fields for Register: SR_AN_ADV1 (Continued)

Bits	Name	Memory Access	Description
14	AN_ADV_ACK	R	Acknowledgement ■ 0: The device did not receive three consecutive and valid DME pages. ■ 1: The device received three consecutive and valid DME pages. **Value After Reset:** 0x0 **Exists:** Always
13	AN_ADV_RF_13	R/W	Remote Fault The management interface sets this bit to provide the Remote Fault information to its link partner: ■ 0: No fault detected ■ 1: Remote fault detected **Value After Reset:** 0x0 **Exists:** Always
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:5	DATA	R/W	Link Code Word Encoding DATA[11:5] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. **Value After Reset:** 0x0 **Exists:** Always
4:0	AN_ADV_SF	R/W	Selector Field This field is set to 5'b0001 to indicate compliance to the IEEE Std 802.3, Annex 28A. **Value After Reset:** 0x1 **Exists:** Always

7.3.10 SR_AN_ADV2

- **Description:** SR AN MMD Advertisement Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x11 : (DWCXLP_APB3_16) ? (0x11*0x2) : (0x11*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

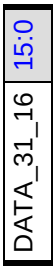


Table 7-396 Fields for Register: SR_AN_ADV2

Bits	Name	Memory Access	Description
15:0	DATA_31_16	R/W	Link Code Word Encoding DATA[31:16] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. DATA[31:21] corresponds to the Ability field A[10:0]. **Value After Reset:** " DWCXLP_KR4_KR ? 0x0380 : !DWCXLP_FEC_EN ? (DWCXLP_KR4_100G ? 0x3380 : DWCXLP_KR4_50G ? 0x0380 : 0x0080) : (DWCXLP_RSFEC_EN? (DWCXLP_KR4_100G ? 0xF380 : DWCXLP_KR4_50G ? 0xC380 : 0xC080) : (DWCXLP_KR4_100G ? 0x7380 : DWCXLP_KR4_50G ? 0x4380 : 0x4080))" **Exists:** Always

7.3.11 SR_AN_ADV3

- **Description:** SR AN MMD Advertisement Register 3
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x12 : (DWCXLP_APB3_16) ? (0x12*0x2) : (0x12*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

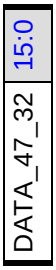


Table 7-397 Fields for Register: SR_AN_ADV3

Bits	Name	Memory Access	Description
15:0	DATA_47_32	R/W	Link Code Word Encoding DATA[47:32] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. DATA[45:32] corresponds to the Ability field A[24:11]. Encoding of ability bits relevant to DWC_xlgpcs are shown below: ■ A[12]: 5GBASE-KR ■ A[13]: 50GBASE-KR or 50GBASE-CR ■ A[14]: 100GBASE-KR2 or 100GBASE-CR2 When DATA[46] (F0) is set to 1, it indicates that the local device supports 10G FEC. When DATA[47] (F1) is set to 1, it indicates that the local device is requesting the link partner to enable 10G FEC. The host should enable the FEC feature by programming the SR PMA MMD BASE-KR FEC Control Register only when both devices advertise FEC capability and alteast one device requests to enable FEC. DATA[44] (F2) indicates request for RS-FEC in 25G Mode. DATA[45] (F3) indicates request for BASE-R FEC in 25G Mode. **Value After Reset:** "(DWCXLP_RS544&&DWCXLP_KR4_100G) ? 0xE: DWCXLP_RS544 ? 0x6 : 0x2" **Exists:** Always

7.3.12 SR_AN_LP_ABL1

- **Description:** SR AN MMD LP Base Page Ability Register 1

  The management uses the SR AN LP Base Page Ability Registers1, 2, and 3 to get the Base Page information received from its link partner. These register contents are valid when bit 6 of the SR AN MMD Status Register is set to 1.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x13 : (DWCXLP_APB3_16) ? (0x13*0x2) : (0x13*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

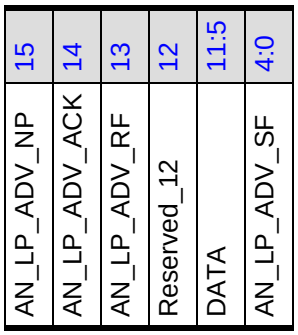


Table 7-398 Fields for Register: SR_AN_LP_ABL1

Bits	Name	Memory Access	Description
15	AN_LP_ADV_NP	R	Next Page ■ 0: The link partner does not want to exchange the Next Page after the Base Page. ■ 1: The link partner wants to exchange the Next Page after the Base Page. **Value After Reset:** 0x0 **Exists:** Always
14	AN_LP_ADV_ACK	R	Acknowledgement ■ 0: The link partner did not receive three consecutive and valid DME pages. ■ 1: The link partner received three consecutive and valid DME pages. **Value After Reset:** 0x0 **Exists:** Always

Table 7-398 Fields for Register: SR_AN_LP_ABL1 (Continued)

Bits	Name	Memory Access	Description
13	AN_LP_ADV_RF	R	Remote Fault ■ 0: No fault. ■ 1: The link partner wants to convey the Remote Fault information. **Value After Reset:** 0x0 **Exists:** Always
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:5	DATA	R	Link Code Word Encoding DATA[11:5] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. **Value After Reset:** 0x0 **Exists:** Always
4:0	AN_LP_ADV_SF	R	Selector Field This field is set to 5'h01 to indicate compliance to the IEEE Std 802.3, Annex 28A. **Value After Reset:** 0x0 **Exists:** Always

7.3.13 SR_AN_LP_ABL2

- **Description:** SR AN MMD LP Base Page Ability Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x14 : (DWCXLP_APB3_16) ? (0x14*0x2) : (0x14*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

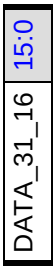


Table 7-399 Fields for Register: SR_AN_LP_ABL2

Bits	Name	Memory Access	Description
15:0	DATA_31_16	R	Link Code Word Encoding DATA[31:16] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. **Value After Reset:** 0x0 **Exists:** Always

7.3.14 SR_AN_LP_ABL3

- **Description:** SR AN MMD LP Base Page Ability Register 3
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x15 : (DWCXLP_APB3_16) ? (0x15*0x2) : (0x15*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

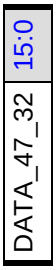


Table 7-400 Fields for Register: SR_AN_LP_ABL3

Bits	Name	Memory Access	Description
15:0	DATA_47_32	R	Link Code Word Encoding DATA[47:32] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. When DATA[46] is set to 1, it indicates that the link partner supports FEC. When DATA[47] is set to 1, it indicates that the link partner is requesting the local device to enable FEC. The host should enable the FEC feature by programming the SR PMA MMD BASE-KR FEC Control Register only when both devices advertise FEC capability and alteast one device requests to enable FEC. **Value After Reset:** 0x0 **Exists:** Always

7.3.15 SR_AN_XNP_TX1

- **Description:** SR AN MMD XNP Transmit Register 1

  This register holds the next page Link Code Word to be transmitted when extended next page is enabled by the host by using Bit 15 of SR AN MMD Advertisement Register 1. On reset, this register contains the Message page with Message code set to NULL PAGE. The host can update this register with any valid Message page that it intends to transmit. Note: Before writing to this register, the host must write to the SR AN XNP Transmit Register 2 and 3.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x16 : (DWCXLP_APB3_16) ? (0x16*0x2) : (0x16*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

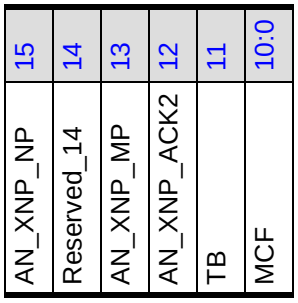


Table 7-401 Fields for Register: SR_AN_XNP_TX1

Bits	Name	Memory Access	Description
15	AN_XNP_NP	R/W	Next Page ■ 0: The management does not intend to exchange the next page after the current page. If the link partner supports the next page (as indicated by Bit 15 of SR AN MMD LP Base Page Ability Register 1 being set to 1), the management performs the next page exchange by programming the SR AN MMD XNP Transmit Register 1, 2, and 3 with a NULL message. ■ 1: The management intends to exchange the next page after the current page. In such a case, the management needs to program the SR AN XNP Transmit Register 1, 2, and 3 with the message to send to its link partner. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-401 Fields for Register: SR_AN_XNP_TX1 (Continued)

Bits	Name	Memory Access	Description
13	AN_XNP_MP	R/W	Message Page This bit indicates that the management contains a formatted message. ■ 0: Unformatted Message Page ■ 1: Formatted Message Page **Value After Reset:** 0x0 **Exists:** Always
12	AN_XNP_ACK2	R/W	Acknowledgement2 ■ 0: Management did not act on the information received from the link partner. ■ 1: Management acted on the information received from the link partner. **Value After Reset:** 0x0 **Exists:** Always
11	TB	R	Toggle Bit The link partner uses this bit to differentiate between two successive pages. The DWC_xlgpcs toggles this bit after sending any new next page. **Value After Reset:** 0x0 **Exists:** Always
10:0	MCF	R/W	Message Code Field or Unformatted Message This bit depends on Bit 13: ■ AN_XNP_MP = 1: Indicates the Message Code as in IEEE Std 802.3, Annex 28C ■ AN_XNP_MP = 0: Indicates an Unformatted Message which is vendor-specific **Value After Reset:** 0x0 **Exists:** Always

7.3.16 SR_AN_XNP_TX2

- **Description:** SR AN MMD XNP Transmit Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x17 : (DWCXLP_APB3_16) ? (0x17*0x2) : (0x17*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



Table 7-402 Fields for Register: SR_AN_XNP_TX2

Bits	Name	Memory Access	Description
15:0	UMCF1	R/W	Unformatted Message Code Field1 **Value After Reset:** 0x0 **Exists:** Always

7.3.17 SR_AN_XNP_TX3

- **Description:** SR AN MMD XNP Transmit Register 3
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x18 : (DWCXLP_APB3_16) ? (0x18*0x2) : (0x18*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



Table 7-403 Fields for Register: SR_AN_XNP_TX3

Bits	Name	Memory Access	Description
15:0	UMCF2	R/W	Unformatted Message Code Field2 **Value After Reset:** 0x0 **Exists:** Always

7.3.18 SR_AN_LP_XNP_ABL1

- **Description:** SR AN MMD LP XNP Ability Register 1

  The management uses the AN LP XNP Ability Register 1, 2, or 3 to get the next page information from the link partner. These register contents are valid when Bit 6 of SR AN MMD Status Register is set to 1 during next page exchange.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x19 : (DWCXLP_APB3_16) ? (0x19*0x2) : (0x19*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

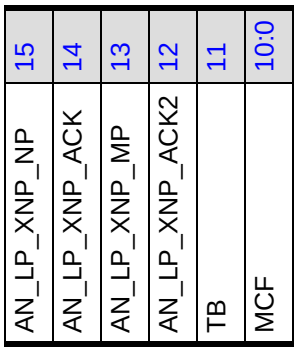


Table 7-404 Fields for Register: SR_AN_LP_XNP_ABL1

Bits	Name	Memory Access	Description
15	AN_LP_XNP_NP	R	Next Page When set, this bit indicates that the link partner wants to exchange the Next Page after the current page. **Value After Reset:** 0x0 **Exists:** Always
14	AN_LP_XNP_ACK	R	Acknowledgement This bit indicates that the link partner is successfully able to receive the Local Device Link Code Word. ■ 0: The link partner not able to receive the Local Device Link Code Word. ■ 1: The link partner is able to receive the Local Device Link Code Word. **Value After Reset:** 0x0 **Exists:** Always

Table 7-404 Fields for Register: SR_AN_LP_XNP_ABL1 (Continued)

Bits	Name	Memory Access	Description
13	AN_LP_XNP_MP	R	Message Page ■ 1: SR AN MMD LP XNP Ability Register 1, 2, and 3 together contain message pages. ■ 0: SR AN MMD LP XNP Ability Register 1, 2, and 3 together contain unformatted message pages. **Value After Reset:** 0x0 **Exists:** Always
12	AN_LP_XNP_ACK2	R	Acknowledgement2 ■ 0: The link partner has not received the page, or it is unable to act on the information received from the local device. ■ 1: The link partner has received the page and it is able to act on the information received from the local device. **Value After Reset:** 0x0 **Exists:** Always
11	TB	R	Toggle Bit This bit is set to make the Management differentiate successive messages: ■ 0: Same Message ■ 1: Next Message **Value After Reset:** 0x0 **Exists:** Always
10:0	MCF	R	Message Code Field or Unformatted Message This field depends on Bit 13 of this register: ■ AN_LP_XNP_MP = 1: This field contains a Message Code described in IEEE Std 802.3, Annexure 28C. ■ AN_LP_XNP_MP = 0: This field contains an unformatted message which is vendor-specific. **Value After Reset:** 0x0 **Exists:** Always

7.3.19 SR_AN_LP_XNP_ABL2

- **Description:** SR AN MMD LP XNP Ability Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x1A : (DWCXLP_APB3_16) ? (0x1A*0x2) : (0x1A*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



Table 7-405 Fields for Register: SR_AN_LP_XNP_ABL2

Bits	Name	Memory Access	Description
15:0	UCF1	R	Unformatted Message Code Field1 **Value After Reset:** 0x0 **Exists:** Always

7.3.20 SR_AN_LP_XNP_ABL3

- **Description:** SR AN MMD LP XNP Ability Register 3
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x1B : (DWCXLP_APB3_16) ? (0x1B*0x2) : (0x1B*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

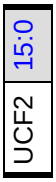


Table 7-406 Fields for Register: SR_AN_LP_XNP_ABL3

Bits	Name	Memory Access	Description
15:0	UCF2	R	Unformatted Message Code Field2 **Value After Reset:** 0x0 **Exists:** Always

7.3.21 SR_AN_COMP_STS

- **Description:** SR AN MMD Completion Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x30 : (DWCXLP_APB3_16) ? (0x30*0x2) : (0x30*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

15	14	13	12	11	10	9:8	7	6	5	4	3	2	1	0
AN_COMP_5G	Reserved_14	XXVG_CRKR	XXVG_CRKRS	AN_COMP_100GCR4	AN_COMP_100GKR4	Reserved_9_8	AN_COMP_RSFEC	AN_COMP_40GCR4	AN_COMP_40GKR4	AN_COMP_FEC_KR_KR4	AN_COMP_KR	AN_COMP_KX4	AN_COMP_KX	BP_AN_ABL

Table 7-407 Fields for Register: SR_AN_COMP_STS

Bits	Name	Memory Access	Description
15	AN_COMP_5G	R	5GBASE-KR Mode Auto-negotiation Completion This bit is valid when the DWC_xpcs is configured for the 5GBASE-KR mode. When this bit is set, it indicates that the auto-negotiation is complete in the 5GBASE-KR mode. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-407 Fields for Register: SR_AN_COMP_STS (Continued)

Bits	Name	Memory Access	Description
13	XXVG_CRKR	R	25GBASE-KR or 25GBASE-CR Mode Auto-negotiation Completion This bit is valid when the DWC_xlgpcs is configured when 25G mode is supported in configuration (when DWCXLP_KR4_KR=0). When this bit is set, it indicates that the auto-negotiation is complete in the 25GBASE-CR/25GBASE-KR mode. **Value After Reset:** 0x0 **Exists:** Always
12	XXVG_CRKRS	R	25GBASE-KR-S or 25GBASE-CR-S Mode Auto-negotiation Completion This bit is valid when the DWC_xlgpcs is configured when 25G mode is supported in configuration (when DWCXLP_KR4_KR=0). When this bit is set, it indicates that the auto-negotiation is complete in the 25GBASE-CR-S/25GBASE-KR-S mode. 25GBASE-CR-S refers to the twin axial copper cable medium without Cause 108 RS-FEC capability. 25GBASE-KR-S refers to the backplane medium without Clause 108 RS-FEC capability. **Value After Reset:** 0x0 **Exists:** Always
11	AN_COMP_100GCR4	R	100GBASE-CR4 Mode Auto-negotiation Completion This bit is valid when the DWC_xlgpcs is configured for multi-lane configuration. When this bit is set, it indicates that the auto-negotiation is complete in the 100GBASE-CR4 mode. **Value After Reset:** 0x0 **Exists:** Always
10	AN_COMP_100GKR4	R	100GBASE-KR4 Mode Auto-negotiation Completion This bit is valid when the DWC_xlgpcs is configured for the Multi lane configurations. When this bit is set, it indicates that the auto-negotiation is complete in the 100GBASE-KR4 mode. **Value After Reset:** 0x0 **Exists:** Always
9:8	Reserved_9_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-407 Fields for Register: SR_AN_COMP_STS (Continued)

Bits	Name	Memory Access	Description
7	AN_COMP_RSFEC	R	RSFEC Mode Auto-negotiation Completion This bit is valid when the DWC_xlgpcs is configured to work with RSFEC mode. When this bit is set, it indicates that the auto-negotiation is complete in the RSFEC mode. **Value After Reset:** 0x0 **Exists:** Always
6	AN_COMP_40GCR4	R	40GBASE-CR4 Mode Auto-negotiation Completion This bit is valid when the DWC_xlgpcs is configured for multi-lane configuration. When this bit is set, it indicates that the auto-negotiation is complete in the 40GBASE-CR4 mode. **Value After Reset:** 0x0 **Exists:** Always
5	AN_COMP_40GKR4	R	40GBASE-KR4 Mode Auto-negotiation Completion This bit is valid when the DWC_xlgpcs is configured for the Multi lane configurations. When this bit is set, it indicates that the auto-negotiation is complete in the 40GBASE-KR4 mode. **Value After Reset:** 0x0 **Exists:** Always
4	AN_COMP_FEC_KR_KR4	R	BASE-R FEC Mode Auto-negotiation Completion This bit is valid when the DWC_xlgpcs is configured to work with BASE-R FEC mode. When this bit is set, it indicates that the auto-negotiation is complete in the FEC mode. **Value After Reset:** 0x0 **Exists:** Always
3	AN_COMP_KR	R	10GBASE-KR Mode Auto-negotiation Completion This bit is valid when the DWC_xlgpcs is configured for the KR mode. When this bit is set, it indicates that the auto-negotiation is complete in the 10GKR mode. **Value After Reset:** 0x0 **Exists:** Always
2	AN_COMP_KX4	R	Reserved Field **Value After Reset:** 0x0 **Exists:** Always
1	AN_COMP_KX	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-407 Fields for Register: SR_AN_COMP_STS (Continued)

Bits	Name	Memory Access	Description
0	BP_AN_ABL	R	Backplane Auto-Negotiation Ability This bit indicates the backplane auto-negotiation ability. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always

7.3.22 SR_AN_COMP_STS2

- **Description:** SR AN MMD Completion Status 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x31 : (DWCXLP_APB3_16) ? (0x31*0x2) : (0x31*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN&&DWCXLP_RS544)

15:7	6	5	4	3	2	1	0
Reserved_15_7	AN_COMP_IRSFEC	AN_COMP_400GKR4	AN_COMP_200GKR2	AN_COMP_100GKR1	Reserved_2	AN_COMP_100GKR2	AN_COMP_50GKR

Table 7-408 Fields for Register: SR_AN_COMP_STS2

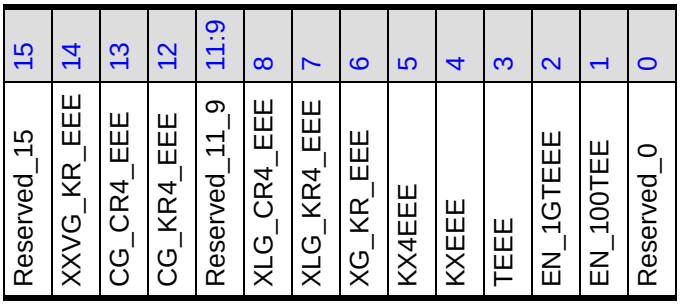
Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	AN_COMP_IRSFEC	R	Interleaved RS-FEC Mode Auto-negotiation Completion. When this bit is set, it indicates that the auto-negotiation is complete in the 100GBASE-KR1 or 100GBASE-CR1 mode with interleaved RSFEC enabled. **Value After Reset:** 0x0 **Exists:** Always
5	AN_COMP_400GKR4	R	400GBASE-KR4 or 400GBASE-CR4 Mode Auto-negotiation Completion. This bit always return 0 as DWC_xlgpcs does not support 400G. **Value After Reset:** 0x0 **Exists:** Always

Table 7-408 Fields for Register: SR_AN_COMP_STS2 (Continued)

Bits	Name	Memory Access	Description
4	AN_COMP_200GKR2	R	200GBASE-KR2 or 200GBASE-CR2 Mode Auto-negotiation Completion. This bit always return 0 as DWC_xlgpcs does not support 200G. **Value After Reset:** 0x0 **Exists:** Always
3	AN_COMP_100GKR1	R	100GBASE-KR1 or 100GBASE-CR1 Mode Auto-negotiation Completion. When this bit is set, it indicates that the auto-negotiation is complete in the 100GBASE-KR1 or 100GBASE-CR1 mode. **Value After Reset:** 0x0 **Exists:** Always
2	Reserved_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	AN_COMP_100GKR2	R	100GBASE-KR2 or 100GBASE-CR2 Mode Auto-negotiation Completion. When this bit is set, it indicates that the auto-negotiation is complete in the 100GBASE-KR2 or 100GBASE-CR2 mode. **Value After Reset:** 0x0 **Exists:** Always
0	AN_COMP_50GKR	R	50GBASE-KR or 50GBASE-CR Mode Auto-negotiation Completion. When this bit is set, it indicates that the auto-negotiation is complete in the 50GBASE-KR or 50GBASE-CR mode. **Value After Reset:** 0x0 **Exists:** Always

7.3.23 SR_AN_EEE_ABL

- **Description:** SR AN MMD EEE Capability Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x3C : (DWCXLP_APB3_16) ? (0x3C*0x2) : (0x3C*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN&&DWCXLP_EEE_EN)



15	14	13	12	11:9	8	7	6	5	4	3	2	1	0
Reserved_15	XXVG_KR_EEE	CG_CR4_EEE	CG_KR4_EEE	Reserved_11_9	XLG_CR4_EEE	XLG_KR4_EEE	XG_KR_EEE	KX4EEE	KXEEE	TEEE	EN_1GTEEE	EN_100TEE	Reserved_0

Table 7-409 Fields for Register: SR_AN_EEE_ABL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14	XXVG_KR_EEE	R/W	25GBASE-R Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 25GBASE-R mode. ■ 0: EEE is not supported in the 25GBASE-R mode ■ 1: EEE is supported in the 25GBASE-R mode Default Value: In 25G Enabled configurations with EEE, this bit always returns 1. Otherwise, this bit returns 0. **Value After Reset:** "(!DWCXLP_KR4_KR) ? \"0x1\" : \"0x0\"" **Exists:** Always

Table 7-409 Fields for Register: SR_AN_EEE_ABL (Continued)

Bits	Name	Memory Access	Description
13	CG_CR4_EEE	R/W	100GBASE-KR4 Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 100GBASE-CR4 mode. ■ 0: EEE is not supported in the 100GBASE-CR4 mode ■ 1: EEE is supported in the 100GBASE-CR4 mode Default Value: In 100G Enabled configurations with EEE, this bit always returns 1. Otherwise, this bit returns 0. **Value After Reset:** "(DWCXLP_KR4_100G) ? \"0x1\" : \"0x0\"" **Exists:** Always
12	CG_KR4_EEE	R/W	100GBASE-KR4 Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 100GBASE-KR4 mode. ■ 0: EEE is not supported in the 100GBASE-KR4 mode ■ 1: EEE is supported in the 100GBASE-KR4 mode Default Value: In 100G Enabled configurations with EEE, this bit always returns 1. Otherwise, this bit returns 0. **Value After Reset:** "(DWCXLP_KR4_100G) ? \"0x1\" : \"0x0\"" **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	XLG_CR4_EEE	R/W	40GBASE-CR4 Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 40GBASE-KR4 mode. ■ 0: EEE is not supported in the 40GBASE-CR4 mode ■ 1: EEE is supported in the 40GBASE-CR4 mode Default Value: In 40G Enabled configurations with EEE, this bit always returns 1. Otherwise, this bit returns 0. **Value After Reset:** "(DWCXLP_25G_ONLY) ? \"0x0\" : \"0x1\"" **Exists:** Always

Table 7-409 Fields for Register: SR_AN_EEE_ABL (Continued)

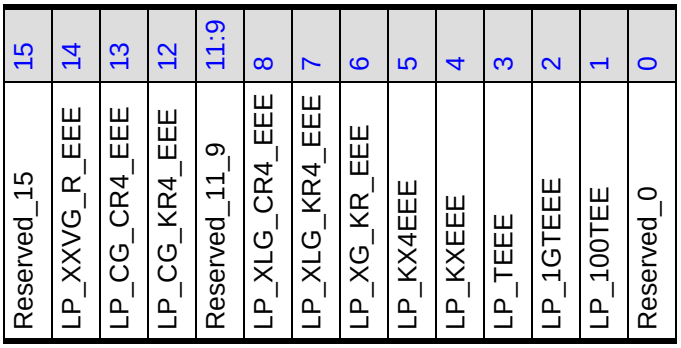
Bits	Name	Memory Access	Description
7	XLG_KR4_EEE	R/W	40GBASE-KR4 Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 40GBASE-KR4 mode. ■ 0: EEE is not supported in the 40GBASE-KR4 mode ■ 1: EEE is supported in the 40GBASE-KR4 mode Default Value: In 40G Enabled configurations with EEE, this bit always returns 1. Otherwise, this bit returns 0. **Value After Reset:** "(DWCXLP_25G_ONLY) ? \"0x0\" : \"0x1\"" **Exists:** Always
6	XG_KR_EEE	R/W	10GBASE-KR Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 10GBASE-KR mode. ■ 0: EEE is not supported in the 10GBASE-KR mode ■ 1: EEE is supported in the 10GBASE-KR mode Default Value: In KR_Present configurations, this bit always returns 1. Otherwise, this bit returns 0. **Value After Reset:** 0x1 **Exists:** Always
5	KX4EEE	R/W	10GBASE-KX4 Mode EEE Support This bit must always be kept at default value of 0. **Value After Reset:** 0x0 **Exists:** Always
4	KXEEE	R/W	1000BASE-KX Mode EEE Support This bit must always be kept at default value of 0. **Value After Reset:** 0x0 **Exists:** Always
3	TEEE	R	10GBASE-T Mode EEE Support This bit indicates the EEE support in the 10GBASE-T mode: ■ 0: EEE is not supported in the 10GBASE-T mode ■ 1: EEE is supported in the 10GBASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-409 Fields for Register: SR_AN_EEE_ABL (Continued)

Bits	Name	Memory Access	Description
2	EN_1GTEEE	R	1000BASE-T Mode EEE Support This bit indicates the EEE support in the 1000BASE-T mode: ■ 0: EEE is not supported in the 1000BASE-T mode ■ 1: EEE is supported in the 1000BASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	EN_100TEE	R	100BASE-T Mode EEE Support This bit indicates the EEE support in the 100BASE-T mode: ■ 0: EEE is not supported in the 100BASE-T mode ■ 1: EEE is supported in the 100BASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.3.24 SR_AN_EEE_LP_ABL

- **Description:** SR AN MMD EEE Link Partner Capability Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x3D : (DWCXLP_APB3_16) ? (0x3D*0x2) : (0x3D*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



15	14	13	12	11:9	8	7	6	5	4	3	2	1	0
Reserved_15	LP_XXVG_R_EEE	LP_CG_CR4_EEE	LP_CG_KR4_EEE	Reserved_11_9	LP_XLG_CR4_EEE	LP_XLG_KR4_EEE	LP_XG_KR_EEE	LP_KX4EEE	LP_KXEEE	LP_TEEE	LP_1GTEEE	LP_100TEE	Reserved_0

Table 7-410 Fields for Register: SR_AN_EEE_LP_ABL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14	LP_XXVG_R_EEE	R	25GBASE-R Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 25GBASE-R mode ■ 0: The link partner supports EEE in the 25GBASE-R mode. ■ 1: The link partner does not support EEE in the 25GBASE-R mode. **Value After Reset:** 0x0 **Exists:** Always

Table 7-410 Fields for Register: SR_AN_EEE_LP_ABL (Continued)

Bits	Name	Memory Access	Description
13	LP_CG_CR4_EEE	R	100GBASE-CR4 Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 100GBASE-CR4 mode ■ 0: The link partner supports EEE in the 100GBASE-CR4 mode. ■ 1: The link partner does not support EEE in the 100GBASE-CR4 mode. **Value After Reset:** 0x0 **Exists:** Always
12	LP_CG_KR4_EEE	R	100GBASE-KR4 Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 100GBASE-KR4 mode ■ 0: The link partner supports EEE in the 100GBASE-KR4 mode. ■ 1: The link partner does not support EEE in the 100GBASE-KR4 mode. **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	LP_XLG_CR4_EEE	R	40GBASE-CR4 Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 40GBASE-CR4 mode ■ 0: The link partner supports EEE in the 40GBASE-CR4 mode. ■ 1: The link partner does not support EEE in the 40GBASE-CR4 mode. **Value After Reset:** 0x0 **Exists:** Always
7	LP_XLG_KR4_EEE	R	40GBASE-KR4 Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 40GBASE-KR4 mode ■ 0: The link partner supports EEE in the 40GBASE-KR4 mode. ■ 1: The link partner does not support EEE in the 40GBASE-KR4 mode. **Value After Reset:** 0x0 **Exists:** Always

Table 7-410 Fields for Register: SR_AN_EEE_LP_ABL (Continued)

Bits	Name	Memory Access	Description
6	LP_XG_KR_EEE	R	10GBASE-KR Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 10GBASE-KR mode ■ 0: The link partner supports EEE in the 10GBASE-KR mode. ■ 1: The link partner does not support EEE in the 10GBASE-KR mode. **Value After Reset:** 0x0 **Exists:** Always
5	LP_KX4EEE	R	KX4 Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 10GBASE-KX4 mode. ■ 0: The link partner supports EEE in the 10GBASE-KX4 mode. ■ 1: The link partner does not support EEE in the 10GBASE-KX4 mode. **Value After Reset:** 0x0 **Exists:** Always
4	LP_KXEEE	R	KX Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 1000BASE-KX mode. ■ 0: The link partner supports EEE in the 1000BASE-KX mode. ■ 1: The link partner does not support EEE in the 1000BASE-KX mode. **Value After Reset:** 0x0 **Exists:** Always
3	LP_TEEE	R	10GBASE-T Mode EEE Support This bit indicates if the link partner supports EEE in the 10GBASE-T mode: ■ 0: EEE is not supported in the 10GBASE-T mode ■ 1: EEE is supported in the 10GBASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Table 7-410 Fields for Register: SR_AN_EEE_LP_ABL (Continued)

Bits	Name	Memory Access	Description
2	LP_1GTEEE	R	1000BASE-T Mode EEE Support This bit indicates if the link partner supports EEE in the 1000BASE-T mode: ■ 0: EEE is not supported in the 1000BASE-T mode ■ 1: EEE is supported in the 1000BASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	LP_100TEE	R	100BASE-T Mode EEE Support This bit indicates if the link partner supports EEE in the 100BASE-T mode: ■ 0: EEE is not supported in the 100BASE-T mode ■ 1: EEE is supported in the 100BASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.3.25 SR_AN_EEE_ABL2

- **Description:** SR AN MMD EEE Capability 2 Register
- **Size:** "(DWCXLP_APB3_32) ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x3E : (DWCXLP_APB3_16) ? (0x3E*0x2) : (0x3E*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN&&DWCXLP_EEE_EN)

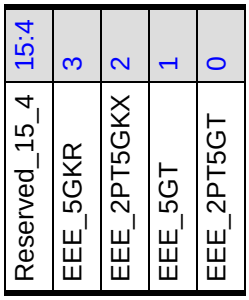


Table 7-411 Fields for Register: SR_AN_EEE_ABL2

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	EEE_5GKR	R/W	5GBASE-KR Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 5GBASE-KR mode ■ 0: EEE is not supported in the 5GBASE-KR mode ■ 1: EEE is supported in the 5GBASE-KR mode **Value After Reset:** "DWCXLP_ADD_5G ? 0x1 : 0x0" **Exists:** Always

Table 7-411 Fields for Register: SR_AN_EEE_ABL2 (Continued)

Bits	Name	Memory Access	Description
2	EEE_2PT5GKX	R/W	2.5GBASE-KX Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 2.5GBASE-KX mode ■ 0: EEE is not supported in the 2.5GBASE-KX mode ■ 1: EEE is supported in the 2.5GBASE-KX mode You must always keep this bit at 0, as DWC_xlgpcs does not support 2.5GBASE-KX Mode. **Value After Reset:** 0x0 **Exists:** Always
1	EEE_5GT	R	5GBASE-T Mode EEE Support This bit indicates if EEE is supported in the 5GBASE-T mode. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
0	EEE_2PT5GT	R	2.5GBASE-T Mode EEE Support This bit indicates if EEE is supported in the 2.5GBASE-T mode. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

7.3.26 SR_AN_EEE_LP_ABL2

- **Description:** SR AN MMD EEE Link Partner Capability 2 Register
- **Size:** "(DWCXLP_APB3_32) ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x3F : (DWCXLP_APB3_16) ? (0x3F*0x2) : (0x3F*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN&&DWCXLP_EEE_EN)

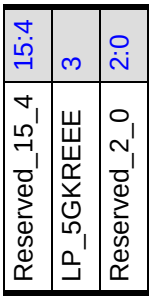
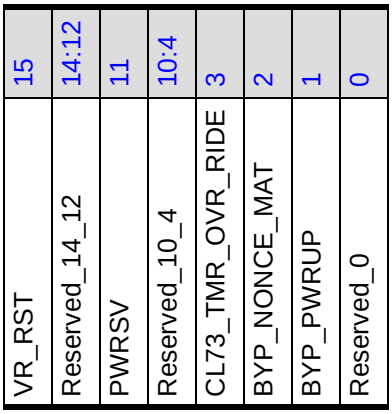


Table 7-412 Fields for Register: SR_AN_EEE_LP_ABL2

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	LP_5GKREEE	R	5GBASE-KR Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 5GBASE-KR mode ■ 0: The link partner does not support EEE in the 5GBASE-KR mode ■ 1: The link partner supports EEE in the 5GBASE-KR mode. **Value After Reset:** 0x0 **Exists:** Always
2:0	Reserved_2_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.3.27 VR_AN_DIG_CTRL1

- **Description:** VR AN MMD Digital Control1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8000 : (DWCXLP_APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



15	14:12	11	10:4	3	2	1	0
VR_RST	Reserved_14_12	PWRSV	Reserved_10_4	CL73_TMR_OVR_RIDE	BYP_NONCE_MAT	BYP_PWRUP	Reserved_0

Table 7-413 Fields for Register: VR_AN_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-Specific Soft Reset (RW,SC Type) When the host sets this bit, the CSR block triggers the vendor-specific software reset process in which all internal blocks are reset, except the Management Interface block and the CSR block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: ■ Synopsys PHY: This bit is self cleared when Bits[4:2] in VR PCS MMD Digital Status Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. ■ Non-Synopsys PHY: This bit is self cleared after the following: - 32 clk_csr_i clocks for the MCI interface, 1 MDC clock period for the MDIO interface. This bit is shared with Bit 15 of the following registers: ■ VR PMA MMD Digital Control1 Register ■ VR PCS MMD Digital Control1 Register **Value After Reset:** 0x0 **Exists:** Always

Table 7-413 Fields for Register: VR_AN_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
14:12	Reserved_14_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	PWRSV	* Varies	For Synopsys PHY configurations: Power Save ■ 0: Normal operation. ■ 1: DWC_xlgpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xlgpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys Multi-protocol PHY, DWC_xlgpcs asserts xlgpcs_tx_lpd_o{lane} and xlgpcs_rx_lpd_o{lane} when this bit is set. When the host clears this bit, DWC_xlgpcs resumes normal operation. After clearing this bit, the host must wait until Bits[4:2] of the VR PCS MMD Digital Status Register indicate that the DWC_xlgpcs is in a normal state. This bit is shared with Bit 11 of the following registers: ■ VR PMA MMD Digital Control1 Register ■ VR PCS MMD Digital Control1 Register For non-Synopsys PHY configurations: ■ Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
10:4	Reserved_10_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	CL73_TMR_OVR_RIDE	R/W	Over-Ride Control for Clause 73 Auto-negotiation Timers. This bit can be set to override the default values of Clause 73 timers used by DWC_xlgpcs for auto-negotiation. If this bit is set, the values programmed in VR_AN_TIMER_CTRL* Registers are used to compute the duration of Auto-neg timers. This bit should be set only after programming the desired values to the timer control registers. **Value After Reset:** 0x0 **Exists:** Always

Table 7-413 Fields for Register: VR_AN_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
2	BYP_NONCE_MAT	R/W	Bypass Nonce-Match Check during Clause 73 auto-negotiation. This bit can be set to 1 if application wishes to perform Clause 73 Auto-negotiation in Loopback mode. As per IEEE specification, auto-negotiation does not work in loopback, as it requires the nonce field of both the devices to be different. When this bit is set, the nonce check is bypassed and the auto-negotiation arbiter FSM can transition from ABILITY_DETECT to ACKNOWLEDGE_DETECT state when ability_match is attained (irrespective of nonce_match). This is a debug-mode bit. **Value After Reset:** 0x0 **Exists:** Always

Table 7-413 Fields for Register: VR_AN_DIG_CTRL1 (Continued)

Bits	Name	Memory Access	Description
1	BYP_PWRUP	* Varies	For Synopsys PHY configurations: Bypass Power-Up Sequence This bit is present only when the DWC_xlgpcs is configured for the Synopsys PHY. In all other configurations, this bit is reserved. When this bit is set to 1, the DWC_xlgpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xlgpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xlgpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xlgpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming the normal transmission and reception.**Note:** This bit is shared with Bit 1 of the following registers: ■ VR PMA MMD Digital Control1 Register ■ VR PCS MMD Digital Control1 Register Access Type: For configurations with Synopsys PHY, this bit is RW. Otherwise, this bit is RO. For non-Synopsys PHY configurations: ■ Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(DWCXLP_SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.3.28 VR_AN_INTR_MSK

- **Description:** VR AN MMD Interrupt Mask Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8001 : (DWCXLP_APB3_16) ? (0x8001*0x2) : (0x8001*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

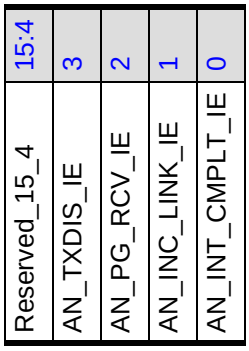


Table 7-414 Fields for Register: VR_AN_INTR_MSK

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	AN_TXDIS_IE	* Varies	AN Transmit Disable Interrupt Enable ■ 0: The AN_TXDIS interrupt is blocked. ■ 1: The AN_TXDIS interrupt is generated. **Note:**This field is present only for configurations with DWCXLP_CL73_ALT parameter enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWCXLP_CL73_ALT ? \"read-write\" : \"read-only\""
2	AN_PG_RCV_IE	R/W	AN Page Received Interrupt Enable ■ 0: The AN_PG_RCV interrupt is blocked. ■ 1: The AN_PG_RCV interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

Table 7-414 Fields for Register: VR_AN_INTR_MSK (Continued)

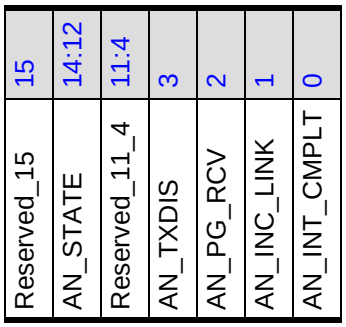
Bits	Name	Memory Access	Description
1	AN_INC_LINK_IE	R/W	AN Incompatible Link Interrupt Enable ■ 0: The AN_INC_LINK interrupt is blocked. ■ 1: The AN_INC_LINK interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
0	AN_INT_CMPLT_IE	R/W	AN COMPLETE Interrupt Enable ■ 0: AN_INT_CMPLT Interrupt is blocked. ■ 1: AN INT_CMPLT Interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

7.3.29 VR_AN_INTR

- **Description:** VR AN MMD Interrupt Register

  Note: The interrupt port sbd_intr_o goes high when any bit of this register is set align with the corresponding bit of VR AN MMD Interrupt Mask Register.

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8002 : (DWCXLP_APB3_16) ? (0x8002*0x2) : (0x8002*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)



15	14:12	11:4	3	2	1	0
Reserved_15	AN_STATE	Reserved_11_4	AN_TXDIS	AN_PG_RCV	AN_INC_LINK	AN_INT_CMPLT

Table 7-415 Fields for Register: VR_AN_INTR

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-415 Fields for Register: VR_AN_INTR (Continued)

Bits	Name	Memory Access	Description
14:12	AN_STATE	R	AN FSM state from which it moved to TX_DISABLE state This field indicate the AN FSM state from which FSM moved to TX_DISABLE state resulting in setting of bit[3] of this register. Encoding is as following: ■ 3'b000: AN_ENABLE ■ 3'b001: ABILITY_DETECT ■ 3'b010: ACKNOWLEDGE_DETECT ■ 3'b011: NEXT_PAGE_WAIT ■ 3'b100: AN_GOOD_CHECK ■ 3'b101: AN_GOOD **Note:** This field is valid only for configurations with PAM4 Mode support. **Value After Reset:** 0x0 **Exists:** Always
11:4	Reserved_11_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	AN_TXDIS	* Varies	AN FSM reached TX_DISABLE (SS,WC Type) The DWC_xlgpcs sets this bit to 1 when the auto-negotiation FSM enters TX_DISABLE state. The source state from which FSM entered TX_DISABLE state is shown in AN_STATE field of this register **Note 1:** The host must clear this bit, by writing 0 to this bit, after recognizing the interrupt. **Note 2:** This field is valid only for configurations with DWCXLP_CL73_ALT parameter enabled. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable **Memory Access:** "DWCXLP_CL73_ALT ? \"read-write\" : \"read-only\""
2	AN_PG_RCV	R/W	AN Page Received (SS,WC Type) The DWC_xlgpcs sets this bit when it receives a new auto-negotiation page (as defined in IEEE Std 802.3, Clause 73). **Note:** The host must clear this bit, by writing 0 to this bit, after recognizing the interrupt. Upon reset, this interrupt is set by default. Hence the Host must clear this interrupt by writing 0 to this bit after a reset. **Value After Reset:** 0x0 **Exists:** Always

Table 7-415 Fields for Register: VR_AN_INTR (Continued)

Bits	Name	Memory Access	Description
1	AN_INC_LINK	R/W	AN Incompatible Link (SS,WC Type) This bit indicates that the DWC_xlgpcs detected an incompatible link during auto-negotiation. ■ 0: No incompatibility detected. ■ 1: Incompatible link. An incompatible link occurs when all of the following conditions are true: ■ The DWC_xlgpcs starts auto-negotiation. ■ The DWC_xlgpcs detects that a DME page is not received by the link partner. ■ The PCS link status for the DWC_xlgpcs remains Not_OK. ■ A time-out for a duration equal to 500 ms occurs when the DWC_xlgpcs autonegotiation arbiter state is in Ability_Detect state. **Note:** The host must clear this bit, by writing 0 to this bit, after recognizing the interrupt. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
0	AN_INT_CMPLT	R/W	AN Complete Interrupt (SS,WC Type) The DWC_xlgpcs sets this bit after the auto-negotiation is complete (as defined in IEEE Std 802.3, Clause 73).**Note:** The host must clear this bit, by writing 0 to this bit, after recognizing the interrupt. **Value After Reset:** 0x0 **Exists:** Always

7.3.30 VR_AN_MODE_CTRL

- **Description:** VR AN MMD KR Mode Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8003 : (DWCXLP_APB3_16) ? (0x8003*0x2) : (0x8003*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (DWCXLP_CL73_AN)

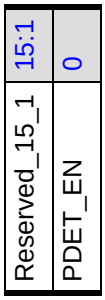


Table 7-416 Fields for Register: VR_AN_MODE_CTRL

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	PDET_EN	R/W	Parallel Detection Enable When set, this bit enables parallel detection in all the BASE-R speed modes (10G/40G/25G/50G/100G) during Clause 73 auto-negotiation. Otherwise, parallel detection is not done for BASE-R speed modes. **Value After Reset:** 0x0 **Exists:** Always

7.3.31 VR_AN_TIMER_CTRL0

- **Description:** VR AN MMD Timer Control 0 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8004 : (DWCXLP_APB3_16) ? (0x8004*0x2) : (0x8004*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** DWCXLP_CL73_AN



Table 7-417 Fields for Register: VR_AN_TIMER_CTRL0

Bits	Name	Memory Access	Description
15:0	BRK_LINK_TIME	R/W	Break Link Timer Duration Control **For configurations with PAM4 Mode support:**If FAST_SIM bit or CL73_TMR_OVR_RIDE bit is programmed to 1, then the value programmed to this field controls the duration of break_link_timer. The value programmed to this field indicates the number of milli-seconds for which break_link_timer should run . Maximum allowed value that can be programmed to this field is 14'h3_FFF. If FAST_SIM bit and CL73_TMR_OVR_RIDE bit are 0, then break_link timer runs for standard duration. **For configurations without PAM4 Mode Support:**DWC_xlgpcs has a 27-bit timer that runs at the frequency of clk_rpcs_tx_i clock. The value programmed to this field is used as the upper 16-bit of the value to be loaded to this timer. The lower bits are hardcoded as 11'h2EE. Application should program appropriate values to this field based on the frequency of clk_rpcs_tx_i clock as per the current operating mode. After programming this field to the desired value, FAST_SIM bit or CL73_TMR_OVR_RIDE bit should be set so that DWC_xlgpcs over-rides the standard value of break_link_timer with the value programmed to this register. **Value After Reset:** 0x0 **Exists:** Always

7.3.32 VR_AN_TIMER_CTRL1

- **Description:** VR AN MMD Timer Control 1 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8005 : (DWCXLP_APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** DWCXLP_CL73_AN



Table 7-418 Fields for Register: VR_AN_TIMER_CTRL1

Bits	Name	Memory Access	Description
15:0	INHBT_TIME	R/W	Link Fail Inhibit Timer Duration Control **For configurations with DWCXLP_CL73_ALT support:**If FAST_SIM bit or CL73_TMR_OVR_RIDE bit is programmed to 1, then the value programmed to this field controls the duration of link_fail_inhibit_timer. The value programmed to this field indicates the number of milli-seconds for which link_fail_inhibit_wait_timer should run (while in LINK_STATUS_CHECK state) . Maximum allowed value that can be programmed to this field is 14'h3_FFF. **For configurations without DWCXLP_CL73_ALT Support:**This field can be programmed to the desired value if application requires to change the duration of Link Fail Inhibit/Autoneg Wait Timer used during CL73 Autonegotiation. DWC_xlgpcs uses same timer values for Link Fail Inhibit Timer and Autoneg Wait Timer DWC_xlgpcs has a 27-bit timer configurations that runs at the frequency of clk_rpcs_tx_i clock. The value programmed to this field is used as the upper 16-bit of the value to be loaded to this timer. The lower bits are hardcoded as 11'h2EE. After programming this field to the desired value, FAST_SIM or CL73_TMR_OVR_RIDE bit should be set so that DWC_xlgpcs over-rides the standard value of this timer with the value programmed to this register. **Value After Reset:** 0x0 **Exists:** Always

7.3.33 VR_AN_TIMER_CTRL2

- **Description:** VR AN MMD Timer Control 2 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8006 : (DWCXLP_APB3_16) ? (0x8006*0x2) : (0x8006*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** DWCXLP_CL73_ALT

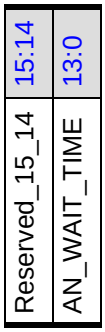


Table 7-419 Fields for Register: VR_AN_TIMER_CTRL2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:0	AN_WAIT_TIME	R/W	Autoneg_Wait_Timer Duration Control If FAST_SIM bit or CL73_TMR_OVR_RIDE bit is programmed to 1, then the value programmed to this field controls the duration of autoneg_wait_timer. The value programmed to this field indicates the number of milli-seconds for which autoneg_wait_timer should run (while in LINK_STATUS_CHECK state) . The autoneg_wait_timer is valid only if vendor-specific parallel-detection feature is enabled. **Value After Reset:** 0x0 **Exists:** Always

7.3.34 VR_AN_TIMER_CTRL3

- **Description:** VR AN MMD Timer Control 3 Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8007 : (DWCXLP_APB3_16) ? (0x8007*0x2) : (0x8007*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** DWCXLP_CL73_ALT

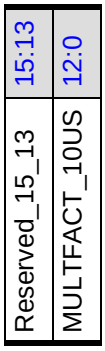


Table 7-420 Fields for Register: VR_AN_TIMER_CTRL3

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:0	MULTFACT_10US	R/W	Multiplication factor to generate a 10 micro-second timer tic Program this field to a suitable value (prior to enabling auto-negotiation) so as to generate a 10 micro-second tic based on the frequency of clk_csr_i. This field should be programmed as follows: MULTFACT_10US = CEIL (10 micro-seconds DIVIDE BY time-period of clk_csr_i). The default value of this field is set to 13'd1000 assuming that clk_csr_i has a typical frequency of 100MHz (10 ns time-period). **Value After Reset:** 0x3e8 **Exists:** Always

7.3.35 VR_AN_DIG_STS

- **Description:** VR AN MMD Digital Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x8010 : (DWCXLP_APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x070000 : (DWCXLP_APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** DWCXLP_CL73_AN

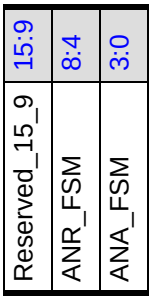


Table 7-421 Fields for Register: VR_AN_DIG_STS

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 7-421 Fields for Register: VR_AN_DIG_STS (Continued)

Bits	Name	Memory Access	Description
8:4	ANR_FSM	R	Auto-negotiation Receive Logic FSM state This FSM is a vendor-specific FSM implementation and encoding is as follows:**When performing AN in modes other than 10G:** ■ 5'd1: IDLE ■ 5'd2: DELIMETER_DETECT ■ 5'd4: DME_DETECT **When performing AN in 10G mode:** ■ 5'd0: RX_IDLE ■ 5'd1: RX_DLD ■ 5'd2: WFC0 ■ 5'd3: WFC1 ■ 5'd4: WFC2 ■ 5'd5: WFC3 ■ 5'd6: WFC4 ■ 5'd7: WFC5 ■ 5'd8: WFC6 ■ 5'd9: WFC7 ■ 5'd10: WFC8 ■ 5'd11: WFC9 ■ 5'd12: WFD0 ■ 5'd13: WFD1 ■ 5'd14: WFD2 ■ 5'd15: WFD3 ■ 5'd16: WFD4 ■ 5'd17: WFD5 ■ 5'd18: WFD6 ■ 5'd19: WFD7 ■ 5'd20: WFD8 ■ 5'd21: WFD9 ■ 5'd22: WFD18 ■ 5'd23: WFD19 ■ 5'd24: WFC21D17 ■ 5'd25: WFC21D18 ■ 5'd26: WFC21D19 ■ 5'd27: WFPT0 **Note:**This field is not reliable when read around the time at which FSM transitions from one state to another. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable

Table 7-421 Fields for Register: VR_AN_DIG_STS (Continued)

Bits	Name	Memory Access	Description
3:0	ANA_FSM	R	Auto-negotiation Arbiter FSM state This field indicates the current status of AN arbiter FSM. Encoding is as follows: ■ 4'b0000: AN_ENABLE ■ 4'b0001: TRANSMIT_DISABLE ■ 4'b0010: ABILITY_DETECT ■ 4'b0011: ACKNOWLEDGE_DETECT ■ 4'b0100: COMPLETE_ACKNOWLEDGE ■ 4'b0101: NEXT_PAGE_WAIT ■ 4'b0110: LINK_STATUS_CHECK ■ 4'b0111: PARALLEL_DETECT_FAULT ■ 4'b1000: AN_GOOD_CHECK ■ 4'b1001: AN_GOOD **Note:**This field is not strictly reliable when read around the time at which FSM transitions from one state to another. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable

7.4 DWC_xlgpcs_map/VS_MMD1 Registers

The Vendor Specific MMD1 Registers address block. Follow the link for the register to see a detailed description of the register.

Table 7-422 Registers for Address Block: DWC_xlgpcs_map/VS_MMD1

Register	Offset	Description
SR_VSMMD_PMA_ID1 on page 1261	* Varies	SR Control MMD PMA Device Identifier Register 1
SR_VSMMD_PMA_ID2 on page 1262	* Varies	SR Control MMD PMA Device Identifier Register 2
SR_VSMMD_DEV_ID1 on page 1265	* Varies	SR Control MMD Device Identifier Register 1
SR_VSMMD_DEV_ID2 on page 1266	* Varies	SR Control MMD Device Identifier Register 2
SR_VSMMD_PCS_ID1 on page 1268	* Varies	SR Control MMD PCS Device Identifier Register 1
SR_VSMMD_PCS_ID2 on page 1269	* Varies	SR Control MMD PCS Device Identifier Register 2
SR_VSMMD_AN_ID1 on page 1271	* Varies	SR Control MMD AN Device Identifier Register 1
SR_VSMMD_AN_ID2 on page 1272	* Varies	SR Control MMD AN Device Identifier Register 2
SR_VSMMD_STS on page 1275	* Varies	SR Control MMD Status Register
SR_VSMMD_CTRL on page 1276	* Varies	SR Control MMD Control Register
SR_VSMMD_PKGID1 on page 1279	* Varies	SR Control MMD Package Identifier Register 1
SR_VSMMD_PKGID2 on page 1280	* Varies	SR Control MMD Package Identifier Register 2 Note: The values written in the fields of this register...

7.4.1 SR_VSMMD_PMA_ID1

- **Description:** SR Control MMD PMA Device Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

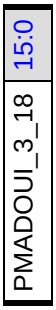


Table 7-423 Fields for Register: SR_VSMMD_PMA_ID1

Bits	Name	Memory Access	Description
15:0	PMADOUI_3_18	* Varies	For configurations with PMA MMD: Organizationally Unique Identifier[3:18] for PMA MMD This field contains Bits[18:3] of 24-bit OUI of device manufacturer. The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The default value of this register is similar to the default value of SR PMA MMD Device Identifier Register 1. The value written in this register is reflected in SR PMA MMD Device Identifier Register 1. Access Type: ■ RW: For PMA MMD configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations For other configurations: Reserved **Value After Reset:** "(DWCXLP_PMA_MMD) ? DWCXLP_IEEE_PMA_DEV_OUI_3_18 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_PMA_MMD&&DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

7.4.2 SR_VSMMD_PMA_ID2

- **Description:** SR Control MMD PMA Device Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x1 : (DWCXLP_APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

15:10	9:4	3:0
PMADOUI_19_24	PMADMMN_5_0	PMADRN_3_0

Table 7-424 Fields for Register: SR_VSMMD_PMA_ID2

Bits	Name	Memory Access	Description
15:10	PMADOUI_19_24	* Varies	For configurations with PMA MMD: Organizationally Unique Identifier[19:24] for PMA MMD The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer of PMA MMD. This field contains Bits[24:19] of 24-bit OUI of device manufacturer. The default value of this field is similar to the default value of the corresponding field in SR PMA MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR PMA MMD Device Identifier Register 2. Access Type: ■ RW: For PMA MMD configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. For other configurations: Reserved **Value After Reset:** "(DWCXLP_PMA_MMD) ? DWCXLP_IEEE_PMA_DEV_OUI_19_24 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_PMA_MMD&&DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\"""
9:4	PMADMMN_5_0	* Varies	For configurations with PMA MMD: Model Number for PMA MMD This field contains the 6-bit Model number of the PMA MMD. The default value of this field is similar to the default value of the corresponding field in SR PMA MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR PMA MMD Device Identifier Register 2. Access Type: ■ RW: For PMA MMD configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. For other configurations: Reserved **Value After Reset:** "(DWCXLP_PMA_MMD) ? DWCXLP_IEEE_PMA_DEV_MODEL_NO_0_5 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_PMA_MMD&&DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

Table 7-424 Fields for Register: SR_VSMMD_PMA_ID2 (Continued)

Bits	Name	Memory Access	Description
3:0	PMADRN_3_0	* Varies	For configurations with PMA MMD: Revision Number for PMA MMD This field contains the 4-bit Revision number of the PMA MMD. The default value of this field is similar to the default value of the corresponding field in SR PMA MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR PMA MMD Device Identifier Register 2. Access Type: ■ RW: For PMA MMD configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. For other configurations: Reserved **Value After Reset:** "(DWCXLP_PMA_MMD) ? DWCXLP_IEEE_PMA_DEV_REV_NO_0_3 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_PMA_MMD&&DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

7.4.3 SR_VSMMD_DEV_ID1

- **Description:** SR Control MMD Device Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x2 : (DWCXLP_APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

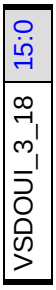


Table 7-425 Fields for Register: SR_VSMMD_DEV_ID1

Bits	Name	Memory Access	Description
15:0	VSDOUI_3_18	* Varies	Organizationally Unique Identifier[3:18] for Vendor-Specific MMD1 This field contains Bits[18:3] of 24-bit OUI of device manufacturer. The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. You can configure the value of each MMD using coreConsultant. Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** DWCXLP_IEEE_VS_DEV_OUI_3_18 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

7.4.4 SR_VSMMD_DEV_ID2

- **Description:** SR Control MMD Device Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x3 : (DWCXLP_APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

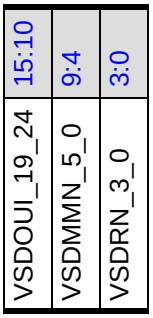


Table 7-426 Fields for Register: SR_VSMMD_DEV_ID2

Bits	Name	Memory Access	Description
15:10	VSDOUI_19_24	* Varies	Organizationally Unique Identifier[19:24] for Vendor-Specific MMD1 This field contains Bits[24:19] of the device manufacturer's 24- bit OUI. The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. You can configure the value of each MMD using coreConsultant. Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** DWCXLP_IEEE_VS_DEV_OUI_19_24 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

Table 7-426 Fields for Register: SR_VSMMD_DEV_ID2 (Continued)

Bits	Name	Memory Access	Description
9:4	VSDMMN_5_0	* Varies	Model Number for Vendor-Specific MMD1 This field contains the 6-bit Model number of the vendor-specific MMD1. You can configure the default value through coreConsultant. Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** DWCXLP_IEEE_VS_DEV_MODEL_NO_0_5 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""
3:0	VSDRN_3_0	* Varies	Revision Number for Vendor-Specific MMD1 This field contains the 4-bit Revision number of the vendor-specific MMD1. You can configure the default value through coreConsultant. Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** DWCXLP_IEEE_VS_DEV_REV_NO_0_3 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

7.4.5 SR_VSMMD_PCS_ID1

- **Description:** SR Control MMD PCS Device Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x4 : (DWCXLP_APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

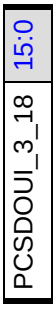


Table 7-427 Fields for Register: SR_VSMMD_PCS_ID1

Bits	Name	Memory Access	Description
15:0	PCSDOUI_3_18	* Varies	Organizationally Unique Identifier[3:18] for PCS MMD This field contains Bits[18:3] of 24-bit OUI of device manufacturer. The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The default value of this register is similar to the default value of SR PCS MMD Device Identifier Register 1. The value written in this register is reflected in SR PCS MMD Device Identifier Register 1. Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** DWCXLP_IEEE_DEV_OUI_3_18 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

7.4.6 SR_VSMMD_PCS_ID2

- **Description:** SR Control MMD PCS Device Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x5 : (DWCXLP_APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

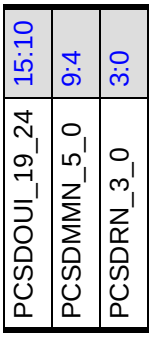


Table 7-428 Fields for Register: SR_VSMMD_PCS_ID2

Bits	Name	Memory Access	Description
15:10	PCSDOUI_19_24	* Varies	Organizationally Unique Identifier[19:24] for PCS MMD This field contains Bits[24:19] of 24-bit OUI of device manufacturer. The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The default value of this field is similar to the default value of the corresponding field in SR PCS MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR PCS MMD Device Identifier Register 2. Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** DWCXLP_IEEE_DEV_OUI_19_24 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

Table 7-428 Fields for Register: SR_VSMMD_PCS_ID2 (Continued)

Bits	Name	Memory Access	Description
9:4	PCSDMMN_5_0	* Varies	Model Number for PCS MMD This field contains the 6-bit Model number of vendor-specific PCS MMD. The default value of this field is similar to the default value of the corresponding field in SR PCS MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR PCS MMD Device Identifier Register 2. Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations **Value After Reset:** DWCXLP_IEEE_DEV_MODEL_NO_0_5 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""
3:0	PCSDRN_3_0	* Varies	Revision Number for PCS MMD This field contains the 4-bit Revision number of vendor-specific PCS MMD. The default value of this field is similar to the default value of the corresponding field in SR PCS MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR PCS MMD Device Identifier Register 2. Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations **Value After Reset:** DWCXLP_IEEE_DEV_REV_NO_0_3 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

7.4.7 SR_VSMMD_AN_ID1

- **Description:** SR Control MMD AN Device Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x6 : (DWCXLP_APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

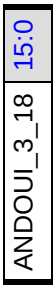


Table 7-429 Fields for Register: SR_VSMMD_AN_ID1

Bits	Name	Memory Access	Description
15:0	ANDOUI_3_18	* Varies	Organizationally Unique Identifier[3:18] for AN MMD This field contains Bits[18:3] of the device manufacturer's 24-bit OUI. The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The default value of this register is similar to the default value of SR AN MMD Device Identifier Register 1. The value written in this register is reflected in SR AN MMD Device Identifier Register 1. Default Value: Configurable for Backplane Ethernet PCS configurations; Otherwise, the default value is 0. Access Type: ■ RW: For Backplane Ethernet PCS configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations **Value After Reset:** "(DWCXLP_CL73_AN) ? DWCXLP_IEEE_AN_DEV_OUI_3_18 : 0x0" **Exists:** Always **Memory Access:** "((DWCXLP_CL73_AN)&&DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

7.4.8 SR_VSMMD_AN_ID2

- **Description:** SR Control MMD AN Device Identifier Register 2
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x7 : (DWCXLP_APB3_16) ? (0x7*0x2) : (0x7*0x4)" + "(DWCXLP_APB3_INTF||DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

15:10	9:4	3:0
ANDOUI_19_24	ANDMMN_5_0	ANDRN_3_0

Table 7-430 Fields for Register: SR_VSMMD_AN_ID2

Bits	Name	Memory Access	Description
15:10	ANDOUI_19_24	* Varies	For Configurations with CL73 Auto-neg support: Organizationally Unique Identifier[19:24] for AN MMD The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. This field contains Bits[24:19] of the 24-bit OUI of device manufacturer. The default value of this field is similar to the default value of the corresponding field in SR AN MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR AN MMD Device Identifier Register 2. Default Value: Configurable for Backplane Ethernet PCS configurations ; Otherwise, the default value is 0. Access Type: ■ RW: For Backplane Ethernet PCS configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations For other configurations: Reserved **Value After Reset:** "(DWCXLP_CL73_AN) ? DWCXLP_IEEE_AN_DEV_OUI_19_24 : 0x0" **Exists:** Always **Memory Access:** "((DWCXLP_CL73_AN)&&DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

Table 7-430 Fields for Register: SR_VSMMD_AN_ID2 (Continued)

Bits	Name	Memory Access	Description
9:4	ANDMMN_5_0	* Varies	For Configurations with CL73 Auto-neg support: Model Number This field contains the 6-bit Model number of AN MMD. The default value of this field is similar to the default value of the corresponding field in SR AN MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR AN MMD Device Identifier Register 2. Default Value: Configurable for Backplane Ethernet PCS configurations; Otherwise, the default value is 0. Access Type: ■ RW: For Backplane Ethernet PCS configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations For other configurations: Reserved **Value After Reset:** "(DWCXLP_CL73_AN) ? DWCXLP_IEEE_AN_DEV_MODEL_NO_0_5 : 0x0" **Exists:** Always **Memory Access:** "((DWCXLP_CL73_AN)&&DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""
3:0	ANDRN_3_0	* Varies	For Configurations with CL73 Auto-neg support: Revision Number This field contains the 4-bit Revision number of AN MMD. The default value of this field is similar to the default value of the corresponding field in SR AN MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR AN MMD Device Identifier Register 2. Default Value: Configurable for configurations with CL73 Support; Otherwise, the default value is 0. Access Type: ■ RW: For Backplane Ethernet PCS configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations For other configurations: Reserved **Value After Reset:** "(DWCXLP_CL73_AN) ? DWCXLP_IEEE_AN_DEV_REV_NO_0_3 : 0x0" **Exists:** Always **Memory Access:** "((DWCXLP_CL73_AN)&&DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

7.4.9 SR_VSMMD_STS

- **Description:** SR Control MMD Status Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x8 : (DWCXLP_APB3_16) ? (0x8*0x2) : (0x8*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

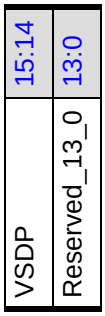


Table 7-431 Fields for Register: SR_VSMMD_STS

Bits	Name	Memory Access	Description
15:14	VSDP	R	Control MMD Device Present This field indicates if the control MMD device is present and responding to this address: ■ 10: Device responding at this address ■ 11, 01, or 00: No device responding at this address **Value After Reset:** 0x2 **Exists:** Always
13:0	Reserved_13_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

7.4.10 SR_VSMMD_CTRL

- **Description:** SR Control MMD Control Register
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x9 : (DWCXLP_APB3_16) ? (0x9*0x2) : (0x9*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

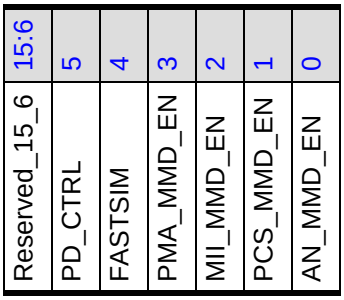


Table 7-432 Fields for Register: SR_VSMMD_CTRL

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	PD_CTRL	* Varies	For Non-Synopsys PHY configurations: Power Down Control This bit is used to control the output port 'xlgpcs_pdown_o'. If this bit is set, xlgpcs_pdown_o port is not asserted when LPM (Low Power Enable) bit is programmed to 1. If this bit is low, xlgpcs_pdown_o port is asserted when LPM (Lowe Power Enable) bit is programmed to 1. For Synopsys PHY configurations: Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!DWCXLP_SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Table 7-432 Fields for Register: SR_VSMMD_CTRL (Continued)

Bits	Name	Memory Access	Description
4	FASTSIM	R/W	Fast Simulation Enable When set, this bit indicates that the Fast simulation is enabled. When this bit is set to 1, all IEEE Std 802.3 defined long timers, implemented in the DWC_xlgpcs are reduced to shorter time period to reduce the simulation time. When it is programmed as 1, the times whose unit is ms are speed down 40 times. The long timers are implemented in Clause 73 auto-negotiation modules and also EEE Tx and Rx modules. For example, the EEE Quiet timeout value in multilane KR4/CR4 modes is 1.7ms-1.8ms. When Fast simulation is set, the Quiet timeout is scaled by 40 so that Quiet interval is from 42.5us - 45us. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
3	PMA_MMD_EN	* Varies	PMA MMD Enable When set, this bit indicates that the PMA MMD device is accessible. When reset, this bit indicates that the PMA MMD device is not accessible. Default Value: If PMA MMD is present, this bit returns 1. Otherwise, this bit returns 0. Access Type: If PMA MMD is present, this bit is RW. Otherwise, this bit is RO. **Value After Reset:** "(DWCXLP_PMA_MMD) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_PMA_MMD) ? \"read-write\" : \"read-only\"""
2	MII_MMD_EN	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	PCS_MMD_EN	R/W	PCS MMD Enable ■ 1: This bit indicates that the PCS MMD device is accessible. ■ 0: This bit indicates that the PCS MMD device is not accessible. **Value After Reset:** 0x1 **Exists:** Always

Table 7-432 Fields for Register: SR_VSMMD_CTRL (Continued)

Bits	Name	Memory Access	Description
0	AN_MMD_EN	* Varies	For Configurations with CL73 Auto-neg support: AN MMD Enable When set, this bit indicates that the AN MMD device is accessible. When reset, this bit indicates that the AN MMD device is not accessible. Default Value: For Backplane Ethernet PCS configurations (configurations with CL73 Auto-neg support enabled), this bit returns 1. Otherwise, this bit returns 0. Access Type: For Backplane Ethernet PCS configurations (configurations with CL73 Auto-neg support enabled),, this bit is RW. Otherwise, this bit is RO. For other configurations: Reserved **Value After Reset:** "(DWCXLP_CL73_AN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(DWCXLP_CL73_AN) ? \"read-write\" : \"read-only\""

7.4.11 SR_VSMMD_PKGID1

- **Description:** SR Control MMD Package Identifier Register 1
- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0xE : (DWCXLP_APB3_16) ? (0xE*0x2) : (0xE*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always



Table 7-433 Fields for Register: SR_VSMMD_PKGID1

Bits	Name	Memory Access	Description
15:0	MMDPOUI_3_18	* Varies	Organizationally Unique Identifier The DWC_xlgpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. This field contains Bits[18:3] of 24-bit OUI of device manufacturer. This is a unique identifier for all MMDs present in the DWC_xlgpcs. The value written in this register is reflected in the following registers: ■ SR PMA MMD Package Identifier Register 1 ■ SR PCS MMD Package Identifier Register 1 ■ SR AN MMD Package Identifier Register 1 Default Value: Configurable Access Type: ■ RW: For configurations DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations **Value After Reset:** DWCXLP_IEEE_PKG_OUI_3_18 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

7.4.12 SR_VSMMD_PKGID2

- **Description:** SR Control MMD Package Identifier Register 2

  Note: The values written in the fields of this register are reflected in the corresponding fields of the following registers:

  - SR PMA MMD Package Identifier Register 2
  - SR PCS MMD Package Identifier Register 2
  - SR AN MMD Package Identifier Register 2

- **Size:** "DWCXLP_APB3_32 ? 32 : 16" bits
- **Offset:** "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0xF : (DWCXLP_APB3_16) ? (0xF*0x2) : (0xF*0x4)" + "(DWCXLP_APB3_INTF | |DWCXLP_MDIO_INTF) ? 0x1E0000 : (DWCXLP_APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

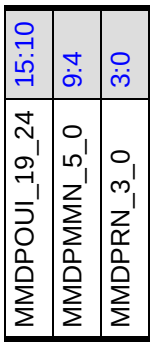


Table 7-434 Fields for Register: SR_VSMMD_PKGID2

Bits	Name	Memory Access	Description
15:10	MMDPOUI_19_24	* Varies	Organizationally Unique Identifier The DWC_xlgpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. This field contains Bits[24:19] of 24-bit OUI of device manufacturer. This is a unique identifier for all MMDs present in the DWC_xlgpcs. Default Value: Configurable Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations **Value After Reset:** DWCXLP_IEEE_PKG_OUI_19_24 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""
9:4	MMDPMMN_5_0	* Varies	Model Number This field contains the 5-bit Model Number of the package in which the MMDs are present in DWC_xlgpcs. Default Value: Configurable Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations **Value After Reset:** DWCXLP_IEEE_PKG_MODEL_NO_0_5 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""
3:0	MMDPRN_3_0	* Varies	Revision Number This field contains the 4-bit revision number of the package in which the MMDs are present in DWC_xlgpcs. Default Value: Configurable Access Type: ■ RW: For configurations with DWCXLP_IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations **Value After Reset:** DWCXLP_IEEE_PKG_REV_NO_0_3 **Exists:** Always **Memory Access:** "(DWCXLP_IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

8 Programming Guidelines

This chapter provides the programming guidelines for switching between various modes and enabling auto-negotiation and Energy Efficient Ethernet features. It contains the following sections:

- "Initializing the DWC_xlgpcs Controller" on page 1284
- "Switching to 10G Speed Mode" on page 1286
- "Switching to 40G Speed Mode" on page 1290
- "Switching to 20G Speed Mode" on page 1295
- "Switching to 25G Speed Mode" on page 1298
- "Switching to 50G Speed Mode" on page 1309
- "Switching to 100G Speed Mode" on page 1316
- "Switching to 5G Speed Mode" on page 1324
- "Programming Guidelines for CR Port Read or Write Operation" on page 1328
- "Programming Guidelines for Clause 72 Start-Up Protocol" on page 1329
- "Programming Guidelines for Clause 73 Auto-Negotiation" on page 1333
- "Programming Guidelines for Energy Efficient Ethernet" on page 1340
- "Programming Guidelines for Test Pattern Mode" on page 1341
- "Programming Guidelines for Automotive Safety Features" on page 1343

Note

- For details of default speed in various configurations, see "Initializing the DWC_xlgpcs Controller".
- If present in the configuration, RS FEC and BASE-R FEC are disabled by default. You can enable them through register programming based on the initialization guidelines in this chapter.

8.1 Initializing the DWC_xlgpcs Controller

Configuration	Default Speed Mode (Speed Mode Supported When out of Reset)
25G	25G
40G	40G
25G-50G (multi-lane)	50G
25G-50G (single-lane)	25G
100G (multi-lane)	100G
100G (single-lane)	25G

Complete the following steps to initialize the DWC_xlgpcs core:

1. Switch on the power supply. This asserts the reset (pwr_on_rst_n=0).
2. Wait for the required amount of time depending on the PHY requirements.
3. De-assert reset (pwr_on_rst_n=1).
4. Perform the following steps for configurations with Synopsys multi-protocol 12G/16G/25G/32G:
   a. Poll bit [0] (INIT_DN) of the `VR_PMA_25G_16G_SRAM` register until it changes to 1.
   b. Based on your configuration, perform the following steps.

- For Synopsys Multi-protocol 12G/16G/25G configurations:

  After waiting for a duration of at least 10 clock cycles of cr_para_clk, software can optionally write to CR registers inside the DWC_xlgpcs to modify the firmware is boot-loaded to the SRAM connected to the PHY.

- For Synopsys Multi-protocol 32G PHY configurations:

  You can by-pass this step if you are enabling automatic boot-loading of firmware and context information.

  After waiting for a duration of at least 10 clock cycles of cr_para_clk, load the SRAM with 'firmware and context restore' content by initiating CR write operation through DWC_xlgpcs

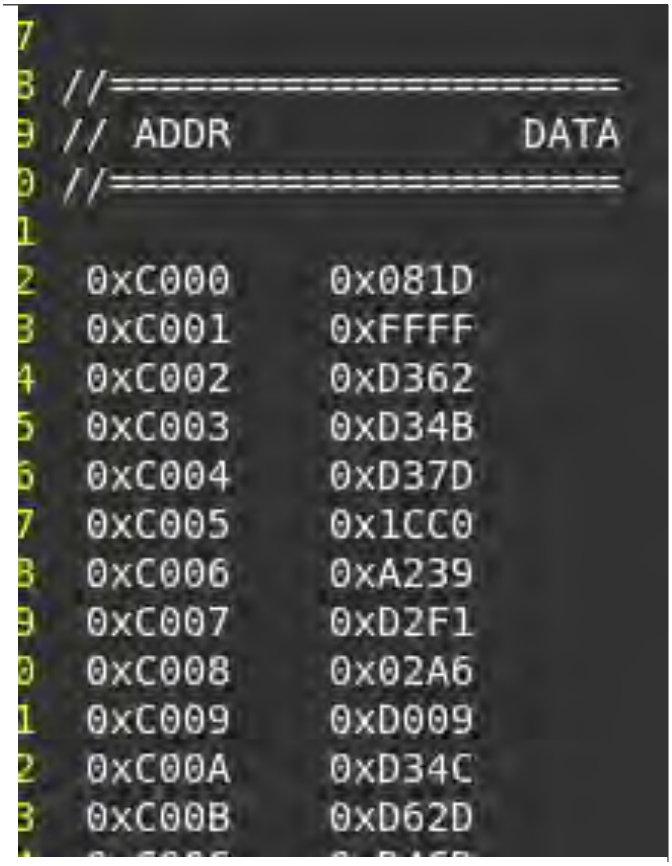
  The data and corresponding addresses for the CR write operation is specified in the sram_firmware file (dwc_e32mp_phy_x4_ns_pcs_raw_mem_sram_cr_para.fw) provided with the PHY package. Re-generate the firmware file by suitably selecting the target operating modes and executing the dwc_e32mp_phy_x4_ns_image_gen.pl perl script provided with the PHY package.

  Figure 8-1 shows the snippet of a sample SRAM firmware file.

  - First column indicates the content (CR address) to be written to the `VR_PMA_MP_25G_16G_CR_ADDR` register of DWC_xlgpcs.
  - Second column indicates the content (CR write data) to be written to the `VR_PMA_MP_25G_16G_CR_DATA` register of DWC_xlgpcs.

c. After SRAM loading is complete, program bit [1] (EXT_LD_DN) of the VR_PMA_25G_16G_SRAM register to 1.

Figure 8-1 Snippet of the Sample SRAM Firmware File



5. Poll bit [15] (RST) of the SR_PCS_CTRL1 register until it becomes 0.
6. During the RTL configuration if you have enabled the 'Add CL73 Auto-negotiation Support' CoreConsultant parameter, program the AN_EN (bit [12]) of the SR_AN_CTRL register to 0 to disable auto-negotiation.
7. Only for the Enterprise 56G PHY configurations, configure DWC_xlgpcs and the PHY to the required operating speed mode by executing instructions in sections 6.2 to 6.7. Steps 1 and 2 of each section can be skipped when coming out of power-on-reset. They are required only when switching from one speed to another without involving a power-on-reset.

Note

Completion of this step indicates that both the PHY and DWC_xlgpcs have successfully completed the reset sequence in the default speed mode for configurations other than Enterprise 56G PHY

For Enterprise 56G PHY configurations, completion of this step indicates that the reset sequence is successfully completed for the chosen speed mode.

8. For Synopsys PHY configurations, program bit [4] (SUPRESS_LOS_DET) and bit [6] (RX_DT_EN_CTL) of the VR_PCS_DEBUG_CTRL register to 1.

9. If you enable 'Add CL73 Auto-negotiation Support' CoreConsultant parameter during RTL configuration and you wish to perform Clause 73 auto-negotiation to negotiate the link capabilities with link-partner, follow the programming flow described in "Programming Guidelines for Clause 73 Auto-Negotiation" on page 1333.

   If you enable the 'Add Link Training Support' CoreConsultant parameter in your RTL configuration, Link training is be performed after the exchange of all the AN pages.

10. Wait for RLU (bit [2]) of the `SR_PCS_STS1` register to indicate 1.

The DWC_xlgpcs controller is now ready to transmit or receive data in the

- Default speed mode if auto-negotiation is skipped,
- Negotiated speed mode if auto-negotiation is performed.

Details of switching the operating mode of DWC_xlgpcs can be found in subsequent sections.

Note

- For Synopsys multi-protocol 32G PHY configurations, the step 5 completes successfully only if you have programmed the values of TX_CNTX_SEL, RX_CNTX_SEL, and CM_CNTX_SEL parameters correctly during RTL configuration.
- Step 10 completes successfully only if you send proper data to the DWC_xlgpcs receiver as per the current operating speed mode.
- Default operating mode is always no-FEC mode.
- For non-Synopsys PHY configurations, keep DWC_xlgpcs in reset until all the clocks stabilize. After all the clocks are stabilized, you can de-assert the reset.

8.2 Switching to 10G Speed Mode

8.2.1 Programming Sequence When Using Multi-Protocol 32G PHY

To switch to the 10G mode Ethernet operation, complete the following steps:

1. Program the bit [0] and bit [1] of the `VR_PCS_DIG_CTRL3` register to 0.
2. Write 4'b0000 to bits [5:2] of the `SR_PCS_CTRL1` register.
3. Write 3'b000 to bits [2:0] of the `SR_PCS_CTRL2` register.
4. To enable BASE-R FEC, program bit [0] in the `SR_PMA_KR_FEC_CTRL` register (if BASE-R FEC is present in the configuration).
5. Configure the PHY to operate at 10.3125Gbps rate by programming the following register fields:
   - CM_CS_0 field of the `VR_PMA_MP_32G_CM_CNTX_SEL0` register to the value indicated in the *context_sel_ID.txt file* for 10G mode.
   - TX_CS_0 field of the `VR_PMA_MP_32G_TX_CNTX_SEL0` register to the value indicated in the *context_sel_ID.txt file* for 10G mode.
   - RX_CS_0 field of the `VR_PMA_MP_32G_RX_CNTX_SEL0` register to the value indicated in the *context_sel_ID.txt file* for 10G mode.

- L0_LN field of the VR_PMA_MP_32G_CNTX_LINK_NUM_CTRL register to a unique value for each instance of the IP when operating in bifurcation mode. Program the LPM bit of SR_PCS_CTRL1 to 1.

6. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
7. Program the LPM bit of the SR_PCS_CTRL register back to 0. Program other bits of this register to values which were present prior to executing step 6.

Note Do not perform read-modified-write.

8. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Good' (3'b100).

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.2.2 Programming Sequence When Using Non-Synopsys PHY/Multi-Protocol 12G/16G/25G PHY

To switch to the 10G mode Ethernet operation, complete the following steps:

1. Program the bit [0] and bit [1] of the VR_PCS_DIG_CTRL3 register to 0.
2. Write 4'b0000 to bits [5:2] of the SR_PCS_CTRL1 register.
3. Write 3'b000 to bits [2:0] of the SR_PCS_CTRL2 register.
4. To enable BASE-R FEC, program bit [0] in the SR_PMA_KR_FEC_CTRL register (if BASE-R FEC is present in the configuration).
5. For Multi-Protocol 25G PHY configurations:
   a. Program the MPLLA_CAL_DISABLE field of the VR_PMA_SNPS_MP_25G_MPLLA_CTRL0 (0x18071) register to 0.
   b. Program the MPLLB_CAL_DISABLE field of the VR_PMA_SNPS_MP_25G_MPLLB_CTRL0 (0x18074) register to 1.
   c. Program various registers as shown in Table 8-1.

Table 8-1 PHY Settings for 10G Mode for Synopsys Multi-Protocol 25G PHY

Field	Values
REF_RANGE	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_REF_CLK_CTRL (0x18091) register to ■ 3'b110 for reference clock frequency 156.25MHz and 312.5 MHz ■ 3'b011 for reference clock frequency 78.125MHz.

Table 8-1 PHY Settings for 10G Mode for Synopsys Multi-Protocol 25G PHY

Field	Values
REF_CLK_DIV2	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_REF_CLK_CTRL (0x18091) register to ■ 1 for reference clock frequency 312.5MHz ■ 0 for reference clock frequency 156.25MHz and 78.125MHz.
REF_MPLLA_DIV	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_REF_CLK_CTRL (0x18091) register to ■ 3'b000 for reference clock frequency 78.125MHz ■ 3'b001 for reference clock frequency 156.25MHz ■ 3'b010 for reference clock frequency 312.5MHz
MPLLA_MULTIPLIER	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL0 (0x18071) register to 12'h84.
MPLLA_DIV_MULT	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL2 (0x18073) register to 8'h8.
MPLLA_TX_CLK_DIV	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL2 (0x18073) register to 3'b010.
MPLLA_LOW_BW	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL3 (0x18077) register to 16'h18.
MPLLA_HIGH_BW	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL4 (0x18079) register to 16'h18.
VCO_REF_LD_0	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_REF0 (0x18096) register to ■ 7'h15 for reference clock frequency 312.5MHz and 156.25 MHz ■ 7'hb for reference clock frequency 78.125MHz.
VCO_LD_VAL_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_LD0 (0x18092) register to ■ 13'h5ac for reference clock frequency 78.125MHz ■ 13'h56a for reference clock frequency 312.5MHz and 156.25MHz.
RX0_CDR_PPM_MAX	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_RX_PPM_CTRL0 (0x18065) register to ■ 5'h12 for reference clock frequency 312.5MHz and 156.25MHz ■ 5'h13 for reference clock frequency 78.125MHz.
MPLLB_SEL_0	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_MPLL_CMN_CTRL (0x18070) register to 0.
TX_ALIGN_WD_XFER_0	Set the value of these fields in the VR_PMA_SNPS_MP_25G_TX_GEN_CTRL5 (0x18046) register to 0.

Table 8-1 PHY Settings for 10G Mode for Synopsys Multi-Protocol 25G PHY

Field	Values
TX0_DCC_CTRL_RNG	Set the value of this field in the VR_PMA_SNPS_MP_25G_TX_DCC_CTRL (0x18047) register to 4'hb
EQ_AFE_RT_0	Set the value of these fields in the VR_PMA_SNPS_MP_25G_RX_AFE_RATE_CTRL (0x1805D) register to 3'b100.
RX0_RATE	Set the value of these fields in the VR_PMA_SNPS_MP_25G_RX_RATE_CTRL (0x18054) register to 3'b001.
RX0_DELTA_IQ	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_IQ_CTRL0 (0x1806A) register to4'b0000.
CTLE_BOOST_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL0 (0x18058) register to5'hc.
CTLE_POLE_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL0 (0x18058) register to 2'h3.
DFE_TAP1_0	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_DFE_TAP_CTRL0 (0x1805E) register to 8'hf.
LF_TH_0	Set the value of these fields in the VR_PMA_SNPS_MP_25G_RX_GENCTRL3 (0x18053) register to 3'h4.

Note For multi-protocol 12G and 16G PHY configurations, default settings configure each line of the PHY to 10.3125Gbps rate. Therefore, no additional programming is required.

6. Program the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
7. Wait for the hardware to clear the value for the VR_RST field (bit [15]).

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.2.3 Programming Sequence When Using Enterprise 56G or 112G PHY

To switch to the 10G KR mode when using Synopsys Enterprise 56G or 112G PHY, complete the following steps:

1. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
2. Wait for the hardware to clear the value for the VR_RST (bit [15]) of the VR_PCS_DIG_CTRL1 register.
3. Write 4'b0000 to bits [5:2] (SS_5_2) of the SR_PCS_CTRL1 register.
4. Write 4'b0000 to bits [3:0] of the SR_PCS_CTRL2 register.

5. Program bits [6:0] (PMA_TYPE) of the SR_PMA_CTRL2 register to 7'b0001011 (10GBASE-KR).
6. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
7. Program bit [2] of the SR_PMA_RS_FEC_CTRL register to 0 (as RS-FEC is not supported in 10G mode).
8. To enable BASE-R FEC (if desired), program bit [0] in the SR_PMA_KR_FEC_CTRL register.
9. Program the Enterprise 56G PHY registers through its own APB interface)
   a. Program the PHY registers as mentioned in Table 8-6 on page 1306 to configure the PHY to 10G mode. For fast-simulation mode, additionally program the registers shown in Table 8-7 on page 1308.
   b. Enable the PMD by programming the pmd_en field in the PMD_CFG[0] (0x1400) register.
10. To program the Enterprise 112G PHY to 10G mode, program the following:
    a. Appropriate control of PHY I/Os.
    b. PHY registers by referring to the SDK app note.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.3 Switching to 40G Speed Mode

8.3.1 Programming Sequence When Using Multi-Protocol 32G PHY

To switch to the 40G mode Ethernet operations, complete the following steps:

1. Write 4'b0011 to bits [5:2] of the SR_PCS_CTRL1 register.
2. Write 4'b0100 to bits [3:0] of the SR_PCS_CTRL2 register.
3. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
4. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd16383 to configure the alignment marker interval.
5. If BASE-R FEC is present in the configuration, to enable BASE-R FEC, program bit [0] in the SR_PMA_KR_FEC_CTRL register.
6. Configure the PHY to operate at 10.3125Gbps rate per lane by programming the following register fields:
   a. CM_CS_0 and CM_CS_1 fields of the VR_PMA_MP_32G_CM_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 10G mode.
   b. CM_CS_2 and CM_CS_3 fields of the VR_PMA_MP_32G_CM_CNTX_SEL1 register to the value indicated in the *context_sel_ID.txt file* for 10G mode.
   c. TX_CS_0 and TX_CS_1 fields of the VR_PMA_MP_32G_TX_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 10G mode.

d. TX_CS_2 and TX_CS_3 fields of the VR_PMA_MP_32G_TX_CNTX_SEL1 register to the value indicated in 'context_sel_ID.txt file' for 10G mode.

e. RX_CS_0 and RX_CS_1 fields of the VR_PMA_MP_32G_RX_CNTX_SEL0 register to the value indicated in 'context_sel_ID.txt file' for 10G mode.

f. RX_CS_2 and RX_CS_3 fields of the VR_PMA_MP_32G_RX_CNTX_SEL1 register to the value indicated in 'context_sel_ID.txt file' for 10G mode.

g. L0_LN, L1_LN, L2_LN and L3_LN field of the VR_PMA_MP_32G_CNTX_LINK_NUM_CTRL register to a common value.

7. Program the LPM bit of the SR_PCS_CTRL1 register to 1.
8. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
9. Program the LPM bit of the SR_PCS_CTRL back to 0. Program other bits of this register to values which were present prior to executing step 7.

   Do not perform read-modified-write.
10. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Good' (3'b100).

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.3.2 Programming Sequence When Using Non-Synopsys PHY/Multi-Protocol 12G/16G/25G PHY

To switch to the 40G mode Ethernet operation, complete the following steps:

1. Write 4'b0011 to bits[5:2] of the SR_PCS_CTRL1 register.
2. Write 4'b0100 to bits[3:0] of the SR_PCS_CTRL2 register.
3. Write 2'b00 to bits[1:0] of the VR_PCS_DIG_CTRL3 register.
4. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd16383 to configure the alignment marker interval.
5. When BASE-R FEC is present in the configuration, to enable BASE-R FEC, program bit [0] in the SR_PMA_KR_FEC_CTRL register.
6. For multi-protocol 25G PHY configurations:
   a. Program the MPLLA_CAL_DISABLE field of the VR_PMA_SNPS_MP_25G_MPLLA_CTRL0 (0x18071) register to 0.
   b. Program the MPLLB_CAL_DISABLE field of the VR_PMA_SNPS_MP_25G_MPLLB_CTRL0 (0x18074) register to 1.

c. Program the various registers as shown in Table 8-2.

Table 8-2 PHY Settings for 40G/20G Mode for Synopsys Multi-Protocol 25G PHY

Field	Value
REF_RANGE	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_REF_CLK_CTRL (0x18091) register to ■ 3'b110 for reference clock frequency 156.25MHz and 312.5 MHz ■ 3'b011 for reference clock frequency 78.125MHz.
REF_CLK_DIV2	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_REF_CLK_CTRL (0x18091) register to ■ 1 for reference clock frequency 312.5MHz. ■ 0 for reference clock frequency 156.25MHz and 78.125MHz.
REF_MPLLA_DIV	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_REF_CLK_CTRL (0x18091) register to ■ 3'b000 for reference clock frequency 78.125MHz ■ 3'b001 for reference clock frequency 156.25MHz ■ 3'b010 for reference clock frequency 312.5MHz
MPLLA_MULTIPLIER	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL0 (0x18071) register to 12'h84.
MPLLA_DIV_MULT	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL2 (0x18073) register to 8'h8.
MPLLA_TX_CLK_DIV	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL2 (0x18073) register to 3'b010.
MPLLA_LOW_BW	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL3 (0x18077) register to 16'h18.
MPLLA_HIGH_BW	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLA_CTRL4 (0x18079) register to 16'h18.
VCO_REF_LD_0, VCO_REF_LD_1	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_REF0 (0x18096) register to ■ 7'h15 for reference clock frequency 312.5MHz and 156.25 MHz ■ 7'hb for reference clock frequency 78.125MHz.
VCO_REF_LD_2, VCO_REF_LD_3	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_REF1 (0x18097) register to ■ 7'h15 for reference clock frequency 312.5MHz and 156.25 MHz. ■ 7'hb for reference clock frequency 78.125MHz.
VCO_LD_VAL_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_LD0 (0x18092) register to ■ 13'h5ac for reference clock frequency 78.125MHz. ■ 13'h56a for reference clock frequency 312.5MHz and 156.25MHz.

Table 8-2 PHY Settings for 40G/20G Mode for Synopsys Multi-Protocol 25G PHY

Field	Value
VCO_LD_VAL_1	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_LD1 (0x18093) register to ■ 13'h5ac for reference clock frequency 78.125MHz. ■ 13'h56a for reference clock frequency 312.5MHz and 156.25MHz.
VCO_LD_VAL_2	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_LD2 (0X18094) register to ■ 13'h5ac for reference clock frequency 78.125MHz ■ 13'h56a for reference clock frequency 312.5MHz and 156.25MHz.
VCO_LD_VAL_3	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_LD3 (0x18095) register to ■ 13'h5ac for reference clock frequency 78.125MHz ■ 13'h56a for reference clock frequency 312.5MHz and 156.25MHz.
RX0_CDR_PPM_MAX, RX1_CDR_PPM_MAX	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_RX_PPM_CTRL0 (0x18065) register to ■ 5'h12 for reference clock frequency 312.5MHz and 156.25MHz ■ 5'h13 for reference clock frequency 78.125MHz.
RX2_CDR_PPM_MAX, RX3_CDR_PPM_MAX	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_RX_PPM_CTRL1 (0x18066) register to ■ 5'h12 for reference clock frequency 312.5MHz and 156.25MHz ■ 5'h13 for reference clock frequency 78.125MHz.
MPLLB_SEL_0, MPLLB_SEL_3_1	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_MPLL_CMN_CTRL (0x18070) register to 0.
TX_ALIGN_WD_XFER_0, TX_ALIGN_WD_XFER_3_1	Set the value of these fields in the VR_PMA_SNPS_MP_25G_TX_GEN_CTRL5 (0x18046) register to 0.
TX0_DCC_CTRL_RNG, TX1_DCC_CTRL_RNG, TX2_DCC_CTRL_RNG, TX3_DCC_CTRL_RNG	Set the value of this field in the VR_PMA_SNPS_MP_25G_TX_DCC_CTRL (0x18047) register to 4'hb.
EQ_AFE_RT_0, EQ_AFE_RT_1, EQ_AFE_RT_2, EQ_AFE_RT_3	Set the value of these fields in the VR_PMA_SNPS_MP_25G_RX_AFE_RATE_CTRL (0x1805D) register to 3'b100.
RX0_RATE, RX1_RATE, RX2_RATE, RX3_RATE	Set the value of these fields in the VR_PMA_SNPS_MP_25G_RX_RATE_CTRL (0x18054) register to 3'b00.
RX0_DELTA_IQ	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_IQ_CTRL0 (0x1806A) register to 4'b0000.

Table 8-2 PHY Settings for 40G/20G Mode for Synopsys Multi-Protocol 25G PHY

Field	Value
RX1_DELTA_IQ	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_IQ_CTRL1 (0x1806B) register to 4'b0000.
RX2_DELTA_IQ	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_IQ_CTRL2 (0x1806C) register to 4'b0000.
RX3_DELTA_IQ	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_IQ_CTRL3 (0x1806D) register to 4'b0000.
CTLE_BOOST_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL0 (0x18058) register to 5'hc.
CTLE_BOOST_1	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL1 (0x18059) register to 5'hc.
CTLE_BOOST_2	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL2 (0x1805A) register to 5'hc.
CTLE_BOOST_3	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL3 (0x1805B) register to 5'hc.
CTLE_POLE_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL0 (0x18058) register to 2'h3.
CTLE_POLE_1	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL1 (0x18059) register to 2'h3.
CTLE_POLE_2	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL2 (0x1805A) register to 2'h3.
CTLE_POLE_3	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL3 (0x1805B) register to 2'h3.
DFE_TAP1_0, DFE_TAP1_1	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_DFE_TAP_CTRL0 (0x1805E) register to 8'hf.
DFE_TAP1_2, DFE_TAP1_3	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_DFE_TAP_CTRL1 (0x1805F) register to 8'hf.
LF_TH_0, LT_TH_1, LF_TH_2, LF_TH_3	Set the value of these fields in the VR_PMA_SNPS_MP_25G_RX_GENCTRL3 (0x18053) register to 3'h4.

Note For multi-protocol 12G and 16G PHY configurations, default settings configure each line of the PHY to 10.3125Gbps rate. Therefore, additional programming is not required.

7. Program the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.

8. Wait for the hardware to clear the value for the VR_RST field (bit [15]).

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.3.3 Programming Sequence when using Enterprise 56G or 112G PHY

To switch to the 40G mode Ethernet operation, complete the following steps:

1. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
2. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.
3. Write 4'b0011 to bits[5:2] of the SR_PCS_CTRL1 register.
4. Write 4'b0100 to bits[3:0] of the SR_PCS_CTRL2 register.
5. Write 2'b00 to bits[1:0] of the VR_PCS_DIG_CTRL3 register.
6. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd16383 to configure the alignment marker interval. To speed-up simulation, program a smaller value to this field.
7. Program bit [2] of the SR_PMA_RS_FEC_CTRL register to 0 if previously 1 (as RS-FEC is not supported in 40G mode).
8. To enable BASE-R FEC (if desired), program bit [0] in the SR_PMA_KR_FEC_CTRL register.
9. Program Enterprise 56G PHY registers through its own APB interface:
   a. Program PHY registers as mentioned in Table 8-6 on page 1306 to configure the PHY to 40G mode. For fast-simulation mode, additionally program, the registers shown in the Table 8-7 on page 1308.
   b. Enable the PMD by programming the pmd_en field in the PMD_CFG[0] (0x1400) register.
10. To program the Enterprise 112G PHY to 40G mode, program the following:
    a. Appropriate control of PHY I/Os.
    b. PHY registers by referring to the SDK app note.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.4 Switching to 20G Speed Mode

8.4.1 Programming Sequence When Using Multi-Protocol 32G PHY

To switch to the 40G mode Ethernet operation, complete the following steps:

1. Write 4'b0011 to bits [5:2] of the SR_PCS_CTRL1 register.

2. Write 4'b0100 to bits [3:0] of the SR_PCS_CTRL2 register.
3. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
4. Write 2'b11 to bits [15:14] and bits [11:10] of the VR_PCS_DIG_CTRL2 register to disable lanes 2 and 3.
5. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd16383 to configure the alignment marker interval.
6. When BASE-R FEC is present in the configuration, to enable BASE-R FEC, program bit [0] in the SR_PMA_KR_FEC_CTRL register.
7. Configure the PHY to operate at 10.3125 Gbps rate per lane by programming the following register fields:
   a. CM_CS_0 and CM_CS_1 fields of the VR_PMA_MP_32G_CM_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 10G mode.
   b. TX_CS_0 and TX_CS_1 fields of the VR_PMA_MP_32G_TX_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 10G mode.
   c. RX_CS_0 and RX_CS_1 fields of the VR_PMA_MP_32G_RX_CNTX_SEL0 register to the value indicated in 'the *context_sel_ID.txt file* for 10G mode.
   d. L0_LN, L1_LN field of the VR_PMA_MP_32G_CNTX_LINK_NUM_CTRL register to a common value.
8. Program the LPM bit of the SR_PCS_CTRL1 register to 1.
9. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
10. Program the LPM bit of the SR_PCS_CTRL register back to 0. Program other bits of this register to values which were present prior to executing step 8.

    Do not perform read-modified-write.
11. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Good' (3'b100).

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.4.2 Programming Sequence when using Non-Synopsys PHY/Multi-protocol 12G/16G/25G PHY

To switch to the 40G mode Ethernet operation, complete the following steps:

1. Write 4'b0011 to bits [5:2] of the SR_PCS_CTRL1 register.
2. Write 4'b0100 to bits [3:0] of the SR_PCS_CTRL2 register.
3. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
4. Write 2'b11 to bits [15:14] and bits [11:10] of the VR_PCS_DIG_CTRL2 register to disable lanes 2 and 3.
5. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd16383 to configure the alignment marker interval.
6. To enable BASE-R FEC, program bit [0] in the SR_PMA_KR_FEC_CTRL register (If BASE-R FEC is present in the configuration).

7. For multi-protocol 25G PHY configurations:
   a. Program the MPLLA_CAL_DISABLE field of the VR_PMA_SNPS_MP_25G_MPLLA_CTRL0 (0x18071) register to 0.
   b. Program the MPLLB_CAL_DISABLE field of the VR_PMA_SNPS_MP_25G_MPLLB_CTRL0 (0x18074) register to 1.
   c. Program various registers (controlling lane 0 and lane 1) as shown in Table 8-2 on page 1292, for lane 0 and lane 1.

   For multi-protocol 12G and 16G PHY configurations, configure each with default settings.
8. Program the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
9. Wait for the hardware to clear the value for the VR_RST field (bit [15]).

Note Before starting packet transfer, you need to monitor for the RLU bit of the SR_PCS_STS1 register to indicate 1.

8.4.3 Programming Sequence when using Enterprise 56G or 112G PHY

1. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
2. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.
3. Write 4'b0011 to bits [5:2] of the SR_PCS_CTRL1 register.
4. Write 4'b0100 to bits [3:0] of the SR_PCS_CTRL2 register.
5. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
6. Write 2'b11 to bits [15:14] and bits [11:10] of the VR_PCS_DIG_CTRL2 to disable lanes 2 and 3.
7. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd16383 to configure the alignment marker interval.

   To speed up the simulation, you can program a smaller value to this field.
8. Program bit [2] of the SR_PMA_RS_FEC_CTRL register to 0 if previously 1 (as RS-FEC is not supported in 40G mode).
9. To enable BASE-R FEC (if desired), program bit [0] in the SR_PMA_KR_FEC_CTRL register.
10. Program Enterprise 56G PHY registers (through its own APB interface):
    a. Program the PHY registers as described in Table 8-6 on page 1306 to configure the PHY to 20G mode. For fast-simulation mode, additionally program the registers shown in the Table 8-7 on page 1308.
    b. Enable the PMD by programming the pmd_en field in the PMD_CFG[0] (0x1400) register.
11. To program the Enterprise 112G PHY to 20G mode, program the following:
    a. Appropriate control of PHY I/Os.

b. PHY registers by referring to the SDK app note.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.5 Switching to 25G Speed Mode

8.5.1 Programming Sequence for Non-Synopsys PHY

To switch to 25G mode Ethernet operation, complete the following steps:

1. Write 4'b0101 to bits[5:2] (SS_5_2 field) of the SR_PCS_CTRL1 register.
2. Write 4'b0111 to bits[3:0] (PCS_TYPE_SEL field) of the SR_PCS_CTRL2 register to select 25GBASE-R PCS.
3. Program bits[6:0] (PMA_TYPE field) of the SR_PMA_CTRL2 register to 7'b0111001 (25GBASE-KR).
4. Program bit[1] (EN_50G field) in the VR PCS MMD Digital Control3 register to 0 (to disable 50G).
5. Program bit[0] (CNS_EN field) in the VR PCS MMD Digital Control3 register to 0 or 1.
6. To enable RS-FEC, perform the following:
   a. Program the AM interval period in field [13:0] of the VR_PCS_AM_CNT register to 14'h400 (16'd1024).

      To speed up the link-up in simulation, program a small value such as 14'h40 (14'd64).
   b. Program the VR_PMA_CWM00 register to 16'h68C1.
   c. Program the VR_PMA_CWM01 register to 16'h3321.
   d. Program the VR_PMA_CWM02 register to 16'h973E.
   e. Program the VR_PMA_CWM03 register to 16'hCCDE.
   f. Set bit [2] in the SR_PMA_RS_FEC_CTRL register to 1.
7. To enable BASE-R FEC, program bit [0] in the SR_PMA_KR_FEC_CTRL register.
8. Configure the PHY to operate at 25.78125 Gbps rate.
9. Program the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
10. Wait for the hardware to clear the value for the VR_RST field (bit [15]).

Note
- Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.
- Do not enable RS-FEC and BASE-R FEC at the same time.

8.5.2 Programming Sequence for Multi-Protocol 32G PHY

To switch to 25G mode Ethernet operation, complete the following steps:

1. Write 4'b0101 to bits [5:2] (SS_5_2) of the SR_PCS_CTRL1 register.
2. Write 4'b0111 to bits [3:0] (PCS_TYPE_SEL) of the SR_PCS_CTRL2 register to select 25GBASE-R PCS.
3. Program bits [6:0] (PMA_TYPE) of the SR_PMA_CTRL2 register to 7'b0111001 (25GBASE-KR).
4. Program bit [1] (EN_50G) in the VR PCS MMD Digital Control3 register to 0 (to disable 50G).
5. Program bit [0] (CNS_EN) in the VR PCS MMD Digital Control3 register to 0 or 1.
6. To enable RS FEC, perform the following:
   a. Program the AM interval period in field [13:0] of the VR_PCS_AM_CNT register to 14'h400 (14'd1024).

      To speed up the link-up in simulation, program a small value such as 14'h40 (14'd64).
   b. Program the VR_PMA_CWM00 register to 16'h68C1.
   c. Program the VR_PMA_CWM01 register to 16'h3321.
   d. Program the VR_PMA_CWM02 register to 16'h973E.
   e. Program the VR_PMA_CWM03 register to 16'hCCDE.
   f. Set bit [2] in the SR_PMA_RS_FEC_CTRL register to 1.
7. To enable BASE-R FEC, program bit [0] in the SR_PMA_KR_FEC_CTRL register.
8. Configure the PHY to operate at 25.78125 Gbps rate by programming the following register fields:
   a. The CM_CS_0 field of the VR_PMA_MP_32G_CM_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
   b. The TX_CS_0 field of the VR_PMA_MP_32G_TX_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
   c. The RX_CS_0 field of the VR_PMA_MP_32G_RX_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
9. Program the LPM bit of the SR_PCS_CTRL1 register to 1.
10. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
11. Program the following values to various bits of the SR_PCS_CTRL1 register:
    a. LPM to 0.
    b. RST to 0.
    c. CS_EN to the default value 0.
    d. SS_5_2 to 4'b0101 (value programmed in step 1).

    Do not perform read-modified-write.
12. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Good' (3'b100).



Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.5.3 Programming Sequence for Multi-Protocol 25G PHY

To switch to 25G mode Ethernet operation, complete the following steps:

1. Write 4'b0101 to bits [5:2] (SS_5_2) of the `SR_PCS_CTRL1` register.
2. Write 4'b0111 to bits [3:0] (PCS_TYPE_SEL) of the `SR_PCS_CTRL2` register to select 25GBASE-R PCS.
3. Program bits [6:0] (PMA_TYPE) of the `SR_PCS_CTRL2` register to 7'b0111001 (25GBASE-KR).
4. Program bit [1] (EN_50G) in the VR PCS MMD Digital Control3 register to 0 (to disable 50G).
5. Program bit [0] (CNS_EN) in the VR PCS MMD Digital Control3 register to 0 or 1.
6. To enable RS FEC, perform the following:
   a. Program the AM interval period in field [13:0] of the `VR_PCS_AM_CNT` register to 14'h400 (14'd1024).

      To speed up the link-up in simulation, program a small value such as 14'h40 (14'd64).
   b. Program the `VR_PMA_CWM00` register to 16'h68C1.
   c. Program the `VR_PMA_CWM01` register to 16'h3321.
   d. Program the `VR_PMA_CWM02` register to 16'h973E.
   e. Program the `VR_PMA_CWM03` register to 16'hCCDE.
   f. Set bit [2] in the `SR_PMA_RS_FEC_CTRL` register to 1.
7. To enable BASE-R FEC, program bit [0] in the `SR_PMA_KR_FEC_CTRL` register.
8. Program various register fields as shown in Table 8-3 on page 1301 to configure PHY common settings.
9. Program the MPLLA_CAL_DISABLE field of the `VR_PMA_SNPS_MP_25G_MPLLA_CTRL0` (0x18071) register to 1.
10. Program the MPLLB_CAL_DISABLE field of the `VR_PMA_SNPS_MP_25G_MPLLB_CTRL0` (0x18074) register to 0.
11. Program various register fields as shown in Table 8-4 on page 1302 to configure pre-lane PHY settings.



Note In Table 8-4 on page 1302, for per-lane settings, register fields controlling all four lanes of the PHY are programmed. However, it is necessary to program only the register fields for lanes 0 and 1. Because values programmed to lanes 2 and 3 are 'don't care', you may skip programming these lanes.

12. Initiate the vendor-specific software reset by writing 1'b1 to the VR_RST field (bit[15]) of the `VR_PCS_DIG_CTRL1` register.
13. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the `VR_PCS_DIG_CTRL1` (0x38000) register.

14. Program the register fields that control lane 0 as shown in Table 8-5 on page 1304.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

Table 8-3 Common PHY Settings for 100G/50G/25G Mode (Per-Lane Rate of 25.78125 Gbps)

Field	Value
Common Settings	
REF_RANGE	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_REF_CLK_CTRL (0x18091) register to ■ 3'b110 for reference clock frequency 156.25MHz and 312.5 MHz ■ 3'b011 for reference clock frequency 78.125MHz.
REF_CLK_DIV2	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_REF_CLK_CTRL (0x18091) register to ■ 1 for reference clock frequency 312.5MHz ■ 0 for reference clock frequency 156.25MHz and 78.125MHz.
REF_MPLLB_DIV	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_REF_CLK_CTRL (0x18091) register to ■ 3'b000 for reference clock frequency 78.125MHz ■ 3'b001 for reference clock frequency 156.25MHz ■ 3'b010 for reference clock frequency 312.5MHz.
MPLLB_MULTIPLIER	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLB_CTRL0 (0x18074) register to 12'ha5.
MPLLB_DIV_MULT	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLB_CTRL2 (0x18076) register to 8'h4.
MPLLB_TX_CLK_DIV	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLB_CTRL2 (0x18076) register to 3'b001.
MPLLB_LOW_BW	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLB_CTRL3 (0x18078) register to 16'h18.
MPLLB_HIGH_BW	Set the value of this field in the VR_PMA_SNPS_MP_25G_MPLLB_CTRL4 (0x1807A) register to 16'h18.

Table 8-4 Per-Lane PHY Settings for 100G/50G/25G Mode

Field	Value
VCO_REF_LD_0, VCO_REF_LD_1	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_REF0 (0x18096) register to ■ 7'h11 for reference clock frequency 312.5MHz and 156.25 MHz ■ 7'h9 for reference clock frequency 78.125MHz.
VCO_REF_LD_2, VCO_REF_LD_3	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_REF1 (0x18097) register to ■ 7'h11 for reference clock frequency 312.5MHz and 156.25 MHz ■ 7'h9 for reference clock frequency 78.125MHz.
VCO_LD_VAL_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_LD0 (0x18092) register to ■ 13'h5cd for reference clock frequency 78.125MHz. ■ 13'h57b for reference clock frequency 312.5MHz and 156.25MHz.
VCO_LD_VAL_1	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_LD1 (0x18093) register to ■ 13'h5cd for reference clock frequency 78.125MHz ■ 13'h5c7b for reference clock frequency 312.5MHz and 156.25MHz.
VCO_LD_VAL_2	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_LD2 (0x18094) register to ■ 13'h5cd for reference clock frequency 78.125MHz ■ 13'h57b for reference clock frequency 312.5MHz and 156.25MHz.
VCO_LD_VAL_3	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_VCO_CAL_LD3 (0x18095) register to ■ 13'h5cd for reference clock frequency 78.125MHz ■ 13'h57b for reference clock frequency 312.5MHz and 156.25MHz.
RX0_CDR_PPM_MAX, RX1_CDR_PPM_MAX	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_RX_PPM_CTRL0 (0x18065) register to ■ 5'h13 for reference clock frequency 312.5MHz and 156.25MHz ■ 5'h14 for reference clock frequency 78.125MHz.
RX2_CDR_PPM_MAX, RX3_CDR_PPM_MAX	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_RX_PPM_CTRL1 (0x18066) register to ■ 5'h13 for reference clock frequency 312.5MHz and 156.25MHz ■ 5'h14 for reference clock frequency 78.125MHz.
MPLLB_SEL_0,	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_MPLL_CMN_CTRL (0x18070) register to 1.

Table 8-4 Per-Lane PHY Settings for 100G/50G/25G Mode

Field	Value
MPLLB_SEL_3_1	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_MPLL_CMN_CTRL (0x18070) register to 7.
TX_ALIGN_WD_XFER_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_TX_GEN_CTRL5 (0x18046) register to 1.
TX_ALIGN_WD_XFER_3_1	Set the value of this field in the VR_PMA_SNPS_MP_25G_TX_GEN_CTRL5 (0x18046) register to 7.
TX0_DCC_CTRL_RNG, TX1_DCC_CTRL_RNG, TX2_DCC_CTRL_RNG, TX3_DCC_CTRL_RNG	Set the value of this field in the VR_PMA_SNPS_MP_25G_TX_DCC_CTRL (0x18047) register to 4'hc.
EQ_AFE_RT_0, EQ_AFE_RT_1, EQ_AFE_RT_2, EQ_AFE_RT_3	Set the value of these fields in the VR_PMA_SNPS_MP_25G_RX_AFE_RATE_CTRL (0x1805D) register to 3'b000.
RX0_RATE, RX1_RATE, RX2_RATE, RX3_RATE	Set the value of these fields in the VR_PMA_SNPS_MP_25G_RX_RATE_CTRL (0x18054) register to 3'b000.
RX0_DELTA_IQ	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_IQ_CTRL0 (0x1806A) register to 4'b0011.
RX1_DELTA_IQ	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_IQ_CTRL1 (0x1806B) register to 4'b0011.
RX2_DELTA_IQ	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_IQ_CTRL2 (0x1806C) register to 4'b0011.
RX3_DELTA_IQ	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_IQ_CTRL3 (0X1806D) register to 4'b0011.
CTLE_BOOST_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL0 (0x18058) register to 5'h0C.
CTLE_BOOST_1	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL1 (0x18059) register to 5'h0C.
CTLE_BOOST_2	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL2 (0x1805A) register to 5'h0C.
CTLE_BOOST_3	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL3 (0x1805B) register to 5'h0C.
CTLE_POLE_0	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL0 (0x18058) register to 2'h3.

Table 8-4 Per-Lane PHY Settings for 100G/50G/25G Mode

Field	Value
CTLE_POLE_1	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL1 (0x18059) register to 2'h3.
CTLE_POLE_2	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL2 (0x1805A) register to 2'h3.
CTLE_POLE_3	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_RX_EQ_CTRL3 (0x1805B) register to 2'h3.
DFE_TAP1_0, DFE_TAP1_1	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_DFE_TAP_CTRL0 (0x1805E) register to 8'hf.
DFE_TAP1_2, DFE_TAP1_3	Set the value of these fields in the VR_PMA_SNPS_MP_25G_16G_DFE_TAP_CTRL1 (0x1805F) register to 8'hf.
LF_TH_0, LF_TH_1, LF_TH_2, LF_TH_3	Set the value of these fields in the VR_PMA_SNPS_MP_25G_RX_GENCTRL3 (0x18053) register to 3'h4.

Table 8-5 Transmit Equalization Settings

Field	Value
TX_EQ_PRE	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_TX_EQ_CTRL0_LN0/1/2/3 (0x18036/0x18038/0x1803A/0x1803C) register to 6'h0.
TX_EQ_MAIN	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_TX_EQ_CTRL0_LN0/1/2/3 (0x18036/0x18038/0x1803A/0x1803C) register to 6'h18.
TX_EQ_POST	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_TX_EQ_CTRL1_LN0/1/2/3 (0x18037/0x18039/0x1803B/0x1803D) register to 6'h0.
TX_EQ_OVER_RIDE	Set the value of this field in the VR_PMA_SNPS_MP_25G_16G_TX_EQ_CTRL1_LN0/1/2/3 (0x18037/0x18039/0x1803B/0x1803D) register to 1 after programming the DWCXLP_LNK_TRAIN_EN parameter to 1.

8.5.4 Programming Sequence for Enterprise 56G or 112G PHY

To switch to 25G mode Ethernet operation, complete the following steps:

1. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
2. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.
3. Write 4'b0101 to bits [5:2] (SS_5_2) of the SR_PCS_CTRL1 register.
4. Write 4'b0111 to bits [3:0] (PCS_TYPE_SEL) of the SR_PCS_CTRL2 register to select 25GBASE-R PCS.

5. Program bits [6:0] (PMA_TYPE) of the SR_PMA_CTRL2 register to 7'b0111001 (25GBASE-KR).
6. Program bit [1] (EN_50G) in the VR PCS MMD Digital Control3 register to 0 (to disable 50G).
7. Program bit [0] (CNS_EN) in the VR PCS MMD Digital Control3 register to 0 or 1.
8. To enable RS FEC, perform the following:
   a. Program the AM interval period in field [13:0] of the VR_PCS_AM_CNT register to 14'h400 (14'd1024).

      To speed up the link-up in simulation, program a small value such as 14'h40 (14'd64).
   b. Program the VR_PMA_CWM00 register to 16'h68C1.
   c. Program the VR_PMA_CWM01 register to 16'h3321.
   d. Program the VR_PMA_CWM02 register to 16'h973E.
   e. Program the VR_PMA_CWM03 register to 16'hCCDE.
   f. Set bit [2] in the SR_PMA_RS_FEC_CTRL register to 1.
9. To enable BASE-R FEC, program bit [0] in the SR_PMA_KR_FEC_CTRL register.
10. Program Enterprise 56G PHY registers through its own APB interface:
    a. Program the PHY registers as mentioned in Table 8-6 on page 1306 to configure the PHY to 25G mode. For fast-simulation mode, additionally program the registers shown in Table 8-7 on page 1308.
    b. Enable the PMD by programming the pmd_en field in the PMD_CFG[0] (0x1400) register.
11. To program the Enterprise 112G PHY to 25G mode, program the following:
    a. Appropriate control PHY I/Os.
    b. PHY registers by referring to the SDK app note.

Table 8-6 Enterprise 56G PHY Register Settings for Various Modes

Register Name	E56 PHY Address Offset	Field Name	Values / Description
PMD_CFG[0]	0x1400	pmd_mode	1'b0 for ETH mode (multi-link)
		tx_en_cfg	For enabling TXS lanes of the PHY ■ 4'b0001: 10G/25G/50GBASE-KR (PAM4) mode ■ 4'b0011: 50G consortium/50GAUI-2 /100G BASE-KR2 mode ■ 4'b1111: 100G NRZ/40G modes
		rx_en_cfg	For enabling RXS Lanes of PHY ■ 4'b0001: 10G/25G/50GBASE-KR (PAM4) mode ■ 4'b0011: 50G consortium/50GAUI-2 /100G BASE-KR2 mode ■ 4'b1111: 100G NRZ/40G modes
		pam4_precode_no_krt_en	■ 4'hF: NRZ ■ 4'h0: PAM4 modes
PLL0_CFG[0]	0x0C10	pll0_clkout_tx_top_left_en	1'b1 Note: Required for 25G/50G/100G mode.
		pll0_clkout_tx_top_right_en	1'b1 Note: Required for 25G/50G/100G mode.
PLL1_CFG[0]	0x0C48	pll1_clkout_tx_top_left_en	1'b1 Note: Required for 10G/40G mode.
		pll1_clkout_tx_top_right_en	1'b1 Note: Required for 10G/40G mode.
PLL0_CFG[2]	0x0C18	pll0_postdiv	Program to 5'h4
PLL0_DIV_CFG[0]	0x0C1C	pll0_prediv_0	Program to 5'h8
		pll0_fbdiv_int_0	■ 11'h2A8: PAM4/100GAUI-4/50GAUI-2 ■ 11'h294: Other modes
PLL1_CFG[2]	0x0C50	pll1_postdiv	Value is programmed to 5'h8
PIN_OVRDEN[0]	0x0C8C	ovrd_en_pll0_signal_type_i	■ 1'b1: PAM4/100GAUI-4/50GAUI-2 ■ 1'b0: Other modes
PIN_OVRDVAL[0]	0x0C94	int_pll0_tx_signal_type_i	■ 1'b1: PAM4/100GAUI-4/50GAUI-2 ■ 1'b0: Other modes

Table 8-6 Enterprise 56G PHY Register Settings for Various Modes

Register Name	E56 PHY Address Offset	Field Name	Values / Description
ANA_OVRDEN[0]	0x0CA4	ovrd_en_ana_refclk_buf_daisy_en_i	Set to 1'b1
ANA_OVRDVAL[0]	0x0CB0	ana_refclk_buf_daisy_en_i	Set to 1'b1
OSC_CAL_N_CDR[4]	0x0014	vco_code_init	■ 10'h5FF: 25G/50G Consortium/100GBASE-KR4 mode ■ 10'h7FF: Other modes
DATAPATH_CFG[0]	0x142C	pll0_tx_symclk_period_ml	■ 3'b111: DATARATE/80 for PAM4 modes ■ 3'b101: DATARATE/40 for NRZ modes
		pll1_tx_symclk_period_ml	3'b101 for DATARATE/40
		pll_tx_symclk_period_sl_0	3'b101: DATARATE/40 for 10G/40G modes Don't care for other modes
		pll_tx_symclk_period_sl_2	3'b101
		pll_tx_symclk_period_sl_3	■ 3'b111: PAM4 modes ■ 3'b101: for 100GAUI-4, 50GAUI-2 mode ■ 3'b111: for other modes
DATAPATH_CFG[1]	0x1430	tx_symdata_width_0	3'b101 for 40UI for 10G/40G modes
		tx_symdata_width_2	3'b101 for 40UI for NRZ modes
		tx_symdata_width_3	■ 3'b111: 80UI for PAM4 modes ■ 3'b101: 40UI for 100GAUI-4/50GAUI-2 mode
		rx_symdata_width_0	3'b101 - for 40UI for 10G/40G modes Don't care for other modes
		rx_symdata_width_2	3'b101 for 40UI for NRZ modes Don't care for other modes
		rx_symdata_width_3	■ 3'b111: 80UI for PAM4 modes ■ 3'b101: 40UI for 100GAUI-4/50GAUI-2 mode

Table 8-6 Enterprise 56G PHY Register Settings for Various Modes

Register Name	E56 PHY Address Offset	Field Name	Values / Description
AN_CFG[1]	0x1438	an0_rate_select_reg	■ 5'd13: 50G-KR (PAM4) and 100GBASE-KR2 (PAM4) ■ 5'd17: 100GAUI-4 and 50GAUI-2 ■ 5'd10: 25G/50G Consortium/100GBASE-KR4 ■ 5'd2: 10G/40G mode
		an1_rate_select_reg	■ 5'd13: 50G-KR (PAM4) and 100GBASE-KR2 (PAM4) ■ 5'd17: 100GAUI-4 and 50GAUI-2 ■ 5'd10: 25G/50G Consortium/100GBASE-KR4 ■ 5'd2: 10G/40G mode
		an2_rate_select_reg	■ 5'd13: 50G-KR (PAM4) and 100GBASE-KR2 (PAM4) ■ 5'd17: 100GAUI-4 and 50GAUI-2 ■ 5'd10: 25G/50G Consortium/100GBASE-KR4 ■ 5'd2: 10G/40G mode
		an3_rate_select_reg	■ 5'd13: 50G-KR (PAM4) and 100GBASE-KR2 (PAM4) ■ 5'd17: 100GAUI-4 and 50GAUI-2 ■ 5'd10: 25G/50G Consortium/100GBASE-KR4 ■ 5'd2: 10G/40G mode
SPARE5	0x1608	spare_5	32'h3F. Note: Required only for PAM4 modes
SPARE52	0x16FC	spare_52	32'hF0: This is required to adjust rx_div66 clock frequency to 402.465 MHz Note: Required only for PAM4 modes

Table 8-7 Enterprise 56G PHY Fast Simulation Register Settings

Register Name	E56 PHY Address Offset	Fast Simulation Value
PMD_CFG[3]	0x140C	32'h19010101
PMD_CFG[4]	0x1410	32'h00011F18
CTRL_FSM_CFG[0]	0x1440	32'hFF010001
CTRL_FSM_CFG[1]	0x1444	32'h00000000
CTRL_FSM_CFG[2]	0x1448	32'h00000000
CTRL_FSM_CFG[3]	0x144C	32'h00000000
CTRL_FSM_CFG[4]	0x1450	32'h00000000

Table 8-7 Enterprise 56G PHY Fast Simulation Register Settings

Register Name	E56 PHY Address Offset	Fast Simulation Value
CTRL_FSM_CFG[5]	0x1454	32'h00030003
CTRL_FSM_CFG[6]	0x1458	32'h00110003
CTRL_FSM_CFG[7]	0x145C	32'hC0110011
CTRL_FSM_CFG[8]	0x1460	32'h40114011
CTRL_FSM_CFG[9]	0x1464	32'h43114311
CTRL_FSM_CFG[10]	0x1468	32'h43114311
CTRL_FSM_CFG[11]	0x146C	32'h43114311
CTRL_FSM_CFG[12]	0x1470	32'h40114311
CTRL_FSM_CFG[13]	0x1474	32'h00010001
CTRL_FSM_CFG[14]	0x1478	32'h00080001
CTRL_FSM_CFG[15]	0x147C	32'h10000000
CTRL_FSM_CFG[16]	0x1480	32'h00000000
CTRL_FSM_CFG[17]	0x1484	32'h00002000
CTRL_FSM_CFG[29]	0x14B4	32'h00000000

8.6 Switching to 50G Speed Mode

8.6.1 Switching to 50G Consortium Mode

8.6.1.1 Programming Sequence for Non-Synopsys PHY

To switch to 50G mode Ethernet operation, complete the following steps:

1. Write 4'b0100 to bits [3:0] of the SR PCS Control2 register to select 40GBASE-R PCS.
2. Set bit [1] (EN_50G) in the VR PCS MMD Digital Control3 register to enable 50G.
3. Set bit [0] (CNS_EN) of the VR PCS MMD Digital Control3 register to enable working in the Consortium mode.
4. Program the PCS_AM_CNT field of the `VR_PCS_AM_CNT` register to 'd1024 (if RS-FEC is enabled) or 'd20479 (if RS-FEC is disabled) to configure the alignment marker interval.

   To speed up the simulation, program a smaller value.
5. To enable RS-FEC (528,514) mode, program bit [2] of the `SR_PMA_RS_FEC_CTRL` register to 1.
6. To enable BASE-R FEC, program bit 0 in the `SR_PMA_KR_FEC_CTRL` register.
7. Configure the two lanes of the PHY to operate at 25.78125 Gbps rate.
8. Program the VR_RST field (bit [15]) of the `VR_PCS_DIG_CTRL1` register to 1.

9. Wait for the hardware to clear the value for the VR_RST field (bit [15]).

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.6.1.2 Programming Sequence for Multi-protocol 32G PHY

To switch to 50G mode Ethernet operation, complete the following steps:

1. Write 4'b0100 to bits [3:0] of the SR PCS Control2 register to select 40GBASE-R PCS.
2. Set bit [1] (EN_50G) in the VR PCS MMD Digital Control3 register to enable 50G.
3. Set bit [0] (CNS_EN) of the VR PCS MMD Digital Control3 register to enable working in the Consortium mode.
4. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd1024 (if RS-FEC is enabled) or 'd20479 (if RS-FEC is disabled) to configure the alignment marker interval.

   To speed up the simulation, program a smaller value.
5. To enable RS-FEC (528,514) mode, program bit [2] of the SR_PMA_RS_FEC_CTRL register to 1.
6. To enable BASE-R FEC, program bit 0 in the SR_PMA_KR_FEC_CTRL register.
7. Configure the PHY to operate at 25.78125 Gbps rate by programming the following register fields:
   - The CM_CS_0 and CM_CS_1 fields of the VR_PMA_MP_32G_CM_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
   - The TX_CS_0 and TX_CS_1 fields of the VR_PMA_MP_32G_TX_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
   - The RX_CS_0 and RX_CS_1 fields of the VR_PMA_MP_32G_RX_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
8. Program the LPM bit of the SR_PCS_CTRL1 register to 1.
9. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
10. Program the following values to various bits of the SR_PCS_CTRL1 register:
    - LPM to 0.
    - RST to 0.
    - CS_EN to the default value 0.
    - SS_5_2 to 4'b0100 (value programmed in step 1).

    Do not perform read-modified-write.
11. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Good' (3'b100).

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.6.1.3 Programming Sequence for Multi-Protocol 25G PHY

To switch to 50G mode Ethernet operation, complete the following steps:

1. Write 4'b0100 to bits [3:0] of the SR PCS Control2 register to select 40GBASE-R PCS.
2. Set bit [1] (EN_50G) in the VR PCS MMD Digital Control3 register to enable 50G.
3. Set bit [0] (CNS_EN) of the VR PCS MMD Digital Control3 register to enable working in the Consortium mode.
4. Program the PCS_AM_CNT field of the `VR_PCS_AM_CNT` register to 'd1024 (if RS-FEC is enabled) or 'd20479 (if RS-FEC is disabled) to configure the alignment marker interval.

   To speed up the simulation, program a smaller value.
5. To enable RS-FEC (528,514) mode, program bit [2] of the `SR_PMA_RS_FEC_CTRL` register to 1.
6. To enable BASE-R FEC, program bit 0 in the `SR_PMA_KR_FEC_CTRL` register.
7. Program various register fields as shown in Table 8-3 on page 1301 to configure PHY common settings.
8. Program the MPLLA_CAL_DISABLE field of the `VR_PMA_SNPS_MP_25G_MPLLA_CTRL0` (0x18071) register to 1.
9. Program the MPLLB_CAL_DISABLE field of the `VR_PMA_SNPS_MP_25G_MPLLB_CTRL0` (0x18074) register to 0.
10. Program various register fields as shown in Table 8-4 on page 1302 to configure per-lane PHY settings.

Note In Table 8-4 on page 1302, for per-lane settings, register fields controlling all four lanes of the PHY are programmed. However, it is necessary to program only the register fields for lanes 0 and 1. Because values programmed to lanes 2 and 3 are 'don't care', you may skip programming these lanes.

11. Initiate the vendor-specific software reset by programming the VR_RST bit [15] of the `VR_PCS_DIG_CTRL1` register to 1'b1.
12. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the `VR_PCS_DIG_CTRL1` (0x38000) register.
13. Program the register fields that control lanes 0 and 1 as shown in Table 8-5 on page 1304.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the `SR_PCS_STS1` register is 1.

8.6.1.4 Programming Sequence for Enterprise 56G or 112G PHY

1. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the `VR_PCS_DIG_CTRL1` register to 1.
2. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the `VR_PCS_DIG_CTRL1` register.

3. Write 4'b0100 to bits [3:0] of the SR PCS Control2 register to select 40GBASE-R PCS.
4. Set bit [1] (EN_50G) in the VR PCS MMD Digital Control3 register to enable 50G.
5. Set bit [0] (CNS_EN) of the VR PCS MMD Digital Control3 register to enable working in the Consortium mode.
6. Program the PCS_AM_CNT field of the `VR_PCS_AM_CNT` register to 'd1024 (if RS-FEC is enabled) or 'd20479 (if RS-FEC is disabled) to configure the alignment marker interval.
7. To enable RS-FEC (528,514) mode, program bit [2] of the `SR_PMA_RS_FEC_CTRL` register to 1.
8. To enable BASE-R FEC, program bit 0 in the `SR_PMA_KR_FEC_CTRL` register.
9. Program Enterprise 56G PHY registers (through its own APB interface):
   a. Program PHY registers as per Table 8-6 on page 1306 to configure the PHY to 50G Consortium mode. For fast-simulation mode, additionally program the registers shown in Table 8-7 on page 1308.
   b. Enable the PMD by programming the pmd_en field in the `PMD_CFG[0]` (0x1400) register.
10. To program the Enterprise 112G PHY to 50G mode, program the following:
    a. Appropriate control of PHY I/Os.
    b. PHY registers by referring to the SDK app note.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the `SR_PCS_STS1` register is 1.

8.6.2 Switching to 50GBASE-KR/CR (PAM4) Mode

8.6.2.1 Programming Sequence for non-Synopsys PHY

1. Program 4'b1000 to bits [3:0] (PCS_TYP_SEL) of the `SR_PCS_CTRL2` register to select 50GBASE-R PCS.
2. Program bits [6:0] (PMA_TYPE) of the `SR_PMA_CTRL2` register to 7'b1000000 (50GBASE-KR).
3. Program 4'b0110 to bits [3:0] of the `SR_PCS_CTRL1` register to select 50Gbps speed.
4. Program the PCS_AM_CNT field of the `VR_PCS_AM_CNT` register to 'd1024 to configure the RS-FEC block interval at which code-word markers must be sent.
5. Program the AM_CNTX40_MOD66 field of the `VR_PMA_CTRL3` register to appropriate value based on the value programmed to the PCS_AM_CNT field.
6. Program the AM_CNTX40_DIV66 field of the `VR_PMA_CTRL4` register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
7. Program bit [9] (BMX_544) of the `VR_PCS_DIG_CTRL3` register to 1 to enable bit-multiplexing.
8. Program bit [10] (GAUI_544) of the `VR_PCS_DIG_CTRL3` register to 0.
9. Write 2'b00 to bits [1:0] of the `VR_PCS_DIG_CTRL3` register.

10. Program bit [2] (RSFEC_EN) of the SR_PMA_RS_FEC_CTRL register to 1.
11. Configure PHY to 50GBASE-KR/CR mode.
12. Initiate the vendor-specific software reset by programming the VR_RST bit [15] of the VR_PCS_DIG_CTRL1 register to 1.
13. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.6.2.2 Programming Sequence for Enterprise 56G or 112G PHY

1. Initiate the vendor-specific software reset by programming the VR_RST bit [15] of the VR_PCS_DIG_CTRL1 register to 1.
2. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.
3. Program 4'b1000 to bits [3:0] (PCS_TYP_SEL) of the SR_PCS_CTRL2 register to select 50GBASE-R PCS.
4. Program bits [6:0] (PMA_TYPE) of the SR_PMA_CTRL2 register to 7'b1000000 (50GBASE-KR).
5. Program 4'b0110 to bits [3:0] of the SR_PCS_CTRL1 register to select 50Gbps speed.
6. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd1024 to configure the RS-FEC block interval at which code-word markers must be sent.
7. Program the AM_CNTX40_MOD66 field of the VR_PMA_CTRL3 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
8. Program the AM_CNTX40_DIV66 field of the VR_PMA_CTRL4 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
9. Program bit [9] (BMX_544) of the VR_PCS_DIG_CTRL3 to 1 to enable bit-multiplexing.
10. Program bit [10] (GAUI_544) of the VR_PCS_DIG_CTRL3 to 0.
11. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
12. Program bit [2] (RSFEC_EN) of the SR_PMA_RS_FEC_CTRL register to 1.
13. Program Enterprise 56G PHY registers (through its own APB interface):
    a. Program PHY registers as mentioned in Table 8-6 on page 1306 to configure the PHY to 50GBASE-KR (PAM4) mode. For fast-simulation mode, additionally program the registers shown in Table 8-7 on page 1308.
    b. Enable the PMD by programming the pmd_en field in the PMD_CFG[0] (0x1400) register.
14. To program the Enterprise 112G PHY to 50G mode, program the following:
    a. Appropriate control of PHY I/Os.

b. PHY registers by referring to the SDK app note.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.6.3 Switching to 50GAUI-2 Mode

8.6.3.1 Programming Sequence for non-Synopsys PHY

1. Program 4'b1000 to bits [3:0] (PCS_TYP_SEL) of the SR_PCS_CTRL2 register to select 50GBASE-R PCS.
2. Program bits [6:0] (PMA_TYPE) of the SR_PMA_CTRL2 register to 7'b1000000 (50GBASE-KR) to select RS544 FEC, instead of RS528 FEC.
3. Optionally, program 4'b0110 to bits [3:0] (SS_5_2) of the SR_PCS_CTRL1 register to select 50Gbps speed.
4. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd1024 to configure the FEC block interval at which code-word markers must be sent.

   To speed up the simulation, program a small value such as 128.
5. Program the AM_CNTX40_MOD66 field of the VR_PMA_CTRL3 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
6. Program the AM_CNTX40_DIV66 field of the VR_PMA_CTRL4 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
7. Program bit [9] (BMX_544) of the VR_PCS_DIG_CTRL3 register to 0 to disable bit-multiplexing.
8. Program bit [10] (GAUI_544) of the VR_PCS_DIG_CTRL3 register to 1 (to indicate GAUI-2 mode).
9. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
10. Program bit [2] (RSFEC_EN) of the SR_PMA_RS_FEC_CTRL register to 1.
11. Configure PHY to 50GAUI-2 mode.
12. Initiate the vendor-specific software reset by programming the VR_RST bit [15] of the VR_PCS_DIG_CTRL1 register to 1.
13. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.6.3.2 Programming Sequence for Enterprise 56G or 112G PHY

1. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
2. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.
3. Program bits [3:0] (PCS_TYP_SEL) of the SR_PCS_CTRL2 register to 4'b1000 to select 50GBASE-R PCS.
4. Program bits [6:0] (PMA_TYPE) of the SR_PMA_CTRL2 register to 7'b1000000 (50GBASE-KR) to select RS544 FEC, instead of RS528 FEC.
5. Optionally, program bits [3:0] (SS_5_2) of the SR_PCS_CTRL1 register to 4'b0110 to select 50Gbps speed.
6. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd1024 to configure the FEC block interval at which code-word markers must be sent.

   To speed up the simulation, program a small value such as 128.
7. Program the AM_CNTX40_MOD66 field of the VR_PMA_CTRL3 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
8. Program the AM_CNTX40_DIV66 field of the VR_PMA_CTRL4 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
9. Program bit [9] (BMX_544) of the VR_PCS_DIG_CTRL3 register to 0 to disable bit-multiplexing.
10. Program bit [10] (GAUI_544) of the VR_PCS_DIG_CTRL3 register to 1 to indicate GAUI-2 mode.
11. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
12. Program bit [2] (RSFEC_EN) of the SR_PMA_RS_FEC_CTRL register to 1.
13. Program Enterprise 56G PHY registers (through its own APB interface):
    a. Program PHY registers as mentioned in Table 8-6 on page 1306 to configure the PHY to 50GAUI-2 mode. For fast-simulation mode, additionally program the registers shown in Table 8-7 on page 1308.
    b. Enable the PMD by programming the pmd_en field in the PMD_CFG[0] (0x1400) register.
14. To program the Enterprise 112G PHY to 50G mode, program the following:
    a. Appropriate control of PHY I/Os.
    b. PHY registers by referring to the SDK app note.



Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.7 Switching to 100G Speed Mode

8.7.1 Switching to 100GBASE-KR4 or 100GBASE-CR4 Mode

8.7.1.1 Programming Sequence for non-Synopsys PHY

To switch to the 100G mode Ethernet operation, complete the following steps:

1. Write 4'b0100 to bits [5:2] of the SR_PCS_Control1 register.
2. Write 4'b0101 to bits [3:0] of the SR_PCS_Control2 register to select 100GBASE-R PCS.
3. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
4. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd4096 (if RS-FEC is enabled) or 'd16383 (if RS-FEC is disabled) to configure the alignment marker interval.

   To speed up the simulation, program a smaller value to this field.
5. For configurations with PAM4 support, program these bits as follows:
   - Bit[3] (FLP) of the SR_PMA_RS_FEC_CTRL register to 0.
   - Bit[9] (BMX_544) of the VR_PCS_DIG_CTRL3 register to 0.
   - Bit[10] (GAUI_544) of the VR_PCS_DIG_CTRL3 register to 0.
6. To enable RS-FEC (528,514) mode, program bit [2] of the SR_PMA_RS_FEC_CTRL register to 1.
7. Configure the four lanes of the PHY to operate at 25.78125 Gbps rate.
8. Program the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
9. Wait for the hardware to clear the value for the VR_RST field.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.7.1.2 Programming Sequence for Multi-Protocol 32G PHY

1. Write 4'b0100 to bits [5:2] of the SR_PCS_Control1 register.
2. Write 4'b0101 to bits [3:0] of the SR_PCS_Control2 register to select 100GBASE-R PCS.
3. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
4. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd4096 (if RS-FEC is enabled) or 'd16383 (if RS-FEC is disabled) to configure the alignment marker interval.

   To speed up the simulation, program a smaller value to this field.
5. For configurations with PAM4 support, program these bits as follows:
   - Bit[3] (FLP) of the SR_PMA_RS_FEC_CTRL register to 0.
   - Bit[9] (BMX_544) of the VR_PCS_DIG_CTRL3 register to 0.
   - Bit[10] (GAUI_544) of the VR_PCS_DIG_CTRL3 register to 0.
6. To enable RS-FEC (528,514) mode, program bit [2] of the SR_PMA_RS_FEC_CTRL register to 1.

7. Configure the PHY (per-lane) to operate at 25.78125G bps rate by programming the following register fields:
   - CM_CS_0 and CM_CS_1 fields of the VR_PMA_MP_32G_CM_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
   - CM_CS_2 and CM_CS_3 fields of the VR_PMA_MP_32G_CM_CNTX_SEL1 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
   - TX_CS_0 and TX_CS_1 fields of the VR_PMA_MP_32G_TX_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
   - TX_CS_2 and TX_CS_3 fields of the VR_PMA_MP_32G_TX_CNTX_SEL1 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
   - RX_CS_0 and RX_CS_1 fields of the VR_PMA_MP_32G_RX_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
   - RX_CS_2 and RX_CS_3 fields of the VR_PMA_MP_32G_RX_CNTX_SEL0 register to the value indicated in the *context_sel_ID.txt file* for 25G mode.
8. Program the LPM bit of the SR_PCS_CTRL1 register to 1.
9. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
10. Program the following values to various bits of the SR_PCS_CTRL1 register:
    - LPM to 0
    - RST to 0
    - CS_EN to default value 0
    - SS_5_2 to 4'b0100 (value programmed in step 1)

    Do not perform read-modified-write.
11. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Good' (3'b100).

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.7.1.3 Programming Sequence for Multi-Protocol 25G PHY

1. Write 4'b0100 to bits [5:2] of the SR_PCS_Control1 register.
2. Write 4'b0101 to bits [3:0] of the SR_PCS_Control2 register to select 100GBASE-R PCS.
3. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
4. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd4096 (if RS-FEC is enabled) or 'd16383 (if RS-FEC is disabled) to configure the alignment marker interval.

   To speed up the simulation, program a smaller value to this field.
5. For configurations with PAM4 support, program these bits as follows:
   - Bit[3] (FLP) of the SR_PMA_RS_FEC_CTRL register to 0.
   - Bit[9] (BMX_544) of the VR_PCS_DIG_CTRL3 register to 0.

- Bit[10] (GAUI_544) of the VR_PCS_DIG_CTRL3 register to 0.

6. To enable RS-FEC (528,514) mode, program bit [2] of the SR_PMA_RS_FEC_CTRL register to 1.
7. Program various register fields as shown in Table 8-3 on page 1301 to configure PHY common settings.
8. Program the MPLLA_CAL_DISABLE field of the VR_PMA_SNPS_MP_25G_MPLLA_CTRL0 (0x18071) register to 1.
9. Program the MPLLB_CAL_DISABLE field of the VR_PMA_SNPS_MP_25G_MPLLB_CTRL0 (0x18074) register to 0.
10. Program various register fields as shown in Table 8-4 on page 1302 to configure per-lane PHY settings.
11. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1'b1.
12. Wait the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 (0x38000) register.
13. Program various register fields as shown in Table 8-5 on page 1304.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.7.1.4 Programming Sequence for Enterprise 56G or 112G PHY

1. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
2. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.
3. Write 4'b0100 to bits [5:2] of the SR_PCS_Control1 register.
4. Write 4'b0101 to bits [3:0] of the SR_PCS_Control2 register to select 100GBASE-R PCS.
5. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
6. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd4096 (if RS-FEC is enabled) or 'd16383 (if RS-FEC is disabled) to configure the alignment marker interval.

   To speed up the simulation, program a smaller value to this field.
7. For configurations with PAM4 support, program these bits as follows:
   - Bit[3] (FLP) of the SR_PMA_RS_FEC_CTRL register to 0.
   - Bit[9] (BMX_544) of the VR_PCS_DIG_CTRL3 register to 0.
   - Bit[10] (GAUI_544) of the VR_PCS_DIG_CTRL3 register to 0.
8. To enable RS-FEC (528,514) mode, program bit [2] of the SR_PMA_RS_FEC_CTRL register to 1.
9. Program Enterprise 56G PHY registers (through its own APB interface):

a. Program PHY registers as mentioned in Table 8-6 on page 1306 to configure the PHY to 100BBASE-KR4 mode (4 lanes of NRZ at 25.78125Gbps). For fast-simulation mode, additionally program the registers shown in Table 8-7 on page 1308.

b. Enable the PMD by programming the pmd_en field in the `PMD_CFG[0]` (0x1400) register.

10. To program the Enterprise 112G PHY to 100GBASE-KR4 or 100GBASE-CR4 mode, program the following:

a. Appropriate control of PHY I/Os.

b. PHY registers by referring to the SDK app note.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the `SR_PCS_STS1` register is 1.

8.7.2 Switching to 100GBASE-KR2 or 100GBASE-CR2 (PAM4) Mode

8.7.2.1 Programming Sequence for non-Synopsys PHY

1. Program 4'b0101 to bits [3:0] (PCS_TYP_SEL) of the `SR_PCS_Control2` register to select 100GBASE-R PCS.
2. Program bits [6:0] (PMA_TYPE) of the `SR_PMA_Control2` register to 7'b1001000 (100GBASE-KR2) to select RS544 FEC, instead of RS528 FEC.
3. Optionally, program 4'b0100 to bits [3:0] (SS_5_2) of the `SR_PCS_CTRL1` register to select 100Gbps speed.
4. Program the PCS_AM_CNT field of the `VR_PCS_AM_CNT` register to 'd4096 to configure the FEC block interval at which code-word markers must be sent.
5. Program the AM_CNTX40_MOD66 field of the `VR_PMA_CTRL3` register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
6. Program the AM_CNTX40_DIV66 field of the `VR_PMA_CTRL4` register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
7. Program bit [3] (FLP) of the `SR_PMA_RS_FEC_CTRL` register to 1.
8. Program bit [9] (BMX_544) of the `VR_PCS_DIG_CTRL3` to 1 to enable bit-multiplexing.
9. Program bit [10] (GAUI_544) of the `VR_PCS_DIG_CTRL3` register to 0.
10. Write 2'b00 to bits [1:0] of the `VR_PCS_DIG_CTRL3` register.
11. Program bit [2] (RSFEC_EN) of the `SR_PMA_RS_FEC_CTRL` register to 1.
12. Program the PHY to 100GBASE-KR2 or 100GBASE-CR2 mode.
13. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the `VR_PCS_DIG_CTRL1` register to 1.

14. Wait the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.



Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.7.2.2 Programming Sequence for Enterprise 56G or 112G PHY

1. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
2. Wait the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.
3. Program 4'b0101 to bits [3:0] (PCS_TYP_SEL) of the SR_PCS_CTRL2 register to select 100GBASE-R PCS.
4. Program bits [6:0] (PMA_TYPE) of the SR_PMA_CTRL2 register to 7'b1001000 (100GBASE-KR2) to select RS544 FEC, instead of RS528 FEC.
5. Optionally, program 4'b0100 to bits [3:0] (SS_5_2) of the SR_PCS_CTRL1 register to select 100Gbps speed.
6. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd4096 to configure the FEC block interval at which code-word markers must be sent.
7. Program the AM_CNTX40_MOD66 field of the VR_PMA_CTRL3 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
8. Program the AM_CNTX40_DIV66 field of the VR_PMA_CTRL4 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
9. Program bit [3] (FLP) of the SR_PMA_RS_FEC_CTRL register to 1.
10. Program bit [9] (BMX_544) of the VR_PCS_DIG_CTRL3 register to 1 to enable bit-muxing.
11. Program bit [10] (GAUI_544) of the VR_PCS_DIG_CTRL3 register to 0.
12. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
13. Program bit [2] (RSFEC_EN) of the SR_PMA_RS_FEC_CTRL register to 1.
14. Program Enterprise 56G PHY registers (through its own APB interface):
    a. Program PHY registers as mentioned in Table 8-6 on page 1306 to configure the PHY to 100GBASE-KR2 (PAM4) mode. For fast-simulation mode, additionally program the registers shown in the Table 8-7 on page 1308.
    b. Enable the PMD by programming the pmd_en field in the PMD_CFG[0] (0x1400) register.
15. To program the Enterprise 112G PHY to 100GBASE-KR2 or 100GBASE-CR2 mode, program the following:
    a. Appropriate control of PHY I/Os.

b. PHY registers by referring to the SDK app note.

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.7.3 Switching to 100GAUI-4 Mode

8.7.3.1 Programming Sequence for Non-Synopsys PHY

1. Program 4'b0101 to bits [3:0] (PCS_TYP_SEL) of the SR_PCS_CTRL2 register to select 100GBASE-R PCS.
2. Program bits [6:0] (PMA_TYPE) of the SR_PMA_CTRL2 register to 7'b1001000 (100GBASE-KR2) to select RS544 FEC, instead of RS528 FEC.
3. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
4. Optionally, program 4'b0100 to bits [3:0] (SS_5_2) of the SR_PCS_CTRL1 register to select 100 Gbps speed.
5. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd4096 to configure the FEC block interval at which code-word markers must be sent.
6. Program the AM_CNTX40_MOD66 field of the VR_PMA_CTRL3 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
7. Program the AM_CNTX40_DIV66 field of the VR_PMA_CTRL4 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
8. Program bit [3] (FLP) of the SR_PMA_RS_FEC_CTRL register to 1.
9. Program bit [9] (BMX_544) of the VR_PCS_DIG_CTRL3 register to 0 to disable bit-multiplexing.
10. Program bit [10] (GAUI_544) of the VR_PCS_DIG_CTRL3 register to 1 to indicate GAUI-4 mode.
11. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
12. Program bit [2] (RSFEC_EN) of the SR_PMA_RS_FEC_CTRL register to 1.
13. Configure the PHY to 100GAUI-4 mode.
14. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
15. Wait the hardware to clear the value for the VR_RST field (bit [15]).

Note Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.7.3.2 Programming Sequence for Enterprise 56G or 112G PHY

1. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register to 1.
2. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.
3. Program 4'b0101 to bits [3:0] (PCS_TYP_SEL) of the SR_PCS_CTRL2 register to select 100GBASE-R PCS.
4. Program bits [6:0] (PMA_TYPE) of the SR_PMA_CTRL2 register to 7'b1001000 (100GBASE-KR2) to select RS544 FEC, instead of RS528 FEC.
5. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
6. Optionally, program 4'b0100 to bits [3:0] (SS_5_2) of the SR_PCS_CTRL1 register to select 100Gbps speed.
7. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd4096 to configure the FEC block interval at which code-word markers must be sent.
8. Program the AM_CNTX40_MOD66 field of the VR_PMA_CTRL3 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
9. Program the AM_CNTX40_DIV66 field of the VR_PMA_CTRL4 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
10. Program bit [3] (FLP) of the SR_PMA_RS_FEC_CTRL register to 1.
11. Program bit[9] (BMX_544) of the VR_PCS_DIG_CTRL3 register to 0 to disable bit-muxing.
12. Program bit[10] (GAUI_544) of the VR_PCS_DIG_CTRL3 register to 1 to indicate GAUI-4 mode.
13. Write 2'b00 to bits [1:0] of the VR_PCS_DIG_CTRL3 register.
14. Program bit [2] (RSFEC_EN) of the SR_PMA_RS_FEC_CTRL register to 1.
15. Program Enterprise 56G PHY registers (through its own APB interface):
    a. Program PHY registers as mentioned in Table 8-6 on page 1306 to configure the PHY to 100GAUI-4 mode. For fast-simulation mode, additionally program the registers shown in Table 8-7 on page 1308.
    b. Enable the PMD by programming the pmd_en field in the PMD_CFG[0] (0x1400) register.
16. To program the Enterprise 112G PHY to 100GAUI-4 mode, program the following:
    a. Appropriate control of PHY I/Os.
    b. PHY registers by referring to the SDK app note.



Before the start of the packet transfer, verify that the value of the RLU bit of the SR_PCS_STS1 register is 1.

8.7.4 Switching to 100GBASE-KR1 or 100GBASE-CR1 (PAM4) Mode

8.7.4.1 Programming Sequence for Non-Synopsys PHY

To configure DWC_xlgpcs to 100G one-lane mode, execute the following steps:

1. Program the PCS_TYP_SEL field (bits [3:0]) of the `SR_PCS_CTRL2` register to 4'b0101 to select 100GBASE-R PCS.
2. Program the PMA_TYP_SEL field (bits [6:0]) of the `SR_PMA_CTRL2` register to 7'b1000111 (100GBASE-KR1).
3. Optionally, program the SS_5_2 field (bits [3:0]) of the `SR_PCS_CTRL1` register to 4'b0100 to select 100Gbps speed.
4. Program the PCS_AM_CNT field of the `VR_PCS_AM_CNT` register to 'd4096 to configure the FEC block interval at which code-word markers are sent.
5. Program the AM_CNTX40_MOD66 field of the `VR_PMA_CTRL3` register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
6. Program the AM_CNTX40_DIV66 field of the `VR_PMA_CTRL4` register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
7. Program the FLP field (bit [3]) of the `SR_PMA_RS_FEC_CTRL` register to 1.
8. Program the BMX_544 field (bit [9]) of the `VR_PCS_DIG_CTRL3` register to 1 to enable bit-multiplexing.
9. If the configuration is multi-lane, program the GAUI_544 field (bit [10]) of the `VR_PCS_DIG_CTRL3` register to 0.
10. Program bits [1:0] of the `VR_PCS_DIG_CTRL3` register to 2'b00.
11. Program bit[5] or bit[6] of the `SR_PMA_RS_FEC_CTRL` register to 1, as follows:
    - Bit[5] for interleaved RS-FEC mode
    - Bit[6] for non-interleaved RS-FEC mode
12. Program the RSFEC_EN field (bit [2]) of the `SR_PMA_RS_FEC_CTRL` register to 1.
13. Program the PHY to 100GBASE-KR1 or 100GBASE-CR1 mode.
14. Initiate the vendor-specific software reset by programming the VR_RST field (bit [15]) of the `VR_PCS_DIG_CTRL1` register to 1.
15. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the `VR_PCS_DIG_CTRL1` register.

8.7.4.2 Programming Sequence for Synopsys Enterprise 112G PHY

1. Program the PCS_TYP_SEL field (bits [3:0]) of the `SR_PCS_CTRL2` register to 4'b0101 to select 100GBASE-R PCS.
2. Program the PMA_TYP_SEL field (bits [6:0]) of the `SR_PMA_CTRL2` register to 7'b1000111 (100GBASE-KR1).

3. Optionally, program the SS_5_2 field (bits [3:0]) of the SR_PCS_CTRL1 register to 4'b0100 to select 100Gbps speed.
4. Program the PCS_AM_CNT field of the VR_PCS_AM_CNT register to 'd4096 to configure the FEC block interval at which code-word markers are sent.
5. Program the AM_CNTX40_MOD66 field of the VR_PMA_CTRL3 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
6. Program the AM_CNTX40_DIV66 field of the VR_PMA_CTRL4 register to an appropriate value based on the definition of this field in the *"Register Descriptions"* chapter and the value programmed to the PCS_AM_CNT field.
7. Program the FLP field (bit [3]) of the SR_PMA_RS_FEC_CTRL register to 1.
8. Program the BMX_544 field (bit [9]) of the VR_PCS_DIG_CTRL3 register to 1 to enable bit-multiplexing.
9. If the configuration is multi-lane, program the GAUI_544 field (bit [10]) of the VR_PCS_DIG_CTRL3 register to 0.
10. Program bits [1:0] of the VR_PCS_DIG_CTRL3 register to 2'b00.
11. Program bit[5] or bit[6] of the SR_PMA_RS_FEC_CTRL register to 1, as follows:
    - Bit[5] for interleaved RS-FEC mode
    - Bit[6] for non-interleaved RS-FEC mode
12. Program the RSFEC_EN field (bit [2]) of the SR_PMA_RS_FEC_CTRL register to 1.
13. Program the Enterprise 112G PHY to 100GBASE-KR1 or 100GBASE-CR1 mode by referring to the following:
    - The SDK app note for PHY configuration.

      For more details, contact the PHY team.
    - The PHY Databook guidelines for PHY I/Os.

8.8 Switching to 5G Speed Mode

8.8.1 Programming Sequence for Multi-Protocol 32G PHY

To switch to the 5G mode Ethernet operation, complete the following steps:

1. Program bit [0] and bit [1] of the VR_PCS_DIG_CTRL3 register to 0.
2. Write 4'b1000 to bits [5:2] of the SR_PCS_CTRL1 register.
3. Write 4'b1111 to bits [2:0] of the SR_PCS_CTRL2 register.
4. Configure the PHY to operate at 5.15625 Gbps rate by programming the following register fields:
    - The CM_CS_0 field of the VR_PMA_MP_32G_CM_CNTX_SEL0 register to the value indicated for 5G mode in the *context_sel_ID.txt* file.

- The TX_CS_0 field of the VR_PMA_MP_32G_TX_CNTX_SEL0 register to the value indicated for 5G mode in the *context_sel_ID.txt* file.
- The RX_CS_0 field of the VR_PMA_MP_32G_RX_CNTX_SEL0 register to the value indicated for 5G mode in the *context_sel_ID.txt* file.
- The L0_LN field of the VR_PMA_MP_32G_CNTX_LINK_NUM_CTRL register to a unique value for each instance of the IP when operating in bifurcation mode.

5. Program the LPM bit of the SR_PCS_CTRL1 register to 1.
6. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
7. Program the bits of the SR_PCS_CTRL register as follows:
   - The LPM bit back to 0.
   - Other bits to values which were present prior to executing step 6.

Note Do not perform read-modified-write.

8. Wait for the PSEQ_STATE field of the VR_PCS_DIG_STS register to indicate 'Power Good' (3'b100).

8.8.2 Programming Sequence for Non-Synopsys PHY or Multi-Protocol 12G, 16G, or 25G PHY

To switch to the 5G mode Ethernet operation, complete the following steps:

1. Program bit [0] and bit [1] of the VR_PCS_DIG_CTRL3 register to 0.
2. Write 4'b1000 to bits [5:2] of the SR_PCS_CTRL1 register.
3. Write 4'b1111 to bits [2:0] of the SR_PCS_CTRL2 register.
4. For Multi-protocol 25G PHY configurations, program the following:
   - The MPLLA_CAL_DISABLE field of the VR_PMA_MP_25G_MPLLA_CTRL0 register to 0.
   - The MPLLB_CAL_DISABLE field of the VR_PMA_MP_25G_MPLLB_CTRL0 register to 1.
   - Various registers as per Table 8-1 on page 1287, with the exception of TX0_RATE of 1 and RX0_RATE of 2.
5. For multi-protocol 16G PHY configurations, program various registers as per Table 8-8.
6. For multi-protocol 12G PHY configurations, program various registers as per Table 8-9 on page 1327.
7. Program the VR_RST field (bit[15]) of the VR_PCS_DIG_CTRL1 register to 1.
8. Wait for the hardware to clear the value for the VR_RST field (bit [15]) of the VR_PCS_DIG_CTRL1 register.

Table 8-8 5G Ethernet for Multi-Protocol 16G PHY Register Settings

Field	Values
MPLLA_MULTIPLIER	Set the values of this field in the VR_PMA_MP_16G_MPLLA_CTRL0 register to 8'd33.
MPLLA_BW	Set the values of this field in the VR_PMA_MP_16G_MPLLA_CTRL3 register to 16'd41022. **Note**: To ensure that these values are accurate, see the latest PHY data book.
VCO_LD_VAL_0	Set the value of this field in the VR_PMA_MP_25G_16G_VCO_CAL_LD0 register as follows: ■ For 78.125 MHz reference clock – 13'd1386 ■ For 156.25 or 312.5 MHz reference clock – 13'd1353
VCO_REF_LD_0	Set the value of these fields in the VR_PMA_MP_25G_16G_VCO_CAL_REF0 register as follows: ■ For 78.125 MHz reference clock – 6'd21 ■ For 156.25 or 312.5 MHz reference clock – 6'd41
CONT_ADAPT_0	Set the value of these fields in the VR_PMA_MP_25G_16G_RX_EQ_CTRL4 register to 1.
TX0_RATE	Set the value of this field in the VR_PMA_MP_25G_16G_TX_RATE_CTRL register to 3'b001.
RX0_RATE	Set the value of this field in the VR_PMA_MP_16G_RX_RATE_CTRL register to 2'b01.
TX0_WIDTH	Set the value of this field in the VR_PMA_MP_16G_TX_WIDTH_CTRL register as follows: ■ For 20-bit gearbox interface – 2'b11 ■ For 16-bit gearbox interface – 2'b10
RX0_WIDTH	Set the value of this field in the VR_PMA_MP_16G_RX_WIDTH_CTRL register as follows: ■ For 20-bit gearbox interface – 2'b11 ■ For 16-bit gearbox interface – 2'b10
MPLLA_DIV16P5_CLK_EN	Set the value of this field in the VR_PMA_MP_16G_MPLLA_CTRL2 register to 1.
MPLLA_DIV10_CLK_EN	Set the value of this field in the VR_PMA_MP_16G_MPLLA_CTRL2 register as follows: ■ For 20-bit gearbox interface – 1 ■ For 16-bit gearbox interface – 0
MPLLA_DIV8_CLK_EN	Set the value of this field in the VR_PMA_MP_16G_MPLLA_CTRL2 register as follows: ■ For 20-bit gearbox interface – 0 ■ For 16-bit gearbox interface – 1
VBOOST_EN_0	Set the value of this field in the VR_ PMA_MP_25G_16G_TX_GENCTRL1 register to 1.

Table 8-8 (Continued)5G Ethernet for Multi-Protocol 16G PHY Register Settings

Field	Values
CTLE_BOOST_0	Set the values of this field in the VR_ PMA_MP_25G_16G_RX_EQ_CTRL0 registers to 5'd10.
VCO_STEP_CTRL_0	Set the values of this field in the VR_ PMA_MP_16G_RX_CDR_CTRL register to 1.
VCO_TEMP_COMP_EN_0	Set the values of this field in the VR_ PMA_MP_16G_RX_CDR_CTRL register to 1.
VCO_FRQBAND_0	Set the value of this field in the VR_ PMA_MP_16G_RX_CDR_CTRL register to 1.
RX0_MISC	Set the value of this field in the VR_ PMA_MP_25G_16G_RX_MISC_CTRL0 register to 2.
RX_DFE_BYP_0	Set the value of this field in the VR_ PMA_MP_25G_RX_GENCTRL4 register to 0.
RX0_DELTA_IQ	Set the value of this field in the VR_ PMA_MP_25G_16G_RX_IQ_CTRL0 register to 3.
RX0_ADPT_MODE	Set the value of this field in the VR_ PMA_MP_25G_16G_ADPT_CTRL0 register to 3.

Table 8-9 5G Ethernet for Multi-Protocol 12G PHY Register Settings

Field	Values
MPLLA_MULTIPLIER	Set the values of this field in the VR_PMA_MP_16G_MPLLA_CTRL0 register to 8'd33.
MPLLA_BW	Set the values of this field in the VR_PMA_MP_16G_MPLLA_CTRL3 register based on the PHY model used: ■ For GF 14 LPP – 11'd1936 ■ For TSMC 16 FFLL – 11'd65 ■ For TSMC 28HPC – 11'd4 **Note**: To ensure that these values are accurate, see the latest PHY data book.
VCO_LD_VAL_0	Set the value of this field in the VR_PMA_MP_25G_16G_VCO_CAL_LD0 register as follows: ■ For 78.125 MHz reference clock – 13'd1386 ■ For 156.25 or 312.5 MHz reference clock – 13'd1353
VCO_REF_LD_0	Set the value of these fields in the VR_PMA_MP_25G_16G_VCO_CAL_REF0 register as follows: ■ For 78.125 MHz reference clock – 6'd21 ■ For 156.25 or 312.5 MHz reference clock – 6'd41
AFE_EN_0, DFE_EN_0	Set the value of these fields in the VR_PMA_MP_12G_RX_GENCTRL5 register to 1.
CONT_ADAPT_0	Set the value of these fields in the VR_PMA_MP_25G_16G_RX_EQ_CTRL4 register to 1.
TX0_RATE	Set the value of this field in the VR_PMA_MP_25G_16G_TX_RATE_CTRL register to 3'b001.

Table 8-9 5G Ethernet for Multi-Protocol 12G PHY Register Settings

Field	Values
RX0_RATE	Set the value of this field in the VR_PMA_MP_16G_RX_RATE_CTRL register to 2'b01.
TX0_WIDTH	Set the value of this field in the VR_PMA_MP_16G_TX_WIDTH_CTRL register as follows: ■ For 20-bit gearbox interface – 2'b11 ■ For 16-bit gearbox interface – 2'b10
RX0_WIDTH	Set the value of this field in the VR_PMA_MP_16G_RX_WIDTH_CTRL register as follows: ■ For 20-bit gearbox interface – 2'b11 ■ For 16-bit gearbox interface – 2'b10
MPLLA_DIV16P5_CLK_EN	Set the value of this field in the VR_PMA_MP_16G_MPLLA_CTRL2 register to 1.
MPLLA_DIV10_CLK_EN	Set the value of this field in the VR_PMA_MP_16G_MPLLA_CTRL2 register as follows: ■ For 20-bit gearbox interface – 1 ■ For 16-bit gearbox interface – 0
MPLLA_DIV8_CLK_EN	Set the value of this field in the VR_PMA_MP_16G_MPLLA_CTRL2 register as follows: ■ For 20-bit gearbox interface – 0 ■ For 16-bit gearbox interface – 1

8.8.3 Programming Sequence for Enterprise 112G PHY

To switch to the 5G mode Ethernet operation, complete the following steps:

1. Program the VR_RST field of the VR_PCS_DIG_CTRL1 register to 1 to initiate vendor soft-reset.
2. Wait for the hardware to clear the value for the VR_RST field.
3. Program bit [0] and bit [1] of the VR_PCS_DIG_CTRL3 register to 0.
4. Write 4'b1000 to bits [5:2] of the SR_PCS_CTRL1 register.
5. Write 4'b1111 to bits [2:0] of the SR_PCS_CTRL2 register.
6. To program the Enterprise 112G PHY to 5G mode, program the following:
   a. Appropriate control of PHY I/Os.
   b. PHY registers by referring to the SDK app note.

8.9 Programming Guidelines for CR Port Read or Write Operation

The following lists provide the programming steps for a read or write operation on the CR port:

8.9.1 Write Operation

To write any register in Synopsys PHY, complete the following:

1. Read the VR_PMA_MP_25G_16G_CR_CTRL register and wait until bit 0 of this register reads as 0.
2. Write the Synopsys PHY register address to the VR_PMA_MP_25G_16G_CR_ADDR register. These addresses are defined in the *Synopsys PHY Databooks*.
3. Write the data to be written to the VR_PMA_MP_25G_16G_CR_DATA register.
4. Write 1 to bit 1 of the VR_PMA_MP_25G_16G_CR_CTRL register.
5. Write 1 to bit 0 of the VR_PMA_MP_25G_16G_CR_CTRL register.
6. Read the VR_PMA_MP_25G_16G_CR_CTRL register and wait until bit 0 of this register reads as 0.

8.9.2 Read Operation

To read any register in Synopsys PHY, complete the following:

1. Read the VR_PMA_MP_25G_16G_CR_CTRL register and wait until bit 0 of this register reads as 0.
2. Write the Synopsys PHY register address to the VR_PMA_MP_25G_16G_CR_ADDR register. These addresses are defined in the *Synopsys PHY Data Books*.
3. Write 0 to bit 1 of the VR_PMA_MP_25G_16G_CR_CTRL register.
4. Write 1 to bit 0 of the VR_PMA_MP_25G_16G_CR_CTRL register.
5. Read the VR_PMA_MP_25G_16G_CR_CTRL register and wait until bit 0 of this register reads as 0.
6. Read the VR_PMA_MP_25G_16G_CR_DATA register, which contains the Read Data.

8.10 Programming Guidelines for Clause 72 Start-Up Protocol

The Clause 72 start-up protocol must be initiated when all pages are exchanged during Clause 73 auto-negotiation and when the auto-negotiation process is waiting for link status to be UP for linkfail_inhibit_timer duration after exchanging all the pages. For information about programming guidelines for Clause 73 auto-negotiation, see "Programming Guidelines for Clause 73 Auto-Negotiation" on page 1333.

The following list explains the start-up protocol process:

1. After power on reset, soft reset, or speed switching, the DWC_xlgpcs initiates the Clause 73 Auto-negotiation by default. The host can also restart the Clause 73 auto-negotiation at any time by programming the SR_AN_CTRL field (bit [9]) of the SR AN MMD Control register.
2. After Clause 73 auto-negotiation is initiated, the host must wait until all the pages are exchanged with the link partner. For more information, see Figure 8-2 on page 1335.
3. For speed mode 25G and 50G, link training based on Clause 72 or Clause 92 or 93 is available. To perform link training using Clause 72, the host must program bit 2 of the VR_PCS_DIG_CTRL3 register. Otherwise, if Clause 92 or Clause 93 is to be used, the host must write 0 into bit 2 of the VR_PCS_DIG_CTRL3 register. When Clause 92- or 93-based link training is selected, the Polynomial identifier 0 for the PRBS pattern is specified using bit [12:11] in the register and the initial seed is programmed in the same register 1.450 using bits [10:0].
4. The host must enable the Clause 72 start-up protocol by programming bit 1 of the SR PMA MMD BASE-KR PMD Control register SR_PMA_KR_PMD_CTRL.

5. The host must restart the Clause 72 start-up protocol by programming bit 0 of the SR PMA MMD BASE-KR PMD Control register SR_PMA_KR_PMD_CTRL.

   At this stage, the host can do any of the following optional steps:

   - Get the status of the start-up protocol (frame lock, local receiver ready, start-up protocol in progress, training failure) by reading the SR PMA MMD BASE-KR PMD Control register SR_PMA_KR_PMD_CTRL.
   - Get the latest received coefficient update or status information from the link partner by reading the following registers:
     - SR PMA MMD 10GBASE-KR LP Coefficient Update register SR_PMA_KR_LP_CEU
     - SR PMA MMD 10GBASE-KR LP Coefficient Status register SR_PMA_KR_LP_CESTS
   - Monitor the coefficient update or status information sent from the local device to the link partner by reading the following registers:
     - SR PMA MMD 10GBASE-KR LD Coefficient Update SR_PMA_KR_LP_CEU
     - SR PMA MMD 10GBASE-KR LD Coefficient Status SR_PMA_KR_LP_CESTS

6. The DWC_xlgpcs performs the start-up KR training as described in "Start-Up Protocol Flow" on page 134. If the training is successfully complete (when the local and the far-end receivers give the Receiver Ready status), the Clause 73 auto-negotiation is resumed (to look for link-okay). The KR training is successful when the DWC_xlgpcs gets the auto-negotiation complete interrupt as indicated by bit 0 of the VR_AN_INTR register.

   If the KR training is unsuccessful (500 msec time out), the link-fault is declared and the Clause 73 AN is automatically restarted.

7. Clear the auto-negotiation complete interrupt bit 0 of VR AN MMD Digital Control1 Register VR_AN_VR_DIG_CTRL.

If the Clause 72 start-up protocol is disabled (bit[1] of the SR PMA MMD BASE-KR PMD Control register SR_PMA_KR_PMD_CTRL is cleared), the host can directly tune the local transmitter filter coefficient. The host can do this by writing to the SR_PMA_KR_LP_CEU register. The host can monitor the update status by reading the SR_PMA_KR_LP_CESTS register before subsequent updates. The following list explains the flow for c(-1) filter which is also applicable for other filters:

1. Read bits[1:0] of the SR PMA MMD 10GBASE-KR LP Coefficient Status register SR_PMA_KR_LP_CESTS and wait until it reads as 2'b00 (NOT_UPDATED status).



This flow is applicable only for increment and decrement requests from the host. For preset and initialize requests, the host must wait until all the Transmit filter status indicates as NOT_UPDATED before issuing the preset or initialize request.

2. Write 2'b01 (to increment) or 2'b10 (to decrement) to bit[1:0] of the SR_PMA_KR_LP_CEU register.
3. Read bits[1:0] of the SR_PMA_KR_LP_CESTS register and wait until it reads as 2'b11 (max), 2'b10 (min) or 2'b01 (updated) as a status.
4. Repeat step 1 through step 3 to further tune the local transmit filters.

DWC_xlgpcs allows software control of Transmit and Receive equalization in which the host can directly program the filter coefficient control and status values for the local device and link partner device.

In Receive equalization, DWC_xlgpcs allows the host to tune the transmit filter coefficients of the link-partner during KR training. Enable this mode by programming bit[15] of VR PMA MMD KRTR Receive equalization Control register. The sequence of steps is as follows:

1. Configure the DUT to the specified speed mode
2. If AN is selected in the configuration, disable AN by programming SR_AN_CTRL(0x70000) bit 12 to 0.
3. Enable Receive S/W equalization flow for each lane by programming `VR_PMA_KRTR_RX_EQ_CFF_CTRL0-3` (0x180BC-0x180BF) bit 15 to 1.
4. Select CL72 or CL92/93 for KRT by programming `VR_PCS_DIG_CTRL3`(0x38003) bit 2 to 0 for CL72 or 1 for CL92/93.
5. If CL92/93 is selected, one can chose polynomial for PRBS pattern ID for each lane by programming bit[12:11] in `SR_PMA_LNK_TRN_PAT`, `SR_PMA_LNK_TRN_PAT_LN1-3`(0x105aa-0x105ad) and initial seed in bit [10:0].
6. Enable link training by programming `SR_PMA_KR_PMD_CTRL`(0x10096) bit 1.
7. Issue restart KRT by programming bit 0 of the `SR_PMA_KR_PMD_CTRL`(0x10096) register.
8. Read the `SR_PMA_KR_LP_CESTS` register and wait until status of 2'b00 (NOT_UPDATED status) for all the three coefficients is obtained.
9. Wait for KRT frames to be locked. Checking `SR_PMA_KR_PMD_STS`(0x10097) bit1,5,9,13 for lane0-3 frame lock status.
10. S/W can send a request to remote end by programming the `VR_PMA_KRTR_RX_EQ_CFF_CTRL(i)` (0x180BC+i) register.

    Issue a coefficient update request to the link-partner by programming any of the fields (PRST, INIT, CFF_UPDT1, CFF_UPDT0, CFF_UPDTM1) of For example, Write 2'b01 to CFF_UPDT0 field for DWC_xlgpcs to transmit a INC request for C(0) coefficient to the link-partner.
11. After local PHY sends a coefficient update request to remote device, the sent coefficient update request is stored in the `VR_PMA_KRTR_RX_EQ_CFF(i)`(0x180C4+i) register.
12. The most recently sent request from DUT is also logged in the `SR_PMA_KR_LD_CEU_LANE(i)` register.
13. After the remote end sends back the status, the most recently received status is stored in the `SR_PMA_KR_LP_CESTS_LANE(i)` register.
14. S/W can send the status to local phy by programming `VR_PMA_KRTR_RX_EQ_STS_CTRL(i)`(180C0+i). This status could be same as the status received from the received frame from the link partner or any other valid status.
15. Issue a 'HOLD' request to the link-partner by writing 2'b00 to the same field that was programmed in step 10.
16. Read the `SR_PMA_KR_LP_CESTS` register and wait until status of 2'b00 (NOT_UPDATED status) for the field for which HOLD was issued. The three coefficients is obtained wait until it indicates NOT_UPDATED for that coefficient.

17. Repeat steps 10-16 for further tuning of the same or other filter coefficients
18. In addition, software can program the 'RR_RDY' field of the VR_PMA_KRTR_RX_EQ_CFF_CTRL(i) (0x180BC-0x180BF) register to 1, once it decides to send no more coefficient update commands to the link-partner. During this mode, DWC_xlgpcs ignores the value of the 'rpcs_rx_trained_i' port of DWC_xlgpcs.

In Transmit equalization, DWC_xlgpcs allows the host to tune the transmit filter coefficients of the local-device during KR training. Enable this mode by programming bit[15] of VR PMA MMD KRTR Receive equalization Control register.

Following these steps:

1. Configure the DUT to the specified speed mode.
2. If AN is selected in the configuration, disable AN by programming SR_AN_CTRL(0x70000) bit 12 as 0.
3. Enable Transmit S/W equalization flow for each lane by programming VR_PMA_KRTR_TX_EQ_STS_CTRL0-3 (0x180B4-0x180B7) bit 15 as 1.
4. Select CL72 or CL92/93 for KRT by programming VR_PCS_DIG_CTRL3(0x38003) bit 2 as 0 for CL72 or 1 for CL92/93.
5. If CL92/93 is selected, one can chose polynomial for PRBS pattern ID for each lane by programming bit[12:11] in SR_PMA_LNK_TRN_PAT, SR_PMA_LNK_TRN_PAT_LN1-3(0x105aa-0x105ad) and initial seed in bit [10:0]
6. Enable KRT by programming SR_PMA_KR_PMD_CTRL(0x10096) bit 1 and reset KRT in bit 0.
7. Wait for KRT frames to be locked. Checking SR_PMA_KR_PMD_STS(0x10097) bit1,5,9,13 for lane0-3 frame lock status.
8. The most recently received request is stored in SR_PMA_MLN_KR_LP_CEU_LANE (i) for each lane.
9. S/W can send a request to local PHY from DUT by writing to VR_PMA_KRTR_TX_EQ_CFF_CTRL(i)(0x180B0+i) for each lane.
10. The status response from PHY for the coefficient update command is stored in VR_PMA_KRTR_TX_EQ_STS(i)(0x180B8+i).
11. S/W can control the status to be sent in the transmit frame. The desired status to be sent is written in VR_PMA_KRTR_TX_EQ_STS_CTRL(i)(0x180B4+i) for each lane.
12. The most recently sent response status by DUT is kept in SR_PMA_KR_LD_CESTS_LANE(i).
13. Before issuing the next coefficient update command, the software need to poll for HOLD command from the link partner in SR_PMA_MLN_KR_LP_CEU_LANE (i).

This section describes Clause 72 Training specific to 10G BASE-KR, similar steps are applicable for 40G BASE-KR (where there are similar registers for each of the 4 lanes).

8.11 Programming Guidelines for Clause 73 Auto-Negotiation

This section describes the hardware-software interaction required for the Clause 73 auto-negotiation process.

8.11.1 Definitions

Base Page: First Page exchange between local device and link partner after the auto-negotiation started or restarted

Next Page: The pages that are exchanged between the local device and the link partner after the Base page

8.11.2 Clause 73 Auto-Negotiation Process

The clause 73 auto-negotiation consists of the following:

1. Auto-negotiation initiation

   The DWC_xlgpcs automatically starts auto-negotiation when you power on, auto negotiation soft reset (bit-15 in the SR_AN_CTRL register), or power cycle (that is, power down and then power up) the DWC_xlgpcs, provided bit [12] of the SR_AN_CTRL register is 1. The host can restart the auto-negotiation by programming bit 9 of the SR_AN_CTRL register to 1.

   The host can disable the auto-negotiation by clearing bit 12 of the SR_AN_CTRL register. When auto-negotiation is disabled, link speed is determined by the speed at which DWC_xlgpcs was powered up.

- The host can change the default values of the Base Page register by writing to the following registers before starting or restarting the Clause 73 auto-negotiation, in the following order:
  - Write to SR_AN_ADV3.
  - Write to SR_AN_ADV2.
  - Write to SR_AN_ADV1.
- If required, the Next page registers must be written in the following order:
  - Write to SR_AN_XNP_TX3.
  - Write to SR_AN_XNP_TX2.
  - Write to SR_AN_XNP_TX1.

This sequence is required because the DWC_xlgpcs considers that the 48-bit Base page and Next Page is ready for transmission once the host writes into SR AN Advertisement Register 1 or SR AN XNP Transmit Register 1, respectively.

2. Speed mode dependency

   DWC_xlgpcs supports only one of the configured speed modes at a time. If DWC_xlgpcs is configured to have multiple speed modes (5G BASE-KR, 10G BASE-CR, 10G BASE-KR, 25G BASE-CR, 25G BASE-KR, 40G BASE-CR4, 40G BASE-KR4, 50G BASE-CR2, 50G BASE-KR2, 50G BASE-CR, 50G BASE-KR, 100G BASE-CR4, 100G BASE-KR4, 100G BASE-CR, and 100G BASE-KR), speed switching is implemented as explained in Sections 8.2 to Sections 8.8 to switch to 10G, 40G, 25G, 50G, 100G, or 5G mode, respectively.

8.11.2.1 Auto-Negotiation Flow for Configurations with DWCXLP_CL73_ALT Disabled

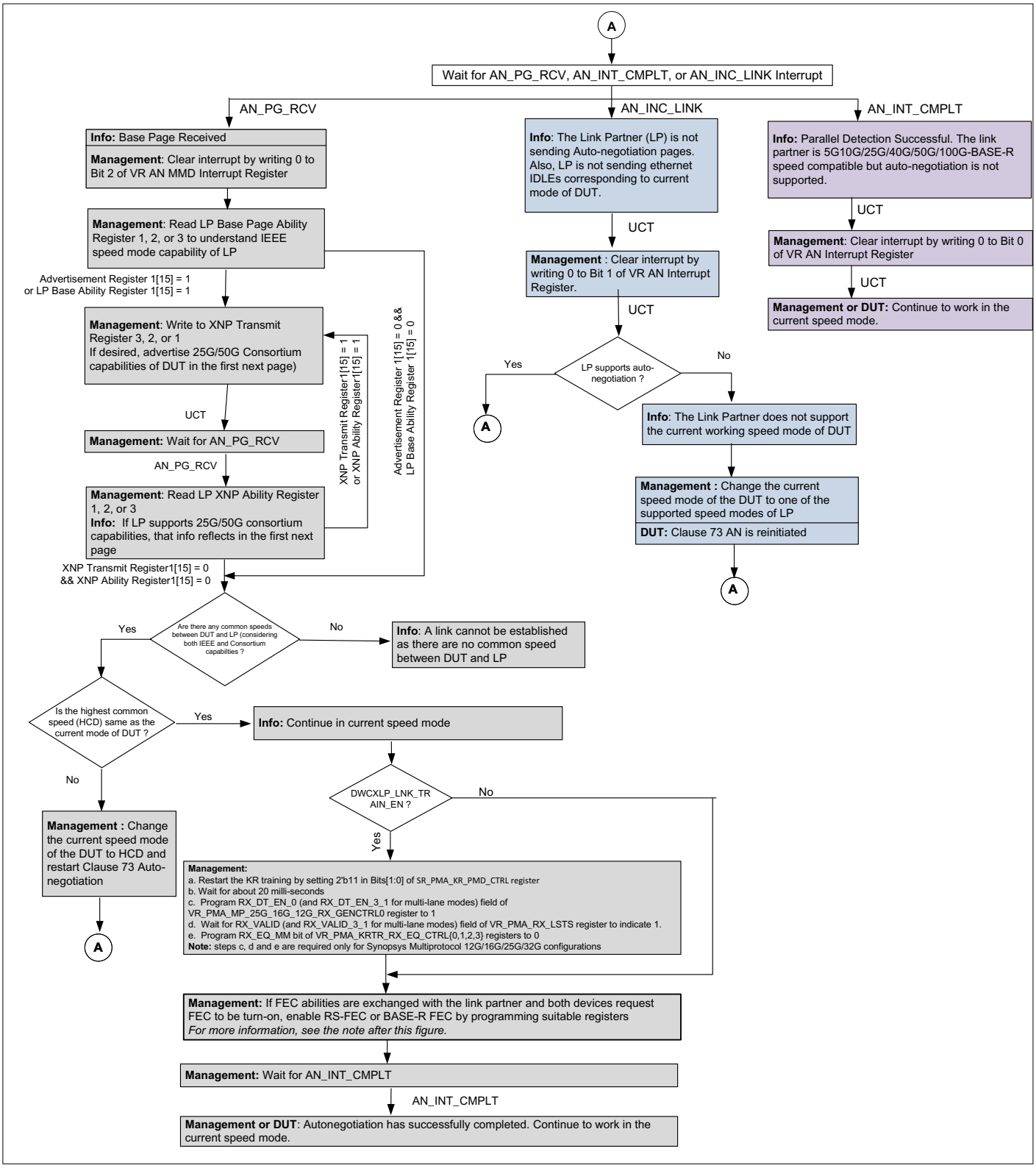
Follow these steps to perform Clause 73 auto-negotiation for configurations that do not have the DWCXLP_CL73_ALT parameter enabled:

1. Disable auto-negotiation by programming the bit[12], AN_EN bit of the `SR_AN_CTRL` register to 0.
2. (Required only for Synopsys Multi-protocol 12G/16G/25G/32G PHY configurations)

   Program the RX_DT_EN_0 and RX_DT_EN_3_1 fields of the `VR_PMA_MP_25G_16G_RX_GENCTRL0` register to 0.
3. (Required only for Synopsys Multi-protocol 12G/16G/25G/32G PHY configurations)

   Program the RX_EQ_MM bit of the `VR_PMA_KRTR_RX_EQ_CTRL0` register to 1.
4. Program 2'b11 to the `SR_PMA_KRT_PMD_CTRL` register to enable Link Training.
5. Enable auto-negotiation by programming the `AN_EN bit of SR_AN_CTRL` register to 1.
6. Follow the instructions in Figure 8-2.

Figure 8-2 Clause 73 Auto-negotiation



Note

- For 10G PHY, if FEC_EN = Enabled and FEC abilities are exchanged with the link partner (bits[47:46] of SR_AN_ADV3 are programmed to 2'b11 and bits[47:46] of SR_AN_LP_ABL3 is programmed to 2'b11), enable the FEC feature by writing 1 to bit[0] of the SR_PMA_KR_FEC_CTRL register. In addition, if FEC_ERROR_FWD = Enabled, enable FEC Error Indicator by writing 1 to bit[1] of the SR_PMA_KR_FEC_CTRL register.
- For 25G PHY, bits[45:44] are used. For more details, see Table 2-21 on page 119.
- If you need to modify the value of the PCS_AM_CNT field to enable RS-FEC mode, a VR_RST is required as per definition of that field. Consequently, another round of auto-negotiation is required to successfully establish the link.

8.11.2.2 Auto-Negotiation Flow for Configurations with DWCXLP_CL73_ALT Enabled

Follow these steps to perform Clause 73 auto-negotiation for configurations with the DWCXLP_CL73_ALT parameter enabled.

Note The DWCXLP_CL73_ALT parameter is automatically enabled for PAM4 configurations.

1. Program the AN_EN bit of the SR_AN_CTRL register to 0 to disable auto-negotiation, if it is already enabled.
2. Program the MULTFACT_10US field of the VR_AN_TIMER_CTRL3 register to an appropriate value based on the frequency of the clk_csr_i signal. The allowed range is between 10 MHz and 400 MHz.
3. Program the AN_EN bit of the SR_AN_CTRL register to 1.
4. Configure the DWC_xlgpcs controller to 10G (NRZ) mode by executing
   a. Steps 1 to 3 in section 8.2.2 (for non-Synopsys PHY)
   b. Steps 1 to 8 in section 8.2.3 (for Enterprise 56G PHY)
5. Configure the PHY to 10GKR (NRZ) mode by executing
   a. For Enterprise 56G PHY, program the following PHY registers:
      i. Program the an0_status_reg bit of the AN_CFG[0] register to 0.
      ii. Program the training_enable_ln0 bit of the BASER_PMD_CONTROL register to 1.
      iii. Program various registers as per Table 8-6 on page 1306 for switching PHY to 10G-KR mode. For fast-simulation mode, additionally program the registers shown in the Table 8-7 on page 1308.
      iv. Program the pmd_en bit of the PMD_CFG[0] register to 1.
   b. For non-Synopsys PHY follow the steps as required.
6. Wait for AN_PG_RCV, AN_INT_CMPLT or AN_INC_LINK interrupt indication in the VR_AN_INTR register.

DWC_xlgpcs asserts sbd_intr_o interrupt, when any bit of the VR_AN_INTR register is 1 and the corresponding bit of the VR_AN_INTR_MSK register is 1.

a. If AN_PG_RCV is received, clear the interrupt bit by writing 0 to it and proceed to step 7.

b. If AN_INT_CMPLT is received, it implies that parallel-detection has occurred successfully in the 10G NRZ mode and the link-partner is 10GBASE-R (and does not support auto-negotiation). Clear the interrupt bit by writing 0 and continue operating DWC_xlgpcs and PHY in 10G mode.

c. If AN_INC_LINK is received, it means that DWC_xlgpcs is neither receiving AN pages nor 10G data from link-partner. Therefore, auto-negotiation fails, and a link cannot be established with the link-partner. Clear interrupt by writing 0.

7. Read SR_AN_LP_ABL1, 2 and 3 and understand the capabilities supported by link-partner.

Note

- If there is no common speed between DUT and link-partner, it implies that a link cannot be established.
- If there is some common speed between DUT and link-partner, software identifies the highest common speed (HCD)

If the link-partner (SR_AN_LP_ABL1 [15] == 1) or the local device (SR_AN_ADV1 [15] == 1) requires to exchange 'Next Pages', proceed to step 8. Otherwise proceed to step 9.

8. Steps to exchange ‘Next Pages’

a. Write to the SR_AN_XNP_TX3, SR_AN_XNP_TX2, and SR_AN_XNP_TX1 registers in sequence.

b. Wait for the AN_PG_RCV interrupt. Then, read the SR_AN_LP_XNP_ABL1, 2 and 3 registers.

If there are more next pages to be exchanged (SR_AN_XNP_TX1 [15] == 1or SR_AN_LP_XNP_ABL1 [15] ==1), proceed to step 8a. Otherwise, proceed to step 9.

9. Wait for the AN arbiter FSM to indicate AN_GOOD_CHECK by polling for bits [3:0] (ANA_FSM) of the VR_AN_DIG_STS register to indicate a value of 4'd8.

10. Configure DWC_xlgpcs and the PHY to the Highest-Common Speed (HCD) identified in Step 7 by executing the steps mentioned in “Configuring DWC_xlgpcs and the PHY to the Highest-Common Speed (HCD)” on page 1338.

11. Wait for the assertion of the AN_INT_CMPLT signal which indicates the successful completion of auto-negotiation.

12. Continue to work on the current speed mode.

If the link goes down, the AN_TXDIS bit of the VR_AN_INTR register is programmed to 1. The sbd_intr_o interrupt pin is asserted, if the corresponding interrupt is enabled in the VR_AN_INTR_MSK register. This

indicates a restart of auto-negotiation. When this event occurs, restart the auto-negotiation by proceeding to step 3.

Note

- When mode-switching is initiated in steps 5 and 10, PHY must de-assert the tx_phy_rdy_i and rx_phy_rdy_i inputs of DWC_xlgpcs. When switching is complete and all clocks are available for stable (at least after 10 stable clocks), assert back the tx_phy_rdy_i and rx_phy_rdy_i. For Synopsys PHY configurations, the control/status signal connections between the DWC_xlgpcs and PHY already ensures this requirement. For non-Synopsys PHY configurations, SoC logic must ensure that this requirement is met.
- After the PHY completes the link-training, it must assert the xlgpcs_sig_ener_det_i signal. Until then, expect xlgpcs_sig_ener_det_i signal to be driven low.

8.11.2.2.1 Configuring DWC_xlgpcs and the PHY to the Highest-Common Speed (HCD)

Configure DWC_xlgpcs and the PHY to the Highest-Common Speed (HCD) by following these steps:

- Non-Synopsys PHY
  a. If the negotiated speed is 40GBASE-KR4 mode, execute the instructions in "Programming Sequence When Using Non-Synopsys PHY/Multi-Protocol 12G/16G/25G PHY" on page 1291.
  b. If the negotiated speed is 25GBASE-KR mode, execute the instructions in "Programming Sequence for Non-Synopsys PHY" on page 1298.
  c. If the negotiated speed is 50GBASE-KR mode, execute the instructions in "Programming Sequence for non-Synopsys PHY" on page 1312.
  d. If the negotiated speed is 100GBASE-KR4 mode, execute the instructions in "Programming Sequence for non-Synopsys PHY" on page 1316.
  e. If the negotiated speed is 100GBASE-KR2/CR2 mode, execute the instructions in "Programming Sequence for non-Synopsys PHY" on page 1319.
  f. If the negotiated speed is 10GBASE-KR, then do not operate a speed switch, as you are already in 10GBASE-KR mode.
  g. If the negotiated speed is 5GBASE-KR, execute the instructions in "Programming Sequence for Non-Synopsys PHY or Multi-Protocol 12G, 16G, or 25G PHY" on page 1325.

  Then, enable Link-training in the PHY by taking suitable steps as required by your PHY.
- Enterprise 56G PHY
  - If the negotiated speed is 40GBASE-KR4 mode:
    - Execute steps 1 to 9a of "Programming Sequence when using Enterprise 56G or 112G PHY" on page 1295.
    - Program training_enable_ln{n}bit of the `BASER_PMD_CONTROL` register to 1.
    - Program restart_training_ln{n} bit to 1'b1.
    - Program the pmd_en field of the `PMD_CFG[0]` register to 1.
  - If the negotiated speed is 25GBASE-KR mode:
    - Execute steps 1 to 10a of "Programming Sequence for Enterprise 56G or 112G PHY" on page 1304.

- Program training_enable_ln{n}bit of the BASER_PMD_CONTROL register to 1.
- Program restart_training_ln{n} bit to 1'b1.
- Program the pmd_en field of the PMD_CFG[0] register to 1.

- If the negotiated speed is 50GBASE-KR mode:
  - Execute steps 1 to 13a of "Programming Sequence for Enterprise 56G or 112G PHY" on page 1313.
  - Program training_enable_ln{n}bit of the BASER_PMD_CONTROL register to 1.
  - Program restart_training_ln{n} bit to 1'b1.
  - Program pmd_en field of the PMD_CFG[0] register to 1.
- If the negotiated speed is 100GBASE-KR4 mode:
  - Execute steps 1 to 11a of "Programming Sequence for Enterprise 56G or 112G PHY" on page 1318.
  - Program training_enable_ln{n}bit of the BASER_PMD_CONTROL register to 1.
  - Program restart_training_ln{n} bit to 1'b1.
  - Program pmd_en field of the PMD_CFG[0] register to 1.
- If the negotiated speed is 100GBASE-KR2 or 100GBASE-CR2 mode:
  - Execute steps 1 to 14a of "Programming Sequence for Enterprise 56G or 112G PHY" on page 1320.
  - Program training_enable_ln{n}bit of the BASER_PMD_CONTROL register to 1.
  - Program restart_training_ln{n} bit to 1'b1.
  - Program pmd_en field of the PMD_CFG[0] register to 1.
- If the negotiated speed is 10GBASE-KR:
  - Execute steps 1 to 9a of section "Programming Sequence When Using Enterprise 56G or 112G PHY" on page 1289.
  - Program training_enable_ln{n}bit of the BASER_PMD_CONTROL register to 1.
  - Program restart_training_ln{n} bit to 1'b1.
  - Program pmd_en field of the PMD_CFG[0] register to 1.

8.12 Programming Guidelines for Energy Efficient Ethernet

To enable the Energy Efficient Ethernet (EEE) feature, the host must complete the following:

1. Check if DWC_xlgpcs supports the EEE feature by reading the SR_PCS_EEE_ABL. Check the bits from the SR_PCS_EEE_ABL register to obtain the EEE abilities under different speed modes.
2. Program various timers used in the EEE mode depending on the clk_eee_i clock frequency:
   - VR MII MMD EEE mode Control register, VR MII MMD EEE Transmit Timer register, VR MII MMD EEE Receive Timer register, VR XS or PCS MMD EEE mode Control register VR_PCS_EEE_MCTRL
   - VR XS or PCS MMD EEE Transmit Timer register VR_PCS_EEE_TXTIMER
   - VR XS or PCS MMD EEE Receive Timer register VR_PCS_EEE_RXTIMER



- The clk_eee_i clock can be independent of Transmit and Receive path clocks. It is free running. The frequency of this clock can be between 30 MHz to 200 MHz.
- For EEE GUAI shut-down mode, VR_PCS_EEE_MISC_TIMER must be written once, even though the default value of Tpq matches the actual value.

3. If FEC is enabled in the KR mode, program bit 5 of the VR XS or PCS MMD EEE mode Control register VR_PCS_EEE_MCTRL to 1 to enable the FEC fast synchronization.
4. Program VR_PCS_DIG_CTRL1(0x38000) bit[13] to enable VSMMD1 registers.
5. Program SR_VSMMD_CTRL(0x1e0009) bit[4] to enable fast_sim.
6. If RS FEC is enabled, program the AM interval period in field [13:0] of the VR PCS MMD Alignment Marker Counter register at address 0x38018.
7. Set bit[2] in the RS FEC control register at address 0x100C8.
8. If FAST_WAKE is needed, program the SR_PCS_EEE_ABL(0x30014) bit[0] to 1'b1. Otherwise, program this bit to 0.
9. Enable the EEE feature on the Transmit path by programming bit 0 of the VR_PCS_EEE_MCTRL register. Similarly, enable the EEE feature on the Receive path by programming bit 1 of the VR_PCS_EEE_MCTRL register.
10. If the PMA service interface is XLAUI or CAUI, program the VR_PMA_DIG_CTRL1(0x18000) register bit[8] to 1'b1. Otherwise, program this bit to 0. If XLAUI or CAUI is permitted to shut down, program SR_PMA_CTRL1 and bit[9] to 1.

    When these bits are programmed, DWC_xlgpcs enables the encoding or decoding of the LPI code groups and also enables the EEE Transmit and Receive controllers.

Note

- Enabling fast_sim in step 4 is optional. It is needed only when simulation time is to be reduced.
- DWC_xlgpcs has recovery time after it comes out of the LPI mode and before it is ready to send the frames. The worst-case recovery time is equal to the following:

`4 Clocks of clk_eee_i period + 3 clocks of clk_xgmii_tx_i, or clk_rpcs_tx_i` (whichever is slower among these three clock)

- The minimum clock period of clk_eee_i can be 30MHz, failing which the behavior of EEE controller is unpredictable.
- To disable EEE on Transmit side, the software must wait for Transmit LPI to enter TX_ACTIVE state by reading the `VR_PCS_DIG_STS` register.

8.13 Programming Guidelines for Test Pattern Mode

Note

- While entering the test pattern mode, ensure that all FEC functions are disabled (Example: BASE-R FEC and RS-FEC functions must be disabled, if they are enabled in the configuration) by programming.
- BER monitoring is disabled by default.
- Link Fault is reported (except in scrambled IDLE mode, where Link up is possible)
- Exercise vendor specific soft reset while entering and exiting Test pattern mode (by programming bit 15 of the `VR_PCS_DIG_CTRL1` register).

8.13.1 Square Wave (PMA/PCS Layer)

1. Configure the DUT into a selected speed mode.
2. If speed mode is 10G/25G, then program bit [3] (TTP_EN) to 1'b1 and bit[1](TP_SEL) to 1'b1 from SR_PCS_TP_CTRL(0x3002A) to enable PCS layer square wave transmit. 3. If speed mode is not 10G mode, program bit [3:0] of SR_PMA_KR_SQUARE_WAVE_CTRL (0x105E6) to enable specific lanes to transmit square wave.
3. Program bits [6:4](NVAL_SEL) from VR_PCS_KR_CTRL (0x38007) to select square wave width. Please refer to VR_PCS_KR_CTRL description for encoding of NVAL_SEL field.

Note

Square wave checker is not defined for the receiver side.

8.13.2 PRBS31/PRBS9: (PMA/PCS layer)

1. Configure the DUT into a selected speed mode.
2. If speed mode is 10G/25G,
   a. Program bit [4] (PRBS31T_EN) to 1'b1 to enable PRBS31 transmit or
   b. Program bit[6] (PRBS9T_EN) to 1'b1 to enable PRBS9 transmit from the `SR_PCS_TP_CTRL`(0x3002A) register.
3. If speed mode is not 10G mode, then perform the following:

a. Program bit [7] for PRBS31 or
b. Program bit[6] for PRBS9 and
c. Program bit [3] to 1'b1 of the SR_PMA_KR_TST_PAT_CTRL (0x105DD) register to enable PRBS transmit in PMA layer.

4. If speed mode is 10G/25G, then
   a. Program the PRBS31R_EN field (bit [5]) to 1'b1 to enable the PRBS31 receiver from the SR_PCS_TP_CTRL(0x3002A) register, or
   b. Program bit[7](PRBS9RXEN) of the VR_PCS_KR_CTRL(0x38007) register to enable the PRBS9 receiver.
5. If speed mode is not 10G mode, then program bit [0] of the SR_PMA_KR_TST_PAT_CTRL (0x105DD) register to enable the test pattern checker in the PMA layer.

Note 5G speed mode can use the test pattern generator of either the PMA or the PCS layer. Therefore, the test pattern checker in the Receive must be from the same layer.

8.13.3 Pseudo-Random (PCS Layer only)

1. Configure the DUT into a selected speed mode.
2. Program bit [3] of the SR_PCS_TP_CTRL(0x3002A) register to 1'b1 to enable the pseudo-random test pattern transmit.
3. Program bit [2] of the SR_PCS_ TP_CTRL(0x3002A) register to 1'b1 to enable the pseudo-random test pattern receiver.
4. Program bit [0] of the SR_PCS_ TP_CTRL(0x3002A) register to select a data pattern for pseudo-random testing. To know the details of the pseudo-random test pattern data supported, refer to the PR_DATA field (bit [3:1]) of the VR_PCS_KR_CTRL register.

8.13.4 Scrambled-IDLE (PCS Layer only)

1. Configure the DUT into a selected speed mode.
2. Program bit [7] and bit [3] of the SR_PCS_TP_CTRL(0x3002A) register to 1'b1 to enable the scrambled-IDLE transmitter.
3. Program bit [2] of the SR_PCS_TP_CTRL(0x3002A) register to 1'b1 to enable the scrambled-IDLE receiver.

Note Because the scrambled-IDLE test pattern mode is not defined, it is not supported for 10G speed mode.

8.14 Programming Guidelines for Automotive Safety Features

8.14.1 Programming Guidelines for ECC Protection for Memories

DWC_xlgpcs enables the ECC protection feature by default. You can disable this feature by programming the ECC_DIS of VR_PCS_SFTY_DISABLE register to 1. When you program the FEC_EN field of the SR_PMA_KR_FEC_CTRL register, or the RSFEC_EN field of the SR_PMA_RS_FEC_CTRL register, ECC protection feature is activated, provided ECC_DIS is 0.

A correctable error interrupt, xlgpcs_sfty_ce_intr_o is generated to indicate to the application that correctable errors are detected. Appropriate status is indicated in the VR_PCS_SFTY_CE_INTR register.

An uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o is generated to indicate to the application that uncorrectable/address errors are detected. Appropriate status is indicated in the VR_PCS_SFTY_UE_INTR1 registers.

Debug information is available in the following registers:

- VR_PCS_RSFECEF_ECC_STS
- VR_PCS_FEC_ECC_STS0
- VR_PCS_FEC_ECC_STS1
- VR_PCS_FEC_ECC_STS2
- VR_PCS_FEC_ECC_STS3
- VR_PCS_FEC_ECC_STS4
- VR_PCS_FEC_ECC_STS5
- VR_PCS_FEC_ECC_STS6
- VR_PCS_FEC_ECC_STS7
- VR_PCS_DSKW_ECC_STS0
- VR_PCS_DSKW_ECC_STS1
- VR_PCS_DSKW_ECC_STS2
- VR_PCS_DSKW_ECC_STS3

To report errors from the BASE-R FEC error forwarding memory, enable the EN_ERR_IND field of the SR_PMA_KR_FEC_CTRL register. Otherwise, errors are not reported.

8.14.1.1 Programming Guidelines for ECC Hardware Error Injection

Program the following fields of the VR_PCS_SFTY_DBG_CTRL register to suitable values to control the error injection mode for 'Error Correction' or 'Error forwarding' memories:

- EI_MODE
- ERR_MODE
- ERR_LOC

Program the VR_PCS_SFTY_ERR_INJ1 register as follows:

1. ECC_SEL fields to select the memory for error injection.
2. ECC_EI bit to 1 to enable error injection in the selected memory.

Note Program the ECC_SEL field and ECC_EI fields in sequence, as two separate APB3 accesses. If you change the ECC_SEL field in the same register write operation in which you program the ECC_EI bit to 1, it is possible that the error is not introduced in the intended memory; instead the error might be introduced in some other memory.

On injecting error, a correctable/uncorrectable error interrupt is asserted. After interrupts are processed, interrupts are cleared by writing 0 to the appropriate bits of the VR_PCS_SFTY_UE_INTR1 and VR_PCS_SFTY_CE_INTR registers.

To report errors from the BASE-R FEC error forwarding memory, the EN_ERR_IND field of the SR_PMA_KR_FEC_CTRL register must be enabled. Otherwise, errors are not reported.

8.14.2 Programming Guidelines for On-chip Data Path Parity Protection

DWC_xlgpcs enables the data path parity protection by default. You can disable this feature by programming the DPP_DIS of VR_PCS_SFTY_DISABLE register to 1.

An uncorrectable error interrupt (xlgpcs_sfty_ue_intr_o) is generated to indicate to the application that a parity mismatch is detected. In addition, the DPP_INT bit of the VR_PCS_SFTY_UE_INTR0 register is programmed to 1.

8.14.2.1 Programming Guidelines for On-Chip Data Path Parity Error Injection

Program the following fields of the VR_PCS_SFTY_DBG_CTRL register to suitable values to control the error injection mode:

- ERR_MODE
- ERR_LOC

Program the DPEI field of the VR_PCS_SFTY_ERR_INJ0 register to inject DPP error. An uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o is generated on injecting errors. In addition, DPP_INT interrupt bit of the VR_PCS_SFTY_UE_INTR0 register is programmed to 1.

To resume normal operation, clear the interrupt bit by writing 0 to the DPP_INT field of the VR_PCS_SFTY_UE_INTR0 register.

8.14.3 Programming Guidelines for FSM Parity Protection

DWC_xlgpcs enables the FSM parity protection by default. You can disable this feature by programming the FP_DIS of VR_PCS_SFTY_DISABLE register to 1.

An uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o is generated when an FSM parity error is detected; that is, FSM does not follow one-hot encoding. Various bits of the VR_PCS_SFTY_UE_INTR0 register indicate the clock domain of the FSM which encountered parity error.

8.14.3.1 Programming Guidelines for FSM Parity Hardware Error Injection

Program the following bits of the VR_PCS_SFTY_ERR_INJ0 register to inject parity error to the FSMs operating in various clock domains:

- TXFPEI
- RXFPEI
- CSRFPEI
- EEEFPEI

An uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o is generated on injecting errors. Various bits of the VR_PCS_SFTY_UE_INTR0 register indicate the clock domain of the FSM which encountered parity error.

To resume normal operation, clear the interrupt bits of the VR_PCS_SFTY_UE_INTR0 register by writing 0.

8.14.4 Programming Guidelines for FSM Timeout Protection

DWC_xlgpcs enables the FSM timeout protection by default. You can disable this feature by programming the FT_DIS field of the VR_PCS_SFTY_DISABLE register to 1.

Program the SFTY_1US_MULT field of the VR_PCS_SFTY_TMR_CTRL register with appropriate number of CSR clock cycles to generate 1 us tic. The default value of this field assumes that clk_csr_i is 100 MHz.

If required, you can modify or scale the timeout values by programming the FSM_TO_SEL field of the VR_PCS_SFTY_TMR_CTRL register.

An uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o i is generated when an FSM timeout error is detected. Various bits of the VR_PCS_SFTY_UE_INTR0 register indicates the clock domain of the FSM which encountered timeout error.

To prevent false timeout interrupts during

- Speed mode switching sequence:

  Program the FT_DIS bit to 1 before the speed-switch sequence. After completing the switching sequence, program the FT_DIS bit to 0.

- Modifying the SFTY_1US_MULT or FSM_TO_SEL fields:

  Program the FT_DIS bit to 1 before changing these fields. Then, program the FT_DIS bit to 0.

8.14.4.1 Programming Guidelines for FSM Timeout Hardware Error Injection

Program the following bits of the VR_PCS_SFTY_ERR_INJ0 register to inject timeout error to the FSMs operating in various clock domains:

- TXFTEI
- RXFTEI
- CSRFTEI
- EEEFTEI

An uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o is generated on injecting errors. Various bits of the VR_PCS_SFTY_UE_INTR0 register indicate the clock domain of the FSM which encountered timeout error. To resume normal operation, clear the interrupt bits of the VR_PCS_SFTY_UE_INTR0 register by writing 0.

8.14.5 Programming Guidelines for CSR Protection

DWC_xlgpcs enables CSR protection by default. You can disable this feature by programming the CSRP_DIS field of the VR_PCS_SFTY_DISABLE register to 1.

An uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o is generated to indicate to the application if a CSR bit has flipped. In addition, the CSR_INT bit of the VR_PCS_SFTY_UE_INTR0 register is programmed to 1.

8.14.5.1 Programming Guidelines for CSR Error Injection

To insert errors in the CSR protection feature, program the CSREI bit of the VR_PCS_SFTY_ERR_INJ0 register to 1.

An uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o is generated on injecting errors. In addition, CSR_INT bit of the VR_PCS_SFTY_UE_INTR0 register is programmed to 1. To resume normal operation, clear the interrupt bits of the VR_PCS_SFTY_UE_INTR0 register by writing 0.

8.14.6 Programming Guidelines for Interface Timeout Protection

DWC_xlgpcs enables interface timeout protection by default. You can disable this feature by programming the IFT_DIS of VR_PCS_SFTY_DISABLE register to 1.

Program the SFTY_1US_MULT field of the VR_PCS_SFTY_TMR_CTRL register with appropriate number of CSR clock cycles to generate 1 us tic. Interface timeout values can be selected by programming the IFT_SEL field of the VR_PCS_SFTY_TMR_CTRL register.

An uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o is generated when an interface timeout error is detected. Various bits of the VR_PCS_SFTY_UE_INTR0 register indicate the interface that encountered a timeout error.

Note To prevent false timeout interrupts while modifying the SFTY_1US_MULT or IFT_SEL fields, program the IFT_DIS to 1 prior to changing these fields. Then, program this bit to 0.

8.14.6.1 Programming Guidelines for Interface Timeout Error Injection

Follow these steps to program the interface timeout error injection:

1. Program the APB3IFEI or PHYIFTEI fields of the VR_PCS_SFTY_ERR_INJ0 register to 1 to inject interface timeout errors.
2. Wait for the assertion of uncorrectable error interrupt, xlgpcs_sfty_ue_intr_o.
3. Read the VR_PCS_SFTY_UE_INTR0 register and verify that the PHY_IFT_INT or APB3_IFT_INT interrupt bit is programmed to 1.
4. To resume normal operation, clear the interrupt bits of the VR_PCS_SFTY_UE_INTR0 register by writing 0.

Note After enabling error-injection bits do not perform register read or write operation until the interrupt is asserted. If you perform register read/write access during this window, it is likely that interrupt is not asserted.

A
Area, Power, and Latency

This appendix specifies the area and power numbers for various features and configurations.

A.1 Area Numbers for Various Features and Configurations

Table A-1 captures the gate-count for the various configurations. Synthesis was performed with industry-standard 7nm library with clock-gating and DFT enabled.

Table A-1 Gate Count for Various Configurations

Configuration	Area (K gates)
Configuration – 25G	
25G without any FEC	72
25G+ BASE-R FEC	94
25G + RS528 FEC	236
25G + BASE-R FEC + RS528 FEC	259
Configuration – 25G-50G	
25G-50G multi-lane without any FEC	265
25G-50G multi-lane + BASE-R FEC	357
25G-50G multi-lane + RS528 FEC	569
25G-50G multi-lane + BASE-R FEC + RS528 FEC	663
25G-50G multi-lane + BASE-R FEC + RS528 FEC + RS544 FEC	872
25G_50G single-lane + RS544 FEC + RS528 FEC	685
25G_50G single-lane + RS544 FEC + RS528 FEC + BASE-R FEC	707
Configuration – 100G	
100G multi-lane without any FEC	490
100G multi-lane + BASE-R FEC	582
100G multi-lane + RS528 FEC	1029
100G multi-lane + RS528 FEC + BASE-R FEC	1122
100G multi-lane + RS544 FEC + RS528 FEC + BASE-R FEC	1510
100G multi-lane + RS544 FEC + RS528 FEC + BASE-R FEC + Int RS-FEC	1234
100G single-lane + RS544 + RS528 FEC	1168
100G single-lane + RS544 + RS528 FEC + Int RS-FEC	891
100G single-lane + RS544 + RS528 FEC + BASE-R FEC	1190
100G single-lane + RS544 + RS528 FEC + BASE-R FEC + Int RS-FEC	914
Configuration – 40G	
40G without any FEC	241

Table A-1 Gate Count for Various Configurations

Configuration	Area (K gates)
40G + BASE-R FEC	330
Miscellaneous Features	
Auto-negotiation	20
Synopsys multi-protocol 12G PHY (four-lane configuration)	10
Synopsys multi-protocol 16G PHY (four-lane configuration)	12
Synopsys multi-protocol 25G PHY (four-lane configuration)	15
Synopsys multi-protocol 32G PHY (four-lane configuration)	10
Link-training per lane (for non-Synopsys PHY configurations)	19
Link-training per lane (for Synopsys PHY configurations)	31
Automotive safety (25G configurations)	72
EEE (25G no-FEC configurations)	7
EEE (25G BASE-R FEC configurations)	15
EEE (25G RS-FEC configurations)	9
EEE (25G BASE-R FEC + RS-FEC configurations)	29
IEEE802.3cx timestamping accuracy	3

Note All area given in NAND2 gates.

A.2 Power Numbers for Various Features and Configurations

Table A-2 lists the power numbers for various speed or FEC modes in different RTL configurations with industry standard 5 nm LVT Libraries with DFT and clock-gating enabled. The power numbers are generated using Prime-Time PX flow.

Table A-2 Power Numbers for Various Speed or FEC Modes in Different Configurations

Speed and FEC Mode	Configuration	Internal Power (W)	Switching Power (W)	Leakage Power (W)	Total Power (W)
10G no FEC mode	25G	3.08e-03	1.79e-04	1.25e-06	3.26e-03
	25G-50G (multi-lane)	8.57e-03	4.21e-04	4.40e-06	9.00e-03
	40G	5.36e-03	3.68e-04	1.65e-06	5.73e-03
	25G-50G (single-lane)	7.42e-03	3.10e-04	3.57e-06	7.74e-03
	100G (single-lane)	1.05e-02	4.18e-04	4.69e-06	1.09e-02
	100G (multi-lane)	1.27e-02	4.81e-04	6.22e-06	1.32e-02
10G BASE-R FEC mode	25G	1.50e-03	7.64e-04	6.55e-04	2.91e-03
	25G-50G (multi-lane)	1.87e-03	1.10e-03	2.36e-03	5.33e-03
	40G	1.42e-03	7.62e-04	8.52e-04	3.03e-03
	25G-50G (single-lane)	1.56e-03	8.39e-04	1.95e-03	4.35e-03
	100G (single-lane)	1.93e-03	1.04e-03	2.40e-03	5.37e-03
	100G (multi-lane)	2.44e-03	1.39e-03	3.17e-03	6.99e-03
25G no FEC mode	25G	1.56e-03	6.65e-04	6.66e-04	2.89e-03
	25G-50G (multi-lane)	3.58e-03	1.78e-03	2.34e-03	7.70e-03
	25G-50G (single-lane)	2.54e-03	1.20e-03	1.91e-03	5.65e-03
	100G (single-lane)	3.19e-03	1.39e-03	2.36e-03	6.94e-03
	100G (multi-lane)	4.88e-03	2.52e-03	3.15e-03	1.05e-02
25G BASE-R FEC mode	25G	1.85e-03	8.64e-04	6.72e-04	3.39e-03
	25G-50G (multi-lane)	3.82e-03	2.01e-03	2.35e-03	8.17e-03
	25G-50G (single-lane)	2.85e-03	1.42e-03	1.92e-03	6.19e-03
	100G (single-lane)	3.63e-03	1.81e-03	2.38e-03	7.82e-03
	100G (multi-lane)	5.03e-03	2.70e-03	3.17e-03	1.09e-02

Table A-2 Power Numbers for Various Speed or FEC Modes in Different Configurations

Speed and FEC Mode	Configuration	Internal Power (W)	Switching Power (W)	Leakage Power (W)	Total Power (W)
25G RS528 FEC mode	25G	2.09e-03	1.10e-03	6.84e-04	3.88e-03
	25G-50G (multi-lane)	4.42e-03	2.62e-03	2.37e-03	9.40e-03
	25G-50G (single-lane)	3.62e-03	2.14e-03	1.94e-03	7.70e-03
	100G (single-lane)	4.78e-03	3.07e-03	2.40e-03	1.03e-02
	100G (multi-lane)	6.14e-03	3.92e-03	3.19e-03	1.33e-02
40G no FEC mode	25G-50G (multi-lane)	3.01e-03	1.62e-03	2.41e-03	7.04e-03
	40G	2.26e-03	1.10e-03	8.88e-04	4.25e-03
	100G (multi-lane)	4.16e-03	2.27e-03	3.24e-03	9.67e-03
40G BASE-R FEC mode	25G-50G (multi-lane)	3.12e-03	1.77e-03	2.40e-03	7.29e-03
	40G	1.67e-03	7.54e-04	7.72e-04	3.19e-03
	100G (multi-lane)	4.14e-03	2.35e-03	3.24e-03	9.73e-03
50G no FEC mode	25G-50G (multi-lane)	4.29e-03	2.11e-03	2.35e-03	8.76e-03
	100G (multi-lane)	5.74e-03	2.86e-03	3.17e-03	1.18e-02
50G BASE-R FEC mode	25G-50G (multi-lane)	4.99e-03	3.02e-03	2.31e-03	1.03e-02
	100G (multi-lane)	7.18e-03	4.75e-03	3.11e-03	1.51e-02
50G RS528 FEC mode	25G-50G (multi-lane)	6.15e-03	3.80e-03	2.38e-03	1.23e-02
	100G (multi-lane)	7.32e-03	4.83e-03	3.22e-03	1.54e-02
50G RS544 FEC mode	25G-50G (multi-lane)	6.60e-03	4.19e-03	2.38e-03	1.32e-02
	25G-50G (single-lane)	5.25e-03	3.54e-03	1.94e-03	1.07e-02
	100G (single-lane)	6.39e-03	4.42e-03	2.42e-03	1.32e-02
	100G (multi-lane)	8.91e-03	5.99e-03	3.24e-03	1.81e-02
100G no FEC mode	100G (multi-lane)	1.04e-02	6.17e-03	3.28e-03	1.99e-02
100G RS528 FEC mode	100G (multi-lane)	1.22e-02	8.41e-03	3.28e-03	2.39e-02
100G RS544 FEC mode	100G (single-lane)	1.13e-02	7.97e-03	2.49e-03	2.17e-02
	100G (multi-lane)	1.51e-02	1.02e-02	3.31e-03	2.86e-02
100G IRSFEC mode	100G (single-lane)	9.88e-03	7.42e-03	2.46e-03	1.98e-02
	100G (multi-lane)	1.30e-02	9.05e-03	3.29e-03	2.54e-02

The RTL simulation waveforms used for generating the SAIF files correspond to the test cases listed in Table A-3.

Table A-3 Test Cases Used for Generating SAIF Files

Speed Mode or FEC Type	Test Case Used
10G no FEC	xlgpcs_txrx_data_frame_TEST_10G_PWR_TEST_QUICK
10G BASE-R FEC 25G no FEC	xlgpcs_fec_txrx_data_frame_TEST_10G_TEST_FEC_PWR_TEST_QUICK
	xxvgpcs_rx_data_frame_TEST_25G_PWR_TEST_QUICK
25G BASE-R FEC	xxvgpcs_fec_txrx_lane0_TEST_25G_TEST_FEC_PWR_TEST_QUICK
25G RS-FEC 40G no FEC	xxvgpcs_cl91_fec_rx_lane0_TEST_25G_PWR_TEST_TEST_RSFEC_QUICK
	xlgpcs_txrx_data_frame_TEST_40G_AM_CNTR_WIDTH_10_PWR_TEST_QUICK
40G BASE-R FEC 50G no FEC	xlgpcs_fec_txrx_data_frame_TEST_40G_TEST_FEC_AM_CNTR_WIDTH_10_PWR_TEST_QUICK
	lgpcs_rx_data_frame_TEST_50G_PWR_TEST_QUICK
50G BASE-R FEC	lgpcs_fec_rx_data_frame_TEST_50G_PWR_TEST_QUICK
50G RS528 FEC	gpcs_cl91_fec_rx_data_frame_TEST_50G_PWR_TEST_TEST_RSFEC_QUICK
50G RS544 FEC 100G no FEC	lgpcs_cl91_fec_rx_data_frame_TEST_50G_PWR_TEST_TEST_RS544_QUICK
	cgpcs_txrx_data_frame_TEST_100G_PWR_TEST_QUICK
100G RS528 FEC	cgpcs_cl91_fec_tx_frame_TEST_100G_PWR_TEST_TEST_RSFEC_QUICK
100G RS544 FEC	cgpcs_cl91_fec_tx_frame_TEST_100G_TEST_RS544_PWR_TEST_QUICK
100G IRSFEC	cgpcs_cl91_fec_tx_frame_TEST_100G_PWR_TEST_TEST_RS544_S100G_RS544_IRSFEC_QUICK

A.3 Latency

Table A-4 lists the latency values in the various speed modes in nanoseconds.

Table A-4 Latency Values in Nanoseconds in Various Speed Modes

Configuration	Speed	FEC Mode	Transmit Nominal (ns)	Receive Nominal (ns) with DWCXLP_CLK_CMP_EN=1	Receive Nominal (ns) with DWCXLP_CLK_CMP_EN=0
100G configuration with 32-bit gearbox interface and DWCXLP_IRSFEC=0	10G	No-FEC	76.8	172.8	115.2
		BASE-R FEC with EF	96	614.4	556.8
		BASE-R FEC without EF	96	409.6	352
	25G	No-FEC	30.72	69.12	46.08
		BASE-R FEC with EF	38.4	245.76	222.72
		BASE-R FEC without EF	38.4	163.84	140.8
		RS528 FEC with EF	56.32	381.44	358.4
		RS528 FEC without EF	56.32	337.92	314.88
	40G	No-FEC	76.8	217.6	204.8
		BASE-R FEC with EF	96	659.2	646.4
		BASE-R FEC without EF	96	454.4	441.6
	50G	No-FEC	30.72	117.76	107.52
		BASE-R FEC with EF	38.4	394.24	384
		BASE-R FEC without EF	38.4	312.32	302.08
		RS528 FEC with EF	56.32	289.28	279.04
		RS528 FEC without EF	56.32	243.04	232.8
		RS544 FEC with EF	54.56	300.08	290.16
		RS544 FEC without EF	54.56	252.96	243.04
	100G	No-FEC	30.72	128	117.76
		RS528 FEC with EF	51.2	230.4	220.16
		RS528 FEC without EF	51.2	186.88	176.64
		RS544 FEC with EF	49.6	233.12	223.2
		RS544 FEC without EF	49.6	188.48	178.56

Table A-4 Latency Values in Nanoseconds in Various Speed Modes (Continued)

Configuration	Speed	FEC Mode	Transmit Nominal (ns)	Receive Nominal (ns) with DWCXLP_CLK_CMP_EN=1	Receive Nominal (ns) with DWCXLP_CLK_CMP_EN=0
100G configuration with 32-bit gearbox interface and DWCXLP_IRSFEC=1	10G	No-FEC	76.8	172.8	115.2
		BASE-R FEC with EF	96	614.4	556.8
		BASE-R FEC without EF	96	409.6	352
	25G	No-FEC	30.72	69.12	46.08
		BASE-R FEC with EF	38.4	245.76	222.72
		BASE-R FEC without EF	38.4	163.84	140.8
		RS528 FEC with EF	56.32	394.24	371.2
		RS528 FEC without EF	56.32	350.72	327.68
	40G	No-FEC	76.8	217.6	204.8
		BASE-R FEC with EF	96	659.2	646.4
		BASE-R FEC without EF	96	454.4	441.6
	50G	No-FEC	30.72	117.76	107.52
		BASE-R FEC with EF	38.4	394.24	384
		BASE-R FEC without EF	38.4	312.32	302.08
		RS528 FEC with EF	56.32	302.08	291.84
		RS528 FEC without EF	56.32	255.84	245.6
		RS544 FEC with EF	54.56	337.28	327.36
		RS544 FEC without EF	54.56	290.16	280.24
	100G	No-FEC	30.72	128	117.76
		RS528 FEC with EF	51.2	243.2	232.96
		RS528 FEC without EF	51.2	199.68	189.44
		Single RS544 FEC with EF	49.6	270.32	260.4
		Single RS544 FEC without EF	49.6	225.68	215.76
		Interleaved RS544 FEC with EF	52.1	374.48	364.56
		Interleaved RS544 FEC without EF	52.1	280.24	270.32

Table A-4 Latency Values in Nanoseconds in Various Speed Modes (Continued)

Configuration	Speed	FEC Mode	Transmit Nominal (ns)	Receive Nominal (ns) with DWCXLP_CLK_CMP_EN=1	Receive Nominal (ns) with DWCXLP_CLK_CMP_EN=0
25G-50G (single-lane) configuration with 32-bit gearbox interface	10G	No-FEC	76.8	172.8	115.2
		BASE-R FEC with EF	96	614.4	556.8
		BASE-R FEC without EF	96	409.6	352
	25G	No-FEC	30.72	69.12	46.08
		BASE-R FEC with EF	38.4	245.76	222.72
		BASE-R FEC without EF	38.4	163.84	140.8
		RS528 FEC with EF	56.32	424.96	401.92
		RS528 FEC without EF	56.32	348.16	325.12
	50G	RS544 FEC with EF	81.84	324.88	314.96
		RS544 FEC without EF	81.84	250.48	240.56
40G configuration with 32-bit gearbox interface	40G	No-FEC	76.8	217.6	204.8
		BASE-R FEC with EF	96	659.2	646.4
		BASE-R FEC without EF	96	454.4	441.6
	10G	No-FEC	76.8	172.8	115.2
		BASE-R FEC with EF	96	614.4	556.8
		BASE-R FEC without EF	96	409.6	352

Note The latencies in Table A-4 are nominal values. Actual latency can vary slightly due to CDC uncertainty, alignment marker insertion or deletion variation, and other.

B
DWC_xlgpcs Automotive Safety

This appendix provides pointers to the Safety Features supported in DWC_xlgpcs and the Automotive Safety Package available. You must have the corresponding license to use these features.

B.1 Overview of Automotive Safety Features

To make an IP better suited for applications in the automotive space, functional safety needs to be considered throughout the IP development process. ISO 26262 addresses functional safety and requires the hardware components to be analyzed with respect to their ability to detect and/or control the effects of random hardware faults. Random hardware faults can occur at any point in time during the life-cycle of the hardware component. They can take the form of permanent faults, or transient faults, and require safety mechanisms to be in place to ensure that severe consequences do not occur as a result of those faults. The DWC_xlgpcs includes the internal safety mechanisms which assist in detecting and handling random hardware faults. These complement the mechanisms already defined in the DWC_xlgpcs protocol.

DWC_xlgpcs is analyzed in the scope of ISO 26262 targeting ASIL-B Random ASIL-D Systematic for the Reference Configuration.

Reference Configuration

As DWC_xlgpcs is a highly configurable IP with several optional features, a reference configuration suited for automotive domain usage is used for the Safety Analysis. This configuration has all the major features and includes all the fault detection and reporting related safety logic in the controller. The details of the reference configuration parameters are provided in the Safety Manual. The internal safety mechanisms monitor the IP logic and report any detected errors to the application. The application should take an action to handle the error conditions and ensure safe operation of the hardware component.

FMEDA

A Failure mode, Effects, and Diagnostic Analysis (FMEDA) has also been performed for this reference configuration. The analysis relies on a per-transistor failure rate derived in accordance to IEC TR 62380. This, in conjunction with area information from design synthesis trials, allows to determine the failure rate for the IP which can then be distributed between the modules of the design according to their relative area.

The list of failure modes and their effects is derived for each module and each of the failure modes is assigned a portion of the failure rate of the module they belong to. If the failure mode has the potential to violate a top level safety requirement (error at a top level output pin) there is a search for a safety mechanism that can prevent that violation, totally or partially. If a safety mechanism exists, for example, ECC protection for RAMs, a diagnostic coverage (DC) value is assigned representing the percentage of the failure rate associated with the specific failure mode that is prevented from violating a top level safety requirement. The process is repeated for all failure modes and finally results are accumulated and IP safety metrics are collected in the FMEDA.

DFMEA

DFMEA document analyses the various failure modes, its effects and root-causes. For each item in the list, the potential failure mode is identified with the corresponding effect of the failure, the severity of the failure (S), the probability of occurrence (O), the detection capability (D), the risk priority number (RPN=S*O*D) and the recommended actions to mitigate the risks of the failure mode.

B.2 Automotive Safety Package Documents

Table B-1 describes the documents that are included in the DWC_xlgpcs AC safety package. These documents are available in the `$DESIGNWARE_HOME/iip_ap/DWC_ac_xlgpcs/latest` directory.

Table B-1 Description of Documents in Automotive Safety AC Package

File Name	Description	Contents
WKPR41_Safety_Manual_DWC_xlgpcs.pdf	Safety Manual	Describes the safety goals, safety mechanisms, assumptions of use and so on. Summarizes the results of various safety analyses.
SG_Quality_Manual.pdf	Quality Manual	Describes the internal Quality Management System and how its specific activities are related to the Synopsys quality process documents that are listed in the Safety Manual.
WKPR22_FMEDA_DWC_xlgpcs.xlsx	FMEDA	Calculates the FIT rates for each module, and for SPFM and LFM. Enumerates diagnostic capabilities for failure modes.
WKPR17_DFMEA_DWC_xlgpcs.xlsx	DFMEA	Documents the DFMEA analysis.
WKPR14_DFA_DWC_xlgpcs.xlsx	DFA	Documents the Dependent Failure Analysis, including common cause failures and cascading failures.
WKPR40_Safety_Case_Report_DWC_xlgpcs.pdf	Safety Case Report	Details the safety life cycle documentation that supports functional safety claims.
Assessment_Report_ASIL-B_Compliant_DWC_xlgpcs.pdf	Assessment Report	Documents the assessment report on the functional safety of the IP.

B.3 Configuring the Automotive Safety Features

When you have a valid automotive safety package license (DWC-AC-ETHER-ENTERPRISE-PCS) you can select the safety features supported by DWC_xlgpcs, from the Automotive Safety Package pane in the Specify Configuration activity dialog box, while configuring the controller in the coreConsultant GUI.

For details of the various automotive safety features supported by DWC_xlgpcs, see "Automotive Safety Features" on page 165.

C
External Memory Requirements

This appendix lists the details of external memories that are required for various configurations.

Table C-1 External Memories Required for Various Configurations

Configuration	External Memory Requirements	
	FEC Memories (instances * depth * width) depth: in locations width: in bits	Deskew Memories (instances * depth * width) depth: in locations width: in bits
DWCXLP_MAIN_MODE=0 (25G) DWCXLP_RSFEC_EN=0 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=0	1 x 32 x 66[1]	N/A
DWCXLP_MAIN_MODE=0 (25G) DWCXLP_RSFEC_EN=0 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	2 x 32 x 66[1]	N/A
DWCXLP_MAIN_MODE=0 (25G) DWCXLP_RSFEC_EN=1 DWCXLP_FEC_EN=0 DWCXLP_FEC_ERROR_FWD=0	1 x 85 x 80[1] 1 x 20 x 257[2]	N/A
DWCXLP_MAIN_MODE=0 (25G) DWCXLP_RSFEC_EN=1 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	1 x 64 x 80[1] 1 x 32 x 80[1] 1 x 20 x 257[2]	N/A
DWCXLP_MAIN_MODE=1 (25G-50G) DWCXLP_SINGLE_LANE=1 DWCXLP_RSFEC_EN=1 DWCXLP_FEC_EN=0 DWCXLP_FEC_ERROR_FWD=0	1 x 64 x 160 1 x 20 x 257	2 x 83 x 66
DWCXLP_MAIN_MODE=1 (25G-50G) DWCXLP_SINGLE_LANE=1 DWCXLP_RSFEC_EN=1 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	2 x 32 x 160 1 x 20 x 257	2 x 83 x 66
DWCXLP_MAIN_MODE=2 (40G) DWCXLP_RSFEC_EN=0 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=0	4 x 32 x 66	4 x 53[3] x 66
DWCXLP_MAIN_MODE=2 (40G) DWCXLP_RSFEC_EN=0 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	8 x 32 x 66	4 x 53[3] x 66

Table C-1 External Memories Required for Various Configurations

Configuration	External Memory Requirements	
DWCXLP_MAIN_MODE=1 (25G-50G) DWCXLP_SINGLE_LANE=0 DWCXLP_RSFEC_EN=0 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=0	4 x 32 x 66	4 x 53 x 66
DWCXLP_MAIN_MODE=1 (25G-50G) DWCXLP_SINGLE_LANE=0 DWCXLP_RSFEC_EN=0 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	8 x 32 x 66	4 x 53 x 66
DWCXLP_MAIN_MODE=1 (25G-50G) DWCXLP_SINGLE_LANE=0 DWCXLP_RSFEC_EN=1 DWCXLP_FEC_EN=0 DWCXLP_FEC_ERROR_FWD=0	1 x 64 x 160 1 x 20 x 257	2 x 64 x 66 2 x 53 x 66
DWCXLP_MAIN_MODE=1 (25G-50G) DWCXLP_SINGLE_LANE=0 DWCXLP_RSFEC_EN=1 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	8 x 32 x 66 1 x 20 x 257	2 x 64 x 66 2 x 53 x 66
DWCXLP_MAIN_MODE=3 (100G) DWCXLP_SINGLE_LANE=0 DWCXLP_IRSFEC=0 DWCXLP_RSFEC_EN=0 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=0	4 x 32 x 66	4 x 95 x 66
DWCXLP_MAIN_MODE=3 (100G) DWCXLP_SINGLE_LANE=0 DWCXLP_IRSFEC=0 DWCXLP_RSFEC_EN=0 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	8 x 32 x 66	4 x 95 x 66
DWCXLP_MAIN_MODE=3 (100G) DWCXLP_SINGLE_LANE=0 DWCXLP_IRSFEC=0 DWCXLP_RSFEC_EN=1 DWCXLP_FEC_EN=0 DWCXLP_FEC_ERROR_FWD=0	1 x 32 x 320 1 x 20 x 257	4 x 95 x 66

Table C-1 External Memories Required for Various Configurations

Configuration	External Memory Requirements	
DWCXLP_MAIN_MODE=3 (100G) DWCXLP_SINGLE_LANE=0 DWCXLP_IRSFEC=0 DWCXLP_RSFEC_EN=1 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	8 x 32 x 66 1 x 20 x 257	4 x 95 x 66
DWCXLP_MAIN_MODE =3 (100G) DWCXLP_SINGLE_LANE=0 DWCXLP_IRSFEC=1 DWCXP_RSFEC_EN=1 DWCXLP_FEC_EN=0 DWCXLP_FEC_ERROR_FWD=0	1 x 67 x 320 1 x 40 x 257	4 x 95 x 66
DWCXLP_MAIN_MODE=3 (100G) DWCXLP_SINGLE_LANE=0 DWCXLP_IRSFEC=1 DWCXP_RSFEC_EN=1 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	8 x 67 x 66 1 x 40 x 257	4 x 95 x 66
DWCXLP_MAIN_MODE=3 (100G) DWCXLP_SINGLE_LANE=0 DWCXLP_IRSFEC=1 DWCXP_RSFEC_EN=1 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	1 x 67 x 320 1 x 40 x 257	4 x 95 x 66
DWCXLP_MAIN_MODE=3 (100G) DWCXLP_SINGLE_LANE=1 DWCXLP_IRSFEC=1 DWCXP_RSFEC_EN=1 DWCXLP_FEC_EN=1 DWCXLP_FEC_ERROR_FWD=1	2 x 67 x 160 1 x 40 x 257	4 x 95 x 66

1. When automotive safety feature is additionally enabled in the configuration, the width of this memory shall be 8 more than the specified value (to accommodate the ECC bits).
2. When automotive safety feature is additionally enabled in the configuration, the width of this memory shall be 10 more than the specified value (to accommodate the ECC bits).
3. When EEE feature is additionally enabled in the configuration, the depth shall be 5 more than the value specified.

D
Internal Parameter Descriptions

Provides a description of the internal parameters that might be indirectly referenced in expressions in the Signals, Parameters, or Registers chapters. These parameters are not visible in the coreConsultant GUI and most of them are derived automatically from visible parameters. **You must not set any of these parameters directly.**

Some expressions might refer to TCL functions or procedures (sometimes identified as **function_of**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the core in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

Table D-1 Internal Parameters

Parameter Name	Equals To
ADPT_MD_W	DWCXLP_E32 ? 3 : 2
DIV_MUL_W	(DWCXLP_E12||DWCXLP_E16) ? 7 : 8
DWCXLP_100G_1LANE	(DWCXLP_MAIN_MODE == 3)&&DWCXLP_SINGLE_LANE
DWCXLP_25G_ONLY	DWCXLP_MAIN_MODE == 0
DWCXLP_32BIT_INTF	DWCXLP_XLGMII_INTF_WIDTH == 2
DWCXLP_50G_25G	(DWCXLP_MAIN_MODE == 1)&&DWCXLP_SINGLE_LANE
DWCXLP_ADD_100G_1LANE	DWCXLP_KR4_100G&&DWCXLP_PAM4
DWCXLP_ADD_5G	1
DWCXLP_ADDRS_WIDTH	DWCXLP_APB3_32 ? (DWCXLP_INDIRECT_ACCESS ? 10: 23) : DWCXLP_APB3_16 ? (DWCXLP_INDIRECT_ACCESS ? 9: 22) : (DWCXLP_INDIRECT_ACCESS ? 8: 21)
DWCXLP_APB3_16	DWCXLP_CSR_INTERFACE == 3
DWCXLP_APB3_32	DWCXLP_CSR_INTERFACE == 4

Table D-1 Internal Parameters (Continued)

Parameter Name	Equals To
DWCXLP_APB3_INTF	DWCXLP_CSR_INTERFACE == 2
DWCXLP_APB_DATA_W	DWCXLP_APB3_32 ? 32 : 16
DWCXLP_CL72_EN	DWCXLP_LNK_TRAIN_EN
DWCXLP_DSKW_BF_AW	(DWCXLP_KR4_100G||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 7 : 6
DWCXLP_E112	DWCXLP_PHY_TYPE ==6
DWCXLP_E12	DWCXLP_PHY_TYPE ==4
DWCXLP_E16	DWCXLP_PHY_TYPE ==2
DWCXLP_E25	DWCXLP_PHY_TYPE ==1
DWCXLP_E32	DWCXLP_PHY_TYPE ==3
DWCXLP_E56	DWCXLP_PHY_TYPE ==5
DWCXLP_EC_MEM_AW	DWCXLP_25G_ONLY ? (DWCXLP_RSFEC_EN ? 6 : (DWCXLP_FEC_EN ? 5 : 0)) : DWCXLP_IRSFEC ? 7 : DWCXLP_50G_25G ? 5 :(DWCXLP_FEC_EN ? ((!DWCXLP_RSFEC_EN||DWCXLP_FEC_ERROR_FWD) ? 5 : (DWCXLP_RSFEC_EN&&!DWCXLP_FEC_ERROR_FWD) ? 5 : 0) : (DWCXLP_RSFEC_EN ? 6 : 0))
DWCXLP_EC_MEM_DW	DWCXLP_25G_ONLY ? (DWCXLP_RSFEC_EN ? 80 : (DWCXLP_FEC_EN ? 66 : 0)) : (DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 160 : (DWCXLP_FEC_EN ? ((!DWCXLP_RSFEC_EN||DWCXLP_FEC_ERROR_FWD) ? 66 : (DWCXLP_RSFEC_EN&&!DWCXLP_FEC_ERROR_FWD) ? 80 : 0) : (DWCXLP_RSFEC_EN ? 160 : 0))
DWCXLP_EF_MEM_AW	DWCXLP_IRSFEC ? 7 : (DWCXLP_FEC_EN&&DWCXLP_FEC_ERROR_FWD)? 5 : 0
DWCXLP_EF_MEM_DW	(DWCXLP_FEC_EN&&DWCXLP_FEC_ERROR_FWD)? ((DWCXLP_25G_ONLY&&DWCXLP_RSFEC_EN) ? 80 : (DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 160 : 66) : 0
DWCXLP_IEEE_AN_DEV_MODEL_NO_0_5	[function_of: DWCXLP_IEEE_AN_DEV_MODEL_NO 0 5]
DWCXLP_IEEE_AN_DEV_OUI_19_24	[function_of: DWCXLP_IEEE_AN_DEV_OUI 19 24]
DWCXLP_IEEE_AN_DEV_OUI_3_18	[function_of: DWCXLP_IEEE_AN_DEV_OUI 3 18]

Table D-1 Internal Parameters (Continued)

Parameter Name	Equals To
DWCXLP_IEEE_AN_DEV_REV_NO_0_3	[function_of: DWCXLP_IEEE_AN_DEV_REV_NO 0 3]
DWCXLP_IEEE_DEV_MODEL_NO_0_5	[function_of: DWCXLP_IEEE_DEV_MODEL_NO 0 5]
DWCXLP_IEEE_DEV_OUI_19_24	[function_of: DWCXLP_IEEE_DEV_OUI 19 24]
DWCXLP_IEEE_DEV_OUI_3_18	[function_of: DWCXLP_IEEE_DEV_OUI 3 18]
DWCXLP_IEEE_DEV_REV_NO_0_3	[function_of: DWCXLP_IEEE_DEV_REV_NO 0 3]
DWCXLP_IEEE_PKG_MODEL_NO_0_5	[function_of: DWCXLP_IEEE_PKG_MODEL_NO 0 5]
DWCXLP_IEEE_PKG_OUI_19_24	[function_of: DWCXLP_IEEE_PKG_OUI 19 24]
DWCXLP_IEEE_PKG_OUI_3_18	[function_of: DWCXLP_IEEE_PKG_OUI 3 18]
DWCXLP_IEEE_PKG_REV_NO_0_3	[function_of: DWCXLP_IEEE_PKG_REV_NO 0 3]
DWCXLP_IEEE_PMA_DEV_MODEL_NO_0_5	[function_of: DWCXLP_IEEE_PMA_DEV_MODEL_NO 0 5]
DWCXLP_IEEE_PMA_DEV_OUI_19_24	[function_of: DWCXLP_IEEE_PMA_DEV_OUI 19 24]
DWCXLP_IEEE_PMA_DEV_OUI_3_18	[function_of: DWCXLP_IEEE_PMA_DEV_OUI 3 18]
DWCXLP_IEEE_PMA_DEV_REV_NO_0_3	[function_of: DWCXLP_IEEE_PMA_DEV_REV_NO 0 3]
DWCXLP_IEEE_VS_DEV_MODEL_NO_0_5	[function_of: DWCXLP_IEEE_VS_DEV_MODEL_NO 0 5]
DWCXLP_IEEE_VS_DEV_OUI_19_24	[function_of: DWCXLP_IEEE_VS_DEV_OUI 19 24]
DWCXLP_IEEE_VS_DEV_OUI_3_18	[function_of: DWCXLP_IEEE_VS_DEV_OUI 3 18]
DWCXLP_IEEE_VS_DEV_REV_NO_0_3	[function_of: DWCXLP_IEEE_VS_DEV_REV_NO 0 3]
DWCXLP_KR4_100G	(DWCXLP_MAIN_MODE == 3)&&!DWCXLP_SINGLE_LANE
DWCXLP_KR4_50G	(DWCXLP_MAIN_MODE == 1||DWCXLP_MAIN_MODE==3)&&!DWCXLP_SINGLE_LANE
DWCXLP_KR4_KR	(DWCXLP_MAIN_MODE == 2)
DWCXLP_LOOPBACK_EN	1
DWCXLP_MDIO_INTF	DWCXLP_CSR_INTERFACE == 0
DWCXLP_MULTI_LANE	((DWCXLP_MAIN_MODE == 1)||(DWCXLP_MAIN_MODE == 2)||(DWCXLP_MAIN_MODE == 3))&&!DWCXLP_SINGLE_LANE
DWCXLP_NUM_LANES	(DWCXLP_25G_ONLY||DWCXLP_50G_25G||DWCXLP_100G_1LANE) ? 1:4

Table D-1 Internal Parameters (Continued)

Parameter Name	Equals To
DWCXLP_PHY_DW	(DWCXLP_R_SRDS_INTF==0) ? 16 : (DWCXLP_R_SRDS_INTF==1) ? 32 : (DWCXLP_R_SRDS_INTF==2) ? 33 : (DWCXLP_R_SRDS_INTF==3) ? 66 : (DWCXLP_R_SRDS_INTF==4) ? 20 : (DWCXLP_R_SRDS_INTF==5) ? 64 : (DWCXLP_R_SRDS_INTF==7) ? 40 : 80
DWCXLP_PMA_MMD	1
DWCXLP_REF_CLK_312	DWCXLP_REF_CLK_FREQ==2
DWCXLP_REF_CLK_78	DWCXLP_REF_CLK_FREQ==1
DWCXLP_RS_EC_MEM_AW	DWCXLP_RSFEC_EN ? ((DWCXLP_KR4_100G||DWCXLP_100G_1LANE)&&!DWCXLP_IRSFEC ? 5 : ((DWCXLP_KR4_50G||DWCXLP_50G_25G)&&!DWCXLP_IRSFEC ? 6 : 7)) : 0
DWCXLP_RS_EC_MEM_DW	DWCXLP_RSFEC_EN ? ((DWCXLP_KR4_100G||DWCXLP_100G_1LANE) ? 320 : ((DWCXLP_KR4_50G||DWCXLP_50G_25G) ? 160 : 80)) : 0
DWCXLP_RSFEC_ERFW_EN	DWCXLP_RSFEC_EN||DWCXLP_RS544
DWCXLP_RSFR_EFR_MEM_AW	DWCXLP_IRSFEC ? 6 : DWCXLP_RSFEC_EN ? 5 : 0
DWCXLP_RSFR_EFR_MEM_DW	257
DWCXLP_SNPS_XAUI_PHY	(DWCXLP_PHY_TYPE !=0)&&(DWCXLP_PHY_TYPE!=5)&&(DWCXLP_PHY_TYPE!=6)
DWCXLP_XLGMII_64BIT	DWCXLP_XLGMII_INTF_WIDTH == 1
ECC_DSKW_MEM_CHKBW0	8
ECC_DSKW_MEM_CHKBW1	8
ECC_DSKW_MEM_CHKBW2	8
ECC_DSKW_MEM_CHKBW3	8
ECC_RPCS_EC_CHKBW0	((DWCXLP_EC_MEM_DW>247)?10:((DWCXLP_EC_MEM_DW>120)?9:8))
ECC_RPCS_EC_CHKBW1	8
ECC_RPCS_EC_CHKBW2	8
ECC_RPCS_EC_CHKBW3	8

Table D-1 Internal Parameters (Continued)

Parameter Name	Equals To
ECC_RPCS_EF_CHKBW0	((DWCXLP_EF_MEM_DW>247)?10:((DWCXLP_EF_MEM_DW>120)?9:8))
ECC_RPCS_EF_CHKBW1	8
ECC_RPCS_EF_CHKBW2	8
ECC_RPCS_EF_CHKBW3	8
ECC_RSFR_EC_CHKBW	((DWCXLP_RS_EC_MEM_DW>247)?10:((DWCXLP_RS_EC_MEM_DW>120)?9:8))
ECC_RSFR_EF_CHKBW	10
FR_W	(DWCXLP_E12) ? 9 : 11
MPLL_BW_W	!(DWCXLP_E12) ? 16 : 11
MPLL_CLK_DIV_W	DWCXLP_E32 ? 4 : DWCXLP_E25 ? 3 : 2
MPLL_MULT_W	(DWCXLP_E16||DWCXLP_E12) ? 8 : 12
MPLL_TH_W	DWCXLP_E32 ? 8 : 2
POL_W	!(DWCXLP_E12) ? 2 : 3
RF_LD_W	(DWCXLP_E12) ? 6 : 7
RX_RATE_W	(DWCXLP_E16||DWCXLP_E12) ? 2 : 3
RX_TAP_W	DWCXLP_E32 ? 9 : 8
RX_WIDTH_W	(DWCXLP_E16||DWCXLP_E12) ? 2 : 3
SNPS_RSVDPARAM_1	0
SNPS_RSVDPARAM_10	0
SNPS_RSVDPARAM_11	0
SNPS_RSVDPARAM_12	0
SNPS_RSVDPARAM_13	0
SNPS_RSVDPARAM_15	0
SNPS_RSVDPARAM_16	DWCXLP_ASP
SNPS_RSVDPARAM_17	DWCXLP_ASP
SNPS_RSVDPARAM_18	0
SNPS_RSVDPARAM_19	DWCXLP_ASP
SNPS_RSVDPARAM_20	DWCXLP_ASP

Table D-1 Internal Parameters (Continued)

Parameter Name	Equals To
SNPS_RSVDPARAM_3	(DWCXLP_ASP&&(DWCXLP_RSFEC_EN||DWCXLP_FEC_EN))||DWCXLP_ADD_ECC_PROT
SNPS_RSVDPARAM_5	0
SR_PMA_KR_PRBS_RX_ERR_CNTR_LANE0	[function_of: 0x6A4]
SR_PMA_KR_PRBS_TX_ERR_CNTR_LANE0	[function_of: 0x640]
SSC_W	(DWCXLP_E12) ? 3 : 1
TX_WIDTH_W	(DWCXLP_E16||DWCXLP_E12) ? 2 : 3
VGA_W	(DWCXLP_E25||DWCXLP_E16) ? 3 : 4

Glossary

Term	Meaning
||A||	Indicates |A| control character in all lanes
||D||	Indicates |D| control character in all lanes
||I||	Indicates |I| control character in all lanes
||K||	Indicates |K| control character in all lanes
||R||	Indicates |R| control character in all lanes
|A|	Control character to indicate PCS Align pattern
|D|	Control character to indicate one of PCS Data pattern
|E|	Control character to indicate PCS Error pattern
|I|	Control character to indicate PCS Idle pattern
|K|	Control character to indicate PCS Comma pattern
|Q|	Control character to indicate PCS Sequence pattern
|R|	Control character to indicate PCS Skip pattern
|S|	Control character to indicate PCS SOP pattern
|T|	Control character to indicate PCS Terminate pattern
10GbE	10-Gigabit Ethernet
AN	Auto-negotiation
ATPG	Automatic Test Pattern Generator
BERT	Bit Error Ratio Test set
BFM	Bus Functional Model
Byte	8 bits
CJTP	Continuous Jitter Test Pattern supported in 10G mode of operation
CR	Control Register

Term	Meaning
CRTP	Continuous Random Test Pattern supported in 10G mode of operation
CSMA/CD	Carrier-Sensing, Multiple-Access with Collision Detection
CSR	Control and Status Register
DFT	Design for Test
DTE	Data Terminal Equipment
DTE XGXS	XGXS Device Interfacing with DTE
DUT	Device Under Test
DWORD	64 bits
EEE	Energy Efficient Ethernet
EMI	Electro-magnetic Interference
FEC	Forward Error Correction
Halfword	16 bits
HFTP	High Frequency Test Pattern
IFG	Inter Frame Gap
LFTP	Low Frequency Test Pattern
LPI	Low Power Idle
LWORD	128 bits
MAC	Media Access Control. Corresponds to OSI Layer 2
MCI	Micro Controller Interface
MDC	MDIO Interface Clock
MDIO	Management Data Input/Output Interface
MII	Media Independent Interface
MFTP	Mixed Frequency Test Pattern
MMD	MDIO Manageable Device
MV	Manchester Violation
MVD	Manchester Violation Delimiter
OUI	Organizationally Unique Identifier
PCS	Physical Coding Sublayer. A type of PHY
PHY	Physical layer of an 802.3-compliant device

Term	Meaning
PMA	Physical Medium Attachment Sublayer
PMD	Physical Medium Dependent Sublayer, a type of PHY
Port ID	Port ID (PORT_ID) of an 802.3-compliant device
PRBS	Pseudo Random Binary Sequence
RS	Reconciliation Sublayer
RXAUI	Reduced XAUI
SerDes	Serializer-Deserializer
SGMII	Serial Gigabit Media Independent Interface
SMA	Station Management Agent
SOF	Start of Frame
Word	32 bits
XAUI	10-Gigabit Attachment Unit Interface. An interface extender for the XGMII
XGMAC	10-Gigabit Media Access Controller
XGMII	10-Gigabit Media-Independent Interface
XGXS	XGMII-Extended Sublayer